Core Algebra

— ***EYNTKA*** **Series** —

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Part I

Groups

The concept of a group is the culmination of multiple branches of mathematics converging on a unifying object that appears in many domains. One of the first mathematical settings in which the ideas leading to the modern notion of a group appeared was the study of polynomials and their solutions. A young[1] mathematician by the name of Évariste Galois contributed a decisive insight to the question of whether the roots of a polynomial could always be expressed by a formula analogous to the quadratic equation using the coefficients, elementary arithmetic operations of $+, \times, -, \div$, and radicals (i.e. $\sqrt[n]{-}$). This is the problem of *solvability of degree n-polynomials*[2]. The degree 3 (cubic) and degree 4 (quartic) cases had been solved in the affirmative roughly 300 years earlier[3]. Abel proved that there is no formula by radicals for the general degree 5 (quintic) polynomial. Galois developed a criterion that determines whether a particular polynomial is solvable by radicals. In particular, there are quintic polynomials $ax^5 + bx^4 + cx^3 + dx^2 + ex + f$ with rational coefficients whose roots cannot be expressed in terms of their coefficients using $+, \times, -, \div$ and radicals. We shall prove this in section 20.2.4.

Around the same time, mathematicians started to discover that there are "more geometries" than just the usual flat geometry of Euclidean space. It was shown that one of Euclid's foundational axioms, the 5th postulate (or parallel postulate), could be replaced by alternative assumptions without producing a contradiction, assuming the consistency of the other axioms. Near the beginning of these developments, a mathematician by the name of Felix Klein saw that the notion of a group is intimately connected with understanding these new geometries, and so he developed the *Erlangen Program*, whose goal was to describe geometries in terms of the transformation groups that preserve their relevant properties[4]. Around the same time, a Norwegian mathematician named Sophus Lie (pronounced "Lee") developed the theory of *Lie groups* which combines group structures and smooth shapes (smooth manifolds) in hopes of extending the techniques of Galois's theory to the study of differential equations [5]. Overall, groups became a major tool in geometry.

While all of this was happening, group-like structures were already appearing in *number theory*. While Gauss was studying binary quadratic forms (expression of the form $ax^2 + bxy + cy^2$), he found that suitable classes of forms with the same "discriminant"[6] can be combined by a rule now called *Gauss composition*: under this rule, the classes form a finite abelian group called the *form class group*. This is a classical example of a *class group*[7]. In another area of number theory, Dirichlet proved that every arithmetic progression $a, a + m, a + 2m, \ldots$, where a and m have no common factor greater than 1, contains infinitely many primes. His proof used characters: group homomorphisms on the invertible remainders modulo m that turn multiplication of remainders into multiplication of complex numbers. We shall cover these results (in modern algebraic language) in [11, Relating Algebra to Number Theory] and [12, Dirichlet's Theorem on Primes in Arithmetic Progressions].

[1] Évariste Galois was shot in a duel on 30 May 1832 and died the following day. The familiar story that he wrote down Galois Theory the night before the duel is an exaggeration: much of his mathematical work predated it.

[2] Here "solvability" means "solvability by radicals": whether the roots can be expressed using arithmetic operations and radicals. It does not mean whether roots exist at all. In modern terms, over a field of characteristic zero, a polynomial is solvable by radicals precisely when its Galois group is a solvable group.

[3] See this link for the cubic formula and this link for the quartic.

[4] The Erlangen Program was later generalized in *Cartan geometry*, which uses Cartan connections on principal bundles to describe geometries beyond the original program's framework. The broader idea of understanding geometry through symmetries also became important in physics; Noether's theorem, for example, relates continuous symmetries of physical systems to conservation laws.

[5] This material belongs in a course on Lie groups, see *Lie Groups* [57].

[6] see definition 19.6.27

[7] More precisely, for a fundamental discriminant D, the form class group is isomorphic to the narrow ideal class group of the quadratic field $\mathbb{Q}(\sqrt{D})$, see [11, Ideal Class Group]. When $D < 0$, the narrow and ordinary ideal class groups agree.

In these early days, different branches of mathematics were developing group-like ideas in slightly different ways. The modern abstract definition emerged gradually through the work of mathematicians such as Camille Jordan, Arthur Cayley, and Walther von Dyck[8]. The formalization of the definition of a group became a huge stepping stone, allowing many disciplines to recognize its usefulness in their domains. Today, groups are a major, if not central, object in almost every branch of mathematics, and they have also been extensively used in physics, chemistry, cryptography, computer science, and many more fields. A central reason groups are fundamental in so many fields is that they can encode the *symmetries* of a system, a notion that comes up naturally a little bit everywhere.

A modern mathematician would now consider groups to be a rather universal and "natural" object. However, groups came late in the development of mathematics, partly due to the mathematical groundwork that needed to happen and partly due to their abstract nature, which can make them seem unnatural to a first-time viewer. A consequence of this is that teaching groups introduces a new level of abstraction that many students have not experienced. The difficulties are further compounded by the fact that visualizations of many groups are scarce or impractical[9]. This, however, is not a bad thing: a whole new way of learning and "experiencing" mathematical objects is the reward for internalizing groups and all the further objects built from them.

In this Part, we shall cover in detail all the fundamental concepts of groups. Most of these concepts shall be instrumental in the rest of the textbook, and the reader shall find that this part of the book is the nexus for most of the algebraic tools that we shall cover in the rest of the book.

For Graduate Student The interpretation of a group representing symmetry is in fact very natural: all groups can be represented as a category with one objects whose morphisms are isomorphisms. In particular, any group G is associated to the category BG. From this perspective, it is not that it is *an* interpretation, but *the* interpretation.

The symmetry is "realized" via a functor to another category. For example, a group action on a set X is equivalent to a covariant functor $BG \to \mathbf{Set}$ that sends the single object of BG to X. A representation is a functor $BG \to \mathbf{Vect}_k$ that sends the single object to a vector space. More generally, a G-action on an object $c \in \mathbf{C}$ is a functor $BG \to \mathbf{C}$ that sends the single object to c[a].

[a] Covariant functors correspond to left group actions, and contravariant functors correspond to right group actions.

There is another way that I (the author of this book) sees groups. They feel like naturally "second-order" objects, in the loose sense that they often encode transformations or invariants associated with an object we first want to study. If we are trying to study some object (numbers, shapes, graphs, etc.), we will want to understand some property of the object. This property can be local (like functions defined on a neighborhood) or global (e.x., homotopy classes of loops based at a point, or symmetries of the whole object). These questions often lead us to consider transformations or equivalences of the object. Groups arise naturally when those transformations are invertible and compose: the automorphisms of an object are its invertible structure-preserving maps to itself, and they form a group under composition[10]. Thus, groups are one way of capturing information about

[8] See [39, 7, 29] for these original works.

[9] We shall see that there are ways of representing any group as a group of symmetries, but sometimes the visual picture is less helpful; for example, some groups are most naturally studied through their actions on very high-dimensional spaces (i.e. 100'000+ dimensional).

[10] In an ordinary category, an automorphism is an isomorphism from an object to itself. In higher categories, analogous notions of equivalence can involve invertibility up to higher morphisms.

an object. There can be other kinds of information about an object (e.g., curvature or twisting) that use additional structures and are not captured by a group alone. Hence, groups are just a part of the puzzle of looking for information about an object.

1

Introduction to Groups

In this chapter, you will cover the basic properties of groups. In the process, we will prove statements that the reader perhaps thought were axiomatic. For example, if $ab = ac$, does that mean $b = c$? We will classify two very important sub-categories: groups with the axiom of *commutativity* ($ab = ba$), and those without it. These two categories of groups will set the narrative for how we study groups, meaning this one axiom plays an incredibly central role in the study of groups! We will cover many examples of groups to get a taste on what groups are, and define the groups which will be referenced indefinitely throughout the rest of the book. We will then cover how to compare two groups, and define the notion of equality between two groups. We will then see how we can use groups to "represent" a system of objects. For example, we will use a group to represent all possible ways of rotating an object.

1.1 Basic Properties and Definitions

As mentioned in [24], algebraic structures are sets along with a function (or functions) which must follow certain rules. The most common type of function used for an algebraic structure is called a *binary operation*:

Definition 1.1.1: Binary Operation

1. A *binary operation* $*$ on a set G is a function $* : G \times G \to G$. For any $a, b \in G$, we shall write $a * b$ or $*(a, b)$. Often the notation is abbreviated to ab.
2. A binary operation $*$ on a set G is *associative* if for every $a, b, c \in G$ we have $a * (b * c) = (a * b) * c$
3. If $*$ is a binary operation on a set G we say elements a and b *commute* if $a * b = b * a$. We say $*$ (or G) is *commutative* if for every $a, b \in G$, $a * b = b * a$

The star of the show in group theory is the binary operation. This function is the key mathematical structure added to a set which we study in group theory. Different group structures on the same set are different binary operations.

Example 1.1.2: Binary Operations

1. Arithmetic operations are a good source of binary operations. As we have discussed in the preliminary chapter, $+$ and $\times$ are functions that take in two numbers as input, and output one number. If you want to practice proving something is a function, you can try showing that $+ : \mathbb{Z} \times \mathbb{Z} \to \mathbb{Z}$ is a function (and since the domain is the cartesian product of two copies of the codomain, it's a binary operation). Similarly, you can do the same with $- : \mathbb{Z} \times \mathbb{Z} \to \mathbb{Z}$.
2. Is $-$ a binary operation with $\mathbb{N}$ (i.e. is $- : \mathbb{N} \times \mathbb{N} \to \mathbb{N}$ a binary operation)? What about $\times$ and $\div$? Which of these binary operations are associative? Are they commutative? What if you switch $\mathbb{N}$ with $\mathbb{Z}$, $\mathbb{Q}$, or $\mathbb{R}$?
3. When working with sets, it is common to take the union or intersection of sets. These are binary operations. If A is a set and $P(A)$ is its powerset, then $\cup : P(A) \times P(A) \to P(A)$ is a binary operation. Similarly for $\cap$
4. The greatest common divisor and the least common multiple of two nonzero integers ([24, Greatest Common Divisor], [24, Least Common Multiple]) are again nonzero integers, so gcd and lcm are binary operations on $\mathbb{Z} \setminus \{0\}$. Is either of them a binary operation on $\mathbb{N}$?
5. If you have taken some multi-variable calculus, then the cross product $\times : \mathbb{R}^3 \times \mathbb{R}^3 \to \mathbb{R}^3$ is an example of a binary operation.
6. Take $E : \mathbb{N} \times \mathbb{N} \to \mathbb{N}$ to be $E(x, y) = x^y$. Check that $E(x, E(y, z))$ is not necessarily equal to $E(E(x, y), z)$ (that is $x^{(y^z)}$ is not necessarily equal to $(x^y)^z$). Hence, exponentiation is a non-assotiative operation.
7. Given two functions f and g over the same domain and codomain ($f : X \to X$ and $g : X \to X$), their composition. Thus composition of functions with the same domain and codomain forms a binary operation.

With these intuitions, we can move on to defining a group:

Definition 1.1.3: Group

A *group* is an ordered pair $(G, *)$, where G is a set and $*$ is a binary operation on G satisfying the following axioms:

1. associativity: $(a * b) * c = a * (b * c)$
2. there exists an element e in G, called the *identity* of G, such that for every $a \in G$, we have $e * a = a$
3. For each $a \in G$ there is an element $b \in G$ called an *inverse* of a, such that $ab = ba = e$. We denote this element b by a^{-1}, so that we can equivalently write $a^{-1} * a = e$

The group $(G, *)$ is called *abelian* (or commutative) if $a * b = b * a$ for every $a, b \in G$.

If $(G, *)$ is a group, we say that G is a group under $*$.

1.1.4 Note Some authors like to make it explicit that it's closed under the operation, meaning if $a, b \in G$, then $a * b \in G$. However, that is implied in the definition of binary operation (i.e. a function), since the codomain of $*$ is G.

Example 1.1.5: First Encounter With Groups

In the following examples, when we are working with numbers, think of them only as a set, and define the operations $+$, $\times$ or others in the way they were intuitively defined when learning arithmetic's.

1. $(\mathbb{Z}, +)$, $(\mathbb{Q}, +)$, $(\mathbb{R}, +)$ are groups. We would say, for example, that $\mathbb{Z}$ is a group under $+$. For each of these examples, we would have the identity to be $e = 0$, and for each element a, $a^{-1} = -a$. Note that technically, the $+$ in each group is not the same $+$, because the domains and co-domains are different. Note that $\mathbb{N}$ is *not* a group under $+$, since not every element has an inverse: There does not exist an $a \in \mathbb{N}$ such that $2 + a = 0$ (i.e. $-2 \notin \mathbb{N}$). [a]
2. $\mathbb{Z}, \mathbb{Q}, \mathbb{R}$ and $\mathbb{C}$ under $\times$ are *not* groups. $\mathbb{Q} - \{0\}$, $\mathbb{R} - \{0\}$ $\mathbb{C} - \{0\}$, $\mathbb{Q}_{>0}$, $\mathbb{R}_{>0}$ *are* groups under $\times$, with $e = 1$, and $a^{-1} = \frac{1}{a}$. Why is that? Take a moment to think about why removing the zero will make those sets groups. Is $\mathbb{Z} - \{0\}$ a group under $\times$? Is $(\mathbb{Z} - \{0\})$ a group under $+$?
3. For any $n \in \mathbb{N}$, $\mathbb{Z}/n\mathbb{Z}$ is a group under modulo $+$ $(\mathbb{Z}/n\mathbb{Z}, +)$. Can you show that 0 is the identity? Can you show that every number has an inverse? Note that $((\mathbb{Z}/n\mathbb{Z}), \times)$ under $\times$ is *not* a group in general! $(\mathbb{Z}/n\mathbb{Z} \setminus \{0\}, \times)$ *is* a group when n is prime. You can pick some non-prime n and try it (ex. $n = 4$). The general proof that $(\mathbb{Z}/n\mathbb{Z} \setminus \{0\}, \times)$ is a group when n is prime will come later (after we introduce theorem 2.3.11)
4. If $(A, \bullet_A)$ and $(B, \bullet_B)$ are groups, we can form a group on their cartesian product $A \times B$ called the *direct product group*, whose elements are those in the Cartesian product

$$A \times B = \{(a, b) | a \in A, b \in B\}$$

and the binary operations is defined point wise:

$$(a, b) * (c, d) = (a \bullet_A c, b \bullet_B d)$$

We can denote this group and its binary operation $(A \times B, (\bullet_A, \bullet_B))$. A common example of a direct product of group is $(\mathbb{R} \times \mathbb{R}, (+, +))$, with the usual notation of vector-addition in $\mathbb{R}^2$. We will dedicate a lot of energy understanding the details of the Cartesian product of groups in chapter 5.

5. Define $+_2 : \mathbb{Z} \times \mathbb{Z} \to \mathbb{Z}$ $x +_2 y = 2 + x + y$. The operation $+_2$ is associative, since

$$(x +_2 y) +_2 z = (2 + x + y) +_2 z = (2 + x + y) + 2 + z = 2 + x + (2 + y + z) = x +_2 (y +_2 z)$$

Furthermore, $e = -2$, since

$$x +_2 -2 = x + 2 - 2 = x$$

and $x^{-1} = -x - 4$ since

$$x^{-1} +_2 x = -x - 4 + 2 + x = -2$$

showing that $(\mathbb{Z}, +_2)$ is indeed a group. Notice that $(\mathbb{Z}, +_2)$ and $(\mathbb{Z}, +)$ are different (for example, their identity is different)[b]. When introducing multiplication to integers, we shall want $+$ and $\cdot$ to behave well together. The usual $+(x, y) = x + y$ will insure this[c].

6. $G = (\mathrm{Mat}_n(\mathbb{R}), \cdot)$ is almost forms a group. A binary operation exists, identity exists, and associativity holds. However, there are non-invertible matrices, meaning that $\forall M \in G$, it's not necessarily the case that $\exists N \in G$ such that $NM = \mathrm{I}$. We will explore this in section 1.2.5⋆

7. Take $\{e\}$ to be the set with a single element, and define the binary operation $\cdot$ to be the only possible function:

$$e \cdot e = e$$

that is, the binary operation that treats e like an identity. Then $(\{e\}, \cdot)$ is a group! This group is called the *trivial group*.

8. Take a set A and its powerset $P(A)$. Is $(P(A), \cup)$ a group? What about $(P(A), \cap)$? answer: [d]

9. Take $(\mathbb{Z}, -)$, where $-$ is the operation of subtraction. Then $(\mathbb{Z}, -)$ is not a group! Can you see which axiom fails?

10. Here is a more complex example which fails for the same reason as the previous one. Take $(\mathbb{R}^3, \times)$ where $\times$ represents the cross product:

$$\begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix} \times \begin{pmatrix} b_1 \\ b_2 \\ b_3 \end{pmatrix} = \begin{pmatrix} a_2 b_3 - a_3 b_2 \\ a_1 b_3 - a_3 b_1 \\ a_1 b_2 - a_2 b_1 \end{pmatrix}$$

or, if you have taken a physics or linear algebra class, you might have seen it defined as

$$a \times b = \det \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix}$$

where we represent the three components of the vector as $\mathbf{i}$, $\mathbf{j}$, $\mathbf{k}$, and we use the determinant to shorten notation. Then $(\mathbb{R}^3, \times)$ is *not* a group! The cross product is not associative! Take

$$\begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}, \begin{pmatrix} 2 \\ 3 \\ 4 \end{pmatrix}, \begin{pmatrix} 3 \\ 4 \\ 5 \end{pmatrix}$$

Then

$$\begin{pmatrix}1\\2\\3\end{pmatrix}\times\begin{pmatrix}2\\3\\4\end{pmatrix}=\begin{pmatrix}-1\\2\\-1\end{pmatrix}\qquad\begin{pmatrix}-1\\2\\-1\end{pmatrix}\times\begin{pmatrix}3\\4\\5\end{pmatrix}=\begin{pmatrix}14\\2\\-10\end{pmatrix}$$

and

$$\begin{pmatrix}2\\3\\4\end{pmatrix}\times\begin{pmatrix}3\\4\\5\end{pmatrix}=\begin{pmatrix}-1\\2\\-1\end{pmatrix}\qquad\begin{pmatrix}1\\2\\3\end{pmatrix}\times\begin{pmatrix}-1\\2\\-1\end{pmatrix}=\begin{pmatrix}-8\\-2\\4\end{pmatrix}$$

meaning $(\mathbb{R}^3,\times)$ is *not* associative, meaning it's not a group. The cross product is a different kind of operation, a *Lie bracket*, which makes $\mathbb{R}^3$ a Lie algebra.

11. If you know some properties of complex numbers, then $G=\{z\in\mathbb{C}|z^n=1 \text{ for some } n\in\mathbb{Z}^+\}$ is a group! Visually, this group looks like points on the unit circle in the complex plane. The main idea here is since $z^n=1$, then elements in G will be of the form $e^{\frac{2\pi i}{n}\cdot k}$ for $1\le k\le n$. If you are comfortable with multiplying complex numbers and trust me that all the elements of G are of this form, then you can check that this is a group.

    More generally, $G=\{z\in\mathbb{C}\mid \|z\|=1\}$ is a group with multiplication (i.e. the unit circle in the complex plane). You can verify that $1\in G$ and for any $z,w\in G$, $\|zw\|=\|z\|\,\|w\|=1\cdot 1=1$. Multiplication on G is associative because multiplication on $\mathbb{C}$ is. Each $z\in G$ has the inverse $z^{-1}=\bar{z}$, since $z\bar{z}=\|z\|^2=1$, and $\bar{z}\in G$ because $\|\bar{z}\|=\|z\|=1$.

12. If you know some complex arithmetic's, show that $G=\{z\in\mathbb{C}\mid 0.1<|z|<10\}$ with the usual multiplication structure of $\mathbb{C}$ satisfies associativity, having an identity, and having an inverse, but is *not* a group.

13. Let A be any set. Consider the collection of every bijective function from A to itself:

    $$G=\{f:A\to A\mid f \text{ is bijective}\}$$

    Then with the binary operation of composition, $(G,\circ)$ is a group. What is the identity, and why do inverses always exist? The collection of bijective functions is a very common group that shows up in many different ways in different areas of mathematics and so has lots of notations:

    $$\mathrm{Bij}(A)\qquad \mathrm{Sym}(A)\qquad \mathrm{Aut}_{\mathbf{Set}}(A)$$

Part of the motivation behind learning group theory is in the abundance of areas groups show up in mathematics. The more domains of mathematics you learn, the more types of groups you will encounter. For example, field theory (Part IV) has *Galois groups* at its heart, Differential Geometry have *lie groups* that allow for a large classification of and construction of "manifolds" (a generalization of the notion of Euclidean space), Algebraic Geometry (see *Classical Algebraic Geometry* [23]) use *algebraic groups* to represent more complicated objects in more familiar terms, Analysis uses *topological groups* (for example, Amenable and Locally Compact groups) in the study of Fourier Analysis and Functional Analysis, Algebraic topology has the *fundamental group* and *homology group* at its heart, and combinatorics uses *geometric group theory* (for example, the *Kleinian group* or *hyperbolic groups* shows up frequently).

Groups can also be thought of as one of the "basic" structures that many other structures will build on or act on. Of the algebraic structures that will be explored in this book, rings, fields, vector-spaces, modules, and (lie and associative) algebras are all also groups.

Overall, it is worth remembering that groups are a very common mathematical object that show up a little bit everywhere.

[a] $\mathbb{N}$ is a weaker structure called a monoid which does not require inverses. More on them in section 7.1
[b] I'm careful in saying they are different instead of saying they have *different group structures*. There is a subtly here on how we define two group structures being the same or different. This is addressed in section 1.4
[c] we shall desire distributivity, or for the expert that $\mathbb{Z}$ is representable in its endomorphism ring
[d] the answer is no! Like $\mathbb{N}$, these are example of monoids

In general, when we write a group, the group operation will be clear from context, so instead of writing $(G, *)$, we will write G. We will take on this convention form here on out unless it's ambiguous. We will also assume the standard binary operation of addition for

$$\mathbb{Z}, \mathbb{Q}, \mathbb{R}, \mathbb{C}, \mathbb{Z}/n\mathbb{Z}$$

unless stated otherwise (ex, if you see $\mathbb{Z}$, assume it is $(\mathbb{Z}, +)$). If we want the multiplicative structure of these numbers, we will add a $\times$ in the superscript and implicitly omit the zero:

$$\mathbb{Z}^\times, \mathbb{Q}^\times, \mathbb{R}^\times, \mathbb{C}^\times (\mathbb{Z}/n\mathbb{Z})^\times$$

These sets are $(\mathbb{Z} - \{0\}, \times)$ [1], $(\mathbb{Q} - \{0\}, \times)$, $(\mathbb{R} - \{0\}, \times)$, and so forth. The sets $\mathbb{R}^+$ and $\mathbb{Q}^+$ shall represent the collection of positive real numbers and positive rational number with the usual multiplication. Furthermore, since we are always working with binary operations, the word "binary" is often omitted for convenience.

1.1.6 Note In many of the previous examples, there is some more structure usually associated with those symbols. For example, in $\mathbb{N}$, $\mathbb{Z}$, $\mathbb{Q}$ and $\mathbb{R}$, there is a notion of some numbers being bigger than others, that is, $1 < 2$, or $n < n + 1$. These are *not* part of the definition of a group, and so cannot be used when proving general statements about groups! It will be very common that you work with groups with *more* structure than their group structure, so be quite careful that the examples you hold when doing exercises *only* represent the structure you want to work with at that moment

You might have noticed that some of the underlying sets of the above groups are subsets of each other: $\mathbb{Z} \subseteq \mathbb{Q} \subseteq \mathbb{R} \subseteq \mathbb{C}$. Does this imply that there is a relation between their group structure? For example does the usual addition in $\mathbb{Z}$ works as in $\mathbb{Q}$ with usual addition? If you think about this particular example, you will see that the answer is yes. But being a subset doesn't always means you will have the same (if any) group structure: $\mathbb{N} \subseteq \mathbb{Z}$, but $\mathbb{N}$ is not even a group! On the other hand, if we take $2\mathbb{Z} \subseteq \mathbb{Z}$, and give $2\mathbb{Z}$ the same operation as $\mathbb{Z}$ (that is $(2\mathbb{Z}, +)$ is our group), then $(2\mathbb{Z}, +)$ acts exactly the same way as $\mathbb{Z}$, with just fewer elements. So, we will give a name to subsets which inherit group properties of their parent group:

Definition 1.1.7: Subgroup

Let G be a group and $H \subseteq G$. Then H is a subgroup of G if it is still a group under the same operation as G. We denote the subgroup by $H \leqslant G$ [a]

[a] In some textbooks, the symbol for subgroup is $H \preceq G$

[1] Note that $\mathbb{Z}^\times$ is not a group under $\times$ as we have pointed out earlier. However, the binary operations $\times$ is still defined on this set

You might have flinched to it saying "the same operation", since it was explicitly mentioned in the definition that for a group G, the operation is $G \times G \to G$, so H cannot have the same operation since the domain and co-domain are different. This is a common convention in algebra for structures which are similar: if we restrict the operation to only the elements of H, then $(H, *|_H)$ should be a group. In particular, if $* : G \times G \to G$ is the binary operation on G, then we say we *restrict* the binary operation to H if we can define $*|_H : H \times H \to H$, where for any two elements $h_1, h_2 \in H$ $*|_H (h_1, h_2) = *(h_1, h_2)$. If $(H, *|_H)$ is a group, we say it's a subgroup of $(G, *)$. In this way, we say that H has "the same" binary operation as G.

Example 1.1.8: Subgroups

1. Take $2\mathbb{Z} \subseteq \mathbb{Z}$, the set of all even integers. The identity is there: $0 \in 2\mathbb{Z}$. If you add two even integers, you get an even integer $(2x + 2y = 2(x + y))$ [a]. Furthermore, if $2x \in 2\mathbb{Z}$, then there exists $-2x \in 2\mathbb{Z}$ such that $2x - 2x = 0$, meaning every element has an inverse. Finally, associativity is implied by associativity in $\mathbb{Z}$: for any $2x, 2y, 2z \in 2\mathbb{Z}$, we have that $2x, 2y, 2z \in \mathbb{Z}$, so $2x + (2y + 2z) = (2x + 2y) + 2z$. Thus $2\mathbb{Z} \leqslant \mathbb{Z}$
2. Right before the previous definition, it was mentioned that $\mathbb{Z} \subseteq \mathbb{Q}$ is an example of a subgroup (i.e. $\mathbb{Z} \leqslant \mathbb{Q}$). This is an exercise worth doing to get comfortable with the concept. Mix an match with other types of numbers to get more practice.
3. Let $\mathbb{Z}^\times \subseteq \mathbb{Q}^\times$ (i.e. $Z - \{0\} \subseteq \mathbb{Q} - \{0\}$). Is $\mathbb{Z}^\times$ a subgroup?
4. Let $H \leqslant G$ and $K \leqslant H$. Is $K \leqslant G$?
5. Here is an exercise in understanding the nuance of binary operations: Is $\mathbb{Z}/5\mathbb{Z} \leqslant \mathbb{Z}$? answer: [b]

[a] Note here how I *did* use extra properties that are not necessarily part of a group (I factored the two out). This is fine, since we are working with a *particular* example, this would not be a proof we can do on general groups

[b] It is no, since the binary operation does not "act" the same way $(3 + 3 = 1)$ in $\mathbb{Z}/5\mathbb{Z}$ but $3 + 3 = 6$ in $\mathbb{Z}$

Groups are missing one key property that numbers usually have: commutativity. Though we already talked about them in the definition of a group, I will add it here to emphasize their importance:

Definition 1.1.9: Abelian Group

Given a group G, if $ab = ba$ for all $a, b \in G$, we call G a *commutative* or *abelian* group

Example 1.1.10: Abelian Groups

$\mathbb{Z}, \mathbb{Q}, \mathbb{R}, \mathbb{C}, \mathbb{Z}/n\mathbb{Z}$ are all Abelian Groups [a]. So are $\mathbb{Q}^\times, \mathbb{R}^\times, \mathbb{C}^\times$. In fact, all examples in the example 1.1.5 were abelian groups with the exception of the bijections on a set with at least three elements. An important non-commutative group will be the Dihedral group (definition 1.2.28)

[a] This is an example where the binary operation is implied. The binary operations shown in example 1.1.5 are the standard binary operations for these sets

As mentioned earlier, being abelian/commutative is insanely powerful. Of all the properties the reader will cover in basic group theory, a lot of them will be properties a group being "almost" or "kind of" abelian. In this text, we will come across many definitions and sections that are centered

around how close they are to being abelian [2]

The reason we relate a lot of conditions to commutativity is because Abelian Groups behave in much more predictable ways as compared to general groups, as shall be developed. The power of commutativity permeates a lot of algebraic structures. Because of this, we often split up domains in mathematics in "commutative" and "not-commutative". As an example of this split, some mathematicians like to reference the set of Abelian groups as **Ab** and the set of all groups as **Group** to show how different a general group versus a general abelian group is [3]. There are entire areas of mathematics where the problem has been solved in the "abelian case", while the "nonabelian case" is an active area of research[4].

For the visual reader, the following is a helpful tool to understand the group operation:

Definition 1.1.11: Cayley Table (Multiplication Table)

Let G be a finite group (that is, a group on a finite set). Label the elements of G as $\{g_1, g_2, ..., g_n\}$, where g_1 represents the identity.

Then *Cayley table* or *multiplication table* of G is the $n \times n$ matrix whose i, j entry is the group element $g_i g_j$.

I personally prefer the term Cayley table since not all binary operations are called "multiplication". It can be confusing to do a multiplication table for $(G, +)$

Example 1.1.12: Cayley Table

Take $G = \mathbb{Z}/4\mathbb{Z}$. Then the Cayley tables for addition and multiplication are:

$+$	0	1	2	3
0	0	1	2	3
1	1	2	3	4
2	2	3	4	0
3	3	4	0	1

$\times$	0	1	2	3
0	0	0	0	0
1	0	1	2	3
2	0	2	0	2
3	0	3	2	1

This can be done for any finite group, and sometimes gives valuable insights instantly. For example, notice that in the Cayley table for multiplication, for row 0, 1, and 3, they contain every element of $\mathbb{Z}/4\mathbb{Z}$, yet the row for 2 does not. Notice further that both tables are symmetrical, that is, position ij is equal to position ji.

In general, it can be quite cumbersome to write out an entire Cayley table, especially when the order the group will be higher than 4, if not infinite. So throughout Part I, we develop algebraic tools to deduce information about groups.

[2] for reference, see definition 3.1.9, definition 2.2.4, and sections 5.3 and 6.1

[3] There are some technicalities here in saying it's the *set* of all groups, due to Russel's paradox. We should instead say the *class* of all groups. This is addressed in [15, Categories, Morphisms, and Size].

[4] There are a huge number of such examples, but to point out one relevant to algebra: class field theory completely classifies abelian extension of fields, while the nonabelian counterpart is a very deeply researched problem and is a special case of the *Langland's Program*. Another example that creates a split between the commutative and non-commutative case is within the study of algebraic geometry where the structures are built out of commutative rings, while the non commutative case would require more machinery from analysis.

1.1.1 Consequences of Group Axioms

With the definition of groups down, we can deduce a lot of direct consequences from the axioms. One thing common with mathematical constructs is an abundance of propositions. These are vital to be able to prove by oneself, and once comfortable should be treated as knowledge as important as the definition of a group. I would even go so far as to say that not knowing the properties of groups means you do not really know what a group is. For example, do you really know what a frying pan is/can do if you have not cooked with it before? Do you know the in's an out of what utensils can you use to not damage it, or what type of food will stick to it or not? You will only know once you have practiced using it.

I would strongly suggest that the reader attempts proving *all* of these propositions before looking at the proof, especially if this is the first time the reader has seen algebraic structures! It is crucial moving forward that these properties are second nature, since everything we build form here on out will constantly be relying on these properties.

The first thing to point out is that if we have $a * e = a$, we can deduce $e * a = a$.

Proposition 1.1.13: one-sided identity implies two-sided identity

Let G be a group and e an identity. If $a * e = a$ then $e * a = a$ (or vice-versa)

***Proof*:**
The identity axiom gives $e * a = a$ for every $a \in G$, so it suffices to prove that $a * e = a$ also holds for every $a \in G$. From the axioms,

$$\begin{aligned} a * e &= a * (a^{-1} * a) && \text{inverse definition (axiom 3)} \\ &= (a * a^{-1}) * a && \text{associativity (axiom 1)} \\ &= e * a && \text{inverse definition (axiom 3)} \\ &= a && \text{left identity (axiom 2)} \end{aligned}$$

as we sought to show.

Some author's put $a * e = e * a = a$ as a definition. This is a matter of preference: since it's a direct consequence of the axioms, it can be put as one of the fundamental rules a group has to follow without worry of constraining the definition. Some authors disagree with this, and would rather keep the definition of group in fuller generality. This preference for generality is a balancing act: the more general you go, the more "complicated" the concept gets. However, if done right, you gain a more "nuanced" perspective with generality. In this book, the more general idea was taken, since some proofs already covered earlier fail for more subtle reasons. For example, $(\mathbb{Z}, -)$ fails to be a group because of associativity, however if we put $a * e = e * a = a$ in the definition, we can also say it fails because $-a = 0 - a \neq a - 0 = a$ (for $a \neq 0$). However, as we have seen in the above proof, this fact *relies on associativity*, and therefore is hiding the fact that $(\mathbb{Z}, -)$ is still failing to be a group because of associativity.

Corollary 1.1.14: one-sided Inverse Implies two-sided Inverse

Let a, a^{-1} be elements of the group G and let e be the identity of G. Then $a * a^{-1} = e$ if and only if $a^{-1} * a = e$

***Proof*:**
Let's say $b = a^{-1}$ so that $a * b = e$. Then $b = b * e = b * a * b$. Multiplying both sies by b^{-1}, we get

$$\begin{aligned} e &= b * b^{-1} \\ &= (b * a * b) * b^{-1} \\ &= b * a * (b * b^{-1}) \\ &= b * a \\ &= a^{-1} * a && b = a^{-1} \end{aligned}$$

as we sought to show.

Proposition 1.1.15: Basic properties of Group elements

1. The identity of G is unique. Furthermore, if for some $x \in G$, $f * x = x$, then is equal to the identity
2. For each $a, \in G$, a^{-1} is uniquely determined
3. $(a^{-1})^{-1} = a$, for all $a \in G$
4. $(a * b)^{-1} = (b^{-1}) * (a^{-1})$.
5. generalized associative law (induction on associativity).

***Proof*:**
Again, I highly encourage you to attempt all of these before looking at these answers!!

1. Let $e \in G$ be the identity, that is, for all $x \in g$ $e * x = x$. Let's say there is another element $f \in G$ that is also the identity. Then

   $$\begin{aligned} e &= e * f && \text{definition of identity} \\ &= f && \text{definition of identity} \end{aligned}$$

   therefore, $e = f$, showing that e is unique.

   Furthermore, if there was an f such that for only *some* $x \in G$, $f * x = x$, then

   $$\begin{aligned} (f * x) * x^{-1} &= (x * x^{-1}) = e \\ \Leftrightarrow f * (x * x^{-1}) &= e \\ \Leftrightarrow f &= e \end{aligned}$$

thus, there cannot be a "partial" identity.

2. Let $a \in G$, and suppose $b, c \in G$ are both inverses of a, that is, $a * b = b * a = e$ and $a * c = c * a = e$. Then

$$\begin{aligned} b &= b * e && \text{proposition 1.1.13} \\ &= b * (a * c) && c \text{ is an inverse of } a \\ &= (b * a) * c && \text{associativity} \\ &= e * c && b \text{ is an inverse of } a \\ &= c && \text{definition of identity} \end{aligned}$$

   Thus $b = c$. Any two inverses of a are equal, so a^{-1} is uniquely determined.

3. Let $a \in G$. By the definition of the inverse, $a^{-1} * a = e$ and $a * a^{-1} = e$. These two equations state exactly that a is an inverse of a^{-1}. By part 2, the inverse of a^{-1} is unique, so $(a^{-1})^{-1} = a$.

4. Let $a, b \in G$. Then

$$\begin{aligned} (a * b) * ((b^{-1}) * (a^{-1})) &= ((a * b) * b^{-1}) * a^{-1} && \text{associativity} \\ &= (a * (b * b^{-1})) * a^{-1} && \text{associativity} \\ &= (a * e) * a^{-1} && \text{inverse definition} \\ &= a * a^{-1} && \text{proposition 1.1.13} \\ &= e && \text{inverse definition} \end{aligned}$$

   By corollary 1.1.14, we also have $((b^{-1}) * (a^{-1})) * (a * b) = e$. Thus $(b^{-1}) * (a^{-1})$ is an inverse of $a * b$. By part 2, it is the only one, so $(a * b)^{-1} = (b^{-1}) * (a^{-1})$.

5. The precise claim is that for any $n \geq 1$ and any $a_1, a_2, \ldots, a_n \in G$, all ways of inserting parentheses into the product $a_1 * a_2 * \cdots * a_n$ (keeping the order of the factors) produce the same element of G. Call a product *left-normed* if it is bracketed as $(\cdots((a_1 * a_2) * a_3)\cdots) * a_n$. We prove, by strong induction on n, that every bracketing of $a_1 * a_2 * \cdots * a_n$ equals the left-normed product of $a_1, \ldots, a_n$. For $n = 1$ and $n = 2$ there is only one bracketing, so there is nothing to prove.

   Now let $n \geq 3$, and consider any bracketing P of $a_1 * \cdots * a_n$. The outermost multiplication of P splits it as $P = Q * R$, where Q is a bracketing of $a_1 * \cdots * a_k$ and R is a bracketing of $a_{k+1} * \cdots * a_n$, for some $1 \leq k \leq n - 1$. If $k = n - 1$, then $R = a_n$, and the induction hypothesis (applied to the $n - 1$ factors of Q) says Q equals the left-normed product of $a_1, \ldots, a_{n-1}$. Hence $P = Q * a_n$ is the left-normed product of $a_1, \ldots, a_n$. If $k < n - 1$, then R has at least two factors, so by the induction hypothesis R equals the left-normed product of $a_{k+1}, \ldots, a_n$, which by its bracketing has the form $S * a_n$, where S is the left-normed product of $a_{k+1}, \ldots, a_{n-1}$. By the associativity axiom,

$$P = Q * (S * a_n) = (Q * S) * a_n$$

   Now $Q * S$ is a bracketing of the $n - 1$ factors $a_1 * \cdots * a_{n-1}$, so by the induction hypothesis it equals the left-normed product of $a_1, \ldots, a_{n-1}$. Thus $P = (Q * S) * a_n$ is the left-normed product of $a_1, \ldots, a_n$, completing the induction. Any two bracketings of $a_1 * a_2 * \cdots * a_n$ therefore both equal the left-normed product of $a_1, \ldots, a_n$, and hence equal each other.

With these in mind, here's some standard notation to simplify writing: One usually write the operation $*$ as $\cdot$ to represent an abstract binary operation, unless otherwise specified. If clear from context, we write $a \cdot b$ as ab. We usually also denote the identity e as 1. We will also implement the following short hand: [5]

$$\underbrace{xxxxxx\cdots x}_{n \text{ times}} = x^n, \qquad \underbrace{x^{-1}x^{-1}\cdots x^{-1}}_{n \text{ times}} = x^{-n}, \qquad x^0 = 1$$

Do not forget that last one is the identity of G. If we are using a group where specific notation is already defined (like with the $+$ operation), we can use it's specific convention. So $a+a+\cdots+a = na$, $-a+-a+\cdots+-a = -na$, and $0a = 0$ [6]. Furthermore, $a+(-a)$ is sometimes shorthanded to $a-a$.

The next result we will cover let's us generalize our intuition of pre-abstract algebra. When learning algebra for the first time, you have probably encountered equations of the form:

$$3x = 5$$

and where asked to solve for x. The way you would do this is by dividing both sides by 3 to get the equation:

$$x = \frac{5}{3}$$

This might seem like the obvious thing to do in any situation. However, you cannot always cancel out a number on one side and get a unique result! Take a moment to review your proof of uniqueness of identity and uniqueness of the inverse, and consider this thought experiment:

1.1.16 Thought Experiment Let's take a set for which every element does *not* have an inverse. Recall (or prove) that $\mathbb{Z}/6\mathbb{Z}$ is not a group under multiplication. In particular, $[2]$ has no multiplicative inverse. Since $[2][2] = [0]$, then:

$$[4] = [2][2] = [2][5] = [10] = [4]$$

However, $[2] \neq [5]$. In other words, I could not "cancel" out the $[2]$ on the left side, i.e. for the equation $[2]x = [4]$ we get $x = [2]$ or $x = [5]$.

This cancelling out of a number on the left [or right] side is something you probably took for granted, but it's the consequence of the fact that every number in a group must have a unique inverse! This fact is proven in the next proposition:

Proposition 1.1.17: Cancellation Property: Unique solution to monomials

Let G be a group and let $a, b \in G$. The equation $ax = b$ and $ya = b$ have unique solutions for $x, y \in G$. In particular, the left and right cancellation properties hold in G:

1. If $au = av$ then $u = v$
2. If $ub = vb$ then $u = v$

[5] $x^0 = 1$ since $x^0 = x^{1-1} = x^1x^{-1} = xx^{-1} = 1$

[6] If you have covered rings, this might look familiar to the first property in proposition 9.1.9. It is quite similar in that this shorthand secretly is a ring-action on the group (but not a $\mathbb{Z}$-module since G is not necessarily abelian). This means that the intuition behind the notation is similar to the proof in proposition 9.1.9: $0a = (1-1)a = a - a = 0$

Sometimes, this proposition is called the *Cancellation Law*

***Proof*:**
Simply use uniqueness of inverse. The point of this theorem is that if $ab = e$, for any element $a \in G$ and some element $b \in G$, then $b = a^{-1}$.

This proposition is very powerful, it means we can bring in our elementary algebra knowledge. In this sense, groups are the structure on which we generalize the idea of doing elementary algebra on. They are limited in that they only allow one operation: you probably have worked with equation with both addition and multiplication (ex. $3x^2 - 5 = 0$). We will work with such equations with 2 binary operations once we study *rings* (see Part II)

With these properties, I will take a moment to explain the power of associativity and commutativity. In all type of numbers systems we saw $(\mathbb{N}, \mathbb{Z}, \mathbb{Q}, \mathbb{R})$, for any two numbers a and b:

$$aaabbbbaaabaaaabbbbbaa$$

is unambiguous. and so, we can write

$$a^3b^3a^3ba^4b^5a^2$$

Which is really powerful. If you did not have associativity, then $aaabbbbaaabaaaabbbbbaa$ would be ambiguous, since it's possible that

$$(aaabbbbaaabaaaabbbbba)a \neq a(aabbbbaaabaaaabbbbbaa)$$

So every pair of elements would have to be bracketed since

$$abc$$

would be ambiguous [7] since it's possible that

$$(ab)c \neq a(bc)$$

and so you would be forced to have an expression like

$$((((aa)b)((bb)(ba)))(((((aa)(ba))a))((a(a(bb)))((b(bb))a))))a$$

with no simpler way of representing the expression, quite a monstrosity. There are more than $2^{22} = 4'194'304$ ways of bracketing this!! The idea is that associativity means that given a sequence of operations that must be performed in a certain order:

$$a_1 * a_2 * a_3 * \cdots * a_n$$

We are allowed to "break up" the problem into smaller problems that can be grouped and operated and then recombined in the right order. For example if we must perform 4 actions in order:

$$a_1 * a_2 * a_3 * a_4$$

Then associativity means that we could have considered $a_3 * a_4 = b_2$ as a new combined actions and $a_1 * a_2 = b_1$ also as a combined action:

$$b_1 * b_2$$

[7] like $\mathbb{R}^3$ with the cross product

Or we could have taken $a_1 * a_2 = b_1$ and then combine it with the action of a_3, $b_1 * a_3 = b_3$ and then with a_4 and get the same result:

$$a_1 * a_2 * a_3 * a_4 = b_1 * b_2 = b_3 * a_4$$

This can also be thought of as "pre-combining" the steps before performing the result in the same order, which if the operation is associative would give you the same result.

Example 1.1.18: Associativity and The Lack of It

1. An example of real world phenomena where this is not applicable rock-paper-scissors. Let's says the (commutative) binary operation takes in two choices and gives the winner (and if the two input are the same return the input). Then:

   $$(\text{rock} * \text{paper}) * \text{scissors} = \text{paper} * \text{rock} = \text{paper} \neq \text{rock} * \text{scissors} = \text{rock} * (\text{paper} * \text{scissors})$$

2. *Divide and Conquer (Parallel Computing)*: Associativity is key in parallel computing. If a sequence of operations is associative, we do not have to wait for a single processor to calculate from left to right:

   $$((((a_1 * a_2) * a_3) * a_4) \dots)$$

   Instead, we can dispatch the problem locally to different processors:

   (a) Processor 1 solves $a_1 * a_2$ locally.

   (b) Processor 2 solves $a_3 * a_4$ locally.

   They then combine their local results[a]. Such "reduction" operations must be associative precisely so they can be solved locally on separate servers and combined later without changing the final outcome.

3. Exponentiation is *not* associative: $(a^b)^c$ is not in general equal to $(a)^{b^c}$.

4. The averaging operation is not associative

5. Convince your self that cooking instruction are usually not associative.

[a] Frameworks like Google's MapReduce rely entirely on this

Since functions are associative, most algebraic structures in mathematics will be associative, however there are entire domain of study which concentrates on a non-associative structure [8] or drop the condition of associativity[9]. Structures without associativity are outside the scope of this book[10]. As a consequence of associativity, we can unambiguously write:

$$x^{a+b} = x^{b+a}$$

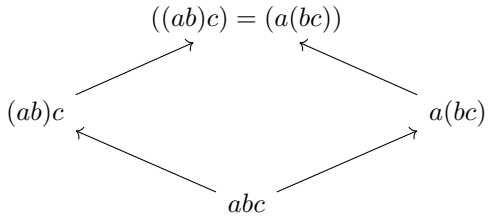
since the order of multiplication does not effect the final result. We can also visualize associativity through the use of directed graphs. If we take abc, then we can visualize which tuple of elements we

[8] the most famous of is called Lie Algebra, with something called the Jacobi Identity replacing associativity. Another is the logical implications: $(a \to b) \to c$ does not always give the same truth value as $a \to (b \to c)$

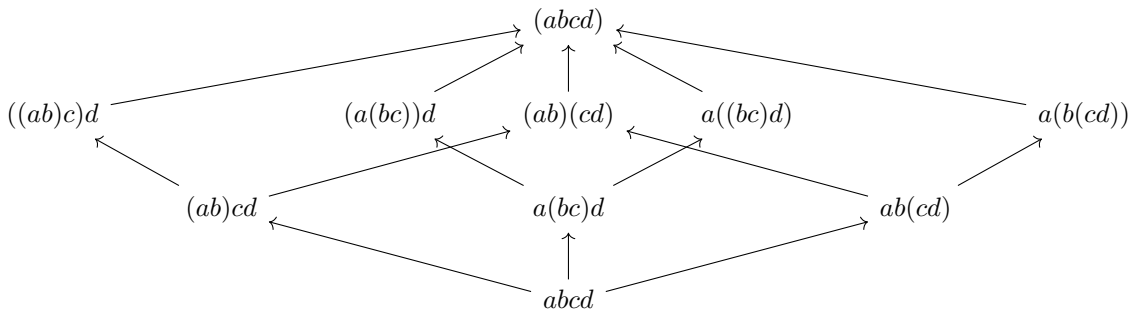
[9] Three such examples are *loops*, *Quasigroups*, and *magmas*

[10] Nonassociative algebras are treated in [54], and quasigroups and loops in [56].

choose to multiply first:



this is more clear if we have 4 elements, any way we choose to bracket our elements leads to the same result:



Note that adding commutativity makes this even more powerful, since

$$aba = aab = a^2b$$

and so we can group all like terms together, and write

$$aaabbbbaaabaaaabbbbbaa \qquad \text{as} \qquad aaaaaaaaaaaabbbbbbbbbb$$

and collapse it to:

$$a^{12}b^{10}$$

That is, the only information we need is the number of times we multiply by a and b, with no regard to the order we multiply them! This added axiom is so powerful that it will turn out that it will constrain the number of possible (finite and finitely generated) groups, which are relatively few[11] (see section 5.1.1)

Exercise 1.1.1

1. Let $G = \mathbb{Q}$, and the binary operation be average : $\mathbb{Q} \times \mathbb{Q} \to \mathbb{Q}$, where average$(x, y) = \frac{x+y}{2}$. Is $(G, \text{average})$ a group? What if $G = \mathbb{Z}/4\mathbb{Z}$?
2. Can a group structure be put on $\emptyset$ (i.e., can there exist a function $* : \emptyset \times \emptyset \to \emptyset$ such that $(\emptyset, *)$ is a group?)

[11] Finitely generated groups are defined in section 2.3, and the result is theorem 5.1.6. As of writing this book, the project of classifying all abelian groups is still a very unsolved problem

3. (If you know some complex analysis) In example 1.1.5(11), it was stated that
$$G = \{z \in \mathbb{C} \mid |z| = 1\}$$
is a group. Prove that G is indeed a group.
4. Let $\text{Bij}(X)$ be the set of all bijections from X to X. Prove that $\text{Bij}(X)$ is a group. In section 1.2.2, we will relabel this as $\text{Sym}(X)$.
5. Show that if G is abelian, then $(ab)^{-1} = a^{-1}b^{-1}$ for all $a, b \in G$.

1.2 Examples of Groups

Now that we covered the basics, it is important to build up a family of common groups to get familiar with the how to recognize groups, and to feel very comfortable manipulating groups. We already showed how many previous mathematical structures were already groups (the canonical examples being $\mathbb{Z}, \mathbb{Q}, \mathbb{R}, \mathbb{C}$). We will now define new groups to show further what is possible with the definition of the group, and to build-up a repertoire of more elaborate examples.

1.2.1 Integers Modulo n and Cyclic Groups

Recall in [24, Modular Arithmetic] that $\mathbb{Z}/n\mathbb{Z}$ is a collection of equivalences classes satisfying $a \sim_n b$ if and only if there exists a $k \in \mathbb{Z}$ st $a - b = kn$. Given that $\mathbb{Z}$ is a group and the results [24, Properties Of Modular Arithmetic], it is reasonable to think that it may form a group:

Proposition 1.2.1: Modular Arithmetic Groups

Let $\mathbb{Z}/n\mathbb{Z}$ be the integesr modulo n. Then $\mathbb{Z}/n\mathbb{Z}$ forms a group under modular addition:
$$(a \bmod n) +_n (b \bmod n) = (a + b) \bmod n$$
with the binary operation denoted as $+_n$, or if the context permits simply as $+$

In almost all cases, the notion $+$ shall be used, for it will be clear from context that the elements will be part of a certain set of modular integers.

***Proof*:**

(Well-defined) By [24, Properties Of Modular Arithmetic](1), the operation is well-defined

(Idenitty) For any n, there is always $\overline{0}_n \in \mathbb{Z}/n\mathbb{Z}$ for $n - n = 0$. Then for any $\overline{a}_n \in \mathbb{Z}/n\mathbb{Z}$:
$$\overline{0}_n + \overline{a})n = \overline{0 + a}_n$$
Then notice that
$$a - (0 + a) = 0 = kn \qquad (\text{with } k = 0)$$
Thus, $\overline{0 + a}_n = \overline{a}_n$, and so $\overline{0}_n$ is the identity for any $n \in \mathbb{N}_{>0}$

(Inverse) For any $\overline{a}_n \in \mathbb{Z}/n\mathbb{Z}$, pick $\overline{-a}_n$ (i.e., the equivalence class that can be represented by $-a$). Then:

$$(a + -a) - 0 = 0$$

implying that $\overline{a}_n = \overline{-a}_n = \overline{a + (-a)}_n = \overline{0}_n$, showing that $\overline{a}_n$ is indeed the inverse. Usually, a representative between $0 \leq t \leq n-1$ so that $\overline{-a}_n = \overline{t}_n$.

(Associaivity) let $\overline{a}_n, \overline{b}_n, \overline{b}_n \in \mathbb{Z}/n\mathbb{Z}$. We want to show that

$$(\overline{a}_n + \overline{b}_n) + \overline{c}_n = \overline{a}_n + (\overline{b}_n + \overline{c}_n)$$

expanding the definition of the binary operation, we get:

$$\overline{(a+b)+c}_n = \overline{a+(b+c)}_n$$

In particular, we want to verify that:

$$[(a+b)+c] - [a+(b+c)] = kn$$

which by the associativity of the integers, we get that their difference is 0, which implies both sides are equal, completing the proof.

Definition 1.2.2: Integers Modulo n

The set $\mathbb{Z}/n\mathbb{Z}$ forms a group under modulo addition called *integers modulo n*.

Hence, for each $n \in \mathbb{N}_{>0}$, there is a group with with n elements. It has yet to be explored whether the modulo multiplication is a viable binary operation, namely if:

$$(a \bmod n) \cdot (b \bmod n) = (ab) \bmod n \tag{1.1}$$

can form a binary operation. Some more care must be put in:

Example 1.2.3: Not Always Group

Take $n = 4$. The reader should verify that $\overline{1}_4$ is the identity given the binary operation defined in equation (1.1), and by proposition 1.1.15(1) it is unique. However, notice that $\overline{2}_4$ has *no* multiplicative inverse:

$$\overline{0 \cdot 2}_4 = \overline{0}_4 \qquad \overline{1 \cdot 2}_4 = \overline{2}_4 \qquad \overline{2 \cdot 2}_4 = \overline{0}_4 \qquad \overline{3 \cdot 2}_4 = \overline{2}_4$$

Hence, $(\mathbb{Z}/n\mathbb{Z}, \cdot_4)$ is *not* a group. However, if we pick $n = 3$, notice that this forms a group!

The reader may have observed that 3 is a prime and may wonder if is the key difference. This is in fact close: the important part is to choose all elements that are *coprime* with n:

Definition 1.2.4: Euler Totient

Define $\varphi\colon \mathbb{N}_{>0} \to \mathbb{N}$ by

$$\varphi(n) := \#\left\{k \in \mathbb{N} \mid 1 \leq k \leq n \text{ and } \gcd(n,k) = 1\right\}$$

that is, $\varphi(n)$ counts the integers between 1 and n that are coprime to n.

For example, $\varphi(1) = 1$, $\varphi(4) = 2$ (counting 1 and 3), and $\varphi(p) = p - 1$ for a prime p, since the only positive divisors of p are 1 and p. The notation φ is often re-used and so should not be considered a "universal" symbol for the Euler Totient.

Definition 1.2.5: Integer Modulo n With Multiplication

Let $n \in \mathbb{N}_{>0}$ and let $\mathbb{Z}/n\mathbb{Z}$ be the integers modulo n. Take $(\mathbb{Z}/n\mathbb{Z})^\times \subseteq \mathbb{Z}/n\mathbb{Z}$ defined to be

$$(\mathbb{Z}/n\mathbb{Z})^\times = \left\{\bar{t}_n \in \mathbb{Z}/n\mathbb{Z} \;\middle|\; 1 \leq t \leq n \text{ and } \gcd(t,n) = 1\right\}$$

Define the binary operation to be:

$$(a \bmod n) \cdot (b \bmod n) = (ab) \bmod n \tag{1.2}$$

Then $(\mathbb{Z}/n\mathbb{Z})^\times$ is an abelian group with identity $\overline{1}_n$ and with $\varphi(n)$ elements. A class $\overline{a}_n$ lies in $(\mathbb{Z}/n\mathbb{Z})^\times$ exactly when $ax + ny = 1$ for some $x, y \in \mathbb{Z}$, and for $a \neq 0$ this holds exactly when $\gcd(a,n) = 1$.

***Proof*:**
Every integer a can be written as $a = qn + r$ with $q \in \mathbb{Z}$ and $1 \leq r \leq n$, by dividing with remainder and replacing a remainder 0 by n. Two distinct integers r, r' in $\{1, \ldots, n\}$ are not congruent modulo n, since $0 < |r - r'| < n$. So each class has exactly one representative in $\{1, \ldots, n\}$, and $(\mathbb{Z}/n\mathbb{Z})^\times$ has $\varphi(n)$ elements by definition 1.2.4.

Let $a = qn + r$ with $1 \leq r \leq n$. If $\gcd(r,n) = 1$, Bézout's identity ([24, Extended Euclidean Algorithm (Bézout's Identity)]) gives $rx + ny = 1$ for some $x, y \in \mathbb{Z}$, and then $ax + n(y - qx) = 1$. Conversely, if $ax + ny = 1$, then $rx + n(y + qx) = 1$, and a positive common divisor of r and n divides 1, so $\gcd(r,n) = 1$. This proves the criterion for membership. For $a \neq 0$ the same two arguments, applied to a in place of r, show that $ax + ny = 1$ for some x, y exactly when $\gcd(a,n) = 1$.

The product of classes is well defined by [24, Properties Of Modular Arithmetic](3). It is associative and commutative, since $\overline{a}_n(\overline{b}_n\overline{c}_n) = \overline{a(bc)}_n = \overline{(ab)c}_n = (\overline{a}_n\overline{b}_n)\overline{c}_n$ and $\overline{ab}_n = \overline{ba}_n$. If $ax + ny = 1$ and $bx' + ny' = 1$, then multiplying the two equations gives

$$ab(xx') + n(axy' + bx'y + nyy') = 1$$

so $\overline{ab}_n \in (\mathbb{Z}/n\mathbb{Z})^\times$, and the operation restricts to $(\mathbb{Z}/n\mathbb{Z})^\times$. The class $\overline{1}_n$ lies in $(\mathbb{Z}/n\mathbb{Z})^\times$, as $1 \cdot 1 + n \cdot 0 = 1$, and it is an identity, since $\overline{1}_n\overline{a}_n = \overline{a}_n$. Finally, if $ax + ny = 1$, then the same equation read as $xa + ny = 1$ shows $\overline{x}_n \in (\mathbb{Z}/n\mathbb{Z})^\times$, and $\overline{a}_n\overline{x}_n = \overline{ax}_n = \overline{1 - ny}_n = \overline{1}_n$. So $(\mathbb{Z}/n\mathbb{Z})^\times$ is an abelian group with identity $\overline{1}_n$, as claimed.

Cyclic Subgroups

Each group G will have something similar to the integers modulo n, which is worth exploring:

Proposition 1.2.6: Creating a Cyclic Subgroup

Let G be a group and take any $x \in G$. Let $\langle x \rangle = \{x^a | a \in \mathbb{Z}\}$. Then $\langle x \rangle$ is a subgroup of G

***Proof*:**
I would recommend you do this as an exercise!

Let $\langle x \rangle = \{x^a | a \in \mathbb{Z}\}$. We will run through the group-axioms:

1. (associativity) Associativity come's from associativity in G. Let $r, s, t \in \langle x \rangle$, and consider
$$(r + s) + t$$
then since $r, s, t \in G$, and G is associative, we get that
$$(r + s) + t = r + (s + t)$$
and since $r, s, t \in \langle x \rangle$ were arbitrary 3 elements, this holds for any elements of $\langle x \rangle$, completing the proof
2. (identity) We need to show that in $\langle x \rangle$, there exists an element e such that $ex^a = x^a$. Since $0 \in \mathbb{Z}$, $x^0 = e \in \langle x \rangle$. And since for any element $y \in x$, $ey = y$, and every x^a is an element of G, $ex^a = x^a e = x^a$
3. (inverse). Let $y \in \langle x \rangle$ be an element of $\langle x \rangle$. Then by definition, for some $a \in \mathbb{Z}$, $y = x^a$. Then since $-a \in \mathbb{Z}$, $x^{-a} \in \langle x \rangle$, and as we have seen earlier:
$$x^a x^{-a} = x^{a-a} = x^0 = e$$
showing that x^a has an inverse for every a.

Thus, $\langle x \rangle$ is a group. Notice that the binary operation is identical to that of G restricted to $\langle x \rangle$, making $\langle x \rangle$ a subgroup of G, completing the proof.

Given any element $x \in G$, we can form a subgroup $\langle x \rangle \leqslant G$. The reason for it being called a cyclic subgroup comes when we consider a finite group G [12]

Example 1.2.7: Why is $\langle x \rangle$ called Cyclic?
Let G be a finite group (i.e. $|G| = n$ for some $n \in \mathbb{N}$), take $x \in G$ and consider $\langle x \rangle \leqslant G$.

1. Show that $\langle x \rangle$ is finite, that is, there is some $k \in \mathbb{N}$ such that $|\langle x \rangle| = k$
2. Show that for any x^{-a}, where $a \in \mathbb{N}$, there exists some number $b \in \mathbb{N}$ such that
$$x^{-a} = x^b$$

[12] More generally, subgroup $\langle x \rangle \leqslant G$ where $|\langle x \rangle|$ is finite

Conclude that $\langle x \rangle$ can be generated by raising number *only* to the power of positive number, i.e.

$$\{x^a \mid a \in \mathbb{Z}\} = \{x^a \mid a \in \mathbb{N}\}$$

3. show that if $|\langle x \rangle| = k$, then $x^k = e$. Use this fact to show that for any $x^a \in \langle x \rangle$, $a \leq k$.

With these facts, we can visualize a finite cyclic subgroup $\langle x \rangle$ using a graph. Let's say 1 is the identity. If we multiply 1 by x $(1 \cdot x)$, we get x. We can represent this visually with

$$1 \xrightarrow{\cdot x} x$$

if we multiply x by x, we get

$$1 \xrightarrow{\cdot x} x \xrightarrow{\cdot x} x^2$$

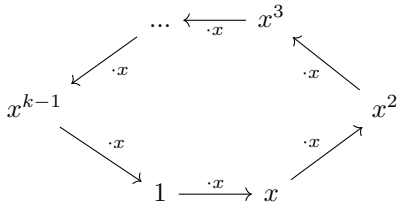
continuing in this fashion, we will make long "chain":

$$1 \xrightarrow{\cdot x} x \xrightarrow{\cdot x} x^2 \xrightarrow{\cdot x} x^3 \xrightarrow{\cdot x} \cdots$$

until we reach 1, at which point we can circle back to it in the graph:

$$1 \xrightarrow{\cdot x} x \xrightarrow{\cdot x} x^2 \xrightarrow{\cdot x} x^3 \xrightarrow{\cdot x} \cdots \longrightarrow x^{k-1} \overset{\cdot x}{\longrightarrow} 1$$

which more conveniently can be written as



producing us a cycle!

Since every element can form a cyclic group, and their structure is characterized by the cardinality of $\langle x \rangle$, we will give a special name to this cardinality:

Definition 1.2.8: Order of an element

For a group G and $x \in G$, define the *order of* x to be the smallest positive integer n such that $x^n = 1$, and denote this integer by $|x|$. In this use, x is said to be of order n. If no positive power of x is the identity, the order of x is defined to be infinity and x is said to be of infinite order

Equivalently, we can also say $|x| = |\langle x \rangle|$, that is, the order of x is the order of the group generated by x

Another word for element with finite order is *torsion elements*. Elements that are not torsion elements

(i.e. with infinite order) are called *torsion-free elements*.

Example 1.2.9: Order of Elements

1. Every element of $\mathbb{Z}, \mathbb{Q}, \mathbb{R}, \mathbb{C}$ under $+$ has infinite except 0. Is the same true for $\mathbb{R}^\times$ (i.e. $\mathbb{R} - \{0\}$ (answer: [a]) Note the identity always has order 1.
2. Can you find the order of the elements in $\mathbb{Z}/p\mathbb{Z}$ for prime p? What if p is not prime? What are the order of the elements of $\mathbb{Z}/60\mathbb{Z}$?

[a] no, take $-1 \in \mathbb{R}^\times$

1.2.10 Note The order of an element depends on *both* the set and the binary operation: -1 has infinite order in $(\mathbb{Z}, +)$ and order 2 in $(\mathbb{R}^\times, \cdot)$ (example 1.2.9).
Also, the word *order* is sometimes used to mean the cardinality of the group.

Every finitely generated abelian group is a direct product of finitely many copies of $\mathbb{Z}$ and of groups $\mathbb{Z}/n\mathbb{Z}$ (theorem 5.1.6).

1.2.2 Symmetric Groups

Let X be any set. As covered in the preliminary chapter, a bijective function $f : X \to X$ always has a (two-sided) inverse function[13], say f^{-1}. Furthermore, function composition is associative, and since the domain and codomain match, the composition of two bijective functions $f, g : X \to X$ gives another bijective function $g \circ f : X \to X$ (as the reader ought to check if they are not convinced). The final property to check for the collection of bijective functions on a set X to form a group is whether there exists a bijective function $\mathrm{id}_X : X \to X$ such that for all bijective functions $f : X \to X$, $\mathrm{id}_X \circ f = f \circ \mathrm{id}_X = f$. The reader should take a moment to think what this function

1.2.11 Moment Taking a moment

The reader may have noticed that this function arises naturally by taking $f \circ f^{-1}$, namely it's the function defined to be $s \mapsto s$. Thus, given any set X, there is a natural group structure on the collection of bijections on X. This set is given a name:

Definition 1.2.12: Symmetric Group

Let X be any set. Then then:

$$S_X := \{f : X \to X \mid f \text{ is a bijection}\}$$

is called the *symmetric group* or *permutation group* of X [with respect to composition $\circ$]. If the cardinality of X is finite, say $|X| = n$, then we denote S_X by S_n.

Other common notations for the symmetric group are:

$$\mathrm{Bij}(A) \qquad \mathrm{Sym}(A) \qquad \mathrm{Aut}_{\mathbf{Set}}(A)$$

[13] This in fact characterizes bijective functions

If X is finite, then the bijections S_n are determined only by the place to which elements map to and not by any property of the elements, making the bijections only dependent on the number of elements and hence the S_n notation being unambiguous. For this reason, it is often assumed that the set over which the bijections are being taken is given by[14]:

$$\{1, 2, ..., n\}$$

A mention on notation is in order: for $f, g \in S_X$, the product fg in S_X is the composite $f \circ g$, so

$$(fg)(x) = f(g(x))$$

A product of permutations is therefore read from right to left: the rightmost factor acts first. This follows the convention for composing functions, which is older than the notation for group operations.

Symmetric groups are rather large. The number of bijections between finite sets grows:

$$|S_n| = n!$$

And hence even for $|X| = 100$, $|S_{100}| \gg 10^{100}$. For $n \in \{1, 2\}$, the bijections are rather simple, namely:

$$f_1 : \{1\} \to \{1\},\ f_1(1) = 1 \qquad g_1 : \{1, 2\} \to \{1, 2\},\ g_1 = \mathrm{id}_{\{1,2\}} \quad g_2(1) = 2\ g_2(2) = 1$$

For larger sets, it may get difficult to track the information a bijection holds. For example, if σ is a bijection on a set with 9 elements:

$$\{1, 2, 3, 4, 5, 6, 7, 8, 9\}$$

then we after the σ permutes, we might get

$$(\sigma(1), \sigma(2), \sigma(3)\sigma(4), \sigma(5), \sigma(6), \sigma(7), \sigma(8), \sigma(9)) = (2, 4, 1, 3, 7, 9, 8, 5, 6)$$

It is cumbersome to write-out the permutation by saying what each value of $\sigma \in S_n$ maps to. Instead, there is the following *matrix notation* as a shorthand:

$$\begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 & 9 \\ 2 & 4 & 1 & 3 & 7 & 9 & 8 & 5 & 6 \end{pmatrix} \tag{1.3}$$

Representing f_1, g_1, g_2 in matrix notation, we would get:

$$\begin{pmatrix} 1 \\ 1 \end{pmatrix} \qquad \begin{pmatrix} 1 & 2 \\ 1 & 2 \end{pmatrix} \qquad \begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix}$$

This is better notation, however we can do even better taking advantage of the fact that a bijection is one-to-one and onto. This implies that every element is only associated with one element and each element must be represented. This allows for the following "cycle" notation. Take equation (1.3), and notice that 1 maps to 2, which maps to 4, which maps to 3, which maps to 1:

$$1 \mapsto 2 \mapsto 4 \mapsto 3 \mapsto 1$$

Furthermore, 5 maps to 7, which maps to 8, which maps to 5:

$$5 \mapsto 8 \mapsto 7 \mapsto 5$$

[14]This set is also sometimes called the *indexing set* and represents labeling each element of some set of size n

Finally, 6 maps to 9, and 9 maps to 6:

$$6 \mapsto 9 \mapsto 6$$

This behavior can be represented via the following *cycle notation*:

$$(1\ 2\ 4\ 3)(5\ 8\ 7)(6\ 9)$$

Each "chain" in the above is called a *cycle*, and the combination of each is called the *cycle representaiton* If a cycle contains n elements, it is called an n-cycle. In the above example, there were not any elements that map to itself. If the element 10 is added to the set and we extend σ to $\overline{\sigma}$so that $\overline{\sigma}(10) = 10$, then we would write $(1\ 2\ 4\ 3)(5\ 8\ 7)(6\ 9)(10)$. By convention, any element that maps to itself is dropped in cycle notation, and hence $\overline{\sigma}$ would in cycle notation also be represented as:

$$(1\ 2\ 4\ 3)(5\ 8\ 7)(6\ 9)$$

Hence, there is an inherent ambiguity in cycle notation on the size of the domain/codomain, since any elements that map to themselves is not taken into accost, however it is usually the element that map to a non-trivial element that are of interest and hence this ambiguity does not effect the theory. Note that:

$$(1\,2\,3) = (3\,1\,2) = (2\,3\,1)$$

Since they all do the same permutation, you are just starting at different positions when converting to the cycle notation.

Example 1.2.13: Cycle Representation

1. Let τ be a bijective function on a set $\{1, 2, 3, 4, 5\}$ such that:

   $$(\tau(1), \tau(2), \tau(3), \tau(4), \tau(5)) = (5, 4, 3, 2, 1)$$

   What is the cycle notation for τ?

2. Cycles can be used to define permutations. Let $X = \{1, 2, 3, 4\}$. Find the bijection τ of X represented by the cycle representation:

   $$(1\ 2\ 3)(2\ 3\ 4)$$

   Notice that the two cycles overlap. To find τ, first find where 1 maps to under the right cycle, and then where the resulting element maps under the left cycle. In function notation, if f is the left cycle and g is the right cycle, then $\tau(1) = f(g(1))$. Proceed with this for all 4 elements. Remember that if an element is not in a cycle, then it is considered to map to itself.

3. Is it true that the functions represented by $(1\,2)(3\,4)$ and $(3\,4)(1\,2)$ are equal, that is:

   $$(1\ 2)(3\ 4) = (3\ 4)(1\ 2)$$

   More generally, do disjoint cycles commute?

To ensure the reader has the right calculation, here is an example of the product of two cycles as reference:

$$(1\ 4\ 5\ 2)(4\ 1\ 3) = (1\ 3\ 5\ 2)$$

Proposition 1.2.14: Disjoint Cycle Decomposition

Let $\sigma \in S_n$. For distinct $a_1, \ldots, a_m \in \{1, \ldots, n\}$ with $m \geq 2$, write $(a_1\ a_2\ \cdots\ a_m)$ for the m-cycle that sends a_i to a_{i+1} for $1 \leq i < m$, sends a_m to a_1, and fixes every other element. Its *support* is $\{a_1, \ldots, a_m\}$. Then σ is a product of cycles whose supports are pairwise disjoint, the empty product being id, and the set of cycles in such a product is determined by σ.

***Proof*:**

Step 1: the orbits of σ. Let $a \in \{1, \ldots, n\}$. The $n+1$ elements $a, \sigma(a), \ldots, \sigma^n(a)$ of an n-element set are not all distinct, so $\sigma^i(a) = \sigma^j(a)$ for some $0 \leq i < j \leq n$, and applying σ^{-i} gives $\sigma^{j-i}(a) = a$. Let m_a be the least positive integer with $\sigma^{m_a}(a) = a$, and put $O_a = \{a, \sigma(a), \ldots, \sigma^{m_a-1}(a)\}$. These m_a elements are distinct, since $\sigma^i(a) = \sigma^j(a)$ with $0 \leq i < j < m_a$ would give $\sigma^{j-i}(a) = a$ with $0 < j - i < m_a$. Because $\sigma^{m_a}(a) = a$, the set O_a contains $\sigma^k(a)$ for every $k \in \mathbb{Z}$, and $\sigma(O_a) = O_a$. If $b \in O_a$, say $b = \sigma^k(a)$, then $O_b \subseteq O_a$ and $a = \sigma^{-k}(b) \in O_b$, so $O_b = O_a$. Hence any two of the sets O_a are equal or disjoint, and together they cover $\{1, \ldots, n\}$. When $m_a \geq 2$, the cycle $c_a = (a\ \sigma(a)\ \cdots\ \sigma^{m_a-1}(a))$ has support O_a, agrees with σ on O_a and fixes every element outside O_a, so $c_b = c_a$ whenever $O_b = O_a$.

Step 2: existence. Let $O_1, \ldots, O_r$ be the distinct sets O_a with at least two elements, and $c_1, \ldots, c_r$ the corresponding cycles. We claim that $\sigma = c_1 c_2 \cdots c_r$. Let $x \in \{1, \ldots, n\}$. If x lies in no O_t, then $O_x = \{x\}$, so $\sigma(x) = x$, and every c_t fixes x. If $x \in O_t$, then every factor c_u with $u \neq t$ fixes both x and $\sigma(x) \in O_t$, since its support is disjoint from O_t, so the product sends x to $c_t(x) = \sigma(x)$.

Step 3: uniqueness. Let $\sigma = d_1 \cdots d_s$ with $d_1, \ldots, d_s$ cycles of pairwise disjoint supports $D_1, \ldots, D_s$. For $x \in D_t$, the argument of Step 2 gives $\sigma(x) = d_t(x) \in D_t$, so $\sigma^k(x) = d_t^k(x)$ for every $k \geq 0$. Writing $d_t = (b_1\ \cdots\ b_m)$, the iterates of d_t carry each b_i through all of $b_1, \ldots, b_m$, so $O_x = D_t$. Then d_t agrees with σ on O_x and fixes everything else, that is, d_t is the cycle attached to O_x in Step 1. An element outside all the D_t is fixed by σ, so its set O_x has one element. Hence the supports $D_1, \ldots, D_s$ are exactly the sets O_a with at least two elements, and $\{d_1, \ldots, d_s\} = \{c_1, \ldots, c_r\}$, completing the proof.

Definition 1.2.15: Cycle Type

Let $\sigma \in S_n$, and write σ as a product of cycles with pairwise disjoint supports (proposition 1.2.14). The *cycle type* of σ is the list of the lengths of these cycles, in increasing order. It is well defined because the cycles are determined by σ.

Example 1.2.16: Cycle-Type

$(1\,2\,4\,3)(5\,7\,8)(6\,9)$ is equivalent to $(6\,9)(5\,7\,8)(1\,2\,4\,3)$ which we can label as a $(2, 3, 4)$-cycle type. If you have a $(1\,2)(3\,4)$ permutation, then it's a $(2, 2)$-cycle type.

A cycle of particular importance are 2-cycles.

Definition 1.2.17: Transposition

2-cycles are usually referred to as ***transpositions***

Using transpositions, any n-cycle can be decomposed into the product of transpositions. For example, take:

$$(1\,2\,4\,3)$$

Then the reader should verify that:

$$(1\,3)(1\,4)(1\,2) = (1\,2\,4\,3)$$

So, the cycle broke up into 3 transpositions. In fact, an n-cycle is a product of $n-1$ transpositions, and of no fewer.

Theorem 1.2.18: n-Cycle Breakdown To Exactly $n-1$ Transpositions

Let σ be an n-cycle in a symmetric group S_N, where $N \geq n$. Then σ can be broken down to the composition of exactly $n-1$ transpositions, and it is not a composition of fewer than $n-1$ transpositions

***Proof*:**
Write $\sigma = (a_1\ a_2\ \cdots\ a_n)$. A decomposition into $n-1$ transpositions is namely:

$$(a_1\ a_2\ \cdots\ a_n) = (a_1\ a_n)(a_1\ a_{n-1})\cdots(a_1\ a_2)$$

To verify this, recall that in a product of cycles the rightmost cycle is applied first. Fix a_i with $2 \leq i \leq n-1$. The factors $(a_1\ a_2), \ldots, (a_1\ a_{i-1})$ fix a_i, the factor $(a_1\ a_i)$ sends a_i to a_1, the factor $(a_1\ a_{i+1})$ sends a_1 to a_{i+1}, and the remaining factors fix a_{i+1}. Hence the product sends a_i to a_{i+1}. Similarly, a_1 is sent to a_2 by the first factor and then fixed by all later factors, and a_n is fixed by every factor except the last, which sends it to a_1. Any other letter appears in no factor and is fixed. The two sides therefore agree, so σ is a product of $n-1$ transpositions.

It remains to show that no decomposition uses fewer. Suppose $\sigma = \tau_k \cdots \tau_1$ with each τ_j a transposition. Form the graph Γ whose vertices are the letters $a_1, \ldots, a_n$ together with every letter appearing in some τ_j, and which has one edge $\{x, y\}$ for each factor $\tau_j = (x\ y)$. We claim that for every vertex x, the letter $\sigma(x)$ lies in the connected component of x. We induct on k. For $k = 0$ the product is the identity and the claim is trivial. If $\sigma = \tau_k \sigma'$ with $\sigma' = \tau_{k-1} \cdots \tau_1$, then $\sigma'(x)$ lies in the component of x by induction, and τ_k either fixes $\sigma'(x)$ or moves it along an edge of Γ, proving the claim. Since σ sends $a_1 \mapsto a_2 \mapsto \cdots \mapsto a_n$, the claim puts all of $a_1, \ldots, a_n$ in a single component. If Γ has m vertices, then with no edges it has m components, and adding an edge decreases the number of components by at most one, so Γ has at least $m-k$ components. On the other hand, since the n distinct letters $a_1, \ldots, a_n$ share one component, Γ has at most $m-n+1$ components. Hence $m-k \leq m-n+1$, that is $k \geq n-1$, as we sought to show.

Corollary 1.2.19: Permutations Are Products Of Transpositions

Every $\sigma \in S_n$ is a product of transpositions, the empty product being id.

***Proof*:**
By proposition 1.2.14, $\sigma = c_1 \cdots c_r$ with each c_t a cycle of some length $m_t \geq 2$. By theorem 1.2.18, each c_t is a product of $m_t - 1$ transpositions. Writing each c_t in this way expresses σ as a product of $(m_1 - 1) + \cdots + (m_r - 1)$ transpositions, as claimed.

The number of transpositions in such a product is not determined by the permutation, since (1 2)(1 2) and the empty product both equal id, but its parity is (theorem 1.2.20(4)).

Theorem 1.2.20: The Sign Of A Permutation

For $\sigma \in S_n$ define the rational number

$$\operatorname{sgn}(\sigma) = \prod_{1 \leq i < j \leq n} \frac{\sigma(j) - \sigma(i)}{j - i}$$

the empty product being 1. Then:

1. $\operatorname{sgn}(\sigma) \in \{1, -1\}$ for every $\sigma \in S_n$
2. $\operatorname{sgn}(\sigma\tau) = \operatorname{sgn}(\sigma)\operatorname{sgn}(\tau)$ for all $\sigma, \tau \in S_n$, and $\operatorname{sgn}(\mathrm{id}) = 1$
3. $\operatorname{sgn}(\tau) = -1$ for every transposition $\tau \in S_n$
4. if $\sigma = \tau_1 \cdots \tau_k$ with transpositions $\tau_1, \ldots, \tau_k$, then $\operatorname{sgn}(\sigma) = (-1)^k$. In particular, any two expressions of σ as a product of transpositions have numbers of factors of the same parity.

***Proof*:**
Let P be the set of two-element subsets of $\{1, \ldots, n\}$. A permutation σ sends each $e = \{a, b\} \in P$ to $\sigma(e) = \{\sigma(a), \sigma(b)\} \in P$, and $e \mapsto \sigma(e)$ is a bijection of P, with inverse $e \mapsto \sigma^{-1}(e)$. For $e = \{a, b\} \in P$ put

$$q_\sigma(e) = \frac{\sigma(b) - \sigma(a)}{b - a}$$

which does not change when a and b are exchanged, so it depends only on e. Each pair $i < j$ gives exactly one $e = \{i, j\} \in P$, so $\operatorname{sgn}(\sigma) = \prod_{e \in P} q_\sigma(e)$.

1. For $e = \{a, b\} \in P$ let $d(e) = |b - a|$, a positive integer, so that $|q_\sigma(e)| = d(\sigma(e))/d(e)$. Reindexing by the bijection $e \mapsto \sigma(e)$ gives $\prod_{e \in P} d(\sigma(e)) = \prod_{e \in P} d(e)$. Hence $|\operatorname{sgn}(\sigma)| = 1$, so $\operatorname{sgn}(\sigma) = \pm 1$.
2. For $e = \{a, b\} \in P$ we have $\tau(a) \neq \tau(b)$, and

$$q_{\sigma\tau}(e) = \frac{\sigma(\tau(b)) - \sigma(\tau(a))}{\tau(b) - \tau(a)} \cdot \frac{\tau(b) - \tau(a)}{b - a} = q_\sigma(\tau(e))\, q_\tau(e)$$

   Taking the product over $e \in P$ and reindexing the first factors by the bijection $e \mapsto \tau(e)$ gives $\operatorname{sgn}(\sigma\tau) = \operatorname{sgn}(\sigma)\operatorname{sgn}(\tau)$. Every factor of $\operatorname{sgn}(\mathrm{id})$ equals 1.
3. Let $\tau = (a\ b)$ with $a < b$. For $e = \{i, j\}$ with $i < j$, the factor $q_\tau(e) = (\tau(j) - \tau(i))/(j - i)$ is negative exactly when $\tau(j) < \tau(i)$. Since τ fixes every $k \neq a, b$, this happens for $e = \{a, b\}$,

and for $e = \{a, k\}$ and $e = \{k, b\}$ with $a < k < b$, because $\tau(a) = b > k = \tau(k)$ and $\tau(k) = k > a = \tau(b)$. It happens for no other e: a pair $\{a, k\}$ or $\{b, k\}$ with $k < a$ or $k > b$ keeps its order, since $\tau(a)$ and $\tau(b)$ lie in $\{a, b\}$, and a pair avoiding a and b is fixed. So exactly $1 + 2(b - a - 1)$ factors are negative, an odd number, and $\operatorname{sgn}(\tau) < 0$. By (1), $\operatorname{sgn}(\tau) = -1$.

4. By (2) and (3) and induction on k, $\operatorname{sgn}(\tau_1 \cdots \tau_k) = \operatorname{sgn}(\tau_1) \cdots \operatorname{sgn}(\tau_k) = (-1)^k$, the case $k = 0$ being $\operatorname{sgn}(\mathrm{id}) = 1$. If also $\sigma = \tau'_1 \cdots \tau'_l$ with transpositions τ'_i, then $(-1)^k = \operatorname{sgn}(\sigma) = (-1)^l$, so k and l have the same parity, completing the proof.

Definition 1.2.21: Parity Of Cycle

Let $\sigma \in S_n$. The *parity* of σ is the parity of the number of factors in any expression of σ as a product of transpositions. Such expressions exist by corollary 1.2.19, and their numbers of factors all have the same parity by theorem 1.2.20(4), so σ has even parity exactly when $\operatorname{sgn}(\sigma) = 1$. A permutation of even parity is called an *even permutation*, and one of odd parity an *odd permutation*.

For example, an m-cycle is a product of $m - 1$ transpositions (theorem 1.2.18), so it is odd when m is even and even when m is odd.

Parity might seem like a weird thing to care about, but it will in fact come in very useful. So useful in fact, that there's a special name for groups based around parity:

Definition 1.2.22: Alternating Group

Let S_n be the permutation group permuting n elements. Then $A_n \subseteq S_n$ is the set of even permutations, that is, $A_n = \{\sigma \in S_n \mid \operatorname{sgn}(\sigma) = 1\}$. By theorem 1.2.20(2), $\mathrm{id} \in A_n$, a product of two even permutations is even, and the inverse of an even permutation σ is even, since $\operatorname{sgn}(\sigma^{-1}) \operatorname{sgn}(\sigma) = \operatorname{sgn}(\mathrm{id}) = 1$. Thus

$$A_n \leqslant S_n$$

Why this group is interesting requires more algebraic tools, so we will leave at defining it for now and get back to it to show it's use.

At the origins of group theory, in the study of polynomial equations, the groups studied were groups of permutations, that is, subgroups of symmetric groups. Cayley proposed abstract axioms for groups in the middle of the nineteenth century, with little impact at the time, and in 1882 Weber gave an axiomatic definition of a finite group [34, Section 27.2]. Later in this book, it shall be shown that every group is isomorphic to a subgroup of a symmetric group (see theorem 4.5.3).

Example 1.2.23: Interesting Exercises on Commutativity

We earlier mentioned how being commutative is a nice property, and that groups which are commutative are quite nice. It was also mentioned that S_n is in a sense the most general group that possible. So, in a sense, if we can establish properties of S_n for particular orders, that can be quite insightful.

Here's one such property worth proving. If we take S_n, $n \geq 3$, then S_n is *not* commutative To prove this, try composing a n-cycles and a 2-cycle which are not disjoint form the left and right and see what you get

Properties of Symmetric Groups

Proposition 1.2.24: Order of element is p iff composition of disjoint p-cycles

Let p be a prime. Then an element of S_n has order p if and only if its cycle decomposition is a product of (commuting) p-cycles.

***Proof*:**

First, if c is an m-cycle $(a_1\ a_2\ \cdots\ a_m)$, then an induction on j shows that $c^j(a_i) = a_{i+j}$, reading subscripts modulo m. In particular $c^m = \mathrm{id}$ while $c^j(a_1) = a_{1+j} \neq a_1$ for $0 < j < m$, so $|c| = m$. Second, if c and d are disjoint cycles, then $cd = dc$. Indeed, injectivity shows each cycle maps the set of letters it moves into itself, so on a letter moved by c both products act as c, on a letter moved by d both act as d, and both fix every other letter.

Now write σ in its cycle representation $\sigma = c_1c_2\cdots c_r$, a product of disjoint cycles with c_t an m_t-cycle and $m_t \geq 2$ (proposition 1.2.14). Since the c_t commute, $\sigma^k = c_1^k c_2^k \cdots c_r^k$ for every k. The permutations $c_1^k, \ldots, c_r^k$ move pairwise disjoint sets of letters, and on every letter that c_t moves the product agrees with c_t^k. Hence $\sigma^k = \mathrm{id}$ if and only if each $c_t^k = \mathrm{id}$, which by the first fact happens if and only if $m_t \big| k$ for every t. Therefore $\sigma^k = \mathrm{id}$ if and only if $\mathrm{lcm}(m_1, \ldots, m_r) \big| k$, that is, $|\sigma| = \mathrm{lcm}(m_1, \ldots, m_r)$.

If the cycle representation of σ consists of p-cycles, then $m_1 = \cdots = m_r = p$, so $|\sigma| = \mathrm{lcm}(p, \ldots, p) = p$. Conversely, if $|\sigma| = p$, then each m_t divides $\mathrm{lcm}(m_1, \ldots, m_r) = p$. Since p is prime and each $m_t \geq 2$, every m_t equals p, so the cycle representation of σ is a product of (commuting) p-cycles.

Note that if p is not prime, the "if and only if" is not necessarily true.

Example 1.2.25: Counter-Examples

1. Take
$$s = (1\,2)(3\,4\,5\,6) = s_1 s_2$$
Then $(s_1 s_2)^4 = 1$, but $|s_1| = 2$.

2. Take $(1\,2)(3\,4\,5) \in S_5$. Then
$$\begin{aligned}(1\,2)(3\,4\,5)^2 &= (3\,5\,4)\\ (1\,2)(3\,4\,5)^3 &= (1\,2)\\ (1\,2)(3\,4\,5)^4 &= (3\,4\,5)\\ (1\,2)(3\,4\,5)^5 &= (1\,2)(3\,5\,4)\\ (1\,2)(3\,4\,5)^6 &= id\end{aligned}$$
but $6 > 5$. Thus, we found a cycle of order 6 in S_5. The trick was to find two disjoint cycles such that the $\mathrm{lcm}(\sigma, \tau) > n$.

Proposition 1.2.26: σ is p-cycle for prime p then σ^k is p-cycle

Let p be a prime and let $\sigma \in S_n$ be a p-cycle. Then σ^k is a p-cycle for every $1 \leq k \leq p-1$.

***Proof*:**
Write $\sigma = (a_0\ a_1\ \cdots\ a_{p-1})$ and read subscripts modulo p. Then $\sigma(a_i) = a_{i+1}$, so induction on k gives $\sigma^k(a_i) = a_{i+k}$, and σ^k fixes every element outside $\{a_0, \ldots, a_{p-1}\}$. Let $1 \leq k \leq p-1$. The elements $a_0, a_k, a_{2k}, \ldots, a_{(p-1)k}$ are distinct: if $a_{ik} = a_{jk}$ with $0 \leq i < j \leq p-1$, then p divides $(j-i)k$, which is impossible because the prime p divides neither $j-i$ nor k, both lying between 1 and $p-1$. So these are all of $a_0, \ldots, a_{p-1}$, and σ^k sends each of them to the next one in the list and $a_{(p-1)k}$ to $a_{pk} = a_0$. Hence $\sigma^k = (a_0\ a_k\ a_{2k}\ \cdots\ a_{(p-1)k})$ is a p-cycle, as claimed.

The proposition fails for cycles of composite length. For the 4-cycle $\sigma = (1\,2\,3\,4)$,

$$\sigma^2 = (1\,3)(2\,4) \qquad \sigma^3 = (1\,4\,3\,2)$$

so σ^3 is again a 4-cycle, while σ^2 is a product of two disjoint 2-cycles. The shape of σ^2 also follows from proposition 1.2.24: $\sigma^2 \neq \mathrm{id}$ and $(\sigma^2)^2 = \sigma^4 = \mathrm{id}$, so σ^2 has order 2 and is a product of disjoint 2-cycles. In general, whenever a power of a cycle has prime order p, proposition 1.2.24 shows that this power is a product of disjoint p-cycles.

Proposition 1.2.27: Conjugation preserves permutation-type

Let $\sigma \in S_n$ and let $\tau = (a_1\ a_2\ \cdots\ a_k)$ be a k-cycle. Then

$$\sigma\tau\sigma^{-1} = (\sigma(a_1)\ \sigma(a_2)\ \cdots\ \sigma(a_k))$$

that is, $\sigma\tau\sigma^{-1}$ is obtained by applying σ to every entry of τ. Consequently, π and $\sigma\pi\sigma^{-1}$ have the same cycle type (definition 1.2.15) for every $\pi \in S_n$.

***Proof*:**
For $1 \leq i \leq k$, $\sigma\tau\sigma^{-1}(\sigma(a_i)) = \sigma(\tau(a_i)) = \sigma(a_{i+1})$, reading a_{k+1} as a_1. If $y \notin \{\sigma(a_1), \ldots, \sigma(a_k)\}$, then $\sigma^{-1}(y) \notin \{a_1, \ldots, a_k\}$ is fixed by τ, so $\sigma\tau\sigma^{-1}(y) = y$. These are the values of the cycle $(\sigma(a_1)\ \cdots\ \sigma(a_k))$, which proves the displayed formula. Now write $\pi = c_1 \cdots c_r$ with cycles c_t of pairwise disjoint supports (proposition 1.2.14). Since the inner factors $\sigma^{-1}\sigma$ cancel,

$$\sigma\pi\sigma^{-1} = (\sigma c_1 \sigma^{-1})(\sigma c_2 \sigma^{-1}) \cdots (\sigma c_r \sigma^{-1})$$

By the formula, $\sigma c_t \sigma^{-1}$ is a cycle of the same length as c_t whose support is the image under σ of the support of c_t. These supports are pairwise disjoint because σ is injective, so by the uniqueness in proposition 1.2.14 the cycle type of $\sigma\pi\sigma^{-1}$ is that of π, as claimed.

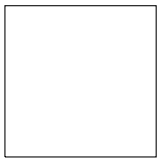
1.2.3 Dihedral Groups

In classical geometry, an n-gon for $n \in \mathbb{N}$ is a 2D shape with all sides of the same length. For example, a triangle is a 3-gon, a square is a 4-gon, a pentagon is a 5-gon, and so forth, while a rectangle and a

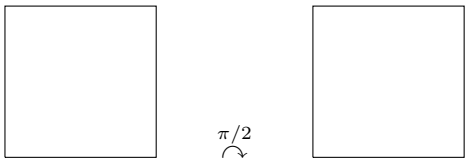
circle are not n-gons. Each of these shapes can be rotated or reflected in such a way that we get back the same shape: these are called the *rigid-motion symmetries* or just *symmetries* of the n-gon. The term *rigid motion* comes from analysis/linear-algebra where a function $f : X \to X$ is called a *rigid motion* if the distance between each point remains the same after the application of the function. Notationally, this would be represented as:

$$\|f(x) - f(y)\| = \|x - y\|$$

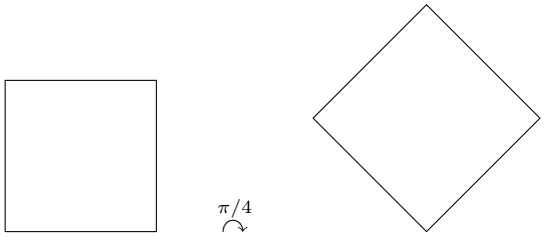
where the $\| - \| : X \to \mathbb{R}_{>0}$ is a special kind of function called the *norm* which defines size and distance information on a space[15]. We shall not use rigid motions explicitly, however we shall insure that when transforming n-gons, their they shall never stretch (ex. a square shall not become a rectangle). We shall further place the restriction that the rigid motions of the n-gons have to map to themselves, this is another way of saying that it is a *symmetry*. For example, if we choose a square (a 4-gon):



and rotated it a quarter turn to the right (90 degrees to the right or $\pi/2$ radians to the right), then we'd get back a square



if we were to rotate it an eighth of a turn, we would not get the exact shape back:



Let's label quarter turns to the right r (for **r**otate). Compounding rotations, the resulting square get's rotated by π, $3\pi/2$, and 2π radians. Each of these can be labeled r^2, r^3, and r^4 respectively. Another symmetry of the square would be reflections, namely there is a vertical, horizontal, and two diagonal reflections: $\sigma_h, \sigma_v, \sigma_{d1}, \sigma_{d_2}$. These are not all the symmetries of a square (for example, we transpose the left-edge with the right edge), however it is the only symmetries that are also *rigid motions* (the reader who is already comfortable with norms form linear algebra is free to prove this fact).

More formally, we may represent any n-gon with n-points, with the restrictions that any symmetry (i.e. bijection) f applied to these n-points must further follow the restrictions that the n-gon these

[15] Norms are introduced formally in *Linear Algebra* [21].

points represent must transform as a rigid-motion (all points must remain equi-distant). This special subset of S_n with this added restriciton is called the *dihedral group*:

Definition 1.2.28: Dihedral Groups

Let S_n be the symmetric group on n elements and take the subset $D_n \subseteq S_n$ for which the permutations further follow the restriction of rigid-motions (so that these points represent an n-gon).Then D_n is called the *dihedral group*.

In order for the set D_n to be a group, one ought to prove this is indeed the csae

***Proof*:**

1. Identity comes from the identity function $\mathrm{id}_n \in D_n$ since it is certainly a rigid motion.
2. The inverse come from the fact that if σ is a rigid motion, it's inverse is as well
3. The set is associative since its a collection of functions
4. The composition of two rigid motions is a rigid motion, and hence the binary operation is well-defined

Proposition 1.2.29: Properties of D_n

Let $G = D_n$. Let r represent rotation by $2\pi/n$, and s represent a reflection about some fixed axis. Then:

1. $1, r, \ldots, r^{n-1}$ are all distinct and $r^n = 1$, so $|r| = n$
2. $|s| = 2$
3. $s \neq r^i$ for any i
4. $sr^i \neq sr^j$ for all $0 \leq i, j \leq n-1$ with $i \neq j$, so

$$D_n = \left\{1, r, r^2, \cdots, r^{n-1}, s, sr, sr^2, \cdots, sr^{n-1}\right\}$$

   i.e. each element can be written uniquely in the form $s^k r^i$ for some $k = 0$ or 1, and $i = 0, \cdots, n-1$

5. $rs = sr^{-1}$. In particular this shows D_n is not abelian!
6. $r^i s = sr^{-i}$ (use induction)

***Proof*:**
Label the vertices of the n-gon $1, 2, \ldots, n$ counterclockwise, and read all labels modulo n, so that vertex $n+1$ means vertex 1. Recall that a product of symmetries applies the rightmost factor first.

In a regular n-gon, the distance between two distinct vertices depends only on how many edges apart they lie along the polygon, and it is smallest exactly when they are adjacent. An element

$g \in D_n$ preserves all distances between vertices, so g sends adjacent vertices to adjacent vertices, that is, $g(k+1) = g(k)+1$ or $g(k+1) = g(k)-1$ for every k. Moreover the sign is the same for every k: if $g(k+1) = g(k)+1$ and $g(k+2) = g(k+1)-1$ for some k, then $g(k+2) = g(k)$, contradicting injectivity (here we use $n \geq 3$). Hence every $g \in D_n$ has the form

$$g(k) = a + \epsilon k \pmod{n} \qquad a \in \{1, \ldots, n\},\ \epsilon \in \{+1, -1\}$$

and there are at most $2n$ such maps. For $n \geq 3$ no map has both forms, since $a + k = a' - k$ for all k would force $n \mid 2$. The sign ϵ of g is therefore well-defined. Composing two such maps gives $g(h(k)) = (a_g + \epsilon_g a_h) + \epsilon_g \epsilon_h k$, so the sign of gh is $\epsilon_g \epsilon_h$.

The rotation r sends each vertex to the next one counterclockwise, so $r(k) = k+1$ and more generally $r^i(k) = k+i$, of sign $+1$. The reflection s produces a mirror image, so it reverses the counterclockwise order of the vertices. Its sign is therefore -1, and by the affine form above, $s(k) = c - k$ for some fixed c.

1. $r^i(n) = i$, so $1, r, \ldots, r^{n-1}$ act differently on the vertex n and are pairwise distinct, while $r^n(k) = k + n = k$, so $r^n = 1$. Hence $|r| = n$.
2. $s^2(k) = c - (c - k) = k$, so $s^2 = 1$, and $s \neq 1$ since s has sign -1 while the identity has sign $+1$. Hence $|s| = 2$.
3. s has sign -1 and every r^i has sign $+1$, so $s \neq r^i$ for any i.
4. If $sr^i = sr^j$, then multiplying by s^{-1} on the left gives $r^i = r^j$, so $i = j$ by (1). Also $sr^i \neq r^j$ for all i, j, since sr^i has sign -1 and r^j has sign $+1$. Hence the $2n$ listed elements are pairwise distinct. Conversely, take any $g \in D_n$, say $g(k) = a + \epsilon k$. If $\epsilon = +1$, then $g = r^a$. If $\epsilon = -1$, then $(sr^{c-a})(k) = s(k + c - a) = a - k = g(k)$, so $g = sr^{c-a}$. Hence the list exhausts D_n, and each element is uniquely of the form $s^k r^i$ with $k = 0$ or 1 and $i = 0, \ldots, n-1$.
5. $(rs)(k) = r(c - k) = c - k + 1$ and $(sr^{-1})(k) = s(k-1) = c - k + 1$, so $rs = sr^{-1}$. Since $n \geq 3$, $r^{-1} \neq r$ by (1), so $rs = sr^{-1} \neq sr$ by the uniqueness in (4), and D_n is not abelian.
6. We induct on i, the case $i = 1$ being (5). If $r^i s = sr^{-i}$, then

$$r^{i+1}s = r(r^i s) = r(sr^{-i}) = (rs)r^{-i} = (sr^{-1})r^{-i} = sr^{-(i+1)}$$

completing the induction.

In particular, we know now any element in D_n can be written as $s^k r^i$, with $k = 0 \vee 1$ and $0 \leq i \leq n-1$. For example, if $n = 12$:

$$(sr^9)(sr^6) = sr^9 sr^6 = sr^9 r^{-6} s = sr^3 s = s^2 r^{-3} = r^{-3} = r^9$$

This group will come up very often since it's a particularly simple.

Proposition 1.2.30: Order of Dihedral Group

Let D_n be a dihedral group. Then:

$$|D_n| = 2n$$

***Proof*:**
By proposition 1.2.29(4), the $2n$ elements $1, r, \ldots, r^{n-1}, s, sr, \ldots, sr^{n-1}$ are pairwise distinct and every element of D_n appears in this list. Hence $|D_n| = 2n$.

Some authors like to put the order of the Dihedral group in the subscript, that is they write D_{2n}. This convention is rather confusing given the original intuition behind Dihedral groups, and so the D_n notation is used instead for the clarity it brings.

1.2.4 Quaternion Group

The Quaternion (or Hamiltonian) group is a group with 2 fundamental interpretation:

1. The Quaternions are a generalization of the complex numbers. We can think of the fact that $i^2 = -1$ to be the key property which distinguishes the complex numbers from other number systems. If we take $C = \{-1, 1, i, -i\} \subseteq \mathbb{C}$, then

   $$\langle C | i^2 = -1 \rangle$$

   will form the group which represents the rotation of $\pi/2$ in the cartesian plane. William Hamilton around the 1840s started a project to try an make 3D numbers $(1, i, j)$, the same way $\mathbb{C}$ can be thought of as 2D $(1, i)$. He could not manage to make them 3D, but realized the key generalization is making them "4D" $(1, i, j, k)$. The story goes that Hamilton had this flash of inspiration near a bridge in 1843 and wrote down the idea under that very bridge. Today, there is a plaque commemorating the moment of inspiration.

2. It got discovered later that Quaternions efficiently capture the idea of rotating a 3D object in 3D space. At the beginning of vector calculus, some mathematician considered using quaternion instead of vectors as the pedagogical tool used to introduce the concept of 3D rotation, but pedagogical history went with the vector notation. In quantum mechanics, the Pauli spin matrices of 1927, multiplied by $-i$, satisfy the quaternion relations, so quaternions also describe *spin*, a fundamental property of sub-atomic particles.

Let $Q_8 = \{1, -1, i, -i, j, -j, k, -k\}$, with the multiplication given by the following rules. The element 1 is the identity. The element -1 commutes with every element and satisfies $(-1)^2 = 1$, and for $a \in \{1, i, j, k\}$ the product $(-1)a$ is written $-a$. Finally,

$$i^2 = j^2 = k^2 = -1 \qquad ij = k \qquad ji = -k \qquad jk = i \qquad kj = -i \qquad ki = j \qquad ik = -j$$

These rules determine every product, for instance $(-i)j = (-1)(ij) = -k$, and fig. 1.1 lists them all. Hamilton summarized them as $i^2 = j^2 = k^2 = ijk = -1$. The multiplication is associative: sending ± 1 to $\pm I$ and

$$i \mapsto \begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix} \qquad j \mapsto \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \qquad k \mapsto \begin{pmatrix} 0 & i \\ i & 0 \end{pmatrix}$$

(and $-a$ to the negative of the image of a) identifies Q_8 with eight complex 2×2 matrices satisfying the same rules, and matrix multiplication is associative.

This group is interesting, for it is the second non-abelian group of order 8 in this book, D_4 being the other. Both come from geometry, D_4 from the symmetries of a square and Q_8 from Hamilton's quaternions, but they have different multiplication tables:

Example 1.2.31: Comparing D_4 and Q_8

Element	1	-1	i	$-i$	j	$-j$	k	$-k$
1	1	-1	i	$-i$	j	$-j$	k	$-k$
-1	-1	1	$-i$	i	$-j$	j	$-k$	k
i	i	$-i$	-1	1	k	$-k$	$-j$	j
$-i$	$-i$	i	1	-1	$-k$	k	j	$-j$
j	j	$-j$	$-k$	k	-1	1	i	$-i$
$-j$	$-j$	j	k	$-k$	1	-1	$-i$	i
k	k	$-k$	j	$-j$	$-i$	i	-1	1
$-k$	$-k$	k	$-j$	j	i	$-i$	1	-1

Figure 1.1: Q_8 Multiplication Table

While D_4 has

Element	e	a	a^2	a^3	x	ax	a^2x	a^3x
e	e	a	a^2	a^3	x	ax	a^2x	a^3x
a	a	a^2	a^3	e	ax	a^2x	a^3x	x
a^2	a^2	a^3	e	a	a^2x	a^3x	x	ax
a^3	a^3	e	a	a^2	a^3x	x	ax	a^2x
x	x	a^3x	a^2x	ax	e	a^3	a^2	a
ax	ax	x	a^3x	a^2x	a	e	a^3	a^2
a^2x	a^2x	ax	x	a^3x	a^2	a	e	a^3
a^3x	a^3x	a^2x	ax	x	a^3	a^2	a	e

Figure 1.2: D_4 Multiplication Table

which the reader will notice is different

We will be able to show that two groups are not the same once we introduce the concept of isomorphisms (section 1.4).

Exercise 1.2.1

1. Show that there exists 4 homomorphisms $\varphi : Q_8 \to \mathbb{R}^\times$

1.2.5⋆ Matrix Groups

⋆ *Optional. This subsection is an aside; nothing later in the book depends on it.*

This section assumes either some linear algebra knowledge or that you have internalized some of [24, Matrices] (however, some of these collection of objects goes beyond that section). All prerequisite knowledge assumed in this section shall be covered in great detail in chapter 13 and section 13.2. For the inner-product background behind the orthogonal and unitary groups, see *Linear Algebra* [21]. Let's first recall the notation we use to describe the collection of matrices:

Definition 1.2.32: Matrix Group Under Addition

Let G be an abelian group. Then The set $M_{n\times m}(\mathbb{F})$ forms a group under element-wise operation of G, namely:

$$A * B \equiv (a_{ij} * b_{ij})_{ij}$$

If the reader is familiar with linear algebra, they may notice this group is not that interesting: it is equivalent to the group G^{nm}[16]. Sticking to $G = k$ is a field (notice that all fields are Abelian groups under addition), the more interesting binary operation that can be performed on matrices is matrix multiplication, since it combines elements in much more interesting ways. As you might recall, matrix multiplication requires dimensions to line up properly, so for associative to hold, we will have the dimensions be the same. Furthermore, the identity must be I, the matrix with 1's on the diagonal, and zero's everywhere else. If there are matrices that multiply and *do not* get to I, then they cannot be in the group. Hence, the subset of invertible matrices from $M_n(k)$ must be selected in order to make a group, which the reader may recall from linear algebra implies we are taking the set for which all matrices have a non-zero determinant

Definition 1.2.33: General Linear Group

Let F be a field. Then

$$GL_n(F) = \{A \mid A \text{ is an } n \times n \text{ matrix with entries from } F \text{ and } \det(A) \neq 0\}$$

We will also name 3 special subgroups of GL_n since it comes up often enough. Check that these are indeed groups:

Definition 1.2.34: Special Linear Group

$$SL_n(F) = \{A \mid A \text{ is an } n \times n \text{ matrix with entries from } F \text{ and } \det(A) = 1\}$$

If $F \in \{\mathbb{R}, \mathbb{C}\}$, then geometrically these can be thought of as the volume and orientation preserving functions.

Definition 1.2.35: Orthogonal Group

$$O_n(F) = \left\{A \;\middle|\; A \text{ is an } n \times n \text{ matrix with entries from } F \text{ and } A^tA = AA^t = I\right\}$$

[16] We can show this once we get to isomorphisms (definition 1.4.12)

In this group, the transformations are *distance-preserving*, which as an exercise the reader can check that for $O_2(\mathbb{R})$, all transformations are rotations and reflections across lines passing through the origin[17]. Limiting to orthogonal matrices with determinant 1 gives us the set of all rotations:

Definition 1.2.36: Special Orthogonal Group

$$SO_n(F) = \{A \in O_n(F) \mid \det(A) = 1\}$$

If the underlying field is clear, we sometimes write $SO(n)$.

For a finite field F with q elements, the group $GL_n(F)$ has $(q^n - 1)(q^n - q) \cdots (q^n - q^{n-1})$ elements (proposition 13.2.20). The count uses the linear algebra of chapter 13.

Exercise 1.2.2

1. for $n > 1$, $\mathbb{Z}/n\mathbb{Z}$ is not a group under multiplication (hint: think of identity)
2. prove that every element in a finite group has finite order
3. if $x \in G$, G a group, and $|x| = n$, then $x^{-1} = x^{n-1}$.
4. $x \in G$, G a group. Then $|x| = |x^{-1}|$. (hint: $(x^{-1})^n = (x^n)^{-1}$ for every $n \in \mathbb{N}$, and the case of infinite order follows from the finite one)
5. $x, g \in G$, G a group. Then $|x| = |g^{-1}xg|$. Deduce that $|ab| = |ba|$ for all $a, b \in G$. (hint: simply assign an order to x, and expand $g^{-1}xg$. things will cancel out nicely. For the second part, start with ab, and multiply like so $babb^{-1}$. By what you have shown, this has the same order).
6. $x \in G$, $|x| = n < \infty$. If $n = st$ for some positive integers s and t, prove that $|x^s| = t$. (hint: how many times do you need to multiply x^s to get to 1)
7. if $x^2 = 1$ for all $x \in G$ then G is abelian.
8. if x is an element of the group G, then $H = \{x^n | n \in \mathbb{Z}\}$ is a subgroup (cf. show for all $h, k \in H$, $hk \in H$, $h^{-1} \in H$) of G (called the *cyclic subgroup* of G generated by x)
9. $A \times B$ is abelian if and only if A and B is abelian
10. Let $A \times B$ be the *direct product* group formed by the cartesian product of the group A, B as shown in example 1.1.5. Show that $(a, 1)$ and $(1, b)$ commute. Deduce that the order of (a, b) is the least common multiple of $|a|$ and $|b|$.
11. any finite group G of even order contains an element of order 2. (hint: let $t(G)$ be the set $\left\{g \in G | g \neq g^{-1}\right\}$. Show that $t(G)$ has even number of elements and every non-identity element of $G - t(G)$ has order 2).
12. Challenging exercise. Let A be a set containing numbers. Define the following

$$\mathcal{A} = \bigoplus_{i=1}^{\infty} A$$

[17] Higher dimensional analogues of "rotation" and "reflection" come from taking this interpretation in lower dimensions and claiming it generalizes to higher dimensions

to be the set of infinitely long tuples in which *all but finitely many* entries are zero. For example, if $A = \{1, 2, 3\}$, then

$$(0, 0, 1, 0, 2, 0, 0, 0, 0, 1, 3, 3, 2, 0, ...)$$

where the ... means there are zeros going on forever. Then this tuples is in $\mathcal{A}$. On the other hand

$$(1, 1, 1, 1, 1, ...)$$

where the ... means the ones are going on forever is not in $\mathcal{A}$.

If G is a group, let

$$\mathcal{G} = \bigoplus_{i=1}^{\infty} G$$

be the set of tuples in which all but finitely many entries are the identity e, and define the operation on $\mathcal{G}$ to be pointwise:

$$(a_1, a_2, a_3, ...) \cdot (b_1, b_2, b_3, ...) = (a_1 b_1, a_2 b_2, a_3 b_3, ...)$$

Show that $\mathcal{G}$ along with this operation is a group.

13. Let p be a prime number. Show that $(\mathbb{Z}/p\mathbb{Z})$ is a cyclic group.

14. Let $n = 8$. What is $(\mathbb{Z}/8\mathbb{Z})^\times$? What if $n = 9$. Conclude that in general, $(\mathbb{Z}/n\mathbb{Z})^\times$ is not necessarily cyclic, but it can be.

15. Show that the rotation groups of the regular tetrahedron, cube and dodecahedron are isomorphic to A_4, S_4 and A_5 respectively, and that a platonic solid and its dual have the same rotation group. Conclude that the five platonic solids give three finite subgroups of $SO(3)$, up to conjugacy, that are neither cyclic nor dihedral.

16. The *Heisenberg group* over a field F is the set of upper triangular 3×3 matrices over F with every diagonal entry equal to 1. Show that it is a subgroup of $GL_3(F)$ and that it is not abelian.

17. (If you know the dot product) Recall that if v, w are vectors then the dot product can be represented as $v \cdot w = v^T w$ where v^T is the transpose. Show that matrices in $O_n(\mathbb{R})$ preserve the dot product (namely $Av \cdot Aw = v \cdot w$).

1.3 Generators and Relations

(When dealing with numbers, they can represent anything, for example 5 apples, 5 cars, 5 textbooks, and so forth. However, when we defined the dihedral group. it was specifically linked to the rotation and reflections of n-gons. In this section, we will show a way to abstract away this dependency on a concrete object to define a group).

Notice that D_n can be written with r and s (reflections and rotations). Similarly, $\langle x \rangle \leqslant G$ is an element from which we can find every other element of $\langle x \rangle$. In a sense, they *generated* the group. This notion will be treated rigorously with free groups later (section 2.3.3), but intuitively this idea makes sense. For now, here how we will define them

Definition 1.3.1: Generators

A subset S of elements of a group G with the property that every element of G can be written as a (finite) product of elements of S and their inverses is called the set of *generators* of G. If $x \subseteq G$ generates x, then we write $\langle x \rangle$. Similarly, if $x, y \in G$ are needed to generate G, we write $\langle x, y \rangle$. More generally, if $A \subseteq G$ generates G we write $\langle A \rangle$.

Example 1.3.2: Generators

1. An easy example would be 1 for $\mathbb{Z}$, since every $x \in \mathbb{Z}$ could be written as adding 1 or it's inverse. So $\mathbb{Z} = \langle 1 \rangle$.
2. By property (4) for D_n, $S = \{r, s\}$ is a set of generators of D_n, so $D_n = \langle r, s \rangle$.

Later, we will see for finite groups that we do not need the inverse operation to generate finite groups. For any set of generators, we have a set of relationships that they satisfy[18]. For D_n, they are

$$r^n = 1, \qquad s^2 = 1, \qquad rs = sr^{-1}$$

Why are these the relations? Because any additional relation between elements of D_n may be derived from these three (example 3.2.4 proves this once presentations are defined formally). Note that there can be more than one set of relations for a group (we will see later that D_n has another set of relations that can be used).

Definition 1.3.3: Presentation of a group

if some group G is generated by a subset S and there is some collection of relations, say $R_1, \cdots, R_m$, which are all equations with elements from $S \cup \{1\}$, such that any relation among the elements of S an be deduced from these, we shall call these generators and relations a *presentation* of G, and write

$$G = \langle S | R_1, \cdots, R_m \rangle$$

Example 1.3.4: Presentation

1. Show that the presentation:

   $$\langle r, s | r^n = 1, s^2 = 1, rs = sr^{-1} \rangle$$

   represents the Dihedral group D_n
2. Let A be any set. Show that

   $$\langle A | ab = 1, \forall a, b \in A \rangle$$

   is group. In particular, show that from any set, we can define the trivial group. Sections 1.2.2 and 2.3.3 give two other, non-trivial, ways of creating groups from arbitrary sets.

[18] With only one exception. When a set of generators has no relationships, we call the group generated by that set a *free group*

1.3.5 Prelude to Word Problem The relations of a presentation can be arbitrary equations, so it may be hard to decide when two elements of the group are equal, or to find the order of the group. For example,

$$\langle x, y \big| x^2 = y^2 = (xy)^2 = 1 \rangle$$

is a presentation of a group of order 4, while the similar-looking presentation

$$\langle x, y \big| x^3 = y^3 = (xy)^3 = 1 \rangle$$

defines an infinite group (example 3.2.5).

It might be tempting to only work with presentations, or to try and come up with every possible group working with generators. A considerable amount of time has been poured into this problem, and absolutely incredibly, it turns out that you can come up with presentations which are *insoluble*:

Example 1.3.6: Insoluble Presentation
The group representation by this monstrous presentation:

$$\begin{array}{lll}
\langle\ a,b,c,d,e,p,q,r,t,k & | & \\
p^{10}a = ap, & pacqr = rpcaq, & ra = ar, \\
p^{10}b = bp, & p^2adq^2r = rp^2daq^2, & rb = br, \\
p^{10}c = cp, & p^3bcq^3r = rp^3cbq^3, & rc = cr, \\
p^{10}d = dp, & p^4bdq^4r = rp^4dbq^4, & rd = dr, \\
p^{10}e = ep, & p^5ceq^5r = rp^5ecaq^5, & re = er, \\
aq^{10} = qa, & p^6deq^6r = rp^6edbq^6, & pt = tp, \\
bq^{10} = qb, & p^7cdcq^7r = rp^7cdceq^7, & qt = tq, \\
cq^{10} = qc, & p^8ca^3q^8r = rp^8a^3q^8, & \\
dq^{10} = qd, & p^9da^3q^9r = rp^9a^3q^9, & \\
eq^{10} = qe, & a^{-3}ta^3k = ka^{-3}ta^3 & \rangle
\end{array}$$

Figure 1.3: Group for which we cannot write out what the elements are

has elements which we *cannot* write out!

The details of this is are well beyond this course, but it's existence is still interesting to know about

https://en.wikipedia.org/wiki/Word_problem_for_groups

Proposition 1.3.7: Generators For Symmetric Group

$$S_n = \langle (i, i+1) \mid 1 \leq i \leq n \rangle$$

***Proof*:**
This proof relies on this key observation:

$$(a\,b)(a\,c)(a\,b)^{-1} = (a\,b)(a\,c)(a\,b) = (b\,c)$$

that is, we shifted a to b (so in a sense if $\sigma = (a\,b)$ and $\tau = (a\,c)$, $\sigma\tau\sigma^{-1} = (\sigma(a), \sigma(b))$).

With that in mind, we will proceed by *induction*. If $n = 2$, then S_2 is generated by this set, so assume it generate S_{n-1} and consider S_n. Take some cycle with n in it:

$$(a_1\, a_2 \ldots a_k\, n)$$

which we have shown earlier can be decomposed as

$$(a_1\, a_2)(a_1\, a_3)\cdots(a_1\, a_k)$$

All the $(a_1\, a_j)$ are in G by the induction hypothesis (they are not in because of the generating set because we do not know if $(a_1\, a_j) = (i\, i+1)$). What's left to show is whether $(a_1\, n)$ is in G. To do this, we use the conjugation fact we brought up earlier:

$$(a_1\, n) = (a_1\, n-1)(n-1\, n)(a_1\, n-1)$$

which since $(n-1\, n) \in S$, and the other two terms are in G by induction hypothesis, it means that $(a_1\, n) \in G$. And Since any cycle can be broken down into the aforementioned form (where if we have a cycle, we can move k to the back, since its the same cycle that represents it), any cycle can be broken down in this fashion, and thus, we completed the proof

As a bonus, if you want to represent such a group:

$$\langle \sigma_i | \sigma_i^2 = 1\,, \sigma_i\sigma j = \sigma_j\sigma i \text{ for } |i-j| > 1\,, (\sigma_i, \sigma_{i+1})^3 = 1 \rangle$$

Though this is not necessary to know, but it can be good to know if you want to show something to be isomorphic to the symmetric group (since if something is isomorphic, it has the same relations).

https://en.wikipedia.org/wiki/Symmetric_group#Generators_and_relations.

There is a way to prove these once we introduce either quotient groups or conjugacy classes (in the group-action section). Thus, we will not prove these here for the nicest way to prove them is once we have those extra tools.

Exercise 1.3.1

1. exercise 7, 16, simply to know how to solve a concrete problem.
2. exercise ,17,18.
3. Do 10 to see that you can find all symmetries
4. Take $(1, i)$ for $2 \leq i \leq n$, that is, the set where you fix one element, then the 2nd position is all other elements.
5. This one is probably the most useful theoretically. You can generate a symmetric group given only 2 cycles!! Take any n-cycles from S_n, and a 2-cycle of adjacent elements in the n-cycle!

1.4 Homomorphisms and Isomorphisms

When comparing D_4 and Q_8 in example 1.2.31, we resolved at looking at the Cayley tables to distinguish the two. However, this becomes very tedious to do for larger groups, and downright

impossible for infinite groups. In this section, we will make precise the notion of comparing groups.

What does it mean to compare groups? If we were comparing tuples, $(a, b) \neq (b, a)$, since the order matters. On the other hand, if we are comparing sets, then $\{a, b\} = \{b, a\}$ since the order does not matter, as long as the elements are the same. For groups, the star of the show is the binary operation. If two groups G and H shared *exactly* the same binary operation, they'd be identical in terms of the binary operation. We often encounter groups with different elements, but the binary operation acts identically in both groups.

Example 1.4.1: "Identical" groups

Let $G = \{0, 1, 2\}$ and $H = \{a, b, c\}$, and define $(G, +_G)$ and $(H, +_H)$ with binary operation's which have Cayley tables:

$+_G$	0	1	2
0	0	1	2
1	1	2	0
2	2	0	1

and

$+_H$	a	b	c
a	a	b	c
b	b	c	a
c	c	a	b

Then notice we can replace 0 with a, 1 with b, and 2 with c, and the two tables would be identical. In other words, the operation $+_H$ acts exactly the same as $+_G$.

It is sometimes restrictive that the two binary operations act *exactly* the same. sometimes that are similar, but the two groups share a different structure. The take-away is that when working with homomorphisms and isomorphissm, what we care about is the binary operation between two groups behave similarly.

1.4.1 Homomorphisms

Definition 1.4.2: Homomorphism

Let $(G, \bullet_G)$, and $(H, \bullet_H)$ be groups. A map $\psi : G \to H$ such that for all $x, y \in G$,

$$\psi(x \bullet_G y) = \psi(x) \bullet_H \psi(y)$$

is called a *homomorphism*

Remember that the left side is the domains operation, and the right side is the co-domains operation. A good way of remembering the rule of a homomorphism is that the resulting image $\varphi(G) = \{\varphi(g) \mid g \in G\}$ is also a group (this is proved in proposition 1.4.4). Another way remembering the rule of a homomorphism is that if there is a homomorphism, then in some way the binary operation must be *preserved*. It is not necessarily perfectly preserved: it's possible that only some of it is preserved. The following examples of homomorphisms may help in gaining an intuition

Example 1.4.3: Homomorphisms

1. Take the integers with the usual addition. Let e represent even integers; o represent odd integers. Then we know that adding even with even is even, even with odd is odd, odd with even is odd, and odd with odd is even:

$$e+e=e,\ o+o=e,\ o+e=e+o=o$$

On the other hand, if we take 1 and -1 and find all possible outcomes of multiplication, we get

$$1\cdot 1=1,\quad (-1)\cdot(-1)=1,\quad 1\cdot(-1)=(-1)\cdot 1=-1$$

Notice how 1 can represent even integers, and -1 represent odd. In this way, multiplication on -1 and 1 "captures" the idea of adding even and odd numbers.

Let's formalize this idea by making an explicit homomorphism. First, $\{-1,1\}$ is a group under usual multiplication $\cdot$. The element 1 is the identity since $1\cdot 1=1$, $-1\cdot 1=-1$. Furthermore -1 it's own inverse ($-1\cdot -1=1$). For associativity, most cases are essentially immediate to check, with the following case being the non-trivial case:

$$(-1\cdot -1)\cdot -1=1\cdot -1=-1\cdot 1=-1\cdot(-1\cdot -1)$$

Any case with 1 in the question be more easily checked since $a\cdot 1=1\cdot a=1$, making the cases of $1\cdot -1\cdot -1$, $-1\cdot 1\cdot -1$, $1\cdot 1\cdot -1$, and so forth easy to check. Thus, $\{-1,1\}$ is a group.

Next, let $\psi:\mathbb{Z}\to\{-1,1\}$ be a function such that

$$\psi(x)=\begin{cases}1 & x \text{ is even}\\ -1 & x \text{ is odd}\end{cases}$$

Then if e represents an arbitrary even number and o an arbitrary odd number:

$$\psi(e+e)=\psi(e)=1=1\cdot 1=\psi(e)\cdot\psi(e)$$
$$\psi(o+o)=\psi(o)=1=(-1)\cdot(-1)=\psi(o)\cdot\psi(o)$$
$$\psi(o+e)=\psi(e+o)=\psi(o)=-1=-1\cdot 1=1\cdot -1=\psi(e)\cdot\psi(o)=\psi(o)\cdot\psi(e)$$

Showing that ψ is indeed a homomorphism.

Generalizing the previous result, notice that our group $\{-1,1\}$ is homomorphic to $\mathbb{Z}/2\mathbb{Z}$ (remember that by default, if we say $\mathbb{Z}/2\mathbb{Z}$ is a group, we mean $(\mathbb{Z}/2\mathbb{Z},+)$ with modulo 2 addition). In particular

$$f:\{-1,1\}\to\mathbb{Z}/2\mathbb{Z}\qquad f(1)=0,\ f(-1)=1$$

Then one should verify that that f is indeed a homomorphism. It is further a bijection, meaning that every element of the group $\mathbb{Z}/2\mathbb{Z}$ represents exactly one element from $\{-1,1\}$, showing we can inter-change the two.

Now, take $\varphi : \mathbb{Z} \to \mathbb{Z}/n\mathbb{Z}$, $\varphi(x) = [x]$. Then φ is a homomorphism. Indeed:

$$\begin{aligned} \varphi(x+y) &= [x+y] \\ &= [x] + [y] && \text{property of } \mathbb{Z}/n\mathbb{Z} \\ &= \varphi(x) + \varphi(y) \end{aligned}$$

If $n = 2$, then this will be exactly the homomorphism we defined earlier.

2. If you have done any calculus or analysis, you might have seen before that:

$$\frac{d}{dx}(f+g) = \frac{d}{dx}(f) + \frac{d}{dx}(g) \qquad \int f + g\,dx = \int f, dx + \int g\,dx$$

so $\frac{d}{dx}$ and $\int$ both satisfy this condition! Are they homomorphisms? In fact, they are, with the group being all differentiable functions for the 1st one, and all integrable functions for the second one. These groups require more build-up within Calculus or Analysis, and so we will not cover these operation in more detail.[a].

3. For those with knowledge of the exponential function e^z (which we will write as exp), then there is a natural way of relating the addition and multiplication operation of $\mathbb{C}$ and $\mathbb{C}^\times$ (as well as $\mathbb{R}$ and $\mathbb{R}^\times$). The map $\exp : \mathbb{C} \to \mathbb{C}^\times$ is surjective group homomorphism. Restricting the domain and co-domain, we also get an injective (but not surjective) map $\exp : \mathbb{R} \to \mathbb{R}^\times$.

4. Verify that there is no bijective homomorphism from D_4 to Q_8. At the moment, this may be a tedious task, it will become easier to show later with some more properties of bijective homomorphisms.

5. Let $G_1 \times G_2$ be the direct product of two groups G_1 and G_2 as defined in example 1.1.5(4). The map $\pi_1 : G_1 \times G_2 \to G_1$ and $\pi_2 : G_1 \times G_2 \to G_2$ defined by

$$\pi_1((g_1, g_2)) = g_1 \qquad \pi_2((g_1, g_2)) = g_2$$

are homomorphisms.

6. Let G, H be any group. Then $\varphi : G \to H$ defined by $\varphi(g) = e_H$ where e_H is the identity of H is a homomorphism. This is called the *trivial homomorphism*.

7. (If you know about $\mathbb{C}$) Show that the subset $\{1, i, -1, -i\} \subseteq \mathbb{C}$ is a subgroup of $\mathbb{C}$ and that there is a bijective group homomorphism $\varphi : \mathbb{Z}/4\mathbb{Z} \to \{1, i, -1, -i\}$.

[a] However, $\frac{d}{dx}(fg)$ is not necessarily equal to $\left(\frac{d}{dx}(f)\right)\left(\frac{d}{dx}(g)\right)$ and $\int fg\,dx$ is not necessarily equal to $\left(\int f\,dx\right)\left(\int g\,dx\right)$. The first is studied in a course on Lie Groups and Lie Algebras, while the other is studied in a course in Measure Theory, in particular L^p spaces and Hölder's inequality

We will often simplify notation since lots of information is usually deduced from context. We will drop the binary operation and simply write $\varphi(ab) = \varphi(a)\varphi(b)$, given that it's clear that when we write ab the binary operation happens in the domain, and that when we werite $\varphi(a)\varphi(b)$ the binary operation happens in the codomain.

One more interpretation of a homomorphism that is a little more abstract but is enlightening in it's own way is that a homomorphism f is a function that "commutes" with the binary operation: If $*$ represent any binary operation (of any group, not just one group), then f is a homomorphism if and

only if

$$f \circ * = * \circ f$$

Concretely, if we let $*$ for a group G be represented as $*_G$ and $*$ for a group H be represented as $*_H$, then

$$f \circ *_G = *_H \circ (f, f)$$

or, if written in terms of elements and with the standard notation for binary operation:

$$f(x *_G y) = f(x) *_H f(y)$$

. This idea can be captured in a diagram like so:

$$\begin{array}{ccc} G \times G & \xrightarrow{\varphi \times \varphi} & H \times H \\ {\scriptstyle *_G(a,b)}\downarrow & & \downarrow{\scriptstyle *_H(\varphi(a),\varphi(b))} \\ G & \xrightarrow[\varphi(ab)]{} & H \end{array}$$

which can be read as "start at $G \times G$, then either go down then right, or go right then down, and both paths should give the same result", i.e. $\varphi(ab) = \varphi(a)\varphi(b)$.

Now that homomorphisms are defined, you might ask yourself "given two groups G, and H, how do I know there is a homomorphism between them?" There is always a trivial answer to this, the homomorphism sending everything to the identity always exist [19]. This is called the *trivial homomorphism* (you should check that it's indeed a homomorphism):

$$\varphi : G \to \{e\} \qquad \varphi(g) = e$$

So the better question would be "How can I find the non-trivial homeomorphisms between these two groups?" This in some cases can be a very difficult question. If we just look at them as set-functions from G to H, then if there are n objects in one category and m in the other, there are minimum n^m possible functions! That's a ton to check. We can deduce properties of homomorphisms which will let us diminish that number:

Proposition 1.4.4: Properties of Homomorphisms

Let G, H be groups and φ be a homomorphism. Then:

1. $\varphi(1_G) = 1_H$
2. $\varphi(g^{-1}) = \varphi(g)^{-1}$
3. $\varphi(g^n) = \varphi(g)^n$
4. the *kernel* $\ker \varphi = \{g \in G \mid \varphi(g) = 1_H\}$ is a subgroup of G (named again in definition 2.2.1)
5. the *image* $\mathrm{Im}(\varphi) = \varphi(G) = \{\varphi(g) \mid g \in G\}$ is a subgroup of H.
6. if $\psi : H \to K$ is also a homomorphism, then $\psi \circ \varphi : G \to K$ is a homomorphism, i.e., the composition of homomorphisms is a homomorphism.

[19] Elaborated in proposition 1.4.5

***Proof*:**

1. $\varphi(1_G) = \varphi(1_G 1_G) = \varphi(1_G)\varphi(1_G)$, so $\varphi(1_G) = \varphi(1_G)\varphi(1_G)$. Then by proposition 1.1.17
$$\varphi(1_G)\varphi(1_G)^{-1} = \varphi(1_G)\varphi(1_G)\varphi(1_G)^{-1}$$
the left hand side becomes 1_H, and the right hand side becomes $\varphi(1_G)1_H = \varphi(1_G)$, giving us
$$\varphi(1_G) = 1_H$$
2. By (1), $\varphi(g)\varphi(g^{-1}) = \varphi(gg^{-1}) = \varphi(1_G) = 1_H$. Multiplying on the left by $\varphi(g)^{-1}$ gives $\varphi(g^{-1}) = \varphi(g)^{-1}$.
3. For $n \geq 0$ we induct on n. The case $n = 0$ is (1), and $\varphi(g^{n+1}) = \varphi(g^n g) = \varphi(g^n)\varphi(g) = \varphi(g)^n\varphi(g) = \varphi(g)^{n+1}$. For $n < 0$, g^n is the inverse of g^{-n}, so (2) and the case $-n > 0$ give $\varphi(g^n) = \varphi(g^{-n})^{-1} = (\varphi(g)^{-n})^{-1} = \varphi(g)^n$.
4. $1_G \in \ker\varphi$ by (1). If $x, y \in \ker\varphi$, then $\varphi(xy) = \varphi(x)\varphi(y) = 1_H$ and $\varphi(x^{-1}) = \varphi(x)^{-1} = 1_H$ by (2). So $\ker\varphi$ contains 1_G and is closed under products and inverses, and associativity holds because it holds in G. Hence $\ker\varphi$ is a subgroup of G (definition 1.1.7).
5. $1_H = \varphi(1_G) \in \varphi(G)$ by (1). For $x, y \in G$, both $\varphi(x)\varphi(y) = \varphi(xy)$ and $\varphi(x)^{-1} = \varphi(x^{-1})$ (by (2)) lie in $\varphi(G)$, and associativity holds because it holds in H. Hence $\varphi(G)$ is a subgroup of H (definition 1.1.7).
6. For $x, y \in G$, $(\psi \circ \varphi)(xy) = \psi(\varphi(x)\varphi(y)) = \psi(\varphi(x))\psi(\varphi(y)) = (\psi \circ \varphi)(x)(\psi \circ \varphi)(y)$.

These propositions can even help us deduce whether [non-trivial] homomorphisms exist. For example, is there a non-trivial homomorphism from $\mathbb{Z}/2\mathbb{Z}$ to $\mathbb{Z}$? Let's say $\varphi : \mathbb{Z}/2\mathbb{Z} \to \mathbb{Z}$ were such a map. By the above proposition we have $\varphi(0) = 0$. Let's say $\varphi(1) = n$. Then
$$0 = \varphi(0) = \varphi(1 + 1) = \varphi(1) + \varphi(1) = n + n$$
giving us $0 = n + n$, which re-arranging is $-n = n$. The only integer equal to its negative is 0, hence $n = 0$. This shows that there is no non-trivial map from $\mathbb{Z}/2\mathbb{Z}$ to $\mathbb{Z}$.

The particular structure of the group that is being preserved depends on the operation and the cardinality of the domain and co-domain. In general, to get an idea of what part of the group is being preserved, you will have to check by doing a couple of example.

Important examples

I will go over some homomorphisms to build a deeper understanding of the concept

The first one is deceptively boring: take $\varphi : G \to \{e\}$ such that $\varphi(g) = e$. The function φ is a homomorphism, since for any $g, h \in G$
$$\varphi(g \cdot h) = e = e \cdot e = \varphi(g) \cdot \varphi(h)$$

This seemingly boring homomorphism has a very important property: for *any* group G, there *always* exists a *unique* homomorphism from G to $\{e\}$. This is also true for homomorphisms with $\{e\}$ as the domain! This fact is important enough to make a proposition:

Proposition 1.4.5: Homomorphism To/From Trivial Group

Let G be any group and I represent the trivial group. Then there always exists a unique homomorphism

$$\varphi : G \to I \qquad \psi : I \to G$$

to and from I

***Proof*:**
Let e_G be the identity of G and e_I the only element of I.

Homomorphisms $G \to I$. The constant map $\varphi(g) = e_I$ is a homomorphism by the computation above. Every function $G \to I$ takes the value e_I at each $g \in G$, so φ is the only function from G to I, and in particular the only homomorphism.

Homomorphisms $I \to G$. Define $\psi(e_I) = e_G$. Then $\psi(e_I e_I) = \psi(e_I) = e_G = e_G e_G = \psi(e_I)\psi(e_I)$, so ψ is a homomorphism. If $\psi' \colon I \to G$ is any homomorphism, then $\psi'(e_I) = e_G$ by proposition 1.4.4(1). A function on I is determined by its value at e_I, so $\psi' = \psi$, completing the proof.

So why is this property important? What can we do with it? In this case with groups, it is not used often [20]. what's more important is the concept that there *always* exists a homomorphism from any group, and that it is a *unique* homomorphism. These properties for the right homomorphisms will come back big-time when we study *quotient groups* (chapter 3): For *every* homomorphism there *always* exists a *unique* group (called the quotient group) which will "represent" the homomorphism.

Thus, we will give a name to groups with this property:

Definition 1.4.6: Initial And Final Groups

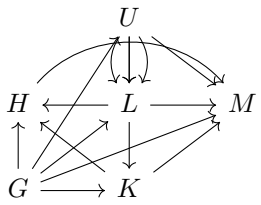
Let G be a group.

1. If for every group H there exists a unique homomorphism $\varphi : G \to H$, then we say that G is an *initial group* (or *initial object*)
2. If for every group H there exists a unique homomorphism $\varphi : H \to G$, then we say that G is a *final group* (or *final object*)
3. A final group is also called a *terminal group* (or *terminal object*), and a group that is both initial and final is called a *zero object*

The reason for this vocabulary is clearer if you think of a graph where the edges are homomorphisms

[20] It is used in the context of Category theory, where $\{e\}$ is both the initial and final group in the category of **Group**, making it an intuitive first step towards introducing universal properties. More on this in chapter 8

and the nodes are groups:



(1.4)

You can see how there is only one homomorphism from G to any other group, and M has only one homomorphism from every other group. Thus, G is initial, and M is final, that is, terminal.

1.4.7 Note An object *can* be both initial and final. The trivial group is one such example

For now, we do not have the tools to properly analyze this result. We will gain those in chapter 2, and revisit this idea in chapter 3.

Redefining Subgroup

Moving back towards things we can prove, I will present an important way of using homomorphisms to *define* subgroups, using the fact that subgroups as a similar structure to the original groups, and homomorphisms preserve some structure of a group. It might already be somewhat intuitive that if $H \leqslant G$, then H is pretty similar to G, especially that it must have the same group operation. With homomorphisms, we can formalize this idea:

Proposition 1.4.8: Defining Sub Group With Homomorphism

Let $(G, \cdot)$ and $(H, \bullet)$ be two groups, and $H \subseteq G$. Then H is a subgroup if and only if there exists a inclusion map $i : H \to G$ which is also a homomorphism

***Proof*:**
We will show that this definition is equivalent to definition 1.1.7, that is, that $(H, \cdot)$ is a group. Let $i : H \to G$ be an homomorphism which is also an inclusion map. All of these will come from proposition 1.4.4.

1. ($e_G = e_H$) We need to how that H has the same identity. This comes for free with proposition 1.4.4
2. (inverse) We need to show that for any $a \in (H, \cdot)$, then there is a $b \in (H, \cdot)$ such that $ab = ba = e_G$. Since $i(a^{-1}) = a^{-1}$, then a^{-1} is a candidate. Indeed, since

$$i(e_H) = i(aa^{-1}) = i(a)i(a^{-1}) = i(a)i(a)^{-1} = aa^{-1} = e_G$$

   then a^{-1} will indeed be our desired inverse
3. (associativity) Similarly to how we have down the other two, but the proof is a little more tedious to carry out. One can do this to try proving associativity at least once in their career.

This is probably one of my favourite proposition using homomorphisms: really showing how homomorphism captures the idea that an algebraic structure is preserved by defining subgroups with an inclusion map that is also a homomorphism.

Collection of Homomorphisms

Next, we will address a little more complicated object. We might ask ourselves in what other way can we use homomorphisms to study groups? If a homomorphism preserves group structure, what possible homomorphisms can we have between a group and itself? What would that give us? Let's first label the set of all such homomorphisms:

Definition 1.4.9: Set of Endomorphisms

1. Given a group G, an endomorphism is a homomorphism from an object to itself: $\varphi : G \to G$.
2. Let $\text{End}(G) := \{\varphi : G \to G \mid \forall a, b \in G,\ \varphi(ab) = \varphi(a)\varphi(b)\}$ represent the set of all endomorphisms on G

The parity map $\psi\colon \mathbb{Z} \to \{-1, 1\}$ of example 1.4.3 is not an element of $\text{End}(\mathbb{Z})$. Its codomain is a different group, and read as a map $\mathbb{Z} \to \mathbb{Z}$ it sends 0 to 1, so it does not preserve addition (proposition 1.4.4(1)). The doubling map $x \mapsto 2x$ is an element of $\text{End}(\mathbb{Z})$, and its image is the set $2\mathbb{Z}$ of even integers, which is also the set of integers that ψ sends to 1. Being the image of a homomorphism, $2\mathbb{Z}$ is a *subgroup* of $\mathbb{Z}$ (proposition 1.4.4(5)).

If we can figure out some structure on $\text{End}(G)$, we can start getting a good idea on what are all sub-structures in G. More excitingly, if $\text{End}(G)$ can be proven to be a group (with composition, $\circ$, as the operation), then we can use all the technics of group theory to study it. However, this turns out not to be the case. The problem comes when trying to prove the existence of an inverse, which does not necessarily exist (try proving it's a group!).

So you might ask if there is still a way to make $\text{End}(G)$ into a group. Perhaps with a nice operation, we can define a [non-trivial] group structure? The one that is usually first attempted is defining addition *pointwise*, that is

$$f, g \in \text{End}(G) \qquad f + g \equiv (f + g)(x) = f(x) + g(x)$$

Where $\equiv$ means we are showing how the ne function "$f + g$" is defined by how it processes every element. With the operation defined, to check that $f + g$ is in $\text{End}(G)$, we need to check if it's a homomorphism, that is

$$(f + g)(x + y) = (f + g)(x) + (f + g)(y)$$

and going through the computation:

$$\begin{aligned}(f + g)(x + y) &= f(x + y) + g(x + y)\\ &= f(x) + f(y) + g(x) + g(y)\end{aligned}$$

and here is were we get stuck, since G is not necessarily commutative.

However, if G is commutative, then $f(y)$ and $g(x)$ commute and the computation continues:

$$\begin{aligned} &= f(x) + f(y) + g(x) + g(y) \\ &= f(x) + g(x) + f(y) + g(y) \\ &= (f+g)(x) + (f+g)(y) \end{aligned}$$

showing that $(f+g)$ is indeed a homomorphism!! Thus, $f+g$ is a well-defined operation. From here, you can check the axioms of a group to check that $\mathrm{End}(G)$ is indeed a group. It's even an abelian group! This is an example of how being commutativity changes how a group effects.

1.4.10 Remark In fact, more is true. If the domain of the functions are arbitrary sets, then $\mathrm{Hom}(A, G)$ is *still* an abliean group! This fact will be very useful to us much further don the line, in Parts II and III, and in *Core Representation Theory* [17].

Dihedral Groups As Groups Of Rotations

The dihedral group D_n of definition 1.2.28 consists of symmetries of a regular n-gon in the plane, and it is also isomorphic to a subgroup of rotations-reflections of 3-dimensional space. A reflection of the polygon is realized in space by turning the polygon over, a rotation by π about an axis in its plane. Let $n \geq 3$ and $\theta = 2\pi/n$, and let

$$R = \begin{pmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{pmatrix} \qquad S = \begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{pmatrix}$$

be the rotation by θ about the z-axis and the rotation by π about the x-axis. Both satisfy $M^tM = MM^t = I$, by $\cos^2\theta + \sin^2\theta = 1$, and both have determinant 1, so they lie in $SO_3(\mathbb{R})$ (definition 1.2.36).

Proposition 1.4.11: D_n Embeds In $SO_3(\mathbb{R})$

Let $n \geq 3$, and let $r, s \in D_n$ be as in proposition 1.2.29. Then $\rho(s^k r^i) = S^k R^i$, for $k \in \{0, 1\}$ and $0 \leq i \leq n-1$, defines an injective homomorphism $\rho\colon D_n \to SO_3(\mathbb{R})$.

***Proof*:**
By the angle addition formulas, R^i is the rotation by $i\theta$ about the z-axis, so $R^n = I$ and $R^i \neq I$ for $0 < i < n$. Also $S^2 = I$, and multiplying out gives $SRS = R^{-1}$, hence $SR^iS = (SRS)^i = R^{-i}$ and $R^iS = SR^{-i}$ for every integer i. By proposition 1.2.29(4), ρ is defined on all of D_n. Since $s^2 = 1$, $r^n = 1$, $S^2 = I$ and $R^n = I$, the formula $\rho(s^a r^b) = S^a R^b$ holds for all integers a and b.

Homomorphism. Let $g = s^k r^i$ and $h = s^l r^j$ with $l \in \{0, 1\}$. By proposition 1.2.29(6), $r^i s^l = s^l r^{(-1)^l i}$, so $gh = s^{k+l} r^{(-1)^l i + j}$. In the same way, $R^iS = SR^{-i}$ gives

$$\rho(g)\rho(h) = S^k R^i S^l R^j = S^{k+l} R^{(-1)^l i + j} = \rho(gh)$$

Injectivity. Every R^i fixes the vector $e_3 = (0, 0, 1)^t$, while $SR^ie_3 = Se_3 = -e_3$, so no R^i equals

any SR^j. If $R^i = R^j$ or $SR^i = SR^j$ with $0 \leq i \leq j \leq n-1$, then $R^{j-i} = I$, so $i = j$. Hence the $2n$ matrices $S^k R^i$ are distinct, and ρ is injective, as claimed.

1.4.2 Isomorphisms

While homomorphisms preserve some algebraic structure, isomorphisms preserve the *entire* algebraic structure. An isomorphism can be thought of as the generalization of the concept of equality: If two groups are isomorphic, they are identical (as groups). They will only differ in the symbols which are used in representing them. The idea of an isomorphism extends far beyond group theory. Here's a fun linguistic example: if I say

one, two, three, four, five

or

un, deux, trois, quatre, cinq

Then both are capturing the idea of $1, 2, 3, 4, 5$, except in one I'm writing it in English, and in the other in French. Those two sentences ostensibly capture the same [numerical] ideas, but with different symbols

When trying to capture this idea in groups, we have to consider what it means for two group to be identical. First, the two groups would need to be of the same size, if not we cannot say they are the same. Therefore, the homomorphism must also be bijective. Next, all the relation between the two groups need to be the same. The parity map of example 1.4.3 preserves the operation but is not injective, since it sends every even integer to 1.

Definition 1.4.12: Isomorphism

The map $\psi : G \to H$ is called an *isomorphism* and G and H are said to be *isomorphic* or of the same *isomorphism* type, written $G \cong H$ if

1. ψ is a homomorphism
2. ψ is a bijection

1.4.13 Note Definition 1.4.12 defines an isomorphism as a bijective homomorphism. In a more categorical framework one would instead ask for a homomorphism $\varphi\colon H \to G$ with $\varphi \circ \psi = \mathrm{id}_G$ and $\psi \circ \varphi = \mathrm{id}_H$. For groups the two definitions agree. Such a φ is an inverse function of ψ, so ψ is a bijection. Conversely, if ψ is a bijective homomorphism, then its inverse function ψ^{-1} is a homomorphism, since $\psi(\psi^{-1}(x)\psi^{-1}(y)) = xy$ for $x, y \in H$, that is, $\psi^{-1}(xy) = \psi^{-1}(x)\psi^{-1}(y)$.[a]

[a] The analogous statement fails in topology, where a continuous bijection need not have a continuous inverse: the identity map from a two-point set with the discrete topology to the same set with the indiscrete topology is continuous, and its inverse is not.

When two groups are isomorphic, it means they only differ by the symbols we choose to use, they are in practice identical. The trivial isomorphism is the identity map $\mathrm{id} : G \to G$. We can also form an

equivalence class of isomorphic groups: take $\mathcal{G}$ to the set of groups and let the equivalence relation be isomorphisms.

Example 1.4.14: Isomorphism Examples

Let V_4 (*Vierergruppe*, meaning 4-group, sometimes called Klein 4-group), is the group that represents the symmetries of a [non-square] rectangle. You can define it as

$$V_4 = \langle a, b | a^2 = b^2 = (ab)^2 = e \rangle$$

Show that:

$$V_4 \cong \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$$

Here $\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$ carries the componentwise operation of the direct product in example 1.1.5.

A less obvious example of a group isomorphism is $\psi : \mathbb{R} \to \mathbb{R}^+$, $\exp(x) = e^x$. e^x is invertible, and thus bijective, and $e^{x+y} = e^x e^y$, preserving group structure.

Proposition 1.4.15: Properties Of Isomorphisms

Let $f\colon G \to H$ and $g\colon H \to K$ be homomorphisms.

1. If f is an isomorphism, then its inverse function $f^{-1}\colon H \to G$ is an isomorphism.
2. If f and g are isomorphisms, then $g \circ f$ is an isomorphism.
3. If $g \circ f$ and f are isomorphisms, then g is an isomorphism.
4. If $g \circ f$ and g are isomorphisms, then f is an isomorphism.

***Proof*:**

1. The inverse function f^{-1} is a bijection. For $x, y \in H$,
$$f(f^{-1}(x)f^{-1}(y)) = f(f^{-1}(x))f(f^{-1}(y)) = xy$$
and applying f^{-1} gives $f^{-1}(x)f^{-1}(y) = f^{-1}(xy)$. So f^{-1} is a homomorphism.
2. The composite $g \circ f$ is a homomorphism by proposition 1.4.4(6), and a composite of bijections is a bijection.
3. By (1), f^{-1} is an isomorphism, so $g = (g \circ f) \circ f^{-1}$ is an isomorphism by (2).
4. By (1), g^{-1} is an isomorphism, so $f = g^{-1} \circ (g \circ f)$ is an isomorphism by (2), completing the proof.

An interesting exercise to show is if A and B are sets of the same cardinality, and S_A and S_B are two symmetric groups, then $S_A \cong S_B$. A bijection $f\colon A \to B$ relabels the elements, and $\sigma \mapsto f \circ \sigma \circ f^{-1}$ is the isomorphism.

Going back for a moment to when we tried to define a group $(\mathrm{End}(G), \circ)$, the fatal flaw was that not all homomorphisms have inverses. However, isomorphisms can be reversed: if $\varphi : G \to H$ is an isomorphism, then its inverse function $\varphi^{-1}\colon H \to G$ is an isomorphism (proposition 1.4.15), with

$\varphi^{-1} \circ \varphi = \mathrm{id}_G$ and $\varphi \circ \varphi^{-1} = \mathrm{id}_H$. In particular, if $\varphi : G \to G$ is an isomorphism, then φ^{-1} is an inverse of φ for composition, and the isomorphisms from G to itself form a group.

Definition 1.4.16: Automorphism

1. if $\varphi : G \to G$ is an isomorphism from a group to itself, then φ is called an ***automorphism***
2. Let $\mathrm{Aut}(G) := \{\varphi : G \to G \mid \varphi \text{ is an automorphism}\}$ be the collection of all automorphisms of G

Proposition 1.4.17: $\mathrm{Aut}(G)$ is a Group

Let G be a group and $\mathrm{Aut}(G)$ be the collection of automorphisms. Then $(\mathrm{Aut}(G), \circ)$ forms a group

***Proof*:**
First, $\circ$ is a binary operation on $\mathrm{Aut}(G)$. If $\varphi, \psi \in \mathrm{Aut}(G)$, then $\varphi \circ \psi$ is a homomorphism by proposition 1.4.4(6), and it is a bijection because a composite of bijections is a bijection. So $\varphi \circ \psi \in \mathrm{Aut}(G)$ (definition 1.4.16), and the axioms of definition 1.1.3 hold:

1. (associativity) For $\varphi, \psi, \chi \in \mathrm{Aut}(G)$ and $x \in G$, both $((\varphi \circ \psi) \circ \chi)(x)$ and $(\varphi \circ (\psi \circ \chi))(x)$ equal $\varphi(\psi(\chi(x)))$, so $(\varphi \circ \psi) \circ \chi = \varphi \circ (\psi \circ \chi)$.
2. (identity) The identity map id_G is a bijective homomorphism, so $\mathrm{id}_G \in \mathrm{Aut}(G)$, and $\mathrm{id}_G \circ \varphi = \varphi \circ \mathrm{id}_G = \varphi$ for every $\varphi \in \mathrm{Aut}(G)$.
3. (inverses) Let $\varphi \in \mathrm{Aut}(G)$ and let $\varphi^{-1} : G \to G$ be its inverse function, which is a bijection. For $x, y \in G$ put $a = \varphi^{-1}(x)$ and $b = \varphi^{-1}(y)$. Then $\varphi(ab) = \varphi(a)\varphi(b) = xy$, so
$$\varphi^{-1}(xy) = ab = \varphi^{-1}(x)\varphi^{-1}(y)$$
and φ^{-1} is a homomorphism. Hence $\varphi^{-1} \in \mathrm{Aut}(G)$, and $\varphi \circ \varphi^{-1} = \varphi^{-1} \circ \varphi = \mathrm{id}_G$.

So $(\mathrm{Aut}(G), \circ)$ is a group, as claimed.

Example 1.4.18: Automorphisms
Simplest example are identity maps. Other examples come form Linear Algebra with bijective linear functions (since $\varphi(a)\varphi(b) = \varphi(ab)$ for a linear function, it *is* an example)

1.4.19 Foreshadowing The concept of automorphisms will not be limited to groups. Many algebraic and non-algebraic structures will have the concept of automorphism defined. Most excitingly, the set of automorphisms for *any* of these objects *always* forms a group! The automorphism groups of field extensions, for example, are the subject of Galois theory (chapter 20).

Exercise 1.4.1

1. In example 1.1.5(5) we showed that $(\mathbb{Z}, +)$ and $(\mathbb{Z}, +_2)$ have different identities and inverses. Show that they are in fact isomorphic. This means that they are not a good example of two different groups on the same set since algebraically, they are identical.
2. Show that $D_4 \ncong \mathbb{Z}/8\mathbb{Z}$. With this knowledge, conclude that we can define two different groups on the set of 8 elements, giving a proper example of two different groups structure on the same underlying set.
3. (Challenging) Try to come up with two different structure on the underlying set of $\mathbb{Z}$ (Hint: recall the structure of $\bigoplus_i^\infty (\mathbb{Z}/2\mathbb{Z})$ and show there is a bijection between $\mathbb{Z}$ and this set, so that you can define a group structure on $\mathbb{Z}$ relative to this bijection. Is this group isomorphic to $\mathbb{Z}$?)

In the following section, we will explore classifying all possible groups on n elements where n has small order (larger orders of n will come when we have more tools)

First Use of Isomorphisms

In mathematics, there's a whole set of theorems that we use to show that classes of mathematical objects, like groups, are isomorphic to each other or not. Such theorems are called *classification theorems*. As a simple example, we can classify all non-abelian groups of order 6, in particular:

1.4.20

any non-abelian group of order 6 is isomorphic to S_3

Holding the theorem to be true for now, we can conclude that $D_3 \cong S_3$, and $GL_2(\mathbb{F}_2) \cong S_3$ without needing to find an explicit map. Not all groups of order 6 are isomorphic to S_3, since $\mathbb{Z}/6\mathbb{Z}$ is abelian and S_3 is not. However, all abelian groups of order 6 are isomorphic to $\mathbb{Z}/6\mathbb{Z}$! Thus, any group of order 6 is isomorphic to either S_3 or $\mathbb{Z}/6\mathbb{Z}$. This classification is the case $p = 2$, $q = 3$ of theorem 4.9.1 (example 4.9.2).

In general, it is difficult, sometime unfeasible, to find an explicit isomorphism mapping. Here are some necessary, but not sufficient, conditions for isomorphisms:

Proposition 1.4.21: Isomorphism Conditions

Let $\psi : G \to H$ be an isomorphism. Then:

1. $|G| = |H|$
2. G is abelian if and only if H is abelian
3. for all $x \in G$, $|x| = |\psi(x)|$

***Proof*:**
By definition 1.4.12, ψ is a bijective homomorphism.

1. The bijection $\psi \colon G \to H$ shows that G and H have the same cardinality, so $|G| = |H|$.

2. Suppose G is abelian and let $h, h' \in H$. Since ψ is surjective, $h = \psi(x)$ and $h' = \psi(y)$ for some $x, y \in G$, and then

$$hh' = \psi(x)\psi(y) = \psi(xy) = \psi(yx) = \psi(y)\psi(x) = h'h$$

so H is abelian. Conversely, suppose H is abelian and let $x, y \in G$. Then $\psi(xy) = \psi(x)\psi(y) = \psi(y)\psi(x) = \psi(yx)$, and since ψ is injective, $xy = yx$. So G is abelian.

3. Let $x \in G$ and $n \in \mathbb{Z}$. By proposition 1.4.4(1) and (3), $\psi(1_G) = 1_H$ and $\psi(x^n) = \psi(x)^n$. Since ψ is injective, $x^n = 1_G$ holds exactly when $\psi(x^n) = \psi(1_G)$, that is exactly when $\psi(x)^n = 1_H$. So the positive integers n with $x^n = 1_G$ are the positive integers n with $\psi(x)^n = 1_H$. If there are none, then x and $\psi(x)$ both have infinite order, and otherwise the smallest such n is both $|x|$ and $|\psi(x)|$ (definition 1.2.8). Hence $|x| = |\psi(x)|$, completing the proof.

Example 1.4.22

This proposition shows that S_3 and $\mathbb{Z}/6\mathbb{Z}$ are not isomorphic, because one is abelian and the other is not. Additionally, $(\mathbb{R} - \{0\}, \times)$ and $(\mathbb{R}, +)$ are not isomorphic, because $-1 \in \mathbb{R}^\times$ has order 2, and no element in $(\mathbb{R}, +)$ has order 2

A homomorphism out of a group given by a presentation is determined by where the generators go, and any choice of images that satisfies the relations gives one. Free groups make this precise (theorem 3.2.3).

Example 1.4.23

1. Take D_n and it's presentation $\langle r, s | r^n = s^2 = 1, sr = r^{-1}s \rangle$. Now take $n = mk$, $k \geq 3$ and consider d_{2k}. Then

$$\psi : d_n \to d_{2k} \qquad \text{by} \qquad \psi(r) = r_1, \text{ and } \psi(s) = s_1$$

is a homomorphism. In other words, the operation between Dihedral groups which are "multiples" are the same (think of a triangle and hexagon. two hexagon rotations is 2 triangle rotation, angle wise).

2. $D_3 = S_3$: $D_3 \subseteq S_3$ (definition 1.2.28), D_3 has order 6 (proposition 1.2.30), and so does S_3.

Proposition 1.4.24: Isomorphism properties

Let $\psi\colon G \to H$ be a homomorphism and $\Psi\colon G \to H$ an isomorphism.

1. $\psi(1_G) = 1_H$
2. $\psi(x^n) = \psi(x)^n$ for all $x \in G$ and $n \in \mathbb{Z}$
3. $|\Psi(x)| = |x|$ for all $x \in G$, so G and H have the same number of elements of order n for each $n \in \mathbb{Z}^+$. If $G \neq 1$, the statement $|\psi(x)| = |x|$ fails for the trivial homomorphism $G \to \{1\}$, which sends every element to 1.
4. G is abelian if and only if H is abelian
5. If S generates G (definition 1.3.1) and $\psi'\colon G \to H$ is a homomorphism with $\psi'(s) = \psi(s)$ for every $s \in S$, then $\psi' = \psi$. That is, a homomorphism is determined by its values on a set of generators.

***Proof*:**

1. This is proposition 1.4.4(1).
2. This is proposition 1.4.4(3).
3. By proposition 1.4.21(3), $|\Psi(x)| = |x|$ for every $x \in G$. So Ψ maps the set of elements of G of order n injectively into the set of elements of H of order n, and onto it, since an element $y \in H$ of order n is $\Psi(x)$ for some $x \in G$, and then $|x| = |y| = n$. Hence the two sets have the same number of elements. If $G \neq 1$, choose $x \neq 1_G$. The trivial homomorphism sends x to 1, which has order 1, while $|x| \neq 1$.
4. This is proposition 1.4.21(2).
5. Let $g \in G$. By definition 1.3.1, $g = s_1^{\epsilon_1} \cdots s_m^{\epsilon_m}$ for some $s_i \in S$ and $\epsilon_i \in \{1, -1\}$, where $m = 0$ gives $g = 1_G$. Since ψ and ψ' are homomorphisms, (2) gives

$$\psi'(g) = \psi'(s_1)^{\epsilon_1} \cdots \psi'(s_m)^{\epsilon_m} = \psi(s_1)^{\epsilon_1} \cdots \psi(s_m)^{\epsilon_m} = \psi(g)$$

so $\psi' = \psi$, completing the proof.

The last point of proposition 1.4.24 is used repeatedly to construct homomorphisms and isomorphisms, for instance between groups and subgroups of S_n once group actions are available.

Example 1.4.25

Prove that the subgroup of $SL_2(\mathbb{F}_3)$ generated by $A = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ and $B = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$ is isomorphic to the quaternion group Q_8 of section 1.2.4.

***Proof*:**
Entries are read modulo 3. Both matrices have determinant 1, so they lie in $SL_2(\mathbb{F}_3)$. Direct computation gives

$$A^2 = B^2 = -I \qquad AB = \begin{pmatrix} -1 & 1 \\ 1 & 1 \end{pmatrix} \qquad BA = \begin{pmatrix} 1 & -1 \\ -1 & -1 \end{pmatrix} = -AB$$

Since $-I$ commutes with every matrix, these give $(AB)^2 = A(BA)B = -A^2B^2 = -I$, $A(AB) = A^2B = -B$, $(AB)A = A(BA) = -A^2B = B$, $B(AB) = (BA)B = -AB^2 = A$ and $(AB)B = AB^2 = -A$. With A, B, AB and $-I$ in the roles of i, j, k and -1, these are exactly the relations of section 1.2.4: $i^2 = j^2 = k^2 = -1$, $ij = k$, $ji = -k$, $ik = -j$, $ki = j$, $jk = i$ and $kj = -i$.

Define $\varphi\colon Q_8 \to SL_2(\mathbb{F}_3)$ by $\varphi(\pm 1) = \pm I$, $\varphi(\pm i) = \pm A$, $\varphi(\pm j) = \pm B$ and $\varphi(\pm k) = \pm AB$. Every element of Q_8 is εu with $\varepsilon \in \{1, -1\}$ and $u \in \{1, i, j, k\}$, and $\varphi(\varepsilon u) = \varepsilon\varphi(u)$. In Q_8 (section 1.2.4) the element -1 commutes with everything and $(-1)^2 = 1$, so $(\varepsilon u)(\varepsilon' u') = \varepsilon\varepsilon'(uu')$, and the relations give $uu' = \delta w$ with $\delta \in \{1, -1\}$ and $w \in \{1, i, j, k\}$. The previous paragraph shows that $\varphi(u)\varphi(u') = \delta\varphi(w)$ for the same δ and w, which is immediate when u or u' is 1. As $-I$ is central,

$$\varphi(\varepsilon u)\varphi(\varepsilon' u') = \varepsilon\varepsilon'\varphi(u)\varphi(u') = \varepsilon\varepsilon'\delta\varphi(w) = \varphi((\varepsilon u)(\varepsilon' u'))$$

so φ is a homomorphism. The top rows of $I, -I, A, -A, B, -B, AB, -AB$ are $(1, 0)$, $(2, 0)$, $(0, 2)$, $(0, 1)$, $(1, 1)$, $(2, 2)$, $(2, 1)$ and $(1, 2)$, which are distinct, so φ is injective.

Let H be the subgroup generated by A and B, the set of finite products of A, B and their inverses (definition 1.3.1). The image of φ is a subgroup of $SL_2(\mathbb{F}_3)$ (proposition 1.4.4(5)) containing A and B, so it contains H. Conversely, every element of the image is a product of copies of A and B, since $-I = A^2$ and $I = A^4$. So φ is a bijective homomorphism from Q_8 onto H, and $H \cong Q_8$, as claimed.

Prove that an isomorphism $\varphi\colon S_n \to S_n$ preserves parity: for every $\sigma \in S_n$, the permutations σ and $\varphi(\sigma)$ are both even or both odd (definition 1.2.21). A homomorphism need not preserve parity, since for $n \geq 2$ the trivial homomorphism $S_n \to S_n$ sends the odd permutation (1 2) to the even permutation id.

***Proof*:**
If $n = 1$, then $S_1 = \{\text{id}\}$ and there is nothing to prove, so let $n \geq 2$. By definition 1.2.21, it suffices to show that $\text{sgn}(\varphi(\sigma)) = \text{sgn}(\sigma)$ for every $\sigma \in S_n$.

The images under φ of all transpositions have one common sign. Indeed, for $g \in S_n$ and a transposition $(a\ b)$, proposition 1.2.27 gives $g(a\ b)g^{-1} = (g(a)\ g(b))$. Given two transpositions $(a\ b)$ and $(c\ d)$, choose $g \in S_n$ with $g(a) = c$ and $g(b) = d$, which is possible because $a \neq b$ and $c \neq d$. Then $(c\ d) = g(a\ b)g^{-1}$, so

$$\varphi((c\ d)) = \varphi(g)\varphi((a\ b))\varphi(g)^{-1}$$

and theorem 1.2.20(2) gives $\text{sgn}(\varphi((c\ d))) = \text{sgn}(\varphi((a\ b)))$, because the signs are the commuting numbers ± 1.

If this common sign were 1, then every $\varphi(\sigma)$ would have sign 1, since σ is a product of transpositions (corollary 1.2.19) and $\varphi(\sigma)$ is the product of their images. This is impossible, because φ is surjective and $\text{sgn}((1\ 2)) = -1$ (theorem 1.2.20(3)). Hence $\text{sgn}(\varphi(\tau)) = -1$ for every transposition τ. Writing $\sigma = \tau_1 \cdots \tau_k$ with transpositions τ_i, theorem 1.2.20 gives $\text{sgn}(\varphi(\sigma)) = \text{sgn}(\varphi(\tau_1)) \cdots \text{sgn}(\varphi(\tau_k)) = (-1)^k = \text{sgn}(\sigma)$, as claimed.

Exercise 1.4.2

1. Find all non-trivial homomorphisms between D_4 and Q_8
2. Let G be an arbitrary group.
    1. Prove that there always exists a homomorphism from $\mathbb{Z}$ to G
    2. Why does this not contradict the fact that I is the only initial group?
3. (If you know some complex analysis) Let S^1 be the circle in the complex plane. show that $f : \mathbb{R} \to S^1$, $f(x) = e^{2\pi ix}$ is a homomorphism.
4. Is there a non-trivial homomorphism from $\mathbb{Z}/n\mathbb{Z}$ to $\mathbb{Z}$?
5. Let $\varphi : G \to H$ be a homomorphism. Show that $\varphi(a + b + c) = \varphi(a) + \varphi(b) + \varphi(c)$. More generally, if $g_1, g_2, ..., g_n \in G$, then $\varphi(g_1 + g_2 + \cdots + g_n) = \varphi(g_1) + \varphi(g_2) + \cdots + \varphi(g_n)$. [hint: recall associativity and induction]
6. Does there exist a surjective homomorphism $\varphi : G \to H$ where G is abelian and H is not?
7. Let S where $|S| = 3$ be a set and let $\text{Sym}(S)$ all bijections on S (i.e. the symmetric group). Show that there exists a homomorphism $\sigma : D_3 \to \text{Sym}(S)$. (it is in fact an isomorphism) (This is a particular example of a *group action*, the subject of chapter 4.)
8. Define the map $\varphi_g : G \to G$, $\varphi(x) = gx$. Is this map in general a homomorphism? If not in general, in a particular case?
9. Define the map $\psi_g : G \to G$, $\psi(x) = gxg^{-1}$. Is this map in general a homomorphism? If not in general, in a particular case?

10. Let $\varphi : G \to H$ be a homomorphism and $\ker(\varphi) = \{e_G\}$. Prove that φ is injective.
11. Show that there exist an injective homomorphism $\varphi_n : D_n \hookrightarrow O_2(\mathbb{R})$.

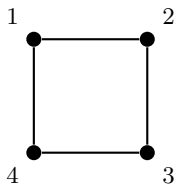
1.5 Symmetry and Transformations: A Prelude to Group Actions

We have so far studied groups as abstract algebraic objects, defined by axioms and explored through subgroups, homomorphisms . But groups did not arise historically from axioms. They arose from *symmetry*. Évariste Galois studied permutations of roots of polynomials. Arthur Cayley studied transformations of space. In every early appearance of what we now call a "group," the central idea was the same: a collection of *reversible transformations* of some object, where transformations can be composed and undone.

In this section, we pause the axiomatic development to revisit this original intuition. The goal is simple: to see that groups *act on things*, and that this perspective is the natural way to think about them. We will make all of this precise in Chapter 4. For now, we build intuition.

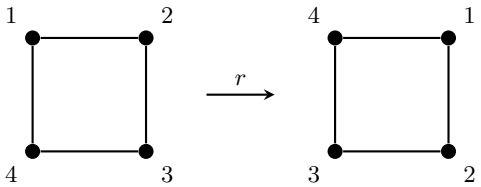
The Symmetries of a Square

Recall from Section 1.2.3 that the dihedral group D_4 consists of the 8 symmetries of a square: four rotations and four reflections. Let us label the corners $1, 2, 3, 4$:



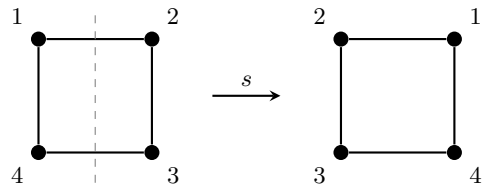
Each symmetry of the square *moves the corners around*. A $90°$ clockwise rotation r sends each corner to the next one clockwise:

$$1 \longmapsto 2, \quad 2 \longmapsto 3, \quad 3 \longmapsto 4, \quad 4 \longmapsto 1.$$



Similarly, the reflection s across the vertical axis swaps the left and right columns:

$$1 \longmapsto 2, \quad 2 \longmapsto 1, \quad 3 \longmapsto 4, \quad 4 \longmapsto 3.$$

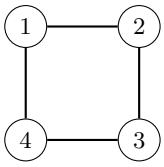


Each element of D_4 determines a specific way of rearranging the four corners, that is, a *bijection* from $\{1, 2, 3, 4\}$ to itself. We can compose these rearrangements: rotating by r twice gives the same result as rotating by r^2. In other words, the group D_4 is *acting* on the set $\{1, 2, 3, 4\}$.

Two observations are worth highlighting. First, every symmetry is *reversible*: if r rotates the square clockwise, then r^{-1} rotates it back. This is why we need a *group* (with inverses) rather than just a monoid. Second, the identity element $e \in D_4$ does nothing to the corners. It is the "trivial symmetry" that leaves everything in place.

Beyond the Square: Symmetry is Everywhere

The group D_4 does not only describe the symmetries of a geometric square. Consider the *cycle graph* C_4 on four vertices:



A *graph automorphism* is a bijection on the vertex set that preserves adjacency. One can verify that the automorphism group $\mathrm{Aut}(C_4)$ is precisely D_4: the same rotations and reflections that preserve the square also preserve which vertices are joined by an edge. The group D_4 captures an abstract *pattern of symmetry* that manifests in both geometry (the square) and combinatorics (the cycle graph). This is the key insight: a single group can describe the symmetries of *many different objects*.

Here are several more instances of this phenomenon, drawn from across mathematics. In each case, a group "acts" on a set by rearranging its elements in a structured, reversible way.

1. **Linear transformations.** The general linear group $GL_n(\mathbb{R})$ acts on $\mathbb{R}^n$: each invertible matrix A transforms a vector $\vec{x}$ to $A\vec{x}$. Composing two transformations corresponds to multiplying the matrices, and the identity matrix leaves every vector fixed. Rotations, reflections, scalings, and shears of n-dimensional space are *all* captured by this single action.

2. **Permutations.** The symmetric group S_n acts on $\{1, 2, \ldots, n\}$ by evaluation: each permutation σ sends i to $\sigma(i)$. This is, in a sense, the "most general" action: every rearrangement of n objects corresponds to some element of S_n. As we will see in Chapter 4, *every* group action can be viewed through the lens of permutations.

3. **The Rubik's Cube.** A standard 3×3 Rubik's Cube has six faces, each consisting of nine colored squares. The center square of each face is fixed (it defines the face's color), leaving $6 \times 8 = 48$ movable *facelets*. Each legal move (a quarter-turn or half-turn of a single face)

permutes these 48 facelets, and the set of all finite sequences of legal moves forms a group $\mathcal{R} \leq S_{48}$. This group is enormous:

$$|\mathcal{R}| \;=\; 43{,}252{,}003{,}274{,}489{,}856{,}000 \;\approx\; 4.3 \times 10^{19},$$

yet it is a *tiny* fraction of $|S_{48}| = 48! \approx 1.2 \times 10^{61}$. Not every conceivable rearrangement of the facelets can be reached by legal moves. The structure of the group $\mathcal{R}$ (its subgroups, generators, and quotients) is precisely what determines which configurations are reachable from a solved cube and which are not.

4. **Translations of polynomials.** Consider the set of all real polynomials. The additive group $(\mathbb{R}, +)$ acts on this set by *translation*: given $t \in \mathbb{R}$ and a polynomial f, define

$$(t \cdot f)(x) \;=\; f(x + t).$$

Translating by t and then by s is the same as translating by $t + s$, and translating by 0 changes nothing. This simple action connects algebra to analysis: Taylor's theorem, for instance, describes exactly how a polynomial changes under translation.

In every example above, three things are true:

1. Each element of the group "does something" to every element of the set: it rearranges, transforms, or moves things around.
2. The **identity element does nothing**: it leaves every element of the set exactly where it was.
3. The group operation is **compatible with composition**: performing one element's transformation followed by another's is the same as performing the transformation of their product.

These three observations are the soul of what we will call a *group action*. Chapter 4 makes them precise as a single homomorphism from the group into a symmetric group, and develops the theory around it. But the underlying philosophy is already clear:

A group is not just an abstract set with a binary operation.
It is a collection of symmetries that act.

Groups Acting on Themselves: A First Glimpse

The examples above involve a group acting on an "external" object: a square, a vector space, a set of facelets. But some of the best applications of group actions come from letting a group act *on its own elements*. There are two natural ways to do this. Given a group G and elements $g, a \in G$:

- Left multiplication: define $g \cdot a = ga$. Each element g "shuffles" the elements of G by multiplying on the left. This will lead us to *Cayley's Theorem*, which asserts that every group is isomorphic to a group of permutations.
- Conjugation: define $g \cdot a = gag^{-1}$. This action detects the internal structure of G: which elements "behave similarly" under the group operation, and how the center $Z(G)$ relates to the rest of the group. It will lead us to the *Class Equation* and, ultimately, to the celebrated *Sylow Theorems*, which describe how a finite group decomposes relative to its prime factors.

Both of these self-actions will be central to Chapter 4. The remarkable fact is that by studying how G transforms *itself*, we can deduce deep structural information about G that is invisible from the axioms alone.

1.6 Cayley Graphs

We have seen that each element g of a group can be thought of as a transformation: it "moves" every element a to ga. If we focus on a set of generators $\{s_1, s_2, \ldots, s_k\}$, then each generator defines a specific *step* that can be taken from any group element. Starting from the identity e and repeatedly taking these steps, we trace out the entire group.

This is precisely the idea behind a *Cayley graph* (Definition 1.6.1): we place a vertex at each element of G and draw a directed edge from a to $s_i a$ for each generator s_i. The resulting picture is a vivid record of the group's structure, and it is nothing other than a *visual record of the left-multiplication action* restricted to the generators. As we explore Cayley graphs, keep this action-based perspective in mind, it is the thread that connects the algebra of groups to the geometry of their visual representations:

Definition 1.6.1: Cayley Graph

Let G be a group and let S be a generating set of G. The *Cayley Graph* $\Gamma = \Gamma(G, S)$ is a colored directed graph constructed as follows:

1. Each element $g \in G$ is assigned a *vertex*: the vertex set $V(\Gamma)$ of Γ bijectivity corresponds with G
2. Each generator $s \in S$ is assigned a colour c_s
3. For any $g \in G$ and $s \in S$, the vertices corresponding to the elements g and gs are joined by a direct edges of colour c_s. Thus, the edge set $E(\Gamma)$ consists of pairs of the form (g, gs), with $s \in S$ providing the colour

some popular, but not necessary, conventions are: [21]

1. The set S is finite
2. $S = S^{-1} = \left\{s^{-1} \middle| s \in S\right\}$
3. $e \notin S$ (i.e., the identity is not in S)

Example 1.6.2: Cayley Graphs

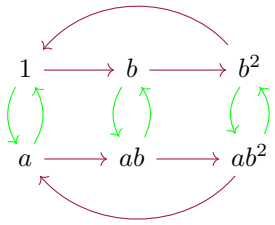
1. Let $G = \mathbb{Z} = \langle 1 \rangle$. Then we can say that $S \cup S = \{1, -1\}$. This gives us the graph:

$$\cdots \quad -3 \longleftarrow -2 \longleftarrow -1 \longleftarrow 0 \longrightarrow 1 \longrightarrow 2 \longrightarrow 3 \quad \cdots$$

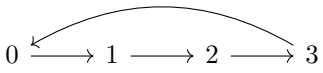
   Take a moment to try and see what the Cayley graph of $\mathbb{Z} \times \mathbb{Z}$ would be (the answer is given in a later example)

[21] These conventions are usually followed in *geometric group theory*

2. Let $G = S_3 = \langle a, b | a^2 = b^3 = 1, ab = b^2 a \rangle$. Then if b is represented by red edges and a is represented by green vertices, we get:



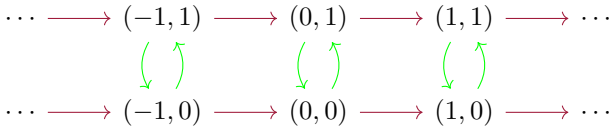
3. Let $G = \mathbb{Z}/4\mathbb{Z}$. Then the Cayley graph is:



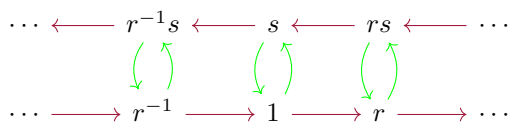
4. Let $G = \mathbb{Z} \times \mathbb{Z}$. Then the Cayley graph is:

$$
\begin{array}{ccccccccc}
 & & & \vdots & & & & & \\
 & & (-2,2) & \longleftarrow & (-1,2) & \longleftarrow & (0,2) & \longrightarrow & (1,2) & \longrightarrow & (2,2) & & \\
 & & \uparrow & & \uparrow & & \uparrow & & \uparrow & & \uparrow & & \\
 & & (-2,1) & \longleftarrow & (-1,1) & \longleftarrow & (0,1) & \longrightarrow & (1,1) & \longrightarrow & (2,1) & & \\
 & & \uparrow & & \uparrow & & \uparrow & & \uparrow & & \uparrow & & \\
\cdots & & (-2,0) & \longleftarrow & (-1,0) & \longleftarrow & (0,0) & \longrightarrow & (1,0) & \longrightarrow & (2,0) & & \cdots \\
 & & \downarrow & & \downarrow & & \downarrow & & \downarrow & & \downarrow & & \\
 & (-2,-1) & \longleftarrow & (-1,-1) & \longleftarrow & (0,-1) & \longrightarrow & (1,-1) & \longrightarrow & (2,-1) & & & \\
 & \downarrow & & \downarrow & & \downarrow & & \downarrow & & \downarrow & & & \\
 & (-2,-2) & \longleftarrow & (-1,-2) & \longleftarrow & (0,-2) & \longrightarrow & (1,-2) & \longrightarrow & (2,-2) & & & \\
 & & & \vdots & & & & & & & & &
\end{array}
$$

5. Let $G = \mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$ and $S = \{(1,0), (0,1)\}$, with red edges for $(1,0)$ and green edges for $(0,1)$. Then the Cayley graph is:



6. Let $G = D_\infty = \langle r, s \mid s^2 = 1,\ rs = sr^{-1} \rangle$ (i.e this is D_n but with no $r^n = 1$ relation) and let $S = \{r, s\}$, with red edges for r and green edges for s. Every element is r^n or $r^n s$ for some $n \in \mathbb{Z}$, and $sr = r^{-1}s$, so $r^n \cdot r = r^{n+1}$ while $r^n s \cdot r = r^{n-1}s$. Then the Cayley graph is:



Just by looking at the graphs, you might be able to see some interesting patterns. For example, by taking the direct product, we found it produces parallel graphs. We see in S_3 and $\mathbb{Z}/n\mathbb{Z}$ that arrow point *to* the identity represent the *order* of the element. Different Cayley graphs can also give different insights: the Cayley graph of $\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$ above consists of two parallel lines with every node connected to the one above it. The Cayley graph of D_∞ has the same shape, except that the red edges of the top line point the other way, a "twist" introduced to it[22].

In most of this textbook, Cayley graphs will not play a prominent role. They come back here and there to give a visual way of looking at new concepts introduced[23]. However, they are the foundation for an area of group theory called *geometric group theory*. For the study of geometric group theory, some knowledge of analysis is needed (ex. metrics). For those interested in such explorations, [43] is an introduction.

[22] Chapter 5 constructs D_∞ from $\mathbb{Z}$ and $\mathbb{Z}/2\mathbb{Z}$ as the semidirect product $\mathbb{Z} \rtimes \mathbb{Z}/2\mathbb{Z}$.
[23] For example, they will come back when working with free groups and semi-direct products

2

Sub Groups and Generating Groups

Now that we have studied some fundamental aspects of group theory and important examples, let's dive deeper in the mechanics used in analyzing and finding groups. The first way we will be studying group is by trying to take smaller groups from the larger group and seeing if we can find information through the pieces of the group.

We will start by trying to find subgroups that exist for *any* group G. These subgroups are sometimes called *natural subgroups*, since they always exist for a group, as though the axioms of the group "naturally" let these subgroups exist. We will then move on to constructing groups given some subset of a group, and show how the basic operations of $\cup$ and $\cap$ interact with groups. Finishing off our exploration of generating groups, we will define the concept of "most general group". It will turn out that the free groups can be thought of as the groups that "generate" all other possible groups. We will end this chapter by exploring a new way of visualizing the subgroups of finite groups through the use of a graph (in particular a lattice) that will become useful when trying to find how to construct larger groups.

2.1 Basic Definition and Properties

First, a reminder of what it means to take a smaller group from the same group:

Definition 2.1.1: Subgroup

Let G be a group with operation $*$. The subset H of G is a *subgroup* of G if H is a nonempty, closed under the restricted operation $*|_H$, that is, it is closed under products and inverse. We shall write $H \leqslant G$.

Recall that the binary operation $*|_H$ on H is a restriction of the binary operation $*$ on G. In particular, if $* : G \times G \to G$ is the binary operation on G, then we say we *restrict* the binary operation to H if we can define $*|_H : H \times H \to H$, where for any two elements $h_1, h_2 \in H$ $*|_H (h_1, h_2) = *(h_1, h_2)$. If $(H, *|_H)$ is a group, we say it's a subgroup of $(G, *)$. In this way, we say that H has "the same" binary operation as H. We will abuse notation and denote the binary operation of the subgroup $H \leqslant G$ as the same operation as the group G, that is, we will write $(G, \cdot)$ and $(H, \cdot)$ instead of $(G, \cdot)$ and $(H, \cdot|_H)$. If we have $H \leqslant G$ and $H \subsetneq G$, then we will denote this as $H < G$

Example 2.1.2: Sub Groups

1. If $1 \in G$ and $H = \{1\}$, $H \leqslant G$. H is called the *trivial subgroup*. Similarly, $G \leqslant G$ always.
2. $\mathbb{Z} \leqslant \mathbb{Q} \leqslant \mathbb{R}$ with the operation of addition.
3. $\mathbb{Z}^+ \not\leqslant \mathbb{Z}$ under addition. Can you prove it?
4. Take the set of all symmetries ($\{1, \sigma\} \subseteq D_4$) and all rotations ($\{1, r, r^2, r^3\} \subseteq D_4$). These are both subgroups.

The subgroup H inherits a lot of properties we have proven in section 1.1.1. For example, for any equation of elements in H, it is also an equation of elements in G, and so will have the same solution(s) or lack thereof. Furthermore, the identity will remain the same, since every group must have a unique identity, and similarly for inverses.

For computational purposes, here is a useful proposition that warps up the conditions of a subgroup to be able to more quickly identify whether a subset is or is not a subgroup:

Proposition 2.1.3: The Subgroup Criterion

A subset H of G is a subgroup if and only if

1. $H \neq \emptyset$
2. for all $x, y \in H$, $xy^{-1} \in H$

Furthermore, if H is finite, then it suffices to check that H is non-empty and closed under its operation.

***Proof*:**
Let $H \subseteq G$

($\Rightarrow$) Let $H \leqslant G$. Then $1 \in H$, so $H \neq \emptyset$. Take $x, y \in H$. If $y \in H$, then $y^{-1} \in H$. Since H is closed under its operation, then $xy^{-1} \in H$

($\Leftarrow$) Let $H \subseteq G$ and consider $(H, \cdot|_H)$.

(a) **(associativity)** Associativity comes for free since the binary operation on H is the same, so if $x, y, z \in H$, then $x(yz) = (xy)z$

(b) **(identity)** Since $H \neq \emptyset$, there must exist some $x \in H$. Then by the 2nd condition, $xx^{-1} = 1 \in H$, so $1 \in H$.

(c) **(inverse)** Let $x \in H$. Since $1 \in H$ then $1x^{-1} = x^{-1} \in H$.

(d) **(well-defined)** Let's make sure that if $x, y \in H$ then $xy \in H$ (making sure that the binary operation is still a function). By what we have just shown, $y^{-1} \in H$. Thus $x(y^{-1})^{-1} = xy \in H$ (where $(x^{-1})^{-1} = x$ was proved in proposition 1.1.15). Thus, $\cdot|_H$ is indeed a function

Thus, $H \leqslant G$

Now, let H be a finite group. Then for any $x \in H$, there can only be finitely many distinct elements $x_1, x_2, ..., x_n, ...$, meaning for some $a, b \in \mathbb{N}$, $x^a = x^b$ with $a < b$. This means that $x^{b-a} = 1$, and so for any $x \in H$, x has finite order. Thus, we get that $x^{-1} = x^{n-1}$, where $x^n = 1$. Thus, we automatically get H is a subgroup if H is closed under its binary operation

Exercise 2.1.1

1. Prove or disprove that the following are subgroups: $\mathbb{Q}^\times \leqslant \mathbb{R}$, $D_3 \leqslant D_4$
2. Let $H, K \leqslant G$. Is $H \cup K$ always a group? Prove or give a counter-example. If not, then identify what condition(s) is needed for $H \cup K$ to be a group.
3. Prove that G cannot have a subgroup H with $|H| = n - 1$, where $n = |G| > 2$.
4. Let $K \leqslant H$ and $H \leqslant G$. Prove that $K \leqslant G$

2.2 Examples of Subgroups

In this section, we will explore "natural" subgroups as well as subgroups that arise given any homomorphism

2.2.1 Kernel and Image

Given any homomorphism $\varphi : G \to H$, are there subsets that can be created using φ which will be subgroups of either G or H? Naturally, the image of a homomorphism is a subgroup of H ($\varphi(G) \subseteq H$), essentially by definition. We might ask ourselves if there exists some subgroup of G that we can construct using φ?

commented old text here:

For the first subgroup, consider the following set

$$\ker(\varphi) \subseteq G \qquad \ker(\varphi) = \{g \in G \mid \varphi(g) = 1\}$$

then $\ker(\varphi) \leqslant G$

Proof. We will use the subgroup criterion. Take $a, b \in \ker(\varphi)$. Then using properties of homomorphisms:

$$1 = \varphi(ab^{-1}) = \varphi(a)\varphi(b^{-1}) = \varphi(a)\varphi(b)^{-1}$$

and since $\varphi(a) = \varphi(b) = 1$, we get

$$\varphi(a)\varphi(b)^{-1} = 11^{-1} = 11 = 1$$

so $ab^{-1} \in \ker\varphi$

□

the set $\ker(\varphi)$ will thus be given a name:

Definition 2.2.1: Kernel

Let $\varphi : G \to H$ be a homomorphism. Then the set

$$\ker(\varphi) = \{g \in G \mid \varphi(g) = 1\}$$

is the *kernel* of φ

2.2.2 Question to Ponder Let's say $\varphi : G \to H$ and $\psi : G \to H$ are two homomorphisms with the same domain and codomain. If $\ker(\varphi) = \ker(\psi)$, does that imply $\varphi = \psi$?

As we have discussed earlier, $\varphi(G)$ is a group:

$$\varphi(G) \subseteq H, \quad \varphi(G) = \{h \in H \mid \varphi(g) = h \text{ for some } g \in G\}$$

For completeness of this section, we put the proof here:

Definition 2.2.3: Image

Let $\varphi : G \to H$ be a homomorphism. Then the set

$$\varphi(G) = \{h \in H \mid \varphi(g) = h \text{ for some } g \in G\}$$

or

$$\text{im}(\varphi) = \{h \in H \mid \varphi(g) = h \text{ for some } g \in G\}$$

is the *image* of G with φ.

Proof. We will use the subgroup criterion as well as proposition 1.4.4. Since $\varphi(e_G) = e_H$, we have that $\varphi(G)$ is non-empty.

Next, take $a, b \in \varphi(G)$, meaning there exists an $x, y \in G$ such that $\varphi(x) = a$ and $\varphi(y) = b$. Since $\varphi(y^{-1}) = \varphi(y)^{-1}$, we have

$$ab^{-1} = \varphi(x)\varphi(y)^{-1} = \varphi(xy^{-1}) \in \varphi(G)$$

thus, $ab^{-1} \in \varphi(G)$, meaning $\varphi(G)$ passes the subgroup criterion, as we sought to show □

Given that $\varphi(G) \subseteq H$, $\varphi(G)$ has structure of H. However, since $\varphi(G)$ also has some of it's structural information based on G, it would not be surprising if we can somehow characterize $\varphi(G)$ through G. The first isomorphism theorem of chapter 3 does this: $\varphi(G)$ is isomorphic to a quotient of G.

2.2.2 Center, Centralizers, and Normalizers

Given any group G, does there exist [non-trivial][1] subsets $H \subseteq G$ which will *always* be subgroups (i.e. $H \leqslant G$), regardless of the operation? Not all subsets will form subgroups. For example, take the set

$$T(G) = \{x \in G \mid x^n = 1, \text{ for some } n \in \mathbb{N}_{>0}\}$$

this set is called the *torsion subset* (i.e. the set of all element of finite order). If $T(G) \leqslant G$, then it must pass the subgroup criterion for all possible groups G. However, take

$$G = \langle x, y \mid x^2 = y^2 = 1 \rangle$$

both $x, y \in T(G)$. However, $xy^{-1} = xy \notin T(G)$ (note that $y^2 = 1 \Rightarrow y = y^{-1}$): the element $(xy)^n = xyxyxyxy\cdots$ will never equal 1, since there are no relations that permit the appropriate cancellation to occur. In particular, note that there is no relation for G which makes $xy = yx$ which would allow the values to commute and cancel. This is rather difficult to prove at this stage in the book. So $T(G)$ need not be a subgroup of G. If G is abelian, then $T(G)$ *is* a subgroup (exercise 2.2.1 (7)).

The question arises then if there are properties for which a subset will *always* be a subgroup for any group G regardless of the operation on G. These types of natural subgroups are in general quite hard to find, however they do exist. The rest of this section will be exploring examples of such subsets.

The first one we will explore is a great example of the power of commutativity: Let

$$Z(G) \subseteq G, \; Z(G) = \{g \in G \mid gx = xg \text{ for all } x \in G\}$$

then $Z(G) \leqslant G$

Proof. We will use the subgroup criterion. For this whole proof, let x be an arbitrary element of G. First note that $1x = x1 = x$, and so $1 \in Z(G)$. Next, take any $a, b \in Z(G)$. Since $bx = xb$, multiplying on the left and right by b^{-1} on both sides of the equation:

$$b^{-1}bxb^{-1} = b^{-1}xbb^{-1} \quad \Rightarrow \quad xb^{-1} = b^{-1}x$$

and since equality is symmetric, $b^{-1}x = xb^{-1}$, meaning $b^{-1} \in Z(G)$. Finally, we have

$$ab^{-1}x = axb^{-1} = xab^{-1}$$

meaning $ab^{-1} \in Z(G)$, proving it's a subgroup by the subgroup criterion □

This subgroup is special enough to be given a name:

[1] the trivial subset being just the identity $\{e_G\}$, which is always the simplest subgroup which has essentially no "interesting" structure, and hence trivial

Definition 2.2.4: Center

Let G be an arbitrary group. Define

$$Z(G) := \{g \in G | gx = xg \text{ for all } x \in G\}$$

to be the set of elements commuting with all the elements of G. This subset is called the *center* of G.

2.2.5 trivia The letter Z is because the German word for center is *Zentrum*.

Example 2.2.6: Center of Groups

The provided examples should be verified by the reader as practice

1. If G is abelian, then $Z(G) = G$. Thus $Z(\mathbb{Z}) = \mathbb{Z}$, $Z(\mathbb{R}) = \mathbb{R}$, and so forth
2. $Z(D_4) = \left\{1, r^2\right\}$
3. $Z(Q_8) = \{1, -1\}$

The Center of other subgroups we have encountered in section 1.2 will be left as exercises.

This should start giving you an idea of the power of commutativity. In fact, it is even more powerful than we need it to be. The notion of the center can be weakened a bit and still remain a group. Take $A \subseteq G$ (notice that A is just a subset). Then define

$$C_G(A) \subseteq G,\ C_G(A) = \left\{g \in G \mid gag^{-1} = a \text{ for all } a \in A\right\}$$

Note that $gag^{-1} = a$ is equivalent to $ga = ag$. The change in equation is due to a further generalization we will shortly make. Before we get to that, we must first show that $C_G(A)$ is indeed a group

Proof. We will use the subgroup criterion. First, since for any $a \in A$, $1a = a1 = a$, $1 \in C_G(A)$

Next, take $x, y \in C_G(A)$. Since $y \in C_G(A)$, $yay^{-1} = a$ for all $a \in A$. Multiplying on the left by y^{-1} and right by y, we get $a = y^{-1}ay = (y^{-1})a(y^{-1})^{-1}$, showing that $y^{-1} \in C_G(A)$. Finally, to show $xy^{-1}a(xy^{-1})^{-1} = xy^{-1}ayx^{-1} = a$, we use associativity and the fact that $x, y, y^{-1} \in C_G(A)$:

$$\begin{aligned} xy^{-1}axy^{-1} &= xay^{-1}yyx^{-1} \\ &= xax^{-1} \\ &= a \end{aligned}$$

and so $xy^{-1} \in C_G(A)$, showing that $C_G(A) \leqslant G$ by the subgroup criterion □

Since it's a subgroup that will always exist, it is special enough to be given a name:

Definition 2.2.7: Centralizer

Let G be an arbitrary group, and $A \subseteq G$[a]. Define

$$C_G(A) := \{g \in G | gag^{-1} = a \text{ for all } a \in A\}$$

This subset of G is called the *centralizer* of A in G. Since $gag^{-1} = a$ if and only if $ga = ag$, $C_G(A)$ is the set of elements of G which commute with every element of A.
If A is a subgroup of G and $C_G(A) = G$, then A is called a *central subgroup*

[a]Notice A is a *subset*, not necessarily a subgroup

If we take, $A = G$ then $C_G(G) = Z(G)$. Equivalently, notice that:

$$\bigcap_{g \in G} C_G(g) = Z(G)$$

since an element lies in either side exactly when it commutes with every element of G.

If $A \leqslant G$ and $C_G(A) = G$, then every element of A commutes with every element of G, so $A \leqslant Z(G)$. This is the reason for the name *central* subgroup.

Example 2.2.8: Centralizer of Groups

Let $A = \{1, r, r^2, r^3\} \subseteq D_4$ and $B = \{1, i\} \subseteq Q_8$

1. $C_{D_4}(A) = A$. Notice how this set is larger than $Z(D_4)$, since $r^3 \in C_{D_4}(A)$ but $r^3 \notin Z(D_4)$
2. $C_{Q_8}(B) = \{1, -1, i, -i\}$. Similar to the previous example, notice that $C_{Q_8}(B)$ is larger than $Z(Q_8)$.

We can generalize this result even further: so far, we have written $ga = ag$ for all $g \in G$. we can re-write this as $g\{a\} = \{a\}g$, where we define $g\{a\} = \{gx \mid x \in \{a\}\}$ and $\{a\}g = \{xg \mid x \in \{a\}\}$ This might seem to be making the situation overly-complex, however the great advantage of this approach is we can now general to taking a general set A

$$gA = Ag \qquad gA = \{ga \mid a \in A\} \;\; Ag = \{ag \mid a \in G\}$$

What this difference does is that it weakens the condition for an element to "directly" commute ($ga = ag$) to "indirectly" commuting, or more precisely commuting with respect to the set A: for every $a \in A$, there exists an $a' \in A$ such that $ga = a'g$. It will not always be the case that if $A \subseteq G$, then $gA = Ag$

Example 2.2.9: Counter-Example and Example

Let $A = \{r, \sigma\} \subseteq D_4$. Take $g = r$. Then:

$$rA = \{r^2, r\sigma\} \qquad Ar = \{r^2, \sigma r\} = \{r^2, r^3\sigma\}$$

notice that $rA \neq Ar$

However, if $A = \{1, r, r^2, r^3\}$ then for any $g \in D_4$, $gA = Ag$. As an example, let $g = \sigma$. Then

$$\sigma A = \{\sigma, \sigma r, \sigma r^2, \sigma r^3\} \qquad A\sigma = \{\sigma, r\sigma, r^2\sigma, r^3\sigma\} = \{\sigma, \sigma r, \sigma r^2, \sigma r^3\}$$

verifying other values of D_4 by hand shows that $gA = Ag$ for all $g \in D_4$

Therefore, we can define the following set of elements that satisfy $gA = Ag$:

$$N_G(A) \subseteq G,\ N_G(A) = \left\{g \in G \mid gAg^{-1} = A\right\}$$

Again, $N_G(A)$ is a subgroup of G. The proof is almost identical to the proof for $C_G(A)$ (change a for A in the proof). We therefore give a name to this special subgroup:

Definition 2.2.10: Normalizer

Take $gAg^{-1} = \left\{gag^{-1} | a \in A\right\}$. Define the *normalizer* of A in G to be the set

$$N_G(A) := \left\{g \in G | gAg^{-1} = A\right\}$$

If A is a group and $N_G(A) = G$ (that is, $gAg^{-1} = A$ for all $g \in G$) we call A a *normal subgroup* of G

If $g \in C_G(A)$, then $gag^{-1} = a$ for every $a \in A$, so $gAg^{-1} = A$ and $g \in N_G(A)$. Hence $C_G(A) \subseteq N_G(A)$, and since both are subgroups of G, $C_G(A) \leqslant N_G(A)$.

Example 2.2.11: Normalizer of Groups

Let $A = \{1, r, r^2, r^3\} \subseteq D_4$ and $B = \{1, i\} \subseteq Q_8$

1. $N_{D_4}(A) = D_4$. Notice how $Z(D_4) \subseteq C_{D_4}(A) \subseteq N_{D_4}(A)$, that is

   $$\{1, r^2\} \subseteq \{1, r, r^2, r^3\} \subseteq D_4$$

   Since A is also a subgroup of D_4, A is also an example of a normal subgroup of D_4

2. $N_{Q_8}(B) = \{1, -1, i, -i\}$ First, $1 \in N_{Q_8}(B)$. To show that $i \in N_{Q_8}(B)$, notice that

   $$iB = \{i, -1\} = Bi$$

   so $i \in N_{Q_8}(B)$. Similarly, for -1 and $-i$. For $j, k \in Q_8$, the relations $ij = k$, $ji = -k$, $ik = -j$ and $ki = j$ (see section 1.2.4) give

   $$jB = \{j, -k\} \neq \{j, k\} = Bj \qquad kB = \{k, j\} \neq \{k, -j\} = Bk$$

   meaning $j, k \notin N_{Q_8}(B)$. Similarly for $-j, -k$. Thus $N_{Q_8}(B) = \{1, -1, i, -i\}$. Notice that $C_{Q_8}(B) = N_{Q_8}(B)$ (example 2.2.8), so the normalizer need not be bigger than the centralizer

3. If $D = \{1, (1\ 2)\} \subseteq S_3$, what is $N_{S_3}(D)$? What about $C_{S_3}(D)$?

The notion of the normalizer is not simply generalization for the sake of generalization, it in fact comes up as a very important property: the kernel of a homomorphism has is always normal!

Proposition 2.2.12: Kernels Are Normal Subgroups

Let $\varphi : G \to H$ be a homomorphism, and take $\ker(\varphi) \leqslant G$. Then $\ker(\varphi)$ is a normal subgroup, that is

$$N_G(\ker(\varphi)) = G$$

In other words, for any $g \in G$, $g(\ker(\varphi)) = (\ker(\varphi))g$

***Proof*:**

Let $\varphi : G \to H$, and for notational convenience, let $K = \ker(\varphi)$.

Take any $g \in G$, and $k \in K$, that is, $k \in G$ such that $\varphi(k) = e_H$. We want to show that $gK = Kg$, or equivalently $gKg^{-1} = K$, for all $g \in G$. Take $gkg^{-1} \in gKg^{-1}$. Then

$$\varphi(gkg^{-1}) = \varphi(g)\varphi(k)\varphi(g^{-1}) = \varphi(g)1\varphi(g^{-1}) = \varphi(g)\varphi(g^{-1}) = \varphi(gg^{-1}) = e_H$$

thus, $gkg^{-1} \in K$, so for some k', $k' = gkg^{-1}$. Since this is true for arbitrary $k \in K$, we get $gKg^{-1} = K$, as we sought to show

The fact that kernels are normal subgroups will become very important in chapter 3 This brings us to the end of our generalizations. This section will be closed with a quick remark about normal subgroups. We did not prove much about them here. However, they are at the heart of chapter 3. For now, you can keep them in the back of your mind.

Exercise 2.2.1

1. $C_G(Z(G)) = G$ and deduce that $N_G(Z(G)) = G$
2. If $A, B \subseteq G$, and $A \subseteq B$, then $C_G(B) \leqslant C_G(A)$
3. For each of S_3, D_4 and Q_8, compute the centralizer of each element and find the center of the group.

   These next 3 subgroups from the next 3 questions will come back when we study group actions (and the conjugacy action, section 4.6)
4. Find the centre of $GL_3(\mathbb{C})$
5. Find the centralizer of $\begin{pmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{pmatrix}$ in $GL(3, \mathbb{C})$
6. Find the centralizer of $\begin{pmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 & -1 \end{pmatrix}$ in $SL(5, \mathbb{C})$, the 5×5 complex matrices having determinant equal to 1
7. Let G be an abelian group and let $T(G) = \{x \in G \mid x^n = 1 \text{ for some } n \in \mathbb{N}_{>0}\}$. Show that $T(G) \leqslant G$. It is called the *torsion subgroup* of G.

8. Show that $C_G(S) \leqslant N_G(S)$ by showing $C_G(S)$ is the kernel of the homomorphsims $N_G(S) \to \text{Sym}(S)$ by $g \mapsto (x \mapsto gxg^{-1})$ (i.e. $T(g)(x) = T_g(x) = gxg^{-1}$)
9. For any group G, and any subset $A \subseteq G$, is it true that $Z(G) \leqslant C_G(A)$?

2.3 Subgroups Generated by Subsets

All groups have an underlying set. We can thus ask ourselves many questions about groups related to it's underlying set. For example, If we have two subgroups $H, K \leqslant G$, is $H \cap K$ or $H \cup K$ also a subgroup? What if we extend to finite or infinite intersection or union? On another hand, we can ask ourselves if there exists subsets $A \subseteq G$ that can in some way generate G using the binary operation. If generating all of G is too strong of a condition, can A always generate a subgroup of G? These are the types of questions that will be asked in this section.

Note that the technics in this section are very common throughout algebra (and many other areas of mathematics). Asking whether a subset of the underlying of an algebraic object "generates" the entire object (or part of the object) will occupy our time in many of the following chapters of this book (especially chapter 13).

2.3.1 Cyclic Groups and Cyclic subgroups

We will start our exploration subsets $A \subseteq G$ which are singletons: $|A| = 1$. The first question we will ask ourselves is if there is a subgroup $H \leqslant G$ which contains A. The answer to this is always yes: $G \leqslant G$ is a subgroup and by definition $A \subseteq G$. Though this is a valid answer, this is a very boring answer (in mathematical lingo, we'd say it is the trivial case). Thus, we look for a more interesting question: what is the *smallest* subgroup of G that contains A[2]. If $H \leqslant G$ is the smallest subgroup containing A, H is called a *cyclic subgroup*. If the entire group G is the smallest subgroup containing A, then G is called a *cyclic group*. To find the cyclic subgroup or groups given a subset $\{a\} = A$, a similar strategy will be used from when the order of an element (definition 1.2.8) was discussed. Cyclic groups they pop-up quite often due to their simplicity.

Definition 2.3.1: Cyclic groups

Let G be a group. The group G is *cyclic* if G can be generated by a single element, i.e., there is some element $x \in G$ such that $G = \{x^n | n \in \mathbb{Z}\}$ (where the operation is usually denoted as multiplication)

Sometimes, the notation C_n used to represent arbitrary cyclic groups of order n (with multiplication notations)

We say G is generated by x (or x generated G), and write $G = \langle x \rangle$. It was proven in proposition 1.2.6 that $\langle x \rangle$ is indeed a group. The existence of a C_n for every n is because there exists the cyclic group $\mathbb{Z}/n\mathbb{Z}$ for every n (or $\langle r_n \rangle$ for $r \in D_n$). Later in this section, we will show that $\mathbb{Z}/n\mathbb{Z}$ is the *only* cyclic group of order n up to isomorphism, meaning we will be able to conclude that $C_n \cong \mathbb{Z}/n\mathbb{Z} \cong \langle r_n \rangle$. Since C_n usually has multiplication notation, for this section we will use 1 to represent the identity.

[2]The notion of "smallest" is clear if $|G|$ is finite, but trickier if $|G|$ is infinite: $2\mathbb{Z} \subsetneq \mathbb{Z}$ although $|2\mathbb{Z}| = |\mathbb{Z}|$. Section 2.3.2 makes the notion precise.

2.3.2 Note The fact that $\langle x \rangle$ is the smallest subgroup containing x forced by construction: if there was a subgroup $H \leqslant G$ such that $H \subsetneq \langle x \rangle$, then there would be number n such that $x^n \in \langle x \rangle$ but $x^n \notin H$. Do you see how this contradicts that H is a subgroup?

Example 2.3.3

1. As mentioned earlier, the usual candidate for this example if $\mathbb{Z}/n\mathbb{Z}$, that is, addition modular n, which can be generated by $\overline{1}$. So $\langle \overline{1} \rangle = \mathbb{Z}/n\mathbb{Z}$. Notice there is some ambiguity here: How can you know if $\overline{1}$ represents the of order $5, 10, 324$? In general, you cannot. It is up using the context to determine the order. If the context requires clarity, a subscript will be added: $\langle \overline{1}_n \rangle = \mathbb{Z}/n\mathbb{Z}$ or $\langle [1]_n \rangle = \mathbb{Z}/n\mathbb{Z}$
2. D_n is *not* cyclic. Try finding a single element in D_n which generates all of D_n
3. Is Q_8 cyclic? What about $\mathbb{Z}$? What about $\mathbb{Z} \times \mathbb{Z}$?

Proposition 2.3.4: $|C_n| = |x|$

If $C_n = \langle x \rangle$, then $|C_n| = |x|$ (where if one side of this equality is infinite so is the other).

***Proof*:**
The idea is to show that that if $|x| < \infty$, then if two elements differ in exponent and are smaller than the group, $a \neq b < n$, then $x^a \neq x^b$, which can be shown by contradiction. For the case where the exponent is greater than or equal to n, $t \geq n$, x^t, such an elements can always equal to an element of order less than n. In particular, if $t \geq n$, then $x^t = x^{t-kn}$ for appropriate k for which $0 \leq t - kn < n$. Thus, in the $t \geq n$ case, the proof comes down to the first case that was proved.

For the $|x| = \infty$ case, assume $x^a = x^b$, $a \neq b$, and proceed by contradiction (i.e. $x^{b-a} = 1$).

2.3.5 Note This does not imply that for *any* $x \in C_n$, $|x| = |C_n|$. For example, $\mathbb{Z}/6\mathbb{Z}$ is cyclic since $\mathbb{Z}/6\mathbb{Z} = \langle \overline{1} \rangle$. However, $|\overline{2}| = 3$, meaning $\overline{2}$ cannot generate $\mathbb{Z}/6\mathbb{Z}$. On the other hand, $\overline{1}$ is not the only element of order 6 in $\mathbb{Z}/6\mathbb{Z}$, can you think of a second? Is there more than two?

Corollary 2.3.6: Cyclic $\Rightarrow$ Abelian

If G is cyclic, then G is abelian

***Proof*:**
Let G be a cyclic group. Take some $g \in G$ that generates it. Take some $a, b \in G$ so $g^n = a$ and $g^m = b$ then

$$ab = g^n g^m = g^{n+m} = g^{m+n} = g^m g^n = ba$$

thus, G is also abelian.

You might not be satisfied with this proof: why in the exponent can we commute n and m? The answer comes the associativity of groups. Since

$$a(aa) = (aa)a$$

it makes no difference to write

$$a^{1+2} \qquad a^{2+1}$$

or more generally $a^{n+m} = a^{m+n}$ due to proposition 1.1.15(5).

Since every element in a cyclic is determined by the exponent, having some information about what power of the generator leads to the identity can give us some information on smaller powers of the generator that also lead to the identity.

Proposition 2.3.7: Powers of an Element

Let G be a group, $x \in G$ and let $m, n \in \mathbb{Z}$ be nonzero. If $x^n = 1$ and $x^m = 1$, then $x^d = 1$, where $d = \gcd(m, n)$. In particular, if $x^m = 1$ for some nonzero $m \in \mathbb{Z}$, then $|x|$ is finite and $|x| \big| m$.

***Proof*:**
By Bézout's identity ([24, Extended Euclidean Algorithm (Bézout's Identity)]) there are integers u, v with $um + vn = d$, so

$$x^d = x^{um+vn} = (x^m)^u (x^n)^v = 1$$

For the second statement, $x^m = 1$ with $m \neq 0$ gives $x^{|m|} = 1$ (if $m < 0$, then $x^{-m} = (x^m)^{-1} = 1$), so x has finite order $k = |x|$. Divide: $m = ck + r$ with $c, r \in \mathbb{Z}$ and $0 \leq r < k$. Then $x^r = x^m (x^k)^{-c} = 1$, and the minimality of k forces $r = 0$, that is, $k \big| m$, as we sought to show.

Corollary 2.3.8

Let G be a group (not necessarily cyclic), let $x \in G$, and let $a \in \mathbb{Z} - \{0\}$.

1. if $|x| = \infty$, then $|x^a| = \infty$
2. if $|x| = n < \infty$, then $|x^a| = \frac{n}{\gcd(n,a)}$. If $a > 0$ divides n, then $|x^a| = \frac{n}{a}$.

***Proof*:**

1. If $|x^a| < \infty$, then there would exist an n such that $(x^a)^n = x^{an} = 1$. (if an is negative, take $-an$). But then x would have finite order, a contradiction
2. Let $d = \gcd(n, a)$, and write $n = dn'$ and $a = da'$, so that $\gcd(n', a') = 1$. For $k \in \mathbb{Z}$, proposition 2.3.7 and its converse ($n \big| m$ implies $x^m = 1$) give

   $$(x^a)^k = 1 \iff n \big| ak \iff n' \big| a'k \iff n' \big| k$$

   the last step because n' and a' are coprime ([24, Extended Euclidean Algorithm (Bézout's Identity)] gives $un' + va' = 1$, so $n' \big| a'k$ implies $n' \big| un'k + va'k = k$). The smallest positive

such k is n', so $|x^a| = n' = n/\gcd(n,a)$. If $a > 0$ divides n, then $\gcd(n,a) = a$, giving $|x^a| = n/a$, completing the proof.

With this idea in mind, let's do what we did before and try to see if we can classify cyclic groups.

For cyclic groups, there's in fact an incredibly simple isomorphism which will be introduced showing that if two cyclic groups have the cardinality (same order), they are in fact isomorphic, and thus can be treated as interchangeable! This means that when dealing with future theorems about cyclic groups, once we found it applies to a cyclic group of a certain order, we know it applies to any cyclic group of the same order! This also means there are only countably many finite cyclic groups, each with a predictable structure.

Theorem 2.3.9: Isomorphism between same-order Cyclic Groups

Any two cyclic groups of the same order are isomorphic. More specifically,

1. if $n \in \mathbb{Z}^+$ and $\langle x \rangle$ and $\langle y \rangle$ are both cyclic groups of order n, then the map

$$\psi : \langle x \rangle \to \langle y \rangle \qquad x^k \mapsto y^k$$

   is well defined and is an isomorphism

2. if $\langle x \rangle$ is an infinite cyclic group, the map

$$\psi : \mathbb{Z} \to \langle x \rangle \qquad k \mapsto x^k$$

   is well defined and is an isomorphism

***Proof*:**

1. By proposition 2.3.4, $|x| = |y| = n$. By proposition 2.3.7, $x^m = 1$ with $m \neq 0$ implies $n \mid m$, and conversely $n \mid m$ gives $x^m = (x^n)^{m/n} = 1$. For $m = 0$ both hold. So $x^m = 1$ exactly when $n \mid m$, and the same holds for y. For $a, b \in \mathbb{Z}$, multiplying by x^{-b} and y^{-b} gives

$$x^a = x^b \iff x^{a-b} = 1 \iff n \mid a-b \iff y^{a-b} = 1 \iff y^a = y^b$$

   The implications from left to right show that ψ is well defined, and those from right to left show that it is injective. It is a homomorphism, since $\psi(x^a x^b) = \psi(x^{a+b}) = y^{a+b} = y^a y^b = \psi(x^a)\psi(x^b)$, and it is surjective, since every element of $\langle y \rangle$ is some $y^k = \psi(x^k)$. Hence ψ is an isomorphism.

2. The map ψ is a homomorphism, since $x^{a+b} = x^a x^b$, and it is surjective, since every element of $\langle x \rangle$ is a power of x. If $\psi(a) = \psi(b)$ with $a > b$, then $x^{a-b} = 1$ with $a - b > 0$, so x has finite order and $\langle x \rangle$ is finite by proposition 2.3.4, a contradiction. Hence ψ is injective, and so an isomorphism.

Two infinite cyclic groups are both isomorphic to $\mathbb{Z}$ by (2), hence to each other, and two finite cyclic groups of the same order are isomorphic by (1), completing the proof.

This let's us properly conclude what was stated at the beginning of this section: if G is s cyclic group

of order n (ex. $\mathbb{Z}/n\mathbb{Z}$ or $\langle r \rangle$, $r \in D_n$) then

$$\mathbb{Z}/n\mathbb{Z} \cong \langle r \rangle$$

Hence, instead of referencing a specific cyclic group, it is sometimes useful to just write C_n (some authors like Z_n) to represent some arbitrary cyclic group of order n, and use $\mathbb{Z}$ for an infinite cyclic group (instead of C_∞ or Z_∞)

Now that we classified all cyclic groups, we can ask ourselves the question of which elements generate one of these groups. this next proposition leads us to the answer

Now that we know how many cyclic sub-groups there are, and how these groups can be generated, we can explore what kind of sub-groups we get from cyclic groups. Quite naturally, every subgroup of a cyclic sub group is cyclic. It will turn out that if you have a cyclic group of infinite order, it has an infinite number of cyclic subgroups, and if its of finite order, lets say n, then every sub groups is of an order that divides n. That is , if $H \leqslant G$, G is cyclic, $|H| = a$, then $a|n$.

Proposition 2.3.10: x relation to cyclic group H

Let $H = \langle x \rangle$.

1. Assume $|x| = \infty$. Then $H = \langle x^a \rangle$ if and only if $a = \pm 1$.
2. Assume $|x| = n < \infty$. Then $H = \langle x^a \rangle$ if and only if $(a, n) = 1$. In particular, the number of generators of H is $\psi(n)$, where ψ is Euler's totient function. (which gives the number of relatively prime numbers to it).

Proof. 1. Suppose $a = \pm 1$. Then $\langle x^a \rangle$ contains x^a and its inverse x^{-a}, hence contains x, hence contains every power of x, so $\langle x^a \rangle = H$.

Conversely, suppose $H = \langle x^a \rangle$. Since $x \in H$, there is a $k \in \mathbb{Z}$ with $x = (x^a)^k = x^{ak}$, and therefore

$$x^{ak-1} = 1$$

Because $|x| = \infty$, the only power of x equal to 1 is x^0, so $ak = 1$. As a and k are integers, this forces $a = \pm 1$.

2. Write $n = |x|$. By proposition 2.3.4 applied to $H = \langle x \rangle$ and to $\langle x^a \rangle$,

$$|H| = |x| = n \qquad |\langle x^a \rangle| = |x^a|$$

and by corollary 2.3.8(2), $|x^a| = n/\gcd(n, a)$. Now $\langle x^a \rangle \leqslant H$, and both are finite, so $H = \langle x^a \rangle$ if and only if the two have the same order, that is if and only if

$$\frac{n}{\gcd(n, a)} = n$$

which happens if and only if $\gcd(a, n) = 1$.

For the count, every element of H is a power of x, and $x^k = x^{k'}$ whenever $k \equiv k' \bmod n$, so $H = \{x^0, x^1, \ldots, x^{n-1}\}$. These n elements exhaust a group of order n and are therefore distinct. By the previous paragraph the generators of H among them are exactly the x^a with $\gcd(a, n) = 1$, so the number of generators is the number of $a \in \{1, 2, \ldots, n\}$ with $\gcd(a, n) = 1$, which is $\psi(n)$.

□

Finally, we will put together all our knowledge into one theorem for easy reference:

Theorem 2.3.11: elements that generate cyclic subgroup

Let $H = \langle x \rangle$ be a cyclic group

1. Every subgroup of H is cyclic. More precisely, if $K \leqslant H$, then either $K = \{1\}$ or $K = \langle x^d \rangle$, where d is the smallest positive integer such that $x^d \in K$
2. If $|H| = \infty$, then for any distinct non negative integers a, b, $\langle x^a \rangle \neq \langle x^b \rangle$. Furthermore, for every integer m, $\langle x^m \rangle = \langle x^{|m|} \rangle$, where $|m|$ denotes the absolute value of m, so that the nontrivial subgroups of h correspond bijectivity with the integers $1, 2, 3, ...$
3. If $|H| = n < \infty$, then for each positive integer a dividing n, there is a unique subgroup of H of order a. This subgroup is the cyclic group $\langle x^d \rangle$, where $d = \frac{n}{a}$. Furthermore, for every integer m, $\langle x^m \rangle = \langle x^{gcd(n,m)} \rangle$, so that the subgroups of H correspond bijectivity with the positive divisors of n.

***Proof*:**
This proof involves a lot of counting.

1. Let $K \leqslant H$. If $K = \{1\}$, then $K = \langle 1 \rangle$ is cyclic. So assume $K \neq \{1\}$. Every element of K lies in $H = \langle x \rangle$, and so has the form x^a for some $a \in \mathbb{Z}$ (definition 2.3.1). Pick an element $x^a \in K$ with $x^a \neq 1$. Then $a \neq 0$, since $x^0 = 1$. Since K is a subgroup, $(x^a)^{-1} = x^{-a} \in K$ as well. One of a and $-a$ is positive, so the set

   $$S = \{b \in \mathbb{Z} | b > 0 \text{ and } x^b \in K\}$$

   is non-empty. By the well-ordering of the positive integers, the set S has a least element d. By construction, d is the smallest positive integer such that $x^d \in K$. We claim that $K = \langle x^d \rangle$.

   For the inclusion $\langle x^d \rangle \subseteq K$: we have $x^d \in K$, and K is closed under products and inverses, so $(x^d)^k \in K$ for every $k \in \mathbb{Z}$.

   For the inclusion $K \subseteq \langle x^d \rangle$: let $x^t \in K$ be arbitrary, with $t \in \mathbb{Z}$. Write $t = qd + r$ with $q, r \in \mathbb{Z}$ and $0 \leq r < d$. Such q and r exist. Indeed, some integers of the form $t - qd$ (for $q \in \mathbb{Z}$) are non-negative: taking $q = -|t|$ gives $t - qd = t + |t|d \geq t + |t| \geq 0$, since $d \geq 1$. By the well-ordering of the non-negative integers we may let $r = t - qd$ be the least non-negative integer of this form. If $r \geq d$, then $r - d = t - (q + 1)d$ would be a smaller non-negative integer of the same form, so $0 \leq r < d$. Now

   $$x^r = x^{t-qd} = x^t (x^d)^{-q} \in K$$

   since $x^t \in K$ and $x^d \in K$, and K is a subgroup. By the minimality of d, the exponent r cannot satisfy $0 < r < d$, so $r = 0$. Thus $x^t = (x^d)^q \in \langle x^d \rangle$. The two inclusions together give $K = \langle x^d \rangle$, so K is cyclic.

2. Assume $|H| = \infty$. Since $H = \langle x \rangle$, proposition 2.3.4 gives $|x| = \infty$. All powers of x are distinct: if $x^s = x^t$ with $s > t$, then $x^{s-t} = x^s (x^t)^{-1} = 1$ with $s - t > 0$, so a positive power of x would equal the identity, contradicting $|x| = \infty$ (definition 1.2.8).

Now let a and b be distinct non-negative integers, and without loss of generality say $a > b \geq 0$. Suppose toward a contradiction that $\langle x^a \rangle = \langle x^b \rangle$. If $b = 0$, then $\langle x^b \rangle = \langle 1 \rangle = \{1\}$, while $x^a \neq x^0 = 1$ because the powers of x are distinct and $a > 0$. Then $x^a \in \langle x^a \rangle$ but $x^a \notin \langle x^b \rangle$, a contradiction. If $b > 0$, then $x^b \in \langle x^a \rangle$ gives $x^b = (x^a)^k = x^{ak}$ for some $k \in \mathbb{Z}$, and since the powers of x are distinct, $b = ak$. Likewise $x^a \in \langle x^b \rangle$ gives $a = b\ell$ for some $\ell \in \mathbb{Z}$. Then $a = ak\ell$, and since $a \neq 0$ we get $k\ell = 1$, so either $k = \ell = 1$ or $k = \ell = -1$. Since a and b are both positive, $b = ak$ forces $k = 1$, hence $a = b$. This contradicts $a \neq b$. Thus $\langle x^a \rangle \neq \langle x^b \rangle$.

For the second claim, let $m \in \mathbb{Z}$. If $m \geq 0$, then $|m| = m$ and there is nothing to prove. If $m < 0$, then $x^{|m|} = x^{-m} = (x^m)^{-1} \in \langle x^m \rangle$, so every power of $x^{|m|}$ lies in $\langle x^m \rangle$, that is, $\langle x^{|m|} \rangle \subseteq \langle x^m \rangle$. In the same way, $x^m = (x^{|m|})^{-1} \in \langle x^{|m|} \rangle$ gives $\langle x^m \rangle \subseteq \langle x^{|m|} \rangle$. Hence $\langle x^m \rangle = \langle x^{|m|} \rangle$.

Finally, consider the map $d \mapsto \langle x^d \rangle$ from the positive integers to the nontrivial subgroups of H. For $d \geq 1$ the subgroup $\langle x^d \rangle$ is nontrivial, since $x^d \neq 1$. The map is surjective by part 1, and it is injective because distinct positive integers $d \neq d'$ give $\langle x^d \rangle \neq \langle x^{d'} \rangle$, by the first claim of this part: distinct non-negative exponents yield distinct subgroups. Thus the nontrivial subgroups of H correspond bijectively with the integers $1, 2, 3, \ldots$

3. Assume $|H| = n < \infty$. Since $H = \langle x \rangle$, proposition 2.3.4 gives $|x| = n$. In particular, $x^n = 1$.

   We first prove that for every integer m, $\langle x^m \rangle = \langle x^{\gcd(n,m)} \rangle$. If $m = 0$, then $\gcd(n, m) = n$, and $\langle x^0 \rangle = \{1\} = \langle x^n \rangle$ since $x^n = 1$. So let $m \neq 0$ and write $g = \gcd(n, m)$. Since g divides m, we have $x^m = (x^g)^{m/g} \in \langle x^g \rangle$, and hence $\langle x^m \rangle \subseteq \langle x^g \rangle$. For the reverse inclusion, corollary 2.3.8 together with proposition 2.3.4 gives

   $$|\langle x^m \rangle| = |x^m| = \frac{n}{\gcd(n, m)} = \frac{n}{g}, \qquad |\langle x^g \rangle| = |x^g| = \frac{n}{\gcd(n, g)} = \frac{n}{g}$$

   where $\gcd(n, g) = g$ because g divides n. Thus $\langle x^m \rangle$ is a subset of the finite set $\langle x^g \rangle$ and has the same cardinality, so $\langle x^m \rangle = \langle x^g \rangle$.

   Existence: let a be a positive divisor of n, and let $d = \frac{n}{a}$. Then d divides n, so by corollary 2.3.8 together with proposition 2.3.4,

   $$|\langle x^d \rangle| = |x^d| = \frac{n}{d} = a$$

   Thus $\langle x^d \rangle$ is a subgroup of H of order a.

   Uniqueness: let $K \leqslant H$ with $|K| = a$. If $a = 1$, then K contains only the identity, so $K = \{1\} = \langle x^n \rangle = \langle x^d \rangle$, since in this case $d = n$. If $a > 1$, then $K \neq \{1\}$, so by part 1, $K = \langle x^k \rangle$, where k is the smallest positive integer such that $x^k \in K$. By corollary 2.3.8 and proposition 2.3.4 again,

   $$a = |K| = |x^k| = \frac{n}{\gcd(n, k)}$$

   so $\gcd(n, k) = \frac{n}{a} = d$. Applying the equality $\langle x^m \rangle = \langle x^{\gcd(n,m)} \rangle$ with $m = k$ gives $K = \langle x^{\gcd(n,k)} \rangle = \langle x^d \rangle$. Thus every subgroup of H of order a equals $\langle x^d \rangle$.

   Finally, every subgroup K of H has order dividing n: this holds for $K = \{1\}$, and otherwise $K = \langle x^k \rangle$ as above has order $\frac{n}{\gcd(n,k)}$, which divides n because $\gcd(n, k)$ divides n. The

map sending a positive divisor a of n to the unique subgroup of H of order a is therefore surjective onto the subgroups of H, and it is injective because subgroups of distinct orders are distinct. Thus the subgroups of H correspond bijectively with the positive divisors of n.

Also, once you have Lagrange's Theorem (theorem 3.3.1), this proof will become much easier.

Note that the finite group examples of theorem 2.3.11 will eventually easily generalize to perhaps the most use theorem of the course: Lagrange Theorem (theorem 3.3.1), which sates that the order of any subgroup divides the order of a group ($H \leqslant G \Rightarrow |H| \big| |G|$).

An abelian subgroup H of G, even one of prime order, need not be contained in the center $Z(G)$:

Example 2.3.12: : $H \notin Z(G)$

Take D_4. Note that $\langle \sigma \rangle \leqslant D_4$ and $|\langle \sigma \rangle| = 2$, and so it's cyclic and abelian. However. $r\sigma = \sigma r^3$ meaning that $\sigma \notin Z(D_4)$. However, it is in the centralizer $C_{D_4}(\sigma) = \{g \in G | g\sigma = \sigma g\}$ for $\sigma\sigma = e = \sigma\sigma$, which can also be "obvious" since $\langle \sigma \rangle$ is abelian, and so it would be in the centralizer.

The intuition behind this is that for something to be in the center, it must commute with *all* of G. It is possible to take a subgroup that only commutes (as we have seen with the idea of the centralizer). This is why the intersection under definition 2.2.4 is true: it must commute with everything.

2.3.13 Note You cannot take $\langle x \rangle \leqslant G$ and do $G - \langle x \rangle$. G is not necessarily a group. Take D_4, and take $K = \langle \sigma \rangle$. Then consider $H = D_4 - K$. Then $r^3 \cdot r\sigma = \sigma$, showing that H is not closed under multiplication. Thus, subtraction of groups is not well-defined. However, it will turn out that under the right circumstances, *dividing* and *multiplying* groups will be well-defined! This will be the subject of the next chapter (chapter 3)

interesting exercise

1. number 9
2. number 13
3. number 16
4. Knowing Cyclic groups, we can look at another great example of a homomorphism: $\psi : \mathbb{Z} \to Z_n$, where $Z_n = \langle x \rangle$,
$$\psi(a) = x^a$$
Prove that it's a homomorphism, then figure out what elements of $\mathbb{Z}$ map to an element of Z_n.
5. the product of two cyclic subgroups is not necessarily cyclic!

2.3.2 Generating Subgroups

We now turn our attention to generating groups or subgroups using more than 1 element. When studying cyclic groups, we asked whether given an element $x \in G$, does there exists a smallest group $H \leqslant G$ such that $x \in H$. It was shown that taking the closure of x by taking all powers of x resulted

in such a group, namely $H = \langle x \rangle$. Phrased differently, the subgroup $\langle x \rangle$ is the smallest subgroup of G that contains the set $\{x\}$. What it means to be the *smallest* is if $L \leqslant G$ is another subgroup where and $x \in L$, then $\langle x \rangle \subseteq L$. We now replace the singleton $\{x\}$ with some arbitrary subset $A \subseteq G$.

As mentioned in the beginning of this chapter, generating subgroups from a subset of a group is not unique to Group Theory. In the following chapters studying other algebraic structures (ex. Ideals, rings, modules, vector-spaces, algebras, fields, etc.), the theme of trying to find minimal sub-object $S \leqslant X$ (subgroup, subspace, subfield, subring, etc.) that contain a subset $A \subseteq X$ from the original object (so that $A \subseteq S \leqslant X$) will repeat itself, and so will the general method for finding the minimal subobject. To find a minimal subobject there are two possible general approach:

1. (Generation Method) If $A \subseteq X$ is a subset of the object, take all possible subobjects containing A and intersect them. In the case of groups, if $A \subseteq G$, then take all possible subgroup $H_A \leqslant G$ containing A (so $A \subseteq H_A \leqslant G$ and intersect them all.
2. (Closure Method) The above approach can be thought as the "global approach". Another way of finding the minimal subobject is instead by the "element approach" (or "local approach"): If A is a subset of the object, then let $\overline{A}$ be the set of all possible combinations of elements of A using the binary operation of G. In the case of groups, of $A \subseteq G$, then $\overline{A}$ will be all the possible combination of elements of A with the binary operation of G.

The main result of this section is that these two methods will coincide and form the same subgroup. In proceeding chapters covering subobjects (subrings, ideals, subspaces, etc.), some details will be omitted due to them being covered in this section, so it is advised that this section be well understood.

We start with the first method. We need show that the intersection of subgroups still forms a group:

Proposition 2.3.14: intersection of Groups

Let G be a group and let $\mathcal{H}$ be any collection of subgroups of G. Then

$$\bigcap_{H \in \mathcal{H}} H$$

Is a subgroup of G.

***Proof*:**
Let $K = \bigcap_{H \in \mathcal{H}} H$. First, $e \in H$ for all $H \in \mathcal{H}$, so $e \in K$. Next, take $a, b \in K$. Since $b \in K$, b is in every subgroup H in $\mathcal{H}$. Thus, b^{-1} is in every subgroup H in $\mathcal{H}$, and so $b^{-1} \in K$. Similarly, ab^{-1} would be in every subgroup H in $\mathcal{H}$ (since each subgroup passes the subgroup test), so $ab^{-1} \in K$. Thus, by the subgroup test, K is a subgroup of G

Therefore, the following is well-defined:

Definition 2.3.15: Minimal Subgroup via Generation

let G be a group and $A \subseteq G$. Then

$$\langle A \rangle = \bigcap_{\substack{H \leqslant G \\ A \subseteq H}} H$$

is called the *closure of* A, or the subgroup *generated by* A. If A is a finite set $\{a_1, a_2, ..., a_n\}$, we can write $\langle a_1, a_2, ..., a_n \rangle$. Furthermore, $\langle A, B \rangle := \langle A \cup B \rangle$ for the union of two sets.

We have already been using this notation when $A = \{x\}$ is a singleton, in particular cyclic subgroups are the closure of singletons. Note that $\langle A \rangle$ always exists, since at minimum $A \subseteq G$ (and naturally $G \leqslant G$), so G is part of the sets in be intersection. This is important to point out as there exists sets where $\langle A \rangle = G$:

Definition 2.3.16: Generating Set and Generators

Let $A \subseteq G$. Then if $\langle A \rangle = G$, then A is called a *generating set* for G, and the elements of A are called *generators*.

If $A = \emptyset$, then A is containing in every group, including the trivial group $\{e_G\}$. As the intersection $\{e_G\}$ with any group is $\{e_G\}$, we see that $\langle \emptyset \rangle = \{e_G\}$.

Proposition 2.3.17: Minimal Subgroup using Generation

Let $A \subseteq G$. Then $\langle A \rangle \leqslant G$ is the unique smallest subgroup containing A.

***Proof*:**
If $H' \leqslant G$ and $A \subseteq H'$, then by definition H' is one of the subgroup that were intersected in the aforementioned definition, since $\langle A \rangle$ is equal to this intersection, $\langle A \rangle \subseteq H'$, so $\langle A \rangle$ is minimal.

Next, $\langle A \rangle$ is the unique smallest subgroup containing A (similarly to how $\langle x \rangle$ is unique in the sense that all cyclic subgroups of the same order are isomorphic to it). If there exists another group K' that was also the smallest subgroup containing A, then K' would be in the intersection above, so $\langle A \rangle \subseteq K'$. But since K' is supposed to be the smallest subgroup containing A, it is also the case that $K' \subseteq \langle A \rangle$. Therefore $\langle A \rangle = K'$, showing uniqueness as well, completing the proof.

The limit with this approach is that all the possible subgroups containing A must be found, and that is not always feasible. There is not a general method to find elements of $\langle A \rangle$ (except, naturally, that $e_G \in \langle A \rangle$). The 2nd approach will give a better insight into this problem by changing perspective away from the high-level and starting with the elements of A and trying to "generate" a subgroup containing A. To that end, define the set

$$\overline{A} := \{a_1^{\sigma_1} a_2^{\sigma_2} \cdots a_n^{\sigma_n} \mid n \geq 1 \text{ and } a_i \in A,\ \sigma_i = \pm 1, 1 \leq i \leq n\}$$

where if $A = \emptyset$ then $\overline{A} = \{e_G\}$. The set $\overline{A}$ contains all possible finite products (using the binary operation of G) of the elements from A. Note that the a_i's need not be distinct, for example a^2

for $a \in A$ would be written a^1a^1. It is also easy to verify that this set is indeed a subgroup of G using the subgroup criterion (consider that n can vary in any possible length, see exercise 2.3.1 (1)). The interesting question is whether this set is also the smallest minimal subgroup containing A, and indeed it is:

Proposition 2.3.18

Let G be a group and $A \subseteq G$ be a subset of G. Then

$$\langle A \rangle = \overline{A}$$

***Proof*:**
If $A = \emptyset$, then $\overline{A} = \{e_G\}$, and $\langle \emptyset \rangle = \{e_G\}$ as computed before proposition 2.3.17. So let $A \neq \emptyset$.

First, $\overline{A}$ is a subgroup of G containing A. It is nonempty, and for $x = a_1^{\sigma_1} \cdots a_n^{\sigma_n}$ and $y = b_1^{\rho_1} \cdots b_m^{\rho_m}$ in $\overline{A}$, the inverse of y is $b_m^{-\rho_m} \cdots b_1^{-\rho_1}$, so

$$xy^{-1} = a_1^{\sigma_1} \cdots a_n^{\sigma_n} b_m^{-\rho_m} \cdots b_1^{-\rho_1}$$

is again a product of elements of A and their inverses, that is, $xy^{-1} \in \overline{A}$. So $\overline{A} \leqslant G$ by proposition 2.1.3. It contains every $a \in A$, as the product a^1 with $n = 1$. Hence $\langle A \rangle \subseteq \overline{A}$ by proposition 2.3.17.

Conversely, let H be a subgroup of G with $A \subseteq H$. Then H contains every $a \in A$ and its inverse, and it is closed under products, so induction on n shows that H contains every product $a_1^{\sigma_1} \cdots a_n^{\sigma_n}$ with $a_i \in A$ and $\sigma_i = \pm 1$. So $\overline{A} \subseteq H$ for every such H, and $\overline{A} \subseteq \langle A \rangle$ by definition 2.3.15. Hence $\langle A \rangle = \overline{A}$, as we sought to show.

From now on, the notation $\langle A \rangle$ will be used to represent the subgroup generated by A.

We can ask some basic question about $\langle A \rangle$ given A and G. If G is abelian, then the order the elements are written in does not matter, and so if $A = \{a_1, ..., a_k\}$, every element can be written as:

$$a_1^{n_1} a_2^{n_2} \cdots a_k^{n_k}$$

This is not the case if G is not abelian. Consider for example:

$$G = \langle a, b \mid a^2 = b^2 = 1 \rangle \tag{2.1}$$

Notably, $ab \neq ba$. Next, if each $a_i \in A$ has finite order, then if G is abelian we have that G is finite. This is because:

$$(ab)^n = a^n = b^n$$

See exercise 2.3.1 (2). This is not necessarily the case if G is non-abelian. Consider again the group in equation (2.1) where $|a| = 2$ and $|b| = 2$. Then we see that:

$$abababa....$$

If G is not abelian, then as mentioned in box 1.3.5 and example 1.3.6, it is exceedingly difficult to determine the structure of G.

We may also ask if there is any "optimal" set of generators for a group, that is a "best" subset $A \subseteq G$ st $\langle A\rangle = G$. Generally, there can be many sets that generate a group: for example $\langle r, \sigma\rangle = \langle r\sigma, \sigma\rangle = D_n$, or $\langle 2, 3\rangle = \langle 7, 5\rangle = \mathbb{Z}/10\mathbb{Z}$. Notice that no proper subset of $\{2, 3\}$ generates $\mathbb{Z}/10\mathbb{Z}$, however there do exist smaller subsets of $\mathbb{Z}/10\mathbb{Z}$ that generate $\mathbb{Z}/10\mathbb{Z}$ (namely, $\{1\}$). Thus, the cardinality of the minimal generating subset cannot be determined by simply finding a minimal generating subset: it would have to be $\min\{|A| \mid A \subseteq G,\ \langle A\rangle = G\}$[3].

Notice also that in the case of D_4 with generators $a = \sigma$ and $b = r\sigma$ have order 2, while D_4 has order 8. Using this show how it is not always the case that all elements of D_4 will be written of the form $a^\alpha b^\beta$ $(\alpha, \beta \in \mathbb{Z})$ like in the abelian case, in particular show that aba cannot be written as $a^\alpha b^\beta$ for any $\alpha, \beta \in \mathbb{Z}$.

2.3.19 Food for Thought If G is finitely generated, is every subgroup of G finitely generated? In other words, if

$$G = \langle S|R\rangle$$

where S is a set of elements and R is the relation on those elements, then can you find a subgroup $H \leqslant G$ such that you need an infinite number of elements to generate H? There is an answer to this question which will be given as exercise 2.3.2 (1), but it can be useful to contemplate this question now as an exercise in the types of questions a mathematicians might ask themselves. You can further ask if $\varphi : G \to H$ is a homomorphsims, and G is finitely generated, is $\varphi(G)$ finitely generated?

Exercise 2.3.1

1. Show that $\overline{A}$ is indeed a subgroup
2. Let G be abelian and $A \subseteq G$ a finite set containing only elements of finite order. Show that $\langle A\rangle$ is finite.
3. Show that every finitely generated subgroup of $\mathbb{Q}$ is cyclic
4. Let (Presentation for V_4. This group is called the Klein 4-group, or *Viergruppe*. Prove that $V_4 \cong \mathbb{Z}_2 \times \mathbb{Z}_2$, showing that a group has a non-unique representation.
5. If $H, K \leqslant G$, is $H \cup K$ a group? If $H \cup K$ is a subgroup in some case, what relation would there be between H and K (does on include the other?)
6. Let $G = k[x, 1/x]$ be the group of polynomials under multiplication.

2.3.3 Free Groups

So far, when we were generating groups from sets, we either took a subset of G (and so took the relation between elements of G), or we defined the set given a presentation. For example, We can make a group from $\{a, b\}$ by defining [4]

$$\langle a, b|a^n = 1, b^2 = 1, ab = b^{-1}a\rangle$$

[3] If G is infinite, it might be that we require the stronger notion of an Infimum: $\inf\{|A| \mid A \subseteq G,\ \langle A\rangle = G\}$
[4] This is D_n

what happens if we get rid of one or two of these relations? What if we get rid of all three, and make a group "free" of relations besides the ones needed to form a group? We explore this question in this section.

Since this group has no restrictions, it can be thought of as the "most general" group on that number of elements. In fact, something more subtle is going on here: since $\{a, b\}$ (or any set A to that matter) is essentially an arbitrary set, we are asking if from *any* arbitrary set, is there a way of defining a group using as little "extra group structure" as possible, or put another way, as free from relations as possible. If we can define such a group given $\{a, b\}$, is there a way of "defining" another group generated by two elements using this more "free group" by adding relations to the "free group"?

The answers to this question lies in exploring how to make a *function* (or to be pedantic a *set-function*) from any set A to a group G

$$f : A \to G$$

be "lifted" to a *group-homomorphism*

$$\varphi : F(A) \to G$$

where $F(A)$ will be our "free group" on A (the notation $F(A)$ is suggestive that this group will indeed be called a free group). The word "lifted" here means that if we restrict $F(A)$ to A, then $\varphi|_A = f$. This can be visualized with the following diagram (called a commutative diagram):

$$\begin{array}{ccc} F(A) & \longrightarrow & G \\ \uparrow & \nearrow & \\ A & & \end{array}$$

Thus, we turn a set function into a group-homomorphism while still keeping the structure of the set-function. More generally, if $f_1 \colon A \to G_1$ and $f_2 \colon A \to G_2$ are functions and $\varphi \colon G_1 \to G_2$ is a homomorphism with $f_2 = \varphi \circ f_1$, then φ is called a lift of f_2 through f_1. Equivalently, φ is such a lift if the following diagram commutes:

$$\begin{array}{ccc} G_1 & \xrightarrow{\varphi} & G_2 \\ {\scriptstyle f_1}\uparrow & \nearrow_{f_2} & \\ A & & \end{array}$$

Why are lifts the key concept? It is because not all set-function can be lifted to group homomorphism:

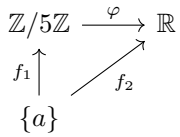
Example 2.3.20: Lifts and non-Lifts

1. Let $A = \{a\}$, $G_1 = \mathbb{Z}/5\mathbb{Z}$, and $G_2 = \mathbb{R}$. Define
   (a) $f_1 : A \to G_1$, $f_1(a) = 0$
   (b) $f_2 : A \to G_2$, $f_2(a) = 4$

   Does A lift to G_1, that is:

$$\begin{array}{ccc} \mathbb{Z}/5\mathbb{Z} & \longrightarrow & \mathbb{R} \\ {\scriptstyle f_1}\uparrow & \nearrow_{f_2} & \\ \{a\} & & \end{array}$$

that is, does there exist a homorphism $\varphi : \mathbb{Z}/5\mathbb{Z} \to \mathbb{R}$ such that $f_2 = \varphi \circ f_1$ If we can find a homomorphism φ such that



then a lift exists. Suppose there were such a φ. Then

$$4 = f_2(a) = (\varphi \circ f_1)(a) = \varphi(0) = 0$$

since a homomorphism sends 0 to 0 (proposition 1.4.4(1)). But $4 \neq 0$ in $\mathbb{R}$, so no such homomorphism exists, and f_2 has no lift through f_1.

2. Now swap the roles of 0 and 4: let $f_1(a) = 4$ in $\mathbb{Z}/5\mathbb{Z}$ and $f_2(a) = 0$ in $\mathbb{R}$. Every homomorphism $\varphi\colon \mathbb{Z}/5\mathbb{Z} \to \mathbb{R}$ is zero, because $5\varphi(x) = \varphi(5x) = \varphi(0) = 0$ in $\mathbb{R}$ for every $x \in \mathbb{Z}/5\mathbb{Z}$ (proposition 1.4.4(3)), so $\varphi(x) = 0$. The zero homomorphism satisfies $\varphi(f_1(a)) = 0 = f_2(a)$, so it is a lift of f_2 through f_1, and it is the only one.

3. Let $f_1'(a) = 1$ in $\mathbb{Z}/5\mathbb{Z}$ and $f_2'(a) = 1$ in $\mathbb{R}$. A lift φ would satisfy $\varphi(1) = 1$, but every homomorphism $\mathbb{Z}/5\mathbb{Z} \to \mathbb{R}$ is zero by the previous item. So f_2' has no lift through f_1'.

4. Let $G_1 = G_2 = \mathbb{Z}/5\mathbb{Z}$ and let $A = \{a\}$. Let

   (a) $f_1 : A \to \mathbb{Z}/5\mathbb{Z},\ f_1(a) = 1$

   (b) $f_2 : A \to \mathbb{Z}/5\mathbb{Z},\ f_2(a) = 2$

   A lift φ of f_2 through f_1 satisfies $\varphi(1) = 2$, hence $\varphi(x) = x\varphi(1) = 2x$ for every x (proposition 1.4.4(3)). Conversely, $x \mapsto 2x$ is a homomorphism of $\mathbb{Z}/5\mathbb{Z}$, since $2(x + y) = 2x + 2y$. So there is exactly one lift. If instead $f_1(a) = f_2(a) = 0$, then every homomorphism $\varphi\colon \mathbb{Z}/5\mathbb{Z} \to \mathbb{Z}/5\mathbb{Z}$ is a lift, since $\varphi(0) = 0$, and the five homomorphisms $x \mapsto mx$ with $0 \leq m \leq 4$ are distinct, because they send 1 to different elements. So there can be several lifts.

Which group $F(A)$, together with a function $i\colon A \to F(A)$, admits a lift through i of every function from A to every group? The trivial group does not, as soon as A has two elements $a \neq b$. If $j\colon A \to \{e\}$ and $f\colon A \to G$ takes distinct values $f(a) \neq f(b)$, then a lift φ of f through j would give $f(a) = \varphi(j(a)) = \varphi(e) = \varphi(j(b)) = f(b)$. The function j imposes the relation $j(a) = j(b)$, and every lift must respect it. So $F(A)$ should satisfy no relations among the elements $i(a)$ beyond those that the group axioms force.

The High-level Idea

The construction below recurs throughout the book in different settings, as the *free objects*[5] and, more generally, as *universal properties*[6]. The *universal property of free groups* is the first universal

[5] *Free monoids*, such as the monoid of strings over an alphabet, appear in section 7.1, and *free modules* in section 12.3.2.

[6] For instance theorem 3.1.15, theorem 5.4.3 (products), theorem 5.4.10 (coproducts), theorem 9.2.4, theorem 9.5.4, theorem 15.3.4 and theorem 15.1.12.

property in this book after that of the trivial group (proposition 1.4.5).

Let's say A is an arbitrary set. Our goal is to construct a group, which we will label $F(A)$, from which every other group which contains A can be derived. First, we will be "relating" A to any group G. Let's take all set-functions:

$$f : A \to G$$

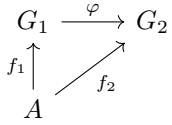
Where A is a set that maps into the group G [7]. Let's label this function along with its co-domain G as (f, G). Choosing all possible f and G will make all possible functions from A to every possible group, essentially telling us all ways A relates to any group G. The fact that there is no restriction on the functions from A to G is a way of saying there is *no* group structure on A which limits the types of maps we can form. Remember that when we lift the map $f : A \to G$ to $\varphi : F(A) \to G$, we need that $\varphi|_A = f$, so the fact that the map can be arbitrary means that $F(A)$ must accommodate for any function from A to G. As we have shown in example 2.3.20, not all groups can accommodate this. Let's label the set of all possible (f, G) as $\mathcal{F}^A$:

$$\mathcal{F}^A = \{(f, G) | f : A \to G\}$$

Next, we want to be able to relate the groups in the co-domain. We want to define $F(A)$ to be able to lift from a set-function to a group-homomorphsims. What we will do is define the following relation between any two pairs (f_1, G_1), (f_2, G_2): a *morphism*

$$(f_1, G_1) \xrightarrow{\varphi} (f_2, G_2)$$

is a homomorphism φ: $G_1 \to G_2$ such that the following diagram commutes:



that is, where $f_2 = \varphi \circ f_1$. You can think of the morphism φ as a homomorphism that is restricted by f_1 and f_2. In fact, every example in example 2.3.20 can be re-written to be morphisms from $\mathcal{F}^A$. Take a moment to go back and translate the previous notation to the new one, and think about what $\mathcal{F}^A$ with its objects and morphisms represent. We have essentially taken any group homomorphism between any two groups $G_1 \xrightarrow{\varphi} G_2$ and also made a relation between A and these groups. In fact, as reader's who have read [24, Categories] and chapter 8 might have guessed, we have defined a category

[7] It might be more intuition for every f to be *injective*, since this way we'd get that every G contains the set A. We can indeed do this, and we will get the same result. For reasons that will be elaborated later in this section, the more general approach is desirable to be able to repeat this trick later in the textbook

Proposition 2.3.21: $\mathcal{F}^A$ is a category

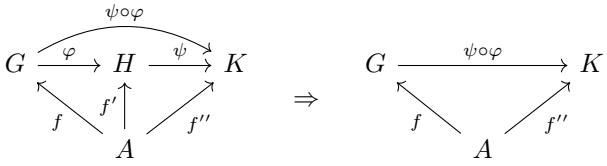
The set $\mathcal{F}^A$ along with the morphisms form a category (we can write $\mathfrak{F}^A S$ as a shorthand to represent $\mathcal{F}^A$ and it's morphisms). In particular:

1. For every $(f, G) \in \mathfrak{F}^A$, there exists an identity morphism $1_{(f,G)} : (f, G) \to (f, G)$
2. If $\varphi, \psi \in \mathfrak{F}^A$ are morphisms

$$\varphi : (f_1, G_1) \to (f_2, G_2) \qquad \psi : (f_2, G_2) \to (f_3, G_3)$$

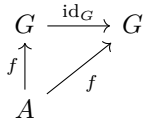
then $\psi \circ \varphi$ is a morphism.

***Proof*:**
First, composition of morphisms will be a morphism in $\mathcal{F}$. If $\varphi, \psi \in \mathcal{F}$ are composable morphisms (i.e. the codomain of φ matches the domain of ψ) then



that is, the morphism $\psi \circ \varphi \in \mathcal{F}^A$ with domain (f, G) and codomain (f'', K) exists

Next, we will check each object $(f, G) \in \mathcal{F}^A$ has an identity morphism. Take an arbitrary $(f, G) \in \mathcal{F}^A$, so that $A \xrightarrow{f} G$. Then the homomorphism $\mathrm{id}_G : G \to G$, $\mathrm{id}_G(x) = x$ satisfies



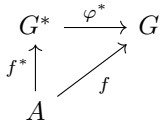
If we take any $\varphi \in \mathcal{F}^A$ where $\varphi : (f, G) \to (f', H)$, then $\mathrm{id}_H \varphi = \varphi \mathrm{id}_G = \varphi$, showing that identities exist.

The fact that any compossible triple is associative comes from function composition.

With the category $\mathfrak{F}^A$ finally defined, we can ask the following question

Does $\mathfrak{F}^A$ have an initial object[8]?

That is, does there exists (f^*, G^*) such that for any object $(f, G) \in \mathfrak{F}^A$, there exists a unique morphism φ^* such that



[8] see definition 1.4.6

Notice that this is the same lifting question we have asked about a theoretical "free group" $F(A)$ near the beginning of this section, but re-phrased! You might ask if such a group even exists? If such a group does exist, then have found a group which let's us relate it to any other group. We will call such a group (if it exists) a *free group*. Essentially, $F(A)$ must be able to relate to any other group, making it the "most general" group. But what is $F(A)$? In the following section, we will construct such a group that will indeed be the initial object of $\mathfrak{F}^A$.

Constructions of $F(A)$

When $A = \{a\}$ the answer is $\mathbb{Z}$, with $j\colon \{a\} \to \mathbb{Z}$ given by $j(a) = 1$. Let (f, G) be an element of $\mathfrak{F}^A$ and put $g = f(a)$. A homomorphism $\varphi\colon \mathbb{Z} \to G$ with $\varphi \circ j = f$ satisfies $\varphi(1) = \varphi(j(a)) = g$. Since $n \in \mathbb{Z}$ is the n-th power of 1 in the additive group $\mathbb{Z}$, proposition 1.4.4(3) forces

$$\varphi(n) = \varphi(1)^n = g^n$$

so there is at most one such φ. Conversely, $n \mapsto g^n$ is a homomorphism because $g^{m+n} = g^m g^n$, and it sends $j(a) = 1$ to $g = f(a)$. So φ exists and is unique.

When $A = \emptyset$ the answer is the trivial group $\{0\}$, which is the initial object of groups (proposition 1.4.5).

For a general set A, the group $F(A)$ is built from formal products of elements of A and of formal inverses of those elements, in which the only cancellation allowed is that of a symbol standing next to its own inverse.

Let A be a set. Choose a set A^{-1} disjoint from A together with a bijection $A \to A^{-1}$, written $a \mapsto a^{-1}$. At this stage a^{-1} is only the name of a new symbol, not an inverse in any group. The elements of $A \sqcup A^{-1}$ are called *letters*. The *formal inverse* of a letter x is the letter x^{-1}, where $x^{-1} = a^{-1}$ if $x = a \in A$ and $x^{-1} = a$ if $x = a^{-1} \in A^{-1}$. Then $(x^{-1})^{-1} = x$ for every letter x, and $x^{-1} \neq x$ because A and A^{-1} are disjoint.

Definition 2.3.22: Word

Let A be a set. A *word* on A is a finite sequence $(x_1, x_2, \ldots, x_n)$ of letters $x_i \in A \sqcup A^{-1}$, written $x_1 x_2 \cdots x_n$. The integer $n \geq 0$ is the *length* of the word, and the word of length 0 is the *empty word*, written 1. The set of all words on A is denoted $W(A)$. The *concatenation* of $w = x_1 \cdots x_n$ and $w' = y_1 \cdots y_m$ is the word $ww' = x_1 \cdots x_n y_1 \cdots y_m$ of length $n + m$.

Concatenation is associative, since $(ww')w''$ and $w(w'w'')$ are the same sequence of letters, and $1w = w1 = w$ for every word w. Concatenation does not make $W(A)$ a group: the length of ww' is the sum of the lengths of w and w', so no nonempty word has an inverse. Inverses appear once adjacent pairs xx^{-1} are allowed to cancel.

Definition 2.3.23: Reduced Word

Let A be a set.

1. If $w = uxx^{-1}v$ for words u, v and a letter x, then passing from w to uv is called an *elementary reduction*, written $w \to uv$. The notation $w \to^* w'$ means that w' is obtained from w by a finite sequence of zero or more elementary reductions.
2. A word is *reduced* if no elementary reduction applies to it, that is, if no two adjacent letters of it are formal inverses of each other. The set of reduced words on A is denoted $F(A)$.

An elementary reduction lowers the length of a word by 2. Every word of length 0 or 1 is reduced.

Example 2.3.24: Reducing a Word

Let $A = \{a, b\}$ and $w = ab^{-1}baa^{-1}b$. The word w contains two adjacent pairs of formal inverses, $b^{-1}b$ and aa^{-1}. Deleting $b^{-1}b$ first gives $aaa^{-1}b$, and deleting aa^{-1} from that gives ab. Deleting aa^{-1} first gives $ab^{-1}bb$, and deleting $b^{-1}b$ from that gives ab again. The word ab is reduced, since $b \neq a^{-1}$, so both orders of cancellation end at the same reduced word. Lemma 2.3.25 shows that the reduced word at the end never depends on the order.

Lemma 2.3.25: Reduced Word Lemma

Let A be a set. For every word $w \in W(A)$ there is exactly one reduced word $\mathrm{red}(w)$ with $w \to^* \mathrm{red}(w)$. In particular, $\mathrm{red}(w) = w$ when w is reduced.

***Proof*:**

Step 1: existence. Let n be the length of w. Apply elementary reductions to w for as long as one applies. Each lowers the length by 2, so the process stops after at most $n/2$ steps, and it stops exactly at a reduced word r, with $w \to^* r$.

Step 2: two elementary reductions of one word. Let $w = x_1 \cdots x_n$, and suppose $w \to w_1$ and $w \to w_2$ delete the adjacent pairs in positions $\{i, i+1\}$ and $\{j, j+1\}$ respectively. Exchanging the names w_1 and w_2 if necessary, we may assume $i \leq j$. If $i = j$, then $w_1 = w_2$. If $j = i + 1$, then $x_{i+1} = x_i^{-1}$ and $x_{i+2} = x_{i+1}^{-1} = x_i$, so deleting $x_i x_{i+1}$ and deleting $x_{i+1}x_{i+2}$ both leave $x_1 \cdots x_{i-1}x_i x_{i+3} \cdots x_n$, and again $w_1 = w_2$. If $j \geq i + 2$, the two pairs do not overlap, and deleting both of them, in either order, gives one word w_3 with $w_1 \to w_3$ and $w_2 \to w_3$. So either $w_1 = w_2$, or some word w_3 satisfies $w_1 \to w_3$ and $w_2 \to w_3$.

Step 3: uniqueness. We show by strong induction on the length n of w that $w \to^* r$ and $w \to^* r'$ with r and r' reduced imply $r = r'$. If w is reduced, then no elementary reduction applies to it, so both sequences of reductions are empty and $r = w = r'$. Otherwise both sequences are nonempty, since w itself is not reduced, and they begin $w \to w_1 \to^* r$ and $w \to w_2 \to^* r'$ with w_1 and w_2 of length $n - 2$. If $w_1 = w_2$, the induction hypothesis for w_1 gives $r = r'$. Otherwise Step 2 gives w_3 with $w_1 \to w_3$ and $w_2 \to w_3$, and Step 1 gives a reduced word r'' with $w_3 \to^* r''$. Then $w_1 \to^* r$ and $w_1 \to^* r''$, so $r = r''$ by the induction hypothesis for w_1, and in the same way $w_2 \to^* r'$ and $w_2 \to^* r''$ give $r' = r''$. Hence $r = r'$.

Steps 1 and 3 give exactly one reduced word $\mathrm{red}(w)$ with $w \to^* \mathrm{red}(w)$. When w is reduced, $w \to^* w$ by the empty sequence, so $\mathrm{red}(w) = w$, completing the proof.

Proposition 2.3.26: Word Group

Let A be a set. The set $F(A)$ of reduced words on A is a group under the operation

$$w \cdot w' = \mathrm{red}(ww')$$

with the empty word as identity. Regard each letter as a reduced word of length 1. Then for every $a \in A$ the letters a and a^{-1} are inverse to each other in $F(A)$, and every reduced word $x_1 x_2 \cdots x_n$ equals the product $x_1 \cdot x_2 \cdots x_n$ in $F(A)$. In particular, every element of $F(A)$ is a product of elements of A and their inverses.

***Proof*:**
By lemma 2.3.25, $\mathrm{red}(ww')$ is a reduced word, so the operation is a binary operation on $F(A)$. Moreover, if $u \to^* u'$ and $v \to^* v'$, then

$$\mathrm{red}(uv) = \mathrm{red}(u'v')$$

Indeed, an elementary reduction inside u or inside v is also an elementary reduction of the concatenation, so $uv \to^* u'v \to^* u'v' \to^* \mathrm{red}(u'v')$. Thus $\mathrm{red}(u'v')$ is a reduced word obtained from uv by elementary reductions, and it equals $\mathrm{red}(uv)$ by the uniqueness in lemma 2.3.25. The axioms of definition 1.1.3 follow:

1. (associativity) Let $u, v, w \in F(A)$. Since $uv \to^* \mathrm{red}(uv)$ and $vw \to^* \mathrm{red}(vw)$, the displayed equality gives

$$(u \cdot v) \cdot w = \mathrm{red}(\mathrm{red}(uv)w) = \mathrm{red}(uvw) = \mathrm{red}(u\,\mathrm{red}(vw)) = u \cdot (v \cdot w)$$

2. (identity) For $w \in F(A)$, $1 \cdot w = \mathrm{red}(w) = w$ and $w \cdot 1 = \mathrm{red}(w) = w$, because w is reduced.

3. (inverses) Let $w = x_1 \cdots x_n \in F(A)$ and $\bar{w} = x_n^{-1} \cdots x_1^{-1}$. Two adjacent letters $x_{k+1}^{-1} x_k^{-1}$ of $\bar{w}$ are formal inverses exactly when $x_{k+1} = x_k^{-1}$, that is, exactly when $x_k x_{k+1}$ is a pair of formal inverses in w. Since w is reduced, this never happens, so $\bar{w} \in F(A)$. In $w\bar{w} = x_1 \cdots x_n x_n^{-1} \cdots x_1^{-1}$, deleting the middle pair n times leaves the empty word, so $w \cdot \bar{w} = \mathrm{red}(w\bar{w}) = 1$. The word obtained from $\bar{w}$ in the same way is w, so the same computation gives $\bar{w} \cdot w = 1$.

So $F(A)$ is a group. For $a \in A$, the case $n = 1$ of (3) says that $a \cdot a^{-1} = a^{-1} \cdot a = 1$. Finally, induction on n shows that a reduced word $x_1 \cdots x_n$ equals the product $x_1 \cdot x_2 \cdots x_n$. For $n = 0$ both sides are the identity. For $n \geq 1$, the word $x_1 \cdots x_{n-1}$ is reduced, being part of a reduced word, and

$$(x_1 \cdots x_{n-1}) \cdot x_n = \mathrm{red}(x_1 \cdots x_n) = x_1 \cdots x_n$$

so the induction hypothesis for $x_1 \cdots x_{n-1}$ completes the proof.

Theorem 2.3.27: Universal Property Of Free Groups

Let A be a set, let $F(A)$ be the group of proposition 2.3.26, and let $i\colon A \to F(A)$ send $a \in A$ to the one-letter word a. For every group G and every function $f\colon A \to G$ there is a unique homomorphism $\varphi\colon F(A) \to G$ such that $f = \varphi \circ i$, that is, such that the following diagram commutes:

$$\begin{array}{ccc} F(A) & \xrightarrow{\varphi} & G \\ {\scriptstyle i}\uparrow & \nearrow_{f} & \\ A & & \end{array}$$

Equivalently, $(i, F(A))$ is an initial object of $\mathfrak{F}^A$: for every object (f, G) of $\mathfrak{F}^A$ there is exactly one morphism $(i, F(A)) \to (f, G)$. The group $F(A)$ is called the *free group* on A.

***Proof*:**

Let G be a group and $f\colon A \to G$ a function. Extend f to the letters by $\hat{f}(a) = f(a)$ and $\hat{f}(a^{-1}) = f(a)^{-1}$ for $a \in A$. Then $\hat{f}(x^{-1}) = \hat{f}(x)^{-1}$ for every letter x: for $x = a$ this is the definition, and for $x = a^{-1}$ it reads $f(a) = (f(a)^{-1})^{-1}$. Define $\Phi\colon W(A) \to G$ by $\Phi(1) = 1_G$ and

$$\Phi(x_1 x_2 \cdots x_n) = \hat{f}(x_1)\hat{f}(x_2)\cdots\hat{f}(x_n)$$

Step 1: Φ respects concatenation and reduction. For words w and w', the definition gives $\Phi(ww') = \Phi(w)\Phi(w')$. If $w = uxx^{-1}v$ and $w \to uv$ is the corresponding elementary reduction, then

$$\Phi(w) = \Phi(u)\hat{f}(x)\hat{f}(x)^{-1}\Phi(v) = \Phi(u)\Phi(v) = \Phi(uv)$$

So Φ is constant along any sequence of elementary reductions, and $\Phi(w) = \Phi(\mathrm{red}(w))$ for every word w (lemma 2.3.25).

Step 2: existence. Let $\varphi\colon F(A) \to G$ be the restriction of Φ to the reduced words. For $w, w' \in F(A)$, Step 1 gives

$$\varphi(w \cdot w') = \Phi(\mathrm{red}(ww')) = \Phi(ww') = \Phi(w)\Phi(w') = \varphi(w)\varphi(w')$$

so φ is a homomorphism. Also $\varphi(i(a)) = \hat{f}(a) = f(a)$ for every $a \in A$, that is, $f = \varphi \circ i$.

Step 3: uniqueness. Let $\varphi'\colon F(A) \to G$ be a homomorphism with $f = \varphi' \circ i$. Then $\varphi'(a) = f(a)$ for $a \in A$. The letter a^{-1} is the inverse of a in $F(A)$ (proposition 2.3.26), so proposition 1.4.4(2) gives $\varphi'(a^{-1}) = f(a)^{-1}$. Hence $\varphi'(x) = \hat{f}(x)$ for every letter x. Every reduced word $w = x_1 \cdots x_n$ is the product $x_1 \cdot x_2 \cdots x_n$ in $F(A)$ (proposition 2.3.26), so

$$\varphi'(w) = \varphi'(x_1)\cdots\varphi'(x_n) = \hat{f}(x_1)\cdots\hat{f}(x_n) = \varphi(w)$$

and $\varphi' = \varphi$.

Finally, a morphism $(i, F(A)) \to (f, G)$ in $\mathfrak{F}^A$ is by definition a homomorphism $\varphi\colon F(A) \to G$ with $f = \varphi \circ i$. Steps 2 and 3 therefore say that there is exactly one such morphism, so $(i, F(A))$ is an initial object of $\mathfrak{F}^A$, as we sought to show.

Corollary 2.3.28: Free Group Is Unique

Let A and B be sets such that $|A| = |B|$. Then

$$F(A) \cong F(B)$$

***Proof*:**
Since $|A| = |B|$, there is a bijection $\beta\colon A \to B$. Let $i_A\colon A \to F(A)$ and $i_B\colon B \to F(B)$ be the maps of theorem 2.3.27. Applying theorem 2.3.27 to the function $i_B \circ \beta\colon A \to F(B)$ gives a homomorphism $\varphi\colon F(A) \to F(B)$ with $\varphi \circ i_A = i_B \circ \beta$. Applying it to $i_A \circ \beta^{-1}\colon B \to F(A)$ gives a homomorphism $\psi\colon F(B) \to F(A)$ with $\psi \circ i_B = i_A \circ \beta^{-1}$. The composite $\psi \circ \varphi\colon F(A) \to F(A)$ is a homomorphism (proposition 1.4.4(6)), and

$$\psi \circ \varphi \circ i_A = \psi \circ i_B \circ \beta = i_A \circ \beta^{-1} \circ \beta = i_A$$

The identity homomorphism of $F(A)$ also satisfies $\mathrm{id}_{F(A)} \circ i_A = i_A$. The uniqueness in theorem 2.3.27, applied to the function $i_A\colon A \to F(A)$, therefore gives $\psi \circ \varphi = \mathrm{id}_{F(A)}$. Exchanging the roles of A and B gives $\varphi \circ \psi = \mathrm{id}_{F(B)}$ in the same way. So φ is a bijective homomorphism, and $F(A) \cong F(B)$ (definition 1.4.12), as claimed.

Since free groups on sets of the same cardinality are isomorphic, we usually write F_n to denote the free group with n generators.

Definition 2.3.29: Freely Generated

Let G be a group and $S \subseteq G$. We say that S *freely generates* G, or that G is *free on* S, if the inclusion of S in G has the universal property of theorem 2.3.27: for every group H and every function $f\colon S \to H$ there is a unique homomorphism $\varphi\colon G \to H$ with $\varphi(s) = f(s)$ for every $s \in S$. By theorem 2.3.27, $F(S)$ is free on its one-letter words, identified with the elements of S.

Example 2.3.30: Free Groups

1. Let $A = \{a\}$. By theorem 2.3.27, the function $f\colon A \to \mathbb{Z}$ with $f(a) = 1$ extends to a homomorphism $\varphi\colon F(A) \to \mathbb{Z}$ with $\varphi(a) = 1$. A reduced word on A has no two adjacent letters aa^{-1} or $a^{-1}a$, so all of its letters are equal: it is the empty word, the letter a repeated $m \geq 1$ times, or the letter a^{-1} repeated $m \geq 1$ times. By proposition 2.3.26, the word a repeated m times is the power a^m in $F(A)$, so φ sends it to $m\varphi(a) = m$ (proposition 1.4.4(3)). In the same way the word a^{-1} repeated m times goes to $-m$, and the empty word goes to 0. So φ is a bijective homomorphism, and

$$F(A) \cong \mathbb{Z}$$

2. The free group on 2 elements, $F(\{x, y\})$, can be pictured through part of its Cayley graph (definition 1.6.1) with respect to $\{x, y\}$:

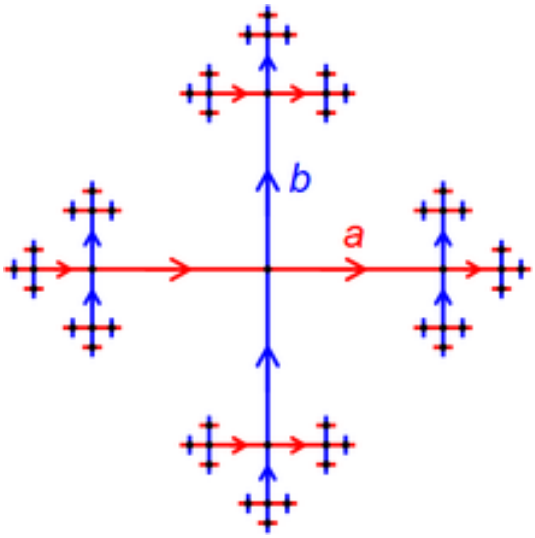


Figure 2.1: Cayley Graph of F_2

The graph is infinite, and the edges are drawn shorter at each step away from the identity so that the picture fits on the page. [a]

[a] If you have done any algebraic topology, this is the fundamental group of the figure 8!

2.3.4 Free Abelian Group

We can repeat this same construction, except limit our attention to abelian groups. In particular, given a set A an any map $f : A \to G$ to an abelian group, we want to lift it to an abelian group $F^{ab}(A)$ in the domain of the homomorphism $\varphi : F^{ab}(A) \to G$.

The construction is identical if we impose a commutativity condition on $F(A)$, that is, for any $a, b \in F^{ab}(A)$, $ab = ba$. If you were uncomfortable with the proof of the universal property of free group, it is worth going over it again, then go over it a 2nd time but with only abliean groups and $F^{ab}(A)$ in mind.

What we will be focusing on this section is finding a nicer group that is isomorphic to $F^{ab}(A)$.

In theorem 2.3.27 the universal map $i\colon A \to F(A)$ is an inclusion. A universal map need not be injective: the bilinear map $M \times N \to M \otimes_R N$ of chapter 15 sends $(m, 0)$ and $(0, n)$ to 0 for all $m \in M$ and $n \in N$.

2.3.5 *Free-Forgetful Adjoint

Is this the right place to ask this question? It's asking whether we have really defined the "right" idea of general group. The way we make this idea rigorous is by asking if we have a functor $F : \mathbf{Group} \to \mathbf{Set}$, that forgets the structure, is there a functor $G : \mathbf{Set} \to \mathbf{Group}$ that "regains" some structure? It cannot regain all of it because sets do not have enough information for that, but is there a way in which we can make sure that our free construction is the "mirror-image" of

forgetting the structure?

Build up to this: If $L(_) : \mathbf{Group} \to \mathbf{Set}$ takes a group G and returns the underlying set: $\mathcal{L}((G, \cdot)) = G$, and $\mathcal{F}(_) : \mathbf{Set} \to \mathbf{Group}$ takes a set and returns a group: $\mathcal{F}(S) = ((F(S), \cdot_{F(S)})$, then

$$\mathrm{Hom}_{\mathbf{Set}}(S, \mathcal{L}(G)) \cong_{\mathbf{Set}} \mathrm{Hom}_{\mathbf{Group}}(\mathcal{F}(S), G)$$

where we there is $\cong_{\mathbf{Set}}$ in the middle because Hom_{Set} does not have a group structure. If it did, we would ask it to preserve that too.

Exercise 2.3.2

1. Let $G = \langle a, b\rangle$ be the free (non-abelian) group generated on two elements. Show that $H = \langle a^n b a^{-n} \mid n \in \mathbb{Z}\rangle$ is an infinitely generated subgroup
2. (harder) Let $G + \mathrm{SL}_2(\mathbb{Z})$. Show it contains a subgroup isomorphic to the free subgroup with two elements.

2.4 Subgroup Lattices

In this section, we will show how to use graphs in order to visually see the subgroup structure of a group.

Before introducing a rigorous definition, let's first do an example: Let's do the subgroup lattice of Q_8. We must first know all the subgroups of Q_8. With some computation, the subgroups that are found are

$$1,\ \langle -1\rangle,\ \langle i\rangle,\ \langle j\rangle,\ \langle k\rangle,\ Q_8$$

where $1 = \{e\}$. By the definition of Q_8, any other set of generators will fall into these groups. For example, $\langle i, j\rangle = Q_8$, since it contains $k = ij$ and hence all eight elements (similarly for i, k and j, k). To construct the lattice, start by imagining Q_8 on the "top" and the identity group on the "bottom"‘

$$Q_8$$

$$1$$

then, apply the first rule of doing the lattice:

1. draw a line if one group is a subgroup of another group.

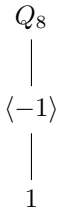
Since $1 \leqslant Q_8$, then

$$\begin{array}{c} Q_8 \\ | \\ 1 \end{array}$$

Next, pick one of the groups, let's say it's $\langle -1\rangle$. Now apply the second rule of drawing a lattice:

2. if $H \subsetneq K \subsetneq G$, then remove the line from H to G (if one is already drawn) and add a line from H to K, and K to G. In other words, only draw a line between subgroups $A \subsetneq B$ if and only if there is no group H such that $A \subsetneq H \subsetneq B$.

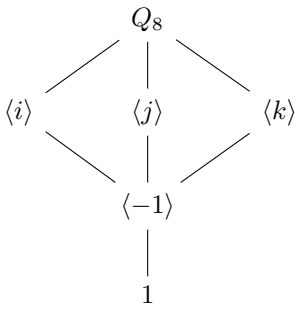
Since $1 \leqslant \langle -1 \rangle \leqslant Q_8$, our diagram will now look like this:



similarly, we can add $\langle i \rangle$ like so:



Next, notice that $\langle i \rangle \not\leqslant \langle j \rangle$, and $\langle j \rangle \not\leqslant \langle i \rangle$, however both $\langle i \rangle$ and $\langle j \rangle$ have $\langle -1 \rangle$ as a subgroup and are both subgroups of Q_8. The same situation occurs for $\langle k \rangle$. Thus, the updated lattice will be:



since we have exhausted our list of possible subgroups, we have finished the subgroup lattice of Q_8.

More generally, the orientation does not matter (we could've had 1 on top and Q_8 on bottom, or going from left to right). What matter's is we know how to translate a line into an appropriate subgroup (i.e. If 1 is on top and Q_8 is on bottom, then a line now becomes a supergroup inclusion $H \geqslant G$). Note also that the particular shape the lattice took does not really matter, a pretty diamond-like shape was chosen to make the lattice more readable.

If G is infinite, the lines of such a diagram need not record which subgroups contain which. In $G = \mathbb{Q}$, every nonzero subgroup H contains a subgroup strictly between 0 and H: if $0 \neq x \in H$, then $\langle 2x \rangle$ is nonzero and does not contain x. So no subgroup of $\mathbb{Q}$ is joined to 0 by a line. Thus, we usually work with finite groups.

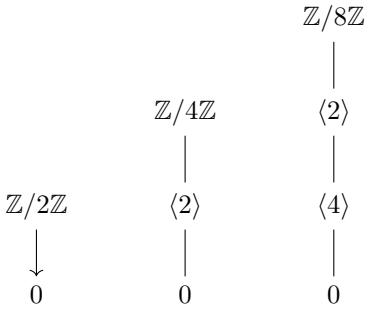
With this build-up, we can formally define a subgroup lattice:

Definition 2.4.1: Group Lattice

Let G be a finite group and $\mathcal{G}$ be the collection of subgroups of G. Then $\Gamma = (\mathcal{G}, E_{\leqslant})$ is the graph where there is an edge between $A, B \in \mathcal{G}$ if and only if $A \subsetneq B$ and there is no subgroup K with $A \subsetneq K \subsetneq B$. When Γ is visualized, we put G on top and the identity group 1 on bottom.

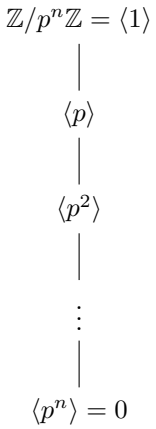
Example 2.4.2: Group Lattices

1. Let $G = \mathbb{Z}/n\mathbb{Z}$. By theorem 2.3.11(3), the subgroups of G correspond to the positive divisors of n. For $n = 2, 4, 8$ the lattices are:



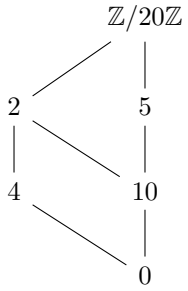
   Note that $\mathbb{Z}/2\mathbb{Z} = \langle 1 \rangle$, or $\mathbb{Z}/4\mathbb{Z} = \langle 1 \rangle = \langle 3 \rangle$, or that $\mathbb{Z}/8\mathbb{Z} = \langle 1 \rangle = \langle 3 \rangle = \langle 5 \rangle = \langle 7 \rangle$

2. More generally, if $G = \mathbb{Z}/p^n\mathbb{Z}$, then

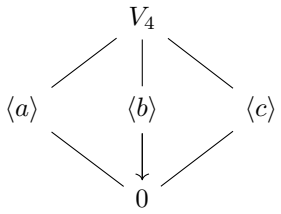


3. If $G = \mathbb{Z}/n\mathbb{Z}$ where n is a non-prime power, we can get some interesting diagram. For

example:



4. Let $G = V_4$, the Klein 4-group of example 1.4.14, and put $c = ab$. Then



This lattice differs from that of Q_8 above, where $\langle -1 \rangle$ lies below all three of $\langle i \rangle$, $\langle j \rangle$ and $\langle k \rangle$. Isomorphic groups have the same lattice, but the converse fails: $\mathbb{Z}/2\mathbb{Z}$ and $\mathbb{Z}/3\mathbb{Z}$ both have the lattice consisting of 0 below the whole group, and they are not isomorphic.

Theorem 2.4.3: Subgroup And Sublattices

Let G be a finite group and $H \leqslant G$. Then the lattice of H is the sublattice of G of groups contained in H: its vertices are the subgroups of G contained in H, and two of them are joined by an edge in the lattice of H exactly when they are joined by an edge in the lattice of G.

***Proof*:**
Since G is finite, so is H, and both lattices are defined (definition 2.4.1). The inclusion $H \to G$ is a homomorphism by proposition 1.4.8, so by proposition 1.4.4(1) and (2) the identity of H is the identity of G, and the inverse in H of each $y \in H$ is its inverse in G.

Step 1: the vertices. Let $K \subseteq H$. The conditions of proposition 2.1.3, that $K \neq \emptyset$ and $xy^{-1} \in K$ for all $x, y \in K$, mean the same thing whether y^{-1} and the product are computed in H or in G. So K is a subgroup of H exactly when it is a subgroup of G. Hence the vertices of the lattice of H are the subgroups of G contained in H.

Step 2: the edges. Let $K \subsetneq L$ be subgroups of H. A subgroup M of G with $K \subsetneq M \subsetneq L$ is contained in H, so it is a subgroup of H by Step 1. Conversely, every subgroup of H is a subgroup of G. So the subgroups lying strictly between K and L are the same in H as in G. By the definition of the edges in definition 2.4.1, K and L are joined in the lattice of H exactly when they are joined in the lattice of G, as we sought to show.

Example 2.4.4: Sublattices Of Subgroups

1. Let $H = \langle i \rangle = \{1, -1, i, -i\} \leqslant Q_8$. Of the subgroups 1, $\langle -1 \rangle$, $\langle i \rangle$, $\langle j \rangle$, $\langle k \rangle$ and Q_8 found at the start of this section, those contained in H are 1, $\langle -1 \rangle$ and $\langle i \rangle$, since $j, k \notin H$. In the lattice of Q_8, the edges among these three join 1 to $\langle -1 \rangle$ and $\langle -1 \rangle$ to $\langle i \rangle$. By theorem 2.4.3, the lattice of $\langle i \rangle$ is the chain $1 \leqslant \langle -1 \rangle \leqslant \langle i \rangle$, which has the shape of the lattice of $\mathbb{Z}/4\mathbb{Z}$ in example 2.4.2.
2. Let $H = \langle 2 \rangle \leqslant \mathbb{Z}/20\mathbb{Z}$, a subgroup of order 10. In the lattice of $\mathbb{Z}/20\mathbb{Z}$ drawn in example 2.4.2, the subgroups contained in $\langle 2 \rangle$ are $\langle 2 \rangle$, $\langle 4 \rangle$, $\langle 10 \rangle$ and 0, while $\langle 5 \rangle$ is not, since $5 \notin \langle 2 \rangle$. The edges among these four join $\langle 2 \rangle$ to $\langle 4 \rangle$ and to $\langle 10 \rangle$, and each of these to 0. By theorem 2.4.3, the lattice of $\langle 2 \rangle$ is a square with $\langle 2 \rangle$ at the top and 0 at the bottom.

If $\varphi\colon G \to H$ is a homomorphism of finite groups, then $\varphi(G) \leqslant H$ (proposition 1.4.4(5)), so by theorem 2.4.3 the lattice of $\varphi(G)$ is the part of the lattice of H below $\varphi(G)$. How it relates to the lattice of G is answered by the fourth isomorphism theorem (theorem 3.5.7).

2.4.5 Foreshadowing The answer is "dual" to theorem 2.4.3: the lattice of $\varphi(G)$ has the same shape as the part of the lattice of G above the kernel $\ker(\varphi)$, that is, the lattice of subgroups of G *containing* $\ker(\varphi)$ (chapter 3).

Exercise 2.4.1

1. Draw the subgroup lattices of $\mathbb{Z}/12\mathbb{Z}$ and of S_3.

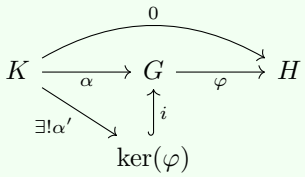
2.5 *Universal Property of the Kernel

No later result in this book depends on this section. The kernel of a homomorphism is characterized by a universal property, and that property describes kernels well beyond groups: kernels and cokernels of module homomorphisms (definition 12.2.12) form the exact sequences of chapter 17⋆, and the cohomology of a complex, a kernel modulo an image ([19, Complexes and (co)Homology]), is what the derived category is built to preserve ([19, The Derived Category]).

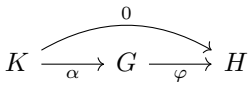
In the following seciton, $0 : G \to H$ will represent the homomorphism $\varphi(G) = e_H [= 0]$. This is indeed a homomorphism (as we have seen in proposition 1.4.5).

Theorem 2.5.1: Universal Property Of The Kernel

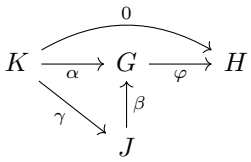
Let $\varphi : G \to H$ be a homomorphism. Then the inclusion map $i : \ker(\varphi) \hookrightarrow G$ is the final object in the category of group homomorphisms $\alpha : K \to G$ such that $\varphi \circ \alpha$ is the trivial map. In other words, every group homomorphism $\alpha : K \to G$ such that $\varphi \circ \alpha$ is trivial factors through i:



Let $\mathcal{C}_\varphi$ be the category whose objects are the pairs (K, α) of a group K and a homomorphism $\alpha : K \to G$ such that the following diagram commutes:



If (K, α) and (J, β) are two objects in this category, define the morphism $\gamma : (K, \alpha) \to (J, \beta)$ to be a homomorphism from K to J such that



commutes.

***Proof*:**
Let $\alpha : K \to G$ such that $\varphi \circ \alpha = 0$, that is

$$\varphi(\alpha(K)) = 0$$

then $\alpha(K) \subseteq \ker(\varphi)$. The inclusion i is itself an object, as $\varphi \circ i$ is trivial. A homomorphism $\alpha' \colon K \to \ker(\varphi)$ with

$$\alpha = i \circ \alpha'$$

must send each k to $\alpha(k)$, so α' is α with its codomain restricted to $\ker(\varphi)$. That map is a homomorphism, which gives existence, and it is the only candidate, which gives uniqueness.

The dual statement, the universal property of the cokernel, needs quotient groups and is theorem 3.2.8.

3

Homomorphisms and Quotient Groups

In this chapter, we take a closer look at homomorphisms and how they relate different groups. We start our exploration by getting a better understanding of the image of a homomorphism. For example, if $\varphi : G \to H$ is a homomorphism, then we saw that $\varphi(G)$ is a group (re-prove this if it is not clear). Since φ is a function, then as we saw in [24] we can define $(G/\sim_\varphi)$ which is bijective to $\varphi(G)$ (in particular, $x \sim_\varphi y \Leftrightarrow \varphi(x) = \varphi(y)$), In particular, using equivalence classes:

$$[g] \mapsto \varphi(g)$$

The collection $(G/\sim_\varphi)$ is called the *fibers* of φ[1]. Since there is a bijection between $\varphi(G)$ and $(G/\sim_\varphi)$, and $\varphi(G)$ is a group, can we define a group structure on $(G/\sim_\varphi)$? Furthermore, is there some "minimal" and "unique" group structure that we can put on $(G/\sim_\varphi)$ so that there is a cannonical identification between $\varphi(G)$ and $(G/\sim_\varphi)$? Those comfortable with the category theory presented so far will recognize this as asking for the universal properties of quotient objects in the category **Grp**. Due to this universal property, the name of the group structure on $(G/\sim_\varphi)$ satisfying the universal property will be the *quotient group*. The exploration of this question is at the center of the first section of this chapter. Due to the axioms of groups, it will turn out that there is a lot of "regularity" in the collection of fibers which allow for an easy representation of $(G/\sim_\varphi)$.

After finishing the first part, we will have developed the appropriate tools to get back to what we started in section 1.3 and define these notions more formally, i.e., how do we impose relations on groups.

For the rest of the chapter, we will use our new theory to give us some very powerful results. We will see that when working with finite groups, the order of the group will give us a lot of information about it's subgroup structure, we will use our understanding of quotient groups to learn how to

[1] Given $\varphi : G \to H$ and an element $h \in H$, the fiber of h is all elements that map to h by φ

build some new types of groups, and we will learn 4 isomorphism theorems which give us lots of information on group structure.

We will end this chapter by introducing one of the most fundamental classification theorem in finite group theory: *Jordan Hölder's Program*. In the last few section, we will show how there are groups that can be considered the "building blocks" of any other finite groups, and show how we can put these groups together to form other groups.

3.1 Quotient Groups

Let $\varphi : G \to H$ be a homomorphism. We start by exploring the structure of $\varphi(G)$. Since $\varphi(G)$ is a set, we know by [24, Surjective And Right Inverse] in the preliminary chapter that there is a bijection between $G/\sim_\varphi$ and $\varphi(G)$ such that for any function $\varphi : G \to H$ where $a \sim b \Leftrightarrow \varphi(a) = \varphi(b)$, we have

$$\begin{array}{ccc} G & \xrightarrow{\varphi} & H \\ {\scriptstyle\pi}\downarrow & \nearrow_{\bar{\varphi}} & \\ G/\sim & & \end{array}$$

We now try to see if there is an equivalent universal property for quotients in **Grp**. To that end, we would then want $G \xrightarrow{\pi} G/\sim$ to be a homomorphism, that is

$$\pi(a \cdot b) = \pi(a) \bullet \pi(b)$$

or, written in terms of equivalence relations,

$$[a \cdot b] = [a] \bullet [b]$$

We would need to have that $\bullet$ is independent of representation. That is, if we have $x, y \in [a]$, then we want

$$x \sim y \Leftrightarrow xb \sim yb \tag{3.1}$$

since we would also want that

$$[b \cdot a] = [b] \bullet [a]$$

we must have that

$$x \sim y \Leftrightarrow bx \sim by \tag{3.2}$$

These are necessary condition in order for $(G/\sim, \bullet)$ to be a group, since if once of these condition's fail, then we will not have a group. For examle, if $x \sim y \not\Rightarrow xb \sim yb$, then $[x][b] \neq [xb]$.

These are necessary conditions, since if $G/\sim$ was a group, then these must hold. We can ask if they are sufficient conditions, that is, condition's such that if they are followed, then $(G/\sim, \bullet)$ must be a group. These two condition's are also sufficient, meaning defining a group structure on $G/\sim$ is equivalent to the equivalence relation respecting these two properties:

Theorem 3.1.1: $(G/\sim,\cdot)$ is the quotient object of Grp

Let G be a group and $G/\sim$ be the set of equivalence relations given $\sim$. Then $G/\sim$ is a group under $\cdot$ if and only if

$$\forall g \in G,\ (x \sim y \Leftrightarrow gx \sim gy \text{ and } xg \sim yg)$$

Furthermore, If $\varphi : G \to H$ is another homomorphism such that $g \sim g' \Rightarrow \varphi(g) = \varphi(g')$, then there exists a homomorphism φ such that the following diagram commutes:

$$\begin{array}{ccc} G/\sim & \xrightarrow{\varphi} & H \\ {\scriptstyle\pi}\uparrow & \nearrow_{\varphi} & \\ G & & \end{array}$$

We say that $\sim$ is *compatible* with the group-structure on G

***Proof*:**
The proof of the $\Rightarrow$ direction we have just showed above the theorem, so we will show that these two equivalence relations are sufficient conditions for to define $(G/\sim,\cdot)$.

First, note that the operation $\cdot : G/\sim \times G/\sim \to G/\sim$ $[a]\cdot[b] = [ab]$ is well-defined, since if we take any other two representative $a' \in [a]$ and $b' \in [b]$, then since

$$b \sim b' \Leftrightarrow ab \sim ab' \text{ and } a'b \sim a'b'$$

and

$$a \sim a' \Leftrightarrow ab \sim a'b \text{ and } ab' \sim a'b'$$

Thus, putting pieces together using transitivity, we get

$$ab \sim ab' \sim a'b' \text{ or } ab \sim a'b \sim a'b'$$

both showing that $ab \sim a'b'$, meaning $\cdot$ is well-defined. Next, we need to show that $(G/\sim,\cdot)$ is indeed a group

1. Let $[a],[b],[c] \in G/\sim$. Then

$$\begin{aligned} ([a][b])[c] &= ([ab])[c] \\ &= [(ab)][c] \\ &= [(ab)c] \\ &= [a(bc)] \\ &= \vdots \\ &= [a]([b][c]) \end{aligned}$$

2. I claim that $[e_G] \in G/\sim$ is the identity. For any $[g] \in G/\sim$,

$$[g][e_G] = [ge_G] = [g] = [e_G g] = [e_G][g]$$

as we sought to show

3. Finally, for any $[g] \in G/\sim$, I claim that $[g]^{-1} = [g^{-1}]$. Indeed:

$$[g][g^{-1}] = [gg^{-1}] = [e_G]$$

showing that $[g^{-1}]$ is indeed the inverse of G

Thus, $(G/\sim, \bullet)$ is indeed a group.

Finally, we must show that G is universal with respect to homomorphisms $\varphi : G \to G'$ such that $g \sim g' \Rightarrow \varphi(g) = \varphi(g')$. Note that we already have that such a function exists, since φ is also a set function, so we already have that there exists a unique set-funciton $\varphi : G/\sim \to G'$ such that $\varphi([g]) = \varphi(g)$. Thus, we only need to check that φ is a homomorphism. We get this pretty much be definition:

$$\varphi([a][b]) = \varphi([ab]) = \varphi(ab) = \varphi(a)\varphi(b) = \varphi([a])\varphi([b])$$

showing that φ is indeed a homomorphism, and must be unique since there is only one possible set function (and hence homomorphism) that satisfies this property (there are no more functions to check).

Since $\bullet$ is the unique structure on $G/\sim$ that respects the universal property, we will drop it without risk of confusion, and write

$$[a][b] = [ab]$$

Definition 3.1.2: Quotient Group Via Congruence Relation

Let G be a group and $\varphi : G \to H$ a group homomorphism giving the induced equivalence relation $\sim_\varphi$. Then the group $G/\sim_\varphi$ from theorem 3.1.1 is called a *quotient group* and the equivalence relation $\sim_\varphi$ is called a *congrunce relation.*

By theorem 3.1.1, the equivalence relations whose quotient is a group are exactly those respecting both (3.2) and (3.1). Property (3.2) alone already forces the class of e_G to be a subgroup whose left cosets are the other classes:

Proposition 3.1.3: Equivalence Classes Using property (3.2)

Let $G/\sim$ be a quotient set given $\sim$ with proprety (3.2). Then

1. The equivalence class $K \in G/\sim$ where $e_G \in K$ is a subgroup of G ($K \leqslant G$)
2. Given any other equivalence calss $[g] \in G/\sim$, for any $a, b \in [g]$

$$a \sim b \iff a^{-1}b \in K \iff aK = bK$$

where the $aK = \{ak \mid k \in K\}$ and $bK = \{bk \mid k \in K\}$

***Proof*:**

1. Let $a, b \in K$, so that $e_G \sim a$ and $e_G \sim b$. Left multiplication by a^{-1} (property (3.2)) turns these into $a^{-1} \sim e_G$ and $a^{-1} \sim a^{-1}b$, so $a^{-1}b \sim e_G$ by symmetry and transitivity, that is,

$a^{-1}b \in K$. Taking $b = e_G$ gives $a^{-1} \in K$, and applying this to a^{-1} and b gives $ab \in K$. Since $e_G \in K$, K is a subgroup of G. With (3.1) in place of (3.2), right multiplication gives $ba^{-1} \in K$ instead.

2. Let $[g]$ be an equivalence class and $a, b \in [g]$ so that $a \sim b$.

   Since $\sim$ respects (3.2) $a^{-1}b \sim e_G$, implying $a^{-1}b \in K$. Next, take any element $x \in a^{-1}bK$. Then $x = a^{-1}bk$. Next, since $a^{-1}b \in K$ and $k \in K$, $x \in K$. Thus, $a^{-1}bK \subseteq K$. This implies that $bK \subseteq aK$, since if $x \in bK$, then $x = bk$, and so $x = a(a^{-1}bk)$, and since $a^{-1}b \in K$ and $k \in k$, we get $x = ak'$ for some $k' = a^{-1}bk \in K$, an so $x \in aK$. Using symmetry of $\sim$, we will get that $aK \subseteq bK$, implying $aK = bK$. Thus, we showed that

   $$a \sim b \Rightarrow a^{-1}b \in K \Rightarrow aK = bK$$

   Note that if we were using property (3.1), then we would get $Ka = Kb$, since we would be multiplying on the right.

   To make this chain of implication go full circle, assume $aK = bK$. Then $a = ae_G \in bK$, so $a = bk$ for some $k \in K$, and $b^{-1}a = k \in K$, that is, $e_G \sim b^{-1}a$. Left multiplication by b (property (3.2)) gives $b \sim a$. So $aK = bK \Rightarrow a \sim b$, completing the loop of implication, thus showing that

   $$a \sim b \Leftrightarrow a^{-1}b \in K \Leftrightarrow aK = bK$$

   as we sought to show

This shows that for any equivalence relation satisfying (3.2) (respectively (3.1)), the equivalence classes are the sets aK (respectively Ka), where a is a representative of the class and K is the class containing e_G. So the subgroup K determines every equivalence class of $G/\sim$.

Corollary 3.1.4: Quotient Set Size

Let G be a group, let $\sim$ be an equivalence relation on G satisfying (3.2) (respectively (3.1)), and let K be the class of e_G. Then every class has as many elements as K:

$$|aK| = |K| \qquad \text{(respectively } |Ka| = |K|\text{)} \qquad \forall a \in G$$

***Proof*:**
For $a \in G$, the map $K \to aK$ given by $k \mapsto ak$ is a bijection, with inverse $x \mapsto a^{-1}x$. Hence $|aK| = |K|$ for every $a \in G$. For the classes Ka the map $k \mapsto ka$ gives $|Ka| = |K|$ in the same way, completing the proof.

Since these cosets represent our equivalence relations, we will give a special name to these quotient sets:

Definition 3.1.5: Cosets

Let $H \leqslant G$. Then for every $g \in G$, we can define

$$gH = \{gh | h \in H\}$$

we call this a [left]-***coset***. We can also define the ***right***-coset:

$$Hg = \{hg | h \in H\}$$

If additive notation is used for G ($(G, +)$), it is common to denote the coset as

$$g + H \qquad H + g$$

The set of left-cosets is represented as G/H, an the set of right-cosets is represented as $G \setminus H$.

Does every subgroup arise as the class of e_G of an equivalence relation satisfying (3.2)? In other words, is there a converse to proposition 3.1.3(1), which states that if $\sim$ respects property (3.2) (or (3.1)), then the equivalence class containing e_G is a subgroup? The answer is yes:

Proposition 3.1.6: Subgroups And Equivalence Classes

Let $H \leqslant G$ be a subgroup of G. Define the equivalence relation

$$a \sim_L b \Leftrightarrow a^{-1}b \in H$$

then $\sim_L$ respects property (3.2)

***Proof*:**
Let's first show that $\sim_L$ is indeed an equivalence relation

1. If $a \sim a$, then $a^{-1}a = e \in H$, which is the case since H is a group
2. Let $a \sim b$. Then $a^{-1}b \in H$, implying $(a^{-1}b)^{-1} = b^{-1}a \in H$, implying $b \sim a$
3. Let $a \sim b$ and $b \sim c$. Then $a^{-1}b, b^{-1}c \in H$. Then $a^{-1}bb^{-1}c = a^{-1}c \in H$, implying $a \sim c$.

Thus, $\sim_L$ is indeed an equivalence relation.
To show it satisfies property (3.2), notice that for any $g \in G$

$$a \sim b \to a^{-1}b \in Ha^{-1}(g^{-1}g)b \in H \Rightarrow (ga)^{-1}gb \in H \Rightarrow ga \sim_L gb$$

showing that $\sim_L$ satisfies property (3.2)

This proposition shows that any subgroup of H will define an equivalence relation $\sim_L$ respecting property (3.2), and thus by proposition 3.1.3(2) partition G (even better, equally partition G due to corollary 3.1.4). Similarly, we can show that the relatoin $a \sim_R b \Leftrightarrow ab^{-1} \in H$ gives an equivalence relation which satisfies property (3.1). Note that since H partitioned G, only $eH = H$ is a subgroup of G since only it contains the identity.

Combining propositions 3.1.3 and 3.1.6 we get the following one to one correspondence:

$$\{H|H \leqslant G\} \leftrightarrow \left\{ \begin{array}{c} \text{equivalence relations } \sim_L \\ \text{on } G \text{ satisfying} \\ \text{property (3.2)} \end{array} \right\}$$

Similarly:

$$\{H|H \leqslant G\} \leftrightarrow \left\{ \begin{array}{c} \text{equivalence relations } \sim_R \\ \text{on } G \text{ satisfying} \\ \text{property (3.1)} \end{array} \right\}$$

This shows that the subgroups of G give us all possible such equivalence relations, that is all one-sided congruence relations. Because of this, we can without ambiguity write G/H to represent $G/\sim$ with an equivalence relation $\sim$ satisfying property (3.2), and similarly for $G \setminus H$.

Is the partition G/H into left cosets always the same as the partition $G \setminus H$ into right cosets, and does $\sim_L$ also satisfy property (3.1)? The answer to both is no:

3.1.7 counter-example Take $G = D_3$. Then $H = \langle\sigma\rangle$ is a subgroup, and has left cosets $eH(=H), rH, r^2H$. giving us

$$\{e, \sigma\}, \{r, r\sigma\}, \{r^2, r^2\sigma\}$$

and the right-cosets are

$$\{e, \sigma\}, \{r, \sigma r\}, \{r^2, \sigma r^2\}$$

using the fact that $r\sigma = \sigma r^2$ and $r^2\sigma = \sigma r$, we can re-write the right cosets as

$$\{e, \sigma\}, \{r, r^2\sigma\}, \{r^2, r\sigma\}$$

showing that they partition G differently than the left cosets!
This means that $\sim_L$ defined by H does not respect property (3.1). If it did, then

$$r \sim_L r\sigma \Rightarrow r(r\sigma)^{-1} = r\sigma r^2 = rr\sigma \in H$$

however, $r^2\sigma \notin H$, showing that it does not satify the propety

Remember that we showed in theorem 3.1.1 that *both* the properties need to be satisfied in order for G/H to be a group. Since every $\sim$ with these properties corresponds to a subgroup H, it will come down to a special subgroup H which has both properties. The property is the normality of definition 2.2.10.

Proposition 3.1.8: When H Induces Both $\sim_L$ And $\sim_R$

Let $H \leqslant G$, and let $a \sim_L b \Leftrightarrow a^{-1}b \in H$ and $a \sim_R b \Leftrightarrow ab^{-1} \in H$. The relations $\sim_L$ and $\sim_R$ induce the same cosets corresponding to H, that is, they have the same equivalence classes, if and only if H is normal in the sense of definition 2.2.10, that is, $gHg^{-1} = H$ for every $g \in G$. Both conditions are equivalent to $gH = Hg$ for every $g \in G$.

***Proof*:**
For $a \in G$, the $\sim_L$-class of a is $\left\{b \in G \;\middle|\; a^{-1}b \in H\right\} = aH$. Since $ab^{-1} = h$ holds exactly when $b = h^{-1}a$, and $h \mapsto h^{-1}$ is a bijection of H onto itself, the $\sim_R$-class of a is $\left\{b \in G \;\middle|\; ab^{-1} \in H\right\} = Ha$.

If $\sim_L$ and $\sim_R$ have the same classes, then for each $g \in G$ the sets gH and Hg are classes of one partition of G, and both contain g, so $gH = Hg$. Conversely, if $gH = Hg$ for every $g \in G$, then every element of G has the same class under $\sim_L$ as under $\sim_R$, so the two relations have the same classes.

If $gH = Hg$, then $gHg^{-1} = (Hg)g^{-1} = H$. Conversely, if $gHg^{-1} = H$, then $gH = (gHg^{-1})g = Hg$. So $gH = Hg$ for every $g \in G$ exactly when $gHg^{-1} = H$ for every $g \in G$, completing the proof.

Definition 3.1.9: Normal Subgroup

Let $H \leqslant G$ such that $aH = Ha$ for every $a \in G$. Then H is a normal subgroup, written $H \trianglelefteq G$

We have already encountered normal subgroups before

Example 3.1.10: Normal Subgroups

1. for any group G, $Z(G) \leqslant G$. Take any $y \in G$ and $x \in Z(G)$. Then by definition of $Z(G)$, $yx = xy \in Z(G)y$, so $yZ(G) \subseteq Z(G)y$, and in the same way $Z(G)y \subseteq yZ(G)$, showing that $Z(G)$ is normal
2. for any homomorphism $\varphi\colon G \to H$, the kernel $\ker(\varphi)$ is normal in G (proposition 2.2.12)

Proposition 3.1.11: properties of normal sub-groups

Let N be a subgroup of a group G. Then the following are equivalent:

1. $N \trianglelefteq G$ (definition 3.1.9)
2. $N_G(N) = G$
3. $gN = Ng$ for all $g \in G$
4. the rule $(aN)(bN) = (ab)N$, which is the operation $\bullet$ of theorem 3.1.1 for the relation $\sim_L$ whose classes are the left cosets, is well defined on the set G/N of left cosets of N (definition 3.1.5), that is, $aN = a'N$ and $bN = b'N$ imply $(ab)N = (a'b')N$, and it makes G/N a group
5. $gNg^{-1} \subseteq N$ for all $g \in G$

***Proof*:**
For $m \in N$ we have $mN = N$, since $mN \subseteq N$ by closure and $N = m(m^{-1}N) \subseteq mN$.

(1) $\Leftrightarrow$ (3) This is definition 3.1.9.

(3) $\Rightarrow$ (5) Let $g \in G$ and $n \in N$. Then $gn \in gN = Ng$, so $gn = n'g$ for some $n' \in N$, and $gng^{-1} = n' \in N$.

(5) $\Rightarrow$ (2) Let $g \in G$. By (5), $gNg^{-1} \subseteq N$, and by (5) for g^{-1}, $g^{-1}Ng \subseteq N$. Hence $N = g(g^{-1}Ng)g^{-1} \subseteq gNg^{-1}$, so $gNg^{-1} = N$ and $g \in N_G(N)$ (definition 2.2.10). So $N_G(N) = G$.

(2) $\Rightarrow$ (3) Let $g \in G$. By (2), $g \in N_G(N)$, so $gNg^{-1} = N$ and $gN = (gNg^{-1})g = Ng$.

(5) $\Rightarrow$ (4) Suppose $aN = a'N$ and $bN = b'N$. Since $a' \in a'N = aN$ and $b' \in b'N = bN$, we may write $a' = an_1$ and $b' = bn_2$ with $n_1, n_2 \in N$. Then

$$a'b' = an_1bn_2 = ab(b^{-1}n_1b)n_2$$

and $m = (b^{-1}n_1b)n_2 \in N$, because $b^{-1}n_1b \in N$ by (5) for $g = b^{-1}$. So $(a'b')N = (ab)mN = (ab)N$, and the operation is well defined. It is associative, since $((aN)(bN))(cN) = ((ab)c)N = (a(bc))N = (aN)((bN)(cN))$. The coset $eN = N$ is an identity, since $(eN)(aN) = aN = (aN)(eN)$, and $a^{-1}N$ is an inverse of aN, since $(aN)(a^{-1}N) = eN = (a^{-1}N)(aN)$. So G/N is a group.

(4) $\Rightarrow$ (5) Let $g \in G$ and $n \in N$. Then $nN = N = eN$, so computing $(nN)(g^{-1}N)$ with the representatives n, g^{-1} and with e, g^{-1} gives $(ng^{-1})N = g^{-1}N$ by well-definedness. Hence $ng^{-1} \in g^{-1}N$, so $ng^{-1} = g^{-1}n'$ for some $n' \in N$, and $gng^{-1} = n' \in N$.

Together with (1) $\Leftrightarrow$ (3), the cycle (3) $\Rightarrow$ (5) $\Rightarrow$ (2) $\Rightarrow$ (3) and the equivalence (4) $\Leftrightarrow$ (5) show that all five conditions are equivalent, as claimed.

Corollary 3.1.12: Subgroups Of Abelian Groups Are Normal

If G is abelian and $H \leqslant G$, then $H \trianglelefteq G$.

***Proof*:**
Let $g \in G$. Since $gh = hg$ for every $h \in H$, $gH = \{gh \mid h \in H\} = \{hg \mid h \in H\} = Hg$. Hence $H \trianglelefteq G$ by definition 3.1.9, as claimed.

Proposition 3.1.8 is very important. The equivalence relatoin $\sim$ corresponding to H has both property (3.2) and (3.1) by proposition 3.1.6 and so by theorem 3.1.1 G/H is a group!

Definition 3.1.13: Quotient Group

Let $H \trianglelefteq G$ be a normal subgroup of G. The *quotient group* G/H, sometimes written G mod H (and read G modulo H or G mod H for short) is the group $(G/\sim, \cdot)$ where $\sim$ is defined in proposition 3.1.6, and $\cdot$ is defined by

$$aHbH := (ab)H$$

An easy way to remember that this operation is defined when is H is normal is by remembering $a(Hb)H = abHH = abH$. For notation, we can write $aH = \bar{a} = [a]$ for a representative a and $G/H = \bar{G}$. The projection $\pi\colon G \to G/\sim$ then reads

$$\pi\colon G \to G/H \qquad \pi(g) = gH$$

and it is a homomorphism, since $\pi(ab) = (ab)H = aH\,bH = \pi(a)\pi(b)$ by the definition of the product in G/H.

Since π is a homomorphism, it's kernel is a subgroup:

$$\ker(\pi) = \{g \in G \mid \pi(g) = eH\}$$

and since $eH = H$, this means that $\ker(\pi) = H$. Since H can be any normal subgroup, this means that H is always the kernel of some map

$$\text{Normal} \Rightarrow \text{Kernel}$$

Is the converse true, that is, the kernel is always normal? If so, that would be really powerful, since the kernel would then always be a normal subgroup of G, and so any homomorphism $\varphi : G \to H$ would induce a quotient group G/K, where $K = \ker(\varphi)$. Kernels are always normal subgroups!

Proposition 3.1.14: Kernel Then Normal

Let $\varphi : G \to H$ be a homomorphism and let $K = \ker(\varphi)$. Then $K \trianglelefteq G$

***Proof*:**
Let K be the kernel of $\varphi : G \to H$, and take any $g \in g$. then

$$\varphi(gkg^{-1}) = \varphi(g)\varphi(k)\varphi(g^{-1}) = \varphi(g)\varphi(k)\varphi(g)^{-1} = \varphi(g)\varphi(g)^{-1} = e_H$$

and so $gkg^{-1} \in K$, showing that $gK = Kg$, meaning $K \trianglelefteq G$

This means that given any homomorphism $\varphi : G \to H$, we can *always* define an equivalence relation on G given the kernel of φ since $\ker(\varphi)$ is normal!

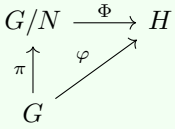
Since any normal subgroup will be the kernel of some homomorphism, and the kernel of any homomorphism will be normal, we get that the two are essentially synonymous in group theory:

$$\text{Normal} \Leftrightarrow \text{Kernel}$$

With quotient groups available, the universal property of quotients (theorem 3.1.1) takes a form stated with a normal subgroup. If a normal subgroup N lies in the kernel of a homomorphism $\varphi\colon G \to H$, then elements of the same coset of N have the same image under φ, so φ factors through G/N:

Theorem 3.1.15: The First Isomorphism Theorem

Let $\varphi : G \to H$ be a homomorphism and let $K = \ker(\varphi)$. If $N \trianglelefteq G$ where $N \subseteq \ker(\varphi)$, then there exists a unique *homomorphism* $\Phi : G/N \to H$ such that the following diagram commutes:



where $\varphi = \Phi \circ \pi$, and π is the natural projection

***Proof*:**
Let $\sim$ be the relation $a \sim b \Leftrightarrow a^{-1}b \in N$ of proposition 3.1.6. The class of a consists of the elements $b \in G$ with $a^{-1}b \in N$, which form the coset aN, and G/N is the group $(G/\sim, \bullet)$ (definition 3.1.13). Let $a, b \in G$ with $a \sim b$. Then $a^{-1}b \in N \subseteq \ker(\varphi)$, so by proposition 1.4.4(2)

$$\varphi(a)^{-1}\varphi(b) = \varphi(a^{-1}b) = e_H$$

that is, $\varphi(a) = \varphi(b)$. By theorem 3.1.1, there is a homomorphism $\Phi\colon G/N \to H$ with $\Phi(aN) = \varphi(a)$ for every $a \in G$, that is, $\varphi = \Phi \circ \pi$. If $\Psi\colon G/N \to H$ is a homomorphism with $\varphi = \Psi \circ \pi$ as well, then $\Psi(aN) = \varphi(a) = \Phi(aN)$ for every $a \in G$, and every element of G/N is a coset aN, so $\Psi = \Phi$, as claimed.

What theorem 3.1.15 (the universal property of quotient groups) shows is that all the information a homomorphism φ gives us can be captured via the quotient group G/K.

Corollary 3.1.16: Canonical Decomposition For Groups

Let $\varphi : G \to H$ be any group homomorphism. Then it may be decomposed into

$$G \xrightarrow[\pi]{} G/\ker(\varphi) \xrightarrow[\Phi]{\cong} \operatorname{im}(G) \xrightarrow[\iota]{} H \qquad (\varphi : G \to H)$$

where Φ is an isomorphism

***Proof*:**
Write $K = \ker(\varphi)$. By proposition 3.1.14, K is normal, so G/K is a group (definition 3.1.13), and the natural projection $\pi\colon G \to G/K$, $a \mapsto aK$, is a homomorphism, since $\pi(ab) = (ab)K = aK\,bK$. The image $\operatorname{im}(G) = \varphi(G)$ is a subgroup of H (proposition 1.4.4(5)), so the inclusion ι is a homomorphism. By theorem 3.1.15 with $N = K$, there is a homomorphism $\Phi_0\colon G/K \to H$ with $\varphi = \Phi_0 \circ \pi$, that is, $\Phi_0(aK) = \varphi(a)$ for every $a \in G$. Its values lie in $\varphi(G)$, so it restricts to a homomorphism $\Phi\colon G/K \to \varphi(G)$ with $\varphi = \iota \circ \Phi \circ \pi$.

The map Φ is surjective, since every element of $\varphi(G)$ is some $\varphi(a) = \Phi(aK)$. It is injective: if $\Phi(aK) = \Phi(bK)$, then $\varphi(a) = \varphi(b)$, so $\varphi(a^{-1}b) = \varphi(a)^{-1}\varphi(b) = e_H$ by proposition 1.4.4(2), that is, $a^{-1}b \in K$, and $aK = bK$ since the coset aK is the class of a for the relation $a \sim b \Leftrightarrow a^{-1}b \in K$ of proposition 3.1.6. So Φ is a bijective homomorphism, that is, an isomorphism (definition 1.4.12), completing the proof.

Corollary 3.1.17: Surjective Homomorphisms

Let $\varphi : G \to H$ be a surjective homomorphism, and let $K = \ker(\varphi)$. Then

$$G/K \cong H$$

***Proof*:**
For $G/K \cong H$, notice that $H = \operatorname{im}(G)$ in corollary 3.1.16.

This last corollary is very useful and one you will use constantly in both theoretical and practical context. It is saying that if we want to know the structure of G/K, if we can find a surjective homomorphism from G to H with kernel K, then $G/K \cong H$, i.e., H will represent the structure!

A surjective homomorphism $\varphi\colon G \to H$ with kernel K, as in corollary 3.1.17, can be recorded as a chain of homomorphisms

$$1 \xrightarrow{\iota_1} K \xrightarrow{\iota_K} G \xrightarrow{\varphi} H \xrightarrow{\pi} 1$$

where 1 is the trivial group, ι_1 and ι_K are the inclusion homomorphisms, and π is the projection of H onto the trivial group.

This representation is called a *short exact sequence*. We will return to short exact sequences soon to flesh out many intersting results when precieving which groups can be put in the 3 non-trivial positions, that is

$$1 \xrightarrow{\iota_1} A \longrightarrow B \longrightarrow C \xrightarrow{\pi} 1$$

Exercise 3.1.1

Throughout these exercises, let $\varphi : G \to H$ be a homomorphism between the two groups G and H, and $K = \ker(\varphi)$

1. Show that $\varphi^{-1}(Q)$ (the pre-image of Q) is a group. If Q was a normal subgroup, $Q \trianglelefteq H$, show that $\varphi^{-1}(Q)$ is also normal. Is it true that if $H \trianglelefteq G$ then $\varphi(H) \trianglelefteq \varphi(G)$?
2. Let $h \in H$. We call $\varphi^{-1}(h)$ a *fiber* of φ over h. Let $F_h = \varphi^{-1}(H)$. Show that
   1. For any $u \in F_h$, $F_h = uK$
   2. For any $u \in F_h$, $F_h = Ku$
3. Let $aKbK = \{akbk' \mid a, b \in G, k, k' \in K\}$ be a set, and $K \trianglelefteq G$. Show that
   $$aKbK = abK$$
   where $abK = \{abk \mid a, b \in G, k \in K\}$ [hint: for $aKbK \subseteq abK$, recall that $aK = Ka$ since K is normal. For $aKbK \supseteq abK$, note that $1 \in K$ so $1 \cdot a \in Ka$. Take advantage of normality to complete the proof]
4. Let $R \leqslant G$. Show that the cosets of G/R are all of equal size.
5. Show that if $aB = 1B$, then $a \in B$
6. Let $H, K \trianglelefteq G$. Show that $H \cap K \trianglelefteq G$
7. Generalizing the previous result, If $\{H_i\}_{i \in I}$ is some collection of normal subroups of G show that $\cap_{i \in I} H_i \trianglelefteq G$
8. Let G be finitely generated and let $N \trianglelefteq G$ be finitely generated. Prove that G/N is finitely generated.
9. For any group G, show that $Z(G) \trianglelefteq G$
10. Let n be a positive integer. Show that $n\mathbb{Q} = \mathbb{Q}$, where $n\mathbb{Q} = \{nq \mid q \in \mathbb{Q}\}$ and nq is the sum of n copies of q in the additive group $\mathbb{Q}$. (Groups with this property for every n are called *divisible* groups. More on them in section 17.3.)
11. Let $G \cong H$, $N \trianglelefteq G$ and $K \trianglelefteq H$. If $N \cong K$ is it true that $G/N \cong H/K$? Prove or show a counter example. hint: [2]
12. (Useful in Lie Groups) Let $GL_n(\mathbb{R})$ be the set of invertible $n \times n$ matrices over $\mathbb{R}$ (more generally, if you know linear algebra, over a field $\mathbb{F}$ with characteristic $\neq 2$ or 3 if $n = 2$). What are the normal subgroups of $GL_n(\mathbb{R})$?
13. We have shown that if we have a map $f : G \to H$ and $N \trianglelefteq H$, we can define a map $\overline{f} : G \to H/N$. Is the converse true? That is, if we have a map $f : G \to H/N$, do we have a map $F : G \to H$ such that $f = \pi \circ F$? [Hint: Consider maps with domains and codomains of $\mathbb{Z}$ and $\mathbb{Z}/n\mathbb{Z}$, $n > 1$]
14. Show that every subgroup of Q_8 is normal. Hence, every subgroup being normal does not imply the group is abelian[3]

[2] Consider $G = H = N = \mathbb{Z}$ and $K = 2\mathbb{Z}$.

[3] On the other hand, Q_8 is special: Every non-abelian group for which all subgroups are normal is of the form

3.2 Presentations

We have seen in section 2.3.3 that if we have any set map from a set S to a group G, $f : S \to G$, then it can be extended it to a homomorphism from the group $F(S)$ to G: $\varphi : F(S) \to G$, $\varphi|_S = f$. We have also seen that if we have a surjective homomorphism $G \to H$, then $G/N \cong H$ (where N is the kenrel of the homomorphism). We can combine these results: Converting to $(G, *)$ notation for groups for a moment, given an identity map $\mathrm{id} : G \to (G, *)$, which is a prefectly valid surjective set function from the set G to the underlying $\varphi : (F(G), *_F) \to (G, *)$. By the first isomorphism theorem, if N is the kernel, then $(F(G)/N, (*_F)_\varphi) \cong (G, *)$. Thus, $\varphi : F(G) \to (G, *)$. By the first isomorphism theorem, if N is the kernel, then $F(G)/N \cong (G, *)$. Thus, every group is the quotient of a free group! We now go back to our usual convention of groups being written as G where we drop the binary operation, where we will consider it clear from context when a map is a set map and when it is a homomorphism.

Let's generalize this result a bit: Let's say $S \subseteq G$ such that $\langle S \rangle = G$, and consider $f : S \to G$. Then by the universal property of free groups, we have that $\varphi : F(S) \to G$ is a group homomorphism. Since $\langle S \rangle = G$, and all possible words are in $F(S)$, then by construction of $F(S)$ and φ, the map φ is a surjective group homomorphism. Thus, for the appropriate kernel N, $F(S)/N \cong G$.

Since $F(S)$ is free from relations, but G has relations, φ somehow imposed relations onto $F(S)$. This means that N captured the idea of the relations imposed on $F(S)$ to make $F(S)/N \cong G$. As observed in section 2.3, any relations of a group can be written as

$$r_1 = r_2 = \cdots = r_k = 1$$

If we let $r_i = \prod_{i \in I_{r_i}} s_i$ to be some arbitrary finite[4] product of elements of S and it's inverses. Then if If we let $r_i = \prod_{i \in I_{r_i}} s_i$ to be some arbitrary product of elements of S and it's inverses. Then if we let $R = \{r_1, r_2, ..., r_k\}$, we can write $G = \langle S | r_1 = r_2 = \cdots = r_k = 1 \rangle$ as

$$G = \langle S | R \rangle$$

for short hand. Since $F(S)/N \cong G$, it must be that $R \subseteq N$. We might be tempted to say $\langle R \rangle \subseteq N$, however the closure *need not be normal*, while N must be a normal subset. Instead, we will define the noraml closure:

Definition 3.2.1: Normal Closure

Let $S \subseteq G$ be a subset of a group G. Then define the intersection of all normal subgroups of G containing S

$$\langle S \rangle^{\trianglelefteq} = \bigcap_{\substack{N_i \trianglelefteq G \\ S \subseteq N_i}} N_i$$

to be *normal closure of* S.

$\langle S \rangle^{\trianglelefteq}$ is indeed a normal subgroup: it is an intersection of subgroups, hence a subgroup (proposition 2.3.14), and for $g \in G$, $g \left(\bigcap_i N_i\right) g^{-1} \subseteq \bigcap_i g N_i g^{-1} = \bigcap_i N_i$. Since $r_1 = r_2 = \cdots = r_k = 1$ in G, then $[r_1] = [r_2] = \cdots = [r_k] = [1] \in F(S)/N$, and so $r_1, r_2 ... r_k \in N$. We thus have that $\langle R \rangle^{\trianglelefteq} \subseteq N$.

$G \cong Q_8 \times B \times D$ where B is something called an elementary abelian 2-group and D is called a torsion abelian group whose elements are of odd order. A group where all subgroups are normal is called a Dedekind group, and a non-abelian Dedekind group is called a Hamiltonian group.

[4] Infinite products would require more care to be well-defined, see EYNTKA real analysis I

Next, let's say we have some $x \in N$. Then that means then $[x] = [1]$, so for some $n' \in N$, $xn' = 1$. But then, xn' must be some combinations of elements $r_1, r_2, \cdots, r_k$, since by the presentation of G, those are the elements that can equal to 1. Thus $N \subseteq \langle R \rangle^{\trianglelefteq}$. Therefore $N = \langle R \rangle^{\trianglelefteq}$

From this, we can formally define a presentation

Definition 3.2.2: Presentation

Let G be any group. Then a *presentation* of G is a pair of sets (S, R) where $\langle R \rangle^{\trianglelefteq}$ is the kernel of the homomorphism $F(S) \to G$. The elements of S are called the *generators* of G, and the elements of R are called the *relations* of G:

$$\langle S | R \rangle := F(S)/\langle R \rangle^{\trianglelefteq}$$

We say G is finitely generated if given a presentation (S, R), $|S|$ is finite. We say G is *finitely presented* if both $|S|$ and $|R|$ are finite.

Notice how this statement is like the dual equivalent of Cayley's theorem (theorem 4.5.3): In Cayley's theorem, every group is the subgroup of a symmetric group. Here, every group is the quotient of a free group.

Theorem 3.2.3: Homomorphisms From A Presentation (von Dyck)

Let $G = \langle S | R \rangle$, let H be a group and $f\colon S \to H$ a function, and let $\varphi\colon F(S) \to H$ be the homomorphism extending f (theorem 2.3.27). If $\varphi(r) = 1$ for every $r \in R$, there is a unique homomorphism $\overline{\varphi}\colon G \to H$ sending the class of each $s \in S$ to $f(s)$. It is surjective when $f(S)$ generates H.

***Proof*:**
$\ker(\varphi)$ is a normal subgroup of $F(S)$ (proposition 3.1.14) containing R, so it contains $\langle R \rangle^{\trianglelefteq}$ (definition 3.2.1). With $\pi\colon F(S) \to G$ the projection, theorem 3.1.15 gives a homomorphism $\overline{\varphi}\colon G \to H$ with $\overline{\varphi} \circ \pi = \varphi$, so $\overline{\varphi}$ sends the class of s to $f(s)$. Every element of G is a product of classes of elements of S and their inverses, since every element of $F(S)$ is such a product and π is surjective, so a homomorphism on G is determined by these values, which gives uniqueness. The image of $\overline{\varphi}$ is a subgroup containing $f(S)$, so it is H when $f(S)$ generates H.

Example 3.2.4

1. $\langle A | \emptyset \rangle$ is the presentation of the free group $F(A)$
2. The dihedral group D_n ($n \geq 3$) of definition 1.2.28 has the presentation $G = \langle r, s | r^n, s^2, rsrs \rangle$. Let $\rho \in D_n$ be the rotation by $2\pi/n$ and σ a reflection. By proposition 1.2.29, $\rho^n = \sigma^2 = 1$ and $\rho\sigma = \sigma\rho^{-1}$, which is $\rho\sigma\rho\sigma = 1$, and every element of D_n is $\sigma^k\rho^i$, so ρ and σ generate D_n, which has order $2n$ (proposition 1.2.30). So theorem 3.2.3 gives a surjective homomorphism $G \to D_n$ with $r \mapsto \rho$, $s \mapsto \sigma$. In G, $rs = sr^{-1}$, so $sr^k = r^{-k}s$ for every k, and every element of G can be written as $r^i s^j$ with $0 \leq i < n$ and $0 \leq j < 2$. So $|G| \leq 2n = |D_n|$, and the surjection is an isomorphism.

Example 3.2.5: A Presentation Of An Infinite Group

The group $G = \langle x, y | x^3, y^3, (xy)^3 \rangle$ is infinite.

***Proof*:**

Let $\omega = e^{2\pi i/3} \in \mathbb{C}$. Then $\omega^3 = 1$ and $\omega \neq 1$, and $t^3 - 1 = (t-1)(t^2+t+1)$ gives $1+\omega+\omega^2 = 0$. Define $a, b \colon \mathbb{C} \to \mathbb{C}$ by $a(z) = \omega z$ and $b(z) = \omega z + 1$. They are bijections, with inverses $z \mapsto \omega^2 z$ and $z \mapsto \omega^2(z-1)$, so they lie in the symmetric group $\mathrm{Sym}(\mathbb{C})$ (definition 1.2.12). For $z \in \mathbb{C}$,

$$a^3(z) = \omega^3 z = z \qquad b^3(z) = \omega^3 z + \omega^2 + \omega + 1 = z \qquad (ab)(z) = \omega^2 z + \omega$$

and applying ab three times gives $(ab)^3(z) = \omega^6 z + \omega^5 + \omega^3 + \omega = z + \omega^2 + 1 + \omega = z$. So $a^3 = b^3 = (ab)^3 = \mathrm{id}$, and theorem 3.2.3 gives a homomorphism $\overline{\varphi} \colon G \to \mathrm{Sym}(\mathbb{C})$ with $x \mapsto a$ and $y \mapsto b$. Its image contains ba^{-1}, and $ba^{-1}(z) = b(\omega^2 z) = z + 1$. The powers $(ba^{-1})^m$ with $m \geq 1$ are the translations $z \mapsto z + m$, which are pairwise distinct, so the image of $\overline{\varphi}$ is infinite. Since $\overline{\varphi}$ maps G onto its image, G is infinite, as claimed.

3.2.6 Note Let $N \trianglelefteq G$. Since N is already normal, $\langle N \rangle^{\trianglelefteq} = N$. However, it is rather difficult to determine whether $S \subseteq N$ is a smallest set that satisfies $\langle S \rangle^{\trianglelefteq} = N$.

Corollary 3.2.7: Finitely Generated Groups

Let G be a group. Then G is finitely generated if and only if G is isomorphic to the quotient group of a finitely generated free group.

***Proof*:**

Let $G \cong F/N$, where F is a free group generated by finitely many elements $t_1, \ldots, t_n$, and let $\pi \colon F \to F/N$ be the canonical projection. Every element of F is a product of the t_i and their inverses, and π is a surjective homomorphism, so every element of F/N is a product of the $\pi(t_i)$ and their inverses. Hence F/N, and so G, is finitely generated.

On the other hand, let G is finitely generated, say $S \subseteq G$ such that $\langle S \rangle = G$ and $|S| = n$. Then there exists an inclusion map $S \hookrightarrow G$. By the universal property of free groups, this extends to a map $\varphi : F(S) \twoheadrightarrow G$. This map is surjective since S is a generating set for G. Then by the 1st isomorphism theorem, $F(S)/\ker(\varphi) \cong G$, fulfilling our requirement.

This book does not develop presentations $F(S)/N \cong G$ further. They are the starting point of *geometric group theory*, which studies a finitely generated group through the geometry of its Cayley graphs. For an introduction see [43], and for a more comprehensive treatment [27].

Theorem 3.2.8: Universal Property Of The Cokernel

Let $\varphi\colon G \to H$ be a homomorphism, let $N = \langle \varphi(G) \rangle^{\trianglelefteq}$ be the normal closure of its image (definition 3.2.1), and let $\pi\colon H \to H/N$ be the projection. Then $\pi \circ \varphi$ is trivial, and every homomorphism $\beta\colon H \to K$ with $\beta \circ \varphi$ trivial factors uniquely as $\beta = \beta' \circ \pi$. The group H/N is the *cokernel* of φ.

***Proof*:**
$\pi \circ \varphi$ is trivial because $\varphi(G) \subseteq N$. If $\beta \circ \varphi$ is trivial, then $\varphi(G) \subseteq \ker(\beta)$, a normal subgroup of H (proposition 3.1.14), so $N \subseteq \ker(\beta)$, and theorem 3.1.15 gives a unique β' with $\beta = \beta' \circ \pi$.

When H is abelian, every subgroup is normal, so $N = \varphi(G)$ and the cokernel is $H/\varphi(G)$. The image need not be normal: for the inclusion of $\langle (1\,2) \rangle$ in S_3, N contains every conjugate of $(1\,2)$, hence every transposition, so $N = S_3$ and the cokernel is trivial.

3.3 Application to Finite Groups

As mentioned earlier, thinking about cosets in themselves is also very useful. In fact, a theorem which you will you to death in this course relies on cosets. With the fact that cosets partition your group, you can deduce this strikingly important result:

Theorem 3.3.1: Lagrange Theorem

If G is a finite group and H is a subgroup of G, then the order of H divides the order of G (i.e., $|H| \big| |G|$), and the number of left cosets of H in G equals $\frac{|G|}{|H|}$

***Proof*:**
Let $H \leqslant G$. By proposition 3.1.6, $a \sim_L b \Leftrightarrow a^{-1}b \in H$ is an equivalence relation on G satisfying (3.2), and the class of a is $\left\{ b \in G \;\middle|\; a^{-1}b \in H \right\} = aH$. The left cosets therefore partition G, and each has $|H|$ elements by corollary 3.1.4. If there are m left cosets, then $|G| = m|H|$, so $|H|$ divides $|G|$ and $m = |G|/|H|$, as we sought to show.

Lagrange's theorem determines every group of prime order up to isomorphism.

Corollary 3.3.2: if G has order p, then it is cyclic

If G is a group of prime order p, then G is cyclic, hence $G \cong Z_p$.

***Proof*:**
Since $|G| = p > 1$, choose $x \in G$ with $x \neq e$. The order of the subgroup $\langle x \rangle$ divides p by theorem 3.3.1, and it is greater than 1 since $\langle x \rangle$ contains e and x. As p is prime, $|\langle x \rangle| = p = |G|$, so $\langle x \rangle = G$ and G is cyclic. By corollary 2.3.6, it is abelian, and since Z_p is cyclic of order p, generated by 1, theorem 2.3.9(1) gives $G \cong Z_p$, completing the proof.

Example 3.3.3
Try going back to theorem 2.3.11 and proving it using Lagrange's Theorem. You will find it much easier, almost trivial even.

As the examples show, Lagrange is incredibly powerful as it let's us relate the order of a group and it's subgroup, which is a fact used extensively.

Definition 3.3.4: Index

If G is a group (possibly infinite) and $H \leq G$, the number of left cosets of H in G is called the index of H in G and is denoted by $|G : H|$.

With this notation, we can re-write Lagrange's Theorem as

$$|G| = |H||G : H|$$

Corollary 3.3.5: x to the order of G

If G is a finite group and $x \in G$, then the order of x divides the order of G. In particular $x^{|G|} = 1$ for all $x \in G$.

Corollary 3.3.6: index of H is 2 $\Rightarrow$ H is normal

If $H \leqslant G$ and the index of H is 2, then H is normal

***Proof*:**
Let $x \in G \setminus H$. The left cosets of H partition G (proposition 3.1.6), and $x \in xH$ while $x \notin H = eH$, so the two left cosets are H and xH, and $xH = G \setminus H$. If $hx \in H$ for some $h \in H$, then $x = h^{-1}(hx) \in H$, a contradiction, so $Hx \subseteq G \setminus H = xH$. Since $x^{-1} \notin H$, the same argument gives $Hx^{-1} \subseteq x^{-1}H$, and replacing every element of these two sets by its inverse gives $xH \subseteq Hx$. Hence $xH = Hx$ for every $x \notin H$, and $hH = H = Hh$ for every $h \in H$. So $gH = Hg$ for every $g \in G$, and $H \trianglelefteq G$ by proposition 3.1.11(3), as claimed.

3.3.7 Why this does not necessarily work when the index is greater than 2 Take $G = D_3$, the triangle group, and $H = \{e, \sigma\}$, a subgroup of index 3. Since $\sigma r = r^{-1}\sigma$, the product $r\sigma$ equals $\sigma r^{-1} = \sigma r^2$, so

$$rH = \{r, \sigma r^2\} \qquad Hr = \{r, \sigma r\}$$

and $\sigma r^2 \neq \sigma r$. So $rH \neq Hr$, and H is not normal.

Because of the power of Lagrange, one might ask themselves if the converse is true: if a number divides the order of a group $n||G|$, does that mean there is a subgroup $H \leqslant G$ such that $|H| = n$? The answer is no:

Example 3.3.8: A_4 Has No Subgroup Of Order 6

The alternating group A_4 has order 12, and it has no subgroup of order 6, although $6\big|12$.

Step 1: the elements of A_4. By proposition 1.2.14, every $\sigma \in S_4$ other than id is a product of cycles whose supports are pairwise disjoint subsets of $\{1,2,3,4\}$ with at least two elements each. So σ is a transposition, a 3-cycle, a 4-cycle, or a product of two disjoint transpositions. By theorem 1.2.18, a 3-cycle is a product of two transpositions and a 4-cycle is a product of three, so by theorem 1.2.20(4) the transpositions and the 4-cycles are odd, while the 3-cycles and the products of two disjoint transpositions are even. Hence A_4 (definition 1.2.22) consists of id, the three permutations

$$(1\,2)(3\,4) \qquad (1\,3)(2\,4) \qquad (1\,4)(2\,3)$$

and the eight 3-cycles, two on each of the four 3-element subsets of $\{1,2,3,4\}$. So $|A_4| = 1+3+8 = 12$.

Step 2: squares lie in a subgroup of order 6. Suppose $H \leqslant A_4$ with $|H| = 6$. The left cosets of H are the classes of the equivalence relation of proposition 3.1.6, so they partition A_4, and by theorem 3.3.1 there are $12/6 = 2$ of them. Every $g \in A_4$ satisfies $g^2 \in H$. This holds if $g \in H$. If $g \notin H$, then $g \in gH$ and $g \notin H = eH$, so the two cosets are H and gH, and $A_4 = H \cup gH$. If moreover $g^2 \notin H$, then $g^2 \in gH$, say $g^2 = gh$ with $h \in H$, and cancelling g (proposition 1.1.17) gives $g = h \in H$, a contradiction.

Step 3: the contradiction. Let c be a 3-cycle. Then c has order 3 by proposition 1.2.24, so $c = c^4 = (c^2)^2$, and Step 2 applied to $g = c^2 \in A_4$ gives $c \in H$. So H contains id and the eight 3-cycles, and $|H| \geq 9 > 6$, a contradiction. Therefore A_4 has no subgroup of order 6.

Are there at least partial converses that are true? There are two famous theorems giving the best converse: *Cauchy's Theorem* and *Sylow's Theorem*:

1. (Cauchy) If G is finite, and $p\big||G|$, then there exists a subgroup $H \leqslant G$ of order p
2. (Sylow) If G is finite and $p^k\Big||G|$, that is p maximally divides $|G|$ ($|G| = p^k m$ for some m where $p \nmid m$), then there exists a subgroup $H \leqslant G$ such that $|H| = p^k$.
3. (Sylow Corollary) If $p^j\big||G|$ for some $j \geq 0$, then there exists a subgroup $H \leqslant G$ such that $|H| = p^j$ (see exercise 4.8.1 (1)).

These theorems rely on prime-number properties to give us a partial converse. Sylow's theorem goes a step further in telling us how many groups of a particular order we can find (so more than there exists one). By the way we will present Sylow's theorem, it will require group-actions, the topic of section 4.8. Cauchy's theorem technically does not need it (see [6]).

For finite abelian groups, the full converse of Lagrange's theorem holds:

Corollary 3.3.9: Cauchy's Theorem For Abelian Groups

If G is a finite abelian group, and n divides $|G|$, then there is a subgroup of G of order n. When $n = p$ is prime, G has an element of order p

***Proof*:**
Step 1: the prime case. We show the sharper statement that a finite abelian group G whose order is divisible by a prime p contains an *element* of order p. The subgroup it generates then has order p by proposition 2.3.4, which is the case $n = p$ of the corollary.

The argument is by induction on $|G|$, so assume the statement holds for every abelian group of order strictly smaller than $|G|$. Since $p \big| |G|$ we have $|G| > 1$, so pick $x \in G$ with $x \neq 1$.

First suppose p divides $|x|$, say $|x| = pm$. By corollary 2.3.8(2),

$$|x^m| = \frac{pm}{\gcd(pm, m)} = \frac{pm}{m} = p$$

and x^m is the element we wanted.

Now suppose p does not divide $|x|$. Put $N = \langle x \rangle$, which is normal in G because G is abelian, so the quotient G/N is defined, and it is again abelian. By theorem 3.3.1,

$$|G/N| = \frac{|G|}{|N|}$$

and $|N| = |x| > 1$ by proposition 2.3.4, so $|G/N| < |G|$. Since p divides $|G| = |N| \cdot |G/N|$ and p is prime and does not divide $|N|$, it must divide $|G/N|$. The induction hypothesis applied to G/N produces $y \in G$ whose coset yN has order p in G/N. Then

$$(yN)^{|y|} = y^{|y|}N = N$$

so by proposition 2.3.7 the order p of yN divides $|y|$. Therefore y falls under the previous case, which produces an element of order p and completes the induction.

Step 2: arbitrary divisors. The argument is by induction on $|G|$, so assume the corollary holds for every abelian group of order strictly smaller than $|G|$. Let $d \big| |G|$. If $d = 1$ then the trivial subgroup serves, so assume $d > 1$ and pick a prime p dividing d.

Since $p \big| d$ and $d \big| |G|$, Step 1 gives a subgroup $P \leqslant G$ with $|P| = p$, and P is normal because G is abelian, so G/P is a group with

$$|G/P| = \frac{|G|}{p} < |G|$$

Now d/p is an integer dividing $|G|/p = |G/P|$, and G/P is abelian, so the induction hypothesis produces a subgroup $\overline{H} \leqslant G/P$ with $|\overline{H}| = d/p$. This $\overline{H}$ pulls back along the projection $\pi \colon G \to G/P$: the preimage $H = \pi^{-1}(\overline{H})$ is a subgroup of G by proposition 2.1.3, since $\pi(1) = P \in \overline{H}$ and $\pi(xy^{-1}) = \pi(x)\pi(y)^{-1}$. An element $g \in G$ lies in H exactly when its coset gP lies in $\overline{H}$, so H is the union of the d/p cosets that are elements of $\overline{H}$. These cosets are pairwise disjoint (proposition 3.1.6) and each has $|P| = p$ elements (corollary 3.1.4), so

$$|H| = \frac{d}{p} \cdot p = d$$

and H has the required order, as claimed.

For finite abelian groups the order forces a subgroup of every order dividing it (corollary 3.3.9),

which fails for A_4 (example 3.3.8). This is a first instance of the structure theory that classifies every finitely generated abelian group in chapter 5. The exponentially more difficult task of the non-abelian case is is why much of finite group theory concerns the non-abelian case.

Be mindful that Lagrange's theorem gives you information about *finite* groups. In general, arithmetics on infinite groups will not be defined in a meaningful way for us. In fact, we can get quite some weird results, like subgroups being isomorphic to their parent group!

Example 3.3.10: Subgroups Of $\mathbb{Z}$
Take the group $\mathbb{Z}$, and any subgroup $n\mathbb{Z} \leqslant \mathbb{Z}$ for $n \in \mathbb{N}_+$. Show that

$$n\mathbb{Z} \cong \mathbb{Z}$$

that is, every (non-trivial) subgroup of $\mathbb{Z}$ is isomorphic to $\mathbb{Z}$!

3.4 Internal Direct Product

Though it might seem inconspicuous at first, the introduction of a normal subgroup will let us be able to define the multiplication of subgroups that from subgroups in themselves!! The idea behind this idea is to decompose a group into smaller subgroups, giving an insight on its structure. Chapter 5 develops this decomposition further.

Definition 3.4.1: HK (Internal Direct Product)

Let H, K be subgroups of a group and define

$$HK = \{hk | h \in H, k \in K\}$$

3.4.2 Note This is *not* necessarily a group! Take $\langle(123)\rangle = H \leqslant S_4$ and $\langle(3\,4)\rangle = K \leqslant S_4$. Then

$$HK = \{\mathrm{id}, (123), (132), (123)(34), (132)(34), (34)\}$$

and $(132)(34) = (3421)$, but $(3421)^2 \notin HK$, meaning HK is not closed!

Proposition 3.4.3: cardinality of HK

Let H, K be finite subgroups of a group. Then

$$|HK| = \frac{|H||K|}{|H \cap K|}$$

***Proof*:**
The set HK is the union of the left cosets hK with $h \in H$, and each of them has $|K|$ elements (corollary 3.1.4). For a subgroup C and elements x, y of the ambient group, $xC = yC$ exactly when $y^{-1}x \in C$: if $xC = yC$, then $x = xe \in yC$, so $y^{-1}x \in C$, and if $y^{-1}x = m \in C$, then $xC = y(mC) = yC$, since $mC = C$. Two cosets h_1K and h_2K are equal or disjoint, because a

common element $h_1k_1 = h_2k_2$ gives $h_2^{-1}h_1 = k_2k_1^{-1} \in K$. For $h_1, h_2 \in H$ the element $h_2^{-1}h_1$ lies in H, so

$$h_1K = h_2K \Leftrightarrow h_2^{-1}h_1 \in K \Leftrightarrow h_2^{-1}h_1 \in H \cap K \Leftrightarrow h_1(H \cap K) = h_2(H \cap K)$$

Hence the number of distinct cosets hK with $h \in H$ is the number of left cosets of $H \cap K$ in H, which is $|H|/|H \cap K|$ by theorem 3.3.1. These cosets are pairwise disjoint, so

$$|HK| = \frac{|H|}{|H \cap K|}|K| = \frac{|H||K|}{|H \cap K|}$$

as we sought to show.

Proposition 3.4.4: When is HK a group

If H and K are subgroups of a group, HK is a subgroup if and only if $HK = KH$

***Proof*:**
Let $H, K \leqslant G$.

($\Rightarrow$) Let $HK \leqslant G$. If $x \in HK$, then $x^{-1} \in HK$, say $x^{-1} = hk$, and $x = k^{-1}h^{-1} \in KH$. So $HK \subseteq KH$. If $kh \in KH$, then $(kh)^{-1} = h^{-1}k^{-1} \in HK$, and since HK is a subgroup, $kh \in HK$. So $KH \subseteq HK$, and $HK = KH$.

($\Leftarrow$) Let $HK = KH$. Then $HK \neq \emptyset$. For $h_1k_1, h_2k_2 \in HK$,

$$(h_1k_1)(h_2k_2)^{-1} = h_1(k_1k_2^{-1}h_2^{-1})$$

and $k_1k_2^{-1}h_2^{-1} \in KH = HK$, say it equals $h'k'$. So $(h_1k_1)(h_2k_2)^{-1} = (h_1h')k' \in HK$, and $HK \leqslant G$ by proposition 2.1.3.

Corollary 3.4.5: Condition on K when Internal Direct Product is Present

If H and K are subgroups of G and $H \leq N_G(K)$, then HK is a subgroup of G. Symmetrically, $HK \leqslant G$ when $K \leq N_G(H)$, in particular when $H \trianglelefteq G$. Furthermore, if H and K are normal subgroups, then $HK \trianglelefteq G$.

***Proof*:**
Suppose $H \leqslant N_G(K)$. For $h \in H$ and $k \in K$, both hkh^{-1} and $h^{-1}kh$ lie in K, because h and h^{-1} lie in $N_G(K)$. Hence $hk = (hkh^{-1})h \in KH$ and $kh = h(h^{-1}kh) \in HK$, so $HK = KH$, and $HK \leqslant G$ by proposition 3.4.4.

If $K \leqslant N_G(H)$, the previous paragraph, with the roles of H and K exchanged, gives $KH = HK$ and $HK \leqslant G$. This applies when $H \trianglelefteq G$, since then $N_G(H) = G$ (proposition 3.1.11).

Finally, let H and K be normal in G. Then $HK \leqslant G$ by the previous paragraph. For $g \in G$, $h \in H$ and $k \in K$, we have $g(hk)g^{-1} = (ghg^{-1})(gkg^{-1})$ with $ghg^{-1} \in H$ and $gkg^{-1} \in K$

(proposition 3.1.11). So $gHKg^{-1} \subseteq HK$ for every $g \in G$, and $HK \trianglelefteq G$ by proposition 3.1.11, completing the proof.

Defining when HK is a group is intersting, but it would be even more useful if we can figure out the structure of HK. For example, Can we say that $HK \cong H \times K$? The answer is not always! In fact, we will find a way of exactly figuring out the structure of HK in chapter 5 once we have learned the appropriate tools in chapter 4

3.4.6 Foreshadowing Thinking about HK like in corollary 3.4.5 will become the defining feature when we discuss *semi-direct* products in chapter 5.

Since we have now defined how we can take the "product" of two groups, we can ask the following intersting question: If $\mathcal{N}$ is some subset of normal subgroup G that is "closed" under multiplication, does it form a group? The reader should fiddle around with this before the answer is given. The same question will be asked in the next part of the book where we will have the equivalent of normal subgroup for rings (an algebraic structure with 2 binary operations). It with the "normal" objects of rings. The answer for groups shall be presented in chapter 7. with the "normal" objects of rings. In the case for groups, such a collection will not form a group.

Thus, we will set internal direct products aside, with only an exception made for the next section, until we have accumulated more tools to discuss them again.

3.5 Isomorphism Theorems

We know turn back to how quotient groups are inseparably related to homomorphisms. Mathematicians who were studying homomorphisms wanted to understand the relation between groups given certain properties, for then we can gain new perspective into problems being solved. For example, if we have a group $HK \leqslant G$, and $K \trianglelefteq G$, can we always determine what HK/K is? Is there *always* an answer to this question, no matter what group HK we pick? Another question is what if we "recusively" take quotients: If $H \trianglelefteq K \trianglelefteq G$, what is $(G/H)/(K/H)$? Is there a simplification that will make this more understandable? Yet another such question is how much can we deduce of the subgroups of G/N (i.e. the lattice of G/N) given what we know about G?

In this section, we will show these results:

1. We will sum up the results of of theorem 3.1.15, but give it the name that is more commonly used
2. We will look at equivalent ways of looking at internal direct products, since they pop up a lot (especially chapter 5)
3. An easier way of thinking of quotient groups whose quotient is a quotient group.
4. the subgroup lattice of G and the subgroup lattice of G/N

We first recall theorem 3.1.15, but given it its classical name:

Theorem 3.5.1: First Isomorphism Theorem

If $\Phi : G \to H$ is a homomorphism of groups, then $\ker(\Phi) \trianglelefteq G$ and $G/\ker\Phi \cong \Phi(G)$.
Similarly, if $N \trianglelefteq G$ and $N \subseteq \ker(\Phi)$, there exists a unique *homomorphism* $\varphi : G/N \to H$ such that the following diagram commutes:

$$\begin{array}{ccc} G & \xrightarrow{\pi} & G/N \\ & \searrow^{\Phi} & \downarrow{\scriptstyle \varphi} \\ & & H \end{array}$$

where $\Phi = \varphi \circ \pi$, and π is the natural projection

The second half of the theorem is the universal property of the quotient group: every homomorphism $\Phi\colon G \to H$ whose kernel contains N factors uniquely through the natural projection $\pi\colon G \to G/N$[5], as $\Phi = \varphi \circ \pi$. Questions about Φ therefore become questions about the group G/N.

Also, remember that if we take $\Phi(G) \leqslant H$, and $N = \ker(\Phi)$ so that

$$\begin{array}{ccc} G & \xrightarrow{\pi} & G/\ker(\Phi) \\ & \searrow^{\Phi} & \downarrow{\scriptstyle \varphi} \\ & & \Phi(G) \end{array}$$

then

$$G/\ker(\Phi) \cong \Phi(G)$$

Example 3.5.2

1. we can get some information about S_n for $n \geq 2$. The set $\{1, -1\}$ is a group under multiplication, and by theorem 1.2.20(2) the sign $\operatorname{sgn}\colon S_n \to \{1, -1\}$ is a homomorphism, which sends σ to 1 if σ is even and to -1 if σ is odd (definition 1.2.21). It is surjective, since $\operatorname{sgn}((1\ 2)) = -1$, and its kernel is the alternating group A_n (definition 1.2.22). By corollary 3.1.17, we then know that

   $$S_n/A_n \cong \{\pm 1\}$$

   and so, using Lagrange, we get that

   $$2 = |S_n/A_n| = \frac{|S_n|}{|A_n|} \qquad \Rightarrow \qquad |A_n| = \frac{1}{2}|S_n|$$

2. Let's say we have two groups G, H with $\varphi : G \to H$ being a surjective homomorphism. Then we know that

   $$G/\ker(\varphi) \cong H$$

   in other words, if we have a surjective homomorphism, we can always view the structure of H through the perspective of the quotient group $G/\ker(\varphi)$!

[5] Here *natural* means that defining π requires no choices. The word also has a technical meaning, the one of definition 18.2.1.

Corollary 3.5.3: properties as consequence

Let $\varphi : G \to H$ be a group homomorphism

1. φ is injective if and only if $\ker \varphi = 1$
2. $|G : \ker \varphi| = |\varphi(G)|$ [a]

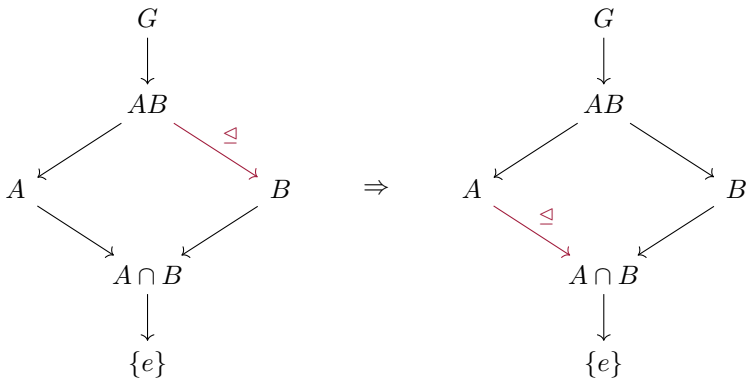
[a] If you have taken linear algebra, this is the Rank-Nullity theorem equivalent for Group Theory

The second Isomorphism theorem let's us view the quotients of internal direct products (definition 3.4.1) form a different angle.

Theorem 3.5.4: The Second or Diamond Isomorphism Theorem

Let G be a group, let A and B be subgroups of G and assume $A \leq N_G(B)$. Then AB is a subgroup of G, $B \trianglelefteq AB$, $A \cap B \trianglelefteq A$, and $AB/B \cong A/A \cap B$.

The name diamond comes form this diagram where the arrow represent subgroups, and a red arrow represents norma subgroup:



and

$$\frac{AB}{B} \cong \frac{A}{A \cap B}$$

For finite A and B, comparing orders on the two sides of $AB/B \cong A/(A \cap B)$ gives $|AB| = |A||B|/|A \cap B|$, recovering proposition 3.4.3 in the case $A \leqslant N_G(B)$.

***Proof*:**

Since $A \leqslant N_G(B)$, $aBa^{-1} = B$ for every $a \in A$. So for $a \in A$ and $b \in B$,

$$ab = (aba^{-1})a \in BA \qquad ba = a(a^{-1}ba) \in AB$$

using $a^{-1} \in A$ in the second. Hence $AB = BA$, and $AB \leqslant G$ by proposition 3.4.4.

For $a \in A$ and $b \in B$, $(ab)B(ab)^{-1} = a(bBb^{-1})a^{-1} = aBa^{-1} = B$, since $bBb^{-1} = B$ for $b \in B$.

So $(ab)B = B(ab)$, and $B \trianglelefteq AB$ (definition 3.1.9).

The quotient AB/B is therefore a group (definition 3.1.13). Define $\varphi\colon A \to AB/B$ by $\varphi(a) = aB$. It is a homomorphism, since $\varphi(aa') = aa'B = (aB)(a'B)$ by the definition of the operation on AB/B. It is surjective: an element of AB/B has the form abB with $a \in A$ and $b \in B$, and $abB = aB = \varphi(a)$ because $bB = B$. Its kernel is $\{a \in A \mid aB = B\}$. If $aB = B$, then $a = ae \in B$. Conversely, if $a \in B$, then $aB \subseteq B$ and $B = a(a^{-1}B) \subseteq aB$, so $aB = B$. Hence $\ker\varphi = A \cap B$, which is normal in A by proposition 3.1.14, and corollary 3.1.17 gives

$$\frac{A}{A \cap B} \cong \frac{AB}{B}$$

as we sought to show.

One way to remember this theorme is through the following mnemonic:

> If we have $\frac{AB}{B}$, then add a $A \cap _$ to the top and bottom, giving us
>
> $$\frac{A \cap AB}{A \cap B} = \frac{A}{A \cap B}$$

Another way to think about the 2nd isomorphism theorem is to consider the case where $A \cap B = \{e\}$. Then this theorem shows that $AB/B \cong A$, and if also $B \leqslant N_G(A)$, that $AB/A \cong B$:

$$\frac{AB}{B} \cong A \qquad \frac{AB}{A} \cong B$$

Heuristically, we can think in this situation that we are "cancelling out" the B's. However, be mindful to take this as a heuristic: this is *not* what is happening!

The third Isomorphism theorem lets quickly simplify quotient groups quotiented *by* a quotient group:

Theorem 3.5.5: The Third Isomorphism Theorem

Let G be a group and let H and K be normal subgroups of G with $H \leq K$. Then $K/H \trianglelefteq G/H$ and

$$(G/H)/(K/H) \cong G/K$$

If we denote the quotient by H with a bar, this can be written

$$\bar{G}/\bar{K} \cong G/K$$

***Proof*:**
Since $K \trianglelefteq G$, $K/H \trianglelefteq G/H$. To show this, first note that K/H is indeed a subgroup of G/H. For any $[x], [y] \in K/H$, $[x][y]^{-1} = [xy^{-1}] \in K/H$ since $xy^{-1} \in K$. Furthermore, for any $[g] \in G/H$, and any coset $[k] \in K/H$, $[g][k] = [k'][g]$, $k' \in K$ (i.e. $[g](K/H) = (K/H)[g]$), since both K and H are normal. Thus $K/H \trianglelefteq G/H$.

Now, define the map

$$\varphi : G/H \to G/K \qquad gH \mapsto gK$$

This map is well-defined: if $g_1H = g_2H$, then $g_2^{-1}g_1 \in H \subseteq K$, so $g_1K = g_2K$.

Now, notice this map is a homomorphism and surjective. It is a homomorphism since

$$\varphi(g_1Hg_2H) \overset{!}{=} \varphi(g_1g_2H) = g_1g_2K \overset{!}{=} g_1Kg_2K = \varphi(g_1H)\varphi(g_2H)$$

where the $\overset{!}{=}$ equalities are the definition of the operations on G/H and G/K (definition 3.1.13). For surjectivity, since $H \leqslant K$, then for any gK, we know there is at least the coset gH that maps to it (there might be more, since $H \leqslant K$). Hence, by corollary 3.1.17

$$(G/H)/\ker(\varphi) \cong G/K$$

to find $\ker(\varphi)$, we put together the information we have so far:

$$\begin{aligned}\ker(\varphi) &= \{gH \in G/H \mid \varphi(gH) = K\} \\ &= \{gH \in G/H \mid gK = K\} \\ &= \{gH \in G/H \mid g \in K\} \\ &= K/H\end{aligned}$$

Thus, replacing $\ker(\varphi)$ with K/H we get

$$(G/H)/(K/H) \cong G/K$$

as we sought to show

An easy heuristic to remember $(G/H)/(K/H) \cong G/K$ is to think about how to simply fraction's of fractions, that is, the "flip and cancel" trick

$$\frac{\frac{1}{2}}{\frac{4}{2}} = \frac{1}{2}\frac{2}{4} = \frac{1}{4}$$

Note however that this is *not* what is happening here. It is just a useful heuristic to remember the 3rd Isomorphism Theorem. Note too that for the simplification to occur, you need that H is in the denominator of both quotients, you cannot have a different value.

Example 3.5.6: Quotients Of Cyclic Groups

Let n and m be positive integers with $n|m$, say $m = nk$. Then $m\mathbb{Z} \subseteq n\mathbb{Z}$, since $mx = n(kx)$, and both are normal in the abelian group $\mathbb{Z}$ (corollary 3.1.12). By theorem 3.5.5,

$$(\mathbb{Z}/m\mathbb{Z})/(n\mathbb{Z}/m\mathbb{Z}) \cong \mathbb{Z}/n\mathbb{Z}$$

For instance, the quotient of $\mathbb{Z}/6\mathbb{Z}$ by its subgroup $\{0, 2, 4\} = 2\mathbb{Z}/6\mathbb{Z}$ has order 2.

Finally, we will relate the subgroup structure (the lattice) of G/N with G. It will be precisely a sublattice of G that contains N! This means that going to G/N will preserve subgroup relations:

Theorem 3.5.7: The Fourth or Lattice Isomorphism Theorem (Correspondence Theorem)

Let G be a group and let N be a normal subgroup of G. Then $H \mapsto \bar{H} = H/N$ is a bijection from the set of subgroups H of G which contain N onto the set of subgroups of $\bar{G} = G/N$.

$$\{H | H \leqslant G,\ H \text{ contains } N\} \leftrightarrow \left\{\overline{H} \leqslant \overline{G}\right\}$$

In particular, every subgroup of $\bar{G}$ is of the form H/N for some subgroup H of G containing N (namely, its pre-image in G under the natural projection homomorphism from G to G/N).

***Proof*:**

Let $\pi\colon G \to \overline{G}$, $\pi(g) = gN$, be the natural projection. It is a surjective homomorphism, since $\pi(ab) = (ab)N = (aN)(bN)$ by the definition of the operation on G/N (definition 3.1.13), and its kernel is N, because $gN = N$ exactly when $g \in N$.

Step 1: the map is defined. Let H be a subgroup of G containing N. Then $H/N = \{hN \mid h \in H\} = \pi(H)$, which is a subgroup of $\overline{G}$ by proposition 1.4.4(5) applied to the restriction of π to H. Moreover $\pi^{-1}(\pi(H)) = H$: if $\pi(g) = \pi(h)$ with $h \in H$, then $gN = hN$, so $g = ge \in hN \subseteq H$, because $N \subseteq H$.

Step 2: injectivity. If H and K are subgroups of G containing N with $H/N = K/N$, then Step 1 gives $H = \pi^{-1}(\pi(H)) = \pi^{-1}(\pi(K)) = K$.

Step 3: surjectivity. Let $\overline{H} \leqslant \overline{G}$ and $P = \pi^{-1}(\overline{H})$. Then $e \in P$, and for $x, y \in P$, $\pi(xy^{-1}) = \pi(x)\pi(y)^{-1} \in \overline{H}$, so $P \leqslant G$ by proposition 2.1.3. It contains N, since $\pi(n) = N$ is the identity of $\overline{G}$ for $n \in N$. Since π is surjective, every element of $\overline{H}$ is $\pi(g)$ for some $g \in G$, and such a g lies in P, so $P/N = \pi(P) = \overline{H}$. Hence the map is a bijection, and the preimage of $\overline{H}$ is the subgroup of G containing N that maps to it, as we sought to show.

This fact has many very nice consequences:

Corollary 3.5.8: Equivalent way of looking at 4th Isomorphism Theorem

Let $N \trianglelefteq G$, let $\pi\colon G \to \overline{G} = G/N$ be the natural projection, and for a subgroup X of G containing N write $\overline{X} = X/N = \pi(X)$. Let A and B be subgroups of G containing N. Then:

1. $A \leqslant B$ if and only if $\overline{A} \leqslant \overline{B}$
2. if $A \leqslant B$ then $|B : A| = |\overline{B} : \overline{A}|$
3. $\overline{\langle A, B\rangle} = \langle \overline{A}, \overline{B}\rangle$
4. $\overline{A \cap B} = \overline{A} \cap \overline{B}$
5. $A \trianglelefteq G$ if and only if $\overline{A} \trianglelefteq \overline{G}$

***Proof*:**
The projection $\pi(g) = gN$ is a surjective homomorphism, since $\pi(ab) = (ab)N = (aN)(bN) = \pi(a)\pi(b)$ by the definition of the operation on G/N (definition 3.1.13). For a subgroup X of G containing N, the set $\overline{X}$ is a subgroup of $\overline{G}$ by proposition 1.4.4(5) applied to the restriction of π to X. Moreover

$$\pi^{-1}(\overline{X}) = X$$

Indeed, $X \subseteq \pi^{-1}(\pi(X))$, and if $\pi(g) \in \overline{X}$, then $gN = xN$ for some $x \in X$, so $g = ge \in gN = xN \subseteq X$ because $N \subseteq X$. We also use that for a subgroup C of any group and elements x, y of that group, $xC = yC$ exactly when $x^{-1}y \in C$: if $xC = yC$, then $y \in yC = xC$, so $x^{-1}y \in C$, and if $x^{-1}y = m \in C$, then $yC = x(mC) = xC$, since $mC = C$.

1. If $A \subseteq B$, then $\overline{A} = \pi(A) \subseteq \pi(B) = \overline{B}$. If $\overline{A} \subseteq \overline{B}$, then $A = \pi^{-1}(\overline{A}) \subseteq \pi^{-1}(\overline{B}) = B$.

2. Let $A \leqslant B$. For $b, b' \in B$,

$$bA = b'A \Leftrightarrow b^{-1}b' \in A \Leftrightarrow \pi(b^{-1}b') \in \overline{A} \Leftrightarrow \pi(b)^{-1}\pi(b') \in \overline{A} \Leftrightarrow \pi(b)\overline{A} = \pi(b')\overline{A}$$

   where the second equivalence holds because $A = \pi^{-1}(\overline{A})$. So $bA \mapsto \pi(b)\overline{A}$ is a well-defined injective map from the left cosets of A in B to the left cosets of $\overline{A}$ in $\overline{B}$. It is surjective, since every element of $\overline{B}$ is $\pi(b)$ for some $b \in B$. Both indices count left cosets, so $|B : A| = |\overline{B} : \overline{A}|$.

3. The subgroup $\langle A, B\rangle = \langle A \cup B\rangle$ contains N (definition 2.3.15). Its image $\overline{\langle A, B\rangle}$ is a subgroup of $\overline{G}$ containing $\overline{A} \cup \overline{B}$, so it contains $\langle \overline{A}, \overline{B}\rangle$ by proposition 2.3.17. Conversely, $P = \pi^{-1}(\langle \overline{A}, \overline{B}\rangle)$ is a subgroup of G by proposition 2.1.3, since $e \in P$ and $\pi(xy^{-1}) = \pi(x)\pi(y)^{-1} \in \langle \overline{A}, \overline{B}\rangle$ for $x, y \in P$. It contains A and B, so $\langle A, B\rangle \subseteq P$ by proposition 2.3.17, and $\overline{\langle A, B\rangle} = \pi(\langle A, B\rangle) \subseteq \pi(P) \subseteq \langle \overline{A}, \overline{B}\rangle$.

4. The subgroup $A \cap B$ contains N (proposition 2.3.14), and $\pi(A \cap B) \subseteq \pi(A) \cap \pi(B)$. Conversely, let $y \in \overline{A} \cap \overline{B}$ and write $y = \pi(g)$ with $g \in G$. Then $g \in \pi^{-1}(\overline{A}) = A$ and $g \in \pi^{-1}(\overline{B}) = B$, so $y = \pi(g) \in \pi(A \cap B)$. Hence $\overline{A \cap B} = \overline{A} \cap \overline{B}$.

5. Every element of $\overline{G}$ is $\pi(g)$ with $g \in G$, and every element of $\overline{A}$ is $\pi(a)$ with $a \in A$. Suppose $A \trianglelefteq G$, and let $g \in G$ and $a \in A$. Then $gag^{-1} \in A$ by proposition 3.1.11, so $\pi(g)\pi(a)\pi(g)^{-1} = \pi(gag^{-1}) \in \overline{A}$. Hence $\overline{A} \trianglelefteq \overline{G}$ by the equivalence of (5) and (1) in proposition 3.1.11. Conversely, suppose $\overline{A} \trianglelefteq \overline{G}$. For $g \in G$ and $a \in A$, $\pi(gag^{-1}) = \pi(g)\pi(a)\pi(g)^{-1} \in \overline{A}$ by proposition 3.1.11, so $gag^{-1} \in \pi^{-1}(\overline{A}) = A$. Hence $A \trianglelefteq G$ by proposition 3.1.11, completing the proof.

Another way of looking at the first part of the Correspondence Theorem is in the following slight generalization: Let G be a group and N a normal subgroup so that G/N is well-defined. If $\mathcal{G}$ contains all subgroups of G, for any $H \in \mathcal{G}$, the map $\pi : H \to G/N$, $\pi(h) = hN$ is well-defined. So, H maps to a subgroup of G/N. If $\mathcal{N}$ represents all subgroups of G/N, then there is a surjective map

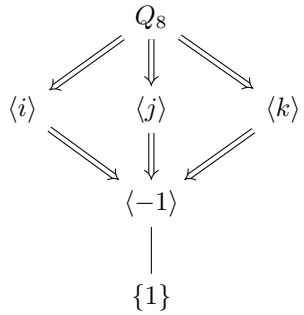
$$\mathcal{G} \twoheadrightarrow \mathcal{N}$$

and if we take $\mathcal{G}|_N$ to be all subgroups that contain N, then there is a bijection between $\mathcal{G}|_N$ and $\mathcal{N}$:

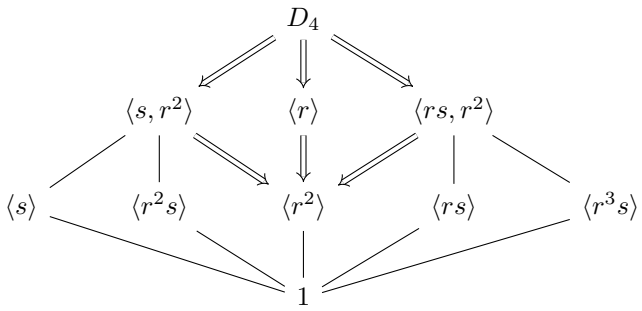
$$\mathcal{G}|_N \leftrightarrow \mathcal{N}$$

Example 3.5.9: Using Correspondence Theorem

1. Let $G = Q_8$, and take $\langle -1 \rangle \trianglelefteq Q_8$. Then in the following lattice, the double lines represent the sublattice of Q_8 which is an isomorphic copy of the lattice of $Q_8/\langle -1 \rangle$



2. Let $G = D_4$ and let $\langle r^2 \rangle \trianglelefteq D_4$. Then with the same definitions as the last example, we get:



Notice that several subgroup of G (for example $\langle s \rangle$) become elements in a larger subgroup (in this case $\langle s, r^2 \rangle$). Thus, every subgroup of $\overline{G}$ might have multiple subgroups of G that map to it, but a unique subgroup which contains N.

A good example how the map of subgroup of G to G/N is surjective is by the following example: Notice that $Q_8/\langle -1 \rangle \cong D_4/\langle r^2 \rangle \cong \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$, and $\langle -1 \rangle \cong \langle r^2 \rangle$, but $Q_8 \ncong D_4$. We will explore this idea in more depth in the next section.

Exercise 3.5.1

1. (The *Butterfly Lemma*, Zassenhaus.) Let G be a group with subgroups $A' \trianglelefteq A$ and $B' \trianglelefteq B$. Show that $A'(A \cap B)$ is a subgroup of A, that $D = (A' \cap B)(A \cap B')$ is a normal subgroup of $A \cap B$, and that $A'(A \cap B')$ is a normal subgroup of $A'(A \cap B)$.

2. In the setting of exercise 3.5.1 (1), show that $a'x \mapsto xD$ ($a' \in A'$, $x \in A \cap B$) is a well-defined surjective homomorphism $A'(A \cap B) \to (A \cap B)/D$ with kernel $A'(A \cap B')$. Conclude that

$$\frac{A'(A \cap B)}{A'(A \cap B')} \cong \frac{A \cap B}{(A' \cap B)(A \cap B')} \cong \frac{B'(A \cap B)}{B'(A' \cap B)}$$

3. A *refinement* of a subnormal series (definition 3.7.3) is a subnormal series containing all of its terms. Use exercise 3.5.1 (2) to show that any two subnormal series of a group have refinements whose factors agree up to order and isomorphism (Schreier), and deduce part (2) of theorem 3.7.9.

3.6 Extensions

Recall (or revisit) that if $H \leqslant G$ (and G is finite), then the subgroup lattice of H is the sublattice of G of all groups *contained* in H (theorem 2.4.3), so *a fortiori* if $K \trianglelefteq G$ is a normal subgroup then the subgroup lattice of K is the sublattice of G of all subgroups contained in K. Even hen G is infinite, the set of subgroup of K is the set o subgroup of G which are contained in K. The Correspondence Theorem has shown us that the "dual" is true for quotient groups (except it orks for *all* groups, not just finite): If $K \trianglelefteq G$, then the subgroup lattice of G/K is the sublattice of G of all groups *containing* K. In this way, K and G/K together capture the subgroup structure of a group! In this section, we will explore how we can find a group structure given *only* the information from K and G/K.

More succintely, Given $K \trianglelefteq G$, the group K and G/K contain the subgroup structure of G. However, K and G/K do not capture the full structure of a group: in the process of quotienting, G/K losses some information:

Example 3.6.1: Subgroup And Quotient Group Limits

Take $G = D_4$ and $K = \{1, r, r^2, r^3\} \cong C_4$. Then $D_4/K \cong C_2$. Label this K'. Then given just K and K', one cannot guarantee that $K' = D_4/K$, other groups can take the place of D_4. For example:

$$H = C_4 \times C_2 \qquad \varphi : C_4 \times C_2 \to C_2,\ \varphi((a,b)) = ((0,b))$$

Here, H is another group here the kernel is C_4 and the quotient is C_2.

It must be noted though that there is still information within cosets: for any $x \in G$, then $[x] = xK \in G/K$, so for any $y \in xK$, $y = xk$ for appropriate $k \in K$ (proposition 3.1.3). This means for every coset, there is some information about how the elements relate. An easy example of this can be seen if $K = Z(G)$.

Example 3.6.2: Relation Within Cosets

Take $G/Z(G)$, which is a group since $Z(G) \trianglelefteq G$ (example 3.1.10). Take any $[x] \in G/Z(G)$, and any $a, b \in [x]$. Then $a = xz_a$ and $b = xz_b$, so $ab = xz_axz_b = xz_bxz_a = ba$ (since $z_a, z_b \in Z(G)$). Thus, within any coset of $G/Z(G)$, the elements commute! This is not nesesarily the case if the elements are *not* within the same coset (ex. $G = D_7$).

However, as we saw in example 3.6.1, there can be more than one group (say G') we can construct from K and K' (where $G'/K = K'$). We will call groups which can be built in this fashion *extension*:

Definition 3.6.3: Extension

Let A and B be groups. If $G/A \cong B$, then we say G *is an extension of* A *by* B.

The idea behind this name is that we want to be able to restrict G to A (i.e. A must be able to map

injectively and homomorphically into G; $A \hookrightarrow G$), similarly to how we can restrict functions (the dual of restricting functions is also called *extending* functions, where the extended function must be exactly the same as the orginal funciton when restricting).

As we saw at the beginning of this section, quotient groups are the perfect candidate for extension of groups since every group can be decomposed into a [normal] subgroup and quotient group.

Since all normal subgroups are also kernels, we can re-write what it means to be an extension in terms of homomorphisms. In the following definition, let I represent the trivial group.

Definition 3.6.4: Short Exact Sequence

Let A, B, C be group. If there exists an injective homomorphism f and a surjective homomorphism g such that

$$I \xrightarrow{\iota} A \xrightarrow{f} B \xrightarrow{g} C \xrightarrow{\pi} I$$

where $\text{im}(f) = \ker(g)$, ι is the trivial injective $(0 \mapsto 0)$ map and π is the total projective map $(c \mapsto 0)$, then we say that B *is an extension of* A *by* C.

Notice that $\text{im}(\iota) = \ker(f)$ since f is injective, $\text{im}(g) = \ker(\pi)$ since g is surjective, and $\text{im}(f) = \ker(g)$ by definition. A sequence in which the image of each map is the kernel of the next is called *exact*[6], whether it is short or a longer chain $X_1 \xrightarrow{f_1} X_2 \xrightarrow{f_2} X_3 \xrightarrow{f_3} X_4 \xrightarrow{f_4} \cdots$ (chapter 17⋆ returns to these). The exact sequence in definition 3.6.4 is the shortest non-trivial sequence with the trivial group at the beginning and end, and so it is called a short exact sequence.

For our purposes, exactness gives $C \cong B/f(A)$ with $f(A) \cong A$, so B is an extension of A by C.

Example 3.6.5: Short Exact Sequences

In the following examples, let 1 represent the trivial group

1. If $\varphi : G \to H$ is a homomorphism and $K = \ker(\varphi)$, we always have

   $$1 \to K \hookrightarrow G \xrightarrow{\varphi} \varphi(G) \to 1$$

   is a short exact sequence. If φ is surjective, then

   $$1 \to K \hookrightarrow G \xrightarrow{\varphi} H \to 1$$

   In this way, every surjective homomorphism represents an extension G of K by H.

2. Let A and B be two arbitrary groups. Then

   $$1 \to A \hookrightarrow A \times B \xrightarrow{(a,b)\mapsto(0,b)} B \to 1$$

   is a short exact sequence. In this way, any group can be extended by any other group to the extensions $A \times B$. This is sometimes called the *trivial extension* [a]

[6] A differential form is called *exact* when it lies in the image of the exterior derivative d, and for $k \geq 1$ the sequence of maps d between spaces of differential forms is exact in the present sense at the k-forms precisely when every closed k-form, one in the kernel of d, is exact. De Rham cohomology [46, De Rham Cohomology] measures the failure of this.

3. Take $A = B = 2$. Then
$$1 \to C_2 \xrightarrow{f} C_4 \xrightarrow{g} C_2 \to 1$$
where $f(x) = 2x$ and $g(0)$ and $g(2)$ both map to 0, while $g(1)$ and $g(3)$ map to 1. It is easy to verify that f and g are indeed homomorphism and that the above sequence is exact.

4. Similarly to the last example, Let $A = C_2$ and $B = C_8$ and $C = C_4$. Can you find the appropriate maps such that
$$1 \to C_2 \to C_8 \to C_4 \to 1$$
is a short exact sequence? Is the following
$$1 \to C_2 \to D_4 \to C_4 \to 1$$
also a short exact sequence with the appropriate maps?

[a] In fact, once we cover homological algebra in chapter 17⋆, the idea of this being the trivial extension will become rigorous

An extension of an X group by an X group is X. When is this statement true when X is:

1. cyclic
2. abelian
3. finite
4. finitely generated
5. Noetherian

It is in general difficult to tell when when given a short exact sequence if the sequence splits. In section 17.2, we will classify which *abelian groups* in a short exact sequence of abelian groups split[7] (abelian groups which always split in a short exact sequence of abelian groups be called *projective groups*). For general groups, the best tool we have is trying to identify when C acts on B.

A normal subgroup H of G gives the short exact sequence
$$1 \to H \to G \to G/H \to 1$$
and every element of G lies in a coset zH for a representative z chosen in each coset. It is natural to ask whether G can be thought of as "$(G/H) \cdot H$". This works when the representatives can be chosen to form a subgroup of G: the extension then splits, and G is a semi-direct product, the subject of chapter 5.

Exercise 3.6.1

1. Let $0 \to \mathbb{Z} \to B \to A \to 0$ be an extension, where B is an abelian group and A is free. Show that B splits. (The converse, that an abelian group A over which every such extension splits is free, is known as Whitehead's problem, and it can be neither proved nor disproved from the ZFC axioms.)

[7] It will turn out that all abelian groups are $\mathbb{Z}$-modules, and so this will be a consequence of classifying when modules split with the space case of when the module is over $\mathbb{Z}$

2. Let $1 \to A \xrightarrow{f} G \xrightarrow{g} B \to 1$ be a short exact sequence of groups with B finite. Show that $f(A)$ is a normal subgroup of G of index $|B|$.

3.7 Classifying all Finite Groups: Hölder Program

With the isomorphism theorems under our belt, we are ready to tackle one of the central questions posed by group-theorists: How can we classify all groups? This sections start by building up the mental models used in order to comprehend the way classification efforts.

We can approach the problem of classification in many different ways. We briefly mentioned the idea trying to construct arbitrary presentation's of groups to try and create groups, but we have mentioned that this can create "insoluble presentation", meaning presentaion in which we cannot represent every element. In general, it is very hard to find groups in that manner. Perhaps instead we can start with the groups we knwo and see if there is a way to "decompose" them. The first thing you might try is decomposing the group into cyclic sub-groups. This is certainly possible (take an element, generate a group from it, remove those elements from your origianl group, continue doing this till you have exhausted the group [8]). But how would you then put them back together? For example, we know that $\langle r \rangle, \langle s \rangle \leqslant D_4$. With what we have learnt, is there a way to represent D_4 using these? In fact, we have just seen that we can but them back together using short exact sequences! The group D_4 is an extension of C_2 by C_4.

However, we must be more careful using extensions when we are trying to classify groups. Are cyclic subgroups really the way to go? In fact, this is not the case: there exists cyclic groups which are an extension of smaller groups. We would want to be able to have some fixed subgroups we can use as our "building puzzles" which are not extensions of any other groups. In this section we will define these "smalles puzzle pieces" or "atoms of groups". Try taking a moment to see if you can come up with a more suitable definition of "smallest puzzle piece" for groups and have an idea of an example of a group following your definition.

Every finite simple group (definition 3.7.1) appears in the list of theorem 3.7.11, the classification of finite simple groups, whose proof occupied group theorists for the better part of the 20th century. For abelian groups the classification is much simpler: theorem 5.1.6 describes every finitely generated abelian group.

Defining our Terms

First, we will have to define what would be our "puzzle piece". Since we are using extensions, we know there is a non-trivial homomorphism from a group G to another group H if G has a non-trivial normal subgroup. If G has *only* trivial subgroups, the structure of G is not "composed" of other possible structure but is as "simple" as possible. This motivates the folloing definition.

Definition 3.7.1: Simple Groups

A (finite or infinite) group G is called *simple* if $|G| > 1$ and the only normal subgroups of G are 1 and G.

[8] This works well in groups of *countable* size. One has to be more careful for groups of *uncountable* size, where the axiom of choice is involved (see appendix A for details on the axiom of choice).

These groups are called "simple" because there are no more group-structures within them [9]. To undersatnd what "no more group-structures within them" means, recall the 4th isomorphism theorem. If N is a noraml subgroup, then ther is a direct correspondence between the subgroups of G/N and the subgroups of G containing N. If there is no $N \trianglelefteq G$ besides $\{1\}$ and G, then the lattice of G cannot be split into G/N and G. So in a sense, that lattice is as small as it gets.

Example 3.7.2: Simple Groups

By Lagrange's theorem, a group of prime order p has no subgroups other than 1 and itself, so groups of prime order are simple. If G is abelian and simple, then every subgroup is normal, so $G = \langle x \rangle$ for any $x \neq e$ and G is cyclic. It is not infinite, since $\mathbb{Z}$ has the proper nontrivial subgroup $2\mathbb{Z}$. So $G \cong \mathbb{Z}/n\mathbb{Z}$, which has a subgroup of order d for every $d \big| n$ (theorem 2.3.11), forcing n to be prime.

There are also non-abelian simple groups, both infinite and finite. The alternating group A_5, of order 60, is simple by theorem 4.9.6.

Looking at simple groups a little more abstractly for a moment, they play an analogous role to primes in arithmetic's of $\mathbb{N}$ as in they cannot be "factored" into smaller [meaningful] pieces[10] (i.e. N and G/N). There is even a "unique factorization theorem" based on this idea. First, we will define what is meant by "factorize" a group

Definition 3.7.3: Subnormal Series

Let G be a group. Then

$$G = G_1 \geqslant G_2 \geqslant G_3 \geqslant \cdots \geqslant G_n = \{1\}$$

is called a *series* (or *tower*) for the group G. If every $G_i \trianglelefteq G$, then G the series is called a *normal series*. If $G_i \trianglelefteq G_{i-1}$ for every $1 \leq i \leq n$, then we call the series a *subnormal series*:

$$G = G_1 \trianglerighteq G_2 \trianglerighteq \cdots \trianglerighteq G_n = \{1\}$$

Sometimes, [normal/subnomral] series are written in the ascending, rather than descending order:

$$\{1\} = G_0 \leqslant G_1 \leqslant \cdots \leqslant G_n = G$$

I chose to write them in descending order since down the line we will be comparing some types of series to other types, and if everything is written in descending order, then we'd have a scendario were we start writing $G_{n-i} \trianglelefteq B_i$ ineated of $G_i \trianglelefteq B_i$, which I feel will make the simple relations a little more notationaly cumbersome than it needs to be.

You might ask whether this a subnormal series would imply a noraml series, that is, that $N_i \trianglelefteq G$? However, that does not need to be the case:

[9] The term simple is one that comes up often enough in matheamtics. It has a sligthly more nuanced meaning than just the simplest object (ex. the trivial group is not a simple group, it's "too simple to be simple". See this nlab article on simple objects

[10] meaningful here meaning there is a notion of combining groups using short exact sequence, unlike general groups cannot be

Example 3.7.4: Normality *not* transitive
Here's an example of how normality is *not* transitive. Let $G = D_4$. Let $H \trianglelefteq G$ such that $H = \langle r\sigma, \sigma r\rangle$. Next, take $K \trianglelefteq H$ such that $K = \langle r\sigma\rangle$. Note that

$$K \trianglelefteq H \trianglelefteq G$$

but $K \not\trianglelefteq G$

thus a subnormal series need not be a noraml series. However, a normal series is a subnormal series: if $N_{i+1} \trianglelefteq G$, then $gN_{i+1}g^{-1} = N_{i+1}$ for every $g \in G$. If we restrict to every $n \in N_i$, then since $N_i \subseteq G$, it is already the cse that $nN_{i+1}n^{-1} = N_{i+1}$. Thus $N_{i+1} \trianglelefteq N_i$. Thus:

$$\text{normal series} \Rightarrow \text{subnormal series} \qquad \text{subnormal series} \not\Rightarrow \text{normal series}$$

Thus, subnormal series can be thought of as the decomposition of a group.

Example 3.7.5: Refining A Subnormal Series
In $G = \mathbb{Z}/8\mathbb{Z}$ every subgroup is normal (corollary 3.1.12), so

$$\mathbb{Z}/8\mathbb{Z} \trianglerighteq \langle 4\rangle \trianglerighteq 0 \qquad \text{and} \qquad \mathbb{Z}/8\mathbb{Z} \trianglerighteq \langle 2\rangle \trianglerighteq \langle 4\rangle \trianglerighteq 0$$

are subnormal series, and the second is obtained from the first by inserting the term $\langle 2\rangle$. The subgroups $\langle 2\rangle$ and $\langle 4\rangle$ have orders 4 and 2, so the terms of the second series have orders $8, 4, 2, 1$. No term can be inserted into the second series: a subgroup M strictly between two consecutive terms would have order divisible by the order of the smaller term and dividing the order of the larger one (theorem 3.3.1), and different from both, which is impossible when the two orders differ by a factor of 2.

In the second series of example 3.7.5 every factor has order 2, hence is simple (example 3.7.2). A series all of whose factors are simple has a name:

Definition 3.7.6: Composition Series

In a group G, a sequence of groups

$$1 = N_0 \leqslant N_1 \leqslant N_2 \leqslant \cdots \leqslant N_{k-1} \leqslant N_k = G$$

is called a *composition series* if $N_i \trianglelefteq N_{i+1}$ and N_{i+1}/N_i is a simple group, for all $0 \leq i \leq k-1$. If the above sequence is a composition series, the quotient group N_{i+1}/N_i are called the *composition factors* of G.

Every finite group $G \neq 1$ has a composition series, as part (1) of theorem 3.7.9 below shows.

Example 3.7.7: Some Composition Series
Take $\mathbb{Z}_p \times \mathbb{Z}_q$. Then

$$\{e\} \trianglelefteq \mathbb{Z}_p \times \{e\} \trianglelefteq \mathbb{Z}_p \times \mathbb{Z}_q$$
$$\{e\} \trianglelefteq \{e\} \times \mathbb{Z}_q \trianglelefteq \mathbb{Z}_p \times \mathbb{Z}_q$$

The quotient of any two will produce simple groups

Take Q_8 Then

$$\{e\} \trianglelefteq \langle -1\rangle \trianglelefteq \langle i\rangle \trianglelefteq Q_8$$
$$\{e\} \trianglelefteq \langle -1\rangle \trianglelefteq \langle j\rangle \trianglelefteq Q_8$$
$$\{e\} \trianglelefteq \langle -1\rangle \trianglelefteq \langle k\rangle \trianglelefteq Q_8$$

Where I figured out the composition series with the help of the lattice of subgroups to trace possible paths and figuring out whether they are normal or not. The quotient so any of these factors will be isomorphic to $\mathbb{Z}_2$, which is simple, and thus there are composition series.

another example is:

$$1 \trianglelefteq \langle s\rangle \trianglelefteq \langle s, r^2\rangle \trianglelefteq D_4 \qquad \text{and} \qquad 1 \trianglelefteq \langle r^2\rangle \trianglelefteq \langle r\rangle \trianglelefteq G$$

Not every infinite group has a composition series.

Example 3.7.8: The Integers Have No Composition Series

Suppose $0 = N_0 \trianglelefteq N_1 \trianglelefteq \cdots \trianglelefteq N_k = \mathbb{Z}$ were a composition series. Then $k \geq 1$ and $N_1 \cong N_1/N_0$ is a nonzero simple group. By theorem 2.3.11, $N_1 = m\mathbb{Z}$ for some $m \geq 1$. But $2m\mathbb{Z}$ is a subgroup of $m\mathbb{Z}$ different from 0 and from $m\mathbb{Z}$, since $m \notin 2m\mathbb{Z}$, and it is normal because $\mathbb{Z}$ is abelian. So $m\mathbb{Z}$ is not simple, a contradiction.

A subnormal series

$$G = G_0 \trianglerighteq G_1 \trianglerighteq G_2 \trianglerighteq \cdots \trianglerighteq G_n = \{1\}$$

breaks into short exact sequences:

$$1 \longrightarrow G_{n-1} \longrightarrow G_{n-2} \longrightarrow G_{n-2}/G_{n-1} \longrightarrow 1$$
$$1 \longrightarrow G_i \longrightarrow G_{i-1} \longrightarrow G_{i-1}/G_i \longrightarrow 1 \quad 1 \longrightarrow G_1 \longrightarrow G_0 \longrightarrow G_0/G_1 \longrightarrow 1$$

so G is built from the quotients $G_0/G_1, G_1/G_2, \ldots, G_{n-1}/G_n = G_{n-1}$ by successive extensions. For a composition series these quotients are simple groups.

One might ask about if a composition series is supposed to be unique factorization, what about there being 3 composition series for Q_8, and what about the re-arranging we have done of $\mathbb{Z}_p \times Z_q$? The *quotients* of each N_{i+1}/N_i will be unique *up to rearrangement*! That is the content of the next theorem:

Theorem 3.7.9: Jordan-Hölder's Theorem

Let G be a finite group with $G \neq 1$. Then

1. G has a composition series
2. The composition factors in a composition series are unique, namely, if $1 = N_0 \leqslant N_1 \leqslant \cdots \leqslant N_r = G$, and $1 = M_0 \leqslant M_1 \leqslant \cdots \leqslant M_s = G$, are two composition series of G, then $r = s$ and there is some permutation π of $\{1, 2, ..., r\}$ such that

$$M_{\pi(i)}/M_{\pi(i)-1} \cong N_i/N_{i-1} \qquad 1 \leq i \leq r$$

***Proof*:**
In a composition series each factor N_i/N_{i-1} is simple, hence nontrivial (definition 3.7.1), so $N_{i-1} \subsetneq N_i$ and the orders $|N_0| < |N_1| < \cdots < |N_r|$ strictly increase.

Step 1: if $N \trianglelefteq G$ and G/N is simple, the only normal subgroups of G containing N are N and G. Let $L \trianglelefteq G$ with $N \subseteq L$. By corollary 3.5.8(5), $L/N \trianglelefteq G/N$, so L/N is N/N or G/N. In the first case $L/N \subseteq N/N$, so $L \subseteq N$ by corollary 3.5.8(1), and $L = N$. In the second case $G/N \subseteq L/N$, so $G \subseteq L$ in the same way, and $L = G$.

Step 2: if N has the largest order among the normal subgroups of G different from G, then G/N is simple. First, $|G/N| > 1$ because $N \neq G$. Let $\overline{M} \trianglelefteq G/N$. By theorem 3.5.7, $\overline{M} = M/N$ for a subgroup M of G containing N, and $M \trianglelefteq G$ by corollary 3.5.8(5). If $M \neq G$, then $|M| \leq |N|$ by the choice of N, and $N \subseteq M$ gives $M = N$, so $\overline{M} = N/N$ is trivial. Otherwise $\overline{M} = G/N$. So G/N is simple.

Step 3: part (1). We use strong induction on $|G|$. Since $G \neq 1$, the trivial subgroup is a normal subgroup different from G, so among the finitely many normal subgroups different from G there is one, N, of largest order, and G/N is simple by Step 2. If $N = 1$, then $1 = N_0 \leqslant N_1 = G$ is a composition series. If $N \neq 1$, then $|N| < |G|$, and the induction hypothesis gives a composition series $1 = N_0 \leqslant \cdots \leqslant N_k = N$ of N. Since $N \trianglelefteq G$ and G/N is simple, putting $N_{k+1} = G$ gives a composition series of G.

Step 4: lists of factors. Call two finite lists of groups $(X_1, \ldots, X_r)$ and $(Y_1, \ldots, Y_s)$ *equivalent* if $r = s$ and there is a permutation π of $\{1, \ldots, r\}$ with $Y_{\pi(i)} \cong X_i$ for $1 \leq i \leq r$. This relation is reflexive, symmetric and transitive: use the identity permutation, the inverse permutation with the inverse isomorphisms, and the composite of two permutations with the composite isomorphisms. Reordering a list, or replacing an entry by an isomorphic group, gives an equivalent list. If X is equivalent to Y and X' to Y', then the list X followed by X' is equivalent to the list Y followed by Y', by using the two permutations side by side. The *list of factors* of a composition series $1 = N_0 \leqslant \cdots \leqslant N_r = G$ is $(N_1/N_0, \ldots, N_r/N_{r-1})$, and part (2) says exactly that any two composition series of G have equivalent lists of factors.

Step 5: part (2). We use strong induction on $|G|$. Let $1 = N_0 \leqslant \cdots \leqslant N_r = G$ and $1 = M_0 \leqslant \cdots \leqslant M_s = G$ be composition series. As $G \neq 1$, $r \geq 1$ and $s \geq 1$. Put $N = N_{r-1}$ and $M = M_{s-1}$. Then $N \trianglelefteq G$ and $M \trianglelefteq G$, the groups G/N and G/M are simple, and in particular $N \neq G$ and $M \neq G$.

Suppose first that $N = M$. If $N = 1$, then $N_{r-1} = N_0$ and $M_{s-1} = M_0$, so $r = s = 1$ because the orders increase strictly, and both lists of factors are $(G/1)$. If $N \neq 1$, then $1 = N_0 \leqslant \cdots \leqslant N_{r-1} = N$ and $1 = M_0 \leqslant \cdots \leqslant M_{s-1} = N$ are composition series of N, and $|N| < |G|$. By the induction hypothesis, $r - 1 = s - 1$ and their lists of factors are equivalent. Appending the common factor G/N to both shows that the two lists of factors of G are equivalent.

Suppose now that $N \neq M$. Since N and M are normal, $N_G(N) = N_G(M) = G$ (proposition 3.1.11), so theorem 3.5.4 applies with $A = M$ and $B = N$, and with $A = N$ and

$B = M$. For $m \in M$ and $n \in N$, $mn = (mnm^{-1})m \in NM$ and $nm = m(m^{-1}nm) \in MN$, so $MN = NM$, a subgroup of G by theorem 3.5.4. It is normal in G by proposition 3.1.11, since $g(mn)g^{-1} = (gmg^{-1})(gng^{-1}) \in MN$ for $g \in G$. By Step 1 applied to N, either $MN = N$ or $MN = G$. If $MN = N$, then $M \subseteq N$, and Step 1 applied to M gives $N = M$ or $N = G$, both of which are excluded. Hence $MN = G$. Put $D = M \cap N$. By theorem 3.5.4, $D \trianglelefteq M$, $D \trianglelefteq N$ and

$$G/N = MN/N \cong M/D \qquad G/M = NM/M \cong N/D$$

A group isomorphic to a simple group is simple, since an isomorphism carries the normal subgroups of one group bijectively onto those of the other. So M/D and N/D are simple, and in particular $M \neq 1$ and $N \neq 1$.

If $D \neq 1$, part (1) gives a composition series $1 = D_0 \leqslant \cdots \leqslant D_t = D$ of D, and if $D = 1$ put $t = 0$ and $D_0 = D$. Let $\Delta = (D_1/D_0, \ldots, D_t/D_{t-1})$, the empty list when $t = 0$. Then $1 = D_0 \leqslant \cdots \leqslant D_t \leqslant N$ is a composition series of N whose list of factors is Δ followed by N/D, and $1 = D_0 \leqslant \cdots \leqslant D_t \leqslant M$ is a composition series of M whose list of factors is Δ followed by M/D. Since $|N| < |G|$, the induction hypothesis for N gives $r - 1 = t + 1$ and shows that $(N_1/N_0, \ldots, N_{r-1}/N_{r-2})$ is equivalent to Δ followed by N/D. In the same way $s - 1 = t + 1$, and $(M_1/M_0, \ldots, M_{s-1}/M_{s-2})$ is equivalent to Δ followed by M/D. Hence $r = s$. Appending the last factors and using $N/D \cong G/M$ and $M/D \cong G/N$, the list of factors of the first series of G is equivalent to Δ followed by $G/M, G/N$, and that of the second series is equivalent to Δ followed by $G/N, G/M$. These two lists differ only in the order of their last two entries, so the two lists of factors of G are equivalent, as we sought to show.

3.7.10 Historical Note: The Hölder Program By theorem 3.7.9, a finite group $G \neq 1$ determines a list of finite simple groups, its composition factors, and G is assembled from them by successive extensions (definition 3.6.3). This splits the classification of finite groups into two problems, known together as the *Hölder program*:

1. classify all finite simple groups
2. determine all ways of assembling finite groups from given composition factors, that is, given groups K and Q, find all groups G with a normal subgroup isomorphic to K whose quotient is isomorphic to Q (the extension problem of section 3.6)

The first problem was solved by the classification of finite simple groups, stated below as theorem 3.7.11, whose proof was completed by the early 2000s. The second problem remains open in general.

Theorem 3.7.11: Types of Finite Groups

There is a list consisting of 18 (infinite) families of simple groups and 26 (or 27 [a]) simple groups not belonging to these families (the *sporadic* simple groups) such that every finite simple group is isomorphic to one of the groups in this list

[a] some include Tits group

The proof of theorem 3.7.11 is spread over hundreds of journal articles totalling tens of thousands

of pages, and it relies on character theory, modular representation theory and the local analysis of finite groups, which this book does not develop. An outline of the whole proof, with the groups of characteristic 2 type treated in detail, is the monograph of Aschbacher, Lyons, Smith and Solomon, *The Classification of Finite Simple Groups: Groups of Characteristic 2 Type* [4].

***Proof Key ideas*:**
By the theorem of Feit and Thompson [30], a nonabelian finite simple group has even order, so it has an element of order 2 by theorem 4.8.7. Brauer and Fowler [5] showed that the order of a finite simple group is bounded in terms of the order of the centralizer of any of its elements of order 2, which suggests identifying simple groups through these centralizers. The bulk of the proof analyses the normalizers of nontrivial p-subgroups, the *local subgroups*. The groups whose elementary abelian 2-subgroups are all small, mostly groups of Lie type of small rank over fields of odd characteristic, are treated separately. The others divide into two kinds: those of *characteristic* 2 *type*, whose local subgroups for the prime 2 resemble those of groups of Lie type over fields of characteristic 2, and those of *component type*, in which the centralizer of some element of order 2 involves a smaller nonabelian simple group, to which induction applies. In each case the local data are shown to match those of a group on the list, and the sporadic groups are the configurations that match no infinite family. The last case, that of the quasithin groups, was settled by Aschbacher and Smith [2, 3] in 2004, completing the classification.

Example 3.7.12
Here are a couple such family of groups:

1. $\{Z_p | p \text{ is prime } \}$
2. the groups $SL_n(\mathbb{F})/Z(SL_n(\mathbb{F}))$ with $n \geq 2$ and $\mathbb{F}$ a finite field, with two exceptions. The groups $SL_2(\mathbb{F}_2)/Z(SL_2(\mathbb{F}_2))$ and $SL_2(\mathbb{F}_3)/Z(SL_2(\mathbb{F}_3))$ have orders 6 and 12, are isomorphic to S_3 and A_4, and are not simple. Every other group in the family is simple, by a theorem of Jordan and Dickson [52, 3.2.9 and p. 78].
3. A_n, $n \geq 5$.

The groups $SL_n(\mathbb{F})/Z(SL_n(\mathbb{F}))$ are the first examples of the *groups of Lie type*, and the remaining families are further groups of Lie type [52, Section 3.2].

Exercise 3.7.1

1. Let $H \trianglelefteq G$ with $H \neq G$. Call H a *maximal normal subgroup* if the only normal subgroups K of G with $H \subseteq K$ are H and G. Show that H is a maximal normal subgroup if and only if G/H is a simple group.

4

Group Actions

In Section 1.5, we saw that groups arise naturally as *symmetries*. The dihedral group captures the symmetries of a polygon, the symmetric group captures rearrangement of a finite set, the general linear group captures invertible linear transformation of a vector space, and so forth. Historically, groups *were* symmetry groups before they were axiomatized. The abstract axioms we introduced in Section 1.1 are a *distillation* of what all these concrete symmetry groups have in common.

In this chapter, we reverse the process. Having built the abstract theory we now reconnect it to its roots by developing the theory of *group actions*: the study of how a group can act on a set (or on itself). We will show:

1. The Orbit-Stabilizer Theorem (section 4.3). If a group G acts on a set X, then X decomposes into orbits, and the size of each orbit is controlled by a subgroup of G (the stabilizer). This is, in spirit, a generalization of Lagrange's Theorem (theorem 3.3.1). We shall see that if the action is *transitive*, then X can be represented as cosets of G, leading the way to motivate why it is natural to study actions of a group on itself

2. Cayley's Theorem (section 4.5). Every group is isomorphic to a subgroup of a symmetric group. In other words, every abstract group *is* a group of permutations.

3. The Class Equation (section 4.6). By letting a group act on itself by conjugation, we obtain an arithmetic identity that splits $|G|$ into contributions from the center $Z(G)$ (definition 2.2.4) and the non-central conjugacy classes. This will immediately imply that every p-group has a nontrivial center.

4. The Sylow Theorems (section 4.8). For any prime p dividing $|G|$, Sylow's three theorems guarantee the *existence* of subgroups of maximal p-power order, prove they are all *conjugate* (hence isomorphic), and tightly constrain their *number*. These theorems are among the most powerful tools in finite group theory.

Along the way, we will see several important *types* of actions (faithful, transitive, free, and regular) and see how each captures a different facet of the relationship between a group and the object it acts on.

A recurring theme will be that the most important applications of group actions come not from letting a group act on some external set, but from letting a group act *on itself* or on its own substructure (elements, subsets, cosets). When a group acts on itself by left multiplication, we get Cayley's Theorem. When it acts on its cosets, we get structural information about normal subgroups. When it acts on itself by conjugation, we get the Class Equation. When it acts on the set of its own Sylow subgroups, we get the Sylow counting theorem. Each of these is an instance of the same machine. What changes is the *choice of action*, and each choice reveals different information.

4.0.1 Applied Aside: Klein's Erlangen Programme In 1872, Felix Klein presented a new vision of geometry at the University of Erlangen. Rather than defining geometries axiomatically (as Euclid had done), Klein proposed that a *geometry* is a set X together with a group G acting on it, and that the "geometric properties" of X are precisely those invariant under the action of G.

Under this lens, different choices of group yield different geometries:

Group G	**Set** X	**Geometry**
Euclidean isometries	$\mathbb{R}^2$	Euclidean geometry
Affine transformations	$\mathbb{R}^2$	Affine geometry
Projective transformations	$\mathbb{RP}^2$	Projective geometry

Euclidean geometry studies properties like distance and angle (invariant under rigid motions); affine geometry studies parallelism (invariant under the larger group of affine maps); projective geometry studies incidence (invariant under the still larger projective group). More symmetry means fewer invariants, hence a "coarser" geometry.

Klein's Programme was one of the first explicit formulations of the idea that *group actions define structure*. It remains a guiding principle: in differential geometry, in topology, and throughout modern mathematics, the question "what is the symmetry group, and how does it act?" is often the first one asked.

4.1 Definitions and First Examples

In Section 1.5, we observed informally that a group can "act" on a set by rearranging its elements. Our goal now is to make this precise. We will give two equivalent formulations of the definition (one emphasizing the homomorphism, the other emphasizing axioms on the action map) and prove that they are interchangeable.

The Homomorphism Definition

Let us return to the motivating example from the prelude. The dihedral group D_n (definition 1.2.28) acts on the set of vertices $\{1, 2, \ldots, n\}$ of a regular n-gon. Each element $g \in D_n$ determines a bijection of the vertex set: a rotation cyclically permutes the labels, a reflection swaps them. Different elements yield different (or sometimes the same) bijections, and, crucially, performing one symmetry followed by another corresponds to composing the associated bijections.

In other words, there is a map ρ that assigns to each group element $g \in D_n$ a permutation $\rho(g) \in \mathrm{Sym}(\{1, \ldots, n\})$, and this map *respects the group structure*: $\rho(gh) = \rho(g) \circ \rho(h)$. That is, ρ is a homomorphism (definition 1.4.2). This observation is the definition.

Definition 4.1.1: Group Action

Let G be a group and let X be a set. A *group action* of G on X is a group homomorphism

$$\rho \colon G \to \mathrm{Sym}(X),$$

where $\mathrm{Sym}(X)$ is the symmetric group on X (definition 1.2.12), i.e. the group of all bijections $X \to X$ under composition.
We write $G \curvearrowright X$ (read "G *acts on* X") to indicate that such a homomorphism has been specified. The set X, together with the action ρ, is called a ***G-set***.

Let us unpack what this definition gives us concretely.

- For each $g \in G$, the image $\rho(g)$ is an element of $\mathrm{Sym}(X)$, that is, a *bijection* from X to X. We can *evaluate* this bijection at any point $x \in X$ to obtain an element $\rho(g)(x) \in X$. One should read "$\rho(g)(x)$" as: *"the result of applying the permutation $\rho(g)$ to the element x."*
- Since ρ is a homomorphism, it respects composition: $\rho(gh) = \rho(g) \circ \rho(h)$ for all $g, h \in G$.
- Since ρ is a homomorphism, it sends the identity to the identity (proposition 1.4.4): $\rho(e) = \mathrm{id}_X$, the identity permutation.

It is convenient to introduce a shorthand that suppresses the homomorphism ρ:

$$g \cdot x \;:=\; \rho(g)(x) \qquad \text{for all } g \in G,\ x \in X.$$

In this notation, the two properties above translate into:

1. Compatibility: $(gh) \cdot x \;=\; g \cdot (h \cdot x)$ for all $g, h \in G$ and $x \in X$.
2. Identity: $e \cdot x \;=\; x$ for all $x \in X$.

These are not new axioms: they are automatic consequences of ρ being a homomorphism. The natural question is: are they *sufficient*? That is, if we have a rule "$g \cdot x$" satisfying just these two properties, does it necessarily come from a homomorphism to $\mathrm{Sym}(X)$? The answer is yes.

The Axiomatic Characterization

Many textbooks take the following as the *definition* of a group action, and then derive the homomorphism as a consequence. Since we have gone the other direction, we state it as a theorem establishing the equivalence.

Theorem 4.1.2: Equivalence of Definitions

Let G be a group and X a nonempty set. There is a bijective correspondence between:

(a) group homomorphisms $\rho\colon G \to \mathrm{Sym}(X)$, and

(b) functions $\cdot\colon G \times X \to X$, written $(g,x) \mapsto g \cdot x$, satisfying

(A1) $(gh)\cdot x = g\cdot(h\cdot x)$ for all $g,h \in G$ and $x \in X$, and

(A2) $e \cdot x = x$ for all $x \in X$.

The correspondence is given by $g \cdot x = \rho(g)(x)$.

***Proof*:**

(a)⇒(b). Given a homomorphism $\rho\colon G \to \mathrm{Sym}(X)$, define $g \cdot x := \rho(g)(x)$. We check the two axioms.

For (A1): since ρ is a homomorphism, $\rho(gh) = \rho(g) \circ \rho(h)$, so

$$(gh)\cdot x \;=\; \rho(gh)(x) \;=\; \big(\rho(g)\circ\rho(h)\big)(x) \;=\; \rho(g)\big(\rho(h)(x)\big) \;=\; g\cdot(h\cdot x).$$

For (A2): since ρ sends identities to identities (proposition 1.4.4), $\rho(e) = \mathrm{id}_X$, so

$$e\cdot x \;=\; \rho(e)(x) \;=\; \mathrm{id}_X(x) \;=\; x.$$

(b)⇒(a). Given a function $\cdot\colon G \times X \to X$ satisfying (A1) and (A2), we must construct a homomorphism $\rho\colon G \to \mathrm{Sym}(X)$. For each $g \in G$, define the function

$$\sigma_g\colon X \to X, \qquad \sigma_g(x) = g\cdot x.$$

We must show two things: that each σ_g is a bijection (hence an element of $\mathrm{Sym}(X)$), and that the map $g \mapsto \sigma_g$ is a group homomorphism.

Step 1: Each σ_g is a bijection. We claim that $\sigma_{g^{-1}}$ is a two-sided inverse for σ_g. For any $x \in X$:

$$(\sigma_g \circ \sigma_{g^{-1}})(x) = \sigma_g(g^{-1}\cdot x) = g\cdot(g^{-1}\cdot x) \overset{\text{(A1)}}{=} (gg^{-1})\cdot x = e\cdot x \overset{\text{(A2)}}{=} x,$$

so $\sigma_g \circ \sigma_{g^{-1}} = \mathrm{id}_X$. An identical computation gives $\sigma_{g^{-1}} \circ \sigma_g = \mathrm{id}_X$. Thus σ_g is a bijection, and $\sigma_g \in \mathrm{Sym}(X)$.

Step 2: $g \mapsto \sigma_g$ is a homomorphism. Define $\rho\colon G \to \mathrm{Sym}(X)$ by $\rho(g) = \sigma_g$. For any $g,h \in G$ and $x \in X$:

$$\rho(gh)(x) = \sigma_{gh}(x) = (gh)\cdot x \overset{\text{(A1)}}{=} g\cdot(h\cdot x) = \sigma_g(\sigma_h(x)) = (\sigma_g\circ\sigma_h)(x) = \big(\rho(g)\circ\rho(h)\big)(x).$$

Since this holds for every $x \in X$, we have $\rho(gh) = \rho(g)\circ\rho(h)$, so ρ is a homomorphism.

Finally, these two constructions are inverse to each other: starting from ρ, forming $g\cdot x = \rho(g)(x)$, and then recovering the homomorphism $g \mapsto \sigma_g$ gives back $\sigma_g = \rho(g)$; conversely, starting from $\cdot$, forming $\rho(g) = \sigma_g$, and then recovering the action gives $\rho(g)(x) = \sigma_g(x) = g\cdot x$. Thus the correspondence is bijective.

4.1.3 Remark: Currying The equivalence above is an instance of a general phenomenon in mathematics (and computer science) called *currying*. A function of two variables $\cdot\colon G \times X \to X$ can be equivalently viewed as a function of one variable $\rho\colon G \to X^X$ that returns a *function* $X \to X$. The group action axioms are exactly the conditions ensuring that ρ lands in $\mathrm{Sym}(X) \subseteq X^X$ and is a homomorphism. This point of view (recasting a "two-variable" structure as a "one-variable" homomorphism) will reappear when we study modules and representations.

Thanks to theorem 4.1.2, we may define group actions using whichever formulation is most convenient. In practice:

- To define an action on a concrete set, it is usually easiest to write down a rule $g \cdot x =$ (some formula) and verify axioms (A1) and (A2).
- To reason abstractly about actions (e.g. to apply the isomorphism theorems), it is usually more powerful to work with the homomorphism $\rho\colon G \to \mathrm{Sym}(X)$.

We give a name to this homomorphism when it arises from an action:

Definition 4.1.4: Permutation Representation

If $G \curvearrowright X$ is a group action, the associated homomorphism $\rho\colon G \to \mathrm{Sym}(X)$ is called the *permutation representation* of the action. We say the action *affords* or *induces* the permutation representation ρ.

Since ρ is a homomorphism, all the tools we developed for homomorphisms apply: its kernel $\ker(\rho)$ is a normal subgroup of G (proposition 2.2.12), its image $\rho(G)$ is a subgroup of $\mathrm{Sym}(X)$ (proposition 1.4.8), and the First Isomorphism Theorem (theorem 3.1.15) gives $G/\ker(\rho) \cong \rho(G) \leq \mathrm{Sym}(X)$. We will exploit these facts heavily starting in Section 4.2.

We close this subsection by recording why we chose the homomorphism formulation as our leading definition. This is a matter of taste, but the reasoning is forward-looking:

- A group action is a homomorphism $G \to \mathrm{Sym}(X)$.
- A linear representation ([17, Representation]) will be a homomorphism $G \to GL(V)$.
- A module (Part III) will be defined via a ring homomorphism $R \to \mathrm{End}(M)$.

Example 4.1.5: Fundamental Group Actions

1. *The tautological action of S_n.* Let $G = S_n$ and $X = \{1, 2, \ldots, n\}$. Define
$$\sigma \cdot i \;=\; \sigma(i).$$
The permutation representation is the identity map $\rho = \mathrm{id}\colon S_n \to \mathrm{Sym}(\{1, \ldots, n\}) = S_n$, which is trivially a homomorphism. This is the most elementary group action: each permutation simply *does what it says*.
2. *Dihedral symmetries of a regular n-gon.* Let $G = D_n$ (definition 1.2.28) and let $X = \{1, 2, \ldots, n\}$ represent the vertices of a regular n-gon, labeled clockwise. The rotation r

acts by $r \cdot i = i + 1 \pmod{n}$, and the reflection s acts by $s \cdot i = n + 1 - i \pmod{n}$ (with an appropriate labeling convention). The permutation representation $\rho\colon D_n \to S_n$ is an injective homomorphism, embedding D_n as a subgroup of S_n.

3. *Matrix groups acting on vectors.* Let $G = GL_n(\mathbb{R})$ (definition 1.2.33) and $X = \mathbb{R}^n$. Define

$$A \cdot \vec{x} \;=\; A\vec{x} \qquad \text{(matrix-vector multiplication).}$$

Axiom (A1) is the associativity of matrix multiplication: $(AB)\vec{x} = A(B\vec{x})$. Axiom (A2) is the identity matrix property: $I_n\vec{x} = \vec{x}$. Thus this defines a group action.

This action is the foundation of linear algebra: every invertible linear transformation $\mathbb{R}^n \to \mathbb{R}^n$ is encoded by an element of $GL_n(\mathbb{R})$, and composition of transformations corresponds to matrix multiplication.

4. *Two actions of $\mathbb{R}^\times$ on $\mathbb{R}^2$.* Let $G = (\mathbb{R}^\times, \cdot)$, the multiplicative group of nonzero reals, and $X = \mathbb{R}^2$. Here are two *different* actions of the same group on the same set:

$$\textbf{Scaling:}\quad r \cdot \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} rx \\ ry \end{pmatrix}, \qquad\qquad \textbf{Hyperbolic:}\quad r \cdot \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} rx \\ r^{-1}y \end{pmatrix}.$$

Both satisfy (A1) and (A2). For the hyperbolic action, $1 \cdot (x, y) = (x, y)$ and $(rs) \cdot (x, y) = (rsx, (rs)^{-1}y) = (r(sx), r^{-1}(s^{-1}y)) = r \cdot (s \cdot (x, y))$, where $(rs)^{-1} = r^{-1}s^{-1}$ uses that $\mathbb{R}^\times$ is commutative, and the scaling action is checked in the same way. The scaling action dilates every vector uniformly, while the hyperbolic action stretches the x-axis and compresses the y-axis (or vice versa), preserving the hyperbolas $\{xy = c\}$.

This example makes an important point: *a group action is not determined by the group and the set alone.* One must specify the rule.

5. *Left multiplication.* Let G be any group, and let $X = G$ (the underlying set of the group itself). Define

$$g \cdot a \;=\; ga \qquad \text{(group multiplication).}$$

Axiom (A1) is exactly the associativity of the group operation: $(gh) \cdot a = (gh)a = g(ha) = g \cdot (h \cdot a)$. Axiom (A2) is the identity property: $e \cdot a = ea = a$.

Note that for a given g, the map $a \mapsto ga$ is *not* a group homomorphism (unless $g = e$). It is only a bijection $G \to G$. This action is our first example of a group acting on *itself*, and it will lead to Cayley's Theorem in Section 4.5.

6. *Conjugation.* Let G be any group, and again let $X = G$. Define

$$g \cdot a \;=\; gag^{-1}.$$

Let us verify the axioms. For (A1):

$$(gh) \cdot a = (gh)\, a\, (gh)^{-1} = g\, h\, a\, h^{-1} g^{-1} = g \cdot (hah^{-1}) = g \cdot (h \cdot a).$$

For (A2): $e \cdot a = eae^{-1} = a$.

Unlike left multiplication, conjugation "preserves" much of the group's internal structure: it fixes the identity, sends subgroups to (isomorphic) subgroups, and preserves the order of elements. This action will lead to the Class Equation in Section 4.6.

7. *Rotations of a tetrahedron.* Let $G = A_4$ (definition 1.2.22), the alternating group on four symbols, and let $X = \{1, 2, 3, 4\}$ represent the vertices of a regular tetrahedron. The 12 elements of A_4 correspond to the 12 rotational symmetries of the tetrahedron:

   - e: the identity (no rotation).
   - 8 rotations by 120° and 240° about axes through a vertex and the center of the opposite face, e.g. $(1\,2\,3)$ and $(1\,3\,2)$ fix vertex 4.
   - 3 rotations by 180° about axes connecting the midpoints of opposite edges, e.g. $(1\,2)(3\,4)$ swaps the two pairs.

   The action is $\sigma \cdot i = \sigma(i)$, i.e. the restriction of the tautological action of S_4 to A_4.

8. *Translation of polynomials.* Let $G = (\mathbb{R}, +)$ and let $X = \mathbb{R}[x]$, the set of all polynomials with real coefficients. Define

$$t \cdot f \;=\; f_t, \qquad \text{where } f_t(x) = f(x + t).$$

   For (A1): $\big((s + t) \cdot f\big)(x) = f(x + s + t)$ and $\big(s \cdot (t \cdot f)\big)(x) = (t \cdot f)(x + s) = f(x + s + t)$. For (A2): $(0 \cdot f)(x) = f(x + 0) = f(x)$.

   This action connects algebra to analysis: Taylor's theorem describes how the coefficients of f_t depend on the coefficients of f and the shift t.

4.1.6 Applied Aside: Symmetry in Chemistry and Crystallography The symmetry group of a molecule (the collection of all rotations and reflections that map the molecule to itself) is called its *point group*. These point groups are finite subgroups of the orthogonal group $O(3)$ (definition 1.2.35). Remarkably, there are only 32 distinct crystallographic point groups compatible with the periodic structure of a crystal lattice.
Group actions enter directly: a point group acts on the set of atoms in a molecule, and the *orbits* of this action (a concept we formalize in Section 4.2) determine which atoms are "chemically equivalent." In spectroscopy, *selection rules*, which govern which transitions between energy levels are allowed, are derived from the representation theory of these point groups. In crystallography, the 32 point groups combine with translational symmetries to yield the *230 space groups*, which classify all possible crystal structures in three dimensions.

A Word on Left and Right Actions

What we have defined above is, more precisely, a *left* group action: the group element appears on the *left* in the expression "$g \cdot x$." There is an equally valid notion of a right action.

Definition 4.1.7: Right Group Action

A *right action* of a group G on a set X is a function $X \times G \to X$, written $(x, g) \mapsto x \cdot g$, satisfying:

(R1) $(x \cdot g) \cdot h = x \cdot (gh)$ for all $g, h \in G$ and $x \in X$, and

(R2) $x \cdot e = x$ for all $x \in X$.

Equivalently, a right action is a group homomorphism $\rho\colon G^{\mathrm{op}} \to \mathrm{Sym}(X)$, where G^{op} is the *opposite group*[a] of G.

[a]The opposite group G^{op} has the same underlying set as G but with the multiplication reversed: $a \cdot_{\mathrm{op}} b = ba$.

Observe the key difference in (R1): the parentheses associate to the *left* $((x \cdot g) \cdot h)$, and the group product is gh in its original order. Compare with the left-action axiom (A1), where the parentheses associate to the *right* $(g \cdot (h \cdot x))$.

Left and right actions are completely interchangeable in the following sense:

Proposition 4.1.8: Converting Between Left and Right Actions

If $\cdot_L\colon G \times X \to X$ is a left action, then

$$x \cdot_R g \;:=\; g^{-1} \cdot_L x$$

defines a right action. Conversely, if $\cdot_R$ is a right action, then $g \cdot_L x := x \cdot_R g^{-1}$ defines a left action.

***Proof*:**
We verify (R1). For any $g, h \in G$ and $x \in X$:

$$\begin{aligned}(x \cdot_R g) \cdot_R h = h^{-1} \cdot_L (x \cdot_R g) = h^{-1} \cdot_L (g^{-1} \cdot_L x) \\ = (h^{-1}g^{-1}) \cdot_L x = (gh)^{-1} \cdot_L x = x \cdot_R (gh).\end{aligned}$$

For (R2): $x \cdot_R e = e^{-1} \cdot_L x = e \cdot_L x = x$. The converse is analogous.

Because of this correspondence, there is no essential loss of generality in working with one type exclusively.

4.1.9 Convention For the remainder of this book, "group action" means *left* group action unless explicitly stated otherwise.

4.2 Orbits, Stabilizers, and the Kernel

Given a group action $G \curvearrowright X$, two fundamental questions arise:

1. From the perspective of X: Given a point $x \in X$, where can x be "moved to" by the elements of G?
2. From the perspective of G: Given a point $x \in X$, which elements of G "leave x in place"?

The answers to these questions, *orbits* and *stabilizers*, are two of the most important concepts in the theory of group actions. As we will see, they are intimately related, and much of the power of group actions comes from playing them off against each other.

Orbits

Consider the special orthogonal group $SO(3)$ (definition 1.2.36) acting on the unit sphere $S^2 = \{(x, y, z) \in \mathbb{R}^3 : x^2 + y^2 + z^2 = 1\}$ by matrix multiplication. If we start at the north pole $N = (0, 0, 1)$ and apply every rotation in $SO(3)$, we reach *every* point on the sphere: $SO(3) \cdot N = S^2$. Now restrict to the subgroup $SO(2) \leq SO(3)$ of rotations about the z-axis. From the north pole, $SO(2)$ goes nowhere: $SO(2) \cdot N = \{N\}$. But from a point P at latitude θ, $SO(2)$ traces out the entire latitude circle through P. The latitude circles, together with the two poles, form a family of non-overlapping subsets that cover the sphere.

This is the geometric prototype of a general phenomenon: under any group action, the set X decomposes into non-overlapping "layers" of points that can be reached from one another.

Definition 4.2.1: Orbit

Let $G \curvearrowright X$. For any $x \in X$, the *orbit* of x under the action of G is the set

$$G \cdot x \;=\; \{g \cdot x \,|\, g \in G\}.$$

In words, $G \cdot x$ is the set of all points in X that x can be "sent to" by some element of G.

The notation $\mathrm{Orb}_G(x)$ or $\mathcal{O}_x$ is also shows up in the literature. We will use $G \cdot x$ throughout.

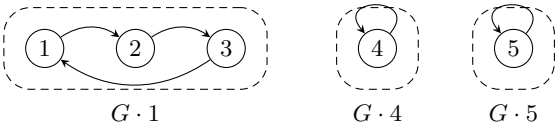
Example 4.2.2: Orbits Under a Cyclic Group

Let $G = \langle(1\ 2\ 3)\rangle = \{e, (1\ 2\ 3), (1\ 3\ 2)\} \cong \mathbb{Z}/3\mathbb{Z}$, viewed as a subgroup of S_5, and let $X = \{1, 2, 3, 4, 5\}$ with the tautological action $\sigma \cdot i = \sigma(i)$. Then:

$$\begin{aligned} G \cdot 1 &= \{e(1),\ (1\ 2\ 3)(1),\ (1\ 3\ 2)(1)\} = \{1,\ 2,\ 3\}, \\ G \cdot 4 &= \{e(4),\ (1\ 2\ 3)(4),\ (1\ 3\ 2)(4)\} = \{4\}, \\ G \cdot 5 &= \{5\}. \end{aligned}$$

Note that $G \cdot 1 = G \cdot 2 = G \cdot 3$ (the three points can all reach each other), and the set X decomposes as the disjoint union

$$X \;=\; \{1, 2, 3\} \sqcup \{4\} \sqcup \{5\}.$$



The decomposition into orbits holds for any group action.

Proposition 4.2.3: Orbits Partition the Set

Let $G \curvearrowright X$. Define a relation $\sim$ on X by

$$x \sim y \quad \iff \quad y = g \cdot x \text{ for some } g \in G.$$

Then $\sim$ is an equivalence relation on X, and the equivalence class of x is precisely the orbit $G \cdot x$. Consequently, the orbits of the action partition X:

$$X = \bigsqcup_{i \in I} G \cdot x_i,$$

where $\{x_i\}_{i \in I}$ is any set of orbit representatives (one element chosen from each orbit).

***Proof*:**
We verify the three properties of an equivalence relation.

- *Reflexive:* $x = e \cdot x$, so $x \sim x$. (Uses axiom (A2).)
- *Symmetric:* If $x \sim y$, then $y = g \cdot x$ for some $g \in G$. Applying g^{-1}:

  $$g^{-1} \cdot y = g^{-1} \cdot (g \cdot x) \overset{(A1)}{=} (g^{-1}g) \cdot x \overset{(A2)}{=} e \cdot x = x,$$

  so $x = g^{-1} \cdot y$, which gives $y \sim x$. (Uses the existence of inverses in G.)
- *Transitive:* If $x \sim y$ and $y \sim z$, then $y = g \cdot x$ and $z = h \cdot y$ for some $g, h \in G$. Then

  $$z = h \cdot y = h \cdot (g \cdot x) \overset{(A1)}{=} (hg) \cdot x,$$

  so $x \sim z$. (Uses closure of the group operation.)

The equivalence class of x is $[x] = \{y \in X : y \sim x\} = \{g \cdot x : g \in G\} = G \cdot x$. Since equivalence classes partition a set, the orbits partition X.

The margin annotations in the proof are worth reflecting on: *reflexivity* uses the identity axiom, *symmetry* uses the existence of inverses, and *transitivity* uses closure. In other words, the group axioms are precisely what is needed to make the orbit relation an equivalence relation. A monoid action (where inverses may not exist) would give reflexivity and transitivity but *not* symmetry, and the orbits would generally fail to partition the set.

We introduce notation for the *set of orbits*:

Definition 4.2.4: Set of Orbits

If $G \curvearrowright X$, the set of all orbits is denoted

$$X/G \;=\; \{G \cdot x : x \in X\}.$$

This is called the *orbit space* (or *set of orbits*) of the action.

Counting orbits will become a central theme once we prove Burnside's Lemma (section 4.3).

Stabilizers and Fixed Points

We now turn to the second question: which group elements "do nothing" to a given point?

Definition 4.2.5: Stabilizer

Let $G \curvearrowright X$ and let $x \in X$. The *stabilizer* (or *isotropy subgroup*) of x in G is

$$G_x \;=\; \{g \in G \,|\, g \cdot x = x\}.$$

The notation $\mathrm{Stab}_G(x)$ is also standard. We use the subscript notation G_x for brevity.

Proposition 4.2.6: The Stabilizer is a Subgroup

For any $x \in X$, the stabilizer G_x is a subgroup of G.

***Proof*:**
We apply the subgroup criterion (proposition 2.1.3).

- *Non-empty:* $e \cdot x = x$ by axiom (A2), so $e \in G_x$.
- *Closed under products and inverses:* Let $g, h \in G_x$. Then

$$(gh^{-1}) \cdot x \overset{(A1)}{=} g \cdot (h^{-1} \cdot x).$$

  Since $h \in G_x$, we have $h \cdot x = x$, and so $h^{-1} \cdot x = h^{-1} \cdot (h \cdot x) = (h^{-1}h) \cdot x = e \cdot x = x$. Therefore $(gh^{-1}) \cdot x = g \cdot x = x$, giving $gh^{-1} \in G_x$.

The fact that G_x is a *subgroup* (not just a subset) is crucial: it means we can bring the full weight of our subgroup theory to bear. In particular, G_x has an index in G given by Lagrange's Theorem (theorem 3.3.1), and this index is precisely the size of the orbit $G \cdot x$ (theorem 4.3.1).

Example 4.2.7: Computing Stabilizers

Consider $G = D_4$ acting on the vertices $X = \{1, 2, 3, 4\}$ of a square, with the labeling from Section 1.5. We compute the stabilizer of vertex 1. An element $g \in D_4$ stabilizes 1 if and only if g maps vertex 1 to itself. Among the 8 elements of D_4:

- The identity e fixes 1.

- The rotations r, r^2, r^3 all move vertex 1.
- Among the four reflections, exactly one fixes vertex 1: the reflection sr^3 across the diagonal through vertex 1 (depending on the labeling convention).

Thus $G_1 = \{e, sr^3\} \cong \mathbb{Z}/2\mathbb{Z}$, a subgroup of order 2. Note that $|G_1| = 2$ and $|G \cdot 1| = |\{1, 2, 3, 4\}| = 4$, so $|G_1| \cdot |G \cdot 1| = 2 \cdot 4 = 8 = |D_4|$. This is the orbit-stabilizer theorem, proved below.

There is a dual notion to the stabilizer that will be useful later:

Definition 4.2.8: Fixed-Point Sets

Let $G \curvearrowright X$.

1. For a single element $g \in G$, the *fixed-point set* of g is
$$X^g = \{x \in X \mid g \cdot x = x\}.$$
2. The *fixed-point set* of the entire group G is
$$X^G = \{x \in X \mid g \cdot x = x \text{ for all } g \in G\} = \bigcap_{g \in G} X^g.$$

Observe the duality: the stabilizer G_x is a subset of G (the group elements that fix a *given point*), while the fixed-point set X^g is a subset of X (the points fixed by a *given group element*). Both notions have the same underlying condition, "$g \cdot x = x$," but viewed from opposite sides:

	Lives in...	**Defined by fixing...**
Stabilizer G_x	G	the point $x \in X$
Fixed-point set X^g	X	the element $g \in G$

4.2.9 A Mnemonic for Orbits and Stabilizers Here is a useful way to keep the two concepts straight. Both G_x (stabilizer) and $G \cdot x$ (orbit) have an element of X in their notation:

- If G_x is *large* (many elements of G fix x), then $G \cdot x$ is *small* (the point x is not moved very far).
- If G_x is *small* (few elements of G fix x), then $G \cdot x$ is *large* (the point x is moved to many different places).

In the extreme: if $G_x = G$ (everything fixes x), then $G \cdot x = \{x\}$ (a singleton orbit). Conversely, if $G_x = \{e\}$ (only the identity fixes x), then the orbit is as large as possible. This inverse relationship will be made precise by the Orbit-Stabilizer Theorem in Section 4.3.

We close this subsection with two remarks connecting stabilizers to concepts from Chapter 2 that will be developed fully later in this chapter.

1. Left multiplication (item 5 of example 4.1.5). When G acts on itself by $g \cdot a = ga$, the stabilizer

of any $a \in G$ is

$$G_a = \{g \in G : ga = a\} = \{e\}.$$

Every stabilizer is trivial: no element other than the identity can "fix" a point under left multiplication.

2. Conjugation (item 6 of example 4.1.5). When G acts on itself by $g \cdot a = gag^{-1}$, the stabilizer of a is

$$G_a = \{g \in G : gag^{-1} = a\} = \{g \in G : ga = ag\} = C_G(a),$$

the centralizer of a (definition 2.2.7). Thus the orbit-stabilizer framework gives us a direct link between orbits (conjugacy classes) and centralizers. We will exploit this connection in Section 4.6.

The Kernel of an Action

The stabilizer G_x captures which group elements fix a *single* point. We now ask: which group elements fix *every* point?

Definition 4.2.10: Kernel of a Group Action

Let $G \curvearrowright X$ with permutation representation $\rho\colon G \to \mathrm{Sym}(X)$. The *kernel* of the action is

$$\ker(G \curvearrowright X) \;=\; \{g \in G \,|\, g \cdot x = x \text{ for all } x \in X\} \;=\; \bigcap_{x \in X} G_x.$$

An element in the kernel "acts trivially": it is a group element that induces the identity permutation on X. This means $\rho(g) = \mathrm{id}_X$, and so the kernel of the *action* is exactly the kernel of the *homomorphism* ρ:

$$\ker(G \curvearrowright X) \;=\; \ker(\rho).$$

Proposition 4.2.11: The Kernel is a Normal Subgroup

$$\ker(G \curvearrowright X) \trianglelefteq G$$

***Proof*:**
Since $\rho\colon G \to \mathrm{Sym}(X)$ is a group homomorphism, its kernel $\ker(\rho)$ is a normal subgroup of G by proposition 2.2.12.

The kernel measures how much "information about G" is lost by the action. If the kernel is large, many group elements are "wasted" as they all induce the same (trivial) permutation. If the kernel is trivial, then distinct group elements always induce distinct permutations. As the kernel is normal, we can always "eliminate" the wasteful elements by taking $G/\ker(\rho)$. This motivates:

Definition 4.2.12: Faithful Action

A group action $G \curvearrowright X$ is called *faithful* (or *effective*) if its kernel is trivial:

$$\ker(G \curvearrowright X) = \{e\}.$$

Equivalently, the action is faithful if and only if the permutation representation $\rho\colon G \to \mathrm{Sym}(X)$ is *injective*.

In a faithful action, distinct group elements always produce distinct permutations of X: if $g \neq h$, then there exists some $x \in X$ with $g \cdot x \neq h \cdot x$. The group G is thus "faithfully represented" as a group of permutations of X.

Example 4.2.13: A Non-Faithful Action

Let $G = \mathbb{Z}/4\mathbb{Z} = \{\bar{0}, \bar{1}, \bar{2}, \bar{3}\}$ and $X = \{1, 2\}$. Define an action by $\bar{n} \cdot i = \tau^n(i)$, where $\tau = (1\ 2) \in S_2$. Then:

$$\bar{0} \cdot i = i, \quad \bar{1} \cdot i = \tau(i), \quad \bar{2} \cdot i = \tau^2(i) = i, \quad \bar{3} \cdot i = \tau^3(i) = \tau(i).$$

The elements $\bar{0}$ and $\bar{2}$ both act as the identity on X, so

$$\ker(\rho) \;=\; \{\bar{0}, \bar{2}\} \;\cong\; \mathbb{Z}/2\mathbb{Z}.$$

The action is *not* faithful. By the First Isomorphism Theorem (theorem 3.1.15),

$$G/\ker(\rho) \;\cong\; \rho(G) \;\leq\; \mathrm{Sym}(X) \;=\; S_2,$$

so $(\mathbb{Z}/4\mathbb{Z})/(\mathbb{Z}/2\mathbb{Z}) \cong \mathbb{Z}/2\mathbb{Z}$ embeds into S_2, consistent with the fact that the *effective* part of the action is just "swap or do not swap."

The example above illustrates a general principle: even if an action is not faithful, we can always *make* it faithful by passing to the quotient.

Proposition 4.2.14: Quotient by the Kernel Acts Faithfully

Let $G \curvearrowright X$ with kernel $K = \ker(\rho)$. Then the quotient group G/K acts faithfully on X via

$$(gK) \cdot x \;=\; g \cdot x.$$

***Proof*:**

We first check well-definedness: if $gK = hK$, then $h^{-1}g \in K$, so $(h^{-1}g) \cdot x = x$ for all x, whence $g \cdot x = h \cdot x$ for all x. Thus the action of gK does not depend on the choice of representative.

The axioms (A1) and (A2) are inherited directly from the action of G. Finally, if gK is in the kernel of this new action, then $g \cdot x = x$ for all $x \in X$, so $g \in K$, meaning $gK = eK$. The kernel of the G/K-action is trivial, so the action is faithful.

In the language of the permutation representation: ρ factors through the quotient as

$$G \;\twoheadrightarrow\; G/K \;\overset{\bar{\rho}}{\hookrightarrow}\; \mathrm{Sym}(X),$$

where $\bar{\rho}$ is injective. This is simply the First Isomorphism Theorem (theorem 3.1.15) applied to ρ.

We summarize the key objects introduced in this section for quick reference.

Object	**Notation**	**Lives in**	**Definition**
Orbit of x	$G \cdot x$	subsets of X	$\{g \cdot x : g \in G\}$
Stabilizer of x	G_x	subgroups of G	$\{g \in G : g \cdot x = x\}$
Fixed-point set of g	X^g	subsets of X	$\{x \in X : g \cdot x = x\}$
Kernel of the action	$\ker(\rho)$	normal subgroups of G	$\bigcap_{x \in X} G_x$

4.3 The Orbit-Stabilizer Theorem

In the previous section, we observed an inverse relationship between orbits and stabilizers: the larger the stabilizer, the smaller the orbit, and vice versa. We also noticed, in the D_4 example (example 4.2.7), that $|G_1| \cdot |G \cdot 1| = |D_4|$. The Orbit-Stabilizer Theorem makes this relationship precise, and reveals it to be a direct consequence of Lagrange's Theorem.

Theorem 4.3.1: The Orbit-Stabilizer Theorem

Let G be a group acting on a set X, and let $x \in X$. There is a bijection

$$G/G_x \longleftrightarrow G \cdot x, \qquad gG_x \longmapsto g \cdot x,$$

between the set of left cosets of the stabilizer G_x in G and the orbit of x. In particular,

$$|G \cdot x| \;=\; [G : G_x].$$

If G is finite, this gives $|G \cdot x| = |G| \,/\, |G_x|$, or equivalently,

$$|G| \;=\; |G \cdot x| \cdot |G_x|.$$

***Proof*:**
Define the map

$$\varphi \colon G/G_x \to G \cdot x, \qquad \varphi(gG_x) = g \cdot x.$$

We verify that φ is a well-defined bijection.

Well-defined. Suppose $gG_x = hG_x$. Then $h^{-1}g \in G_x$, so $(h^{-1}g) \cdot x = x$. By axiom (A1):

$$g \cdot x = h \cdot \big((h^{-1}g) \cdot x\big) = h \cdot x.$$

Thus $\varphi(gG_x) = \varphi(hG_x)$, and the map does not depend on the choice of coset representative.

Injective. Suppose $\varphi(gG_x) = \varphi(hG_x)$, i.e. $g \cdot x = h \cdot x$. Then

$$(h^{-1}g) \cdot x = h^{-1} \cdot (g \cdot x) = h^{-1} \cdot (h \cdot x) = (h^{-1}h) \cdot x = e \cdot x = x,$$

so $h^{-1}g \in G_x$, hence $gG_x = hG_x$.

Surjective. Every element of $G \cdot x$ has the form $g \cdot x$ for some $g \in G$, and $g \cdot x = \varphi(gG_x)$.

Since φ is a bijection, $|G \cdot x| = |G/G_x| = [G : G_x]$. When G is finite, Lagrange's Theorem (theorem 3.3.1) gives $[G : G_x] = |G|/|G_x|$.

4.3.2 Remark The formula $|G| = |G \cdot x| \cdot |G_x|$ has exactly the same shape as Lagrange's Theorem: $|G| = [G : H] \cdot |H|$. It *is* Lagrange's Theorem, applied to the subgroup $H = G_x$. The content of the Orbit-Stabilizer Theorem is the bijection φ that identifies the abstract index $[G : G_x]$ with the concrete, countable object $G \cdot x$.

When X is finite, we can combine the Orbit-Stabilizer Theorem with the fact that orbits partition X (proposition 4.2.3) to obtain a useful counting formula.

Corollary 4.3.3: The Orbit-Counting Formula

Let G be a finite group acting on a finite set X, and let $x_1, \ldots, x_k$ be representatives from the distinct orbits. Then

$$|X| \;=\; \sum_{i=1}^{k} |G \cdot x_i| \;=\; \sum_{i=1}^{k} [G : G_{x_i}].$$

***Proof*:**

Since the orbits partition X (proposition 4.2.3), we have $|X| = \sum_{i=1}^{k} |G \cdot x_i|$. Replacing each $|G \cdot x_i|$ by $[G : G_{x_i}]$ via theorem 4.3.1 yields the result.

This formula will be the key ingredient in the Class Equation (Section 4.6), where we apply it to the conjugation action and split the sum according to whether orbit representatives lie in the center.

Example 4.3.4: Orbits of Polynomials Under S_4

Let $R = \mathbb{Z}[x_1, x_2, x_3, x_4]$, the ring of polynomials in four variables with integer coefficients, and let S_4 act on R by permuting the indices:

$$\sigma \cdot p(x_1, x_2, x_3, x_4) \;=\; p(x_{\sigma(1)}, x_{\sigma(2)}, x_{\sigma(3)}, x_{\sigma(4)}).$$

We compute the orbit of $f = x_1 + x_2$ under this action.

Rather than applying all 24 elements of S_4 to f, we first identify the stabilizer. A permutation σ fixes $x_1 + x_2$ if and only if the set $\{\sigma(1), \sigma(2)\} = \{1, 2\}$ (since addition is commutative, the order within the pair does not matter). This means σ must permute $\{1, 2\}$ among themselves and $\{3, 4\}$ among themselves. The stabilizer is therefore

$$(S_4)_f \;=\; \langle (1\ 2),\ (3\ 4) \rangle \;\cong\; \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z},$$

a copy of the Klein four-group, with $|(S_4)_f| = 4$. By the Orbit-Stabilizer Theorem,

$$|S_4 \cdot f| = \frac{|S_4|}{|(S_4)_f|} = \frac{24}{4} = 6.$$

We can now enumerate the orbit with confidence that we are looking for exactly 6 polynomials. Choosing one representative from each coset of $(S_4)_f$ yields:

$$S_4 \cdot (x_1 + x_2) = \{x_1 + x_2,\ x_1 + x_3,\ x_1 + x_4,\ x_2 + x_3,\ x_2 + x_4,\ x_3 + x_4\}.$$

The Orbit-Stabilizer Theorem saved us from checking all 24 permutations: knowing the stabilizer immediately told us the orbit size, and we only needed to find 6 distinct images.

Example 4.3.5: S_n Acting on $\{1, \ldots, n\}$

Consider S_{10} acting on $\{1, \ldots, 10\}$ by $\sigma \cdot i = \sigma(i)$. For any $i, j \in \{1, \ldots, 10\}$, there exists a permutation sending i to j (for instance, the transposition $(i\ j)$), so the action has a single orbit: $S_{10} \cdot i = \{1, \ldots, 10\}$ for every i.

The stabilizer of, say, 1 consists of all permutations that fix 1:

$$(S_{10})_1 = \{\sigma \in S_{10} : \sigma(1) = 1\} \cong S_9.$$

The Orbit-Stabilizer Theorem confirms: $|S_{10} \cdot 1| = |S_{10}|/|(S_{10})_1| = 10!/9! = 10$.

Example 4.3.6: D_4 Acting on Vertices, Revisited

The group D_4 acts on $X = \{1, 2, 3, 4\}$ (the vertices of a square). Since any vertex can be sent to any other by a suitable rotation, the action has a single orbit of size 4. From example 4.2.7, the stabilizer of vertex 1 has order 2. The Orbit-Stabilizer Theorem gives $|G \cdot 1| = 8/2 = 4$, as expected.

Now consider D_4 acting on the set of two diagonals, $\mathcal{D} = \{d_1, d_2\}$, where $d_1 = \{1, 3\}$ and $d_2 = \{2, 4\}$. A rotation by 90° swaps the two diagonals, while a rotation by 180° preserves both. Here $|G \cdot d_1| = 2$ and $|G_{d_1}| = 8/2 = 4$. The stabilizer G_{d_1} consists of the symmetries that preserve the diagonal d_1 as a set: $G_{d_1} = \{e,\ r^2,\ sr,\ sr^3\}$, a copy of the Klein four-group.

Burnside's Lemma

The Orbit-Stabilizer Theorem gives the size of each individual orbit. But in many applications, particularly in combinatorics, what we really want to know is the *number* of orbits. The following classical result answers this question.

Theorem 4.3.7: Burnside's Lemma

Let G be a finite group acting on a finite set X. Then the number of orbits is

$$|X/G| \ = \ \frac{1}{|G|} \sum_{g \in G} |X^g|,$$

where $X^g = \{x \in X : g \cdot x = x\}$ is the fixed-point set of g (definition 4.2.8).

In words: the number of distinct orbits equals the average number of fixed points, averaged over all group elements.

4.3.8 Historical Note Despite its name, this result was known to Cauchy (1845) and proved in full generality by Frobenius (1887). Burnside included it in his influential 1897 textbook, citing Frobenius, and subsequent authors attributed it to Burnside. For this reason, some authors call it the *Cauchy-Frobenius Lemma*. This book will follow the (ahistorical but widespread) convention of calling it Burnside's Lemma.

***Proof*:**
The proof is a double-counting argument. Consider the set of pairs

$$\mathcal{S} \;=\; \{(g,x) \in G \times X \big| g \cdot x = x\}.$$

We count $|\mathcal{S}|$ in two ways.

Counting by group elements (summing over $g \in G$). For each $g \in G$, the number of $x \in X$ with $g \cdot x = x$ is $|X^g|$. Therefore

$$|\mathcal{S}| = \sum_{g \in G} |X^g|. \tag{4.1}$$

Counting by set elements (summing over $x \in X$). For each $x \in X$, the number of $g \in G$ with $g \cdot x = x$ is $|G_x|$. Therefore

$$|\mathcal{S}| = \sum_{x \in X} |G_x|. \tag{4.2}$$

We now simplify (4.2) by grouping the sum according to orbits. Let $\mathcal{O}_1, \ldots, \mathcal{O}_k$ be the distinct orbits. For any x in an orbit $\mathcal{O}_i$, the Orbit-Stabilizer Theorem (theorem 4.3.1) gives $|G_x| = |G|/|\mathcal{O}_i|$. Summing over the $|\mathcal{O}_i|$ elements of the orbit:

$$\sum_{x \in \mathcal{O}_i} |G_x| \;=\; |\mathcal{O}_i| \cdot \frac{|G|}{|\mathcal{O}_i|} \;=\; |G|.$$

Each orbit contributes exactly $|G|$ to the sum, and there are $k = |X/G|$ orbits, so

$$\sum_{x \in X} |G_x| \;=\; |X/G| \cdot |G|.$$

Equating (4.1) and (4.2):

$$\sum_{g \in G} |X^g| \;=\; |X/G| \cdot |G|,$$

and dividing by $|G|$ yields the result.

The double-counting technique at the heart of this proof is worth internalizing: whenever a condition involves both a group element and a set element (here, "g fixes x"), counting the pairs from both sides often yields a clean identity.

Let us see Burnside's Lemma in action.

Example 4.3.9: Counting Colorings of a Square

How many distinct ways can we color the vertices of a square using 2 colors (say, black and white), if two colorings are considered "the same" when one can be obtained from the other by a symmetry of the square?

Let X be the set of all 2-colorings of 4 labeled vertices, so $|X| = 2^4 = 16$. The symmetry group $G = D_4$ acts on X by permuting the vertices (and hence the colors assigned to them). The distinct colorings "up to symmetry" are precisely the orbits, so we seek $|X/G|$.

We compute $|X^g|$ for each $g \in D_4$. Using the vertex labeling 1 (top-left), 2 (top-right), 3 (bottom-right), 4 (bottom-left), the eight elements act as the following permutations:

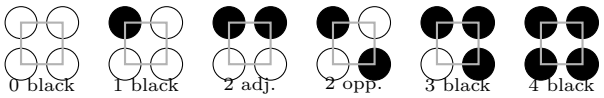
g	Permutation	Cycle structure	$\lvert X^g \rvert$
e	()	four fixed points	$2^4 = 16$
r	(1 2 3 4)	one 4-cycle	$2^1 = 2$
r^2	(1 3)(2 4)	two 2-cycles	$2^2 = 4$
r^3	(1 4 3 2)	one 4-cycle	$2^1 = 2$
s	(1 2)(3 4)	two 2-cycles	$2^2 = 4$
sr	(2 4)	one 2-cycle, two fixed	$2^3 = 8$
sr^2	(1 4)(2 3)	two 2-cycles	$2^2 = 4$
sr^3	(1 3)	one 2-cycle, two fixed	$2^3 = 8$

The rule for each row: a coloring is fixed by g if and only if all vertices within each cycle of g receive the same color. So $|X^g| = 2^{(\text{number of cycles of } g)}$, counting fixed points as 1-cycles.

Burnside's Lemma gives:

$$|X/G| \;=\; \frac{1}{8}(16 + 2 + 4 + 2 + 4 + 8 + 4 + 8) \;=\; \frac{48}{8} \;=\; 6.$$

The six distinct colorings are:



4.3.10 Applied Aside: Counting Molecular Structures The same technique applies in chemistry. Consider a benzene ring C_6H_6, where each hydrogen atom can be replaced by a substituent (say, chlorine). How many distinct chlorinated benzene derivatives exist? The symmetry group of the hexagonal ring is D_6, acting on the six positions. Burnside's Lemma (or its refinement, Pólya's Enumeration Theorem) answers this question systematically. For instance, with two possible substituents (H or Cl) at each of 6 positions, there are

$$\frac{1}{12}(2^6 + \cdots) = 13$$

distinct molecules, a count that matches experimental chemistry. This interplay between symmetry and enumeration is foundational in chemical graph theory.

4.4 Types of Actions

Some actions allow different group elements to "look the same" on X. Others ensure that every group element acts distinctly. Some actions can move any point to any other. Others split X into multiple unreachable regions. In this section, we introduce a taxonomy of four properties that classify the most important qualitative features of an action: faithful, transitive, free, and regular.

Faithful Actions

We have already defined this notion in definition 4.2.12: an action $G \curvearrowright X$ is *faithful* if $\ker(\rho) = \{e\}$, i.e. if the only element that fixes every point of X is the identity. Equivalently, the permutation representation $\rho\colon G \to \mathrm{Sym}(X)$ is injective, so G embeds into $\mathrm{Sym}(X)$.

Faithfulness is a *global* condition: it says nothing about any individual stabilizer G_x, only about their intersection $\bigcap_{x\in X} G_x$. Individual stabilizers may well be nontrivial.

Example 4.4.1: Faithful Actions

1. The tautological action of S_n on $\{1, \ldots, n\}$ is faithful: any nontrivial permutation must move at least one element.
2. The action of D_n on the vertices of a regular n-gon is faithful for $n \geq 3$: distinct symmetries of the polygon induce distinct permutations of the vertices. Note, however, that stabilizers are nontrivial: the stabilizer of a vertex is the order-2 subgroup generated by the reflection fixing that vertex.
3. The conjugation action of G on itself (item 6 of example 4.1.5) has kernel

   $$\ker(\rho) = \{g \in G : gag^{-1} = a \text{ for all } a \in G\} = Z(G),$$

   the center of G (definition 2.2.4). This action is faithful if and only if $Z(G) = \{e\}$, i.e. if and only if G has trivial center.

As we saw in proposition 4.2.14, any action can be made faithful by passing to the quotient $G/\ker(\rho)$. In this sense, the "faithful part" of an action is always available. The kernel measures the redundancy.

Transitive Actions

Definition 4.4.2: Transitive Action

A group action $G \curvearrowright X$ is called *transitive* if it has exactly one orbit, i.e. $G \cdot x = X$ for every $x \in X$. Equivalently, for any $x, y \in X$, there exists some $g \in G$ such that $g \cdot x = y$.

In a transitive action, every point can be reached from every other point. No point is "isolated" from the rest.

Example 4.4.3: Transitive and Non-Transitive Actions

1. The action of S_n on $\{1, \ldots, n\}$ is transitive: for any i, j, the transposition $(i\ j)$ sends i to j.
2. The action of D_4 on the four vertices of a square is transitive: r cyclically permutes all four vertices.
3. The action of D_4 on the set X of unordered distinct pairs $\{i, j\}$ of the 4 vertices that permutes like the above action is not transitive: show there are 2 orbits.
4. The action of $\langle(1\ 2\ 3)\rangle$ on $\{1, 2, 3, 4, 5\}$ from example 4.2.2 is *not* transitive: it has three orbits.
5. The conjugation action of G on itself is transitive if and only if every pair of elements is conjugate. For finite groups, this can only happen when $|G| = 1$ (since the identity is conjugate only to itself, so if $|G| > 1$, the singleton $\{e\}$ is always an orbit separate from any non-identity element).

Transitive actions enjoy a useful structural property: the stabilizers of any two points are conjugate subgroups.

Proposition 4.4.4: Stabilizers of Points in the Same Orbit

Let $G \curvearrowright X$. If $y = g \cdot x$ for some $g \in G$, then

$$G_y = g\, G_x\, g^{-1}.$$

In particular, if the action is transitive, then the stabilizers of any two points in X are conjugate subgroups of G.

***Proof*:**
Let $G_x = \{g \in G \mid g \cdot x = x\}$ and $y = g \cdot x$. Then act both sides by g^{-1} so that $g^{-1} \cdot y = g^{-1} \cdot (g \cdot x) = (g^{-1}g) \cdot x = x$. Now, for any $h \in G_x$ (constantly using group-action associativity):

$$ghg^{-1} \cdot y = gh \cdot x = g \cdot x = y$$

showing $gG_xg^{-1} \subseteq G_y$, equivalently that $G_x \subseteq g^{-1}G_yg$. Working the argument in reverse, for any $k \in G_y$, we get:

$$g^{-1}kg \cdot x = g^{-1} \cdot (k \cdot y) = g^{-1} \cdot y = x$$

showing $g^{-1}G_yg \subseteq G_x$, equivalently $G_y \subseteq gG_xg^{-1}$. Both containments give $G_y = gG_xg^{-1}$.

This proposition has an important consequence: in a transitive action, knowing the stabilizer of *any single point* determines (up to conjugacy) the stabilizer of every point.

Even more is true. A transitive action is completely determined, up to a natural notion of equivalence, by the stabilizer of any chosen point. To make this precise, we need a brief definition.

Definition 4.4.5: Isomorphism of G-sets

Let $G \curvearrowright X$ and $G \curvearrowright Y$ be two actions of the same group. A bijection $\varphi\colon X \to Y$ is an *isomorphism of G-sets* (or a *G-equivariant* bijection) if

$$\varphi(g \cdot x) = g \cdot \varphi(x) \qquad \text{for all } g \in G,\ x \in X.$$

If such a bijection exists, we write $X \cong_G Y$ and say the two G-sets are *isomorphic*.

In other words, φ is a relabeling of X as Y that is compatible with the group action: it does not matter whether we act first and then relabel, or relabel first and then act.

Theorem 4.4.6: Structure Theorem for Transitive Actions

Let $G \curvearrowright X$ be a transitive action, and fix any point $x_0 \in X$. Then the bijection

$$\varphi\colon G/G_{x_0} \xrightarrow{\sim} X, \qquad gG_{x_0} \mapsto g \cdot x_0,$$

from the Orbit-Stabilizer Theorem (theorem 4.3.1) is an isomorphism of G-sets, where G acts on G/G_{x_0} by left multiplication: $h \cdot (gG_{x_0}) = (hg)G_{x_0}$.

In particular, every transitive G-set is isomorphic to G/H for some subgroup $H \leq G$.

***Proof*:**
We already know from theorem 4.3.1 that φ is a well-defined bijection (using transitivity so that $G \cdot x_0 = X$). It remains to verify equivariance. For any $h \in G$ and any coset gG_{x_0}:

$$\varphi(h \cdot gG_{x_0}) = \varphi((hg)G_{x_0}) = (hg) \cdot x_0 = h \cdot (g \cdot x_0) = h \cdot \varphi(gG_{x_0}).$$

Thus φ respects the action, and the two G-sets are isomorphic.

This theorem is one of the most clarifying results in the theory: it says that, up to relabeling, *every* transitive action is a coset action. We will study coset actions in detail in Section 4.5. If the action is furthermore faithful, then as there is one coset $G_x = \{e\}$ and:

$$G \cong_G (X, \rho)$$

where ρ represents the faithful transitive action.

4.4.7 Geometric Interpretation In differential geometry, is is very common to have group actions on a manifold or a bundle on the manifold. The action can always be made faithful as explained above, and often the action is transitive on fibers. Hence, each fiber of a bundle has the "structure" of a group in the sense of the above G-set isomorphism. This motivates shifting our perspective from an action on a set to an action on a group, as in many important cases we can study what happens to geometric objects by studying the left-multiplication or conjugation actin on G!
Note that a fiber bundle where each fiber is G-set isomorphic to a group G is called a G-torsor, or a Principal homogeneous space.

Free Actions

Faithful actions require the *intersection* of all stabilizers to be trivial. Free actions require something much stronger: every *individual* stabilizer must be trivial.

Definition 4.4.8: Free Action

A group action $G \curvearrowright X$ is called *free* if

$$G_x = \{e\} \qquad \text{for every } x \in X.$$

Equivalently, if $g \cdot x = x$ for some $x \in X$, then $g = e$.

There is a useful reformulation: an action is free if and only if distinct group elements never produce the same result at any point.

Proposition 4.4.9: Characterization of Free Actions

An action $G \curvearrowright X$ is free if and only if for all $g, h \in G$ and all $x \in X$,

$$g \cdot x = h \cdot x \quad \Longrightarrow \quad g = h.$$

***Proof*:**
If $g \cdot x = h \cdot x$, then $(h^{-1}g) \cdot x = x$, so $h^{-1}g \in G_x$. If the action is free, $G_x = \{e\}$, so $h^{-1}g = e$, giving $g = h$. Conversely, if the implication holds for all g, h, x, then taking $h = e$: $g \cdot x = x$ implies $g = e$, so every stabilizer is trivial.

Example 4.4.10: Free Actions

1. The left-multiplication action $G \curvearrowright G$ (item 5 of example 4.1.5) is free: if $ga = a$, then $g = e$ by cancellation (proposition 1.1.17). This holds for any group G.
2. The action of $(\mathbb{Z}, +)$ on $\mathbb{R}$ by $n \cdot x = x + n$ is free: if $x + n = x$, then $n = 0$.
3. The action of D_4 on the vertices of a square is *not* free: the stabilizer of each vertex has order 2.

4. The action of $\mathbb{Z}/2\mathbb{Z} = \{e, \sigma\}$ on $\{1, 2, 3, 4\}$ via $\sigma = (1\ 2)(3\ 4)$ is free: σ has no fixed points. It is not transitive, however (the orbits are $\{1, 2\}$ and $\{3, 4\}$).

Free actions interact cleanly with the Orbit-Stabilizer Theorem.

Corollary 4.4.11: Orbit Sizes in Free Actions

If G is a finite group acting freely on X, then every orbit has size $|G|$. In particular, $|G|$ divides $|X|$.

***Proof*:**
For every $x \in X$, $|G_x| = 1$, so $|G \cdot x| = |G|/|G_x| = |G|$ by theorem 4.3.1. Since the orbits partition X (proposition 4.2.3), $|X|$ is a sum of copies of $|G|$.

Note that every free action is faithful: if $G_x = \{e\}$ for all x, then $\ker(\rho) = \bigcap_x G_x = \{e\}$. The converse is false, as the example of D_4 on the vertices of a square shows (faithful, but not free).

Regular Actions

The strongest condition arises when we ask for an action that is simultaneously free and transitive.

Definition 4.4.12: Regular Action

A group action $G \curvearrowright X$ is called *regular* if it is both free and transitive.

Regular actions have a particularly elegant characterization.

Proposition 4.4.13: Characterization of Regular Actions

An action $G \curvearrowright X$ is regular if and only if for every pair $x, y \in X$, there exists a *unique* $g \in G$ such that $g \cdot x = y$.

***Proof*:**
Transitivity says "at least one such g exists." Freeness says "at most one such g exists" (by proposition 4.4.9: if $g \cdot x = y = h \cdot x$, then $g = h$). Together, they give exactly one.

Conversely, if for every x, y there exists a unique g with $g \cdot x = y$, then existence gives transitivity, and uniqueness gives freeness.

Example 4.4.14: Regular Actions

1. The left-multiplication action $G \curvearrowright G$ is regular: it is free (example 4.4.10) and transitive (given $a, b \in G$, the element $g = ba^{-1}$ satisfies $g \cdot a = ba^{-1}a = b$). This is the *canonical* regular action.
2. The action of $(\mathbb{Z}, +)$ on $\mathbb{R}$ by translation ($n \cdot x = x + n$) is free but not transitive ($\frac{1}{2}$ is not in the orbit of 0), hence not regular.

3. The action of D_4 on the four vertices of a square is transitive but not free, hence not regular.

By the Orbit-Stabilizer Theorem, a regular action of a finite group G on a set X satisfies $|X| = |G \cdot x| = |G|$ for every x: the set and the group have the same cardinality. In fact, the Structure Theorem for Transitive Actions (theorem 4.4.6) shows precisely what every regular action looks like:

Corollary 4.4.15: Regular Actions are Left Multiplication

If G acts regularly on X, then $X \cong_G G$ as G-sets. That is, up to isomorphism, the left-multiplication action of G on itself is the *only* regular G-action.

***Proof*:**
By theorem 4.4.6, any transitive G-set is isomorphic to G/G_{x_0} for some $x_0 \in X$. Since the action is also free, $G_{x_0} = \{e\}$, so $X \cong_G G/\{e\} = G$.

4.5 Action by Left Multiplication

An action of G on a set X (definition 4.1.1) needs two pieces of data besides G, the set X and the homomorphism $G \to \mathrm{Sym}(X)$. Every group carries both without further choices: its underlying set, and the actions defined by its own operation, such as left multiplication and conjugation. By theorem 4.4.6, every transitive action is isomorphic to the action of G by left multiplication on a set of cosets G/H, so actions built from G itself already describe all transitive G-sets. For an action of G on itself, the orbits are subsets of G and the stabilizers are subgroups of G, so the Orbit-Stabilizer Theorem (theorem 4.3.1) applied to it gives statements about G alone.

The simplest self-action is left multiplication: each group element shuffles the entire group by multiplying on the left.

Let G be a group and let $X = G$ (the underlying set). The action is

$$g \cdot a = ga \qquad \text{for all } g, a \in G.$$

We verified in item 5 of example 4.1.5 that this is a valid group action. We can now classify it using the taxonomy of Section 4.4:

Proposition 4.5.1: Left Multiplication is a Regular Action

The left-multiplication action $G \curvearrowright G$ is regular (free and transitive).

***Proof*:**
The action is free: if $g \cdot a = a$, then $ga = a$, so $g = e$ by cancellation (proposition 1.1.17). The action is transitive: given any $a, b \in G$, taking $g = ba^{-1}$ gives $g \cdot a = ba^{-1}a = b$.

Since the action is faithful (every free action is), the permutation representation $\rho\colon G \to \mathrm{Sym}(G)$ is injective. Let us see what ρ looks like concretely.

Example 4.5.2: Permutation Representation of the Klein Four-Group

Let $G = \{e, a, b, c\}$ be the Klein four-group, with multiplication defined by $a^2 = b^2 = c^2 = e$ and $ab = c$, $ac = b$, $bc = a$. Label $e \leftrightarrow 1$, $a \leftrightarrow 2$, $b \leftrightarrow 3$, $c \leftrightarrow 4$. For each $g \in G$, we compute the permutation $\sigma_g(i) = j$ where $g \cdot g_i = g_j$:

g	$g \cdot e$	$g \cdot a$	$g \cdot b$	$g \cdot c$	σ_g
e	e	a	b	c	$()$
a	a	e	c	b	$(1\ 2)(3\ 4)$
b	b	c	e	a	$(1\ 3)(2\ 4)$
c	c	b	a	e	$(1\ 4)(2\ 3)$

The image $\rho(G) = \{(), (1\ 2)(3\ 4), (1\ 3)(2\ 4), (1\ 4)(2\ 3)\}$ is a subgroup of S_4 isomorphic to the Klein four-group. One can verify directly that $g \mapsto \sigma_g$ is a homomorphism, and injectivity is visible from the table: distinct rows give distinct permutations.

The example above embeds a group of order 4 into S_4. The same construction works for any group.

Theorem 4.5.3: Cayley's Theorem

Every group is isomorphic to a subgroup of a symmetric group. More precisely, if G is a group, then the permutation representation $\rho\colon G \to \mathrm{Sym}(G)$ of the left-multiplication action is an injective homomorphism. If $|G| = n$, then G is isomorphic to a subgroup of S_n.

***Proof*:**

The left-multiplication action $G \curvearrowright G$ is free (proposition 4.5.1), hence faithful (definition 4.2.12). Thus the permutation representation $\rho\colon G \to \mathrm{Sym}(G)$ has trivial kernel, so ρ is injective. By the First Isomorphism Theorem (theorem 3.1.15), $G \cong \rho(G) \leq \mathrm{Sym}(G)$.

If $|G| = n$, we may label the elements of G as $\{g_1, \ldots, g_n\}$ and identify $\mathrm{Sym}(G) \cong S_n$, so G embeds into S_n.

Cayley's Theorem shows that the abstract axioms (closure, associativity, identity, inverses) do not create anything genuinely "new" beyond permutation groups. Every group, no matter how abstractly defined, *is* a group of symmetries: a subgroup of a symmetric group. The Erlangen Programme's vision (see the aside in box 4.0.1) is vindicated: groups and symmetries are one and the same.

4.5.4 Historical Note When group theory began in the early 19th century, a "group" simply *meant* a group of permutations. Cayley was among the first to propose the abstract axiomatic definition. His 1854 theorem then showed that the abstraction lost nothing: the original permutation-based intuition remains fully applicable to every abstract group.

That said, the embedding $G \hookrightarrow S_n$ is rarely practical. The symmetric group S_n has order $n!$, which is vastly larger than n for even modest values: if $|G| = 100$, then $|S_{100}| = 100! \approx 9.3 \times 10^{157}$. Working inside S_n is typically no easier than working directly with G. The theorem's value is *conceptual*: it guarantees that any general statement proved about subgroups of symmetric groups automatically applies to all groups.

The following example illustrates a subtler limitation: Cayley's bound $n = |G|$ is sometimes optimal (you genuinely cannot embed the group into any smaller symmetric group).

Example 4.5.5: The Minimum Faithful Degree of Q_8

The quaternion group $Q_8 = \{\pm 1, \pm i, \pm j, \pm k\}$ has order 8, so Cayley's Theorem gives $Q_8 \hookrightarrow S_8$. We claim that Q_8 cannot act faithfully on any set X with $|X| \leq 7$.

Suppose $Q_8 \curvearrowright X$ with $|X| \leq 7$. For any $x \in X$, the Orbit-Stabilizer Theorem (theorem 4.3.1) gives

$$|G_x| = \frac{|Q_8|}{|Q_8 \cdot x|} = \frac{8}{|Q_8 \cdot x|} \geq \frac{8}{7} > 1,$$

so $G_x \neq \{e\}$ for every x. Now, a particular property of Q_8 is that every nontrivial subgroup contains -1 (one can verify this by inspecting the subgroup lattice of Q_8, cf. example 1.2.31). Since G_x is a nontrivial subgroup for every x, we have $-1 \in G_x$ for every x. Therefore $-1 \in \bigcap_{x \in X} G_x = \ker(\rho)$, and the action is not faithful.

Thus, the Cayley embedding into S_8 is the smallest symmetric group that can faithfully contain Q_8.

Action on the Coset Space

Left multiplication acts on the full group G. We now generalize by letting G act on the set of left cosets of a subgroup.

Let $H \leq G$, and let $G/H = \{gH : g \in G\}$ denote the set of left cosets (this is a set, not necessarily a group, since H need not be normal). Define

$$g \cdot (aH) = (ga)H \qquad \text{for all } g \in G,\ aH \in G/H.$$

This is well-defined: if $aH = bH$, then $b^{-1}a \in H$, so $(gb)^{-1}(ga) = b^{-1}a \in H$, giving $(ga)H = (gb)H$. The two axioms are immediate: $(gh) \cdot aH = (gh)aH = g \cdot (haH)$ and $e \cdot aH = aH$.

Note that when $H = \{e\}$, the cosets of H are simply the elements of G, and this action reduces to left multiplication. In this sense, the coset action *generalizes* the construction of the previous subsection.

The following theorem extracts the structural information that the coset action provides.

Theorem 4.5.6: Properties of the Coset Action

Let $H \leq G$, and let G act on G/H by left multiplication. Let $\pi_H \colon G \to \mathrm{Sym}(G/H)$ be the associated permutation representation. Then:

1. The action is transitive.
2. The stabilizer of the coset eH is H:
$$G_{eH} = H.$$
3. The kernel of the action is the largest normal subgroup of G contained in H:
$$\ker(\pi_H) = \bigcap_{g \in G} gHg^{-1}.$$

***Proof*:**

1. Given any two cosets $aH, bH \in G/H$, take $g = ba^{-1}$. Then $g \cdot aH = (ba^{-1})aH = bH$. Since a and b were arbitrary, the action is transitive.

2. The stabilizer of eH is $\{g \in G : g \cdot eH = eH\} = \{g \in G : gH = H\}$. Now $gH = H$ if and only if $g \in H$ (recall that $gH = H \Leftrightarrow g \in H$). Thus $G_{eH} = H$.

3. The kernel is $\ker(\pi_H) = \bigcap_{aH \in G/H} G_{aH}$. The stabilizer of an arbitrary coset aH is:

$$\begin{aligned} G_{aH} &= \{g \in G : g \cdot aH = aH\} = \{g \in G : gaH = aH\} \\ &= \{g \in G : a^{-1}ga \in H\} = \{g \in G : g \in aHa^{-1}\} = aHa^{-1}. \end{aligned}$$

   Therefore

$$\ker(\pi_H) = \bigcap_{a \in G} aHa^{-1}.$$

   It remains to show this is the *largest* normal subgroup of G contained in H. Since $\ker(\pi_H) \leq G_{eH} = H$ (part 2), the kernel is contained in H. It is normal in G since it is the kernel of a homomorphism (proposition 2.2.12).

   Conversely, let $N \trianglelefteq G$ with $N \leq H$. For any $a \in G$: $N = aNa^{-1} \leq aHa^{-1}$ (since $N \leq H$ implies $aNa^{-1} \leq aHa^{-1}$). Thus $N \leq \bigcap_{a \in G} aHa^{-1} = \ker(\pi_H)$, confirming maximality.

Example 4.5.7: D_4 Acting on Cosets of $\langle s \rangle$

Let $G = D_4$ and $H = \langle s \rangle = \{e, s\}$. The index $[D_4 : H] = 8/2 = 4$, so there are four left cosets:

$$C_1 = \{e,\ s\}, \quad C_2 = \{r,\ sr^3\}, \quad C_3 = \{r^2,\ sr^2\}, \quad C_4 = \{r^3,\ sr\}.$$

The coset action gives a homomorphism $\pi_H \colon D_4 \to S_4$. To find the image, we compute where r and s send each coset:

$$\begin{aligned} r\colon\ & C_1 \mapsto C_2,\ C_2 \mapsto C_3,\ C_3 \mapsto C_4,\ C_4 \mapsto C_1 && \Longrightarrow \quad \sigma_r = (1\ 2\ 3\ 4), \\ s\colon\ & C_1 \mapsto C_1,\ C_2 \mapsto C_4,\ C_3 \mapsto C_3,\ C_4 \mapsto C_2 && \Longrightarrow \quad \sigma_s = (2\ 4). \end{aligned}$$

The image $\pi_H(D_4) = \langle (1\ 2\ 3\ 4), (2\ 4) \rangle$ is a subgroup of S_4 isomorphic to D_4. In particular, $\ker(\pi_H) = \{e\}$, so the action is faithful: D_4 embeds into S_4. This is smaller than the Cayley embedding into S_8, and it recovers the natural representation of D_4 as the symmetry group of a square.

Theorem 4.5.6 predicted this: the kernel is $\bigcap_{g \in G} g\langle s \rangle g^{-1}$. Since the only conjugates of $\langle s \rangle$ are $\{e, s\}$ and $\{e, sr^2\}$, their intersection is $\{e\}$.

Example 4.5.8: Recovering the Index-2 Theorem

If $[G : H] = 2$, the coset action gives a homomorphism $\pi_H \colon G \to S_2 \cong \mathbb{Z}/2\mathbb{Z}$. Since $|S_2| = 2$, the kernel has index at most 2 in G. But $\ker(\pi_H) \leq H$ and $[G : H] = 2$, so $\ker(\pi_H) = H$. Since kernels are normal, $H \trianglelefteq G$.

We have thus re-derived corollary 3.3.6 as a special case of the coset action.

Subgroups of Smallest Prime Index

The observation in the previous example generalizes beyond index 2. The key insight is that if $[G : H] = p$ for a small prime p, then the coset action maps G into S_p, and S_p is "small enough" to force the kernel to be large.

Theorem 4.5.9: Subgroups of Smallest Prime Index are Normal

Let G be a finite group and let p be the smallest prime dividing $|G|$. If $H \leq G$ with $[G : H] = p$, then $H \trianglelefteq G$.

***Proof*:**
The coset action gives a homomorphism $\pi_H \colon G \to \mathrm{Sym}(G/H) \cong S_p$. Let $K = \ker(\pi_H)$. By theorem 4.5.6, $K \trianglelefteq G$ and $K \leq H$. We will show $K = H$.

By the First Isomorphism Theorem (theorem 3.1.15), $G/K \cong \pi_H(G) \leq S_p$, so $|G/K|$ divides $|S_p| = p!$. Since $K \leq H$, the index decomposes as

$$[G : K] = [G : H] \cdot [H : K] = p \cdot [H : K].$$

Therefore $p \cdot [H : K]$ divides $p!$, which gives $[H : K]$ divides $(p-1)!$.

On the other hand, $[H : K]$ divides $|H|$ by Lagrange's Theorem (theorem 3.3.1), and $|H|$ divides $|G|$, so every prime factor of $[H : K]$ is also a prime factor of $|G|$. Since p is the *smallest* prime dividing $|G|$, every prime factor of $[H : K]$ is at least p.

But every prime factor of $(p-1)!$ is strictly less than p. The only positive integer all of whose prime factors are simultaneously $\geq p$ and $< p$ is 1. Therefore $[H : K] = 1$, so $K = H$, and $H = \ker(\pi_H) \trianglelefteq G$.

Example 4.5.10: Applications of the Index-p Theorem

1. If $|G| = 2m$ with m odd, the smallest prime dividing $|G|$ is 2. Any subgroup of index 2 is normal. This recovers corollary 3.3.6.
2. Let $|G| = 2 \cdot 3 \cdot 5 = 30$. The smallest prime dividing $|G|$ is 2. If G has a subgroup of index 2 (i.e. order 15), it is automatically normal. Any subgroup of index 3 need not be normal, since 3 is not the smallest prime dividing 30.
3. Let $|G| = 3 \cdot 7 = 21$. The smallest prime is 3. If G has a subgroup of index 3 (i.e. order 7), it is normal.
4. Let $|G| = 5 \cdot 11 \cdot 13 = 715$. The smallest prime is 5. A subgroup of index 5 (order 143), should one exist, is normal.

4.5.11 Remark A group of order n need not have a subgroup of every possible index. For instance, A_4 has order 12 and the smallest prime dividing 12 is 2, but A_4 has no subgroup of order 6 (cf. the counterexample to the converse of Lagrange's Theorem, example 3.3.8). The theorem says that *if* a subgroup of index p exists, it must be normal. It does not guarantee existence. The question of existence will be addressed by the Sylow Theorems in Section 4.8.

We pause to take stock. Starting from the simple idea of a group multiplying its own elements, we have obtained:

- Cayley's Theorem: every group embeds into a symmetric group.
- The coset action: a versatile tool that converts subgroup information ($H \leq G$) into homomorphism information ($G \to S_p$).
- The index-p theorem: a powerful normality criterion that requires nothing more than knowing $[G : H]$ and the prime factorization of $|G|$.

4.6 Action by Conjugation and the Class Equation

In Section 4.5, a group acted on itself by left multiplication, and the payoff was Cayley's Theorem: every group is a group of permutations. We now let a group act on itself by *conjugation*, and the payoff will be of a completely different character: arithmetic information about the internal structure of the group, encoded in an identity called the *Class Equation*.

Conjugation on Elements

Let G be a group and let $X = G$. The *conjugation action* is defined by

$$g \cdot a \;=\; gag^{-1} \qquad \text{for all } g, a \in G.$$

We verified in item 6 of example 4.1.5 that this is a valid group action. Let us now identify its orbits and stabilizers.

Definition 4.6.1: Conjugacy Class

Two elements $a, b \in G$ are *conjugate* in G if $b = gag^{-1}$ for some $g \in G$, i.e. if they lie in the same orbit of the conjugation action. The orbit

$$\mathrm{cl}(a) \;=\; G \cdot a \;=\; \{gag^{-1} : g \in G\}$$

is called the *conjugacy class* of a.

Since orbits partition the set (proposition 4.2.3), conjugacy classes partition G: every element belongs to exactly one conjugacy class.

The stabilizer of the conjugation action has a familiar name:

$$G_a = \{g \in G : gag^{-1} = a\} = \{g \in G : ga = ag\} = C_G(a),$$

the centralizer of a in G (definition 2.2.7). The Orbit-Stabilizer Theorem (theorem 4.3.1) therefore gives:

$$|\operatorname{cl}(a)| \;=\; [G : C_G(a)]. \tag{4.3}$$

An element a has a singleton conjugacy class, $\operatorname{cl}(a) = \{a\}$, if and only if $gag^{-1} = a$ for all $g \in G$, i.e. if and only if a commutes with every element of G. This is precisely the condition $a \in Z(G)$ (definition 2.2.4). Thus:

The singleton conjugacy classes are exactly the elements of the center.

Example 4.6.2: Conjugacy Classes of Small Groups

1. If G is abelian, then $gag^{-1} = a$ for all g, a, so every conjugacy class is a singleton: $\operatorname{cl}(a) = \{a\}$ for all a. The conjugation action is trivial.
2. Let $G = Q_8 = \{\pm 1, \pm i, \pm j, \pm k\}$. The center is $Z(Q_8) = \{1, -1\}$ (example 1.2.31). The non-central elements form three conjugacy classes:

   $$\operatorname{cl}(i) = \{i, -i\}, \qquad \operatorname{cl}(j) = \{j, -j\}, \qquad \operatorname{cl}(k) = \{k, -k\}.$$

   To verify: $jij^{-1} = ji(-j) = ji \cdot j^{-1}$. Using $ji = -k$ and $(-k)(-j) = kj = -i$, so $jij^{-1} = -i$. Each conjugacy class has size $[Q_8 : C_{Q_8}(i)] = 8/4 = 2$, since $C_{Q_8}(i) = \langle i \rangle$ has order 4.
3. Let $G = D_4$. The center is $Z(D_4) = \{e, r^2\}$. The non-central conjugacy classes are:

   $$\operatorname{cl}(r) = \{r, r^3\}, \qquad \operatorname{cl}(s) = \{s, sr^2\}, \qquad \operatorname{cl}(sr) = \{sr, sr^3\}.$$

   Each has size 2, consistent with $[D_4 : C_{D_4}(r)] = 8/4 = 2$, etc.

The Class Equation

We now translate the orbit decomposition of the conjugation action into an arithmetic identity.

Theorem 4.6.3: The Class Equation

Let G be a finite group, and let $g_1, \ldots, g_r$ be representatives of the distinct conjugacy classes of G that are *not* contained in $Z(G)$. Then

$$|G| \;=\; |Z(G)| \;+\; \sum_{i=1}^{r} [G : C_G(g_i)].$$

Each term $[G : C_G(g_i)]$ in the sum is strictly greater than 1 and divides $|G|$.

***Proof*:**
The conjugacy classes partition G, so $|G| = \sum |\operatorname{cl}(a)|$ as a ranges over a set of orbit representatives. We split this sum into two parts:

- Elements $a \in Z(G)$ have $|\operatorname{cl}(a)| = 1$. There are $|Z(G)|$ such elements.
- Elements $a \notin Z(G)$ have $|\operatorname{cl}(a)| = [G : C_G(a)] > 1$ (since $C_G(a) \neq G$ when $a \notin Z(G)$). Let $g_1, \ldots, g_r$ be representatives of these non-central classes.

Combining:

$$|G| = \sum_{a \in Z(G)} 1 + \sum_{i=1}^{r} |\operatorname{cl}(g_i)| = |Z(G)| + \sum_{i=1}^{r} [G : C_G(g_i)].$$

Each $[G : C_G(g_i)]$ divides $|G|$ by Lagrange's Theorem (theorem 3.3.1), and is > 1 since $g_i \notin Z(G)$.

Example 4.6.4: The Class Equation in Practice

1. For $G = Q_8$: $|Z(Q_8)| = 2$ and the three non-central classes each have size 2, giving $8 = 2 + 2 + 2 + 2$.
2. For $G = D_4$: $|Z(D_4)| = 2$ and the three non-central classes each have size 2, giving $8 = 2 + 2 + 2 + 2$.
3. For $G = S_3$: $Z(S_3) = \{e\}$, and the conjugacy classes are $\{e\}$, $\{(1\ 2), (1\ 3), (2\ 3)\}$, $\{(1\ 2\ 3), (1\ 3\ 2)\}$, giving $6 = 1 + 3 + 2$.
4. If G is abelian, the Class Equation reads $|G| = |G|$, which is vacuously true but uninformative. The Class Equation is a tool for *non-abelian* groups.

p-Groups Have Nontrivial Center

The Class Equation becomes especially powerful when $|G|$ is a prime power. The following theorem is the first result that students typically cannot prove *without* group actions.

Theorem 4.6.5: p-Groups Have Nontrivial Center

Let p be a prime and let G be a group of order p^α for some $\alpha \geq 1$. Then $Z(G) \neq \{e\}$. More precisely, p divides $|Z(G)|$.

***Proof*:**
The Class Equation gives

$$p^\alpha = |Z(G)| + \sum_{i=1}^{r} [G : C_G(g_i)].$$

For each i, the centralizer $C_G(g_i)$ is a proper subgroup of G (since $g_i \notin Z(G)$), so $[G : C_G(g_i)] > 1$. Since $C_G(g_i) \leq G$ and $|G| = p^\alpha$, Lagrange's Theorem gives $|C_G(g_i)| = p^{\beta_i}$ for some $\beta_i < \alpha$. Therefore $[G : C_G(g_i)] = p^{\alpha - \beta_i}$, which is divisible by p.

The left-hand side p^α is divisible by p, and every term in the sum is divisible by p. Therefore $|Z(G)|$ is divisible by p, so $|Z(G)| \geq p > 1$.

The argument is clean and short, but notice how much machinery is involved beneath the surface: the conjugation action, the Orbit-Stabilizer Theorem (to get $|\operatorname{cl}(g_i)| = [G : C_G(g_i)]$), Lagrange's Theorem (to deduce $p \big| [G : C_G(g_i)]$), and the orbit partition (to write the Class Equation). This is a vivid demonstration of the power of group actions as a *proof technique*.

Corollary 4.6.6: Groups of Order p^2 Are Abelian

If $|G| = p^2$ for some prime p, then G is abelian. More precisely, $G \cong \mathbb{Z}/p^2\mathbb{Z}$ or $G \cong \mathbb{Z}/p\mathbb{Z} \times \mathbb{Z}/p\mathbb{Z}$.

***Proof*:**

By theorem 4.6.5, $p \big| |Z(G)|$. Since $Z(G) \leq G$ and $|G| = p^2$, Lagrange gives $|Z(G)| \in \{p, p^2\}$.

If $|Z(G)| = p^2$, then $Z(G) = G$ and G is abelian.

If $|Z(G)| = p$, then $|G/Z(G)| = p^2/p = p$, so $G/Z(G)$ has prime order and is therefore cyclic (corollary 3.3.2). We claim this forces G to be abelian, contradicting $Z(G) \neq G$.

Indeed, let $G/Z(G) = \langle gZ(G) \rangle$. Then every element of G can be written as $g^k z$ for some $k \in \mathbb{Z}$ and $z \in Z(G)$. For any two elements $g^{k_1} z_1$ and $g^{k_2} z_2$:

$$(g^{k_1} z_1)(g^{k_2} z_2) = g^{k_1} g^{k_2} z_1 z_2 = g^{k_2} g^{k_1} z_2 z_1 = (g^{k_2} z_2)(g^{k_1} z_1),$$

where we used that powers of g commute with each other, that $z_1, z_2 \in Z(G)$ commute with everything, and that $Z(G)$ is abelian. Thus G is abelian, a contradiction.

Therefore $|Z(G)| = p^2$, and G is abelian of order p^2. By the classification of finite abelian groups (or by direct argument using corollary 3.3.2), the only abelian groups of order p^2 are $\mathbb{Z}/p^2\mathbb{Z}$ and $\mathbb{Z}/p\mathbb{Z} \times \mathbb{Z}/p\mathbb{Z}$.

4.6.7 Remark The key step ("if $G/Z(G)$ is cyclic, then G is abelian") is worth remembering in its own right. More generally, if $G/Z(G)$ is cyclic, the same argument shows every pair of elements commutes, so G is abelian and $Z(G) = G$. This means $G/Z(G)$ is trivial, a stronger conclusion than cyclic. In other words: $G/Z(G)$ *can never be a nontrivial cyclic group.*

Conjugation on Subsets and Subgroups

So far, conjugation has acted on *elements* of G. There is a natural extension to *subsets*: let $\mathcal{P}(G)$ denote the power set of G, and define

$$g \cdot S = gSg^{-1} = \{gsg^{-1} : s \in S\} \qquad \text{for } g \in G,\ S \subseteq G.$$

This defines a group action $G \curvearrowright \mathcal{P}(G)$. (Verification of the axioms is identical to the element case.)

Definition 4.6.8: Conjugate Subsets

Two subsets $S, T \subseteq G$ are *conjugate* in G if $T = gSg^{-1}$ for some $g \in G$.

When we restrict attention to the set of all *subgroups* of G, conjugation sends subgroups to (isomorphic) subgroups, so this restriction is also a valid action.

The stabilizer and orbit of a subset S under this action connect to familiar concepts:

- The *stabilizer* of S is
$$G_S = \{g \in G : gSg^{-1} = S\} = N_G(S),$$
the normalizer of S in G (definition 2.2.10).
- The *orbit* $G \cdot S = \{gSg^{-1} : g \in G\}$ is the set of all conjugates of S.

The Orbit-Stabilizer Theorem immediately yields:

Corollary 4.6.9: Number of Conjugates

The number of distinct conjugates of a subset $S \subseteq G$ is $[G : N_G(S)]$. In particular, S is the only member of its conjugacy class if and only if $N_G(S) = G$, i.e. if and only if $S \trianglelefteq G$ (when S is a subgroup) or S is a union of conjugacy classes (when S is an arbitrary subset).

This corollary will be indispensable in the proof of Sylow's Theorems (Section 4.8), where we count the number of conjugates of a Sylow p-subgroup.

Example 4.6.10: Conjugates of a Subgroup

Let $G = S_4$ and $H = \langle(1\ 2)\rangle = \{e, (1\ 2)\}$. The normalizer $N_{S_4}(H)$ consists of all $\sigma \in S_4$ with $\sigma H \sigma^{-1} = H$, i.e. $\sigma(1\ 2)\sigma^{-1} \in \{e, (1\ 2)\}$. Since conjugation preserves cycle type (proposition 1.2.27), $\sigma(1\ 2)\sigma^{-1}$ is always a transposition. It equals $(1\ 2)$ precisely when σ fixes the set $\{1, 2\}$, so

$$N_{S_4}(H) = \{\sigma \in S_4 : \{\sigma(1), \sigma(2)\} = \{1, 2\}\} = \{e,\ (1\ 2),\ (3\ 4),\ (1\ 2)(3\ 4)\}.$$

Therefore H has $[S_4 : N_{S_4}(H)] = 24/4 = 6$ conjugates. These are the six subgroups $\langle(i\ j)\rangle$ for $1 \leq i < j \leq 4$, one for each transposition.

Conjugacy in S_n and Cycle Types

Since S_n plays a distinguished role throughout this chapter, we take a moment to characterize its conjugacy classes completely. The answer is as clean as one could hope: conjugacy in S_n is determined entirely by cycle type.

Theorem 4.6.11: Conjugacy Classes in S_n

Two permutations in S_n are conjugate if and only if they have the same cycle type (definition 1.2.15).

***Proof*:**

One direction was established in proposition 1.2.27: if $\tau = \sigma\pi\sigma^{-1}$, then τ has the same cycle type as π, because conjugation by σ sends the cycle $(a_1\ a_2\ \cdots\ a_k)$ to $(\sigma(a_1)\ \sigma(a_2)\ \cdots\ \sigma(a_k))$, preserving the cycle length.

For the converse, suppose π and τ have the same cycle type. Write them in disjoint cycle notation (including 1-cycles for fixed points):

$$\pi = (a_1\ \cdots\ a_{k_1})(b_1\ \cdots\ b_{k_2})\cdots(z_1\ \cdots\ z_{k_m}),$$
$$\tau = (c_1\ \cdots\ c_{k_1})(d_1\ \cdots\ d_{k_2})\cdots(w_1\ \cdots\ w_{k_m}),$$

where the cycle lengths $k_1, k_2, \ldots, k_m$ are the same in both (this is what "same cycle type" means). Define $\sigma \in S_n$ by

$$\sigma(a_i) = c_i, \quad \sigma(b_j) = d_j, \quad \ldots, \quad \sigma(z_l) = w_l$$

for all indices. Since both decompositions account for every element of $\{1, \ldots, n\}$ exactly once, σ is a well-defined bijection. Then

$$\sigma\pi\sigma^{-1} = (\sigma(a_1)\ \cdots\ \sigma(a_{k_1}))(\sigma(b_1)\ \cdots\ \sigma(b_{k_2}))\cdots = (c_1\ \cdots\ c_{k_1})(d_1\ \cdots\ d_{k_2})\cdots = \tau.$$

So π and τ are conjugate.

Example 4.6.12: Conjugacy Classes of S_4

The partitions of 4 are $4 = 4 = 3+1 = 2+2 = 2+1+1 = 1+1+1+1$, giving five cycle types and hence five conjugacy classes:

Cycle type	*Example*	*Description*	*Class size*	$\lvert C_{S_4}(\cdot)\rvert$
(1^4)	e	identity	1	24
$(2, 1^2)$	(1 2)	transpositions	6	4
(2^2)	(1 2)(3 4)	double transpositions	3	8
$(3, 1)$	(1 2 3)	3-cycles	8	3
(4)	(1 2 3 4)	4-cycles	6	4

The class sizes sum to $1+6+3+8+6 = 24 = |S_4|$. The centralizer sizes in the last column satisfy $|\operatorname{cl}(\pi)| \cdot |C_{S_4}(\pi)| = 24$ by (4.3).

Since $Z(S_4) = \{e\}$ (for $n \geq 3$, the center of S_n is trivial), the Class Equation reads:

$$24 = 1 + 6 + 3 + 8 + 6.$$

More generally, there is a formula for the size of each conjugacy class in S_n. If a permutation has cycle type $(1^{a_1}\, 2^{a_2}\, \cdots\, n^{a_n})$ (meaning a_k cycles of length k, with $\sum k\, a_k = n$), then the size of its centralizer is

$$|C_{S_n}(\pi)| \;=\; \prod_{k=1}^{n} k^{a_k} \cdot a_k! \tag{4.4}$$

and the conjugacy class has size $n!/|C_{S_n}(\pi)|$. (The factor k^{a_k} accounts for rotations within each k-cycle, and $a_k!$ accounts for permuting cycles of the same length.) We leave the verification as an exercise.

4.7 Automorphisms and Inner Automorphisms

In Section 4.6, the conjugation action $g \cdot a = gag^{-1}$ gave us the Class Equation by viewing each g as a *permutation* of the underlying set of G. But conjugation does more than permute elements: it preserves the group operation. That is, the map $a \mapsto gag^{-1}$ is not just a bijection $G \to G$ but an *automorphism*. Upgrading the permutation representation from $\mathrm{Sym}(G)$ to $\mathrm{Aut}(G)$ will yield one of the most important structural results in the chapter.

Fix an element $g \in G$ and consider the map

$$\varphi_g \colon G \to G, \qquad \varphi_g(x) = gxg^{-1}.$$

Proposition 4.7.1: Conjugation by g is an Automorphism

For each $g \in G$, the map φ_g is an automorphism of G.

***Proof*:**
We verify that φ_g is a bijective homomorphism.

Homomorphism. For any $x, y \in G$:

$$\varphi_g(xy) = g(xy)g^{-1} = (gxg^{-1})(gyg^{-1}) = \varphi_g(x)\,\varphi_g(y).$$

Bijection. The map $\varphi_{g^{-1}}$ is a two-sided inverse:

$$(\varphi_g \circ \varphi_{g^{-1}})(x) = g(g^{-1}xg)g^{-1} = x = g^{-1}(gxg^{-1})g = (\varphi_{g^{-1}} \circ \varphi_g)(x).$$

Thus $\varphi_g \in \mathrm{Aut}(G)$ (definition 1.4.16).

Since each φ_g is an automorphism, we can ask whether the assignment $g \mapsto \varphi_g$ is compatible with the group structure.

Proposition 4.7.2: The Conjugation Map is a Homomorphism

The map

$$\Phi \colon G \to \mathrm{Aut}(G), \qquad \Phi(g) = \varphi_g,$$

is a group homomorphism, with $\ker(\Phi) = Z(G)$.

***Proof*:**
For any $g, h \in G$ and $x \in G$:

$$\Phi(gh)(x) = \varphi_{gh}(x) = (gh)x(gh)^{-1} = g(hxh^{-1})g^{-1} = \varphi_g(\varphi_h(x)) = (\varphi_g \circ \varphi_h)(x).$$

Since this holds for all x, we have $\Phi(gh) = \Phi(g) \circ \Phi(h)$, so Φ is a homomorphism.

The kernel consists of those g for which φ_g is the identity automorphism:

$$\ker(\Phi) = \{g \in G : gxg^{-1} = x \text{ for all } x \in G\} = \{g \in G : gx = xg \text{ for all } x \in G\} = Z(G).$$

Let us pause to compare this with the conjugation action from Section 4.6. There, we had a homomorphism $\rho\colon G \to \mathrm{Sym}(G)$ (the permutation representation of the conjugation action). Now we have a homomorphism $\Phi\colon G \to \mathrm{Aut}(G)$. Since $\mathrm{Aut}(G) \leq \mathrm{Sym}(G)$ (proposition 1.4.17), these are compatible: Φ is simply ρ with its codomain refined from $\mathrm{Sym}(G)$ to $\mathrm{Aut}(G)$. The kernel is the same in both cases, i.e. $Z(G)$, but the refined codomain gives us access to the richer structure of $\mathrm{Aut}(G)$.

Definition 4.7.3: Inner Automorphism

The image of the conjugation homomorphism $\Phi\colon G \to \mathrm{Aut}(G)$ is called the group of *inner automorphisms* of G:

$$\mathrm{Inn}(G) \;=\; \Phi(G) \;=\; \{\varphi_g : g \in G\} \;\leq\; \mathrm{Aut}(G).$$

An automorphism that is *not* inner is called an *outer automorphism*.

Since Φ is a homomorphism with kernel $Z(G)$ and image $\mathrm{Inn}(G)$, the First Isomorphism Theorem (theorem 3.1.15) immediately yields:

Theorem 4.7.4: $G/Z(G) \cong \mathrm{Inn}(G)$

For any group G,

$$G/Z(G) \;\cong\; \mathrm{Inn}(G).$$

***Proof*:**
Apply the First Isomorphism Theorem to $\Phi\colon G \to \mathrm{Aut}(G)$:

$$G/\ker(\Phi) \;\cong\; \Phi(G), \qquad \text{i.e.,} \qquad G/Z(G) \;\cong\; \mathrm{Inn}(G).$$

Short as it is, this theorem encodes a great deal of structural information. It says that the "non-abelian part" of G, measured by the quotient $G/Z(G)$, is faithfully captured by the inner automorphisms. The larger the center, the fewer inner automorphisms there are, and vice versa.

Example 4.7.5: Inner Automorphisms of Familiar Groups

1. If G is abelian, then $Z(G) = G$, so $\mathrm{Inn}(G) \cong G/G = \{e\}$. Every automorphism of G is outer.
2. For $G = S_3$: $Z(S_3) = \{e\}$, so $\mathrm{Inn}(S_3) \cong S_3/\{e\} \cong S_3$. Since $|\mathrm{Aut}(S_3)| = 6 = |S_3|$ (one can verify that every automorphism of S_3 is inner), we have $\mathrm{Inn}(S_3) = \mathrm{Aut}(S_3)$.
3. For $G = D_4$: $Z(D_4) = \{e, r^2\}$, so $\mathrm{Inn}(D_4) \cong D_4/\{e, r^2\}$, a group of order 4. Since $D_4/\{e, r^2\}$ is non-cyclic (it is generated by two elements of order 2), $\mathrm{Inn}(D_4)$ is isomorphic to the Klein four-group. In fact, $|\mathrm{Aut}(D_4)| = 8$, so $[\mathrm{Aut}(D_4) : \mathrm{Inn}(D_4)] = 2$: there are automorphisms of D_4 that cannot be realized by conjugation.

The construction above has a natural generalization. Instead of conjugating G by itself, we can conjugate any subgroup H by elements that normalize it.

Theorem 4.7.6: The N/C Theorem

Let $H \leq G$. The map

$$\Psi\colon N_G(H) \to \mathrm{Aut}(H), \qquad \Psi(g)(h) = ghg^{-1},$$

is a well-defined homomorphism with $\ker(\Psi) = C_G(H)$. Consequently,

$$N_G(H) \,/\, C_G(H) \;\hookrightarrow\; \mathrm{Aut}(H).$$

In particular, $N_G(H)/C_G(H)$ is isomorphic to a subgroup of $\mathrm{Aut}(H)$.

***Proof*:**
The map $\Psi(g) = \varphi_g|_H$ (conjugation by g, restricted to H) is well-defined because $g \in N_G(H)$ means $gHg^{-1} = H$ (definition 2.2.10), so φ_g sends H to H. That $\varphi_g|_H$ is an automorphism of H follows from proposition 4.7.1 by restriction.

The verification that Ψ is a homomorphism is identical to the proof of proposition 4.7.2. The kernel is

$$\ker(\Psi) = \{g \in N_G(H) : ghg^{-1} = h \text{ for all } h \in H\} = C_G(H) \cap N_G(H) = C_G(H),$$

where the last equality uses $C_G(H) \leq N_G(H)$ (if g commutes with every element of H, then certainly $gHg^{-1} = H$). The embedding follows from the First Isomorphism Theorem (theorem 3.1.15).

When $H \trianglelefteq G$, we have $N_G(H) = G$, so the theorem gives $G/C_G(H) \hookrightarrow \mathrm{Aut}(H)$. When $H = G$ (which is always normal in itself), we recover $G/Z(G) \cong \mathrm{Inn}(G)$ as a special case.

4.7.7 Foreshadowing: Semi-Direct Products The N/C Theorem has an important special case. Suppose $H \trianglelefteq G$ and $K \leq G$. Then $K \leq N_G(H) = G$, so restricting Ψ to K gives a homomorphism

$$\psi\colon K \to \mathrm{Aut}(H), \qquad k \mapsto (h \mapsto khk^{-1}).$$

This homomorphism ψ encodes how K "twists" H by conjugation, and it will be the key ingredient in the definition of the *semi-direct product* (chapter 5). Knowing H, K, and ψ is enough to reconstruct the group G (under suitable conditions).

To define the quotient $\mathrm{Aut}(G)/\mathrm{Inn}(G)$, we first verify that $\mathrm{Inn}(G)$ is normal in $\mathrm{Aut}(G)$.

Proposition 4.7.8: $\mathrm{Inn}(G) \trianglelefteq \mathrm{Aut}(G)$

For any group G, $\mathrm{Inn}(G)$ is a normal subgroup of $\mathrm{Aut}(G)$.

***Proof*:**
Let $\sigma \in \mathrm{Aut}(G)$ and $\varphi_g \in \mathrm{Inn}(G)$. For any $x \in G$:

$$(\sigma \circ \varphi_g \circ \sigma^{-1})(x) = \sigma\big(g \cdot \sigma^{-1}(x) \cdot g^{-1}\big) = \sigma(g) \cdot x \cdot \sigma(g)^{-1} = \varphi_{\sigma(g)}(x).$$

Thus $\sigma \circ \varphi_g \circ \sigma^{-1} = \varphi_{\sigma(g)} \in \mathrm{Inn}(G)$, and $\mathrm{Inn}(G) \trianglelefteq \mathrm{Aut}(G)$.

Definition 4.7.9: Outer Automorphism Group

The *outer automorphism group* of G is the quotient

$$\mathrm{Out}(G) \;=\; \mathrm{Aut}(G) \,/\, \mathrm{Inn}(G).$$

The outer automorphism group measures how much of $\mathrm{Aut}(G)$ is *not* explained by conjugation. A group with $\mathrm{Out}(G) = \{e\}$ is called *complete* (when it also has trivial center). In such a group, every automorphism is inner.

Example 4.7.10: Outer Automorphisms

1. If G is abelian, then $\mathrm{Inn}(G) = \{e\}$, so $\mathrm{Out}(G) \cong \mathrm{Aut}(G)$. All automorphisms are outer.
2. $\mathrm{Out}(S_n) = \{e\}$ for $n \neq 6$: every automorphism of S_n is inner. The case $n = 6$ is exceptional: $|\mathrm{Out}(S_6)| = 2$, giving S_6 a unique outer automorphism (up to composition with inner ones).
3. $\mathrm{Out}(D_4) \cong \mathbb{Z}/2\mathbb{Z}$: there is one nontrivial coset of $\mathrm{Inn}(D_4)$ in $\mathrm{Aut}(D_4)$.

4.7.11 Foreshadowing: Matrix Groups Have Outer Automorphisms As $\mathrm{Aut}(S_n) = \mathrm{Inn}(S_n)$, and $S_n \hookrightarrow M_n(F)$ by permuting the columns (or rows), it is tempting to guess that this must imply enough rigidity so that the group of matrices has only inner automorphisms, i.e. conjugation. As it turns out, this is not the case! On $G = GL_n(F)$ for a field F (definition 1.2.33), both

$$\tau(A) = (A^T)^{-1}$$

and, for $F = \mathbb{C}$,

$$\kappa(A) = \overline{A} \text{ (conjugate every entry)}$$

are automorphisms (check that each respects products and has an inverse). Neither is inner: the key is that conjugation preserves the determinant, since $\det(BAB^{-1}) = \det A$. But for $A = \mathrm{diag}(2, 1, \ldots, 1) \in GL_n(\mathbb{R})$ we get $\det \tau(A) = \frac{1}{2} \neq 2$, and for $A = \mathrm{diag}(i, 1, \ldots, 1) \in GL_n(\mathbb{C})$ we get $\det \kappa(A) = -i \neq i$. So $\mathrm{Out}(GL_n(\mathbb{R}))$ and $\mathrm{Out}(GL_n(\mathbb{C}))$ are nontrivial.

However, these are essentially the only sources. Schreier and van der Waerden [55] proved that for $n \geq 3$, every automorphism of the quotient of $SL_n(F)$ by its scalar matrices comes from conjugation by an invertible matrix, an automorphism of the field F applied to the entries, and possibly τ. Why exactly these appear is explained in *Geometric Representation Theory* [22].

For later use, the following short exact sequence that encodes the relationship between the center, the group, and its inner automorphisms is written out explicitly:

$$1 \longrightarrow Z(G) \hookrightarrow G \xrightarrow{\Phi} \mathrm{Inn}(G) \longrightarrow 1 \tag{4.5}$$

In particular, $Z(G)$ measures how far G is from being faithfully represented by its inner automorphisms. $\mathrm{Inn}(G)$ captures the "non-central" part of G. The center is precisely the part that acts trivially by conjugation.

4.7.1 Characteristic Subgroups

Normal subgroups are defined by invariance under *inner* automorphisms: H is normal in G if $\varphi_g(H) = H$ for every $\varphi_g \in \mathrm{Inn}(G)$. A natural strengthening is to require invariance under *all* automorphisms.

Definition 4.7.12: Characteristic Subgroup

A subgroup $H \leq G$ is *characteristic* in G, written $H \operatorname{char} G$, if

$$\sigma(H) = H \qquad \text{for every } \sigma \in \mathrm{Aut}(G).$$

Proposition 4.7.13: Properties of Characteristic Subgroups

1. If $H \operatorname{char} G$, then $H \trianglelefteq G$.
2. If H is the unique subgroup of G of a given order, then $H \operatorname{char} G$.
3. If $H \operatorname{char} K$ and $K \trianglelefteq G$, then $H \trianglelefteq G$.

***Proof*:**

1. Since $\mathrm{Inn}(G) \leq \mathrm{Aut}(G)$, invariance under all automorphisms implies invariance under inner automorphisms: $gHg^{-1} = \varphi_g(H) = H$ for all $g \in G$.
2. Let $\sigma \in \mathrm{Aut}(G)$. Since σ is an isomorphism, $\sigma(H)$ is a subgroup of G with $|\sigma(H)| = |H|$. By uniqueness, $\sigma(H) = H$.
3. Let $g \in G$. Since $K \trianglelefteq G$, the inner automorphism φ_g restricts to an automorphism $\varphi_g|_K \in \mathrm{Aut}(K)$ (because $gKg^{-1} = K$). Since $H \operatorname{char} K$, this automorphism preserves H:

   $$gHg^{-1} = \varphi_g(H) = (\varphi_g|_K)(H) = H.$$

   Since this holds for every $g \in G$, $H \trianglelefteq G$.

Item 3 is the principal reason characteristic subgroups are useful. Recall that normality is *not* transitive: $H \trianglelefteq K$ and $K \trianglelefteq G$ does *not* imply $H \trianglelefteq G$ in general (example 3.7.4). Characteristic subgroups provide a partial remedy: if the inner link is characteristic (rather than only normal), transitivity is restored.

Example 4.7.14: Characteristic Subgroups

1. The center $Z(G)$ is characteristic in G: if $\sigma \in \mathrm{Aut}(G)$ and $z \in Z(G)$, then for any $x \in G$,

   $$\sigma(z)\,x = \sigma(z)\,\sigma(\sigma^{-1}(x)) = \sigma(z \cdot \sigma^{-1}(x)) = \sigma(\sigma^{-1}(x) \cdot z) = x\,\sigma(z),$$

   so $\sigma(z) \in Z(G)$. Thus $\sigma(Z(G)) \subseteq Z(G)$, and applying the same argument to σ^{-1} gives equality.
2. The commutator subgroup $G' = \langle xyx^{-1}y^{-1} : x, y \in G \rangle$ is characteristic in G: any automorphism σ sends commutators to commutators ($\sigma(xyx^{-1}y^{-1}) = \sigma(x)\sigma(y)\sigma(x)^{-1}\sigma(y)^{-1}$), so

$\sigma(G') = G'$.

3. Every Sylow p-subgroup of a group G is normal if and only if it is the *unique* Sylow p-subgroup. By item 2 of proposition 4.7.13, such a subgroup is automatically characteristic. We will use this in Section 4.8.

4.7.15 Remark: Normality vs. Characteristic The logical chain is:

$$\text{characteristic} \implies \text{normal} \implies \text{subgroup},$$

and neither implication reverses. A normal subgroup that is not characteristic: consider $G = \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$ and $H = \{(0,0),(1,0)\}$. Then $H \trianglelefteq G$ (since G is abelian), but the automorphism swapping the two coordinates sends H to $\{(0,0),(0,1)\} \neq H$. So H is normal but not characteristic.

We summarize the automorphism-related groups and their relationships in a single diagram:

$$\begin{array}{ccccccccc} 1 & \longrightarrow & Z(G) & \hookrightarrow & G & \xrightarrow{\Phi} & \mathrm{Inn}(G) & \hookrightarrow & \mathrm{Aut}(G) & \twoheadrightarrow & \mathrm{Out}(G) & \longrightarrow & 1 \\ & & & & & & \rotatebox{90}{$\hookrightarrow$} & & & & & & \\ & & & & & & \mathrm{Aut}(G) & & & & & & \end{array}$$

The top row is exact: $Z(G)$ is the kernel of Φ, $\mathrm{Inn}(G)$ is its image (hence the kernel of the projection to $\mathrm{Out}(G)$), and $\mathrm{Out}(G)$ is the cokernel. This picture captures the hierarchy at a glance: $Z(G)$ measures commutativity, $\mathrm{Inn}(G)$ measures conjugation, and $\mathrm{Out}(G)$ measures the "exotic" automorphisms that conjugation cannot produce.

4.8 Sylow's Theorems

Lagrange's Theorem (theorem 3.3.1) shows that if $H \leq G$, then $|H|$ divides $|G|$. The converse is false in general: A_4 has order 12 but no subgroup of order 6 (example 3.3.8). Sylow's Theorems provide the best *partial converse*: for every prime power p^α dividing $|G|$, a subgroup of order p^α not only exists but is essentially unique up to conjugation, and the number of such subgroups is tightly constrained. These are among the most powerful and frequently used results in finite group theory, and their proofs are an applications of everything we have built in this chapter: the Orbit-Stabilizer Theorem, the Class Equation, and the coset action.

Definition 4.8.1: p-Group

Let p be a prime. A group P is called a *p-group* if $|P| = p^k$ for some $k \geq 0$. A subgroup $H \leq G$ that is a p-group is called a *p-subgroup* of G.

Definition 4.8.2: Sylow p-Subgroup

Let G be a finite group and write $|G| = p^\alpha m$ where $\alpha \geq 0$ and $p \nmid m$ (so p^α is the largest power of p dividing $|G|$). A subgroup $P \leq G$ of order p^α is called a *Sylow p-subgroup* of G. We write $\mathrm{Syl}_p(G)$ for the set of all Sylow p-subgroups of G, and $n_p = n_p(G) = |\mathrm{Syl}_p(G)|$ for their number.

Example 4.8.3: Sylow Subgroups of Small Groups

1. $|S_3| = 6 = 2 \cdot 3$. Sylow 2-subgroups have order 2: these are $\langle(1\ 2)\rangle$, $\langle(1\ 3)\rangle$, $\langle(2\ 3)\rangle$. Sylow 3-subgroups have order 3: only $\langle(1\ 2\ 3)\rangle$. So $n_2 = 3$ and $n_3 = 1$.
2. $|A_4| = 12 = 2^2 \cdot 3$. Sylow 2-subgroups have order 4. Sylow 3-subgroups have order 3.
3. If $|G| = p^\alpha$ (a p-group), then G itself is its only Sylow p-subgroup, so $n_p = 1$.
4. $|\mathbb{Z}/n\mathbb{Z}| = n$. For each prime p dividing n, the unique subgroup of order p^α (where $p^\alpha \| n$) is the unique Sylow p-subgroup. (Here uniqueness follows from the fact that $\mathbb{Z}/n\mathbb{Z}$ has exactly one subgroup of each order dividing n.)

Before proving Sylow's Theorems, we isolate a lemma that will be used repeatedly. It says that when a p-group acts on a finite set, the number of fixed points is congruent to the total size of the set modulo p.

Lemma 4.8.4: Fixed-Point Congruence for p-Group Actions

Let P be a finite p-group acting on a finite set X. Then

$$|X| \equiv |X^P| \pmod{p},$$

where $X^P = \{x \in X : g \cdot x = x \text{ for all } g \in P\}$ is the set of fixed points (definition 4.2.8).

***Proof*:**
The orbits of P on X partition X. By the Orbit-Stabilizer Theorem (theorem 4.3.1), each orbit has size $[P : P_x] = |P|/|P_x|$, which divides $|P| = p^k$, so every orbit size is a power of p. An orbit has size 1 (i.e. p^0) if and only if it is a fixed point. Every other orbit has size p^j with $j \geq 1$, hence divisible by p. Therefore

$$|X| = |X^P| + \sum_{\substack{\text{non-fixed} \\ \text{orbits}}} p^{j_i} \equiv |X^P| \pmod{p}.$$

This lemma is the mechanism behind many counting arguments in p-group theory. For instance, theorem 4.6.5 (p-groups have nontrivial center) is a special case: applying the lemma to the conjugation action $G \curvearrowright G$ with $P = G$ gives $|G| \equiv |Z(G)| \pmod{p}$, so $p \big| |Z(G)|$. This lemma also gives an immediate proof of Cauchy's theorem corollary 3.3.9 worth pointing out:

Corollary 4.8.5: Cauchy Theorem using Fixed Point Theorem

Let G be a finite abelian group and p a prime dividing $|G|$. Then there exists a subgroup $P \leqslant G$ of order p.

***Proof*:**
Take the set

$$X = \{(g_1, ..., g_p) \in G^p \mid g_1 g_2 \cdots g_p = 1\}$$

This set has order $|G|^{p-1}$, for after choosing the elements $g_1, ...g_{p-1}$, the choice of g_p is forced:

$$g_p = (g_1 \cdots g_{p-1})^{-1}$$

Next, the shift action $\sigma : X \to X$ given by $\sigma(g_1, g_2, ..., g_p) = (g_2, g_3, ..., g_p, g_1)$ is well defined, since if $a = g_1$ and $b = g_2 \cdots g_p$ then $ab = 1$ and $b^{-1} = a$ meaning $ba = 1$ corollary 1.1.14. Then as $\sigma^k = \mathrm{id}$, this defines a $\mathbb{Z}/p\mathbb{Z}$-action on X. As $|\mathbb{Z}/p\mathbb{Z}| = p$, by lemma 4.8.4:

$$\left|X^{(\mathbb{Z}/p\mathbb{Z})}\right| \equiv |X| \equiv |G|^{p-1} \equiv 0 \bmod p$$

Hence $p \big| |X^{(\mathbb{Z}/p\mathbb{Z})}|$ implying there is a non-trivial element g such that $g^p = 1$, showing g has order p, and hence $\langle p \rangle \leqslant G$, completing the proof.10425

Theorem 4.8.6: Sylow's Theorems

Let G be a finite group with $|G| = p^\alpha m$, where p is prime and $p \nmid m$.

1. *(Existence.)* G has a Sylow p-subgroup, i.e. a subgroup of order p^α.
2. *(Conjugacy.)* Any two Sylow p-subgroups of G are conjugate. More generally, if P is a Sylow p-subgroup and Q is any p-subgroup of G, then $Q \leq gPg^{-1}$ for some $g \in G$.
3. *(Counting.)* The number n_p of Sylow p-subgroups satisfies:
   (a) $n_p = [G : N_G(P)]$ for any $P \in \mathrm{Syl}_p(G)$,
   (b) n_p divides m,
   (c) $n_p \equiv 1 \pmod p$.

We prove each part in turn.

***Proof of Sylow I*:**
We proceed by strong induction on $|G|$. If $|G| = 1$ (so $\alpha = 0$), the trivial subgroup has order $p^0 = 1$ and there is nothing to prove. Suppose $|G| > 1$ and that the result holds for all groups of order less than $|G|$. Write $|G| = p^\alpha m$ with $\alpha \geq 1$ and $p \nmid m$.

Consider the Class Equation (theorem 4.6.3):

$$|G| = |Z(G)| + \sum_{i=1}^{r} [G : C_G(g_i)].$$

Case 1: $p \big| |Z(G)|$. Since $Z(G)$ is abelian, Cauchy's Theorem for abelian groups (corollary 3.3.9) gives an element $z \in Z(G)$ of order p. Let $N = \langle z \rangle$. Since $N \leq Z(G)$, we have $N \trianglelefteq G$, and $|G/N| = p^{\alpha-1}m < |G|$.

By the induction hypothesis, G/N has a subgroup $\overline{P}$ of order $p^{\alpha-1}$. By the Correspondence Theorem (theorem 3.5.7), there exists $P \leq G$ with $N \leq P$ and $P/N = \overline{P}$. Then

$$|P| = |P/N| \cdot |N| = p^{\alpha-1} \cdot p = p^{\alpha}.$$

Case 2: $p \nmid |Z(G)|$. Since $p \big| |G|$ and the Class Equation gives $|Z(G)| = |G| - \sum [G : C_G(g_i)]$, if p divided every term $[G : C_G(g_i)]$, then p would divide $|Z(G)|$, a contradiction. So there exists some g_j with

$$p \nmid [G : C_G(g_j)].$$

Since $|G| = |C_G(g_j)| \cdot [G : C_G(g_j)]$ and $p^{\alpha} \big| |G|$ while $p \nmid [G : C_G(g_j)]$, we conclude $p^{\alpha} \big| |C_G(g_j)|$. But $g_j \notin Z(G)$, so $C_G(g_j) \neq G$ and $|C_G(g_j)| < |G|$.

By the induction hypothesis, $C_G(g_j)$ has a subgroup P of order p^{α}. Since $P \leq C_G(g_j) \leq G$ and $|P| = p^{\alpha}$, P is a Sylow p-subgroup of G. □

***Proof of Sylow II*:**
Let $P \in \mathrm{Syl}_p(G)$ and let Q be any p-subgroup of G. Consider the set of left cosets G/P, and let Q act on G/P by left multiplication: $q \cdot gP = (qg)P$.

Since $|G/P| = [G : P] = m$ and $p \nmid m$, the Fixed-Point Congruence (lemma 4.8.4) gives

$$m = |G/P| \equiv |(G/P)^Q| \pmod{p}.$$

Since $p \nmid m$, we have $(G/P)^Q \neq \varnothing$: there is at least one fixed point.

A coset gP is fixed by all $q \in Q$ if and only if $qgP = gP$ for all $q \in Q$, i.e. $g^{-1}qg \in P$ for all $q \in Q$, i.e. $g^{-1}Qg \leq P$, i.e. $Q \leq gPg^{-1}$.

So there exists $g \in G$ with $Q \leq gPg^{-1}$. If Q is itself a Sylow p-subgroup, then $|Q| = p^{\alpha} = |gPg^{-1}|$, and the inclusion forces $Q = gPg^{-1}$: all Sylow p-subgroups are conjugate. □

***Proof of Sylow III*:**
Let $\mathcal{S} = \mathrm{Syl}_p(G)$ and $n_p = |\mathcal{S}|$.

Part (a): $n_p = [G : N_G(P)]$. The group G acts on $\mathcal{S}$ by conjugation: $g \cdot P_i = gP_ig^{-1}$. By Sylow II, this action is transitive (any two Sylow p-subgroups lie in the same orbit). The stabilizer of P under this action is

$$G_P = \{g \in G : gPg^{-1} = P\} = N_G(P).$$

By the Orbit-Stabilizer Theorem (theorem 4.3.1), $n_p = |\mathcal{S}| = [G : N_G(P)]$.

Part (b): $n_p \big| m$. Since $P \leq N_G(P) \leq G$, the index decomposes as

$$[G : P] = [G : N_G(P)] \cdot [N_G(P) : P].$$

Thus $n_p = [G : N_G(P)]$ divides $[G : P] = m$.

Part (c): $n_p \equiv 1 \pmod{p}$. Fix $P_1 \in \mathcal{S}$ and let P_1 act on $\mathcal{S}$ by conjugation. By the Fixed-Point Congruence (lemma 4.8.4),

$$n_p \equiv |\mathcal{S}^{P_1}| \pmod{p},$$

where $\mathcal{S}^{P_1}$ is the set of Sylow p-subgroups fixed by every element of P_1.

We claim $\mathcal{S}^{P_1} = \{P_1\}$. Suppose $P_j \in \mathcal{S}^{P_1}$, so $P_1 \leq N_G(P_j)$. Since $P_j \trianglelefteq N_G(P_j)$ and $P_1 \leq N_G(P_j)$, the product P_1P_j is a subgroup of $N_G(P_j)$ (proposition 3.4.4). Its order is

$$|P_1P_j| = \frac{|P_1| \cdot |P_j|}{|P_1 \cap P_j|} = \frac{p^\alpha \cdot p^\alpha}{|P_1 \cap P_j|} = \frac{p^{2\alpha}}{|P_1 \cap P_j|},$$

which is a power of p (since $P_1 \cap P_j$ is a p-group). So P_1P_j is a p-subgroup of G containing P_1. Since P_1 is a *maximal* p-subgroup (it has order p^α), we must have $P_1P_j = P_1$, which gives $P_j \leq P_1$. Since $|P_j| = |P_1| = p^\alpha$, it follows that $P_j = P_1$.

Thus $|\mathcal{S}^{P_1}| = 1$, and $n_p \equiv 1 \pmod{p}$. □

With Sylow's Theorems in hand, we can immediately prove the full version of Cauchy's Theorem for arbitrary finite groups, completing the proof that was deferred in section 3.3:

Theorem 4.8.7: Cauchy's Theorem

If G is a finite group and p is a prime dividing $|G|$, then G has an element of order p.

***Proof*:**
Write $|G| = p^\alpha m$ with $\alpha \geq 1$ and $p \nmid m$. By Sylow I, G has a Sylow p-subgroup P of order p^α. Let $x \in P$ with $x \neq e$. Then $|\langle x \rangle|$ divides $|P| = p^\alpha$ (theorem 3.3.1), so $|x| = p^\beta$ for some $1 \leq \beta \leq \alpha$. The element $y = x^{p^{\beta-1}}$ satisfies

$$|y| = \frac{|x|}{(|x|,\, p^{\beta-1})} = \frac{p^\beta}{p^{\beta-1}} = p,$$

where we used theorem 2.3.11. Since $y \in P \leq G$, G has an element of order p.

Note the logical structure: the abelian case of Cauchy's Theorem (corollary 3.3.9) was used *in the proof of Sylow I*, and the full Cauchy's Theorem is now derived *from* Sylow I. There is no circular reasoning.

Example 4.8.8
Let $G \neq 1$ be a finite abelian group whose only normal subgroups are 1 and G. Then $G \cong Z_p$ for a prime p.

***Proof*:**
Since $|G| > 1$, some prime p divides $|G|$, and theorem 4.8.7 gives an element $x \in G$ of order p. The subgroup $\langle x \rangle$ has order p (proposition 2.3.4) and is normal by corollary 3.1.12, so $\langle x \rangle = G$. Thus G is cyclic of order p, and $G \cong Z_p$ by theorem 2.3.9, as claimed.

Consequences

The following corollary collects the most frequently used structural consequences.

Corollary 4.8.9: Characterizations of a Unique Sylow Subgroup

Let $P \in \mathrm{Syl}_p(G)$. The following are equivalent:

1. $n_p = 1$ (i.e. P is the unique Sylow p-subgroup of G).
2. $P \trianglelefteq G$.
3. $P \operatorname{char} G$.

***Proof*:**
Item 1 $\Rightarrow$ item 3: If P is the unique subgroup of G of order p^α, then P is characteristic by item 2 of proposition 4.7.13.

Item 3 $\Rightarrow$ item 2: Characteristic implies normal (item 1 of proposition 4.7.13).

Item 2 $\Rightarrow$ item 1: If $P \trianglelefteq G$ and Q is any other Sylow p-subgroup, then by Sylow II, $Q = gPg^{-1}$ for some $g \in G$. Since P is normal, $gPg^{-1} = P$, so $Q = P$. Thus $n_p = 1$.

In particular, to show that a Sylow subgroup is normal, it suffices to show $n_p = 1$. The congruence and divisibility conditions from Sylow III often force this conclusion.

Example 4.8.10: Using Sylow's Theorems

1. Let $|G| = 15 = 3 \cdot 5$. Then $n_3 \mid 5$ and $n_3 \equiv 1 \pmod 3$, giving $n_3 \in \{1\}$. Similarly, $n_5 \mid 3$ and $n_5 \equiv 1 \pmod 5$, giving $n_5 \in \{1\}$. Both Sylow subgroups are normal and unique, each cyclic of prime order.
2. Let $|G| = 12 = 2^2 \cdot 3$. Then $n_3 \mid 4$ and $n_3 \equiv 1 \pmod 3$, giving $n_3 \in \{1, 4\}$. Also, $n_2 \mid 3$ and $n_2 \equiv 1 \pmod 2$, giving $n_2 \in \{1, 3\}$. Both possibilities genuinely occur: for A_4, $n_3 = 4$ and $n_2 = 1$; for $\mathbb{Z}/12\mathbb{Z}$, $n_3 = 1$ and $n_2 = 1$.
3. Let $|G| = 99 = 9 \cdot 11$. Then $n_{11} \mid 9$ and $n_{11} \equiv 1 \pmod{11}$. The divisors of 9 are $1, 3, 9$, and only $1 \equiv 1 \pmod{11}$. So $n_{11} = 1$ and the Sylow 11-subgroup is normal.

4.8.11 A Strategy for Sylow Problems Many problems in finite group theory reduce to the following systematic procedure:

1. Write $|G| = p^\alpha m$ with $p \nmid m$ for each prime p dividing $|G|$.
2. For each p, list the candidates for n_p: they must simultaneously divide m and be congruent to $1 \pmod{p}$.
3. Use the candidates to deduce structural information: if $n_p = 1$, the Sylow p-subgroup is normal; if $n_p > 1$, count the elements of order p^k across all Sylow subgroups and check for contradictions.

We will apply this strategy systematically in Section 4.9.

4.8.12 Applied Aside: Noether's Theorem in Physics While Sylow's Theorems concern finite groups, the underlying philosophy (that symmetries constrain structure) extends far beyond algebra. In 1918, Emmy Noether proved that every continuous symmetry of a physical system corresponds to a conserved quantity:

Symmetry (group action)	*Conservation law*
Time translation	Energy
Spatial translation	Momentum
Rotational symmetry	Angular momentum

Noether's Theorem uses *Lie groups* (continuous groups acting smoothly on manifolds), but the conceptual root is identical to everything in this chapter: a group acts on a space, and the structure of the action (its orbits, its stabilizers, its kernel) reveals deep invariants. The conservation laws of physics are, in a precise sense, consequences of group actions.

Exercise 4.8.1

1. Let p be prime, P a p-group, and $N \trianglelefteq G$.
   a) Prove that if $N \neq \{e\}$, then $N \cap Z(P) \neq \{e\}$
   b) Prove that for all $i \geq 0$ where $p^i \big| |N|$, then N contains a subgroup of order p^i that is normal in P. Show these groups can form a central series.
   c) Let G be any finite group. Show that for any integer $i \geq 0$, if $p^i \big| G$, then there exists a subgroup $H \leqslant G$ where $|H| = p^i$. By example 3.3.8, this is the strongest converse to Lagrange's Theorem (theorem 3.3.1).

4.9 Applications of Sylow's Theorems

We now put the Sylow machinery to work. The goal is to demonstrate, through a sequence of increasingly sophisticated examples, how the existence, conjugacy, and counting theorems combine to

constrain, and sometimes completely determine, the structure of a finite group from its order alone.

Groups of Order pq

Let $p < q$ be distinct primes and let $|G| = pq$. This is the simplest case beyond prime and prime-squared orders, and Sylow's Theorems give a nearly complete classification.

Theorem 4.9.1: Classification of Groups of Order pq

Let $p < q$ be primes and $|G| = pq$.

1. G has a unique (hence normal) Sylow q-subgroup.
2. If $p \nmid (q-1)$, then G is cyclic: $G \cong \mathbb{Z}/pq\mathbb{Z}$.
3. If $p \mid (q-1)$ and G is not cyclic, then G is non-abelian and $n_p = q$. For $p = 2$ there is, up to isomorphism, exactly one non-abelian group of order $2q$, the dihedral group D_q (definition 1.2.28).

***Proof*:**

1. By Sylow III, $n_q \mid p$ and $n_q \equiv 1 \pmod{q}$. The divisors of p are 1 and p. Since $p < q$, we have $p \not\equiv 1 \pmod{q}$ (as $1 < p < q$). Therefore $n_q = 1$, and the unique Sylow q-subgroup Q is normal in G (corollary 4.8.9).

2. By Sylow III, $n_p \mid q$ and $n_p \equiv 1 \pmod{p}$. Since q is prime, $n_p \in \{1, q\}$. If $n_p = q$, then $q \equiv 1 \pmod{p}$, i.e. $p \mid (q-1)$. Contrapositively, if $p \nmid (q-1)$, then $n_p = 1$, and the Sylow p-subgroup P is also normal.

   Now $P \cong \mathbb{Z}/p\mathbb{Z}$ and $Q \cong \mathbb{Z}/q\mathbb{Z}$ (both have prime order, hence are cyclic by corollary 3.3.2). Since $P \cap Q$ is a subgroup of both P and Q, its order divides $\gcd(p, q) = 1$, so $P \cap Q = \{e\}$. By proposition 3.4.3, $|PQ| = |P||Q|/|P \cap Q| = pq = |G|$, so $PQ = G$.

   We claim G is abelian. For any $x \in P$ and $y \in Q$, the commutator $xyx^{-1}y^{-1}$ lies in Q (since $Q \trianglelefteq G$ gives $xyx^{-1} \in Q$, so $xyx^{-1}y^{-1} \in Q$) and also in P (since $P \trianglelefteq G$ gives $yx^{-1}y^{-1} \in P$, so $xyx^{-1}y^{-1} \in P$). Thus $xyx^{-1}y^{-1} \in P \cap Q = \{e\}$, giving $xy = yx$.

   Since every element of $G = PQ$ is a product of an element of P and an element of Q, and these all commute, G is abelian. Let x generate P and y generate Q. Then $|xy| = \operatorname{lcm}(p, q) = pq$ (since $\gcd(p, q) = 1$), so $G = \langle xy \rangle \cong \mathbb{Z}/pq\mathbb{Z}$.

3. Suppose $p \mid (q-1)$. If G is abelian, then P and Q are both normal, and the argument of (2) shows $G \cong \mathbb{Z}/pq\mathbb{Z}$. So if G is not cyclic, it is non-abelian, the same argument shows that P is not normal, and $n_p = q$ by the count in (2). For $p = 2$ a non-abelian group of order $2q$ exists, the dihedral group D_q, of order $2q$ (proposition 1.2.30) and non-abelian (proposition 1.2.29(5)), with $D_3 = S_3$ for $q = 3$.

   For $p = 2$ the non-abelian group is unique up to isomorphism. Let G be non-abelian of order $2q$, let $Q = \langle y \rangle$ be its normal Sylow q-subgroup from (1), and let $x \in G$ have order 2 (theorem 4.8.7). Then $xyx^{-1} \in Q$, say $xyx^{-1} = y^s$, and

$$y = x^2yx^{-2} = xy^sx^{-1} = y^{s^2}$$

so q divides $s^2 - 1 = (s-1)(s+1)$, and since q is prime it divides $s-1$ or $s+1$. If $q \mid s-1$, then x commutes with y, and $G = \langle x, y\rangle$ (its order is divisible by 2 and by q) would be abelian. Hence $xyx^{-1} = y^{-1}$, and $xy^i = y^{-i}x$ for every i. Since $\langle x\rangle \cap Q = 1$, the $2q$ elements $y^i x^j$ with $0 \le i < q$ and $0 \le j < 2$ are distinct, so they are all of G, and

$$(y^i x^j)(y^k x^l) = y^{i+(-1)^j k} x^{j+l}$$

determines every product. Two non-abelian groups of order $2q$ therefore have the same multiplication in these coordinates, and $y^i x^j \mapsto y'^i x'^j$ is an isomorphism between them.

For odd p with $p \mid q-1$ the same classification holds: a non-abelian group of order pq exists (corollary 5.3.5, using semi-direct products) and is unique up to isomorphism (corollary 11.3.16, using the cyclicity of $(\mathbb{Z}/q\mathbb{Z})^\times$).

Example 4.9.2: Concrete Cases

1. $|G| = 15 = 3 \cdot 5$. Here $p = 3, q = 5$, and $3 \nmid 4$. So $G \cong \mathbb{Z}/15\mathbb{Z}$: there is only one group of order 15.
2. $|G| = 35 = 5 \cdot 7$. Here $p = 5, q = 7$, and $5 \nmid 6$. So $G \cong \mathbb{Z}/35\mathbb{Z}$.
3. $|G| = 6 = 2 \cdot 3$. Here $p = 2, q = 3$, and $2 \mid 2$. There are two groups: $\mathbb{Z}/6\mathbb{Z}$ and the non-abelian group S_3.
4. $|G| = 21 = 3 \cdot 7$. Here $p = 3, q = 7$, and $3 \mid 6$. There are two groups: $\mathbb{Z}/21\mathbb{Z}$ and a non-abelian group of order 21 (corollaries 5.3.5 and 11.3.16).

Groups of Order p^2q: A Worked Example

We illustrate the general strategy for $|G| = p^2q$ ($p \neq q$ primes) through the concrete case $|G| = 12 = 2^2 \cdot 3$.

Example 4.9.3: Sylow Analysis of Groups of Order 12

Let $|G| = 12$. The Sylow constraints give:

$$n_3 \mid 4 \text{ and } n_3 \equiv 1 \pmod 3 \implies n_3 \in \{1, 4\},$$
$$n_2 \mid 3 \text{ and } n_2 \equiv 1 \pmod 2 \implies n_2 \in \{1, 3\}.$$

Case $n_3 = 4$. There are 4 Sylow 3-subgroups, each cyclic of order 3 with 2 non-identity elements. Since distinct Sylow 3-subgroups intersect trivially (their intersection has order dividing 3 and is a proper subgroup of each, so it is $\{e\}$), there are $4 \times 2 = 8$ elements of order 3. The remaining $12 - 8 = 4$ elements (including e) must form the unique Sylow 2-subgroup. Thus $n_2 = 1$ and the Sylow 2-subgroup P is normal.

Since P has order 4, $P \cong \mathbb{Z}/4\mathbb{Z}$ or $P \cong V_4 \cong \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$ (corollary 4.6.6). The group A_4 is an example of this case, with $P \cong V_4$ and $n_3 = 4$.

Case $n_3 = 1$. The unique Sylow 3-subgroup $Q \cong \mathbb{Z}/3\mathbb{Z}$ is normal. The Sylow 2-subgroup P (order 4) acts on Q by conjugation, giving a homomorphism $P \to \mathrm{Aut}(Q) \cong \mathbb{Z}/2\mathbb{Z}$ (by theorem 4.7.6).

The structure of G depends on this homomorphism: different choices yield the cyclic group $\mathbb{Z}/12\mathbb{Z}$, the dihedral group D_6, and the dicyclic group Dic_3 (among others, depending on whether $P \cong V_4$ or $P \cong \mathbb{Z}/4\mathbb{Z}$).

The key technique on display is *element counting*: when n_p is large, the Sylow p-subgroups account for many elements, leaving fewer elements for Sylow subgroups at other primes. This often forces $n_q = 1$ at a different prime, yielding a normal subgroup.

The same strategy applies to any order p^2q. In general:

1. Compute the Sylow constraints for n_p and n_q.
2. If both $n_p = 1$ and $n_q = 1$ are forced, both Sylow subgroups are normal, and G is a semi-direct product (or direct product) of a p-group and a q-group.
3. If $n_q > 1$ is possible, use element counting to check whether $n_p = 1$ is forced. If so, the normal Sylow p-subgroup provides structural leverage.
4. The full classification typically requires the semi-direct product machinery of chapter 5.

4.9.1 Simplicity of A_5

The alternating group A_5 is simple. Through Galois theory, this is the group-theoretic reason that the general quintic equation cannot be solved by radicals.

Lemma 4.9.4: Splitting Of Conjugacy Classes In A_n

Let $n \geq 2$, let $\sigma \in A_n$, and let K be the conjugacy class of σ in S_n.

1. If $C_{S_n}(\sigma)$ contains an odd permutation, then K is the conjugacy class of σ in A_n.
2. If every element of $C_{S_n}(\sigma)$ is even, then K is the union of two conjugacy classes of A_n, each of size $|K|/2$, namely the classes of σ and of $\tau\sigma\tau^{-1}$ for any odd $\tau \in S_n$.

***Proof*:**
Write $C = C_{S_n}(\sigma)$, and write $\mathrm{cl}_{A_n}(\sigma)$ for the conjugacy class of σ in A_n. By theorem 1.2.20(2), $\mathrm{sgn}(\pi^{-1}) = \mathrm{sgn}(\pi)^{-1}$, so $\mathrm{sgn}(\pi\rho\pi^{-1}) = \mathrm{sgn}(\rho)$ for all $\pi, \rho \in S_n$. In particular $K \subseteq A_n$ and $\mathrm{cl}_{A_n}(\sigma) \subseteq K$. The centralizer of σ in A_n is $C \cap A_n$, and $|A_n| = n!/2$ by example 3.5.2(1). By (4.3), applied in S_n and in A_n, and theorem 3.3.1,

$$|K| = \frac{n!}{|C|} \qquad |\,\mathrm{cl}_{A_n}(\sigma)| = \frac{n!}{2\,|C \cap A_n|}$$

1. Let $\tau \in C$ be odd. If $x \in C$, then $\tau x \in C$, since $\tau x \sigma = \tau\sigma x = \sigma\tau x$, and $\mathrm{sgn}(\tau x) = -\mathrm{sgn}(x)$. So $x \mapsto \tau x$ maps the even elements of C to the odd ones, and $y \mapsto \tau^{-1}y$ maps the odd ones back, because τ^{-1} also commutes with σ. Hence exactly half of the elements of C are even, $|C \cap A_n| = |C|/2$, and $|\,\mathrm{cl}_{A_n}(\sigma)| = n!/|C| = |K|$. Since $\mathrm{cl}_{A_n}(\sigma) \subseteq K$ and both sets are finite, $\mathrm{cl}_{A_n}(\sigma) = K$.
2. Now $C \cap A_n = C$, so $|\,\mathrm{cl}_{A_n}(\sigma)| = |K|/2$. Let $\tau \in S_n$ be odd, for instance $\tau = (1\ 2)$, and put $\sigma' = \tau\sigma\tau^{-1} \in K$. An element $x \in S_n$ commutes with σ' exactly when $\tau^{-1}x\tau$ commutes

with σ, so $C_{S_n}(\sigma') = \tau C \tau^{-1}$, whose elements are even. The same count, applied to σ', whose conjugacy class in S_n is also K, gives $|\operatorname{cl}_{A_n}(\sigma')| = |K|/2$. If $\sigma' = \alpha\sigma\alpha^{-1}$ with $\alpha \in A_n$, then $\tau\sigma\tau^{-1} = \alpha\sigma\alpha^{-1}$, so $\alpha^{-1}\tau \in C$, which is impossible because $\operatorname{sgn}(\alpha^{-1}\tau) = -1$. So the classes of σ and σ' in A_n are distinct, hence disjoint, since conjugacy classes partition A_n. Both lie in K, and together they have $|K|/2 + |K|/2 = |K|$ elements, so their union is K, as claimed.

Lemma 4.9.5: Conjugacy Classes of A_5

The group A_5 has five conjugacy classes, with sizes 1, 12, 12, 15, 20.

***Proof*:**
By theorem 1.2.18 and theorem 1.2.20(4), a permutation whose disjoint cycles have lengths $m_1, \ldots, m_r$ has sign $(-1)^{(m_1-1)+\cdots+(m_r-1)}$. Of the seven cycle types in S_5, the four even ones are:

Cycle type	*Example*	*Count in A_5*
(1^5)	e	1
$(3, 1^2)$	(1 2 3)	$\binom{5}{3} \cdot 2 = 20$
$(2^2, 1)$	(1 2)(3 4)	$\frac{1}{2}\binom{5}{2}\binom{3}{2} = 15$
(5)	(1 2 3 4 5)	$5!/5 = 24$

Total: $1 + 20 + 15 + 24 = 60 = |A_5|$.

In S_5, permutations of the same cycle type form a single conjugacy class (theorem 4.6.11), and lemma 4.9.4 decides whether such a class is one class of A_5 or two.

- 3-*cycles:* $C_{S_5}((1\ 2\ 3))$ contains the odd permutation (4 5), which is disjoint from (1 2 3). One A_5-class of size 20.
- $(2^2, 1)$-*type:* $C_{S_5}((1\ 2)(3\ 4))$ contains the odd permutation (1 2). One A_5-class of size 15.
- 5-*cycles:* let $\sigma = (1\ 2\ 3\ 4\ 5)$. Its class in S_5 consists of the 24 five-cycles, so $|C_{S_5}(\sigma)| = 120/24 = 5$ by (4.3). The subgroup $\langle\sigma\rangle$ has order 5 (proposition 1.2.24) and commutes with σ, so $C_{S_5}(\sigma) = \langle\sigma\rangle$, whose elements are powers of the even permutation σ and hence even. The class splits into two A_5-classes of size $24/2 = 12$, represented by σ and by $(4\ 5)\sigma(4\ 5)^{-1} = (1\ 2\ 3\ 5\ 4)$ (proposition 1.2.27).

The five class sizes are therefore $1, 12, 12, 15, 20$, as claimed.

Theorem 4.9.6: A_5 is Simple

The alternating group A_5 has no nontrivial proper normal subgroups.

***Proof*:**
Suppose $N \trianglelefteq A_5$ with $N \neq \{e\}$. Since N is normal, it is a union of conjugacy classes and must include $\{e\}$. The class sizes are $1, 12, 12, 15, 20$. Since $|N|$ divides $|A_5| = 60$ (theorem 3.3.1), we check all possible unions:

Sum including 1	*Divides* 60*?*
$1 + 12 = 13$	No
$1 + 15 = 16$	No
$1 + 20 = 21$	No
$1 + 12 + 12 = 25$	No
$1 + 12 + 15 = 28$	No
$1 + 12 + 20 = 33$	No
$1 + 15 + 20 = 36$	No
$1 + 12 + 12 + 15 = 40$	No
$1 + 12 + 12 + 20 = 45$	No
$1 + 12 + 15 + 20 = 48$	No
$1 + 12 + 12 + 15 + 20 = 60$	Yes (all of A_5)

No proper nontrivial union of conjugacy classes (including $\{e\}$) has order dividing 60. Therefore $N = \{e\}$ or $N = A_5$, and A_5 is simple.

4.9.7 Remark: The Quintic and Simplicity The simplicity of A_5 is the key obstruction in Galois's proof that there is no formula (using radicals) for the roots of a general polynomial of degree ≥ 5. The rough idea: the "solvability" of a polynomial equation by radicals corresponds to a property of its Galois group called *solvability* (which we will define in Section 6.1.1), and A_5 is the smallest non-solvable group. The chain of normal subgroups required by solvability cannot exist in a simple group with more than one element. Since A_5 arises as the Galois group of certain quintic polynomials, those polynomials are not solvable by radicals.

Exercise 4.9.1

1. Let $|G| = pqr$ where $p < q < r$ are distinct primes. Show that G has a normal Sylow subgroup for at least one of the primes p, q, r.

   Hint: Apply Sylow III to all three primes. If $n_r > 1$ and $n_q > 1$, count the elements of orders r and q (they contribute $(n_r)(r-1) + (n_q)(q-1)$ non-identity elements from distinct cyclic subgroups). Show that if all three $n_p, n_q, n_r > 1$, the total exceeds pqr.

2. Let $|G| = 30 = 2 \cdot 3 \cdot 5$.

   1. Show that $n_5 \in \{1, 6\}$ and $n_3 \in \{1, 10\}$.
   2. Using exercise 4.9.1 (1), conclude that G has a normal Sylow 5-subgroup or a normal Sylow 3-subgroup (or both).
   3. Deduce that G has a normal subgroup of order 15.

Hint: If $Q \trianglelefteq G$ is a Sylow 5-subgroup, consider G/Q and use the classification of groups of order 6 (theorem 4.9.1). If the Sylow 3-subgroup R is also normal, argue using $|QR| = 15$.

3. Let $|G| = 36 = 2^2 \cdot 3^2$.

   1. Show that $n_3 \in \{1, 4\}$.
   2. If $n_3 = 4$, show that the action of G on $\mathrm{Syl}_3(G)$ by conjugation gives a homomorphism $\pi\colon G \to S_4$. Conclude that $\ker(\pi)$ is a nontrivial normal subgroup of G.
      Hint: $|G| = 36 > 24 = |S_4|$, so π cannot be injective.

   Deduce that every group of order 36 has a nontrivial proper normal subgroup (and hence is not simple).

4. Show that the only simple group of order 60 is A_5 (up to isomorphism).

   Hint: Let G be simple of order 60. Show $n_5 = 6$, giving $[G : N_G(P)] = 6$ for $P \in \mathrm{Syl}_5(G)$. The action of G on the 6 cosets of $N_G(P)$ gives an injective homomorphism $G \hookrightarrow S_6$. Argue that the image lies in A_6, and identify it with A_5.

5. Let p be an odd prime. Show that every Sylow p-subgroup of D_n is cyclic and normal.

   Hint: The cyclic subgroup $\langle r \rangle \trianglelefteq D_n$ has index 2.

4.10 Summary

This chapter developed the theory of group actions from first principles and applied it to obtain structural results about finite groups. We collect the main threads here for reference.

The Core Framework

A *group action* of G on a set X is a homomorphism $\rho\colon G \to \mathrm{Sym}(X)$ (definition 4.1.1), equivalently a map $G \times X \to X$ satisfying the compatibility and identity axioms (theorem 4.1.2). Every action decomposes X into *orbits* (definition 4.2.1), and the size of each orbit is governed by the *stabilizer* via the Orbit-Stabilizer Theorem (theorem 4.3.1):

$$|G \cdot x| = [G : G_x].$$

The number of orbits can be computed by Burnside's Lemma (theorem 4.3.7): $|X/G| = \frac{1}{|G|} \sum_{g \in G} |X^g|$.

Types of Actions

Type	*Defining condition*	*Consequence*
Faithful	$\ker(\rho) = \{e\}$	$G \hookrightarrow \mathrm{Sym}(X)$
Transitive	One orbit: $G \cdot x = X$	$X \cong_G G/G_{x_0}$ for any x_0
Free	All stabilizers trivial: $G_x = \{e\}$	Every orbit has size $\lvert G\rvert$
Regular	Free + Transitive	$X \cong_G G$ (left multiplication)

The Key Self-Actions

The most powerful applications came from letting G act on itself or on structures derived from itself. The following table summarizes the three principal self-actions and their consequences.

Action	*Set X*	*Orbit of x*	*Stabilizer G_x*	*Kernel*	*Key result*
Left mult. $g \cdot a = ga$	G	G (one orbit)	$\{e\}$	$\{e\}$	Cayley's Thm
Conjugation $g \cdot a = gag^{-1}$	G	$\mathrm{cl}(a)$	$C_G(a)$	$Z(G)$	Class Equation
Coset action $g \cdot aH = gaH$	G/H	G/H (one orbit)	H (at eH)	$\bigcap_g gHg^{-1}$	Index-p Thm

Two further actions played essential roles:

- *Conjugation on subgroups* ($g \cdot S = gSg^{-1}$): the stabilizer is the normalizer $N_G(S)$, and the number of conjugates of S is $[G : N_G(S)]$ (corollary 4.6.9).
- *Conjugation on Sylow subgroups*: transitivity (Sylow II) gives $n_p = [G : N_G(P)]$, and the Fixed-Point Congruence (lemma 4.8.4) gives $n_p \equiv 1 \pmod{p}$.

The Major Theorems

1. *Orbit-Stabilizer Theorem* (theorem 4.3.1): $|G \cdot x| = [G : G_x]$.
2. *Burnside's Lemma* (theorem 4.3.7): $|X/G| = \frac{1}{|G|} \sum_g |X^g|$.
3. *Cayley's Theorem* (theorem 4.5.3): Every group embeds into a symmetric group.
4. *The Class Equation* (theorem 4.6.3): $|G| = |Z(G)| + \sum [G : C_G(g_i)]$.
5. *p-Groups have nontrivial center* (theorem 4.6.5).
6. $G/Z(G) \cong \mathrm{Inn}(G)$ (theorem 4.7.4).
7. *The N/C Theorem* (theorem 4.7.6): $N_G(H)/C_G(H) \hookrightarrow \mathrm{Aut}(H)$.
8. *Sylow's Theorems* (theorem 4.8.6): Existence, conjugacy, and counting of Sylow p-subgroups.
9. *Cauchy's Theorem* (theorem 4.8.7): If $p \big| |G|$, then G has an element of order p.

Looking Ahead

The idea that a group *acts* on something, and that the structure of the action reveals the structure of the group, is not a technique that ends with this chapter. In the chapters ahead, group actions will reappear in new forms:

- In the study of *semi-direct products* (chapter 5), the N/C Theorem and the conjugation action will be the key ingredients for constructing new groups from old ones.
- In *Galois theory*, a Galois group acts on the roots of a polynomial and on intermediate field extensions. The Fundamental Theorem of Galois Theory is, at its core, a correspondence between subgroups (stabilizers) and orbits (field extensions).
- In *module theory* (Part III), the group action framework generalizes to rings acting on abelian groups. The single condition "ρ is a homomorphism" will become the defining property of a module.

The unifying philosophy of this chapter bears repeating one last time:

> *Groups are not static objects. They are agents of symmetry, and the way to understand them is to watch how they act: on sets, on themselves, and on each other.*

4.11⋆ Group Actions in Advanced Mathematics

⋆ *Optional. This section is an aside; nothing later in the book depends on it.*

This section is addressed to the mathematically curious reader who wishes to see where the ideas of this chapter lead. No proofs are given, and nothing here is prerequisite for the rest of the book.

The theory of group actions developed in this chapter (homomorphisms to symmetric groups, orbits, stabilizers, counting) is the finite, discrete case of a theme that pervades nearly every branch of modern mathematics. In each new setting, the pattern is the same: a group acts on a structured object, and the interplay between the action and the structure produces deep theorems. We sketch three such settings.

4.11.1 Representation Theory

In this chapter, a group action was a homomorphism $\rho\colon G \to \mathrm{Sym}(X)$, where X was only a *set*. What happens if we upgrade X to a *vector space*?

A *(linear) representation* of a group G is a homomorphism

$$\rho\colon G \to GL(V),$$

where V is a vector space over a field $\mathbb{F}$ and $GL(V)$ is the group of invertible linear maps $V \to V$. Each group element acts on V not as a bijection alone, but as an *invertible linear transformation*: it respects addition and scalar multiplication.

Every group action $G \curvearrowright X$ on a finite set gives rise to a representation in a canonical way. Let $V = \mathbb{F}^X$ be the vector space with basis $\{e_x : x \in X\}$, one basis vector per element of X. Then G acts on V by permuting the basis: $g \cdot e_x = e_{g\cdot x}$. This is the *permutation representation* in the linear sense, and it converts our set-theoretic theory into linear algebra.

The power of representation theory lies in the tools of linear algebra that become available: eigenvalues, traces, direct-sum decompositions. The *character* of a representation is the function $\chi\colon G \to \mathbb{F}$ defined by $\chi(g) = \mathrm{tr}(\rho(g))$. Characters satisfy remarkable orthogonality relations, and for finite groups over $\mathbb{C}$, the irreducible characters form a basis for the space of class functions (functions constant on conjugacy classes). The number of irreducible representations equals the number of conjugacy classes, a striking connection to the Class Equation of Section 4.6.

Two landmark results illustrate the depth of the theory:

- *Maschke's Theorem ([17, Maschke's Theorem]):* If $\mathrm{char}(\mathbb{F}) \nmid |G|$, then every representation decomposes as a direct sum of irreducible representations. This is the analogue of diagonalizability for group actions.
- *Burnside's p^aq^b Theorem ([17, Burnside's Theorem]):* Every group of order p^aq^b (two prime factors) is solvable. Remarkably, the only known proof for over a century used *representation theory*, not pure group theory: a vivid demonstration that "linearizing" group actions can unlock results inaccessible by other means. (A purely group-theoretic proof was found by Thompson and others only much later.)

4.11.2 Algebraic Topology

Group actions enter topology through the theory of *covering spaces*.

Let X be a topological space and $\widetilde{X} \to X$ a covering map. The *deck transformations* (or covering transformations) are the homeomorphisms $\varphi\colon \widetilde{X} \to \widetilde{X}$ that commute with the projection to X. These form a group $\mathrm{Deck}(\widetilde{X}/X)$ that acts on $\widetilde{X}$ by homeomorphisms. When the covering is *normal* (or *regular*, note the terminology!), the deck group acts *freely and transitively* on each fiber. This is precisely a regular action in the sense of definition 4.4.12, and the base space is recovered as the orbit space:

$$X \;\cong\; \widetilde{X} \,/\, \mathrm{Deck}(\widetilde{X}/X).$$

The fundamental group $\pi_1(X, x_0)$ acts on the fiber $p^{-1}(x_0)$ by *monodromy*: lifting a loop in X to a path in $\widetilde{X}$ and tracking where the endpoint lands. This action encodes the topology of the covering:

- The *orbits* of the monodromy action correspond to the connected components of $\widetilde{X}$ (if the covering is connected, there is one orbit: the action is transitive).
- The *stabilizer* of a point in the fiber corresponds to the image of $\pi_1(\widetilde{X})$ in $\pi_1(X)$, the "surviving" loops.
- The *orbit-stabilizer correspondence* becomes the classical bijection between connected coverings of X (up to isomorphism) and conjugacy classes of subgroups of $\pi_1(X)$, a topological analogue of the Structure Theorem for Transitive Actions (theorem 4.4.6).

More broadly, whenever a group G acts on a topological space X, one can study *equivariant topology*: homology and cohomology theories that respect the action. The *equivariant cohomology* $H^*_G(X)$ combines the topology of X with the algebra of G, and its computation often reduces to understanding the fixed points and stabilizers of the action, exactly the tools developed in this chapter.

4.11.3 Category Theory

Category theory provides the most general and conceptually clean framework for understanding group actions.

A group G can be viewed as a category $\mathbf{B}G$ with a single object $*$ and one morphism $* \to *$ for each element of G, with composition given by the group operation. (The invertibility of morphisms makes $\mathbf{B}G$ a *groupoid*, a category in which every morphism is an isomorphism.)

From this viewpoint:

Algebraic concept	*Categorical translation*	*Target category*
Group action on a set	Functor $\mathbf{B}G \to \mathbf{Set}$	Sets and functions
Linear representation	Functor $\mathbf{B}G \to \mathbf{Vect}_{\mathbb{F}}$	Vector spaces
Module over a group ring	Functor $\mathbf{B}G \to \mathbf{Ab}$	Abelian groups
Topological G-space	Functor $\mathbf{B}G \to \mathbf{Top}$	Topological spaces

In each row, the pattern from definition 4.1.1 is visible: a group action is a homomorphism into the automorphism group of some object. The categorical perspective replaces "automorphism group" with "endomorphism monoid" and "homomorphism" with "functor," but the essential idea is identical.

The collection of all G-sets itself forms a category, denoted G-**Set**, whose morphisms are the G-equivariant maps (definition 4.4.5). The Structure Theorem for Transitive Actions (theorem 4.4.6) becomes the statement that every transitive G-set is isomorphic to a coset space G/H in this category. Burnside's Lemma can be reinterpreted as a statement about the *colimit* of the functor $\mathbf{B}G \to \mathbf{Set}$.

Perhaps most strikingly, the passage from group actions to module theory, the subject of much of Part III, is simply the passage from functors into **Set** to functors into **Ab**. The group ring $\mathbb{Z}[G]$ arises as the *free abelian group* on the morphisms of $\mathbf{B}G$, and a $\mathbb{Z}[G]$-module is exactly a functor $\mathbf{B}G \to \mathbf{Ab}$. In this way, the entire theory of modules over group rings (and by extension, much of representation theory) is a natural continuation of the ideas introduced in this chapter.

A group action is a functor from a one-object groupoid.
Change the target category, and the whole of algebra unfolds.

5

Building Groups: Direct and Semi-Direct Products

A group G with a normal subgroup H and $G/H \cong K$ is an extension of H by K (definition 3.6.3), and a composition series (section 3.7) assembles every finite group from simple groups by repeated extensions. The groups H and K alone do not determine G: both D_4 and $C_4 \times C_2$ are extensions of C_4 by C_2 (example 3.6.1). This chapter studies two constructions that produce an extension from given data. The *direct product* $H \times K$ is built from H and K alone, and the *semi-direct product* $H \rtimes K$ from H, K and an action of K on H by automorphisms. Inside a group G, the direct product appears when $G = HK$ for normal subgroups H and K with $H \cap K = 1$, and the semi-direct product when only H is required to be normal. Every extension of H by K is isomorphic to a subgroup of a single group built from H and K, the complete wreath product, by a theorem of Kaloujnine and Krasner [52, Exercises 11.1, no. 12]. This book does not develop wreath products.

5.1 Classical Definitions And Properties

You might recall example 1.1.5(4), we showed we can create a group-structure given two groups. We revisit this definition here in generality:

Definition 5.1.1: Direct Product

The *direct product* of $G_1 \times G_2 \times \cdots$ of groups $G_1, ..., G_n, ...$ with the operations $*_1, ..., *_n, ...$ respectively is the set of sequences $(g_1, ..., g_n, ...)$ (or n-tuples if the number of groups is finite) where $g_9 \in G_i$ with the operation defined componentwise:

$$(g_1, ..., g_n, ...) * (h_1, ..., h_n, ...) = (g_1 *_1 h_1, ..., g_n *_n h_n, ...)$$

Sometimes, the direct product is called the *external direct product* to differentiate it from the *internal direct product* (see definition 3.4.1)

Example 5.1.2

1. Think of $\mathbb{R}^2 = \mathbb{R} \times \mathbb{R}$ with component-wise addition: $(\mathbb{R} \times \mathbb{R}, (+, +))$ or $(\mathbb{R}^2, (+, +))$.
2. The groups can also be arbitrary. Let $G_1 = \mathbb{Z}$, $G_2 = S_3$ and $G_3 = GL_2(\mathbb{R})$. Then $G_1 \times G_2 \times G_3$ with the operation defined component wise is indeed a group

For the direct product, there are a few general properties you can prove as an exercise:

Proposition 5.1.3: properties of Direct Products

Let every G_i be a group.

1. $|\prod_i G_i| = \prod_i |G_i|$
2. For each fixed i, the set of elements of G which have the identity of G_j in the jth position for all $j \neq i$ and arbitrary elements of G_i in position i is a subgroup of G isomorphic to G_i:
$$G_i \cong (1_{g_1}, ..., g_i, 1_{g_{i+1}}, ...)$$
3. For each fixed i, define $\pi_i : G \to G_i$ by
$$\pi_i((g_1, ..., g_n)) = g_i$$
then π_i is a surjective homomorphsim with
$$\begin{aligned}\ker(\pi_i) &= \{(g_1, ..., g_{i-1}, 1, g_{i+1}, ..., g_n) | g_j \in G_j \text{ for all } j \neq i\} \\ &\cong G_1 \times \cdots \times G_{i-1} \times G_{i+1} \times \cdots \times G_n\end{aligned}$$
4. Under the identifications in part (2), if $x \in G_i$ and $y \in G_j$ for some $i \neq j$, then $xy = yx$

***Proof*:**
Throughout, write $G = \prod_i G_i$, whose elements are the tuples $(g_1, ..., g_n, ...)$ with $g_i \in G_i$ and whose operation is componentwise. Note that G is a group: the tuple $(1_{G_1}, ..., 1_{G_n}, ...)$ is a two-sided identity, the componentwise inverse $(g_1^{-1}, ..., g_n^{-1}, ...)$ is a two-sided inverse, and associativity holds in each component.

1. As a set, G is the Cartesian product of the underlying sets of the G_i: an element of G is

exactly a choice of one element of G_i for each i. Hence $|\prod_i G_i| = \prod_i |G_i|$.

2. Fix i and let $H_i \subseteq G$ be the described set. Define $\iota_i : G_i \to G$ by letting $\iota_i(g)$ be the tuple with g in position i and 1_{G_j} in every position $j \neq i$. Since the operation of G is componentwise and $1_{G_j}1_{G_j} = 1_{G_j}$, we get $\iota_i(g)\iota_i(h) = \iota_i(gh)$, so ι_i is a homomorphism. Its image is exactly H_i, which is therefore closed under products and, since $\iota_i(g)^{-1} = \iota_i(g^{-1})$, under inverses. As H_i also contains the identity, $H_i \leqslant G$. Finally, ι_i is injective, for if $\iota_i(g) = \iota_i(h)$ then comparing position i gives $g = h$. Hence ι_i restricts to an isomorphism $G_i \cong H_i$.

3. For tuples $g = (g_1, ..., g_n, ...)$ and $h = (h_1, ..., h_n, ...)$, the componentwise operation gives $\pi_i(gh) = g_i h_i = \pi_i(g)\pi_i(h)$, so π_i is a homomorphism. It is surjective since $\pi_i(\iota_i(g)) = g$ for every $g \in G_i$. A tuple lies in $\ker(\pi_i)$ if and only if its ith position is 1_{G_i}, which is exactly the displayed set. The map sending such a tuple to the tuple obtained by deleting position i is a bijection onto $G_1 \times \cdots \times G_{i-1} \times G_{i+1} \times \cdots$, and it is a homomorphism since both operations are componentwise. Hence $\ker(\pi_i) \cong G_1 \times \cdots \times G_{i-1} \times G_{i+1} \times \cdots$.

4. Identify x with $\iota_i(x)$ and y with $\iota_j(y)$, where $i \neq j$. In the product $\iota_i(x)\iota_j(y)$, position i is $x \cdot 1_{G_i} = x$, position j is $1_{G_j} \cdot y = y$, and every other position is the identity. The product $\iota_j(y)\iota_i(x)$ has position i equal to $1_{G_i} \cdot x = x$, position j equal to $y \cdot 1_{G_j} = y$, and every other position the identity. The two products agree, so $xy = yx$.

In the last part of proposition 5.1.3, elements of different factors commute. The factors do not interact under the componentwise operation, and this forces the commutation.

Exercise 5.1.1

1. Does there exist a nontrivial group G such that $G \times G \cong G$? hint: [1]
2. Let G be a finitely generated abelian group. Show that every subgroup is also a finitely generated abelian group.
3. If $H \leqslant G_1 \times G_2$, are there always subgroups $H_1 \leqslant G_1$ and $H_2 \leqslant G_2$ such that $H \cong H_1 \times H_2$? Prove it or give a counter-example.

In definition 5.1.1, the factors need not be the same. When they are, G^n denotes the direct product of n copies of G, for instance $G^4 = G \times G \times G \times G$. An element $(g_0, \ldots, g_{n-1})$ of G^n is the same data as the function $f\colon \{0, \ldots, n-1\} \to G$ with $f(i) = g_i$, and the componentwise product of two tuples corresponds to the pointwise product $(ff')(i) = f(i)f'(i)$ of the functions. So G^n is isomorphic to the group of functions $\{0, \ldots, n-1\} \to G$ under pointwise multiplication. The same description works for any set B and any group A: the functions $B \to A$ form a group A^B under pointwise multiplication, the direct product of copies of A indexed by B. For example, $(\mathbb{Z}/2\mathbb{Z})^{\mathbb{R}}$ is the group of all functions $\mathbb{R} \to \mathbb{Z}/2\mathbb{Z}$.

5.1.1 Fundamental theorem of Finitely Generated Abelian Groups

This theorem is stated, but not proved till till much later (chapter 14, theorem 14.1.9). The main idea that is trying to be presented here is the power of of being abelian: If we have an abelian, it is *such* a nice group that we can classify them all with isormophism to $\mathbb{Z}$ and $\mathbb{Z}_{p^\alpha}$ for appropriate power-prime.

[1] Think about $G = \prod_{n=1}^{\infty} \mathbb{Z}$

It will also help in our indeavour to classify groups. With this representation, it becomes much much easier. Here are two terms we introduce now to present the theorem:

Definition 5.1.4: Finitely Generated Group

A group G is *finitely generated* if $\langle A \rangle = G$ for some finite subset $A \subseteq G$.

Definition 5.1.5: Free Abelian Group

$\mathbb{Z}^r = \mathbb{Z} \times \cdots \times \mathbb{Z}$. $\mathbb{Z}^r$ is called the *free abelian group of rank r*.

Theorem 5.1.6: Fundamental Theorem of Finitely Generated Abelean Groups

Let G be a finitely generated abelian group. Then:

$$G \cong \mathbb{Z}^r \times \mathbb{Z}_{n_1} \times \cdots \mathbb{Z}_{n_s}$$

for some integers $r, n_1, ..., n_s$ with the following conditions:

1. $r \geq 0$ and $n_j \geq 2$ for all j, and
2. $n_{i+1} | n_i$ for $1 \leq i \leq s-1$

Furthermore, r and $n_1, \ldots, n_s$ are uniquely determined by G.

***Proof*:**
Existence is corollary 14.1.18, the case $R = \mathbb{Z}$ of theorem 14.1.9, and uniqueness is the case $R = \mathbb{Z}$ of theorem 14.1.16.

The rest of this section gives the vocabulary for describing abelian groups.

Notice also that it's for *finitely generated abelian groups*. What about infinitely generated abelian groups? We can simplify for a moment and ask about countably generated abelian groups. We have solved this for p-groups, but we have yet to solve it in the general case:

https://math.stackexchange.com/questions/3927834/classification-of-countably-infinite-abelian-groups

Definition 5.1.7: free rank and invariant factors

r is called the *free rank* or *betti number*, while $n_1, ..., n_s$ are called the *invariant factors*. The whole expression is called the *invariant factor decomposition of G*.

Exercise 5.1.2

1. Let $A \subseteq B$ be two finitely generated free abelian groups, each of rank n. Show that B/A has rank 0.
2. Let $G = (\mathbb{Z}/n\mathbb{Z})^\times$. Decompose G into a product of cyclic groups with additive notation.

5.2 Recognizing Direct Products

Let's say we have a (not necessarily abelian) group G. Can G be decomposed into the direct product of two smaller groups? This question is what we will be exploring in this section.

The 4th identity of proposition 5.1.3 gives us a very valuable insight on how to think of direct products. If we have two groups G_1 and G_2, we showed that creating a set of 2-tulpes with component-wise operation creates a new group. With the fact that components commute ($xy = yx$, if $x \in G_1$, $y \in G_2$), we can also define the direct product in terms of the following:

$$G_1 \times G_2 = \langle x, y | xy = yx, x \in G_1, y \in G_2 \rangle$$

where the operation is done when everything is "moved to the appropriate place"

Example 5.2.1: presentation example
Let $G_1 = S_3$ and $G_2 = \mathbb{Z}_5$. Take $x = (1\ 2)4$ and $y = 4(2\ 3)2$. Then

$$xy = (1\ 2)44(2\ 3)2 = (1\ 2)(2\ 3)442 = (1\ 2\ 3)2$$

You might ask yourself what's the point of this 2nd perspective? The idea is that we can now ask ourselves when can a group be decomposed into smaller groups. In other words, if we had a group G, can we split it into two parts G_1, G_2 such that $G \cong G_1 \times G_2$?

Thus, we will explore exactly how far a two subgroups (or more generally subsets) of a group are from being commutative, and then use that information to find out when we can decompose groups.

We start with defining a way to capture *how commutative* a groups/subgroups/elements are:

Definition 5.2.2: Commutator and Commutator Group

Let G be a group, let $x, y \in G$, and let A, B be nonemtpy subset of G

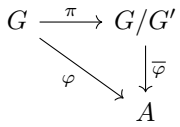
1. Define $[x, y] = x^{-1}y^{-1}xy$. $[x, y]$ is called the *commutator* of x and y
2. Define $[A, B] = \langle [a, b] \big| a \in A,\ b \in B \rangle$, the group geenrated by the commutators of elements from A and B
3. Define $G' = \langle [x, y] | x, y \in G \rangle$ to be the subgroup of G generated by commutators of all element sfrom G. G' is called the *commutator subgroup* of G

Notice that if x and y commute, then $[x, y] = 1$. If all the element of G commute, then $G' = \{1\}$. Thus, the commutator captures an idea of how much the elements of G commute

Proposition 5.2.3: Commutator Subgroup Properties

Let G be a group, $x, y \in G$, and $H \leqslant G$. Then

1. $xy = yx[x, y]$
2. $H \trianglelefteq G$ if and only if $[H, G] \leqslant H$
3. for every automorphism σ of G, $\sigma([x, y]) = [\sigma(x), \sigma(y)]$ and $\sigma(G') = G'$. Hence G' char G, so $G' \trianglelefteq G$, and G/G' is abelian
4. G/G' is the largest abelian quotient of G. That is, if $H \trianglelefteq G$ and G/H is abelian, then $G' \leqslant H$. Conversely, if $G' \leqslant H$, then $H \trianglelefteq G$ and G/H is abelian
5. (Universal Abelian Quotient) let $\pi \colon G \to G/G'$ be the natural projection. If $\varphi \colon G \to A$ is a homomorphism into an abelian group A, then $G' \leqslant \ker(\varphi)$, and there is a unique homomorphism $\overline{\varphi} \colon G/G' \to A$ with $\varphi = \overline{\varphi} \circ \pi$, that is, the following diagram commutes:



***Proof*:**

1. $yx[x, y] = yxx^{-1}y^{-1}xy = xy$.
2. For $h \in H$ and $g \in G$,
$$g^{-1}hg = h[h, g]$$
since $h[h, g] = hh^{-1}g^{-1}hg$. Suppose $H \trianglelefteq G$. Then $g^{-1}hg \in H$ by proposition 3.1.11(5), applied to g^{-1}, so $[h, g] = h^{-1}(g^{-1}hg) \in H$. Thus the subgroup H contains every generator of $[H, G]$, and $[H, G] \leqslant H$ by proposition 2.3.17. Conversely, suppose $[H, G] \leqslant H$. Then $g^{-1}hg = h[h, g] \in H$ for all $h \in H$ and $g \in G$. Replacing g by g^{-1} gives $gHg^{-1} \subseteq H$ for every $g \in G$, so $H \trianglelefteq G$ by proposition 3.1.11.
3. Let $\sigma \in \mathrm{Aut}(G)$. By proposition 1.4.4(2),
$$\sigma([x, y]) = \sigma(x)^{-1}\sigma(y)^{-1}\sigma(x)\sigma(y) = [\sigma(x), \sigma(y)]$$
so σ sends every commutator to a commutator. The set $P = \{g \in G \mid \sigma(g) \in G'\}$ is a subgroup of G by proposition 2.1.3: it contains 1, since $\sigma(1) = 1$, and if $g, h \in P$, then $\sigma(gh^{-1}) = \sigma(g)\sigma(h)^{-1} \in G'$. It contains every commutator, so $G' \subseteq P$ by proposition 2.3.17, that is, $\sigma(G') \subseteq G'$. The inverse σ^{-1} is also an automorphism of G, so the same argument gives $\sigma^{-1}(G') \subseteq G'$, and applying σ gives $G' \subseteq \sigma(G')$. Hence $\sigma(G') = G'$ for every $\sigma \in \mathrm{Aut}(G)$, so G' char G (definition 4.7.12) and $G' \trianglelefteq G$ by proposition 4.7.13(1). Finally, for $x, y \in G$, (1) and $[x, y] \in G'$ give
$$(xG')(yG') = xyG' = yx[x, y]G' = yxG' = (yG')(xG')$$
so G/G' is abelian.

4. Suppose $H \trianglelefteq G$ and G/H is abelian, and let $x, y \in G$. Then $xyH = (xH)(yH) = (yH)(xH) = yxH$, so $xy \in yxH$, say $xy = yxh$ with $h \in H$. Then $h = (yx)^{-1}xy = x^{-1}y^{-1}xy = [x, y]$, so H contains every commutator, and $G' \leqslant H$ by proposition 2.3.17. Conversely, suppose $G' \leqslant H$. Every generator $[h, g]$ of $[H, G]$ is a commutator, so $[H, G] \leqslant G' \leqslant H$ by proposition 2.3.17, and $H \trianglelefteq G$ by (2). For $x, y \in G$, the commutator $[x, y]$ lies in H, so $[x, y]H = H$, and by (1)

$$(xH)(yH) = xyH = yx[x, y]H = yxH = (yH)(xH)$$

Thus G/H is abelian.

The converse also follows from the isomorphism theorems. By (3), G/G' is abelian, so $H/G' \trianglelefteq G/G'$ (corollary 3.1.12), and $H \trianglelefteq G$ by corollary 3.5.8(5). By theorem 3.5.5,

$$G/H \cong (G/G')/(H/G')$$

and a quotient Q/N of an abelian group Q is abelian, since $(aN)(bN) = abN = baN = (bN)(aN)$.

5. Let $\varphi\colon G \to A$ be a homomorphism into an abelian group A. For $x, y \in G$, proposition 1.4.4(2) and the commutativity of A give

$$\varphi([x, y]) = \varphi(x)^{-1}\varphi(y)^{-1}\varphi(x)\varphi(y) = 1$$

so the subgroup $\ker(\varphi)$ (proposition 1.4.4(4)) contains every commutator, and $G' \leqslant \ker(\varphi)$ by proposition 2.3.17. Since $G' \trianglelefteq G$ by (3), theorem 3.1.15 with $N = G'$ gives a unique homomorphism $\overline{\varphi}\colon G/G' \to A$ with $\varphi = \overline{\varphi} \circ \pi$, namely $\overline{\varphi}(gG') = \varphi(g)$, which completes the proof.

Example 5.2.4

If G is abelian, then for any $x, y \in G$, $[x, y] = x^{-1}y^{-1}xy = x^{-1}xy^{-1}y = 1$, and so $G' = 1$, and $G/G' \cong G$.

With this, we can answer now give criterion to recognize direct products. First, a lemma:

Lemma 5.2.5: Representation's of HK

Let H and K be subgroups of a group G. The number of distinct ways of writing each element of the set HK in the form hk, for some $h \in H$ and $k \in K$ is $|H \cap K|$. In particular, if $H \cap K = 1$, then each element of HK can be written uniquely as a product of hk, for some $h \in H$ and $k \in K$.

***Proof*:**

Let $x \in HK$, say $x = hk$ with $h \in H$ and $k \in K$, and let $R_x = \{(h', k') \in H \times K \mid h'k' = x\}$ be the set of ways of writing x. Define $\theta\colon H \cap K \to R_x$ by $\theta(t) = (ht, t^{-1}k)$. It takes values in R_x, since $ht \in H$, $t^{-1}k \in K$ and $(ht)(t^{-1}k) = hk = x$. It is injective, since $ht = ht'$ forces $t = t'$. It is surjective: if $h'k' = x = hk$, then $t = h^{-1}h' = kk'^{-1}$ lies in $H \cap K$, and $\theta(t) = (h', (kk'^{-1})^{-1}k) = (h', k')$. So $|R_x| = |H \cap K|$ for every $x \in HK$, and when $H \cap K = 1$

each element of HK has exactly one expression hk, as claimed.

Theorem 5.2.6: Recognition Theorem For Direct Products

Let G be a group, and let H, K be subgroups such that

1. H and K are normal in G, and
2. $H \cap K = \{1\}$

Then HK is a subgroup of G, and $\mu\colon H \times K \to HK$, $\mu(h,k) = hk$, is an isomorphism, so

$$HK \cong H \times K$$

Conversely, if HK is a subgroup of G and μ is an isomorphism, then $H \cap K = \{1\}$ and both H and K are normal subgroups of HK.

***Proof*:**
Suppose first that H and K are normal in G and $H \cap K = 1$. For $h \in H$ and $k \in K$, the element $hkh^{-1}k^{-1}$ equals $(hkh^{-1})k^{-1}$, which lies in K because $K \trianglelefteq G$, and it equals $h(kh^{-1}k^{-1})$, which lies in H because $H \trianglelefteq G$ (proposition 3.1.11). So it lies in $H \cap K = 1$, and $hk = kh$. The set HK is a subgroup of G by corollary 3.4.5. For $(h,k), (h',k') \in H \times K$,

$$\mu((h,k)(h',k')) = hh'kk' = hkh'k' = \mu(h,k)\mu(h',k')$$

because k commutes with h', so μ is a homomorphism. It is surjective by the definition of HK, and injective by lemma 5.2.5, since $H \cap K = 1$. Hence μ is an isomorphism.

Conversely, suppose that HK is a subgroup and μ is an isomorphism. If $t \in H \cap K$, then $\mu(t, t^{-1}) = 1 = \mu(1,1)$, so $t = 1$ by injectivity, and $H \cap K = 1$. For $h \in H$ and $k \in K$,

$$hk = \mu((1,k)(h,1)) = \mu(1,k)\mu(h,1) = kh$$

Now let $x = h'k' \in HK$ and $h \in H$. Then $xhx^{-1} = h'k'hk'^{-1}h'^{-1} = h'hh'^{-1} \in H$, since k' commutes with h. So $xHx^{-1} \subseteq H$ for every $x \in HK$, and in the same way $xKx^{-1} \subseteq K$. By proposition 3.1.11, applied in the group HK, both H and K are normal in HK, completing the proof.

In particular, if H and K are normal subgroups of G with $H \cap K = 1$ and $HK = G$, then $G \cong H \times K$. The converse cannot conclude normality in G when $HK \neq G$: for $G = S_3$, $H = \langle (1\ 2) \rangle$ and $K = 1$, the map μ is an isomorphism, while H is not normal in S_3, since $(1\ 3)(1\ 2)(1\ 3)^{-1} = (2\ 3)$ (proposition 1.2.27).

The propety that $H \cap K = 1$ has a name:

Definition 5.2.7: Complement

Let H be a subgroup of G. A subgroup K of G is called a *complement* for H in G if $G = HK$ and $H \cap K = 1$.

Thus, we can rephrase the previous theorem: if a normal subgroup H of G has a complement K that is also normal, then $G \cong H \times K$.

Another way of thinking about this theorem is that the *internal direct product* HK (definition 3.4.1) of two subgroups H and K of G agrees with their *external direct product* $H \times K$ (definition 5.1.1) when the two conditions in theorem 5.2.6 hold. If $H \cap K \neq \{1\}$, or one of them is not normal in HK, then the map μ is not an isomorphism. In the next section, we will explore a generalization of the direct product, the *semi-direct product*, which describes HK when H is normal and $H \cap K = 1$ (theorem 5.3.7).

Note that if G is abelian, finitely generated or not, and $H, K \leqslant G$ satisfy $H \cap K = 1$, then $HK \cong H \times K$, since every subgroup of an abelian group is normal (corollary 3.1.12). This is another example of the niceties of being commutative.

5.2.1 A word on Perfect Groups

We saw that if a group G is abelian, then $G' = \{1\}$. What if we had the complete opposite: if $G = G'$? Such a group has a name:

Definition 5.2.8: Perfect Group

Let G be a group and G' be the commutator subgroup. If $G = G'$, then G is called a *perfect group*

Example 5.2.9: Perfect Groups

The alternating group A_5 is perfect. Its commutator subgroup A_5' is normal in A_5 by proposition 5.2.3(3), so $A_5' = 1$ or $A_5' = A_5$ by theorem 4.9.6. If $A_5' = 1$, then $x^{-1}y^{-1}xy = 1$, that is, $xy = yx$, for all $x, y \in A_5$. But the 3-cycles $(1\,2\,3)$ and $(3\,4\,5)$ are products of two transpositions (theorem 1.2.18), so they lie in A_5, and

$$(1\,2\,3)(3\,4\,5) = (1\,2\,3\,4\,5) \neq (1\,2\,4\,5\,3) = (3\,4\,5)(1\,2\,3)$$

Hence $A_5' = A_5$.

Perfect groups are linked to solvable groups, which section 6.1.1 defines through the repeated commutator subgroups $G' \supseteq (G')' \supseteq \cdots$. For a perfect group every term of this sequence equals G, so a perfect group $G \neq 1$ is not solvable.

5.3 Semi-Direct Products

As we mentioned earlier, not all groups can be broken down into [non-trivial] direct products:

Example 5.3.1: $G \ncong H \times K$

Let $G = D_3 = \langle \sigma, r | \sigma^2 = r^3 = 1,\ \sigma r = r^{-1}\sigma \rangle$, whose six elements $\sigma^k r^i$ with $k \in \{0, 1\}$ and $0 \leq i \leq 2$ are distinct (proposition 1.2.29(4)). The subgroup $\langle \sigma \rangle = \{1, \sigma\}$ has order 2, and $\langle r \rangle$ has order 3. Since $\langle r \rangle$ has index 2, $\langle r \rangle \trianglelefteq D_3$ (corollary 3.3.6). So $\langle r \rangle \langle \sigma \rangle$ is a subgroup (corollary 3.4.5), and since $\langle r \rangle \cap \langle \sigma \rangle = 1$, its order dividing both 2 and 3 (theorem 3.3.1), proposition 3.4.3 gives

it 6 elements, so $\langle r\rangle\langle\sigma\rangle = D_3$. However, $\langle\sigma\rangle$ is not normal in D_3: from $\sigma r = r^{-1}\sigma$ and $\sigma^2 = 1$ we get $r\sigma = \sigma r^{-1}$, so $r\sigma r^{-1} = \sigma r^{-2} = \sigma r$, which is not in $\{1, \sigma\}$. By theorem 5.2.6, the map $\langle\sigma\rangle \times \langle r\rangle \to D_3$, $(a, b) \mapsto ab$, is therefore not an isomorphism. In fact no isomorphism exists:

$$D_3 \ncong \langle\sigma\rangle \times \langle r\rangle$$

because $\langle\sigma\rangle \times \langle r\rangle$ is abelian, both factors being cyclic (corollary 2.3.6) and the operation componentwise, while D_3 is not abelian (proposition 1.2.29(5)), and isomorphic groups are both abelian or both not (proposition 1.4.21(2)).

Since, $\langle\sigma\rangle \cong \mathbb{Z}/2\mathbb{Z}$ and $\langle r\rangle \cong \mathbb{Z}/3\mathbb{Z}$, we can swtich over to that notation. Then we can say that

$$D_3 \ncong \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/3\mathbb{Z}$$

The question might arise on whether there is a suitable generalization of the direct product to classify the above group. More generally, is there a way to capture the structure of HK, where H and K are *arbitrary* groups (i.e. not subgroups of a larger group G). That question is explored in this section. Note that if G was abelian then automatically, $K \trianglelefteq G$, since every subgroup of an abelian group is normal. Thus, this distinction becomes interesting only when G is non-abelian.

The way we will do this is by trying to fix the fact that elements of the group H and K will no longer always commute with each other. Note that we always have that $HK = KH$ (proposition 3.4.4), but that did not imply $hk = kh$. In the dihedral example, we had $rs = sr^{-1}$. If we were to try to write this as a product, we'd get

$$(r, 1)(1, s) = (1, s)(r^{-1}, 1)$$

Another example would be S_3. The group S_3 is another group of order 6 with subgroups of order 2 and 3, for example: (1 2) and (1 2 3)). Since $|S_3| = 6$, we again have $|(1\ 2\ 3)| \trianglelefteq S_3$ and $\langle(1\ 2\ 3)\rangle\langle(1\ 2)\rangle \cong S_3$. However, one can easily check that $(1\ 2) \ntrianglelefteq S_3$ (try taking $g = (1\ 2\ 3)$ and see if $g\langle(1\ 2)\rangle g^{-1} = \langle(1\ 2)\rangle$). We again get that $ab \neq ba$, and instead, they commute in a different manner. let $a = (1\ 2)$ and $b = (1\ 2\ 3)$. Then you can verify that:

$$ab = b^2a$$

meaning when a and b comute, b becomes b^2. Thus, we can write

$$S_3 = \langle a, b | a^2 = b^3 = 1,\ ab = b^2a\rangle$$

If were to try to write this as a product, we'd get that

$$(a, 1)(1, b) = (1, b^2)(a, 1)$$

So the question is "Is there a more general way to capture how elements in a group commute"?

As a way to motivate the coming definition, Let's start by having a larger group G, and recall that to define HK as a group, we need $H \trianglelefteq G$ and $K \leqslant G$.

Since HK is a group, we already have that $(h_1k_1)(h_2k_2) = h_3k_3$ for some $h_3k_3 \in HK$ (similarly to how in a direct product, $(a_1, b_1)(a_2b_2) = (a_1a_2, b_1b_2) = (a_3, b_3)$). More precicely, we take $h_1k_1, h_2k_2 \in HK$, we can re-write $h_1k_1h_2k_2$ as $h_3k_3 \in HK$. We showed this in proposition 3.4.4.

Doing it again here, notice exactly how we get to write $h_1k_1h_2k_2$ as h_3k_3:

$$\begin{aligned}(h_1k_1)(h_2k_2) &= h_1k_1h_2(k_1^{-1}k_1)k_2\\ &= h_1(k_1h_2k_1^{-1})k_1k_2\\ &= (h_1(k_1h_2k_1^{-1}))k_1k_2\\ &= h_3k_3\end{aligned}$$

where $h_3 = h_1(k_1h_2k_1^{-1})$ (since $H \trianglelefteq G$, $gh_1g^{-1} \in H$) and $k_3 = k_1k_3$.

To get to this construction, we heavily relied on *starting* with the group G and all the properties of the subgroups H, and K (ex. $H \cap K = \{e\}$, $H \trianglelefteq G$). Relying so heavily on a bigger group G does not help in our generalization of the direct product: when defining an arbitrary direct product $G_1 \times G_2$, we did not rely on some bigger group G from which G_1 and G_2 were normal subgroups. Thus, we have to figure out a way to do this construction by only relying on information on H and K.

To do this, we must be able to describe the operation between $(h_1k_1) \cdot (h_2k_2)$ without using the normality of $H \trianglelefteq G$. It is easy for the k elements since $k_3 = k_1k_2$, so the question about about defining $k_1h_2k_1^{-1}$.

We have in fact already encountered a solution to this problem. Since H is normal in G, the group K *acts* on H by conjugation:

$$k \cdot h = khk^{-1} \qquad \text{for } h \in H, k \in K$$

Since H is normal, $khk^{-1} = h' \in H$, meaning if we know how every k acts on every h, we can drop the necessity for a larger group G. In fact, all this information *can* be captured with a group-action:

$$\varphi : K \to \text{Aut}(H)$$

meaning we now only rely on the product of K, H, and the homomorphism φ which captures how H and K interact. Importantly for defining a group with this, $\text{Aut}(H) = \text{Aut}_{\mathbf{Grp}}(H)$, not $\text{Aut}_{\mathbf{Set}}(H)$, and so

$$\varphi(x)(ab) = \varphi(x)(a)\varphi(x)(b)$$

Thus, we have eliminated the need for a larger group G!

With this idea, we can define a our new product:

Definition 5.3.2: Semi-Direct Product

Let H and K be groups and let φ be a homomorphism form K into $\text{Aut}(H)$. Let $\cdot$ denote the (left) action of K on H determined by φ. Let G be the set of ordered pairs (h, k) with $h \in H$ and $k \in K$ and define the following multiplication on G:

$$(h_1, k_1)(h_2, k_2) = (h_1k_1 \cdot h_2, k_1k_2)$$

we usually denote $G = H \rtimes_\varphi K$, or $H \rtimes K$ if the context permits it

Notice how we are not using a bigger group, but a lot of intuition can be derived from thinking about H and K as subgroups of a larger group, showing how subgroups still play an important role in the analysis and conception of semi-direct products, but it is certainly not necessary as we will point out in the examples.

The $\rtimes$ notation is to remember that H is the normal subgruop of $H \rtimes K$ (as we will prove soon). It might be evident that in most cases $H \rtimes K \ncong K \rtimes H$, since we'd be defining different actions. More on this when we show some examples.

Another way of remembering which group is acting on which is with a pop-culture reference (curtosy of one of my professors): if you are aware of what a "Kamehameha" is from dragon ball, then the symbol "$\rtimes$" can be thought of as Goku's hand preforming the Kamehameha.



If you are unaware of the cultural reference, here's a youtube video I found (or you can google it):

https://www.youtube.com/watch?v=_ct3ccrU4AI

As silly as it may be, this turned out to be the most efficient way for me to remember which groups acts on which.

Before we get too hasty, we should prove that this is indeed a group and that all the properties stated earlier hold:

Proposition 5.3.3: Properties Of Semi-Direct Product

1. This multiplication makes G into a group of order $|G| = |H||K|$
2. The sets $\{(h,1)|h \in H\}$ and $\{(1,k)|k \in K\}$ are subgroups of G and the maps $h \mapsto (h,1)$ and $k \mapsto (1,k)$ for $k \in K$ are isomorphisms of these subgroups with the groups H and K respectively:
$$H \cong \{(h,1)|h \in H\} \qquad K \cong \{(1,k)|k \in K\}$$
3. $H \trianglelefteq G$
4. $H \cap K = \{e\}$
5. for all $h \in H$ and $k \in K$, $khk^{-1} = k \cdot h = \varphi(k)(h)$

***Proof*:**

1. To show $H \rtimes K$ is a group, we need to show it obeys the axioms of a group

(a) **associativity**. Let $(a,x),(b,y),(c,z) \in H \rtimes K$. Then

$$\begin{aligned}((a,x)(b,y))(c,z) &= (a\ x\cdot b, xy)(c,z)\\ &= (a\ x\cdot b\ (xy)\cdot c, xyz)\\ &= (a\ x\cdot b\ x\cdot(y\cdot c), xyz)\\ &= (a\ \varphi(x)(b)\ \varphi(x)(y\cdot c), xyz)\\ &= (a\ \varphi(x)(b\ y\cdot c), xyz)\\ &= (a\ x\cdot(b\ y\cdot c), xyz)\\ &= (a,x)(by\cdot c, yz)\\ &= (a,x)((b,y)(c,z))\end{aligned}$$

where I switched over to homomorphism notation to show that since $\varphi(x)$ is a homomorphism (note that $\varphi(x)$ is an automorphism of H), then $\varphi(x)(st) = \varphi(x)(s)\varphi(x)(t)$.

(b) **Identity**: I claim that $(1,1)$ is an identity. In particular, for any $(x,y) \in H \rtimes K$

$$(1,1)(x,y) = (1\ 1\cdot x, 1y) = (x,y) = (x,y)(1,1)$$

(c) **Inverse**: I claim that for any $(h,k) \in H \rtimes K$, $(h,k)^{-1} = (k^{-1}\cdot h^{-1}, k^{-1})$ is an inverse. Indeed:

$$\begin{aligned}(h,k)(k^{-1}\cdot h^{-1}, k^{-1}) &= (hk\cdot(k^{-1}\cdot h^{-1}), kk^{-1})\\ &= (h(kk^{-1})\cdot h^{-1}, 1)\\ &= (h\ 1\cdot h^{-1}, 1) = (hh^{-1}, 1)\\ &= (1,1)\end{aligned}$$

and similarly for $(h,k)^{-1}(h,k)$.

Thus, $H \rtimes K$ is a group.

2. Let $\hat{H} = \{(h,1) | h \in H\}$ and $\hat{K} = \{(1,k) | k \in K\}$. Since $\varphi\colon K \to \mathrm{Aut}(H)$ is a homomorphism, $\varphi(1) = \mathrm{id}_H$, so $1\cdot b = b$ for $b \in H$. Since each $\varphi(k)$ is a homomorphism, $k\cdot 1 = \varphi(k)(1) = 1$ for $k \in K$ (proposition 1.4.4). Hence, for $a,b \in H$ and $a',b' \in K$,

$$(a,1)(b,1) = (a\ 1\cdot b, 1) = (ab,1) \qquad (1,a')(1,b') = (1\ a'\cdot 1, a'b') = (1,a'b')$$

so $h \mapsto (h,1)$ and $k \mapsto (1,k)$ are injective homomorphisms $H \to G$ and $K \to G$. Their images $\hat{H}$ and $\hat{K}$ are subgroups of G by proposition 1.4.4, isomorphic to H and K.

3. Identify h with $(h,1)$ and k with $(1,k)$. By (2), $(1,k)^{-1} = (1,k^{-1})$, and

$$\begin{aligned}khk^{-1} &= (1,k)(h,1)(1,k^{-1})\\ &= (k\cdot h, k)(1,k^{-1})\\ &= (k\cdot h\ k\cdot 1, kk^{-1})\\ &= (k\cdot h, 1)\end{aligned}$$

which lies in $\hat{H}$. So $(1,k)\hat{H}(1,k)^{-1} \subseteq \hat{H}$, and the same computation with k^{-1} gives $(1,k)^{-1}\hat{H}(1,k) \subseteq \hat{H}$, hence equality. So every element of $\hat{K}$ normalizes $\hat{H}$, and so does every element of $\hat{H}$. Every element of G is $(h,k) = (h,1)(1,k)$, and $N_G(\hat{H})$ is a subgroup of G, so $N_G(\hat{H}) = G$ and $\hat{H} \trianglelefteq G$.

4. If $(h,1)=(1,k)$, then $h=1$ and $k=1$, so $\hat{H}\cap\hat{K}=\{(1,1)\}$.
5. This is the computation in (3): $khk^{-1}=(k\cdot h,1)$, which is identified with $k\cdot h=\varphi(k)(h)$.

Point (5) is where the semi-direct product departs from the direct product. In the coming example, play close attention to exactly how the action commute elements.

Remember that the semi-direct products are a generalization of direct products. Exactly how is what this next theorem identifies:

Proposition 5.3.4: When The Semi-Direct Product is a Direct Product

Let H and K be groups and let $\varphi: K\to \mathrm{Aut}(H)$ be a homomorphism. Then the following are equivalent:

1. The identity (set) map between $H\rtimes K$ and $H\times K$ is a group homomorphism hence an isomrophism
2. φ is the trivial homomorphism form K into $\mathrm{Aut}(H)$
3. $K\trianglelefteq H\rtimes K$

***Proof*:**
Write $G=H\rtimes K$ and identify H and K with the subgroups $\{(h,1)\}$ and $\{(1,k)\}$ of G as in proposition 5.3.3(2).

$(1\Rightarrow 2)$: By definition 5.3.2 the product in $H\rtimes K$ is

$$(h_1,k_1)(h_2,k_2)=(h_1\ k_1\cdot h_2,\ k_1k_2)$$

while the product in $H\times K$ is (h_1h_2,k_1k_2). If the identity set map is a homomorphism, these two agree for all choices, and comparing first coordinates gives $k_1\cdot h_2=h_2$ for every $h_2\in H$ and every $k_1\in K$. That says $\varphi(k_1)=\mathrm{id}_H$ for every $k_1\in K$, which is (2).

$(2\Rightarrow 3)$: If φ is trivial then $k\cdot h=h$ for all $h\in H$ and $k\in K$, so proposition 5.3.3(5) gives $khk^{-1}=h$, that is $hk=kh$. Hence every $h\in H$ normalizes K, and every element of K normalizes K, so $N_G(K)$ contains both H and K. Every element of G is of the form $(h,k)=(h,1)(1,k)$, so $G=HK$ and therefore $N_G(K)=G$, which says $K\trianglelefteq G$.

$(3\Rightarrow 1)$: Assume $K\trianglelefteq G$ and take $h\in H$, $k\in K$. Since K is normal, $hkh^{-1}\in K$, so

$$hkh^{-1}k^{-1}=(hkh^{-1})k^{-1}\in K$$

and since H is normal by proposition 5.3.3(3), $kh^{-1}k^{-1}\in H$, so the same element $hkh^{-1}k^{-1}=h(kh^{-1}k^{-1})$ lies in H. By proposition 5.3.3(4), $H\cap K=1$, so $hkh^{-1}k^{-1}=1$ and $hk=kh$. Feeding this back into proposition 5.3.3(5) gives $k\cdot h=khk^{-1}=h$, so the product in $H\rtimes K$ reads

$$(h_1,k_1)(h_2,k_2)=(h_1h_2,k_1k_2)$$

which is the product in $H\times K$. The identity set map is therefore a homomorphism, and being a bijection it is an isomorphism, which is (1).

Corollary 5.3.5: A Non-Abelian Group Of Order pq

Let $p < q$ be primes with $p \big| q-1$. Then there is a non-abelian group of order pq.

***Proof*:**
The nonzero classes modulo q form a group $(\mathbb{Z}/q\mathbb{Z})^\times$ of order $q-1$ under multiplication, since each is invertible by Bézout ([24, Extended Euclidean Algorithm (Bézout's Identity)]). As $p \big| q-1$, it contains an element s of order p (theorem 4.8.7). Multiplication by s is an automorphism of the additive group $\mathbb{Z}/q\mathbb{Z}$, and $\varphi\colon \mathbb{Z}/p\mathbb{Z} \to \mathrm{Aut}(\mathbb{Z}/q\mathbb{Z})$, $k \mapsto (h \mapsto s^k h)$, is a well-defined homomorphism because $s^p = 1$. By proposition 5.3.3, $G = \mathbb{Z}/q\mathbb{Z} \rtimes_\varphi \mathbb{Z}/p\mathbb{Z}$ has order pq, and for the classes $h = \overline{1}$ of $\mathbb{Z}/q\mathbb{Z}$ and $k = \overline{1}$ of $\mathbb{Z}/p\mathbb{Z}$, $khk^{-1} = \varphi(k)(h) = s \neq h$. So G is non-abelian.

Example 5.3.6: Semi-Direct Products

1. Let H be any abelian group (even of infinite order) and let $K = \langle x \rangle \cong Z_2$. be the group of order 2. Define $\varphi : K \to \mathrm{Aut}(H)$ by mapping x to the automorphism of inversion on H, so that the associated action is $x \cdot h = h^{-1}$, or, written in homomorphism form:

$$\varphi(x)(h) = h^{-1} \qquad \varphi(e)(h) = h$$

Note that this is an automorphism, for

$$f_x : H \to H, \qquad f_x(h) = h^{-1} \qquad f_x(xy) = (xy)^{-1} = x^{-1}y^{-1} = f_x(x)f_x(y)$$

since H is abelian, the map is it's own inverse (making it bijective).

Thus, we can define $H \rtimes_\varphi K$. To distinguish between the group-action and the group operation, I will ues the $+$ symbol to represent the operation for both groups. To get comfortable with this, let's do a computation:

$$(a_1, x)(a_2, x) = (a_1 + x \cdot a_2, x + x) = (a_1 a_2^{-1}, 0)$$

$$\begin{aligned} xhx^{-1} &= (0, x)(h, 0)(0, x^{-1}) \\ &= (0 + x \cdot h, 0 + x)(0, x^{-1}) = (-h, x)(0, x) \\ &= (-h + x \cdot 0, x + x) \\ &= (-h - h, 0) \\ &= (-h, 0) \\ &= h^{-1} \end{aligned}$$

Thus, we have:

$$xhx^{-1} = h^{-1} \Rightarrow xh = h^{-1}x$$

in the special case where $H = Z_n$, then $\mathbb{Z}/n\mathbb{Z} \rtimes \mathbb{Z}/2\mathbb{Z} \cong D_n$!!! if $H = \mathbb{Z}$, then we'd say $G = D_\infty$.

2. Let $H = \mathbb{R}^2$ and $K = SO(2)$.

3. Since simple groups have no normal subgroups, simple groups cannot be decomposed into semi-direct products, showing further how they are like "prime numbers" for groups.

With the idea of the semi-direct product in mind and a couple of examples, we end this section by given sufficient and necessary conditions to recognize semi-direct products:

Theorem 5.3.7: Recognition Theorem For Semi-Direct Products

Suppose G is a group with subgrosup H and K such that

1. $H \trianglelefteq G$
2. $H \cap K = 1$

Let $\varphi : K \to \text{Aut}(H)$ be the homomorphism define dby mapping $k \in K$ to the automrophis mof left conjugation by k on H. Then

$$HK \cong H \rtimes K$$

In particualr, if $G = HK$, with H and K satisfying the aforementioend conditions, then

$$G \cong H \rtimes K$$

Put another way, a group is a semi-derict product if there exists a complement for some proper normal subgroup of G.

***Proof*:**
First, φ is well defined and is a homomorphism. For $k \in K$ the map $c_k : h \mapsto khk^{-1}$ sends H into H because $H \trianglelefteq G$, it is a homomorphism of H, and $c_{k^{-1}}$ is a two-sided inverse for it, so $c_k \in \text{Aut}(H)$. For $k_1, k_2 \in K$ and $h \in H$,

$$c_{k_1k_2}(h) = (k_1k_2)h(k_1k_2)^{-1} = k_1(k_2hk_2^{-1})k_1^{-1} = (c_{k_1} \circ c_{k_2})(h)$$

so $\varphi : k \mapsto c_k$ is a homomorphism $K \to \text{Aut}(H)$.

Next, HK is a subgroup of G. Since $H \trianglelefteq G$ we have $hk = k(k^{-1}hk) \in KH$ and $kh = (khk^{-1})k \in HK$ for all $h \in H$ and $k \in K$, so $HK = KH$, and $HK \leqslant G$ by proposition 3.4.4.

Each element of HK is hk for exactly one pair (h, k): if $h_1k_1 = h_2k_2$ then $h_2^{-1}h_1 = k_2k_1^{-1}$ lies in $H \cap K = 1$, so $h_1 = h_2$ and $k_1 = k_2$. Hence

$$\Psi : HK \to H \rtimes K \qquad hk \mapsto (h, k)$$

is a well defined bijection. It remains to see that it is a homomorphism. Using $k_1h_2k_1^{-1} = \varphi(k_1)(h_2) = k_1 \cdot h_2$,

$$(h_1k_1)(h_2k_2) = h_1(k_1h_2k_1^{-1})k_1k_2 = (h_1\ k_1 \cdot h_2)(k_1k_2)$$

and the right hand side is already in the form (element of H)(element of K), so

$$\Psi\left((h_1k_1)(h_2k_2)\right) = (h_1\ k_1 \cdot h_2,\ k_1k_2) = (h_1, k_1)(h_2, k_2) = \Psi(h_1k_1)\Psi(h_2k_2)$$

by definition 5.3.2. Therefore $HK \cong H \rtimes K$, and when $G = HK$ this reads $G \cong H \rtimes K$.

A semi-direct product $G = H \rtimes_\varphi K$ is an extension of H by K. By definition 5.3.2, the map $G \to K$

sending (h, k) to k is a surjective homomorphism, and its kernel is the subgroup $\{(h, 1) \mid h \in H\}$, which is isomorphic to H (proposition 5.3.3(2)). So $G/H \cong K$ by corollary 3.1.17. Moreover, the homomorphism $K \to G$ sending k to $(1, k)$ composed with this map is the identity of K. Extensions with such a homomorphism are called split, and every split extension is a semi-direct product (proposition 5.3.9).

Definition 5.3.8: Split Sequence

A short exact sequence $1 \to A \xrightarrow{f} B \xrightarrow{g} C \to 1$ (definition 3.6.4) *splits* if there is a homomorphism $\psi\colon C \to B$ with $g \circ \psi = \mathrm{id}_C$, that is, if the following diagram commutes:

$$\begin{array}{ccccccccc} 1 & \longrightarrow & A & \xrightarrow{f} & B & \xrightarrow{g} & C & \longrightarrow & 1 \\ & & & & & \nwarrow^{\psi} & \uparrow{\scriptstyle \mathrm{id}_C} & & \\ & & & & & & C & & \end{array}$$

Proposition 5.3.9: Split Extensions Are Semi-Direct Products

Let $1 \to A \xrightarrow{f} B \xrightarrow{g} C \to 1$ be a short exact sequence that splits by $\psi\colon C \to B$. Then $N = f(A)$ is a normal subgroup of B isomorphic to A, the subgroup $Q = \psi(C)$ is isomorphic to C, $N \cap Q = 1$ and $B = NQ$. Hence $B \cong N \rtimes Q$, with Q acting on N by conjugation.

***Proof*:**
By exactness $N = \ker(g)$, which is normal in B (proposition 3.1.14), and f is an injective homomorphism onto N, so $N \cong A$. If $\psi(c) = \psi(c')$, then $c = g(\psi(c)) = g(\psi(c')) = c'$, so ψ is an injective homomorphism onto Q, and $Q \cong C$. If $\psi(c) \in N$, then $c = g(\psi(c)) = 1$, so $N \cap Q = 1$. For $b \in B$ put $q = \psi(g(b)) \in Q$. Then $g(bq^{-1}) = g(b)g(\psi(g(b)))^{-1} = g(b)g(b)^{-1} = 1$, so $bq^{-1} \in N$ and $b = (bq^{-1})q \in NQ$. Hence $B = NQ$, and theorem 5.3.7 gives $B \cong N \rtimes Q$, as claimed.

When the action of Q on N is trivial, the multiplication of definition 5.3.2 is componentwise and $N \rtimes Q$ is the direct product $N \times Q$. For modules, split sequences return in section 17.1. Not every short exact sequence splits:

Example 5.3.10: Counter-Example
Write $C_2 = \mathbb{Z}/2\mathbb{Z}$ and $C_4 = \mathbb{Z}/4\mathbb{Z}$ additively, and let $f\colon C_2 \to C_4$ send 1 to 2 and $g\colon C_4 \to C_2$ send x to its residue modulo 2. Then

$$1 \longrightarrow C_2 \xrightarrow{f} C_4 \xrightarrow{g} C_2 \longrightarrow 1$$

is exact, since f is injective, g is surjective and $\mathrm{im}(f) = \{0, 2\} = \ker(g)$. It does not split: a homomorphism $\psi\colon C_2 \to C_4$ sends 1 to an element x with $x + x = \psi(1 + 1) = \psi(0) = 0$, so $x \in \{0, 2\}$ and $g(\psi(1)) = 0 \neq 1$. A semi-direct product of two groups of order 2 contains the two subgroups $\{(h, 1)\}$ and $\{(1, k)\}$ of order 2, which meet trivially (proposition 5.3.3(2) and (4)), while 2 is the only element of order 2 in C_4. So C_4 is not such a semi-direct product.

5.4 A Categorical approach of the Product

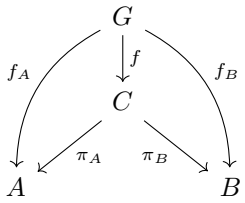
The idea of being able to take the "product" of two objects is not unique to sets and groups. As we continue to explore more and more algebraic structures, we will keep asking ourselves a similar question: Does algebraic structure X have the flexibility/restriction to define products? [2] What we will be doing then is finding a property which we can use to define what a product is in different algebraic settings.

Let's pick two arbitrary groups A and B. We defined product in the classical definition as the new group $(A \times B, (\cdot_A, \cdot_B))$. Let's denote this new group C. The key idea is that C is a group that contains groups A and B, and you can "recover" A and B from C. The way we can recover these two groups from C is simply by projcting to them:

$$\pi_A : C \to A,\ (a,b) \mapsto a \qquad \pi_B : C \to B,\ (a,b) \mapsto b$$

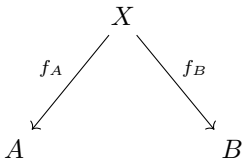
It should be checked that π_A and π_B are both surjective homomorphisms. Notice that $\pi_A(C) \cong A$ and $\pi_A(C) \cong B$. In this way, we have "recovered" A and B from C. Furthermore, C contains no more information than information about A and B. So essentially, if there exists a group C that most efficienly stores A and B, in particular if there are two functions (in our case π_A, π_B) which let's us recover A and B, then we can say C is the product of A and B. let's write this group and these two functions (C, π_A, π_B)

But is C the most efficient group for storing information about the group A and B? We can answ3re this by relating C to any other group, as is common in category theory. Let's say we have a group $G \in \text{Obj}(Grp)$ were we have a homomorphism $G \xrightarrow{f_A} A$ and $G \xrightarrow{f_B} B$ (so some structe of G is in A and B). Then does there exist a homomorphism $f : G \to C$ which can compose with π_A and π_B and produce f_1 and f_2? In other words: (Rushed)



In a sense, it's saying that whatever structure G has of A and B, C must have it too, since we can pass through C wihtout loosing that structure.

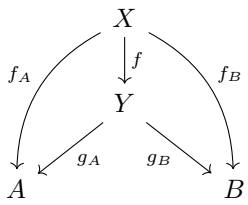
We can now formalize this idea with categories. The objects of $C_{A,B}$ are



and such an object can be represented as (X, f_A, f_B), where X is a group and $f_A \colon X \to A$ and $f_B \colon X \to B$ are homomorphisms. If (X, f_A, f_B) and (Y, g_A, g_B) are objects, then a morphism

[2] As usual with category theory, the question of whether the product exists goes beyond algebra: Topological spaces, uniform spaces, graphs, ordered sets, and other domains of mathematics will also have products!

$f : (X, f_A, f_B) \to (Y, g_A, g_B)$ is a homomorphism $f\colon X \to Y$ such that the following diagram commutes, that is, $g_A \circ f = f_A$ and $g_B \circ f = f_B$:



Example 5.4.1: Morphisms In $C_{A,B}$

Let $A = B = \mathbb{Z}$, and consider the objects $(\mathbb{Z}, \mathrm{id}_\mathbb{Z}, \mathrm{id}_\mathbb{Z})$ and $(\mathbb{Z} \times \mathbb{Z}, \pi_1, \pi_2)$, where π_1 and π_2 are the projections, which are homomorphisms (proposition 5.1.3(3)). The diagonal homomorphism $f\colon \mathbb{Z} \to \mathbb{Z} \times \mathbb{Z}$, $f(n) = (n, n)$, is a morphism from the first object to the second, since $\pi_1 \circ f = \mathrm{id}_\mathbb{Z}$ and $\pi_2 \circ f = \mathrm{id}_\mathbb{Z}$. The homomorphism $f'(n) = (n, 0)$ is not, since $\pi_2(f'(1)) = 0 \neq 1$.

Proposition 5.4.2: $C_{A,B}$ Forms A Category

Let $C_{A,B}$ be defined as above. Then $C_{A,B}$ forms a category

***Proof*:**

The requirements of a category hold, as in the proof of proposition 2.3.21:

1. (composition) Let $f\colon (X, f_A, f_B) \to (Y, g_A, g_B)$ and $g\colon (Y, g_A, g_B) \to (Z, h_A, h_B)$ be morphisms. The composite $g \circ f\colon X \to Z$ is a homomorphism by proposition 1.4.4(6), and

$$h_A \circ (g \circ f) = (h_A \circ g) \circ f = g_A \circ f = f_A$$

   and in the same way $h_B \circ (g \circ f) = f_B$. So $g \circ f$ is a morphism $(X, f_A, f_B) \to (Z, h_A, h_B)$.

2. (identities) For an object (X, f_A, f_B), the identity homomorphism id_X satisfies $f_A \circ \mathrm{id}_X = f_A$ and $f_B \circ \mathrm{id}_X = f_B$, so it is a morphism from (X, f_A, f_B) to itself. For morphisms f out of (X, f_A, f_B) and f' into it, $f \circ \mathrm{id}_X = f$ and $\mathrm{id}_X \circ f' = f'$.

3. (associativity) Composition of morphisms is composition of functions, and for composable morphisms f, g, k both $(k \circ (g \circ f))(x)$ and $((k \circ g) \circ f)(x)$ equal $k(g(f(x)))$.

So $C_{A,B}$ is a category, as claimed.

We can ask whether $C_{A,B}$ has *final* objects, that is, an object $(G, f_A, f_B) \in C_{A,B}$ where for any $(H, g_A, g_B) \in C_{A,B}$, $\mathrm{Hom}((H, g_A, g_B), (G, f_A, f_B))$ is a singleton. In fact, (C, π_A, π_B) is such an object. In other words, we will show that (C, π_A, π_B) must be the only group with this property since the axioms of groups and homomorphisms forces it to be.

Theorem 5.4.3: Universal Property Of The Product For Groups

Let A and B be groups. Then $A \times B$ is a group, and $\pi_A \colon A \times B \to A$ and $\pi_B \colon A \times B \to B$ are homomorphisms. For every group H and all homomorphisms $f_A \colon H \to A$ and $f_B \colon H \to B$, there exists a unique homomorphism $f \colon H \to A \times B$ such that $f_A = \pi_A \circ f$ and $f_B = \pi_B \circ f$. The homomorphism f is denoted $f_A \times f_B$.

***Proof*:**
The set $A \times B$ with the componentwise operation is a group, and π_A and π_B are homomorphisms (proposition 5.1.3). Next, let $(A \times B, \pi_A, \pi_B)$ be as above and let $(H, f_A, f_B) \in C_{A,B}$. Define

$$f : H \to A \times B \qquad f(h) = (f_A(h), f_B(h))$$

This f is a homomorphism because f_A and f_B are and the operation on $A \times B$ is componentwise. Also $(\pi_A \circ f)(h) = \pi_A(f_A(h), f_B(h)) = f_A(h)$, and similarly for f_B, showing that the diagram indeed comutes, and so f is indeed a morphism in $C_{A,B}$.

To show that f is unique, let's assume there is another morphism f'. Then

$$f_A(h) = (\pi_A \circ f')(h) = \pi_A(f'(h)) = \pi_A(f'(h)_A, f'(h)_B) = f'(h)_A$$

where $f'(h)_A$ and $f'(h)_B$ represent the components of $f'(h)$. Similarly, $f'(h)_B = f_B(h)$. Therefore, $f'(h) = (f_A(h), f_B(h))$. However, that is exactly what f does:

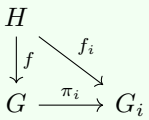
$$f(h) = (f_A(h), f_B(h)) = f'(h)$$

thus $f = f'$, showing uniqueness. Thus (C, π_A, π_B) is the final object of $C_{A,B}$. In other words, $A \times B$ respects the universal property of the product in Group.

We can further generalize this for an arbitrary direct product, that is if $(G_i)_{i \in I}$ is a family of groups, then their direct product $G = \prod_{i \in I} G_i$ along with $\pi_i : G \to G_i$ for every $i \in I$ will be a final object in $C_{(G_i)_{i \in I}}$.

Theorem 5.4.4: Universal Property For Arbitrary Direct Product Of Groups

Let $(G_i)_{i \in I}$ be a family of groups, let $G = \prod_{i \in I} G_i$ be their direct product, whose elements are the families $(g_i)_{i \in I}$ with $g_i \in G_i$ and whose operation is componentwise, and let $\pi_i \colon G \to G_i$ be the projections $\pi_i((g_j)_{j \in I}) = g_i$. For every group H, and every I-indexed family of homomorphisms $f_i : H \to G_i$, there exists a unique homomorphism $f \colon H \to G$ such that the following diagrams commute for every $i \in I$:



***Proof*:**
As in the proof of proposition 5.1.3, G is a group: the family $(1_{G_i})_{i\in I}$ is an identity, the family $(g_i^{-1})_{i\in I}$ is an inverse of $(g_i)_{i\in I}$, and associativity holds in each component. Each π_i is a homomorphism, since $\pi_i((g_j)_j(h_j)_j) = g_ih_i = \pi_i((g_j)_j)\pi_i((h_j)_j)$.

Define $f\colon H \to G$ by $f(h) = (f_i(h))_{i\in I}$. Since each f_i is a homomorphism and the operation of G is componentwise, $f(hh') = (f_i(h)f_i(h'))_{i\in I} = f(h)f(h')$, so f is a homomorphism. Also $\pi_i(f(h)) = f_i(h)$ for all $h \in H$, that is, $\pi_i \circ f = f_i$ for every $i \in I$.

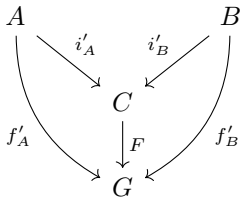
Let $f'\colon H \to G$ be a homomorphism with $\pi_i \circ f' = f_i$ for every $i \in I$. For $h \in H$, the ith component of $f'(h)$ is $\pi_i(f'(h)) = f_i(h)$, which is the ith component of $f(h)$. So $f'(h) = f(h)$ for every h, and $f' = f$, completing the proof.

Exercise 5.4.1

1. Given that you have seen the universal property of products for groups, what do you think it would be for sets? What would be the generalization for an aribtary category? (For a hint, see the preliminary chapter or section 5.4)
2. let $1 \to A \to B \to C \to 1$ be an exact sequence. Show there exists an action $\rho : C \to \mathrm{Out}(A)$

5.4.1 The dual of the Product

What happens if you flip the arrows? Instead of looking at $C \xrightarrow{\pi_A} A$ and $C \xrightarrow{\pi_B} B$, we look at $A \xrightarrow{i'_A} C$ and $B \xrightarrow{i'_A} C$, and for all groups G, if we have $A \to G$ and $B \to G$, then there exists a homomorphism $f : C \to G$ such that the following diagram commutes



Does $(A \times B, i_A, i_B)$ where $i_A(a) = (a, 1)$ and $i_B(b) = (1, b)$ work? Let's break down the restrictions of this commutative diagram to check. By the above commutative diagram, we need that $f'_A(a) = F(i_A(a)) = F((a, 1))$ and that $f'_B(b) = F(i_B(b)) = F((1, b))$. So we have

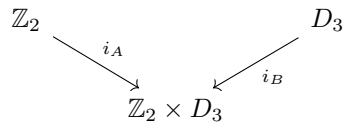
$$F((a, 1))F((1, b)) = f'_A(a)f'_B(b)$$

Since F must be a homomorphism, then:

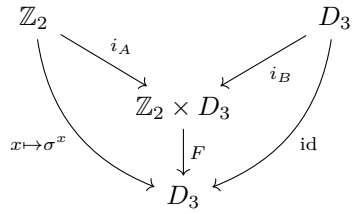
$$F((a, 1))F((1, b)) = F((a, 1)(1, b)) = F((a, b)) = F((1, b)(a, 1)) = F((1, b))F((a, 1))$$

that is $f'_A(a)f'_B(b) = f'_B(b)f'_A(a)$, that is, the image of A and B must always commute. However, we can see that there will exist a group G and homomorphisms f'_A f'_B where $f'_A(a)f'_B(b) \neq f'_B(b)f'_A(a)$:

Example 5.4.5: Counter-example
Let $A = \mathbb{Z}_2$ and $B = D_3$ so that we get



Let $G = D_3$, $f'_A : \mathbb{Z}_2 \to D_3$, $f'_A(x) = \sigma^x$, $f'_B : D_3 \to D_3$, $f'_B(x) = x$ (i.e. f'_B is the identity map). Then is it true that there exists a homomorphism F such that:



By definition $F((a,1)) = F(i_A(a)) = f'_A(a) = \sigma^a$ and $F((1,b)) = F(i_B(b)) = f'_B(b) = b$, so

$$F((a,b)) = \sigma^a b$$

and by what we just saw, we must also have that $F((a,b)) = b\sigma^a$ for every $a \in A$ and $b \in B$. However

$$F((1,2)) = \sigma r^2 \neq r^2 \sigma$$

showing that F is *not* a well-defined homomorphism!

Restricting to abelian groups removes this problem. When G is abelian, $F((a,b)) = f'_A(a)f'_B(b)$ is a homomorphism, since $F((a,b)(a',b')) = f'_A(a)f'_A(a')f'_B(b)f'_B(b') = f'_A(a)f'_B(b)f'_A(a')f'_B(b')$, and by the computation above it is the only homomorphism with $F \circ i_A = f'_A$ and $F \circ i_B = f'_B$. So $(A \times B, i_A, i_B)$ is the initial object in the category whose objects are triples (G, f_A, f_B) with G abelian and whose morphisms are the commutative diagrams above.

The problem here came from the extra structure of $A \times B$: by proposition 5.1.3(4), the copies of A and B inside $A \times B$ commute with each other, $(a,1)(1,b) = (1,b)(a,1)$.

So does there exist a group and maps will be the initial object in this category? The answer to this question is yes! It is a group we have not yet encountered. The main idea here is that this group is supposted to be the most generic way of having A and B (so as little relations btween A and B) so that the homomorphism from this group to any G does not preserve these properties. We have encountered a similar situation like this before when defining the *free group* (see section 2.3.3). Something analogous is defined called the *free product*.

Definition 5.4.6: Free Product

Let H and K be groups with presentations (S_H, R_H) and (S_K, R_K) (definition 3.2.2), where the generating sets S_H and S_K are disjoint. Then

$$H * K = \langle S_H \cup S_K | R_H \cup R_K \rangle$$

Example 5.4.7: Free Products

The free product $\mathbb{Z}/2\mathbb{Z} * \mathbb{Z}/2\mathbb{Z} = \langle x, y \big| x^2 = y^2 = 1 \rangle$ is the group of section 2.2.2 whose elements of finite order do not form a subgroup: x and y have order 2, but xy does not have finite order.

Exercise 5.4.2

1. Show that $(H * K, i_H, i_K)$ respects the universal property of the co-product in **Grp**.

Infinite abelian case

It should be mentioend again that if we restrict to **Ab**, then the universal property of the coproduct is fulfilled by $H \times K$, which differs from $H * K$ unless H or K is trivial. Is this also true in the infinite case: given abelian groups $G_1, G_2, \dots$, does $\prod_i G_i$ respect the universal property of the coproduct? The answer is no!

Example 5.4.8: The Product Is Not A Coproduct In Ab

Let $G = \prod_{i=1}^{\infty} \mathbb{Z}$, and let $\iota_i \colon \mathbb{Z} \to G$ send x to the sequence with x in position i and 0 elsewhere. Let $D \subseteq G$ be the set of sequences with only finitely many nonzero entries. It is a subgroup: it contains 0, and a difference of two such sequences is nonzero only where one of them is. Since G is abelian, $D \trianglelefteq G$ (corollary 3.1.12), and the quotient $H = G/D$ is an abelian group, with the projection $\pi \colon G \to H$ a homomorphism (definition 3.1.13). Take $f_i = 0 \colon \mathbb{Z} \to H$ for every i. Both the zero homomorphism $G \to H$ and π satisfy $F \circ \iota_i = 0 = f_i$ for every i, because $\iota_i(\mathbb{Z}) \subseteq D$. They are different: the sequence $(1, 1, 1, \dots)$ has infinitely many nonzero entries, so it does not lie in D, and $\pi((1, 1, 1, \dots)) \neq 0$. So the homomorphism F with $F \circ \iota_i = f_i$ for all i is not unique, and G with the maps ι_i does not have the universal property of the coproduct in **Ab**.

In the subgroup D of this example every element is a finite sum of elements $\iota_i(x_i)$, so a homomorphism out of D is determined by its composites with the ι_i. This is what makes D the coproduct, as theorem 5.4.10 below shows for any family.

A group still exists which can fix our scenario and respect the universal property for coproducts in the infinite case:

Definition 5.4.9: Direct Sum Of Groups

Let $\{A_i\}_{i\in I}$ be a family of groups. Then the *direct sum*

$$A = \bigoplus_{i\in I} A_i$$

is a subset of $\prod_{i\in I} A_i$ consisting of all elements $(x_i)_{i\in I}$ with $x_i \in A_i$ such that $x_i = 0$ for all but finitely many indices i (cofinitely many indices).

It is a subgroup of $\prod_{i\in I} A_i$: it contains the identity, and if (x_i) and (y_i) have $x_i = 0$ outside a finite set S and $y_i = 0$ outside a finite set T, then $(x_i - y_i)$ has $x_i - y_i = 0$ outside $S \cup T$. Note that in the finite case, $\bigoplus_{i=1}^n G_i = \prod_{i=1}^n G_i$. It only differs once we reach the infinite case

Theorem 5.4.10: Universal Property Of Coproducts In Ab

Let $(G_i)_{i\in I}$ be a family of abelian groups, written additively, let $G = \bigoplus_{i\in I} G_i$ (definition 5.4.9), and for $j \in I$ let $\iota_j \colon G_j \to G$ send z to the family with z in position j and 0 in every other position. Then G respects the universal property of the coproduct in **Ab**: for every abelian group H and every family of homomorphisms $f_i \colon G_i \to H$, there is a unique homomorphism $F \colon G \to H$ with $F \circ \iota_i = f_i$ for every $i \in I$.

***Proof*:**
By the remark after definition 5.4.9, G is a subgroup of $\prod_{i\in I} G_i$, so it is a group under the componentwise operation, and it is abelian since each G_i is. Each ι_j lands in G and is a homomorphism, since the operation is componentwise.

Step 1: definition of F. The *support* $\{i \in I \mid x_i \neq 0\}$ of $x = (x_i)_{i\in I} \in G$ is finite. For a finite set $S \subseteq I$ containing the support of x, put

$$F_S(x) = \sum_{i\in S} f_i(x_i)$$

a finite sum in the abelian group H, whose value does not depend on the order of its terms. It does not depend on S either. If $S \subseteq S'$, the extra terms $f_i(x_i)$ with $i \in S' \setminus S$ have $x_i = 0$, so they vanish because $f_i(0) = 0$ (proposition 1.4.4(1)). Any two admissible sets are contained in their union, so all the $F_S(x)$ agree, and we put $F(x) = F_S(x)$.

Step 2: F is the required homomorphism. Let $x, y \in G$, and let S be the union of their supports, which also contains the support of $x + y$. Since each f_i is a homomorphism and H is abelian,

$$F(x+y) = \sum_{i\in S} f_i(x_i + y_i) = \sum_{i\in S} f_i(x_i) + \sum_{i\in S} f_i(y_i) = F(x) + F(y)$$

For $j \in I$ and $z \in G_j$, the support of $\iota_j(z)$ is contained in $\{j\}$, so $F(\iota_j(z)) = f_j(z)$, that is, $F \circ \iota_j = f_j$.

Step 3: uniqueness. Let $x \in G$ have support contained in the finite set S. Then $x = \sum_{i\in S} \iota_i(x_i)$,

since both sides have ith component x_i for $i \in S$ and 0 for $i \notin S$. If $F' \colon G \to H$ is a homomorphism with $F' \circ \iota_i = f_i$ for every i, then

$$F'(x) = \sum_{i \in S} F'(\iota_i(x_i)) = \sum_{i \in S} f_i(x_i) = F(x)$$

so $F' = F$, as we sought to show.

5.4.2 Universal Property of Semi-Direct Products

Every element (h, k) of $H \rtimes_\varphi K$ is the product $(h, 1)(1, k)$, so a homomorphism out of $H \rtimes_\varphi K$ is determined by its values on the two factors.

Proposition 5.4.11: Universal Property Of The Semi-Direct Product

Let H and K be groups, let $\varphi \colon K \to \mathrm{Aut}(H)$ be a homomorphism, let $G = H \rtimes_\varphi K$ (definition 5.3.2), and let $\iota_H(h) = (h, 1)$ and $\iota_K(k) = (1, k)$. Let L be a group.

1. If $F \colon G \to L$ is a homomorphism, then $f = F \circ \iota_H$ and $g = F \circ \iota_K$ are homomorphisms with
$$g(k)f(h)g(k)^{-1} = f(\varphi(k)(h)) \qquad \text{for all } h \in H,\ k \in K$$
2. Conversely, if $f \colon H \to L$ and $g \colon K \to L$ are homomorphisms satisfying this condition, then there is a unique homomorphism $F \colon G \to L$ with $F \circ \iota_H = f$ and $F \circ \iota_K = g$, namely $F(h, k) = f(h)g(k)$.

***Proof*:**
The maps ι_H and ι_K are homomorphisms by proposition 5.3.3(2). Every automorphism of H fixes 1 (proposition 1.4.4(1)), so the multiplication of definition 5.3.2 gives, for $h \in H$ and $k \in K$,

$$(h, 1)(1, k) = (h, k) \qquad (1, k)(h, 1) = (\varphi(k)(h), k) = (\varphi(k)(h), 1)(1, k)$$

1. The maps f and g are composites of homomorphisms, hence homomorphisms (proposition 1.4.4(6)). Applying F to the second identity gives $g(k)f(h) = f(\varphi(k)(h))g(k)$, which is the condition.
2. If F is such a homomorphism, the first identity gives $F(h, k) = F(h, 1)F(1, k) = f(h)g(k)$, which proves uniqueness. For existence, put $F(h, k) = f(h)g(k)$. Then $F(h, 1) = f(h)$ and $F(1, k) = g(k)$, and for $(h_1, k_1), (h_2, k_2) \in G$ the condition gives
$$F((h_1, k_1)(h_2, k_2)) = f(h_1)f(\varphi(k_1)(h_2))g(k_1)g(k_2) = f(h_1)g(k_1)f(h_2)g(k_2)$$
which is $F(h_1, k_1)F(h_2, k_2)$. So F is a homomorphism, as claimed.

When φ is trivial, G is the direct product $H \times K$, and the condition says that every $f(h)$ commutes with every $g(k)$. So the homomorphisms out of $H \times K$ correspond to the pairs of homomorphisms out of H and K whose images commute elementwise. This is a property of maps out of $H \times K$, while theorem 5.4.3 describes maps into it.

A group can be a semi-direct product in several ways. In D_6, with the rotation r and reflection s of proposition 1.2.29, the rotations $\langle r \rangle \cong C_6$ have index 2, hence form a normal subgroup (corollary 3.3.6), with complement $\langle s \rangle$, so $D_6 \cong C_6 \rtimes C_2$ (theorem 5.3.7). Also, r^2 has order 3 and $sr^2s^{-1} = r^{-2}$ (proposition 1.2.29), so $\langle r^2, s \rangle = \{1, r^2, r^4, s, sr^2, sr^4\}$ is a copy of D_3 of index 2, hence normal. The center $\langle r^3 \rangle \cong C_2$ is normal, the two meet trivially, and $6 \cdot 2 = 12$, so $D_6 \cong D_3 \times C_2$ (theorem 5.2.6), a semi-direct product with trivial action. In general, $G \cong H \rtimes K$ exactly when H is a normal subgroup with a complement K (theorem 5.3.7 and proposition 5.3.3).

Exercise 5.4.3

1. Show that $\mathrm{GL}_n(F) \cong \mathrm{SL}_n(F) \rtimes F^\times$ for every field F and every $n \geq 1$.

6

Further Topics in Group Theory

Two subgroups measure how far a group is from being abelian: the commutator subgroup, which is trivial exactly when the group is abelian, and the center, which is the whole group exactly when the group is abelian. Iterating the commutator subgroup leads to metabelian and solvable groups, iterating the center leads to nilpotent groups, and subnormal series with cyclic factors lead to polycyclic groups.

6.1 Generalization of Abelian Groups

So far we saw that

$$\text{cyclic} \Rightarrow \text{Abelian}$$

We found a way to find how close agroup is to being abelian using commutator groups. If $G' = \{1\}$, then G is abelian. Otherwise, G is not abelian. We will develop further generalization's from this concept.

Definition 6.1.1: Metabelian Group

Let G be a group. If $G'' = (G')' = \{1\}$, we say that G is a *metabelian* group

Proposition 6.1.2: Properties Of Metabelian Groups

Let G be a metabelian group. Then

1. any subgroup is metabelian
2. If $\varphi : G \to H$ is a homomorphism, then $\varphi(G)$ is metabelian

***Proof*:**

1. Let $H \leqslant G$. Every commutator of elements of H is a commutator of elements of G, so it lies in G', and $H' \leqslant G'$ by proposition 2.3.17. Applying this to the subgroup H' of the group G' gives $H'' \leqslant G'' = \{1\}$, so H is metabelian.
2. Let $\varphi : G \to H$ be a homomorphism and G a metabelian group, and consider $\varphi(G) \leqslant H$. We will show that homomorphisms preserve commutator subgroups to show that $\varphi(G)$ is metabelian. In particular, if $N \leqslant G$, then $\varphi(N') = \varphi(N)'$. Indeed, $\varphi([x, y]) = [\varphi(x), \varphi(y)]$ for $x, y \in N$, so the commutators of elements of $\varphi(N)$ are exactly the images of the commutators of elements of N. The image $\varphi(N')$ is a subgroup (proposition 1.4.4(5)) containing all of them, so $\varphi(N)' \leqslant \varphi(N')$ by proposition 2.3.17. The preimage $\varphi^{-1}(\varphi(N)')$ is a subgroup of G by proposition 2.1.3, since $\varphi(xy^{-1}) = \varphi(x)\varphi(y)^{-1}$, and it contains every commutator of elements of N, so it contains N', which gives $\varphi(N') \leqslant \varphi(N)'$. Thus

$$\varphi(G)'' = \varphi(G')' = \varphi(G'') = \varphi(\{1\}) = \{1\}$$

and so $\varphi(G)$ is also metabelian.

Example 6.1.3: Metabelian Groups

1. Any abelian group is always metabelian
2. if A and B are abelian groups, and $A \rtimes B$ is well-defined, then $A \rtimes B$ is metabelian.

Proposition 6.1.4: Extension of Abelian Groups

A group G is metabelian if and only if it has an abelian normal subgroup A with G/A abelian, that is, if and only if G is an extension of an abelian group by an abelian group:

$$1 \longrightarrow A \longrightarrow G \longrightarrow M \longrightarrow 1$$

with A and M abelian.

***Proof*:**
Let $A \trianglelefteq G$ be abelian with G/A abelian. By proposition 5.2.3(4), $G' \subseteq A$, so G' is abelian and $G'' = 1$. Conversely, let $G'' = 1$, so that G' is abelian. It is normal: for $g \in G$ and $h \in G'$, $ghg^{-1} = (ghg^{-1}h^{-1})h \in G'$. By proposition 5.2.3(4), G/G' is abelian, so $A = G'$ works, completing the proof.

6.1.1 Solvable Groups

Metabelian groups are a simple example of a group called a *solvable* group. Instead of looking at the 2nd commutator subgroup being trivial, we can ask if the nth commutator subgroup will be trivial:

Definition 6.1.5: nth Commutator Group

Let G be a group. Put $G_{(0)} = G$, and for $n \geq 1$ let

$$G_{(n)} = [G_{(n-1)}, G_{(n-1)}] = (G_{(n-1)})'$$

the subgroup generated by the commutators $[a, b]$ with $a, b \in G_{(n-1)}$ (definition 5.2.2). It is called the nth commutator group, and $G_{(1)} = G'$.

Definition 6.1.6: Solvable Groups

A group G is *solvable* if there is a subnormal series with commutator subgroup:

$$G = G_{(0)} \trianglerighteq G_{(1)} \trianglerighteq G_{(2)} \trianglerighteq \cdots \trianglerighteq G_{(n)} = \{1\}$$

such a subnormal series is called the *derived series*. If $G_{(n)} = 1$, we say G has *solvable length* n.

Example 6.1.7: Solvable Groups

1. Every abelian group is solvable, since $1 \trianglelefteq G$ is a normal series.
2. Any metabelian group is solvable, with the series is called the *derived series of* G.

$$1 = G'' \trianglelefteq G' \trianglelefteq G$$

Since G/G' is abelian, and $G'/G'' \cong G'$ which is abelian.

There are equivalent definitions of solvability that are worth keeping in mind to have a variety of intuitions.

Lemma 6.1.8: characteristic of nth commutator

Let G be a group. Then $G_{(i+1)}$ char $G_{(i)}$ for any $i \in \mathbb{N}$. Consequently, $G_{(i+1)} \trianglelefteq G_{(i)}$.

***Proof*:**
By definition 6.1.5, $G_{(i+1)}$ is the commutator subgroup of the group $G_{(i)}$, so proposition 5.2.3(3), applied to the group $G_{(i)}$, gives $G_{(i+1)}$ char $G_{(i)}$, and $G_{(i+1)} \trianglelefteq G_{(i)}$ by proposition 4.7.13(1).

Normality can also be proved directly. Every commutator of elements of $G_{(i)}$ lies in $G_{(i)}$, so $G_{(i+1)} \leqslant G_{(i)}$ by proposition 2.3.17. Let $g \in G_{(i)}$ and $h \in G_{(i+1)}$. Then $h \in G_{(i)}$, so

$ghg^{-1}h^{-1} = [g^{-1}, h^{-1}]$ is a commutator of elements of $G_{(i)}$ and lies in $G_{(i+1)}$, and hence

$$ghg^{-1} = (ghg^{-1}h^{-1})h \in G_{(i+1)}$$

Thus $gG_{(i+1)}g^{-1} \subseteq G_{(i+1)}$ for every $g \in G_{(i)}$, and $G_{(i+1)} \trianglelefteq G_{(i)}$ by proposition 3.1.11(5), as claimed.

Theorem 6.1.9: Equivalent Definitions Of Solvable group

Let G be a group. The following are equivalent

1. G is solvable
2. There exists a subnormal series

$$G = A_0 \trianglerighteq A_1 \trianglerighteq A_2 \trianglerighteq \cdots \trianglerighteq A_n = \{1\}$$

such that A_i/A_{i+1} is abelian for $0 \leq i < n$. Such a subnormal series is called a *subnormal series with abelian factors.*

***Proof*:**
For a group A and a normal subgroup M of A, the quotient A/M is abelian if and only if $A' \leqslant M$. Indeed, for $x, y \in A$ the cosets xM and yM commute exactly when $(yx)^{-1}xy = [x, y] \in M$, and M contains every commutator of elements of A exactly when it contains the subgroup A' that they generate (proposition 2.3.17).

(1) $\Rightarrow$ (2) Let $G_{(n)} = 1$. By lemma 6.1.8, $G = G_{(0)} \trianglerighteq G_{(1)} \trianglerighteq \cdots \trianglerighteq G_{(n)} = 1$ is a subnormal series. Each factor $G_{(i)}/G_{(i+1)}$ is abelian by the first paragraph, applied to $A = G_{(i)}$ and $M = G_{(i+1)} = A'$.

(2) $\Rightarrow$ (1) Let (A_i) be a subnormal series with abelian factors. We show $G_{(i)} \leqslant A_i$ by induction on i. For $i = 0$ both are G. If $G_{(i)} \leqslant A_i$, then every commutator of elements of $G_{(i)}$ is a commutator of elements of A_i, so $G_{(i+1)} \leqslant (A_i)'$ by proposition 2.3.17. The first paragraph, applied to $A = A_i$ and $M = A_{i+1}$, gives $(A_i)' \leqslant A_{i+1}$, since A_i/A_{i+1} is abelian. So $G_{(i+1)} \leqslant A_{i+1}$. Hence $G_{(n)} \leqslant A_n = 1$, and G is solvable, as claimed.

The use of the 2nd definition is to show that finding any solvable series for a group is enough to show a group is solvable. The advantage of the 1st definition is that it is the most refined solvable series of a group, since the commutator subgroup is smallest subgroup which makes the quotients A_i/A_{i+1} abelian.

Proposition 6.1.10: Properties Of Solvable Groups

Let G be solvable. Then

1. subgroups of solvable groups are solvable
2. if $\varphi : G \to H$ is a homomorphism, then $\varphi(G)$ is solvable
3. if M is a group with a normal subgroup N such that N and M/N are solvable, that is, if M is an extension of a solvable group by a solvable group, then M is solvable.

***Proof*:**
Let $G_{(n)} = 1$.

1. Let $H \leqslant G$. We show $H_{(i)} \leqslant G_{(i)}$ by induction on i. For $i = 0$ this is $H \leqslant G$. If $H_{(i)} \leqslant G_{(i)}$, then every commutator of elements of $H_{(i)}$ is a commutator of elements of $G_{(i)}$, so $H_{(i+1)} \leqslant G_{(i+1)}$ by proposition 2.3.17. Hence $H_{(n)} \leqslant G_{(n)} = 1$, and H is solvable.
2. First, $\psi(A') = \psi(A)'$ for every homomorphism ψ and every subgroup A of its domain. Indeed, $\psi([x, y]) = [\psi(x), \psi(y)]$, so the commutators of elements of $\psi(A)$ are the images of the commutators of elements of A. The image $\psi(A')$ is a subgroup (proposition 1.4.4(5)) containing all of them, so $\psi(A)' \leqslant \psi(A')$ by proposition 2.3.17. The preimage of $\psi(A)'$ is a subgroup by proposition 2.1.3, since $\psi(xy^{-1}) = \psi(x)\psi(y)^{-1}$, and it contains every commutator of elements of A, so it contains A', which gives $\psi(A') \leqslant \psi(A)'$. Induction on i now gives $\varphi(G_{(i)}) = \varphi(G)_{(i)}$, so $\varphi(G)_{(n)} = \varphi(1) = 1$, and $\varphi(G)$ is solvable.
3. Let $N \trianglelefteq M$ with $N_{(l)} = 1$ and $(M/N)_{(k)} = 1$, and let $\pi\colon M \to M/N$ be the projection, a surjective homomorphism with kernel N (definition 3.1.13). By the first sentence of (2), induction on i gives $\pi(M_{(i)}) = (M/N)_{(i)}$, so $\pi(M_{(k)}) = 1$ and $M_{(k)} \leqslant N$. Induction on j from definition 6.1.5 gives $(M_{(k)})_{(j)} = M_{(k+j)}$. The argument of (1), applied to the subgroup $M_{(k)}$ of N, gives $M_{(k+l)} = (M_{(k)})_{(l)} \leqslant N_{(l)} = 1$. So M is solvable, completing the proof.

Corollary 6.1.11: Finite P-Groups

Let G be a finite p-group. Then G is solvable.

***Proof*:**
We use strong induction on $|G|$. If $|G| = 1$, then $G_{(0)} = G = 1$. Let $|G| > 1$. By theorem 4.6.5, $Z(G) \neq 1$, and $Z(G)$ is normal in G (example 3.1.10). The quotient $G/Z(G)$ is a p-group of order $|G|/|Z(G)| < |G|$ (theorem 3.3.1), so it is solvable by the induction hypothesis. The group $Z(G)$ is abelian, so all its commutators equal 1, $Z(G)' = 1$, and $Z(G)$ is solvable. By proposition 6.1.10(3), G is solvable, as claimed.

One more theorem gives a sufficient condition for solvability:

Theorem 6.1.12: Feit-Thompson's Theorem

Let G be a finite group of odd order. Then G is solvable.

The proof of theorem 6.1.12 occupies a journal article of 255 pages and combines the local analysis of finite groups with character theory, which this book does not develop. It is due to W. Feit and J. G. Thompson, *Solvability of groups of odd order* [30].

***Proof Key ideas*:**
A counterexample of smallest order is a nonabelian simple group G of odd order in which every proper subgroup is solvable. Indeed, every proper subgroup has smaller odd order, and a proper nontrivial normal subgroup N would make N and G/N solvable, hence G solvable by proposition 6.1.10(3). Abelian groups are solvable, so G is nonabelian. The proof determines the possible structure of the maximal subgroups of such a G by local analysis, the study of the normalizers of nontrivial p-subgroups, and combines this with character theory to show that no such G exists, completing the outline.

Consequently, a finite group that is not solvable has even order, so it has an element of order 2 by theorem 4.8.7.

Theorem 6.1.13: Solvability Condition

A finite group G is solvable if and only if for every divisor n of $|G|$ such that $\left(n, \frac{|G|}{n}\right) = 1$, G has a subgroup of order n.

Both directions of theorem 6.1.13, which is due to P. Hall, need tools this book does not develop. That a solvable group has these subgroups rests on the Schur-Zassenhaus theorem, which gives a complement to a normal subgroup whose order is coprime to its index. The converse rests on Burnside's theorem that groups of order $p^a q^b$, for primes p and q, are solvable, and the standard proof of Burnside's theorem uses character theory. Both directions are proved in Robinson, *A Course in the Theory of Groups* [52, Theorems 9.1.7 and 9.1.8]. Robinson's Theorem 9.1.8 and [28, Section 19.2, Theorem 8] both prove the converse in a stronger form, which assumes only the subgroups whose index is the order of a Sylow subgroup.

***Proof Key ideas*:**
Suppose G is solvable and $(n, |G|/n) = 1$, and argue by induction on $|G|$. A minimal normal subgroup M of a solvable group is a p-group for some prime p, say of order p^k. If p divides n, then p^k divides n, and induction gives a subgroup of G/M of order n/p^k, whose preimage in G has order n. If p does not divide n, induction gives a subgroup K/M of G/M of order n, and the Schur-Zassenhaus theorem gives a complement to M in K, which is a subgroup of order n.

Conversely, suppose G has all these subgroups, in particular, for every prime p, a subgroup whose index is the largest power of p dividing $|G|$. These subgroups pass to normal subgroups and to quotients, so a counterexample of smallest order is simple. If at most two primes divide $|G|$, Burnside's theorem shows that G is solvable. Otherwise, the intersection of these subgroups for all but two of the primes has order divisible by only two primes, so it is solvable by Burnside's theorem, and a minimal normal subgroup of it produces a proper nontrivial normal subgroup of

G, contradicting simplicity and completing the outline.

6.1.2 Nilpotent Groups

We have defined solvable groups using nth commutator groups to give a notion of generalization from being commutative. What if we used the center instead? Since $Z(G) \trianglelefteq G$, the quotient $G/Z(G)$ is a group, and the elements of G whose cosets lie in its center form a subgroup of G containing $Z(G)$. Repeating this gives the *upper central series*: $Z^0(G) = 1$, and $Z^{i+1}(G)$ is the set of $g \in G$ whose coset $gZ^i(G)$ lies in $Z(G/Z^i(G))$. Since $gZ^i(G)$ commutes with $xZ^i(G)$ exactly when $[g,x] = g^{-1}x^{-1}gx \in Z^i(G)$ (definition 5.2.2),

$$Z^{i+1}(G) = \left\{ g \in G \;\middle|\; [g,x] \in Z^i(G) \text{ for all } x \in G \right\}$$

So $Z^1(G) = Z(G)$ and $Z^i(G) \subseteq Z^{i+1}(G)$. By induction each $Z^i(G)$ is a normal subgroup of G, being the preimage P of the normal subgroup $Z(G/Z^{i-1}(G))$ (example 3.1.10) under the canonical projection $\pi\colon G \to G/Z^{i-1}(G)$: for $g \in G$ and $p \in P$, $\pi(gpg^{-1}) = \pi(g)\pi(p)\pi(g)^{-1}$ lies in that normal subgroup, so $gpg^{-1} \in P$. So the next quotient is defined. A group whose upper central series reaches G is called *nilpotent*:

Definition 6.1.14: Nilpotent Groups

Let G be a group. If $Z^m(G) = G$ for some $m \in \mathbb{N}$, then G is said to be nilpotent. The smallest such m is called the *(nilpotency) class of* G.

Example 6.1.15: Nilpotent Groups

1. If G is abelian, then $Z(G) = G$, meaning all abelian groups are nilpotent
2. A metabelian group need not be nilpotent. In S_3, every commutator $\sigma^{-1}\tau^{-1}\sigma\tau$ is an even permutation, since by theorem 1.2.20(2) its sign is $\operatorname{sgn}(\sigma)^{-1}\operatorname{sgn}(\tau)^{-1}\operatorname{sgn}(\sigma)\operatorname{sgn}(\tau) = 1$ (definition 1.2.21). So $S_3' \subseteq A_3$, which has order 3 and is abelian (corollary 3.3.2), and $S_3'' = 1$, so S_3 is metabelian. But $Z(S_3) = 1$, since each transposition fails to commute with the other two transpositions and each 3-cycle fails to commute with (1 2). So $Z^i(S_3) = 1$ for every i, and S_3 is not nilpotent.
3. Every finite p-group G is nilpotent. If $Z^i(G) \neq G$, then $G/Z^i(G)$ is a nontrivial p-group, so its center $Z^{i+1}(G)/Z^i(G)$ is nontrivial by theorem 4.6.5, and $Z^i(G) \neq Z^{i+1}(G)$. The orders $|Z^i(G)|$ therefore increase strictly until $Z^i(G) = G$, which happens after at most a steps if $|G| = p^a$.
4. This is not so much an example as future motivation: Nilpotent groups make an appearance in the classification of Lie Groups.

Proposition 6.1.16: Equivalent Definition

Let G be a group, and define its *lower central series* by $G_0 = G$ and $G_i = [G, G_{i-1}]$ for $i \geq 1$, the subgroup generated by the commutators $[g, h]$ with $g \in G$ and $h \in G_{i-1}$ (definition 5.2.2). The following are equivalent:

1. G is nilpotent (its upper central series reaches G)
2. The lower central series reaches the trivial group, that is, for some n
$$G = G_0 \trianglerighteq G_1 \trianglerighteq G_2 \trianglerighteq \cdots \trianglerighteq G_n = \{1\}$$
3. There exists a *central series*, that is, a chain of subgroups
$$G = A_0 \geqslant A_1 \geqslant \cdots \geqslant A_n = \{1\}$$
with $[G, A_i] \leqslant A_{i+1}$ for $0 \leq i < n$, or equivalently, with every A_i normal in G and $A_i/A_{i+1} \leqslant Z(G/A_{i+1})$ for $0 \leq i < n$

***Proof*:**

First, by the description of $Z^{i+1}(G)$ before definition 6.1.14, an element y lies in $Z^{i+1}(G)$ exactly when $[y, x] \in Z^i(G)$ for all $x \in G$, and since $[x, y] = [y, x]^{-1}$ and $Z^i(G)$ is a subgroup, exactly when $[x, y] \in Z^i(G)$ for all $x \in G$. Second, a subgroup containing a set contains the subgroup that the set generates (proposition 2.3.17). Third, each G_i is normal in G and $G_{i+1} \leqslant G_i$, which we prove by induction on i. For $i = 0$, $G_0 = G \trianglelefteq G$. Suppose $G_i \trianglelefteq G$ and let $x \in G$. Conjugation by x is an automorphism of G (proposition 4.7.1), so $x[g, h]x^{-1} = [xgx^{-1}, xhx^{-1}]$ by proposition 5.2.3(3), and $xhx^{-1} \in G_i$ when $h \in G_i$. So the preimage of G_{i+1} under conjugation by x, which is a subgroup by proposition 2.1.3 because conjugation is a homomorphism, contains every generator of G_{i+1}, hence contains G_{i+1} by the second fact. Thus $xG_{i+1}x^{-1} \subseteq G_{i+1}$ for every $x \in G$, and $G_{i+1} \trianglelefteq G$ by proposition 3.1.11. Each generator $[g, h] = (g^{-1}h^{-1}g)h$ with $h \in G_i$ lies in G_i because $G_i \trianglelefteq G$, so $G_{i+1} \leqslant G_i$, and then $G_{i+1} \trianglelefteq G_i$.

(1) ⇒ (2) Let $Z^m(G) = G$. We show $G_i \leqslant Z^{m-i}(G)$ for $0 \leq i \leq m$ by induction on i. For $i = 0$ both sides are G. If $G_i \leqslant Z^{m-i}(G)$ with $i < m$, then each generator $[g, h]$ of G_{i+1} has $h \in Z^{m-i}(G)$, so $[g, h] \in Z^{m-i-1}(G)$ by the first fact, and $G_{i+1} \leqslant Z^{m-i-1}(G)$ by the second. For $i = m$ this gives $G_m \leqslant Z^0(G) = 1$, so (2) holds with $n = m$.

(2) ⇒ (3) If $G_n = 1$, then $A_i = G_i$ is a central series, since $G_{i+1} \leqslant G_i$ by the third fact and $[G, G_i] = G_{i+1}$.

(3) ⇒ (1) Let (A_i) be a central series. We show $A_{n-j} \leqslant Z^j(G)$ for $0 \leq j \leq n$ by induction on j. For $j = 0$, $A_n = 1 = Z^0(G)$. If $A_{n-j} \leqslant Z^j(G)$ with $j < n$, let $y \in A_{n-j-1}$. For every $x \in G$, $[x, y] \in [G, A_{n-j-1}] \leqslant A_{n-j} \leqslant Z^j(G)$, so $y \in Z^{j+1}(G)$ by the first fact. Hence $A_{n-j-1} \leqslant Z^{j+1}(G)$. For $j = n$ this gives $G = A_0 \leqslant Z^n(G)$, so G is nilpotent.

It remains to see that the two descriptions of a central series in (3) agree. Suppose $[G, A_i] \leqslant A_{i+1}$ for $0 \leq i < n$. For $g \in G$ and $h \in A_i$ with $i < n$, $ghg^{-1} = h[h, g^{-1}]$ and $[h, g^{-1}] = [g^{-1}, h]^{-1} \in [G, A_i] \leqslant A_{i+1} \leqslant A_i$, so $ghg^{-1} \in A_i$. Hence every A_i is normal in G by proposition 3.1.11, the

last one being trivial. Now let $A_{i+1} \trianglelefteq G$. The cosets gA_{i+1} and hA_{i+1} commute exactly when $(hg)^{-1}gh = [g,h] \in A_{i+1}$, so $A_i/A_{i+1} \leqslant Z(G/A_{i+1})$ holds exactly when $[g,h] \in A_{i+1}$ for all $g \in G$ and $h \in A_i$. By the second fact this is the condition $[G, A_i] \leqslant A_{i+1}$, completing the proof.

With the lower central series, we can gain an intuition on why the group is called nilpotent. Let G be any group and $g \in G$ be any element. Define the *adjoint action*

$$\mathrm{ad}_g : G \to G \qquad \mathrm{ad}_g(x) = [g,x]$$

If G is nilpotent of class c, then $(\mathrm{ad}_g)^c(x) = e$ for all $g, x \in G$. Indeed, by the description of $Z^{i+1}(G)$ before definition 6.1.14, $[g,y] = [y,g]^{-1}$ lies in $Z^i(G)$ whenever $y \in Z^{i+1}(G)$, so ad_g maps $Z^{i+1}(G)$ into $Z^i(G)$. Starting from $x \in G = Z^c(G)$ and applying ad_g c times therefore lands in $Z^0(G) = 1$. In this sense each ad_g is nilpotent. A group in which every pair g, x satisfies $(\mathrm{ad}_g)^n(x) = e$ for some n, where n may depend on g and x, is called an *Engel group*. Every nilpotent group of class c is an Engel group, with the single exponent $n = c$ for all pairs, but an Engel group need not be nilpotent.

Example 6.1.17: An Engel Group That Is Not Nilpotent

For $k \geq 2$ let $P_k = D_{2^k}$, the dihedral group of the regular 2^k-gon, with the rotation r_k of order 2^k and a reflection s_k (proposition 1.2.29). Let G be the set of sequences $(y_k)_{k\geq 2}$ with $y_k \in P_k$ and $y_k = 1$ for all but finitely many k. The argument after definition 5.4.9, written multiplicatively, shows that G is a subgroup of the direct product $\prod_{k\geq 2} P_k$, so its operation and its commutators are computed componentwise.

1. G is an Engel group. Let $g, x \in G$, and choose $K \geq 2$ with $g_k = x_k = 1$ for every $k > K$. The elements of G equal to 1 in every position $k > K$ form a subgroup G_K containing g and x. An element of G_K is determined by its entries in positions $2, \ldots, K$, so by proposition 1.2.30

$$|G_K| = \prod_{k=2}^{K} |P_k| = \prod_{k=2}^{K} 2^{k+1}$$

   Thus G_K is a finite 2-group, hence nilpotent of some class c (example 6.1.15), and the computation above, carried out inside G_K, gives $(\mathrm{ad}_g)^c(x) = e$.

2. G is not nilpotent. In P_k, proposition 1.2.29 gives $s_k^{-1} = s_k$ and $r_k^i s_k = s_k r_k^{-i}$, so $s_k r_k^i s_k = r_k^{-i}$ for every $i \geq 0$, hence for every integer i since $r_k^{2^k} = 1$. Therefore $\mathrm{ad}_{s_k}(r_k^j) = s_k r_k^{-j} s_k r_k^j = r_k^{2j}$, and by induction on m, $(\mathrm{ad}_{s_k})^m(r_k) = r_k^{2^m}$ for every $m \geq 0$. Suppose G were nilpotent of class c, and choose $k \geq 2$ with $k > c$. Let g and x be the elements of G with s_k, respectively r_k, in position k and 1 in every other position. The entry of $(\mathrm{ad}_g)^c(x)$ in position k is $r_k^{2^c}$, which is not 1, because $0 < 2^c < 2^k$ and r_k has order 2^k. This contradicts $(\mathrm{ad}_g)^c(x) = e$, so G is not nilpotent.

Proposition 6.1.18: Nilpotent Then Solvable

Let G be a nilpotent group. Then G is solvable

***Proof*:**
Let $G_0 = G, G_1, G_2, \dots$ be the lower central series of G (proposition 6.1.16), and let $G_{(0)} = G$ and $G_{(i+1)} = (G_{(i)})'$ be the derived series (definition 6.1.5), so that $G_{(i+1)}$ is generated by the commutators $[a, b]$ with $a, b \in G_{(i)}$ (definition 5.2.2). We show $G_{(i)} \leqslant G_i$ by induction on i. For $i = 0$ both are G. If $G_{(i)} \leqslant G_i$, then every generator $[a, b]$ of $G_{(i+1)}$ has $a \in G$ and $b \in G_{(i)} \subseteq G_i$, so it is one of the generators of $G_{i+1} = [G, G_i]$, and $G_{(i+1)} \leqslant G_{i+1}$ by proposition 2.3.17. Since G is nilpotent, $G_n = 1$ for some n by proposition 6.1.16. Then $G_{(n)} = 1$, so G is solvable (definition 6.1.6), as claimed.

Note that the converse is not true

Example 6.1.19: Solvable, but not Nilpotent
Let

$$G = \left\{ \begin{pmatrix} a & b \\ 0 & c \end{pmatrix} \;\middle|\; a, b, c \in \mathbb{R},\ ac \neq 0 \right\} \subseteq GL_2(\mathbb{R})$$

The product and the inverse of elements of G are

$$\begin{pmatrix} a & b \\ 0 & c \end{pmatrix} \begin{pmatrix} a' & b' \\ 0 & c' \end{pmatrix} = \begin{pmatrix} aa' & ab' + bc' \\ 0 & cc' \end{pmatrix} \qquad \begin{pmatrix} a & b \\ 0 & c \end{pmatrix}^{-1} = \begin{pmatrix} a^{-1} & -b(ac)^{-1} \\ 0 & c^{-1} \end{pmatrix}$$

so $I \in G$ and $xy^{-1} \in G$ for all $x, y \in G$, and $G \leqslant GL_2(\mathbb{R})$ by proposition 2.1.3.

The diagonal map. By the product formula, the map $\delta \colon G \to \mathbb{R}^\times \times \mathbb{R}^\times$ sending the matrix above to (a, c) is a homomorphism into an abelian group. It is surjective, since it sends the diagonal matrix with diagonal entries a, c to (a, c), and its kernel is

$$H = \left\{ \begin{pmatrix} 1 & b \\ 0 & 1 \end{pmatrix} \;\middle|\; b \in \mathbb{R} \right\}$$

So $H \trianglelefteq G$ (proposition 3.1.14) and $G/H \cong \mathbb{R}^\times \times \mathbb{R}^\times$ (corollary 3.1.17). By the product formula, $b \mapsto \begin{pmatrix} 1 & b \\ 0 & 1 \end{pmatrix}$ is an isomorphism from $(\mathbb{R}, +)$ onto H, so H is abelian. By proposition 5.2.3(5), $G' \leqslant \ker(\delta) = H$.

The center. Let $g = \begin{pmatrix} a & b \\ 0 & c \end{pmatrix} \in Z(G)$. By the product formula,

$$g \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} a & a + b \\ 0 & c \end{pmatrix} \qquad \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} g = \begin{pmatrix} a & b + c \\ 0 & c \end{pmatrix}$$

so $a = c$. In the same way, g times the diagonal matrix d with diagonal entries $1, 2$ has upper right entry $2b$, while dg has upper right entry b, so $b = 0$. Conversely, the scalar matrices aI with $a \neq 0$ commute with every matrix. Hence

$$Z(G) = \{aI \mid a \in \mathbb{R},\ a \neq 0\}$$

G is not nilpotent. The only scalar matrix in H is I, so $Z(G) \cap H = \{I\}$. Let $g \in Z^2(G)$. For every $x \in G$, the commutator $[g, x]$ lies in $Z(G)$ by the description of $Z^2(G)$ before definition 6.1.14, and in $G' \leqslant H$, so $[g, x] = I$ and $gx = xg$. Hence $Z^2(G) = Z(G)$. If $Z^i(G) = Z(G)$ for some $i \geq 1$, then $Z^{i+1}(G)$ is the set of $g \in G$ with $[g, x] \in Z(G)$ for all $x \in G$, which is $Z^2(G) = Z(G)$. By

induction, $Z^i(G) = Z(G)$ for every $i \geq 1$. Since $\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ is not scalar, $Z(G) \neq G$, so no $Z^m(G)$ equals G, and G is not nilpotent (definition 6.1.14).

G is solvable. The subgroup G' lies in the abelian group H, so it is abelian, and $G'' = (G')' = \{I\}$ by example 5.2.4. So the derived series $G \trianglerighteq G' \trianglerighteq G'' = \{I\}$ reaches the identity, and G is solvable (definition 6.1.6).

Another example is S_3, which is solvable but not nilpotent by example 6.1.15(2) and example 6.1.7(2). The subgroup A_3 has order 3 by example 3.5.2(1), so it has index 2 and is normal (corollary 3.3.6), and it meets $\langle(1\ 2)\rangle$ trivially because $(1\ 2)$ is odd. By lemma 5.2.5, $A_3\langle(1\ 2)\rangle$ has $3 \cdot 2 = 6$ elements, so it is S_3, and $S_3 \cong A_3 \rtimes \langle(1\ 2)\rangle$ by theorem 5.3.7. Both factors have prime order, so they are abelian (corollary 3.3.2) and hence nilpotent (example 6.1.15(1)). So a semi-direct product of nilpotent groups need not be nilpotent. More generally, an extension of a nilpotent group by a nilpotent group need not be nilpotent.

Thus, we have the following chain of implications

$$\text{cyclic} \Rightarrow \text{abelian} \Rightarrow \text{nilpotent} \Rightarrow \text{solvable}$$

by corollary 2.3.6, example 6.1.15(1) and proposition 6.1.18. None of the last three properties makes a group finitely generated (definition 5.1.4). For instance $\mathbb{Q}$ is abelian but not finitely generated: if $q_1, \ldots, q_m \in \mathbb{Q}$ have a common denominator $d \geq 1$, then the subgroup $\{x/d \mid x \in \mathbb{Z}\}$ contains each q_i, hence contains $\langle q_1, \ldots, q_m\rangle$ (proposition 2.3.17), but it does not contain $1/(2d)$.

So nilpotent groups do not interact well with extensions. However, the fac that their slightly stronger than extensions, but more general then abelian groups, brings us some useful properties:

Proposition 6.1.20: Properties Of Nilpotent Groups

Let G be a nilpotent group. Then

1. If H is a proper subgroup of G, then H is a proper normal subgroup of $N_G(H)$. We call this property the *normalizer property*, and we sometimes say that the "normalizer grows"
2. If G is finite, every Sylow subgroup of G is normal
3. If G is finite, G is the direct product of its Sylow subgroups
4. If G is finite and d divides the order of G, then G has a normal subgroup of order d

In fact, a finite group with property (4) is nilpotent, giving us another definition of nilpotency for finite groups.

***Proof*:**

1. Let H be a proper subgroup of G. By the definition of the normalizer, $H \trianglelefteq N_G(H)$, so it remains to find an element of $N_G(H)$ outside H. Let $Z^m(G) = G$. Since $Z^0(G) = 1 \subseteq H$ and $Z^m(G) \not\subseteq H$, there is a largest $i < m$ with $Z^i(G) \subseteq H$. Choose $x \in Z^{i+1}(G)$ with $x \notin H$. For $h \in H$, $xhx^{-1}h^{-1} = [x^{-1}, h^{-1}]$ lies in $Z^i(G) \subseteq H$ because $x^{-1} \in Z^{i+1}(G)$,

so $xhx^{-1} \in H$. Thus $xHx^{-1} \subseteq H$. The same argument with x^{-1} in place of x gives $x^{-1}Hx \subseteq H$, and so $xHx^{-1} = H$. Hence $x \in N_G(H)$ and $x \notin H$.

2. Let G be finite, let P be a Sylow p-subgroup of G (definition 4.8.2), and let $N = N_G(P)$. Then $|P|$ is the largest power of p dividing $|G|$, and $|P|$ divides $|N|$, which divides $|G|$ (theorem 3.3.1), so P is also a Sylow p-subgroup of N. Every element of N normalizes P, so $P \trianglelefteq N$, and corollary 4.8.9, applied to the group N, gives P char N. In the same way $N \trianglelefteq N_G(N)$, so proposition 4.7.13(3) gives $P \trianglelefteq N_G(N)$, that is, $N_G(N) \subseteq N_G(P) = N$. So $N_G(N) = N$. If $N \neq G$, part (1) would make N a proper subgroup of $N_G(N)$, which is false. Hence $N_G(P) = G$, and $P \trianglelefteq G$ by proposition 3.1.11.

3. Let G be finite with $|G| = p_1^{a_1} \cdots p_k^{a_k}$ for distinct primes p_i and $a_i \geq 1$, and let P_i be a Sylow p_i-subgroup (theorem 4.8.6), which is normal by (2). For $i \neq j$, $P_i \cap P_j = 1$, since its order divides both $p_i^{a_i}$ and $p_j^{a_j}$ (theorem 3.3.1). For $x \in P_i$ and $y \in P_j$ with $i \neq j$, the element $xyx^{-1}y^{-1}$ lies in P_j, being $(xyx^{-1})y^{-1}$, and in P_i, being $x(yx^{-1}y^{-1})$, so it equals 1 and $xy = yx$. Define

$$\mu\colon P_1 \times \cdots \times P_k \to G \qquad \mu(x_1, \ldots, x_k) = x_1 x_2 \cdots x_k$$

Moving each y_i to the right past the elements x_j with $j > i$ gives

$$\mu((x_1, \ldots, x_k)(y_1, \ldots, y_k)) = x_1 y_1 \cdots x_k y_k = x_1 \cdots x_k \, y_1 \cdots y_k$$

so μ is a homomorphism. Its image is a subgroup of G (proposition 1.4.4(5)) containing every P_i, so its order is divisible by every $p_i^{a_i}$ (theorem 3.3.1), hence by $|G|$, and μ is surjective. Since $|P_1 \times \cdots \times P_k| = |G|$ (proposition 5.1.3(1)), the surjection μ is a bijection, so $G \cong P_1 \times \cdots \times P_k$.

4. Let G be finite, with P_i and μ as in (3), and let $d = p_1^{b_1} \cdots p_k^{b_k}$ divide $|G|$, where $0 \leq b_i \leq a_i$. First, a group P of order p^a, for a prime p, has a normal subgroup of order p^b for each $0 \leq b \leq a$. We induct on a. For $b = 0$ take 1, which settles $a = 0$. Let $1 \leq b \leq a$. By theorem 4.6.5, p divides $|Z(P)|$, so $Z(P)$ has an element z of order p by theorem 4.8.7. The subgroup $Z = \langle z \rangle$ has order p and is normal in P, because $gz^l g^{-1} = z^l$ for $g \in P$. By the induction hypothesis, P/Z, of order p^{a-1} (theorem 3.3.1), has a normal subgroup of order p^{b-1}. By theorem 3.5.7 it is M/Z for a subgroup M of P containing Z. Then $M \trianglelefteq P$ by corollary 3.5.8(5), and $|M| = |M/Z||Z| = p^b$ by theorem 3.3.1.

   Now choose $N_i \trianglelefteq P_i$ with $|N_i| = p_i^{b_i}$, and put $N = \mu(N_1 \times \cdots \times N_k)$, a subgroup of G (proposition 1.4.4(5)). Since μ is injective, $|N| = |N_1| \cdots |N_k| = d$. The subgroup $N_1 \times \cdots \times N_k$ is normal in $P_1 \times \cdots \times P_k$, because conjugation is computed componentwise and each $N_i \trianglelefteq P_i$. Since μ is surjective, every $g \in G$ is $\mu(u)$ for some u, and for $v \in N_1 \times \cdots \times N_k$, $g\mu(v)g^{-1} = \mu(uvu^{-1}) \in N$. So $N \trianglelefteq G$ by proposition 3.1.11.

Finally, let G be a finite group with property (4), and factor $|G|$ as in (3). For each i, property (4) gives a normal subgroup P_i of order $p_i^{a_i}$, which is a Sylow p_i-subgroup. The argument of (3) used only that the P_i are normal of these orders, so it gives an isomorphism $\mu\colon P_1 \times \cdots \times P_k \to G$. Each P_i is nilpotent (example 6.1.15), so by proposition 6.1.16 it has a central series, and repeating its last term 1 we may take all of them of one length n, say $P_i = A_{i,0} \geqslant \cdots \geqslant A_{i,n} = 1$ with $[P_i, A_{i,j}] \leqslant A_{i,j+1}$. Put $B_j = \mu(A_{1,j} \times \cdots \times A_{k,j})$, a subgroup of G. Then $B_0 = G$, $B_n = 1$ and $B_{j+1} \leqslant B_j$. For $g = \mu(u)$ and $b = \mu(v)$ with $v \in A_{1,j} \times \cdots \times A_{k,j}$, the commutator

$[g, b] = \mu([u, v])$ lies in B_{j+1}, because the components $[u_i, v_i]$ of $[u, v]$ lie in $A_{i,j+1}$. Hence $[G, B_j] \leqslant B_{j+1}$ by proposition 2.3.17, so (B_j) is a central series of G, and G is nilpotent by proposition 6.1.16, completing the proof.

Exercise 6.1.1

1. Prove that any nilpotent group of class 3 or lower is metabelian
2. Show that if $\mathrm{Aut}(G)$ is abelian, then G is nilpotent of class at most 2

6.1.3 Polycyclic Groups

We have extended a abelian group using *central extensions* and *abelian extensions*. In this section, we will once again be extending an abelian group, but this time we will ask what happens if we extend a cyclic group in such a way that G_i/G_{i+1} is cyclic?

Definition 6.1.21: Polycyclic Group

Let G be a group. Then if

$$G = G_0 \trianglerighteq G_1 \trianglerighteq \cdots \trianglerighteq G_n = \{1\}$$

st G_i/G_{i+1} is cyclic, Then G is said to be polycyclic. The series G admits is called a *subnormal series with cyclic factors*. If $G_n = \{1\}$, then we say that G is polycyclic with length n.
If $n = 2$, then we say that G is a *metacyclic group*.

Example 6.1.22: polycyclic groups
By theorem 5.1.6, every finitely generated abelian group is polycyclic: the group $\mathbb{Z}^r \times \mathbb{Z}_{n_1} \times \cdots \times \mathbb{Z}_{n_s}$ has the series of its partial products, whose factors are cyclic.

Proposition 6.1.23: Polycyclic Then Finitely Generated

Let G be a polycyclic group. Then G is finitely generated.

***Proof*:**
Let G be polycyclic, so that

$$G = G_0 \trianglerighteq G_1 \trianglerighteq \cdots \trianglerighteq G_n = \{1\}$$

is the subnormal series with cyclic factors. For $0 \leq i < n$ choose $g_i \in G_i$ whose coset $g_i G_{i+1}$ generates the cyclic group G_i/G_{i+1}. We show $G_i = \langle g_i, \ldots, g_{n-1} \rangle$ by downward induction on i, the case $i = n$ being $G_n = 1 = \langle \emptyset \rangle$. Suppose $G_{i+1} = \langle g_{i+1}, \ldots, g_{n-1} \rangle$ with $i < n$. Every generator $g_i, \ldots, g_{n-1}$ lies in G_i, so $\langle g_i, \ldots, g_{n-1} \rangle \subseteq G_i$ (proposition 2.3.17). Conversely, for $x \in G_i$ we have $xG_{i+1} = (g_i G_{i+1})^\alpha = g_i^\alpha G_{i+1}$ for some $\alpha \in \mathbb{Z}$, so $x = g_i^\alpha y$ with $y \in G_{i+1} = \langle g_{i+1}, \ldots, g_{n-1} \rangle$, and $x \in \langle g_i, \ldots, g_{n-1} \rangle$. For $i = 0$ this gives $G = \langle g_0, \ldots, g_{n-1} \rangle$, showing that G is finitely generated.

Note that the converse is false:

Example 6.1.24: Finitely Generated, But Not Polycyclic
Let $G = A_5$, which is a simple non-cyclic group. Since it simple, it's subnormal series is only

$$A_5 \trianglerighteq \{1\}$$

and so, it cannot be polycyclic.

For finitely generated groups the implications combine as follows:

$$\text{cyclic} \Rightarrow \text{abelian} \Rightarrow \text{polycyclic} \Rightarrow \text{solvable}$$

Here a finitely generated abelian group is polycyclic by example 6.1.22, and a polycyclic group is solvable by theorem 6.1.9, because cyclic groups are abelian. Every polycyclic group is finitely generated (proposition 6.1.23). A finitely generated nilpotent group is polycyclic as well: it satisfies the maximal condition on subgroups [52, 5.2.17], a tool this book does not develop, and refining its lower central series gives a central series with cyclic factors [52, 5.2.18].

6.2 Non-finitely generated abelian groups

The additive group $\mathbb{Q}$ is abelian and not finitely generated, as the paragraph after example 6.1.19 shows. Nor is $\mathbb{Q}$ isomorphic to a direct sum of cyclic groups (definition 5.4.9). Any two nonzero elements a/b and c/d of $\mathbb{Q}$ have the nonzero common multiple $ac = (bc)(a/b) = (ad)(c/d)$, whereas two nonzero elements of a direct sum that are supported in different positions have no nonzero common multiple. So such a direct sum would have only one nonzero summand, and $\mathbb{Q}$ would be cyclic.

Prüfer theorems give two conditions under which an abelian group, written additively, is a direct sum of cyclic groups, without the assumption that it is finitely generated. An abelian group G has *bounded order* if there is an integer $n \geq 1$ with $nx = 0$ for every $x \in G$, and every abelian group of bounded order is a direct sum of cyclic groups. For a prime p, let G be an abelian group in which every element has order a power of p. An element $x \in G$ has *infinite height* if for every $k \geq 1$ there is $y \in G$ with $p^k y = x$, and if G is countable and has no nonzero elements of infinite height, then G is a direct sum of cyclic groups. Prüfer proved both statements for countable groups. Both are proved with pure subgroups and heights [40, Theorems 6 and 11], which this book does not develop. The first holds for every abelian group of bounded order, while the second fails for some uncountable groups [40, p. 74].

6.3 Linear Groups

A *linear group* is a subgroup of $GL_n(F)$ (definition 1.2.33) for a field F and an integer $n \geq 1$. The groups $SL_n(F)$, $O_n(F)$ and $SO_n(F)$ of definitions 1.2.34 to 1.2.36 are linear groups, and so is the group of invertible upper triangular matrices of example 6.1.19. Lie theory studies linear groups over $\mathbb{R}$ and $\mathbb{C}$ through their Lie algebras, and invariant theory studies the polynomial functions on F^n that a linear group leaves unchanged.

7

Generalization of Groups

In this chapter, I will cover common algebraic objects which are generalizations of groups.

7.1 Monoid

Having a unique inverse is one of the more powerful axioms you can add to an algebraic structure. Recall how the uniqueness of a linear solution is bassed off of the fact that the invere exists and is unique (proposition 1.1.17). This property is also called the *cancellation property*. There are many mathematical structures which do not have this ($\mathbb{N}$ would be a group if inverses existed). For this reason, the study of this structure is also of interest to many

Definition 7.1.1: Monoid

A *Monoid* is an ordered pair $(M, *)$ where M is a set and $*$ is a binary operation on M satisfying the following axioms

1. associativity: $(a * b) * c = a * (b * c)$
2. there exists an element e in G, called the *identity* of G such that for every $a \in G$, we have $a * e = e * a = a$ [a]

The Monoid $(M, *)$ is called *abelian* (or commutative) if $a * b = b * a$ for every $a, b \in M$

[a] it can in fact be proven that if $a * e = a$ then $e * a = a$ (or vice-versa); an indispensable exercise worth doing

Example 7.1.2
Naturally, all groups are monoids. I will give non-group example of monoids:

1. The natural numbers $\mathbb{N}$
2. Given a set A and it's powerset $P(A)$ (set of all substs), then $(P(A, \cup))$ is a monoid with $\emptyset$ as the identity.
3. Given a set A and it's powerset $P(A)$, then $(P(A), \cap)$ is a monoid with A as the identity.
4. The set $\{0, 1\}$ with any boolean operation ($\Rightarrow$, NOR, AND, etc.) is a monoid. More commonly, the set is relabled as $\{\text{False}, \text{True}\}$
5. The set of strings, given an alphabet is a free monoid.
6. Given a set A, the set of all functions $f : A \to A$ is a monoid! This is incredibly powerful, since we can now put an algebraic structure on sets themselves!! Any set has a default algebraic structure! Further wil be true soon: When we study rings, rings will be an abelian group, along with a second operation which will be a monoid!!

   If A is an abelian group, the homomorphisms $A \to A$ even form a ring, with pointwise addition and composition as multiplication.
7. The normal subgroups of a group G form a monoid under the product of subgroups, the product of N and M being the set NM of all products nm with $n \in N$ and $m \in M$, and the identity being $\{e\}$.

Recall that $\text{Aut}(G)$ is a group, but $\text{End}(G)$ was not since not every element has an inverse (section 1.4)? This then makes $\text{End}(G)$ a monoid

Proposition 7.1.3: $\text{End}(G)$ Is a Monoid

Let G be a group. Then $\text{End}(G)$ is a monoid under composition, with identity id_G.

***Proof*:**
A composite of homomorphisms $G \to G$ is a homomorphism (proposition 1.4.4(6)), so composition is a binary operation on $\text{End}(G)$. It is associative, since $((f \circ g) \circ h)(x) = f(g(h(x))) = (f \circ (g \circ h))(x)$ for all $x \in G$. The identity map id_G is a homomorphism, and $f \circ \text{id}_G = \text{id}_G \circ f = f$ for every $f \in \text{End}(G)$, completing the proof.

Example 7.1.4
For $G = D_3$, the monoid $\text{End}(G)$ is not a group: the trivial homomorphism $x \mapsto e$ has no inverse, since $f \circ (x \mapsto e)$ is the trivial homomorphism for every $f \in \text{End}(G)$, and never id_G.

There is a lot of intereting theory for monoids which I will simply sum up for now to get to the main reason I wanted to bring them up

1. Recall that I said that groups have the cancellation property because of the unique inverse of every element. It is possible that there are monoids with this property that are not groups.

Such monoids are called monoids *with the cancellation property*. The natural numbers is an example of such a monoid. Monoids with the cancellation property can be extended to form groups (called the Grothendieck construction). If the monoid is finite, then it implies that it's a group

2. The automorphisms of a group form a group, and its endomorphisms only a monoid. Dropping the condition of being a homomorphism, the functions $f\colon S \to S$ on any set S form a monoid under composition, with identity id_S.

3. The same way we defined group-actions with groups, we can define monoid-actions with the same axioms. These pop-up in computer-science in the study of automata.

4. In functional programming, monads is a central tool in programming. Monads are monoids! The objects on which they are monoids is a more complicated subject, and will be left for the appendix on category theory [1]

5. free-monoids play a pivitol role in modeling the behavior of machines (ex. Turing machines).

Though all these reaon are good reasons to look into monoids, I bring them up for their use in Category Theory. More particularly, we can define a monoid to be a category with a single element!

The Grothendieck construction named in (1) above is worth carrying out in full, because it is the mechanism by which a number of later invariants are built. Whenever a class of objects carries a notion of direct sum, its isomorphism classes form an abelian monoid, and the invariant one wants is the group this construction produces: the Grothendieck group of vector bundles, of finitely generated projective modules, and of a C^*-algebra are each defined this way. The construction is stated for an arbitrary abelian monoid, since cancellation is exactly what it does *not* need.

Definition 7.1.5: Grothendieck Completion

Let $(M, +)$ be an abelian monoid. Define a relation on $M \times M$ by

$$(m, n) \sim (m', n') \quad \text{if and only if} \quad m + n' + p = m' + n + p \ \text{ for some } p \in M$$

The *Grothendieck completion* of M is the quotient $M^{-1}M = (M \times M)/\sim$ with addition induced coordinatewise, together with the monoid homomorphism

$$\iota\colon M \to M^{-1}M, \qquad m \mapsto [(m, 0)]$$

The class of (m, n) is written $[m] - [n]$, which the group structure below justifies.

The auxiliary p in the relation is what makes $\sim$ transitive without assuming cancellation, and it is exactly the obstruction measured in part (3) below.

[1] For those who are impatient or know some Category theory, a monoid on a category of functors which are derived given by a particular adjoint

Proposition 7.1.6: Properties of the Grothendieck Completion

Let M be an abelian monoid. Then:

1. $M^{-1}M$ is an abelian group, and every element of it has the form $[m]-[n]$ for some $m,n\in M$.
2. $\iota(m)=\iota(n)$ if and only if $m+p=n+p$ for some $p\in M$.
3. ι is injective if and only if M has the cancellation property.
4. For every abelian group A, composition with ι is a bijection
$$\mathrm{Hom}_{\mathbf{Grp}}(M^{-1}M,A)\xrightarrow{\sim}\mathrm{Hom}_{\mathbf{Mon}}(M,A)$$
natural in M and A. That is, $M\mapsto M^{-1}M$ is left adjoint to the forgetful functor from abelian groups to abelian monoids.

***Proof*:**
Step 1: $\sim$ is an equivalence relation. Reflexivity and symmetry are immediate. For transitivity, suppose $m+n'+p=m'+n+p$ and $m'+n''+q=m''+n'+q$. Adding the two equations gives
$$m+n''+(n'+m'+p+q)=m''+n+(n'+m'+p+q)$$
so $(m,n)\sim(m'',n'')$ with the witness $r=n'+m'+p+q$. Note that no cancellation was used, which is the point of carrying p at all.

Step 2: the group structure. Coordinatewise addition respects $\sim$, since adding a fixed pair to both sides of the defining equation preserves it. Associativity and commutativity are inherited from M, the class $[(0,0)]$ is an identity, and $[(n,m)]$ is inverse to $[(m,n)]$ because $(m+n,\ n+m)\sim(0,0)$ with $p=0$. So $M^{-1}M$ is an abelian group, and $[(m,n)]=\iota(m)-\iota(n)$, which is (1).

Step 3: parts (2) and (3). By definition $\iota(m)=\iota(n)$ means $(m,0)\sim(n,0)$, that is $m+p=n+p$ for some p, which is (2). Then ι is injective exactly when $m+p=n+p$ forces $m=n$, which is the cancellation property, giving (3).

Step 4: the universal property. Let $f\colon M\to A$ be a monoid homomorphism into an abelian group. If $(m,n)\sim(m',n')$, say $m+n'+p=m'+n+p$, then applying f and cancelling in A, which is legitimate because A is a group, gives $f(m)-f(n)=f(m')-f(n')$. So
$$\tilde{f}([m]-[n]):=f(m)-f(n)$$
is well defined, and it is a homomorphism because addition in $M^{-1}M$ is coordinatewise. It satisfies $\tilde{f}\circ\iota=f$. For uniqueness, any homomorphism g with $g\circ\iota=f$ is determined on the elements $\iota(m)$, and these generate $M^{-1}M$ by (1), so $g=\tilde{f}$. Naturality in M and A is immediate from the formula, completing the proof.

Example 7.1.7: Grothendieck Completions

1. $\mathbb{N}$ under addition is cancellative, so ι is injective, and $\mathbb{N}^{-1}\mathbb{N}=\mathbb{Z}$. This is the motivating case named at the start of the section.

2. Let $M = \{0, 1\}$ with $1 + 1 = 1$. Cancellation fails, since $1 + 1 = 0 + 1$ while $1 \neq 0$, and indeed $M^{-1}M = 0$: taking $p = 1$ shows $(1, 0) \sim (0, 0)$, so $\iota(1) = 0$.

3. More generally, if M has an absorbing element ω, meaning $\omega + m = \omega$ for every m, then $M^{-1}M = 0$: for any m, taking $p = \omega$ gives $m + \omega = 0 + \omega$. This is why the construction is applied to *finitely generated* objects rather than arbitrary ones, an infinite direct sum being absorbing.

Exercise 7.1.1

1. In the following exercises, we will test your understanding of the idea of defining quotient objects by building up to defining a quotient monoid

   a) Recall the difference between an equivalence relation and a congruence relation. What condition do we need in order that $[x][y] = [xy]$ in a monoid?

7.2 Groupoid

Recall how at the very begining under definition 1.1.3, I remarked how some authors prefer to say groups are not closed under multiplication? Some authors introduce this convention to be able to later drop this axiom in order to define the more general notion of *groupoid*. A groupoid is a group which is not closed under the binary operation! [2]

Definition 7.2.1: Groupoid

A groupoid is a set G with a unary operation $^{-1} : G \to G$ and a patial function $* : G \times G \rightharpoonup G$ where the following axioms hold:

1. Where $a * b$ and $b * c$ are defined, $(a * b) * c = a * (b * c)$. Conversaly, if $(a * b) * c$ and $a * (b * c)$ is defined, then so are both $a * b$ and $b * c$ and $(a * b) * c = a * (b * c)$
2. $a^{-1} * a$ and $a * a^{-1}$ are always well-defined
3. if $a * b$ is defined, then $a * b * b^{-1} = a$ and $a^{-1} * a * b = b$

[2] Technically, this means it's not a binary operation, because binary operations are functions, which need to be closed, but the spirit of the sentence is correct

8

Category of Group

Throughout the book, we will be introducing more and more algebraic structures of varying levels of similarity and differences[1]. It would be very convenient to be able to have a way to compare these algebraic structures. In this section, we elaborate more on the concept of *Categories* which were introduced in [24, Categories]. When we introduce new algebraic structures, we will show how the algebraic structure also forms a *category*, and by doing so we will open up the ability to compare and contrast all sorts of properties. For example, we saw in example 1.1.5 that if G and H are groups we can define $G \times H$ to be a group under point-wise operations. Will the same type of construction be possible for algebraic structures in coming chapters? For rings? Group-actions? Fields? Vector-space? Will all of these objects have a notion similar to subgroup? Will they all have a notion of homomorphism/isomorphism, or a similar notion to group action? The answer will be yes to some and no to others! In order for these structure to have or not have a property, say for example the ability to do a "direct product", there must be a way of saying "the direct product of groups follows the same principle as the direct product of rings/group-actions/fields/vector spaces/etc." This will be accomplished with the language of category theory.

As mentioned in [24, Categories], the core concept of categories is that we want to be able to have a structure that holds *objects* (ex. groups), and a particular way to *relate* these objects (ex. homomorphisms). See the analogy in the aforementioned section for an analogous point of reference. Different categories can have the same objects but different relations, or the same relations but different objects, or different objects and relations. Commonly, a relation between two objects will be a way to change from one object to another while preserving some of the original structure of the object. These types of relations are called *structure preserving* relations. As an example of relations between objects that preserve some structure, take homomorphism, which relates two groups if there is some common group-structure. The fact that homomorphisms preserve (some or all of) the structure can also be found in the etymology of the word: the word *homomorphism* come from

[1] In this book, there will be 15 algebraic objects that can be considered "key objects" to algebra, while we also work with 6 others non-algebraic objects throughout the book (ex. sets, graphs, ordered sets, etc.)

homo- meaning "same" [2] and *morphism* meaning shape. Since homomorphsims preserve (some or all) group structure, they in a sense change a shape from one group to another while keeping some of the "sameness" between the two. Many different mathematical objects have a relation between them that preserve some of it's structure (topologies with topology-preserving functions, graphs and graph-preserving functions, or ordered sets with order-preserving functions to name few), which is why when categories were first introduced, the word *morphism* came to be the name used to describe relations between objects (i.e. if A and B are related in a category, there is a morphism from A to B). The word morphism is a generalization of the word homomorphism to capture the idea of relation's between objects.

I was purposefully being vague in the start by saying that a category has a "particular way to relate objects" instead of just jumping to saying that morphisms preserve some of the structure of the objects. At the beginning of category theory in the 1940s, the types of relations morphisms represented were relations that preserved (some or all of) the structure of the objects. As time went on, more mathematical objects were shown to form categories using morphisms that did not preserve structure in a way we saw with homomorphisms, and the idea that some of the structure is preserved can be a stretch. A partially ordered set $(S, \leq)$ is a category whose objects are the elements of S, with a single morphism $x \to y$ when $x \leq y$ and none otherwise, and these morphisms are not maps between sets at all. This is good to know early on to not get the wrong idea of categories, however in the undergraduate level, 99% of the time morphism in categories will be about preserving some/all the structure of a particular object.

With this very high-level idea, let's give some concrete examples before introducing the definition of a category. If we take all the groups and all the possible homomorphism between groups; that will in fact from the category of groups, sometimes denoted **Grp** or **Group**. Another example would be the collection of all sets and all function between sets. That will form the category of sets, sometimes denoted **Set**. We will soon bring in the definition of a category to prove that it will follow the axioms of a category, for now we will build up some terminology for describing these collections.

Definition 8.0.1: Category Of Grp and Set

1. Let **Grp** represent the collection of all groups and homomorphism between Groups and **Set** the collection of all sets and functions between sets.
2. Let Obj(**Grp**) represent all the objects in the category of **Grp**. Similarly for Obj(**Set**).
3. For any $G, H \in \text{Obj}(\mathbf{Grp})$, Let $\text{Hom}_{\mathbf{Grp}}(G, H)$ represent all homomorphisms from G to H. Similarly, if $A, B \in \text{Obj}(\mathbf{Set})$, let $\text{Hom}_{\mathbf{Set}}(A, B)$ represent all function form A to B.

In many ways, properties of these collections was the starting point for the first category theorists definition of a category, so it is worth taking a closer look at the properties. To not clutter the analysis, we will stick with **Grp** to build-up to the definition of a category.

When we relate two group using homomorphsims, we always have a domain and a co-domain:

$$\varphi : G \to H$$

This means that relations are *directional*: a homomorphism from G to H is different from a homomorphism from H to G. In some structure, direction of the relation does not matter (for

[2] Other words with homo- would be homeostasis (same state), homochromatic (same colour), or homogeneous (same kind/nature, or uniform structure). It's antonym would be hetero- (heterostatis, heterochromatic, heterogenous)

example, in equivalence relations, $x \sim y$ if and only if $y \sim x$). In categories, we pay careful attention at the direction of the arrow, and so when generalizing, a morphism from an object A to B is not the same from a morphism from an object B to A.

When we work with homomorphism, we showed in proposition 1.4.4 that the composition of two homomorphisms is a homomorphism. Particularly, if have a homomorphism from G to H, and a homomorphism from H to K, then there will always exist a homomorphism from G to K:

$$G \xrightarrow{f} H \xrightarrow{g} K \qquad G \xrightarrow{g \circ f} K$$

So, if there is a way to get from G to K through homomorphism, there must be a way to "get there directly" too. Morphisms will have the same property.

Finally, notice that for every group there always exists a "trivial" homomorphism:

$$1_G : G \to G \qquad 1_G(g) = g$$

that is, the homomorphism takes G to itself. In a sense, what is being said here is that every object trivially relates to itself. Another way of thinking that 1_G is trivial is that when composed with other homomorphisms (with appropriate domain or co-domain), it does not change them. If:

$$\alpha : G \to H, \ \beta : F \to G$$

were homomorphsims, then

$$\alpha \circ 1_G = \alpha \qquad 1_G \circ \beta = \beta$$

We want this same property to exist for categories: That every object in a category has a *trivial morphism* which can compose with morphisms (with appropriate domain or co-domain) and not affect it

You might have noticed that all of these properties are about the morphisms of a category. This is a common philosophy in category theory: What's important is not the objects, but the how an objects relates to other object! It is arguable that the more mathematics you will encounter, the more it will be clear that it is the way objects can relate to object objects that i the defining feature of many objects. In fact, many algebraic objects that we will see will be defined by how they relate to other objects.

With this, we will finally define a category:

Definition 8.0.2: Category

A ***category*** C consists of[a]

1. A collection of *objects*, denoted $\text{Obj}(C)$
2. A collection of *morphisms*, denoted $\text{Mor}C$

where

- Each morphism has a specified ***domain*** and ***codomain*** object. Sometimes, this is written as $f : X \to Y$, where f is the morphism with domain X and codomain Y [b]. For any $A, B \in \text{Obj}(C)$, $\text{Hom}_C(A, B)$ denotes all morphisms form A to B.
- Each object has an ***identity morphism*** $1_X : X \to X$
- For each pair of morphisms f, g with the codomain of f equal to the domain of g (ex. $f \in \text{Hom}C(A, B)$, $g \in \text{Hom}_C(B, C)$), there exists a specified ***composite morphism*** gf (or $g \cdot f$)[c] whose domain is equal to the domain of f and whose codomain is equal to the codomain of g, i.e.

$$f : X \to Y \qquad g : Y \to Z \qquad \Rightarrow \qquad gf : X \to Z$$

such that

1. For any $f : X \to Y$, the composites $1_Y f$ and $f 1_X$ are both equal to f (i.e. unital with the identity homomorphism)
2. For any composable triple of morphisms f, g, h, the composites $h(gf)$ and $(hg)f$ are equal (i.e. associative), and so we can write hgf unambiguously.

[a] The word *collection* being used instead of *set* was purposeful. We will get to this
[b] Note that f **need not be a function**! Namely, the morphism need not have a unique codomain
[c] the notation is purposefully not $\circ$ so as to not draw too much of an analogy of gf with the process of composing functions to demonstrate that in some cases gf is *not* function composition

Example 8.0.3: Examples of Categories

1. It was mentioned that groups along with homomorphisms from a category. Indeed, by proposition 1.4.4 the composition of two homomorphsims is a homomorphism, so $gf = g \circ f$. Homomorphsims are associative since function composition is associative ([24, Composition of Functions]), and for any $G \in \text{Obj}(Grp)$, the map $\text{id} :\to G$, $\text{id}(g) = g$ is the identity morphisms since for any $f : F \to G$ and $h : G \to H$, $f \circ \text{id} = f$ and $\text{id} \circ h = h$.
2. If we take all groups, and restrict to taking only the isomorphisms, we will again have a category. Check that all the axioms are still satisfied. A category in which every morphism is an isomorphism is called a *groupoid* (section 7.2 describes the same notion as a set with a partially defined operation). A groupoid with a single object x is a group, namely the morphisms $x \to x$ under composition.
3. If we take all abelian groups and use homomorphsims, this will again be a category. This category is significant enough to deserve it's own notation: this category will denoted **Ab**.

4. If we take all sets and all functions between sets, we will get the category of **Set**.

   Notice that every morphism in **Grp** can be thought of as a morphism in **Set** if we "forget" the group structure of the domain and co-domain, and the homomorphism condition of the morphism of **Grp**. In This way, the category of **Grp** and the category of **Set** are related if we "forget" some of the additional structure of **Grp**. Is it possible to have the reverse? That is, starting with **Set**, is it possible to somehow "upgrade" it to **Grp**? Is there more than one way of doing this? Is there a "natural" way of doing it, that is, is there a way of performing this upgrade in such a way that it follows from the properties of groups and sets without needing to add any more conditions? These questions will be further explored in section 2.3.3. generally.

5. If we take the collection partially ordered sets, that the collection of $(S, \leq)$ where $S \in \text{Obj}(Set)$ and $\leq$ is a possible partial order on S, then we can define a notion similar to homomorphisms that preserve (some or all of) the structure of the order, i.e. we can define a $f : (S, \leq_S) \to (R, \leq_R)$ where f is the appropriate morphism. Can you think of what this morphism should be?

6. Graphs and graph isomorphisms ([24, Graphs]) also form a category. A *graph homomorphism* is a map on vertices sending adjacent vertices to adjacent vertices, and a colouring of a graph with k colours is the same as a homomorphism to the complete graph K_k (see [37]).

Before continuing with the language of category, we will add one more example of a category which is a little more abstract than the previous categories but one we will constantly use. Namely, we can take categories themselves to be objects and define a notion of morphism between categories! This category will be denoted **Cat**. Since morphisms relate objects, the morphisms in the category of **Cat** allows us to find commonalities between different categories! We have in fact already described a category morphism in example 8.0.3(4): If we take **Grp**, we can forget the group structure of every group and the homomorphism property of every homomorphsims and we will end up with **Set**. In effect, we did $(G, *) \to G$ and mapped the homomorphism f where $f : (G, *_G) \to (H, *_H)$ to the function f $f : G \to H$ where the function f is identical to the homomorphsims f except it no longer requires the homomorphism property (in fact, it is impossible for it to have the homomorphism property since sets do not have a binary operation). We will call category morphisms *Functors* (as a generalization of the word function)

Definition 8.0.4: [Covariant] Functor

Let C, D be two categories. Then a *Functor* F will consist of:

1. a function between $\text{Obj}(C)$ and $\text{Obj}(D)$. If $a \in C$ is an object, then $F(a) \in D$ denotes the object in D.
2. a function between Mor$C$ and Mor$D$. If $f \in C$ is a morphism in C, then $F(f)$ denotes the morphism in D.

such that

1. F must preserves the domain and codomain of each morphism: $f : A \to B$ must map to $F(f) : F(A) \to F(B)$
2. F must map identities to identities: $F(\text{id}_f) = \text{id}_{F(f)}$
3. F must preserve composition: $F(g \cdot_C f) = F(g) \cdot_D F(f)$.

Example 8.0.5: Functor

1. For any category C, there is always a trivial functor sending objects to themselves and functions to themselves
2. The relation above between **Group** and **Set** is indeed a functor: $F : \textbf{Group} \to \textbf{Set}$ Such a functor is called a *forgetful functor* since it forgets some of the structure of the category. In the coming chapters, many of the objects are augmentation of sets. All of these categories will have a forgetful functor $F : \textbf{C} \to \textbf{Set}$.

Some authors prefer to be more specific about the second condition and define it instead as the following: for any two object $a, b \in \text{Obj}(C)$ there exists a function $\text{Hom}_C(a, b) \to \text{Hom}_D F(a), F(b)$. We will see many more examples of functors throughout this book. For now, it is left as an optional exercise to prove that categories and functors together form a category[3].

So far, all the categories we have dealt were all in some fundamental sense based on sets: each object was either a set with an augmentation (ex. $(G, *)$ or $(S, \leq)$) or a collection of sets following certain restrictions (ex. graphs). Because of this, every object in every category so far has the notion of having "elements". However, there is nothing in the definition of an object in a category that restricts objects to needing elements:

Example 8.0.6: Non-Set Category

Let $\mathbb{N}$ be the set of all natural numbers and $\leq$ be the usual order. Then we define a category C whose objects are the numbers of $\mathbb{N}$ and whose morphism is $\leq$, that is for every $x \in \mathbb{N}$, $\leq$ is the morphism that relates x to numbers greater than or equal to number y, which we will label $x \leq y$. Notice that $\leq$ is not a function, since x is not "mapped" to a single element but to every element greater than it. Furthermore, $\leq$ is the only morphism in in this category. Composition will return $\leq$ (since it's the only morphism) and we see that if $x \leq y$ and $y \leq z$ then their composition results

[3]There is a technicality about the fact we are taking "the category of all categories" in the sense that this is self-referential creating a contradiction usually called Russel's Paradox. There is a definition of *small category* which circumnavigates this problem which was not introduced so as to no overload new terminology

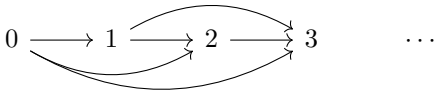
in $x \leq z$.

This is indeed a category: $\leq$ has a domain and co-domain (if you like you can picture $\leq: x \to y$ as long as it is kept in mind that $\leq$ is not a function). For any $x \in \mathbb{N}$, x has an identity morphism, namely the only morphism $\leq$, (you can picture $\leq \cdot \leq = \leq$).

More generally, any set S with some partial order forms a category. For example, if X is a set, then $(P(X), \subseteq)$ is a category whose objects are all subsets of X and the morphism is the partial order induced by $\subseteq$ (i.e. $U \subseteq V$ then $U \to V$).

(Another would be to simply draw diagrams!!)

The set $\mathbb{N}$ was an arbitrary choice in the previous construction: we could have chosen $\mathbb{R}$ or $\{0\}$ or an arbitrary set with an arbitrary ordering. We can visualize the category in the example by imagining a directed graph where every edge represents the morphism between the domain and codomain:



Such a directed graph is called a *commutative diagram*. The full reason behind the name is a little abstract without the proper build-up[4], however we can still understand the etymology given the following illustrative example: imagine we start at the number 1, and every horizontal arrow represents multiplying on the right by x, and every vertical arrow represents multiplying on the right by y, and we form a square like so:

$$\begin{array}{ccc} 1 & \xrightarrow{\cdot x} & x \\ \downarrow{\scriptstyle \cdot y} & & \downarrow{\scriptstyle \cdot y} \\ y & \xrightarrow{\cdot x} & yx = xy \end{array}$$

Similarly, we could have started with $G \times G$ in the top left, let the right arrow be the swap operation $(x, y) \mapsto (y, x)$, the bottom arrow being the identity $(x, y) \mapsto (x, y)$, and then both arrow to the bottom right being the group multiplication arrows.

This square shape is so common that the any directed graph represented morphisms came to be called a commutative diagram. In section 8.2, we shall further explore the use of commutative diagram to understand properties of categories.

8.1 Basic properties of Morphisms

I clearly do not have enough knowledge to write this yet:

https://en.wikipedia.org/wiki/Monomorphism.

Read this and come back.

When we studied function, we started by classifying injective and surjective functions due to their simplicity and universality. It would be nice if there are some nice equivalent, or at least similar, notions of injectivity and subjectivity for morphisms in general categories. Since objects need not

[4]We will often care about morphisms between categories commuting in some "natural" way. This will lead to the study of natural transformation from which the name commutative diagram will takes it's name

have elements, the definition has to take this generality into account. Fortunately, there is already a property of injective and surjective functions that was characterized in terms of functions in [24, Injective And Left Inverse] and [24, Surjective And Right Inverse]: it was shown that all injective functions have a left inverse and all surjective functions have a right-inverse. These notions can be generalized in the following way:

Definition 8.1.1: Monomorphism

Let $f : A \to B$ be a morphism. Then if for any morphism $\alpha : Z \to A$, $\beta : Z \to A$, if

$$f \circ \alpha = f \circ \beta \Rightarrow \alpha = \beta$$

then we call f a monomorphism

Intuitively, monomorphisms should take you to a "unique place" similarly to how injective functions have a single element to represent them.

Proposition 8.1.2: In Set, monomorphism if and only if Injective

Let $f : A \to B$ be a set function. Then f is injective if and only if it is a monomorphism

***Proof*:**

($\Rightarrow$) Let's say f is injective, and that $f \circ \alpha = f \circ \beta$. We want to show that $\alpha = \beta$. By [24, Injective And Left Inverse], there exists a left-inverse f^{-1}. Now, notice that

$$\begin{aligned}
(f^{-1} \circ f) \circ \alpha &= f^{-1} \circ (f \circ \alpha) && \text{associativity of composition} \\
&= f^{-1} \circ (f \circ \beta) && \text{by assumption} \\
&= (f^{-1} \circ f) \circ \beta && \text{associativity of composition}
\end{aligned}$$

meaning we get $\mathrm{id}_A \circ \alpha = \mathrm{id}_A \circ \beta$, and so

$$\alpha = \beta$$

as we sought to show

($\Leftarrow$) Let's say f is a monomorphism so that $f \circ \alpha = f \circ \beta$ implies $\alpha = \beta$. Let's say f was not injective, that is, for some $a, b \in A$, $f(a) = f(b)$ but $a \neq b$. Choose $B' = f(B)$ to be the domain of α and β where α and β both agree where to send elements with the exception of α sending $f(a)$ to a and β sending $f(a)$ to b. Then

$$f \circ \alpha = f \circ \beta$$

but $\alpha \neq \beta$ a contradiction. Since those elements were arbitrary, it must be the case that,

$$f(a) = f(b) \Rightarrow a = b$$

showing injectivity.

One can also prove this directly by letting $Z = \{p\}$ (i.e., a singleton) and using the monomorphism property to conclude that

$$f(a) = f(b) \Rightarrow a = b$$

Similarly, we can define an epimorphism:

Definition 8.1.3: Epimorphism

Let $f : A \to B$ be a morphism.

If for any function $\alpha : B \to Z$, $\beta : B \to Z$, if

$$\alpha \circ f = \beta \circ f \Rightarrow \alpha = \beta$$

then we call f a epimorphism

Proposition 8.1.4: In Set, epimorphism if and only if Surjective

Let $f : A \to B$ be a set function. Then f is surjective if and only if it is a epimorphism

***Proof*:**

($\Rightarrow$) Let f be surjective, and let $\alpha, \beta : B \to Z$ satisfy $\alpha \circ f = \beta \circ f$. Take any $b \in B$. By surjectivity, there exists $a \in A$ with $f(a) = b$, and so

$$\alpha(b) = \alpha(f(a)) = \beta(f(a)) = \beta(b)$$

Since b was arbitrary, $\alpha = \beta$, showing f is an epimorphism.

($\Leftarrow$) Let f be an epimorphism, and suppose toward a contradiction that f is not surjective, that is, there exists $b_0 \in B$ with $b_0 \notin f(A)$. Let $Z = \{0, 1\}$ and define $\alpha, \beta : B \to Z$ by $\alpha(b) = 0$ for every b, while $\beta(b) = 0$ for $b \in f(A)$ and $\beta(b) = 1$ otherwise. For every $a \in A$ we have $f(a) \in f(A)$, so $\alpha(f(a)) = 0 = \beta(f(a))$, that is, $\alpha \circ f = \beta \circ f$. Since f is an epimorphism, $\alpha = \beta$. But $\alpha(b_0) = 0$ while $\beta(b_0) = 1$, a contradiction. Hence f is surjective.

We have mentioned earlier that in category theory, we define objects based on how they relate to other objects. We have in fact done this before, when subgroups were characterized through homomorphisms (proposition 1.4.8).

Definition 8.1.5: Subobject

Let C be a category and $X \in C$ be an object in this category. We say $Y \in C$ is a *suboject* of X is there exists a monomorphism

$$Y \to X$$

To demonstrate what I mean, Let's take an arbitrary set X, and an arbitrary set Y. When can I say that Y is a subset of X, or vice versa? You probably guessed it: if one can be part of included in another, one is a subset of the other. Written in terms of functions: if there exists an inclusion map $i : Y \to X$, then Y is a subset of X ($Y \subseteq X$). We then can define a set being a subset as follows:

If there exists an inclusion map $i : X \to Y$, then X is a subset of Y ($X \subseteq Y$).

If both inclusion maps exist, then there is two injections, and so by the Schröder-Bernstein theorem

(see appendix A), there exists a bijection between X and Y ($|X| = |Y|$). In this way, we can *define* a set to be a *subset* of another set if there exists an inclusion map from Y *to* the bigger set. If we weaken our need for i to be an inclusion map, to i being an injective map, we can start comparing sets with different elements [5]

We can apply this now to groups. By proposition 1.4.8, a subset $H \subseteq G$ which is a group is a subgroup exactly when the inclusion map $H \to G$ is a homomorphism.

Since the same technique is used in defining both concepts, we can say each are similar concepts or types of "objects". Thus, we can say what we have here is applied the more general idea of *sub-object* to both situations.

For sets, injections in both directions give a bijection (Schröder-Bernstein). For groups, injective homomorphisms in both directions give an isomorphism when the groups are finite (proposition 8.1.9), and can fail for infinite groups.

Example 8.1.6: sub-objects

1. **subobjects in Grp**: Recall in proposition 1.4.8 where we showed that if $(H, \bullet)$ is a group and $H \subseteq G$, then H is a subgroup if and only if the inclusion map $i : H \to G$ is a homomorphism. This proposition was the proof that **Grp** had subojects!
2. **subobjects in Set**: by proposition 8.1.2, a monomorphism $Y \to X$ in **Set** is an injective map, which is a bijection from Y onto its image, a subset of X. So the subobjects of X are the sets in bijection with subsets of X.

Definition 8.1.7: Isomorphism In C

Let C be a category, and $f\colon X \to Y$ be a morphism. Then f is said to be an *isomorphism* if there is a morphism $g\colon Y \to X$ with $gf = 1_X$ and $fg = 1_Y$.

Example 8.1.8: Isomorphisms in Different Categories

1. In **Grp**, the isomorphisms are the bijective homomorphisms of definition 1.4.12: the inverse function of a bijective homomorphism ψ is a homomorphism, since $\psi(\psi^{-1}(a)\psi^{-1}(b)) = ab$.
2. In **Set**, the isomorphisms are the bijections, whose inverse functions are again morphisms.
3. In the category of partially ordered sets and order-preserving maps, a bijective morphism need not be an isomorphism. The identity map from $\{a, b\}$ with no relation between a and b to $\{a, b\}$ ordered by $a \leq b$ preserves order, but its inverse does not, as $a \leq b$ holds in the target and fails in the source.

Proposition 8.1.9: Mutual Injections Between Finite Groups

Let G, H be finite groups. If there exist injective homomorphisms $G \to H$ and $H \to G$, then $G \cong H$.

[5] It does not matter what the elements of a set are. What matters is that there *are* arbitrary distinct elements in a set. So being an inclusion or injective is "equivalent" in a broader sense.

***Proof*:**
The injections give $|G| \leq |H| \leq |G|$, so an injective homomorphism $G \to H$ is an injective map between finite sets of the same size. It is therefore bijective, and an isomorphism (definition 1.4.12).

For infinite groups this fails. Let B be the group of sequences $(x_n)_{n\in\mathbb{N}}$ in $\mathbb{Z}/4\mathbb{Z}$ with finitely many nonzero terms, under termwise addition, and let $A = \mathbb{Z}/2\mathbb{Z} \times B$. Then $b \mapsto (0, b)$ is an injective homomorphism $B \to A$, and $(x, (x_n)_n) \mapsto (\iota(x), x_0, x_1, \dots)$ is an injective homomorphism $A \to B$, where $\iota\colon \mathbb{Z}/2\mathbb{Z} \to \mathbb{Z}/4\mathbb{Z}$ sends 1 to 2. But every element of order 2 in B has all its terms in $\{0, 2\}$, so it is twice an element of B, while $(1, 0) \in A$ has order 2 and is not twice any element of A. So $A \ncong B$.

The great advantage of defining subgroup/subset in this way is that we can now define what it means to be a sub-object in many other areas of mathematics! We can even ask whether it is *possible* to define the notion of sub-object. There are some areas in mathematics where the concept of an injective map cannot be defined!![6]

For an object X of a category C, the isomorphisms $X \to X$ (definition 8.1.7) form a group $\mathrm{Aut}_C(X)$ under composition. Composition is associative and 1_X is an identity. If f and f' are isomorphisms with inverses g and g', then $(f \circ f') \circ (g' \circ g) = 1_X = (g' \circ g) \circ (f \circ f')$, so $f \circ f'$ is an isomorphism, and the inverse g of f is an isomorphism with inverse f. For $C = \mathbf{Set}$ the isomorphisms are the bijections (example 8.1.8), and $\mathrm{Aut}_{\mathbf{Set}}(X) = \mathrm{Sym}(X)$. The groups $\mathrm{Aut}_C(X)$ come back when studying *group actions*.

8.2 Commutative diagrams

As mentioned countless times, categories try to capture the idea of relations between objects. Very often, this is done by forcing some relation between morphisms. We have already seen an example of this in the category of **Set** and **Grp** by saying the composition of two functions (or homomorphisms) is again a function (or homomorphism). We represented this by the diagram:

$$X \xrightarrow{f} Y \xrightarrow{g} Z \qquad X \xrightarrow{h} Z$$

where we usually denote h as $g \circ f$. We can form all sorts of relations between objects and enforce the fact that compositions of functions must lead to other arrows. We call this a commutative diagram

Forgetting composition and identities, and keeping only the objects and the arrows, turns a category into a directed graph that may have several arrows between two vertices, called a *quiver*.

Definition 8.2.1: Commutative Diagram

A *diagram* in a category C is a directed graph whose vertices are labelled by objects of C and whose arrows are labelled by morphisms of C between the corresponding objects. The diagram *commutes* if any two directed paths of positive length with the same starting vertex and the same ending vertex give the same morphism when their arrows are composed.

[6] The category of posets $(P, \leq)$ is an example.

Example 8.2.2: Homomorphisms As Commutative Squares
Let G and H be groups, and write $m_G\colon G\times G\to G$ and $m_H\colon H\times H\to H$ for their multiplications. For a function $\varphi\colon G\to H$, let $\varphi\times\varphi\colon G\times G\to H\times H$ send (a,b) to $(\varphi(a),\varphi(b))$. In the category of sets, the square

$$\begin{array}{ccc} G\times G & \xrightarrow{\varphi\times\varphi} & H\times H \\ {\scriptstyle m_G}\downarrow & & \downarrow{\scriptstyle m_H} \\ G & \xrightarrow{\varphi} & H \end{array}$$

commutes exactly when $\varphi(m_G(a,b))=m_H(\varphi(a),\varphi(b))$ for all $a,b\in G$, that is, when $\varphi(ab)=\varphi(a)\varphi(b)$. So φ is a homomorphism exactly when this square commutes.

8.3 Summary

A category (definition 8.0.2) gives objects, morphisms between them, and a composition law; a functor (definition 8.0.4) is the corresponding map between categories; and a commutative diagram (definition 8.2.1) is the notation in which an equality between composites is asserted. Against that backdrop, each notion below replaces a statement about *elements* by a statement about *morphisms*. What permits the replacement in each case is that the two agree in **Set**, where the element-level notion is the familiar one.

Notion in ***Set***	*Categorical form*	*Agreement in* ***Set***
Injective function	Monomorphism (definition 8.1.1)	proposition 8.1.2
Surjective function	Epimorphism (definition 8.1.3)	proposition 8.1.4
Subset $Y\subseteq X$	Subobject (definition 8.1.5)	inherited, the definition being via a monomorphism Y -

The pattern is the one every later categorical definition in this book follows: a property phrased by quantifying over elements is re-phrased by quantifying over morphisms into or out of the object, and the re-phrasing is checked against **Set** before being adopted elsewhere.

Part II

Rings

Now that we have the essentials covered groups, we can move on to structures which have 2 binary operations on the set. These structures are called *rings* [7] As with commutative groups, commutative rings will be very different from rings, so much so that one can think of ring theory as really being 2 seperate domains put under one label: commutative rings and non-commutative rings. However, similarly to how all groups share some common structure (a notion of homomorphism, isomorphism theorem, direct product, free group, group-action, etc.), so do rings, and so we will be building on the foundation of ring theory.

We may wonder how to interpret adding a second binary operation to an abelian group? Does this restrict our structure? Does this limit what we can prove? Does it give more tools to work with when proving? The way I like to perceive adding this second operation is that there are more properties that we have at our disposal to work with. The "type" of new property we add also determines what properties we work with (for example, as is briefly addressed in appendix C.4, we can add a topological structure to groups which gives us different tools we can use to study groups). In the case of adding a new binary operations, we gain more "algebraic" tools to work with. For example, we have an understanding of how the prime numbers of $\mathbb{N}$ interact when we add them[8] or when we multiply them, but we have little knowledge of the interaction of prime numbers between these two binary operations. Furthermore, for any abelian group, we can always add at least one new binary operation that satisfies the axioms of a ring[9] (which we will present near the end of section 9.1.1). In that way, all groups can be made into rings, and we can study them as rings.

For the 1st operation, we will always use additive notation. For example, if we are taking the internal direct product of two groups in this notation, we will write $H + K$, since we will reserve the multiplicative notation for the 2nd binary operation.

[7] The term *ring* was probably a contraction of David Hilbert's term *Zahlring*, literally translated as *number ring*. The word ring was probably coined because of the cyclic nature of the numbers Hilbert was working with. More details on this discussion can be found here: https://math.stackexchange.com/questions/61497/why-are-rings-called-rings

[8] All be it to a very limited degree

[9] Similarly to how all abelian groups are topological groups with the discrete topology

9

Introduction To Rings

With Groups extensively covered, what's "basic" has now changed. We need not so extensively elaborate intuitions for homomorphisms, quotients, products, or actions. Instead, we will show how these concept grow when considering *rings*. Fundamentally, a ring is a set with two operators (hence, is a triple $(R, +, \cdot)$ where $+$ and $\cdot$ satisfy certain axioms[1]).

As with groups homomorphisms, "ring homomorphisms" will play a central role in the study of rings. There will be an equivalent to "normal groups" which we will extensively study: the equivalent for rings is called *ideals*. Manipulating and combining ideals is at the core of ring theory. After exploring ideals, we will take a moment to look at a way to embedd any ring R into a larger ring that permits the notion of "division" (as we will see, unlike groups, not all elements of a ring will have an inverse with the second operation). Through this construction, we will show how to take $\mathbb{Z}$ and create $\mathbb{Q}$.

[1] One source of the axioms: the endomorphisms of an abelian group, under pointwise addition and composition, satisfy all of them (section 9.1.1).

9.1 Basic Properties and Definitions

Definition 9.1.1: Ring

A *ring* R is a set together with two binary operations $+$ and $\cdot$ (called addition and multiplication), $(R, +, \cdot)$ satisfying the following axioms:

1. $(R, +)$ is an *abelian* group, where the identity is denoted by 0
2. **(Mult. Associativity)** $(a \cdot b) \cdot c = a \cdot (b \cdot c)$ for all $a, b, c \in R$
3. **(Mult. Identity)** There exists an $e. \in R$ such that for every $r \in R$, $e. \cdot r = r \cdot e. = r$. The element $e.$ is usually labeled 1.
   - Sometimes, we use *rng* to denote a ring without multiplicative identity.
4. **(Distributivity)** For all $a, b, c \in R$

   $$a \cdot (b + c) = (a \cdot b) + (a \cdot c) \qquad \text{and} \qquad (a + b) \cdot c = (a \cdot c) + (b \cdot c)$$

   The two equations separately are called left and right distributively respectively.

If $a \cdot b = b \cdot a$ for all $a, b \in R$, we call the ring a *commutative ring*.

We will usually write $a \cdot b$ as ab for convenience. The convention that the multiplicative identity is 1 and additive identity is 0 is essentially universal, and so it is safe enough to have it as part of the definition of a ring.

9.1.2 Ring always has identity? Some author's do not include 1 in the definition of a ring (we called rings without a (multiplicative) identity *rng*). In this case, a ring with a multiplicative identity is usually called a unital ring. Any *rng* R can be extended to a ring by taking $\mathbb{Z} \oplus R$ with identity $(1, 0)$ (What is the multiplication here that makes $\mathbb{Z} \oplus R$ into a ring?)[a]. The most common place in mathematics where mathematicians usually do not want rings to have identities is in analysis. For those inclined towards examples see [47]. In Algebra, it is now pretty rare to find a textbook that define a ring without a multiplicative identity, however some classical textbooks have done so (for example [28] and [36])

[a] If R is an algebra over a commutative ring k, the same construction with k in place of $\mathbb{Z}$ adjoins an identity and keeps the k-algebra structure.

Just like associativity was a powerful axiom, so is distributivity for linking the $+$ operation and the $\times$ operation. If you have any arbitrary expression:

$$(a + b)(c + d + e(f + g(i + j)kl(m + n)op))q$$

It can be simplified to the sum of monomials:

$$ac + bc + ad + bd + aefq + befq + aegiklmopq + begiklmopq + aegjklmopq$$
$$+begjklmopq + aegiklnopq + begiklnopq + aegjklnopq + begjklnopqq$$

There are non-distributive algebraic structures. However, the study of non-distributive structures seems to be much scarser than that of non-associative structures or structures without inverse. [2]. Distribuitivity seems to be how most structure link the two operations. At the minimum it usually would be distributive on only one side [3].

9.1.3 Note R being an abelian group is unavoidable given the distributive property

$$a+b+a+b=(1+1)(a+b)=a+a+b+b \Rightarrow a+b=b+a$$

Another way to think about G being abelian is that we want to concentrate on the *ring* operation now, so if we can make the group-operation very-well behaved (as we have shown commutativity does in chapter 5), it will not obtrude our study of the ring operation. This is a common technic.

Example 9.1.4: Ring Examples

1. Taking any abelian group G, and define $a \cdot b = 0$. This is the *trivial ring*. If If $G = \{0\}$, then we call this the *zero ring*. In fact, all $ab = 0$ for all $a, b \in R$ if and only if R is the zero ring:

   Proof. If R is the zero ring, then we already have that $0 \cdot 0 = 0$. Thus, we must prove the converse

   Let's say $\forall a, b \in R$, $ab = 0$. Then $1 \cdot 1 = 0$, meaning $1 = 0$

   Then, notice that

   $$\begin{aligned} &\Leftrightarrow\ 0 \cdot a = (0+0) \cdot a = 0 \cdot a + 0 \cdot a \\ &\Leftrightarrow\ 0 \cdot a = 0 \cdot a + 0 \cdot a \\ \Leftrightarrow\ 0 \cdot a - 0 \cdot a &= 0 \cdot a + 0 \cdot a - 0 \cdot a \qquad \text{additive inverse} \\ &\Leftrightarrow 0 = 0 \cdot a \end{aligned}$$

   Thus, we have $0 = 0 \cdot a = 1 \cdot a = a$ for all a, and so all element equal 0, showing it's a zero ring. □

   In fact, this proof has a very powerful corollary: if $1 = 0$, then we have the zero ring! This property usually breaks many parts of many proofs, so we usually say that $1 \neq 0$ in most theorems and propositions. In some textbooks, this is even part of the axioms of a ring

2. The integers $\mathbb{Z}$ is the canonical example of a ring. When ring theory started, a lot of study was with trying to characterize properties of integers. Note that $ab = ba$, making $\mathbb{Z}$ a *commutative ring*.

3. $\mathbb{Z}/n\mathbb{Z}$ is a [commutative] ring with identity

4. $\mathbb{R}, \mathbb{Q}, \mathbb{C}$ are all [commutative] rings with identity. Naturally, $\mathbb{N}$ is not [a]

5. If R and S are rings, then $R \times S$ is also a ring with operation component wise. [b]. What is the identity of $R \times S$?

[2] I found this paper: generalized De Morgan algebra
[3] Something called a *Quasifields* is an example

6. If you have taken Linear Algebra, then all fields are an example of [commutative] rings

7. Recall the group Q_8 (see section 1.2.4). Using it, we will define the set $\mathbb{H} = \{a + bi + cj + dk \mid a, b, c, d \in \mathbb{R}\}$ where the elements bi, cj, an dk are distinct elements of $\mathbb{H}$, and are not another element in $\mathbb{R}$ or Q_8 (think of a, b, c, d as coefficients). Then we can make $\mathbb{H}$ into a ring by defining addition to be ponitwise:

$$(a + bi + cj + dk) + (a' + b'i + c'j + d'k) = (a + a') + (b + b')i + (c + c')j + (d + d')k$$

and multiplication through repeated use of the distributive property, which distributes to:

$$\begin{aligned}(a + bi + cj + dk)(a' + b'i + c'j + d'k) = &(aa' - bb' - cc' - dd') \\ &+ (ab' + ba' + cd' - dc')i \\ &+ (ac' - bd' + ca' + db')j \\ &+ (ad' + bc' - cb' + da')k\end{aligned}$$

Then $\mathbb{H}$ is called the ring of *quaternions*. It is a non-commutative ring containing the complex numbers $\mathbb{C} = \{a + bi \mid a, b \in \mathbb{R}\}$, and in fact infinitely many copies of $\mathbb{C}$: each $u = bi+cj+dk$ with $b^2 + c^2 + d^2 = 1$ satisfies $u^2 = -1$, so $\{a + a'u \mid a, a' \in \mathbb{R}\}$ is a subring isomorphic to $\mathbb{C}$.

8. If X is a set and A is a ring, then the collection of all function $f : X \to A$ are a ring with function addition and multiplication being done point-wise: $(f + g)(x) = f(x) + g(x)$, $(fg)(x) = f(x)g(x)$. All ring structure on this function space comes from A. A concrete example of this is if $X = [0, 1]$, $A = \mathbb{R}$, and f is continuous $f : [0, 1] \to \mathbb{R}$. These types of rings show up a lot in analysis with the collection of function from a space X (a topological space, or measurable space, a manifold, etc.) to $\mathbb{R}$ or $\mathbb{C}$ giving many properties of the domain of the functions.

9. The set $2\mathbb{Z}$ is not a ring since it does not have an identity. It is thus a *rng*.

10. From section 1.2.5⋆, we saw that $(M_n(G), \cdot)$ is not necessarily a group since inverses were not not always defined. However, this is not a requirements for ring multiplication! It is very tedious and a bit hard at this stage to check that associativity and distributivity hold (there will be a simple proof of this in corollary 13.2.7), but they are verifiable and show that $(M_n(R), +, \cdot)$ is a ring. Note that $M_n(R)$ is not commutative. For example

$$\begin{pmatrix}1 & 0\\ 1 & 0\end{pmatrix} \cdot \begin{pmatrix}1 & 1\\ 0 & 0\end{pmatrix} \neq \begin{pmatrix}1 & 1\\ 0 & 0\end{pmatrix} \begin{pmatrix}1 & 0\\ 1 & 0\end{pmatrix}$$

Matrix rings will be explored in more detail soon, and will have a whole chapter (in particular [17, Representation of Finite Groups]) dedicated to the study of when matrix rings "act" on finite groups (in a similar way groups acted on sets in group-actions).

11. Recall when I asked if $P(X)$ is a group under $\cup$? Here is another similar question with rings. Take $P(X)$ again, and define addition and multiplication as

$$A + B = (A - B) \cup (B - A) \qquad A \cdot B = A \cap B$$

prove that it's a [commutative ring]. Another thing you can show is that for any $Y \in P(X)$, $Y^2 = Y$. If a ring has this property, it is said to be a *boolean ring*. [c]

12. Let G be any abelian group. Then $(\text{End}_{\mathbf{Grp}}(G), +, \circ)$ is a ring. In this way, any ring is formed from an abelian group. [d]

[a] $\mathbb{N}$ is a semi-ring with $+$ and $\cdot$

[b] Though this might feel "trivially true", the fact that taking the direct product of two rings will produce the same algebraic structure, in this case a ring, is not always true! See example 9.1.27 for a counter-example

[c] This ring comes up in *measure theory*, where the set of measurable sets can be made into a boolean ring

[d] The general term for taking [abelian] objects (ex. abelian groups) from an category C and working with $\text{End}(X)$ as the new object and morphisms that also preserve $\circ$ is **Ab**-enriched category given the abelian category C

When working with rings, we will assume the usual ring structure for $\mathbb{Z}, \mathbb{Q}, \mathbb{R}$ and $\mathbb{Z}/n\mathbb{Z}$, and use the same notation if it's clear in context. The notation $\mathbb{Z}^\times, \mathbb{Q}^\times, ...$ will soon be repurposed for another use later in this section.

Note that every ring is naturally an abelian group as well as a monoid (see section 7.1). When working with homomorphisms for rings, both structure need to be preserved in some way for the image of a funciton to be a ring. There is in fact one more sneaky criterion that needs to be added in order for a the image of a function be be a ring, and hence a *ring homomorphism*:

Definition 9.1.5: Ring Homomorphism

Let R and S be rings. Then:

1. A *ring homomorphism* is a map $\varphi\colon R \to S$ satisfying, for all $a, b \in R$,
   (a) $\varphi(a+b) = \varphi(a) + \varphi(b)$
   (b) $\varphi(a \cdot b) = \varphi(a) \cdot \varphi(b)$
   (c) $\varphi(1_R) = 1_S$
2. If $\varphi\colon R \to S$ is a ring homomorphism, and there exists a ring homomorphism $\psi\colon S \to R$ with $\psi \circ \varphi = \text{id}_R$ and $\varphi \circ \psi = \text{id}_S$, then φ is a ring isomorphism

Note that just like for group homomorphisms, a ring homomorphism φ is a ring isomorphism exactly when it is bijective. A ring isomorphism has the inverse map ψ, so it is bijective. Conversely, if φ is bijective, its inverse map ψ satisfies $\varphi(\psi(x)\psi(y)) = \varphi(\psi(x))\varphi(\psi(y)) = xy$, so $\psi(xy) = \psi(x)\psi(y)$ by injectivity of φ, and in the same way $\psi(x+y) = \psi(x) + \psi(y)$ and $\psi(1_S) = 1_R$.

Note that the reason we add the condition that $\varphi(1_R) = 1_S$ is so that the image $\varphi(R)$ is a subring of S. If a ring is not defined to have a multiplicative identity (as is done for *rngs*), then the condition $\varphi(1_R) = 1_S$ is not necessary. Furthermore, where $\psi(e_G) = e_H$ is a provable property given that ψ is a group-homomorphism and e_G, e_H are the respective identities of the domain and codomain, the same is not true for rings: the first 2 conditions of a ring homomorphism can hold while $\varphi(1_R) \neq 1_S$. The map $\mathbb{Z} \to \mathbb{Z} \times \mathbb{Z}$, $n \mapsto (n, 0)$, preserves sums and products, and its image $\mathbb{Z} \times \{0\}$ is a ring with identity $(1, 0)$, but it is not a subring of $\mathbb{Z} \times \mathbb{Z}$, whose identity is $(1, 1)$.

Many times, mathematical objects will have multiples different structures defined on them, but we sometimes want to only work with some of them[4]. Thus, a group homomorphism between rings will only preserve group structure, while a ring homomorphism preserves both. Section 9.3 relates the structure of $\varphi(R) \subseteq S$ to that of R, as was done with quotient groups.

[4] An example with full of different structures is $\mathbb{R}$ which can come with a group, ring, and order and structure, but also structures that we have not covered, some of which are non-algebraic (ex. completeness (see uniformly continuous functions and uniform spaces), or topological)

From definition 9.1.5, we can define a sub-ring:

Definition 9.1.6: Sub-Ring

Let R be a ring and $S \subseteq R$ a subset. Then S is a *subring* of R if it is a ring under the operations of R and $1_R \in S$, that is, if the inclusion map $i : S \to R$ is a ring homomorphism[a].

[a] See proposition 1.4.8

Corollary 9.1.7: sub-ring condition

A subset S of a ring R is a subring if and only if S is an additive subgroup of R that is closed under multiplication and contains 1_R.

***Proof*:**
If S is a subring, it is a ring under the operations of R, so it is an additive subgroup closed under multiplication, and $1_R = i(1_S) = 1_S \in S$ because the inclusion i is a ring homomorphism. Conversely, such an S is an abelian group under $+$, multiplication restricts to an associative operation on S that distributes over $+$ because it does on R, and $1_R \in S$ is an identity for it. So S is a ring, and the inclusion preserves sums, products and 1, completing the proof.

Notice that there is no special symbol for a subring like there as for subgroup. This will be a recurring pattern as we continue to explore algebra: there is no special symbol for "subobjects". We will always say "Let S be a subring of R" or "Let $S \subseteq R$ be a subring of R".

One subring is available in every ring and measures how far the multiplication is from being commutative. It is the ring-theoretic analogue of definition 2.2.4, and unlike the group case it is a subring rather than merely a subgroup.

Definition 9.1.8: [Ring] Center

Let R be a ring. The *center* of R is

$$Z(R) := \{c \in R \mid cx = xc \text{ for all } x \in R\}$$

the set of elements commuting with every element of R. A ring is commutative exactly when $Z(R) = R$.

That $Z(R)$ is a subring is immediate from corollary 9.1.7: it contains 0 and 1, it is closed under subtraction since $(c - c')x = cx - c'x = xc - xc' = x(c - c')$, and it is closed under multiplication since $(cc')x = c(c'x) = c(xc') = (cx)c' = x(cc')$. It is commutative, being a set of pairwise commuting elements, so the center of any ring is a commutative ring.

Proposition 9.1.9: Properties of Rings

Let R be a ring. Then:

1. $0a = a0 = 0$ for all $a \in R$
2. $(-a)b = a(-b) = -(ab)$ for all $a, b \in R$ (recall $-a$ is the additive inverse)
3. $(-a)(-b) = ab$ for all $a, b \in R$
4. The multiplicative identity is unique and $-a = (-1)a$.
5. Generalized Distribuitivity: for all $a, b_1, \ldots, b_n \in R$

$$a(b_1 + \cdots + b_n) = ab_1 + \cdots + ab_n \qquad (b_1 + \cdots + b_n)a = b_1a + \cdots + b_na$$

and hence $(a_1 + \cdots + a_m)(b_1 + \cdots + b_n) = \sum_{i=1}^{m} \sum_{j=1}^{n} a_i b_j$ for all $a_i, b_j \in R$.

***Proof*:**
All of these proofs take advantage of the abelian structure of R and distributivity. Sums of several terms need no brackets by the generalized associative law in the group $(R, +)$ (proposition 1.1.15(5)).

1. Using the fact that 0 is an additive identity and distribuitivity, we get:

$$0 \cdot a = (0 + 0)a = 0 \cdot a + 0 \cdot a$$

Since $0 \cdot a \in R$, there exists an additive inverse, thus

$$0 \cdot a - 0 \cdot a = 0 \cdot a + 0 \cdot a - 0 \cdot a$$

implying $0 = 0 \cdot a$. In the same way $a \cdot 0 = a(0 + 0) = a \cdot 0 + a \cdot 0$, and adding $-(a \cdot 0)$ to both sides gives $0 = a \cdot 0$.

2. By distributivity and (1), $ab + (-a)b = (a + (-a))b = 0b = 0$, so $(-a)b$ is an additive inverse of ab, and inverses in the group $(R, +)$ are unique (proposition 1.1.15(2)), so $(-a)b = -(ab)$. Likewise $ab + a(-b) = a(b + (-b)) = a0 = 0$ gives $a(-b) = -(ab)$.

3. By (2), applied first to a and $-b$ and then to a and b, $(-a)(-b) = -(a(-b)) = -(-(ab))$, and $-(-(ab)) = ab$ by proposition 1.1.15(3).

4. If e and e' are both multiplicative identities, then $e = ee' = e'$. By (2), $(-1)a = -(1a) = -a$.

5. By induction on n, the case $n = 1$ being trivial. If the first identity holds for n, then distributivity gives

$$a(b_1 + \cdots + b_{n+1}) = a(b_1 + \cdots + b_n) + ab_{n+1} = ab_1 + \cdots + ab_n + ab_{n+1}$$

and the second identity follows in the same way from right distributivity. For the last claim, the second identity with $a = b_1 + \cdots + b_n$ gives $(a_1 + \cdots + a_m)a = a_1a + \cdots + a_ma$, and the first identity expands each a_ia into $a_ib_1 + \cdots + a_ib_n$, completing the proof.

In particular, $(-1)^2 = (-1)(-1) = 1 \cdot 1 = 1$ by proposition 9.1.9(3).

Consequences of no Multiplicative Inverse

Now that we know what is needed to be a ring, let's put names to common rings with properties added on. For each of these properties, I would recommend to go back and see which rings we covered have these properties. If you want to verify your intuition, you can check the glossary (see Part XIV).

The first thing is that the multipication operation is missing one of the axioms of a group: elements need not have a multiplicative inverse. If every element has a multiplicative inverse, we will give a name to such a ring:

Definition 9.1.10: Division Ring

A ring R with identity 1, where $1 \neq 0$ is called a *division ring* (or a *skew field*) if every non-zero element $a \in R$ has a multiplicative inverse, $b \in R$ such that $ab = 1$.

The inverse b of a nonzero a in a division ring also satisfies $ba = 1$. Here $b \neq 0$, since $a0 = 0 \neq 1$ (proposition 9.1.9(1)), so $bc = 1$ for some $c \in R$, and then $a = a(bc) = (ab)c = c$, that is, $ba = bc = 1$.

If furthermore, multipication is commutative, the resulting ring is a commutative division ring, whcih is called *field*:

Definition 9.1.11: Field

Let F be a divison ring. If F is also *commutative*, then F is called a *field*.

Sometimes, the symbol k (from German Körper) or $\mathbb{F}$ will be used for a field. The field is the "smallest" mathematical structure in which we can perform all the standard arithmetic operations $+, -, \times, \div$ in the usual way. If R is a field, then it is as though there is two compatible abelian group structures (both are associatve, have an identity, have an inverese, both commute), the two operations being linked through distributivity. In particular, it's like $(R, +)$ and $(R - \{0\}, \cdot)$ are combined through distributivity. . If R is a field, we can call the sub-rings of R a sub-field. For example, $\mathbb{Q}$ is a subfield of $\mathbb{R}$.

Example 9.1.12: Subfields Of $\mathbb{R}$ Contain $\mathbb{Q}$

Every subfield K of $\mathbb{R}$ contains $\mathbb{Q}$. The identity 1_K of K satisfies $1_K 1_K = 1_K \neq 0$, and multiplying by its inverse in $\mathbb{R}$ gives $1_K = 1$. As K is closed under addition and additive inverses, it contains $n = 1 + \cdots + 1$ and $-n$ for every $n \in \mathbb{N}$, so $\mathbb{Z} \subseteq K$. As K is closed under multiplication and inverses of nonzero elements, it contains mn^{-1} for all $m, n \in \mathbb{Z}$ with $n \neq 0$, so $\mathbb{Q} \subseteq K$.

Zero divisors and Units

If a ring is not a division ring, then the cancellation property does not necessarily hold (see proposition 1.1.17). In groups, we constantly take advantage of the fact that $ab = ac \Rightarrow b = c$. This proof relied on the [unique] inverse of elements. However, there are rings in which this does not happen

Example 9.1.13

Let $R = \mathbb{Z}/6\mathbb{Z}$ with the usual addition and multiplication. First, show that it's a ring.

Then, notice that [2] times any number in $\mathbb{Z}/6\mathbb{Z}$ is not [1] (i.e. there does not exist $[a]$ such that $[2][a] = [1]$):

$$[2][1] = [2]\,, [2][2] = [4]\,, [2][3] = [0]\,, [2][4] = [2]\,, [2][5] = [4]$$

Furthermore, there exists an element ([3]) such that [2] times it gives the additive identity: $[2][3] = [0]$

So it breaks down! You might wonder if for any $a \in R$ if there exists an element b such that $ab = 0$ or $ab = 1$, but that need not be the case:

Example 9.1.14: No Multiplicative Inverses

In $\mathbb{Z}$ with the usual addition and multiplication, take any $a \in \mathbb{Z}$, $a \notin \{0, 1\}$. Then there does not exist an element b such that $ab = 1$, which is shown by $\mathbb{Z}$ not having multiplicative inverse, but also $ab = 0$ cannot happen in $\mathbb{Z}$ for any $b \in \mathbb{Z}$

Thus, one cannot imply the other in general

You might further wonder if for any $a \in R$ that only $ab = 0$ happens or only $av = 1$ happens, that is, if there exists an element $b \in R$ such that $ab = 0$, then there does *not* exist an element $v \in R$ such that $av = 1$ (or vice versa). However, this again is not the case! What if $ab = 0$, or $ab = 1$, does that means $ba = 0$ or $ba = 1$? Also no (in the non-commutative case)!

Example 9.1.15

1. Take S to be the set of all infinite sequences. This set, under component-wise addition, is a group. Then take $\text{End}(S)$, making function composition multiplication. Then this is a ring with identity.

2. Prove that

$$P(a_1, a_2, ..., a_n, ...) = (a_1, 0, ..., 0, ...) \tag{9.1}$$
$$L(a_1, a_2, ..., a_n, ...) = (a_2, a_3, ..., a_{n-1}, ...) \tag{9.2}$$
$$R(a_1, a_2, ..., a_n, ...) = (0, a_1, ..., a_{n+1}, ...) \tag{9.3}$$
$$0(a_1, a_2, ..., a_n, ...) = (0, 0, ..., 0, ...) \tag{9.4}$$
$$1(a_1, a_2, ..., a_n, ...) = (a_1, a_2, ..., a_n, ...) \tag{9.5}$$
$$\tag{9.6}$$

   are all elements in $\text{End}(S)$ $(L, R, P, 0, 1 \in \text{End}(S))$

3. Show that $LP = 0$, but $LR = 1$. Then, show that $PL \neq 0$ (so $LP = 0$, but $PL \neq 0$)

4. You just showed that $LR = 1$. Show that $RL \neq 1$.

Since it's not commutative, it's not imediately clear that since we found an example where $ab = 0$ but $ac = 1$, then we can find the same but multiplying from the left side. However, we can construct exactly such an example from the previous example:

1. Define $\text{End}(S)^{op}$ to be the object where function composition is reversed:

$$f \cdot g = g \circ f$$

show that $\text{End}(S)^{op}$ is still a ring. Convince yourself that all the previous elements $(1) - (5)$ are in $\text{End}(S)^{op}$.

2. Show that $PL = 0$, but $RL = 1$. Then, show that $PL \neq 0$ (so $LP = 0$, but $PL \neq 0$)

So, we cannot carry over our intuition of the cancellation property into general rings! However, thanks to associatvity, there is *one* relation we can prove: if $ab = 0$, then there does not exist a c such that $ca = 1$. That is, if there exists an element b such that $ab = 0$, then there does not exist a c such that $ca = 1$:

Proposition 9.1.16: Partial Result for ab and ca

Let $a \in R$ such that there exists an element $b \in R$ such that $ab = 0$ (or $ba = 0$). Then there does not exist an element c such that $ca = 1$ (or $ac = 1$), that is:

$$\begin{matrix} ab = 0 \\ ca \neq 1 \end{matrix} \quad \text{and} \quad \begin{matrix} ab = 1 \\ ca \neq 0 \end{matrix}$$

***Proof*:**
Take $a \in R$ as in the proposition. Let's say there exists a $b \in R$, $b \neq 0$, such that $ab = 0$. Let's say there is a c such that $ca = 1$, $c \neq 0$. Then:

$$b = 1b = (ca)b = c(ab) = c(0) = 0$$

a contradiction since $b \neq 0$

Let's say there exists a $b \in R$, $b \neq 0$, such that $ba = 1$. Let's say there is a c such that $ac = 0$, $c \neq 0$.

$$c = 1c = (ba)c = b(ac) = b \cdot 0 = 0$$

a contradiction since $c \neq 0$

Note that as a consequence of this theorem, if an element a has a left-sided inverse l and ar right-sided inverse r, then $l = r$, since:

$$l = l1 = l(ar) = (la)r = 1r = r$$

Since multiplicative inverses can either multiply to 1, to 0 or to neither, we will take a moment to name elements that can multiply into 1, 0, or neither:

Definition 9.1.17: Zero Divisors and Units

Let R be a ring.

1. Let $a \in R$ be a nonzero element. Then a is called a *left* (resp. *right*) *zero divisor* if there is a nonzero element $b \in R$ such that ab (resp. ba) equals to 0. If $ab = ba = 0$, then a is called a *two-sided zero divisor*, or simply a *zero-divisor*
2. Assume $1 \neq 0$. An element u of R is called a left (resp. right) *unit* in R if there is some v in R such that vu (resp. uv) equals to 1. If $uv = vu = 1$. , then u is called a *two-sided unit*, or simply *unit* for short

If R is commutative, then there is no distinction between left and right zero-divisors/units.

The term zero-divisors comes from the following observation: if $ab = n$ in a commutative ring for simplicity, then we may say that $a|n$ and $b|n$ (i.e. a, b divide n). Then if $ab = 0$, then "$a\big|0$" and "$b\big|0$", so a, b divide 0 (they are *zero-divisors*). This notion of divisibility shall further be explored in chapter 10. Note that if u is a unit of R so that there is some v such that $uv = vu = 1$, then v is unique (think of the same proof for groups). The same is not the case for zero-divisors (try playing around in $\mathbb{Z}/10\mathbb{Z}$). Since any unit must be a two-sided inverse, we can define the following:

Proposition 9.1.18: Group of Units

Let R be a commutative ring, and let $R^\times$ denote the set of all units of R. Then $R^\times$ is a group under $\cdot$, and is called the *group of units of* R

***Proof*:**
Associativity comes from associativity of multiplication, multiplicative identity comes from the fact that 1 is a unit, and inverses come from the definition of units.

It only remains to show that $\cdot$ is indeed a well-defined operation. Does it map units of R to units of R. Take $a, b \in R^\times$. Since both are units, then there exists a u and v st $au = ua = bv = vb = 1$. Then I claim that vu is the element such that $(vu)(ab) = 1$. Indeed:

$$(vu)(ab) = v(ua)b = v \cdot 1 \cdot b = v \cdot b = 1$$

Thus $ab \in R^\times$, showing it is indeed well-defined as we sought to show

Sometimes, the group of units of R is denoted $U(R)$, R^*, or $E(R)$ (from German *Einheit*, meaning "oneness") Notice how this let's us characterize division rings and fields in a new way: if every non-zero element of a ring (resp. commutative ring) is a unit, the ring is then a division ring (resp. field). Note too that it is not always obvious when a ring can be characterized as a field in this way:

Example 9.1.19: Every Element is a Unit

Let D be a not-a-perfect square rational number [a] Then take $\mathbb{Q}(\sqrt{(D)}) \subseteq \mathbb{C}$,

$$\mathbb{Q}(\sqrt{(D)}) := \left\{ a + b\sqrt{(D)} \;\middle|\; a, b \in \mathbb{Q} \right\}$$

Note that if $D > 0$, then $\mathbb{Q}(\sqrt{D}) \subseteq \mathbb{R}$. This set is indeed a ring since

$$\forall x, y \in \mathbb{Q}(\sqrt{5}) \qquad a + b\sqrt{D} - c - d\sqrt{D} = (a - c) + (b - d)\sqrt{D}$$
$$\forall x, y \in \mathbb{Q}(\sqrt{5}) \qquad (a + b\sqrt{D})(c + d\sqrt{D}) = (ac + bdD) + (ad + bc)\sqrt{D}$$

showing it's indeed a [commutative] ring.Note that D being not-a-square implies every element can be written uniquely. If it were a square, say 9, then

$$2\sqrt{9} = 6$$

meaning two elements were written the same way! Because of this, if a, b are non-zero, then $a^2 - b^2D$ is non-zero. For example, if $D = 4$, then

$$4^2 - 2^2 4 = 16 - 16 = 0$$

but Since D is not-a-square,

$$a^2 - b^2D = 0 \Rightarrow a^2 = b^2D \Rightarrow \frac{a^2}{b^2} = D \Rightarrow \frac{a}{b} = \sqrt{D}$$

which since D is not-a-square, it's square-root is not a rational number [b], hence a contradiction. Thus, $a^2 - b^2D$ is non-zero if a, b are non-zero. The point of this is to notice that

$$(a + b\sqrt{D})(a - b\sqrt{D}) = a^2 - b^2D$$

then $a + b\sqrt{D}$, $a - b\sqrt{D}$ must both be non-zero! Thus, we can define

$$\frac{a - b\sqrt{D}}{a^2 - b^2D}$$

to be the inverse of $a + b\sqrt{D}$! This shows that every non-zero element of $\mathbb{Q}(\sqrt{D})$ is a unit, and hence it's a field!

[a] perfect square number: $x \in X$ such that $\sqrt{x} \in X$
[b] A similar proof to that of $\sqrt{2}$ is not rational!

This also means that since every element in a field has a multiplicative inverse, a field has *no* zero-divisors.

As we saw, if x is *not* a zero divisor, this does not imply that x *is* a unit, and vice-versa, as we have seen with the integers. However, in the finite case, this *does* hold

Proposition 9.1.20: In finite ring, not zero divisor ⇒ unit

If R is a finite ring, and $x \in R$ is *not* a zero-divisor, then x is a unit

***Proof*:**
The key idea of finiteness is that we can take advantage of forcing injective maps being surjective. Withought loss of generality, when there are zero divisors, we take the left zero divisors. The proof for right zero-divisors works identically by switching the order of the elements.

Let's say $x \in R$ is not a left zero divisor so that for no element $y \in R$ there is $xy = 0$. If $x = 1$, we are done, so let's say $x \neq 1$. Define the map $y \mapsto xy$. [a]. Since for no y there is $xy = 0$, then this map is injective. Since R is finite and the map is from R to R, then this map is also surjective. Thus, there exists a y' such that $xy' = 1$. It remains to show that $y'x = 1$.

The element x can either be a right zero-divisor or it can not be. If it is not a right-zero divisor, then the map the map $y \mapsto yx$ would be injective, and hence surjective, and we can find an element y'' such that $y''x = 1$. So, let's say x was a right zero-divisor. If it was a right zero-divisor, then there would exist a $z \in R$ such that $zx = 0$, and x would be a right, but not a left zero-divisor. However, there exists a y' such that $xy' = 1$, which is a contradiction by proposition 9.1.16. Therefore, in the finite case, if x is a right zero-divisor, it would also have to be a left zero-divisor

Therefore, x is not a zero-divisor, so we can again define the map $y \mapsto yx$ which is also injective, and so there exists a y'' such that $y''x = 1$. Finally, notice that:

$$y' = 1 \cdot y' = (y''x) \cdot y' = y'' \cdot 1 = y''$$

showing that $y' = y''$ and so $xy' = y'x = 1$, fulfilling the condition for x being a unit, as we sought to show.

[a] Tangentially, note that this function is not in general a ring homomorphism. Can you see why?

No Zero-Divisors: Integral Domains

Numbers as they are generally known do not have zero divisors (ex. $\mathbb{N}, \mathbb{Z}, \mathbb{Q}, \mathbb{R}, \mathbb{C}$). A large part of the study of rings is the study of number theory, and so defining rings which do not have zero divisors is important for ease of communication:

Definition 9.1.21: Integral Domain

A commutative ring with identity $1 \neq 0$ is called *integral domain* if it has a no zero divisors

Example 9.1.22: Integral Domains

1. $\mathbb{Z}$ is the canonical example, even the name-sake
2. $\mathbb{Q}$ and $\mathbb{R}$ are integral domains. Note that every (nonzero) elements are also units
3. $(\mathrm{GL}_n(\mathbb{F}), +, \cdot))$ is an integral domain
4. $M_n(F), +, \cdot)$ is not an integral domain, since

$$\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \cdot \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$$

5. Let $F = \mathrm{Hom}_{\mathbf{Set}}(X, R)$ be a collection of functions where $|X| > 2$ $R \neq \{0\}$. As noted in example 9.1.4 this is a ring. This ring is not an integral domain. Let $a, b \in X$, $a \neq b$, and $x \in R$, $x \neq 0$. Then define a function where $f(a) = 0, f(b) = x$, and the rest of the elements of f map to zero, and define $g(a) = x, g(b) = 0$, and the rest of the elements of g map to

zero. Note that f and g are not the zero function since they are not identically zero, but $fg = 0$. Importantly

$$fg(a) = f(a)g(a) = 0x = 0 \qquad fg(b) = f(b)g(b) = x0 = 0$$

The set of functions from a set X to a ring R is rarely an integral domain when $|X| > 0$ and $R \neq \{0\}$, since it would have to be the case that $fg = 0$ implies that either f is zero everywhere or g is zero everywhere. However, they do exist (an important example in complex analysis is the ring of power-series, or more generally the ring of analytic functions).

Sometimes the term *domain* used for the same type of ring without the commutative property:

Definition 9.1.23: Domain

A ring in which there are no zero-divisors is called a domain

The nicest property of integral domains is that the cancellation law holds!

Proposition 9.1.24: Cancellation Law for Integral Domains

Assume a, b, c are elements of any ring with a not a zero divisor. If $ab = ac$, then either $a = 0$ or $b = c$.

***Proof*:**

$$\begin{aligned} & ab = ac \\ \Leftrightarrow\ & a(b-c) = 0 \\ \Leftrightarrow\ & a = 0 \text{ or } b - c = 0 \\ \Leftrightarrow\ & a = 0 \text{ or } b = c \end{aligned}$$

With this property of integral domains, proposition 9.1.20 can be expanded upon:

Corollary 9.1.25: Finite Integral Domain is a Field

If R is a finite integral domain, then R is a field

Just like in proposition 9.1.20, the trick in this proof is a neat use of function to capture the idea of how an element acts on every other element in an object, not unlike how we can think when working with group actions:

***Proof*:**
Let R be a finite integral domain. and let a be a nonzero element of R. By the cancellation law the map $x \mapsto ax$ is an injective function. To see this, let's say it was not injective so $ax = ay$ but $x \neq y$. Then

$$ax - ay = 0 \quad \Rightarrow \quad a(x - y) = 0$$

Since a is non-zero and there are no zero divisors, it must be that $x - y = 0$, so $x = y$.

Now, since the map is injective, its image has $|R|$ elements, so it is surjective[a]. In particular, there is some $b \in R$ such that $ab = 1$. Since R is commutative, also $ba = 1$, so a is a unit in R.

Since a was arbitrary nonzero element, R is a field.

[a] For infinite sets an injective map from a set to itself need not be surjective, as $n \mapsto 2n$ on $\mathbb{Z}$ shows.

9.1.26 Note If commutativity is dropped and every nonzero element is required to be a unit, giving a division ring, the conclusion survives: every finite division ring is commutative, hence a field. This is Wedderburn's little theorem, proved in theorem 19.7.20 from the class equation and cyclotomic polynomials.

Though integral domains are nice, they loose a every important property: If R is an integral domain, then $R \times R$ is *not* an ingeral domain

Example 9.1.27: product of integrals domains is not an integral domain
Let R be an arbitrary integral domain. Then take $R \times R$, with addition and multiplication done point-wise. Then

$$(1,0) \cdot (0,1) = (0,0)$$

meaning there are zero-divisors, and so $R \times R$ is not a integral domain A particular consequence of this is that the direct product of fields is not a field (since fields are integral domains) More generally, integral domains do not satisfy the universal property of products.

A summary of all these properties is in section 9.1.2.

Exercise 9.1.1

1. An element $e \in R$ is called *idempotent* if $e^2 = e$. Does there exist a ring R that has exactly 3 idempotent elements? Find one or show it's not possible.
2. We will prove that every finite integral domain is a ring using the pigeonhole principle
   a) Show... click here for this proof
3. Let $\mathbb{C}$ be the set of expressions $a+bi$ with $a, b \in \mathbb{R}$, with $(a+bi)+(c+di) = (a+c)+(b+d)i$ and $(a+bi)(c+di) = (ac-bd)+(ad+bc)i$, the product obtained by expanding and using $i^2 = -1$. Prove that $\mathbb{C}$ is a ring. (Example 19.2.7 realizes $\mathbb{C}$ as the quotient ring $\mathbb{R}[x]/(x^2+1)$.)
4. Let' say $s \neq 1$ and $s^2 = s$. This propety is called idempotence. Show that the only idempotent element in an integral domain is 0 and 1
5. Let R be a ring with $1 \neq 0$. Is $\varphi : R \to R$, $\varphi(r) = 0$ a ring-homomorphism? If so, prove it, and if not, would it be a rng-homomorphism (i.e. would the image be a rng)?
6. Show that the only idempotent elemnt in a field' is 0 or 1 (there exists a proof using properties of a field but not properties of integral domain)
7. A field is said to be characteristic n if there exists a number n st $\underbrace{1+1+1\cdots+1}_{n \text{ times}} = 0$. If no such number exists, we say that the field has characteristic 0. Give an example (or examples) of fields with characteristic 0, 2, and 5.
8. Let $u \in R$ be a unit and $R \to S$ a ring homomorphism. Then $\varphi(u)$ is a unit. Is the same true

for zero-divisors? What about non-zero-divisors?

9. Show that if $\mathbb{F}$ is a field of characteristic 0, then $\mathbb{Q} \subseteq \mathbb{F}$.

10. (For those who know some Analysis and want examples of *rngs*) In the process of defining the Fourier series, it is necessary to define the convolution product between two continuous functions (on compact support, or more generally L^2)

11. (ibid last comment) Let X be a locally compact Hausdorff space (every $x \in X$ has a neighbourhood contained in a compact set) that is not compact. Let $C_0(X)$ be the set of continuous complex functions $f : X \to \mathbb{C}$ that vanish at infinity (for every $\epsilon > 0$, there exists a compact $K \subseteq X$ such that $|f(x)| < \epsilon$ for all $x \in X - K$). Then $(C_0(X), +, \cdot)$ with function addition and multiplicatoin is a rng, but not a ring (namely, the function $1(x)$ does not vanish at infinity). This rng is used quite often in analysis: it comes up in Fourier analysis, in defining Radon Measures, and in the study of C^*-algebras to name a few examples (see [31] and *C*-Algebras and K-Theory* [13]).

    Loosely speaking, rngs correspond to locally compact spaces, while rings correspond to compact spaces. For analysts, working over compact spaces is much too restrictive ($\mathbb{R}^n$ is not compact), while locally compact spaces are much more natural to work with ($\mathbb{R}^n$ is locally compact). Furthermore, recall we have mentioned we can turn a rng into a ring by doing $\mathbb{Z} \oplus R$ to some rng R. If we were interested in keeping nice analytical properties, for a $\mathbb{C}$-algebra R we would instead take $\mathbb{C} \oplus R$ (for instance $\mathbb{C} \oplus C_0(X)$). This corresponds to the one-point compactification of X: $C(X \cup \{\infty\}) \cong \mathbb{C} \oplus C_0(X)$.

12. Show that $(F, +, \cdot)$ be a ring. Show that $(F, +, \cdot)$ is a field if and only if $(F, +)$ is an abelian group and the set $F^\times$ is equal to $F - \{0\}$.

13. Let R be a ring and R^{op} denote the ring where the underlying set and addition operation is the same, and the multiplication is defined as $x * y = yx$ where yx happens in R. Show that $\text{id} : R \to R^{\text{op}}$ is not necessarily a ring isomorphism (it is sometimes called an *anti-isomorphism*). When is it a ring isomorphism? Show that for a commutative ring k, transposition $A \mapsto A^T$ is a ring isomorphism $M_n(k) \to M_n(k)^{\text{op}}$.

14. show that if $x^n = 0$, then $(x - 1)$ is an invertible element. An element where $x^n = 0$ is called a *nilpotent element*.

15. recall that $(x + y)^n = \sum_{k=1}^{n} \binom{n}{k} x^n y^{n-k}$. Show that the binomial formula holds if and only if $xy = yx$.

16. I the set of all left (resp. right) invertible elements a group? How that $Z(R)$ is a subring of R.

17. let G be a cyclic group of order n. For each $k \in \mathbb{Z}$, define $f : G \to G$, $f_k(g) = kg$. The n the map $k \mapsto f_k$ defines a ring isomorphism $\mathbb{Z}/n\mathbb{Z} \cong_{\mathbf{Ring}} \text{End}_{\mathbf{Ring}}(G)$ and a group isomorphism $(\mathbb{Z}/n\mathbb{Z})^* \cong_{\mathbf{Grp}} \text{Aut}_{\mathbf{Grp}}(G)$

9.1.1 Category of Ring

We now approach rings a 2nd time, but this time, we will take a more birds-eye view of the category or rings. We will see if this category has some properties different than the category of groups, and whether there is a general way of extending groups to form rings.

Proposition 9.1.28: Category of Ring

The collection of Rings along with ring homomorphisms form a category. This category will be denoted **Ring**.

***Proof*:**
If $\varphi\colon R \to S$ and $\psi\colon S \to T$ are ring homomorphisms, then $\psi \circ \varphi$ preserves sums and products because each map does, and $(\psi \circ \varphi)(1_R) = \psi(1_S) = 1_T$. Composition of functions is associative, and the identity map of a ring is a ring homomorphism that is an identity for composition, completing the proof.

Since **Ring** is a category, it will be verified what are the initial/final objects, and whether there are product, sub-, and quotient objects, and whether quotient objects can be classified through kernels. We can further ask about the relation between **Group** and **Ring**, and ask whether there is an analogous concept of ring-actions. In this section we will drop the subscript and write $\mathrm{Hom}(R, S)$ or $\mathrm{End}(R)$ instead of $\mathrm{Hom}_{\mathbf{Ring}}(R, S)$ or $\mathrm{End}_{\mathbf{Ring}}(R)$

The fact that sub-objects and product objects exist in rings (that is, subrings and product rings) has already been explored in definition 9.1.6 and example 9.1.4(5), though not in categorical terms. We will mention some more about product of rings and the consequences of integral domains not having products as an interesting example of category with no products. Quotient objects will be revisited in more detail in section 9.3.

Rings and $\mathbb{Z}$

The first thing to check is whether **Ring** has initial and final objects. In definition 1.4.6, we introduced the concept of initial and final group. We found there to not really be an intersting initial and final group: only the identity group. The same is true for the final object of rings. However, the zero ring cannot be the initial: if R is the zero ring and $\varphi\colon R \to S$ is a homomorphism, then $1_S = \varphi(1_R) = \varphi(0_R) = 0_S$, so S is the zero ring. There is still an initial object in the category or rings: we have our first example of a non-trivial initial object!

Proposition 9.1.29: $\mathbb{Z}$ is an Initial Ring

The ring $\mathbb{Z}$ is an intial ring

***Proof*:**
Let R be any ring (with identity) and let $\varphi : \mathbb{Z} \to R$ such that

$$\varphi(z) = 1 + 1 + 1 \ (z \text{ times}) \ = z \cdot 1$$

this is a ring homomorphism. For

$$\varphi(a{+}b) = (a{+}b){\cdot}1 = a{\cdot}1{+}b{\cdot}1 = \varphi(a){+}\varphi(b) \qquad \varphi(ab) = (ab){\cdot}1 = a(b{\cdot}1) = (a{\cdot}1)(b{\cdot}1) = \varphi(a)\varphi(b)$$

Furthermore, if $\varphi' : \mathbb{Z} \to R$ was another ring homomorphism, then by proposition 9.1.9

$$\varphi'(1) = 1\,, \varphi'(2) = 1 + 1\,, \cdots$$

that is, by the fact that the identity *must* be distributed in this fashion, we are forced that

$$\varphi = \varphi'$$

making it unique

If rings are not required to have a multiplicative identity, then $\mathbb{Z}$ is no longer initial: the identity map $\mathbb{Z} \to \mathbb{Z}$ and the zero map $\mathbb{Z} \to \mathbb{Z}$ both preserve sums and products. The initial object of that category is the zero ring.

So how can we use this result? The way I think about it is something "integer-like" in every ring [with identity]. It might not be commutative, it might not be injective (the codomain might even be finite), or it might be uncountably big (like $\mathbb{R}$ or $M_n(\mathbb{C})$) but there is still something integer like. For this reason, asking questions like "what properties of the integers do rings have" is a very sensible question! In fact, that will be essentially what we do in chapter 10!

For example, if we take $x \in \mathbb{Z}$, then $x = p_1 p_2 \cdots p_n$, that is x has a *unique prime factorization*. In the next chapter, we will study rings which have this similar property (they will be called *unique factorization domains*). Similarly, to find the $gcd(a, b)$, we can use the euclidean algorithm which algorithmically shows how much the set of divisors of the two numbers intersect. This is very handy since we can in very few steps find the common divisors of two numbers of extremely large numbers. Rings in which the notion of the euclidean algorithm is well-dfined will be called *euclidean domains*. Finally, a little more difficult but important concept, any number can be decomposed into *unique* prime numbers in $\mathbb{Z}$. We will find the equivalent of "numbers" in rings to be "ideals" (the original name was "ideal numbers" of rings). If every ideal can be generated by a single element $x \in R$, the ideal is called a *principal ideal*. Thus, a ring which is has "numbers" (ideals) which are generated by one element (just like a number just needs itself to describe, well, itself) will be called a *principal ideal domain*.

Thus, in the folloing chapters, some basic theory will extensively be used. Take a moment to review [24, Number Theory] for a refresher on some basic tools of number theory.

Rings and $\mathrm{End}(G)$

You might ask how should we choose which of the axioms are "nice" axioms to add to our structure. One way of doing so is by thinking about the multiplication action being permutations of the elements, not unlike ho the group operation can be thought of as the permutation of elements (i.e. any group $(G, +)$ is a subgroup of $\mathrm{Sym}(G)$) [6]

We saw in definition 1.4.9 that if G is an abelian group that $\mathrm{End}(G)$ is an agelian group under pointwise addition. Notice that there is already a 2nd operation, since function can compose:

$$(\mathrm{End}(G), +, \circ)$$

Then, it can be checked that composition here will make this into a ring (as shown in example 9.1.4)

Conversely, every ring $(R, +, \cdot)$ embeds in $\mathrm{End}((R, +))$: the map $r \mapsto (x \mapsto rx)$ is an injective ring homomorphism, injective because r is recovered as the image of 1. This is the ring analogue of Cayley's theorem (theorem 4.5.3). Cayley's theorem is the Yoneda lemma ([15, The Yoneda Lemma]) for the one-object category of a group G. Applied to the monoid $(R, \cdot)$, the Yoneda lemma gives the multiplicative half of this embedding, and distributivity places its image in $\mathrm{End}((R, +))$.

[6] Research **Ab**-enriched categories for information about this process

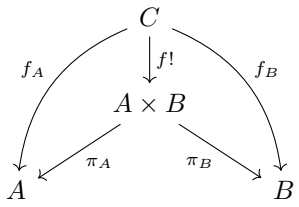
:exercse

Let $\mathrm{Aut}_{\mathbf{Ab}}(R)$ be the set of group-automorphisms on R. Show that $U(R) \cong_{\mathbf{Ab}} \mathrm{Aut}_{\mathbf{Ab}}(R)$

let R be a commutative ring. Show that $\lambda : R \to \mathrm{End}_{\mathbf{Ab}}(R)$ by $r \mapsto (y \mapsto ry)$ is indeed a ring homomorphism. If R is not commutative, show that it's almost abelian.

Products of Rings

Rings have products: Given any to rings A and B, there exists a ring $A \times B$ along ring homomorphisms $\pi_A : A \times B \to A$ and $\pi_B : A \times B \to B$ such that for any C here $f_A : C \to A$ and $f_B : C \to B$, there exists a unique $f! : C \to A \times B$ such that the folloing diagram commutes:



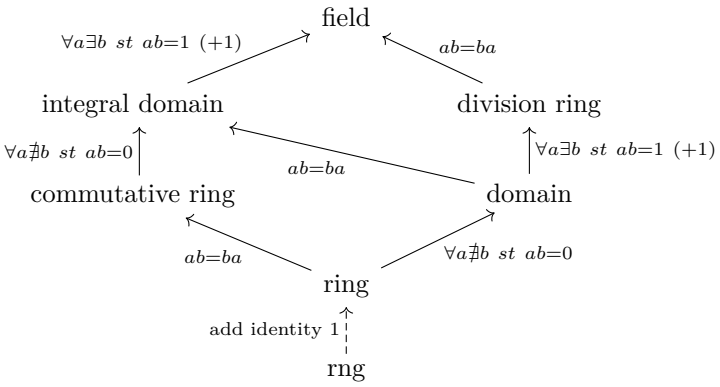
In this way, products in **Ring** are like products in **Grp**. A similar situation occurs for coproducts.

However, as mentioned earlier, if we retirct to *integral domains* and take the category whose objects are integral domains (this is indeed still a category as can be verified), they are no longer products, as demonstrated in example 9.1.27!

Since since all fields are integral domains, this means that products are not defined for fields! As a consequence of this, we cannot repeat the same trick as in chapter 5 in creating larger groups using products. Thus, extensions of fields (similar to how they were defined for groups in section 3.6) will have to be used to study the creation of larger fields. To study field extensions, we must first study ring homomorphisms more extensively. The particular way in which ring homomorphissm will be used in constructing larger fields requires more build-up, and so the study of fields is relegated to Part IV.

9.1.2 Summary

Keeping track of these properties and their relation to everything can be a bit to take in. It might help to have a visual too:



The *rng* can have more arrows extending from it to commutative ring, domain, and integral domain, but the diagram would be cumbersome, so I opted to make it a dashed line from *rng* to *ring*, since, it is not needed to "upgrade" those rings until we get to division ring and fields. I added a (+1) where we really need the multiplicative identity to define the next concept.

Notice that if a commutative ring only has units, it is then a commutative division ring, and hence a field. The line as not drawn so as to not clutter the diagram.

look at comment:

The following categorical results are also noted:

1. $\mathbb{Z}$ is the initial object of rings
2. All rings are subrings of $\mathrm{End}((R, +))$
3. Products exist for rings, but not for integral domains (and so not for fields).

Exercise 9.1.2

1. Since $\mathbb{Z}$ is initial, for any ring R and natural homomorphism $\mathbb{Z} \to \mathbb{R}$, $\mathrm{im}(\mathbb{Z})$ exists is a subring. Show that $\mathrm{im}(\mathbb{Z})$ is the *smallest* subring of any ring. Show further that $\mathrm{im}(\mathbb{Z})$ is either $\mathbb{Z}_n$ for any $n > 0$ or $\mathbb{Z}$. If $|\mathrm{im}(\mathbb{Z})| < \infty$, then we say the ring R has *characteristic* n, and if $\mathrm{im}(\mathbb{Z}) \cong \mathbb{Z}$ then R has *characteristic 0*.
2. Let R be a ring and define R^{op} to be the ring with the same underlying set and addition operation, and multiplication defined by $x * y = yx$, where yx happens in R. Show that R^{op} is a ring, and that $(_)^{\mathrm{op}} : \mathbf{Ring} \to \mathbf{Ring}$ is a functor (where do ring homomorphisms map to?). Is it covariant or contravariant?
3. Let $a, b \in R$ such that $ab = ba$. Show that ab is invertible if and only if a and b are invertible. Generlize your result to show that if $ab \neq ba$, then if ab invertible, then a is left-invertible, and b is right-invertible.
4. Show that $\mathrm{End}_{\mathbf{Ab}}(\mathbb{Z}) \cong_{\mathbf{Ring}} \mathbb{Z}$.

9.2 Examples of Rings

The following are rings that will show up time and time again, and being aquainted with them is very useful. Many of these rings will be the starting point to a new field of study in mathematics. For example, Polynomial rings are pivotal in commutative algebra (*Commutative Algebra* [16]), and when the polynomial ring takes on coefficients from a field (i.e. a polynomial ring over a field), then they form the foundation for field theory (see Part IV). Matrix rings are a common the "representation objects", the same way $\mathrm{Sym}(X)$ was an object that can be used to "represent" groups (see [17, Semisimple Rings and Modules]). Group rings can be thought of as the dual of the group of unity of rings, and also appear frequently in representation theory (see [17, Representation of Finite Groups]) and group cohomology (see [19, Group Cohomology]).

There is also an optional section on formal power series covering the elementary operations that can be performed on formal power series. These results are important for a class in complex analysis, however the formal power series ring will come up as examples for certain concepts introduced later in the book as well. These examples of formal power series will not be critical in understanding any material, and so they can be comfortably skipped.

9.2.1 Polynomial Rings

Polynomial Rings are one of the first examples we will give of a [commutative] extension of a ring. Let R be a commutative ring [7]. What we will be doing is extending R by an arbitrary unkown element x. In a sense, what we are doing is saying x is now an element of R, an then we figure out how to define this new structure (which we label $R[x]$) so that it's also a ring. Since $x \in R[x]$, then $x \cdot x \in R[x]$. Since we know nothing of x, we can only say that $x \cdot x = x^2$ is another element. We can continue this process inductively getting us every x^k for all $k \in \mathbb{N}$. For any other element r: $r \cdot x = rx$ where rx is a new element of $R[x]$. A similar reasoning to creating new elements applies to adding: $r + x$ is a new element. However, it need not be the case that an element has an inverse, so we do not define an x^{-1} such that $xx^{-1} = 1$[8]. All of these properties together give us the following definition:

[7] In the exercises, there are questions concerning the study of non-commutative R forming a polynomial ring and why it is often preferred for R to be commutative

[8] polynomial rings where the indeterminate x has an inverse are called the *ring of fractions of R[x].* see section 9.5

Definition 9.2.1: Polynomial Ring

Let R be a ring and x be an indeterminante. Then the sum

$$a_n x^n + a_{n-1}x^{n-1} + \cdots + a_1 x + a_0$$

with $n \geq 0$ and each $a_i \in R$ is called a *polynomial* in x with coefficients a_i in R. The set of all such polynomials is called the ring of *polynomial in the variable x with coefficients in R* (or *polynomial ring* if context permits) and will be denoted $R[x]$. Addition is defined as

$$(a_n x^n + \cdots + a_1 x + a_0) + (b_n x^n + \cdots + b_1 x + b_0) = (a_n + b_n)x^n + \cdots (a_1 + b_1)x + (a_0 + b_0)$$

where a_n or b_n may be zero in order for addition of polynomials of different degrees to be defined. Multiplication is perfomed by defining $(a_i x^i)(b_j x^j) = a_i b_i x^{i+j}$ and then extending to all polynomials by the distributive law. This is called the *convolution product*
The following are pertinent definitions with regards to polynomials.

1. If $a_n \neq 0$, then the polynomial is said to be of *degree* n, and $a_n x^n$ is called the *leading coefficient* (where the leading coefficeint of the zero polynomial is taken to be 0).
2. The polynomial is *monic* if $a_n = 1$.

For reference, here is the general formula for polynomial multiplication (i.e. the convolution product)

$$\left(\sum_{i=0}^{\deg(p)} p_i x^i\right) \cdot \left(\sum_{i=0}^{\deg(q)} q_j x^j\right) = \sum_{k=0}^{\deg(p)+\deg(q)} \sum_{l=0}^{k} p_l q_{k-l} x^k$$

9.2.2 Remark If you have done Fourier Analysis (or intend to in the future), then the convolution of two continuous (more generally measurable) functions follow's the exact same properties as polynomial multiplication. In fact, polynomial multiplication would be a discrete convolution.

Note that R is always a subring of $R[x]$, which we can see through the natural inclusion map[9] $R \hookrightarrow R[x]$ of $r \mapsto r$.

Usually, when we manipulate polynomials we treat them as functions. It must be stressed that we are *not* treating the elements of $R[x]$ as functions. The study of the elements of $R[x]$ as functions will be done in *Classical Algebraic Geometry* [23], which will establish the largest connection between algebra and geometry in the 20th century.

If $R = F$ is a field, then every coefficient is a unit, meaning we can divide each coefficient a polynomial to make it *monic*, that is,

$$a_n x^n + a_{n-1}x^{n-1} + \cdots + a_1 x + a_0 \overset{1/a_n}{\mapsto} x^n + a'_{n-1}x^{n-1} + \cdots + a'_1 x + a'_0$$

where $a'_i = \frac{a_i}{a_n}$. We shall see that often-times many results *up to multiplication by unit* (meaning if the result holds on p, it will hold on cp, where c is a unit). For this reason, it is worth keeping the term *monic* in mind.

[9] Natural in the sense that defining it requires nothing besides R and $R[x]$. The technical meaning of the word is the one of definition 18.2.1.

We can extend the definition of a polynomial ring to a polynomial ring over several variables by simply making $R[x]$ a polynomial ring over y, that is

$$R[x,y] = (R[x])[y] = R([y])[x] = R[y,x]$$

Continuing inductively, we can define a ring over a finite number of variable this way:

$$R[x_1, x_2, ..., x_{n+1}] = (R[x_1, x_2, ..., x_n])[x_{n+1}]$$

This process can be extended "indefinitely"[10] to define $R[x_1, x_2, ..., x_n, ...]$, in which every polynomial (i.e. elements of $R[x_1, x_2, ...,]$) has finitely many terms, and each term can have multiples of any of variables of $R[x_1, x_2, ..., x_n, ...]$, for example:

$$x_{30}^2 x_7^{23523} x_2 - r x_{100} x^{39} \in R[x_1, x_2, ...]$$

Multiplication in $R[x_1, x_2, ...]$ is the same as the multiplication in polynomial rings over finitely many variables, where we know the value for any term be expanding out the multiplication far enough to group like-terms. If $x_1^{d_1} x_2^{d_2} \cdots x_k^{d_k}$ is a term, then the *total degree* of the term is

$$d = d_1 + d_2 + \cdots + d_k$$

and the tuple $(d_1, d_2, ..., d_k)$ is the multi-degree of the term.

Properties of Polynomial Rings

To motivate the next theorem, let's say we have two polynomials of degree 1, $ax + b, cx + d \in R[x]$. We can ask what is the degree of $(ax + b)(cx + d) = acx^2 = (ad + bc)x + bd$. We might expect the degree to be 2. However, it is possible that a and c are zero-divisors: take $4x + 1, 4x + 1 \in (\mathbb{Z}/8\mathbb{Z})[x]$. In fact, in the example just stated, $4x + 1$ is in fact a unit in $(\mathbb{Z}/8\mathbb{Z})[x]$! If R is an integral domain, then some of its structure will be simpler:

Proposition 9.2.3: Degree Of Polynomials

Let R be an integral domain and let $p(x), q(x)$ be nonzero elements of $R[x]$. Then

1. degree $p(x)q(x)$ = degree $p(x)$ + degree $q(x)$
2. the units of $R[x]$ are the units of R
3. $R[x]$ is an integral domain

***Proof*:**
Write $p(x) = \sum_{i=0}^{n} a_i x^i$ and $q(x) = \sum_{j=0}^{m} b_j x^j$ with $a_n \neq 0$ and $b_m \neq 0$, so that $\deg p(x) = n$ and $\deg q(x) = m$, and put $a_i = 0$ for $i > n$ and $b_j = 0$ for $j > m$. By the formula for the

[10] For those uncomfortable with how the fact that the above process of defining a polynomial ring will always only produce polynomial rings with finitely many variables, the ring $R[x_1, x_2, ...]$ is the union of the chain $R[x_1] \subseteq R[x_1, x_2] \subseteq \cdots$, a *colimit* ([15, Cocone and Colimit])

product of polynomials, the coefficient of x^k in $p(x)q(x)$ is

$$c_k = \sum_{l=0}^{k} a_l b_{k-l}$$

1. Let $k \geq n+m$. A term $a_l b_{k-l}$ of c_k with $l > n$ vanishes because $a_l = 0$. A term with $l \leq n$ has $k - l \geq k - n \geq m$, with equality only when $k = n+m$ and $l = n$, so it vanishes because $b_{k-l} = 0$, except the term $a_n b_m$ when $k = n+m$. Hence $c_k = 0$ for $k > n+m$ and $c_{n+m} = a_n b_m$, which is nonzero because R has no zero divisors. Therefore $\deg p(x)q(x) = n+m$.

2. A unit u of R is a unit of $R[x]$, with the same inverse, since R is a subring of $R[x]$. Conversely, let $u \in R[x]$ and $v \in R[x]$ satisfy $uv = 1$. Neither is 0, since $0 \cdot v = 0 \neq 1$, so (1) gives $\deg u + \deg v = \deg 1 = 0$. Hence $\deg u = \deg v = 0$, so u and v are constants, that is, elements of R with $uv = 1$, and u is a unit of R.

3. The ring $R[x]$ is commutative: reindexing by $l \mapsto k - l$ and using that R is commutative turns c_k into $\sum_{l=0}^{k} b_l a_{k-l}$, the coefficient of x^k in $q(x)p(x)$. Its identity 1 is nonzero because $1 \neq 0$ in R. If $p(x)$ and $q(x)$ are nonzero, then by (1) the coefficient $c_{n+m} = a_n b_m$ of $p(x)q(x)$ is nonzero, so $p(x)q(x) \neq 0$. So $R[x]$ has no zero divisors and is an integral domain (definition 9.1.21), as we sought to show.

A ring homomorphism out of $R[x]$ is determined by its restriction to R and its value at x, and every such pair of data occurs:

Theorem 9.2.4: Universal Property of Polynomial Rings

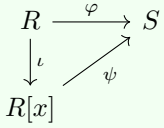
Let R and S be commutative rings and let

$$\varphi : R \to S$$

be any ring homomorphism and let $s \in S$ be any element of S. Then there is a unique ring homomorphism

$$\psi : R[x] \to S$$

such that $\psi(x) = s$ and which makes the following diagram commute, where ι is the inclusion



A nuance that must be mentioned is that polynomial rings may be over different rings. We thus may thus ask how a ring S is generated with respect to a ring R. As demonstrated earlier, $\mathbb{Z}$ is the initial object of **Ring**, and so in the arbitrary case $f : X \to S$ for some set X, we shall have a polynomial ring on the elements X over $\mathbb{Z}$. For those who are categorically inclined, try figuring out what is the category involved behind the scenes (this is similar to the construction for free groups in section 2.3.3)

***Proof*:**
Every $p \in R[x]$ is $p = \sum_{i\geq 0} a_i x^i$ for unique coefficients $a_i \in R$, all but finitely many zero.

Existence. Define

$$\psi\left(\sum_{i\geq 0} a_i x^i\right) = \sum_{i\geq 0} \varphi(a_i) s^i$$

Then $\psi(a) = \varphi(a)$ for $a \in R$, so $\psi \circ \iota = \varphi$, and $\psi(x) = s$ and $\psi(1) = \varphi(1) = 1$. Coefficients of a sum are the sums of the coefficients and φ is additive, so ψ is additive. Let $q = \sum_{j\geq 0} b_j x^j$. The coefficient of x^k in pq is $\sum_{l=0}^{k} a_l b_{k-l}$, so

$$\psi(pq) = \sum_{k\geq 0} \sum_{l=0}^{k} \varphi(a_l)\varphi(b_{k-l}) s^k$$

On the other hand, generalized distributivity (proposition 9.1.9(5)) and the commutativity of S give

$$\psi(p)\psi(q) = \sum_{i,j\geq 0} \varphi(a_i)\varphi(b_j) s^{i+j}$$

and grouping the terms with $i + j = k$ turns this into the expression for $\psi(pq)$. So ψ is a ring homomorphism.

Uniqueness. If $\psi' \colon R[x] \to S$ is a ring homomorphism with $\psi' \circ \iota = \varphi$ and $\psi'(x) = s$, then, as ψ' preserves sums and products,

$$\psi'\left(\sum_{i\geq 0} a_i x^i\right) = \sum_{i\geq 0} \psi'(a_i)\psi'(x)^i = \sum_{i\geq 0} \varphi(a_i) s^i$$

so $\psi' = \psi$, completing the proof.

A particular case of this is when $\varphi : R \to R$ is an identity map. In this case, we may link the notion of a [formal] polynomial to a polynomial function:

Definition 9.2.5: Evalutation Map

Let R be a ring and let $\alpha \in R$. The natural ring homomorphism

$$\varphi : R[x] \to R$$

which acts as the identity on R and which sends x to α is called the *evaluation at* α and is often denoted ev_α.
We say that α is a *zero* (or root) of $f(x)$ if $f(x)$ is in the kernel of ev_α.

The evaluation map can come in quite handy in many proofs that will be seen later (for instance in example 9.4.28, once prime ideals are available).

When working with functions, they are determined by the value at each point of evaluation. It must

be stressed that in general, a polynomial is *not* determined by it's evaluation

Example 9.2.6: Polynomial Not Determined By Evaluation
[Polynomial Not Determined By EvaluationPolynomial Not Determined By Evaluation] Take $x^2 - x \in (\mathbb{Z}/2\mathbb{Z})[x]$. Does the evaluation of this polynomial uniquely determine $x^2 - x$?

9.2.2⋆ Formal Power Series

⋆ *Optional. This subsection is an aside; nothing later in the book depends on it.*

Any polynomial is a finite linear combination of the terms x^n. If we allow for infinitely many terms, we call such an object a *formal power series*, and write it like so:

$$f(x) = \sum_{i=0}^{\infty} a_i x^i \qquad a_i \in R$$

The reader may ask what does it mean to take an infinite linear combination, as we have defined the + notation to only take in two inputs, and if we recurse over the inputs we may have a finite linear combination. These are valid concerns which will be addressed in *Commutative Algebra* [16] (completion). For now, we may consider $f(x)$ in the above equation as being an infinite tuple[11]

$$(a_0, a_1, a_2, \dots) \in \prod_{i=0}^{\infty} R$$

however we should take this identification as simply insuring that $f(x)$ is well-defined. We shall not write $f(x)$ in the notation of tuples and from now on shall assume we can take infinite sums. The collection of all formal power series are certainly closed under termwise addition. We must make sure multiplication is still well-defined. For multipilcation to be well-defined, each coefficient must be well-defined. Fortunately, this is the case. If $p(x)$ and $q(x)$ where

$$\begin{aligned} p(x)q(x) &= \left(\sum_{i=0}^{\infty} a_i x^i\right)\left(\sum_{j=0}^{\infty} b_j x^j\right) \\ &= a_0b_0 + (a_0b_1 + a_1b_0)x + (a_0b_2 + a_1b_1 + a_2b_0)x^2 + \cdots \\ &= \sum_{n=0}^{\infty}\left(\sum_{i=0}^{n} a_i b_{n-i}\right)x^n \end{aligned}$$

Each coefficient $\sum_{i=0}^{n} a_i b_{n-i}$ is a finite sum, so the product is well defined, and the sequence of these coefficients is called the *convolution* of the sequences (a_i) and (b_j). The ring axioms hold. Addition is termwise, so $(R[[x]], +)$ is an abelian group. The coefficient of x^n in both $(pq)r$ and $p(qr)$, where $r(x) = \sum_k c_k x^k$, is the finite sum of the products $a_i b_j c_k$ over all $i + j + k = n$, so multiplication is associative. The coefficients of $p(q + r)$ are $\sum_{i=0}^{n} a_i(b_{n-i} + c_{n-i})$, the sums of those of pq and pr, and similarly on the other side, so multiplication distributes over addition. The constant series 1 is an identity. Thus $R[[x]]$ is a ring:

[11] this may push the concern in asking how $\prod_{i=0}^{\infty}$ is defined. For those with this concern, they may skip to [16, Completion] or they may take for granted that we may always make a countable number of choices and remember the order

Definition 9.2.7: Formal Power Series

The collection of all infinite R-linear combinations of $\{1, x, x^2, ...\}$ forms a ring known as the ring of *formal power series* over R, and is denoted $R[[x]]$

The term "formal" comes from how we often associate to an object $f \in R[[x]]$ the evaluation function. If we had a function with a domain and codomain for which $r \mapsto \sum_{i=0}^{\infty} a_i r^i$, we would call this function a *power series*. However, for such a function to be defined, we require the ability to unambiguously define summing infinitely many elements of a ring. Such considerations require the introduction of a topology on our ring, something we shall not cover. We shall remark that it is very commmon in compex analysis to work with power series and choose $R \in \{\mathbb{R}, \mathbb{C}\}$ with the usual euclidean topology, leading to many very important results about complex-differentiable functions. There is one value we may unambiguously define without the need for any convergence considerations:

$$f(0) := a_0$$

Proposition 9.2.8: Formal Power Series Integral Domain

Let R be an integral domain. Then the ring $R[[x]]$ of formal power series over R is an integral domain

***Proof*:**
Write $f = \sum_{i\geq 0} a_i x^i$ and $g = \sum_{j\geq 0} b_j x^j$ for elements of $R[[x]]$. By the product formula above, the coefficient of x^n in fg is $\sum_{i=0}^{n} a_i b_{n-i}$. Reindexing by $i \mapsto n - i$ and using that R is commutative turns this into $\sum_{i=0}^{n} b_i a_{n-i}$, the coefficient of x^n in gf, so $R[[x]]$ is commutative. Its identity is the constant series 1, which is nonzero because $1 \neq 0$ in R.

Now let f and g be nonzero, let m be the least index with $a_m \neq 0$ and let k be the least index with $b_k \neq 0$. The coefficient of x^{m+k} in fg is

$$\sum_{i=0}^{m+k} a_i b_{m+k-i} = a_m b_k$$

because $a_i = 0$ for $i < m$, and for $i > m$ the index $m + k - i$ is less than k, so that $b_{m+k-i} = 0$. Since R is an integral domain, $a_m b_k \neq 0$, hence $fg \neq 0$. So $R[[x]]$ is a commutative ring with $1 \neq 0$ and no zero divisors, that is, an integral domain (definition 9.1.21), as we sought to show.

Given that $R[[x]]$ is the extension of $R[x]$, it may see intriguing to think that these two rings share very similar properties. However, the "infinitude" of $R[[x]]$ will make the ring act rather differently. The following may be the most striking change when extending to infinite polynomials:

Lemma 9.2.9: Existence of More Units

Let $R[[x]]$ be the formal power series over a ring R. Then $(1 - x)$ has an inverse in $R[[x]]$

***Proof*:**
Notice that

$$(1-x)(1+x+x^2+\cdots+x^n+\cdots)=1$$

and hence by definition that means that $1-x$ has an inverse, namely $\sum_{n=0}^{\infty} x^n$.

By convention, we denote inverses using quotient notation, and hence we would write:

$$\frac{1}{1-x} := \sum_{n=0}^{\infty} x^n$$

For those who have taken calculus, they may be familiar with an "analysis" version of this proof relying on absolute convergence. The above shows that this result is purely algebraic. In fact, we can fully classify all units in a formal power series

Proposition 9.2.10: Units In Formal Power Series

Let $f \in k[[x]]$. Then f is a unit (i.e. f has an inverse) if and only if $a_0 \neq 0$

***Proof*:**
The condition is certainly necessary since if

$$g(x)=\sum_n b_n x^n \qquad f(x)g(x)=1$$

then we must have $a_0 b_0 = 1$, so $a_0 \neq 0$. If conversely we have $a_0 \neq 0$, we shall show that the formal power series $(a_0)^{-1}f(x) = f_1(x)$ has an inverse $g_1(x)$, and hence $(a_0)^{-1}g_1(x)$ is the inverse of $f(x)$. Decompose $f_1(x)$ into:

$$f_1(x) = 1 - u(x)$$

then substitute $u(x)$ for y in the proof of lemma 9.2.9 to get that $1-u(x)$ has an inverse. But then we may conclude that $f(x)$ has an inverse, thus completing the proof.

Another question we may ask is the following: Given two polynomials $f(x), g(x)$, we may take $f(g(x))$, (replace x with $g(x)$). Can the same be done for power series? The following shows when this is possible:

Definition 9.2.11: Formal Derivative

Let $f \in k[[x]]$ be a formal power series. Then if $f(x) = \sum_{k=0}^{\infty} a_k x^k$, the formal derivative of f is the value

$$\frac{d}{dx}(f) = \sum_{k=0}^{\infty} k a_k x^{k-1}$$

we usually denote the derivative of $f(x)$ as $f'(x)$, and for higher derivatives we write $f^{(n)}(x)$ to represent taking recursively the derivative of f n times.

We may use the formal derivative recursively to find the coefficients of f. Namely, $f(0) = a_0$, $f'(0) = a_1$, ..., $f^{(n)}(0) = a_n$, and so forth.

Let f, g be power series. We may wonder when $g(f(z))$ is well-defined:

Proposition 9.2.12: Composing Power Series

Let f, g be formal power series. Then $g(f(z))$ is well-defined if $b_0 = 0$.

***Proof*:**
When composing power series, we get:

$$a_0 + a_1(b_1 z + b_2 z^2 + \cdots) + a_2(b_1 z = b_2 z^2 + \cdots)^2 + \cdots$$

where each coefficient is an infinite sum, and so we must ask for their convergence conditions. One thing we can do to guarantee this is by setting $b_0 = 0$, which will guarantee that all the coefficients are finite sums and hence well-defined.

We often denote the composition as $g \circ f$.

Proposition 9.2.13: Associativity Of Composition

Let $f, g, h \in k[[x]]$ be power series with $f(0) = 0$ and $g(0) = 0$. Then

$$(h \circ g) \circ f = h \circ (g \circ f)$$

***Proof*:**
The composites $h \circ g$ and $g \circ f$ are defined by proposition 9.2.12, since $g(0) = 0$ and $f(0) = 0$. Writing $g = \sum_{n \geq 0} b_n x^n$, the constant term of $g \circ f = \sum_{n \geq 0} b_n f^n$ is $b_0 = g(0) = 0$, because f has no constant term, so $h \circ (g \circ f)$ is defined as well, and $(h \circ g) \circ f$ is defined because $f(0) = 0$.

Fix $v \in k[[x]]$ with $v(0) = 0$ and consider substitution of v. For any $u_1, u_2 \in k[[x]]$,

$$(u_1 + u_2) \circ v = u_1 \circ v + u_2 \circ v \qquad (u_1 u_2) \circ v = (u_1 \circ v)(u_2 \circ v)$$

because each coefficient of each side is the same finite polynomial expression in the coefficients of u_1, u_2 and v. Taking $u_1 = u_2 = \cdots = u$ and inducting on n, the second identity gives

$$u^n \circ v = (u \circ v)^n$$

for every $n \geq 0$.

Now write $h = \sum_{n \geq 0} c_n x^n$, so that $h \circ g = \sum_{n \geq 0} c_n g^n$ by definition of composition. Substituting f,

$$(h \circ g) \circ f = \sum_{n \geq 0} c_n \, (g^n \circ f) = \sum_{n \geq 0} c_n (g \circ f)^n = h \circ (g \circ f)$$

where the middle equality is the displayed identity with $u = g$ and $v = f$, and the last is again the definition of composition, applied to h and $g \circ f$.

The first equality above interchanges substitution with an infinite sum. Since $g(0) = 0$ the series g^n has no terms of degree below n, and likewise for $(g \circ f)^n$ since $(g \circ f)(0) = 0$. So for each fixed p only the indices $n \leq p$ contribute to the coefficient of x^p on either side, and the interchange is an identity between two finite sums.

Define $I(x) = x$. Notice that for any $f \in k[[x]]$, $I \circ f = f \circ I = f$.

Proposition 9.2.14: Composition Has Inverse

Let $f \in k[[x]]$ and $I(x) = x$. Then a right inverse exists, $f \circ g$ if and only if $f(0) = 0$ and $f'(0) \neq 0$ (i.e. $a_1 \neq 0$). For g to also be a left-inverse, we require $g(0) = 0$, in which case

$$f \circ g = g \circ f = I$$

9.2.15 remark (If you know some calculus) Notice how the condition $f'(0) \neq 0$ bears resemblence to the inverse function theorem

***Proof*:**
We want to a g such that $f(g(z)) = z$. We want:

$$a_0 + a_1(b_1 z + b_2 z^2 + \cdots) + a_2(b_1 z = b_2 z^2 + \cdots)^2 = z$$

expanding and using the method of undetermined coefficients expanding the first two terms we get:

$$a_0 = 0 \qquad a_1 b_1 = 1$$

This shows us that $a_0 = 0$ and $a_1 = f'(0) \neq 0$ are necessary conditions. To show they are sufficient conditions, we can see that we may deduce all other coefficeints given these conditions.For example, we may deduce the coefficient of z^n as $a_0 + a_1 g(z) + \cdots + a_n g(z)^n$, so

$$a_1 b_n + P_n(a_2, ..., a_n, b_1, ..., b_{n-1}) = 0$$

Thus, we would start by doing $b_1 = 1/a_1$, and $b_2, b_3, ...$ are defined recursively (this resulting polynomials is called *Bells polynomial*). With our construction, we have that g satisfies $b_0 = 0$ and $b_1 \neq 0$. Thus, we can do the same thing to g, that is $g(f_1(w)) = w$. These inverses are equal:

$$f_1 = \mathrm{id} \circ f_1 = (f \circ g) \circ f_1 = f \circ (g \circ f_1) = f \circ \mathrm{id} = f$$

where we assumed the associativity of composition, something that can be checked. Finally, for the radius of convergence, we shall delay such considerations for now, noting that when we show that holomorphic functions are analytic we can use the inverse function theorem to estimate the values.

Example 9.2.16: Formal Power Series
Some classical examples of power series include:

1. $\exp(x) = e^x = \sum_{k=0}^{\infty} \frac{x^k}{k!}$
2. $\cos(x) =$
3. $\sin(x) =$
4. $\log(x) =$

9.2.3 Matrix Rings

As was briefly mentioned in the preliminary chapter ([24, Matrices]), many structures can be "represented as matrices", that is, instead of studying all these structure as separate algebraic objects, they all follow the same rules as matrix rings.

Definition 9.2.17: Matrix Ring

Let R be a ring. Then $M_n(R)$ is the collection of all $n \times n$ matrices with coefficients in R. Addition is defined term-wise and multiplication is defined via matrix-multiplication as given in [24, Matrices]. A matrix may be represented by:

$$(a_{ij})_{ij} \in M_n(R)$$

where a_{ij} represents the value at position ij of the matrix. The identity of $M_n(R)$ is denoted I_n.

In the above notation, matrix multiplication AB of two matrices can be represented as

$$(c)_{ij} = \left(\sum_{k=1}^{n} a_{ik} b_{kj} \right)_{ij}$$

Note that $M_1(R) \cong R$, and $M_2(R)$ is *not* commutative, and always has zero divisors, for:

$$\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \cdot \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \neq \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \cdot \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$$

R can be embedded into R by taking by taking mapping it to the diagonal:

$$a \mapsto (a)_{ii}$$

where the matrix $(a)_{ii}$ has zero in all other entries. It should be check that the subring of diagonal entries with all the same value is a subring, sometimes denoted $\iota(R) \subseteq M_n(R)$ to emphasize that $\iota(R)$ is an embedding, or RI_n. It should be check that $I_n = \iota(1)$. If R is commutative, then:

$$Z(M_n(R)) = RI_n$$

that is, the center of the matrix ring is the diagonals with the same coefficients. Slightly more generally, the subset of diagonal matrices also form a subring., espec Diagonal matrices are rather common, and so they are often denoted

$$\text{diag}(a_1, a_2, ..., a_n) = (a_i)_{ii}$$

Using the diagonal embedding, we can define a functor $M_n(-) : \mathbf{Ring} \to \mathbf{Ring}$ mapping R to the diagonal and φ to $M_n(\varphi)$ that maps $(a_{ij})_{ij} \mapsto (\varphi(a_{ij}))_{ij}$ (the reader should feel free to check this is indeed a functor). Thus, there is the following diagram:

$$\begin{array}{ccc} R & \xrightarrow{\varphi} & S \\ \downarrow & & \downarrow \\ M_n(R) & \xrightarrow[M_n(\varphi)]{} & M_n(S) \end{array}$$

Thus, if $S \subseteq R$, then $M_n(S) \subseteq M_n(R)$.

Matrix rings shall appear in many context, notably in the case when $R = k$ is a field in the theory of vector-spaces and when R is a division ring in representation theory.

9.2.4 Group Rings

We have seen how to take a set and turn it into a group in many ways (ex. trivial group $T(X)$, permutation group S_X, free group $F(X)$). Of these methods the free-group construction gave us a way to lift a set map to a group map. If you have read section 2.3.5, **Set** and **Grp** form a free-forgetful adjoint, which tells us that than the "lifting" of set functions to group homomorphisms happened in a "natural way". We can repeat this process with groups and rings, forming a "free ring" with respect to groups. This "free ring" will not be free with respect to *sets*, and so will differ from free groups[12]. However, after going through free rings with respect to groups, the reader might start seeing why both constructions are called *free*.

Definition 9.2.18: Group Ring

Let R be a commutative ring and let G be any group, where $g_0 = e$ is the identity, with the operation written multiplicity. Then define the *group ring* $R[G]$ (or RG) of G with respect to R to be the set of all formal sums:

$$\sum_{g_i \in G} a_i g_i$$

where only finitely many a_i are non-zero where addition is done point wise and the product is defined per monomial as $(ag_i)(bg_j) = (ab)g_k$ where ab is performed in R and $g_i g_j = g_k$, and then extended this is extended to multiplication of the formal sums using convolution.

Since $g_0 = e$, we often write $a_0 g_0 = a_0$. In this way, we can identify $R \subseteq R[G]$, and R in as a subring of $R[G]$ has it's original ring operations. Similarly, taking $a_{ij} = \delta_{ij}$[13] where

$$\delta_{ij} = \begin{cases} 0 & i \neq j \\ 1 & i = j \end{cases}$$

we can identify $G \subseteq R[G]$, and G as a subgroup of $R[G]$ has it's original group operation. Even better, $G \leqslant R[G]^\times$. It is an easy exercise to check that $R[G]$ is indeed a ring, and that $R[G]$ is commutative if and only if G is commutative. Another simple exercise to check is that R (identified as a subring of $R[G]$) is in the center of $R[G]$ ($R \subseteq Z(G)$), and that $1 \in R$ (identified as a subring of $R[G]$) is the identity of $R[G]$

The reader may see that group rings are indeed the free objects with respect to the forgetful functor **Ring** $\to$ **Grp** by repeating the construction given for free groups: Take the collection of set maps $f : G \to R$ where only finitely many $f(g)$ are nonzero (such maps are said to have *finite support*[14]). Then two elements are added pointwise, $h \mapsto f(h) + g(h)$, and multiplication has the following clever interpretation:

$$h \mapsto \sum_{ab=h} f(a)g(b) = \sum_{a \in G} f(a)g(a^{-1}h)$$

[12] We will cover free rings later

[13] this is called the Kronecker delta

[14] For those who know some calculus/analysis, this is similar to how many smooth functions are defined on compact support

Notice this sum is indeed finite since f, g have finite support.

Example 9.2.19: Group Rings

1. Let $R = \mathbb{Z}$ and $G = D_4$. Then, for example $3r^2 - s, rs + 9 \in \mathbb{Z}[D_4]$.
2. If G is any finite subgroup, and $S \subseteq R$ is a subring, then $S[G] \subseteq R[G]$ is a subring.
3. (If you know about the quaternions $\mathbb{H}$), then $\mathbb{R}[Q_8] \neq \mathbb{H}$, even though they both contain Q_8. Can you see why?
4. Let $R = \mathbb{Q}$ and $G = S_3$. Let $a = 3(2\ 3) - 4(1\ 2\ 3)$ and $b = 1/2(1\ 2) + 5(3\ 2\ 1)$. Compute:

$$a + b \qquad ab$$

Note that if G has an element of finite order, then the group ring always has zero divisors: let $g \in G$ such that $m = |g| > 1$. Then

$$(1 - g)(1 + ... + g^{m-1}) = 1 - g^m = 1 - 1 = 0$$

Exercise 9.2.1

1. Is $R[x]$ isomorphic to $\oplus_{i=1}^{\infty} R$? Is $R[[x]]$ isomorphic to $\prod_{i=1}^{\infty} R$?)
2. Let $f|g$. Does this imply $\deg(f)\big|\deg(g)$?
3. Let R be a **non**-commmutative ring. Define the appropriate ring so that for any set function $f : A \to S$, there is a unique map $\iota : A \to R$ to the ring $F : R \to S$ such that the diagram commutes. That is, define the free object in the general case so that it is left-adjoint to the forgetful functor **Ring** $\to$ **Set**.
4. Take the full sub-category **Poly-Ring** $\subseteq$ **Ring**. What is the inital object of **Poly-Ring**?
5. Let $p(x) \in R[x]$ be a polynomial such that for all $r \in R$,

$$p(r) = a_n r^n + \cdots + a_1 r + a_0 \neq 0$$

   Show that there is an extension $S \supseteq R$ such that there exists $s \in S$ where $p(s) = 0$. In particular, there exists an S such that there exists a natural inclusion $\iota : R \hookrightarrow S$, $p(x)$ naturally includes in S and some element $s \in S$ is a root to $p(x)$:

$$p(s) = a_n s^n + \cdots + a_1 s + a_0 = 0$$

6. Show that $R[\mathbb{N}] \cong R[x]$. Show that $R[\mathbb{Z}] \cong R[x, x^{-1}]$
7. Is $\mathbb{Z}[x, x^{-1}] \cong \mathbb{Z}[[x]]$

9.3 Ring Homomorphisms and Quotient Rings

In this section, we explore the structure of the quotient objects of rings, that is, if $\varphi : R \to S$ is a ring-homomorphism, we aim to describe the structure of $\varphi(R) \subseteq S$ given structure for R. We first

recall the definition of ring homomorphisms from the first section:

Definition 9.3.1: Ring Homomorphism Remainder

Let R and S be rings. Then:

1. *ring homomorphism* is a map $\psi : R \to S$ satisfying
   (a) $\varphi(a+b) = \varphi(a) + \varphi(b)$
   (b) $\varphi(a \cdot b) = \varphi(a) \cdot \varphi(b)$
   (c) $\varphi(1_R) = 1_S$
2. The *kernel* of the ring homomorphism φ, denoted $\ker(\varphi)$ is the set of elements of R that map to 0 in S
3. A ring homomorphism with an inverse that is a ring homomorphism is a *ring isomorphism.*

If the context is clear, "homomorphism" will be used instead of "ring homomorphism". Also, if A, B are rings, then $A \cong B$ will always be interpreted as an isomorphism between the two rings.

Example 9.3.2

1. $\pi : Z \to \mathbb{Z}/2\mathbb{Z}$, here $\pi(x) = [x]$ is the natural projection map, is a ring homomorphism
2. $\varphi_n : \mathbb{Z} \to \mathbb{Z}$, $\varphi_n(x) = nx$ is *not* a ring-homomorphism.
3. $\varphi : \mathbb{Q}[x] \to \mathbb{Q}$, $\varphi(p(x)) = p(0)$ is a ring homomorphism, here $\ker\varphi = \{a_n x^n + \cdots + a_1 x\}$. Notice that every elements in the kernel has an x. We can also think of this as $p(x) \cdot x$ for any $p(x) \in \mathbb{Q}[x]$. This way of looking at it will become the main way we look at the kernel of a ring homomorphism shortly.
4. As shown earlier, if $S \subseteq R$ has a ring-structure, then if the inclusion map $i : S \to \mathbb{R}$ is a [ring] homomoprhism, then S is a subring.
5. The inclusion map $\mathbb{Z} \hookrightarrow \mathbb{Q}$ is a ring homomorphism. Is there a non-trivial homomorphim from $\mathbb{Q}$ to $\mathbb{Z}$?
6. Embedd $k[x] \hookrightarrow k[[x]]$ by mapping $k[x]$ to be all formal power series with only finitely many nonzero elements. Then by proposition 9.2.10, any polynomial $p(x) \in k[x]$ such that $p(0) \neq 0$ under the evaluation map is invertible in $k[[x]]$. Hence, there is a power series that can be identified by $1/p(x)$. Thus, we may define polynomial quotients in $k[[x]]$
$$\frac{p(x)}{q(x)} \qquad q(0) \neq 0$$
7. Is the derivative map $\frac{d}{dx} : k[[x]] \to k[[x]]$ defined by
$$\frac{d}{dx}\left(\sum_{k=0}^{\infty} a_k x^k\right) = \sum_{k=0}^{\infty} k a_k x^{k-1}$$
a ring homomorphism? what is $\frac{d}{dx}(fg)$?

8. Let k be a field and $T = \sum_{n\geq 0} b_n y^n \in k[[y]]$ with $b_0 = 0$. For $f = \sum_{n\geq 0} a_n x^n \in k[[x]]$, define $f \circ T = \sum_{n\geq 0} a_n T^n$, where each coefficient is a finite sum because T^n has no terms of degree below n. Show that $f \mapsto f \circ T$ is a ring homomorphism $k[[x]] \to k[[y]]$. What goes wrong if $b_0 \neq 0$?

Just as the study of group-homomorphisms is equivalent to finding a normal subgroup H and studying G/H, so too will there be a link between ring-homomorphisms and a special subset I of R for which R/I represents the homomorphism. Just like for group-homomorphisms, the set I is based on the kernel of the ring-homomorphism, and R/I will be isomorphic to the image of φ:

Proposition 9.3.3: Image and Kernel of Ring Homomorphism

Let R, S be rings and $\psi : R \to S$ be a homomorphism.

1. The image of ψ is a sub-ring of S
2. the kernel of ψ is, in general, *not* a subring of R. Rather, the kernel $I = \ker(\psi)$ retains the abelian structure of R (i.e. $a - b \in I$ for all $a, b \in I$), and is closed under multiplication (left/right multiplication)[a] from elements of R (for all $r \in R$ and $i \in I$, $ri \in \ker(\psi)$ and $ir \in \ker(\psi)$).

[a] note that it was specified that it was both left and right multiplication because the multiplication is not necessarily commutative, while addition is

As an example of a kernel, take $\varphi : \mathbb{Z}[x] \to \mathbb{Z}[x]$, $p(x) \mapsto p(0)$. Then $\ker(\varphi) = x\mathbb{Z}[x]$, which is not a ring as $1 \notin x\mathbb{Z}[x]$. Kernels are the more general object *rng*. Note that if we define a ring to not have an identity, then the kernel would indeed be a sub-ring of R. Some authors define rings to not have an identity for this very reason. However, as a consequence, almost every theorem will have to be prefaced with "let R be a ring with identity", meaning the study of ring theory will ultimately be the study of a special case of rings with identity, which begs the question whether broadening the definition to not have an identity is worth the trade-off. However one sees it, it is simply a mathematical reality that kernels are different objects than rings (as the proposition suggests), and this author belives it is better to treat them as such[15]. Kernels will soon be defined as *ideals* (definition 9.3.6) since they have different properties from rings.

***Proof*:**
The map ψ is a homomorphism of the additive groups, so $\psi(a-b) = \psi(a) - \psi(b)$ for all $a, b \in R$ (proposition 1.4.4(2)).

1. The image $\psi(R)$ is an additive subgroup of S (proposition 1.4.4(5)). It is closed under multiplication, since $\psi(a)\psi(b) = \psi(ab)$, and it contains $1_S = \psi(1_R)$. By corollary 9.1.7, $\psi(R)$ is a subring of S.
2. Let $I = \ker \psi$ and take $a, b \in I$. Then

$$\psi(a - b) = \psi(a) - \psi(b) = 0 \Rightarrow a - b \in I$$

showing that I is closed under subtraction, and in particular is an additive subgroup of R.

Now take $r \in R$. We want to show that for any $i \in I$, $ri, ir \in I$. The key to this proof is

[15] The fact that the kernel is not a ring means that the category of **Ring** does not have kernel objects

proposition 9.1.9(1). Then

$$\psi(ri) = \psi(r)\psi(i) = \psi(r) \cdot 0 = 0 \Rightarrow ri \in I$$

and similarly

$$\psi(ir) = \psi(i)\psi(r) = 0 \cdot \psi(r) = 0 \Rightarrow ir \in I$$

Thus, I is closed under left-right multiplication of R, completing the proof.

9.3.4 Remark As a consequence of kernels not being subrings, the concept of extensions as discussed in section 3.6 do not apply so simply to rings! A group extension relied on the fact that both N and G/N were groups. Since ideals are not rings, the same concept does not apply for rings! There is an analogous concept of a "ring extension"[a], which generalize the notion of an extension by just requiring a ring extension S of R to be that there exists an injective ring homomorphism $\varphi : R \to S$ (similarly to how if $N \trianglelefteq G$ then we always have that $N \hookrightarrow G$). This notion of ring extension will be central to us in [16, Ring Extension and Integral Extension] [b], but the "block-building" story that happened for groups does not play out in the same way for rings. However, any ring R is naturally an R-module (Part III), and modules have extensions defined on them [c]. Thus, we may study the extensions of rings in a group-theoretic way in the context of module theory.

[a] not to be confused with ring extensions that extend a ring by an abelian group as described in Nagata Masayoshi's book *Local Rings* (see Wikipedia article)
[b] It will be the fundamental way we look at varieties and number fields
[c] The category of R-module is closed under kernels

This does not mean that we cannot define $R/\ker(\varphi)$. In fact, $R/\ker(\varphi)$ will be a well-defined ring, as we explore now. Note that since R is an abelian group, and φ is also always a group-homomorphism, $\ker\varphi$ is always be a normal group. If we label $I = \ker(\varphi)$ then R/I already has a group structure, that is:

$$(r + I) + (s + I) = (r + s) + I$$

is well defined. The question becomes if we can also add a multiplicative structure to the quotient group, that is:

$$(r + I) \cdot (s + I) = (rs) + I \qquad \text{(in equivalence class notation: } [x \cdot y] = [x] \cdot [y])$$

With I the kernel of a ring homomorphism, the multiplication is well-defined. For $i, k \in I$ and $r, s \in R$:

$$\varphi((r+i)(s+k)) = \varphi(r+i)\varphi(s+k) = (\varphi(r)+\varphi(i))(\varphi(s)+\varphi(k)) = \varphi(r)\varphi(s)+0+0+0 = \varphi(r)\varphi(s)$$

where we used both the additive and multiplicative linearity condition. Labelling $[x] = x + I$, we call $(R/I, +_\sim, \cdot_\sim)$ with $+_\sim$ and $\cdot_\sim$ being defined through φ the *quotient ring* [16]. Thus, I being the kernel of a ring is a sufficient condition to make R/I into a ring. Is it a necessary condition? That is, can there be weaker structures that still preserves the multiplicative structure as shown above? For example, can I simply be a normal subgroup and always preserve the multiplicative structure? The answer is no:

[16] as before we will delay defining the quotient object till we find the sufficient conditions on I independent of a ring homomorphism

Example 9.3.5: : Quotient group without preserving multiplicative property
Take $\mathbb{Q}/\mathbb{Z}$ as a quotient group. Since $\mathbb{Q}$ is abelian, $\mathbb{Q}/\mathbb{Z}$ is a well-defined quotient group. Suppose the rule $(r+\mathbb{Z})(s+\mathbb{Z}) = rs+\mathbb{Z}$ were a well-defined multiplication. Since $1+\mathbb{Z} = 0+\mathbb{Z}$, every $s \in \mathbb{Q}$ would satisfy

$$s+\mathbb{Z} = (1+\mathbb{Z})(s+\mathbb{Z}) = (0+\mathbb{Z})(s+\mathbb{Z}) = 0+\mathbb{Z}$$

using $0 \cdot s = 0$ (proposition 9.1.9(1)). But $\frac{1}{2}+\mathbb{Z} \neq 0+\mathbb{Z}$, since $\frac{1}{2} \notin \mathbb{Z}$. So the multiplication of $\mathbb{Q}$ does not pass to the quotient group $\mathbb{Q}/\mathbb{Z}$.

So we must figure out what properties I requires for closure of multiplication. Let's say that I is a normal subgroup, and let's work with the cosets R/I. Let's say that multiplication of cosets is well defined. That would imply that

$$[r+i][s+j] = [rs] \qquad i,j \in I$$

that is, multiplication is independent of representation. If $r = s = 0$, then that means that

$$(i+I)(j+I) \in I \Rightarrow ij \in I$$

meaning that I is closed under multiplication. Next, what if $s = 0$ and r is arbitrary. Then that would lead to $[r][0] = [r0] = [0]$ by proposition 9.1.9(1), which, written in terms of cosets with $j \in I$ representing the class $[0] = I$, will be:

$$\begin{aligned} &(r+I)\cdot(j+I) = I \\ \Rightarrow &rj+I = I \\ \Rightarrow &rj \in I \end{aligned}$$

and since r is any arbitrary element $r \in R$, then I must be closed under left multiplication of any element of R. Similarly if $r = 0$, meaning $js \in I$.

Conversely, let's say I is closed under multiplication from the left and right-side: $\forall i \in I$, $ri \in I$, $ir \in I$ $\forall r \in R$. Then

$$\begin{aligned} \big((r+i)+I\big)\cdot\big((s+j)+I\big) &= (r+i)(s+j)+I \\ &= (rs+rj+is+ij)+I \\ &= (rs+I)+(rj+I)+(is+I)+(ij+I) \end{aligned}$$

and since I is closed under multiplication, $ij \in I$, and since I is closed under left and right multiplication by any element of R, $rj, is \in I$, meaning we are left with:

$$= (rs)+I$$

meaning we are once again representative independent. Thus, we showed that left and right R-closure of I is a necessary *and* sufficient condition for the cosets defined by I to have a 2nd operation.

As with cosets, the equivalence classes induced by ring homomorphisms are given a special name:

Definition 9.3.6: Ideals

Let R be a ring, and let I be a subset of R, and let $r \in R$. Then:

1. $rI = \{ra | a \in I\}$, and $Ir = \{ar | a \in I\}$
2. a subset I of R is a *left ideal* if I is an additive subgroup of R that is closed under *left* multiplication by elements from R ($rI \subseteq I$ for all $r \in R$)
3. Similarly, I is a *right* ideal if I is an additive subgroup of R that is closed under *right* multiplication by elements from R ($Ir \subseteq I$ for all $r \in R$)
4. A subset I that is *both* a left and right ideal is called an *ideal* (or, for added emphasis, a *two-sided* ideal) of R

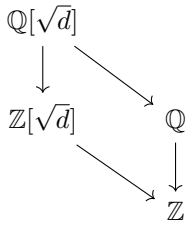
Example 9.3.7

1. Take $m \in \mathbb{Z}$. Then $m\mathbb{Z} \subseteq \mathbb{Z}$ is an ideal. Similarly for $\mathbb{Z}m$.
2. Take $m \neq 0$ and $m\mathbb{Z} \subseteq \mathbb{R}$. Then $m\mathbb{Z}$ is not an ideal (since $\frac{1}{2m} \cdot m = \frac{1}{2} \notin m\mathbb{Z}$).
3. In $\mathbb{Z}_4 \times \mathbb{Z}_4$, $\langle (1,1) \rangle$ is a sub-ring, but not an ideal
4. Take any product of two rings $R \times S$. Then $R \times 0 \subseteq R \times S$ is an ideal, since for any $(a, 0) \in R \times 0$, $(R \times S) \cdot (a, 0) \subseteq R \times 0$. It is a ring with identity $(1, 0)$, but it is not a subring of $R \times S$ when $S \neq 0$, as it does not contain $(1, 1)$. No proper ideal contains 1, since an ideal containing 1 contains $r \cdot 1 = r$ for every r.

9.3.8 Trivia Why is it called an ideal? In number theory, people tried finding more information about polynomials like

$$x^2 - d = 0 \Rightarrow x = \sqrt{d}$$

And we would construct the field $\mathbb{Q}(\sqrt{d}) = \left\{a + b\sqrt{d} | a, b \in \mathbb{Q}\right\}$ which was called $\mathbb{Q}$ with $\sqrt{d}$ adjoint. This ring (in fact field) has a sub-ring $\mathbb{Z}[\sqrt{d}]$ and Q, making the diagram:



So, with this construction, we have the question of developping the concept of prime number in $\mathbb{Z}$ to it in $\mathbb{Z}[\sqrt{d}]$. If you try to take the same idea (a element that has no divisors in the ring except the element and ± 1). But it turns that if you try to define an element of that, elements do not necessarily have a unique factorization!
Example
This is so fundamental that this was dropped as a good definition as prime element in $\mathbb{Z}[\sqrt{d}]$. Insead, the proper generalization of this concept will turn out to be ideals in $\mathbb{Z}$ which will be called *prime ideals*, and then look at their natural extension into $\mathbb{Z}[\sqrt{d}]$. The word "ideal" here comes from how these elements would be the *ideal numbers*. The word number will eventaully get dropped, leaving us with ideal.

9.3.9 Computational Shortcut To prove that I is an ideal, we must show that I is non-empty, that it is closed under subtraction (so that it is an additive subgroup, normal since $(R, +)$ is abelian), and that $rI \subseteq I$ and $Ir \subseteq I$ for all $r \in R$. Since R has a 1, you can replace "closed under subtraction" by "closed under addition", since $(-1)a = -a$ (proposition 9.1.9(4)) will cover the negative case

Another reason to treat ideals differently than rings is the following:

> Another way of how ideals are different form rings is that being a ideal is "ring-dependent". For example $(x) = \mathbb{R}[x]x$ is an ideal of $\mathbb{R}[x]$, but not an ideal of $\mathbb{C}[x]$, since $i \cdot x = ix \notin \mathbb{R}[x]x$. This points again to how ideals are different.

This discussion is summarized in the following proposition:

Definition 9.3.10: Quotient Ring

When I is an ideal of R. The ring R/I with the natural multiplicative operation is called the *quotient ring* of R by I.

Proposition 9.3.11: Quotient groups and Ring Homomorphism

Let R be a ring and let I be an ideal of R. Then the (additive) quotient group R/I is a ring under the binary operation:

$$(r+I)+(s+I)=(r+s)+I \qquad \text{and} \qquad (r+I)\cdot(s+I)=(r\cdot s)+I$$

for all $r, s \in R$. Conversely, if I is any additive subgroup of R such that the above operations are well defined, then I is an ideal of R.

***Proof*:**
The additive group $(R,+)$ is abelian, so I is a normal subgroup and R/I is an abelian group under the stated addition, which is well defined. If $r+I=r'+I$ and $s+I=s'+I$, write $r'=r+i$ and $s'=s+j$ with $i,j\in I$. Then

$$r's'=rs+(rj+is+ij)$$

and $rj, is, ij \in I$ because I is an ideal, so $r's'+I=rs+I$ and the multiplication is well defined. The associative and distributive laws hold in R/I because they hold for representatives in R, for instance $\big((r+I)(s+I)\big)(t+I)=(rs)t+I=r(st)+I=(r+I)\big((s+I)(t+I)\big)$, and $1+I$ is an identity. So R/I is a ring.

Conversely, let I be an additive subgroup for which the multiplication is well defined, and let $r\in R$ and $i\in I$. Since $i+I=0+I$, well definedness and proposition 9.1.9(1) give

$$ri+I=(r+I)(i+I)=(r+I)(0+I)=0+I \qquad ir+I=(i+I)(r+I)=(0+I)(r+I)=0+I$$

so $ri, ir\in I$, and I is an ideal of R, as claimed.

Just like for groups, we can ask if every ideal is the kernel of a ring homomorphism, essentially asking for the 1st isomorphism theorem for rings (i.e. the universal property of quotient rings). This turns out to be the case:

Theorem 9.3.12: The first isomorphism theorem for Rings

1. If $\varphi : R\to S$ is a homomorphism of rings, then the kernel of φ is an ideal of R, the image of φ is a subring of S and $R/\ker(\varphi)$ is isomorphic as a ring to $\varphi(R)$, by the isomorphism $r+\ker(\varphi)\mapsto\varphi(r)$.
2. If I is any ideal of R, then the map

   $$R\to R/I \qquad \text{defined by} \qquad r\mapsto r+I$$

   is a surjective ring homomorphism with kernel I (this homomorphism is called the *natural projection* of R onto R/I). Thus every ideal is the kernel of a ring homomorphism and vice versa.

***Proof*:**

1. Let $K = \ker(\varphi)$. By proposition 9.3.3(2), K is an additive subgroup of R closed under left and right multiplication by elements of R, that is, an ideal (definition 9.3.6), and by proposition 9.3.3(1), $\varphi(R)$ is a subring of S. So R/K is a ring (proposition 9.3.11). The additive group of R is abelian, so K is a normal subgroup of it contained in the kernel of φ, and theorem 3.1.15 applied to the additive groups gives a homomorphism of additive groups $\Phi\colon R/K \to S$ with $\Phi(r+K) = \varphi(r)$. It is multiplicative, since

$$\Phi\big((r+K)(s+K)\big) = \Phi(rs+K) = \varphi(rs) = \varphi(r)\varphi(s) = \Phi(r+K)\Phi(s+K)$$

   and $\Phi(1+K) = \varphi(1) = 1$, so Φ is a ring homomorphism. Its image is $\varphi(R)$. It is injective: if $\Phi(r+K) = \varphi(r) = 0$, then $r \in K$ and $r + K = 0 + K$. So Φ is a bijective ring homomorphism from R/K onto $\varphi(R)$, hence a ring isomorphism by the remark following definition 9.1.5.

2. Let $\pi(r) = r + I$. By the definition of the operations of R/I (proposition 9.3.11), $\pi(r+s) = \pi(r) + \pi(s)$, $\pi(rs) = \pi(r)\pi(s)$ and $\pi(1) = 1 + I$, so π is a ring homomorphism, and it is surjective because every element of R/I is a coset $r + I$. Its kernel is $\{r \in R \mid r + I = 0 + I\} = I$. So every ideal is the kernel of a ring homomorphism, and by (1) every kernel of a ring homomorphism is an ideal, completing the proof.

As with groups, the bar notation $\overline{r}$ or r mod I notation is used to represent the equivalence class $[r]$.

Example 9.3.13

1. R and $\{0\}$ are ideals. An ideal is proper if $I \neq R$. The set with only zero is called the *trivial* ideal.

2. $n\mathbb{Z}$ is an ideal of $\mathbb{Z}$.

3. Take $R = \mathbb{Z}[x]$, and let I be the subset whose polynomials are of degree at least 2. Note that I is closed under addition and multiplication, and thus an ideal. The cosets consist of the elements with the same 0th and 1st term, so

$$\overline{1 + 2x + x^2} = \overline{1 + 2x - 5x^3} = \overline{1 + 2x + x^6 + 87x^{1231}}$$

   Note that $\overline{x}\,\overline{x} = \overline{x^2} = \overline{0}$, thus R/I has a zero divisor, even though $\mathbb{Z}[x]$ does not .

4. We saw $\mathbb{Q}[x]/(x)$. Let's try another polynomial: $\mathbb{Q}[x]/(x^2-2)$. Since we can divide polynomials, we can divide any polynomial by $x^2 - 2$. We can in fact keep dividing it till we get a polynomial of lesser degree. We then get that

$$\mathbb{Q}[x]/(x^2-2) = \{ax + b | a, b \in \mathbb{Q}\}$$

   Note that $x^2 - 2$, so $x^2 - 2 = 0 \Rightarrow x = \pm\sqrt{2}$. So, since $[x^2 - 2] = [0]$, this will make

$$\mathbb{Q}[x]/(x^2-2) \cong \mathbb{Q}(\sqrt{2}) = \left\{x + y\sqrt{2} | x, y \in \mathbb{Q}\right\}$$

   Or, more specifically, if we take the ring-homomorphism $\varphi : \mathbb{Q}[x] \to \mathbb{Q}(\sqrt{2})$, $p(x) \mapsto p(\sqrt{2})$, then This is a isomorphism.

5. If we take $k[x, y]$, then $k[x, y]/(x) \cong k[y]$
6. Let $G = \langle g \rangle$ be a cyclic group and k be any field. Define:

$$\varphi : k[G] \to k[x] \qquad \varphi(ag^k) = ax^k \qquad \forall a \in k$$

Show that:

$$k[G] \cong \frac{k[x]}{x^n - 1}$$

Example 9.3.14
Let's say you want to figure out all integer solutions to

$$x^2 + y^2 = 3z^2$$

Suppose such integers exist. Observe first that we may assume x, y and z have no factors in common, since otherwise we could divide through this equation by the square of this common factor and obtain another set of integer solutions smaller than the initial ones. This equation simply states a relation between these elements in the ring $\mathbb{Z}$. As such, the same relation must also hold in any quotient ring as well. In particular, this relation must hold in $\mathbb{Z}/n\mathbb{Z}$ for any integer n. The choice $n = 4$ is particularly efficacious, for the following reason: the squares mod 4 are just 0^2, 1^2, 2^2, 3^2, i.e., 0, 1 (mod 4). Reading the above equation mod 4 (that is, considering this equation in the quotient ring $\mathbb{Z}/4\mathbb{Z}$), we must have

$$\begin{Bmatrix} 0 \\ 1 \end{Bmatrix} + \begin{Bmatrix} 0 \\ 1 \end{Bmatrix} = + \begin{Bmatrix} 0 \\ 3 \end{Bmatrix} \qquad (\text{mod} 4)$$

where the $\begin{Bmatrix} 0 \\ 1 \end{Bmatrix}$ indicate that either a 0 or a 1 may be taken. The only possibility is if they are all even, but we assumed they were not, thus there is no solution.

Note that this technique has limitation. The following equation has solution integers modulo any n, but no integer solutions (which is hard to prove).

$$3x^3 + 4x^3 + 5z^3 = 0$$

Eventually, we will be using a similar technique to factor polynomial.

There is also the 2nd, 3rd, and 4th isomorphisms theorems. Each of these are quite easy to show: We already know they are true because R is an *additive group* (or *correspondence of groups*, in the case of the Fourth Isomorphism theorem). It only needs to be show that the group isomorphism is also a multiplicative map, and thus a ring-isomorphism.

Beforehand, we had $H, K \leqslant G$ and defined HK. Since we are now writing the group operation additively, we will convert our notation to saying $H + K$. If I, J are subrings of R, then since they are both normal (even abelian) subgroups, then $I + J$ is well-defined.

Theorem 9.3.15: 2nd, 3rd, 4th Isomorphism Theorem for Rings

Let R be a ring.

(2) Let A be a subring and B be an ideal of R. Then $A+B$ is a subring of R, B is an ideal of $A+B$, $A \cap B$ is an ideal of A, and $\frac{A+B}{B} \cong \frac{A}{A \cap B}$

(3) Let I, J be ideals of R with $I \subseteq J$. Then J/I is an ideal of R/I and $(R/I)/(J/I) \cong R/J$

(4) Let I be an ideal of R. The correspondence $A \leftrightarrow A/I$ is an inclusion preserving bijection between the set of subrings A of R that contain I and the set of subrings of R/I, and it restricts to an inclusion preserving bijection $J \leftrightarrow J/I$ between the set of ideals J of R that contain I and the set of ideals of R/I. In both cases the inverse sends a subset of R/I to its preimage under the natural projection $R \to R/I$.

***Proof*:**
In (3) and (4), $\pi\colon R \to R/I$ is the natural projection $r \mapsto r+I$, a surjective ring homomorphism with kernel I (theorem 9.3.12(2)), and for a subset $X \subseteq R$ we write $X/I = \{x+I \mid x \in X\} = \pi(X)$.

(2) The set $A+B$ contains $0 = 0+0$ and $1 = 1+0$, and it is an additive subgroup of R, since $(a+b)-(a'+b') = (a-a')+(b-b')$ for $a, a' \in A$ and $b, b' \in B$. It is closed under multiplication, since

$$(a+b)(a'+b') = aa' + (ab' + ba' + bb')$$

where $aa' \in A$ and the bracket lies in B because B is an ideal of R. By corollary 9.1.7, $A+B$ is a subring. It contains B, and B absorbs multiplication by every element of R, so B is an ideal of $A+B$, and $(A+B)/B$ is a ring by proposition 9.3.11. Define

$$\varphi\colon A \to \frac{A+B}{B} \qquad \varphi(a) = a+B$$

For $a, a' \in A$, $\varphi(a+a') = (a+B)+(a'+B)$ and $\varphi(aa') = (a+B)(a'+B)$ by the definition of the operations on the quotient ring, and $\varphi(1) = 1+B$, so φ is a ring homomorphism. It is surjective, since $(a+b)+B = a+B = \varphi(a)$ for $a \in A$ and $b \in B$. Its kernel is $\{a \in A \mid a+B = B\} = A \cap B$. By theorem 9.3.12(1) applied to φ, $A \cap B$ is an ideal of A and $A/(A \cap B) \cong \varphi(A) = (A+B)/B$.

(3) The set $J/I = \pi(J)$ is an additive subgroup of R/I, being the image of the additive group J under the restriction of π to J (proposition 1.4.4(5)). For $r \in R$ and $j \in J$, $(r+I)(j+I) = rj+I$ and $(j+I)(r+I) = jr+I$ lie in J/I because $rj, jr \in J$, so J/I is an ideal of R/I. Define

$$\psi\colon R/I \to R/J \qquad r+I \mapsto r+J$$

If $r+I = r'+I$, then $r-r' \in I \subseteq J$, so $r+J = r'+J$, and ψ is well defined. It sends sums to sums, products to products and $1+I$ to $1+J$, by the definition of the operations on the two quotient rings, and it is surjective. Its kernel is $\{r+I \mid r \in J\} = J/I$, because $r+J = J$ exactly when $r \in J$. By theorem 9.3.12(1), $(R/I)/(J/I) \cong R/J$.

(4) The additive group of R is abelian, so I is a normal subgroup of it. By theorem 3.5.7 applied to the additive groups, $X \mapsto X/I$ is a bijection from the additive subgroups X of R containing I onto the additive subgroups of R/I, and it preserves inclusions in both directions by corollary 3.5.8(1). Its inverse is $Y \mapsto \pi^{-1}(Y)$, because $\pi^{-1}(Y)$ is an additive subgroup containing I with $\pi(\pi^{-1}(Y)) = Y$, as π is surjective. It therefore suffices to show that this bijection matches subrings with subrings and ideals with ideals. Let A be an additive subgroup of R containing I. Then $\pi^{-1}(A/I) = A$: every $a \in A$ has $\pi(a) = a + I \in A/I$, and conversely, if $\pi(r) = a + I$ with $a \in A$, then $r - a \in I \subseteq A$, so $r \in A$.

If A is a subring, then A/I contains $1 + I$ and is closed under multiplication, since $(a + I)(a' + I) = aa' + I$, so A/I is a subring of R/I by corollary 9.1.7. Conversely, if A/I is a subring, then $\pi(1) = 1 + I \in A/I$ gives $1 \in \pi^{-1}(A/I) = A$, and for $a, a' \in A$, $\pi(aa') = \pi(a)\pi(a') \in A/I$ gives $aa' \in A$. So A is a subring.

If A is an ideal of R, then A/I is an ideal of R/I by the argument of (3). Conversely, if A/I is an ideal of R/I, then for $r \in R$ and $a \in A$ the elements $\pi(ra) = \pi(r)\pi(a)$ and $\pi(ar) = \pi(a)\pi(r)$ lie in A/I, so $ra, ar \in \pi^{-1}(A/I) = A$, and A is an ideal of R. This completes the proof.

9.3.16 Remark We can define a *simple ring* to be a nonzero ring whose only two-sided ideals are 0 and R[a], which mirrors the definition of simple groups. The simple commutative rings are exactly the fields[b]. For a field k, the matrix ring $M_n(k)$ is a simple ring that is not commutative for $n \geq 2$: if a two-sided ideal I contains $A \neq 0$ with $a_{ij} \neq 0$, then I contains $a_{ij}^{-1} E_{pi} A E_{jq} = E_{pq}$ for all p, q, where E_{pq} is the matrix with 1 in position (p, q) and 0 elsewhere, so I contains $\sum_p E_{pp} = 1$ and $I = M_n(k)$.

[a] the reason for sticking to two-sided ideals is because we can quotient R by two-sided ideals to get a quotient ring, but we cannot quotient by 1-sided ideals and get a quotient ring

[b] This fails for *rngs*: $\mathbb{Z}/p\mathbb{Z}$ with the zero multiplication has no ideals other than 0 and itself, and it is not a field.

Exercise 9.3.1

1. Let R, S be rings and consider the product ring $R \times S$. What are the ideals of $R \times S$?
2. Let I be an ideal of the ring R and let $(I) = I[x]$ denote the ideal of $R[x]$ generated by I (the set of polynomials with coefficient I). Then

   $$R[x]/(I) \cong (R/I)[x]$$

   In particular, if I is a prime ideal of R then (I) is a prime ideal of $R[x]$. Conclude that $\mathbb{Z}[x]/(n\mathbb{Z}[x]) \cong (\mathbb{Z}/n\mathbb{Z})[x]$
3. Show that the set of upper-triangular matrices is a subring.

9.4 Properties and Further Structures of Ideals

Since the kernel of ring homomorphisms are ideals, they form the core of much of ring theory. Recall how when working with groups, they can be thought of as $F(A)/N$ for an appropriate normal set N which represents the *relations* on the generators A. Ideals will play a role of similar importance: Ideals will represent the relations. There will be an analogous notion of free ring, however the study of generation of free rings shall be delayed to Part III since the two constructions are identical.

This is not to say that thinking of ideals as adding relations is un-useful! For example, since $\mathbb{Z}$ is initial in rings, it is of great interest to see how what relations ideals of $\mathbb{Z}$ can give us, what properties do they have, and how they translate with ring homomorphisms. (We will focus a lot on this in number theory).

The set generated by a subset $A \subseteq R$ under addition, negation and multiplication need not be a subring, since it need not contain 1. In $\mathbb{Z}$, the set generated by $\{2\}$ is $(2) = 2\mathbb{Z}$. In $\mathbb{Z}[x]$, the set generated by $\{1 + x\}$ is $\{p(1 + x) \mid p \in \mathbb{Z}[y],\ p(0) = 0\} = (1 + x)\,\mathbb{Z}[1 + x] = (1 + x)$, since $\mathbb{Z}[1 + x] = \mathbb{Z}[x]$, and it does not contain 1, because $(1 + x)g(x)$ has degree at least 1 for every nonzero g. Both sets are rngs and not rings. Since every rng embeds in a ring (box 9.1.2), this book works with rings.

9.4.1 Operations and Generation on Ideals

The first thing we might think of is way of extending how to manipulates ideals through binary operations (in essense, we find different ways of combining what relations we impose on a ring). We give a list of common operations of ideals

Definition 9.4.1: Operations on Ideals

Let R be a ring and $I, J \subseteq R$ be (left, right, or two-sided) ideals. Then:

1. $I \cap J$ is the intersection of two ideals. More generally,

$$\bigcap_{i \in \mathcal{I}} I_i$$

is the intersection of ideals I_i indexed by $\mathcal{I}$.

2. Define the *sum* of I and J by $I + J = \{a + b | a \in I, b \in J\}$. More generally,

$$\sum_{i \in \mathcal{I}} I_i = \left\{ \sum_{i \in N} a_i \;\middle|\; N \subseteq \mathcal{I},\ |N| \in \mathbb{N},\ a_i \in I_i \right\}$$

is the sums of ideals I_i indexed by $\mathcal{I}$.

3. Define the *product* of I and J, denoted IJ, to be the set of all finite sums of elements of he form ab wit $a \in I$ and $b \in J$.

$$\left\{ \sum_{i=1}^{n} a_i b_i \middle| a_i \in I,\ b_i \in J \right\}$$

Note that we must take finite sums since the ideal must be closed under addition. More generally:

$$\prod_{i \in \mathcal{I}} I_i = \left\{ \sum_{i=1}^{m} a_{i_1} \cdots a_{i_n} \;\middle|\; a_{i_1}, ..., a_{i_n} \in \mathcal{I}, n, m \in \mathbb{N} \right\}$$

4. For any $n \geq 1$, define the nth power of I, denoted I^n to be the set consisting of all finite sums of elements of the form $a_1 a_2 \cdots a_n$ with $a_i \in I$ for all $i>$ Equivalently, I^n is defined inductively by defining $I^1 = I$, and $I^n = II^{n-1}$ for $n = 2, 3, ...$ By convention $I^0 = R$.

5. Let R be commutative. Define the *ideal quotient* $(I : J)$ to be:

$$(I : J) := \{x \in R \mid xI \subseteq J\}$$

6. Let R be commutative. Define the *radical* of I to be

$$\sqrt{I} := \{r \in R \mid r^n \in I \text{ for some } n \geq 1\}$$

An ideal satisfying $I = \sqrt{I}$ is called a *radical ideal*.

Note that if I, J are *both* ideals, then $I + J$ is an ideal (this set is closed under sums and $r(a + b) = ra + rb \in I + J$). It should be emphasized that the set IJ is the *finite sum* of elements of the form ab. This condition is required so that IJ is closed under addition, which is necessary for IJ to be an ideal. Just like groups and rings, the union of two ideals is not an ideal (for example, $2\mathbb{Z} \cup 3\mathbb{Z}$ does not contain 5, however $2 \in 2\mathbb{Z}$, $3 \in 3\mathbb{Z}$). Can you show that the quotient ideal $(I : J)$ is indeed an

ideal? In some very lose way, if $a \in (x)$, then that means that x is a "factor" of a in the sense that $a = rx$ for some $r \in R$)

Example 9.4.2

1. Let $I = 6\mathbb{Z}$ and $J = 10\mathbb{Z}$. Then $I + J$ is the ideal of all integers of the form $6x + 10y$ for $x, y \in \mathbb{Z}$. Since 6 an 10 are divisible by 2, meaning that $6x + 10y = 2(3x + 5y)$, so that $2\mathbb{Z} \supseteq 6\mathbb{Z} + 10\mathbb{Z}$. Conversely, we see that $2 = 6(2) + 10(-1)$, so $2x = 6(2x) + 10(-x)$, meaning that every element of $2\mathbb{Z}$ can be written of the form $6\mathbb{Z} + 10\mathbb{Z}$, so $2\mathbb{Z} \subseteq 6\mathbb{Z} + 10\mathbb{Z}$. Thus $2\mathbb{Z} = 6\mathbb{Z} = 10\mathbb{Z}$. We can generalize this to $n\mathbb{Z} + m\mathbb{Z} = d\mathbb{Z}$ where $d = (m, n)$, in the next section. Similarly, it is worth checking that $6\mathbb{Z}10\mathbb{Z} = 60\mathbb{Z}$.
2. Take $I \subseteq \mathbb{Z}[x]$ to be the ideal whose constant term is even. Then (2) and (x) are in subsets of I. Thus, $4 = 2 \cdot 2$ and $x^2 = x \cdot x$ are in I^2, as is their sum $x^2 + 4$. However, $x^2 + 4$ cannot be written as a product $p(x)q(x)$ of two elements of I.
3. the ideal $((0) : I)$ known as
4. Let $(2), (6) \subseteq \mathbb{Z}$. Then $((2) : (6)) = (3)$, showing how the quotient ideal generlizes some notion of division. *annihilator of* I and consists of all elements of R such that $rx = 0$ for all $x \in I$. Usually, for any ideal $I = (r)$ generated by a single element, we write $(r : I)$ instead of $((r) : I)$. Using this, can you deduce what $(0 : R)$ is equal to?

The radical needs a word of its own, since unlike the other five operations it is not built from sums and products of elements already in the ideals, and that it is an ideal at all is a small argument rather than an observation.

Proposition 9.4.3: The Radical Is an Ideal

Let R be a commutative ring and $I \subseteq R$ an ideal. Then $\sqrt{I}$ is an ideal of R containing I, and $\sqrt{\sqrt{I}} = \sqrt{I}$.

***Proof*:**

Every $r \in I$ satisfies $r^1 \in I$, so $I \subseteq \sqrt{I}$. In particular $0 \in \sqrt{I}$.

For closure under the ring action, let $r \in \sqrt{I}$ with $r^n \in I$ and let $a \in R$. Since R is commutative, $(ar)^n = a^n r^n$, which lies in I because I absorbs multiplication, so $ar \in \sqrt{I}$.

For closure under addition, let $r^n \in I$ and $s^m \in I$. By the binomial theorem, valid in any commutative ring,

$$(r+s)^{n+m-1} = \sum_{i=0}^{n+m-1} \binom{n+m-1}{i} r^i s^{n+m-1-i}$$

Consider a single term. If $i \geq n$ then $r^i = r^n r^{i-n} \in I$. And if $i \leq n-1$ then $n+m-1-i \geq m$, so $s^{n+m-1-i} \in I$ by the same argument. Either way the term lies in I, and I is closed under addition, so $(r+s)^{n+m-1} \in I$ and $r + s \in \sqrt{I}$. Together with the previous paragraph this makes $\sqrt{I}$ an ideal.

Finally, if $r \in \sqrt{\sqrt{I}}$ then $r^n \in \sqrt{I}$ for some n, so $(r^n)^m = r^{nm} \in I$ for some m, giving $r \in \sqrt{I}$. The reverse inclusion is the first paragraph applied to $\sqrt{I}$, completing the proof.

We can also ask oursevles about the generation of ideals:

Definition 9.4.4: Generating Ideals

1. (A) is the smallest ideal containing A. Then
$$(A) = \cap_{A \subseteq I} I$$
where I is an ideal containing A.
2. RA is he set of all finite sums of elements of the form ra between R and A.
$$\{r_1 a_1 + \cdots + r_k a_k | r_i \in R\,, a_i \in A\}$$
Similarly for AR and RAR.
3. An ideal generated by a single element is a *principal ideal*.
4. an ideal generated by a finite set is called a *finitely generated ideal*.

If R is commutative, then $RA = AR = RAR = (A)$.

9.4.5 Moment To Ponder You might ask yourself, with addition and multiplication defined on the ideals, does the set of ideals form a ring? Try it out! Is this a way to study ideals? More on studying the ideal structure of a ring when studying Modules in Part III

In the following chapter's, ideals generate by a singleton (a) will be of keen interest. As a prelude, notice that if $b \in (a)$, then $b = ra$ for some $r \in R$. Phrased differently, a divides b. Also, $b \in (a)$ if and only if $(b) \subseteq (a)$. In other words, being an element inside the ideal gives some idea of "divisibility" or "factorization".

Example 9.4.6

1. (0) and R are both principal ideals, since (0) is generated by 0, and $R = (1)$ (since we assumed R has an identity).
2. Every ideal of $\mathbb{Z}$ is a principal ideal

Proof. We will show that $(m) + (n) = gcd(m, n)$

($\Rightarrow$) Let $x \in (m) + (n)$. Let's show that $x \in (m, n)$. Since $x \in (m) + (n)$, then:
$$x = ma + nb \qquad a, b \in \mathbb{Z}$$
letting $p = (m, n)$, we see that
$$ma + nb = (m'a + n'b)p = cp \in (m, n)$$
where m', n' are the remainders, and $c = m'a + n'b$, showing that $cp \in (m, n)$

($\Leftarrow$) Let $p = gcd(m, n)$. Without loss of generality, let's say $m > n$. We will take advantage of the euclidean algorithm. First, start with:

$$m = a_1 n - r_1$$

where r_1 is the remaidner by dividng m by n and a is the number of times n can be divided into m. This is the extended euclidean algorithm. We can work our way down till we get

$$r_k = a_{k+1} p - 0$$

where $p = gcd(m, n)$. Then, we can use Bézout's identity from the results to get that:

$$p = k_1 m - k_2 n$$

with appropriate k_1, k_2. This then implies that

$$cp = ck_1 m - ck_2 n$$

for any $c \in \mathbb{Z}$, and so cp can be written of the form $am + bn$, meaning $cp \in (m) + (n)$, as we sought to show.

We can use this to show that *any* (A) is a principle ideal in $\mathbb{Z}$. Since $\mathbb{Z}$ is commutative, we can think of $(A) = RA$. Then we can do the same greatest common denominator trick to get down to our desied number such that $d\mathbb{Z}$ is equal to $\mathbb{Z}A$. We can use this to show that every ideal of $\mathbb{Z}$ are principal ideals. Since $\mathbb{Z}$ is also an integral domain, meaning it's also a domain, we will end up calling $\mathbb{Z}$ a *principal ideal domain.*

However, it's not always true that if (a, b) can be reduced to (c) that a ring is a principal ideal domain. There is a sub-class of rings called *Bézout Domains* for which every pair of number can be reduced to a principal ideal, but the ring itself is *not* a principal ideal domain. More on that in chapter 10. □

3. Let $R = \mathbb{Z}[x]$. Then $(2, x)$ is *not* a principal ideal. Let

$$(2, x) = \{2p(x) + xq(x) | p(x), q(x) \in \mathbb{Z}[x]\}$$

that is, all polynomials with *even* constants (and hence, a proper ideals, since it does not containt 1). Let's say for some $a(x) \in \mathbb{Z}[x].(2, x) = (a(x))$. Since $2 \in (a(x))$, there must be some $p(x)$ such that $2 = p(x)a(x)$. By proposition 9.2.3, we know that degree $p(x)a(x) =$ degree $p(x)+$ degree$a(x)$, and since degree(2) is 0, then $a(x)$ and $p(x)$ must *both* be constant polynomials. Since 2 is a prime number, the only possibilities are $p(x), a(x) \in \{\pm 1, \pm 2\}$. If $a(x) = \pm 1$, then $1 \in (a(x))$, and so by proposition 9.4.10, $(a(x)) = \mathbb{Z}[x]$. But that's a contradiction to $(2, x)$ being proper. Thus, it must be that $a(x) = \pm 2$. But then, $x \in (\pm 2)$, and so $x = p(x)(\pm 2)$. But this is impossible. We'd need $\frac{1}{2}x$ to be $p(x)$, but $\frac{1}{2} \notin \mathbb{Z}$. Thus, $(2, x)$ is *not* a principal ideal. Thus $\mathbb{Z}[x]$ is *not* a principal ideal domain.

I like to thinke of elements of (p, q) for some $R[x]$ as elements in which every term has p or q as a factor. This is because when taking $\alpha p + \beta q$ for $\alpha, \beta \in R[x]$, you can have α or β be 0, meaning you just need elements which have p or q as a factor.

4. If we take $\mathbb{Q}[x]$ instead of $\mathbb{Z}[x]$, then this would be solved since $(2) = \mathbb{Q}[x]$ since $\frac{1}{2}2 = 1$, maning (2) has a unit, meaning $(2, x)$ has a unit. In fact, $\mathbb{Q}[x]$ will become a principal ideal domain because of this! Try playing around with the following ideals to see they are indeed principal

- $(2, x)$
- $(x, x^2 + 1)$
- $(p(x), q(x))$ for $p, q \in \mathbb{Q}[x]$.

5. Take $R = k[x]$ and take $I = \{p(x) \in k[x] | p(q) = 0\}$ for some arbitrarry $q \in \mathbb{F}$. One can check that it is an ideal. More interestingly though, we know from high-school math that if $p(x) = 0$, then we can factor out $p(x) = (x - q)r(x)$ for any polynomial $r(x) \in \mathbb{F}[x]$. This will thus make $I = ((x - q))$, making these ideals principal ideals! However, if $I = \{p(x) \in \mathbb{F}[x] | p(a) = p(b) = 0, a \neq b\}$ for two arbitrary $a, b \in F$, $a \neq b$, then I is *not* a principal ideal.

As the intuition for ideals suggest, they are a sort of generalization of properties of $\mathbb{Z}$. For example, if $a \in R$ and (a) is an ideal, then $R/(a)$ is well-defined. Now, take $\overline{b} \in R/(a)$ and generate $(\overline{b})$. Then it should be verified that

$$(R/(a))/(\overline{b}) = R/(a, b)$$

Furthermore, notice that in $\mathbb{Z}$, the ideal (a, b) is equal to the ideal $(\gcd(a, b))$. For this reason, ideals capture some more general sense of divisibility for rings! Exploring the notion of divisibility further will be the goal of chapter 10.

Since ideals are so important, it is often the case that studying rings with simpler ideal structure are of much interest. Two particular cases are especially important due to the abundance of finiteness:

Definition 9.4.7: Noetherian rings

Let R be a commutative ring. Then if every ideal[a] $I \subseteq R$ is finitely generated, then R is called a *Noetherian ring.*

[a] Be sure not to mistake ideal for ring, for finitely generated rings have not been defined

Noetherian rings are particularly nice in that they are finite locally and globally. If we just require that the ring is finitely generated, it may be that a subring is non-finitely generated. Conversely. We shall have lots of experience with Noetherian rings later, for now it is useful to know these rings since since being Noetherian is stable under taking the quotient, and is also stable under the "converse" of taking the quotient (if you have covered section 3.6, the Noetherian condition is stable under extensions). For the next proposition, note that a finitely generated ideal is defined identically to a finitely generated ring (that it is generated by finitely many elements):

Proposition 9.4.8: Noetherian Rings Under Quotients And Extensions

Let R be a commutative ring and I an ideal of R.

1. If R is Noetherian, then R/I is Noetherian.
2. Conversely, if R/I is Noetherian and every ideal of R contained in I is finitely generated, then R is Noetherian.

***Proof*:**
Let $\pi\colon R \to R/I$ be the natural projection, a surjective ring homomorphism with kernel I (theorem 9.3.12(2)). The ring R/I is commutative because R is, and in a commutative ring the ideal generated by $a_1, \dots, a_k$ is $\{r_1a_1 + \cdots + r_ka_k \mid r_1, \dots, r_k \in R\}$ (definition 9.4.4).

1. Let $\overline{J}$ be an ideal of R/I and $J = \pi^{-1}(\overline{J})$. Then J is an ideal of R: it is an additive subgroup, being the preimage of one under the additive homomorphism π, and $\pi(rx) = \pi(r)\pi(x) \in \overline{J}$ for $r \in R$ and $x \in J$. As R is Noetherian, J is generated by some $a_1, \dots, a_k \in J$. Every element of $\overline{J}$ is $\pi(x)$ for some $x \in J$, since π is surjective, and writing $x = r_1a_1 + \cdots + r_ka_k$ gives $\pi(x) = \pi(r_1)\pi(a_1) + \cdots + \pi(r_k)\pi(a_k)$. So $\overline{J}$ is contained in the ideal generated by $\pi(a_1), \dots, \pi(a_k)$, and it contains that ideal because it is an ideal containing each $\pi(a_i)$. Hence every ideal of R/I is finitely generated, and R/I is Noetherian.

2. Let J be an ideal of R. Its image $\pi(J)$ is an ideal of R/I: it is an additive subgroup, and $\pi(r)\pi(x) = \pi(rx) \in \pi(J)$ for $r \in R$ and $x \in J$, where every element of R/I has the form $\pi(r)$. As R/I is Noetherian, $\pi(J)$ is generated by finitely many of its elements, which we write as $\pi(a_1), \dots, \pi(a_k)$ with $a_1, \dots, a_k \in J$. The intersection $J \cap I$ is an ideal of R, being an additive subgroup that absorbs multiplication by R, and it is contained in I, so by hypothesis it is generated by some $b_1, \dots, b_l$. We claim that J is generated by $a_1, \dots, a_k, b_1, \dots, b_l$. These elements lie in J, so the ideal they generate is contained in J. Conversely, let $x \in J$. Then $\pi(x) \in \pi(J)$, so $\pi(x) = \pi(r_1)\pi(a_1) + \cdots + \pi(r_k)\pi(a_k)$ for some $r_i \in R$, and $y = x - (r_1a_1 + \cdots + r_ka_k)$ lies in $\ker \pi = I$. It also lies in J, since x and the a_i do. So $y \in J \cap I$, that is, $y = s_1b_1 + \cdots + s_lb_l$ for some $s_j \in R$, and

$$x = r_1a_1 + \cdots + r_ka_k + s_1b_1 + \cdots + s_lb_l$$

Hence every ideal of R is finitely generated, and R is Noetherian, as we sought to show.

One of the simplest Noetherian rings are those for which every ideal is not only finitely generated, but also principal. If the ideal is furthermore an integral domain (to avoid pathologies from zero-divisors) then it is a principal ideal domain:

Definition 9.4.9: Principal Ideal Domain

Let R be an integral domain. Then if every ideal $I \subseteq R$ is a principal ideal, then R is called a *Principal Ideal Domain*.

As a simple example, $\mathbb{Z}$ is a PID (can you see why?). It should be noted that *every* ideal is principal, even those that may seem infinitely-generated. In fact, there are domain in which only finitely-generated ideals are principle (such rings are called *Bézout* rings).

Another way of looking at principal ideal domains is that they are "locally free". Here, locality is taken to mean that any proper submodule is free. Since R is an integral domain, it is torsion free, and since R is principle, if $I \subseteq R$ is a submodule (i.e. ideal), then

$$I = (a) \qquad a \in R$$

This fact shall make PID's very powerful, and in fact the idea that PID's have "local freeness" shall be further explore in chapter 17⋆[17].

[17] A projective module over a PID is in fact a free module

Principal Ideal Domains (PIDs) will have the property of having "prime elements" well-defined and lend itself useful in many classification theorems (for example, study of Jordan Canonical Forms in chapter 14 will heavily rely on this type of ring).

Exercise 9.4.1

1. $(I \cap J)(I + J) = IJ$

9.4.2 Ideals and Units

As briefly mentioned, some ideals seem to be related to a notion of divisibility. This intuition more clear with the following theorem:

Proposition 9.4.10: units in Ideals and ideals in fields

Let R be a ring.

1. A left or right ideal I of R satisfies $I = R$ if and only if I contains a unit
2. Let R be nonzero. Then R is a division ring if and only if its only left- and right-sided ideals are 0 and itself. If R is commutative, then the only ideals of R are 0 and itself if and only if R is a field.

***Proof*:**

1. one inclusion is easy. The key part to the other inclusion is that I must be closed under multiplication from all elements of R. If $I = R$ then I contains the unit 1. If I contains a unit u, then given its inverse $v \in R$ and any element $r \in R$

$$r = r \cdot 1 = r(vu) = (rv)u \qquad r = 1 \cdot r = (uv)r = u(vr)$$

and the first expression lies in I when I is a left ideal, the second when I is a right ideal. So $I = R$.

2. If R is a division ring, then every nonzero element is a unit, so every nonzero left or right ideal contains a unit and equals R by (1). Conversely, suppose the only left and right-sided ideals are R and 0, and let $u \in R$ be nonzero. The set $Ru = \{ru \mid r \in R\}$ is a left ideal containing $u = 1u$, so $Ru = R$ and $1 = vu$ for some $v \in R$. Likewise uR is a right ideal containing u, so $1 = uw$ for some $w \in R$. Then $v = v(uw) = (vu)w = w$, so u is a unit. Since this holds for every nonzero element and $1 \neq 0$, R is a division ring. If R is also commutative, then Ru is an ideal, and the same argument shows that a nonzero commutative ring whose only ideals are 0 and itself is a commutative division ring, and thus a field. Conversely, a field is a division ring, so by the first part its only left ideals, and in particular its only ideals, are 0 and itself, completing the proof.

9.4.11 Note As a consequence, if $1 \in I$, then $I = R$, since 1 is a unit!! This means that in a proper ideal, $1 \notin I$, and so, they cannot be sub-rings with identity $1 \neq 0$.

It should be pointed out it was necessary to have the condition that both let and right-sided ieals are trivial, instead of just picking one or saying that only two-sided ideals are trivial.

Example 9.4.12: Two-Sided Ideals Trivial, but non-trivial left/right-sided ideals

Let k be a field and $M_2(k)$ the matrix ring over k. Then $M_2(k)$ has non-trivial left and right-sided ideals (think of columns and rows of $M_2(k)$): the matrices whose second column is zero form a left ideal, since multiplying on the left keeps a zero column zero, and the matrices whose second row is zero form a right ideal, and both are nonzero and proper. However, $M_2(k)$ has *no* two-sided non-trivial ideals (box 9.3.16).

Now, $M_2(k)$ is not a division ring: the element

$$\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$$

does not have an inverse, since

$$\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix} = 0$$

and an invertible A with $AB = 0$ forces $B = A^{-1}AB = 0$.

Since any two-sided ideal is a left and right ideal, then division ring's naturally have no non-trivial two-sided ideals.

Corollary 9.4.13: Division Ring and Ring-Homomorphism

If R is a division ring, then any non-zero ring-homomorphism $\varphi : R \to S$ to any other ring must be injective.

***Proof*:**

Let φ be a ring-homomrophism. The kernel of φ is a two-sided ideal of R, but R is a divison ring, meaning the only ideals of R are 0 and R. Since φ is non-trivial, $\ker \varphi \neq R$, and so $\ker \varphi = 0$, making φ injective.

This might make one think that studying division rings, or more particularly fields, in terms of homomorphisms is less useful. However, we will *still* use ring-homomorphisms on fields, since mapping a field injectively into another field will turn out to be quite useful: we can study how fields *embedd* into one another (this will be called a *field extensions*, which will be defined in Part IV, and an *integral extension* for more general commutative rings, which will be defined in [16, Integral Elements]).

If a ring does only have 0 and R as two-sided ideals, we have the analogous concept to simple groups

Definition 9.4.14: Simple Rings

Let R be a nonzero ring. Then R is said to be *simple* if its only two-sided ideals are the 0 ideal and itself.

One might think that from here, there will be a similar story as in group theory: A lot of time and effort spent on classifying all rings (and finite rings) given this idea of a simple ring. It turns out though that the classification of finite rings are in fact much much easier: Every finite simple ring is

isomorphic to the ring $M_n(\mathbb{F}_q)$ over a finite field of order q (i.e., intimately related to matrix rings)! In fact, the study of simple rings is the setting of representation theory! Simple rings are covered in [17, Semisimple Rings and Modules].

9.4.3 Maximal Ideals

Many times, we might want if an ideals is *as large as possible* in a ring. That is, if I is an ideal, then there does not exist another J such that $I \subseteq J$ unless $J = R$

Definition 9.4.15: Maximal ideal

An ideal M in an arbitrary ring S is called a *maximal ideal* if $M \neq S$ and the only ideals containing M are M and S.

Example 9.4.16: Ring with maximal ideal

Take $\mathbb{Z}$ and $6\mathbb{Z}$. Since $6 = 2(x)$, $6\mathbb{Z} \subseteq 2\mathbb{Z}$. However, since 2 is a prime, we cannot repeat this same trick twice. Thus, $2\mathbb{Z}$ is a maximal ideal. In general, $p\mathbb{Z}$ for prime p is a maximal ideal

All rings have a maximal ring with the exception of a ring with the trivial ring structure:

Example 9.4.17: Ring without maximal ideal

Take any group with a non-maximal group (ex $\mathbb{Q}$) and take the trivial ring structure ($ab = 0$).

The only restriction on a ring that requires it to have a maximal ideal is that $1 \neq 0$. The way we will prove this will require a form of "super-induction". The use of this proof technique requires the axiom of choice. For those who want to have an idea of Zorn's lemma, see appendix A.3, and for those who want to see how it links to the axiom of choice, see appendix A.

Proposition 9.4.18: Ring with $1 \neq 0$ have Maximal ideals

Let R be a non-trivial ring and $I \subsetneq R$ be a proper ideal. Then I is contained in a maximal ideal

***Proof*:**

Since $I \subsetneq R$, $1 \neq 0$), and $1 \notin I$ (or else $I = R$), a maximal ideal can exist. To show that it is the case, we will us Zorn's Lemma (theorem A.2.3). Let $\mathcal{I}$ be the collection of all proper ideals of R containing I, and let it be ordered by inclusion. Then $(\mathcal{I}, \subseteq)$ is a partial order. If e take any chain $\mathcal{C} \subseteq \mathcal{S}$, then

$$J = \bigcup_{A \in \mathcal{C}} A$$

is the ideal which is the upper-bound for $\mathcal{C}$. For this to be the case, J must be an ideal.

1. J is non-empty, since any ideal must contain 0, thus $0 \in J$
2. Take $a, b \in J$ and consider $a - b$. Then for some $A, B \in \mathcal{C}$, $a \in A$ and $b \in B$. Since $\mathcal{C}$ is a chain, $A \subseteq B$ or $B \subseteq A$. Witout loss of generality, let's assume $A \subseteq B$ and say $a, b \in B$ (if

e pick $B \subseteq A$ do $a, b \in A$). Then since B is an ideal, $a - b \in B$, so $a - b \in J$, showing that J is closed under subtraction

3. Take $a \in J$ and consider $r \cdot a$ and $a \cdot r$. Then since $a \in J$, a is in some ideal, let's say $a \in A$. Then $r \cdot a \in A$ and $a \cdot r \in A$, so $r \cdot a, a \cdot r \in J$, showing it's closed uner left and right multiplication

Thus, J is an ideal. It is further a proper ideal. If it ere not, then $1 \in J$, meaning for some ideal A, $1 \in A$. But then $A = R$, showing it is *not* a proper ideal, a contradiction to the definition of J. Thus, J must also be a proper ideal.

Therefore, by Zorn's Lemma, $\mathcal{I}$ must have a maximal element M, that is, if there is an element M' such that $M \subseteq M'$, then it must be that $M' \subseteq M$, so $M = M'$, which fulfills the definition of a maximal ideal, as e sought to show.

Note that we took the union of ideals within the proof. In general, if $I, J \subseteq R$, then $I \cup J$ is not an ideal (consider $2\mathbb{Z} \cup 3\mathbb{Z}$). The reason it worked for this proof is because the ideals are in a chain:

$$I_1 \subseteq I_2 \subseteq \cdots \subseteq I_n \subseteq \cdots$$

It should be pointed out that it is possible to have an infinitely long chain of the form $(x_1) \subsetneq (x_1, x_2) \subsetneq \cdots$ of which the union is not a maximal ideal, hence being maximal is a different condition that such a chain terminating in finite itterations, or even countably-infinite many iterations. This is part of the importance of using Zorn's lemma in the proof.

Since Zorn's Lemma was used, there is no way to computationally figure out what the maximal ideal is by just using this proposition. If a ring is also commutative, we can more easily characterize it with the structure of their quotient ring

Proposition 9.4.19: Maximal Ideals in Commutative Rings

Let R be a commutative ring with $1 \neq 0$. The ideal M is a maximal ideal if and only if the quotient R/M is a field

***Proof*:**
This follow's from the Correspondence Theorem (theorem 9.3.15(4)) and proposition 9.4.10. Both conditions force $M \neq R$: a maximal ideal is proper (definition 9.4.15), and a field is nonzero, while R/M is the zero ring exactly when $1 \in M$, that is, exactly when $M = R$ (proposition 9.4.10(1)). So let $M \neq R$. Then R/M is a nonzero ring, and it is commutative because R is. By theorem 9.3.15(4), applied with $I = M$, the map $J \mapsto J/M$ is an inclusion preserving bijection from the ideals J of R that contain M onto the ideals of R/M, and it sends M to the zero ideal and R to R/M. Hence the only ideals of R containing M are M and R if and only if the only ideals of R/M are (0) and R/M. The first condition says that M is maximal, and by proposition 9.4.10(2), applied to the nonzero commutative ring R/M, the second says that R/M is a field. Thus, M is a maximal ideal if and only if R/M is a field, as we sought to show.

The use of the Correspondence theorem in proposition 9.4.19 can be visualized with such a diagram:

$$\begin{array}{cc} R & R/M \\ | & | \\ M & 0 \end{array}$$

Notice that if R was not commutative, then left ideals are not necessarily right ideals, and so the theorem does not apply: if we pick any one-sided ideal (say a left-ideal), and perform proposition 9.4.18 to show the existence of a maximal left-ideal (and hence, we have a non-empty collection of maximal left-ideals), it is not guarnateed that the set of maximal left-ideals will contain any proper right-ideals[18]. A different approach must be taken to study R/I when R is not commutative and I is a maximal left/right ideal, which will lead to many fruitful results in *Core Representation Theory* [17]. This case will require some more module theory, and so will be left alone for now.

In the commutative case, this let's us study properties of R by studying the field R/M for all maximal ideal M. If R has a single maximal ideal (say $\mathfrak{m}$) then $R/\mathfrak{m}$ is called the *residue field*. An example of a local ring would be $\mathbb{Z}/4\mathbb{Z}$, with single maximal ideal $\{\overline{0}, \overline{2}\}$. Such are called local rings for a reason. They show up intensively in commutative algebra, and will be central in [23, Smooth Points and Regular Local Rings].

This last characterization of maximal ideals through quotienting for commutative rings will come to be very useful in the construction of some fields. For example this will be used to construct all finite fields by taking quotient of the ring $\mathbb{Z}[x]$ by maximal ideals. In a more specific case, we will be construction the "extension" of a field k by first defining a polynomial ring $k[x]$, and quotienting it by a maximal ideal $M \subseteq k[x]$. This will be the foundation of *field Theory*, and will be covered in Part IV

Example 9.4.20

1. We have hand-waved a bit before how $p\mathbb{Z}$ is a maximal field. Since $\mathbb{Z}$ is commutative, we can now rigorously state this by noticing that $\mathbb{Z}/p\mathbb{Z}$ is a field
2. The ideal $(2, x)$ is a *maximal* in $\mathbb{Z}[x]$, since $\mathbb{Z}[x]/(2, x) \cong \mathbb{Z}/2\mathbb{Z}$, and $\mathbb{Z}/2\mathbb{Z}$ is a field
3. Let $R = \mathcal{F}([0, 1], \mathbb{R})$ be the ring of all functions from $[0, 1]$ to $\mathbb{R}$, and let M_a be the kernel of evaluation at a ($M_a = \{f \in R | f(a) = 0\}$). Show that M_a is maximal.

Note that a ring can have multiple maximal ideals. For example, $\mathbb{Z} \times \mathbb{Z}$ is a ring, and $I = ((1, 0), (0, 2))$ is an ideal generated by $(1, 0)$ and $(0, 2)$. Then $(\mathbb{Z} \times \mathbb{Z})/I \cong \mathbb{Z}/2\mathbb{Z}$, meaning I is a maximal ideal. Similarly, $J = ((2, 0), (0, 1))$ is an ideal and $(\mathbb{Z} \times \mathbb{Z})/J \cong \mathbb{Z}/2\mathbb{Z}$. Thus, both I and J are maximal. The quotients by two maximal ideals need not be isomorphic: in $\mathbb{Z}$ the ideals (3) and (5) are maximal, and $\mathbb{Z}/3\mathbb{Z} \ncong \mathbb{Z}/5\mathbb{Z}$ since they have different orders.

9.4.4 Prime Ideals

The first definition given of prime numbers is that a number is prime if and only if it's only divisors are 1 and itself, that is the only numbers such that $a|p$ are 1 and p. This definition works well for integers, but does it generalize well when we try to define divisibility over more general rings. This is due to a sneaky "global" assumption when working with primes, namely that an element is prime if

[18] Indeed, we can modify example 9.4.12 to see that $M_2(k)/(0)$ is not a field, and hence a counter-example

any factorization that may include the prime element *has* to include the element as a factor. The following example should illustrate how this is not necessarily the case:

Example 9.4.21: The Need For Globalness

[The Need For Globalness] Take $R = \mathbb{Z}[2i]$. Then $4 = (-2i)(-2i)$, and $-2i$ can no longer be factor[a]. However, we also have $4 = 2 \cdot 2$, and 2 can no longer factor (which in this case can be verified by taking a few cases). Thus, even though 2 was a factor of 4, we may split 4 differently, showing that there are "local" factorizations.

[a] up to multiplying by a unit, but factorizaiton is only defined up to unt, think $2 \cdot 3 = (-2) \cdot (-3)$

This subtly becomes very important, and shall be explored in depth in chapter 10.

To capture this idea, consider a prime number p. Then p is a prime number and if p divides ab, $a, b \in \mathbb{Z}$, then p divides a or p divides b. For example, if $3|15$, then either $3|3$ or $3|15$. If p was not a prime number, this is not the case: $8|16[= 4 \cdot 4]$, but $8 \nmid 4$ and $8 \nmid 4$. The prime ideals for the integers are the sets that contain all the multiples of a given prime number

Definition 9.4.22: Prime Ideal

Assume R is commutative. An ideal P is called a ***prime ideal*** if $P \neq R$ and whenver the product ab of two element $a, b \in R$ is an element of P, then at least one of the a an b is an element of P.

Since R is commutative, another way of saying the condition is that if A, B where ideals and $AB \subseteq P$, then $A \subseteq P$ or $B \subseteq P$. The non-commutative case does not generalize well, however we discuss a remedy for this case in [11, Relating Algebra to Number Theory][19].

Example 9.4.23: Prime Ideals

The origins of the idea come from generalizing the notion of primes in $\mathbb{Z}$, so we will do the same. The prime numbers p generate the prime-ideals (p).

Let's say $ab \in (p)$, meaning $ab = rp$ for some $r \in \mathbb{Z}$ (note that $\mathbb{R}$ is commutative so $(p) = Rp$). Since $ab = rp$ that impleis that $p|ab$ (p divides ab): If you think about what it means for a number x to divide y ($x|y$), then that means there exists another number a such that $ax = y$. For example $3|6$, so $2\dot{3} = 6$.

Now, since $p|ab$, then $p|a$ or $p|b$. If $p \nmid a$ and $p \nmid b$, then ab would not have it as a prime number under their factorization, but then $p \nmid ab$, a contradiction. So, $p|a$ or $p|b$. But that would mean $r_a p = a$ or $r_b p = b$ for appropriate r_a or r_b. But then, $a \in (p)$ or $b \in (p)$.

Here's a concrete exmpale. Take $3\mathbb{Z}$, an ideal of $\mathbb{Z}$. (can think of $\varphi : \mathbb{Z} \to \mathbb{Z}/3\mathbb{Z}$, $\varphi(x) = [x]$ to convince yourself it's an ideal). Now let's say $ab \in 3\mathbb{Z}$. That means that $3|ab$, meaning either $3|a$ or $3|b$. In either case, that would imply $a \in 3\mathbb{Z}$ or $b \in 3\mathbb{Z}$, showing that $3\mathbb{Z}$ is a prime ideal. On the other hand, if we have $6\mathbb{Z}$, Then $2 \cdot 3 \in 6\mathbb{Z}$, but $2 \notin 6\mathbb{Z}$ and $3 \notin 6\mathbb{Z}$, meaning $6\mathbb{Z}$ is *not* a prime ideal.

Note that this is an *inclusvie* or, not exlusive. If $p = 2$, and $a = b = 2$, then $2|4$, and both $2|2$ and $2|2$.

[19] namely, the ideal group is abelian for certain non-abelian

Another way I like to think about is is that we already know that if $a \in I$ and $b \in I$, then $ab \in I$. However, it's not clear the converse is true $ab \in I$, $a \in I$ and $b \in I$. This condition is a little too strong: Since $1 \cdot x \in I$ then $1 \in I$, meaning $I = R$. So what if we loosen the condition a bit , and say $ab \in I$ then $a \in I$ *or* $b \in I$. In other words, we can to a limited degree work backwards from *any* combination of elements $ab \in I$ to find one of them keeps needing to be in I. For example, if $abc \in I$ (say we braketed $(ab)c$) then either $ab \in I$ or $c \in I$. Since $ab \in I$, then either $a \in I$ or $b \in I$. This idea will come back in section 10.1, when we factor elements in an integral domain, and will be the key idea to a couple of proofs[20].

This interpretation shows that a prime element is irreducible (both notions are defined in definition 10.1.5). Let R be an integral domain, $p \in R$ a prime element, and $p = rr'$. Then $p \mid rr'$, so $p \mid r$ or $p \mid r'$. If $r = ps$, then $p = psr'$, and cancelling $p \neq 0$ gives $sr' = 1$, so r' is a unit. Divisibility both ways does not force equality: it only makes r and p associates (definition 10.1.2), as $1 \mid 2$ and $2 \mid 1$ in $\mathbb{R}$ while $1 \neq 2$. If $p \mid r'$ instead, the same argument makes r a unit. Proposition 10.1.7 records this.

Another thought: Take (x) and consider $ab \in (x)$. Then $ab = cx$ for appropriate $c \in R$. Then we no get the question $cx \in (x)$ too. This though is "obvious" since $x \in (x)$. But ab is not obvious. So the "representation" of the value that both ab and cx represents is important: it's a question about hether the constituent components of a represetnation is in there (a or b or c or x)

Here are some more example of prime ideals:

Example 9.4.24: Further Examples of Prime Ideals

1. Take $C([0,1])$, that is, all continuous function to/from $[0,1]$. Let $I \subseteq C([0,1])$ be all polynomials with roots at $\frac{1}{2}$ and $\frac{1}{3}$ ($f(\frac{1}{2}) = f(\frac{1}{3}) = 0$). Is I an ideal? Is it a prime ideal? Hint [a].

2. As an exercise to show that you can no longer "go backards", prove that if R is a commutative ring and P is a prime ideal of R, then R/P is an integral domain. (Hint [b]).

[a] Think about two polynomials with single roots such that when multiplied, you get an element of I
[b] Assume R/P has a zero-divisor

This last example is important enough to be referenced many times, and so we make it a proposition:

Proposition 9.4.25: Prime Ideals And Integral Domains

Assume R is commmutative. Then the ideal P is a prime ideal in R if and only if the quotient ring R/P is an integral domain

***Proof*:**
The quotient ring R/P (proposition 9.3.11) is commutative, since $(a+P)(b+P) = ab+P = ba+P = (b+P)(a+P)$, and its identity is $1+P$. First, $1+P \neq 0+P$ exactly when $1 \notin P$, that is, exactly when $P \neq R$ (proposition 9.4.10(1)). Second, the product $(a+P)(b+P) = ab+P$ is the zero class P exactly when $ab \in P$, and $a+P = P$ exactly when $a \in P$, so R/P has no zero divisors exactly when $ab \in P$ implies $a \in P$ or $b \in P$. Hence R/P is an integral domain (definition 9.1.21) exactly when P is a prime ideal (definition 9.4.22), as claimed.

[20] for example, proposition 10.1.32 and theorem 10.1.30

Corollary 9.4.26: Maximal Ideal Then Prime Ideal

Assume R is commutative. Then every maximal ideal of R is a prime ideal

***Proof*:**
If M is a maximal idea, then by proposition 9.4.19 R/M is a field. A field is an integral domain, so the corollary folllows from proposition 9.4.25

Example 9.4.27

1. The principal ideals generated by (p) where p is prime are both prime and maximal. Not all prime ideals are maximal: (0) is a prime ideal, but not maximal.
2. The ideal (x) is a prime ideal in $\mathbb{Z}[x]$ since $\mathbb{Z}[x]/(x) \cong \mathbb{Z}$, but not maximal since $\mathbb{Z}$ is not a field. Or, you can diretly see that $(x) \subseteq (2, x)$, meaning that (x) is not maximal.

Example 9.4.28: Evaluation At Zero
The ideal (y) of $\mathbb{Q}[x, y] = (\mathbb{Q}[x])[y]$ is a prime ideal. Write each $p \in \mathbb{Q}[x, y]$ as $p = \sum_{j\geq 0} p_j y^j$ with $p_j \in \mathbb{Q}[x]$. By theorem 9.2.4, applied to the identity of $\mathbb{Q}[x]$ and the element $0 \in \mathbb{Q}[x]$, there is a ring homomorphism $\varphi\colon \mathbb{Q}[x, y] \to \mathbb{Q}[x]$ that is the identity on $\mathbb{Q}[x]$ and sends y to 0, namely $\varphi(p) = p_0$, the evaluation at $y = 0$. It is surjective, since $\varphi(f) = f$ for $f \in \mathbb{Q}[x]$. Its kernel consists of the p with $p_0 = 0$, which are exactly the multiples $p = y\sum_{j\geq 1} p_j y^{j-1}$ of y, so $\ker\varphi = (y)$. By theorem 9.3.12(1), $\mathbb{Q}[x, y]/(y) \cong \mathbb{Q}[x]$, which is an integral domain by proposition 9.2.3(3), and so (y) is a prime ideal by proposition 9.4.25.

Exercise 9.4.2

1. Let R, S be rings and $R \times S$ be the product ring. What are the prime ideals of $R \times S$(answer: $R \times P$ and $Q \times S$ for primes P and Q. In general, only one component can be the ideal)
2. Let $a, b \subseteq R$ be ideals. Show that $(a + b)(a \cap b) \subseteq ab$ and $ab \subseteq a \cap b$. Show that if $(a, b) = (1)$, then $a \cap b = ab$.
3. exercise on quotient ideals $(a : b)$
4. Let $f : R \to S$ be a ring homomorphism and $p \subseteq S$ a prime ideal. Show that $f^{-1}(p)$ is a prime ideal. If $m \subseteq S$ is maximal, is $f^{-1}(m)$ maximal? (Consider $R = \mathbb{Z}$ and $S = \mathbb{Q}$)
5. Perhaps, if possible at this stage how if if $a_1, a_2, ..., a_n$ are ideals and $\cap_i a_i$ is contained in the prime ideal p, then p contains one of the a_i, and if $p = \cap_i a_i$, then $p = a_i$ for on of them (shows how prime genelize numbers, so to speak).
6. Notice that we have defined what I^k is. Let's say we started with an ideal J. Is there a rasonable way of defining $\sqrt{J}$? Define $\sqrt{J} = \{a \in R \mid \exists n \in \mathbb{N}_{>0},\ a^n \in J\}$. Show that $\sqrt{J}$ is an ideal.

9.5 Ring of Fractions: Localization

In this section, we will take a moment to discuss how we can take any ring with no zero divisors, and extend the ring to one in which every element has a multiplicative inverse! More generally, given any non zero-divisor element $d \in R$ (where R will have to be commutative), we can construct a ring R_d which will where $d^{-1} \in R_d$. This concept was applied in the field of *algebraic geometry* for the purposes of looking at a ring "locally". What this means and how to think about this consruction "geometrically" will be discussed much later in [16, Localization]. In this section, the full generality of this construction is not particularly useful. What's more interesting at the moment is that this gives us a way to add units to our ring!

We will start with a practical example by using $\mathbb{Z}$ to construct $\mathbb{Q}$! Taking $\mathbb{Z}$, we can make an equivalence class

$$(a, b) \sim (c, d) \Leftrightarrow ad = bc$$

where $b, c \neq 0$. Then $[(a, b)]$ represents a fraction $\frac{a}{b}$, with the point of the equivalence relation is to make all fractions with similar divisors be the same:

$$\frac{1}{2} = \frac{2}{4} = \frac{3}{6} = \cdots$$

I will stick with fraction notation for these equivalence relationships. We can see that we can define

$$\frac{a}{b} + \frac{c}{d} = \frac{ad + bc}{bd} \qquad \text{and} \qquad \frac{a}{b} \cdot \frac{c}{d} = \frac{ac}{bd}$$

It should be checked that these operations are independent of representative. With this construction, we can identity $\mathbb{Z}$ with $\left\{\frac{a}{1} | a \in \mathbb{Z}\right\}$ [21], showing how $\mathbb{Z}$ is a sub-ring of $\mathbb{Q}$.

With that intuition, we can ask about general commutative rings R. The problem comes when we allow $d = 0$ or be a zero divisor. Say $d \neq 0$ $bd = 0$. Then:

$$d = \frac{d}{1} = \frac{bd}{b} = \frac{0}{b} = 0$$

We will avoid this by taking a set $R^\times$ which represents all the units of R

[21] meaning construct a isomorphism

Theorem 9.5.1: Ring of Fractions

Let R be a commutative ring R (not necessarily with identity). Let D be any nonempty subset of R that does not contain 0, does not contain any zero-divisors and is closed under multiplication (i.e. $ab \in D$, for every $a, b \in D$)[a]. Then there is a commutative ring Q with 1 such that Q contains R as as a subring and every element of D is a unit in Q. The ring Q has the following additional properties

1. Every element of Q is of the form rd^{-1} for some $r \in R$ nd $d \in D$. In particualr if $D = R - \{0\}$ then Q is a field
2. (uniqueness of Q) The ring Q is the "smallest" ring containing R in which al elements of D become units. That is, if S is any commutative ring with identity and let $\varphi : R \to S$ be any injective ring homomorphism such that $\varphi(d)$ is a unit in S for every $d \in D$. Then there is an injective homomorphism $\Phi : Q \to S$ such that $\Phi|_R = \varphi$. In other words, any ring containing an isomorphic copy of R in which which all the elements of D become units must also contain an isomrophic copy of Q.

[a] such a set is called a *semi-group*

***Proof*:**
We will start like we started with $\mathbb{Z}$. Let $F = \{(r, d) | r \in R, d \in D\}$ (d in the 2nd component because we do not want zero-divisors in the bottom, and we need the denominator to be closed to define multiplication). Then, we define the equivalence relation

$$(r, d) \sim (s, e) \text{ if and only if } re = sd$$

It is easy to show that $\sim$ is an equivalence relation [a]

We commonly denote (r, d) by $\frac{r}{d}$. Now, let

$$Q = \left\{ [(a, b)] \middle| a \in R, d \in D \right\}$$

be the set of equivalence relations. Note that $\frac{r}{d} = \frac{re}{de}$ for any $e \in D$, as we would expect.

We can define addition and multiplication as

$$\begin{aligned} [(a, b)] + [(c, d)] = [(ad + bc, bd)] \quad &\Rightarrow \quad \frac{a}{b} + \frac{c}{d} = \frac{ad + bc}{bd} \\ [(a, b)] \cdot [(c, d)] = [(ac, bd)] \quad &\Rightarrow \quad \frac{a}{b} \cdot \frac{c}{d} = \frac{ac}{bd} \end{aligned}$$

Then the following properties need to be checked. I will show that addition is well-defined, and the rest is not hard to show that it follows:

1. The operations are well-defined (not dependent on representatives)

   Proof. We will show addition is well-defined. Let

   $$\frac{a}{b} = \frac{a'}{b'} \ (ab' = a'd) \qquad \frac{c}{d} = \frac{c'}{d'} \ (c'd = cd')$$

then

$$\frac{ad+bc}{bd} = \frac{a'd'+b'c'}{b'd'}$$

which equivalently means:

$$\begin{aligned}
(ad+bc)(b'd') &= adb'd' + bcb'd' && \text{distributivity in } R \\
&= ab'dd' + cd'bb' && \text{commutativity in } R \\
&= ab'dd' + cd'bb' && ab' = a'b,\ cd' = c'd \\
&= (a'd' + b'c')(bd)
\end{aligned}$$

thus, the two are equivalent. A similar procedure works for multiplication □

2. Q is an abliean group under addition, where the aditive identity is $\frac{0}{d}$ for any $d \in D$ and the additive inverse of $\frac{a}{d}$ is $\frac{-a}{d}$
3. Multiplication is associative, distributive, commutative,
4. Q has an identity $\frac{d}{d}$ for any $d \in D$

Next, we show that R can indeed be embedded into Q. Let

$$i : R \to Q \qquad r \mapsto \frac{rd}{d}$$

Since $\frac{rd}{d} = \frac{re}{e}$, this function is independent of choice of d. Since d is closed under multiplication:

$$i(rs) = \frac{rsd}{d} = \frac{rs}{1} = \frac{r}{1}\frac{s}{1} = \frac{rd}{d}\frac{sd}{d} = i(r)i(s)$$

finally, for injectivity of the inclusion, notice that

$$i(r) = 0 \Leftrightarrow \frac{rd}{d} = \frac{0}{d} \Leftrightarrow rd^2 = 0 \Leftrightarrow r = 0$$

meaning the kernel is trivial, and i is injective.

Next, we show that every element $q \in Q$ can be written as rd^{-1}. Since $d = \frac{d}{1}$, then $d^{-1} = \frac{1}{d}$ since $\frac{d}{1}\frac{1}{d} = \frac{1}{1}$. Then by definition of an element in Q, we define an elemnent as $\frac{r}{d} \in Q$, and so we have that $q = rd^{-1}$.

Note that if $D = R - \{0\}$, then every element has a multiplicative inverse, and so that makes Q into a field.

Finally, we must show this extension is unique. To do this, let's say $\varphi : R \to S$ is an injective ring-homomorphism such that $\varphi(d)$ is a unit in S. In this way, S acts as "another" extension of R (since we shows that R can be imbedded into Q). Since $q = rd^{-1}$ for every $q \in Q$, extend φ to Φ by defining

$$\Phi(rd^{-1}) = \varphi(s)\varphi(d)^{-1}$$

for all $r, d \in R$. Note that $\varphi(d)^{-1}$ is defined since $\varphi(d)$ is a unit in S. This map is well-defined. If we have $rd^{-1} = re^{-1}$, then multiplying around we get $re = rs$, so by the property of ring homomrphisms $\varphi(r)\varphi(e) = \varphi(s)\varphi(d)$ and so

$$\Phi(rd^{-1}) = \varphi(r)\varphi(d)^{-1} = \varphi(s)\varphi(e)^{-1} = \Phi(se^{-1})$$

showing it is indeed well-defined. Checking Φ is a ring homomorphism is strait forward using commutativity of Q.

Last step is to show Φ is injective. There is a sneaky trick you can do to pull this off. Let's say $rd^{-1} \in \ker(\Phi)$. Then that means that $r \in \ker(\varphi)$ since we'd need $\Phi(rd^{-1}) = \varphi(r)\varphi(d)^{-1} = 0$ and $\varphi(d)^{-1}$ cannot be zero since d is a unit. However, $\ker(\varphi)$ is trivial, so $r = 0$, but then $rd^{-1} = 0d^{-1} = 0$, meaning Φ is also injective! Thus, we forced the S to have Q embedded inside of it, meaning *any* extension of R to S would inevitably have Q. This makes Q the smallest extension of R and the unique extension of R.

[a] Reflexivity and symmetry come immediately. For transitivity, combining the two equations appropriately and realise you can factor by $d \in D$, which is not zero or a zero-divisor, which will force the equation you want to be zero, forcing the equivalence for transitivity

We will give Q a name:

Definition 9.5.2: Ring Of Fraction

Let R be a commutative ring, D be the appropriate subset and Q be the extension as in the above theorem. Then

1. The ring Q is called the *ring of fractions* of D with respect to R and is dneoted $D^{-1}R$
2. If R is an integral domain an $D = R - \{0\}$, Q is called the *field of fractions* or *quotient field* of R.

we often want all units of a ring R. We denote all units as $\mathbb{R}^\times$.

Here are some examples of ring of fractions

Example 9.5.3

1. the trivial example would a a field $\mathbb{F}$, which would be it's own field of fractions.
2. the construction we have done at the beginning is a special case of our construction. So, if we take $R = \mathbb{Z}$, and take $D = \mathbb{Z} - \{0\}$, then $D^{-1}\mathbb{Z} = \mathbb{Q}$
3. Let $R = \mathbb{Z}$ and take $D = \{1\}$. Then the quotient ring would be $\mathbb{Z}$ again!
4. $R = \mathbb{Z}$, and take $D = \{19, 19^2, ...\}$. This is a valid D, since it does not contain 0, does not contain zero divisors, and is closed under multiplicationso what is $D^{-1}\mathbb{Z}$?
5. The ring $2\mathbb{Z}$ is alsmot an integral domain. It doens't have an identity. However, $(R^\times)^{-1}(2\mathbb{Z}) = \mathbb{Q}$ as well! If you think about it, being able to divide will give us the identity, as well as $\frac{1}{2}$, which will produce the rest
6. $Z[p]$ should be elaborated on
7. $\mathbb{F}[x]$ becomes $\mathbb{F}(x)$, that is, elements of the form $\frac{p(x)}{q(x)}$ where $q(x)$ is not the zero-polynomial (Since $\mathbb{F}[x]$ is an integral domain, it's the only value we would need to worry about). Notice the square-brakets became parenthesis. In this way, we might see the motivation to write $\mathbb{Q}(\sqrt{D})$, since we are allowed to divide those values as well! The difference here being that $x, x^2, x^3, ..., x^n, ...$ are all distinct elements, non of them being an element of their original

field. However, this same chain doens't appear for $Q(\sqrt{D})$: $\sqrt{D}, \sqrt{D}^2 = D$. This is because $\sqrt{D}$ is not an arbitrary indeterminate value, giving us this relation, similar to what we saw with group-rings.

8. Take $\mathbb{Z}[i]$ and $D = R^\times = R - \{0\}$. Then the ring of fractions is $\mathbb{Q}[i]$! Notice it's not something like $\mathbb{Z}(i)$. This comes down to how multiplication is defined in this ring, which is much more akin to multiplying complex numbers.
9. what is the field of fractions of $\mathbb{Q}[\sqrt{2}]$?
10. What if we have $R = \mathbb{Z}/4\mathbb{Z}$ and $D = \{1, 3\}$?

Theorem 9.5.4: Universal Property Of Field Of Fractions

If $h : R \to F$ is an injective homomorphism from R into a field F, then there exists a unique ring homomorphism $g : Q \to F$ which extends h

***Proof*:**
Use the uniqueness of the extension to show that $\Phi(Q)$ will be in any sub-field.

With the relation $(r, d) \sim (s, e)$ if and only if $u(re - sd) = 0$ for some $u \in D$, the construction of $D^{-1}R$ extends to multiplicative sets D that contain zero-divisors, and $D^{-1}R$ is universal among ring homomorphisms $R \to S$ sending D into $S^\times$. The map $R \to D^{-1}R$ need not be injective, since $r/1 = 0$ whenever $dr = 0$ for some $d \in D$. This is localization, developed in [16, Localization].

(Another perspective would be that we are giving a solution to all equations of the form $bx - a = 0$, for $a, b \in R$. Thus, for $R = \mathbb{Z}$, we are doing:

$$\frac{\mathbb{Z}[x_{1/2}, x_{1/3}, ..., x_{1/p}, ...]}{(2x_{1/2} - 1, 3x_{1/3} - 1, ...,)} =: \mathbb{Q}$$

Though this way of looking at it seems more complicated, it fits better with the idea that we are adding relations to $\mathbb{Z}$. In particular, it shows that we are adding all *linear* relations (i.e. solutions) to $\mathbb{Z}$. We will in later section's show how we add non-linear solution.

Note that we construct $\mathbb{Z}$ from $\mathbb{N}$ by adding all monomial-linear polynomials. We then construct $\mathbb{Q}$ by taking all linear equations. We shall further explore in *Commutative Algebra* [16] how to think of non-monic polynomials with non-unit leading coefficients.)

9.5.5 Foreshadowing By taking the "infinite" version of polynomials, we managed to make all the non-zero-divisors become units. In this section, another way of producing using by formally inverting them. These two processes shall seem very different, however they are related, and the connection will be made in [16, Completion].

9.5.1 $\mathbb{R}$ Is Not Algebraic over $\mathbb{Q}$

The Grothendieck completion turns the monoid $\mathbb{N}$ into the group $\mathbb{Z}$ (example 7.1.7), the field of fractions turns $\mathbb{Z}$ into $\mathbb{Q}$, and adjoining a root of $x^2 + 1$ turns $\mathbb{R}$ into $\mathbb{C}$ (example 19.2.7). The step

from $\mathbb{Q}$ to $\mathbb{R}$ is of a different kind. Adjoining to $\mathbb{Q}$ roots of polynomials, even of all of them, gives only algebraic numbers, and the real algebraic numbers $\overline{\mathbb{Q}}^{\mathbb{R}}$ (section 21.1) form a countable field, since there are countably many polynomials over $\mathbb{Q}$ and each has finitely many roots. As $\mathbb{R}$ is uncountable, it is not reached this way. Its standard constructions, by Dedekind cuts or by Cauchy sequences, use the order or the absolute value of $\mathbb{Q}$ ([18, The Real Numbers]).

Exercise 9.5.1

1. Show that $\mathbb{Q}(\sqrt{2}) = \left\{a + b\sqrt{2} \;\middle|\; a, b \in \mathbb{Q}\right\}$ is a subfield of $\mathbb{R}$ and that it is the field of fractions of its proper subring $\mathbb{Z}[\sqrt{2}] = \left\{a + b\sqrt{2} \;\middle|\; a, b \in \mathbb{Z}\right\}$.

9.6 Chinese Remainder Theorem

Let's say we have a ring R and a collection of ideals $I_1, I_2, ..., I_k$. Then given

$$a_1 \bmod I_1, \cdots, a_k \bmod I_k$$

does there exist a $a \in R$ such that

$$a \equiv a_1 \bmod I_1, \cdots, a \equiv a_k \bmod I_k$$

Think about it for a moment before looking at the answer

9.6.1 Taking a moment aking a moment to think about the answer.

The answer depends on the choice of I_i. The reader should have tried this with $R = \mathbb{Z}$ and noticed it is not always possible, for example if $R = \mathbb{Z}$, $a \equiv 1 \bmod 2$, $a \equiv 2 \bmod 4$ does not have a solution.

The Chinese Remainder Theorem (CRT) is a way of finding an to this question. It's name sake most likely comes from a book *Sun Tzu Suan Ching*. It is said that it was originally developed either for accounting purposes or for dividing troops.

Before we get into the actual theorem, we will first explore a direct application of Chinese Remainder Theorem:

Example 9.6.2: Concrete Use
simple form

$$\text{if } gcd(m, n) = 1 \text{ and } 0 \leq a < m\,, 0 \leq b < n$$

then

$$\exists!\ k,\ 0 \leq k < mn \text{ such that } k \equiv a \bmod m\,, k \equiv b \bmod n$$

Let's say: $m = 6$, $n = 5$, $a = 4$, $b = 2$. Then $k = 22$.

Proof. We have

$$k \equiv 4 \bmod 6,\ k \equiv 2 \bmod 5$$

so using the classical method, we first find

$$6 \cdot 5 = 30$$

from this, we find $30/6 = 5$ and $30/5 = 6$ to be the values such that and kno that we get the solution by solving the equivalent problem of:

$$k \equiv 5 \cdot 1 \bmod 30,\ k \equiv 6 \cdot 1 \bmod 30$$

meaning the solution is $(4 \cdot 5 \cdot 5) + (2 \cdot 6 \cdot 6) =$ □

When generalizing this result, we can no longer ask for numbers to be co-prime. The equivalent concept in rings are co-maximal ideals [22]

Definition 9.6.3: Comaximal Ideals and Elements

The ideals A and B of the ring R are *comaximal* if $A + B = (1)$. When R is commutative, two elements $a, b \in R$ are *comaximal* (also called *strictly coprime*) if the ideals (a) and (b) are comaximal, that is, if $(a, b) = (1)$. Equivalently, there exist $x, y \in R$ with $xa + yb = 1$.

The model case is $\mathbb{Z}$. If $a, b \in \mathbb{Z}$ share no prime factor, the Extended Euclidean Algorithm ([24, Extended Euclidean Algorithm]) provides $x, y \in \mathbb{Z}$ with $xa + yb = 1$, so $1 \in (a) + (b)$ and $(a) + (b) = \mathbb{Z}$ by proposition 9.4.10. For integers, "coprime" and "comaximal" are therefore the same condition. In a general commutative ring they separate, and the Chinese Remainder Theorem needs the stronger of the two.

9.6.4 Convention: Coprime Versus Strictly Coprime Call a, b in a commutative ring R *coprime* if their only common divisors are units. Comaximal elements are coprime: if $d \mid a$, $d \mid b$ and $xa + yb = 1$, then $d \mid 1$. The converse fails (example 9.6.5), so the ideas need distinct terminology. Two nomenclatures are common:

- Atiyah and Macdonald call ideals with $A + B = (1)$ *coprime* (or comaximal), so for them the ideal-theoretic is the basic idea.
- Milne, in his treatment of Hensel's lemma for étale cohomology, keeps *coprime* for the "no common factor" notion over a field and writes *strictly coprime* for $(f, g) = (1)$ over a ring.

Thus ,we say *comaximal*, which has only one meaning.

Example 9.6.5: Coprime But Not Comaximal

In $\mathbb{Z}[x]$, the elements 2 and x are coprime. If $d \mid 2$, then by proposition 9.2.3 d is a constant, so $d \in \{\pm 1, \pm 2\}$, and $\pm 2 \nmid x$. Hence only the units ± 1 divide both. They are not comaximal, since $(2, x)$ is the proper ideal of polynomials with even constant term (example 9.4.6).

The Chinese Remainder map $\mathbb{Z}[x] \to \mathbb{Z}[x]/(2) \times \mathbb{Z}[x]/(x)$ is therefore not surjective. A preimage $f(x)$ of $(1, 0)$ would satisfy $f(x) \in 1 + (2)$ and $f(x) \in (x)$, so its constant term $f(0)$ would be both odd and 0. The kernel $(2) \cap (x) = (2x)$ is still the product of the two ideals. The conclusion of lemma 9.6.9 can hold without its hypothesis, but surjectivity cannot.

The most useful test for comaximality in a polynomial ring involves a linear factor.

[22] why not co-prime ideals? We will show how prime and maximal ideals will match in special rings in [11, Ring Structure and Dedekind Domains], here we would usually apply such a result. However, we stay general since we can prove it in more general contexts

Proposition 9.6.6: Comaximality With A Linear Polynomial

Let B be a commutative ring, $a \in B$ and $f(x) \in B[x]$. Then

1. $(f(x), x-a) = (f(a), x-a)$ as ideals of $B[x]$
2. $f(x)$ and $x-a$ are comaximal if and only if $f(a) \in B^\times$
3. if B is an integral domain, $f(x)$ and $x-a$ are coprime if and only if $f(a) \neq 0$

***Proof*:**

For (1), each $k \geq 1$ gives $x^k - a^k = (x-a)(x^{k-1} + ax^{k-2} + \cdots + a^{k-1})$. Writing $f(x) = \sum_k c_k x^k$, we get

$$f(x) - f(a) = \sum_k c_k (x^k - a^k) = (x-a)\, q(x)$$

for some $q(x) \in B[x]$. Hence $f(a) = f(x) - (x-a)q(x) \in (f(x), x-a)$ and $f(x) = f(a) + (x-a)q(x) \in (f(a), x-a)$, so the two ideals coincide.

For (2), if $f(a) \in B^\times$ then $1 \in (f(a), x-a) = (f(x), x-a)$. Conversely, suppose $1 = u(x)f(x) + v(x)(x-a)$. Applying the evaluation homomorphism ev_a (definition 9.2.5), which fixes B, gives $1 = u(a)f(a)$, so $f(a) \in B^\times$.

For (3), degrees add in $B[x]$ (proposition 9.2.3). So a unit of $B[x]$ has degree 0, and $x-a$ is not a unit. If $f(a) = 0$, the identity from (1) reads $f(x) = (x-a)q(x)$, and $x-a$ is a common non-unit divisor. Conversely, let d be a common non-unit divisor and write $x - a = d\,e$, so $\deg d + \deg e = 1$. If $\deg d = 0$, then $d = c$ is a constant, and comparing leading coefficients in $x - a = c\,e$ shows $c \in B^\times$, contradicting that d is not a unit. So $\deg e = 0$, $e = c$ is a constant, the same comparison gives $c \in B^\times$, and $d = c^{-1}(x-a)$. Then $x - a \mid f(x)$, and applying ev_a gives $f(a) = 0$.

Example 9.6.7: Nonzero Is Not Enough

Take $B = \mathbb{Z}$, $f(x) = x$ and $a = 2$. Then $f(2) = 2 \neq 0$, so x and $x-2$ are coprime by proposition 9.6.6(3). But $2 \notin \mathbb{Z}^\times$, so they are not comaximal. Indeed $(x, x-2) = (2, x-2) = (2, x)$, the ideal of example 9.6.5. Over $\mathbb{Q}$, the same polynomials are comaximal, since $1 = \frac{1}{2}x - \frac{1}{2}(x-2)$. Over a field every nonzero element is a unit, so parts (2) and (3) give the same condition.

9.6.8 Foreshadowing: Hensel's Lemma The two notions meet in Hensel's lemma for a local ring A with maximal ideal $\mathfrak{m}$ and residue field $k = A/\mathfrak{m}$. Its hypothesis is a factorization $\bar{f} = g_0 h_0$ with g_0, h_0 coprime in $k[x]$, the field notion. Its conclusion is a lift $f = gh$ in $A[x]$ with g, h comaximal, the ring notion. The bridge is that for g monic, g and h are comaximal in $A[x]$ exactly when $\bar{g}$ and $\bar{h}$ are coprime in $k[x]$. Its proof applies Nakayama's lemma to the finite A-module $A[x]/(g,h)$, and it belongs with the commutative algebra of local rings.

Lemma 9.6.9: Characterizing Products of Coprime Ideals

Let $I_1, I_2, ..., I_n$ be pairwise comaximal ideals. Then:

$$\bigcap_{i=1}^{n} I_i = \prod_{i=1}^{n} I_i$$

***Proof*:**
We shall do induction so consider first when $n = 2$ so that $I, J \subseteq R$ such that $I + J = (1)$. It is always the case that $IJ \subseteq I \cap J$ so we have to show that $I \cap J \subseteq IJ$ when $I + J = (1)$. Given $I + J = (1)$, there exists $a \in I$, $b \in J$, such that $a + b = 1$. Now, if $r \in I \cap J$, then

$$r = r \cdot 1 = r(a + b) = ra + rb \in IJ$$

Since $r \in J$ and $a \in I$, $ra \in IJ$, an similarly $rb \in IJ$ since $r \in I$ and $b \in J$.

Then the inductive step follows immediately.

Lemma 9.6.10: Recursive Comaximal Of Products

Let $I_1, I_2, ..., I_n \subseteq R$ be ideals such that $I_i + I_n = (1)$ for all $i \in \{1, ..., n-1\}$. Then $(I \cdots I_{n-1}) + I_n = (1)$

***Proof*:**
By assumption, there exists $a_i \in I_k$ and $1 - a_k \in I_i$ such that

$$(1 - a_1) \cdots (1 - a_{n-1}) \in I_1 \cdots I_{n-1}$$

And

$$1 - (1 - a_1) \cdots (1 - a_{n-1}) \in I_n$$

since it is a comination of the elements $a_1, a_2, ..., a_{n-1} \in I_n$

Theorem 9.6.11: Chinese Remainder Theorem

Let $A_1, A_2, ..., A_k$ be ideals in R. The map

$$R \to R/A_1 \times \cdots \times R/A_k \qquad \text{defined by} \qquad r \mapsto (r + A_1, \cdots, r + A_k)$$

is a ring homomorphism with kernel $A_1 \cap A_2 \cap \cdots \cap A_k$. If for each $i, j \in \{1, 2, ..., k\}$ with $i \neq j$ the ideals A_i and A_j are comaximal, then this map is surjective and $A_1 \cap \cdots \cap A_k = A_1 A_2 \cdots A_k$, so

$$R/(A_1 A_2 \cdots A_k) = R/(A_1 \cap A_2 \cap \cdots \cap A_k) \cong R/A_1 \times R/A_2 \times \cdots \times R/A_k$$

***Proof*:**
Take

$$R \to \bigoplus_{i=1}^{n} R/A_i \qquad a \mapsto (a \bmod A_1, ..., a \bmod A_n)$$

Then this is easily checked to be a ring homomorphism with kernel $A = \bigcap_i A_i$. Assuming now the ideals are comaximal, we shall show surjectivity. We shall do a proof by induction. If $n = 1$ where done, so let for let's say

$$R/(A_1 \cdots A_{n-1}) \cong R/A_1 \times \cdots \times R/A_{n-1}$$

Then by lemma 9.6.10, $(A_1 \cdots A_{n-1}) + A_n = (1)$, and hence this is reduced to two ideals. By comaximality there exists $a_1 \in A_1$, $a_2 \in A_2 \cdots A_{n-1}$ such that $1 = a_1 + a_2$. Then writting

$$x = x_2 a_1 + x_1 a_2$$

we get that $x \equiv x_1 \bmod A_1$ and $x \equiv x_2 \bmod A_2 \cdots A_n$. But then $x \mapsto (x_1 \bmod A_1, x_2 \bmod A_2 A_3 \cdots A_{n-1})$, completing the proof.

A useful consequence to remember is that if there are elements $r_i \in R$ where $r_i \bmod A_i$, then there exists an element a

$$a \equiv r_i \bmod A_i$$

The ability to solve such a system mod A_i is sometimes called the *Weak Approximation Theorem*.

Corollary 9.6.12: Chinese Remainder Theorem For Integers

Let $R = \mathbb{Z}$ $a_1, a_2, ..., a_n \in \mathbb{Z}$ be elements such that $\gcd(a_i, a_j) = 1$ for all $i \neq j$. Then if $a = a_1 \cdots a_n$, The function

$$\varphi : \mathbb{Z}/a\mathbb{Z} \to \mathbb{Z}/a_1\mathbb{Z} \times \cdots \times \mathbb{Z}/a_k\mathbb{Z}$$

given in the natural way is an isomorphism

***Proof*:**
This is an immediate consequene of the chinese remainder theorem since $\gcd(a, b) = 1$ if and only if $(a, b) = (a) + (b) = 1$

Exercise 9.6.1

1. Show that if $P + Q = R$, then for any $n, m > 0$, $P^n + Q^m = R$. Conclude the CRT can be generalized to ideals of the form $A_1^{n_1}, ..., A_k^{n_k}$.

9.7⋆ Monoidal Objects

⋆ *Optional. This section is an aside; nothing later in the book depends on it.*

Maybe have this here to be able to introduce R-algebras as the monoidal objects in the R-**Mod** Category?

10

Exploring Integral Domains: Factorization and Valuation

This chapter will assume basic number theory knowledge from the preliminary chapter ([24, Number Theory]).

Integral domains are a huge overarching category of rings, being in a meaningful way the generalization of integers. That they have no zero-divisors makes it possible to partially invert elements (by proposition 9.1.24). There are many different consequences of this result especially in the right specializations, for example divisibility, size, and even the euclidean algorithm shall all be definable on the appropriate integral domains[1]. These topics all interact greatly with each other and there is no linear way of addressing these ideas. However, books are inherently written with a total ordering and we must learn content in some order, and hence this chapter has been written with the idea of exploration the notion of divisibility (which the reader can think of the best that can be done to check invertibility if an element is not a unit). As this exploration naturally progresses, all other pertinent notions about integral domains shall become natural questions and will be answered.

The following diagram sums up the results of this chapter:

[1]There is also a more geometric interpretation, which is that integral domains will correspond to "irreducible" geometric objects (think of how $R \times S$ is no longer integral. This idea is explored in [23, Irreducibility and the Decomposition Theorem]

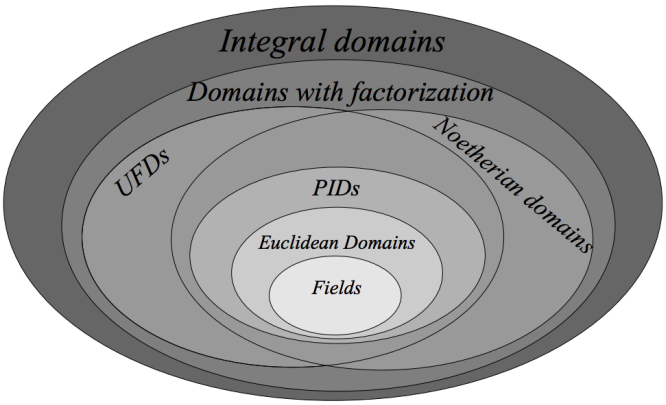


Figure 10.1: Image form Alufi's textbook

At the end of this chapter, we shall explore how these abstract notions were in fact developed to solve number-theoretic problems. We shall see our first example of finding the solution to a Diophantine equation, namely

$$x^2 + y^2 = p$$

In other words, which primes can be written as sums of two squares. As equations are the fundamental way in which we determine the properties of objects, this will be the first glimpse into the properties of prime numbers. This exploration is greatly generalized in [11, Application: Quadratic Extensions and Quadratic Reciprocity].

10.0.1 Note Some results in this chapter apply beyond integral domains. It will be made extra-clear when that is.

Goal

In fields, every element has an inverse. This is not so for general rings, as we have seen when studying units and zero-divisors. If not every element is a unit, there is a new idea that is introduced when learning some basic number theory: *factorization*. In this way, we can "decompose" an element into some basic building blocks. This idea of factorizing is one the reader is already familiar with: any number $x \in \mathbb{Z}$ can be factored, even uniquely, into prime numbers:

$$x = up_1^{k_1} p_2^{k_2} \cdots p_n^{k_n}$$

where $u \in \{1, -1\}$ and each p_i is a prime. Is there a generelizable notion of being factorizable in more general rings? Is this idea related to prime ideals? Some types of rings do admit an idea of factorization, or even some stronger conditions (like the euclidean algorithm being well-defined, or the factorization being unique), while other rings have a weaker condition (like the factorization not being unique or even existing). The first major focus of this chapter are these ideas.

The overarching idea in factorization is centered around the idea of divisibility like we have seen in $\mathbb{Z}$. For example:

$$2|8 \qquad 32|17728 \qquad \text{but} \qquad 3 \not| 8 \qquad 5 \not| 7$$

When an element is divisible, let's say $2|8$, then there exists some $x \in \mathbb{Z}$ such that $2x = 8$, that is, 8 got subdivided into other elements. In this way, factorization can be thought of as a "partial division" or "partial inverse" condition. This idea will be at the heart of our following exploration.

The result will give 3 new classes of rings and develop on 2 previously mentioned classes or rings which are very often studied.

1. *Atomic Domains* (or Domains with Factorization) are will be rings in which element can be factored into elements that can no longer be factored (they will be called irreducible elements), though not necessarily uniquely
2. *Noetherian domains* are similar to Unique Factorization domain as in every element can be factorzied, but it need not be unique. The reader may recall these as rings in which every ideal may be finitely generated.
3. *Unique Factorization Domains* (UFDs) will allow for every element to have a unique factorization, and let's us define the notion of there being a greatest common divisor between elements.
4. *Principal Ideal Domains* (PIDs) will be stronger than PIDs in that they allow for unique factorization and will have a more rigid structure. The reader may recall these as rings in which every ideal may be generated by a single element.
5. *Euclidean Domains* will have the property of the euclidean division algorithm applying to them to quickly find the greatest common divisor. These will allow for algorithmic decomposition of elements into prime elements.

Since factorization is heavily inspired by the properties of $\mathbb{Z}$, all rings in this chapter will be *integral domains* unless specified otherwise.

10.1 Factorization

We will first define more rigorously the notion of elements being factorizable

Definition 10.1.1: Divisibility

Let R be a commutative ring and $a, b \in R$ be elements of R. Then we say that a *divides* b (or b *is divisible by* a) if there exists an element c such that

$$ca = b$$

Equivalently, $ca = b$ if and only if $b \in (a)$. Notationally, we write $a|b$ to mean "a divides b".

If R is not commutative, one can define left and right divisibility separately, but this book does not treat the non-commutative case. When a divides b, one also says that b is a *multiple* of a. This terminology is used much less often.

The equivalent condition with ideals is a very insightful way of understanding divisibility. Here are the four equivalent conditions of divisibility for ease of reference:

1. $b|a$ if and only if

2. $a = bb'$ if and only if
3. $a \in (b)$ if and only if
4. $(a) \subseteq (b)$

It is unfortunate that the notation $b|a$ stuck, since you will have to swap at some point the order of the a and the b given that it's read "b divides a" and not "a is divided by b". It's a matter of getting used to.

The notation for divisibility using ideals is particularly insightful here. Notice that if $u \in R$ is a unit, then for any element $r \in R$ $(r) = (ur)$. Both inclusion are easy and should be proven before continuing to read. Because of this property, it is immediate that $a\big|b$ if and only if $a\big|ub$, that is, if an element is divided (or not divided) by another, then multiplying it by a unit let's it remain divisible (or not divisible) by the other element. This is in fact something the reader is already is familiar with. Notice that $6 = -1 \cdot -6$. Then $2\big|6$, and $2\big|-6$, but also $5 \nmid 6$ and $5 \nmid -6$. Thus, when we think of divisibility, it is important to think of a number being divisible "up to a unit":

Definition 10.1.2: Associative

Let R be a commutative ring and let $a, b \in R$ be two elements. Then we say that a and b are *associates* if the ideals generated by a and b are equal, that is:

$$(a) = (b)$$

It is important to check that this is equivalent to saying that a and b are associates if $a\big|b$ and $b\big|a$, showing how this links to divisibility. Notice that in the previous definition, R was not an integral domain, but just a commutative ring, which allows zero-divisors. This in fact breaks the intuition of associativity we just showed:

Example 10.1.3: associativity in non-integral domains

Let $R = \mathbb{Z}/6\mathbb{Z}$. Then $(\overline{2}) = (\overline{4})$, since

$$(\overline{2}) = \{\overline{2}, \overline{4}, \overline{0}\} = \{\overline{4}, \overline{2}, \overline{0}\} = (4)$$

Furthermore, $\overline{4} = \overline{2} \cdot \overline{2}$ but $\overline{2}$ is *not* a unit.

For this reason, restricting to integral domains let's us keep this "up to a unit" intuition for for divisibility using ideals:

Proposition 10.1.4: Associativity in Integral Domains

Let R be an integral domain. Then a and b are associates if and only if $a = ub$ for some unit $u \in R$

***Proof*:**

Let $(a) = (b)$, so that for some $r, s \in R$, $ra = b$ and $a = sb$. If $a = 0$, then $b = ra = 0$ and $a = 1 \cdot b$. Otherwise

$$sra = sb = a$$

and cancelling a (proposition 9.1.24) gives $sr = 1$, so s is a unit and $a = sb$.

Conversely, if $a = ub$ for a unit $u \in R$, then $(a) \subseteq (b)$ since $u \cdot b = a$, and $(b) \subseteq (a)$ since $u^{-1} \cdot a = b$.

The conclusion of proposition 10.1.4 can still hold in a ring with zero divisors. In example 10.1.3, the units of $\mathbb{Z}/6\mathbb{Z}$ are $\overline{1}$ and $\overline{5}$. The principal ideals of $\mathbb{Z}/6\mathbb{Z}$ are $(\overline{0}) = \{\overline{0}\}$, $(\overline{3}) = \{\overline{0}, \overline{3}\}$, $(\overline{2}) = (\overline{4}) = \{\overline{0}, \overline{2}, \overline{4}\}$ and $(\overline{1}) = (\overline{5}) = \mathbb{Z}/6\mathbb{Z}$, so the only distinct associates are $\overline{1}, \overline{5}$ and $\overline{2}, \overline{4}$, and $\overline{5} = \overline{5} \cdot \overline{1}$ and $\overline{4} = \overline{5} \cdot \overline{2}$ differ from their partners by the unit $\overline{5}$. Most work on divisibility in rings takes place in integral domains, where the cancellation property holds, and this book restricts to integral domains as well.

With a notion of divisibility up to unit established, we can start thinking of what are the building blocks, the "fundamental elements", into which we will factorize an element:

Definition 10.1.5: Irreducible and Prime Elements

Let R be an intergral domain

1. Let $r \in R$ be a nonzero element that is not a unit. Then r is called *irreducible* in R if whenever $r = ab$, at least one of a or b must be a unit in R. Otherwise, r is said to be *reducible*.

2. The nonzero element $p \in R$ is called *prime* in R if the ideal (p) generated by p is a prime ideal. In other words, a nonzero element p is a prime if it is not a unit and whenver $p|ab$ for any $a, b \in R$ then either $p|a$ or $p|b$.

Example 10.1.6: Irreducible and Prime elements

Let $R = \mathbb{Z}$, and take $6 \in \mathbb{Z}$. Then we can see that $2 \cdot 3 = 6$. Since 2 and 3 are both not units, that means that 6 is a reducible element in $\mathbb{Z}$. On the other hand, 2 and 3 are not reducible elements. If there existed a a, b such that $2 = ab$, then by our knowledge that 2 is a prime number (or by some elementary number theory argument), the only number that divide it are 1 and 2, so it must be that either

$$1 \cdot 2 = 2 \quad \text{or} \quad -1 \cdot -2 = 2$$

In both cases, a is a unit, and so 2 is irreducible (similarly for 3). Furthermore, 2 and -2 are associates, since $2 = -1 \cdot -2$, i.e., the differ by a unit. in $\mathbb{Z}$, every n are associates to it's negative $-n$.

As pointed out in example 9.4.23, 2 is also a prime element since (2) is a prime ideal. In fact, in the previous exercise, we used the fact that 2 is a prime integer (which corresponds to the prime ideals of $\mathbb{Z}$) in order to show it's irreducible. Though 2 is both an irreducible and a prime element in $\mathbb{Z}$, it need not be the case in general integral domains that an element is both!

As an example, take $\mathbb{F}[x]$ to be the polynomials over a field, and take $F_{0x} \subseteq \mathbb{F}[x]$ to be all the polynomials such that the coefficient of x is zero. As an example of elements in F_{0x}:

$$ax^3 + b \in F_{0x} \qquad cx^3 + dx \notin F_{0x} \qquad \forall a, b, c, d \in \mathbb{F}$$

It is easy to check that F_{0x} is an integral domain (as you should check), and so we can try to find irreducible and prime elements. Take x^6. It has two factorizations: $x^6 = x^2x^2x^2$ and $x^6 = x^3x^3$. The elements x^2 and x^3 are irreducible, but not prime.

To show x^2 and x^3 are irreducible, let's say

$$x^2 = p(x)q(x)$$

then since we are working with polynomials over an integral domain, by proposition 9.2.3:

$$\deg(x^2) = \deg(p(x)) + \deg(q(x))$$

meaning $2 = \deg(p(x)) + \deg(q(x))$. Since both $p(x)$ and $q(x)$ cannot be one (since $ax \notin F_{0x}$, for all $a \in \mathbb{F}$) then one of them has to be of degree zero, but since x^2 is monic this implies it must be 1, meaning it's a unit, meaning x^2 is irreducible. Similarly for x^3. Notice that this means that x^6 had two decompositions into irreducible factors, meaning the decomposition was not unique.

Now, take (x^2), i.e. the ideal generated by x^2. Notice that $(x^3)(x^3) \in (x^2)$ because $x^4 \cdot x^2 \in (x^2)$. However, $x^3 \notin (x^2)$. If it were, then there would exist a polynomal $p(x)$ such that $p(x)x^2 = x^3$. By the degree formula, $p(x)$ would have to be of degree 1. However, no degree 1 polynomial exists in F_{0x}. Therefore, (x^2) is not prime! This harkens back to prime ideals (definition 9.4.22), where being prime in some way means we can work backwards. This "working backwards" let's us guarantee we decompose our number into unique elements, as we will soon see.

Though not all irreducible elements are prime, the converse is true, that is, being "globally divisible"‘ is a strong enough condition that it guarantee's irreducibility:

Proposition 10.1.7: Prime Then Irreducible In Integral Domain

Let R be an integral domain. Then prime elements in R are always irreducible

***Proof*:**
Supose $(p) \subseteq R$ is a nonzero prime ideal, and that $p = ab$. Since (p) is a prime ideal, then either $a \in (p)$ or $b \in (p)$. If $b \in (p)$. Then $b = rp$ for some $r \in R$, and if $a \in (p)$ then $a = pr$ for some $r \in R$. Without loss of generality, let's say $a \in (p)$. Then

$$p = ab = prb$$

and since R is an integral domain, the cancellation property holds (proposition 9.1.24), and we get

$$1 = rb$$

implying r, b are units, and so b is a unit. The same argument applies if we chose b. Thus, p is irreducible.

Thus, being prime is stronger than being an irreducible. The natural question to ask now is whether all elements in an integral domain can be decomposed into irreducible or prime factors (if it is factored into prime factors, then automatically it is factored into irreducible factors). It might already be clear that some Integral domains we have explored have this property (ex. $\mathbb{Z}$, and if you are familiar with some manipulation of polynomials, then $\mathbb{F}[x]$). This is however not the case for all integral domains:

Example 10.1.8: Domain with no Factorization

1. Let $R = \mathbb{Z} + x\mathbb{Q}[x]$ be the polynomials with rational coefficients and integer constant term. It is closed under addition, negation and multiplication and contains 1, so it is a subring of $\mathbb{Q}[x]$ (corollary 9.1.7), hence an integral domain. Its units are ± 1: if $fg = 1$, then f and g have degree 0, so they are integers. The element x has no factorization into irreducibles. In

a product $x = f_1 f_2 \cdots f_k$ of elements of R, degrees add, so exactly one factor has degree 1 and the others are nonzero integers. The factor of degree 1 is $a + qx$ with $a \in \mathbb{Z}$ and $q \in \mathbb{Q}$, and comparing constant terms gives $a = 0$. But $qx = 2 \cdot \frac{q}{2}x$, where neither 2 nor $\frac{q}{2}x$ is a unit, so qx is not irreducible.

2. Let $\overline{\mathbb{Z}} \subseteq \mathbb{C}$ be the set of *algebraic integers*: the $r \in \mathbb{C}$ with

$$a_0 + a_1 r + a_2 r^2 + \cdots + r^n = 0$$

for some $a_0, \ldots, a_{n-1} \in \mathbb{Z}$, that is, the roots of monic polynomials with integer coefficients. It is studied at length in [11, Relating Algebra to Number Theory]. That $\overline{\mathbb{Z}}$ is closed under addition and multiplication is a theorem about integral extensions, [16, Integral Elements Form Rings], and it is not proved here. Granting it, $\overline{\mathbb{Z}}$ is a subring of $\mathbb{C}$, hence an integral domain. It is not a field: $2 \in \overline{\mathbb{Z}}$, but $1/2 \notin \overline{\mathbb{Z}}$, since multiplying $(1/2)^n + a_{n-1}(1/2)^{n-1} + \cdots + a_0 = 0$ by 2^n gives $1 = -2(a_{n-1} + 2a_{n-2} + \cdots + 2^{n-1}a_0)$, which is impossible.

It has no irreducible elements at all. Let $r \in \overline{\mathbb{Z}}$ be nonzero and not a unit, and let $\sqrt{r} \in \mathbb{C}$ be a square root of r (write $r = \rho e^{i\theta}$ and take $\sqrt{\rho}\, e^{i\theta/2}$). If $f(r) = 0$ with f monic in $\mathbb{Z}[x]$, then $f(y^2)$ is monic and vanishes at $y = \sqrt{r}$, so $\sqrt{r} \in \overline{\mathbb{Z}}$. It is not a unit, since $1/\sqrt{r} \in \overline{\mathbb{Z}}$ would give $1/r = (1/\sqrt{r})^2 \in \overline{\mathbb{Z}}$. So $r = \sqrt{r} \cdot \sqrt{r}$ is a product of two non-units and is not irreducible. In particular 2 has no factorization into irreducibles.

For this reason, we must define a special subclass of integral domains which do admit a factorization into irreducibles for every element:

Definition 10.1.9: Atomic Domains

Let R be an integral domain. Then R is an *atomic domain* (or *domain with factorization*) if for every non-unit nonzero element $r \in R$, there exists irreducible elements $q_1, q_2, ..., q_n$ such that

$$r = q_1 q_2 \cdots q_n$$

that is, every element has a *factorization* (or *decomposition*) into irreducible

The name atomic is due to every element being decomposable into element that can no longer be subdivided, similar to atoms, where the word atom comes from the Greek *atomos* meaning "not cuttable". Note that the q_i's need not be distinct (for example, we will show that $\mathbb{Z}$ is an atomic domain, and that $4 = 2 \cdot 2$). Usually, we are more interested in domains where the factorization into irreducible elements is unique:

Definition 10.1.10: Unique Factorization Domain (UFD)

Let R be an integral domain. Then R is a *Unique Factorization Domain* (commonly abbreviated to UFD) if for every non-unit nonzero element $r \in R$, there exists irreducible elements $q_1, q_2, ..., q_n$ and a unit u such that

$$r = uq_1q_2 \cdots q_n$$

and if there is another set of irreducible element $q_1', q_2', ..., q_m'$ and a unit u' such that $r = u'q_1'q_2' \cdots q_m'$, then $n = m$, and after some reshuffling the factors agree up to units (i.e. they are associates), that is, for a permutation σ of $\{1, \ldots, n\}$ and units $u_1, \ldots, u_n \in R$

$$q'_{\sigma(i)} = u_i q_i \qquad \text{for } 1 \leq i \leq n$$

The following corollary is stated for the sake of reference:

Corollary 10.1.11: Unique Factorization Domain ⇒ Atomic Domain

Let R be a Unique Factorization Domain. Then R is an Atomic domain

***Proof*:**
By definition, every element in a Unique Factorization Domain has a factorization into irreducibles, and so is an Atomic Domain

Note that "unique" means unique up to unit, as we pointed out earlier. The factorization in both definitions could have had the unit dropped and just have been written $r = q_1, q_2, ..., q_n$ since it is a the result is an associate to $u^{-1}r$, and so it is harmless to drop the unit. However, I chose to keep the unit in the definition since it may lead to certain unfortunate mistakes when factoring concrete elements.

Example 10.1.12: Unique Factorization Domains

1. Every field is trivially a Unique Factorization Domain, since every nonzero element is a unit, so there are no elements for which the two properties must be verified. In some textbooks, fields are purposefully omitted as Unique Factorization Domains to omit mentioning them as exceptions in proofs. They are still considered Unique Factorization Domains in this book since they fit well in the diagram presented at the beginning of this chapter. However, in a sense, this means that in a field there is no "unique factorization", or at least it's vacuously true that all elements can be uniquely factorized since no nonzero elements are non-units.

2. As we saw in example 11.2.5, $F_{0x} \subseteq \mathbb{F}[x]$ which has x coefficient of 0 is not a Unique Factorization Domain. We saw that x^2 and x^3 are irreducible. If we take x^6, then has two factorizations: $x^6 = x^2x^2x^2$ and $x^6 = x^3x^3$, and hence the factorization is not unique.

3. The ring $\mathbb{Z}[2i]$ is *not* a Unique Factorization Domain. $4 = 2 \cdot 2 = ((-2i) \cdot 2i)$. The elements 2 and $2i$ are irreducible, but not associates, since $2 = 2 \cdot i$, but $i \notin \mathbb{Z}[2i]$. They are furthermore not prime, since $\mathbb{Z}[2i]/(2i) \cong \mathbb{Z}/4\mathbb{Z}$ which is not an integral domain (see proposition 9.4.25)

4. In the ring $\mathbb{Z}[i]$, 2 and $2i$ *are* associates. We have $i \in \mathbb{Z}[i]$, and $i \cdot i^3 = 1$ showing that i is a

unit, and hence 2 and $2i$ are indeed associated. $\mathbb{Z}[i]$ is also a Unique Factorization Domain (section 10.3).

5. If R is a UFD, then $R[x]$ is a UFD (chapter 11). Note how this contrasts with PID's ($\mathbb{Z}$ is a PID but $\mathbb{Z}[x]$ is not, see example 9.4.6(3)).

All of these are examples of Unique Factorization Domains. Atomic Domains which are not Unique Factorization Domains are rather difficult to find. Fortunately, there is an easy criterion that allows us to find (a subset of) common atomic domains

10.1.1 Example of Atomic Domains: Noetherian Domains

The goal of this section is to show that Noetherian domains are atomic domains, and that there exists noetherian domains which are not Unique Factorization Domain's. Let's first remember the definition of a Noetherian Domain, or more generally a Noetherian ring:

Definition 10.1.13: Noetherian Rings

Let R be ring. Then if every ideal $I \subseteq R$ is finitely generated, then R is called a *Noetherian Ring*. If R is an integral domain, we call R a *Noetherian Domain*

Most results that will be presented apply to Noetherian rings in general, and so we will work with Noetherian rings. The first thing to show is how to relate Noetherian rings to the idea of factorization. Emmy Noether introduced the following condition in 1921 [50], and it characterizes Noetherian rings:

Proposition 10.1.14: Ascending Chain Condition

Let R be a commutative ring. The following are equivalent:

1. R is a Noetherian ring
2. for every chain of ascending ideals of R

$$I_1 \subseteq I_2 \subseteq \cdots \subseteq I_n \subseteq \cdots$$

there exists an number m such that for all $n > m$, $I_n = I_{n+1} = \cdots$

3. Every nonempty family of ideals of R partially ordered by inclusion has a maximal element

For a noncommutative ring the same proof shows that every left ideal is finitely generated exactly when the left ideals satisfy (2) and (3), and likewise on the right. The left and right conditions can differ.

***Proof*:**

(1) $\Rightarrow$ (2) Let R be a Noetherian ring. If $I_1 = R$ we are done, so assume I_1 is a proper ideal of R, and consider any ascending chain of left ideals

$$I_1 \subseteq I_2 \subseteq I_3 \subseteq \cdots \subseteq I_n \subseteq \cdots$$

Take

$$I = \bigcup_{i=1}^{\infty} I_i$$

which is an ideal for a similar reasoning as presented in the proof that every proper ideal is contained in a maximal ideal (see proposition 9.4.19). Since R is a Noetherian domain and $I \subseteq R$, then I is finitely generated. Let's say $I = (a_1, a_2, ..., a_n)$. Then these generators must each be in some ideal I_m, meaning that for all $n > m$, $I_{n+1} = I_{n+2} = \cdots$, showing that R respects the ascending chain condition

(2) $\Rightarrow$ (3) Let R respect the ascending chain condition on it's left ideals, and let $\mathcal{F}$ be a family of left-ideals of R. Pick some random $I_1 \in \mathcal{F}$. If I_1 is maximal in $\mathcal{F}$, then we'd be done, so let's say I_1 is not maximal in $\mathcal{F}$. Then there exists an I_2 such that $I_1 \subset I_2$. If I_2 is maximal in $\mathcal{F}$, then we are done. If not, then we can continue this process. If every element we pick is not maximal, we can create an infinitely long chain[a]

$$I_1 \subsetneq I_2 \subsetneq I_3 \cdots \subsetneq I_n \subsetneq \cdots$$

However, that contradicts the ascending chain condition, meaning for some n, the chain must stabilize. Thus, $\mathcal{F}$ must have a maximal element

(3) $\Rightarrow$ (1) Let $I \subseteq R$ be a left-ideal. Take $\mathcal{I}$ to be the set of all finitely-generated left-ideals contained in I. The set $\mathcal{I}$ is nonempty since $\{0\} \in \mathcal{I}$. It is thus a family of left-ideals of R and so by the 3rd condition, $\mathcal{I}$ admits a maximal,/Then if say I' [b]. If $I \neq I'$, then there exists an element $x \in I$ such that $x \notin I'$. But then $(I' \cup x)$ is a finitely generated left-ideal larger than I', contradicting the maximality of I'. Hence, $I = I'$, showing that I is finitely generated.

[a] technically, to create this infinitely long chain, we used the axiom of choice

[b] Note that Zorn's lemma cannot be used here since it is not known whether every chain will have an upper bound in this set, their union potentially no longer being finitely generated

10.1.15 Remark This proof will become a special case of the equivalent proof for modules in section 12.3.4.

For our purposes, this let's us think of Noetherian domains in terms of the ascending chain condition. You probably already see how this might start related to Factorization domains: If every element in the ring is finitely generated, it must be decomposable into a finite number of factors. This is essentially true:

Proposition 10.1.16: A.C.C. and Factorization

Let R be an integral domain, and let r be a nonzero nonunit element of R. Then if every ascending chain of *principal ideals*

$$(r) \subseteq (r_1) \subseteq (r_2) \subseteq \cdots \subseteq (r_n) \subseteq$$

stabilizes (there exists an m such that for all $l > m$, $(r_l) = (r_{l+1}) = \cdots$), then r can be factored into irreducible elements. In particular, if R respects the A.C.C. on principal ideals, then R is an atomic domain.

Re-writing these ideal using the divisibility notation, it may be more evident: if $r_1 | r$, $r_2 | r_1$, and so on eventually stabalizes, then r can be factored into irreducible elements.

***Proof*:**
Let's prove this using the contrapositive ($\neg B \Rightarrow \neg A$), so that we will show if r does not factor into irreducible then there exists an ascending chain of principal ideals that does not stabalize.

Since r is not irreducible, then there exists two nonunit non zero element r_1, s_2 suc that $r = r_1 s_2$. Since r does not have a factorization, either r_1 or s_1 are reducible. Without loss of generality, let's say r_1 is reducible. Then $(r) \subseteq (r_1)$ since $s_2 \cdot r_1 = r$, and $r_1 = r_2 s_2$. The same being applied to r_1, we get $(r) \subseteq (r_1) \subseteq (r_2)$. Continuing on indefinitely, we get

$$(r) \subseteq (r_1) \subseteq (r_2) \subseteq \cdots \subseteq (r_n) \subseteq \cdots$$

which never stabilizes since r does not factor, as we sought to show.

Notice how similar this is to Noetherian rings: in a Noetherian ring the ascending chain condition holds for all ideals (proposition 10.1.14), so in particular for its *principal ideals*.

Corollary 10.1.17: Noetherian Domain ⇒ Atomic Domain

Let R be a Noetherian Domain. Then it is an Atomic Domain

***Proof*:**
By proposition 10.1.14, R respects the ascending chain condition on its ideals. Since any element $r \in R$ forms a principal ideal, any ascending chain of principal ideals must stabalize. Thus, by proposition 10.1.16 r must have a factorization. Since r was arbitrary, every element has a factorization, and so R is an Atomic Domain as we sought to show

An atomic domain need not satisfy the ascending chain condition on principal ideals. P. M. Cohn asserted that it does [25], and Grams [33] constructed an atomic domain in which it fails. Proposition 10.1.16 still links a finiteness condition on ideals to a factorization condition.

A counter-example to the converse of corollary 10.1.17 is simplest to verify after Hilbert's Basis Theorem in [16, Noetherian Rings, Hilbert's Basis Theorem, and Localization]. For the reader who wants to know the counter-example, they can either skip to Hilbert's basis theorem in *Commutative Algebra* [16, Hilbert's Basis Theorem (HBT)] (you have everything you need to know to understand it now), or take on faith that "if R is a Noetherian Domain then $R[x_1, x_2, ..., x_n]$ is a Noetherian Domain". The following is a counter-example:

Example 10.1.18: Atomic Domain $\not\Rightarrow$ Noetherian Domain

Let $R = \mathbb{Z}[x_1, x_2, ..., x_n, ...]$ be a polynomial ring over an infinite number of variables (see section 9.2). Notice that $\mathbb{Z}$ is a Noetherian Domain: As we have shown, $\mathbb{Z}$ a Principal Ideal Domain (example 9.4.6, and so any ideal of $\mathbb{Z}$ is generated by one element, and so it is Noetherian. By Hilbert's Basis Theorem, $\mathbb{Z}[x_1, x_2, ..., x_n]$ is a Noetherian Domain as well.

Now, any polynomial $f \in R$ has finitely many terms by definition, and so is also an element of the subring $\mathbb{Z}[x_{\sigma(1)}, x_{\sigma(2)}, ..., x_{\sigma(n)}]$ where $\sigma(i)$ is some appropriate permutation function so that the terms of f are in the aforementioned subring. But then it is an element of a Noetherian Domain, which means it could be factored by proposition 10.1.16. Since $\mathbb{Z}[x_{\sigma(1)}, x_{\sigma(2)}, ..., x_{\sigma(n)}] \hookrightarrow R$, the same factorization happens in R, and since f was arbitrary, R is an Atomic Domain

However, R is not a Noetherian Domain. The ideal $(x_1, x_2, ..., x_n, ...)$ of all of the terms of R is not finitely generated (it does not respect the ascending chain condition).

This means that in the image presented in the beginning of the chapter, the strict inclusion of Noetherian Domains in Atomic Domains is justified. Similarly, Being an Atomic Domain does not mean the factorization is unique

Example 10.1.19: Atomic Domain $\not\Rightarrow$ Unique Factorization Domain

Take the ring F_{0x} from example 10.1.6. Take any element $f \in F_{0x}$, and form a ascending chain of principal ideals. For the sake of contradiction, let's say the chain never stabilizes: $(f) \subsetneq (f_1) \subseteq (f_2) \subsetneq \cdots \subseteq (f_n) \subsetneq \cdots$. Then since $(f_i) \subsetneq (f_j)$, and every degree zero element is a unit (recall that $F_{0x} \subseteq \mathbb{F}[x]$ for a field $\mathbb{F}$) then the degree of f_j must be *less* than that of f_i. This means we have a decreasing chain:

$$\deg(f) > \deg(f_1) > \deg(f_2) > \cdots > \deg(f_n) > \cdots$$

Notice importantly that this chain *must* terminate since there are no polynomials of degree 0. Since f was arbitrary, every element has a factorization and so it is an Atomic Domain.

However, as we have shown in example 10.1.6, $x^6 = x^2x^2x^2 = x^3x^3$ are two factorizations of x^6 where x^2 and x^3 are not associates, showing it is not a Unique Factorization Domain.

Notice the ring F_{0x} is not a Noetherian Domain in the previous example. Using Hilbert's Basis Theorem, we can also find a Noetherian Domain which is not a UFD:

Example 10.1.20: Noetherian Domain $\not\Rightarrow$ Unique Factorization Domain

Let

$$R = \frac{\mathbb{C}[x, y, z, w]}{(xw - yz)}$$

be the polynomial ring over 4 variables quotiented by the ideal $(xw - yz)$ [a]. By Hilbert's Basis Theorem, $\mathbb{C}[x, y, z, w]$ is a Noetherian ring. By proposition 9.4.8, $\frac{\mathbb{C}[x,y,z,w]}{(xw-yz)} = R$ is a noetherian ring. It is also an integral domain, since $(xw - yz)$ is a prime ideal. Indeed $\mathbb{C}[x, y, z, w]$ is a Unique Factorization Domain (theorem 11.2.3), and $xw - yz$ is irreducible: as a polynomial in w over $\mathbb{C}[x, y, z]$ it has degree 1, so any factorization has a factor of w-degree 0, which divides both coefficients x and yz and is therefore a unit. Irreducible elements of a Unique Factorization Domain are prime (proposition 10.1.29). It is therefore a Noetherian Domain (and hence an Atomic Domain).

However, it is not a Unique Factorization Domain. The element x, y, z, w are all irreducible, and not associates. Since $xw - yz = 0$, we have $r = xw = yz$ has two distinct factorization, showing that R is not a Unique Factorization Domain.

[a] In [23, Algebraic Sets and the Zariski Topology], we will see this ring be called the *quadratic cone in* $\mathbb{A}^4$

The converse is not true either: being a Unique Factorization Domain does not imply it is a Noetherian Domain. In fact, the ring $\mathbb{Z}[x_1, x_2, ..., x_n, ...]$ from example 10.1.18 is also a Unique Factorization Domain that is not a Noetherian Domain. A polynomial involves only finitely many variables, say $x_1, \ldots, x_n$, and so does every factor of it, since the degrees in each other variable add up to 0. So its factorizations in $\mathbb{Z}[x_1, x_2, \ldots]$ are exactly its factorizations in $\mathbb{Z}[x_1, \ldots, x_n]$, where the units are also ± 1, and $\mathbb{Z}[x_1, \ldots, x_n]$ is a Unique Factorization Domain by theorem 11.2.3 applied n times, starting from the Unique Factorization Domain $\mathbb{Z}$ (corollary 10.1.34).

Thus, we have shown (or claimed) that

$$\begin{array}{rcl@{\qquad}rcl} \text{Noetherian Domain} & \Rightarrow & \text{Atomic Domain} & \text{Atomic Domain} & \not\Rightarrow & \text{Noetherian Domain} \\ \text{UFD} & \Rightarrow & \text{Atomic Domain} & \text{Atomic Domain} & \not\Rightarrow & \text{UFD} \\ \text{UFD} & \not\Rightarrow & \text{Noetherian Domain} & \text{Noetherian Domain} & \not\Rightarrow & \text{UFD} \end{array}$$

This completes some of the layers in the diagram presented at the beginning of this chapter.

Exercise 10.1.1

1. Let R be a commutative ring generated as a ring by finitely many elements $r_1, \ldots, r_n$, that is, every element of R is a polynomial expression with integer coefficients in $r_1, \ldots, r_n$. Must R be Noetherian?
2. Show that $\mathbb{C}[x, y, z, w]/(xw - yz)$ is an integral domain.
3. Let R be a commutative ring in which every prime ideal is finitely generated. Show that R is Noetherian. (Hint: otherwise, Zorn's lemma gives an ideal I maximal among the ideals of R that are not finitely generated. If $ab \in I$ with $a, b \notin I$, consider the ideals $I + (a)$ and $\{r \in R \mid ra \in I\}$.)

10.1.2 Unique Factorization Domains

We know continue exploring Atomic Domains in which the factorization of each element is unique (up to unit). As a reminder, here is the definition of a UFD:

Definition 10.1.21: Unique Factorization Domain (UFD)

Let R be an integral domain. Then R is a *Unique Factorization Domain* (commonly abbreviated to UFD) if for every non-unit nonzero element $r \in R$, there exists irreducible elements $q_1, q_2, ..., q_n$ such that

$$r = q_1 q_2 \cdots q_n$$

and if there is another set of irreducible element $q_1', q_2', ..., q_m'$ such that $r = q_1' q_2' \cdots q_m'$, then $n = m$, and after some reshuffling the factors agree up to units (i.e. they are associates), that is, for a permutation σ of $\{1, \ldots, n\}$ and units $u_1, \ldots, u_n \in R$

$$q'_{\sigma(i)} = u_i q_i \qquad \text{for } 1 \leq i \leq n$$

Perhaps the most intuitive way of thinking about elements in Unique Factorization Domains is by linking to every nonzero element of the UFD a multiset (a set that allow's multiplicity) of irreducible factors, in which associate irreducible elements are identified. Units are associated to the empty multiset. The multiplication structure can also be thought of as forming a free abelian monoid structure with all prime elements as generators.

Lemma 10.1.22: Irreducible and Multisets

Let R be a Unique Factorization Domain. For a nonzero $r \in R$, write $r = uq_1 q_2 \cdots q_n$ with u a unit and the q_i irreducible ($n = 0$ when r is a unit), and let the *multiset of irreducible elements* of r be the multiset $M(r)$ of the classes of $q_1, \ldots, q_n$ under the relation of being associates. Then $M(r)$ depends only on r, and for nonzero elements $a, b, c \in R$

1. $(a) \subseteq (b)$ if and only if the multiset of irreducible elements of b is contained in the multiset of irreducible elements of a
2. a and b are associates if and only if their corresponding multisets are equal
3. if $a = bc$, then the corresponding multiset of a is the sum of the multisets of b and c, in which multiplicities add

***Proof*:**

A multiset M is contained in a multiset M' when every class occurs in M at most as often as in M'. An associate vq of an irreducible element q is irreducible: it is nonzero, since v and q are nonzero in an integral domain, and not a unit, since otherwise $q = v^{-1}(vq)$ would be a unit, and if $vq = xy$, then $q = (v^{-1}x)y$, so $v^{-1}x$ or y is a unit, and then x or y is a unit.

$M(r)$ depends only on r. An expression $r = uq_1 \cdots q_n$ exists: with $u = r$ and $n = 0$ if r is a unit, and with $u = 1$ by definition 10.1.21 otherwise. If r is a unit, then it has no such expression with $n \geq 1$, for then $q_1(uq_2 \cdots q_n r^{-1}) = 1$ would make q_1 a unit, while irreducible elements are not units (definition 10.1.5). So $M(r)$ is empty for every unit r. If r is not a unit and $r = uq_1 \cdots q_n = u'q_1' \cdots q_m'$, then $n, m \geq 1$, and $r = (uq_1)q_2 \cdots q_n = (u'q_1')q_2' \cdots q_m'$ are two factorizations into irreducible elements, so definition 10.1.21 gives $n = m$ and a matching of the factors in which matched factors are associates. As uq_1 is an associate of q_1 and $u'q_1'$ one of q_1', the two expressions give the same multiset of classes.

1. By definition 10.1.1, $(a) \subseteq (b)$ exactly when $a = bc$ for some $c \in R$, and then $c \neq 0$ because $a \neq 0$. If $a = bc$, then $M(b) \subseteq M(b) + M(c) = M(a)$ by (3), whose proof below does not use (1). Conversely, suppose $M(b) \subseteq M(a)$, and write $a = uq_1 \cdots q_n$ and $b = vp_1 \cdots p_m$ as in the statement. Every class occurs in $M(b)$ at most as often as in $M(a)$, so after renumbering the q_i we have $m \leq n$ and q_i an associate of p_i for $i \leq m$, say $q_i = w_i p_i$ with w_i a unit (proposition 10.1.4). Then

$$a = uw_1 \cdots w_m \, p_1 \cdots p_m \, q_{m+1} \cdots q_n = b \cdot \left(v^{-1} uw_1 \cdots w_m \, q_{m+1} \cdots q_n\right)$$

   so $a \in (b)$, that is, $(a) \subseteq (b)$.

2. By definition 10.1.2, a and b are associates exactly when $(a) \subseteq (b)$ and $(b) \subseteq (a)$, which by (1) holds exactly when $M(b) \subseteq M(a)$ and $M(a) \subseteq M(b)$, that is, when $M(a) = M(b)$.

3. If $b = uq_1 \cdots q_n$ and $c = vq'_1 \cdots q'_m$, then $a = bc = (uv)q_1 \cdots q_n q'_1 \cdots q'_m$ with uv a unit, so $M(a) = M(b) + M(c)$, completing the proof.

This lemma essentially shows that working with elements of a Unique Factorization Domain comes down to working with their corresponding multiset, making the analysis of their structure more often than not being more set theoretical or number theoretical. As an example of this, Unique Factorization Domains make it very easy to define the concept of *greatest common divisor*.

Greatest Common Divisor

The set of sets of multi-sets thus has a natural lattice paralleled on rings given by the divisibility. In fact, meet and joins are well-defined[2]:

Definition 10.1.23: Greatest Common Divisors In Commutative Rings

Let R be a commutative ring and let $a, b \in R$ with $b \neq 0$. Then the *greatest common divisor* of a and b is a nonzero element d such that

1. $d|a$ and $d|b$,
2. if $d'|a$ and $d'|b$ then $d'|d$

A greatest common divisor of a and b will be denoted $\gcd(a, b)$.

Notice that in the definition, R was a commutative ring, since divisibility was defined for commutative rings in general. It should be noted that the gcd of two elements is unique *up to associativity*. it is helpful to put the definition of gCD in ideal-interpretation of divisibility:

Definition 10.1.24: GCD Using Principal Ideals

Let R be a commutative ring and $a, b \in R$, not both zero. An element $d \in R$ is a gcd of a and b if (d) is the smallest principal ideal containing $(a) + (b)$.

In a Principal Ideal Domain, $(a) + (b)$ is itself principal, so $(\gcd(a, b)) = (a) + (b)$.

[2]See appendix D.1.

Since $(d) \supseteq (a) + (b)$ exactly when d divides both a and b, this matches the previous definition of gcd. The ideal $(a) + (b)$ need not be principal: in $\mathbb{Z}[x]$, $\gcd(2, x) = 1$, but $(2) + (x) \neq (1)$, since every element of $(2) + (x)$ has an even constant term. There do exist rings for which the gcd between two elements do not exist (in example 10.1.6, take the ring F_{0x} from the second part of the example. A divisor of x^m in F_{0x} is a nonzero constant times x^k with $x^{m-k} \in F_{0x}$, that is $m - k \neq 1$. So the common divisors of x^5 and x^6 are, up to units, 1, x^2 and x^3, and x^2 does not divide x^3 because $x \notin F_{0x}$, while x^3 does not divide x^2 by degree. Hence $\gcd(x^5, x^6)$ does not exist). For Unique Factorization Domains, every element has a gcd:

Proposition 10.1.25: GCD in Unique Factorization Domain

let R be a Unique Factorization Domain. Then for any $a, b \in R$ with $b \neq 0$, $\gcd(a, b)$ exists. Furthermore, if $a \neq 0$ and

$$a = up_1^{n_1} p_2^{n_2} \cdots p_k^{n_k} \qquad b = vp_1^{m_1} p_2^{m_2} \cdots p_k^{m_k}$$

is the factorization of a and b, that is, every p_i is irreducible, u and v are units, p_i and p_j are not associates for $i \neq j$, and some n_i's or m_i's can be zero. Then the gcd (up to a unit) is:

$$\gcd(a, b) = p_1^{\min(n_1, m_1)} p_2^{\min(n_2, m_2)} \cdots p_k^{\min(n_k, m_k)}$$

***Proof*:**
If $a = 0$, then b is a gcd of a and b (definition 10.1.23): it divides $0 = 0 \cdot b$ and b, and every common divisor of 0 and b divides b. So let $a \neq 0$, and let $d = p_1^{\min(n_1, m_1)} p_2^{\min(n_2, m_2)} \cdots p_k^{\min(n_k, m_k)}$. Since $\min(n_i, m_i) \leq n_i$ and $\min(n_i, m_i) \leq m_i$ for each i, $d \mid a$ and $d \mid b$. Write $M(-)$ for the multisets of lemma 10.1.22. As the p_i are pairwise non-associate, $M(a)$ contains the class of p_i exactly n_i times, $M(b)$ contains it exactly m_i times, and $M(d)$ contains it exactly $\min(n_i, m_i)$ times, with no other classes. If there is some c that also divides a and b ($c \mid a$ and $c \mid b$), then $c \neq 0$ since $a \neq 0$, and $(a) \subseteq (c)$ and $(b) \subseteq (c)$. By lemma 10.1.22(1), $M(c)$ is contained in both $M(a)$ and $M(b)$, so every class occurs in $M(c)$ at most $\min(n_i, m_i)$ times if it is the class of p_i, and not at all otherwise. Hence $M(c) \subseteq M(d)$, so $(d) \subseteq (c)$ by lemma 10.1.22(1), that is, $c \mid d$. So d is a gcd of a and b, as we sought to show.

10.1.26 Relation Between Lattice's and UFDs It should be noted that there are integral domains which are not Unique Factorization Domain's but do have the concept of gcd defined for every element. The class of all integral domains which have the concept of gcd defined for every element is broader than Unique Factorization Domain's and are not always Noetherian, and so such domains need their own name and are called *gcd domains*. In a sense, gcd domains are a generalization of Unique Factorization Domains: A gcd domain is Unique Factorization Domain if and only if the gcd domain satisfies the ascending chain condition on principal ideals (similarly to how Atomic domains do). However, a gcd domain is not always an Atomic domain, and hence fits a bit awkwardly in our current hierarchy. In essence, this is because gcd domains hold information on information between elements, while atomic domains hold element-wise information.
For future purposes, gcd domains do fit into an important hierarchy of domains: *normal domains* shall be introduced in *Commutative Algebra* [16, Integral Elements], and all gcd domains are normal domains:

$$\text{normal domains} \supseteq \text{gcd domains} \supseteq \text{UFD's}$$

The definition of GCD's using ideals suggests that the GCD, when it exists, ought to be unique in general, and indeed it is:

Proposition 10.1.27: Uniqueness Of Greatest Common Divisor (In Integral Domain)

Let R be an integral domain. If (d) and (d') are both greatest common divisors of then $(d) = (d')$.

***Proof*:**
If d and d' are greatest common divisors, then they must each divide each other (condition 2 of definition 10.1.23), and so we will get $(d') \subseteq (d)$ and $(d') \supseteq (d)$, meaning $(d') = (d)$,

Just like for the integers, we may also define the dual of the GCD, the least common multiple. In the following, the LCM will be defined in terms of principle ideals, the similar definition should be translated into elements:

Definition 10.1.28: Least Common Multiple (lCM)

Let R be a commutative ring and $(a), (b) \subseteq R$. Then the *least common muliple* is the ideal I such that $I \subseteq (a)$ and $I \subseteq (b)$ and is the largest such ideal. Namely

$$\text{lcm}((a), (b)) = (a) \cap (b)$$

Properties of UFD's

As we have shown in examples 10.1.6 and 10.1.20 being an irreducible element in an integral domain does not imply that the element is a prime element. This was because prime elements are "stronger" than irreducible in that they not only have to be able to be divisible, but they have to be "globally divisible" for any factorization. However, since factorization is unique in a Unique Factorization

Domain, it should come as no surpise that there is no distinction between prime elements and irreducible elements:

Proposition 10.1.29: Prime If And Only If Irreducible In UFD

Let R be a Unique Factorization Domain. Then a nonzero element is a prime if and only if it is irreducible

***Proof*:**
The fact that prime elements are irreducible was proven in proposition 10.1.7, so we will show the converse. Let a be an irreducible element. Then a is nonzero and not a unit. Now, assume $bc \in (a)$ so that $d \cdot a = bc$. If $b = 0$ or $c = 0$, then that element lies in (a). Otherwise $bc \neq 0$ and $(bc) \subseteq (a)$, so, with $M(-)$ the multisets of lemma 10.1.22, parts (1) and (3) of that lemma give

$$M(a) \subseteq M(bc) = M(b) + M(c)$$

As $a = 1 \cdot a$ with a irreducible, $M(a)$ consists of the class of a alone. That class therefore occurs in $M(b)$ or in $M(c)$, that is, $M(a) \subseteq M(b)$ or $M(a) \subseteq M(c)$, and lemma 10.1.22(1) gives $(b) \subseteq (a)$ or $(c) \subseteq (a)$. In any case, at least one of b or c are in (a), showing that a is prime (definition 10.1.5), as we sought to show.

Thus, if we see every irreducible element being prime as representing "global" irreducibility (or divisibility), then if we further assume the A.C.C. on principle ideals, we see that this ought to fully characterize UFDs. The following theorem demonstrate exactly this:

Theorem 10.1.30: Unique Factorization Domain Equivalence

Let R be an integral domain. Then R is a Unique Factorization Domain if and only if

1. The ascending chain condition for principal ideal holds in R
2. Every irreducible element is a prime element

***Proof*:**

($\Rightarrow$) Let R be a Unique Factorization Domain. Then by proposition 10.1.29 every irreducible element is a prime element, and by corollary 10.1.11 R is an atomic domain, and so by proposition 10.1.16 R respects the ascending chain condition (this can also be proven directly using the corresponding multisets to elements)

($\Leftarrow$) Let R satisfy the A.C.C. for principal ideals and let every irreducible be prime. By proposition 10.1.16, R is an Atomic domain. It remains to show that R factors uniquely. To show this, let n be a nonunit and

$$n = p_1 p_2 \cdots p_n = q_1 q_2 \cdots q_m$$

be two factorizations of n into irreducibles. Then since $q_2 q_3 \cdots q_n \cdot q_1 = p_1 p_2 \cdots p_n$, we have $p_1 p_2 \cdots p_n \in (q_1)$. Since q_1 is irreducible, by the 2nd condition it is a prime, and so for some i, $p_i \in (q_1)$. Without loss of generality, we can say $p_1 \in (q_1)$ by just rearranging and relabelling the factors. Since

$p_1 \in (q_1)$, then $u_1 \cdot q_1 = p_1$. Since p_1 is irreducible and q_1 is not a unit, necessarily u_1 is a unit. Then, we can re-write the equation as

$$u_1 q_1 p_2 \cdots p_n = q_1 q_2 \cdots q_m$$

Cancelling out q_1 from each side and replacing p_2 by $u_1 p_2$ produces us

$$p_2 p_3 \cdots p_n = q_2 q_3 \cdots q_m$$

from here, we can continue this process for the rest of the p_i's, where we will end up with

$$u_1 u_2 \cdots u_n q_1 q_2 \cdots q_m = q_1 q_2 \cdots q_m$$

leting $u = u_1 u_2 \cdots u_n$, we get $u q_1 q_2 \cdot q_n = q_1 q_2 \cdots q_n$. Note that $n = m$ since if not then at some point the remaininng part of the left hand side will be equal to 1, a contradiction to n not being a unit. Thus, the two factorization are equal up to associates, as we sought to show.

Notice how the uniqueness of the factorization allowed us to make the A.C.C. for principal ideals a defining feature of Unique Factorization Domains, in contrast to ho the A.C.C. for principal ideals was insufficient to define Atomic Domains.

This theorem is also very useful in easily remembering how Unique Factorization Domain's and Noetherian Domains are different: If we start with a domain with the A.C.C. for principal ideals (i.e. the types of Atomic domains we ill work with) strengthen the A.C.C. condition to work on all ideals, we get a Noetherian domain. Conversely, if we are working with a Noetherian domain and we loosen the A.C.C. condition to principal ideals but add that all irreducible are primes, then we get a Unique Factorization Domain.

10.1.3 Important UFD: Principle Ideal Domains

Principal Ideal Domains are stronger than Noetherian Domains in the sense that every ideal is generated by a single element (rather than finitely many element for Noetherian Domains). They are therefore immediately Atomic Domains by proposition 10.1.16. By theorem 10.1.30, if a Principal Ideal Domain also has that every irreducible element is prime, then it is automatically a Unique Factorization Domain. Are irreducible elements prime in a Principal Ideal Domain? The answer is yes. Hence PIDs form a subclass of UFDs. Take a moment to think about why irreducible elements are prime in PIDs before proceeding to the next proposition (hint: can you prove prime ideals *must* be maximal ideals in a PID?). From this proof, there will now be a new set of examples for Unique Factorization Domains similarly to how all Noetherian domains are an easy set of examples for Atomic domains

10.1.31 Taking a moment

As the hint suggests, we first prove the following important intermediary result:

Proposition 10.1.32: In PID, Prime Ideal iff Maximal Ideal

Let R be a Principal Ideal Domain and take a nonzero ideal $(r) \subseteq R$. Then (r) is a prime ideal if and only if it is a maximal ideal.

Intuitively, since

***Proof*:**
Let R be a Principal Ideal Domain. If an ideal of R is maximal, by corollary 9.4.26 that ideal is prime. Thus we will prove the converse.

Let $(p) \subseteq R$ be a prime ideal, and assume that $I \supseteq (p)$ is an ideal that contains (p). We must show that $I = (p)$ or $I = R$. Since R is a Principal Ideal Domain, there exists an $m \in R$ such that $(m) = I$. Since $(m) \supseteq (p)$, $p \in (m)$, and so $p = rm$. Since p is a prime element, then either $r \in (p)$ or $m \in (p)$. If $m \in (p)$, then $(m) = (p)$, fulfilling the $I = (p)$ case. If $r \in (p)$ then $r = sp = ps$ and so

$$p = rm = psm$$

And by the cancellation property of integral domains, $1 = sm$, and so $1 \in (m)$ implying that $(m) = R$, fulfilling the $I = R$ condition. That means that (p) fulfills the conditions of being a maximal ideal, as we sought to show.

The main idea is that any ideal requries only 1 element to repersent it, which as it will turn out will be key to why elements will factor uniquely.

Proposition 10.1.33: Prime If And Only If Irreducible In PID

Let R be a Principal Ideal Domain. Then a nonzero element is a prime if and only if it is irreducible. Along with the fact that Principal Ideal Domain are Noetherian Domains (and hence respecting the A.C.C. on principal ideals) we have

$$\text{PID} \Rightarrow \text{UFD}$$

***Proof*:**
Since a Principal Ideal Domain is an integral domain, we have just shown in proposition 10.1.7 that prime elements are irreducible, so it remains to prove the converse.

Let's say p is an irreducible element of R. To show it's prime, we must how (p) is a prime ideal. Since we are working in a Principal Ideal Domain, by proposition 10.1.32, it's equivalent to show that (p) is a maximal ideal. So, for the sake of contradiction, let's assume (p) is not maximal, and so there exists a strictly larger ideal that is the maximal ideal containing (p) $(p) \subsetneq (m) \neq R$ for appropriate $m \in R$. Then

$$p = am$$

for some $a \in R$. Since p is irreducible, then either a or m is a unit in R. If M is a unit, then $m^{-1}m = 1 \in (m)$ implying $(m) = (1) = R$, but then it's not a maximal ideal, since maximal ideals are proper ideals. Conversely, if a is a unit, then by proposition p and m are associates, meaning $(p) = (m)$. But since (m) was maximal, that makes (p) maximal. Thus, (p), by proposition 10.1.32, (p) is a prime ideal.

Therefore, since all Principal Ideal Domain are Noetherian Domains and Noetherian respect

the A.C.C. on principal ideals, by theorem 10.1.30, all Principal Ideal Domain are Unique Factorization Domains, as we sought to show.

There is also a very simple proof I like I found in Eisenbud: (Eisenbud) We know that if R is a Unique Factorization Domain, then it respects the A.C.C. on principal ideals. Use that to find a maximal ideal, which must be generated by one element since it's a Principal Ideal Domain, but then it must be that for one of the elements in the chain, it plateau's, and so it fulfills the ascending chain condition, and so it is a Unique Factorization Domain.

Corollary 10.1.34: $\mathbb{Z}$ is a UFD (Fundamental Theorem of Arithmetics)

The integers ($\mathbb{Z}$) as a ring are a Unique Factorization Domain.

***Proof*:**
$\mathbb{Z}$ is a Principal Ideal Domain: the zero ideal is (0), and a nonzero ideal $I \subseteq \mathbb{Z}$ contains a least positive element n, and for $a \in I$, dividing gives $a = qn + r$ with $0 \leq r < n$ and $r = a - qn \in I$, so $r = 0$ and $I = (n)$. By proposition 10.1.33, every Principal Ideal Domain is a Unique Factorization Domain, completing the proof.

The converse of this theorem is not true, that is if an integral domain is a Unique Factorization Domain, this does not imply that it is a PID, justifying the strict inclusion of the image at the very beginning. Recall from example 9.4.6 that the ideal $(2, x)$ from the ring $\mathbb{Z}[x]$ is not a principal ideal. However, the ring $\mathbb{Z}[x]$ is a Unique Factorization Domain. This is rather difficult to show directly: if we have any any polynomial $p(x) \in \mathbb{Z}[x]$, how can we guarantee it factors and that it's unique up to units, that is that $\mathbb{Z}[x]$ respects the ascending chian conditions on principal ideals and that all irreducibles are primes? On page 397, we shall show this result.

Proposition 10.1.35: Sufficient Condition for Greatest Common Divisor

If a and b are nonzero elements in the commutative ring R such that the ideal gnerated by a and b is a principal ideal (d), then d is a greatest common divisor of a and b

***Proof*:**
The ideal generated by a and b in the commutative ring R is $(a, b) = \{ra + sb \mid r, s \in R\}$, and by hypothesis $(a, b) = (d)$.

Since $a \in (a, b) = (d)$, there is an $r \in R$ with $a = rd$, that is $d \big| a$, and the same argument gives $d \big| b$. Also $d \neq 0$, since otherwise $(d) = 0$ would force $a = 0$.

Now suppose $d' \big| a$ and $d' \big| b$, say $a = d'u$ and $b = d'v$. Since $d \in (d) = (a, b)$, we may write $d = xa + yb$ for some $x, y \in R$, and then

$$d = xd'u + yd'v = d'(xu + yv)$$

so $d' \big| d$. Both requirements of definition 10.1.23 hold, so d is a greatest common divisor of a and b.

Corollary 10.1.36: Coprime Equals Comaximal In A PID

Let R be a principal ideal domain and $a, b \in R$ nonzero. Then 1 is a greatest common divisor of a and b if and only if a and b are comaximal (definition 9.6.3).

***Proof*:**
Write $(a, b) = (d)$. By proposition 10.1.35, d is a greatest common divisor of a and b, and by proposition 10.1.27 every greatest common divisor generates (d). So 1 is a greatest common divisor exactly when $(d) = (1)$, that is, when $(a, b) = (1)$.

Being a unique factorization domain is not enough: $\mathbb{Z}[x]$ is one (theorem 11.2.3, proved later), yet $\gcd(2, x) = 1$ while $(2, x) \neq (1)$ (example 9.6.5).

10.1.37 Warning: Two Meanings of (a, b) Some texts write (a, b) both for $\gcd(a, b)$ and for the ideal generated by a and b. In a PID the clash is harmless by corollary 10.1.36. In $\mathbb{Z}[x]$ it is not: "$(2, x) = 1$" is true for the gcd and false for the ideal. We reserve (a, b) for the ideal.

This proposition explains the (a, b) notation for the gcd of a and b.

Note that an integral domain in which every ieal generated by 2 elements (a, b) is a principal ideal is called a *Bézout Domain* (more generally, if every *finitely generated* ideal is principal, then we have a Bézout Domain. For Principal Ideal Domains, every ideal is principal). Note that it's not a *necessary* condition.

For future reference, we also prove the following[3]:

Proposition 10.1.38: UFD And Maximal, Then PID

Let R be a Unique Factorization Domain in which every nonzero prime ideal is maximal. Then R is a Principal Ideal Domain

***Proof*:**
Step 1: every prime ideal is principal. Let P be a prime ideal of R. If $P = (0)$, it is principal. Otherwise choose $0 \neq r \in P$. It is not a unit, since otherwise $1 = r^{-1}r \in P$ and $P = R$, while a prime ideal is proper (definition 9.4.22). By definition 10.1.21, $r = q_1 q_2 \cdots q_n$ with $n \geq 1$ and each q_i irreducible. As P is prime and $q_1(q_2 \cdots q_n) \in P$, either $q_1 \in P$ or $q_2 \cdots q_n \in P$, and repeating the argument with the shorter product gives $q_i \in P$ for some i. By proposition 10.1.29, q_i is a prime element, so (q_i) is a nonzero prime ideal (definition 10.1.5), hence maximal by hypothesis. Since $(q_i) \subseteq P \neq R$, maximality gives $P = (q_i)$.

Step 2: a maximal non-principal ideal. Let N be the set of ideals of R that are not principal, partially ordered by inclusion, and suppose for a contradiction that $N \neq \emptyset$. Every chain in N has an upper bound in N. For the empty chain, any member of N is one. For a nonempty chain $\mathcal{C}$, the union $U = \bigcup_{A \in \mathcal{C}} A$ is an ideal: two elements $x, y \in U$ lie in a common member of the chain, which contains $x - y$, and rx lies in the member containing x for every $r \in R$. If

[3]This will be useful in number theory

$U = (x)$ were principal, then $x \in A$ for some $A \in \mathcal{C}$, so $(x) \subseteq A \subseteq U = (x)$ and $A = (x)$ would be principal, which is impossible. So $U \in N$, and it contains every member of $\mathcal{C}$. By Zorn's lemma (theorem A.2.3), N has a maximal element $\mathfrak{n}$.

Step 3: the contradiction. The ideal $\mathfrak{n}$ is proper, since $R = (1)$ is principal, and it is not prime by Step 1. So there are $a, b \in R$ with $ab \in \mathfrak{n}$, $a \notin \mathfrak{n}$ and $b \notin \mathfrak{n}$. The ideal $\mathfrak{n} + (a)$ strictly contains $\mathfrak{n}$, so it does not lie in N by the maximality of $\mathfrak{n}$, that is, $\mathfrak{n} + (a) = (u)$ for some $u \in R$. Let

$$J = \{r \in R \mid ru \in \mathfrak{n}\}$$

This is an ideal of R: it contains 0, and for $r, r' \in J$ and $s \in R$, $(r - r')u = ru - r'u \in \mathfrak{n}$ and $(sr)u = s(ru) \in \mathfrak{n}$. It contains $\mathfrak{n}$, because $\mathfrak{n}$ is an ideal. It contains b: writing $u = x + ta$ with $x \in \mathfrak{n}$ and $t \in R$, we get $bu = bx + t(ab) \in \mathfrak{n}$. Since $b \notin \mathfrak{n}$, the ideal J strictly contains $\mathfrak{n}$, so $J \notin N$ and $J = (v)$ for some $v \in R$.

We claim $\mathfrak{n} = (uv)$. Since $v \in J$, $uv \in \mathfrak{n}$, so $(uv) \subseteq \mathfrak{n}$. Conversely, let $x \in \mathfrak{n}$. Then $x \in \mathfrak{n} + (a) = (u)$, so $x = ru$ for some $r \in R$, and $ru = x \in \mathfrak{n}$ gives $r \in J = (v)$, say $r = sv$. Hence $x = suv \in (uv)$. So $\mathfrak{n} = (uv)$ is principal, contradicting $\mathfrak{n} \in N$. Therefore $N = \emptyset$, every ideal of R is principal, and R, being an integral domain, is a Principal Ideal Domain (definition 9.4.9), completing the proof.

Exercise 10.1.2

1. Let $R[x]$ be a Principal Ideal Domain. Then R is a UFD. Conclude that if R is not a field, then $R[x]$ is not a Principal Ideal Domain!

   Since $\mathbb{Z}$ is not a field, then $\mathbb{Z}[x]$ is not a Principal Ideal Domain! In example 9.4.6(3) it was manually shown that $(2, x)$ is not a principal ideal in $\mathbb{Z}[x]$. This theorem makes it easier to prove.

   The converse is true too: if F is a field, $F[x]$ is a Principal Ideal Domain, and in fact a Euclidean Domain (section 11.1).

2. (Challenging) Let R be an integral domain. Then if every prime ideal is principle, R is a PID (In this way, UFDs are almost PIDs, however not all ideals are principle)

3. (If you know Complex Analysis) Show that the ring of entire functions is a gcd domain. Show that the ring of real analytic functions is a Bézout domain.

10.2 Euclidean Domains

So far, we have been exploring divisibility in terms of being able to directly factor elements. However, it is in general difficult to establish when an element is divisible by another (try it out yourself with relatively large elements of some of the more complicated UFDs that were presented). If an element is not divisible by another element is there some way to measure to what "degree" an element is not divisible by another? Can we quantify the lack of divisibility given by the gcd.

In the integers, there is the *euclidean algorithm* that gives the answer. The existence of an algorithm to find the GCD cannot be under-estimated: If given two integers with over a thousand digits, it will

be impossible to find the greatest common divisor in a reasonable amount of time, that is, before the heat death of the universe (until a faster way of factorizing is found). The problem is about our initial choice for divisors. If I say R is a Unique Factorization Domain and $x \in R$, then how would you start finding it's prime factorization? If you have some a and b such that $ab = x$, How do you start decomposing them? Even for the integers, it is in general very difficult to find prime factors of an element x[5].

For the integers, Euclid in approximately the year 300BC discovered that there is a simple algorithm for finding the GCD that quantifies exactly how divisible two numbers are. The key observation is that if $a, b \in \mathbb{Z}$, where $a < b$ (notice the user of the total order, $<$, of $\mathbb{Z}$) then we may write

$$b = ka + r$$

where $r < a$. If $a|b$, then $r = 0$. If $a \nmid b$, then r is the "torsion" or "remainder" that is left-over. In terms of the GCD, if $a|b$, then $\gcd(a, b) = a$, and if $a \nmid b$, then $\gcd(a, b) < a$ and in the "worst case scenario" $\gcd(a, b) = 1$. The problem when generalizing this to general rings is that we require a notion of "size", namely it was crucial to use the fact that $a < b$ and it is imperative that $r < a$ to have a reasonable notion of such a factorization. In this section, we shall endavour to find a function, say $v : R \to \mathbb{N}_{\geq 0}$ that will allow us to assign a natural number to each ring element so that we may recapture this notion from the integers.

Definition 10.2.1: [Ring] Valuation

Any function $v : R \to \mathbb{Z}$ with $v(0) = 0$ is called a *valuation* on the ring R. If $v(a) > 0$ for $a \neq 0$, then v is said to be a *positive valuation*.

10.2.2 Note The notion of valuation shall be defined later in these notes on fields as *real [Archimedean] valuation* or *absolute value* (definition 21.2.1) and generalized to *G-valuation* ([16, $\mathbb{R}$-Valuation])[a]. The notion of a G-valuation is a generalization of the absolute value, however it is often presented with the axiom that $v(0) = \infty$. Valuations can just as well be defined with $v(0) = 0$, and hence the use of the term "valuation" here is justified.

[a] More specifically non-Archimedean valuations shall be generalized to G-valuations

10.2.3 Note There are at least two common uses of the word "valuation" in contemporary abstract algebra: an *absolute-valuation* or more commonly an *absolute value* and a *G-valuation* ([16, $\mathbb{R}$-Valuation]). These two concepts are non-compatible[a], notably G-valuations have axiom dictating that $v(0) = \infty$ while absolute values has an axiom dictating that $v(0) = 0$. The above use of the term valuation is closer to an *absolute value*.

[a] Though an absolute value may induce a G-valuation, this is shown when the concepts are introduced

If you know some analysis, you may wonder if the valuation is related to the one introduced on $\mathbb{R}$ or $\mathbb{C}$ (or normed vector spaces more generally). These concepts are indeed related, and we shall even see more explicit connection in chapter 21⋆.

[5] RSA encryption is based off of the fact that prime factorization is a every difficult task if you do not know a priori what primes you are working with

Returning to the issue at hand, we shall require a valuation N to be defined on an integral domain R that satisfies the following property:

Definition 10.2.4: Euclidean Valuation and Euclidean Domain

Let R be an integral domain and $N : R \to \mathbb{N}_{\geq 0}$ a valuation. Then v is a *euclidean valuation* on R if for any two elements a, b of R with $b \neq 0$ there exists elements q, r in R such that

$$a = qb + r \qquad \text{with } r = 0 \text{ or } N(r) < N(b)$$

The element q is called the *quotient* and the element r the *remainder* of the division
if a euclidean valuation can be defined on a ring, then the ring is said to be a *Euclidean Domain* (or posses a *division algorithm*)

The main thing we can do in euclidean domains is run the division algorithm to keep "dividing" our numbers

$$\begin{aligned} a &= q_0 b + r_0 \\ b &= q_1 r_0 + r_1 \\ r_0 &= 2_0 r_1 + r_2 \\ &\vdots \\ r_{n-1} &= q_{n+1} r_n \end{aligned}$$

Such a sequence will terminate since $N(b) > N(r_0) > \cdots > N(r_n)$, and it will eventually hit 0.

Example 10.2.5: Euclidean Domains

1. Every field with the euclidean valuation is a euclidean domain! In fact, any valuation will work!

   Proof. Let $x \in \mathbb{F}$ be an element of a field. Then since every element in a field has a unit, then
   $$x = xb + 0$$
   where $b = x^{-1}$.

   □

   Thus, a commutative division ring is a field, and hence a Euclidean domain.

2. In general, every unit will trivially satisfy this condition, since if $u \in R$ is a unit, then
   $$u = uu^{-1} + 0$$
   and so terminates instantly. So in a sense, Euclidean domains say that if we have elements that are *not* units, how many properties of divisible things can we put on them?

   Note importantly that euclidean domains must be domains (meaning no zero-divisors)! So oddities with multiplying not increasing the "size" of elements will not happen.

3. Ring of fractions in general are not euclidean domains, since they might have zero-divisors. So even though rings of fractions can be thought of as a way of enlarging a ring to make it's elements more divisible, it cannot do so if it has zero-divisors.

4. The integers $\mathbb{Z}$ are a euclidean domain with $N(a) = |a|$. The division algorithm is your usual euclidean algorithm. If you take $a, b \in \mathbb{Z}$, and $a < b$ then $b = aq + r$. We see that $N(r) = |r| < |b| = N(b)$. This will always be the case with $b \neq 0$! Thus, it is a Euclidean Domain.

5. If $\mathbb{F}$ is a field, then $\mathbb{F}[x]$ is also a Euclidean domain with $N(p(x)) = deg(p(x))$. The division algorithm is your usual polynomial division. The reason is simple for valuation: polynomial division *must* produce a remainder r with degree *less* than b, by definition! This makes $\mathbb{F}[x]$ into a Euclidean Domain, as section 11.1 proves in full.

6. Let
$$\mathbb{Z}[i] = \{a + bi | a, b \in \mathbb{Z}\}$$
And let
$$N(a + bi) = (a + bi)(a - bi) = a^2 + b^2$$
which is the usual valuation on $\mathbb{C}$. Intuitively, this is because we have subrings $\mathbb{Z}[i] \subseteq \mathbb{C}$. These are called the *Gaussian Integers*. We will show that Gaussian integers ($\mathbb{Z}[i]$) is a Euclidean Domain.

   Consider $a, b \in \mathbb{Z}[i]$. We need to show that
$$a = kb + r \qquad N(r) < N(b)$$
for appropriate k, r. Since $\mathbb{Z}[i] \subseteq \mathbb{Q}(i)$, take
$$\frac{a}{b} = \frac{ac + bd}{c^2 + d^2} + \left(\frac{bc - ad}{c^2 + d^2}\right) = r + si$$
this gives us an idea of how close a is to b by having by looking at their fraction. Let p be the closest (or one of the 2 closest) integers to r and q be the closest (or one of the 2 closest) integers to s so that $|r - p|, |s - q| \leq \frac{1}{2}$. Then we will show that
$$a = (p + qi)b + r$$
Whatis r? Notice that $t = (r - p) + (s - q)i$, so that $r = tb$, giving us $r = a - (p + qi)b$, showing that $r \in \mathbb{Z}[i]$. With this, we can calcualte the valuations quite easily! Notice that $N(t) = (r - p)^2 + (s - q)^2$ would at most
$$N(t) \leq \frac{1}{4} + \frac{1}{4} = \frac{1}{2}$$
and so, since $N(\alpha\beta) = \alpha\beta\,\overline{\alpha\beta} = (\alpha\overline{\alpha})(\beta\overline{\beta}) = N(\alpha)N(\beta)$ for all $\alpha, \beta \in \mathbb{C}$, and $N(b) > 0$ for $b \neq 0$,
$$N(r) = N(tb) = N(t)N(b) \leq \frac{1}{2}N(b) < N(b)$$
showing that $\mathbb{Z}[i]$ is indeed a Euclidean Domain.

7. Another class of Euclidean domains come from polynomial rings over fields. The proof is a little more elaborate and so we shall leave it till page 392.

Notice how central defining a valuation is to showing a ring is a euclidean domain. This algorithm *almost* produces a divisibility condition, making Euclidean domains "close" to fields. However, they are not fields, since $\mathbb{Z}$ is a euclidean domain. So instead of every element having an inverse, for every pair of elements (a, b), we can find a number(s) c such that we have a notion of $c|a$ and $c|b$. If every pair of numbers has this condition, we have a Euclidean domain!

We have not seen many examples of Euclidean domains. The following is a class of rings that shall provide a testing ground for finding more Euclidean domains and will come up a lot in [11, Relating Algebra to Number Theory]:

Example 10.2.6: Rings of Integers of Quadratic Extenions

Let D be a squarefree integer (it's prime-factorization has exactly one factor for each prime that appears in it, so $10 = 2 \cdot 5$ is square free, but $18 = 2^2 \cdot 3$ is not). Then we can define $\mathbb{Z}[\sqrt{D}] = \left\{a + b\sqrt{D} | a, b \in \mathbb{Z}\right\}$, which is a ring, even a subring, of the quadratic field $\mathbb{Q}(\sqrt{D})$ (that we defined in the first examples). If we take $D \equiv 1 \bmod 4$, then consider the slighty larger subset

$$\mathbb{Z}\left[\frac{1+\sqrt{D}}{2}\right] = \left\{a + b\frac{1+\sqrt{D}}{2} | a, b \in \mathbb{Z}\right\}$$

is also a subring. With this, define

$$\mathcal{O} = \mathcal{O}_{\mathbb{Q}(\sqrt{(D)})} = \mathbb{Z}[\omega] = \{a + b\omega | a, b \in \mathbb{Z}\}$$

where

$$\omega = \begin{cases} \sqrt{D} & \text{If } D \equiv 2, 3 \bmod 4 \\ \frac{1+\sqrt{D}}{2} & \text{If } D \equiv 1 \bmod 4 \end{cases}$$

called the *ring of integers* in the quadratic field $\mathbb{Q}(\sqrt{D})$. The idea being that the elements of $\mathcal{O}$ of the field $\mathbb{Q}(\sqrt{D})$ have many properties analogous to those of the subring of integers $\mathbb{Z}$ in the field of rational numbers. Or, if we have a "field extension", which in a diagram we can think of as:

$$\begin{array}{ccc} \mathbb{F} & \supseteq & O_{\mathbb{F}} \\ \uparrow & & \uparrow \\ \mathbb{Q} & \supseteq & \mathbb{Z} \end{array}$$

where the arrows represent inclusions, then we want $\mathbb{F}$ to share properties that $\mathbb{Z}$ has Given an extension field $\mathbb{F}$ for $\mathbb{Q}$, we identity rings of integers $O_{\mathbb{F}}$ a subrging of F analogous to $\mathbb{Z} \subset \mathbb{Q}$

Note also that $\mathbb{Z}[\omega] = \mathcal{O}_{\sqrt{(D)}}$ is an integral domain, and that it's field of fraction is $\mathbb{Q}(\sqrt{D})$, giving us another way of thinking of ho $\mathbb{Z}[\omega]$ is sort of the "natural" integral domain to choose.

In the special case where $= -1$, we get $\mathbb{Z}[i]$, called the *Gausian integers*, used to study powers of numbers in the complex plane.

The reason for defining this is because we can define the concept of a *field norm*. Let $N : \mathbb{Q}(\sqrt{D}) \to Q$ by

$$N(a + b\sqrt{D}) = (a + b\sqrt{D})(a - b\sqrt{D}) = a^2 - Db^2 \in \mathbb{Q}$$

which is non-zero if $a + b\sqrt{D} \neq 0$. This gives an idea of "size" in the field $\mathbb{Q}(\sqrt{D})$. For instance, when $D = -1$, then the valuation of $a + bi$ is $a^2 + b^2$ (as we know it from Complex Analysis!),

which is the square of the length of this complex number considered as a vector in the compelx plane.

We can see that N is multiplicative, that is, $N(\alpha\beta) = N(\alpha)N(\beta)$ for all $\alpha, \beta \in \mathbb{Q}(\sqrt{D})$ given by

$$N(a+b\omega) = (a+b\omega)(a-b\overline{\omega}) = \begin{cases} a^2 - b^2 D & \text{if } D \cong 2,3 \bmod 4 \\ a^2 + ab + \frac{1-\sqrt{D}}{2} & \text{if } D \cong 1 \bmod 4 \end{cases}$$

where

$$\overline{\omega} = \begin{cases} -\sqrt{D} & \text{if } D \equiv 2,3 \bmod 4 \\ \frac{1-\sqrt{D}}{2} & \text{if } D \equiv 1 \bmod 4 \end{cases}$$

It follows that $N(\alpha) \in \mathbb{Z}$, meaning we can relate properties of $\mathbb{Z}$ to the field norm!

For example, we can use this norm to charactetrize the units of $\mathbb{Z}[\omega]$. If $N(\alpha) = \pm 1$, then $(a+b\omega)^{-1} = (a+b\overline{\omega})$, which is an element of $\mathbb{Z}[\omega]$, and so α is a unit. Suppose, conversely, that α is a unit in $\mathbb{Z}[\omega]$, so $\exists \beta$ such that $\alpha\beta = 1$. Then since N is multiplicative,

$$N(\alpha)N(\beta)N(\alpha\beta) = N(1) = \pm 1$$

and since $N(\alpha)$, $N(\beta)$ must be integers, they must be ± 1, and so

$$\text{the element } \alpha \text{ is a unit in } \mathcal{O} \text{ if and only if } N(\alpha) = \pm 1$$

Thus, the field norm also gives us an easy way to find the units of a given ring of integers.

We will be using the field norm to how for certain values of D, $\mathbb{Z}[\sqrt{D}]$ will be a Euclidean Domain, and not for others (ex. we have already shown that $\mathbb{Z}[\sqrt{-1}] = \mathbb{Z}[i]$ is a Euclidean Domain).

Euclidean domain ⇒ PID

As mentioned earlier, Euclidean Domains will be useful to factorizing elements in UFDs in which they are defined. Not all UFDs may have a valuation satisfying the euclidean algorithm, namely since being able to define such a va implies a good deal of structure on the ring. One important consequence of the existence of such a norm is that every Eucildean domain is a PID

Proposition 10.2.7: Euclidean Domain ⇒ Pid

Given a Euclidean domain R, then every ideal of R is a principal ideal, meaning R is also a PID. In particular, if I is an ideal, then $I = (d)$, where d is any nonzero element of I of minimum norm

***Proof*:**
If I is the zero ideal, then we are done, so let $I \neq (0)$. This means that we can choose an element $d \in I$ such that $N(d)$ is a minimum element. This exists since $\{N(a) | a \in I,\ a \neq 0\}$ has a minimum element by the well-ordering of the non-negative integers. We now show that $(d) = I$

($\subseteq$) it's clear that $(d) \subseteq I$ by the definition of ideals

($\supset$) Let $a \in I$ be any element of our ideal. Then since R is a euclidean domain, we can write a as :

$$a = qd + r$$

where $r = 0$ or $N(r) < N(d)$. Since $r = a - qd$, and $a, qd \in I$, then $r \in I$. But then, it's impossible that $N(r) < N(d)$ since $N(d)$ is already the minimal norm!! Thus $r = 0$, implying $a = qd$. But then $a = qd \in (d)$, as we sought to show

As usual, the contra-positive should be kept in mind: R is *not* an Euclidean Domain if it is *not* a PID.

Example 10.2.8: Implications

1. Since $\mathbb{Z}$ is a Euclidean domain, it is A Principal Ideal Domain. Note that we did not show that PID $\Rightarrow$ Euclidean domain, we will present a counter-example soon, so this was only a hint in the right direction
2. $\mathbb{Z}[x]$ is *not* a PID since $(2, x)$ is a non-principal ideal, and so not a euclidean domain.
3. $\mathbb{Q}[x]$ is a PID, and can be shown to be the euclidean domain with $N(p(x)) = \deg(p(x))$
4. $\mathbb{Z}[\sqrt{5}]$ is not a PID, and so not a Euclidean Domain

You might wonder if *all* PID's are euclidean domains. That need not be the case. To prove this, we will first prove a another way of finding Euclidean domains:

Definition 10.2.9: Side Divisor

Let $\overline{R} = R^\times \cup \{0\}$ be the collection of units of R together with 0. An element $u \in R - \overline{R}$ is called a *universal side divisor* if for evry $x \in R$ there is some $z \in \overline{R}$ such that u divides $x - z$ in R, i.e., there is a type of "division algorithm" for u: every x may be written as $x = qu + z$ where z is either zero or a unit.

The existence of universal side divisors is weakening of the Euclidean condition, since we are saying that for any element $x \in R$, we can "divide" it by u and have a remainder of a unit z. These conditions are not as general as the conditions for the euclidean algorihtm (where z does not need to be constrained to be $N(z) = 1$). With this, we prove the following equivalent condition:

Proposition 10.2.10: Euclidean Domain Implies Universal Side Divisors

Let R be an integral domain that is not a field. If R is a euclidean domain, then there are universal side divisors in R

***Proof*:**
Suppose R is Euclidean with respect to some norm N and let u be an element $R - \overline{R}$ (which is non-empty since F is not a field), of minimal norm. For any $x \in R$ write $x = qu + r$ where r is etiher 0 or $N(r) < N(u)$. In either case the minimality of u implies $r \in \overline{R}$. Hence u is a univerasl side divisor in R

With this, we can find

Example 10.2.11: : PID but not euclidean domain

Let $\alpha = \frac{1+\sqrt{-19}}{2}$, and take the following ring of integers.

$$R = \mathcal{O}_{\mathbb{Q}(\sqrt{-19})} = \mathbb{Z}[\alpha] = \{a + b\alpha \mid a, b \in \mathbb{Z}\}$$

Since $\alpha^2 = \alpha - 5$, a product $(a + b\alpha)(c + d\alpha) = (ac - 5bd) + (ad + bc + bd)\alpha$ is again of this form, so R is a subring of $\mathbb{C}$ (corollary 9.1.7) and an integral domain. The complex conjugate of $a + b\alpha$ is $a + b\overline{\alpha}$, and $\alpha + \overline{\alpha} = 1$, $\alpha\overline{\alpha} = 5$, so the field norm is

$$N(a + b\alpha) = (a + b\alpha)(a + b\overline{\alpha}) = a^2 + ab + 5b^2$$

Being the restriction of $z \mapsto z\overline{z} = |z|^2$, it is multiplicative, and it takes values in $\mathbb{N}$. This ring is a PID, a fact shown in example 10.2.15. However, it is not a Euclidean domain, for there does not exist a universal side divisor.

Note that if $b \neq 0$, then

$$a^2 + ab + 5b^2 = \left(a + \frac{b}{2}\right)^2 + \frac{19b^2}{4} \geq \frac{19}{4}$$

so $N(a + b\alpha) \geq 5$, while $N(a) = a^2$. Hence the norm never takes the values 2 and 3, the value 1 is taken exactly at ± 1, and the value 4 exactly at ± 2. A unit u has $N(u)N(u^{-1}) = N(1) = 1$, so $N(u) = 1$, and $\overline{R} = \{0, \pm 1\}$ are the units of R together with 0. In particular R is not a field. If $2 = \beta\gamma$, then $4 = N(\beta)N(\gamma)$, so one of β, γ has norm 1 and is ± 1: the divisors of 2 are ± 1 and ± 2. If $3 = \beta\gamma$, then $9 = N(\beta)N(\gamma)$, and since 3 is not a value of N, again one factor is ± 1: the divisors of 3 are ± 1 and ± 3.

Now suppose $u \in R$ is a universal side divisor (definition 10.2.9). Taking $x = 2$, u divides one of $2 - 0$, $2 - 1$ and $2 + 1$, and it does not divide 1, since it is not a unit. So u is a nonunit divisor of 2 or of 3, that is, $u \in \{\pm 2, \pm 3\}$. Taking $x = \alpha$, u divides one of α, $\alpha - 1$ and $\alpha + 1$, whose norms are 5, 5 and 7. But a multiple $w = qu$ of ± 2 has $N(w) = 4N(q)$ divisible by 4, and a multiple of ± 3 has norm divisible by 9, while neither 4 nor 9 divides 5 or 7. So R has no universal side divisor, and by proposition 10.2.10 it is not a Euclidean domain.

It is tempting to think that PIDs are close to being Euclidean domains as they share a notion of gcd and lcm, unique factorization. The following measure the degree of seperation of a Principal Ideal Domain being a Euclidean Domain:

Definition 10.2.12: Dedikend-Hasse Valuation

Define N to be the *Dedekind-Hasse valuation* if N is a positive norm and for every nonzero $a, b \in R$ either a is an element of the ideal (b) or there is a nonzero elemtn of the ideal (a, b) of norm strictly smaller than the norm of b (i.e., either b divides a in R or their exists $s, t \in R$ such that $0 < N(sa - tb) < N(b)$

Note that R is Euclidean with respect to a positive norm N if it is always possible to satisfy the Dedekind-Hasse norm with $s = 1$, so this is indeed a weakening of our condition for Euclidean Domains.

Proposition 10.2.13: Principal Ideal Domain If And Only If Dedekind-Hasse Norm

Let R be an integral domain.

1. If R has a Dedekind-Hasse valuation, then R is a Principal Ideal Domain.
2. If R is a Principal Ideal Domain, let $n(a)$ be the number of irreducible factors of a nonzero $a \in R$, counted with multiplicity. Then $N(0) = 0$ and $N(a) = 2^{n(a)}$ for $a \neq 0$ define a Dedekind-Hasse valuation on R, and $N(ab) = N(a)N(b)$ for all $a, b \in R$.

In particular, R is a Principal Ideal Domain if and only if it has a Dedekind-Hasse valuation.

***Proof*:**

1. Suppose N is a Dedekind-Hasse valuation on R and let $I \subseteq R$ be an ideal. If $I = 0$ then $I = (0)$, so assume $I \neq 0$ and choose a nonzero $b \in I$ with $N(b)$ minimal, which is possible because N takes positive integer values on nonzero elements and $\mathbb{N}$ is well ordered. Since I is an ideal, $(b) \subseteq I$. For the reverse inclusion take a nonzero $a \in I$. By definition 10.2.12, either $a \in (b)$, or there are $s, t \in R$ with

$$0 < N(sa - tb) < N(b)$$

The second alternative cannot occur: $sa - tb$ lies in I, and $N(sa - tb) > 0$ forces $sa - tb \neq 0$ since $N(0) = 0$, so $sa - tb$ would be a nonzero element of I of norm smaller than $N(b)$, against the choice of b. Hence $a \in (b)$, so $I = (b)$. Every ideal of R is principal, and R is an integral domain by hypothesis, so R is a Principal Ideal Domain.

2. Suppose R is a Principal Ideal Domain. By proposition 10.1.33, R is a Unique Factorization Domain, so every nonzero $a \in R$ may be written $a = up_1p_2 \cdots p_n$ with u a unit, the p_i irreducible, and $n \geq 0$ (the case $n = 0$ being a a unit). By definition 10.1.21 the integer n depends only on a, so

$$N(0) = 0 \qquad N(a) = 2^n \quad \text{for } a \neq 0$$

is well defined, and it is a positive valuation in the sense of definition 10.2.1, since $N(a) \geq 1$ for $a \neq 0$. Counting irreducible factors on both sides of a product shows

$$N(ab) = N(a)N(b)$$

for all $a, b \in R$. In particular a nonzero x is a unit if and only if $N(x) = 1$, and $N(x) \geq 2$ for a nonzero nonunit.

It remains to check the Dedekind-Hasse condition for nonzero $a, b \in R$. The ideal (a, b) is principal, say $(a, b) = (r)$, and $r \neq 0$ because $a \in (r)$ is nonzero. Assume $a \notin (b)$. Since $b \in (r)$ we may write $b = xr$. Here x is not a unit, for otherwise $(b) = (r) \ni a$. So $N(x) \geq 2$ and

$$N(b) = N(x)N(r) \geq 2N(r) > N(r)$$

Finally $r \in (a, b)$ means $r = sa - tb$ for suitable $s, t \in R$, and $0 < N(r) < N(b)$. Thus N is a Dedekind-Hasse valuation on R, completing the proof.

Corollary 10.2.14: Multiplicative Dedekind-Hasse Norm

Let R be a Principal Ideal Domain. Then R has a multiplicative Dedekind-Hasse norm.

***Proof*:**
The valuation N of proposition 10.2.13(2) is a Dedekind-Hasse valuation on R with $N(ab) = N(a)N(b)$ for all $a, b \in R$, completing the proof.

Example 10.2.15: : $\mathbb{Z}[\frac{1+\sqrt{-19}}{2}]$ is a PID

Let $R = \mathbb{Z}[\alpha]$ with $\alpha = \frac{1+\sqrt{-19}}{2}$ and norm $N(a + b\alpha) = a^2 + ab + 5b^2$, as in example 10.2.11. The norm N is a Dedekind-Hasse valuation (definition 10.2.12), so R is a Principal Ideal Domain by proposition 10.2.13(1), and by example 10.2.11 it is a Principal Ideal Domain that is not a Euclidean Domain. The norm takes positive integer values on nonzero elements and $N(0) = 0$, so it remains to check the condition on nonzero $u, v \in R$ with $u \notin (v)$.

Step 1: elements of R and of $K = \{p + q\sqrt{-19} \mid p, q \in \mathbb{Q}\}$. As $a + b\alpha = \frac{(2a+b)+b\sqrt{-19}}{2}$, the ring R consists of the numbers $\frac{m+n\sqrt{-19}}{2}$ with $m, n \in \mathbb{Z}$ of the same parity. In particular $R \subseteq K$, and R contains $\mathbb{Z}[\sqrt{-19}]$. The formula $N(p + q\sqrt{-19}) = p^2 + 19q^2 = |p + q\sqrt{-19}|^2$ extends N to a multiplicative function on K. Every $\gamma \in K$ can be written $\gamma = \frac{a+b\sqrt{-19}}{c}$ with $a, b, c \in \mathbb{Z}$ and $c \geq 1$, and choosing c minimal, a, b and c have no common divisor $d > 1$, since dividing by it would give a smaller denominator.

Step 2: reduction. Let $u, v \in R$ be nonzero with $u \notin (v)$, and put $\gamma = \frac{u}{v} = \frac{u\overline{v}}{N(v)}$. The conjugate $\overline{v}$ lies in R, since $\overline{a + b\alpha} = (a + b) - b\alpha$, so $\gamma \in K$, and $\gamma \notin R$, since $u = \gamma v$ and $u \notin (v)$. Suppose $s, t \in R$ satisfy $0 < N(s\gamma - t) < 1$. Then $su - tv = v(s\gamma - t)$ has

$$0 < N(su - tv) = N(v)N(s\gamma - t) < N(v)$$

which is the Dedekind-Hasse condition. So it suffices to find such s, t for every $\gamma = \frac{a+b\sqrt{-19}}{c} \in K \setminus R$ written as in Step 1. Then $c \neq 1$, since $a + b\sqrt{-19} \in R$.

Step 3: the case $c \geq 5$. The set $I = \{ax + by + cz \mid x, y, z \in \mathbb{Z}\}$ is an additive subgroup of $\mathbb{Z}$ containing $c > 0$. Let d be its least positive element. Dividing an element $e \in I$ by d gives $e = q_1 d + r_1$ with $0 \leq r_1 < d$ and $r_1 = e - q_1 d \in I$, so $r_1 = 0$ and $d \mid e$. In particular d divides a, b and c, so $d = 1$, and $ax + by + cz = 1$ for some $x, y, z \in \mathbb{Z}$. Divide $ay - 19bx$ by c with a remainder of least absolute value, $ay - 19bx = cq + r$ with $q, r \in \mathbb{Z}$ and $|r| \leq c/2$, and put $s = y + x\sqrt{-19}$ and $t = q - z\sqrt{-19}$, both in $\mathbb{Z}[\sqrt{-19}] \subseteq R$. Then

$$s\gamma - t = \frac{(ay - 19bx) + (ax + by)\sqrt{-19}}{c} - q + z\sqrt{-19} = \frac{r + (ax + by + cz)\sqrt{-19}}{c} = \frac{r + \sqrt{-19}}{c}$$

so $N(s\gamma - t) = \frac{r^2+19}{c^2} > 0$. For $c = 5$ the integer r satisfies $|r| \leq 2$, and $N(s\gamma - t) \leq \frac{23}{25} < 1$. For $c \geq 6$, $r^2 \leq \frac{c^2}{4}$ and $19 \leq \frac{19c^2}{36}$, so $N(s\gamma - t) \leq \frac{1}{4} + \frac{19}{36} = \frac{7}{9} < 1$.

Step 4: the cases $c = 2, 3, 4$. Squares of integers are 0 or 1 modulo 3, with 0 exactly for multiples of 3, and the square of an odd integer $2k + 1$ is $4k(k + 1) + 1 \equiv 1 \bmod 8$.

- $c = 2$: a and b are not both even, and not both odd, since $\gamma \notin R$ (Step 1). So $a - 1$ and b have the same parity, $t = \frac{(a-1)+b\sqrt{-19}}{2}$ lies in R, and $s = 1$ gives $s\gamma - t = \frac{1}{2}$, of norm $\frac{1}{4}$.

- $c = 3$: a and b are not both multiples of 3, so $a^2 + 19b^2 \equiv a^2 + b^2 \bmod 3$ is not divisible by 3. Write $a^2 + 19b^2 = 3q + r$ with $r \in \{1, 2\}$, and put $s = a - b\sqrt{-19}$ and $t = q$. Then $s\gamma = \frac{a^2+19b^2}{3}$, so $s\gamma - t = \frac{r}{3}$, of norm $\frac{r^2}{9} \leq \frac{4}{9}$.
- $c = 4$: a and b are not both even. If exactly one of them is odd, then $a^2 + 19b^2 = 4q + r$ with $r \in \{1, 3\}$, and $s = a - b\sqrt{-19}$, $t = q$ give $s\gamma - t = \frac{r}{4}$, of norm at most $\frac{9}{16}$. If both are odd, then $a^2 + 19b^2 \equiv 1 + 19 \equiv 4 \bmod 8$, say $a^2 + 19b^2 = 8q + 4$, and $s = \frac{a-b\sqrt{-19}}{2} \in R$, $t = q$ give $s\gamma = \frac{a^2+19b^2}{8} = q + \frac{1}{2}$, so $s\gamma - t = \frac{1}{2}$, of norm $\frac{1}{4}$.

In every case $0 < N(s\gamma - t) < 1$, so N is a Dedekind-Hasse valuation and R is a Principal Ideal Domain.

10.3 Application: Sums of Primes and Gaussian Integers

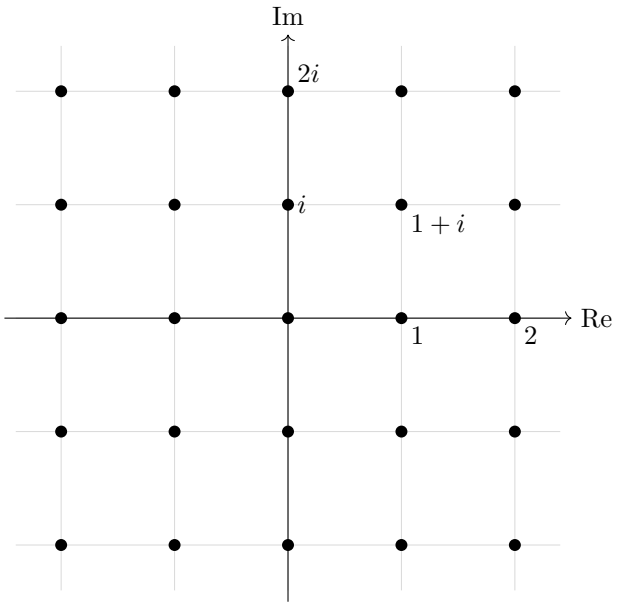
Let us use the theory we have developed to get our first result about prime numbers, namely let us classify all prime numbers that can be written as the sum of two squares $p = x^2 + y^2$. [6]

Recall that the Gaussian integers are of the form:

$$\mathbb{Z}[i] \subseteq \mathbb{C}$$

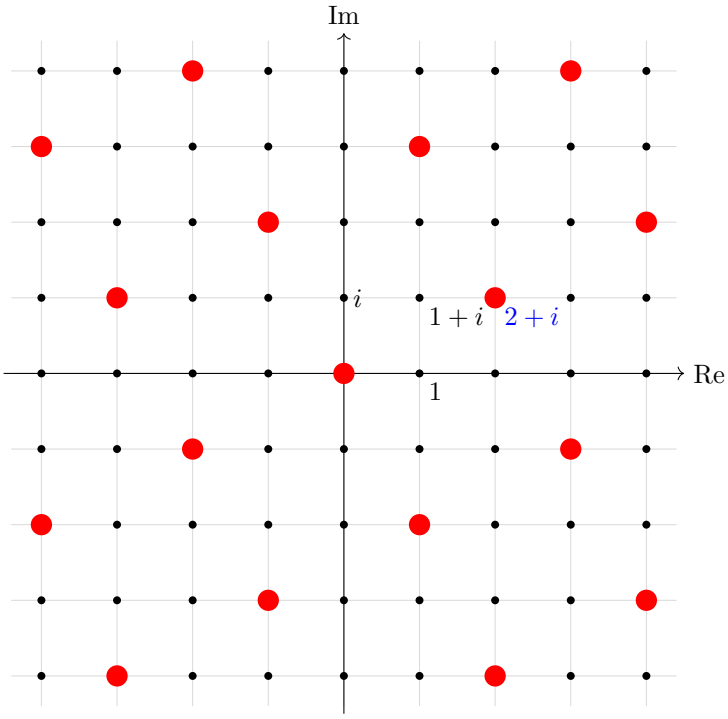
which can be thought of as the smallest ring containing $\mathbb{Z}$ and i. The Gaussian integers have a pretty geometric picture as a lattice:

[6] A prime p is never the square of an integer, but it may be congruent to a square modulo another prime q, in which case p is called a *quadratic residue* modulo q. For distinct odd primes p and q, the law of *quadratic reciprocity* relates whether p is a square modulo q to whether q is a square modulo p ([11, Application: Quadratic Extensions and Quadratic Reciprocity]). This section needs a single such residue: the proof of theorem 10.3.3 reduces the question to whether -1 is a square modulo p.



The Gaussian Integers $\mathbb{Z}[i]$

This gives us a geometric picture of this ring! For example, we may visualize principal ideals in this grid, for example the ideal $(2 + i)$ is the point:



The Gaussian Integers $\mathbb{Z}[i]$ with $(2+i)$ highlighted

Recall how we introduced the field-norm on on quadratic integers, where $\mathbb{Z}[i] = \mathbb{Z}[\sqrt{-1}]$ is an example of quadratic integers. With the above visuals, it is more clear to see that the field norm has similar geometric information to the norm presented in real or complex analysis, namely:

$$N(a+bi) = (a+bi)(a-bi) = a^2 + b^2 = \|a+bi\|^2$$

Using the fact that the codomain of the field norm is $\mathbb{N}_{\geq 0}$, we can see that the norm is in fact a euclidean valuation

Lemma 10.3.1: Gaussian Integers Are Euclidean Domain

Let $R = \mathbb{Z}[i]$ and N be the field norm. Then N is a Euclidean valuation. Furthermore, N is multiplicative so that for every $z, w \in \mathbb{Z}[i]$,

$$N(zw) = N(z)N(w)$$

***Proof*:**
For $\alpha \in \mathbb{C}$ put $N(\alpha) = \alpha\overline{\alpha} = |\alpha|^2$, which extends the norm of $\mathbb{Z}[i]$ since $(a+bi)(a-bi) = a^2+b^2$. Complex conjugation satisfies $\overline{\alpha\beta} = \overline{\alpha}\,\overline{\beta}$, so for all $\alpha, \beta \in \mathbb{C}$

$$N(\alpha\beta) = \alpha\beta\,\overline{\alpha}\,\overline{\beta} = (\alpha\overline{\alpha})(\beta\overline{\beta}) = N(\alpha)N(\beta)$$

and the case $\alpha, \beta \in \mathbb{Z}[i]$ is the multiplicativity claim.

The ring $\mathbb{Z}[i]$ is an integral domain, being a subring of the field $\mathbb{C}$, and N takes values in $\mathbb{N}_{\geq 0}$ with $N(0) = 0$, so it is a valuation as required in definition 10.2.4. Let $z, w \in \mathbb{Z}[i]$ with $w \neq 0$, so that $N(w) > 0$. Write $z\overline{w} = m + ni$ with $m, n \in \mathbb{Z}$. In $\mathbb{C}$,

$$\frac{z}{w} = \frac{z\overline{w}}{w\overline{w}} = \frac{m}{N(w)} + \frac{n}{N(w)}\, i = x + yi$$

with $x, y \in \mathbb{Q}$. Choose integers s and t nearest to x and y, so that $|x - s| \leq \frac{1}{2}$ and $|y - t| \leq \frac{1}{2}$, and put $q = s + ti \in \mathbb{Z}[i]$ and $r = z - qw \in \mathbb{Z}[i]$. Then $z = qw + r$, and in $\mathbb{C}$

$$r = w\left(\frac{z}{w} - q\right) = w\big((x - s) + (y - t)i\big)$$

so that, by multiplicativity,

$$N(r) = N(w)\left((x - s)^2 + (y - t)^2\right) \leq N(w)\left(\frac{1}{4} + \frac{1}{4}\right) = \frac{N(w)}{2} < N(w)$$

Hence $r = 0$ or $N(r) < N(w)$, and N is a Euclidean valuation on $\mathbb{Z}[i]$ (definition 10.2.4), completing the proof.

What could be more insightful is to understand using the lattice picture the significant of the field norm on $\mathbb{Z}[i]$ (this was demonstrated in Aluffi's textbook see [1]). Take $z, w \in \mathbb{Z}[i]$ with $w \neq 0$ nd consider the lattice given by (w). Then either z lies on a point in this lattice, in which case $z = qw$ for appropriate q, or z is inside a "box" of the lattice. Pick some points qw that is a vertex of the box containing z, and set $r = z - qw$ so that $z = qw + r$. By visual inspection, we see that the length of r is ells than that of r, namely:

$$N(r) < N(w)$$

This trick shall not work for all quadratic fields, as we shall demonstrates after finishing elaborating on Gaussian integers.

If u is a unit, then $N(u)N(u^{-1}) = N(1) = 1$ with both factors non-negative integers, so $N(u) = 1$, and conversely $N(u) = u\overline{u} = 1$ exhibits $\overline{u}$ as an inverse of u. As $a^2 + b^2 = 1$ forces $(a, b) = (\pm 1, 0)$ or $(0, \pm 1)$, the units are exactly $1, -1, i, -i$. We can use the norm to also detect irreducible elements. To see this, let' say $p \in \mathbb{Z}$ is a prime. Then since $\mathbb{Z}$ is an integral domain, p is irreducible, and so $p = ab$ if and only if a or b is a unit. Let's now take $\pi \in \mathcal{O}$ such that $p = N(\pi)$. If $\pi = \alpha \cdot \beta$, then:

$$p = N(\pi) = N(\alpha)N(\beta)$$

meaning either $N(\alpha)$ or $N(\beta)$ is a unit, meaning it's value is ± 1. Thus, we have that

$$\text{If } N(\alpha) \text{ is } \pm p \text{ for a prime } p \in \mathbb{Z}, \text{ then } \alpha \text{ is irreducible (even prime) in } \mathbb{Z}[i]$$

The converse is not necessarily true: we can have a prime in $\mathbb{Z}[i]$ but it's norm is not p. Fortunately, this is a feature of primes in $\mathbb{Z}[i]$ and demonstrates a deeper insight on the structure of primes in $\mathbb{Z}[i]$:

Lemma 10.3.2: Gaussian Integers Norm of Primes

Let $\pi \in \mathbb{Z}[i]$ be a prime element. Then

$$N(\pi) \in \{p, p^2\}$$

for appropriate prime $p \in \mathbb{Z}$, which can be chosen so that π divides p in $\mathbb{Z}[i]$.

Fermat's theorem of sum of square (theorem 10.3.3) shall show that both p and p^2 show up as norms of primes.

***Proof*:**

Let $\pi \in \mathbb{Z}[i]$ be a prime element, that is, (π) is a prime ideal in $\mathbb{Z}[i]$. As π is not a unit, $N(\pi) \neq 1$, so the integer $N(\pi) = \pi\overline{\pi} \geq 2$ is a product of prime numbers (corollary 10.1.34). The prime element π divides this product, so by induction on the number of factors it divides one of them, say p.

Now, take $(p) \subseteq \mathbb{Z}[i]$. Then $(p) \subseteq (\pi)$ so that $p = x\pi$ for some $x \in \mathbb{Z}[i]$ (which may be a unit, as p may be an associate of π). Then $N(p) = p^2$, and since the norm is multiplicative, $p^2 = N(x)N(\pi)$ and

$$N(\pi)\big|N(p) = p\overline{p} = pp = p^2$$

so $N(\pi) \in \{1, p, p^2\}$, and $N(\pi) \neq 1$ gives $N(\pi) \in \{p, p^2\}$, completing the proof.

Irreducible does not imply prime. In $\mathbb{Z}[\sqrt{-5}]$, with norm $N(a + b\sqrt{-5}) = a^2 + 5b^2$, the element $1 + \sqrt{-5}$ has norm 6. It is irreducible, since a proper factor would have norm 2 or 3, and $a^2 + 5b^2$ takes neither value. It is not prime: it divides $2 \cdot 3 = (1 + \sqrt{-5})(1 - \sqrt{-5})$ but neither 2 nor 3, as 6 divides neither $N(2) = 4$ nor $N(3) = 9$. The argument above applies to a *prime* element π of any quadratic ring $\mathbb{Z}[\sqrt{d}]$ with norm $N(a + b\sqrt{d}) = a^2 - db^2$: π divides $N(\pi) = \pi\overline{\pi}$, hence some prime p of $\mathbb{Z}$, and so $N(\pi) = \pm p$ or $\pm p^2$. For irreducible elements that are not prime it can fail, as $N(1 + \sqrt{-5}) = 6$ shows. Under the right framework, this connection can be made more precise, which will be part of the aim of [11, Relating Algebra to Number Theory].

Let us find some examples of primes with norms p and p^2. Let's take $\alpha = a + bi \in \mathbb{Z}[i]$ to be prime. Then $N(\alpha) = \alpha\overline{\alpha} = a^2 + b^2 \in \{p, p^2\}$. If p is reducible in $\mathbb{Z}[i]$,then

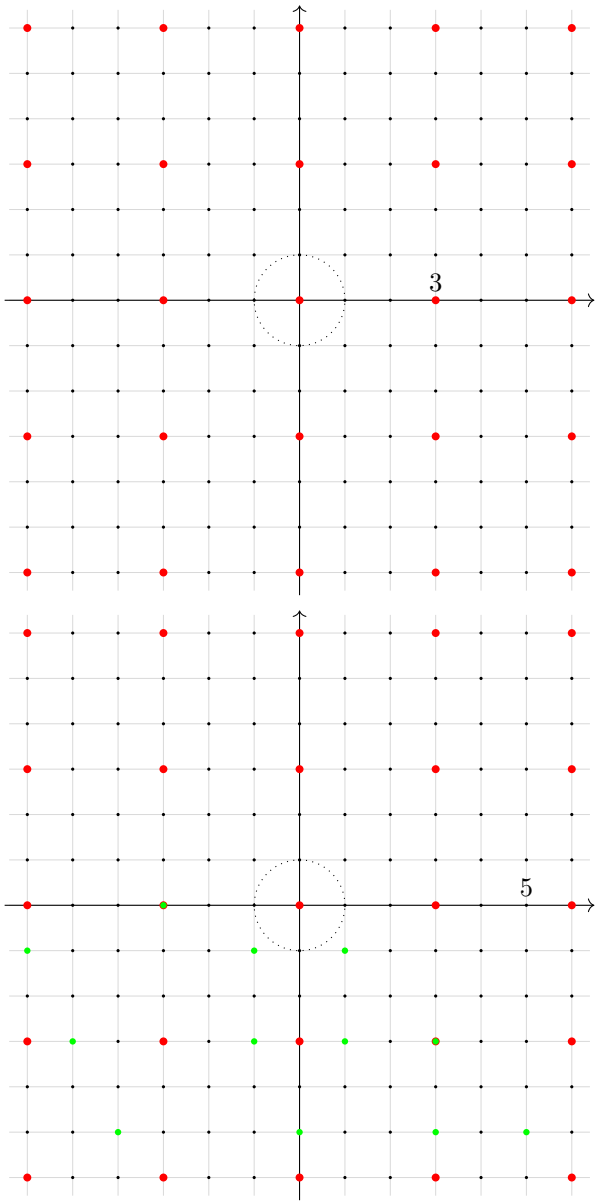
$$p = \pi\pi' \qquad \Rightarrow \qquad N(p) = N(\pi)N(\pi') = p^2$$

Thus, p will factor in $\mathbb{Z}[i]$ into two irreducibles if and only if $p = a^2 + b^2$, that is, p is the sum of two integers squares. If p is not the sum of two integer squares, it will remain irreducible. Take or example $3 \in \mathbb{Z}[i]$. Then $N(3) = 9$, and if 3 factored we would need an element $a + bi$ such that $N(a+bi) = a^2+b^2 = 3$ for natural numbers $a, b \in \mathbb{N}$. However, it can be quickly verified that no such numbers exist, and so 3 is prime in $\mathbb{Z}[i]$. For another example, take 2. Then as $2 = 1^2 + 1^2$, 2 factors in $\mathbb{Z}[i]$! In particular, $2 = (1 + i)(1 - i) = -i(1 + i)^2$ (note that $N(-i) = 1$ and so these two are associates). This will be the only case were conjugate irreducibles $a + bi$ and $a - bi$ can be associate. Let us next consider 5, so we need to see if there is an element $a + bi$ such that $N(a + bi) = 5$, and indeed:

$$N(2 + 1 \cdot i) = 2^2 + (1)^2 = 5$$

from which we find $5 = (2 + i)(2 - i)$. This factorization picture can be seen more clearly when viewing this elements via their ideals in the lattice. In the case of 3, there is no possible further

refinement that can be done to the lattice, while for 5 we may further refine the lattice (without it being the entire lattice):



Let us now find a condition that shall let us immediately tell if a prime can splits in $\mathbb{Z}[i]$. First, any number in $\mathbb{Z}$ is congruent to $0, 1, 2, 3 \bmod (4)$. Therefore, the square of any number is congruent to either 0 or 1. The possible sums of square are thus $0, 1, 2 \bmod 4$. If the sum is $0, 2 \bmod 4$, then it is even, so a prime $p = a^2 + b^2$ of this kind is $p = 2$. Thus an odd prime (so $p > 2$) in $\mathbb{Z}$ that is the sum of two squares must be congruent to 1 mod 4 (ex. 5). Therefore, if $p \equiv 3 \bmod 4$ in $\mathbb{Z}$, then p is *not* the sum of squares, and remains irreducible (and thus prime) in $\mathbb{Z}[i]$.

Theorem 10.3.3: Fermat's Theorem on Sums Of Squares

The prime p is the sum of two integer squares, $p = a^2 + b^2$, $a, b \in \mathbb{Z}$ if and only if $p = 2$ or $p \equiv 1 \bmod 4$. Except for interchanging a and b or changing the sign of a and b, the representation of p as a sum of two squares is unique.

Furthermore, up to multiplication by the units $\pm 1, \pm i$, the prime elements in the Gaussian integers $\mathbb{Z}[i]$ are as follows:

1. $1 + i$ (which has norm 2)
2. the primes $p \in \mathbb{Z}$ with $p \equiv 3 \bmod 4$ (which have norm p^2)
3. $(a+bi)$, $(a-bi)$, the two non-associate irreducible factors of $p = a^2+b^2 = (a+bi)(a-bi)$ for the primes $p \in \mathbb{Z}$ with $p \equiv 1 \bmod 4$ (both of which have norm p).

In other words, each prime $p \equiv 1 \bmod 4$ can be written as the sum of two squares:

$$p = a^2 + b^2$$

***Proof*:**

The ring $\mathbb{Z}[i]$ is a Euclidean domain (lemma 10.3.1), hence a Principal Ideal Domain (proposition 10.2.7), in which irreducible and prime elements coincide (proposition 10.1.33). Its units are the elements of norm 1, since $uv = 1$ gives $N(u)N(v) = 1$ in $\mathbb{N}$ and $N(u) = u\overline{u} = 1$ exhibits an inverse of u, and the solutions of $a^2 + b^2 = 1$ give the units $\pm 1, \pm i$. An element π whose norm is a prime number q is irreducible: it is not a unit, and if $\pi = \alpha\beta$, then $q = N(\alpha)N(\beta)$, so α or β has norm 1 and is a unit.

Step 1: a prime $p = a^2 + b^2$ is 2 or $\equiv 1 \bmod 4$. The squares $(2k)^2 = 4k^2$ and $(2k+1)^2 = 4k(k+1)+1$ are 0 or 1 modulo 4, so $a^2 + b^2$ is 0, 1 or 2 modulo 4. An odd sum is therefore 1 modulo 4, and an even prime is 2. Conversely $2 = 1^2 + 1^2$.

Step 2: if $p \equiv 1 \bmod 4$, then p divides $n^2 + 1$ for some $n \in \mathbb{Z}$. Write $p = 4k + 1$. The group $(\mathbb{Z}/p\mathbb{Z})^\times$ of definition 1.2.5 consists of the classes of $1, 2, \ldots, p-1$, all coprime to the prime p. A class $\overline{x}$ with $1 \le x \le p-1$ equals its own inverse exactly when p divides $x^2 - 1 = (x-1)(x+1)$. This happens for $x = 1$ and $x = p-1$, and for no $2 \le x \le p-2$, since then $x-1$ and $x+1$ lie between 1 and $p-1$, so p divides neither, nor their product ([24, Uniqueness Of Prime Factorization]). The classes $\overline{1}$ and $\overline{p-1} = \overline{-1}$ are distinct as $p > 2$, and the remaining classes fall into pairs $\{\overline{x}, \overline{x}^{-1}\}$ of distinct classes with product $\overline{1}$. As $(\mathbb{Z}/p\mathbb{Z})^\times$ is abelian, the product of all its elements is therefore $\overline{1} \cdot \overline{-1} = \overline{-1}$, that is, $(p-1)! \equiv -1 \bmod p$. Since $p - j \equiv -j \bmod p$,

$$(p-1)! = \prod_{j=1}^{2k} j \cdot \prod_{j=1}^{2k} (p-j) \equiv (-1)^{2k} \big((2k)!\big)^2 = \big((2k)!\big)^2 \bmod p$$

So $n = (2k)!$ satisfies $n^2 \equiv -1 \bmod p$, that is, $p \big| n^2 + 1$.

Step 3: if $p \equiv 1 \bmod 4$, then p is a sum of two squares. With n from Step 2, p divides

$n^2+1=(n+i)(n-i)$ in $\mathbb{Z}[i]$, but it divides neither factor, since $\frac{n}{p}\pm\frac{1}{p}i\notin\mathbb{Z}[i]$. So p is not prime, hence not irreducible, in $\mathbb{Z}[i]$. It is nonzero and not a unit, as $N(p)=p^2$, so $p=\alpha\beta$ with α,β not units. Then $p^2=N(\alpha)N(\beta)$ with $N(\alpha),N(\beta)\neq 1$, so $N(\alpha)=p$, and writing $\alpha=a+bi$ gives $p=a^2+b^2$. With Step 1, this proves the first claim.

Step 4: uniqueness. Let $p=a^2+b^2=c^2+d^2$. The element $\pi=a+bi$ has norm p, so it is irreducible and hence prime. It divides $p=(c+di)(c-di)$, so it divides $c+di$ or $c-di$, say $c+\epsilon di=\pi\gamma$ with $\epsilon=\pm1$. Comparing norms, $p=pN(\gamma)$, so γ is a unit, and $c+\epsilon di$ is one of $\pm(a+bi)$ and $\pm i(a+bi)=\pm(-b+ai)$. Hence (c,d) is obtained from (a,b) by interchanging the two entries and changing signs.

Step 5: the prime elements of $\mathbb{Z}[i]$. The elements listed are prime. The norm of $1+i$ is 2, and the elements $a\pm bi$ in (3) have norm p, so they are irreducible, hence prime. If $p\equiv 3 \bmod 4$ and $p=\alpha\beta$ with α,β not units, then $N(\alpha)=p$ as in Step 3, and p would be a sum of two squares, against Step 1. So p is irreducible, hence prime. In (3), the associates of $a+bi$ are $\pm(a+bi)$ and $\pm(-b+ai)$, and $a-bi$ equals one of them only if $b=0$, $a=0$ or $a=\pm b$, which would make the odd prime $p=a^2+b^2$ a square or twice a square. So $a+bi$ and $a-bi$ are not associates.

Conversely, let π be a prime element. By lemma 10.3.2, $N(\pi)\in\{p,p^2\}$ for a prime p with $\pi\big|p$. If $N(\pi)=p$, then $\pi=a+bi$ with $a^2+b^2=p$, so $p=2$ or $p\equiv 1 \bmod 4$ by Step 1. For $p=2$, π is one of $\pm1\pm i$, the associates $1+i$, $i(1+i)$, $-(1+i)$ and $-i(1+i)$ of $1+i$. For $p\equiv 1 \bmod 4$, $\pi=a+bi$ is one of the elements in (3). If $N(\pi)=p^2$, write $p=\pi\gamma$. Then $N(\gamma)=1$, so π is an associate of p, and p is prime in $\mathbb{Z}[i]$. Then p is not a sum of two squares, for $p=(a+bi)(a-bi)$ would be a product of two elements of norm p, which are not units. So $p\neq 2$ and $p\not\equiv 1 \bmod 4$ by Step 3, that is, $p\equiv 3 \bmod 4$, completing the proof.

Lemma 10.3.4: Writing p as n^2+1

The prime number $p\in\mathbb{Z}$ divides an integer of the form n^2+1 if and only if p is either 2 or is an odd prime congruent to 1 mod 4.

***Proof*:**
The prime 2 divides 1^2+1. If $p\equiv 1 \bmod 4$, then $p=a^2+b^2$ by theorem 10.3.3. Here a and b are nonzero, since a prime is not a square, so $0<b^2<p$ and $p\nmid b$. Hence $\gcd(b,p)=1$, and the class of b lies in the group $(\mathbb{Z}/p\mathbb{Z})^\times$ (definition 1.2.5). Choose b' with $bb'\equiv 1 \bmod p$ and put $n=ab'$. Then

$$n^2+1\equiv a^2b'^2+b^2b'^2=b'^2p\equiv 0 \bmod p$$

so $p\big|n^2+1$.

Conversely, let p be an odd prime dividing n^2+1. In $\mathbb{Z}[i]$, p divides $n^2+1=(n+i)(n-i)$ but neither factor, since $\frac{n}{p}\pm\frac{1}{p}i\notin\mathbb{Z}[i]$, so p is not a prime element of $\mathbb{Z}[i]$. By theorem 10.3.3(2), the primes congruent to 3 modulo 4 are prime elements of $\mathbb{Z}[i]$, so $p\not\equiv 3 \bmod 4$, and the odd prime p is congruent to 1 modulo 4, as we sought to show.

Though this condition is equivalent to Fermat's Theorem on Sums of Squares, it is not as informative:

$5|3^2+1=10$, however $3+i$ and $3-i$ are *not* prime (for example, $3+i=(1+i)(2-i)$). There is however something very interesting that was demonstrated in this entire section: we have answered a purely number theoretic question about which primes can be expressible as sums of squares (a fact that should at first not be so obvious) by taking advantage of a lot of simple theoretical properties built-up about factorization and of the group $(\mathbb{Z}/p\mathbb{Z})^\times$!

10.3.5 Foreshadowing Fermat's Theorem of sums of two squares will be generalized to the factorization of any polynomial over any finite extension of a ring ([11, Dedekind-Kummer Theorem]), though the strategy of writing a prime p as some polynomial expression is in general more difficult.

Exercise 10.3.1

1. Find all integer solutions of $y^2=x^3-1$ by factoring y^2+1 in $\mathbb{Z}[i]$.

11

Exploring Polynomial Rings

When first studying algebra, we often looked to find solutions to linear and quadratic equations:

$$ax + b = 0 \qquad ax^2 + bx + c = 0$$

This naturally generalizes to polynomial of higher degree. In pre-abstract algebra, solutions to such equations where usually limited to $\mathbb{R}$, probably $\mathbb{C}$. We have now developed the abstraction to consider the value (or root) of such a polynomial for an arbitrary ring R, that is for an arbitrary polynomial in $R[x]$, does there exist an element $r \in R$ such that

$$p(r) = 0$$

As is the usual strategy in algebra, we generalize a problem into an algebraic object, in this case the polynomial ring $R[x]$, and look for interesting properties of the objects and tools we can use to manipulate the object. Such polynomials rings are paramount to study. To give an idea of their importance:

1. they arbitrarily approximate continuous functions on compact neighborhoods
2. all complex differentiable function can be locally be represented as infinite polynomials (i.e. they are analytic)
3. All relations in rings are represented by polynomial (similarly to how all relations in a group is represented by a monomial)

The last point shows that polynomial rings are used to "locally" represent rings, namely through the universal property of [commutative] universal rings (see theorem 9.2.4). It may thus seem natural to consider studying polynomials rings themselves to see what properties we may deduce from them.

One necessary limitation of the exploration of polynomial rings in chapter is that we shall not look at finding solutions to polynomials, we shall rather look at the structure of polynomial rings. The

reason for this is that it is in fact a much more difficult task to find solutions requiring more build-up that starts venturing into the domain of algebraic geometry. The beginnings of this effort shall be presented in [23, Algebraic Sets and the Zariski Topology].

It must be noted that we shall further develop on polynomial rings in *Commutative Algebra* [16]. This exploration is left to that book since it is not necessary to cover some results until much later.

Note that this chapter shall use some exercises from exercise set 9.3.1, most importantly:

Proposition 11.0.1: Ideals of Polynomial Ring wrt. R

Let $I \subseteq R$ be an ideal. Then:

$$(R[x])/(IR[x]) \cong (R/I)[x]$$

***Proof*:**
Let $\pi\colon R[x] \to (R/I)[x]$ reduce every coefficient modulo I, $\sum_i a_i x^i \mapsto \sum_i (a_i + I)x^i$. It preserves sums, and products because the coefficients of a product, $\sum_j a_j b_{i-j}$, are sums of products of coefficients and $R \to R/I$ is a ring homomorphism. It sends 1 to 1 and is surjective. Its kernel is the set of polynomials with every coefficient in I, which is $IR[x]$: every element $\sum a \cdot f$ ($a \in I$, $f \in R[x]$) of $IR[x]$ has its coefficients in I, and $\sum_i a_i x^i$ with every $a_i \in I$ is the sum of the elements $a_i \cdot x^i$ of $IR[x]$. Theorem 9.3.12 gives the isomorphism.

11.1 Polynomial Rings over Fields

We will now explore polynomial rings with coefficients from a field $\mathbb{F}$. The first question we may ask if what type of ring is $\mathbb{F}[x]$? It's not a field, since we already saw that x is not invertible (meaning it's also not a division ring). The next strongest ring we know is Euclidean domains. If $\mathbb{F}[x]$ is a euclidean domain, there must be a norm we define, which we will use $N(p(x)) = \deg(p(x))$ (the usual polynomial norm). Then $\mathbb{F}[x]$ is indeed a Euclidean Domain:

Theorem 11.1.1: Field Then $\mathbb{F}[x]$ Is Euclidean Domain

Let $\mathbb{F}$ be a field. The $\mathbb{F}[x]$ is a Euclidean Domain. In particular, if $a(x)$ and $b(x)$ are two polynomials in $\mathbb{F}[x]$ with $b(x)$ nonzero, then there are *unique* $q(x)$ and $r(x)$ such that

$$a(x) = q(x)b(x) + r(x_0) \qquad \text{with} \qquad r(x) = 0 \text{ or } \deg(r(x)) < \deg(b(x))$$

***Proof*:**
First, the edge cases:

If $a(x)$ is the zero polynomial, then let $q(x) = r(x) = 0$. So let $a(x)$ be a nonzero polynomial. Let $b(x)$ be an arbitrary other polynomial. If $\deg(a(x)) < \deg(b(x))$, then let $q(x) = 0$ and $r(x) = a(x)$.

Now, take the nonzero polynomial $a(x)$ such that $\deg(a(x)) > \deg(b(x))$. We will do a proof by induction on the degree of $a(x)$ ($n = \deg(a(x))$).

For visualization, we can denote $a(x)$ and $b(x)$ as:

$$a(x) = a_n x^n + a_{n-1}x^{n-1} + \cdots + a_0 \qquad b(x) = b_m x^m + b_{m-1}x^{m-1} + \cdots + b_0$$

We want to apply the induction hypothesis. Let $a'(x) = a(x) - \frac{a_n}{b_m}x^{n-m}b(x)$, which purposely defined to eliminate the monomial of degree n from $a(x)$. Then by the induciton hypothesis:

$$a'(x) = q'(x)b(x) + r(x) \qquad r(x) = 0 \text{ or } \deg(r(x)) < \deg(p(x))$$

Then, we "reverse" the step of taking $a'(x)$ by doing $q(x) = q'(x) + \frac{a_n}{b_n}x^{n-m}$, which will give us

$$a(x) = q(x)b(x) + r(x) \qquad r(x) = 0 \text{ or } \deg(r(x)) < \deg(p(x))$$

showing we can factor two polynomials.

It remains to show that this factorization is unique. Suppose $q(x), r(x)$, and $q_0(x)$, $r_0(x)$ are both polynomials that satisfy the euclidean condition. Then

$$a(x) - q(x)b(x) \qquad a(x) - q_0(x)b(x)$$

are both of degree less than m. The difference between these two polynomials then:

$$b(x)(q(x) - q_0(x))$$

is also less than m. But the degree of product of polynomials over an integral domain is the sum of their degree ($m > \deg(b(x)(q(x) - q_0(x))) = \deg(b(x) + \deg(q(x) - q_0(x))$. So, $q(x) - q_0(x) = 0$, implying $q(x) = q_0(x)$. This then implies $r(x) = r_0(x)$, completing the proof of uniqueness

Corollary 11.1.2: Field Then $\mathbb{F}[x]$ PID And UFD

Let $\mathbb{F}$ be a field. The $\mathbb{F}[x]$ is a Principal Ideal domain, and a Euclidean Domain

***Proof*:**
By the previous theorem (theorem 11.1.1), we know $\mathbb{F}[x]$ is a euclidean domain. By proposition 10.2.7, it is a Principal Ideal Domain, and since PID's are UFD's, it is also a Unique Factorization Domain

Example 11.1.3

1. $\mathbb{Q}[x]$ is a Principal Ideal Domain since $\mathbb{Q}$ is a field.
2. $\mathbb{Q}[x, y]$ is not a Principal Ideal Domain, since $\mathbb{Q}[x]$ is not a field (using the contra-positive).

One final consequence of $\mathbb{F}[x]$ being a Principal Ideal Domain:

Corollary 11.1.4: Kernel Of Evaluation Map

Let $\mathbb{F}$ be a field and let α be an element of $\mathbb{F}$. Then the kernel of ev_α is the ideal $(x - \alpha)$

***Proof*:**
Let $I = \ker(\text{ev}_\alpha)$. By definition of ev_α, $x - \alpha \in I$. Now, since $\mathbb{F}$ is a field, $\mathbb{F}[x]$ is a Euclidean Domain, and thus a Principal Ideal Domain, so I is a principal ideal. Let's say $I = (p)$ for some $p \in \mathbb{F}[x]$. Then by definition, $x - \alpha = up$, meaning $p|x - \alpha$. Note that p cannot have degree higher than 2, since then $x - \alpha \notin I$. So, if p has degree one, then the two elements are associates, meaning $I = (p) = (up) = (x - \alpha)$. If p has degree 0, then it must be a constant. However, p has a root at α, so if p were a constant, it would have to be 0. But then $x - \alpha \notin (0)$, a contradiction. Therefore $(x - \alpha) = \ker(\text{ev}_\alpha)$.

Here is also a fun result I want to put down:

Example 11.1.5: generalization of Famous infinite primes proof
Let F be a finite field. Then there are infinitely many prime elements in $F[x]$

Proof. Let's assume there is a finite amount. Then $p_1, ..., p_n$ is our list of irreducible polynomials. Note that $\deg(p_i) \geq 1$ for each polynomial. Then take $P = p_1 p_1 \cdots p_n + 1$. Then $\deg(P) \geq 1$, and so P is not a unit or 0. Therefore, P must be divisible by some p_i, meaning $p_i|P$, which implies $qp_i = P$, or equivalently $P \in (p_i)$. Next, by closure of ideals, $-p_1 p_2 \cdots p_n \in (p_i)$. Then, by closure under addition:

$$P - p_1 p_2 \cdots p_n = p_1 p_2 \cdots p_n + 1 - p_1 p_2 \cdots p_n = 1 \in (p_i)$$

meaning $up_i = 1$ for appropriate $u \in F[x]$, implying p_i is a unit! However, prime elements are *not* units. Since p_i was arbitrary, *no* p_i can divide P. But then, P will have non of the factor's in it's factorization, a contradiction to deg > 0 and the p_i's being the only irreducibles! Therefore, P must be a new prime number!

Therefore, there is an infinite number of prime numbers. □

We finish off with an application of the Chinese remainder theorem with the new fact that $k[x]$ is a Euclidean domain, and hence a UFD:

Proposition 11.1.6: Chinese Remainder On Polynomials

Let $g(x)$ be a nonconstant monic element of $k[x]$ and let

$$g(x) = f_1(x)^{n_1} f_2(x)^{n_2} \cdots f_k(x)^{n_k}$$

be its factorization into irreducibles, where the $f_i(x)$ are distinct. Then we have the following isomorphism of rings:

$$k[x]/(g(x)) \cong k[x]/(f_1(x)^{n_1}) \times k[x]/(f_2(x)^{n_2}) \times \cdots \times k[x]/(f_k(x)^{n_k})$$

***Proof*:**
This follows from the Chinese Remainder Theorem (theorem 9.6.11) and corollary 10.1.36, but we will re-state it here. Since $(f_i(x)^{n_i})$ and $(f_i(x)^{n_i})$ are comaximal if $f_i(x)$ and $f_j(x)$ are distinct (they are relatively prime in the Euclidean Domain $k[x]$, hence $gcd(f_i^{n_i}, f_j^{n_j}) = 1$, and so $p(x)f_i(x)^{n_i} + q(x)f_j(x)^{n_j} = 1$, so $1 \in (f_i^{n_i}) + (f_j^{n_j})$ implying $(f_i^{n_i}) + (f_j^{n_j}) = k[x]$ by proposition 9.4.10).

Exercise 11.1.1

1. Let $R[x]$ be a Principal Ideal Domain. Show that $R[x]$ is a Euclidean Domain.

11.2 Polynomial Rings over UFD's

We have seen R is an integral domain then $R[x]$ is an integral domain. We have also seen that if R is an integral domain, than we can embed R into a field of fraction F (see section 9.5). This means that $R[x] \subseteq F[x]$. Furthermore, since F is a field, then $F[x]$ is a euclidean domain. With this, we can do lots of our computation in $F[x]$ instead of $R[x]$, at the expense of allowing fractional coefficients. The question then arises if we can use information from $F[x]$ to gain information about $R[x]$!

For example, take a polynomial $p(x) \in R[x]$. Since $p(x) \in k[x]$, and $k[x]$ is a Unique Factorization Domain, then $p(x)$ factors in $k[x]$ into irreducibles. We can then ask if $p(x)$ will factor in $R[x]$. The answer is no in general, since $R[x]$ is not always a Unique Factorization Domain, since that would imply R is a Unique Factorization Domain since the coefficients of $R[x]$ must also factor uniquely. However, if R were a unique factorization domain, then this will imply that $R[x]$ is a Unique Factorization Domain! Hence, factoring in $k[x]$ gives information on how we factor in $R[x]$. The way we will prove this is first by uniquely factoring in $k[x]$, then "clear denominators" to obtain unique factorization in $R[x]$!

The first thing to do is make precise the notion of comparing factorization of a polynomial in $k[x]$ and a factorization in $R[x]$

Proposition 11.2.1: Gauss's Lemma

Let R be a Unique Factorization Domain with field of fraction K and let $p(x) \in R[x]$. If $p(x)$ is reducible in $K[x]$ then $p(x)$ is reducible in $R[x]$. More precisely, if $p(x) = A(x)B(x)$ for some non constant polynomials $A(x), B(x) \in K[x]$, then there are nonzero elements $r, s \in F$ such that $rA(x) = a(x)$ and $sB(x) = b(x)$, such that $a(x), b(x) \in R[x]$ and $p(x) = a(x)b(x)$ is a factorization in $R[x]$

A concrete case that can be kept in mind is with the polynomials $p(x), q(x) \in \mathbb{Q}[x]$, then $rp(x), sq(x) \in \mathbb{Z}[x]$ for appropriates $r, s \in \mathbb{Q}$. Note that it is possible that $q(x) \in \mathbb{Q}$, i.e. $q(x)$ is a constant polynomial[1].

***Proof*:**
Let's say $p(x) \in k[x]$ and is reducible $p(x) = A(x)B(x)$. Then the coefficients of $p(x)$ are quotients from R. Since R is a Unique Factorization Domain, let's take d a number common betewen all denominators for all the coefficients of $A(x)B(x)$, so that we can obtain

$$dp(x) = a'(x)b'(x)$$

where $a'(x), b'(x)$ are now in $R[x]$ since we multiplied away the denominators with d! If d is a unit, then

$$a(x) = d^{-1}a'(x) \qquad b(x) = b'(x)$$

[1] In textbooks where the contra-positive of this statement is used, it is often specified that $p(x)$ is *primitive*, meaning for all principal prime $\mathfrak{p} \subseteq R$, $p(x) \notin \mathfrak{p}R[x]$. In the UFD case, this amounts to the gcd of the coefficients being 1

which completes the proof, so let's assume d is not a unit. Since $d \in R$, and R is a Unique Factorization Domain, then there is a unique factorization into irreducibles:

$$d = p_1 \cdots p_n$$

Since p_1 is an irreducible element, and R is a Unique Factorization Domain, then $(p_1) = p_1 R[x]$ (since $R[x]$ is commutative $(a) = aR$) is prime (exercise), and so $R[x]/p_1 R[x]$ is an integral domain (proposition 9.4.25) and $R[x]/p_1 R[x] \cong (R/p_1 R)[x]$ (exercise).

Now, reducing $dp(x) = a'(x)b'(x)$ modulo p_1, we get

$$\overline{0} = \overline{a'(x)} \cdot \overline{b'(x)}$$

Since $(R/p_1 R)[x]$ is an integral domain, then either $\overline{a'(x)}$ or $\overline{b'(x)}$ is zero, say $\overline{a'(x)} = 0$. This would imply $p_1 | a'(x)$, and so every coefficient of $a'(x)$ is divisible by p_1, meaning $\frac{1}{p_1} a'(x)$ also has coefficients from R! So, we can divide both sides by p_1 (that is d by p_1 and either $a'(x)$ or $b'(x)$ by p_1). But then, the factor d on the left hand side has one fewer irreducible element! We can continue this procedure till we cancel out all of the factors, leaving us with $p(x) = a(x)b(x)$, with $a(x), b(x) \in R[x]$, $a(x)$ and $b(x)$ being F-mutiples of $A(x)$ and $B(x)$!

Note that $A(x)$ and $B(x)$ are not necessarily R-multiples, nor are they necessarily unique. For example, $x^2 \in Q[x]$ is reducible, and so reducible in $\mathbb{Z}[x]$. One factorization would be $A(x) = 2x$ and $B(x) = \frac{1}{2}x$. However, $\frac{1}{2} \notin \mathbb{Z}$: no integer multiple gives us a factorization of $A(x)$ and $B(x)$. What the theorem will tell us is that $\frac{1}{2}A(x) \in \mathbb{Z}[x]$, and $2B(x) \in \mathbb{Z}[x]$.

Since k is a field, the elements of R becomes units in $k[x]$ (the units in $k[x]$ being the non-zero constants). For example, $7x$ factors in $\mathbb{Z}[x]$ and produces two irreducible 7 and x, so $7x$ is *not* irreducible in $\mathbb{Z}[x]$. However, in $\mathbb{Q}[x]$, $7x$ is a unit 7 times an irreducible, so $7x$ *is* irreducible in $\mathbb{Q}[x]$. However, this is the *only* difference between the irreducible elements in $R[x]$ and $k[x]$

Corollary 11.2.2: Irreducibility In $R[x]$ And $k[x]$

Let R be a UFD, let $k = K(R)$ be its field of fractions, and let $p(x) \in R[x]$. Suppose the greatest common divisor of the coefficients of $p(x)$ is 1. Then $p(x)$ is irreducible in $R[x]$ if and only if it is irreducible in $K[x]$. In particular, if $p(x)$ is a monic polynomial that is irreducible in $R[x]$, then $p(x)$ is irreducible in $K[x]$.

***Proof*:**
We will proceed by the contrapositive: $p(x)$ is reducible in $R[x]$ if and only if $p(x)$ is reducible $K[x]$. The ($\Leftarrow$) direction is Gauss's Lemma, so let's go ($\Rightarrow$). Let's say $p(x)$ is reducible in $R[x]$ and the greatest comon divisor of the coefficients is 1. That means that when we reduce $p(x) = a(x)b(x)$, neither $a(x)$ or $b(x)$ can be constants! Note that polynomials which are not constants in $K[x]$ are not units, meaning $p(x) = a(x)b(x)$ is also a factorization in $K[x]$, completing the proof!

This theorem is already incredibly powerful, it means that the task of finding the irreducible elements of $R[x]$ reduces to finding the irreducible elements of $k[x]$ (which we know is a euclidean domain) up to a multiple of R. We shall later find more techniques for finding solutions to polynomials over fields, which by our above results immediately give us a way to translate these results for integral

domains whose field of fraction is the given polynomial ring over a field.

Yet another result is the following which shows that extending a ring to form a polynomial ring preserves UFD!

Theorem 11.2.3: R UFD If And Only If $R[x]$ UFD

R is a Unique Factorization Domain if and only if $R[x]$ is a Unique Factorization Domain

***Proof*:**
If $R[x]$ is a Unique Factorization Domain, then the constants of $R[x]$ must uniquely factor, meaning R is a Unique Factorization Domain.

So let's suppose R is a Unique Factorization Domain, and take $p(x) \in R[x]$. If $p(x)$ is a constant, then it's uniquely factorizable, so let's say $\deg(p(x)) > 0$. Let d be the g.c.d. of the coefficients of $p(x)$, so that $p(x) = dp'(x)$, meaning the g.c.d. of the coefficients of $p'(x)$ is 1. Note that this factorization is unique up to a unit multiple of d (i.e. up to a unit in R by proposition 9.2.3), and since d can be factored uniquely in R (and these are also irreducibles in $R[x]$ too), it suffices now to prove that $p'(x)$ can be uniquely factored into irreducibles in $R[x]$.

Let k be the field of fractions of R. Then since $k[x]$ is a Unique Factorization Domain (theorem 11.1.1), $p(x)$ can be factored uniquely into irreducibles in $k[x]$. By Gauss's Lemma, such a factorization implies a factorization in $R[x]$ whose factors are F-multiples of the factors in $k[x]$. Since the g.c.d. of $p'(x)$ is 1, the g.c.d. of the coefficients of each of these factors in $R[x]$ must be 1. By corollary 11.2.2, each of these factors are irreducible in $R[x]$! Hence $p(x)$ can be written as the product of irreducibles in $R[x]$.

The uniqueness of the factorization of $p(x)$ follows from the uniqueness in $k[x]$. Suppose

$$p'(x) = q_1(x)q_2(x)\cdots q_n(x) = q_1'(x)q_2'(x)\cdots q_m'(x)$$

Since the g.c.d. of the coefficients of $p'(x)$ is 1, so to it is for both factorizations. By corollary 11.2.2, each $q_i(x)$ and $q_j'(x)$ is an irreducible in $k[x]$. By uniqueness of factorization on $k[x]$, $n = m$, and the up to rearrangement, the two factorizations are the same up to associates, that is $q_i(x) = uq_i'(x)$ for a unit $u \in k[x]$. It remains to show they are associates in $R[x]$. By proposition 9.2.3, the untis of $k[x]$ are the units of F, so

$$q_i(x) = \frac{a}{b}q_i'(x)$$

for $a, b \in R$. Thus, $bq_i(x) = aq_i'(x)$. Thinking about it another way, the g.c.d. of the coefficients of the polynomial on the left hand side is b, and a on the right hand side respectively. Since in a Unique Factorization Domain, the coefficients of a nonzero polynomial is unique up to units ($a = ub$ for a unit in R), then $q_i(x) = uq_i'(x)$, and so $q_i(x)$ and $q_i'(x)$ are associates in R as well, completing the proof!

Corollary 11.2.4: Polynomial Rings in Several Variables over a UFD are UFDs

If R is a Unique Factorization Domain, then a polynomial ring in an arbitrary number of variables with coefficients in R is also a UFD

***Proof*:**
For the finitely many case, apply induction on theorem 11.2.3, since $R[x,y] = R[x][y]$, and $R[x]$ is a Unique Factorization Domain. The general case follwos fromthe definition of a polynomial ring in an arbitrar ynumber of variables as the union of polynomial rings in finitely many variables.

Example 11.2.5

1. $\mathbb{Z}[x], \mathbb{Z}[x,y]$, and so on are Unique Factorization Domains. The ring $\mathbb{Z}[x]$ now gives a good example of a Unique Factorization Domain that is *not* a Principal Ideal Domain!
2. Similarly, $\mathbb{Q}[x]$, $\mathbb{Q}[x,y]$, and so forth are Unique Factorization Domains.

Finally, we will ask if we can weaken the condition's and still get the same result. What if R was just an integral domain? Can we follow a similar construction? Generalizing in a different direction, if R is a Unique Factorization Domain, does that imply that $R[[x]]$ is a Unique Factorization Domain? The answer to both is in general no

Example 11.2.6: Counter-Example

1. Let $R = \mathbb{Z}[2i]$, and let $p(x) = x^2 + 1$. Then the field of fractions of R is $F = \mathbb{Q}[2i]$. The polynomial $p(x)$ factors uniquely into a product of two linear facrors in $k[x] : x^2 = (x-i)(x+1)$. so in particular $p(x)$ *is* reducible in $\mathbb{Q}[2i]$. However, Neither of these factors lies in $R[x]$ (because $i \notin R$) so $p(x)$ is *irreducible* in $R[x]$. In particualr, by corollary 11.2.2, $\mathbb{Z}[2i]$ is *not* a Unique Factorization Domain! If it were, then it should've been reducible!
2. It is a rather technical project to show that if R is a UFD, then $R[[x]]$ need not be a UFD. The somewhat canonical counter-example is

$$R = \frac{k[x,y,z]}{(x^2 + y^3 + z^7)}$$

See [53] for an in-depth explanation[a].

[a] this paper does require some algebraic knowledge introduced in *Commutative Algebra* [16]

Exercise 11.2.1

1. Let $x^5 - x + 1 \in \mathbb{Q}[x]$. Is this polynomial irreducible? (Hint: use Gauss's Lemma)
2. Recall that if R Is a UFD, it is not necessarily the case that $R[[x]$. Show that $\mathbb{R}[[x]]$ is a UFD

11.3 Irreducibility of Polynomials

Now that we have shown R, $R[x]$, $R[x,y]$ and so forth are Unique Factorization Domains if and only if R is a Unique Factorization Domain and that irreducibility over $R[x]$ is symmetrically reflected to irreducibility over $k[x]$, we now raise the question about finding irreducible elements in such a ring (i.e. the prime elements, the "building blocks"). A polynomial $p(x)$ is reducible if we can find

two polynomials of smaller degree such that $p(x) = a(x)b(x)$. A polynomial is irreducible if no such polynomials exist. In general this is difficult, which is why we look for nicer irreducibility criteria.

Throughout these proofs, we shall be taking advantage of the fact that the field of fractions is a UFD, hence the polynomial ring over the field is a UFD. We shall not even require the ring that we are using to construct the field of fraction to be a UFD, simply an integral domain. To demonstrate, consider the following key fact:

Proposition 11.3.1: Finding Linear Factors

Let R be an integral domain, $p(x) \in R[x]$ and $r \in R$. Then $p(r) = 0$ if and only if $x - r$ divides $p(x)$ in $R[x]$.

***Proof*:**
If $p(x) = (x - r)q(x)$ with $q(x) \in R[x]$, then $p(r) = (r - r)q(r) = 0$. Conversely, let $p(x) = \sum_{k=0}^{n} a_k x^k$ with $p(r) = 0$. For $k \geq 1$, $x^k - r^k = (x - r)\sum_{j=0}^{k-1} x^j r^{k-1-j}$, so

$$p(x) = p(x) - p(r) = \sum_{k=1}^{n} a_k(x^k - r^k)$$

is divisible by $x - r$ in $R[x]$, completing the proof.

Corollary 11.3.2: Maximum Number Of Roots

[Maximum Number Of Roots] Let R be an integral domain and $p(x) \in R[x]$ of degree n. Then $p(x)$ has *at most* n roots.

***Proof*:**
Let $k = K(R)$ be the field of fraction of R. Then the number of roots of $p(x) \in [R]x$ is certainly less than or equal to the number of roots in $p(x) \in k[x]$. Now, *a fortiori* k is a UFD, hence by theorem 11.2.3 $k[x]$ is a UFD, and so each root corresponds to an (irreducible) factor of degree 1. But then, f can have at most n factors of degree 1 by unique factorization, completing the proof.

Note that this statement can fail if R is not an integral domain!

Example 11.3.3: Polynomial With More Roots And Its Degree

[Polynomial With More Roots And Its Degree] Let $x^2 + x \in (\mathbb{Z}/6\mathbb{Z})[x]$. Show that this polynomial of degree 2 has the 4 roots 0, 2, 3 and 5.

However, using proposition 11.3.1, we may show that polynomials over infinite integral domains are determined by their roots!

Corollary 11.3.4: Polynomials Detremiend By Evaluation

[Polynomials Detremiend By Evaluation] Let R be an infinite integral domain and $f, g \in R[x]$. Then $f = g$ if and only if

$$r \mapsto f(r) \qquad \text{and} \qquad r \mapsto g(r)$$

agree for all $r \in R$

***Proof*:**
Two functions agree if every $r \in R$ is a root of $f - g<$ but a nonzero polynomial over R cannot have infinitely many roots by corollary 11.3.2, hence they must be equal as we sought to show.

Continuing from this, we may ask about the irreducibility of degree 2 and 3 polynomials, which is an immediate consequence of the above results:

Proposition 11.3.5: Degree 2 Or 3 Polynomials

A polynomial of degree two or three over a field F is reducible if and only if it has a root in F.

***Proof*:**
Let $f \in F[x]$ have degree 2 or 3. If $f = gh$ with g, h not units, then g and h have positive degree, since the units of $F[x]$ are the nonzero constants, and $\deg g + \deg h = \deg f \leq 3$. So one factor has degree 1, say $g = ax + b$ with $a \neq 0$, and $-b/a$ is a root of f. Conversely, if c is a root of f, then $f = (x - c)h$ by proposition 11.3.1, with $\deg h = \deg f - 1 \geq 1$, so f is reducible, completing the proof.

The reason for this is that if we have a degree higher than 4, then we can reduce to two irreducible polynomials of degree 2, like $(x^2 + 1)(x^2 + 1)$ in $\mathbb{Q}[x]$. For higher degree polynomials, we may have more solutions depending on what type of field we are working over. For example, we shall introduce in section 19.5.1 the existence of field for which only linear polynomials are irreducible ($\mathbb{C}$ being a canonical example of such a field). Yet another example is that any polynomial of degree ≥ 3 over $\mathbb{R}$ is reducible[2]. For more general rings, it is in general much more difficult to determine if a polynomial is reducible. There are however some tricks that help in arbitrary situations, especially if R is a UFD (for then we may use the gcd)

This next proposition let's us limit the number of "rational" solutions of a polynomial over a UFD.

Proposition 11.3.6: Rational Roots Test

Let R be a UFD and $k = K(R)$. Let $p(x) = a_n x^n + a_{n-1} x^{n-1} + \cdot + a_1 x + a_0$ be a polynomial of degree n with coefficients in the UFD R. If $r/s \in k$ with $gcd(r, s) = 1$, and r/s is a root of $p(x)$ ($p(r/s) = 0$), then r divides the constant term and s divides the leading coefficient of $p(x)$: $r \big| a_0$ and $s \big| a_n$. In particular, if $p(x)$ is a monic polynomial with integer coefficients and $p(d) \neq 0$ for all integers d dividing the constant term of $p(x)$, then $p(x)$ has no roots in $k[x]$.

[2]This is exercise ref:HeRE (closure of fields, 3rd exercise). An astute student would be able to work out this exercise

***Proof*:**
Let's say $p(r/s) = 0$. Then

$$p(r/s) = 0 = a_n(r/s)^n + a_{n-1}(r/s)^{n-1} + \cdots + a_1(r/s) + a_0$$

multiplying all by s^n, we get:

$$0 = a_n r^n + a_{n-1} r^{n-1} s + \cdots + a_1 r s^{n-1} + a_0 s^n$$

so

$$a_n r^n = s(-a_{n-1} r^{n-1} - \cdots - a_1 r s^{n-1} - a_0 s^{n-1})$$

thus, $s \big| a_n r^n$. $gcd(r, s) = 1$, then it must be that $s | a_n$. Doing the same trick for a_0:

$$a_0 s^n = r(-a_n r^{n-1} - a_{n-1} r^{n-2} - \cdots - a_1 s^{n-1})$$

then $r | a_0$.

If $a_n = 1$, then we'd have $s|1$, so $r/s = r/1 = r$. This means we only need to check the values r which divide a_0. If every value that divdse a_0 is not a root of $p(x)$, then $p(x)$ has no roots.

This gives us a really easy way of testing irreducibility of low-degree polynomials:

Example 11.3.7: Irreducibility Of Low-Degree Polynomials

1. The polynomial $p(x) = x^3 - 3x - 1$ is irreducible in $\mathbb{Z}[x]$. Since the polynomial is monic, by corollary 11.2.2, it is irreducible in $\mathbb{Z}[x]$ if it is irreducible in $\mathbb{Q}[x]$. By the rational number test, we are given the candidates ± 1. Plugging both of them in, we get no roots of $\mathbb{Q}[x]$, and so by proposition 11.3.5, this polynomial is irreducible, and so is irreducible in $\mathbb{Z}[x]$.
2. For any prime p, $x^2 - p$ and $x^3 - p$ are irreducible in $\mathbb{Q}[x]$. We have ± 1 and $\pm p$ as candidates. Both of them are not roots, and therefore, these polynomials are not reducible!
3. The polynomial $x^2 + 1$ is *reducible* in $(\mathbb{Z}/2\mathbb{Z})[x]$, namely $(1)^2 + 1 = 1 + 1 = 0$. It's factors are $(x + 1)(x + 1) = x^2 + 1$.
4. The polynomial $x^2 + x + 1$ is *irreducible* in $(\mathbb{Z}/2\mathbb{Z})[x]$, since there are no roots.
5. Similarly for $x^3 + x + 1$ in $(\mathbb{Z}/2\mathbb{Z})[x]$

All the polynomials in the aforementioned examples are all of degree less than or equal to 3. This is because when factoring polynomials of degree greater than or equal to 4, it's possible that we get 2nd degree polynomials as irreducible factors (like $(x^2 + 1)(x^2 + 1)$ in $\mathbb{Q}[x]$, which has no factors).

We use another trick for these type of polynomials: We can use proposition 11.0.1 ($R[x]/I[x] \cong (R/I)[x]$) to generalize the rational root test:

Proposition 11.3.8: Reduction Test

Let I be a proper ideal in the integral domain R and let $p(x)$ be a non-constant *monic* polynomial in $R[x]$. If the image of $p(x)$ in $(R/I)[x]$ cannot be factored in $(R/I)[x]$ into two polynomials of smaller degree, then $p(x)$ is irreducible in $R[x]$.

***Proof*:**
Suppose $p(x)$ is reducible in $R[x]$, so $p(x) = a(x)b(x)$ with neither factor a unit. Since R is an integral domain, $\deg a + \deg b = \deg p$, and the leading coefficients of $a(x)$ and $b(x)$ multiply to 1, so both are units of R. A factor of degree 0 would then be a unit, so both factors have positive degree, and hence degree smaller than $\deg p$. Multiplying $a(x)$ by the inverse u^{-1} of its leading coefficient u, and $b(x)$ by u, makes both factors monic without changing their product.

Reducing every coefficient modulo I is a ring homomorphism $R[x] \to (R/I)[x]$, the map behind proposition 11.0.1. It sends $p(x) = a(x)b(x)$ to a product of two monic polynomials of the same degrees as $a(x)$ and $b(x)$, since $1 \neq 0$ in R/I for the proper ideal I. So the image of $p(x)$ factors in $(R/I)[x]$ into two polynomials of smaller degree, contradicting the hypothesis.

The power here comes from the fact that R/I might have much simpler structure for factoring (ex. $\mathbb{Z}$ vs $\mathbb{Z}/n\mathbb{Z}$!) For example, in exercise set 11.2.1, we may prove that $x^5 - x + 1$ using this trick rather quickly! So, If we reduce a polynomial into $(R/I)[x]$, and show it cannot be factored, then it is irreducible in $R[x]$ as well! However, the converse (if $p(x)$ is irreducible in $R[x]$, then $p(x)$ is irreducible in every $(R/I)[x]$) is not true.

Example 11.3.9: Limitation of reduction method

Take $x^4 + 1 \in \mathbb{Z}[x]$. This polynomial is irreducible in $\mathbb{Z}[x]$, (hint: recall that $\sqrt{2}$ is irrational), but is reducible for every prime. Verify this by considering cases of

$$p \equiv 1, 3, 5, 7 \bmod 8$$

since the case of $p = 2$ can quickly be manually verified.

So, if you get that a polynomial is reducible in $(R/I)[x]$, that unfortunately does not tell you it's reducible in $R[x]$.

A special case of proposition 11.3.8 that is quite useful is called Eisenstein's Criterion (or Eisenstein-Schönenmann Criterion, since Schönenmann discovered it first)

Proposition 11.3.10: Eisenstein's Criterion

Let P be a prime ideal of the commutative ring R and let $f(x) = x^n + a_{n-1}x^{n-1} + \cdots + a_1x + a_0$ be a polynomial in $R[x]$ (where $n \geq 1$). Suppose $a_{n-1}, ..., a_0 \in P$, but $a_0 \notin P^2$. Then $p(x)$ is irreducible.

***Proof*:**
Let's take $p(x)$ like in the proposition. For the sake of contradiction, let's say $p(x)$ is reducible: $p(x) = a(x)b(x)$, where $a(x)$ and $b(x)$ are non-constant.Then by proposition 11.3.8 (the

contrapositive of it), it is reducible in $(R/P)[x]$. Since every non-leading coefficient is in P:

$$\overline{p(x)} = \overline{x^n} = \overline{a(x)} \cdot \overline{b(x)}$$

Since P is prime, R/I is an integral domain, and so the constant terms of $a(x)$ and $b(x)$ must be in P, or else we will get $\overline{x} \cdot \overline{y} = \overline{0}$. But then, multiplying those terms means $a_0 \in P^2$, a contradiction!

For the purposes of this book, the following is the most immediate use of Eisenstein's Criterion:

Corollary 11.3.11: Eisenstein's Criterion For $\mathbb{Z}[x]$

Let p be a prime in $\mathbb{Z}$ and let $f(x) = x^n + a_{n-1}x^{n-1} + \cdots a_1 x + a_0$. If every a_i is divisible by p, but a_0 is not divisible by p^2, then $f(x)$ is irreducible in both $\mathbb{Z}[x]$ and $\mathbb{Q}[x]$.

***Proof*:**
This is proposition 11.3.10 and using corollary 11.2.2.

Here are some examples using Eisenstein's criterion, and some more tricks when finding irreducible polynomials:

Example 11.3.12: Irreducibility Tricks

1. When solving polynomials of degree 3 or less, it'd suggest to always start with checking roots: it's the simpliest criterion.
2. The polymoial $x^4 + 10x + 5$ in $\mathbb{Z}[x]$ is irreducible, if we take $p = 5$. Since $5|10$, $5|5$, but $5^2 \nmid 5$, then Eisntein's criterion holds.
3. For any integer a such that $p|a$ but $p^2 \nmid a$, then $x^n - a$ is irreducible in $\mathbb{Z}[x]$. If $a = p$ a prime, then $x^n - p$ is irreducible for all positive integers n, so for $n \geq 2$, the nth root of p are not rational numbers (i.e. has no roots in $\mathbb{Q}[x]$)
4. We cannot apply the same tricks we did before to the polynomial x^4+1 in $\mathbb{Z}[x]$, since nothing divides prime, and any proper ideal will not contain 1. Instead, we will use a trick called *substitution*.

   Consider the polynomial $f(x+1) = (x+1)^4 + 1 = x^4 + 4x^3 + 6x^2 + 4x + 2$. On this polynimal, Eisntein's criterion works, since all non-leading coefficients are divisible by 2, and $2^2 \nmid a_0$. From this, it follows that $f(x)$ is irreducible, since any factorization of $f(x)$ will be a factorization for $f(x+1)$ (replace x by $x+1$ in the factorization!) Remember this trick for polynomials which seem at first glance to not get an application of Eisntein's criterion.
5. (cyclotomic polynomials) Perhaps the most famous application of Eisenstein's criterion is on polynomials of the following form:

   $$1 + x + x^2 + \cdots + x^{p-1} \in \mathbb{Z}[x]$$

   Eisenstein's criterion cannot be directly applied, however as before we can do the $x \mapsto x+1$ trick to get

   $$p(x+1) = 1 + (x+1) + \cdots + (x+1)^{p-1} = x^{p-1} + \binom{p}{p-1}x^{p-2} + \cdots + \binom{p}{2}x + \binom{p}{1}$$

where $\binom{p}{1} = p$.It suffices to show that $p \Big| \binom{p}{k}$ for $1 \leq k \leq p-1$ to conclude the result.

6. Notice that in the proof, we reduced to $(R/P)[x]$. If $R = \mathbb{F}$ was a field, then the only ideals are 0 and $\mathbb{F}$, meaning $(\mathbb{F}/P)[x] \cong 0[x] \cong 0$. Therefore, Einsteins criterion does not work will with such polynomials.

   For example, is $x^4 + 1$ reducible in $\mathbb{F}_5[x]$ ($\mathbb{F}_5$ is the field with 5 elements, and is isomorphic to $\mathbb{Z}/5\mathbb{Z}$)? Over $\mathbb{Q}$ it is irreducible, since substituting $x + 1$ gives $x^4 + 4x^3 + 6x^2 + 4x + 2$, which satisfies Eisenstein's criterion at 2 (corollary 11.3.11). It also has no roots, as can be verified by plugging in $0, ..., 4$.

   We cannot use einstein's criteron, but since the degree is 4, we know if a factorization did exist, they would have to be of degree 2, that is

   $$x^4 + 1 = (x^2 + ax + b)(x^2 + cx + d) = x^4 + (a + c)x^3 + (b + ac + d)x^2 + (ad + bc)x + bd$$

   note that the coefficients of x^2 are one since the coeficient of x^4 is one. We can then deduce that

   $$a + c = 0, b + ac + d = 0, ad + bc = 0, bd = 1$$

   Manipulating a bit, we can get the following information:

   $$\begin{aligned} c &= -a \\ b + a(-a) + d &= 0 \\ ad + b(-a) &= 0 \\ a^2 &= b + d \\ ad &= ab \end{aligned}$$

   If $a \neq 0$, then $ad = ab$ gives $b = d$, so $b^2 = 1$ and $a^2 = 2b = \pm 2$. But the squares in $\mathbb{F}_5$ are $0, 1, 4$, so this is impossible. If $a = 0$, then $b = -d$ and $bd = 1$ give $b^2 = -1 = 4$, so $b = 2$ and $d = 3$ (or the reverse), and $c = 0$. This gives

   $$p(x) = x^4 + 1 = (x^2 + 2)(x^2 + 3)$$

   Thus, $p(x)$ is reducible in $\mathbb{F}_5[x]$.

Over finite fields, irreducibility can also be decided by an algorithm that factors polynomials, for example Berlekamp's algorithm.

Finally, we give a few important results that shall be very handy tools. The first tells us that polynomial division is insensitive to extending the field: for $f(x), g(x) \in F[x]$, the quotient and remainder, and with them the question of whether $g(x)$ divides $f(x)$, come out the same in $F[x]$ and in $E[x]$ for any extension E of F:

Proposition 11.3.13: Invariant Under Field Extensions

Let $F \subseteq E$ be an extension of fields and let $f(x), g(x) \in F[x]$ with $g(x)$ nonzero. Then the quotient and remainder of $f(x)$ upon division by $g(x)$ are the same whether computed in $F[x]$ or in $E[x]$. In particular, $g(x) \Big| f(x)$ in $F[x]$ if and only if $g(x) \Big| f(x)$ in $E[x]$.

***Proof*:**
By theorem 11.1.1 applied in $F[x]$, write

$$f(x) = q(x)g(x) + r(x) \qquad \text{with} \qquad r(x) = 0 \text{ or } \deg(r(x)) < \deg(g(x))$$

where $q(x), r(x) \in F[x]$. Since $F[x] \subseteq E[x]$, this same equation holds in $E[x]$, and the condition on $r(x)$ is unchanged. By the uniqueness part of theorem 11.1.1 applied in $E[x]$, the pair $(q(x), r(x))$ is therefore also the quotient and remainder of $f(x)$ by $g(x)$ in $E[x]$, proving the first claim.

For the second claim, note that $g(x)\big|f(x)$ in $F[x]$ if and only if $r(x) = 0$. Indeed, if $r(x) = 0$ then $f(x) = q(x)g(x)$, and conversely if $f(x) = h(x)g(x)$ with $h(x) \in F[x]$, then $f(x) = h(x)g(x) + 0$ is a division with remainder 0, so $r(x) = 0$ by uniqueness. The same equivalence holds in $E[x]$, and by the first claim the remainder is the same polynomial $r(x)$ in both rings. Hence $g(x)\big|f(x)$ in $F[x]$ if and only if $r(x) = 0$ if and only if $g(x)\big|f(x)$ in $E[x]$.

This next one is of particular interest for the structure of the multiplicative group of a field!

Theorem 11.3.14: Subgroup of Multiplicative Group is Cyclic

A finite subgroup of the multiplicative group of a field is cyclic. In particular, if F is a finite field, then the multiplicative group $F^\times$ of nonzero elements of F is a cyclic group.

***Proof*:**
We will use the fundamental theorem for finitely generated abliean groups!

Let $A \subseteq F^\times$ be a finite subgroup. It is abelian, so by theorem 5.1.6 (with $r = 0$, as A is finite):

$$A \cong \mathbb{Z}/n_1\mathbb{Z} \times \mathbb{Z}/n_2\mathbb{Z} \times \cdots \times \mathbb{Z}/n_k\mathbb{Z}$$

where $n_k\big|n_{k-1}, n_{k-1}|n_{k-2}, ..., n_2\big|n_1$. Since n_k divides the order of each of the cyclic groups in the direct product, it follows that each direct factor contains n_k elements g with $g^{n_k} = 1$: in $\langle x\rangle$ of order m with $n_k\big|m$, $(x^j)^{n_k} = 1$ exactly when $m\big|jn_k$, that is when $\frac{m}{n_k}\big|j$, which holds for n_k values of j modulo m (proposition 2.3.7). If k were greater than 1, there would therefore be a total of more than n_k such elements. But then there would be more than n_k roots of the polynomial $x^{n_k} - 1$ in the field F, contradicting corollary 11.3.2. Hence $k \leq 1$, and the group is cyclic.

Corollary 11.3.15: $(\mathbb{Z}/p\mathbb{Z})^\times$ is cyclic

Let p be a prime. The multiplicative group $(\mathbb{Z}/p\mathbb{Z})^\times$ of nonzero residue classes mod p is cyclic

***Proof*:**
Since $\mathbb{Z}/p\mathbb{Z}$ is a field, apply theorem 11.3.14

Corollary 11.3.16: Non-Abelian Groups Of Order pq Are Unique

Let $p < q$ be primes with $p \mid q-1$. Any two non-abelian groups of order pq are isomorphic.

***Proof*:**
Let G be non-abelian of order pq. Its Sylow q-subgroup $Q = \langle y \rangle$ is normal (theorem 4.9.1(1)), and some $x \in G$ has order p (theorem 4.8.7). Then $xyx^{-1} = y^s$ for an integer s, so $x^j y^k x^{-j} = y^{s^j k}$ for all $j \geq 0$ and k, and $y = x^p y x^{-p} = y^{s^p}$, so $s^p \equiv 1 \pmod{q}$. If $s \equiv 1$, then x and y commute, and $\langle x, y \rangle$, whose order is divisible by p and by q, is all of G and abelian, so the class of s has order p in $(\mathbb{Z}/q\mathbb{Z})^\times$. That group is cyclic (corollary 11.3.15), so it has exactly one subgroup of order p (theorem 2.3.11). Since $\langle x \rangle \cap Q = 1$, the pq elements $y^i x^j$ with $0 \leq i < q$ and $0 \leq j < p$ are distinct and form G, and

$$(y^i x^j)(y^k x^l) = y^{i+s^j k} x^{j+l}$$

Let G' be another such group, with y', x' and s'. Both s and s' generate the unique subgroup of order p of $(\mathbb{Z}/q\mathbb{Z})^\times$, so $s \equiv s'^t$ for some t with $p \nmid t$. Replacing x' by x'^t, which also has order p, replaces s' by s'^t, so the two multiplication rules coincide, and $y^i x^j \mapsto y'^i x'^j$ is an isomorphism $G \to G'$.

Corollary 11.3.17: Isomorphisms Of Multiplicative Groups

Let $n \geq 2$ be an integer with factorization $n = p_1^{\alpha_1} p_2^{\alpha_2} \cdots p_r^{\alpha_r}$ in $\mathbb{Z}$, where $p_1, ..., p_r$ are distinct primes. Then we have the following isomorphisms of (multiplicative) groups:

1. $(\mathbb{Z}/n\mathbb{Z})^\times \cong (\mathbb{Z}/p_1^{\alpha_1}\mathbb{Z})^\times \times (\mathbb{Z}/p_2^{\alpha_2}\mathbb{Z})^\times \times \cdots \times (\mathbb{Z}/p_r^{\alpha_r}\mathbb{Z})^\times$
2. $(\mathbb{Z}/2^\alpha\mathbb{Z})^\times$ is the direct product of a cyclic group of order 2 and a cyclic group of order $2^{\alpha-2}$, for all $\alpha \geq 2$
3. $(\mathbb{Z}/p^\alpha\mathbb{Z})^\times$ is a cyclic group of order $p^{\alpha-1}(p-1)$ for all odd primes p and $\alpha \geq 1$

***Proof*:**
For (1), the ideals $(p_i^{\alpha_i})$ are pairwise comaximal, as $\gcd(p_i^{\alpha_i}, p_j^{\alpha_j}) = 1$ for $i \neq j$, so theorem 9.6.11 gives a ring isomorphism $\mathbb{Z}/n\mathbb{Z} \cong \prod_i \mathbb{Z}/p_i^{\alpha_i}\mathbb{Z}$. A ring isomorphism restricts to an isomorphism of unit groups, and the units of a product of rings are the tuples of units.

For (2) and (3), let p be a prime, $k \geq 1$, with $k \geq 2$ if $p = 2$, and let $u \in \mathbb{Z}$. Then

$$(1 + p^k u)^p = 1 + p^{k+1} u + \sum_{j=2}^{p} \binom{p}{j} p^{jk} u^j \equiv 1 + p^{k+1} u \pmod{p^{k+2}}$$

since for $2 \leq j < p$ the prime p divides $\binom{p}{j}$ and $jk + 1 \geq k + 2$, and $pk \geq k + 2$ (for $p = 2$ this is where $k \geq 2$ is used). Applying this with $k + i$ in place of k gives, by induction on $i \geq 0$,

$$(1 + p^k)^{p^i} = 1 + p^{k+i} u_i \qquad \text{with } u_i \equiv 1 \pmod{p}$$

So if $\alpha > k$, the class of $1 + p^k$ in $(\mathbb{Z}/p^\alpha\mathbb{Z})^\times$ satisfies $(1 + p^k)^{p^{\alpha-k}} = 1$ and $(1 + p^k)^{p^{\alpha-k-1}} \neq 1$, and has order exactly $p^{\alpha-k}$.

For (3), let p be odd. The group $(\mathbb{Z}/p^\alpha\mathbb{Z})^\times$ has $p^\alpha - p^{\alpha-1} = p^{\alpha-1}(p-1)$ elements, the classes prime to p. Reduction modulo p is a surjective homomorphism onto $(\mathbb{Z}/p\mathbb{Z})^\times$, which is cyclic of order $p-1$ (corollary 11.3.15). If g maps to a generator, its order is a multiple $(p-1)t$ of $p-1$, and g^t has order $p-1$. The element $1+p$ has order $p^{\alpha-1}$ (the case $k=1$ above, or trivially if $\alpha = 1$). The orders are coprime and the group is abelian, so $g^t(1+p)$ has order $p^{\alpha-1}(p-1)$ and generates the group.

For (2), $(\mathbb{Z}/4\mathbb{Z})^\times = \{1,3\}$ is cyclic of order 2, which is the case $\alpha = 2$. Let $\alpha \geq 3$. The group has $2^{\alpha-1}$ elements, and $5 = 1+2^2$ has order $2^{\alpha-2}$ (the case $p=2$, $k=2$ above). Every power of 5 is 1 modulo 4, while -1 is not, so $\langle -1\rangle \cap \langle 5\rangle = \{1\}$. Both subgroups are normal, the group being abelian, and $|\langle -1\rangle||\langle 5\rangle| = 2^{\alpha-1}$, so the group is the direct product $\langle -1\rangle \times \langle 5\rangle$ (theorem 5.2.6).

This description of $(\mathbb{Z}/n\mathbb{Z})^\times$ is used in the proof of the Kronecker-Weber theorem ([11, Kronecker Weber's Theorem]).

Exercise 11.3.1

1. Show that there are irreducible polynomial in $\mathbb{Z}[x]$ and $\mathbb{Q}[x]$ for every degree.
2. Let $f(x)$ be an irreducible monic polynomial of $\mathbb{Z}[x]$. Show that $f(x)$ mod p has a root for infinitely many primes. The opposite is true too when $\deg f \geq 2$: there are infinitely many primes for which $f(x)$ mod p does *not* have a root, though that direction lies deeper, through the Chebotarev density theorem[3]
3. Show that $t^2+t+1 \in \mathbb{F}_2[t]$ is irreducible. Use this to show that $\mathbb{F}_2[t]/(t^2+t+1)$ is a field with 4 elements.

[3] see [45]

Part III

Modules

When we explored groups, we saw that the kernel of groups were groups (in particular a special subset of a group called the normal subgroup), and if we have an abelian group, every abelian group is the kernel of some group.

The story for rings is more complicated. As we mentioned at the beginning of the part on rings, some authors include an identity in the definition, others do not. If we include an identity, then the kernel of ring homomorphisms are not rings. Even if we take exclude identities as part of the definition, notice that practically none of the ideals we have encountered are among the rings previously mentioned, essentially because almost every ring we have mentioned has an identity, but if an ideal has an identity, it is the entire ring. The cokernels is also not as well-behaved (recall that $i : \mathbb{Z} \hookrightarrow \mathbb{Q}$ is an epimorphism). In this way, the kernel of rings and rings are very different objects, and so a lot goes into studying ideals as though they are a seperate object in ring theory. In particular, we do not focus a lot of attention of constructing rings based on other rings (similar to how we constructed polyabelian or poly-cyclic groups). Overall, the category of **Ring** and **Rng** are not the best behaved categories.

The concept of Modules can be thought of as a fix to this. Modules are more similar to abliean groups than they are to rings, for example if you have any sub-module, then the quotient will always be well-defined. Furthermore, all Rings can be thought as a module, making the study of module subsumes the study of rings. As we shall see,

We shall see that an R-module M will be an abelian group M with an ring action by R. The reason M is an abelian group and not just a set like in group actions is because it is a generalization of vector-spaces, not an addition to group actions. What I mean by this is that you have to think of the reasoning from the point of view that vector-spaces came first, and then there were many instances where the field acting on the group was too restrictive, which lead to module.

Over the 19th and 20th century, many mathematical objects were studied that all turn out to be modules. To those who want the punchline, I will point out a few right now:

1. All abelian groups are $\mathbb{Z}$-modules
2. All linear transformations are $F[x]$-modules
3. All rings are modules (If R is a ring, then it is an R-module over itself)
4. All ideals are modules (if $I \subseteq R$ is an ideal, it is an R-module)
5. All vector-spaces over k are k-modules
6. All vector-bundles are projective modules

12

Introduction to Modules

A lot of the algebraic structures we have covered so far share a similar story: we ask how maps between them can be studied (leading us to explore sub-objects and quotient-objects), and we ask how to generate them. Modules will have the same story at the beginning. When we will be building up modules, they will more closely resemble groups and rings. In groups, simple groups and free groups played a bigger role than they did in ring theory, and free modules return in section 12.3.2.

12.1 Basic Definitions and Examples

Definition 12.1.1: [left]-Module

Let R be a ring (more generally a rng). A *left R-module* or a *left module over R* is a set M together with

1. a binary operation $+$ on M under which M is an **abelian group**, and
2. an action of R on M (that is, a map $R \times M \to M$) denoted by rm, for all $r \in R$ and for all $m \in M$ which satisfies
   (a) $(r+s)m = rm + sm$, for all $r, s \in R$, $m \in M$
   (b) $(rs)m = r(sm)$, for all $r, s \in R$, $m \in M$
   (c) $r(m+n) = rm + rn$ for all $r \in R$, $m, n \in M$

   If the rng R is a ring (i.e. has a 1), we impose the additional axiom

   (d) $1m = m$, for all $m \in M$

   And we sometimes call such a module a *unital* R-module.

Right modules are defined similarly. If the R ring is commutative, and M is an left $R-$*module*, we can turn it into a right $R-$*module* by defining $rm = mr$. To distinguish when different rings may be acting on an abelian group M, the symbol $\curvearrowright$ (read *acts on*) may be used to show that a ring R is acting on M: $R \curvearrowright M$

Unless stated otherwise, the term "module" will reference "left-module" Note that for a module M, we usually denote what ring it is over (or what ring acts upon it) by saying it is an *[ring symbol here]-module* (ex. R-module, $F[x]$-module, $[M_{m\times n}(\mathbb{F})]$-module). We might also say R acts on M where R is the ring and M is the abelian group. We will always be working *rings* actions, however as noted in the definition, we can also work with *rng* actions. Note too that we will often say that "R acts on the module M" to mean that we define a ring-action of R on the abelian group M.

When working over rings, the distributivity condition forces M to be abelian: let $r = s = 1$:

$$(1+1)\cdot(m+n) = (1+1)\cdot m + (1+1)\cdot n = 1\cdot m + 1\cdot m + 1\cdot n + 1\cdot n = m+m+n+n$$

$$(1+1)\cdot(m+n) = 1\cdot(m+n) + 1\cdot(m+n) = 1\cdot m + 1\cdot n + 1\cdot m + 1\cdot n = m+n+m+n$$

which implies $m+n = n+m$ for all $m, n \in M$. Notice how this when working with group actions, the group acting on the set did not effect any strucutre of set, namely since the set has no additional structure. Be mindeful that when referring to a module, we will usually call M the module. For example we might say "Let $\mathbb{Z}$ act on $\mathbb{Q}$ by $r \cdot a = ra$. Then $\mathbb{Q}$ has the following properties". Here $\mathbb{Q}$ is being regarded as a module, namely a $\mathbb{Z}$-module, not just as an abelian group or a ring.

If R is commutative then we defined a two-sided action $r \cdot m = m \cdot r$. If R is not commutative, then defining this property will have problems with $2(b)$: Let R be non-commutative, let $r \cdot m = m \cdot r$, and take $r, s \in R$ such that $rs \neq sr$ and $rs \cdot m \neq sr \cdot m$ (an example of such a module will be presented in the example 12.1.7). Then:

$$(rs)m = m(rs) = (mr)s = (rm)s = sr(m) = (sr)m$$

however, $sr \cdot m \neq rs \cdot m$ so this statement is not necessarily always true. Note how it may be possible that even though $rs \neq sr$ that $rs \cdot m = sr \cdot m$, for example, the trivial module structure $r \cdot m = m$ for any $r \in R$ will produce this result, however, as example 12.1.7 shows, every module induces a *faithful* module (the word faithful being defined equivalently to the group-action case[1]), and so making a distinction between left and right modules is very important over non-commutative rings.

We now have many operators to keep track of: the group operation of M, the two rings operations of R, and the ring action of R on M. When working with modules, we often focus on the ring-action, and use group and rings structures that were previously covered.

One might ask themselves if modules and group-rings might be similar concept, boths somehow being a ring "acting" on a group. If this suspicions crossed your mind, it is good idea to check how these two structure's differ (for example, if $|G| > 1$, recall a group-ring always has zero-divisors. Find an example in the list of examples bellow that does not have this property).

Since a module is an action on an abelian group, it would make sense if there is an equivalent to a permutation representation for module's as there is for group-actions. Recall that any group action can be seen as a group-homomorphism:

$$\rho : G \to \mathrm{Sym}(S)$$

The exact same idea works for defining an R-module as an action. In order to do so, we need an a notion of an R-module homomorphism that preserves some of the structure. Before looking at the definition bellow, see if you can come up with the definition of an R-module homomorphism.

Definition 12.1.2: R-module homomorphism

Let R be a ring and let M and N be R-module A map $\varphi : M \to N$ is an *R-module homomorphism* if it respects the R-module structure of M and N, i.e.

1. $\varphi(x + y) = \varphi(x) + \varphi(y)$ for all $x, y \in M$
2. $\varphi(r \cdot x) = r\varphi(x)$, for all $r \in R$, $m \in M$

Since M and N are both abelian groups, R-module homomorphisms are sometimes called *R-linear group homomorphisms*, where the map is R-linear since the ring actions passes through the homomorphism. Notice that both M and N both have the same ring acing on them: it is imperative that an R-module homomorphism be betewen R-modules.

Returning to the discussion on treating R-modules as an action, it is a good exercise to check ones understanding of the axioms of modules that the "axiomatic representation" of an R-module is equivalent to defining an R-module as a ring-homomorphism

$$\rho : R \to \mathrm{End}_{\mathbf{Ab}}(M)$$

where $\mathrm{End}_{\mathrm{Ab}}(M)$ is the collection of all group-homomorphisms from M to itself. Notice that $\mathrm{End}(M)$ is an [abelian] group because M is an abelian group, while the ring operation of composition, $\circ$, automatically satisfies the condition to make $\mathrm{End}_{\mathbf{Ab}}(M)$ a ring, which should be checked (for example, $\mathrm{id} : M \to M$ is an element of $\mathrm{End}_{\mathbf{Ab}}(M)$, hence it has a multiplicative identity). It will sometimes be useful to define an R-module as a ring homomorphism $\rho\colon R \to \mathrm{End}_{\mathbf{Ab}}(M)$. For instance, *restricting scalars* (definition 15.3.1) along a ring homomorphism $f\colon S \to R$ is the composition $\rho \circ f\colon S \to \mathrm{End}_{\mathbf{Ab}}(M)$.

[1] That is, if $r \cdot m = s \cdot m$ for every $m \in M$, then $r = s$.

As with rings and and with groups, we can define a category: given a ring R, the collection of R-modules and the collection of R-module homomorphisms forms a category:

Definition 12.1.3: R-Mod

The collection of all R-modules and R-module homomorphisms forms a category, which we will denote R-**Mod** or $\mathbf{Mod}_R$.

This category has some of the nice properties that the other categories that were extensively worked in so far have:

Definition 12.1.4: Classical definition of Submodule

Let R be a ring and let M be an R-module. An R-submodule of M is a subgroup N of M which is closed under the action of ring elements, i.e., $rn \in N$, for $r \in R$, $n \in N$

Definition 12.1.5: Categorical Definition of Submodule

Let $N \subseteq M$. Then if there exists a R-module inclusion map $\varphi : N \to M$ that is also an R-module homomorphism, then N is a submodule of M

Proposition 12.1.6: The sub-module Criterion

Let R be a ring and let M be a R-module. A subset of N of M is a submodule of M if and only if

1. $N \neq \emptyset$, and
2. $x + ry \in N$ for all $r \in R$, and for all $x, y \in N$

***Proof*:**
Easy to prove both ways (recall that M is an abelian group, and for the other way, simply set the values of r and x to some appropriate values)

Example 12.1.7: Modules

1. Let R be any ring. Then $M = R$ is a left R-module with the ring multiplication as the ring action: $r \cdot s = rs$. Take a moment to think about what are the submodules of R? The answer will be revieled in the next sentence. the sub-modules of M are the left-ideals of R! This is because ideals of R are closed under multiplication of R, and the axioms of modules require closure under "scalar multiplication" by the ring elements. Is R/I for any (two-sided) ideal of R also an R-module? Take a moment to think about it. The answer is that R/I is indeed an R-module! We will soon show that the kernel of module-homomorphisms will be submodules, showing that the axioms of a module "fix" the category of **Ring** by being closed under kernels!

   From now on, if we say that R is a [left/right] R-module over itself, it will be taken that the R-action on R is [left/right] multiplication by R, with a bias towards *left* R-modules if R is

non-commutative and which side of the action was not specified. We will also sometimes say "let R have the natural R-module structure" to mean the one induced by the ring operation of R.

2. Let R be a ring with 1 and let $n \in \mathbb{Z}^+$. Then since R is a group, we can define the direct product:
$$R^n = \{(a_1, ..., a_n) | a_i \in R, \text{ for all } i\}$$
We can turn R^n to an R-module by defining addition in R^n component wise, and the action being componentwise:
$$r(a_1, ..., a_n) = (ra_1, ..., ra_n)$$
Such a module R^n is called a *free module* [a]. More on these in section 12.3.2. If $R = \mathbb{R}$, then we have the usual algebraic structure of $\mathbb{R}^n$ common in analysis and linear algebra.

3. If M is an R-module, and S is a sub-ring of R, then M is automatically an S-module (i.e. the set M is also a module under the ring action with S). Since if the module axioms hold for R, i.e. the ring being closed under the ring action and the rules holding, then it will still hold for a sub-ring. In this way $\mathbb{R}$ is an $\mathbb{R}$-module, a $\mathbb{Q}$-module, and a $\mathbb{Z}$-module with the ring action being defined by regular multiplication. Notice though that the $\mathbb{R}$-submodules, $\mathbb{Q}$-submodules, and $\mathbb{Z}$-submodules differ, showing that having a different ring act on $\mathbb{R}$ greatly changes its module structure.

4. Let M be an R-module. If $I \subseteq R$ is a two-sided ideal, is it always true that M is an R/I-module? Under what condition's is it an R/I-module? The following is an exploration of when this is the case: Let I be a 2-sided ideal of R such that $am = 0$ for all $a \in I$ and $m \in M$, that is, $I \subseteq \text{Ann}_R(M)$ (definition 12.3.21), the largest such ideal. Then we say that M is *annihilated* by I. In this situation, we can make M into an (R/I)-module by defining an action of the quotient ring R/I on M as follows: for each $m \in M$, and $(r + I) \in R/I$, let
$$(r + I)m = rm$$
This is well defined since for any $a \in I$, $(r + a)m = rm + am = rm$. Check that the axioms of a module still hold. When $I = \text{Ann}_R(M)$, M is a *faithful* R/I-module: if $(r + I)m = 0$ for all $m \in M$, then $r \in \text{Ann}_R(M) = I$.

   If, conversely, if $I = \text{Ann}_R(M)$, and we have defined an R/I-action on a module M, can we define an R-action on M? More generally, given two rings R and S acting on M, when can we say that the module structure they induce on M is "comparable"‘ in some way? One way two types of modules can be equivalent is if they are something called "Morita equivalence", which is settled in chapter 18$\star$ [b])

5. What is and is not a well defined R-module is not always intuitive. Take for example the group ring $R = \mathbb{Z}[G]$ where $G = \{1, \sigma\}$ is a group of order 2. For the abelian group, take the module $M = \mathbb{Z}e_1 + \mathbb{Z}e_2$, where e_1, e_2 are generators for $\mathbb{R}^2$ (for example, $(1, 0)$ and $(0, 1)$ generate $\mathbb{R}^2$ as an abelian group), and the product $\mathbb{Z}e_i = \{ze_i | z \in \mathbb{Z}\}$.

   Define the action as:
$$\sigma(e_1) = e_1 + 2e_2 \qquad \sigma(e_2) = -e_2$$
Then M is a $\mathbb{Z}[G]$-module! The big one to check is $r_1(r_2m) = (r_1r_2)m$. The other conditions follow more easily. This is an example of how the limit on what can be defined as a module is not easy, since an example like this one exists, checking when a module is defined can be quite useful.

6. Let R be any ring and let $F(A,R)$ be the set of all [set]-functions from A to R:

$$f \in F(A,R) \Rightarrow f : A \to R$$

Then define the action by R to be

$$r \cdot f = rf \qquad r \cdot f(a) := rf(a) \qquad \forall a \in A$$

since $f(a) \in R$ Band $r \in R$. Define the group operation to be the new function $f+g$ where

$$(f+g)(a) := f(a) + g(a) \qquad \forall a \in A$$

Note that these functions are *not* R-module homomorphisms, since the set A has no structure! This R-module is is often dentoed as R^A (to symbolise all functions from A to R)[c].

Very often in Analysis and Algebraic Geometry, the set A will be all continuous functions on a topological space X, denoted $C(X)$, and the ring R would be the real or complex numbers $\mathbb{R}$ or $\mathbb{C}$. The study of these spaces is done commonly in functional analysis and classical algebraic geometry, usually requiring some analysis knowledge. In this textbook, these types of R-module will come back when discussing free-modules in theorem 12.3.11.

7. Let X be any set and $P(X)$ the powerset. Sending $A \subseteq X$ to its indicator function $\chi_A \colon X \to \mathbb{Z}/2\mathbb{Z}$, with $\chi_A(x) = \overline{1}$ for $x \in A$ and $\chi_A(x) = \overline{0}$ otherwise, is a bijection from $P(X)$ onto the module $(\mathbb{Z}/2\mathbb{Z})^X$ of the previous item, since a subset is recovered as the set of points where its indicator function is $\overline{1}$. Transporting the module structure along this bijection makes $P(X)$ a $\mathbb{Z}/2\mathbb{Z}$-module, in which $A+B$ is the symmetric difference $(A \setminus B) \cup (B \setminus A)$, since $\chi_A + \chi_B$ takes the value $\overline{1}$ exactly at the points lying in exactly one of A and B.

[a] not dissimilar to free-groups we studied in section 2.3.3

[b] For the categorically inclined, R-mod and S-mod are Morita equivalent if they are equivalent as categories. Are R-mod and R/I-mod equivalent?

[c] The notation matches exponentiation of numbers: for finite A and R there are $|R|^{|A|}$ functions $A \to R$.

When we worked with group-actions, we can take any group G and make it act on any set S. This is not the case with modules (i.e. given some fixed ring R and abelian group M, there does not always exist a ring-homomorphism $\varphi : R \to \mathrm{End}_{\mathbf{Ab}}(M)$):

Example 12.1.8: Impossible Module

Consider $M = \mathbb{Z}$ and $R = \mathbb{Q}$. Is there always a $\mathbb{Q}$ action over $\mathbb{Z}$? If it was, then:

$$\frac{1}{2} \cdot 1 = z$$

must be an element of $\mathbb{Z}$ ($z \in \mathbb{Z}$). Then:

$$z + z = \frac{1}{2} \cdot 1 + \frac{1}{2} \cdot 1 = (\frac{1}{2} + \frac{1}{2}) \cdot 1 = 1 \cdot 1 = 1$$

as a direct consequence of the module axioms. However, this implies

$$z + z = 1$$

which is not possible for any $z \in \mathbb{Z}$, and thus a contradiction! This is also an example where you can have a $\mathbb{Z} \subseteq \mathbb{Q}$ and a defined $\mathbb{Z}$-module, but there is no extension of M to a $\mathbb{Q}$-module for

which the previous module structure is embedded within it (opposite to how if we had a $\mathbb{Q}$-module, it would automatically be a $\mathbb{Z}$-module)[a]

[a]There is in fact still a notion of extending a module along a ring map, however the original module need not embed in the result. See section 15.3, and corollary 15.3.8 for exactly when it does

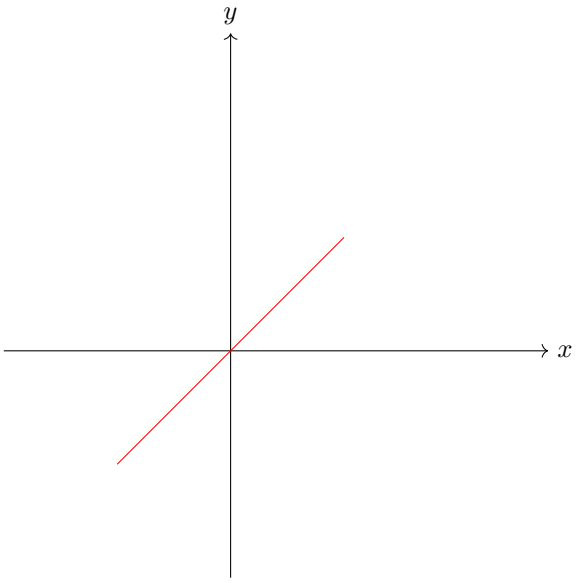
12.1.1 Vector-spaces

If the ring is a field, we give a special name to such a module:

Definition 12.1.9: Vector Space

Let M be an R-module. If $R = k$ is a field, then we call M a *vector-space* over k. An arbitrary field is often denoted by k or $\mathbb{F}$.

Due to historical reasons $\mathbb{F}$-modules homomorphisms are called *linear transformations*. The motivation for this will be seen in chapter 13, but the gist is that when $\mathbb{F} = \mathbb{R}$, the geometric intuition of the images and kernels of linear transformations will all be hyper-planes in the domain and codomain. A simple example would be $f : \mathbb{R} \to \mathbb{R} \times \mathbb{R}$ with $f(x) = (x, x)$. The resulting hyper-plane (simply called a line since it's 1 dimensional), looks like so:



If you are comfortable with linear algebra, then you are probably familiar of how simple the structure of vector-spaces are (if not, you may skip this remark: all that will be discussed here is covered in chapter 13). Essentially, a vector space V over k can be completely determined by it's dimension. For modules over a general ring R, this is sometimes far from the case: sometimes R will not even embed itself in V! However, there are ways in which we can systematically loosen the conditions on the ring k acting on V and get similar, but weaker, results. Thus, if it is intuitive to think of modules as generlizations of vector-spaces, which historically is how many mathematicians thought

of modules, then these chapters on module theory can be thought of as what happens if the action on a vector-space no longer forces to give a basis.

Example 12.1.10: Vector Spaces

1. The most common example is when $R = \mathbb{R}$ and $M = R^n = \mathbb{R}^n$. This vector-space is commonly reffered to as *euclidean space* due to the classical geometrical properties that can be defined on it(ex. a notion of distance, length, angle, translation and rotation)
2. If I is a maximal ideal in a commutative ring, and $IM = 0$, then M is a vector space over R/I (since R/I is a field)
3. Take $\mathbb{F}[x]$ to be a polynomial ring over a field $\mathbb{F}$, an let $f(x)$ be any polynomial of degree $n \geq 1$. Then $V = \mathbb{F}[x]/(f(x))$ is another polynomial ring. Let $\mathbb{F}$ act on V by $a \cdot \overline{f} = \overline{af}$. This will be a vector-space. If $\mathbb{F}$ is a finite field of order p, then we can show that V has p^n.
4. (If you know some calculus). Consider $\mathbb{R}[x]$. Then it is a vector-space over $\mathbb{R}$ ($\mathbb{R}[x]$ is an abelian group, and the action is simply the multiplication in $\mathbb{R}[x]$, since $\mathbb{R} \subseteq \mathbb{R}[x]$). Recall that the derivative has the following two properties:
$$\frac{d}{dx}(p(x) + q(x)) = \frac{d}{dx}p(x) + \frac{d}{dx}q(x)$$
$$\frac{d}{dx}(cf(x)) = c\left(\frac{d}{dx}f(x)\right)$$
This shows that the derivative is in fact a linear transformation (i.e. $\mathbb{R}$-module homomorphism) from $\mathbb{R}[x]$ to itself![a]
5. If R is a nonzero ring where the only two-sided ideals are (0) and R, then R is called a *simple ring* (similarly to how simple groups are groups whose only normal subgroups are $\{e\}$ and the entire group). Then $Z(R)$ (the center of the ring) *has* to be a field (can you show this?). If $k = Z(R)$, then any simple ring is a vector-space over its center.

[a]For those who know some Linear algebra and Fourier Analysis, the "diagonalization" of the derivative is the Fourier transform of a function! For details, see the Gelfand Transformation of unbounded operators

If $R = \mathbb{F}$ is a field, we give a special name to sub-modules:

Definition 12.1.11: Subspace

If $W \subseteq V$ is a submodule of a vector space, then we call it a *subspace*

12.1.12 Remark (If you know linear algebra) When determining where a linear transformation maps, it is sufficient to know where the basis maps to. General R-modules need not have a basis well-defined on them. However, *cyclic* modules will always have a basis, since $M = R \cdot x$ for some $x \in M$. Thus, R is always a cyclic R-module over itself since $R \cdot 1 = R$. General R-modules need not have a basis, and so R as an R-module a particularly nice module.
As a consequence of this, for any R-module homomorphism: $\varphi : R \to N$, it suffices to know where $\varphi(1)$ maps to. In general, it is much harder to determine a map $\varphi : M \to N$ given where only certain elements are mapped to.

Exercise 12.1.1

1. Let $\langle m \rangle = M$ be an R-module generated by a single element (i.e. if $x \in M$, $x = a_i m$ for some $a_i \in R$)[2]. Show that $\text{Ann}_R(M)$ (definition 12.3.21) is a two-sided ideal of R.
2. Let M be as in the previous question, with R commutative. Show that $R/\text{Ann}_R(M) \cong M$ as R-modules, so that ideals can be studied through the cyclic modules R/I.
3. (practice with categories). Let A be a set and $F(A)$ be the group freely generated by A (definition 2.3.29).
    1. Prove that the collection of all vector-spaces along with linear transformations form a category. Label it k-**Vect**.
    2. Show that $F(A)$ can have a vector-space structure over k put on it such that $F : \textbf{Set} \to k\textbf{-Vect}$ is a functor. (Hint: See the functor defined in section 2.3.5)
4. What is the initial object of Mod_R (if any)?

12.1.2 Algebras

Recall that if S is a ring, then it is always an abelian group, and there is always a unique homomorphism $\mathbb{Z} \to S$. Using this homomorphism, we can define a $\mathbb{Z}$-action given by $z \cdot s = \varphi(z)s$, and so every ring is automatically a $\mathbb{Z}$-module. More generally, any ring map $R \to S$ where R is commutative produces an R-module structure on the ring S that is compatible with the multiplication of S. To encapsulate this extra structure, we shall define a new term known as an *R-algebra* that represents a ring that also has a ring action on it.

The definition is simple if constrained to commutative rings, but there are many important cases where we must work with a non-commutative codomain (particularly in *Core Representation Theory* [17]), and so the above intuition must be made more precise. Given any ring homomorphism $\varphi : R \to S$, we can always define an R-module structure on S by defining

$$r \cdot s = \varphi(r)s$$

that is, the action is multiplication in S. The axioms of an R-module is immediate given the axioms of a ring and that φ is a homomorphism. Since we can multiply, we run into situations like the following: If $s_1 = r_1 \cdot x$ and $s_2 = r_2 \cdot y$, then:

$$s_1 s_2 = (r_1 \cdot x)(r_2 \cdot y)$$

[2] Module generation is the subject of section 12.3.

Such an element is not necessarily equal to $(r_1r_2)\cdot xy$ (similarly to how with rings, if the underlying set is not abelian, distributivity would not hold). This would be desirable, as in we would want $\varphi : R \to S$ to be a ring homomorphism that was also compatible with the R-module structure in the following way:

$$(r_1 \cdot x)(r_2 \cdot y) = (r_1r_2) \cdot xy$$

As a commutative diagram:

$$\begin{array}{ccc} (R \times S) \times (R \times S) & \xrightarrow{a \times a} & S \times S \\ \downarrow{\scriptstyle \ell} & & \downarrow{\scriptstyle m} \\ R \times S & \xrightarrow[a]{} & S \end{array}$$

where the left map is

$$\ell\big((r_1, x), (r_2, y)\big) = (r_1r_2,\ xy), \qquad \ell = (\mu_R \times m) \circ \sigma,$$

with σ the shuffle $\big((r_1, x), (r_2, y)\big) \mapsto \big((r_1, r_2), (x, y)\big)$.

Getting back to the homomorphism $\varphi : R \to S$ and the R-module structure on S, if we further require that R be *commutative* and that $\varphi(R) \subseteq Z(S)$ (the multiplicative center), then the left and right R-module structure will coincide (i.e. left and right multiplication produce the same module structure), and the ring operation of S will be compatible with that of R, that is

$$(r_1r_2) \cdot (s_1s_2) = \varphi(r_1r_2)(s_1s_2) = \varphi(r_1)\varphi(r_2)(s_1s_2) = \varphi(r_1)s_1\varphi(r_2)s_2 = (r_1 \cdot s_1)(r_2 \cdot s_2)$$

Boxing these properties we define an algebra:

Definition 12.1.13: R-algebra

Let S be a ring, R a commutative ring. Then S is called an (associative) R-algebra if S is also an R-module satisfying:

$$r \cdot (xy) = (r \cdot x)y = x(r \cdot y)$$

Equivalently, S is an R-algebra if there exists ring homomorphism $\varphi : R \to S$ where $\varphi(R) \subseteq Z(S)$. Furthermore:

1. If S is a division ring, then the R-algebra S is called a *division algebra*
2. If S is commutative, then the R-algebras S is called a *commutative algebra*

Note that S has has a natural R-module structure by forgetting the multiplication so that the action

$$r \cdot a[= a \cdot r] = \varphi(r)a$$

is the R-module structure. Unless otherwise stated, this will be the assumed R-module structure of an R-algebra S.

Algebra homomorphisms are defined similarly to how they have been defined before (can you figure it out without looking at the following definition?)

Definition 12.1.14: R-Algebra Homomorphism and Isomorphism

If A and B are R-algebra, then $\varphi : R \to S$ is an R-algebra homomorphism if φ is a ring-homomorphism and

$$r \cdot \varphi(x) = \varphi(r \cdot x)$$

If there exists an inverse function φ that is also an R-algebra homomorphism, then A and B are *algebra isomorphic*, or just *isomorphic* to each other, and we write $A \cong B$ if the context permits.

With this definition, the collection of R-algebra's along with R-algebra homomorphism forms the category R-**Alg**.

12.1.15 For the Categorically Inclined: Monoidal Object in R-module

Another way of saying that an R-algebra adds the notion of multiplication to an R-module in a "consistent" way is by noticing that R-algebra are the monoidal objects in the category of R-mod. Since all rings are $\mathbb{Z}$-modules, the discussion in section 9.7⋆ shows that rings are the monoidal objects in $\mathbb{Z}$-Mod (Hence, the category $\mathbb{Z}$-Alg is the category **Ring**).

$$\begin{array}{ccc}
(R \otimes S) \otimes (R \otimes S) & \xrightarrow{a \otimes a} & S \otimes S \\
\cong \downarrow {\scriptstyle 1_R \otimes \tau_{S,R} \otimes 1_S} & & \\
(R \otimes R) \otimes (S \otimes S) & & \downarrow {\scriptstyle m} \\
\mu_R \otimes m \downarrow & & \\
R \otimes S & \xrightarrow[a]{} & S
\end{array}$$

The word "associative" was put in parenthesis in the definition of an algebra since there is a notion of an algebra that is not associative: some author's say the object in definition 12.1.13 is an *associative algebra*, and define algebras more generally as follows: Take an R-module M, and equip it with a binary operation such that the binary operation is left and right distributive (for all $a, b, c \in M$, $a \cdot (b + c) = a \cdot b + a \cdot c$, an $(b + c) \cdot a = b \cdot a + c \cdot a$), and that it is compatible with scalars (for all $a, b \in R$ and $x, y \in M$, $(ab) \cdot (xy) = (a \cdot x)(b \cdot y)$), then M is called an R-algebra. If the binary operation is associative and an identity exists, then it would be a ring operation it will be called an associative algebra.

There is a good reason adding the prefix "associative" in the greater context of mathematics: a large domain of study in mathematics is the study of *lie groups* and *lie algebras*, which focus on understanding the structure of smooth manifolds and representations of algebraic groups. Notably, Lie algebras are *not associative*, satisfying instead a relation called the *Jacobi identity*. However, since they are non-associative, they are not *rings*. They are thus quite different objects from R-algebras as we have defined them, and so do not encompass our current discussion. Their study takes up many books (among them *Lie Groups* [57] and [38]) which the reader with a good background in linear algebra has the algebraic background to understand[3].

From now on, when we say algebra or R-algebra, we will mean an associative algebra as defined in

[3]Though a good basis in differential geometry would go a long way for intuition

definition 12.1.13.

If $R = k$ is a field, then using the terminology of vector-spaces, a k-algebra would be defining a way of multiplying vectors. It is non-trivial to always define a multiplication structure on an R-module (I suggest you try it on $\mathbb{R}^n$!). Even harder is to ask whether there is a "natural" way of creating a multiplication structure, or more broadly to make an R-module M big enough so that we can define an R-algebra in a "natural" way[4]. We will in fact define such a notion in section 16.3.

Example 12.1.16: R-Algebras

1. If R has positive characteristic p ($\#(\text{im}\mathbb{Z}) = p < \infty$), then it is more natural to define a $\mathbb{Z}/p\mathbb{Z}$-algebra, namely since as a $\mathbb{Z}$-algebra the kernel of the map $\mathbb{Z} \to R$ would be $p\mathbb{Z}$, and hence by the 1st isomorphism theorem it homomorphism commutes with $\mathbb{Z}/p\mathbb{Z} \to R$.

2. If S is commutative, then any subring of $R \subseteq S$ will define R-algebra on S (take $\iota : R \hookrightarrow S$). If S is not commutative, any subring of the center will define an R-algebra.

3. Let $R[x_1, x_2, ..., x_n]$ be the polynomial ring over several variables. Then it is an R-algebra through $r \cdot p(x) = rp(x)$.

4. Let $k[G]$ be a group ring over the field k. Then $k[G]$ is naturally a k-algebra.

5. If D is a divisor ring, then $Z(D)$ is always a *field*. If $k = Z(D)$, then any division ring D has a k-algebra structure on it, which also implies it is a vector-space over k.

6. Let $R = k^n$ for some field k. Then R is a k-algebra, with multiplication done point-wise.

7. Let $R = M_n(k)$ over a field k. Then R is a k-algebra. More generally, If A is a k-algebra, then $M_n(A)$ is a k-algebra.

   Compare this example to the previous one, which we will label $S = k^{n^2}$. As vector-spaces over k, $R \cong_k S$. However, as k-algebras, $R \ncong_{k\textbf{-Alg}} S$, since the multiplication structure is different between the two (one has nilpotent elements, the other does not).

8. If A, B are k-algebra, so is $A \times B$. By the previous part, that implies

   $$M_n(A) \times M_n(B)$$

   is also a k-algebra.

9. At the heart of real analysis is the $\mathbb{R}$-algebra $\mathbb{R}$, and at the heart of complex analysis is the $\mathbb{C}$-algebra $\mathbb{C}$. Note that $\mathbb{C}$ is also a $\mathbb{R}$-algebra.

10. Let $R = \mathbb{R}^2$. Then we know that $\mathbb{R}^2$ has point-wise multiplication, but we can also define the following:

    $$(a, b) \cdot (c, d) = (ac - bd, ad + bc)$$

    the multiplication of complex numbers $a + bi$ and $c + di$ written in coordinates, then this also forms an $\mathbb{R}$-algebra. In fact, this is $\mathbb{R}$-algebra isomorphic to $\mathbb{C}$ (over $\mathbb{R}$), by $(a, b) \mapsto a + bi$!

[4] natural meaning it satisfies the appropriate universal property

11. One might ask themselves how many $\mathbb{R}$-algebra's exist (i.e. classify all $\mathbb{R}$-algebras). A little more generally, one might ask how many finite-dimensional associatvive divison algebras there are, that is, algebra on which you can has division and asociativity, and the notion of finite-dimension is well-defined. Only $\mathbb{R}$, $\mathbb{C}$, and $\mathbb{H}$ exist, with dimension 1, 2, and 4! This is called *Frobenius' Theorem*. In this way, the real, compelx, and quaternions are kind of special. [a]

There are a large number of areas that algebras appear in, to name a notable few:

1. commutative algebras in algebraic geometry
2. Banach-Algebra's in real analysis
3. C^*-algebra or $*$-algebras more generally in functional analysis
4. Lie algebras in Lie theory, which are not algebras in the present sense, as their multiplication need not be associative. The cross product on $\mathbb{R}^3$ is an example: $(e_1 \times e_1) \times e_2 = 0$, while $e_1 \times (e_1 \times e_2) = e_1 \times e_3 = -e_2$. Every algebra A becomes a Lie algebra with the product $ab - ba$, see *Lie Groups* [57].

[a] https://math.stackexchange.com/questions/368456/frobenius-from-hurwitzs-theorem

Theorem 12.1.17: Universal Property of Free R-algebras

$R[A]$ is a free commutative R-algebra over a [finite set] A

Notice how this is similar to the universal property of commutative polynomial rings. In fact, that is a special case of this theorem by taking $R = \mathbb{Z}$

***Proof*:**
Let S be any commutative R-algebra and $\alpha : A \to S$ be any set-function. Since S is an commutative R-algebra, it is accompanied by a function $f : R \to S$ defining its multiplication.

Since $A = \{a_1, a_2, ..., a_n\}$ is finite, $R[A] = R[x_1, x_2, ..., x_n]$. Similarly to how the universal property of polynomial rings was proven (theorem 9.2.4), we will extend $f : R \to S$ to $f_1 : R[x_1] \to S$ by mapping $f_1(x_1) = \alpha(a_1)$ and extending linearly. f_1 is still an R-algebra homomorphism since $f_1(n_0 + n_1x + n_2x^2 + \cdots + n_kx^k) = f(n_0) + f(n_1)\alpha(a_1) + f(n_2)\alpha(a_1)^2 + \cdots + f(n_k)\alpha(a_1)^k$, and Similarly for multiplying two polynomials (using the binomial theorem and both the additive and multiplicative linearity of f_1). We can again extend f_1 to f_2 by defining $f_2(a_2) = \alpha(a_2)$ so that $f_2 : (R[x_1])[x_2] \to S$ (usually written as $f_2 : R[x_1, x_2] \to S$) is an R-algebra homomorphism. Continuing inductively will get us an R-algebra homomorphism $F : R[x_1, x_2, ..., x_n] \to S$. Thus, picking $\iota : A \to R[x_1, x_2, ..., x_n]$ will show that $\alpha : A \to S$ lifts uniquely to $F : R[A] \to S$, as we sought to show.

As a consequence of this, all R-algebras are quotient of polynomial rings!! That is, if S is a finitely generearted S algebra

$$R[x_1, x_2, ..., x_n] \twoheadrightarrow S$$

12.1.3 Abelian Groups as Modules

Some entire domains of study are done on particular modules. We have already studied a particular collection modules extensively! The most immediate example is that all rings are R-modules over themselves with the action defined by:

$$r \cdot x = rx$$

Then submodules are precisely the ideals of R, and so the study of all R-modules includes the study of the ring R.

In this section, we show another important classes of R-modules: let $R = \mathbb{Z}$, and let $M = A$ be *any* abelian group (finite or infinite), and write the operation of A as $+$. Make A into a $\mathbb{Z}$-module by defining the ring action as :

$$na = \begin{cases} a + a + \cdots + a \ (n \text{ times}) & \text{if } n > 0 \\ 0 & \text{if } n = 0 \\ -a - a - \cdots - a \ (n \text{ times}) & \text{if } n < 0 \end{cases}$$

Then M is a $\mathbb{Z}$-module. In fact, this ring action is the only possible ring action on M (i.e. abelian groups):

Proposition 12.1.18: Abelian Group iff $\mathbb{Z}$-module

Every abelian group is a $\mathbb{Z}$-module, and every $\mathbb{Z}$-module is an abelian group

***Proof*:**
for $\Rightarrow$, we can always define the previous ring-action and so it's always a $\mathbb{Z}$-module

For $\Leftarrow$, it is simply a consequence of the existence of 1 and the distributivity property, i.e., the ring action $\mathbb{Z}$ performs on abelian groups is "stuck" since we must play out the consequence of the axioms.

By definition

$$n \cdot 1 = n$$

and therefore

$$n \cdot 2 = n \cdot (1 + 1) = n \cdot 1 + n \cdot 1 = n + n$$

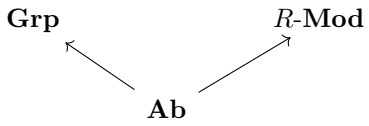
and continuity inductively

$$n \cdot k = n \cdot (1 + \cdots + 1) = n + \cdots + n$$

k times. This result is exactly the ring action we defined earlier. And so, forced by the axioms of groups and modules, it must be that every $\mathbb{Z}$ action on an abelian foms this particular $\mathbb{Z}$-module.

This proposition has a very intersting consequence: for every R-module, $R \neq \mathbb{Z}$, then it always has at least one other module structure, namely the natural $\mathbb{Z}$-module structure! An intuitive way to remember this fact is that numbers as we have learnt them earlier in our life usually have a $\mathbb{Z}$-module structure ($\mathbb{Z}$, $\mathbb{Q}$, $\mathbb{R}$, $\mathbb{C}$). The natural numbers form an abelian monoid that is not a group, so this intuition does not apply to $\mathbb{N}$.

This example can also make us think of the class of all abelian groups as a subclass of modules. Abelian groups are also a subclass of groups. However, not every group is a module (and vice versa). The following diagram I'm about to draw should not be taken as a commutative diagram, but as an "analogy":



So in a sense, we are "switching gears" when studying modules. We are taking a different approach to the study of algebraic structures as compared to studying groups.

Exercise 12.1.2

1. Recall that for any $\alpha \in R$ for a ring R, $f_\alpha : R[x] \to R$ mapping $p(x) \mapsto p(a)$ is a ring-homomorphism. Is it an R-module homomorphism?

12.2 Homomorphism and Quotients of Modules

In this section we explore R-module homomorphisms and how the image of R-module homomorphisms can be represented by a particular quotient of the domain (as we have seen with groups and rings). As R-module homomorphsism are explored, they should be constrasted to rings in that they will have many nicer properties. They are in fact comparable to abelian groups. Every subgroup of an abelian group is normal, and so too every submodule of a R-module will be "normal".

R-Module Homomorphisms

Definition 12.2.1: R-module homomorphism Terminology

Let R be a ring, M and N be R-module, and $\varphi : M \to N$ be a function. Then:

1. φ is called an *R-module homomorphism* if:
   (a) $\varphi(x+y) = \varphi(x) + \varphi(y)$ for all $x, y \in M$
   (b) $\varphi(r \cdot x) = r\varphi(x)$, for all $r \in R$, $m \in M$

   If furthermore there exists a φ^{-1} such that $\varphi \circ \varphi^{-1} = \varphi^{-1} \circ \varphi = \text{id}$ and φ^{-1} is an R-module homomorphism, then φ is called an *R-module isomorphism* and we write $M \cong_{R\textbf{-Mod}} N$, or $M \cong N$ if it is unambiguous

   If $R = k$ is a field, then k-module homomorphisms are called *linear maps*, *linear functions*, or *linear transfomration*, and k-module isomorphisms are called *linear isomorphisms* (due to historical precedence)

2. Denote
   (a) $\ker(\varphi) = \{m \in M \mid \varphi(m) = 0\}$
   (b) $\text{im}(\varphi) = \varphi(M) = \{n \in N | \varphi(m) = n\}$

3. Define $\text{Hom}_R(M, N)$ be the set of all R-module homomorphisms from M into N, $\text{End}_R(M)$ the set of all R-module homomorphism from M to itself, and $\text{Aut}_R(M)$ the set of all all R-module isomorphisms from M to itself. A function $\varphi \in \text{End}_R(M)$ is called an *R-module enormophism*, and a function $\varphi \in \text{Aut}_R(M)$ is called an *R-module automorphism*.

Notice how it's key that the two structures be R-modules. If M is an R-module and N is an S-module, then $\varphi : M \to N$ is not an R-module or S-module homomrophsim unless either R has a compatible S-module structure of N has a compatible R-module structure that is preserved under φ (this is similar to how G-equivariant functions are betwen two G-sets)

Many of the properties demonstrated of group and ring homomorphisms apply for R-modules

Lemma 12.2.2: R-Module Homomorphism Equivalence

Let $\varphi : M \to N$ be a function. Then φ is an R-module homomorphism if and only if for every $r \in R$ and $x, y \in M$,

$$\varphi(x + ry) = \varphi(x) + r\varphi(y)$$

***Proof*:**
First suppose φ is an R-module homomorphism. Then for every $r \in R$ and $x, y \in M$,

$$\varphi(x + ry) = \varphi(x) + \varphi(ry) = \varphi(x) + r\varphi(y)$$

using additivity and then compatibility with the ring action.

Conversely, suppose $\varphi(x+ry) = \varphi(x) + r\varphi(y)$ for every $r \in R$ and $x, y \in M$. Taking $r = 1$ and using the unital axiom in N gives

$$\varphi(x+y) = \varphi(x) + 1 \cdot \varphi(y) = \varphi(x) + \varphi(y)$$

so φ is additive. In particular $\varphi(0) = \varphi(0+0) = \varphi(0) + \varphi(0)$, and cancelling in the abelian group N gives $\varphi(0) = 0$. Taking $x = 0$ in the hypothesis now gives

$$\varphi(ry) = \varphi(0+ry) = \varphi(0) + r\varphi(y) = r\varphi(y)$$

so φ is compatible with the ring action, and φ is an R-module homomorphism.

Lemma 12.2.3: R-Module Isomorphism Equivalence

Let $\varphi : M \to N$ be an R-module homomorphism. Then φ is an R-module isomorphism if and only if φ is an R-module homomorphism and bijective

***Proof*:**
An isomorphism has an inverse function, so it is bijective. Conversely, let φ be a bijective R-module homomorphism with inverse function ψ. For $x, y \in N$ and $r \in R$,

$$\psi(x+y) = \psi(\varphi(\psi(x)) + \varphi(\psi(y))) = \psi(\varphi(\psi(x) + \psi(y))) = \psi(x) + \psi(y)$$

and $\psi(rx) = \psi(r\varphi(\psi(x))) = \psi(\varphi(r\psi(x))) = r\psi(x)$. So ψ is an R-module homomorphism and φ is an isomorphism, completing the proof.

Lemma 12.2.4: Composition Of R-Module Homomorphism/Isomorphism

Let M, N, L be R-modules. If $\varphi \in \mathrm{Hom}_R(L, M)$ and $\psi \in \mathrm{Hom}_R(M, N)$ then $\psi \circ \varphi \in \mathrm{Hom}_R(L, N)$

***Proof*:**
For every $r \in R$ and $x, y \in L$,

$$(\psi \circ \varphi)(x + ry) = \psi\big(\varphi(x) + r\varphi(y)\big) = \psi(\varphi(x)) + r\psi(\varphi(y)) = (\psi \circ \varphi)(x) + r(\psi \circ \varphi)(y)$$

where the first equality applies lemma 12.2.2 to φ and the second applies it to ψ. By lemma 12.2.2 again, $\psi \circ \varphi$ is an R-module homomorphism, that is, $\psi \circ \varphi \in \mathrm{Hom}_R(L, N)$.

Lemma 12.2.5: R-module Automorphism Group

Let $\mathrm{Aut}_R(M)$ be the set of R-module automorphisms on M. Then $(\mathrm{Aut}_R(M), \circ)$ forms a group under composition.

***Proof*:**
For closure, let $\varphi, \psi \in \mathrm{Aut}_R(M)$. By lemma 12.2.4, $\psi \circ \varphi$ is an R-module homomorphism, and by the same lemma so is $\varphi^{-1} \circ \psi^{-1}$. Since

$$(\psi \circ \varphi) \circ (\varphi^{-1} \circ \psi^{-1}) = \mathrm{id} = (\varphi^{-1} \circ \psi^{-1}) \circ (\psi \circ \varphi)$$

the homomorphism $\psi \circ \varphi$ has a two-sided inverse which is itself an R-module homomorphism, so $\psi \circ \varphi \in \mathrm{Aut}_R(M)$ by definition 12.2.1.

Associativity holds because composition of functions is associative.

For the identity, $\mathrm{id}_M(x+ry) = x+ry = \mathrm{id}_M(x)+r\mathrm{id}_M(y)$, so id_M is an R-module homomorphism by lemma 12.2.2, and it is its own two-sided inverse, so $\mathrm{id}_M \in \mathrm{Aut}_R(M)$. Moreover $\mathrm{id}_M \circ \varphi = \varphi \circ \mathrm{id}_M = \varphi$ for every φ.

Finally, for inverses, let $\varphi \in \mathrm{Aut}_R(M)$. By definition 12.2.1, φ^{-1} is an R-module homomorphism, and φ is a two-sided inverse of φ^{-1} which is an R-module homomorphism, so $\varphi^{-1} \in \mathrm{Aut}_R(M)$, and $\varphi \circ \varphi^{-1} = \varphi^{-1} \circ \varphi = \mathrm{id}_M$.

Example 12.2.6: Module Homomorphisms

1. if a ring is an R-module over itself with the natural ring action, then R-module homomorphisms need not be ring homomorphisms (even from R to itself), and ring homomorphisms need not be R-module homomorphisms. For example, $\mathbb{Z}$ act on itself in the natural way, and let $\varphi(x) = 2x$. φ is not a ring homomorphism [a], but φ is an R-module homomorphism. For the example going the other way, $F[x]$ act on itself in the natural way, and let $\varphi : f(x) \mapsto f(x^2)$. Then φ is not an $F[x]$-module homomorphism [b], but it is a ring homomorphism.

2. Let R be a ring and $M = R^n$ be an R-module with the natural ring action. Then the projective map is naturally an R-module homomorphism.

3. If we take $\mathbb{R}$ to be an $\mathbb{R}$ module over $\mathbb{R}$, then $\mathbb{R}$ has no non-trivial submodules (since $\mathbb{R}$ is a field, it's only ideals are 1 and itself). We call such a module a simple or an irreducible module. Thus, the quotient structure of $\mathbb{R}$ as an $\mathbb{R}$-module is trivial. However, if we consider $\mathbb{R}$ as a $\mathbb{Q}$-module (which we can). Then $\mathbb{Q}$ is a submodule of $\mathbb{R}$ (since $\mathbb{Q}$ is closed under the ring action of $\mathbb{Q}$)! $\mathbb{R}$ as a $\mathbb{Q}$-module will be an infinite-dimensional vector-space [c]. Therefore, it becomes meaningful in this situation to look at $\mathbb{R}/\mathbb{Q}$ under the $\mathbb{Q}$ action!!

4. It should be done as an exercise that if $\varphi : M \to N$ is an R-module homomorphism, then the image and kernel are both R-modules.

[a] $\varphi(xy) = 2xy \neq 2x2y = \varphi(x)\varphi(y)$
[b] $x^2 = \varphi(x) = \varphi(x \cdot 1) = x\varphi(1) = x$, a contradiction
[c] In fact, it will even be *uncountably* infinite-dimensional

Recall in remark HomGrpIfAbRmk that $(\mathrm{Hom}_{\mathbf{Set}}(A, G), +)$ (where $+$ is defined pointwise: $f+g(x) = f(x) + g(x)$) can be given a group structure if G is abelian, and is not a group if G was not abelian (i.e. it not important what the domain was for the structure of the hom-set). Since every R-module is always over an abelian group, $\mathrm{Hom}_R(M, N)$ is *always* an abelian group. In the following proposition we further the structure of $\mathrm{Hom}_R(M, N)$

Proposition 12.2.7: Properties of $\mathrm{Hom}_R(M,N)$

Let M, N and L be R-module. Then

1. $(\mathrm{Hom}_R(M,N),+)$ is an abelian group with $+$ being defined pointwise:

$$(\varphi+\psi)(m)=\varphi(m)+\psi(m) \qquad \forall \varphi,\psi\in \mathrm{Hom}_R(M,N),\ \forall m\in M$$

   In particular $\varphi+\psi\in\mathrm{Hom}_R(M,N)$

2. If R is a commutative ring, then for $r\in R$ define $r\varphi$ by

$$(r\varphi)(m)=r(\varphi(m)) \qquad \text{for all } m\in M$$

   Then $r\varphi\in\mathrm{Hom}_R(M,N)$, and with this action $\mathrm{Hom}_R(M,N)$ is an R-module.

3. With addition defined as above, $(\mathrm{End}_R(M),+,\circ)$ is a ring. When R is commutative, $\mathrm{End}_R(M)$ is an R-algebra.

***Proof*:**
The first statement was proven above remark HomGrpIfAbRmk.

For an R-action to be defined on $\mathrm{Hom}_R(M,N)$ we would need the following to be equal:

$$(rs\varphi)(m)=rs\cdot\varphi(m)$$
$$(r(s\varphi))(m)=r\cdot(s\varphi)(m)=sr\cdot\varphi(m)$$

showing R must be commutative. The other axioms follow quickly

Finally, $(\mathrm{End}_R(M),+,\circ)$ follows the ring axioms. If R is commutative then for ring homomorphism $f:R\to\mathrm{End}_R(M)$, since f is commutative, $f(R)$ is always abelian, and hence $f(R)\subseteq Z(\mathrm{End}_R(M))$, and so $\mathrm{End}_R(M)$ is always an R-algebra, completing the proof.

Note that $(\mathrm{Aut}_R(M),+)$ is *not* a group structure, since the identity *does not exist* (the zero homomorphism is not an automorphism)![5]

Like with groups and rings, we want to know what conditions are needed on a sub-module so that:

$$(x+N)+(y+N)=(x+y)+N \qquad r(x+N)=(rx)+N$$

are both well defined. Since Modules are already abelian groups, than M/N is already well-defined as an abelian group! What's left is to see what condition's are needed so that a submodule N can be used to define an R-module structure on the abelian group M/N. Like abelian groups, every sub-module is "normal"!

[5] semi-groups are groups without an identity. Thus, $(\mathrm{Aut}_R(M)+)$ is a natural example of a semi-group(!), with a compatible group structure with $\circ$

Proposition 12.2.8: sub-modules are "normal" sub-modules

Let R be a ring and let M be an R-module and let N be a submodule of M. The (additive, abeliean) quotient group M/N can be made into an R-module by definin an action of elements of R by

$$r(x+N) = (rx) + N, \qquad \text{for all } r \in R,\ x+N \in M/N$$

The natural projection $\pi : M \to M/N$ defined by $\pi(x) = x+N$ is an R-module homomorphism with kernel N.

***Proof*:**
First, M/N is already an abelian quotient group. We want the ring action to be well defined, that is

$$x + N = y + N$$

which equivalently means that $x - y \in N$. Since N is a submodule, $r(x-y) \in N$, thus

$$rx + N = ry + N$$

therefore, the R-action on M naturally induces an action on M/N. The rest of the module axioms can now e easily verified.

Next, we must show the implication goes the other way, that is, if the elements of the quotient groups are formed by the cosets $x+N$, then N is the kernel of some R-module homomorphis (i.e. the natural projection). The natural projection map π described above is already the natural projection of the abelian group M into the abelian group M/N. It only remains to show that it is a R-module homomorphism:

$$\begin{aligned} \pi(rm) &= rm + N \\ &= r(m+N) && \text{(by definition of the action of } R \text{ on } M/N) \\ &= r\pi(m) \end{aligned}$$

Thus, the natural projection an R-module homomorphism, completing the proof.

As with groups and rings, R-modules also have the isomorphism theorems. Since R-modules are already abelian groups, half the proof is already done. What is left to check would be if the action still preserved under the isomorphisms. To define the 2nd isomorphism theorem, we define what it means to take sum of two modules:

Definition 12.2.9: Sum Of Modules

Let A, B be submodules of an R-module M. Then the *sum* of A and B is the set

$$A + B = \{a + b | a \in A, b \in B\}$$

As before, $A + B$ is the smallest module containing A and B (as you sould check)

Theorem 12.2.10: Isomorphism Theorems For Modules

1. Let M, N be R-modules and let $\varphi : M \to N$ be an R-module homomorphism. Then $\ker(\varphi)$ is a submodule of M and
$$M/\ker(\varphi) \cong \varphi(M)$$
2. Let A, B be submodules of the R-module M. Then
$$\frac{A+B}{B} \cong \frac{A}{A \cap B}$$
3. Let M be an R-module, and let A and B be submodules of M, with $A \subseteq B$. Then
$$(M/A)/(B/A) \cong M/B$$
4. Let N be a submodule of the R-module M. There is a bijection beetween the submodules of M which contain N and the submodules of M/N. The correspondence is give nby $A \leftrightarrow A/n$, an for all $A \supseteq N$. The correspondence commutes with the processes of taking sums and intersection (i.e., is a lattice isomorphism between the lattice of submodules of M/N an dthe lattice of submodules of M which contain N

***Proof*:**
Quotients of R-modules by submodules are R-modules by proposition 12.2.8, and a bijective R-module homomorphism is an isomorphism by lemma 12.2.3.

1. The kernel is a subgroup of M, and it is closed under the action, since $\varphi(rm) = r\varphi(m) = 0$ for $m \in \ker(\varphi)$. Likewise $\varphi(M)$ is a submodule of N, as $r\varphi(m) = \varphi(rm)$. The map $\Phi\colon M/\ker(\varphi) \to \varphi(M)$, $m + \ker(\varphi) \mapsto \varphi(m)$, is well defined and injective, since $\varphi(m) = \varphi(m')$ exactly when $m - m' \in \ker(\varphi)$, and it is surjective. It is R-linear, since $\Phi((m + \ker(\varphi)) + (m' + \ker(\varphi))) = \varphi(m + m') = \varphi(m) + \varphi(m')$ and $\Phi(r(m + \ker(\varphi))) = \varphi(rm) = r\varphi(m)$. So Φ is an isomorphism.
2. The sum $A+B$ is a submodule of M containing B, since $(a+b)+(a'+b') = (a+a')+(b+b')$ and $r(a+b) = ra+rb$. The map $A \to (A+B)/B$, $a \mapsto a+B$, is R-linear, and it is surjective, since $(a + b) + B = a + B$. Its kernel is $A \cap B$, so (1) gives $A/(A \cap B) \cong (A + B)/B$.
3. The subset $B/A = \{b + A \mid b \in B\}$ of M/A is a submodule, since $r(b + A) = rb + A$. The map $M/A \to M/B$, $m + A \mapsto m + B$, is well defined because $A \subseteq B$, and it is R-linear and surjective, with kernel B/A. So (1) gives $(M/A)/(B/A) \cong M/B$.
4. Let $\pi\colon M \to M/N$ be the natural projection, which is R-linear (proposition 12.2.8). Applied to the additive groups, theorem 3.5.7 shows that $A \mapsto A/N = \pi(A)$ is a bijection from the subgroups of M containing N onto the subgroups of M/N, with inverse $\overline{A} \mapsto \pi^{-1}(\overline{A})$. The image of a submodule under the R-linear map π is a submodule, and so is the preimage of a submodule, so the bijection restricts to a bijection between the submodules of M containing N and the submodules of M/N. Both directions preserve inclusions. The sum of two submodules is the smallest submodule containing both, since by (2) it is a submodule and

every submodule containing both contains each sum of an element of one and an element of the other, and the intersection is the largest submodule contained in both. Write β for the bijection and let $A, B \supseteq N$ be submodules. As $A, B \subseteq A+B$, also $\beta(A), \beta(B) \subseteq \beta(A+B)$, so $\beta(A)+\beta(B) \subseteq \beta(A+B)$. Conversely $\beta^{-1}(\beta(A)+\beta(B))$ is a submodule containing A and B, so it contains $A+B$, and applying β gives $\beta(A+B) \subseteq \beta(A)+\beta(B)$. For intersections, $A \cap B \subseteq A, B$ gives $\beta(A \cap B) \subseteq \beta(A) \cap \beta(B)$, and $\beta^{-1}(\beta(A) \cap \beta(B))$ is a submodule contained in A and in B, so it lies in $A \cap B$, and applying β gives $\beta(A) \cap \beta(B) \subseteq \beta(A \cap B)$, completing the proof.

12.2.1 Extensions and Exactness

We conclude this section with a quick word on extensions and where they will show up. Similarly to groups (and rings), we may wonder under what conditions is a module M an extension of two modules A and B, that is $M \cong B/A$, or in terms of a short exact sequence:

$$0 \to A \to M \to B \to 0$$

Section 3.7 showed that mathematicians managed to find the building blocks of finite group extensions, known as simple groups, and classify all finite simple groups, and section 5.4 showed that semi-direct products can be represented by split exact sequences. The classification of simple modules depends on the ring that's acting, R, and is much simpler since the underlying group is abelian. Similar to groups, there exist classification project for R-modules for some particular rings R, at least four of which will be covered in this book[6].

We also endeavour to study R-modules in which the extension property works nicely. This is further explored in [17, Semisimple Rings and Modules]. Since all modules are abelian groups, the semi-direct product is simply the direct product. However, split exact sequences are still an important concept since it gives us a way of recognizing direct products using homomorphisms. A particularly interesting collection of modules are those for which every short exact sequence is a split exact sequence, or phrased differently that every short exact sequence splits. This endeavour is taken up in chapter 17⋆, and explored in full length in *Homological Algebra* [19].

The vocabulary for a sequence of R-module homomorphisms that represent extensions is quite important, and so we define the term here for future use:

Definition 12.2.11: Exact Chain Complex

Let $M_0, M_1, M_2, ..., M_n, ...$ is a collection of R-modules $d_i : M_i \to M_{i-1}$ be a collection of R-module homomorphisms where

$$\cdots \xleftarrow{d_0} M_0 \xleftarrow{d_1} M_1 \xleftarrow{d_2} \cdots \xleftarrow{d_n} M_n \xleftarrow{d_{n+1}} \cdots$$

and $\ker(d_i) = \text{im}(d_{i-1})$. Then the tuple $(M_i, d_i)_{i=-\infty}^{\infty}$ is called an *exact chain complex*, *exact chain*, or *exact sequence*. Often, we abbreviate and write $(M_\bullet, d_\bullet)$. If the arrows are pointing the opposite direction and $\ker(d_i) = \text{im}(d_{i+1})$ then the resulting tuple $(M_i, d_i)_{i=-\infty}^{\infty}$ is called an *exact cochain*. If the chain has only finite length, (i.e. only finitely many of the modules are non-zero), then the resulting exact (co)chain is called a *finite exact (co)chain*.

[6] When R is a Field, a Principal Ideal Domain, a Dedekind Domain, and Semi-Simple

A single homomorphism already sits inside such a chain, once one more module is named.

Definition 12.2.12: Cokernel

Let $\varphi\colon M \to N$ be a homomorphism of R-modules. The *cokernel* of φ is the quotient module

$$\operatorname{coker}(\varphi) := N/\operatorname{im}(\varphi)$$

The kernel measures how far φ is from being injective and the cokernel how far it is from being surjective: φ is injective exactly when $\ker(\varphi) = 0$, and surjective exactly when $\operatorname{coker}(\varphi) = 0$. Both statements are packaged in the one exact sequence

$$0 \to \ker(\varphi) \xhookrightarrow{\iota} M \xrightarrow{\varphi} N \xrightarrow{\pi} \operatorname{coker}(\varphi) \to 0$$

which is exact at $\ker(\varphi)$ because ι is injective, at M because $\ker(\varphi)$ is by definition $\operatorname{im}(\iota)$ and is what φ kills, at N because $\ker(\pi) = \operatorname{im}(\varphi)$ is exactly what the quotient divides out, and at $\operatorname{coker}(\varphi)$ because π is surjective.

Example 12.2.13: Kernels And Cokernels

1. Let $n \geq 2$ and let $\varphi\colon \mathbb{Z} \to \mathbb{Z}$ be multiplication by n, a homomorphism of $\mathbb{Z}$-modules. As $nx = 0$ forces $x = 0$ in $\mathbb{Z}$, $\ker(\varphi) = 0$, and $\operatorname{im}(\varphi) = n\mathbb{Z}$, so the sequence reads

   $$0 \to 0 \to \mathbb{Z} \xrightarrow{\cdot n} \mathbb{Z} \to \mathbb{Z}/n\mathbb{Z} \to 0$$

   The map φ is injective and, as $\mathbb{Z}/n\mathbb{Z} \neq 0$, not surjective, although its source and target are the same module.

2. Let $\varphi\colon \mathbb{Z}^2 \to \mathbb{Z}^2$ be $\varphi(x, y) = (x + y, 2x + 2y)$, the map given by the matrix with rows $(1, 1)$ and $(2, 2)$. Then $\varphi(x, y) = 0$ exactly when $y = -x$, so $\ker(\varphi) = \{(t, -t) \mid t \in \mathbb{Z}\} \cong \mathbb{Z}$, and $\operatorname{im}(\varphi) = \{(s, 2s) \mid s \in \mathbb{Z}\}$. The homomorphism $\psi\colon \mathbb{Z}^2 \to \mathbb{Z}$, $\psi(a, b) = b - 2a$, is surjective, since $\psi(0, b) = b$, and its kernel is $\{(a, 2a) \mid a \in \mathbb{Z}\} = \operatorname{im}(\varphi)$, so $\operatorname{coker}(\varphi) \cong \mathbb{Z}$ by theorem 12.2.10(1). The sequence becomes

   $$0 \to \mathbb{Z} \xrightarrow{t \mapsto (t,-t)} \mathbb{Z}^2 \xrightarrow{\varphi} \mathbb{Z}^2 \xrightarrow{\psi} \mathbb{Z} \to 0$$

   and φ is neither injective nor surjective, with kernel and cokernel both isomorphic to $\mathbb{Z}$.

3. Let R be a ring and let $\pi\colon R^2 \to R$ be the projection $\pi(a, b) = a$. It is surjective, so $\operatorname{coker}(\pi) = 0$, and $\ker(\pi) = \{(0, b) \mid b \in R\} \cong R$. The sequence becomes

   $$0 \to R \xrightarrow{b \mapsto (0,b)} R^2 \xrightarrow{\pi} R \to 0 \to 0$$

   For the inclusion $\iota\colon R \to R^2$, $a \mapsto (a, 0)$, the roles are exchanged: $\ker(\iota) = 0$, and $(a, b) \mapsto b$ is a surjective homomorphism $R^2 \to R$ with kernel $\operatorname{im}(\iota)$, so $\operatorname{coker}(\iota) \cong R$ by theorem 12.2.10(1).

A presentation $R^n \xrightarrow{\varphi} R^m \xrightarrow{\pi} M \to 0$ of a module M (definition 12.3.26) gives $M \cong \operatorname{coker}(\varphi)$, since π is surjective with kernel $\operatorname{im}(\varphi)$ (theorem 12.2.10(1)). So modules given by generators and relations are cokernels, and this is why the sequence recurs in chapter 13 and chapter 14. Section 13.2.5 computes such cokernels from a matrix. For a linear map φ between vector spaces of the same finite

dimension, the Rank-Nullity Theorem (corollary 13.1.18) gives $\ker(\varphi) = 0$ exactly when $\operatorname{coker}(\varphi) = 0$ (corollary 13.1.20), and the first example above shows that this fails over $\mathbb{Z}$. Over a PID, every finitely generated module is the cokernel of an injective map of free modules (corollary 14.1.8) whose matrix is diagonal in suitable bases (corollary 14.1.7), and the cokernel of a diagonal matrix with diagonal entries $a_1, \ldots, a_m$ is the direct sum of the cyclic modules $R/(a_i)$ and a free module, which is the form in theorem 14.1.9. Further on, proposition 17.1.10 joins the kernels and cokernels of the three maps of a homomorphism of short exact sequences into one exact sequence, and proposition 17.6.7 computes Ext and Tor of $\mathbb{Z}/n\mathbb{Z}$ against an abelian group A as the kernel $A[n]$ and the cokernel A/nA of multiplication by n on A.

Exercise 12.2.1

1. Let M be an R-module. Prove that $\operatorname{Hom}_R(R, M) \cong M$
2. Let R^{op} be the opposite ring of R (exercise 9.1.1 (13)), and let $\operatorname{End}_R(R)$ be all R-module homomorphisms from R to itself. Considering $(\operatorname{End}_R(R), +, \circ)$ as a ring show that $R^{\text{op}} \cong_{\mathbf{Ring}} \operatorname{End}_R(R)$ [Hint ; think of Yoneda's lemma]

12.3 Generation of Modules

We now extend the notion of generating algebraic structures to subsets of a module. We will start by generalizing the results of section 2.3.2: we take a subset of an R-module, and permit all possible combinations and R-actions of those elements to form an R-module, or equivaelntly we take the intersection of all R-submodules containing the elements of the set.

Then, as with groups and rings, the easiest way of creating a larger module from smaller ones is to take the direct sum or direct product. Due to their simplicity, it is worth taking the time to find out under what condition's can a module be decomposed into direct products or sums (just like for groups in section 5.2). We then take the approaching of trying to define the "most free" module we can put on an arbitrary set, similar to what we have done in section 2.3.4, that is we are looking to find the universal property of free-modules. As compared to groups, we will spend more time looking at minimal generating sets. Recall that for groups, it is possible that minimal generating sets can differ in cardinatliy. With modules, we will ask if there are modules where there exists a subset $A \subseteq M$ such that A generates M and if any other subset B generated A, then $|A| \leq |B|$. It will turn out that some modules have this property, some will be close, and other will not have this property at all.

After having tackled the question of generating modules "freely" and how to recognize free modules, we will then turn our attention to finitely generated modules, and modules where all submodules are finitely generated. Similalrly to how we can classify all finitely generated abelian groups, finitely generated and Noetherian modules have more "regularly", or equivalently have less "pathological" properties.

A new approach to studying structure's of objects we will take is considering the collection of homomorphisms on modules. As we saw in proposition 12.2.7, $\operatorname{Hom}_R(M, N)$ is always an abelian group, and if R is commutative it is always R-module. Thus, the "symmetries" of an R-module are themselves and R-module, meaning we would like to explore what properties M or N can have given the properties of $\operatorname{Hom}_R(M, N)$ have as an R-module, or even better that $\operatorname{End}_R(M)$ has as an R-algebra. This exploration is very fruitful, and will be of key importance in the theory of

representation (of finite groups, lie algebras, etc.). In fact, if this intrigues the reader, after reading chapter 13 and chapter 14, the reader can skip to [17, Semisimple Rings and Modules] (only requiring knowledge from section 19.5 for the last 2 propositions of that chapter). A particular example of the above is when $N = R$ so that we consider $\mathrm{Hom}_R(M, R)$. This module is called the *dual module* of M, and will be very important in the study of M, even outside the field of algebra[7].

How does the generation of modules compare to the generation of groups and rings? With general groups, we saw that it was very difficult to show what the elements in an arbitrarily generated group look like (it is considered one of the hardest problems in mathematics, known as the *word problem for groups*). However, if G is abelian and finitely generated, then every element can be written in a very simple form due to the fundamental theorem of finitely generated abelian groups (theorem 5.1.6) With rings, we had that the objects that we quotient by (ideals) were no longer rings, and so took the time to study generation of ideals, but we have not studied any structure theorems on rings.

With modules, the story of the exploration of module structures is much closer to that of abelian groups, namely because every module *is* an abelian group. One major advantage of studying module is that many of the previous structure's we have studied can be represented as modules: since all abelian groups are $\mathbb{Z}$-modules, the study of the structure of abelian group is subsumed in the study of the structure of modules (we prove the Fundamental Theorem of Finitely Generated Abelian Groups in greater generality in chapter 14), and since all rings are module over themselves, the study of the structure of modules is also the study of the structure of rings! Due to the fact that all module are abelian groups, we can always collect "like terms":

$$r_1a_1 + r_2a_2 + s_1a_1 = (r_1 + s_1)a_1 + a_2$$

due to this, the order of the generating elements does not affect their sum! It becomes meaningful to ask if the ring acting on a module somehow determines some (or even all!) of the structure of the module. This question is a very meaningful question, and the answer will vary depending on what type of ring is acting on the module: chapters 13 and 14 and [17, Representation of Finite Groups] will each focus modules over particular types of rings and show how the answer to this question changes depending on what type of ring is acting on the module.

12.3.1 Direct Sum of Modules

We first update the terminology from the generation of groups and rings for the generation of modules:

[7] dual modules are central to Functional analysis and Harmonic Analysis

Definition 12.3.1: Generating Modules

Let M be an left R-module and let $(N_i)_{i\in I}$ be the submodules of M where I is an indexing set.

1. The *sum* of $(N_i)_{i\in I}$ is the smallest submodule of M containing each N_i, and is the set of all finite sums of elements from N_i:

$$\sum_i N_i := \{a_1 + \cdots + a_k | a_i \in N_i \text{ for all } i\}$$

If I is finite with $|I| = k$, then it is usually written as

$$N_1 + \cdots + N_k$$

2. For any subset $A \subseteq M$, then RA is the set:

$$RA = \left\{r_1a_1 + \cdots + r_ma_m | r_1, ..., r_m \in R, a_1, ..., a_m \in A, m \in \mathbb{Z}^+\right\}$$

(where by convention $RA = \{0\}$ if $A = \emptyset$) is the smallest [left] submodule of M generated by the elements of A, an is called the *[left] submodule generated by* A. If A is finite the set $\{a_1, a_2, ..., a_n\}$ we shall write

$$Ra_1 + \cdots Ra_n$$

If N is a submodule of M (possibly $N = M$) and $N = RA$, for some subset, then A is called the *set of generators* or *generating set for* N, and we say N is generated by A and is sometimes written as $\langle A\rangle = N$.

3. A submodule N of M (possibly $N = M$) is *finitely generated* if there is some finite subset A of M such that $N = RA$.

4. A submodule N of M (possibly $N = M$) is *cyclic* if there exists an element $a \in M$ such that $N = Ra$, that is, if N is generated by one element

$$N = Ra = \{ra|\ r \in R\}$$

and is sometimes written as $\langle a\rangle = N$.

It is an easy starter exercise to show that the structures in definition 12.3.1 are indeed modules. If R is a *rng*, then it is not necessarily the case that $A \subseteq RA$.

Example 12.3.2: Example

1. Let $R = \mathbb{Z}$ and let M be any $\mathbb{Z}$-module. Take any $a \in M$. Then $\mathbb{Z}a$ is a cyclic sub-module. It is in fact a cyclic subgroup! More generally, A generates M as a $\mathbb{Z}$-module if and only if A generates M as an [abelian] group. That is, generating M through A as a group (using $+$ and $-$ in groups) will produce as many elements as generating M through A as a module (which also allows $\mathbb{Z}$-actions). Thus, the definition of a finitely generated $\mathbb{Z}$-module is identical to the definition of a finitely generated group (definition 5.1.4).

2. Let R be a ring and let $M = R$ be the [left] R-module of R acting on itself. Then R is finitely generated as a module, even if R is infinitely generated as a ring! Even better: it is cyclic, since $R \cdot 1 = R$. Thus, in terms of generation, R is the "simplest" R-module.

3. If R is an R-module over itslef, then $I \subseteq R$ is a finitely generated R-module if and only if it is a finitely generated *left-ideal*. If R was a right-module, then I would be a finitely generated *right-ideal*, and if R was commutative, I would be a finitely generated *two-sided ideal*.

4. Let M be a finitely generated R-module. Then it is not the case that every submodule of M is finitely generated! Take $P[x_1, x_2, ...] = P[X]$ (i.e. the polynomial ring over countably many coefficients) which is a $P[X]$-module over itself. Then $P[X]$ is cyclic as a $P[X]$-module, and hence finitely generaed, however $(x_2, x_4, x_6, ...) \subseteq P[X]$ is a $P[X]$-submodule of $P[X]$ (being a left-ideal), however, it is an *infinitely generated* left-ideal. Finitely generated modules in which all submodules are finitely generated will be called *Noetherian*.

5. Let R^k be an R-module with the action being pointwise multiplication. Then R^k is finitely generaed by
$$e_i = (\underbrace{0, ..., 0}_{i-1 \text{ times}}, 1, \underbrace{0, ..., 0}_{k-i \text{ times}})$$
hence, for any $(r_1, r_2, ..., r_k) \in R^k$,
$$(r_1, ..., r_k) = \sum_i r_i e_i$$

6. Here is a neat way of writing down M if $A = M/B$. Then $M \cong A' + B$, since any element in M is some element of A (namely A' is a set of coset representative when taking $b = 0$), and an element of B:
$$\frac{M}{B} = \frac{A' + B}{B} \cong A$$
Note that is in general smmimply a *subset* of M, we can define the operation on A' by taking:
$$a_1 + a_2 = a_3 \qquad a_3 \text{ is the appropriate representative from } [a_1 + a_2] \text{ that belongs in } A'$$
In this way, $A' \not\subseteq M$ for as an R-module since the binary operation is different. Note too that $M = A' + B$ does not imply that $M \cong A \oplus B$. For example, consider $A = \mathbb{Z}/6\mathbb{Z}$ with $M = \mathbb{Z}$, and $B = 6\mathbb{Z}$. Letting $A' = \{0, 1, 2, 3, 4, 5\}$, we get that as sets:
$$M = A' + B = \{0, 1, 2, 3, 4, 5\} + 6\mathbb{Z} = \mathbb{Z}$$
for example, $10 = 4 + 6$, and $101 = 5 + 96$. However, giving A' the group structure of $\mathbb{Z}/6\mathbb{Z}$:
$$\mathbb{Z} \ncong_R (\mathbb{Z}/6\mathbb{Z}) + 6\mathbb{Z}$$
For the right hand side has torsion elements, while the left does not. For the $+$ to be upgraded to a $\oplus$, the operation of the quotient must in some natural way be compatible with the operation in M, that is A must in not only be the quotient M/B, but must also be considered as $A \subseteq M$. This is explore in chapter 17$\star$.

Naturally, we would want to find when direct sums of modules behave like direct products, or more specifically external direct sums

Definition 12.3.3: Direct Product and Sum of Modules

Let $(M_i)_{i\in I}$ be a collection of R-modules where I is an indexing set. Then the collection of tuples $(m_i)_{i\in I}$ where $m_i \in M_i$ with addition and action of R defined componentwise is called the *direct product* of $(M_i)_{i\in I}$, denoted

$$\prod_{i\in I} M_i$$

If I is finite where $|I| = n$, then it is usually denoted:

$$M_1 \times \cdots \times M_k$$

If we take the subset of $\prod_{i\in I} M_i$ where only finitely many of the terms are nonzero, then this this set is called the *external direct sum*, and is written as:

$$\bigoplus_{i\in I} M_i$$

If I is finite where $|I| = n$, then it is usually denoted:

$$M_1 \oplus M_2 \oplus \cdots \oplus M_n$$

If I is infinite and $R \neq 0$, then $\bigoplus_{i\in I} M_i \subsetneq \prod_{i\in I} M_i$.

12.3.4 Remark It might be tempting to say that:

$$M \cong N_1 \oplus N_2 \quad \Leftrightarrow \quad M = N_1 + N_2$$

however that is not the case: Take $\mathbb{Z}$ as a $\mathbb{Z}$-module over itself, $N_1 = 2\mathbb{Z}$ an $N_2 = (0)$. Then $2\mathbb{Z} + 0 = 2\mathbb{Z}$, and:

$$\mathbb{Z} \cong 2\mathbb{Z} \qquad \text{but} \qquad \mathbb{Z} \neq 2\mathbb{Z}$$

which we can see since, $1 \notin 2\mathbb{Z}$

If $M \cong A \times B$ for some modules M, A, and B, then for any $m \in M$, there exist unique pair of elements $a \in A, b \in B$ such that $m \mapsto (a, 0) + (0, b)$. Thus, any generating set (or more specifically minimal generating set) of M breaks down to a question of a generating set (or minimal generating gset) of A and B. The key here is that any element $m \in M$ can be uniquely represented by elements a and b (namely since these two modules are isomorphic).

Let's say conversely that we want to find submodules $A, B \subseteq M$ for which $M = A + B$. If a module M is finitely generated, by Zorn's lemma there must exist a minimal nonnegative integer d such that M is generated by d elements (and no less)[8]. If M was not finitely generated, this need not be the case. If M can be generated by d elements, it need not be the case that there is a unique set of generating elements, nor the case that the R-linear combination of these elements be unique:

[8] Note that minimal does not mean smallest! There can be multiple different $d \in \mathbb{N}$ that satisfy this property, see the discussion in section 2.3

Example 12.3.5: Non-Unique Sets And R-Linear Combinations

It will turn out that we almost never get a unique minimal generating set [a]. Take for example R over itself. Let $u \in R$ be a unit of R. Since u is a unit, let $v \in R$ such that $uv = 1$. Then $R \cdot 1 = R \cdot (vu) = (Rv)u = R \cdot u$ (since $u \in R$, $uv = 1 \in Rv$, and from that we can show that $(Rv)u = Ru$).

[a] If you have already taken a course in linear algebra, this is equivalent to saying that there is no unique basis

Though unique minimal generated sets are rare to come accorss, uniqe R-linear combinations are much more common. For example, taking R over itself so that $\{1\}$ generated R as an R-module. Then every R-linear combination is unique, that is

$$r \cdot 1 = s \cdot 1 \Leftrightarrow r = s$$

We can see this by taking the homomorphism $\varphi : R \to M$, $r \mapsto r \cdot 1$ and look at the kernel. This property is very desirable, since it will allow us to show that M can split into a direct sum of submodules. If the R-linear combinations are unique, then we need only concern ourselves with any minimal generating set when considering the structure of such a module!

For the rest of the chapter, we identify equivalent properties to saying R-linear combinations are unique, and identify modules which have this property.

Proposition 12.3.6: Equivalent Conditions for Linear Combinations

Let $N_1, ..., N_k$ be (not necessarily distinct) submodules of R-module M. Then the following are equivelent

1. the map $\pi : \bigoplus_{i \in I} N_i \to \sum_i N_i$ defined by

$$(n_i)_{i \in I} \mapsto \sum_i n_i$$

   is an R-module isomorphim. If $|I| < k$ is finite, we write:

$$N_1 + N_2 + \cdots + N_k \cong N_1 \times N_2 \times \cdots \times N_k$$

2. $N_j \cap (N_1 + N_2 + \cdots + N_{j-1} + N_{j+1} + \cdots + N_k) = 0$ for all $j \in \{1, 2, ..., k\}$
3. Every $x \in N_1 + N_2 + \cdots + N_k$ can be written *uniquely* in the form $a_1 + a_2 + \cdots + a_k$ with $a_i \in N_i$

***Proof*:**

For (1) implies (2). let's assume that the intersection is not empty. Then there exists some a_j such that

$$a_j = a_1 + a_2 + \cdots + a_{j-1} + a_{j+1} + \cdots + a_k$$

However, this implies that

$$\pi(0, ..., a_j, ..., 0) = \pi(a_1, a_2, ..., a_{j-1}, a_{j+1}, ..., a_k)$$

but

$$(0, ..., a_j, ..., 0) \neq (a_1, a_2, ..., a_{j-1}, a_{j+1}, ..., a_k)$$

contradiction injectivy, since π is an isomorphism

For (2) implies (3). Let's say there are two linear combinations

$$a_1 + a_2 + \cdots + a_k = b_1 + b_2 + \cdots + b_k$$

then we can isolate any jth components:

$$b_j - a_j = (b_1 - a_1) + \cdots + (b_{j-1} - a_{j-1}) + (b_{j+1} - a_{j+1}) + \cdots + (b_k - a_k)$$

which implies by our previous proposition

$$b_j - a_j \in N_j \cap (N_1 + N_2 + \cdots + N_{j-1} + N_{j+1} + \cdots + N_k) = 0$$

thus, $b_j - a_j = 0$ and so $b_j = a_j$ for every j, showing that the linear combination is unique!

For (3) implies (1), Notice that the function is surjective construction: for any element $a_1 + a_2 + \cdots + a_k$, the element $(a_1, a_2, ..., a_k)$ would map to it. What (2) does is show injectivity, since if two sums are the same, then the direct product of elements must also be the same!. This map is also easily checked to be an R-module homomorphism. For example

$$\begin{aligned} \pi(r \cdot (a_1, a_2, ..., a_k) &= \pi(ra_1, ra_2, ..., ra_k) && (12.1)\\ &= ra_1 + ra_2 + \cdots + ra_k && (12.2)\\ &= r \cdot (a_1 + a_2 + \cdots + a_k) && (12.3)\\ &= r\pi(a_1, a_2, ..., a_k) && (12.4) \end{aligned}$$

where line 3 comes from induction on $r \cdot (m + n) = r \cdot m + r \cdot n$, and linearity comes easily too.

This show's us how we may split up a module into "direct products" which satisfy the universal property of direct products. We now introduce a notion that is slightly stronger than the external direct product and of sums of modules:

Definition 12.3.7: Internal Direct Sum

Let $(N_i)_{i \in I}$ be a collection of submodules of M modules respecting one of the 3 condition's in proposition 12.3.6. If:

$$\sum_i N_i = M$$

then we will replace the summation symbol $\sum_i$ with $\bigoplus_i$. If the indexing set is finite, then:

$$N_1 + \cdots + N_k = N_1 \oplus \cdots \oplus N_k$$

Thus, we are *overloading* the notation of $\oplus$ to represent the fact that the sum $\sum_i N_i$ is isomorphic to $\oplus_i N_i$ *and* that $\sum_i N_i = M$

We have already shown in remark where M is an R-module that is isomorphic to a direct sum of submodules $(N_i)_{i \in I}$ but $\sum_i N_i \neq M$, namely by taking $\mathbb{Z}$ as a $\mathbb{Z}$-module and $N_1 = 2\mathbb{Z}$, $N_2 = (0)$. Then:

$$\mathbb{Z} \cong 2\mathbb{Z} \oplus (0) \qquad \mathbb{Z} \neq 2\mathbb{Z} \oplus (0)$$

In chapter 17⋆ and [17, Semisimple Rings and Modules], we will further explore how to find when a

module M can be decomposed into direct products.

One case is settled immediately, and is used often enough to record. When the *ring* is a product, every module over it splits, and the splitting is forced by the two idempotents $(1,0)$ and $(0,1)$. The upshot is that module theory over $R \times S$ is nothing more than module theory over R and over S carried out side by side.

Proposition 12.3.8: Modules over a Product Ring

Let R and S be rings and put $e = (1,0)$ and $f = (0,1)$ in $R \times S$, so that e and f are central idempotents with $e + f = 1$ and $ef = 0$. Then every $(R \times S)$-module M decomposes as an internal direct sum

$$M = eM \oplus fM$$

where eM is an R-module and fM is an S-module. The assignments $M \mapsto (eM, fM)$ and $(M_1, M_2) \mapsto M_1 \oplus M_2$ are mutually inverse, and give an equivalence of categories

$$(R \times S)\textbf{-Mod} \;\simeq\; R\textbf{-Mod} \times S\textbf{-Mod}$$

Under it, a module is finitely generated, or projective, exactly when both of its components are. Freeness is *not* componentwise: $M_1 \oplus M_2$ is free over $R \times S$ if and only if M_1 and M_2 are free of the *same* rank, since a free $(R \times S)$-module is $(R \times S)^n \cong R^n \oplus S^n$.

***Proof*:**

Step 1: the decomposition. For $m \in M$ we have $m = 1 \cdot m = (e + f)m = em + fm$, so $M = eM + fM$. If $x \in eM \cap fM$, write $x = em' = fm''$. Then $x = ex = e(fm'') = (ef)m'' = 0$, using that e is idempotent and $ef = 0$. So the sum is direct.

Step 2: the components are modules over the factors. The subgroup eM is stable under $R \times S$, and f acts on it as $fe = 0$, so the action factors through $(R \times S)/(0 \times S) \cong R$, making eM an R-module. Symmetrically fM is an S-module.

Step 3: the two constructions are inverse. Given R-module M_1 and S-module M_2, let $R \times S$ act on $M_1 \oplus M_2$ componentwise. Then $e(M_1 \oplus M_2) = M_1$ and $f(M_1 \oplus M_2) = M_2$, recovering the pair. Conversely Step 1 rebuilds M from (eM, fM). Both assignments send a homomorphism to its restrictions, and a map commutes with the action of e and f exactly when it preserves the two summands, so the correspondence is functorial in both directions and is an equivalence.

Step 4: the listed properties. Finite generation and projectivity are checked componentwise because the equivalence carries $R \times S$ to the pair (R, S) and direct sums to direct sums. A generating set for M projects to generating sets of eM and fM and conversely, giving finite generation. And M is a summand of a free $(R \times S)$-module exactly when each component is a summand of a free module over its factor, giving projectivity.

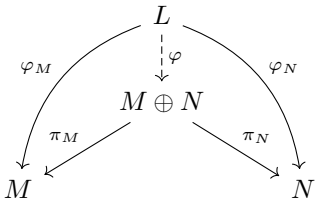
Freeness is the exception, and the equivalence is what shows why. A free $(R \times S)$-module of rank n corresponds to the pair (R^n, S^n), so M is free exactly when its two components are free *of a common rank*. For $R = S = k$ a field, $M = k \oplus k^2$ has both components free, of ranks 1 and 2, and is projective. But it is not free, since that would force $1 = 2$. This completes the proof.

To conclude, we re-phrase the above results in the language of universal properties. The properties of $M \oplus N$ can be described in terms of maps $\varphi : L \to M \oplus N$ where this map is uniquely determined

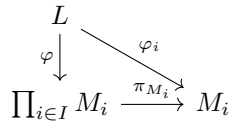
by the maps $\varphi_M : L \to M$ and $\varphi_N : L \to N$, that is:

Proposition 12.3.9: Universal Property of Products, Modules

Let M, N be two R-modules and let $\pi_M \colon M \oplus N \to M$ and $\pi_N \colon M \oplus N \to N$ be the projections $(m, n) \mapsto m$ and $(m, n) \mapsto n$. Then $M \oplus N$ as constructed above satisfies the universal property of products: for any R-module L, if $\varphi_M \colon L \to M$ and $\varphi_N \colon L \to N$ are two R-module homomorphisms, then there exists a unique R-module homomorphism $\varphi \colon L \to M \oplus N$ such that the following diagram commutes:



More generally, if $(M_i)_{i \in I}$ is a collection of R-modules, then for any R-module L and collection of R-module homomorphisms $\varphi_i \colon L \to M_i$, there exists a unique R-module homomorphism $\varphi \colon L \to \prod_{i \in I} M_i$ such that all of the following diagrams commute, where π_{M_i} is the projection onto the ith component:



When I is finite, $\prod_{i \in I} M_i = \bigoplus_{i \in I} M_i$.

***Proof*:**
When I is finite, every tuple $(m_i)_{i \in I}$ has finitely many nonzero entries, so $\prod_{i \in I} M_i = \bigoplus_{i \in I} M_i$ (definition 12.3.3), and the statement for $M \oplus N$ is the case $I = \{1, 2\}$, $M_1 = M$, $M_2 = N$ of the general one. So let $(M_i)_{i \in I}$, L and the φ_i be as in the general statement. Each projection $\pi_{M_i} \colon \prod_j M_j \to M_i$, $(m_j)_j \mapsto m_i$, is an R-module homomorphism, since addition and the action of R on the product are defined componentwise.

Existence. Define $\varphi(\ell) = (\varphi_i(\ell))_{i \in I}$ for $\ell \in L$. For $\ell, \ell' \in L$ and $r \in R$,

$$\varphi(\ell + \ell') = (\varphi_i(\ell) + \varphi_i(\ell'))_{i \in I} = \varphi(\ell) + \varphi(\ell') \qquad \varphi(r\ell) = (r\varphi_i(\ell))_{i \in I} = r\varphi(\ell)$$

because each φ_i is R-linear and the operations on $\prod_i M_i$ are componentwise. So φ is an R-module homomorphism, and $\pi_{M_i}(\varphi(\ell)) = \varphi_i(\ell)$ for every i and ℓ, that is, $\pi_{M_i} \circ \varphi = \varphi_i$.

Uniqueness. Let $\psi \colon L \to \prod_i M_i$ be an R-module homomorphism with $\pi_{M_i} \circ \psi = \varphi_i$ for every i. For $\ell \in L$, the ith component of $\psi(\ell)$ is $\pi_{M_i}(\psi(\ell)) = \varphi_i(\ell)$, which is also the ith component of $\varphi(\ell)$. A tuple is determined by its components, so $\psi(\ell) = \varphi(\ell)$ for every ℓ, and $\psi = \varphi$, as claimed.

For infinite I the product cannot be replaced by the direct sum. Take $R = L = \mathbb{Z}$, $M_i = \mathbb{Z}$ and $\varphi_i = \mathrm{id}$ for $i \in \mathbb{N}$. If some homomorphism $\psi\colon \mathbb{Z} \to \bigoplus_{i\in\mathbb{N}} \mathbb{Z}$ satisfied $\pi_{M_i} \circ \psi = \mathrm{id}$ for every i, then its composite with the inclusion $\bigoplus_{i\in\mathbb{N}} \mathbb{Z} \subseteq \prod_{i\in\mathbb{N}} \mathbb{Z}$ would satisfy the same equations, so by uniqueness $\psi(1) = (1, 1, 1, \dots)$, which has infinitely many nonzero entries and does not lie in $\bigoplus_{i\in\mathbb{N}} \mathbb{Z}$.

12.3.2 Free Modules

Given a ring R and a an arbitrary abelian group A, we can always define the trivial module via the action $r \cdot a = 0$: let's label this module $T(A)$ and call it the trivial R-module. Just like trivial groups, trivial modules offer little in structure[9]. It would be desirable if we can find a better struture on A, in particular if A represents the generating set of some R-module M, then we would like to define a "free R-module" $F(A)$ (or more particularly $F^R(A)$ to remember the ring R) which represents an R-module with no relationship, and then find a R-submodule $N \subseteq F(A)$ such that $F(A)/N \cong M$ (i.e. the submodule N woule represent adding relationships to $F(A)$). This mirrors the development of the free group from sections 2.3.3 and 3.2.

Let's say we were to define a R-module $F^R(A)$ on the generating set A of M with only the bare minimum relations to make it into an R-module ($1 \cdot m = m$, $(rs) \cdot m = r \cdot (s \cdot m)$, and so forth). Then similarly to free groups, we would want that for any element of $x \in F^R(A)$, which is some linear combination:

$$x = \sum_{a\in B\subseteq A} r_a \cdot a \quad \text{for some subset } B \subseteq A$$

to be equal to 0 if and only if $r_a = 0$ for all a. This is similar to free groups, where any relation between generating elements can be move to the left hand side and the right hand side equals to the identity element. By proposition 12.3.6, this means that $F^R(A) = R^{|A|}$ is our desired free R-module. To see why, suppose some $r_a \neq 0$. Then that means we have added some relation into $F^R(A)$, that is the function $\varphi : R^{|A|} \to F^R(A)$ would have a non-trivial kernel, going against the idea that $F^R(A)$ is a free-module.

If $F^R(A)$ is a free module on A, just like for free groups, any set-function from A to any R-module N, $f : A \to N$, should lift to an R-module homomorphism $F : F^R(A) \to N$, i.e. there is a map $\iota : A \hookrightarrow F^R(A)$ that injects A into $F^R(A)$ and $f = F \circ \iota$. In particular, given any arbitrary map $f : A \to N$, no matter where f maps A, whether f is injective or not, we can find a map F such that $F|_A = f$, so F must map any element of the set A exactly as how f would map it, hence F is an extension of f that has the restriction of being an R-module homomorphism. Furthermore, N is any R-module, meaning any $f : A \to N$ can be lifted to $F : F^R(A) \to N$ (see how this is not the case for an arbitrary domain M, or check out section 2.3.3 for the parallel problem in group theory). A consequence of this result is that we can determine how $F^R(A)$ maps to an R-module N simply by determining where A maps to, that is we have the universal property of free modules.

[9]When dealing with groups, we also defined $\mathrm{Sym}(A)$. As a thought exercise, can you see why we cannot parallel the same idea with module? Hint: Think of $\mathbb{Z} \hookrightarrow \mathbb{Q}$

12.3.10 On Recovering A We would have to already know of the set A in order to do the above strategy. The notation $F^R(A)$ makes it seem obvious that we know about the set A, namely since we will soon construct $F^R(A)$ with respect to A, but it is not so obvious that if we are given a module M that we say is a free R-module, can we find a subset A such that $F^R(A) \cong_R M$. In fact, depending on the ring R acting on M, it will sometimes be possible to "recover" A, while other times it will be impossible[a]. Chapter 13 takes this up for vector spaces.

[a] It will be possible to say there exists such a set using the axiom of choice, but it will be impossible to give a description of the elements

We now prove that $F^R(A) = R^{|A|}$ does indeed respect the universal property of free R-modules.

Theorem 12.3.11: Universal Property of Free Modules

For any set A, there exists a free R-module $F^R(A)$ on the set A where $F^R(A)$ satisfies the following *universal property*: If M is any R-module and $\varphi : A \to M$ is any map of sets, then there is a unique R-module homomorphism $\psi : F(A) \to M$ such that $\psi|_A = \varphi$, that is, the following diagram commutes

$$\begin{array}{ccc} S & \xhookrightarrow{\text{incl.}} & F(S) \\ & \searrow^{\varphi} & \downarrow{\scriptstyle \psi} \\ & & M \end{array}$$

When A is a finite set $\{a_1, a_2, ..., a_n\}$, $F^R(A) = Ra_1 \oplus \cdots + \oplus RA_n \cong R^n$, and if A has any cardinality, then:

$$F^R(A) = \bigoplus_{a \in A} R_a$$

where A is treated as an indexing set.

***Proof*:**
if $A = \emptyset$, then let $F(A) = \{0\}$. Then this trivially satisfies:

$$\begin{array}{ccc} \emptyset & \xhookrightarrow{\text{incl.}} & \{0\} \\ & \searrow^{\varphi} & \downarrow{\scriptstyle \Phi} \\ & & M \end{array}$$

since $\emptyset$ is included in any set, φ is the vacuous map, and $\Phi(0) = 0$ makes Φ into an R-module homomorphism which restircted to $\emptyset$ is itself again.

So let $A \neq \emptyset$. What we will do is construct funtions with the domain as a set and the co-domain as the ring R. These functions will be the mechanism by which set A is given an R-module structure! Let $F^R(A)$ be the collection of all set functions $f : A \to R$ such that $f(a) = 0$ for all but finitely many $a \in A$:

$$F^R(A) = \{f \in \text{Hom}_{\mathbf{Set}}(A, R) \mid f(a) \neq 0 \text{ for only finitely many } a \in A\}$$

Make $F^R(A)$ into an R-module by defining pointwise addition and pointwise multiplication for

the action:

$$(f+a)(a) = f(a) + g(a)$$
$$(rf)(a) = r(f(a))$$

for all $a \in A$, $r \in R$, $f, g \in F^R(A)$. Notice how $F^R(A)$ is isomorphic to $R^{|A|}$, meaning this construction matches definition 12.3.12.

Let $\iota : A \to F^R(A)$ vis $a \mapsto f_a$, where f_a is the function which is 1 at a and zero everywhere else. We can thus identify a set $A \leftrightarrow A' \subseteq F^R(A)$ as the subset containing all of f_a. Notice that since any f has only finitely many non-zero elements, we can write any $f \in F^R(A)$ as

$$f = r_1 f_{a_1} + r_2 f_{a_2} + \cdots + r_n f_{a_n}$$

which by our identification of A' with A, represents:

$$r_1 a_1 + r_2 a_2 + \cdots + r_n a_n$$

where f takes on the value of r_i at a_i and is zero at all other elements of A. Even better, by construction, every element of $F^R(A)$ has a *unique* expression as such a formal sum, since by construction, non of the a_i have any relation to each other, there cannot be a way of cancelling them out!

To establish the universal property, of $F^R(A)$, Let's say we have a function $\varphi : A \to M$, where M is an R-module. We need to show this function raises to an R-module homomorphism $\Phi : F^R(A) \to M$. Let Φ be defined by

$$\sum_{i=1}^{n} r_i a_i \mapsto \sum_{i=1}^{n} r_i \varphi(a_i)$$

by the uniqueness fo the expression of the elements of $F^R(A)$ as linear combinations of a_i, we see that Φ is indeed a well dfiend R-module homomrophsim!! Indeed, two elements of $F^R(A)$ can be written over a common finite list $a_1, \ldots, a_n$ of distinct elements of A, allowing zero coefficients, and then

$$\Phi\left(\sum_i r_i a_i + \sum_i s_i a_i\right) = \sum_i (r_i + s_i)\varphi(a_i) = \Phi\left(\sum_i r_i a_i\right) + \Phi\left(\sum_i s_i a_i\right)$$
$$\Phi\left(r \sum_i r_i a_i\right) = \sum_i r r_i \varphi(a_i) = r\Phi\left(\sum_i r_i a_i\right)$$

By definition, the restriction of $\Phi|_A$ is φ. For uniqueness of $F^R(A)$, since $F^R(A)$ is generated by A, once we know the values of an R-module homomorphism on A, tis value on every element of $F^R(A)$ are uniquely determined (by force of construction), so Φ is the unique extension of Φ to all of $F^R(A)$.

If A is finite ($\{a_1, a_2 ..., a_n\}$), proposition 12.3.6

$$F^R(A) = Ra_1 \oplus Ra_2 \oplus \cdots \oplus Ra_n$$

Since $R \cong Ra_i$ (under $r \mapsto ra_i$) by proposition 12.3.6 again, the direct sum is isomorphic to R^n.

Definition 12.3.12: Free Module

An R-module F is said to be *free* on the subset A of F if for every nonzero element x of F, there exists a unique nonzero elements $r_1, ..., r_n$ of R and unique $a_1, ..., a_n$ in A such that $x = r_1a_1 + \cdots r_na_n$, for some $n \in \mathbb{Z}^+$, which by proposition 12.3.6 implies that:

$$F = \bigoplus_{i \in I} R$$

If R is a commutative ring, the cardinality of A is called the *free rank* of F, or just *rank* if unambiguous.

Note that A need not be finite, but the linear combinations of elements of A need to be finite (notice that we say $a_1, ..., a_n \in A$, and not $a_1, ..., a_n, ... \in A$). From now on, we will write R^n to mean the free R-module $R \oplus R \oplus \cdots \oplus R$. So $(\mathbb{Z}/2\mathbb{Z})^3$ is a free $(\mathbb{Z}/2\mathbb{Z})$-module. R^1 is a free R-module of rank 1 (to be distinguished from R which we will treat as a ring). If the superscript is infinite, say $N = |\mathbb{N}|$ or $N = |\mathbb{R}|$, then R^N is considered an infinite direct sum, not a direct product (unlike the R^A of example 12.1.7, which denotes all functions $A \to R$). The direct product $\prod_{n \in \mathbb{N}} \mathbb{Z}$, by contrast, is not a free $\mathbb{Z}$-module (the Baer-Specker theorem, [32]). Furthermore, the notion of rank is not necessarily well-defined if R is not commutative. We shall explore more of this in chapter 13.

Example 12.3.13: Free Modules

1. Let R be any ring with identity. Then R over itself is a free-module! The element 1 will always generated all of R since 1 is a unit (and so $ab \neq 0$ for any $b \neq 0$) and $r \cdot 1 = r$, and so it's a linearly independent generating set!

2. The "freeness" is dependent on the ring you choose, as well as the module (i.e., the abelian group). So $\mathbb{Z}/n\mathbb{Z}$ over itself is free (by the logic provided above), but not over $\mathbb{Z}$ (which we already know is a well-defined module), since

   $$n \cdot 3 = 0$$

   In general, one criterion for freeness is that $r \cdot a \neq 0$ for any r. This criterion is called being *torsion-free*, and if $r \cdot a = 0$, then r is said to be a *torsion element* of the R-module M. However, a module being torsion free does not imply the module is free.

3. Since $\mathbb{Z}/n\mathbb{Z}$ overt $\mathbb{Z}$ is not a free-module, it does not have a basis. If we try to choose $A \subseteq \mathbb{Z}/n\mathbb{Z}$ to be a basis, it will always have torsion elements. $\mathbb{Z}/n\mathbb{Z}$ over $\mathbb{Z}$ is also an example of a module *without* any linearly independent sets!

4. Let $\mathbb{C}[x, y]$ act on itself. Then $\mathbb{C}[x, y]$ is a free module. As stated in example 12.1.7, ideals of modules over themselves are sub-modules. Take $(x, y) \subseteq \mathbb{C}[x, y]$. Any two distinct nonzero $f, g \in (x, y)$ satisfy $g \cdot f - f \cdot g = 0$, so a basis of (x, y) has one element h, and $(x, y) = (h)$. Then h divides x and y, so h is a nonzero constant, but every element of (x, y) has zero constant term. So (x, y) is *not* free. However, (x, y) is certainly torsion-free, since $\mathbb{C}[x, y]$ is an integral domain.

   Thus, freeness is not preserved by taking sub-modules

5. Let $\mathbb{Z}$ act on itself. Then $\mathbb{Z}/n\mathbb{Z}$ is a quotient $\mathbb{Z}$-module as we have seen in section 12.2. However, as we have seen in the 2nd example, $\mathbb{Z}/n\mathbb{Z}$ over $\mathbb{Z}$ is not free.

Thus, freeness is not preserved when taking quotient modules.

6. Let's say $S = \{1, 2, 3\}$, and $M = \mathbb{R}^3$ (over $\mathbb{R}$). Take any $f : S \to M$. For our concrete example, let's say:

$$f(1) = (\varphi, \sqrt{2}, 4) \qquad f(2) = (0, 0, 0) \qquad f(3) = (0, 5753, 9)$$

I chose particularly random values to show that f is indeed any set function map. Now if you consider $F^R(S)$ (over $\mathbb{R}$), we'd have some uncountable set of elements, which most importantly are all distinct! It's that last part that let's us say that

$$\psi : F^R(S) \to \mathbb{R}^3$$

will be the unique homomorphism where $\psi|_S = f$. Consider

$$\psi(x) = \psi(\sum_{i=1}^{3} r_i s_i) = \sum_{i=1}^{3} r_i \varphi(s_i)$$

This function is indeed well defined since every x is *unique* in it's represetnation from S, since if it was not, and we consider

$$x = y(\sum_{i=1}^{3} r_i s_i = \sum_{i=1}^{3} r_i' s_i)$$

then it is not true in general that

$$\psi(x) = \psi(y)$$

what let's use insure that this will always be true is precicely the linear independence, or free property, of the module!

Finally, this is indeed a [unique] homomorphism. it is easy to check that

$$\psi(x + ry) = \psi(x) + r\psi(y)$$

and the uniqueness will come from free modules elements having no relation to each other (if they did, you might be able to have a second homomrophism).

7. You can have a module with linearly independent sets, but that always fail to generate the module, meaning it is not a free module. For example, $\prod_{n\in\mathbb{N}} \mathbb{Z}$ has infinite linearly independent subsets, but it is not a free $\mathbb{Z}$-module ([32]).

8. Take $\mathbb{C} \times \mathbb{C}$ over itself. This is free module (its basis is $(1, 1)$). Then $\mathbb{C} \times \{0\}$ is a submodule, but not a free module: every element $(a, 0)$ satisfies $(0, 1) \cdot (a, 0) = (0, 0)$ with $(0, 1) \neq 0$, so no nonempty subset of $\mathbb{C} \times \{0\}$ is linearly independent, and the nonzero module $\mathbb{C} \times \{0\}$ has no basis.

9. Let R be a Principal Ideal Domain acting over itself. Then any submodule N of R is an ideal, and since it is principal, $N = (a)$ for some $a \in R$. If $a = 0$, then $N = 0$ is free on the empty set. If $a \neq 0$, then $r \mapsto ra$ is an isomorphism $R \to N$, injective because R is an integral domain, so N is free of rank 1. Thus every submodule of R is free.

10. $\mathbb{R}$ has a basis as a $\mathbb{Q}$-vector space (theorem 13.1.8), and the basis is uncountable, since the $\mathbb{Q}$-span of a countable set is countable while $\mathbb{R}$ is not! Note that even though the basis is uncountable, any element $x \in \mathbb{R}$ will still be a $\mathbb{Q}$-linear combination of *finitely many* basis elements.

11. On the other hand $\mathbb{Q}$ over $\mathbb{Z}$ is *not* a free module (i.e. $\mathbb{Q}$ is not a free abelian group). To see this, notice that for any $\frac{a}{b}, \frac{c}{d} \in \mathbb{Q}$, there exists $m, n \in \mathbb{Z}$ (namely $m = bc,\ n = -da$) such that $(bc)\frac{a}{b} + (-da)\frac{c}{d} = 0$, and hence any two elements are linearly dependent, meaning if $\mathbb{Q}$ was a free $\mathbb{Z}$-module, it would have to be cyclic. As we have seen in section 2.3.4, the only free cyclic $\mathbb{Z}$-module is $\mathbb{Z}$. Can you see why $\mathbb{Z} \not\cong_{\mathbb{Z}} \mathbb{Q}$?

Take note of the fact that whether a module is free or not is highly dependent on what ring is acting on it. In $\mathbb{Z}$-**Mod**, $\mathbb{Z}/n\mathbb{Z}$ is certainly not a free (it is not a free group). However, we have just seen that $\mathbb{Z}/n\mathbb{Z}$ is a free-module over itself! The intuition here behind being "free" comes down to the ring-action being "free".

Corollary 12.3.14: Consequences Of Universal Property

1. If F_1 and F_2 are free modules on the same set A, there is a unique isomorphism between F_1 and F_2 which is the identity map on A

2. if F is any free R-module on a free-generating set A, then $F \cong F^R(A)$. In particular, F enjoys the same universal property with respect to A as $F^R(A)$ does in theorem 12.3.11.

***Proof*:**
Any module F which is free on a subset A satisfies the universal property of theorem 12.3.11: for every R-module M and every set map $\varphi : A \to M$ there is a unique R-module homomorphism $\psi : F \to M$ with $\psi|_A = \varphi$.

For uniqueness, note that by definition 12.3.12 every nonzero $x \in F$ can be written as $x = r_1a_1 + \cdots + r_na_n$ with $r_i \in R$ and $a_i \in A$, so any homomorphism ψ restricting to φ must satisfy $\psi(x) = r_1\varphi(a_1) + \cdots + r_n\varphi(a_n)$, and $\psi(0) = 0$. Hence ψ is determined by φ.

For existence, we first observe that representations over a common finite subset are unique even when zero coefficients are allowed: if $\sum_{a\in B} r_a a = \sum_{a\in B} s_a a$ for a finite $B \subseteq A$, then dropping the terms with zero coefficient on each side yields two expressions of the same element with nonzero coefficients, which coincide by the uniqueness in definition 12.3.12 (for the zero element, the expression with all coefficients zero is the only one, since an expression with a nonzero coefficient would violate uniqueness for that element). Hence $r_a = s_a$ for all $a \in B$. We may therefore define $\psi\left(\sum_{a\in B} r_a a\right) := \sum_{a\in B} r_a\varphi(a)$, which is well-defined by the observation. Given $x = \sum_{a\in B} r_a a$ and $y = \sum_{a\in B'} s_a a$, we may enlarge both representations by zero coefficients to the common finite set $B \cup B'$, and then

$$\psi(x+y) = \sum_a (r_a + s_a)\varphi(a) = \psi(x) + \psi(y) \qquad \psi(rx) = \sum_a (rr_a)\varphi(a) = r\psi(x)$$

so ψ is an R-module homomorphism, and $\psi(a) = \varphi(a)$ for $a \in A$ by construction. This proves the claim, and with it the second sentence of part (2).

1. By definition 12.3.12, $A \subseteq F_1$ and $A \subseteq F_2$. Applying the universal property of F_1 to the inclusion $A \hookrightarrow F_2$ gives a homomorphism $\psi_1 : F_1 \to F_2$ restricting to the identity on A,

and symmetrically we get $\psi_2 : F_2 \to F_1$ restricting to the identity on A. The composite $\psi_2 \circ \psi_1 : F_1 \to F_1$ restricts to the identity on A, as does id_{F_1}, so by the uniqueness in the universal property of F_1 we get $\psi_2 \circ \psi_1 = \mathrm{id}_{F_1}$. Similarly $\psi_1 \circ \psi_2 = \mathrm{id}_{F_2}$. Hence ψ_1 is an isomorphism which is the identity on A, and it is the only one: any homomorphism restricting to the identity on A equals ψ_1 by the same uniqueness.

2. Let $\iota : A \to F^R(A)$ be the inclusion of A into $F^R(A)$ from theorem 12.3.11. By the universal property of F applied to ι, there is a homomorphism $\psi : F \to F^R(A)$ with $\psi|_A = \iota$, and by the universal property of $F^R(A)$ applied to the inclusion $A \hookrightarrow F$, there is a homomorphism $\chi : F^R(A) \to F$ with $\chi \circ \iota$ the inclusion. Then $\chi \circ \psi : F \to F$ restricts to the identity on A, so $\chi \circ \psi = \mathrm{id}_F$ by the uniqueness in the universal property of F, and $\psi \circ \chi$ and $\mathrm{id}_{F^R(A)}$ both restrict to ι on A, so $\psi \circ \chi = \mathrm{id}_{F^R(A)}$ by the uniqueness in theorem 12.3.11. Hence $F \cong F^R(A)$.

12.3.15 Foreshadowing Though a free R-module has no relations, it is not the case that any submodule has no relation, that is the submodule of a free module *need not be free*! In other words, though it may at first seem counter-intuitive, the process of taking a submodule can add relations: $\mathbb{C} \times \{0\}$ is a submodule of the free module $\mathbb{C} \times \mathbb{C}$ and is not free (example 12.3.13). This contrasts to how the subgroup of a free group is always free (see appendix B). There are certain rings for which the R-submodule are always free, while there are other rings where submodules are far from being free (ex. some submodules will be torsion modules, which will be far away from being free). We will explore these concepts further in chapters 13 and 17⋆.

Measuring degree of Freeness

As we have seen, not all modules are free modules. However, we may still like to find collection of elements that are linearly independent (i.e. that are "free" of relations). We explore these notions here.

Definition 12.3.16: [Free] Rank

For any ring R, the *rank* of an R-module M is the maximum number of R-linearly independent elements of M, that is if M has rank n, there exists elements $y_1, y_2, ..., y_n$ such that

$$r_1 y_1 + r_1 y_2 + \cdots + r_n y_n = 0 \quad \Leftrightarrow \quad r_1 = r_2 = \cdots = r_n = 0$$

If for all $n \in \mathbb{N}$, there exists a R-linearly independent set, then M is said to be of infinite rank.

Due to the importance of the notion of rank for modules in geometry, the rank is also sometimes called the *Betti Number*[10]. The rank of a module is similar to the size of the generating set of a module, and it will in fact match exactly the idea of rank for modules over fields, but this is not necessarily the case for modules over weaker rings:

[10]The kth Betti number loosely refers to the number of k-dimensional holes ona a topological surface

Example 12.3.17: Different Rank and Size of Generators

Let $R = \mathbb{Z}[x]$ and let $M = (2, x)$ be an R-module. If we take $2 \in M$, then since R is an R an integral domain and M is an ideal of R, there does not exist a r such that

$$ra = 0$$

thus, $\{2\}$ is a linearly independent set from M, meaning it's at least rank 1. What if we two elements? Let's take $a, b \in M$. Since $a, b \in R$, in particular $-a \in R$

$$ba - ab = 0$$

meaning 2 elements will never be linearly independent. This will naturally generalize to n elements, so M is of rank 1.

However, M is generated by 2 elements, showing that the rank and the number of generators does not match.

Sometimes, authors define the minimal set of generators to be the rank of the matrix, and rank as was defined above would be called the *free rank*. Since the size of the generating sets rarely comes up outside the context of free generating sets, this book decided to stick with "rank" to mean the maximal number of linearly independent elements.

Not that it is yet to be established if there is a unique maximal rank (just like in a tree, there may be many maximal heights). For modules over integral domains, the maximal height will alwaays be unique, and under appropriate finiteness conditions it shall also be unique, however there are counter-examples over more general rings. This shall be further explore in chapter 13.

Using the notion of rank, we can see how a module might fail to be free. In the "worst case scenario", we can single out elements fail to be linearly independent:

Definition 12.3.18: Torsion Element

Let M be an R-module and $m \in M$. Then m is called a *torsion element* if m is linearly *de*pendent, in particular there exists a nonzero $r \in R$ such that

$$r \cdot m = 0$$

The collection of all torsion elements of M is denoted $\text{Tor}_R(M)$ or $\text{Tor}(M)$ is the context is clear. An element that is not torsion is called *torsion-free*.

Elements that are torsion-free can be thought of as being "locally" free. Being is a property of modules, while being torsion-free is a property of elements. If every element has this property, then we give a name to such a module:

Definition 12.3.19: Torsion-Free Modules

Let M be an R-module. Then if $\text{Tor}_R(M) = \{0\}$, then M is said to be *torsion-free* or to be a *torsion-free module*.

Being torsion free is an interesting example of the fact that it is not always enough that every element have a property for the global module to have the property. In this case, even if every element itself is free, this does not imply the module itself is free

Example 12.3.20: Torsion-free, But Not Free Module

Take $\mathbb{C}[x, y]$ over itself. Then since it's an integral domain, it is torsion free and each submodule is torsion free. However, (x, y) is not free.

Looking instead at the ring action instead of the module element, we can also find all elements of the ring that fail to be linearly dependent with *all* module elements:

Definition 12.3.21: Annihilator

Let M be an R-module. Then the set

$$\text{Ann}_R(M) = \{r \in R \mid r \cdot m = 0 \ \forall m \in M\}$$

is called the *annihilator* of M.

Recall that this set appeared in example 12.1.7(4), in the discussion of modules where the annihilator is trivial. Such a module is given a name:

Definition 12.3.22: Faithful Module

Let M be an R-module. Then if $\text{Ann}_R(M) = \{0\}$, M is said to be a *faithful R-module*, or that the ring R is acting faithfully on M

If R does not act faithfuully on M, when we may always induce a faithful action along the lines of example 12.1.7(4). The same cannot be said about non-torsion modules, instead we may ask if a module M can be split into a torsion-free part and a torsion part. Being trosion free is much more flexible than being free:

Proposition 12.3.23: Extensions And Submodules Of Torsion-Free Modules

Let M be an R-module. Then:

1. If M is torison-free, then any submodule is torsion free
2. the direct sum of torsion free modules is torsion-tree
3. if M is an extension of torsion-free modules, that is $0 \to A \to M \to B \to 0$ is a short exact sequence with A and B being torsion-free, then M is torsion-free.
4. If R is an integral domain M is free, then M is torsion-free (i.e. free modules over integral domains are torsion-free)

***Proof*:**

1. This is immediate after realising $\text{Tor}_R(N) \subseteq \text{Tor}_R(M)$
2. this is also almost immediate givn a similar type of reasoning as in part (1)
3. If $m \in M$, where $r \cdot m = 0$, then $r \cdot \overline{m} = \overline{0}$, but since $r \in\in \text{Tor}_R(B) = \{0\}$ $r = 0$, and so $\text{Tor}_R(M) = 0$.

4. Since the direct of free modules is free, and the integral domain R as a module over itself is torsion free, M is torsion-free.

In chapter 14, we shall further explore the decomposition of the torsion and torsion-free part of a module over nicer enough rings so that the decomposition is possible.

If N is not a torsion module, then $\text{Ann}(N) = 0$. It is possible that N is a torsion module, but $\text{Ann}(N) = 0$:

Example 12.3.24: Torsion Module With No Annihilators
Take

$$Z = \bigoplus_k \mathbb{Z}/2^k\mathbb{Z}$$

then Z is a $\mathbb{Z}$-torsion module, since for every $z \in \mathbb{Z}$, we can take 2^k, where we pick k relative to the largest nonzero component. However,

$$\text{Ann}(Z) = 0$$

Since if $r \in \mathbb{Z}$ was an annihilator, then $r \cdot z = 0$ for every z, but z can take any value of $\mathbb{N}$, leading us to a contradiction

If the module is finitely generated the annihilator will always be non-empty, since if we annihilate the generators, we annihilate the entire module.

Exercise 12.3.1

1. Let M be an R-module, and define $\text{Ann}_R(M) = \{r \in R \mid r \cdot m = 0,\ \forall m \in M\}$.
    1. Show that $\text{Ann}_R(M)$ is a two-sided ideal
    2. If $\text{Ann}_R(M) = 0$, then M is said to be a faithful module. Show that all free modules are faithful, but there are faithful non-free modules
2. Let R be a commutative ring and an R-module over itself. Then show that an ideal $I \subseteq R$ is a free R-module if and only if it is principal and generated by a nonzero divisor.
3. Show that $\mathbb{Z}^{\mathbb{N}}$ (a direct product of countably many $\mathbb{Z}$ is *not* a free abelian group

12.3.3 Presentations and Resolutions

We claimed that all R-modules are a quotient of a free R-module, just how all groups are quotient of free groups. We will demonstrate this here, and then explore a bit furter how we can think of relations and presentations of modules:

Proposition 12.3.25: Constructing Modules using Free Modules

Any R-module M is the quotient of a free module $F^R(A)$ for appropriate set A:

$$F^R(A)/K \cong_R M$$

***Proof*:**
Let M be an R-module and $\mathcal{F} = F^R(M)$. Define the map:

$$\varphi : \mathcal{F} \to M \qquad (r_i m_i)_{i \in M} \mapsto \sum_{i \in M} r_i m_i$$

where any element on the left hand side is a tulpe indexed by M where all but finitely many of the terms are nonzero, and hence the right hand side is well-defined. This map is evidentely surjective (since the tulpe of m in the first element maps to m), and so if K is the kernel of this map, by the first isomrophism theorem:

$$F^R(M)/K \cong_R M$$

as we sought to show.

Hence, the kernel of a free module plays the role of codifying all the relations in a module. When working with groups, we encoded these relations writting a group as either, for example:

$$\langle x, y \ : \ xy - yx = 0 \rangle \qquad \text{or} \qquad \frac{F(x,y)}{xy - yx}$$

The former notation is easier to read before quotients are introduced. Since the kernel is the central object defining the relations, it enters the definition of a presentation of a module:

Definition 12.3.26: Presentation Of Module

Let M be an R-module. Then a *presentation* of M is a an exact sequence:

$$R^n \to R^m \to M \to 0$$

If m and n are finite (i.e. if $m, n \in \mathbb{N}$), then M is said to be *finitely presented*, or a *finitely presented module*.

Note that m and n need not be natural numbers, and that the map $R^n \to R^m$ in a presentation need not be injective, so a presentation is in general *not* a short exact sequence

$$0 \to R^n \to R^m \to M \to 0 \tag{12.5}$$

For instance, the map $\mathbb{Z}[x]^2 \to \mathbb{Z}[x]$ given by $(f, g) \mapsto 2f + xg$ is the first map of a presentation of $\mathbb{Z}[x]/(2, x)$, and its kernel contains $(x, -2) \neq 0$. Over a PID a presentation of the form (12.5) always exists: the kernel of a surjection $R^m \to M$ is a submodule of a free module, hence free, as theorem 14.1.6 proves. Conversely, if R is an integral domain and the kernel of every surjection from a finitely generated free R-module is free, then R is a PID (proposition 12.3.29). The observations below assume these results.

It is informative to the reader to see how presentations of modules compares to presentations of groups. Note that $\mathbb{Z}$ is a Principal Ideal Domain, and so for any finitely generated abelian group (i.e. any finitely generated $\mathbb{Z}$-module), we have the exact sequence:

$$0 \to \mathbb{Z}^n \to \mathbb{Z}^m \to G \to 0$$

Furthermore, any subgroup of a free group is free (see appendix B), and so more generally for any group G and appropriate sets A and B:

$$0 \to F(A) \to F(B) \xrightarrow{\varphi} G \to 0$$

where $F(A) = \ker(\varphi)$. Over some other rings some modules admit no chain like (12.5) (examples are in *Homological Algebra* [19]), and this distinguishes groups from modules. In fact, there are even modules for which there is no finite length chain of free modules. If we interpret equation (12.5) as saying that the relations on a module is free, then if we extend the length by 1:

$$0 \to R^n \to R^m \to R^p \to M \to 0$$

can interpret the extra homomorphism as the relations of the relations, where the "relations of the relations" are free. Some modules do not have a finite length chain of this sort!! We give a name to such a chain

Definition 12.3.27: Free Resolution

Let R be a ring and M an R-module. A *free resolution* of M is an *exact* chain complex (definition 12.2.11)

$$\cdots \to F_2 \to F_1 \to F_0 \to M \to 0$$

in which every F_i is a free R-module. The resolution has *finite length* k if $F_i = 0$ for every $i > k$, so that it reads

$$0 \to F_k \to F_{k-1} \to \cdots \to F_1 \to F_0 \to M \to 0$$

and M is then said to have a *finite length free resolution* of length k.

Exactness is what makes the resolution say something about M: at F_0 it says that the relations among the chosen generators are exactly the image of F_1, at F_1 that the relations among *those* relations are the image of F_2, and so on.

By definition, all free modules have a finite length free resolution of length zero since

$$0 \to R^{n_0} \to M \to 0$$

i.e. $R^{n_0} \cong M$. By the universal property of free modules, any R-module M has a free resolution (can you see why). The resolution is called a *free* resolution since we are using free-modules. If we use a different type of module, we will get a different type of resolution[11].

Example 12.3.28: Free Resolution

Let G be a finitely generated abelian group. Then by the fundamental theorem for such groups (theorem 5.1.6),

$$G \cong \mathbb{Z}^n \oplus \mathbb{Z}/n_1 Z \oplus \mathbb{Z}/n_2\mathbb{Z} \oplus \cdots \oplus \mathbb{Z}/n_k\mathbb{Z}$$

As pointed out, all free-modules have a resolution of length 0, namely

$$0 \to \mathbb{Z}^n \to \mathbb{Z}^n \to 0$$

[11] Projective, injective and flat modules are introduced in chapter 17⋆, and the corresponding resolutions in *Homological Algebra* [19].

On the other hand, any cyclic torsion group has a resolution of length 1:

$$0 \to \mathbb{Z} \xrightarrow{\cdot n} \mathbb{Z} \xrightarrow{\pi} \mathbb{Z}/n\mathbb{Z} \to 0$$

These results can be combined to show that G always has a resolution of length 1. Modules whose free resolutions all have length at least 2 will be given in *Homological Algebra* [19].

Proposition 12.3.29: Free Resolutions Of Length 1, Then PID

Let R be an integral domain such that for every finitely generated R-module M and every surjective homomorphism $R^n \to M$, the kernel is a free R-module, so that

$$0 \to R^m \to R^n \to M \to 0$$

is exact for some m. Then R is a PID.

***Proof*:**
Let $I \subseteq R$ be an ideal. The projection $R \to R/I$ is surjective with kernel I, so I is free. Two elements $a \neq b$ of R are linearly dependent over R: if both are nonzero, $b \cdot a + (-a) \cdot b = 0$ is a nontrivial relation, and a set containing 0 is dependent. So a basis of I has at most one element, and $I = 0 = (0)$ or $I = Ra = (a)$. Since I was arbitrary, every ideal of R is principal, making R a PID, as we sought to show.

12.3.4 Finitely Generated Modules and Noetherian Modules

As with many structures in mathematics, the "infinite" case (in our case infinitely generated objects) have different behavior than the finite case and require some different tools to analyze (possibly some analysis to use the notion of limits). We thus study the case where modules are finitely generated or all submodules are finitely generated.

Let R be a ring and let M be a left R-module. We first define what types of modules are *finitely generated*, and some consequences of being finitely generated,

Definition 12.3.30: Finitely Generated R-Module

Let M be an R-module. Then M is said to be finitely generated if there exists an $n \in \mathbb{N}_{>0}$ such that $M \cong_R R^n/K$ for some R-module K, that is there exists a map φ where

$$R^n \xrightarrow{\varphi} M \to 0$$

is exact so that $\ker(\varphi) = K$ and $R^n/K \cong_R M$.

Though we have defined finitely generated modules, we usually require the stronger condition that all submodules of a finitely generated R-module to be finitely generated (which is not true for every finitely generated modules)

Example 12.3.31: Finitely Generated But A Non-finitely Generated Submodule
Take $R = k[x_1, x_2, ..., x_n, ...]$ be the ring with infinitely many indeterminates over a field. Then R is cyclic over itself, generated by 1, but the R-submodule, i.e. ideal, $(x_1, x_2, ..., x_n, ...) \subseteq R$ is not finitely generated as an R-module.

The main inhibitor to this is We will phrase this condition a little differently and show it's equivalent to the previous statement:

Definition 12.3.32: Noetherian R-Module and Ring

Let M be a left R-module Then:

1. M is said to be *Noetherian R-module* or to satisfy the *ascending chain condition on submodules* (or A.C.C. On submodules), that is there are no infinite increasing chains of submodules, i.e., whenever
$$M_1 \subseteq M_2 \subseteq M_3 \subseteq \cdots$$
is an increasing chain of submodules of M, then there is a positive integer m such that for all $k \geq m$, $M_k = M_m$ (so the chain becomes stationary at stage m: $M_m = M_{m+1} = M_{m+2} = \cdots$)
2. The ring R is said to be Neotherian if it is Neotherian as a left-module over itself, i.e., if there are no infinite increasing chains of left-ideals in R.

One can have an analogous definition of A.C.C. for right and two-sided ideals in (possibly noncommutative) ring R. For noncommutative rings, left and right A.C.C. need not agree. In the ring of matrices $\begin{pmatrix} a & b \\ 0 & c \end{pmatrix}$ with $a \in \mathbb{Z}$ and $b, c \in \mathbb{Q}$, the matrices with $a = c = 0$ and $b \in \frac{1}{2^n}\mathbb{Z}$ form a left ideal for each n, and these left ideals increase strictly, so the ring is not left Noetherian. It is right Noetherian ([41]).

Theorem 12.3.33: Equivalent Conditions To Noetherian

Let R be a ring and let M be a left R-module. Then the following are equivalent

1. M is a Noetherian R-module
2. Every nonempty set of submodules of M contains a maximal element under inclusion
3. Every submodule of M (including M) is finitely generated

***Proof*:**
(1) $\Rightarrow$ (2). Let S be any collection of submodules. Let's pick some sub-module $M_1 \in S$. If S is maximal, then we are done. If not, then M_1 is not maximal, meaning there exists an M_2 such that $M_1 \subseteq M_2$. If M_2 is maximal, we are done. If not, pick M_3 in a similar fashion. Keep picking sub-modules M_i. If none of them are maximal, then by the axiom of choice, we can pick an infinitely long chain of submodules
$$M_1 \subseteq M_2 \subseteq \cdots \subseteq M_k \subseteq \cdots$$
which is a contradiction to the statement that M is Noetherian.

For (2) $\Rightarrow$ (3), Let M be a module. Pick $x \in M$. If $\langle x_1 \rangle = M$, then M is finitely generated. If $\langle x_1 \rangle = N_1 \subsetneq M$, then pick another element $x_2 \in M - N_1$. If $\langle x_1, x_2 \rangle = M$, we are done. If $\langle x_1, x_2 \rangle = N_2 \subsetneq M$, then pick another element $x_3 \in M - N_2$. Either this stabalizes, or we can craete an infinite chain:

$$N_1 \subseteq N_2 \subseteq \cdots \subseteq N_k \cdots$$

This is a collection of submomdules of M, and so by the 2nd condition there must exist a maximal module, say N. Since every module in this chain is finitely generated, N mut be finitely generated, and since it's maximal adding one more element to N, $N + x$ and taking it's closure, $\overline{N + x} = N'$ must make the new larger module N' equal to M. Hence, M is finitely genearted.

For (3) $\Rightarrow$ (1), Take any ascending chain of modules

$$N_1 \subsetneq N_2 \subsetneq \cdots \subsetneq N_k \subsetneq \cdots$$

then $N = \bigcup_{i=1}^{\infty} N_i$ is a module, and by assumption it is finitely generated, say by $\{x_1, x_2, ..., x_n\}$. By definition of N, for each x_i, there exists a j such that $x_i \in N_j$. Since N is the union of the modules N_j, it must be that at some finite distance into the chain, a module contains all the generators. But that means the chain stabalizes.

Note that if M is a Noetherian R-module, then R need not be Noetherian. If R is noetherian, then M is a Noetherian R-module? (Use the fact that the direct sum and quotient of a noetherian ring is noetherian, an the uni prop of free modules).[12]. In proposition 9.4.18, it was shown that rings where $1 \neq 0$ have maximal ideals. Since all rings are modules over themsleves, it might seem that all rings are Noetherian modules due to the second conditions. However, this is certainly not the case, consider $R[x_1, x_2, ..., x_n, ...]$ over itself and the chain $(x_1) \subsetneq (x_1, x_2,) \subsetneq \cdots$. The subtly lies in the fact that the maximal element must lie in the collection of submodules, while the maximal element in the previous example is the union of all the submodules.

Many textbooks introduce Noetherian modules as modules where each submodule is finitely generated. The reason the A.C.C. was chosen for this textbook was for it's greater applicability in many other fields. Notions of being "Noetherian" are pervasive in mathematics, but not all structure are "Noetherian" by having every substructure be finitely generated (sometimes, the notion of substructure is not even well-defined)[13]. The A.C.C. allows us to still consider those structures.

Since Principal Ideal Domain are *a fortiori* Noetherian rings, finitely generated modules over PID's are are also Noetherian, and all the results we discussed earlier apply to such modules (including PID's over themsleves).

Finite Generation vs. Finite Type

Being finitely generated as an R-module and finitely generated as an R-algebra are very differnet:

Example 12.3.34: Finite Generation: Module vs. Algebra
Take $R = k[x]$ as an k-algebra and consider the natural k-module structure. Then it is an *infinitely generated* k-module generated (for example by $\{1, x, x^2, x^3, ...\}$ however it is a *finitely generated* k-algebra generated by $\{1, x\}$.

[12]This mirrors a more general theme in which a property of the underlying ring "becomes" the property of all modules over that ring. Can you see how we can (in a sense) call a field a "free ring"?

[13]See Noetherian topologies for an interesting example

Mo generally, contrast an finitely generated R-module M to a finitely generated R-algebra M:

$$R^{\oplus A} \twoheadrightarrow M \quad R[A] \twoheadrightarrow M$$

Definition 12.3.35: Finite Type

Let A be an R-algebra. Then A is of *finite type* (over R) if A is finitely generated as an R-algebra.

Often, instead of finite type the terms "finitely generated as an R-algebra" or even "finitely generated R'algebra" are often used interchangeably.

Exercise 12.3.2

1. Is a subring of a Noetherian ring Noetherian?
2. (If you know some complex analysis) In this exercise, we try to show how the Noetherian property is not stable by taking subrings and show demonstrate a correspondence between being Noetherian and how the zero of functions are in some way "well-behaved". Consider the following chain of subrings:

$$\begin{gathered}\mathbb{R}[x] \\ \subseteq \\ \text{Analytic functions on } \mathbb{R} \\ \subseteq \\ \text{Analytic functions on } [0,1] \\ \subseteq \\ \text{Analytic functions on } (0,1) \\ \subseteq \\ \text{Analytic functions at } 0 \\ \subseteq \\ \text{Smooth functions near } 0 \\ \subseteq \\ \text{Former Power series}\end{gathered}$$

   which of these rings are Noetherian, and which are not?
3. Show that a finitely generated R-module over a Noetherian ring R is itself Noetherian
4. Let M be an R-module over an integral domain R. Then if M is of rank n, any submodule $N \subseteq M$ can be of rank at most $n \leq m$ (hint: take the ring of fractions).

12.3.5 Direct Sums and Homomorphisms

For the first time in this book, we will pay closer attention to hom-sets. The structure of hom-sets of modules will allo us to determine many important properties of modules.

Proposition 12.3.36: Direct Sums And Homomorphisms

Let $(M_i)_{i\in I}$ and $(N_j)_{j\in J}$ be a collection of modules with I and J indexing sets. Then:

$$\mathrm{Hom}_R\left(\bigoplus_{i\in I} M_i, N\right) = \prod_{i\in I} \mathrm{Hom}_R(M_i, N)$$

and

$$\mathrm{Hom}_R\left(M, \prod_{j\in J} N_j\right) = \prod_{j\in J} \mathrm{Hom}_R(M, N_j)$$

***Proof*:**
let $\varphi : \oplus_{i\in I} M_i \to N$. Then for any $(\alpha_i)_{i\in I} \in \oplus_{i\in I} M_i$, there are only finitely many non-zero entries. For ease of visualization, lkets represent the element $(\alpha_i)_{i\in I}$ as

$$(\alpha_1, \alpha_2, ..., \alpha_n, 0, ...)$$

This represetnation can be confusing on two terms: we have implicitly chosen an "order" of the elements and re-labelled the indices of the finite eleents accordingly, and it makes the set I look like it is *countable*, which is not necessarily the case. These two remarks should be kept in mind while working with this represetnation of the element.

Since there are n elements, write each of them as:

$$\alpha_i = \underbrace{0 + \cdots + 0}_{i-1 \text{ times}} + \alpha_i + \underbrace{0 + \cdots + 0}_{n-i \text{ times}}$$

Now, the finiteness condition was key here so that we may use linearity inductively, so that for any $\varphi \in \mathrm{Hom}_R\left(\bigoplus_{i\in I} M_i, N\right)$

$$\varphi((\alpha_i)_{i\in I}) = \varphi(\alpha_1) + \cdots + \varphi(\alpha_n)$$

Now, it is immediate that each function on the right hand side is naturally isomorphic to a function $\varphi_i \mathrm{Hom}_R(M_i, N)$. Labelling eahc of thes functions a φ_i, we get:

$$\varphi \to (\varphi_i)_{i\in I}$$

which is an isomorphism.

The direct product case is done almost identically, except we do not need to worry about finite additivity, since for any

$$\varphi(\alpha) = (\beta_i)_{i\in I}$$

we can define:

$$\varphi_i(\alpha) = \beta_i$$

which will give us the desired isomorphism, completing the proof.

As a consequence of this, we can simplify:

$$\mathrm{Hom}_R\left(\bigoplus_i M_i, \bigoplus_j N_j\right) \cong \prod_{i,j} \mathrm{Hom}_R(M_i, N_j)$$

If you are comfortable with linear algebra, then in the case where both direct sums have only finitely many components (also known as *summands*), then if φ_{ji} represent's the component function taking in the ith component in the domain and is in the jth component of the domain, then: the there is an isomrophism mapping $\varphi \mapsto (\varphi_{ij})$ where the maps is defined as:

$$\varphi \begin{pmatrix} s_1 \\ s_2 \\ \vdots \\ s_n \end{pmatrix} = \begin{pmatrix} \varphi_{11} & \varphi_{12} & \cdots & \varphi_{1n} \\ \varphi_{21} & \varphi_{22} & \cdots & \varphi_{2n} \\ \vdots & \vdots & \ddots & \cdots \\ \varphi_{n1} & \varphi_{n2} & \cdots & \varphi_{nn} \end{pmatrix} \begin{pmatrix} s_1 \\ s_2 \\ \vdots \\ s_n \end{pmatrix}$$

Showing that there is a natural relation between $\mathrm{Hom}_R(\oplus_i^m M_i, \oplus_j^n N_j)$ and matrix rings, especially if $M_i = N_j$. More on that in [17, Semisimple Rings and Modules].

Notice how important linearity is to split up the homomorphsim in direct products in the *domain*. To make sure that no misconceptions are being formed, here is a counter-example when a function is not linear:

Example 12.3.37: Hom's And Products Not Commuting in Domain

Let R be a nonzero commutative ring and $f\colon R \times R \to R$, $f(x,y) = xy$. Then $f(x,0) = 0$ and $f(0,y) = 0$ for all $x, y \in R$, while $f(1,1) = 1$. So $f(x,y) \neq f(x,0) + f(0,y)$ for $x = y = 1$, and f is not determined by its restrictions to the two factors, while an R-linear $\varphi\colon A \oplus B \to N$ satisfies $\varphi(a,b) = \varphi(a,0) + \varphi(0,b)$.

On the other hand, every function $f\colon M \to A \oplus B$, linear or not, is determined by its two components $\pi_A \circ f$ and $\pi_B \circ f$, since $f(m) = (\pi_A(f(m)), \pi_B(f(m)))$. For R-linear f, the components are the homomorphisms of the universal property of the product (proposition 12.3.9).

Exercise 12.3.3

1. Show that if M is an R-module, then $\mathrm{Hom}_R(R, M) \cong M$
2. Let M and N be R-modules, let ι_j and π_i be the inclusions and projections of the summands of $M \oplus M$ and of $N \oplus N$, and for $\varphi \in \mathrm{Hom}_R(M \oplus M, N \oplus N)$ put $\varphi_{ij} = \pi_i \circ \varphi \circ \iota_j \in \mathrm{Hom}_R(M, N)$, $i, j \in \{1, 2\}$. Show that $\varphi \mapsto (\varphi_{ij})$ is a bijection from $\mathrm{Hom}_R(M \oplus M, N \oplus N)$ onto the 2×2 arrays of elements of $\mathrm{Hom}_R(M, N)$, and that for $N = M$ it is a ring isomorphism from $\mathrm{End}_R(M \oplus M)$ onto the ring of 2×2 matrices with entries in $\mathrm{End}_R(M)$, multiplied by the usual formula with composition in place of multiplication of entries.
3. Let $\varphi : A \to B$ be an R-module isomorphism. Show that $\Phi : \mathrm{End}_R(A) \to \mathrm{End}_R(B)$ where $\Phi(f) = \varphi \circ f \circ \varphi^{-1}$ is a ring isomorphism, and if R is commutative an R-algebra isomrophism.
4. show that if $I \subsetneq R$ is a proper ideal, $I \cong S$ as an R-module isomorphism $\varphi(e) = 1$ a,d $e^2 = e$, then φ is in fact a ring-isomrophism, so I has the ring structure of S (though it cannot have the ring structure of R, since it cannot contain a unit).

13

Modules over Fields: Vector Spaces and Linear Algebra I

This chapter focuses on the study of modules where the ring are fields (i.e. [unital] commutative division ring). We will use the letter k (from German Körper) or the symbol $\mathbb{F}$ to represent an arbitrary field. As we have mentioned in definition 12.1.9, a module over a field is called a vector-space due to the historical precedence of the terminology which is now widely accepted in and beyond mathematics (for example, the term vector-space is used in physics, biology, and chemistry). This precedence of modules over fields comes from the fact that historically, fields were studied before general rings. Many problems in physics in the 19th century turned out to requires vector-spaces (ex. solutions to ODEs formed vector spaces, many important spaces of functions formed vector-spaces, the Cartesian plane formed a vector space), leading to precedence in nomenclature for modules over fields. Mathematically, giving another name to modules over fields is not unreasonable: as we will see, having a field act on a module is so restrictive (or powerful) that we will only need to keep track of two pieces of information to fully understand the structure and the classification of vector-spaces: the underlying field k, and a new number called the *dimension* of the vector-space[1]. No algebraic structure we have encountered requires so little data to be able to classify them (for example, the classification of finite simple groups required tens of thousands of pages over almost a century, while the classification of vector-space is regularly taught in a 1st or 2nd year undergraduate class).

Due to the simplicity of vector-space structures, vector space homomorphisms (k-module homomorphisms) will be completely classified. (A further word on this before talking a bit about dual theory and geometry)

As a consequence of the historical precedence of vector-spaces, a separate vernacular was developed for modules over fields that is now considered standard:

[1]This holds in infinite dimension too, where the dimension is a cardinal: any two bases of a vector space have the same cardinality [1, Proposition VI.1.9]. This book proves it for spaces with a finite basis (corollary 13.1.13).

Terminology over any Ring	Terminology over Fields
M is an R-module	M is a vector space over R
m is an element of M	m is a vector in M
N is a submodule of M	N is a subspace of M
M/N is a quotient module	M/N is a quotient space
M is a free module of rank n	M is a vector space of dimension n
M is a nonzero cyclic module	M is a 1-dimensional vector space
$\varphi : M \to N$ is an R-module homomorphism	$\varphi : M \to N$ is a (R) linear transformation
M and N are isomorphic as R-modules	M and N are isomorphic vector spaces
the subset A of M generates M	The subset A of M spans M
$M = RA$	each element of M is a linear combination of elements of A, i.e., $M = \text{Span}(A)$

13.1 Vector Spaces and their Properties

Vector spaces are famous for having a very simple structure: Given a vectorspace V over k, then V is isomorphic to k^N for appropriate N. In a linear algebra class, we say that every vector space has a basis. This N will be called the *dimension* of the vector-space. In many linear algebra classes, the attention is limited to $N \in \mathbb{N}$. However, this section will allow N to be of any cardinality (ex. $|\mathbb{N}|$ or $|\mathbb{R}|$). This is to show some of the power of vector-spaces: the (algebraic) structure does not get more complicated when dealing with infinite dimensions[2]

Before these results are shown, for those who gain an intuition via examples, here are a couple more example's of vector-spaces

Example 13.1.1: Vector Spaces
We already gave 4 examples in example 12.1.10. We give a few more here

1. We can generlize k^n as follows: notice that we can define k^n as

$$k^n = \{f \mid f : \{1, ..., n\} \to \mathbb{R}\}$$

   where addition is defined pointwise, and since k is commutative we can define a natural k-action on k^n (can you see it?). Any function $f \in k^n$ will tell us what value we should have in the component i. For example, if $f(1) = \pi$, $f(2) = 3$, an $f(3) = \sqrt{10}$, then this function is equivalent to:

$$(\pi, 3, \sqrt{10})$$

   Thus, We can consider:

$$k^{\mathbb{N}} = \{f \mid f : \mathbb{N} \to \mathbb{R}\}$$

   To be the $\mathbb{N}$-dimensional vector-space, or $k^{\mathbb{R}} = \{\mathbb{R} \mid f : \mathbb{R} \to \mathbb{R}\}$ to be the $\mathbb{R}$-dimensional vector space

2. Generalizing the above example, let k be a field, X a set, and $V = \text{Hom}_{\mathbf{Set}}(X, k)$ (another common notation is k^X). Then we can make V into a vector-space via point-wise operation

[2] On the other hand, the set-theoretical and analytic structure (if there is a topology present) gets more complicated, and arguably non-trivial, in infinite dimensions. For example, in infinite dimensions, surjective does not imply injective (similar to the proof for infinite sets), and there will exist linear transformations that are not continuous (for the appropriate topology).

in the codomain: for any $f, g \in V$, $c \in k$, $f + g$ is defined to be:

$$(f + g)(x) = f(x) + g(x) \qquad \forall x \in X$$

while fg and cf is defined as:

$$(fg)(x) = f(x)g(x) \qquad (cf)(x) = cf(x) \qquad \forall x \in X$$

These definitions are possible since the image of any function in V lands in a field, where all the above operations are well-defined. If we take the collection of lienar transformation from V to k (where V is a vector space), then we often denote that as V^*. Note that since k is also a ring, we can also define $(fg)(x) = f(x)g(x)$, making $\mathrm{Hom}_{\mathbf{Set}}(X, k)$ into a k-algebra.

Often, the set X is given its own properties (a popular example are X is a locally compact Hausdorff space, or X is a smooth manifold). Then we often try to deduce more information about X given the structure of hom-set $\mathrm{Hom}_{\mathbf{Set}}(X, k)$.

3. Let $R = k[x]$ for some field x. Then $k[x]$ is a k vector-space with action $c \cdot p(x) = cp(x)$. More generally, if $R = k[x_1, x_2, ..., x_n, ...]$ is some polynomial ring over several variables, then R is a k vector-space with $c \cdot p(x_1, x_2, ..., x_n, ...) = cp(x_1, x_2, ..., x_n, ...)$.

4. Let $k = \mathbb{C}$ an $V = \mathbb{C}^n$. Then $\mathbb{C}^n$ is a complex vector space with the action:

$$z \cdot (z_1, ..., z_n) = (zz_1, ..., zz_n)$$

There is a second natural action, known as the *conjugate action*:

$$z \cdot (z_1, ..., z_n) = (\overline{z}z_1, ..., \overline{z}z_n)$$

More generally, if V is a vector-space over $\mathbb{C}$ (also known as a complex vector space), then $\overline{V}$ denote the same space with the element being conjugated before the action is applied.

5. (If you know some ODE's) Given a linear ordinary differential equation, the set of all solutions form's a vector-space. Take for example $\varphi : C^\infty(\mathbb{R}) \to C^\infty(\mathbb{R})$ where:

$$f \mapsto f'' + f$$

Consider the subspace V where

$$V := \{f \in C^\infty(\mathbb{R}) \mid f'' + f \equiv 0\}$$

Then using elementary ODE theory, we find that:

$$V = \mathrm{span}_{\mathbb{R}}(\{e^t, e^{-t}\})$$

The fact that all solutions to the ODE $f'' + f = 0$ is a linear combination of these two solutions is a remarkable property of *linear* ODE's. This simple property of linearity can be taken very far: in the world of PDE's (Parital Differential Equations), if you have something called a *linear PDE*, which means that if you have two solutions, then add them and/or scale the terms, the result is another solution. The power we get from this is that we can then see if this is generalizable to taking *limits* of some very simple solutions to the PDE. As a common example, let's say the solution is of the form $\cos(\alpha x)$ or $\sin(\beta x)$ for any α, β. Then under the right conditions[a], the collection of these (after being closed under addition) is *dense* in $L^2(S^1)$, meaning we can use limits to construct *any* $L^2(S^1)$ function and find another solution to our PDE. A common place this particular example is used is *Fourier Analysis*[b]

6. (If you know Differential Geometry) Let M be a smooth n-manifold and consider the collection of all $\varphi : C^\infty(M) \to \mathbb{R}$. Then the collection of all *derivations at* p[c] for some point $p \in M$, denoted $\text{der}_p(M) \subseteq \text{Hom}_{\mathbf{Smooth}}(C^\infty(M), \mathbb{R})$ forms a vector-space of dimension n, and is a common definition of the *tangent space* of the manifold M at the point p[d]

[a] See a textbook covering Fourier Analysis, ex. Pugh or Folland
[b] See [51]
[c] linear function where $\varphi(fg) = \varphi(f)g + f\varphi(g)$
[d] See [46, Tangent Vector as Derivation, Dimension of the Tangent Space]

Now, we shall develop a more sophisitaced notion of generating set:

Definition 13.1.2: Linear Independence And Basis

Let M be an R-module.

1. A subset $S \subseteq M$ is said to be *linearly indepenent*, or a set of *linearly independent elements*, if for any equation

$$a_1 m_1 + a_2 m_2 + ... + a_n m_n = 0$$

with $a_1, a_2, ..., a_n \in R$ and $m_1, m_2, ..., m_n \in S$, then $a_1 = a_2 = \cdots = a_n = 0$. If this is not the case, the elements are called *linearly dependent*. In other words, if $I = \{m_1, m_2, ..., m_n\} \subseteq M$, so that $\iota : I \to M$ is injective, then:

$$\begin{array}{ccc} F^R(I) & \xrightarrow{\varphi} & M \\ {\scriptstyle j}\uparrow & \nearrow_{\iota} & \\ I & & \end{array} \tag{13.1}$$

commutes[a] and φ is injective. Note that I can be an arbitrary indexing set, that is for every $i \in I$, $m_i \in M$ corresponds to it, and then we would have an injection $\iota : I \to M$ which lifts to $\varphi : F^R(I) \to M$. If $R = k$, then the elements are called *linearly independent vectors*.

2. A *basis* for M is an *set* $\{a_1, a_2, ..., a_n\}$ of elements that is linearly independent and which generate sM (i.e. span M).

3. An *ordered basis* for M is an *ordered set* (a tuple) $(a_a, a_2, ..., a_n)$ of linearly independent elements which generates M (i.e. span M).

[a] The diagram commuting is the universal property of free modules

The point of an *ordered basis* is so that $A \subseteq M$ gives a *unique* isomorphism:

$$\varphi : F^R(A) \cong M$$

i.e. The φ in equation (13.1) is both injective and surjective. In particular, two ordered bases will be considered different even if one is a rearrangement of the other (hence why it's ordered).

13.1.3 Convention In this section, all bases shall implicitly be ordered basis. The convention shall be dropped when working with tensor products of vector spaces.

Note that the φ from the commutative diagram in equation (13.1) has as domain all *finite* combinations of elements of $\{m_i\}_{i\in I}$, so that when writing:

$$\sum_{i\in I} r_i m_i$$

only finitely many r_i are nonzero. If φ is injective, then $\sum_{i\in I} r_i m_i = 0$ if and only if $r_i = 0$ for all $i \in I$. Sometimes to emphasize the fact that it's ordered, a basis is called an *ordered basis*. In terms of indexing sets, the ordered elements are generalized to elements having different indexes. Notice how being linearly independent is different from proposition 12.3.6(3), where if $x \in \oplus_{i\in I} N_i$ implies there exists unique elements $v_i \in N_i$ such that $x = \sum_i v_i$. We are adding another requirement on $\bigoplus_i v_i$, namely that the function φ in the above definition is injective.

A module M having a basis is in fact a very restricting condition:

Lemma 13.1.4: Basis iff Free

Let M be an R-module. Then M is free if and only if it admits a basis, in paritcular $B \subseteq M$ is a basis if and only if the natural homomorphism $R^{\oplus B} \to M$ is an isomorphism

***Proof*:**
This is immedaite from the definition: if $B \subseteq M$ is linearly independent and generates M, then $\varphi : R^{\oplus B} \to M$ is injective and surjective, and if $\varphi : R^{\oplus B} \to M$ is an isomorphism, then $\varphi(B)$ is a subset of M which generates it and is linearly independent.

Example 13.1.5: Linearly Dependent and Independent Sets

1. Let $M = \mathbb{Z}/n\mathbb{Z}$ be a $\mathbb{Z}$-module. Then:

   $$n \cdot [1]_n = 0$$

   but naturally $n \neq 0$. More generally, if R is a ring with a zero divisor, and $(z) \subseteq R$ is the sub-module generated by the zero-divor (z), then if $zz' = 0$, $z' \cdot z = 0$,showing that the singleton z is linearly dependent.

2. Let $R = \mathbb{C}[x, y]$ be a module over itself. This ring has no zero divisors, hence the above does not apply. However, that there are still linearly dependent elements, for example:

   $$y \cdot x + (-x) \cdot y = xy - xy = 0$$

   showing that x, y are *not* linearly independnet over $\mathbb{C}[x, y]$. However, if we consider $\mathbb{C}[x, y]$ over $\mathbb{C}$, then $\{x, y\}$ *are* linearly independent, since $\mathbb{C}[x, y]$ is the free $\mathbb{C}$-algebra. Hence:

   $$F^{\mathbb{C}}(\{x, y\}) = \mathbb{C}^2 \to \mathbb{C}[x, y]$$

   is an injective function. However, $\{x, y\}$ is not a basis since the above homomorphsim is not surjective, since nothing maps to xy. Is there any finite that will create a surjective map?

3. If R is any ring (which includes rings with torsion elements), then $(1) = R$, and $r \cdot 1 = r1$ if and only if $r = 0$. Hence,
$$R \mapsto R$$
is an injective homomorphism (even surjective, making it a basis). This shows that the choice of subset of the module effects whether the set if linearly independent or a basis.

   Though R has a basis when over itself, show that $\mathbb{Z}/2\mathbb{Z} \oplus \mathbb{Z}/4\mathbb{Z}$ over $\mathbb{Z}$ does *not* have a basis, or even a linearly independent set.

As we saw in the above example, some modules have linearly independent sets, some do not, and others have a basis. By a simple applications of Zorn's lemma and some manipulation of definition, we can show that every module has a maximally linearly independent set:

Lemma 13.1.6: Maximally Linearly Independent Set

Let M be an R-module and let $S \subseteq M$ be a lienarly independnet set. Then there exsits a maximal linearly independent subset of M containing S.

***Proof*:**
This is an application of Zorn's lemma: Let $\mathcal{L}$ be the collection of linearly independent subset of M containing S ordered by inclusion. Since $S \in \mathcal{L}$, $\mathcal{L} \neq \emptyset$. If we take any chain $\mathcal{C} \subseteq \mathcal{L}$, and take $C = \cup\mathcal{C}$, then C is linearly independent, since it it were linearly dependent, then only finitely many linear element would be used in the equation, but then they all belong to some $C_i \in \mathcal{C}$, which by assumption is linearly independnet. Hence, by Zorn's lemma $\mathcal{L}$ contains a maximally linearly independent set containing S.

13.1.7 Trivia The Axiom of Choice and the above lemma are in fact equivalent, hence it is impossible to prove the above statement without using Zorn's lemma.

If S is a maximally linearly independent set containing $\emptyset$, we shall simply say that S is maximally linearly independent. Note that being maximally linearly independent does not guarantee a unique size or that it generates the entire module, for example $\{2\} \subseteq \mathbb{Z}$ is maximally linearly independent but does not generate $\mathbb{Z}$. In the special case where $R = k$ is a field so that we have a vector-space V, then a maximally linearly independent set is indeed a basis!

Theorem 13.1.8: Vector Space has Basis

Let V be a vector space over k and let B be a maixmally lienarly independnet set. Then B is a basis of V, hence V is a free k-module. Furthermore, if S is a liearly independent set, then S is contained in a basis B.

***Proof*:**
We need to show that B generates V. Let $v \in V$, $v \notin B$. Then by maximality of B, $B \cup \{v\}$ is linearly dependent, hence there exists $v_1, v_2, ..., v_n$ and $c_0, c_1, ..., c_n$ such that
$$c_0 v + c_1 v_1 + \cdots + c_n v_n = 0$$
with not all $c_i = 0$. Note that $c_0 \neq 0$ or else we would get $c_1 v_1 + \cdots + c_n v_n = 0$ with not all

$c_i \neq 0$, contradicting linearly independence of elements in B. Since k is a field, c_0 is a unit, hence:

$$v = (-c_0^{-1}c_1)v_1 + \cdots + (-c_0^{-1}c_n)v_n$$

But then, $v \in \text{span}(B)$, showing that B generated V. By lemma 13.1.4, V is a free k-module.

Furthermore, if S is lienarly independent, then S is contained in a maximally linearly independent set, say B. But by our above reasoning B is a basis, hence S is contained in a basis, as we sought to show.

Hence, every vector-space is free. Before continuing our exploration of bases, a quick comment must be made: we can define the dual notion, where a *minimally spanning set* is a spanning set no proper subset of which spans V, and the same statement holds in the reverse direction, namely:

Proposition 13.1.9: Vector space, Minimally Spanning then Basis

Let V be a k-vector space and let B be a minimally spanning set. Then B is a basis for V, making V a free k-module. Furthermore, if S is any spanning set, then S contains a minimal spanning set B.

***Proof*:**
For the first claim, suppose toward a contradiction that the minimally spanning set B is linearly dependent, so there exist distinct $v_1, ..., v_n \in B$ and scalars $c_1, ..., c_n \in k$, not all zero, with

$$c_1v_1 + c_2v_2 + \cdots + c_nv_n = 0$$

Reordering, we may take $c_1 \neq 0$. Since k is a field, c_1 is a unit, and so

$$v_1 = (-c_1^{-1}c_2)v_2 + \cdots + (-c_1^{-1}c_n)v_n \in \text{span}(B \setminus \{v_1\})$$

Now take any $v \in V$. Since B spans, v is a linear combination of elements of B. Replacing every occurrence of v_1 in that combination by the displayed expression yields a linear combination of elements of $B \setminus \{v_1\}$, so $\text{span}(B \setminus \{v_1\}) = V$, contradicting minimality of B. Hence B is linearly independent and spanning, that is, a basis, and by lemma 13.1.4, V is a free k-module.

For the second claim, order the linearly independent subsets of S by inclusion. The union of a chain of linearly independent sets is again linearly independent, since a linear dependence involves only finitely many elements, which all lie in a single member of the chain. By Zorn's lemma, there is a maximal linearly independent subset $B \subseteq S$. We claim B spans V. Take any $s \in S$. If $s \in B$, then $s \in \text{span}(B)$. Otherwise, $B \cup \{s\}$ is linearly dependent by maximality, so there is a dependence $c_0s + c_1v_1 + \cdots + c_nv_n = 0$ with $v_i \in B$ and not all coefficients zero, and $c_0 \neq 0$ since B itself is linearly independent. As c_0 is a unit, $s \in \text{span}(B)$. Hence $S \subseteq \text{span}(B)$, and since every $v \in V$ is a linear combination of elements of S, each of which is a linear combination of elements of B, we get $\text{span}(B) = V$. Finally, B is minimally spanning: if $\text{span}(B \setminus \{b\}) = V$ for some $b \in B$, then $b = c_1v_1 + \cdots + c_nv_n$ with $v_i \in B \setminus \{b\}$, and $b - c_1v_1 - \cdots - c_nv_n = 0$ is a linear dependence in B, contradicting the linear independence of B. Hence $B \subseteq S$ is a minimal spanning set.

Corollary 13.1.10: Subspaces Of Vector-Spaces Are Free

Let V be a vector-space and $W \subseteq V$ be a subspace. Then W is free

***Proof*:**
Let $W \subseteq V$ be a subspace. Since W is a k-module, by theorem 13.1.8, it is free.

Note that this is not true for arbitrary free-modules: there exists submodule of free modules which are *not* free[3]. In fact, since any ring R is a module over itself, then it is easy to find counter-examples. For instance, the ideal (x, y) of $k[x, y]$ is not free: two elements f, g are never linearly independent, since $g \cdot f - f \cdot g = 0$, so a basis would have one element, and (x, y) is not principal, since a generator would divide both x and y and so be a nonzero constant, while every element of (x, y) has constant term 0. Another simpler example is as follows

Example 13.1.11: Non Free Submodule
Take $\mathbb{Z}_6$ over itself, and consider the ideal (2). Show that it is not free[a].

[a]Hint: it's torsion

In later chapters, we shall see that submodules of free modules over slightly more general rings, namely PIDs, will still be free[4], however this generlization is limited in scope and does not further generlize over more general rings, as we shall discuss later.

The key in theorem 13.1.8 is that every element in a field is a unit. Commutativity was not important. Thus, this same proof can work for division rings more generally[5]. Over a general ring, minimal generating sets can have different sizes: $\{1\}$ and $\{2, 3\}$ both generate $\mathbb{Z}$ as a $\mathbb{Z}$-module, and no proper subset of either does. We thus ask the question of when do we get bases with different cardinalites, or if there are conditions for which different basis always have the same cardinality. For vector-spaces, every basis *always* has the same cardinality! The following theorem is the key result for the proof:

Theorem 13.1.12: Replacement Theorem

Let $A = (a_1, a_2, \ldots, a_n)$ be a basis of V and let $b_1, b_2, ..., b_m$ be linearly independent vectors in V. Then $m \leq n$, and after renumbering the a_i, for each $k \in \{0, 1, ..., m\}$ the family $(b_1, b_2, ..., b_k, a_{k+1}, a_{k+2}, ..., a_n)$ is a basis of V. In other words, the elements $b_1, b_2, ..., b_m$ can be used to successively replace elements of the basis A, and the result is still a basis.

***Proof*:**
We proceed by induction on k. For $k = 0$ the family is A. Let $0 \leq k < m$, and suppose that $(b_1, \ldots, b_k, a_{k+1}, \ldots, a_n)$ is a basis of V. It spans V, so

$$b_{k+1} = \beta_1 b_1 + \cdots + \beta_k b_k + \alpha_{k+1} a_{k+1} + \cdots + \alpha_n a_n \tag{13.2}$$

for some scalars. Not all the α_i are zero, since otherwise $b_{k+1} - \beta_1 b_1 - \cdots - \beta_k b_k = 0$ would contradict the linear independence of $b_1, \ldots, b_{k+1}$. In particular $k < n$, and after renumbering

[3]Though it is true for free groups, see appendix B
[4]see theorem 14.1.6
[5]And will be used in [17, Semisimple Rings and Modules]

$a_{k+1}, \dots, a_n$ we may assume $\alpha_{k+1} \neq 0$. Then

$$a_{k+1} = \alpha_{k+1}^{-1} \left(b_{k+1} - \beta_1 b_1 - \cdots - \beta_k b_k - \alpha_{k+2} a_{k+2} - \cdots - \alpha_n a_n \right)$$

lies in the span of $b_1, \dots, b_{k+1}, a_{k+2}, \dots, a_n$, so this family spans every element of the old basis, hence V. It is linearly independent: suppose

$$\gamma_1 b_1 + \cdots + \gamma_{k+1} b_{k+1} + \delta_{k+2} a_{k+2} + \cdots + \delta_n a_n = 0$$

Substituting equation (13.2) for b_{k+1} gives a relation in the old basis, in which the coefficient of a_{k+1} is $\gamma_{k+1}\alpha_{k+1}$, the coefficient of b_i is $\gamma_i + \gamma_{k+1}\beta_i$ for $i \leq k$, and the coefficient of a_i is $\delta_i + \gamma_{k+1}\alpha_i$ for $i \geq k+2$. All of them vanish. As $\alpha_{k+1} \neq 0$, $\gamma_{k+1} = 0$, and then every γ_i and every δ_i is 0. So $(b_1, \dots, b_{k+1}, a_{k+2}, \dots, a_n)$ is a basis of V. Since $k < n$ for every $k < m$, also $m \leq n$, completing the proof.

The proof uses only that nonzero scalars are invertible, so it applies verbatim to modules over a division ring, with the scalars acting on the left. The Replacement Theorem relates the sizes of linearly independent sets and spanning sets:

Corollary 13.1.13: Steinitz exchange lemma

1. Suppose V has a finite basis with n elements. Any set of linearly independent vectors has at most n elements. Any spanning set has at least n elements
2. If V has some finite basis then any two bases of V have the same cardinality

***Proof*:**

1. Let A be a basis with n elements. A linearly independent set S has at most n elements: otherwise it contains $n+1$ linearly independent vectors, which theorem 13.1.12 forbids. A spanning set S contains a basis B (proposition 13.1.9), which has at most n elements by the previous sentence, so B is finite. Applying theorem 13.1.12 to the basis B and the linearly independent vectors of A gives $n \leq |B| \leq |S|$.
2. Every basis of V is linearly independent and spanning, so by (1) it has exactly n elements, where n is the size of the given finite basis, completing the proof.

A finitely generated vector space has a finite basis, contained in any finite spanning set (proposition 13.1.9), and by corollary 13.1.13(2) all of its bases have the same number of elements, so the following is well defined:

Definition 13.1.14: Dimension

Let V be a vector space over F. Then the cardinality of the basis is called the *dimension* of V, and is denoted $\dim_F(V)$ or $\dim(V)$ if context permits. If V is not finitely generated, then we say that V is infinite dimensional, and we write $\dim V = \infty$.

As we saw in example 13.1.5, a singleton set with a single element which has torsion is always linearly dependent. In an integral domain, we have some more power:

Corollary 13.1.15: Module Linearity

Let R be an integral domain and let M be a free R-module of rank $n < \infty$. Then any $n+1$ elements of M are R-linearly dependent, i.e., for any $y_1, y_2, ..., y_{n+1} \in M$, there are elements $r_1, r_2 ..., r_{n+1} \in R$ not all nonzero such that

$$r_1 y_1 + r_2 y_2 + \cdots + r_{n+1} y_{n+1} = 0$$

***Proof*:**
There is a trick similar to Gauss's lemma. We can embed R in its quotient field (since R is an integral domain, this is indeed an embedding), say k is its quotient field. Since M is a free module,

$$M \cong \underbrace{R \oplus R \oplus \cdots R \oplus R}_{n \text{ times}}$$

and so

$$M \subseteq \underbrace{k \oplus k \oplus \cdots \oplus k}_{n \text{ times}}$$

Since the latter is an n dimensional vector space over F, so any $n+1$ dimensional elements of M are F-linearly dependent. By clearing the denominators of the scalars (ex. multiplying through by the product of all the denominaotrs), we get an R-linear dependence realtion among $n+1$ elements of M.

An important consequence of the existence of a basis is that linear transformation $f : V \to W$ can be defined by first determining where a basis of V maps to, and then extending linearly (that is, if $v = \sum_{\beta_i \in \mathcal{B}} b_i \beta_{\in} W$, then $f(v) = f\left(\sum_{\beta_i \in \mathcal{B}} b_i \beta_i\right) = \sum_{\beta_i \in \mathcal{B}} b_i f(\beta_i)$). Note however that the choice basis is important, we shall get back to this in section 13.2, and that vector spaces may have many different bases (of the same cardinality by the replacement theorem):

Example 13.1.16: Different Basis
As we have defined in example 12.3.2 the set

$$e_i = (\underbrace{0, ..., 0}_{i-1 \text{ times}}, 1, \underbrace{0, ..., 0}_{k-i \text{ times}}) \qquad 1 \leq i \leq n$$

is a basis for $\mathbb{Q}^n$. There are countably many different basis for $\mathbb{Q}^n$. Can you think of a few? If you know about inner product spaces, can you think of an orthonormal [hamel][a] basis?

[a] The term orthonormal basis is usually reserved for *countable* linear combination for *Hilbert space*. Since the linear combination is not finite, it requires some analysis to define. This would be covered in a text on *Functional Analysis*. If you know about Hilbert spaces, then taking $\mathbb{C}[x]$ as the closure of $\mathbb{C}[x]$, is there an orthonormal basis for it?

13.1.1 Quotient Spaces and Subspaces

Now that we have established the struture of a vector-space, we can some questions about subspaces the structure of quotient spaces, with particular interest in finite vector-spaces to be able to take advantage of counting arguments. We already saw that subspace of vector spaces are free k-module.

Proposition 13.1.17: Dimension Of Quotient Space

Let V be a finite dimensional vector space over k and let $W \subseteq V$ be a subspace. Then:

1. W and the quotient space V/W are finite dimensional, and
$$\dim V = \dim W + \dim V/W$$
2. $\dim W \leq \dim V$, with equality only if $W = V$
3. if $V_1, \ldots, V_r$ are finite dimensional vector spaces over k, then $V_1 \times \cdots \times V_r$ is finite dimensional and $\dim(V_1 \times \cdots \times V_r) = \dim V_1 + \cdots + \dim V_r$

***Proof*:**
Let $n = \dim V$. The quotient module V/W is a vector space over k, and we write $\overline{v} = v + W$ for the class of $v \in V$.

1. The subspace W has a basis B (theorem 13.1.8). It is a linearly independent subset of V, so it has at most n elements (corollary 13.1.13(1)), say $B = (w_1, \ldots, w_m)$. So W is finite dimensional, and $\dim W = m$ by corollary 13.1.13(2). By theorem 13.1.8, B is contained in a basis of V, which has n elements by corollary 13.1.13(2), so this basis can be written $(w_1, \ldots, w_m, u_1, \ldots, u_{n-m})$. The classes $\overline{u}_1, \ldots, \overline{u}_{n-m}$ span V/W: every $v \in V$ is $\sum_i a_i w_i + \sum_j b_j u_j$ for some scalars, and $\overline{w}_i = 0$ since $w_i \in W$, so $\overline{v} = \sum_j b_j \overline{u}_j$. They are linearly independent: if $\sum_j b_j \overline{u}_j = 0$, then $\sum_j b_j u_j \in W$, say $\sum_j b_j u_j = \sum_i a_i w_i$, and the relation $\sum_i a_i w_i - \sum_j b_j u_j = 0$ in the basis of V forces every $b_j = 0$. So V/W has a basis with $n - m$ elements, and
$$\dim V/W = n - m = \dim V - \dim W$$
2. By (1), $\dim W = m \leq n$. If $m = n$, the basis of V chosen in the proof of (1) is $(w_1, \ldots, w_n)$, whose elements lie in W, so $V = \operatorname{span}(w_1, \ldots, w_n) \subseteq W$ and $W = V$.
3. For each i let $\iota_i \colon V_i \to V_1 \times \cdots \times V_r$ send v to the tuple with v in position i and 0 elsewhere, and choose a basis B_i of V_i, which has $\dim V_i$ elements. The vectors $\iota_i(b)$, for $1 \leq i \leq r$ and $b \in B_i$, span the product, since $(v_1, \ldots, v_r) = \sum_i \iota_i(v_i)$ and each v_i is a linear combination of elements of B_i. They are linearly independent: if $\sum_i \sum_{b \in B_i} c_{i,b}\, \iota_i(b) = 0$, then the ith component gives $\sum_{b \in B_i} c_{i,b} b = 0$ in V_i, so every $c_{i,b} = 0$. Hence they form a basis of the product with $\dim V_1 + \cdots + \dim V_r$ elements, completing the proof.

Naturally, since we are working with quotients, we have an associated linear transformation The following is the "homomorphism-equivalent" of the previous proposition:

Corollary 13.1.18: Rank-Nullity Theorem

Let $\varphi : V \to W$ be a linear transformation of vector spaces over k with $\dim(V) < \infty$. Then $\ker \varphi$ is a subspace of V, $\varphi(V)$ is a finite dimensional subspace of W, and
$$\dim_k(V) = \dim_k(\ker(\varphi)) + \dim_k(\varphi(V))$$

***Proof*:**
By theorem 12.2.10(1), applied to the k-module homomorphism φ, the kernel $\ker\varphi$ is a subspace of V and there is an isomorphism $\psi\colon V/\ker\varphi \to \varphi(V)$. The image is a subspace of W, since it contains $0=\varphi(0)$ and $\varphi(v)+c\varphi(v')=\varphi(v+cv')$ for $v,v'\in V$ and $c\in k$. By proposition 13.1.17(1), $\ker\varphi$ and $V/\ker\varphi$ are finite dimensional and

$$\dim V = \dim\ker\varphi + \dim V/\ker\varphi$$

Let $(\overline{v}_1,\dots,\overline{v}_r)$ be a basis of $V/\ker\varphi$. The family $(\psi(\overline{v}_1),\dots,\psi(\overline{v}_r))$ spans $\varphi(V)$, since ψ is linear and surjective. It is linearly independent, since $\sum_i c_i\psi(\overline{v}_i)=\psi(\sum_i c_i\overline{v}_i)=0$ forces $\sum_i c_i\overline{v}_i=0$ by injectivity of ψ, hence every $c_i=0$. So $\varphi(V)$ is finite dimensional with $\dim\varphi(V)=r=\dim V/\ker\varphi$, and $\dim V=\dim\ker\varphi+\dim\varphi(V)$, completing the proof.

Since the dimensions are of the subspaces are invariant to basis, we can give a name to the dimension of the image and null space of a linear transformation

Definition 13.1.19: Rank And Nullity

If $\varphi: V\to W$ is a linear transformation of vector spaces over k, $\ker(\varphi)$ is called the *null space* of φ ,and the dimension of $\ker(\varphi)$ is called the *nullity* of φ. The dimension of $\varphi(V)$ is called the *rank* of φ. If $\ker(\varphi)=0$, the transformatoin is said to be *nonsingular*.

The reasoning behind the name *nonsingular* is due to the fact that very often in (differentil) geometry, we associated to shapes a vector-space at each point (such an object is called a *vector bundle*) and if the curve "drops rank" this would correspond to a particular function having $\dim_k(\ker(\varphi))>0$[6]. For example, the curve $x^3=y^2$ "drops rank" at $(0,0)$.

Corollary 13.1.20: Properties of [finite dim.] Linear Transformations

Let $\varphi: V\to W$ be a linear transformation of vector spaces of the same finite dimension n. Then the following are equivalent

1. φ is an isomorphism
2. φ is injective, i.e., $\ker(\varphi)=0$
3. φ is surjective, i.e. $\varphi(V)=W$
4. φ sends some basis $(v_1,\dots,v_n)$ of V to a basis $(\varphi(v_1),\dots,\varphi(v_n))$ of W
5. φ sends every basis of V to a basis of W

Notice that this proof is saying that if $f: V\to W$ is a k-homomorphism where $\dim_k(V)=\dim_k(W)=n$, then given f is injective, it is surejctive. This fails to be the case even for maps between free moduels over PIDs, for example $f:\mathbb{Z}\to\mathbb{Z}$ mapping $z\mapsto 2z$ is an injective $\mathbb{Z}$-module homomorphism between free $\mathbb{Z}$-modules, but is certainly not surjective! This also fails for infinite dimensional vectors spaces as will be demonsarted by a counter-example after the proof. Hence, the fact that it's sufficient to check injectivity shows how linear maps between finite vector-spaces are akin to maps between

[6] In particular, if T_pf is the tangent map at p, then T_pf would drop rank

finite sets[7].

***Proof*:**
The map φ is injective exactly when $\ker(\varphi) = 0$, since $\varphi(v) = \varphi(v')$ means $v - v' \in \ker(\varphi)$. By corollary 13.1.18, $n = \dim \ker(\varphi) + \dim \varphi(V)$.

(1)⇒(5) Let $(v_1, \dots, v_n)$ be a basis of V. Every $w \in W$ is $\varphi(v)$ for some $v = \sum_i c_i v_i$, so $w = \sum_i c_i \varphi(v_i)$, and $(\varphi(v_1), \dots, \varphi(v_n))$ spans W. If $\sum_i c_i \varphi(v_i) = 0$, then $\varphi(\sum_i c_i v_i) = 0$, so $\sum_i c_i v_i = 0$ by injectivity and every $c_i = 0$. Hence $(\varphi(v_1), \dots, \varphi(v_n))$ is a basis of W.

(5)⇒(4) The space V has a basis with n elements, since $\dim V = n$.

(4)⇒(3) If $(\varphi(v_1), \dots, \varphi(v_n))$ is a basis of W, then every $w \in W$ is $\sum_i c_i \varphi(v_i) = \varphi(\sum_i c_i v_i)$.

(3)⇒(2) If $\varphi(V) = W$, then $\dim \ker(\varphi) = n - n = 0$. As $\{0\} \subseteq \ker(\varphi)$ also has dimension 0, proposition 13.1.17(2) gives $\ker(\varphi) = \{0\}$.

(2)⇒(1) If $\ker(\varphi) = 0$, then $\dim \varphi(V) = n = \dim W$, so $\varphi(V) = W$ by proposition 13.1.17(2). Thus φ is a bijective linear map, an isomorphism by lemma 12.2.3, completing the proof.

Note that finiteness of dimension is important in the previous corrollaries: these need not hold in infinite dimension

Example 13.1.21: Infinite Dimensinoal Linear Transformations
Take $V = \ell^2$, the set of all infinite sequences of real numbers satisfying the property:

$$\left(\sum_{i \in \mathbb{N}}^{\infty} |x_i|^2 \right)^{1/2} < \infty$$

Then define $f : V \to V$ via:

$$(x_1, x_2, ..., x_n, ...) \mapsto (x_1, \frac{x_2}{2}, ..., \frac{x_n}{n})$$

Then this function is injective, but not surjective. Even more is true. Here V and f belong to some of the nicest infinite dimensional spaces and functions (V is a Hilbert space and f is a bounded self-adjoint linear operator), and hence this property is not easily rectified.

We now take a look at subspaces. We have already established that all subspaces of a vector-spaces are free. Another importnant property is that every subspace always has a *compliment*, that is if $W \subseteq V$, then there exists a $U \subseteq V$ such that $U \cap W = \{0\}$ and $W + U = V$. This is certainly *not* true for general R-modules. In the $\mathbb{Z}$-module $M = \mathbb{Z}/4\mathbb{Z}$, take $W = \{0, 2\}$. A subgroup U with $W + U = M$ contains an element u with $1 = u + w$ for some $w \in W$, so $u \in \{1, 3\}$, and u generates M. Then $U = M$ and $U \cap W = W \neq \{0\}$, so W has no complement.[8] However, we may always find such a compliment for subspaces of vector-spaces, given the space is finite dimensional:

[7] In fact, this parallels is runs more deeply than it may at first seem, see a text on K-theory for some interesting elaboration on this notion

[8] In **Grp**, the corresponding question is which short exact sequences split. See section 17.2 for a generalized notion of splitting and proposition 5.3.9 for groups.

Proposition 13.1.22: Existence Of Compliment

Let V be a finite dimensional vector space and $W \subseteq V$ be a subspace. Then there exists a subspace $U \subseteq V$ such that

1. $W \cap U = \{0\}$
2. $W + U = V$

If these two conditions are satisfied, we will write $V = W \oplus U$ and call V the direct sum of W and U (as in definition 12.3.3).

***Proof*:**
Let $n = \dim V$. By proposition 13.1.17(1), W has a finite basis $(w_1, \dots, w_m)$. By theorem 13.1.8 it is contained in a basis of V, which has n elements (corollary 13.1.13(2)), so this basis can be written $(w_1, \dots, w_m, u_1, \dots, u_{n-m})$. Let $U = \operatorname{span}(u_1, \dots, u_{n-m})$.

1. If $v \in W \cap U$, then $v = \sum_i a_i w_i = \sum_j b_j u_j$ for some scalars, and the relation $\sum_i a_i w_i - \sum_j b_j u_j = 0$ in the basis of V forces all coefficients to vanish, so $v = 0$.
2. Every $v \in V$ is $\sum_i a_i w_i + \sum_j b_j u_j$ for some scalars, with $\sum_i a_i w_i \in W$ and $\sum_j b_j u_j \in U$, so $W + U = V$, completing the proof.

Finite dimension only keeps the bases finite: by theorem 13.1.8, a basis of W extends to a basis of V in any dimension, and the same argument produces a complement of W in every vector space V.

Corollary 13.1.23: Finite Generation of Subspace

Let $W \subseteq V$ be a subspace of a finite dimensional vector space. Then W is finitely generated

***Proof*:**
By proposition 13.1.17(1), W is finite dimensional, so it has a finite basis, and a basis generates W (definition 13.1.2), completing the proof.

If $V = W \oplus U$, every $v \in V$ is $w + u$ for unique $w \in W$ and $u \in U$: if $w + u = w' + u'$, then $w - w' = u' - u \in W \cap U = \{0\}$. So $\pi_W(w + u) = w$ defines a map $\pi_W : V \to V$, which is linear, with image W and kernel U. Such maps are important enough to be given a name:

Definition 13.1.24: Projection Map

Let $\pi : V \to V$ be a linear transformation $U \subseteq V$. Given $V = U \oplus W$, then the *projection map* (with respect to U) is the map $\pi_U : V \to V$ such that $\pi(v) = \pi(u + w) = u$.

Notice $\pi \circ \pi = \pi$, that is, applying the map twice gives back the same result: $\pi(\pi(v)) = \pi(v)$. This in fact characterizes such a map:

Proposition 13.1.25: Characterizing Projection Maps

Let π be linear operator over a [not necessarily finite dimensional] vector space such that $\pi \circ \pi = \pi$. Then π is a projection map.

***Proof*:**
Let $U = \text{im}(\pi)$ and $W = \text{im}(I - \pi)$. Then for any $v \in V$:

$$\pi(v) + v - \pi(v) = v$$

hence $U + W = V$. If $x \in U \cap W$. Since $x \in \text{im}(\pi)$, $x = \pi(y)$ for some $y \in V$, and so:

$$\pi(x) = \pi(\pi(y)) = \pi(y) = x$$

hence, $\pi(x) = x$. Then:

$$\begin{aligned}
x &= \pi(x) && x \in U \\
&= (I - \pi)(x) && x \in W \\
&= \pi((I - \pi)(x)) && (I - \pi)(x) = x \in U \\
&= \pi(x - \pi(x)) \\
&= \pi(x - x) && \pi(x) = x \\
&= \pi(0) \\
&= 0
\end{aligned}$$

Hence $x = 0$, and so $U \cap W = \{0\}$. With the above, we get that

$$V = U \oplus W = \ker(\pi) \oplus \text{im}(\pi)$$

as we sought to show.

Notice how the above proof did not rely on any structure of V (notably, it was not important that V had a basis). Thus, if $\varphi : M \to M$ is any R-module homomorphism such that $\varphi \circ \varphi = \varphi$, then

$$M = \ker(\varphi) \oplus \text{im}(\varphi)$$

For this reason, it is much more common to define projection maps as maps that satisfy the property of $\varphi^2 = \varphi$ (i.e., they are *idempotent*).

Exercise 13.1.1

1. Let $a \in \mathbb{R}$ and let $\mathbb{R}_n[x]$ be the $\mathbb{R}$-vector space of real polynomials of degree at most n, together with 0. Show that $1, (x-a), (x-a)^2, \ldots, (x-a)^n$ is a basis of $\mathbb{R}_n[x]$.
2. Let R be a commutative ring that is *not* a field. Show that not all R-modules are free, i.e. find a counter-example [Hint: Look back at the examples given in section 12.1]
3. (More challenging) Let R be any ring (not necessarily commutative). Show that if any left (or right) R-module is free, then R is a divison ring.

4. Show that $(x, y) \subseteq \mathbb{C}[x, y]$ is not a free $\mathbb{C}[x, y]$-submodule.
5. Show that any uncountable set of $\mathbb{C}[x]$ is linearly dependent.
6. Let $R = k[x_1, ..., x_n]$ and $I = (x_1, ..., x_n)$, and let $m \geq 0$. Show that the ideal I^m is the k-subspace of R spanned by the monomials $x_1^{r_1} \cdots x_n^{r_n}$ of total degree $r_1 + \cdots + r_n \geq m$, and that the classes of the monomials of total degree exactly m form a k-basis of the quotient I^m / I^{m+1}.

13.2 Linear Transformations and Matrix Representation

As mentioned at the beginning of the chapter, k-module homomorphisms are called linear transformation in the theory of vector-spaces. In this section, we will show how every linear transformation between finite dimensional vector-spaces can be represented as a matrix. Since all vector-spaces are free with an invariant basis number, when defining a linear transformation we only need to be concerned with where the basis get's mapped to. If $\varphi : V \to W$, where V and W are finite dimensional vector-spaces over k, and $\varphi \in \mathrm{Hom}_k(V, W)$. Let $\beta_V = \{v_1, v_2, ..., v_n\}$ and $\beta_W = \{w_1, w_2, ..., w_m\}$ be (ordered) bases for V and W respectively. Then we are only concerned where the v_i's map to, that is, what $\varphi(v_j) = w_j$ is. Since they map into the vector-space W, $w_j = \sum_{i=1}^{m} b_{ij} w_i$, $b_{ij} \in k$,

$$\varphi(v_j) = \sum_{i=1}^{m} b_{ij} w_i$$

Let $M_{\beta_V}^{\beta_W}(\varphi)$ be the $m \times n$ matrix whose ij entry is b_{ij} (i.e. the coefficients of w_i given v_j). This matrix is given a name:

Definition 13.2.1: Matrix Representation

Let $\varphi : V \to W$ be a linear transformation between two finite dimensional vector spaces with dimension n and M and basis β_V and β_W respecitvely. Then the *matrix representation* $M_{\beta_V}^{\beta_W}$ of φ (also written as $M_{\beta_V}^{\beta_W}(\varphi)$) is

$$\begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{pmatrix}$$

where each entry ij is the coefficient of the ith basis element of β_W in the linear combination of where the element v_j maps.
If a matrix A is a matrix representation of a linear transformation φ, we say that A *represents* φ *with respect to the basis* β_V, β_W. Similarly, φ would be the linear transformation represented by A with respect to the bases β_V, β_W.

With the matrix representation of φ, we can re-write $\varphi(v) = w$ as

$$\begin{pmatrix} c_{11} & c_{12} & \cdots & c_{1n} \\ c_{21} & c_{22} & \cdots & c_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ c_{m1} & c_{m2} & \cdots & c_{mn} \end{pmatrix} \begin{pmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{pmatrix} = \begin{pmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{pmatrix} \tag{13.3}$$

where $v = \sum_{i=1}^{n} a_i v_i$ and $\varphi(v) = \sum_{i=1}^{m} b_i w_i$. Sometimes, the following notation is used as a shorthand for (13.3): $[\varphi(v)]_{\beta_W} = M_{\beta_V}^{\beta_W} [v]_{\beta_V}$.

Example 13.2.2: Matrix Representation

1. Let $V = \mathbb{R}^3$, $W = \mathbb{R}^2$, and take $\beta_V = \{(1,0,0),(0,1,0),(0,0,1)\}$ and $\beta_W = \{(1,0),(0,1)\}$ (i.e., the standard basis for $\mathbb{R}^3$ and $\mathbb{R}^2$). Let

   $$\varphi(x,y,z) = (4x + 2z, x - y + 5z)$$

   Then $\varphi(1,0,0) = (4,1)$, $\varphi(0,1,0) = (0,-1)$, and $\varphi(0,0,1) = (2,5)$. Thus, the matrix representation of φ with respect to β_V and β_W is

   $$\begin{pmatrix} 4 & 0 & 2 \\ 1 & -1 & 5 \end{pmatrix}$$

Theorem 13.2.3: Matrix Representation Correspondence

Let V and W be a finite dimensional vector space over F with dimension n and m, as well as basis β_V and β_W respectively. Then the map $\Psi : \text{Hom}_F(V,W) \to M_{mn}(F)$, where $\varphi \mapsto M_{\beta_V}^{\beta_W}(\varphi)$ is a [vector-space] isomorphism.

Note that the operation on $\text{Hom}_F(V,W)$ if point-wise function addition and $M_{mn}(F)$ is matrix addition.

***Proof*:**

To show Ψ is F-linear, it comes down to the fact that every $\varphi \in \text{Hom}_F(V,W)$ is F-linear. Since every column of $M_{\beta_V}^{\beta_W}$ represents $\varphi(v_i)$, and $(\varphi + \psi)(v) = \varphi(v) + \psi(v)$, then we get that the map Ψ is also F-linear.

The map Ψ is surjective, since given a matrix $A \in M_{mn}(F)$, the map φ that maps to it can be defined by reversing the process that was done in equation (13.3). This map is injective since if $A = B$, then $A = M_{\beta_V}^{\beta_W}(\varphi) = M_{\beta_V}^{\beta_W}(\psi) = B$. Since a basis uniquely defines a linear transformation and both representations are with respect to the same bases, it must be that $\varphi = \psi$, proving injectivity.

Thus, Ψ is an isomorphism, as we sought to show.

13.2.4 Note Different choices of bases produces different isomorphisms. There is no "natural choice" in the same way that there is no natural choice for a basis for a vector-space

Corollary 13.2.5: Dimension Of $\mathrm{Hom}_F(V,W)$

The dimension of $\mathrm{Hom}_F(V,W)$ is $\dim(V)\cdot\dim(W)$

***Proof*:**
Since $\mathrm{Hom}_F(V,W)$ is isomorphic to $M_{mn}(F)$, and $\dim(M_{mn}(F)) = mn$, then $\dim(\mathrm{Hom}_F(V,W)) = mn = \dim(V)\cdot\dim(W)$

We have shown that function addition is compatible with matrix addition. When it comes to function composition, the story becomes a little trickier, since we need the co-domain of one function to line-up with the domain of the next. Thus, to that end, let U, V, W be finite dimensional vector-spaces with dimension k, n, m and bases $\beta_U = \{u_1, u_2, ..., u_k\}$, $\beta_V = \{v_1, v_2, ..., v_n\}$, and $\beta_W = \{w_1, w_2, ..., w_m\}$ respectively. Let $\varphi : U \to V$ and $\psi : V \to W$ be two linear transformations. Their composition $\psi \circ \varphi$ is a linear fransformation from U to W, and so there exists a representation $M_{\beta_U}^{\beta_W}(\psi \circ \varphi)$ given by

$$(\psi \circ \varphi)(u_j) = \sum_{i=1}^{m} c_{ij} w_i \qquad c_{ij} \in F$$

Since φ and ψ are linear transfomation, they too have a matrix representation:

$$\varphi(u_j) = \sum_{p=1}^{n} a_{pj} v_p \qquad \psi(v_p) = \sum_{i=1}^{m} b_{ip} w_i$$

from this, we can find c_{ij} in terms of the coefficients a_{pj} and b_{ip}:

$$\begin{aligned}
(\psi \circ \varphi)(u_j) &= \psi\left(\sum_{p=1}^{n} a_{pj} v_p\right) \\
&= \sum_{p=1}^{n} a_{pj} \psi(v_p) \\
&= \sum_{p=1}^{n} a_{pj} \sum_{i=1}^{m} b_{ip} w_i \\
&= \sum_{p=1}^{n} \sum_{i=1}^{m} a_{pj} b_{ip} w_i \\
&= \sum_{i=1}^{m} \sum_{p=1}^{n} a_{pj} b_{ip} w_i
\end{aligned}$$

Thus, if we let $c_{ij} = \sum_{p=1}^{n} a_{pj} b_{ip}$, we get

$$= \sum_{i=1}^{m} c_{ij} w_i$$

Take a moment to look over this and think about how this would be represented as multiplying the matrix $M_{\beta_U}^{\beta} V(\varphi)$ and $M_{\beta_V}^{\beta_W}(\psi)$. In fact, the coefficents of $M_{\beta_U}^{\beta_W}(\psi \circ \varphi)$ are exactly what you get by multiplying the other two matrices! This is summed up in the following theorem:

Theorem 13.2.6: Matrix Representation And Composition

Let U, V, W be finite dimensional vector-spaces with bases β_U, β_V, β_W, of size k, m, n respectively. Let $\varphi : U \to V$ and $\psi : V \to W$ be two linear transformations. Then

$$M_{\beta_U}^{\beta_W}(\psi \circ \varphi) = M_{\beta_V}^{\beta_W}(\psi) M_{\beta_U}^{\beta_V}(\varphi)$$

Another way of seeing this theorem is by saying the following diagram commutes:

$$\begin{array}{ccc} M_{mp}(F) \times M_{mp}(F) & \longrightarrow & M_{mn}(F) \\ \downarrow & & \downarrow \\ \mathrm{Hom}_F(F^p, F^m) \times \mathrm{Hom}_F(R^p, R^m) & \longrightarrow & \mathrm{Hom}_F(F^n, F^m) \end{array}$$

Corollary 13.2.7: Matrix Multiplication Properties

Let $M_{mn}(F)$ be the set of $m \times n$ matrices over F. Then given appropriate matrices, matrix multiplication is associative and distributive

Note how this is different from the associativity and distributivity of $GL_n(F)$. Namely, all the matrices of $GL_n(F)$ are square matrices. We have not yet proven that non-square matrices are associative and distributive, particularly because they do not always line up properly, and their result can pass from one set of matrices to another (for example $A \in M_{ab}(F)$, $B \in M_{bc}(F)$, $AB \in M_{ac}(F)$)

***Proof*:**
Distributivity follows by the same technique as associativity, which is proved here. Let A, B, C be matrices such that $(AB)C$ and $A(BC)$ are defined. By theorem 13.2.6, (AB) represents $S \circ T$ for appropriate linear transformations S, T. Multiplying on the right side by C, $(AB)C$, the resulting matrix represents $(S \circ T) \circ R$ for appropriate R. Similarly, $A(BC)$ is represented by $S \circ (T \circ R)$. Since the composition of functions is associative,

$$\Psi((AB)C) = (S \circ T) \circ R = S \circ (T \circ R) = \Psi(A(BC))$$

and since Ψ is injective, $(AB)C = A(BC)$.

Corollary 13.2.8: Matrix Representation Isomorphisms

1. If β is a basis of the n-dimensional vector space V, the map $\varphi \mapsto M_\beta^\beta(\varphi)$ is a ring isomorphism and a vector-space isomorphism of $\mathrm{Hom}_F(V, V)$ (i.e. $\mathrm{End}(V)$) onto $M_n(F)$:

$$\mathrm{End}(V) \cong_{\mathbf{Ring}} M_n(F) \qquad \mathrm{End}(V) \cong_{F\text{-}\mathbf{Vect}} M_n(F)$$

2. If we restrict to automorphisms, then $\mathrm{Aut}(V) \cong GL_n(F)$ as groups[a]. In particular

$$\mathrm{Aut}_{\mathbf{Vect}}(V) \cong_{\mathbf{Grp}} GL_n(F)$$

[a] $\mathrm{Aut}(V)$ is not a ring

***Proof*:**
In both cases, we have already shown that it is isomorphic as F-vector-spaces, so we will show they are isomorphic as rings

(1) Using corollary 13.2.7, we see that $M_n(F)$ is a ring (since multiplying two $n \times n$ matrices get's us an $n \times n$ matrix). We have shown this map is bijective in theorem 13.2.3. Theorem 13.2.6 shows that this map also preserves the ring structure of $\text{End}(V)$, showing that the map is a ring-isomorphism.

(2) The units of $\text{End}(V)$ are the automorphisms of V, and the units of $M_n(F)$ form $GL_n(F)$. A ring isomorphism maps units bijectively onto units, since it preserves products and the identity, and so does its inverse. So the isomorphism of (1) restricts to a bijection $\text{Aut}(V) \to GL_n(F)$ that preserves products, a group isomorphism.

Finally, we convert the powerful result of corollary 13.1.20 into for matrices, in particular we showed that if f is injective between two finite dimensional vector space of the same dimension, then it is surjective. This result is important enough that the matrix representation of such an f should be given a name

Definition 13.2.9: Nonsingular Matrix

An $m \times n$ matrix A is called *nonsingular* if $Ax = 0$ with $x \in F^n$ implies $x = 0$

The paragraph after definition 13.1.19 explains the name.

Proposition 13.2.10: Nonsingular Square Matrix

Let A be an $n \times n$ matrix. Then A is invertible if and only if A is nonsingular

***Proof*:**
If A is invertible and $Ax = 0$, then $x = A^{-1}(Ax) = A^{-1}0 = 0$, so A is nonsingular.

Conversely, let A be nonsingular. Then A represents some linear transformation $\varphi : V \to V$ for some finite dimensional vector-space V with respect to basis β and γ for the domain and codomain. Since $\varphi(x) = 0$ if and only if $x = 0$, then φ is injective, and so by corollary 13.1.20, φ is an isomorphism. Thus, there exists a linear transfomration φ^{-1} such that $\varphi^{-1} \circ \varphi = \text{id}$. By using theorem 13.2.3, let B be the representation of φ^{-1} with respect to γ and β. Using theorem 13.2.6 we have

$$AB = M_\beta^\gamma(\varphi)M_\gamma^\beta(\varphi^{-1}) = M_\gamma^\gamma(\varphi \circ \varphi^{-1}) = M_\gamma^\gamma(I) = I$$

thus $AB = I$. similarly $BA = I$. Thus, $B = A^{-1}$, as we sought to show.

Exercise 13.2.1

1. We know that in general, if R is a ring and $x \in R$ is not a unit, that does not imply that x is a zero-divisor (ex. take the integers). Show that for the matrix ring $M_n(k)$, if $A \in M_n(k)$ is not a unit, then A is a zero-divisor.

2. Show that a polynomial of degree at most n over a subfield of $\mathbb{C}$ is uniquely determined by its values at $n+1$ distinct points (You may freely use the fact that a polynomial of degree n can be broken down into a product of n linear terms).

3. (If you know some complex arithmetics) Let $f : \mathbb{C} \to \mathbb{C}$ be a $\mathbb{C}$-linear function. Recall that $\mathbb{C}$ is a 2 dimensional $\mathbb{R}$-vector space, and so f may be represented as a real matrix. Show that if $f(z) = \alpha z = (a + bi)z$, then the matrix representation of f in the basis $(1, i)$ will be of the form:
$$[f] = \begin{pmatrix} a & -b \\ b & a \end{pmatrix}$$

13.2.1 Change of Basis

While representing vectors in a vector space in terms of coordinates simplifies their representation, and linear transformations and their compositions can be seen as matrices and matrix multiplication, we are *always* dependent on a choice of basis (i.e. a choice of isomorphism $\varphi : k^n \to V$). As pointed out earlier, there is no "canonical" choice for a basis: there are many choices of bases which will produce isomorphic vector spaces[9]. Thus, in this section we will explore how to change from one basis to another to be able to compare representations. We will be working with vector-spaces, but we will only be using properties of free-modules meaning the theory is true for free-modules in general.

Let V be an F-vector space of dimension n. Let β_A and β_B be two basis for V, meaning we have the two isomorphissm
$$F^{\oplus\beta_A} \xrightarrow{\varphi} V \qquad F^{\oplus\beta_B} \xrightarrow{\psi} V$$

Then we will get the following isomorphism:
$$F^{\oplus\beta_A} \xrightarrow{\psi^{-1}\circ\varphi} F^{\oplus\beta_B}$$

Then by theorem 13.2.3, there is a matrix $N_{\beta_A}^{\beta_B}$ that corresponds to $(\psi^{-1} \circ \varphi)$. In particular, the jth column of $N_{\beta_A}^{\beta_B}$ is the image of the jth element of A viewed as a column vector in $f^{\oplus\beta_B}$.
If $\beta_A = (a_1, a_2, ..., a_n)$ and $\beta_B = (b_1, b_2, ..., b_n)$ then
$$(\psi^{-1} \circ \varphi)(a_j) = \sum_{i=1}^{n} n_{ij} b_i$$

We give this matrix a name:

Definition 13.2.11: Change Of Basis

Let V be a F-vector space of dimension n, and let β_A and β_B be two bases corresponding to the isomorphisms φ and ψ. Then the *change of basis matrix* is the matrix representation of $(\psi^{-1} \circ \varphi)$.
We can represent this matrix as $N_{\beta_A}^{\beta_B}$ or $N_{\beta_A}^{\beta_B}(V)$

[9]There are ways to make it "canonical" by augmenting the vector-space to have further structure which may naturally induce a basis (for example, an inner product). For such examples, see *Linear Algebra* [21].

Corollary 13.2.12: Properties Of Change Of Basis

Let β_A, β_B and β_C be bases of an n-dimensional vector space V. Then

1. $N_{\beta_A}^{\beta_B}$ is invertible and $(N_{\beta_A}^{\beta_B})^{-1} = N_{\beta_B}^{\beta_A}$
2. $N_{\beta_B}^{\beta_C} N_{\beta_A}^{\beta_B} = N_{\beta_A}^{\beta_C}$

***Proof*:**
For a basis β of V let $\varphi_\beta \colon F^n \to V$ be the isomorphism sending the jth standard basis vector e_j to the jth element of β. By definition 13.2.11, N_β^γ is the matrix of $\varphi_\gamma^{-1} \circ \varphi_\beta \colon F^n \to F^n$ with respect to the standard basis $e = (e_1, \ldots, e_n)$ in domain and codomain, and the matrix of the identity map in this basis is I_n, since $\mathrm{id}(e_j) = e_j$ (definition 13.2.1).

1. By theorem 13.2.6,

$$N_{\beta_A}^{\beta_B} N_{\beta_B}^{\beta_A} = M_e^e\left((\varphi_{\beta_B}^{-1} \circ \varphi_{\beta_A}) \circ (\varphi_{\beta_A}^{-1} \circ \varphi_{\beta_B})\right) = M_e^e(\mathrm{id}) = I_n$$

$$N_{\beta_B}^{\beta_A} N_{\beta_A}^{\beta_B} = M_e^e\left((\varphi_{\beta_A}^{-1} \circ \varphi_{\beta_B}) \circ (\varphi_{\beta_B}^{-1} \circ \varphi_{\beta_A})\right) = M_e^e(\mathrm{id}) = I_n$$

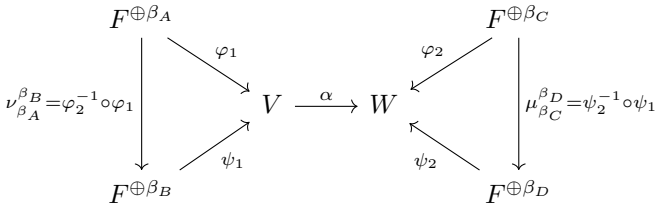
   Hence $N_{\beta_A}^{\beta_B}$ is invertible with inverse $N_{\beta_B}^{\beta_A}$.

2. By theorem 13.2.6,

$$N_{\beta_B}^{\beta_C} N_{\beta_A}^{\beta_B} = M_e^e\left((\varphi_{\beta_C}^{-1} \circ \varphi_{\beta_B}) \circ (\varphi_{\beta_B}^{-1} \circ \varphi_{\beta_A})\right) = M_e^e(\varphi_{\beta_C}^{-1} \circ \varphi_{\beta_A}) = N_{\beta_A}^{\beta_C}$$

   as claimed.

With the change of basis for a vector space taken care of, we know check how to perform a change of basis for a matrix representation of a linear transformation. The idea behind this comes down to the following visual:



Let N_A^B be the matrix representation of $\varphi_2^{-1} \circ \varphi_1$ and M_C^D be the matrix representation of $\psi_2^{-1} \circ \psi_1$. If we choose the basis β_A for V and β_C for W, then we have:

$$\varphi_2^{-1} \circ \alpha \circ \varphi_1 : F^{\oplus\beta_A} \to F^{\oplus\beta_C} \tag{13.4}$$

Let P be the matrix representation of $\varphi_2^{-1} \circ \alpha \circ \varphi_1$. If we choose the basis β_B for V and β_D for W, then we have:

$$\psi_2^{-1} \circ \alpha \circ \psi_1 : F^{\oplus\beta_B} \to F^{\oplus\beta_D} \tag{13.5}$$

now with some manipulation, we can transform equation (13.5) into equation (13.4) like so

$$\psi_2^{-1} \circ \alpha \circ \psi_1 = (\mu_{\beta_C}^{\beta_D} \circ \varphi_2^{-1}) \circ \alpha \circ (\varphi_1 \circ (\nu_{\beta_A}^{\beta_B})^{-1}) = \mu_{\beta_C}^{\beta_D} \circ (\varphi_2^{-1} \circ \alpha \circ \varphi_1) \circ (\nu_{\beta_A}^{\beta_B})^{-1}$$

And using the matrix correspondence, this is equivalent to

$$M_{\beta_C}^{\beta_D} P (N_{\beta_A}^{\beta_B})^{-1} = M_{\beta_C}^{\beta_D} P N_{\beta_B}^{\beta_A} = Q$$

This should be able to be read quite intuitively: $N_{\beta_B}^{\beta_A}$ converts our vector represented by a linear combination of vectors in β_B into linear combination of vectors in β_A, then P transforms the result and outputs it as a linear combination of vectors in β_C, and finally $M_{\beta_C}^{\beta_D}$ converst the result into a linear representation of vectors in β_D. W

We sum up these results into a proposition:

Proposition 13.2.13: Change Of Basis For Linear Transformation

Let $\varphi : V \to W$ be a linear transformation from two finite dimensional vector spaces, and let P be the matrix representation of φ with respect to bassi β_A and β_C for V and W respectively. Then the matrix representing linear transformation φ with respect to basis β_B and β_D is

$$M_{\beta_C}^{\beta_D} P N_{\beta_B}^{\beta_A}$$

with the notation given above.

More generally, if M and N are any invertible matrix, they represent a change of basis, so MPN would be a change of basis (given the appropriate dimensions).

Definition 13.2.14: Equivalence

Let R be a commutative ring and $P, Q \in M_{m,n}(R)$ be $m \times n$ matrices over R. Then P and Q are *equivalent* if there exist invertible matrices $M \in M_m(R)$ and $N \in M_n(R)$ such that

$$Q = MPN$$

Read through bases, this says that P and Q represent the same homomorphism between free R-modules of ranks n and m, with respect to two choices of bases of the domain and of the codomain.

Products of matrices over a commutative ring R are associative, $(AB)C = A(BC)$, since by distributivity and associativity in R the (i,j) entry of both sides is $\sum_{k,l} a_{ik} b_{kl} c_{lj}$. With this, equivalence of matrices is an equivalence relation on $M_{m,n}(R)$. It is reflexive, since $P = I_m P I_n$. It is symmetric, since $Q = MPN$ gives $P = M^{-1} Q N^{-1}$. It is transitive, since $Q = MPN$ and $S = M'QN'$ give $S = (M'M)P(NN')$, where $M'M$ and NN' are invertible with inverses $M^{-1}M'^{-1}$ and $N'^{-1}N^{-1}$.

For the reading through bases, let φ: $R^n \to R^m$ be $x \mapsto Px$, so that P is the matrix of φ in the standard bases, and let M and N be invertible. The columns $\beta_j = Ne_j$ of N form a basis of R^n: they span, since $x = N(N^{-1}x) = \sum_j (N^{-1}x)_j \beta_j$, and they are linearly independent, since $\sum_j c_j \beta_j = Nc = 0$ forces $c = N^{-1}Nc = 0$. Applied to the invertible matrix M^{-1}, the same two

computations show that its columns $\gamma_i = M^{-1}e_i$ form a basis of R^m, and

$$\varphi(\beta_j) = PNe_j = M^{-1}(MPN)e_j = \sum_{i=1}^{m}(MPN)_{ij}\gamma_i$$

so MPN is the matrix of φ with respect to the bases (β_j) and (γ_i). Conversely, let Q be the matrix of φ with respect to bases (β_j) of R^n and (γ_i) of R^m, and let N and C be the matrices whose columns are the β_j and the γ_i. Comparing $\varphi(\beta_j) = PNe_j$ with $\varphi(\beta_j) = \sum_i Q_{ij}\gamma_i = CQe_j$ gives $PN = CQ$. A matrix $B \in M_r(R)$ whose columns $b_1, \dots, b_r$ form a basis of R^r is invertible: writing $e_k = \sum_i b'_{ik}b_i$ defines $B' = (b'_{ik})$ with $BB' = I_r$, so $B(B'B - I_r) = 0$, and $B'B = I_r$ because the columns of B are linearly independent. Applied to N and to C, this shows that both are invertible, and $Q = C^{-1}PN$ is equivalent to P. Thus there is an entire equivalence class of matrices representing a linear transformation, where any representative of the equivalence class equates to a choice of basis for the domain and codomain. This brings up an interesting question: is there a particular representative (i.e. choices of bases for P, Q) that will make it "easier" to see what the linear transformation is doing? This is a very interesting question, and in the next section shall be explored to great success, in particular, all linear transformation can be represented in "reduced form" and which can be algorithmically found.

13.2.2 Similarity

Before proceeding, there is one more important concept that must be covered, namely the special case where $V = W$ is the same vector-space:

$$\mathrm{Hom}_R(V, V) = \mathrm{End}_R(V)$$

that is, we shall look at linear transformations from V to itself. Such endomorphisms are in fact extremely common. They can be thought of as representing the "change in state" of a system. As the reader may surmise, this means they have applications all other the physical science, however they shall amply show up in pure mathematics. In fact, chapter 14 is essentially the theoretical build-up for studying such functions.

In this subsection R is a commutative ring, and the main case is a field $R = k$.

We shall simply start by asserting that the vector-space V in $\mathrm{Hom}_R(V, V)$ shall be considered to be the exact same space, not two isomorphic copies of these spaces as generally is considered. In this situation, an element $\alpha \in \mathrm{Hom}_R(V, V) = \mathrm{End}_R(V)$ is usually called a *linear operator* to emphasize the fact that the domain and codomain are "identical". With this in mind, there are new questions that could be asked, such as if there exists $v \in V$ such that $\alpha(v) = \lambda v$ for some $\lambda \in R$, or if a subspace $W \subseteq V$ maps into itself.

Another important consequence of identifying the domain and codomain is that when choosing a basis, the *same* basis must be chosen for both of them (as α is considered to be mapping V to itself). This means that there is a more specific notion of equivalence as only one matrix is chosen:

Definition 13.2.15: Similarity

Let R be a commutative ring. Two matrices $A, B \in M_n(R)$ are *similar*, written $A \sim B$, if $B = PAP^{-1}$ for some $P \in \mathrm{GL}_n(R)$. Two endomorphisms α, β of an R-module M are *similar*, written $\alpha \sim \beta$, if $\beta = \varphi \circ \alpha \circ \varphi^{-1}$ for some automorphism φ of M.

The two notions match through matrix representations, and both are equivalence relations:

Lemma 13.2.16: Similarity Of Matrices And Operators

Let R be a commutative ring and let M be a free R-module with a basis $\beta = (b_1, \dots, b_n)$. For $\alpha \in \mathrm{End}_R(M)$ let $[\alpha]_\beta \in M_n(R)$ be its matrix, defined by $\alpha(b_j) = \sum_i ([\alpha]_\beta)_{ij} b_i$.

1. Similarity is an equivalence relation on $M_n(R)$ and on $\mathrm{End}_R(M)$.
2. $[\alpha \circ \alpha']_\beta = [\alpha]_\beta [\alpha']_\beta$ for $\alpha, \alpha' \in \mathrm{End}_R(M)$, and $[\mathrm{id}_M]_\beta = I_n$. In particular, if α is an automorphism, then $[\alpha]_\beta$ is invertible with inverse $[\alpha^{-1}]_\beta$.
3. For $\alpha, \alpha' \in \mathrm{End}_R(M)$, $\alpha \sim \alpha'$ if and only if $[\alpha]_\beta \sim [\alpha']_\beta$.
4. If γ is another basis of M with n elements, then $[\alpha]_\gamma \sim [\alpha]_\beta$. Conversely, every matrix similar to $[\alpha]_\beta$ is $[\alpha]_\gamma$ for some basis γ of M with n elements.

***Proof*:**
Matrix multiplication over R is associative (see the discussion after definition 13.2.14), and an endomorphism of M is determined by its values on the basis β, hence by its matrix, while every matrix is the matrix of some endomorphism (corollary 12.3.14).

1. For matrices, $A = I_n A I_n^{-1}$. If $B = PAP^{-1}$, then $A = P^{-1}B(P^{-1})^{-1}$. If $B = PAP^{-1}$ and $C = QBQ^{-1}$, then $C = (QP)A(QP)^{-1}$, where $(QP)^{-1} = P^{-1}Q^{-1}$. The same three computations, with composition in place of multiplication and automorphisms in place of invertible matrices, prove the claim for endomorphisms.
2. Write $A = [\alpha]_\beta$ and $A' = [\alpha']_\beta$. Since R is commutative,
$$(\alpha \circ \alpha')(b_j) = \alpha\Big(\sum_l a'_{lj} b_l\Big) = \sum_l a'_{lj} \sum_i a_{il} b_i = \sum_i (AA')_{ij} b_i$$
and $\mathrm{id}_M(b_j) = b_j$ gives $[\mathrm{id}_M]_\beta = I_n$. If α is an automorphism, then $[\alpha]_\beta [\alpha^{-1}]_\beta = [\alpha \circ \alpha^{-1}]_\beta = I_n$, and in the same way $[\alpha^{-1}]_\beta [\alpha]_\beta = I_n$.
3. If $\alpha' = \varphi \circ \alpha \circ \varphi^{-1}$, then $[\alpha']_\beta = [\varphi]_\beta [\alpha]_\beta [\varphi]_\beta^{-1}$ by (2). Conversely, let $[\alpha']_\beta = P[\alpha]_\beta P^{-1}$ with P invertible, and let φ and ψ be the endomorphisms with $[\varphi]_\beta = P$ and $[\psi]_\beta = P^{-1}$. By (2), $[\varphi \circ \psi]_\beta = [\psi \circ \varphi]_\beta = I_n = [\mathrm{id}_M]_\beta$, so $\varphi \circ \psi = \psi \circ \varphi = \mathrm{id}_M$, and φ is an automorphism with inverse ψ. Then $[\varphi \circ \alpha \circ \varphi^{-1}]_\beta = P[\alpha]_\beta P^{-1} = [\alpha']_\beta$, so $\alpha' = \varphi \circ \alpha \circ \varphi^{-1}$.
4. Let $\gamma = (c_1, \dots, c_n)$, write $c_j = \sum_i n_{ij} b_i$ and $b_j = \sum_i n'_{ij} c_i$, and let $N = (n_{ij})$ and $N' = (n'_{ij})$. Then $b_j = \sum_i n'_{ij} \sum_l n_{li} b_l = \sum_l (NN')_{lj} b_l$, so $NN' = I_n$ by the uniqueness of coordinates in the basis β, and in the same way $N'N = I_n$. With $A = [\alpha]_\beta$ and $B = [\alpha]_\gamma$,
$$\alpha(c_j) = \sum_l n_{lj} \alpha(b_l) = \sum_i (AN)_{ij} b_i \qquad \alpha(c_j) = \sum_k b_{kj} c_k = \sum_i (NB)_{ij} b_i$$
so $AN = NB$ and $B = N^{-1}AN$, which is similar to A. Conversely, let $B = PAP^{-1}$ with P invertible, put $N = P^{-1} = (n_{ij})$ and $c_j = \sum_i n_{ij} b_i$. The c_j span M, since $\sum_i p_{ij} c_i = \sum_l (NP)_{lj} b_l = b_j$, and they are linearly independent, since $\sum_j x_j c_j = \sum_l (Nx)_l b_l = 0$ forces $Nx = 0$ and then $x = PNx = 0$. So $\gamma = (c_1, \dots, c_n)$ is a basis, and the computation above gives $[\alpha]_\gamma = N^{-1}AN = PAP^{-1} = B$, completing the proof.

In the language of group theory, $\mathrm{GL}_n(R)$ acts on $M_n(R)$ by conjugation[10], and $A \sim B$ if and only if A and B lie in the same orbit of this action.

Just like for equivalence, we may ask for when two matrices are similar, or if there is a nice representation in the orbit. This question is a bit trickier because the domain and codomain have are the same, and so there is less flexibility in choosing representatives. Chapter 14 finds a basis giving a nice representation of α.

13.2.3 Gaussian Elimination: Finding a Nice Basis

Let $\varphi : V \to W$ be a linear transformation, and P be some matrix representation given by choosing a basis for V and W. Then P belongs to some equivalence class representing all possible representatives of φ. If we multiply P on the left or right by an invertible matrix, then by by proposition 13.2.13, P remains in the same equivalence class. It would thus be nice if given P, we may simplify P as much as possible to get an "nice" looking represetnative for φ. To do this, we will find nice sets of invertible matrices that do "elementary" operations to a marix which will allow the matrix P to be put in as simple a form as possible. In particular, we will find invertible matrice that will allow us to perfom the following "elementary" operations:

1. switching two rows (resp. columns) or P. For example, let's say we want to switch the 1st and 2nd row of a matrix. This can be done by multiplying on the left (resp. right) by:

$$\begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

   and the identity on the right (resp. left). In particular, we swapped the rows of the above matrix.

2. multiply all elements in a row (resp. column) of P by a unit of R by multiplying on the left (resp. right) by a matrix of the form:

$$\begin{pmatrix} u & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

   This particular matrix will scale the first row (resp. column) by a factor of u when multiplying on the left (resp. right).

3. add a multiple of one row (resp. column) to another row (resp. column) of P. For example, to add a multiple of the second row to the first, we would multiply on the left by:

$$\begin{pmatrix} 1 & 3 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

   This particular matrix will add 3 copies of the 2nd row to the 1st row when multiplying on the left, and 3 copies of the 1st column to the 2nd column when multiplying on the right.

[10] Recall that conjugation is *the* natural group-action

Over any commutative ring R, these matrices are invertible, and multiplying by one of them on the left acts on rows while multiplying on the right acts on columns:

Proposition 13.2.17: Equivalence And Elementary Matrices

Let R be a commutative ring and $P, Q \in M_{m,n}(R)$. The *elementary row operations* are interchanging two rows, multiplying a row by a unit of R, and adding r times one row to a different row, for some $r \in R$. The *elementary column operations* are defined in the same way with columns. For $1 \leq k, l \leq m$ let $E_{kl} \in M_m(R)$ be the matrix with 1 in position (k, l) and 0 elsewhere, and for $k \neq l$, a unit u and $r \in R$ put

$$S_{kl} = I_m - E_{kk} - E_{ll} + E_{kl} + E_{lk} \qquad D_k(u) = I_m + (u-1)E_{kk} \qquad T_{kl}(r) = I_m + rE_{kl}$$

The matrices of the same names in $M_n(R)$ are defined in the same way with n in place of m.

1. For $X \in M_{m,n}(R)$, the matrices $S_{kl}X$, $D_k(u)X$ and $T_{kl}(r)X$ are X with rows k and l interchanged, with row k multiplied by u, and with r times row l added to row k. The matrices XS_{kl}, $XD_k(u)$ and $XT_{kl}(r)$ are X with columns k and l interchanged, with column k multiplied by u, and with r times column k added to column l.

2. These matrices are invertible:

$$S_{kl}S_{kl} = I \qquad D_k(u)D_k(u^{-1}) = D_k(u^{-1})D_k(u) = I$$

$$T_{kl}(r)T_{kl}(-r) = T_{kl}(-r)T_{kl}(r) = I$$

3. If Q is obtained from P by a finite sequence of elementary row and column operations, then P and Q are equivalent (definition 13.2.14).

***Proof*:**

1. For $X \in M_{m,n}(R)$, the kth row of $E_{kl}X$ is the lth row of X, and its other rows are 0. So $S_{kl}X = X - E_{kk}X - E_{ll}X + E_{kl}X + E_{lk}X$ is X with rows k and l first replaced by 0 and then by the rows l and k of X, $D_k(u)X = X + (u-1)E_{kk}X$ is X with row k replaced by u times itself, and $T_{kl}(r)X = X + rE_{kl}X$ is X with r times row l added to row k. For the matrices $E_{kl} \in M_n(R)$, the lth column of XE_{kl} is the kth column of X and its other columns are 0, and the same expansions show that XS_{kl}, $XD_k(u)$ and $XT_{kl}(r)$ are X with columns k and l interchanged, with column k multiplied by u, and with r times column k added to column l.

2. By (1), $S_{kl}S_{kl}$ is S_{kl} with rows k and l interchanged, which is the identity matrix, and $D_k(u)D_k(u^{-1})$ is $D_k(u^{-1})$ with row k multiplied by u, which is the identity matrix, and in the same way $D_k(u^{-1})D_k(u) = I$. Since $E_{kl}E_{kl} = 0$ for $k \neq l$, $T_{kl}(r)T_{kl}(-r) = I - r^2E_{kl}E_{kl} = I$, and in the same way $T_{kl}(-r)T_{kl}(r) = I$.

3. By (1) and (2), every elementary row operation is left multiplication by an invertible matrix in $M_m(R)$, and every elementary column operation is right multiplication by an invertible matrix in $M_n(R)$. If Q is obtained from P by a finite sequence of these operations, associativity of matrix multiplication (see the discussion after definition 13.2.14) lets the left factors and the right factors be collected separately, so $Q = E_s \cdots E_1 P F_1 \cdots F_t$ with

invertible $E_i \in M_m(R)$ and $F_j \in M_n(R)$. The products $M = E_s \cdots E_1$ and $N = F_1 \cdots F_t$ are invertible, with inverses $E_1^{-1} \cdots E_s^{-1}$ and $F_t^{-1} \cdots F_1^{-1}$, and $Q = MPN$. Hence P and Q are equivalent, as claimed.

Due to these matrices being the transition matrices between two equivalent matrices, we give matrices that perform elementary operations a name:

Definition 13.2.18: Elementary Matrices

The matrices S_{kl}, $D_k(u)$ and $T_{kl}(r)$ of proposition 13.2.17 are called *elementary matrices*. By proposition 13.2.17(1), multiplying a matrix by an elementary matrix on the left performs an elementary row operation, and on the right an elementary column operation. By proposition 13.2.17(2), the inverse of an elementary matrix is again an elementary matrix: $S_{kl}^{-1} = S_{kl}$, $D_k(u)^{-1} = D_k(u^{-1})$ and $T_{kl}(r)^{-1} = T_{kl}(-r)$.

Notice that proposition 13.2.17(3) is not an "if and only if", only an "if". This is because it is possible that P and Q are equivalent, but the invertible matrices that "connects" the two are not the product of elementary matrices. If R has certain extra properties, then it will be the case that all invertible matrices can be written as a product of elementary matrices, as explored below. The invertible matrices are given a name

Definition 13.2.19: General Linear Matrices

Let R be a ring and $M_n(R)$ the set of all matrices of of dimension n over R. Then:

$$\mathrm{GL}_n(R) := (M_n(R))^\times$$

that is, $\mathrm{GL}_n(R)$ is the set of all units of $M_n(R)$, i.e. the set of all invertible matrices of dimension n over R.

The word "dimension" is used to indicate the number of rows and columns (which are both equal in this case, and hence why a single number is named). The term "general" is used since there other set of invertible matrices commonly studied, usually given a different prefix ("special", "orthogonal", "orthonormal", "unitary", and so forth). This set of invertible matrices is called "general" since it ill encompass *all* of these other sets of invertible matrices. For who may have come across these terms before, notice that this set was already mentioned in definition 1.2.33. One thing that as difficult to prove directly with matrices but easy to prove now is that $(\mathrm{GL}_n(R), \cdot)$ is associative: every invertible matrix represents a function, and function composition is associative, and hence $\cdot$ is an associative operation.

Proposition 13.2.20: Order of GL_n Over A Finite Field

Let F be a finite field with q elements and let $n \geq 1$. Then

$$|\mathrm{GL}_n(F)| = \prod_{i=0}^{n-1} (q^n - q^i) = (q^n - 1)(q^n - q)(q^n - q^2) \cdots (q^n - q^{n-1})$$

***Proof*:**
Call a sequence $(v_1, \dots, v_m)$ of vectors in F^n *independent* if $c_1v_1 + \cdots + c_mv_m = 0$ with $c_1, \dots, c_m \in F$ forces $c_1 = \cdots = c_m = 0$. Let A be an $n \times n$ matrix over F with columns $a_1, \dots, a_n \in F^n$. For $x \in F^n$ we have $Ax = x_1a_1 + \cdots + x_na_n$, so A is nonsingular (definition 13.2.9) exactly when $(a_1, \dots, a_n)$ is independent. By proposition 13.2.10, A is invertible exactly when it is nonsingular, so $|\mathrm{GL}_n(F)|$ is the number of independent sequences $(a_1, \dots, a_n)$ in F^n.

Let $0 \leq i \leq n-1$, let $(a_1, \dots, a_i)$ be independent, and let $W = \{c_1a_1 + \cdots + c_ia_i \mid c_1, \dots, c_i \in F\}$. The map $F^i \to W$ sending $(c_1, \dots, c_i)$ to $c_1a_1 + \cdots + c_ia_i$ is surjective by definition and injective by independence, so $|W| = q^i$. The sequence $(a_1, \dots, a_{i+1})$ is independent exactly when $a_{i+1} \notin W$. Indeed, if $a_{i+1} = c_1a_1 + \cdots + c_ia_i$, then $c_1a_1 + \cdots + c_ia_i - a_{i+1} = 0$ is a relation with a nonzero coefficient. Conversely, if $a_{i+1} \notin W$ and $c_1a_1 + \cdots + c_{i+1}a_{i+1} = 0$, then $c_{i+1} = 0$, since otherwise $a_{i+1} = -c_{i+1}^{-1}(c_1a_1 + \cdots + c_ia_i) \in W$, and then $c_1 = \cdots = c_i = 0$ by the independence of $(a_1, \dots, a_i)$. So each independent sequence of length i extends in exactly $q^n - q^i$ ways, and multiplying over $i = 0, 1, \dots, n-1$ gives the formula, as claimed.

As mentioned, there can be matrices in $\mathrm{GL}_n(R)$ which are not a product of elementary matrices, that is, $\mathrm{GL}_n(R)$ need not be generated by elementary matrices. However, if $R = k$ is a field, then this is in fact the case, and the "if" in proposition 13.2.17(3) becomes an "iff"

Proposition 13.2.21: General Linear Group Of Field

Let k be a field and $A \in \mathrm{GL}_n(k)$. Then there are elementary matrices $E_1, \dots, E_s \in M_n(k)$ with

$$E_s \cdots E_1 A = I_n$$

so $A = E_1^{-1} \cdots E_s^{-1}$ is a product of elementary matrices, and $\mathrm{GL}_n(k)$ is generated by the elementary matrices. Consequently, two matrices $P, Q \in M_{m,n}(k)$ are equivalent if and only if Q is obtained from P by a finite sequence of elementary row and column operations.

***Proof*:**
Step 1: row reduction. We show by induction on j, for $0 \leq j \leq n$, that there is a product F_j of elementary matrices such that the first j columns of F_jA are the standard basis vectors $e_1, \dots, e_j$ of k^n. For $j = 0$ take $F_0 = I_n = D_1(1)$. Let $1 \leq j \leq n$ and $B = F_{j-1}A$, with columns $b_1, \dots, b_n$, so that $b_i = e_i$ for $i < j$. The matrix B is invertible, as a product of invertible matrices (proposition 13.2.17(2)), hence nonsingular (proposition 13.2.10). Some entry B_{ij} with $i \geq j$ is nonzero: otherwise $b_j = \sum_{i<j} B_{ij}e_i = \sum_{i<j} B_{ij}b_i$, and the vector $x = e_j - \sum_{i<j} B_{ij}e_i$, whose jth entry is 1, satisfies $Bx = b_j - \sum_{i<j} B_{ij}b_i = 0$. Choose such an i. In the first $j-1$ columns, every row of index at least j has zero entries, since those columns are $e_1, \dots, e_{j-1}$. Interchange rows i and j if $i \neq j$, then multiply row j by the inverse of its jth entry, and then, for each $l \neq j$, add $-c_l$ times row j to row l, where c_l is the current (l, j) entry. The jth column becomes e_j, and the first $j-1$ columns do not change, since in those columns the rows i and j have zero entries. By proposition 13.2.17(1) these operations are left multiplications by elementary matrices, and F_j is their product times F_{j-1}. For $j = n$, the columns of F_nA are $e_1, \dots, e_n$, that is, $F_nA = I_n$, and we write $F_n = E_s \cdots E_1$.

Step 2: generation. Since $E_s \cdots E_1A = I_n$ and A is invertible, $E_s \cdots E_1 = (E_s \cdots E_1A)A^{-1} = A^{-1}$, so $A = (E_s \cdots E_1)^{-1} = E_1^{-1} \cdots E_s^{-1}$. Each E_i^{-1} is an elementary matrix (definition 13.2.18),

so A is a product of elementary matrices. Conversely, every elementary matrix lies in $\mathrm{GL}_n(k)$ by proposition 13.2.17(2), so the subgroup of $\mathrm{GL}_n(k)$ generated by the elementary matrices is $\mathrm{GL}_n(k)$.

Step 3: equivalence. If Q is obtained from P by elementary operations, then P and Q are equivalent by proposition 13.2.17(3). Conversely, let $Q = MPN$ with $M \in \mathrm{GL}_m(k)$ and $N \in \mathrm{GL}_n(k)$. By Step 2, $M = M_1 \cdots M_a$ and $N = N_1 \cdots N_b$ with elementary matrices M_i and N_j, so $Q = M_1(\cdots(M_a(PN_1)\cdots)\cdots)N_b$ by associativity of matrix multiplication, and by proposition 13.2.17(1) each multiplication by an elementary matrix performs an elementary row or column operation, completing the proof.

Step 1 also computes inverses:

Corollary 13.2.22: Finding Inverse Of Matrix

Let k be a field and $A \in \mathrm{GL}_n(k)$. If a finite sequence of elementary row operations transforms A into I_n, then the same operations, applied to I_n, produce A^{-1}. Equivalently, if elementary row operations transform the $n \times 2n$ matrix $(A|I_n)$ into a matrix of the form $(I_n|B)$, then $B = A^{-1}$. Such a sequence of operations exists by proposition 13.2.21.

***Proof*:**
By proposition 13.2.17(1), the row operations are left multiplications by elementary matrices $E_1, \ldots, E_s$, applied in this order. If they transform A into I_n, then $E_s \cdots E_1 A = I_n$, so $E_s \cdots E_1 = (E_s \cdots E_1 A)A^{-1} = A^{-1}$, and the same operations applied to I_n give $E_s \cdots E_1 I_n = A^{-1}$. For the second statement, the jth column of EX is E times the jth column of X, for all $E \in M_n(k)$ and $X \in M_{n,2n}(k)$, so $E(A|I_n) = (EA|E)$. If the operations, with product $E = E_s \cdots E_1$, transform $(A|I_n)$ into $(I_n|B)$, then $EA = I_n$ and $E = B$, so $B = A^{-1}$ by the first part, completing the proof.

Reducing a matrix by elementary row operations, as in Step 1 of the proof of proposition 13.2.21, is called *Gaussian Elimination.*[11]

Over a field, every matrix is equivalent to a matrix of a simple shape, and the shape is determined by a single number:

[11] Though Gauss has most likely invented the above method when working with matrices, it has been known under slightly different symbols and theory at least as far back as Han China

Corollary 13.2.23: Equivalence Of Matrices

Let K be a field, let $P \in M_{m,n}(K)$, and let k be the dimension of the subspace $P(K^n) = \{Px \mid x \in K^n\}$ of K^m. For $0 \leq l \leq \min(m,n)$ write

$$B_l = \left(\begin{array}{c|c} I_l & 0 \\ \hline 0 & 0 \end{array}\right) \in M_{m,n}(K)$$

where each 0 is a zero matrix of the appropriate size, so that B_0 is the zero matrix. Then:

1. $0 \leq k \leq \min(m,n)$, and P is equivalent to B_k
2. P is equivalent to B_l only for $l = k$, and two matrices $P, Q \in M_{m,n}(K)$ are equivalent if and only if $\dim P(K^n) = \dim Q(K^n)$
3. $P(K^n)$ is the subspace of K^m spanned by the columns of P, and the subspace of K^n spanned by the rows of P, written as columns, also has dimension k

***Proof*:**
Let $\varphi\colon K^n \to K^m$ be the linear map $x \mapsto Px$, whose matrix with respect to the standard bases is P, since $\varphi(e_j) = Pe_j = \sum_i P_{ij} e_i$.

1. By corollary 13.1.18, $n = \dim \ker \varphi + k$, so $k \leq n$, and $k \leq m$ by proposition 13.1.17(2), since $\varphi(K^n) \subseteq K^m$. The kernel has a basis $(u_{k+1}, \ldots, u_n)$ (proposition 13.1.17(1)), which is contained in a basis of K^n with n elements (theorem 13.1.8, corollary 13.1.13(2)), and we write this basis as $(u_1, \ldots, u_n)$. The vectors $\varphi(u_1), \ldots, \varphi(u_k)$ span $\varphi(K^n)$, since $\varphi(u_j) = 0$ for $j > k$. They are linearly independent: if $\sum_{i \leq k} c_i \varphi(u_i) = 0$, then $\sum_{i \leq k} c_i u_i \in \ker \varphi$ is a combination of $u_{k+1}, \ldots, u_n$, and the independence of $(u_1, \ldots, u_n)$ forces every $c_i = 0$. Extend $(\varphi(u_1), \ldots, \varphi(u_k))$ to a basis $(w_1, \ldots, w_m)$ of K^m with $w_i = \varphi(u_i)$ for $i \leq k$ (theorem 13.1.8, corollary 13.1.13(2)). Then $\varphi(u_j) = w_j$ for $j \leq k$ and $\varphi(u_j) = 0$ for $j > k$, so the matrix of φ with respect to the bases (u_j) and (w_i) is B_k. By proposition 13.2.13, $B_k = MPN$ for change of basis matrices M and N, which are invertible by corollary 13.2.12(1). Hence P is equivalent to B_k.

2. Let $Q = MPN$ with $M \in \mathrm{GL}_m(K)$ and $N \in \mathrm{GL}_n(K)$. Then $N(K^n) = K^n$, since $x = N(N^{-1}x)$, so $Q(K^n) = M(P(K^n))$. The linear map $\rho\colon P(K^n) \to K^m$, $y \mapsto My$, is injective, since $My = 0$ forces $y = M^{-1}My = 0$, and its image is $Q(K^n)$. By corollary 13.1.18, $\dim P(K^n) = 0 + \dim Q(K^n)$. So equivalent matrices give subspaces of the same dimension. As $B_l(K^n) = \mathrm{span}(e_1, \ldots, e_l)$ has dimension l, P can be equivalent to B_l only for $l = k$. If $\dim P(K^n) = \dim Q(K^n) = k$, then by (1) $B_k = MPN = M'QN'$ for invertible M, N, M', N', so $Q = (M'^{-1}M)P(NN'^{-1})$ is equivalent to P.

3. For $x \in K^n$, $Px = x_1 p_1 + \cdots + x_n p_n$, where p_j is the jth column of P, so $P(K^n)$ is the span of the columns. The rows of P, written as columns, are the columns of the transpose P^t. By (1), $P = M^{-1}B_k N^{-1}$ with M and N invertible. Over the commutative field K, $(XY)^t = Y^t X^t$, both having (a,b) entry $\sum_c x_{bc} y_{ca}$, so $P^t = (N^{-1})^t B_k^t (M^{-1})^t$, where $(N^{-1})^t$ is invertible with inverse N^t, since $N^t(N^{-1})^t = (N^{-1}N)^t = I$ and $(N^{-1})^t N^t = (NN^{-1})^t = I$, and likewise for $(M^{-1})^t$. So $P^t \in M_{n,m}(K)$ is equivalent to B_k^t, the matrix of the same shape in $M_{n,m}(K)$, whose image $\mathrm{span}(e_1, \ldots, e_k) \subseteq K^n$ has dimension k. By

(2) applied to P^t, the span of the columns of P^t has dimension k, completing the proof.

So there are exactly $\min(m,n)+1$ equivalence classes of $m \times n$ matrices over a field, one for each k with $0 \leq k \leq \min(m,n)$. This also shows that if we look at a linear transformation in the right way (i.e. pick the right basis), a linear transformation *always* looks like a projection eliminating certain dimensions followed by an embedding of the rest of the space into the codomain! See proposition 13.1.22 to see how this intuition and the existence of complements of subspaces work together.

Some authors define the *row rank* and the *column rank* of a matrix as the dimensions of the spaces spanned by its rows and by its columns, and show that the two are the same number, which is corollary 13.2.23(3):

Definition 13.2.24: [matrix] Rank

Let A be an $m \times n$ matrix over a field. Then the *row rank* is the dimension of the space spanned by its rows, and the *column rank* is the dimension of the space spanned by its columns. By corollary 13.2.23(3), the row rank and the column rank are equal, and the single number is called the *rank* of the matrix. If the rank is $\min(m,n)$, the matrix is said to be of *full rank*.

The map $x \mapsto Ax$ represented by an $m \times n$ matrix A of rank r has image of dimension r by corollary 13.2.23(3), so it is surjective when $r = m$ (proposition 13.1.17(2)) and injective when $r = n$, its kernel then having dimension $n - r = 0$ (corollary 13.1.18).

To finish off this section, we can ask if the same logic can be done for the notion of similarity. Unfortunately, similarity is more complicated: since only one matrix is being picked (i.e. only one basis is being picked) the rows and column operations must be done "simultaneously". Not all hope is lost: using properties of PID's will show us how to find a good matrix representation for similar matrices in the next chapter.

Gaussian Elimination over Euclidean Domain

From the rings we explored earlier, the euclidean domain is a good start, and indeed with some cleverness a matrix over a euclidean domain can be reduced, though not intirely to an identity matrix in a sub-matrix of the original.

To demonstrate the idea, let R be a euclidean domain with valuation ν, and take any 2×2 matrix:

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix}$$

then use elementary matrices to put the element with the lowest valuation at position $(1,1)$. Then use the euclidean algorithm to find:

$$b = aq + r$$

where either $r = 0$ or $\nu(r) < \nu(a)$. Then do the appropriate elementary operation to get:

$$\begin{pmatrix} a & r \\ c & d - aq \end{pmatrix}$$

if $r = 0$, then we have eliminated one of the components. If $r \neq 0$, then repeat the same process, moving the term with the lowest valuation to the $(1,1)$ position using elementary matrices. Since

each term has a finite valuation, it must be that eventually $r = 0$. With minor modifications, the same thing can be done in the $(2, 1)$ position, getting us:

$$\begin{pmatrix} a' & 0 \\ 0 & d' \end{pmatrix}$$

Next, notice that we may find an a', d' such that $a' \big| d'$. For if not, we may consider:

$$\begin{pmatrix} a' & d' \\ 0 & d' \end{pmatrix}$$

and repeat the process until we get to the same state, and keep repeating until either $d' = 0$ or $a' \big| d'$. Thus, we produce a 2×2 matrix where the diagonal divides each other.

From this, it is not too hard to see that this is easily generelizable to an $n \times n$ matrix:

Theorem 13.2.25: Smith Normal Form

Let R be a euclidean domain and $P \in M_{m,n}(R)$. Then P is equivalent to a matrix of the form

$$\left(\begin{array}{ccc|c} [ccc|c]d_1 & \cdots & 0 & 0 \\ \vdots & \ddots & 0 & 0 \\ 0 & \cdots & d_k & 0 \\ \hline 0 & \cdots & 0 & 0 \end{array}\right)$$

where

$$d_1 \big| \cdots \big| d_k$$

The resulting matrix is called the *smith normal form*.

***Proof*:**

Throughout, all matrices produced from P by elementary row and column operations are equivalent to P by proposition 13.2.17(3). We prove the theorem itself by induction on $\min(m, n)$. If $P = 0$, then P is already of the stated form with $k = 0$, so assume P is nonzero.

The following clearing claim holds for every nonzero P, with no induction: P is equivalent to a block matrix whose $(1, 1)$ entry is some nonzero d_1, whose remaining first row and first column are zero, and whose lower right block P' (possibly empty) has every entry divisible by d_1.

Consider the set of valuations $\nu(a)$ of all nonzero entries a of all matrices equivalent to P. This is a nonempty set of non-negative integers, so by well-ordering it has a least element. Choose a matrix Q equivalent to P having a nonzero entry a with $\nu(a)$ least. Swapping rows and columns, we may assume a is in position $(1, 1)$.

We claim a divides every entry b of the first row of Q. Say b is in position $(1, j)$ with $j \geq 2$, and divide: $b = qa + r$ with $r = 0$ or $\nu(r) < \nu(a)$. Subtracting q times the first column from the jth column produces a matrix equivalent to P with r in position $(1, j)$. If $r \neq 0$, then $\nu(r) < \nu(a)$ contradicts the minimality of $\nu(a)$. Hence $r = 0$ and $a \big| b$. The same argument with rows in place of columns shows a divides every entry of the first column.

Consequently, subtracting the appropriate multiple of the first column from each other column clears the rest of the first row, and then subtracting the appropriate multiple of the (new) first

row from each other row clears the rest of the first column. We arrive at an equivalent matrix with $(1,1)$ entry a, zeros elsewhere in the first row and column, and some lower right block P'.

Next, a divides every entry c of P'. Say c is in position (i,j) with $i,j \geq 2$. Adding the ith row to the first row leaves the $(1,1)$ entry equal to a (the ith row has first entry 0) and places c in position $(1,j)$. Dividing $c = qa + r$ and subtracting q times the first column from the jth column as before, minimality of $\nu(a)$ forces $r = 0$, so $a \big| c$. This proves the clearing claim with $d_1 = a$.

In the base case $\min(m,n) = 1$ of the induction, the block P' is empty, so the clearing claim already puts P in the stated form with $k = 1$. For the inductive step, apply the clearing claim to P. If P' is zero, the matrix is in the stated form with $k = 1$. Otherwise, the induction hypothesis, that is, the theorem for the smaller matrix P', provides a sequence of elementary operations on P' bringing it to the form $\operatorname{diag}(d_2, \ldots, d_k)$ bordered by zeros, with $d_2 \big| \cdots \big| d_k$. These operations involve only the last $m-1$ rows and last $n-1$ columns, so they do not disturb the first row and column, and applying them yields a matrix of the stated shape, equivalent to P.

It remains to verify $d_1 \big| d_2$. Each elementary operation replaces every entry of a matrix by an R-linear combination of its previous entries, so by induction along the sequence of operations, every entry of the reduced form of P' is an R-linear combination of entries of P'. Since d_1 divides every entry of P', it divides every such combination, and in particular $d_1 \big| d_2$. Hence $d_1 \big| d_2 \big| \cdots \big| d_k$, completing the proof.

Using smith normal forms, we may prove that $\mathrm{GL}_n(R)$ is generated by elementary matrices for a Euclidean domain R. This fails for some Principal Ideal Domains: for the ring of integers $\mathbb{Z}[\frac{1+\sqrt{-19}}{2}]$ of $\mathbb{Q}(\sqrt{-19})$, which is a Principal Ideal Domain but not Euclidean, GL_2 is not generated by elementary matrices [26]. On the other hand, we can put any matrix in $\mathrm{GL}_n(R)$ into smith normal form using the gcd and some non-elementary matrix operations. This shall be further explored in chapter 14.

13.2.4 Geometric Result: Systems of Linear Equations

Vector spaces and linear maps have part of their origins in geometry, where the term *linear* originates. The rigorous interpretation of a concept being "geometrical" is something that would be explored in a class on [differential] geometry, and so for the uses of this book geometrical shall mean "locally or globally looks like the solutions to equations":

$$\begin{aligned} f_1(\vec{x}) &= a_1 \\ f_2(\vec{x}) &= a_2 \\ &\vdots \\ f_n(\vec{x}) &= a_n \end{aligned}$$

Often, the constant is moved to the left modifying the equations to be:

$$\begin{aligned} f_1(\vec{x}) - a_1 &= 0 \\ f_2(\vec{x}) - a_2 &= 0 \\ &\vdots \\ f_n(\vec{x}) - a_n &= 0 \end{aligned}$$

This understanding of geometrical is in fact very potent, for example the reader may have already encountered many geometrical objects that are defined in this way, for example:

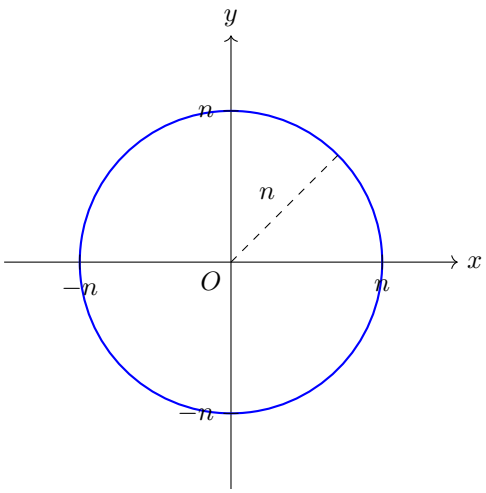


Figure 13.1: solution to $x^2 + y^2 - n^2 = 0$

For the particular explorations of this section, we shall show that vector spaces can be interpreted as the of solutions to *linear equations*:

Definition 13.2.26: Linear Equation and System of Linear Equations

Let V be a k-vector space of dimension n and $k[x_1, x_2, ..., x_n]$ be a polynomial ring over the field k with n variables. Then a *linear equation* is a polynomial, possibly over several variables, where the degree of each variable is at most 1:

$$a_1x_1 + a_2x_2 + \cdots + a_nx_n$$

A collection of linear equations is called a *system of linear equations*.

Given a (system of) linear equation(s), we can define *linear functions*: if $\sum_i a_i x_i$ is a linear equation then we can define:

$$L_1 : k^n \to k \qquad L(x_1, x_2, ..., x_n) = a_1x_1 + a_2x_2 + \cdots a_nx_n$$

and similarly a system of linear functions is a collection of linear functions $L_1, ..., L_m$. Since these functions are linear, they can be represented by a matrix, namely given the standard basis:

$$A = [L_1] = \begin{pmatrix} a_1 & a_2 & \cdots & a_n \end{pmatrix}$$

so that if $\vec{x}$ is an $n \times 1$ matrix with with x_i coefficients, then plugging in a value from k^n is represented as matrix multiplication by

$$A\vec{x} = \begin{pmatrix} a_1 & a_2 & \cdots & a_n \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix} = a_1x_1 + a_2x_2 + \cdots + a_nx_n$$

Equivalently, any system of linear equation can be combined to form a linear equation:

$$L : k^n \to k^m \qquad L = \begin{bmatrix} L_1 \\ L_2 \\ \vdots \\ L_m \end{bmatrix} \tag{13.6}$$

Let's now say we want to find the solution to the system:

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + \cdots a_{1n}x_n &= b_1 \\ a_{21}x_1 + a_{22}x_2 + \cdots a_{2n}x_n &= b_2 \\ &\vdots \\ a_{m1}x_1 + a_{m2}x_2 + \cdots a_{mn}x_n &= b_m \end{aligned}$$

Then if $\vec{b} = (b_{k1})$, then what we are solving for is:

$$A\vec{x} = \vec{b}$$

If A is invertible and $n = m$, then the answer is simple:

$$\vec{x} = A^{-1}\vec{b}$$

where A^{-1} is found using corollary 13.2.22. In general, corollary 13.2.23 gives invertible matrices $M \in \mathrm{GL}_m(k)$ and $N \in \mathrm{GL}_n(k)$ with $MAN = B_r$, where r is the rank of A. Writing $\vec{x} = N\vec{y}$ and $\vec{c} = M\vec{b}$, the system $A\vec{x} = \vec{b}$ is equivalent to $(MAN)\vec{y} = \vec{c}$, since M and N are invertible, that is, to the system

$$y_1 = c_1 \qquad \cdots \qquad y_r = c_r \qquad 0 = c_{r+1} \qquad \cdots \qquad 0 = c_m$$

So $A\vec{x} = \vec{b}$ has a solution if and only if $c_{r+1} = \cdots = c_m = 0$, and then its solutions are the vectors $N\vec{y}$ with $y_i = c_i$ for $i \leq r$ and $y_{r+1}, \ldots, y_n$ arbitrary. Over a Euclidean domain R such as $\mathbb{Z}$, theorem 13.2.25 gives invertible matrices M and N over R with MAN diagonal with entries $d_1 \big| \cdots \big| d_r$ followed by zeros, and the same substitution turns the system into $d_iy_i = c_i$ for $i \leq r$ and $0 = c_i$ for $i > r$, which has a solution $\vec{y} \in R^n$ if and only if $d_i \big| c_i$ for $i \leq r$ and $c_i = 0$ for $i > r$.

Where this links to geometry is when considering the space of solution to systems of linear equations. First, notice that without loss of generality the linear equations may equal to 0, for this would simply translate the resulting set of solutions in the vector-space, hence it suffices to consider:

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + \cdots a_{1n}x_n &= 0 \\ a_{21}x_1 + a_{22}x_2 + \cdots a_{2n}x_n &= 0 \\ &\vdots \\ a_{m1}x_1 + a_{m2}x_2 + \cdots a_{mn}x_n &= 0 \end{aligned}$$

That is, it suffices to consider the kernel of a system of linear equations, equivalently the kernel of L as defined in equation (13.6).

Now, to endeavor exploring the shape of the kernel of linear functions, let's start with $k = \mathbb{R}$, $L : \mathbb{R}^3 \to \mathbb{R}$, and always use the standard basis. It is easy to visualize $\mathbb{R}, \mathbb{R}^2, \mathbb{R}^3$, and higher dimensional analogues can be fathomed. Naturally, if $L(x) = 0$, then $\ker = \mathbb{R}^3$. It is more interesting if L is non-trivial, which for linear functions implies L is surjective. Then by the Rank-Nullity Theorem $\dim \ker = 3 - 1 = 2$, i.e. it is a plane. A plane that is one-dimension less than the ambient space (in this case $\mathbb{R}^3$) is called a *hyper-plane*. Since it is a subspace, it must contain 0 and hence passes through the origin. Since the plane is two dimension, it is spanned by two vectors. Using Gaussian elimination, we can deduce it is spanned by two vectors:

$$\left\{ \begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix}, \begin{pmatrix} b_1 \\ b_2 \\ b_3 \end{pmatrix} \right\}$$

Example 13.2.27: Finding A Hyperplane

Take $L(x, y, z) = 3x + 2y - 3z$, which is surjective since $L(c/3, 0, 0) = c$ for every $c \in \mathbb{R}$. By the Rank-Nullity Theorem (corollary 13.1.18) its kernel has dimension $3 - 1 = 2$. The vectors $(2, -3, 0)$ and $(1, 0, 1)$ lie in the kernel, since $3 \cdot 2 + 2 \cdot (-3) = 0$ and $3 - 3 = 0$, and they are linearly independent, since neither is a multiple of the other. So they form a basis of the kernel, since a linearly independent set lies in a basis (theorem 13.1.8) and every basis of the kernel has 2 elements (corollary 13.1.13(2)). The kernel when embedded in $\mathbb{R}^3$ looks like so:

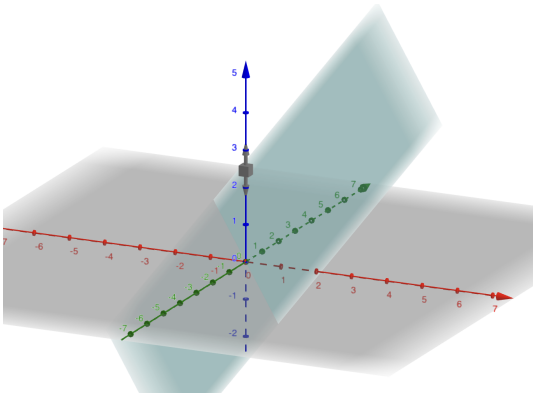


Figure 13.2: $3x + 2y - 3z = 0$

Notice that since the image is one dimensional, the hyper-plane is defined by only one equation, however it is a two dimensional object. More generally, for a surjective linear function $L : \mathbb{R}^n \to \mathbb{R}$, the hyperplane shall be $(n-1)$ dimensional and be defined by a single equation.

Next, let's consider the case where $L : \mathbb{R}^3 \to \mathbb{R}^2$, that is there are two linear equations $L_1 : \mathbb{R}^3 \to \mathbb{R}$ and $L_2 : \mathbb{R}^3 \to \mathbb{R}$. Let's again assume that L is surjective, or full-rank, for if L was one dimension less it would be equivalent to the case of $L : \mathbb{R}^3 \to \mathbb{R}$. Then by the Rank-Nullity theorem we have that $\ker L = 3 - 2 = 1$. Intuitively, this is because there are more restrictions for $L(x) = 0$, namely both L_1 and L_2 must equal 0. Geometrically, this can be thought of as the *intersection* of the kernel of L_1 and the kernel of L_2. Since L is full rank, the intersection is not parallel (the geometric word for this is that the intersection is *transversal*). More generally, $L : \mathbb{R}^n \to \mathbb{R}^2$ would have $\dim \ker L = n - 2$, and would be defined by two linear equations.

Extended the pattern further, given a full-rank linear function $L : \mathbb{R}^n \to \mathbb{R}^m$, the resulting plane given by the kernel would be $n - m$ dimensional and would represent the intersection of m hyper-plane of dimension $n - 1$, each new hyper-plane restricting the dimension of the kernel. The orientation of the kernel is completely dependent on the linear map in question. Nothing in these results required to work specifically with $k = \mathbb{R}$, it was simply convenient for visual purposes. These results can be summarized into the following:

Theorem 13.2.28: Classification Of Planes

Let V be a k-vector-space of dimension n and let $0 \leq m \leq n$, where $k^0 = \{0\}$.

1. If $L\colon V \to k^m$ is a surjective (that is, full-rank) linear map, then $\ker L$ is a subspace of V of dimension $n - m$
2. Every subspace W of V of dimension $n - m$ is $\ker L$ for some surjective linear map $L\colon V \to k^m$
3. Two surjective linear maps $L, L'\colon V \to k^m$ have the same kernel if and only if $L' = g \circ L$ for some linear automorphism g of k^m

Thus the sub-spaces of V of dimension $n-m$ correspond to the full-rank linear maps $V \to k^m$, taken up to composition with automorphisms of k^m. When $V = k^n$, every subspace is the set of solutions of a system of linear equations $a_{i1}x_1 + \cdots + a_{in}x_n = 0$, $1 \leq i \leq m$.

***Proof*:**

1. By the Rank-Nullity Theorem (corollary 13.1.18), $\ker L$ is a subspace of V and $\dim \ker L = \dim V - \dim L(V) = n - m$, since $L(V) = k^m$ has dimension m.
2. The subspace W has a basis $w_1, \ldots, w_{n-m}$ (theorem 13.1.8), with $n - m$ elements since $\dim W = n - m$. This linearly independent set is contained in a basis B of V (theorem 13.1.8), which has n elements by corollary 13.1.13(2). So we may write $B = \{w_1, \ldots, w_{n-m}, u_1, \ldots, u_m\}$. Let $L\colon V \to k^m$ be the linear map with $L(w_i) = 0$ for all i and $L(u_j) = e_j$, the jth standard basis vector of k^m (corollary 12.3.14). For $v = \sum_i c_i w_i + \sum_j d_j u_j$,
$$L(v) = d_1 e_1 + \cdots + d_m e_m = (d_1, \ldots, d_m)$$
so L is surjective, and $L(v) = 0$ exactly when every $d_j = 0$, that is, when $v \in W$, the coefficients of v with respect to B being unique. Hence $\ker L = W$.
3. If $L' = g \circ L$ with g an automorphism, then $L'(v) = 0$ exactly when $L(v) = 0$, since g is injective, so $\ker L' = \ker L$. Conversely, suppose $\ker L = \ker L' = W$, and define $g\colon k^m \to k^m$ by $g(L(v)) = L'(v)$. Every element of k^m has the form $L(v)$, as L is surjective, and g is well defined: if $L(v) = L(v')$, then $v - v' \in W = \ker L'$, so $L'(v) = L'(v')$. It is linear, since $g(L(v) + cL(v')) = g(L(v + cv')) = L'(v + cv') = L'(v) + cL'(v')$ for $c \in k$. It is surjective, because L' is, and injective, because $g(L(v)) = 0$ means $v \in \ker L' = \ker L$, that is, $L(v) = 0$. By lemma 12.2.3, g is a linear automorphism of k^m, and $L' = g \circ L$ by construction.

Finally, let $V = k^n$ and write $L = (L_1, \ldots, L_m)$ with components $L_i\colon k^n \to k$. For $x = \sum_j x_j e_j$ and $a_{ij} = L_i(e_j)$, linearity gives $L_i(x) = a_{i1}x_1 + \cdots + a_{in}x_n$, so $\ker L$ is the set of solutions of the system $a_{i1}x_1 + \cdots + a_{in}x_n = 0$, $1 \leq i \leq m$. By (2), every subspace of k^n has this form, completing the proof.

Given that L is surjective, notice that as m increases in size, the dimension of the kernel decreases in

size. In a sense it is the "co-dimension" of the domain. This observation is not lost on mathematicians, and the notion of codimension is quite important in geometry: the dimension of an object represents the degree of freedom an object may have within a space (ex. a hyper-plane is $n-1$ dimensional) while the codimension represents the degree of restriction an object may have within a space (ex. a hyper-plane is 1 codimensional).

13.2.5 Use Case: Reading Presentations

Recall from section 12.3.3 that a presentation of a [finitely generated] R-module can be represented as an exact sequence:

$$R^n \to R^m \to M \to 0$$

Using the above results, we can more easily read the presentation. Let $\varphi : R^n \to R^m$. Then by exactness, we have that $M \cong \text{coker}(\varphi)$. Hence, if we find the image of φ, we found the relations of M, and and if R is a field of euclidean domain, this is task is simplified using Gaussian elimination:

Example 13.2.29: Finding A Presentation
Let $R = \mathbb{Z}$ and let φ be represented by the matrix:

$$\begin{pmatrix} 1 & 3 \\ 2 & 3 \\ 5 & 9 \end{pmatrix}$$

which means that $\varphi : \mathbb{Z}^2 \to \mathbb{Z}^3$ and this matrix determines an abelian group G (since finitely generated abelian groups have free resolutions of length at most 1 as mentioned in section 12.3.3). To find G, let us first simplify the above matrix. The reader should check that the matrix reduces to:

$$\begin{pmatrix} 1 & 0 \\ 0 & 3 \\ 0 & 0 \end{pmatrix}$$

Hence, G is isomorphic to the cokernel of $\varphi : \mathbb{Z}^2 \to \mathbb{Z}^3$ that maps $(1,0)$ to $(1,0,0)$ and $(0,1)$ to $(0,3,0)$. In particular, since the image is $\mathbb{Z} \oplus 3\mathbb{Z} \oplus 0$, we get

$$G \cong \text{coker}\varphi \cong \frac{\mathbb{Z} \oplus \mathbb{Z} \oplus \mathbb{Z}}{\mathbb{Z} \oplus 3\mathbb{Z} \oplus 0} \cong (\mathbb{Z}/3\mathbb{Z}) \oplus \mathbb{Z}$$

Showing how a matrix can uniquely represent an abelian group! In the next chapter, we shall show how a matrix can more generally represent a finitely generated module over a PID.

13.2.30 Remark Notice that we may eliminate first row and column if the leading term is 1 (or more generally a unit) since in the quotient the summand shall be completely quotiented.

13.3 Dual Spaces and Forms

Let's say you are studying some mathematical object, say the set of linear transformations. We may want to find numerical properties. For example of linear transformations, we can quantify the size of

a matrix, or give some numerical value to similarity, or the dimension of the image. We can define a function from $L(V)$ to a field, say $\mathbb{R}$, that shall give us this quantification. The collection of all these functions are called the *functionals* on $L(V)$. Depending on what type of property we want want the invariances that are being sought after, we will have different types of functionals. For example, when working in analysis, we shall often want our functional to be continuous or perhaps bounded in some way [12]. In Algebra, we shall want our functionals to be homomorphisms, and depending on what other property we want we shall add more properties to our functionals (ex. it shall be important in [23, Affine Space and Regular Functions] that the functional preserves solutions to a special set of polynomial equations).

The set of functionals on a vector space V can be thought of as containing all possible quantifiable "information" about a space (with the limits of the information provided by the properties of the functionals). Most of the time, the space of functionals is larger than the original space[13]. In some special cases, the spaces of all functionals is isomorphic to the space over which we are taking the functionals (as we shall see, all linear functionals on a finite dimensional vector space V is isomorphic to V). In this special case, we can think of our space being characterized by the property the functionals represent! This realization was the origin of *dual theory*, which we shall start exploring in this section. Dual theory is normally explored in earnest in Real and Functional analysis, however it shall show up in some important ways in [23, Coordinate Rings], and certain key functionals shall give a lot of information on linear transformations, namely the trace and the determinant.

13.3.1 Dual Spaces

In mathematics, there is frequently two "opposing" or "dual" ways of describing objects: Let's say X is our object of study

1. internally: Using properties internal to your object to describe that object. Often, maps *to* the object will give us internal information ($f : A \to X$ for some appropriate object A). This can may also be thought of as local information
2. externally: Using some exterior property (ex. be it by putting it an ambient space, or a sapce in which the object X "resonates" or "interacts" with[14]). Ofen, maps *from* the object will give us external information ($f : X \to A$ for osme appropriate object A). This can also often be thought as global information

As an example of this, when we were working with group theory, we found two way of looking at all possible groups:

1. Using quotients of free groups, we managed to make normal subgroups of free groups represent relations between elements. In the process of doing this, we defined a surjective homomorphism:
$$\varphi : F(A) \to G$$
2. Using Cayley's theorem, we manage to embed any group into a subgroup of S_A. In the process of doing this, we defined an invective map
$$\psi : G \to S_A$$

[12] See Lebesgue Spaces in [47] for more details

[13] This is a guiding principle behind many generalizations in Mathematics, Distribution Theory being the prototypical example

[14] If you know Fourier Analysis, think of phase space

Notice how the first one defined us what's happening between elements in a sort of "local" way, while in the 2nd example we defined it by using properties of the larger group S_A and how elements of G embed into S_A. Another example of this principle can be seen with the universal property of products and coproducts of groups. In a sense, the universal property of products only requires the product group to have groups A and B "locally" in the most efficient way, leading to the direct product of A and B, while the corpoduct group require it to have A and B "globally" in the most efficient way, leading to the free product of A and B.

Often when studying an object X, we will study if from these two perspectives: from maps going *into* X and maps going *out of* X. One remarkable phenomena that occurs often in mathematics is that there is often a [contravariant] *dual object* in which the maps *into* and *out of* X are naturally associated to maps *out of* and *into* a dual object X^*. Note the reversal in the directions of the maps! It is often very fruitful to look at objects from the perspective of both X and X^* to gain more information.

To demonstrate a simple but powerful example of "dualness" in mathematics (or the pervasiveness of dualness), we require knowledge of fields outisde of algebra. Usually, I (the author of this book) would decide to leave out any such example to insure no prerequisites are required to learn the material. However, I have decided to break this rule here for those who know a bit of topology since it is an example so powerful that it changed my view of duality, and requires elementary topology knowledge. The following example is *not* necessary for understanding any material of this book and can be considered motivation for dual spaces and dual theory in general for those who know some topology (if you need a refresher on topology, see appendix C). In this example, we will take a moment and venture outisde of algebra and into the realm of analysis and and geometry to give an interesting example.We will give a common phenomena in which "algebraic" object have "geometric" objects as their dual. In particular, unlike the duality of a group in its represetnation in terms of the quotient of a free group or an embedding into S_G, both of which are algebraic ways of looking at groups, we will show that dual objects are sometimes found in other fields of mathematics, allowing us to connect these two fields of mathematics proof-strategies and techniques!

> First, define a *Boolean Algebra* to be a $\mathbb{F}_2$-algebra (where $\mathbb{F}_2 = \mathbb{Z}/2\mathbb{Z}$). Since $\mathbb{F}_2$ is a field, a Boolean Algebra B is module-isomorphic to $\mathbb{F}_2^n$ for some cardinality n. For $0, 1$ in a Boolean algebra B:
>
> $$0+0=0 \quad 0+1=1+0=0 \quad 1+1=0 \qquad 0\cdot 0=0\cdot 1=1\cdot 0=0 \quad 1\cdot 1=0$$
>
> We usually also define the additional operation of $\neg$:
>
> $$\neg(x) = 1 - x$$
>
> If you have done some computer science or logic, then you might notice that 1 and 0 can represent the values of *true* and *false* respectively, while $+$ and $\cdot$ represent XOR and AND respectively (hence the name Boolean to this algebra). This algebra underlines most of the logic behind computer science, making us want to find properties of boolean algebras. One way we may study these properties is by looking for maps going *into* a boolean algebra B. Dually, by a result known as *Stone Duality*, we can instead study all the maps *out of* a totally disconnected[15] Hausdorff spaces B^*! Thus, the "dual" of Boolean Algebra is all continuous maps from B^* to $\mathbb{R}$! More generally, locally compact[16]

[15] A topological space in which the only connected sets are singletons

[16] a topological space in which for any point x, there exists a compact neighbourhood

topological spaces (which we can consider as a geometric object) are the dual of a type of algebra known as a C^*-algebras[17] (which we consider as algebraic objects): this is known as *Gelfand Duality*[18]. The study of these objects that are related to each other by "duality" is called *Dual Theory*.

Coming back to algebra and the study of vector-spaces, we will see the very beginnings of Dual Theory by studying the simplest case of dual objects, *dual vector-spaces*. The way we will define the dual of a vector-space will be through "external" means. When we defined vector spaces using a basis, that is a "local" property. In a sense, we showed that the map $f : F(A) \to V$ can define a vector space where A is a basis. What we will now do is the *dual* equivalent: we will define a vector space by mapping V into a different space.

The question arrises as to what space might we might we map to? When working with groups, it was not immediately obvious that S_A would be the right group to map. Fortunately for us, there is an "obvious" space to map to: for any vector-sapce over k, we are always given two algebraic objects, namely the abelian group V and the field k. Since k is always a vector-sapce over itself,we can consider linear maps $f : V \to k$! The collection of all such maps is called the *dual space* of V.

Definition 13.3.1: Dual Space

Let V be a vector space over k. Then $V^* = \mathrm{Hom}_k(V, k)$ is the space of all linear transformations from V to f, and is called the *dual space* of V.

By proposition 12.2.7, $\mathrm{Hom}_k(V, k)$ is a vector-space over k (if you are not convinced, it is recommended to reprove this before continuing). The notation $V^\vee$ is also often used to define the dual space and will be used interchangibly. Elements inside the dual space V^* are sometimes called *covectors*, paralleling to how elements inside a vector space V is called *vectors*. In physics and analysis, there is seomtimes a different convention for how to write the evalutation of if $f \in V^*$ at $x \in V$: instead of writing $f(x)$, to mean "evaluate f at the point x", it is sometimes written $\langle f, x\rangle$.

Elements of V^* are given a name:

Definition 13.3.2: Linear Functionals

Let V^* be the dual space of V. Then $f \in V^*$ is called a *linear functional* or *linear form*.

Example 13.3.3: Linear Functional and Dual Spaces

1. (If you know some calculus) Perhaps the most famous linear functional is $\int_a^b ___ dx : L^1(\mathbb{R}) \to \mathbb{R}$, where $L^1(\mathbb{R})$ will denote the set of all (either Riemann or Lebesgue)[a] integrable functions $f : \mathbb{R} \to \mathbb{R}$ where $\int_{\mathbb{R}} |f| dx < \infty$ (so that the integral is well-defined). Note that $L^1(\mathbb{R})$ is a vector-space with addition and scalar multiplication defined pointwise since the codomain is

[17] A Banach Algebra over $\mathbb{C}$ with a map known as an involution satisfying the properties of an adjoins

[18] This duality requires a good deal more topology and analysis than is part of the scope of this book. However, in [23, The Zariski Topology], we will see the Zariski Duality which studies a geometric-algebraic duality without the need for much analysis or topology

$\mathbb{R}$:

$$f + g \Leftrightarrow (f+g)(a) = f(a) + (g)$$
$$c \cdot f \Leftrightarrow (c \cdot f)(a) = cf$$

Now, $\int_a^b ___ dx$ takes in a function $f : \mathbb{R} \to \mathbb{R}$, an gives th number $\int_a^b f dx \in \mathbb{R}$. By basic properties of integarls, we know that:

$$\int_a^b (f+g)dx = \int_a^b f dx + \int_a^b g dx$$
$$\int_a^b cf dx = c\int_a^b f dx \qquad \forall a \in \mathbb{R}$$

and hence $\int_a^b ___ dx$ is indeed a linear functional! Hence $\int_a^b ___ dx \in L^1(\mathbb{R})^*$, for any $a, b \in \mathbb{R}$[b] The study of these functionals is part of functional analysis (*Functional Analysis* [47]).

2. Let $C([a,b])$ be the collection of all continuous maps from $[a,b]$ to $\mathbb{R}$, $f : [a,b] \to \mathbb{R}$, with addition and scalar multiplication defined pointwise. Then define: $\text{ev}_x : C([a,b]) \to \mathbb{R}$ to be

$$f \mapsto f(x)$$

Then it can be shown that ev_x is a linear transformation, called the *evaluation map*, and is thus a linear functional. More generally, as long as the codomain is a field $\mathbb{R}$, then the evaluation map is a linear functional.

3. A more advanced case of the above example occurs if we consider the set $C^1(\mathbb{R})$ of all differentiable functions[c] forms a vector-space with point-wise addition and scalar multiplicaiton. The operator $D_a : C^1(\mathbb{R}) \to \mathbb{R}$ takes in a differentiable function f and gives back $Df(a)$ (which in first year calculus is more commonly denoted $f'(a)$).

4. (If you know geometry or analysis) The following examples are not examples of dual spaces as we have defined them, but the generlization of them. In this generlization, the dual space we have defined would be called a *linear* dual space, or the dual space of in the category of $\mathbf{Vect}_k$. More generally, if X is an object in a category, then given a *dualizing* object D (often it is $\mathbb{R}$, $\mathbb{C}$, $\mathbb{C}^\times$, $\mathbb{Z}/n\mathbb{Z}$, or $\mathbb{R}/\mathbb{Z}$ depending on the context), then $\text{Hom}_C(X, D)$ is the dual space. In the following example, let D be either $\mathbb{R}$ or $\mathbb{C}$. Let's say X is either a vector-space over $\mathbb{R}$ or $\mathbb{C}$ (you should have function spaces in mind) or a manifold over $\mathbb{R}$ or $\mathbb{C}$. Then $C_0(X)$, $C^k(X)$, $L^p(X)$, an so forth are all subspace of the dual space X^* in the appropriate category.

Another place in which [ring] funcitonals shall become of paramount importance is in [23, Affine Space and Regular Functions] and [23, The Dictionary], where a duality is set up between algebraic and geometric objects.

[a] For the sake of this example, it does not matter, though L^1 usually denote *Lebesuge* measurable functions where $\int |f| < \infty$

[b] (if you know some analysis) Every *bounded* linear functional on $L^1(\mathbb{R})$ has the form $f \mapsto \int fg$ for a unique $g \in L^\infty(\mathbb{R})$ [47, Description Of Dual Of Lp]. The algebraic dual $L^1(\mathbb{R})^*$ used here also contains unbounded functionals.

[c] in particular, continuously differentiable functions. This distinction is of no importance for this example

In the above examples, all the vector-spaces are infinite dimensional. The vector-space being infinite dimensional will in fact greatly complicate the structure of dual spaces[19]. In general, more structure is added to a vector-space (ex. a norm) to have better "control" of infinite dimensional vector-spaces. Such discussion is left to a course on functional analysis[20]. We will mainly focus on the theory of finite dimensional vector spaces, since the results of this theory is in it of itself quite fruitful and provides a good basis for its study of dual spaces more generally.

Given a basis for V, it straight-forward to find a basis for $V^\vee$:

Definition 13.3.4: Dual Basis

Let $\beta = \{v_1, v_2, ..., v_n\}$ be a basis for the finite dimensional vector-space V. Then define $v_i^* \in V^*$ to be

$$v_i^*(v_j) = \begin{cases} 1 & \text{if } i = j \\ 0 & \text{if } i \neq j \end{cases}$$

for $1 \leq i, j \leq n$. Let β^* be the set of dual-basis elements

Since β^* has the word basis in it's name, you might guess that it is ineed a basis for the dual space:

Proposition 13.3.5: Dual Basis

Let β^* be defined as in definition 13.3.4. Then β^* is a basis for V^*

***Proof*:**

Note first that each v_i^* is well-defined: V is free on the basis β, so by corollary 12.3.14 a linear map out of V may be prescribed arbitrarily on β.

We show $\beta^* = \{v_1^*, \dots, v_n^*\}$ spans V^*. Let $f \in V^*$ and set $g := \sum_{i=1}^n f(v_i)\, v_i^*$. For each basis vector v_j,

$$g(v_j) = \sum_{i=1}^{n} f(v_i)\, v_i^*(v_j) = f(v_j)$$

since $v_i^*(v_j)$ is 1 when $i = j$ and 0 otherwise. Hence for any $v = \sum_j c_j v_j \in V$, linearity gives $g(v) = \sum_j c_j g(v_j) = \sum_j c_j f(v_j) = f(v)$, so $f = g \in \operatorname{span}(\beta^*)$.

For linear independence, suppose $\sum_{i=1}^n c_i v_i^* = 0$ in V^*. Evaluating at v_j gives

$$0 = \sum_{i=1}^{n} c_i v_i^*(v_j) = c_j$$

for every j, so all coefficients vanish. Hence β^* is linearly independent and spans V^*, that is, a basis.

As a consequence, V^* has the same dimension as V

[19] This is one of the analytic properties that was discussed in the footnote in page 464

[20] See *Functional Analysis* [47] and [31].

Proposition 13.3.6: Reflectivity Of Finite Dimensional Vector Spaces

Let V be a finite dimensional vector space. Then $V \cong V^*$.

***Proof*:**
By proposition 13.3.5, since V and V^* are of the same dimension, $V \cong V^*$.

An intuitive way of viewing the reflectively product to reference the intuition presented in the introduction of this section mentioning how functional represent *information*. Each linear function provides some information about a vector-space (prototypical examples would be the projection functions $\pi_{\beta,i} : V \to k$ picking a component of a vector given a choice of basis β). The type of information provided is restricted by the properties of the functions on V, in this case the co-domain is always k, and the functional is always linear. In finite dimension, these functions can be completely characterized by elements in V by the existence of the above isomorphism. The choice of the isomorphism is still an important factor to consider, since there is not natural basis or extra information that allows for favoring one isomorphism over another.

Since the structure of V isomorphic to V^*, it would be interesting if each map $f : V \to W$ can be associated to an appropriate "dual" function f^* where composition is respected, hopefully giving the existence of the functor $(-)^*$ from Vect_k to itself. To start, there is a notion of a dual function:

Definition 13.3.7: Dual map (Transpose)

Let $f : V \to W$ be a linear transformation. Then the *transpose* of f, denoted f^*, is defined as:

$$f^* : W^* \to V^* \qquad f^*(\varphi) = \varphi \circ f$$

A basis of a finite-dimensional V gives an isomorphism $V \cong V^*$ sending each basis vector to its dual functional, and changing the basis can change it: on $V = \mathbb{Q}$ the bases $\{1\}$ and $\{2\}$ give $v \mapsto (w \mapsto vw)$ and $v \mapsto (w \mapsto vw/4)$. So none of them is canonical. The transpose involves no choice. Note that the transpose of a function insures that maps that used to "out of" the space V will now "go into" the space V^* (and similarly for maps that go into V). Thus, this definition matches with our goal for V^* to be the dual of V, and furthermore is independent of any choice of basis. The name transpose comes from the matrix representation of f^*

Proposition 13.3.8: Matrix Representation Of Dual Map

Let $f : V \to W$ be a linear transformation and $A = (a_{ij})$ be the matrix representation of f given some basis β and γ for V and W respectively. Then if $f^* : W^* \to V^*$ is the dual map of f, where W^* and V^* have as basis γ^* and β^*, then the matrix representation A^* of f^* is:

$$A^* = [f^*]_{\gamma^*}^{\beta^*} = (a_{ji}) = A^t$$

Intuitively, in the matrix representation, each entry a_{ij} transposed with entry a_{ji}. For this reason, the map f^* is sometimes called the *transpose* of f.

***Proof*:**
Write $\beta = \{v_1, \ldots, v_n\}$ and $\gamma = \{w_1, \ldots, w_m\}$. By definition 13.2.1, the entries of A are determined by

$$f(v_i) = \sum_{k=1}^{m} a_{ki}\, w_k$$

Fix j and consider the functional $f^*(w_j^*) = w_j^* \circ f \in V^*$. Evaluating on each basis vector v_i gives

$$f^*(w_j^*)(v_i) = w_j^*(f(v_i)) = w_j^*\left(\sum_{k=1}^{m} a_{ki}\, w_k\right) = a_{ji}$$

since $w_j^*(w_k)$ is 1 when $k = j$ and 0 otherwise. The functional $\sum_{i=1}^{n} a_{ji}\, v_i^*$ takes the same value a_{ji} on each v_i, and two linear maps agreeing on the basis β agree on all of V by linearity. Hence

$$f^*(w_j^*) = \sum_{i=1}^{n} a_{ji}\, v_i^*$$

By definition 13.2.1 applied to f^* with domain basis γ^* and codomain basis β^*, the (i, j) entry of $[f^*]_{\gamma^*}^{\beta^*}$ is the coefficient of v_i^* in $f^*(w_j^*)$, which is a_{ji}. Hence $A^* = (a_{ji}) = A^t$.

As a corollary, if A is a nonsingular map $f : V \to W$, then the matrix representation for the map $(f^{-1})^* : V^* \to W^*$ is $(A^{-1})^t$.

Similarly to how the kernel of a linear transformation f provides the needed information to deduce the structure of the image of f, we get the dual equivalent for f^*, given an appropriate "dual" for the kernel:

Definition 13.3.9: Annihilator Of Dual Map

$U \subseteq V$ be a subspace of V. Then the *annihilator* of U is:

$$U^o := \{f \in V^* \mid f(U) = 0\}$$

Note that $V^0 = \{0\}$ and $\{0\}^0 = V$. Another common notation is $U^\perp$, which is pronounced "U perp", which shows how U^0 is related to the orthogonal complement of U. That relation needs an inner product, and is carried out in *Linear Algebra* [21].

Lemma 13.3.10: Properties Of Annihilator

Let $U \subseteq V$ be a subspace. Then:

1. $U^0 \subseteq V^*$ is a subspace
2. if $\dim(V) < \infty$, then $\dim(V) = \dim(U) + \dim(U^0)$

***Proof*:**

1. The zero functional vanishes on U, so $0 \in U^0$ and U^0 is nonempty. If $f, g \in U^0$ and $c \in k$, then for every $u \in U$,

$$(f + cg)(u) = f(u) + cg(u) = 0 + c \cdot 0 = 0$$

so $f + cg \in U^0$. Hence U^0 is closed under addition and scalar multiplication, and is a subspace of V^*.

2. Let $n = \dim(V)$ and consider the restriction map

$$\rho : V^* \to U^* \qquad \rho(f) = f|_U$$

which is linear since restriction commutes with addition and scalar multiplication of functionals. Its kernel is exactly the set of functionals vanishing on U, that is, $\ker(\rho) = U^0$.

We claim ρ is surjective. First, U is finite dimensional: a basis $u_1, \dots, u_k$ of U exists by theorem 13.1.8, and it is a linearly independent subset of V, so $k \leq n$ by corollary 13.1.13. By theorem 13.1.8 again, the linearly independent set $\{u_1, \dots, u_k\}$ is contained in a basis B of V. Given $g \in U^*$, define $f : V \to k$ on the basis B by $f(u_i) = g(u_i)$ for $1 \leq i \leq k$ and $f(b) = 0$ for the remaining $b \in B$, extended linearly, which is possible by corollary 12.3.14. Then $f|_U$ and g are linear maps agreeing on the basis $u_1, \dots, u_k$ of U, so $\rho(f) = g$, proving surjectivity. Hence the image of ρ is U^*.

By proposition 13.3.5, $\dim(V^*) = n$ and $\dim(U^*) = \dim(U)$. Applying the Rank-Nullity theorem (corollary 13.1.18) to ρ gives

$$\dim(V) = \dim(V^*) = \dim(\ker(\rho)) + \dim(\text{im}(\rho)) = \dim(U^0) + \dim(U)$$

as we sought to show.

Proposition 13.3.11: Properties Of Dual Maps

Let V and W be (not necessarily finite dimensional) vector space over k, $f \in \text{Hom}_k(V, W)$ so that $f^* \in \text{Hom}_k(W^*, V^*)$ is its dual map. Then:

1. $\ker(f^*) = (\text{im}(f))^0$ (implying f^* injective if and only if f surjective)
2. $\text{im}(f^*) \subseteq (\ker(f))^0$ is a subspace
3. If $\dim(V), \dim(W) < \infty$, then $\text{im}(f^*) = (\ker(f))^0$ (implying f^* surjective if and only if f injective)

***Proof*:**
Let X be a vector space over k, let $U \subseteq X$ be a subspace, and choose a basis B_U of U (theorem 13.1.8).

Fact (a): every $h \in U^*$ extends to X. Extend B_U to a basis B of X (theorem 13.1.8), and let $\tilde{h} \colon X \to k$ be the linear map with $\tilde{h}(b) = h(b)$ for $b \in B_U$ and $\tilde{h}(b) = 0$ for $b \in B \setminus B_U$

(corollary 12.3.14). Then $\tilde{h}|_U$ and h are linear maps agreeing on the basis B_U of U, so $\tilde{h}|_U = h$.

Fact (b): if $x \in X \setminus U$, then some $\lambda \in U^0$ has $\lambda(x) = 1$. The set $B_U \cup \{x\}$ is linearly independent: in a relation $cx + \sum_b c_b b = 0$ with $b \in B_U$, the coefficient c is 0, since otherwise $x = -c^{-1} \sum_b c_b b \in U$, and then every c_b is 0. Extend $B_U \cup \{x\}$ to a basis B of X, and let λ be the linear map with $\lambda(x) = 1$ and $\lambda(b) = 0$ for the other $b \in B$. Then λ vanishes on the basis B_U of U, so $\lambda \in U^0$.

1. For $\psi \in W^*$, $f^*(\psi) = \psi \circ f$ is the zero functional exactly when $\psi(f(v)) = 0$ for every $v \in V$, that is, when ψ vanishes on $\mathrm{im}(f)$. So $\ker(f^*) = (\mathrm{im}(f))^0$. If f is surjective, then $\ker(f^*) = W^0 = \{0\}$ and f^* is injective. If f is not surjective, Fact (b) with $X = W$, $U = \mathrm{im}(f)$ and some $x \in W \setminus \mathrm{im}(f)$ gives a nonzero element of $(\mathrm{im}(f))^0 = \ker(f^*)$, and f^* is not injective.
2. The image of f^* is a subspace of V^*, since $f^*(\psi)+cf^*(\psi') = f^*(\psi+c\psi')$ for $\psi, \psi' \in W^*$ and $c \in k$. For $\psi \in W^*$ and $v \in \ker(f)$, $f^*(\psi)(v) = \psi(f(v)) = \psi(0) = 0$, so $f^*(\psi) \in (\ker(f))^0$.
3. By (2) it suffices to show $(\ker(f))^0 \subseteq \mathrm{im}(f^*)$. Let $\lambda \in (\ker(f))^0$ and define $h\colon \mathrm{im}(f) \to k$ by $h(f(v)) = \lambda(v)$. It is well defined: if $f(v) = f(v')$, then $v - v' \in \ker(f)$, so $\lambda(v) - \lambda(v') = \lambda(v - v') = 0$. It is linear, since $h(f(v) + cf(v')) = h(f(v + cv')) = \lambda(v + cv') = h(f(v)) + ch(f(v'))$. By Fact (a) with $X = W$ and $U = \mathrm{im}(f)$, h extends to some $\psi \in W^*$, and then $f^*(\psi)(v) = \psi(f(v)) = h(f(v)) = \lambda(v)$ for every $v \in V$, so $\lambda = f^*(\psi)$. Hence $\mathrm{im}(f^*) = (\ker(f))^0$. If f is injective, this is $\{0\}^0 = V^*$, and f^* is surjective. If f is not injective, choose $0 \neq v \in \ker(f)$. Fact (b) with $X = V$, $U = \{0\}$ and $x = v$ gives $\mu \in V^*$ with $\mu(v) = 1$, so $\mu \notin (\ker(f))^0 = \mathrm{im}(f^*)$, and f^* is not surjective, completing the proof.

The first point in proposition 13.3.11 is particularly useful in that in infinite dimensions, f being injective *does not* imply f is surjective. In the right situation, checking the injectivity of a linear transformation is *much* easier than checking surjectivity, and so checking the injectivity of f^* is often advantageous.

The next topic we will cover might at first seem abstract, however it is an important demonstration of the concept of *naturality*. Since we have taken $\mathrm{Hom}_k(V, k)$ and have shown it is a vector-space, we can take the dual of it again:

Definition 13.3.12: Double Dual

Let V be a vector-space and V^* it's dual. Then the *double-dual* of V is the dual of V^*, denoted V^{**}

The double dual has the property of being *naturally isomorphic* to V. To explain the idea of a natural isomorphism, we will show how V is *not* naturally isomorphic to V^*:

Example 13.3.13: Double Dual Motivation

Let V be a finite-dimensional vector space over k and $V^* = \mathrm{Hom}(V, \mathbb{F})$. If $\beta = \{e_1, ..., e_n\}$ is a basis for V, and $\beta^* = \{e_1^*, ..., e_n^*\}$ is the dual basis for V^* where:

$$e_j^*(e_i) = \begin{cases} 1 & i = j \\ 0 & i \neq j \end{cases}$$

Then the maps $e_i \mapsto e_i^*$ are bijective and linear, meaning V and V^* are isomorphic ($V \cong V^*$).

We can further consider $V^{**} = \mathrm{Hom}(\mathrm{Hom}(V, \mathbb{F}), \mathbb{F})$ being the set of all linear maps mapping all elements form $\mathrm{Hom}(V, \mathbb{F})$ to $\mathbb{F}$. We can in similar fashion make an isomorphism $V^* \cong V^{**}$ letting us conclude that $V \cong V^{**}$.

This whole construction started with a choice of basis form V, which need not be unique, and so it's not a "property" of V. However, we can still define an isomorphism between V and V^{**} *without* any reference to a basis. For any $v \in V$, we can define a map

$$(f \mapsto f(v)) : V^* \to \mathbb{F} \qquad (f \mapsto f(v))(g) = g(v)$$

that is, every functional $g \in V^*$ evaluates the value and gets us $g(v)$. This map is sometimes called the evaluation function. Let's switch up the notation for simplicity:

$$ev_v : V^* \to \mathbb{F} \qquad ev_v(g) = g(v)$$

where ev_v stands for evaluate at v. Now, the map $ev \colon V \to \mathrm{Hom}(V^*, \mathbb{F})$, $v \mapsto ev_v$, is an *isomorphism*. For linearity, let $k \in \mathbb{F}$ and $v, w \in V$:

$$ev(k \cdot v) = ev(kv) = ev_{kv} = k \cdot ev_v = kev(v)$$

$$ev(v + w) = ev_{v+w} = ev_v + ev_w = ev(v) + ev(w)$$

where kv represents the vector we get when applying the ring action on v. Note that we essentially used the linearity of the object we are mapping to in order to prove linearity. The map ev is injective: if $ev_v = 0$ and $v = \sum_i c_i e_i$, then $c_j = e_j^*(v) = ev_v(e_j^*) = 0$ for every j, so $v = 0$. By proposition 13.3.5, applied to V and then to V^*, $\dim V^{**} = \dim V^* = \dim V$, so the injective linear map ev is bijective (corollary 13.1.20).

Thus, we defined this isomorphism with *no* reference to a basis on V. We cannot do the same trick for $V \to V^*$, since if we try to do $v \mapsto f(v)$, it is ambiguous which f is meant.

The intuition behind the double-dual having a natural isomorphism is that there is more information to work with. The reader should ponder on what extra information has been provided.

13.3.2 Prelude to Adjoints of Vector Space

Now that we have the basic structure of (finite dimensional) dual spaces, we may return to the discussion of defining V internally or externally. Let's say that you have been given $\mathrm{Hom}_k(V, k)$, but not the structure of V. Is the information in $\mathrm{Hom}_k(V, k)$ sufficient to recover the structure of V? The answer is almost: the following proposition shows how much information we can recover:

Proposition 13.3.14: Recovering V From Its Dual

Let V be some vector-space and V^* its dual.

1. If $v, w \in V$ satisfy $f(v) = f(w)$ for all $f \in V^*$, then
$$v = w$$
that is, the linear map $\mathrm{ev}\colon V \to V^{**}$, $v \mapsto \mathrm{ev}_v$ of example 13.3.13 is injective
2. If V is finite dimensional, then $\mathrm{ev}\colon V \to V^{**}$ is an isomorphism

***Proof*:**

1. Let $u = v - w$ and suppose $u \neq 0$. The set $\{u\}$ is linearly independent, since $cu = 0$ with $c \neq 0$ would give $u = c^{-1}cu = 0$. Extend it to a basis B of V (theorem 13.1.8) and let $f\colon V \to k$ be the linear map with $f(u) = 1$ and $f(b) = 0$ for $b \in B \setminus \{u\}$ (corollary 12.3.14). Then $f(v) - f(w) = f(u) = 1 \neq 0$, contradicting the hypothesis. Hence $u = 0$, that is, $v = w$. Since $\mathrm{ev}_v = \mathrm{ev}_w$ means exactly that $f(v) = f(w)$ for every $f \in V^*$, this is the injectivity of ev.
2. The map ev is linear: for $v, w \in V$, $c \in k$ and $f \in V^*$, $\mathrm{ev}_{v+cw}(f) = f(v) + cf(w) = (\mathrm{ev}_v + c\mathrm{ev}_w)(f)$. It is injective by (1). Applying proposition 13.3.5 to V and then to V^* gives $\dim V^{**} = \dim V^* = \dim V < \infty$, so ev is an injective linear map between vector spaces of the same finite dimension, hence an isomorphism (corollary 13.1.20), completing the proof.

13.3.3 Functor of Points for Vector Spaces

Another way to bring in the intuition from the introduction is to take the dual of a functional $V \to k$, namely to think of elements of V as functions $k \to V$. Every linear transformation $g\colon k \to V$ is of the form
$$t \mapsto tx$$
with $x = g(1)$, since $g(t) = g(t \cdot 1) = tg(1)$, and for every $x \in V$ this formula defines a linear map. Represent these transformations as $g_x : k \to V$. Taking the dual of g_x we get:

$$\begin{aligned} g_x^*(\varphi) &= \varphi \circ g_x \\ &\equiv \varphi(g_x(t)) && \forall t \in k \\ &= \varphi(tx) && \forall t \in k \\ &= t\varphi(x) && \forall t \in k \\ &\equiv g_{\varphi(x)} \end{aligned}$$

Hence g_x^* maps φ to $g_{\varphi(x)} : k \to k$ where $t \mapsto t\varphi(x)$. With the identification between elements of V and the linear transformations $k \to V$, we may interpret this as a functional $V^* \to k$ where:
$$\varphi \mapsto \varphi(x)$$

Given $V \cong V^*$, then if β is the basis defining the isomorphism, each $\varphi \in V^*$ is associated to a vector $v \in V$, namely the vector for which:

$$\begin{pmatrix} \beta_1 \\ \beta_2 \\ \vdots \end{pmatrix} \mapsto \sum_i v_i \beta_i$$

where $\sum_i$ is a finite sum. Then the linear functional $V \to k$ is defined by:

$$\begin{pmatrix} v_1 \\ v_2 \\ \vdots \end{pmatrix} \mapsto \sum_i v_i x_i$$

where again $\sum_i$ is a finite sum since only finitely many v_i (and x_i) are nonzero Let's label this functional f_x. The element $x \in V$ may be interpreted, functionally, as $t \to tx$ or $v \to \sum_i v_i x_i$. Geometrically, the first function can loosely be thought of as defining a magnitude and direction. This is *not* a correct interpretation in infinite dimensions in general, since a magnitude requires a norm and a bare vector space carries none, however it is valid in finite dimension[21]. The second function can is the "dual" notion, where instead of focusing on x, the focus is on how x manipulates every other element. The reader may want to ponder how this relates to the "internal" and "external" way of describing object mentioned at the beginning of section 13.3. The first description, $x \mapsto (\varphi \mapsto \varphi(x))$, is the evaluation map $\mathrm{ev}\colon V \to V^{**}$, which is injective and involves no choice (proposition 13.3.14(1)). The second, $x \mapsto f_x$, is an injective linear map $V \to V^*$, since $f_x(\beta_j) = x_j$ vanishes for every j only when $x = 0$, but it depends on the basis β.

As mentioned, it is not always the case that $V \cong V^*$. This immediately falls apart if V is infinite dimensional. To demonstrate this, let us consider the duality presented above where each $x \in V$ is associated with the functional f_x. The key is that given any basis β for V, $[x]_\beta$ has only finitely many components. Hence, the following valid function $F \in V^*$ is not associated to any f_x:

$$[v]_\beta \to \sum_i v_i$$

where the sum is finite since v when represented as an infinite tuple must only have finitely many nonzero components. The functional F is associated to the sequence $(1, 1, ..., 1, ...)$, namely the sequence where 1 is in each position. No f_x can be associated to F since by construction f_x, when represented as an infinite tuple, must have all but finitely position zero. This immediately shows that V^* is much larger, and hence contains much more information than the elements of V can represent. In fact, V^* contains even more interesting functionals. Consider:

$$(v_1, v_2, ..., v_n, ...) \mapsto (0, v_1, ..., v_{n-1}, ...)$$

Notationally, this may be represented as $f(\beta_i) = \beta_{i+1}$ given a basis β for V. This is called he *right shift operator*. This linear transformation can only be defined since there are infinite dimensions, hence each component may shift over by 1.

There are in fact so many linear functionals on infinite dimensional vector spaces that the study of such spaces is almost always accompanied with additional structure to bring some more regularity to the elements. The core problem stems from the extra flexibility working with infinitely many dimensions entails. Hence, some form of "finiteness" is usually imposed via the introduction of a *norm*,

[21] This interpretation holds true in Hilbert Spaces

for instance one induced by an *inner product*. The transpose makes $(-)^*$ a contravariant functor, since $(g \circ f)^* = f^* \circ g^*$, and it involves no basis. For a finite-dimensional real inner product space V, $v \mapsto \langle -, v\rangle$ is an isomorphism $V \cong V^*$ that depends on no basis. For a linear map $L\colon V \to W$ between finite-dimensional real inner product spaces, transporting L^* through these isomorphisms gives a linear map $L^\dagger\colon W \to V$, the *adjoint* of L, whose matrix in orthonormal bases is $[L]^t$. The exploration of these notions is very fruitful even in finite dimensions, and it is carried out in *Linear Algebra* [21].

Exercise 13.3.1

1. Let V be finite dimensional and $U \subseteq V$ a subspace. Show that $U^{00} \cong U$. If $\widehat{(___)}$ is the natural embedding of V into V^{**}, show that $\hat{U} = U^{00}$.
2. Let V be a finite dimensional vector space and $U_1, U_2 \subseteq V$ subspaces. Show that:
    1. $(U_1 + U_2)^0 = U_1^0 \cap U_2^0$
    2. $(U_1 \cap U_2)^0 = U_1^0 + U_2^0$
    3. if $W \subseteq V$ is a subspace, then $(V/W)^* \cong W^0$
3. Show that $(-)^*$ is a contravariant functor from the category of vector spaces over k to itself, that is, $(g \circ f)^* = f^* \circ g^*$ for linear maps $f\colon U \to V$ and $g\colon V \to W$, and $(\mathrm{id}_V)^* = \mathrm{id}_{V^*}$.

13.3.4 Multilinear Forms

Studying linear functionals allow us to give some scalar value to vectors. However, this is sometimes not enough: it is very common in mathematics that one of the two scenarios occur (we will see examples of these in this and coming section[22])

1. We want to study many functional on spaces simultaneously (ex. both V and V^*)
2. We require many copies of the same space to have enough "data" to define the desired functional

In this section, we elaborate on the theory allowing us to do just that, give some examples elaborating the above intuitions, and in the coming section we will see how these are of great use in defining some invariants for linear transformations and bringing in geometry into vector-spaces:

[22]There are also many examples in differential geometry, such as covariant/contravariant/mixed tensor fields on a manifold

Definition 13.3.15: Bilinear and Multilinear Maps

Let V and W be vector spaces over k. Then a map $\varphi : V \times W \to k$ is called *bilinear* if

1. for any $v \in V$, the map $\varphi(v, \cdot) : W \to k$ mapping $w \mapsto \varphi(v, w)$ is linear
2. for any $w \in W$, the map $\varphi(\cdot, w) : V \to k$ mapping $v \mapsto \varphi(v, w)$ is linear.

More generally, if $V_1, V_2, ..., V_n$ are vector-space sover k, then $\varphi : V_1 \times V_2 \times \cdots \times V_n \to k$ is *multilinear* (or n-linear if the number of variables is known) if for all i, $1 \leq i \leq n$, given fixed $v_1, v_2, ..., v_{i-1}, v_{i+1}, ..., v_n$, the map

$$\varphi(v_1, v_2, ..., v_{i-1}, \cdot, v_{i+1}..., v_n)$$

is linear. Two important types of multilinear maps are:

- (Symmetric) for any $v_i, v_j \in V$:

$$\varphi(v_1, ..., v_i, ..., v_j, ..., v_n) = \varphi(v_1, ..., v_j, ..., v_i, ..., v_n)$$

- (Alternating) for any $v_i, v_j \in V$:

$$\varphi(v_1, ..., v_i, ..., v_j, ..., v_n) = -\varphi(v_1, ..., v_j, ..., v_i, ..., v_n)$$

13.3.16 WARNING Since we are now dealing with multilinear maps, these are *no longer morphisms in the category of* Vect_k! Hence, multilinear maps are a different object of study. In chapter 15, we will show how we can study multilinear maps via linear algebra.

Example 13.3.17: Bilinear and Multilinear Maps

1. Let $\varphi : V \times V^* \to k$ be defined by $(v, f) \mapsto f(v)$. Then φ is bilinear
2. Let $V = \mathbb{R}^n$ be a vector-space over $\mathbb{R}$ and take $\varphi : \mathbb{R}^n \times \mathbb{R}^n \to \mathbb{R}$ to be defined as:

$$\varphi\left(\begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix}, \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{pmatrix}\right) = \sum_{i=1}^{n} x_i y_i$$

   Then φ is bilinear, and is called the *dot product* or *scalar product* of two vectors $x, y \in \mathbb{R}^n$
3. Let $\mathbb{R}$ be a vector-space over itself, and take $V : \mathbb{R} \times \mathbb{R} \times \cdots \times \mathbb{R} \to \mathbb{R}$. Then the map that multiplies all the elements:

$$V(x_1, x_2, ..., x_n) = x_1 x_2 \cdots x_n$$

   is a n-linear. As a paticualr example, consider $\varphi : \mathbb{R} \times \mathbb{R} \times \mathbb{R} \to \mathbb{R}$, and imagine any 3-tulpe (x_1, x_2, x_3) represents the length, width, and height of a shape. Then $V(x_1, x_2, x_3) = x_1 x_2 x_3$ will give the volume of a rectangular prism (or a cube if $x_1 = x_2 = x_3$). The fact that

multiplication produces a natural multi-linear map means that multilienar maps show up a lot in geometry. it is for this reason that multilinear maps are sometimes called *volume forms*. We will see more of these in section 16.3.

4. (If you know some calculus) Let $V = C([0,1], \mathbb{R})$ be the set of all continuous functions from $[0,1]$ to $\mathbb{R}$. Then define $\varphi : V \times V \to \mathbb{R}$ to be:

$$\varphi(f,g) = \int_0^1 f(t)g(t)\,dt$$

Then φ is bilinear. This is the infinite dimensional equivalent of the dot product.

In the rest of this section, we will focus on Bilinear forms: the results can easily be generlized using induction to Mutilinear forms. The first thing worth trying to find is wether there is still a reasonable notion of matrix representaiton of bilinear forms, since they are no longer linear maps. The following definition and lemma give us jut that:

Definition 13.3.18: Bilinear Form Matrix Representation

Let V and W be finite dimensional vector spaces over k of dimension n and m and with basis β, γ respectively. Let $\varphi : V \times W \to k$ a bilinear form. Then define the *matrix representation* of φ to be:

$$[\varphi]_{\beta,\gamma} := (\varphi(\beta_i, \gamma_j))_{\substack{1 \le i \le n \\ 1 \le j \le m}}$$

Lemma 13.3.19: Bilinear Form Matrix Representation

Let V and W be finite dimensional vector spaces over k of dimension n and m and with basis β, γ respectively. Let $\varphi : V \times W \to k$ a bilinear form. Then

$$\varphi(v,w) = [v]_\beta^t [\varphi]_{\beta,\gamma} [w]_\gamma$$

***Proof*:**
Write $v = \sum_{i=1}^n a_i \beta_i$ and $w = \sum_{j=1}^m b_j \gamma_j$, so that $[v]_\beta$ has entries a_i and $[w]_\gamma$ has entries b_j. Bilinearity of φ expands each argument separately:

$$\varphi(v,w) = \varphi\Big(\sum_i a_i\beta_i,\ \sum_j b_j\gamma_j\Big) = \sum_{i=1}^n \sum_{j=1}^m a_i b_j\, \varphi(\beta_i, \gamma_j)$$

On the other side, the (i,j) entry of $[\varphi]_{\beta,\gamma}$ is $\varphi(\beta_i, \gamma_j)$, so the matrix product is $[v]_\beta^t [\varphi]_{\beta,\gamma} [w]_\gamma = \sum_i \sum_j a_i\, \varphi(\beta_i, \gamma_j)\, b_j$. The two sums agree term by term.

Definition 13.3.20: Orthogonal

Let $\varphi : V \times V \to k$ be a bilinear form and $S \subseteq V$ a *subset*. Then the set:

$$S^{\perp} := \{y \in V \mid \varphi(x, y) = 0 \text{ for every } x \in S\}$$

is called the *orthogonal complement* of S. If $\varphi(x, y) = 0$, then x is *perpendicular* to y, written $x \perp y$. If $x \perp y$ for all $y \in S$, then $x \perp S$.

The orthogonal complement is always a subspace, whatever the subset S was.

Lemma 13.3.21: The Orthogonal Complement Is a Subspace

Let $\varphi\colon V \times V \to k$ be a bilinear form and $S \subseteq V$ any subset. Then $S^{\perp}$ is a subspace of V, and $S^{\perp} = (\operatorname{span} S)^{\perp}$.

***Proof*:**

Linearity of φ in its second argument gives $\varphi(x, 0) = 0$ for every x, so $0 \in S^{\perp}$. If $y, y' \in S^{\perp}$ and $a \in k$, then for every $x \in S$

$$\varphi(x, y + ay') = \varphi(x, y) + a\,\varphi(x, y') = 0 + a \cdot 0 = 0$$

so $y + ay' \in S^{\perp}$, and $S^{\perp}$ is a subspace.

For the second claim, $S \subseteq \operatorname{span} S$ forces $(\operatorname{span} S)^{\perp} \subseteq S^{\perp}$. Conversely let $y \in S^{\perp}$ and let $x = \sum_{i=1}^{m} a_i x_i$ with $x_i \in S$. Linearity in the first argument gives $\varphi(x, y) = \sum_i a_i \varphi(x_i, y) = 0$, so $y \in (\operatorname{span} S)^{\perp}$, as we sought to show.

Taking S to be all of V produces the subspace on which the form carries no information at all, and the form deserves a name according to whether that subspace is trivial.

Definition 13.3.22: The Radical and Nondegeneracy

Let $\varphi\colon V \times V \to k$ be a bilinear form. The *radical* of φ is the subspace

$$\operatorname{rad}(\varphi) := V^{\perp} = \{y \in V \mid \varphi(x, y) = 0 \text{ for every } x \in V\}$$

The form φ is *nondegenerate* if $\operatorname{rad}(\varphi) = 0$, and *degenerate* otherwise.

For a symmetric form the radical is the same computed on either side, so no ambiguity arises from having used the second argument. The radical is detected by the matrix of the form in any single basis, which is the form in which nondegeneracy is checked in practice.

Lemma 13.3.23: Nondegeneracy and the Matrix of the Form

Let V be finite dimensional with basis β, let $\varphi\colon V \times V \to k$ be a bilinear form, and let $A = [\varphi]_{\beta,\beta}$ be its matrix (definition 13.3.18). Then

$$\operatorname{rad}(\varphi) = \left\{ y \in V \;\middle|\; A^t[y]_\beta = 0 \right\}$$

and consequently φ is nondegenerate if and only if A is invertible. The criterion does not depend on which basis β was chosen, since the left-hand side does not.

***Proof*:**
By lemma 13.3.19, $\varphi(x,y) = [x]_\beta^t A[y]_\beta$ for all $x, y \in V$. Fix y. Then $y \in \operatorname{rad}(\varphi)$ means $[x]_\beta^t A[y]_\beta = 0$ for every $x \in V$, and as x ranges over V the coordinate vector $[x]_\beta$ ranges over all of k^n. A vector $u \in k^n$ satisfies $w^t u = 0$ for every $w \in k^n$ if and only if $u = 0$, so the condition is $A[y]_\beta = 0$. Transposing the defining relation gives the displayed form. Hence $\operatorname{rad}(\varphi) = 0$ if and only if A has trivial kernel, which for a square matrix is invertibility.

The final sentence needs no comparison of bases: $\operatorname{rad}(\varphi)$ was defined without reference to any basis, so if A is invertible for one choice of β the radical is zero, and then the same equivalence run backwards makes the matrix invertible for every other choice, completing the proof.

Exercise 13.3.2

1. Let $f\colon V \times W \to k$ be a bilinear form on a pair of spaces, and write $W^\perp := \{v \in V \mid f(v,w) = 0 \text{ for all } w \in W\}$ and $V^\perp := \{w \in W \mid f(v,w) = 0 \text{ for all } v \in V\}$ for its two kernels. Show that f descends to a well-defined bilinear form $\bar{f}\colon V/W^\perp \times W/V^\perp \to k$, and that $\bar{f}$ is non-degenerate in each variable.

13.4 Linear Operators Invariants: Trace and Determinant

In this section, we will go over to common functionals on $M_n(k)$ (equivalently, $\operatorname{End}_k(V)$ for an n-dimensional vector-space) that give more insight on linear transformations. A good chunk of the following theory works well over more general rings, and so we will sometimes work over a general commutative ring R.

The first functional on $M_n(k)$ we will define is called the *trace*, and it takes the same value on similar matrices (that is, on matrices related by a change of basis):

Definition 13.4.1: Trace

Let R be a commutative ring and $M_n(R)$ the collection of all $n \times n$ matrices over R. Then $\operatorname{tr} : M_n(R) \to R$ is the map:

$$\operatorname{tr}(A) = \sum_{i=1}^{n} a_{ii}$$

The trace is studied for the following very useful properties:

Proposition 13.4.2: Properties of Trace

Let R be a commutative ring and $A, B \in M_n(R)$. Then:

1. $\text{tr}(A) = \text{tr}(A^t)$
2. $\text{tr}(AB) = \text{tr}(BA)$

***Proof*:**
Write $A = (a_{ij})$ and $B = (b_{ij})$.

1. The (i,i) entry of $A^t = (a_{ji})$ is a_{ii}, so $\text{tr}(A^t) = \sum_{i=1}^n a_{ii} = \text{tr}(A)$.
2. The (i,i) entry of AB is $\sum_{l=1}^n a_{il}b_{li}$ and the (l,l) entry of BA is $\sum_{i=1}^n b_{li}a_{il}$, so

$$\text{tr}(AB) = \sum_{i=1}^n \sum_{l=1}^n a_{il}b_{li} \qquad \text{tr}(BA) = \sum_{l=1}^n \sum_{i=1}^n b_{li}a_{il}$$

Since R is commutative, $a_{il}b_{li} = b_{li}a_{il}$ for all i and l, and the two finite double sums have the same terms. Hence $\text{tr}(AB) = \text{tr}(BA)$, as claimed.

As a consequence, if A and B are similar so that there exists a P such that $A = PBP^{-1}$, then:

$$\text{tr}(A) = \text{tr}(PBP^{-1}) = \text{tr}(P(BP^{-1})) = \text{tr}((BP^{-1})P) = \text{tr}(B(P^{-1}P)) = \text{tr}(B)$$

Hence the trace is invariant under change of basis, making it a good tool to study linear maps without needing to pick a basis. Let V be a finite dimensional vector space over k and $f\colon V \to V$ a linear map. If β and γ are bases of V, then the matrices $[f]_\beta^\beta$ and $[f]_\gamma^\gamma$ of f with respect to β and to γ are similar by lemma 13.2.16(4), so they have the same trace, and the following definition does not depend on the basis:

Definition 13.4.3: Trace Of Linear Map

Let V be a finite dimensional vector space and $f : V \to V$ a linear map, and let β be a basis of V. Then define $\text{tr}(f)$ to be

$$\text{tr}(f) := \text{tr}([f]_\beta^\beta)$$

the trace of the matrix of f with respect to β in both the domain and the codomain.

Corollary 13.4.4: Trace Of Dual

Let V be a finite dimensional vector space, $f : V \to V$ a linear map and $f^* : V^* \to V^*$ its dual map. Then:

$$\text{tr}(f) = \text{tr}(f^*)$$

***Proof*:**
Let β be a basis of V, let β^* be the dual basis of V^* (proposition 13.3.5), and let $A = [f]_\beta^\beta$, so that $\text{tr}(f) = \text{tr}(A)$ by definition 13.4.3. By proposition 13.3.8, applied with $W = V$ and $\gamma = \beta$, the

matrix of f^* with respect to β^* in both the domain and the codomain is A^t. So $\operatorname{tr}(f^*) = \operatorname{tr}(A^t)$ by definition 13.4.3, computed in the basis β^*, and $\operatorname{tr}(A^t) = \operatorname{tr}(A)$ by proposition 13.4.2(1). Hence $\operatorname{tr}(f^*) = \operatorname{tr}(f)$, completing the proof.

The next map is a very common multilinear form that will be used extensively: this form will also be independnet of basis, but with the extra information giving a very clear geometric picture of how a linear transformation effects a 1-cube in $\mathbb{R}^n$:

Definition 13.4.5: Determinant

Let R be a commutative ring and $A = (a_{ij}) \in M_n(R)$ a square matrix. then the *determinant* of A is a map $\det : M_n(R) \to R$:

$$\det(A) := \sum_{\sigma \in S_n} (-1)^\sigma \prod_{i=1}^{n} a_{i\sigma(i)} \in R$$

Proposition 13.4.6: Properties Of The Determinant

Let R be a commutative ring and $A \in M_n(R)$. Then:

1. $\det(I_n) = 1$.
2. $\det(A) = \det(A^t)$, here $(a_{ij})^t = (a_{ji})$.
3. (multilinearity) det is R-linear in each column separately: if the j-th column of A is $cu + c'u'$ for $c, c' \in R$ and column vectors u, u', and if A_u, $A_{u'}$ denote the matrices agreeing with A outside column j and having u, respectively u', in column j, then

   $$\det(A) = c \det(A_u) + c' \det(A_{u'})$$

   By (2) the same holds for rows.
4. (alternating) If two columns of A are equal then $\det(A) = 0$. The same holds for rows. Combined with (3), if one column is a scalar multiple of another then $\det(A) = 0$.
5. (permuting rows) If A_τ denotes the matrix whose i-th row is the $\tau(i)$-th row of A, for $\tau \in S_n$, then $\det(A_\tau) = (-1)^\tau \det(A)$.

***Proof*:**
All five are read off the defining sum.

1. For $A = I_n$ the product $\prod_i \delta_{i\sigma(i)}$ vanishes unless $\sigma(i) = i$ for every i, so only $\sigma = \mathrm{id}$ contributes, and it contributes 1.
2. By definition $\det(A^t) = \sum_\sigma (-1)^\sigma \prod_i a_{\sigma(i)i}$. Re-indexing the product by $j = \sigma(i)$ gives $\prod_i a_{\sigma(i)i} = \prod_j a_{j\sigma^{-1}(j)}$, and $(-1)^\sigma = (-1)^{\sigma^{-1}}$ since a permutation and its inverse have the same parity. As σ runs over S_n so does σ^{-1}, so the two sums agree term for term.
3. In the term indexed by σ, exactly one factor is drawn from column j, namely $a_{\sigma^{-1}(j)\,j}$.

Each term is therefore R-linear in column j with the other columns held fixed, and a sum of such terms is too.

4. Suppose columns $j \neq k$ of A are equal, and let $\tau = (j\ k)$. Since $a_{ij} = a_{ik}$ for all i, one has $a_{i\,\tau(m)} = a_{im}$ for every i and m, and hence

$$\prod_i a_{i\,\tau\sigma(i)} = \prod_i a_{i\sigma(i)}$$

while $(-1)^{\tau\sigma} = -(-1)^{\sigma}$. The assignment $\sigma \mapsto \tau\sigma$ is an involution of S_n without fixed points, so it partitions S_n into pairs whose two terms are negatives of one another. The sum is therefore 0. Note that this argument cancels terms exactly and so does not require 2 to be invertible in R. The statement for rows follows by (2), and the statement for proportional columns by pulling the scalar out with (3).

5. By definition $\det(A_\tau) = \sum_\sigma (-1)^\sigma \prod_i a_{\tau(i)\sigma(i)}$. Substituting $j = \tau(i)$ turns the product into $\prod_j a_{j\,\sigma\tau^{-1}(j)}$. Writing $\rho = \sigma\tau^{-1}$, so that $(-1)^\sigma = (-1)^\rho(-1)^\tau$, and letting ρ run over S_n as σ does,

$$\det(A_\tau) = (-1)^\tau \sum_{\rho \in S_n} (-1)^\rho \prod_j a_{j\rho(j)} = (-1)^\tau \det(A)$$

completing the proof.

The determinant is multiplicative, and this holds for all square matrices over a commutative ring, not only invertible ones. The proof is a direct expansion of the defining sum, with proposition 13.4.6(4) killing every term that is not indexed by a permutation.

Proposition 13.4.7: Multiplicativity of the Determinant

Let R be a commutative ring and let $A, B \in M_n(R)$. Then

$$\det(AB) = \det(A)\det(B)$$

***Proof*:**
Write $A = (a_{ij})$ and $B = (b_{ij})$, so that $(AB)_{ij} = \sum_k a_{ik}b_{kj}$. Expanding the defining sum,

$$\det(AB) = \sum_{\sigma \in S_n} (-1)^\sigma \prod_{i=1}^{n} \left(\sum_{k=1}^{n} a_{ik} b_{k\sigma(i)} \right)$$

Expanding the product over i amounts to choosing an index k for each i, that is, a function $f\colon \{1,\dots,n\} \to \{1,\dots,n\}$, so

$$\det(AB) = \sum_f \left(\prod_{i=1}^{n} a_{if(i)} \right) \left(\sum_{\sigma \in S_n} (-1)^\sigma \prod_{i=1}^{n} b_{f(i)\sigma(i)} \right)$$

the sum being over all n^n such functions. For fixed f, let B_f be the matrix whose i-th row is the $f(i)$-th row of B, so that $(B_f)_{ij} = b_{f(i)j}$. The inner sum is exactly $\det(B_f)$.

If f is not injective, say $f(i) = f(i')$ with $i \neq i'$, then B_f has two equal rows and $\det(B_f) = 0$ by proposition 13.4.6(4). Only the injective f survive, and an injective self-map of a finite set is a

permutation, so the sum runs over $f = \tau \in S_n$. For such an f, B_τ is B with its rows permuted by τ, so $\det(B_\tau) = (-1)^\tau \det(B)$ by proposition 13.4.6(5). Therefore

$$\det(AB) = \sum_{\tau \in S_n} \left(\prod_{i=1}^{n} a_{i\tau(i)} \right) (-1)^\tau \det(B) = \left(\sum_{\tau \in S_n} (-1)^\tau \prod_{i=1}^{n} a_{i\tau(i)} \right) \det(B) = \det(A)\det(B)$$

as we sought to show.

A block diagonal matrix has the determinant one would hope for. This is the computation that makes determinants of matrices assembled out of smaller ones tractable.

Proposition 13.4.8: Determinant of a Block Diagonal Matrix

Let R be a commutative ring and let $M \in M_n(R)$ be block diagonal with square diagonal blocks $B_1, \ldots, B_m$, so that $M_{uv} = 0$ whenever u and v lie in different blocks. Then

$$\det(M) = \prod_{t=1}^{m} \det(B_t)$$

***Proof*:**

Index the rows and columns of M by pairs (t, u), where t names the block and u the position inside it, so that $M_{(t,u),(t',u')}$ is $(B_t)_{uu'}$ when $t = t'$ and 0 otherwise. A permutation σ of the index set contributes a nonzero term to the defining sum only if σ maps each block into itself, since otherwise some factor $M_{(t,u),\sigma(t,u)}$ has $t' \neq t$ and vanishes. Such a σ is precisely a tuple $(\sigma_1, \ldots, \sigma_m)$ of permutations of the individual blocks, acting as $\sigma(t, u) = (t, \sigma_t(u))$.

These block permutations move disjoint sets of indices, so σ is the product of the commuting permutations they determine and $(-1)^\sigma = \prod_t (-1)^{\sigma_t}$, the sign being a homomorphism. Hence

$$\det(M) = \sum_{(\sigma_1,\ldots,\sigma_m)} \prod_{t=1}^{m} (-1)^{\sigma_t} \prod_{u} (B_t)_{u\sigma_t(u)} = \prod_{t=1}^{m} \left(\sum_{\sigma_t} (-1)^{\sigma_t} \prod_{u} (B_t)_{u\sigma_t(u)} \right) = \prod_{t=1}^{m} \det(B_t)$$

the middle step being the distributive law, since the tuples range independently over the factors, completing the proof.

Determinants detect nonsingular matrices. For a linear map $f\colon \mathbb{R}^2 \to \mathbb{R}^2$, $|\det f|$ is the area of the image of the unit square, a parallelogram, and when $\det f \neq 0$ its sign records whether f preserves orientation. The image of the square is flattened to a segment or a point exactly when f is not invertible, that is, when $\det f = 0$ (see also *Linear Algebra* [21]).

Proposition 13.4.9: Elementary Matrices And Determinant

Let R be a commutative ring, let $A \in M_n(R)$, and let A' be the matrix obtained from A by the row operation named in each item. Then

1. (switching two rows) $\det(A') = -\det(A)$
2. (adding r times one row to a different row, $r \in R$) $\det(A') = \det(A)$
3. (multiplying one row by $c \in R$, for instance by a unit) $\det(A') = c\det(A)$

The same three formulas hold for the corresponding column operations.

***Proof*:**
By proposition 13.4.6(3) and (2), det is R-linear in each row when the other rows are held fixed.

1. If A' is A with rows $k \neq l$ interchanged, then the ith row of A' is the $\tau(i)$th row of A for the transposition $\tau = (k\ l)$. By proposition 13.4.6(5), $\det(A') = (-1)^\tau \det(A) = -\det(A)$, since a transposition has sign -1 (theorem 1.2.20(3)).
2. Let A' be A with its kth row ρ_k replaced by $\rho_k + r\rho_l$, where $l \neq k$ and ρ_l is the lth row. Linearity in row k gives $\det(A') = \det(A) + r\det(A'')$, where A'' is A with ρ_k replaced by ρ_l. The matrix A'' has two equal rows, so $\det(A'') = 0$ by proposition 13.4.6(4), and $\det(A') = \det(A)$.
3. If A' is A with its kth row multiplied by c, linearity in row k gives $\det(A') = c\det(A)$.

If A' is obtained from A by a column operation, then $(A')^t$ is obtained from A^t by the corresponding row operation, and det is unchanged by transposition (proposition 13.4.6(2)), so the three formulas hold for column operations as well, completing the proof.

Over any commutative ring, the inverse of an invertible matrix A is $\det(A)^{-1}$ times a matrix whose entries are determinants of matrices built from A (proposition 13.4.12).

Definition 13.4.10: Adjugate Matrix

Let R be a commutative ring and $A \in M_n(R)$, and let $e_1, \ldots, e_n$ be the standard basis of R^n. For a column vector $u \in R^n$ and $1 \leq j \leq n$, write $A_j(u)$ for the matrix obtained from A by replacing its jth column with u. The *cofactors* of A are $c_{ij}(A) = \det(A_j(e_i))$ for $1 \leq i, j \leq n$, and the *adjugate* of A is the matrix $\operatorname{adj}(A) \in M_n(R)$ whose (j, i) entry is $c_{ij}(A)$.

Proposition 13.4.11: The Adjugate Identity

Let R be a commutative ring and $A \in M_n(R)$. Then

1. (expansion along a column) $\det(A_j(u)) = \sum_{i=1}^n u_i c_{ij}(A)$ for every $u \in R^n$ and $1 \leq j \leq n$
2. $\operatorname{adj}(A)A = A\operatorname{adj}(A) = \det(A)I_n$

***Proof*:**

1. Since $u = \sum_i u_i e_i$ and det is R-linear in the jth column (proposition 13.4.6(3)),

$$\det(A_j(u)) = \sum_{i=1}^{n} u_i \det(A_j(e_i)) = \sum_{i=1}^{n} u_i c_{ij}(A)$$

2. *Step 1:* $\operatorname{adj}(A)A = \det(A) I_n$. Let a_l be the lth column of A. By (1), the (j,l) entry of $\operatorname{adj}(A)A$ is

$$\sum_{i=1}^{n} c_{ij}(A) a_{il} = \det(A_j(a_l))$$

For $l = j$ this is $\det(A_j(a_j)) = \det(A)$. For $l \neq j$ the matrix $A_j(a_l)$ has its jth and lth columns both equal to a_l, so the entry is 0 (proposition 13.4.6(4)).

Step 2: a cofactor computed by rows. Fix i and j, and let X be A with its jth row replaced by the row vector e_i^t. We claim $\det(X) = \det(A_i(e_j))$. Let Z be the matrix that agrees with A outside row j and column i, has jth row e_i^t and has ith column e_j (the two prescriptions agree at position (j,i), where both give 1). The ith column of X is $e_j + \sum_{k \neq j} a_{ki} e_k$, so linearity in column i (proposition 13.4.6(3)) gives $\det(X) = \det(X_i(e_j)) + \sum_{k \neq j} a_{ki} \det(X_i(e_k))$. Here $X_i(e_j) = Z$, and for $k \neq j$ the jth row of $X_i(e_k)$ is zero, so $\det(X_i(e_k)) = 0$ by linearity in that row. Hence $\det(X) = \det(Z)$. Next, the jth row of $Y = A_i(e_j)$ is $e_i^t + \sum_{l \neq i} a_{jl} e_l^t$, so linearity in row j gives $\det(Y) = \det(Y') + \sum_{l \neq i} a_{jl} \det(Y_l)$, where Y' and Y_l are Y with its jth row replaced by e_i^t and by e_l^t. Here $Y' = Z$, and for $l \neq i$ the ith column of Y_l is zero, so $\det(Y_l) = 0$ by linearity in that column. Hence $\det(Y) = \det(Z)$, which proves the claim.

Step 3: $\operatorname{adj}(A^t) = \operatorname{adj}(A)^t$. The transpose of $(A^t)_j(e_i)$ is the matrix X of Step 2, so by proposition 13.4.6(2) and Step 2, the (j,i) entry of $\operatorname{adj}(A^t)$ is

$$c_{ij}(A^t) = \det(X) = \det(A_i(e_j)) = c_{ji}(A)$$

which is the (i,j) entry of $\operatorname{adj}(A)$, that is, the (j,i) entry of $\operatorname{adj}(A)^t$.

Step 4: $A \operatorname{adj}(A) = \det(A) I_n$. Step 1 for A^t gives $\operatorname{adj}(A^t)A^t = \det(A^t) I_n = \det(A) I_n$. Over the commutative ring R, $(XY)^t = Y^t X^t$, since both have (a,b) entry $\sum_c x_{bc} y_{ca}$. Transposing and using Step 3 gives $A \operatorname{adj}(A) = \det(A) I_n$, completing the proof.

Proposition 13.4.12: Invertibility And Determinants

Let R be a commutative ring. Then:

1. $A \in M_n(R)$ is invertible if and only if $\det(A)$ is a unit in R, and in that case $A^{-1} = \det(A)^{-1}\operatorname{adj}(A)$
2. the determinant is a homomorphism $\det : \mathrm{GL}_n(R) \to R^\times$, that is, for any $A, B \in \mathrm{GL}_n(R)$:
$$\det(A \cdot B) = \det(A) \cdot \det(B)$$

***Proof*:**

1. If A is invertible, then $\det(A)\det(A^{-1}) = \det(AA^{-1}) = \det(I_n) = 1$ by proposition 13.4.7 and proposition 13.4.6(1), so $\det(A)$ is a unit. Conversely, let $\det(A)$ be a unit and $B = \det(A)^{-1}\operatorname{adj}(A)$. By proposition 13.4.11(2), $AB = \det(A)^{-1}A\operatorname{adj}(A) = I_n$ and $BA = \det(A)^{-1}\operatorname{adj}(A)A = I_n$, so A is invertible with $A^{-1} = B$.
2. By (1), $\det(A)$ is a unit for every $A \in \mathrm{GL}_n(R)$, so det maps $\mathrm{GL}_n(R)$ into $R^\times$, and $\det(AB) = \det(A)\det(B)$ for all $A, B \in M_n(R)$ by proposition 13.4.7, completing the proof.

Over an integral domain, a vanishing determinant is the same as a nonzero solution of $Ax = 0$:

Proposition 13.4.13: Determinant And Kernel Over An Integral Domain

Let R be an integral domain and $A \in M_n(R)$. Then $\det(A) = 0$ if and only if $Ax = 0$ for some nonzero $x \in R^n$.

***Proof*:**

Suppose $Ax = 0$ with $x \neq 0$, and choose i with $x_i \neq 0$. By proposition 13.4.11(2), $\det(A)x = \operatorname{adj}(A)Ax = 0$, so $\det(A)x_i = 0$, and $\det(A) = 0$ as R is an integral domain.

Conversely, suppose $\det(A) = 0$, and let K be the field of fractions of R (definition 9.5.2), so that $A \in M_n(K)$ with the same determinant. As 0 is not a unit of K, the matrix A is not invertible in $M_n(K)$ (proposition 13.4.12(1)), so it is not nonsingular (proposition 13.2.10): $Ay = 0$ for some nonzero $y \in K^n$. Write $y_i = r_i/s_i$ with $r_i, s_i \in R$ and $s_i \neq 0$, and let $s = s_1 \cdots s_n \neq 0$. Then $x = sy$ lies in R^n, it is nonzero since $sy_i \neq 0$ whenever $y_i \neq 0$, and $Ax = sAy = 0$, completing the proof.

We will finish off this section with an application of determinants to solve systems of linear equations. This will become particularly useful for a few proofs in the book[23].

[23]The adjugate identity (proposition 13.4.11), which is the form of Cramer's rule that holds without assuming that $\det(A)$ is a unit, is the form used in *Commutative Algebra* [16, Integral Element Equivalences].

Proposition 13.4.14: Cramer's Rule

Let R be a commutative ring, let $A \in M_n(R)$ with $\det(A)$ a unit of R, and let $b \in R^n$. For $1 \leq j \leq n$ let $A^{(j)}$ be the matrix obtained by replacing the jth column of A by the column vector b. Then the system $Ax = b$ has exactly one solution $x \in R^n$, given by

$$x_j = \det(A)^{-1} \det(A^{(j)}) \qquad 1 \leq j \leq n$$

***Proof*:**

By proposition 13.4.12(1), A is invertible with $A^{-1} = \det(A)^{-1} \operatorname{adj}(A)$. If $Ax = b$, then $x = A^{-1}Ax = A^{-1}b$, and $x = A^{-1}b$ is a solution, since $A(A^{-1}b) = b$. So the system has exactly one solution, $x = \det(A)^{-1} \operatorname{adj}(A)b$. Its jth entry is

$$x_j = \det(A)^{-1} \sum_{i=1}^{n} c_{ij}(A) b_i = \det(A)^{-1} \det(A_j(b)) = \det(A)^{-1} \det(A^{(j)})$$

by proposition 13.4.11(1), since $A^{(j)} = A_j(b)$, as we sought to show.

A determinant also measures the size of the parallelepiped spanned by n vectors of $\mathbb{R}^n$:

Definition 13.4.15: Volume

Let $e = (e_1, e_2, \ldots, e_n)$ be the standard basis of $\mathbb{R}^n$. For vectors $v_1, \ldots, v_n \in \mathbb{R}^n$ write $v_j = \sum_{i=1}^{n} a_{ij} e_i$ and let $A = (a_{ij}) \in M_n(\mathbb{R})$, the matrix whose jth column is v_j. Then the *volume* of $v = (v_1, \ldots, v_n)$ is

$$\operatorname{vol}(v) = |\det(A)|$$

The absolute value around $\det(A)$ is to eliminate orientation information. Geometrically, when $v_1, v_2, ..., v_n$ are linearly independent, $\operatorname{vol}(v)$ is the volume of the parallelepiped

$$P = \left\{ \sum_{i=1}^{n} c_i v_i \;\middle|\; 0 \leq c_i < 1 \right\}$$

spanned by them, and the intuitive notion of volume of P agrees with this definition. The case $v = e$ gives $A = I_n$, so the standard basis e has volume $\det(I_n) = 1$ (proposition 13.4.6(1)). The volume depends on the family and not only on its members: for $n = 2$ the family $v = (e_1, e_1)$ has volume 0, since its matrix A has two equal columns (proposition 13.4.6(4)). Since $\det(A^t) = \det(A)$ (proposition 13.4.6(2)), the volume is also the absolute value of the determinant of the matrix whose rows are the v_j.

Exercise 13.4.1

1. Let $A \in M_n(k)$, $k \subseteq \mathbb{C}$. Show that $\det(\overline{A}) = \overline{\det(A)}$ where we define $\overline{A}$ component-wise to be $(\overline{A})_{ij} = \overline{a_{ij}}$ with $\overline{a_{ij}}$ being the complex conjugate ($\overline{x + iy} = x - iy$)
2. Let R be a commutative ring and $A \in M_n(R)$, and for $1 \leq i, j \leq n$ let $A^{(ij)} \in M_{n-1}(R)$ be the matrix obtained from A by deleting row i and column j. Show that the cofactors

of definition 13.4.10 are $c_{ij}(A) = (-1)^{i+j}\det(A^{(ij)})$, and deduce the expansion $\det(A) = \sum_{i=1}^{n}(-1)^{i+j}a_{ij}\det(A^{(ij)})$ along the jth column.

3. (If you know multi-variable calculus) Treating $M_n(\mathbb{R})$ as $\mathbb{R}^{n^2}$, show that the derivative of the determinant map $\det : M_n(\mathbb{R}) \to \mathbb{R}$ at the identity I is the trace tr: $(\det)'(I) = \text{tr}$. This gives a surprisingly interesting link between the two concepts, and is important in the beginning of the study of *Lie Algebras* of many matrix groups such as $\text{GL}_n(k)$, $\text{SL}_n(k)$, $O_n(k)$, and so forth, where $k \in \{\mathbb{R}, \mathbb{C}\}$.

4. More generally, show that $\det : M_n(\mathbb{R}) \to \mathbb{R}$ has a derivative at any point A, namely: $(D_A \det)(X) = \text{tr}(\text{adj}(A)X)$, with $\text{adj}(A)$ as in definition 13.4.10.

5. Let R be an integral domain with field of fractions K, and let $A \in M_n(R)$. Show that A is invertible in $M_n(K)$ if and only if $\det(A) \neq 0$, and find a matrix $A \in M_2(\mathbb{Z})$ with $\det(A) \neq 0$ that is not invertible in $M_2(\mathbb{Z})$.

13.5 Eigenvectors and Characteristic Polynomial

After having found the trace and determinant, the reader may wonder if there are more invariant functionals on $\text{End}_k(V)$ for an n-dimensional vector space. As it turns out, these are both special cases of a collection of n invariant given by the coefficients of the following polynomial:

Definition 13.5.1: Characteristic Polynomial

Let R be a commutative ring and $A \in M_n(R)$. The *characteristic polynomial* of A is

$$P_A(t) := \det(tI_n - A) \in R[t]$$

the determinant (definition 13.4.5) of the matrix $tI_n - A$, whose entries lie in the commutative ring $R[t]$ of R-polynomials with indeterminate t. If M is a free R-module with a basis of n elements and $\alpha \in \text{End}_R(M)$, the *characteristic polynomial* of α is $P_\alpha(t) := P_A(t)$, where A is the matrix of α with respect to a basis of M with n elements.

The origins of this seemingly complicated looking polynomial comes from its link to eigenvectors and eigenvalues (namely scalars and vector of a linear operator that satisfy $\alpha(v) = \lambda v$ for appropriate $\lambda \in R$ and $v \neq 0$) which shall be explored soon. Some important properties are established first:

Proposition 13.5.2: Properties Of The Characteristic Polynomial

Let R be a commutative ring, let $n \geq 1$ and $A \in M_n(R)$, and write $P_A(t) = \sum_{i \geq 0} c_i t^i$. Then:

1. $P_A(t)$ is monic of degree n
2. the coefficient of t^{n-1} is $c_{n-1} = -\mathrm{tr}(A)$
3. the constant term is $c_0 = (-1)^n \det(A)$
4. if $B = PAP^{-1}$ for some $P \in \mathrm{GL}_n(R)$, then $P_B = P_A$. In particular, the characteristic polynomial of an endomorphism α of a free R-module with a basis of n elements does not depend on the basis used to compute it, and similar endomorphisms have the same characteristic polynomial.

***Proof*:**
Write $X = tI_n - A \in M_n(R[t])$, so that $X_{ij} = t\delta_{ij} - a_{ij}$ and, by definition 13.4.5,

$$P_A(t) = \sum_{\sigma \in S_n} (-1)^\sigma \prod_{i=1}^{n} X_{i\sigma(i)}$$

1. The term of $\sigma = \mathrm{id}$ is $\prod_{i=1}^{n}(t - a_{ii})$. A permutation $\sigma \neq \mathrm{id}$ moves at least two indices: if $\sigma(i) \neq i$, then $\sigma(\sigma(i)) \neq \sigma(i)$ since σ is injective, so $\sigma(i)$ is moved as well. For a moved index i the factor $X_{i\sigma(i)} = -a_{i\sigma(i)}$ is a constant, and every factor has degree at most 1 in t, so the term of $\sigma \neq \mathrm{id}$ has degree at most $n-2$. Expanding $\prod_i (t - a_{ii})$ gives t^n plus terms of degree less than n, so P_A is monic of degree n.

2. By the proof of (1), only the term of $\sigma = \mathrm{id}$ contributes to the coefficient of t^{n-1}. By induction on n, $\prod_{i=1}^{n}(t - a_{ii}) = t^n - (a_{11} + \cdots + a_{nn})t^{n-1} + h$ with $h = 0$ or $\deg h < n-1$: the case $n = 1$ is $t - a_{11}$, and multiplying $t^{n-1} - st^{n-2} + h'$, where $h' = 0$ or $\deg h' < n-2$, by $t - a_{nn}$ gives $t^n - (s + a_{nn})t^{n-1}$ plus terms of degree less than $n-1$. Hence $c_{n-1} = -\sum_i a_{ii} = -\mathrm{tr}(A)$.

3. The evaluation $R[t] \to R$, $t \mapsto 0$, is a ring homomorphism (theorem 9.2.4), and it sends P_A to its constant term c_0. The defining sum of $\det(X)$ is built from the entries of X by sums and products, so applying the evaluation entrywise gives

$$c_0 = \det(-A) = \sum_{\sigma \in S_n} (-1)^\sigma \prod_{i=1}^{n}(-a_{i\sigma(i)}) = (-1)^n \det(A)$$

4. Let $B = PAP^{-1}$. In $M_n(R[t])$ the scalar matrix tI_n commutes with P, so $tI_n - B = P(tI_n - A)P^{-1}$. By proposition 13.4.7, over the commutative ring $R[t]$, and proposition 13.4.6(1),

$$P_B = \det(P) P_A \det(P^{-1}) = P_A \det(PP^{-1}) = P_A \det(I_n) = P_A$$

   If A and B are the matrices of an endomorphism α with respect to two bases with n elements, then A and B are similar by lemma 13.2.16(4), so $P_A = P_B$. If $\alpha' \sim \alpha$, then their matrices with respect to one basis are similar by lemma 13.2.16(3), so $P_{\alpha'} = P_\alpha$, completing the proof.

For an endomorphism α of a free R-module with a basis of $n \geq 1$ elements, parts (2), (3) and (4) show that the trace and the determinant of the matrix of α do not depend on the basis. They are called the *trace* $\mathrm{tr}(\alpha)$ and the *determinant* $\det(\alpha)$ of α, the trace agreeing with definition 13.4.3 for vector spaces. Hence, the characteristic polynomial is of the form:

$$t^n - (\mathrm{tr}\alpha)t^{n-1} + \cdots + (-1)^n(\det \alpha)$$

The trace and the determinant are special among these coefficients: they coincide when $n = 1$, over a field the determinant is zero exactly when α is not surjective (proposition 13.4.12(1), corollary 13.1.20), and the trace is zero if $\alpha = \beta\gamma - \gamma\beta$ is a commutator, since $\mathrm{tr}(\beta\gamma) = \mathrm{tr}(\gamma\beta)$ (proposition 13.4.2). Furthermore, note that all coefficients are invariant under similarity, meaning that if the trace or determinant between two linear operators differ, they are *not* similar! Unfortunately, the converse does not hold:

Example 13.5.3: Similar Matrices

Let:

$$A = \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix} \qquad B = \begin{pmatrix} 2 & 1 \\ 0 & 2 \end{pmatrix}$$

These each have the characteristic polynomial $(t-2)^2 = t^2 - 4t + 4$. However, they are not similar: $A = 2I$, and conjugating A by any invertible matrix P gives $P^{-1}(2I)P = 2I$, and hence never B.

As mentioned earlier, the characteristic polynomial naturally arises when studying eigenvector and values. Let us start by formally defining these terms:

Definition 13.5.4: Eigenvalue and Spectrum

Let M be a free R-module over a commutative ring R and let $\alpha \in \mathrm{End}_R(M)$ be a linear operator on M. Then a scalar $\lambda \in R$ is called an *eigenvalue* of α if there exists a nonzero $v \in M$ such that

$$\alpha(v) = \lambda v$$

The collection of all eigenvalues is called the *spectrum* of α.

The existence of eigenvalues of an operator α means that in some subspace $N \subseteq M$ the transformation $\alpha|_N$ is simply scaling. Over $\mathbb{C}$[24], a suitable choice of basis splits the behavior of a linear operator into two kinds: it can only scale or *shear*. In example 13.5.3, matrix B scaled the first basis vector and sheared the second basis vector. Scaling is the easier of the two behaviours as the algebraic invariant for detecting scaling is the characteristic polynomial, while detecting and characterizing shearing requires more complicated invariants: shearing shall be explored in section 14.3.

Eigenvalues are extremely important, having a wide range of applications in quantum mechanics (the term *spectrum of an atom* comes from finding the spectrum of an endomorphism) functional analysis (K-theory has its origin in the spectrum of infinite linear operators), differential equations (in finding a basis to the solutions of an ODE and in defining the Fourier Transform), to number theory and Field theory (as we shall see later in the book), and so much more! Due to the prominence of eigenvalues, some may claim that the study of endomorphisms of vector-spaces is one of the fundamental concepts of mathematics or the sciences. In *Algebraic Geometry* [9] it is shown that any

[24] so that there is no "algebraic obstruction", an intuition that shall be covered in great detail in chapter 19

commutative ring has a spectrum associated to (though by the definition of a spectrum of a ring, the connection shall not be obvious[25]).

The eigenvalues are invariant under similarity. Let α, β be similar using ρ ($\alpha = \rho \circ \beta \circ \rho^{-1}$) and let $\beta(v) = \lambda v$ with $v \neq 0$. Then $\alpha \circ \rho = \rho \circ \beta$ and

$$\alpha(\rho(v)) = (\rho \circ \beta \circ \rho^{-1})(\rho(v)) = \rho \circ \beta(v) = \rho(\lambda v) = \lambda \rho(v)$$

with $\rho(v) \neq 0$, since ρ is injective and $v \neq 0$. So every eigenvalue of β is an eigenvalue of α, and the converse follows by exchanging the roles, as $\beta = \rho^{-1} \circ \alpha \circ \rho$.

Now, as promised, the following shows why the characteristic polynomial is a natural concept based on its invariance in similarity:

Proposition 13.5.5: Eigenvalues And Characteristic Polynomial

Let R be an integral domain, let M be a free R-module with a basis of n elements, and let $P_\alpha(t)$ be the characteristic polynomial of the linear operator $\alpha \in \text{End}_R(M)$. Then $\lambda \in R$ is an eigenvalue if and only if λ is a root of $P_\alpha(t)$. In other words, the set of roots of $P_\alpha(t)$ in R is exactly the spectrum of α.

***Proof*:**
Let A be the matrix of α with respect to a basis β of M with n elements, and let $\lambda \in R$. A vector $v \in M$ with coordinate vector $x \in R^n$ satisfies $\alpha(v) = \lambda v$ exactly when $(\lambda I_n - A)x = 0$, and $v \neq 0$ exactly when $x \neq 0$. So λ is an eigenvalue if and only if $(\lambda I_n - A)x = 0$ for some nonzero $x \in R^n$, that is, by proposition 13.4.13, if and only if $\det(\lambda I_n - A) = 0$. The evaluation $R[t] \to R$, $t \mapsto \lambda$, is a ring homomorphism (theorem 9.2.4), and applying it entrywise to the defining sum of $\det(tI_n - A)$ gives $P_\alpha(\lambda) = \det(\lambda I_n - A)$. Hence λ is an eigenvalue if and only if $P_\alpha(\lambda) = 0$, as we sought to show.

Example 13.5.6: Eigenvalues And Characteristic Polynomials

1. If α is not injective, then 0 is an eigenvalue, otherwise 0 is not an eigenvalue.
2. Take
$$\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$$
This matrix has a characteristic polynomial of $t^2 + 1$. If $R = \mathbb{R}$, then this characteristic polynomial has *no* roots. Similarly, the linear operator this matrix represents has no eigenvalues. Geometrically, this matrix can be seen as rotating $\mathbb{R}^2$ by $\pi/2$ degrees, which is why it has no eigenvalues over $\mathbb{R}$. If $R = \mathbb{C}$, then the polynomial has two roots, namely $i, -i$. The reader should verify that these are eigenvalues of a linear operator represented by this matrix. Intuitively, multiplying by i can be thought of as representing rotation by $\pi/2$ degree's which is why it is an eigenvalue.

[25]The curious reader may checkout EYNKTA functional analysis where this connection is fully fleshed out, as the connection is fundamentally a functional analysis result and would be too far out of the scope of an algebra textbook to be covered

3. Take
$$\begin{pmatrix} 0 & 2 \\ 1 & 0 \end{pmatrix}$$
This matrix has a characteristic polynomial of $t^2 - 2$. If $R = \mathbb{R}$, then this polynomial has roots $\pm\sqrt{2}$, while if $R = \mathbb{Q}$, this polynomial has no roots, which equivalently implies that there is no eigenvalues when $R = \mathbb{Q}$. Geometrically, this transformation can be seen as reflecting the plane across a line passing through the origin with an irrational slope. Since the slope is irrational, there is no nonzero irrational scalar to a vector, and hence cannot be a rational eigenvalue.

Geometrically, eigenvectors correspond to the linear operator scaling in the appropriate direction. In the simplest case, an n-dimensional linear operator α can be represented in a basis for which $[\alpha]$ just scales appropriately in each direction. We take a moment to explore these type of matrices.

Since roots have the notion of multiplicity (the number of roots sharing the same value), it is natural to contemplate the multiplicity of eigenvalues and see what information this gives us:

Definition 13.5.7: Algebraic Multiplicity

Let λ be an eigenvalue of a linear operator α over a finitely generated free module. Then the *algebraic multiplicity* is its multiplicity as a root of $P_\alpha(t)$.

Example 13.5.8: Algebraic Multiplicity
Let M be a finitely generated free module over an integral domain R.

1. the identity operator I on M has a single eigenvalue, namely 1, which has algebraic multiplicity equal to the rank of M.
2. for $M = R^2$ and the operator with matrix $A = \begin{pmatrix} 1 & 1 \\ 0 & 2 \end{pmatrix}$, $P_A(t) = (t-1)(t-2)$, so the eigenvalues 1 and 2 each have algebraic multiplicity 1.

Notice that the sum of the algebraic multiplicities is bounded by the rank of the free module M (resp. the dimension of the vector-space V). If R is *algebraically closed* (meaning every polynomial of degree n over R has exactly n roots[26]), then the sum of the algebraic multiplicities of the eigenvalues always equals the rank of M. The prototypical algebraically closed field is $\mathbb{C}$, the complex numbers.

The definition has been careful to call the multiplicity "algebraic". There is another notion of multiplicity which corresponds to the geometric behavior of multiplicity, namely there is a special subspace that is linked to each eigenvalue:

[26] See definition 19.5.1 and the lemma under it for more details

Definition 13.5.9: Eigenspace And Eigenvector

Let λ be an eigenvalue associated to the linear operator α over a finitely generated free R-module M. Then the sub-module

$$E_\lambda = \ker(\lambda I - \alpha)$$

is called the *eigenspace* associated to λ, and the elements of E_λ are called *eigenvectors* and represent the elements of M such that

$$\alpha(v) = \lambda v$$

Technically, it is more correct to call these spaces "eigen-modules" rather than eigenspace, however the study of eigen-modules over vector-spaces greatly precedes the generalization to eigen-modules over R-modules, giving us our current vocabulary.

Definition 13.5.10: Geometric Multiplicity

Let E_λ be an eigenspace corresponding to an eigenvalue λ. Then the rank of E_λ is the *geometric multiplicity* of λ.

The eigenspace is mapped into itself by α: if $\alpha(v) = \lambda v$, then $\alpha(\alpha(v)) = \lambda\alpha(v)$, so $\alpha(E_\lambda) \subseteq E_\lambda$ (we shall later call submodules mapped into themselves by an operator α α-stable). The geometric multiplicity is invariant under similarity: if $\alpha = \rho \circ \beta \circ \rho^{-1}$ with ρ an automorphism, the computation above shows that ρ maps the eigenspace of β for λ into the eigenspace of α for λ, and ρ^{-1} maps it back in the same way. So the two eigenspaces are isomorphic, and an isomorphism carries linearly independent elements to linearly independent elements, so they have the same rank (definition 12.3.16).

Example 13.5.11: Geometric And Algebraic Multiplicity

Take

$$\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$$

Then the characteristic polynomial is $(t-1)^2$, and hence 1 is an eigenvalue with algebraic multiplicity 2. To find it's geometric multiplicity, by considering

$$\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} v_1 \\ v_2 \end{pmatrix} = \begin{pmatrix} v_1 + v_2 \\ v_2 \end{pmatrix}$$

we see that the right hand side is equal to $1v$ if and only if $v_2 = 0$, hence the eigenspace is

$$\text{span} \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

which is 1 dimensional, hence the geometric multiplicity is 1.

A result that should stand out to the reader given the above example is the following

Proposition 13.5.12: Algebraic And Geometric Multiplicity

Let R be an integral domain and let λ be an eigenvalue associated to the linear operator α over a finitely generated free R-module M. Then the geometric multiplicity is always less than or equal to the algebraic multiplicity

***Proof*:**

Let n be the rank of M. Fix a basis of M, identifying M with R^n, and let $A \in M_n(R)$ represent α, so that $P_\alpha(t) = \det(tI - A) \in R[t]$. Let K be the field of fractions of R (definition 9.5.2), so $R \subseteq K$ and $R^n \subseteq K^n$, and let α_K be the linear operator on K^n given by the same matrix A.

Let g be the geometric multiplicity of λ and choose g elements $v_1, \dots, v_g$ of E_λ which are linearly independent over R, as the rank provides. They are also linearly independent over K: if $\sum_i c_i v_i = 0$ with $c_i \in K$ not all zero, multiplying through by the product of the denominators of the c_i (nonzero, as R is a domain) gives a dependence with coefficients in R, not all zero, a contradiction. Since each v_i lies in $\ker(\lambda I - A)$ computed over K, we conclude

$$d := \dim_K \ker(\lambda I - \alpha_K) \geq g$$

Next we show $(t-\lambda)^d$ divides $P_\alpha(t)$ in $K[t]$. By theorem 13.1.8, choose a basis $w_1, \dots, w_d$ of $\ker(\lambda I - \alpha_K)$ and extend it, again by theorem 13.1.8, to a basis $w_1, \dots, w_n$ of K^n. In this basis, $\alpha_K(w_j) = \lambda w_j$ for $j \leq d$, so the representing matrix has the block form

$$A' = \begin{pmatrix} \lambda I_d & B \\ 0 & C \end{pmatrix}$$

for some blocks B and C. Since A and A' represent the same operator α_K, we have $\det(tI - A') = P_\alpha(t)$ by proposition 13.5.2(4). Now consider

$$tI - A' = \begin{pmatrix} (t-\lambda) I_d & -B \\ 0 & tI_{n-d} - C \end{pmatrix}$$

as a matrix over the commutative ring $K[t]$. In the defining sum of definition 13.4.5, a permutation σ contributes a nonzero term only if every one of the last $n-d$ indices is sent to one of the last $n-d$ indices, since the entries in those rows and the first d columns vanish. Being a bijection, such a σ then permutes the first d indices and the last $n-d$ indices separately, and every factor of its term comes from the two diagonal blocks. The nonzero terms therefore coincide with those of the block diagonal matrix obtained by replacing $-B$ with 0, and proposition 13.4.8 gives

$$P_\alpha(t) = \det\big((t-\lambda) I_d\big) \det(tI_{n-d} - C) = (t-\lambda)^d \det(tI_{n-d} - C)$$

in $K[t]$. In particular $(t-\lambda)^g$ divides $P_\alpha(t)$ in $K[t]$.

Finally, we transfer the divisibility to $R[t]$. The polynomial $h(t) = (t-\lambda)^g$ is monic with coefficients in R. Dividing P_α by h within $R[t]$ is possible with quotient and remainder in $R[t]$: at each step one subtracts the leading term of the current remainder times a power of t times h, and since h is monic no denominators appear. This produces $P_\alpha = q_0 h + r_0$ with $q_0, r_0 \in R[t]$ and $r_0 = 0$ or $\deg r_0 < \deg h$. Read in $K[t]$, this must coincide with the division of P_α by h in $K[t]$ by the uniqueness in theorem 11.1.1, and the latter has remainder 0 since $h \big| P_\alpha$ in $K[t]$. Hence $r_0 = 0$ and $P_\alpha(t) = (t-\lambda)^g\, q_0(t)$ with $q_0 \in R[t]$. The multiplicity of λ as a root of $P_\alpha(t)$ is therefore at least g, that is, the geometric multiplicity is less than or equal to the algebraic multiplicity.

One of the main results comes in the algebraic multiplicities add up to the rank of the free module and when the algebraic multiplicity matches the geometric multiplicity. Then there exists a basis P making α a bunch of scalars, namely:

Theorem 13.5.13: Diagonalizable

Let α be a linear operator on an n-dimensional vector space V over a field k. Then if the algebraic multiplicities of the eigenvalues sum to n and the algebraic multiplicities of each eigenvalue equal to their respective geometric multiplicities, then for any representation A of α there exists an invertible matrix $P \in \mathrm{GL}_n(k)$ such that:

$$A = P \begin{pmatrix} \lambda_1 & 0 & \cdots & 0 \\ 0 & \lambda_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \lambda_n \end{pmatrix} P^{-1}$$

where $\lambda_1, \lambda_2, ..., \lambda_n$ are the eigenvalues of α counted with algebraic multiplicity, and α (resp. A) is said to be *diagonalizable*. Moreover, such a P exists if and only if there exists a basis for V consisting of eigenvectors.

***Proof*:**
Eigenvectors belonging to distinct eigenvalues are independent in the following sense: if $\mu_1, \ldots, \mu_r$ are distinct eigenvalues and $u_i \in E_{\mu_i}$ satisfy $u_1 + \cdots + u_r = 0$, then every $u_i = 0$. We induct on r, the case $r = 1$ being trivial. Applying the operator $\alpha - \mu_r I$ to the relation kills u_r and gives

$$(\mu_1 - \mu_r)u_1 + \cdots + (\mu_{r-1} - \mu_r)u_{r-1} = 0$$

where $(\mu_i - \mu_r)u_i \in E_{\mu_i}$. By the induction hypothesis, $(\mu_i - \mu_r)u_i = 0$ for each $i < r$, and since $\mu_i \neq \mu_r$ and k is a field, $u_i = 0$ for $i < r$. The relation then forces $u_r = 0$ as well.

Now let $\mu_1, \ldots, \mu_r$ be the distinct eigenvalues of α, with algebraic multiplicities $m_1, \ldots, m_r$ equal to the respective geometric multiplicities, and $m_1 + \cdots + m_r = n$. Choose a basis B_i of each eigenspace E_{μ_i}, so $|B_i| = m_i$, and let $B = B_1 \cup \cdots \cup B_r$, a set of n vectors. We claim B is linearly independent. In a linear dependence among the elements of B, group the terms by eigenspace: the dependence reads $u_1 + \cdots + u_r = 0$ with $u_i \in E_{\mu_i}$ the part of the combination using B_i. By the previous paragraph each $u_i = 0$, and then all coefficients vanish since each B_i is linearly independent. By theorem 13.1.8, B is contained in a basis of V, and by corollary 13.1.13 every basis of V has n elements, so B is itself a basis of V.

Order $B = \{b_1, \ldots, b_n\}$ and let λ_j denote the eigenvalue with $\alpha(b_j) = \lambda_j b_j$. The representation of α in the basis B is then the diagonal matrix D with entries $\lambda_1, \ldots, \lambda_n$, the eigenvalues counted with algebraic multiplicity. If A represents α with respect to a basis β, then A and D represent the same operator in two bases, so by proposition 13.2.13 there is an invertible change of basis matrix $P \in \mathrm{GL}_n(k)$ with $A = PDP^{-1}$.

For the final claim, one direction is the argument above: a basis of eigenvectors produces the diagonal representation D and the matrix P. Conversely, suppose $A = PDP^{-1}$ with D diagonal with entries $d_1, \ldots, d_n$, where A represents α with respect to a basis $v_1, \ldots, v_n$. Define $b_j := \sum_i P_{ij} v_i$, so that the coordinate vector of b_j is $[b_j] = Pe_j$. Since P is invertible, $b_1, \ldots, b_n$

form a basis of V. Moreover

$$[\alpha(b_j)] = A[b_j] = APe_j = PDe_j = d_jPe_j = d_j[b_j]$$

so $\alpha(b_j) = d_jb_j$, and $b_1, \ldots, b_n$ is a basis of V consisting of eigenvectors.

Definition 13.5.14: Diagonalizable

Let α be a linear operator over a finitely generated free R-module M. Then if α admits a representation whose only nonzero entries lie on the diagonal, α is said to be *diagonalizable*. Similarly, a matrix that is similar to a diagonal matrix is said to be *diagonalizable*.

If α is diagonalizable, then M can be expressed in the following useful decomposition:

Theorem 13.5.15: Spectral Decomposition

Let M be a finitely generated free module over an integral domain R and let $\alpha : M \to M$ be diagonalizable. Then M admits the following decomposition into direct sums

$$M \cong \bigoplus_{\text{eigenvalue } \lambda} E_\lambda$$

***Proof*:**
By definition 13.5.14, α admits a diagonal representation: there is a basis $e_1, \ldots, e_n$ of M and scalars $d_1, \ldots, d_n \in R$ with $\alpha(e_i) = d_ie_i$. Each d_i is an eigenvalue of α, witnessed by the nonzero vector e_i.

Conversely, every eigenvalue λ of α is one of the d_i. Indeed, let $\alpha(v) = \lambda v$ with $v \neq 0$ and write $v = \sum_i v_ie_i$. Comparing coefficients of e_i in $\alpha(v) = \lambda v$ gives $d_iv_i = \lambda v_i$, that is, $(d_i - \lambda)v_i = 0$ for every i. Some v_i is nonzero, and since R is a domain, $\lambda = d_i$ for that i.

For an eigenvalue λ, the eigenspace is exactly the span of the basis eigenvectors carrying λ:

$$E_\lambda = \bigoplus_{i\,:\,d_i=\lambda} Re_i$$

The right hand side is contained in E_λ, since each generator e_i with $d_i = \lambda$ satisfies $\alpha(e_i) = \lambda e_i$ and E_λ is a submodule. Conversely, if $v = \sum_i v_ie_i \in E_\lambda$, then as above $(d_i - \lambda)v_i = 0$ for every i, so $v_i = 0$ whenever $d_i \neq \lambda$ (again since R is a domain), and v lies in the right hand side.

Finally, partition the index set $\{1, \ldots, n\}$ by the value of d_i. Since $e_1, \ldots, e_n$ is a basis, every element of M is uniquely an R-linear combination of the e_i, so grouping the terms by eigenvalue expresses every element of M uniquely as a sum of components in the submodules $\bigoplus_{d_i=\lambda} Re_i = E_\lambda$. By proposition 12.3.6, this exhibits

$$M \cong \bigoplus_{\text{eigenvalue } \lambda} E_\lambda$$

as we sought to show.

For future reference, let's give a name to the above decomposition:

Definition 13.5.16: Spectral Decomposition

Let $\alpha : M \to M$ be diagonalizable. Then the decomposition in the above theorem is called the *spectral decomposition* of M.

Given the spectral decomposition of M, α can simply be thought of as scaling each eigenspace E_λ by the eigenvalue, making the "action" of α on the space M completely transparent!

If the algebraic multiplicity does not sum to the rank of M or the geometric multiplicity does not match the algebraic multiplicity, then the matrix is *not* diagonalizable. If the algebraic multiplicity does not sum to the rank of M, then if we are working over field[27] we can take something called the *algebraic closure* of the field, which as the name implies means it is *algebraically closed*. To rigorously go over the algebraic closure of a field is too much work at this stage, and so will just be taken for granted when it is brought up, the curious reader may checkout section 19.5.1. With the reader's current knowledge, most of the section can be parsed.

If the geometric multiplicity does not match the algebraic multiplicity, then it is a harder to find a form. However, in the next chapter, we shall see that there exists a representation that is *almost* diagonalizable: it shall be diagonalizable with possibly a 1 in the super-diagonal (the position right above the diagonal). In fact, *all* square matrices will be able to have this almost diagonal form! This shall be the aforementioned *shearing* behaviour that is the only other behaviour besides scaling that a matrix shall represent when choosing an appropriate basis.

13.6 Geometry and Inner Product

A vector space over an arbitrary field carries no notion of length and no notion of angle, and none can be manufactured from the field axioms alone. Producing one takes two things a general field does not have: an ordering, so that a quantity can be called nonnegative, and a conjugation, so that $\langle v, v\rangle$ can be forced to be real. The real and complex numbers give both, and over them the whole geometric apparatus follows, namely inner products and the Cauchy-Schwarz inequality, orthonormal bases and the Gram-Schmidt process, orthogonal complements and projections, the Riesz representation of a linear functional, adjoints, and the spectral theorem for normal and self-adjoint operators.

That development belongs to a book about $\mathbb{R}$ and $\mathbb{C}$ rather than to one about modules over a general ring, and it is carried out in *Linear Algebra* [21]. Nothing later in this book depends on it.

[27] more generally, over integral domains, since the ring of fractions of an integral domains is a field

14

Modules Over PIDs: Linear Algebra II

Modules over fields have some of the simplest possible structures. Because they are always free, they admit a trivial short exact sequence:

$$0 \to R^n \to M \to 0$$

In the language of free resolutions (definition 12.3.27), every module over a field admits a free resolution of length 0 (recall indexing starts at 0). It is natural to want to study modules that admit a free resolution of length 1. Specifically, if R is a ring and M is a finitely generated module with a surjection

$$R^n \to M \to 0$$

we ask when the kernel is also free, meaning there exists an R^m such that

$$0 \to R^m \to R^n \to M \to 0$$

is exact. As was foreshadowed in section 12.3.3, all finitely generated modules over PIDs admit free resolutions of length 1, and hence they shall be the focus of this chapter. Having a free resolution of length 0 (i.e., being free) implied the existence of a basis, and even over PIDs, matrix representations of linear transformations admitted Smith Normal Forms. A free resolution of length 1 offers slightly less rigid structural results. However, it remains incredibly powerful, introducing us to *Jordan Canonical Forms* and *Rational Canonical Forms* and expands the number of modules for which we have a structure theorem. These forms give a new way of representing linear transformations and invariants of similarity (section 14.3).

The reader may naturally ask if it is interesting to further push the length of the free resolution and explore modules that admit a free resolution of length 2, 3, or n. While finitely generated

modules over a polynomial ring in finitely many variables over a field always have finite projective dimension (Hilbert's Syzygy Theorem, [19, Hilbert's Syzygy Theorem]), most interesting examples of such modules become a good deal more complex and require a build-up in algebraic geometry and homological algebra. For the interested reader, these lengths are intimately linked to the *Krull dimension* of a commutative ring[1]. The essential concepts for understanding this link will be covered later (specifically, [23, Dimension] and [19, Homological Dimension]), and further reading can be done in *Homological Algebra* [19, Auslander-Buchsbaum-Serre Theorem].

Goal

The goal of this section is to show finitely generated modules over PID's have the following structure:

$$M \cong R^k \oplus R/(a_1) \oplus \cdots \oplus R/(a_m)$$

We will define the terms a_m and how to find k soon. Essentially, M shall be split into a part which is "as linearly dependent as possible", and the rest of M will be the elements which cannot be linearly independent (over R). These "rest of elements" of M can be identified with the simplest quotient rings of R, in particular a cyclic module.

Since PID's are UFD's, we will be able to find a 2nd representation given appropriate prime elements:

$$M \cong R^k \oplus R/(p_1) \oplus \cdots \oplus R/(p_n)$$

where p_i will turn out to represent prime elements of R (and the k will be the same).

A consequence of this would be the proof of the Fundamental Theorem of finitely Generated abelian groups (theorem 5.1.6). Since $\mathbb{Z}$ is a finitely generated Principal Ideal Domain, if M is a $\mathbb{Z}$-module (and so an abelian group), it will be a consequence of the following theoretical build-up that

$$M \cong \mathbb{Z}^k \oplus \mathbb{Z}/n_1\mathbb{Z} \oplus \cdots \oplus \mathbb{Z}/n_k\mathbb{Z}$$

Another consequence will be the ability to classify all linear maps from a vector-space to itself (up to similarity). This means that the structure of $M_n(k)$ and $\mathrm{GL}_n(k)$ will be eminently understandable, having huge consequence in representation theory (*Core Representation Theory* [17]) and many areas beyond algebra (Differential Geometry, K-theory, Optimization, ODEs, and so forth). In particular, if V is a vector space over $\mathbb{F}$ and T is a linear transformation, then (section 14.2 gives the details) V becomes an $F[x]$-module on which x acts by T:

$$p(x) \cdot v = (a_n x^n + a_{n-1}x^{n-1} + \cdots + a_1 x + a_0) \cdot v = a_n T^n(v) + \cdots + a_1 T(v) + a_0 v$$

then we can use the fact that $F[x]$ is a Principal Ideal Domain to decompose T into a nicer representation. Since every linear transformation can be represented as a matrix, then we can also decompose matrices in this fashion (which will be called Rational Canonical form and Jordan Canonical Form).

Note that finiteness is an important condition. The $\mathbb{Z}$-module $\mathbb{Q}$ is torsion-free but not free: two elements of $\mathbb{Q}$ are linearly dependent over $\mathbb{Z}$ (example 12.3.13), so a basis would consist of a single $q \neq 0$, while $q/2 \notin \mathbb{Z}q$. So torsion-free modules over a PID need not be free when they are not finitely generated.

[1]The link is not perfect. Often some additional conditions must be placed on the ring, such as R needing to be a regular local ring or Cohen-Macaulay

14.1 Over PID, Submodule of Free Module is Free

Theorem 13.1.12 used the fact that a field k was composed of only units (and 0) to show dimension of a vector-space was well-defined. Not all free R-modules have this property. Fortunately, this will not be an issue for PID's as the following much more general result shows:

Lemma 14.1.1: Invariant Basis Well-defined Over Commutative Ring

Let R be a nonzero commutative ring and let $n, m \in \mathbb{N}$. If $R^n \cong R^m$ as R-modules, then $n = m$

This result can be proven for arbitrary cardinalities, which is left as an exercise. In particular a vector space over a field has a well-defined dimension, finite or not.

***Proof*:**
As R is nonzero, the zero ideal is proper, so it lies in a maximal ideal $\mathfrak{m}$ (proposition 9.4.18), and $k = R/\mathfrak{m}$ is a field (proposition 9.4.19). For $t \in \mathbb{N}$ let $\mathfrak{m}R^t$ be the submodule of tuples with all entries in $\mathfrak{m}$. The map $R^t \to k^t$, $(r_1, \dots, r_t) \mapsto (r_1 + \mathfrak{m}, \dots, r_t + \mathfrak{m})$, is surjective with kernel $\mathfrak{m}R^t$, so $R^t/\mathfrak{m}R^t \cong k^t$ (theorem 12.2.10(1)), and this is an isomorphism of k-vector spaces for the action $(r + \mathfrak{m})(v + \mathfrak{m}R^t) = rv + \mathfrak{m}R^t$. Let $\varphi\colon R^n \to R^m$ be an isomorphism. A tuple with entries in $\mathfrak{m}$ is $\sum_i c_i e_i$ with every $c_i \in \mathfrak{m}$, and φ maps it to $\sum_i c_i \varphi(e_i)$, whose entries lie in the ideal $\mathfrak{m}$. The same holds for φ^{-1}, so $\varphi(\mathfrak{m}R^n) = \mathfrak{m}R^m$, and $v + \mathfrak{m}R^n \mapsto \varphi(v) + \mathfrak{m}R^m$ is a bijective k-linear map. Then:

$$(R/\mathfrak{m})^n \cong (R/\mathfrak{m})^m$$

as vector spaces, and an isomorphism carries the standard basis of k^n to a basis of k^m with n elements, so $n = m$ by corollary 13.1.13(2), completing the proof.

Definition 14.1.2: Invariant Basis Number

Let R be a ring. Then R has the *invariant basis number property* (IBN property) if $R^n \cong R^m$ if and only if $n = m$.

A free R-module where R has the IBN property has the notion of *rank* well-defined.

If R is not commutative, this need not be the case:

Example 14.1.3: Ring Without IBN Property
Let $V = k^{\oplus \mathbb{N}}$ and $R = \mathrm{End}_k(V)$. Prove that:

$$R \cong R^4$$

Thus, there is no ambiguity for the notion of rank when working with free-modules over commutative rings.

Recall that a sub-module of a free module need not be free. For example, if we take $\mathbb{C}[x, y]$ over itself, then it is free with rank 1, but the sub-module (ideal) (x, y) is not free. If it were of rank 1[2],

[2]Why cannot the rank be ≥ 1? hint: use field of fractions

then (x, y) would be generated by a single element, but it has been shown earlier this is not possible. However, if R is a PID, then just like for modules over fields, submodules of free modules over a PID are free. For free modules, this was because every module over a field is free, however there do exist modules over PID's that are not free (consider any torsion abelian group). To prove this result, we shall show that if F is a free module over a PID and $M \subseteq F$, we can iteratively split off a cyclic direct summand $\langle y \rangle$ from $M \subseteq F$, which shall result in separating the free part from the torsion part. Furthermore, y can be chosen in such a way that it is a scalar multiple of a basis element of F: $y = ax$ for appropriate $a \in R$. Think of submodules of $\mathbb{Z}^n$ as a canonical example of this behavior:

Lemma 14.1.4: Splitting Modules Over PID

Let R be a PID, F a finitely generated free module over R, and $M \subseteq F$ be a nonzero submodule. Then there exist a nonzero element $a \in R$, elements $x \in F$, $y \in M$, submodules $F' \subseteq F$ and $M' \subseteq M$, and a homomorphism $\alpha \colon F \to R$ such that

1. $y = ax$
2. $M' = F' \cap M$
3. $F = \langle x \rangle \oplus F'$, $M = \langle y \rangle \oplus M'$
4. $\alpha(x) = 1$, $\ker(\alpha) = F'$, and $\alpha(M) = (a)$ is maximal among the ideals $\varphi(M)$ with $\varphi \in \mathrm{Hom}_R(F, R)$, that is, if $(a) \subseteq \varphi(M)$ for such a φ, then $\varphi(M) = (a)$

***Proof*:**
Since $F \cong R^n$, it is perhaps tempting to think that our factor may be one of the components of R^n, however this implicitly chooses a particular representation (i.e. a basis for the isomorphism $\cong$). Since we do not know that M is free yet, we cannot know how to pick a basis representation of F that matches well with a basis representation of M, we cannot pick a basis β for F in such a way that the free submodule M is a projection. In that case, the splitting in the lemma would be obtained by choosing a component, or in terms of functions using the projection map:

$$\pi(r_1, r_2, ..., r_n) = r_1$$

Instead, we consider all homomorphisms $\varphi \colon F \to R$ at once and choose one for which $\varphi(M)$ is maximal.

Thus, define the following collection of ideals

$$\mathscr{F} = \{\varphi(M) \mid \varphi \in \mathrm{Hom}_R(F, R)\} = \{\varphi(M) \mid \varphi \in F^*\}$$

Each $\varphi(M)$ is a submodule of the R-module R, that is, an ideal. Every ideal of R is principal, hence finitely generated, so theorem 12.3.33, (3) $\Rightarrow$ (2), applied to the R-module R, gives a maximal element of the nonempty set $\mathscr{F}$, of the form $\alpha(M)$ for some $\alpha \colon F \to R$. Fix an isomorphism $F \cong R^n$ and let $\pi_k \colon F \to R$ send an element to its kth coordinate. A nonzero element of M has a nonzero coordinate, so $\pi_k(M) \neq 0$ for some k. If $\alpha(M) = 0$, then $\alpha(M) \subseteq \pi_k(M)$, and maximality gives $\alpha(M) = \pi_k(M) \neq 0$, which is impossible. So $\alpha(M) \neq 0$. Since R is a PID, $\alpha(M) = (a)$ for some nonzero $a \in R$. Thus, there exists a $y \in M$ such that $\alpha(y) = a$.

We shall see that this y and a are the desired values. To find the desired x, we shall show that a

divides $\varphi(y)$ for all $\varphi \in F^*$. To see this, fix $\varphi \in F^*$ and let $b \in R$ be a generator of the ideal

$$(b) = (a, \varphi(y))$$

Then there exists $r, s \in R$ such that

$$b = ra + s\varphi(y)$$

Using this, define the homomorphism $\psi := r\alpha + s\varphi$. Then $\psi(y) = ra + s\varphi(y) = b$, so $\alpha(M) = (a) \subseteq (b) \subseteq \psi(M)$, and by maximality $\psi(M) = \alpha(M)$. But then $b \in (a)$, and

$$(a) = (b) = (a, \varphi(y))$$

namely, $\varphi(y) \in (a)$, hence a divides $\varphi(y)$. Now, with the isomorphism $F \cong R^n$ fixed above, we may write:

$$y = (s_1, s_2, ..., s_n)$$

so that $\pi_i(y) = s_i$. Since $\pi_i \in F^*$, a divides s_i for each s_i, namely there exists $r_1, ..., r_n$ such that $s_i = ar_i$. Letting x be the element associated to the element

$$(r_1, ..., r_n)$$

in the isomorphism, we get that

$$y = ax$$

giving the desired x.

Let's now show that x, y have the desired properties. First, notice that

$$a = \alpha(y) = \alpha(ax) = a\alpha(x)$$

Since R is an integral domain and $a \neq 0$, $\alpha(x) = 1$. Next, let $F' = \ker(\alpha)$ and $M' = F' \cap M$. Notice that for every $z \in F$,

$$z = \alpha(z)x + (z - \alpha(z)x)$$

which by linearity:

$$\alpha(z - \alpha(z)x) = \alpha(z) - \alpha(z)\alpha(x) = \alpha(z) - \alpha(z) = 0$$

Thus,

$$z - \alpha(z)x \in \ker \alpha$$

implying that $F = \langle x \rangle + F'$. Next, take $rx \in \langle x \rangle \cap F'$. Then $\alpha(rx) = 0$, and by linearity $r\alpha(x) = 0$, but $\alpha(x) = 1$ so $r = 0$, and so $\langle x \rangle \cap F' = 0$, which by definition implies:

$$F = \langle x \rangle \oplus F'$$

Giving part of the 3rd claim. Finally, if $z \in M$, then $a \big| \alpha(z)$, and indeed

$$\alpha(z) \in \alpha(M) = (a)$$

Writing $\alpha(z) = ca$, we get $\alpha(z)x = cax = cy$. Splitting z as we have done before, notice that:

$$z - \alpha(z)x = z - cy \in M \cap F' = M'$$

so $z = cy + (z - cy)$ and $M = \langle y \rangle + M'$. As $\langle y \rangle \subseteq \langle x \rangle$ and $M' \subseteq F'$, also $\langle y \rangle \cap M' \subseteq \langle x \rangle \cap F' = 0$, and so we get:

$$M = \langle y \rangle \oplus M'$$

This proves the third claim, and the fourth holds by the choice of α, a and F', completing the proof.

With this, we may prove one of the main results:

Theorem 14.1.5: Submodule Of Free Module Over PID

Let R be a PID, let F be a free R-module of finite rank n, and let $M \subseteq F$. Then M is free, of rank at most n

***Proof*:**
If $M = 0$, we are done, so let $M \neq 0$. By lemma 14.1.4 there exists a $y_1 \in M$ and a $M^1 \subseteq M$ such that

$$M = \langle y_1 \rangle \oplus M^1$$

If $M^1 = 0$, we are done. Otherwise, since $M^1 \subseteq F$, we may keep applying lemma 14.1.4 producing a y_2 and M^2, and so on. After j steps, $M = \langle y_1 \rangle \oplus \cdots \oplus \langle y_j \rangle \oplus M^j$, and each y_i is nonzero, since $\alpha(y_i) = a\alpha(x) = a \neq 0$ for the elements a, x and the homomorphism α of lemma 14.1.4 at step i. The family $(y_1, \ldots, y_j)$ is linearly independent: if $\sum_i r_i y_i = 0$, then every $r_i y_i = 0$ by the uniqueness in proposition 12.3.6(3), and $r_i = 0$, since the free module F over the integral domain R is torsion-free (proposition 12.3.23(4)). This forces $j \leq n$. Indeed, let K be the field of fractions of R (definition 9.5.2) and view $F \cong R^n$ inside K^n. A relation $\sum_i c_i y_i = 0$ with $c_i \in K$, multiplied by a common denominator $d \neq 0$ of the c_i, becomes a relation with coefficients $dc_i \in R$, so every $dc_i = 0$ and every $c_i = 0$. So $y_1, \ldots, y_j$ are linearly independent in the K-vector space K^n, which has a basis with n elements, and $j \leq n$ by corollary 13.1.13(1). Hence the process terminates, leaving us with a $M^m = 0$ for some $m \leq n$, at which point:

$$M = \langle y_1 \rangle \oplus \langle y_2 \rangle \oplus \cdots \oplus \langle y_m \rangle$$

and $(y_1, \ldots, y_m)$ is a basis of M, as we sought to show.

The finite rank hypothesis is not needed for the conclusion, only for this proof: what makes the argument stop is that a linearly independent subset of F cannot have more than n elements. Removing the hypothesis costs exactly one thing, namely a well-ordering in place of the counter m, and the following proof is the same argument with that substitution made. It is worth reading the finite case first, because the well-ordering conceals how simple the mechanism is: peel off one basis vector, split what the submodule contributes in that coordinate, and recurse.

Theorem 14.1.6: Submodule Of Free Module Over PID, General Case

Let R be a PID, let F be a free R-module of any rank, and let $M \subseteq F$ be a submodule. Then M is free

***Proof*:**
Let $(e_\lambda)_{\lambda \in \Lambda}$ be a basis of F and well-order Λ, which is possible by theorem A.2.4. For each λ write

$$F_{<\lambda} = \bigoplus_{\mu<\lambda} Re_\mu, \qquad F_{\leq\lambda} = \bigoplus_{\mu\leq\lambda} Re_\mu,$$

and set $M_{<\lambda} = M \cap F_{<\lambda}$ and $M_{\leq\lambda} = M \cap F_{\leq\lambda}$.

Step 1: one coordinate at a time. Every $x \in F_{\leq\lambda}$ has a unique expression $x = y + re_\lambda$ with

$y \in F_{<\lambda}$ and $r \in R$, so

$$\pi_\lambda \colon F_{\leq\lambda} \to R, \qquad \pi_\lambda(y + re_\lambda) = r$$

is a well defined R-module homomorphism with $\ker(\pi_\lambda) = F_{<\lambda}$. Its image on $M_{\leq\lambda}$ is a submodule of R, that is an ideal, and R is a principal ideal domain, so

$$\pi_\lambda(M_{\leq\lambda}) = (a_\lambda) \qquad \text{for some } a_\lambda \in R$$

Let $\Lambda' = \{\lambda \in \Lambda \mid a_\lambda \neq 0\}$, and for $\lambda \in \Lambda'$ choose $x_\lambda \in M_{\leq\lambda}$ with $\pi_\lambda(x_\lambda) = a_\lambda$.

Step 2: the splitting at λ. If $a_\lambda = 0$ then $M_{\leq\lambda} \subseteq \ker(\pi_\lambda)$, so $M_{\leq\lambda} = M_{<\lambda}$. If $a_\lambda \neq 0$ then

$$M_{\leq\lambda} = M_{<\lambda} \oplus Rx_\lambda$$

For the sum: given $m \in M_{\leq\lambda}$, write $\pi_\lambda(m) = ra_\lambda$. Then $\pi_\lambda(m - rx_\lambda) = 0$, so $m - rx_\lambda \in M \cap F_{<\lambda} = M_{<\lambda}$. For directness: if $rx_\lambda \in M_{<\lambda}$ then $0 = \pi_\lambda(rx_\lambda) = ra_\lambda$, and R is a domain with $a_\lambda \neq 0$, so $r = 0$. The same computation shows $Rx_\lambda \cong R$ is free of rank one.

Step 3: the x_λ span M. Put $N = \sum_{\lambda\in\Lambda'} Rx_\lambda \subseteq M$ and suppose $N \neq M$. Every element of F has finite support, so every $m \in M$ lies in $M_{\leq\lambda}$ for λ the largest index in its support. Hence the set

$$S = \{\lambda \in \Lambda \mid M_{\leq\lambda} \not\subseteq N\}$$

is nonempty, and being a subset of a well-ordered set it has a least element λ_0. Choose $m \in M_{\leq\lambda_0}$ with $m \notin N$. By Step 2, $m = m' + rx_{\lambda_0}$ with $m' \in M_{<\lambda_0}$, where $r = 0$ in the case $a_{\lambda_0} = 0$. Now m' has finite support contained in $\{\mu \mid \mu < \lambda_0\}$, so $m' \in M_{\leq\mu}$ for some $\mu < \lambda_0$, and $M_{\leq\mu} \subseteq N$ by minimality of λ_0. Then $m = m' + rx_{\lambda_0} \in N$, a contradiction. So $N = M$.

Step 4: the x_λ are independent. Suppose $r_1x_{\lambda_1} + \cdots + r_nx_{\lambda_n} = 0$ with $\lambda_1 < \cdots < \lambda_n$ in Λ'. Each x_{λ_i} with $i < n$ lies in $F_{\leq\lambda_i} \subseteq F_{<\lambda_n} = \ker(\pi_{\lambda_n})$, so applying π_{λ_n} leaves $0 = \pi_{\lambda_n}(r_nx_{\lambda_n}) = r_na_{\lambda_n}$, whence $r_n = 0$. Repeating with λ_{n-1}, and so on down, gives $r_i = 0$ for every i.

So $(x_\lambda)_{\lambda\in\Lambda'}$ is a basis of M, and M is free.

For finitely generated F, the maximality in lemma 14.1.4(4) gives a more precise result, the analogue over a PID of the Smith normal form (theorem 13.2.25):

Corollary 14.1.7: Basis For Submodules Of Free Modules

Let R be a PID, F a finitely-generated free R-module, and $M \subseteq F$ a submodule. Then there exists a basis $(x_1, x_2, ..., x_n)$ for F and nonzero elements $a_1, a_2, ..., a_m \in R$ with $m \leq n$ such that

$$(a_1x_1, ..., a_mx_m)$$

is a basis for M. Furthermore, $a_1|a_2|\cdots|a_m$

***Proof*:**

The rank n of F is well defined by lemma 14.1.1, and we argue by induction on n. If $M = 0$, then any basis $(x_1, \ldots, x_n)$ of F works, with $m = 0$. This covers $n = 0$, where $F = 0$. So let $n \geq 1$, assume the statement for free modules of rank $n - 1$, and let $M \neq 0$.

Step 1: splitting off one basis vector. Lemma 14.1.4 gives a nonzero $a_1 \in R$, elements $x_1 \in F$ and $a_1x_1 \in M$, submodules $F' \subseteq F$ and $M' = F' \cap M$, and a homomorphism $\alpha \colon F \to R$ such

that $F = \langle x_1 \rangle \oplus F'$, $M = \langle a_1 x_1 \rangle \oplus M'$, $\alpha(x_1) = 1$, $\ker(\alpha) = F'$, and $\alpha(M) = (a_1)$ is maximal among the ideals $\varphi(M)$, $\varphi \in \mathrm{Hom}_R(F, R)$.

Step 2: F' is free of rank $n-1$. By theorem 14.1.5, the submodule F' of F is free with a basis of some $k \leq n$ elements, so $F' \cong R^k$. The map $R \to \langle x_1 \rangle$, $r \mapsto r x_1$, is surjective, and injective since $r x_1 = 0$ gives $r = \alpha(r x_1) = 0$. As $\langle x_1 \rangle \cap F' = 0$, proposition 12.3.6 identifies F with $\langle x_1 \rangle \times F'$, so

$$R^n \cong F \cong R \times R^k \cong R^{k+1}$$

and $k = n - 1$ by lemma 14.1.1.

Step 3: the two bases. By the induction hypothesis, applied to the submodule M' of F', there are a basis $(x_2, \ldots, x_n)$ of F' and nonzero $a_2, \ldots, a_m \in R$, with $m - 1 \leq n - 1$ and $a_2 \big| a_3 \big| \cdots \big| a_m$, such that $(a_2 x_2, \ldots, a_m x_m)$ is a basis of M'. Then $(x_1, \ldots, x_n)$ is a basis of F. It spans, since $F = \langle x_1 \rangle + F'$. If $\sum_i r_i x_i = 0$, then $r_1 x_1 = -\sum_{i \geq 2} r_i x_i$ lies in $\langle x_1 \rangle \cap F' = 0$, so $r_1 = \alpha(r_1 x_1) = 0$, and then $r_2 = \cdots = r_n = 0$ by the independence of $(x_2, \ldots, x_n)$. Likewise $(a_1 x_1, \ldots, a_m x_m)$ spans $M = \langle a_1 x_1 \rangle + M'$, and if $\sum_i r_i a_i x_i = 0$, then $r_1 a_1 x_1 \in \langle x_1 \rangle \cap F' = 0$, as $M' \subseteq F'$, so $r_1 a_1 = \alpha(r_1 a_1 x_1) = 0$ and $r_1 = 0$, because $a_1 \neq 0$ and R is an integral domain. The independence of $(a_2 x_2, \ldots, a_m x_m)$ then gives $r_2 = \cdots = r_m = 0$. So $(a_1 x_1, \ldots, a_m x_m)$ is a basis of M, and $m \leq n$.

Step 4: $a_1 \big| a_2$. If $m = 1$ there is nothing left to prove, so let $m \geq 2$. Every element of F is $\sum_i r_i x_i$ with unique coefficients $r_i \in R$, so $\varphi\left(\sum_i r_i x_i\right) = r_1 + r_2$ is a well defined homomorphism $\varphi \colon F \to R$. Then $a_1 = \varphi(a_1 x_1) \in \varphi(M)$, so $(a_1) \subseteq \varphi(M)$, and the maximality of $\alpha(M) = (a_1)$ forces $\varphi(M) = (a_1)$. Hence $a_2 = \varphi(a_2 x_2) \in (a_1)$, that is, $a_1 \big| a_2$. With $a_2 \big| \cdots \big| a_m$ this gives $a_1 \big| a_2 \big| \cdots \big| a_m$, completing the proof.

Corollary 14.1.8: If PID, Then Free Resolutions Of Length 1

Let R be a PID. Then for every finitely generated R-module M and every surjective homomorphism:

$$R^{m_0} \xrightarrow{\pi_0} M \to 0$$

there exists a free R-module R^{m_1} and a homomorphism $\pi_1 : R^{m_1} \to R^{m_0}$ such that the sequence

$$0 \to R^{m_1} \xrightarrow{\pi_1} R^{m_0} \xrightarrow{\pi_0} M \to 0$$

is exact. Along with proposition 12.3.29, an integral domain R is a PID if and only if, for every finitely generated R-module M, the kernel of every surjection $R^n \to M$ is free.

***Proof*:**
Since R^{m_0} is an R-module over a PID, $\ker(\pi_0)$ is is free by theorem 14.1.5, and since R^{m_0} is finitely generated, $\ker(\pi_0) \cong R^{m_1}$ for some $m_1 \in \mathbb{N}$. Hence choose the natural embedding $\pi_1 : \ker(\pi_0) \hookrightarrow R^{m_0}$ giving the desired exact sequence, completing the proof.

Since every finitely generated module over a PID has a free resolution of length 1, it is characterized by a matrix (up to equivalence), namely if

$$0 \to R^m \xrightarrow{\varphi} R^n \xrightarrow{\pi_0} M \to 0$$

Then $M = \mathrm{coker}(\varphi)$, and by exactness we know that φ injects into R^n (or, if the reader rather not

use exactness, φ injects since $R^m = \ker(\pi_0)$). Then by corollary 14.1.7, applied to the submodule $\ker(\pi_0)$ of R^n, there exist a basis $(x_1, x_2, ..., x_n)$ of R^n and nonzero elements $a_1, ..., a_m \in R$ where $a_1 \big| a_2 \big| \cdots \big| a_m$ and $(a_1x_1, a_2x_2, ..., a_mx_m)$ forms a basis for $\ker(\pi_0) \cong R^m$, the number of these elements being m by lemma 14.1.1. Hence, in the basis $(\varphi^{-1}(a_1x_1), \ldots, \varphi^{-1}(a_mx_m))$ of R^m and the basis $(x_1, \ldots, x_n)$ of R^n, M is presented by the matrix:

$$\begin{pmatrix} a_1 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & a_m \\ 0 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & 0 \end{pmatrix}$$

By box 13.2.30, any row/column which contains an a_i that is a unit may be removed. As $a_1 \big| a_2 \big| \cdots \big| a_m$, the a_i that are units come first, and removing them and renumbering leaves non-units $a_1, a_2, ..., a_m$, with $n - m$ unchanged. Then:

$$M \cong R/(a_1) \oplus \cdots \oplus R/(a_m) \oplus R^{n-m}$$

Notice what just happened: a finitely generated module over a PID is isomorphic to a rather simple structure, that is such modules lend themselves to a classification theorem! The following theorem summarizes this result and proves a couple more important facts:

Theorem 14.1.9: Invariant Factor Form of Modules over PID's

Let R be a Principal Ideal Domain and let M be a finitely generated R-module. Then:

1. M is isomorphic to a direct sum of finitely many cyclic modules. More precisely,

   $$M \cong_R R^r \oplus R/(a_1) \oplus R/(a_2) \oplus \cdots \oplus R/(a_m)$$

   for some integer $r \geq 0$ and nonzero element $a_1, a_2, ..., a_m$ of R which are not units in R and which satisfy the divisibility relations

   $$a_1 \mid a_2 \mid \cdots \mid a_m$$

2. M is torsion-free if and only if M is free

3. In the decomposition of (1)

   $$\mathrm{Tor}(M) \cong_R R/(a_1) \oplus R/(a_2) \oplus \cdots \oplus R/(a_m)$$

   in particular M is a torsion module if and only if $r = 0$, and in this case, if $M \neq 0$, the annihilator of M is the ideal (a_m).

Theorem 14.1.16 proves the uniqueness of this decomposition up to isomorphism, allowing us to say it is *the* direct sum of finitely many cyclic modules.

***Proof*:**

1. This is the above analysis, stated more succinctly. Let $x_1, x_2, ..., x_n$ be a generating set of M of minimal cardinality, which exists as M is finitely generated. Then given R^n with basis $b_1, b_2, ..., b_n$:
$$\pi : R^n \to M \qquad \pi(b_i) = x_i$$
is a surjective R-module homomorphism. By corollary 14.1.7, applied to the submodule $\ker(\pi)$ of R^n, there are a basis $y_1, y_2, ..., y_n$ for R^n and nonzero $a_1 \big| a_2 \big| \cdots \big| a_m$ in R, $m \leq n$, such that $a_1 y_1, a_2 y_2, ..., a_m y_m$ is a basis for $\ker(\pi)$. Then:
$$M \cong R^n / \ker(\pi) = (y_1 R \oplus y_2 R \oplus \cdots \oplus y_n R)/(a_1 y_1 R \oplus a_2 y_2 R \oplus \cdots \oplus a_m y_m R)$$
The homomorphism $\sum_i r_i y_i \mapsto (r_{m+1}, \ldots, r_n, r_1 + (a_1), \ldots, r_m + (a_m))$ is well defined, the coefficients r_i being unique, and surjective. Its kernel consists of the $\sum_i r_i y_i$ with $a_i \big| r_i$ for $i \leq m$ and $r_i = 0$ for $i > m$, that is, $a_1 y_1 R \oplus \cdots \oplus a_m y_m R$, so by theorem 12.2.10(1) this quotient is isomorphic to:
$$R^{n-m} \oplus R/(a_1) \oplus R/(a_2) \oplus \cdots \oplus R/(a_m)$$
No a_i is a unit. Otherwise a_1, which divides a_i, is a unit, so $y_1 = a_1^{-1}(a_1 y_1) \in \ker(\pi)$, and $M = \pi(R^n)$ is generated by the $n-1$ elements $\pi(y_2), \ldots, \pi(y_n)$, contradicting the minimality of n. This proves (1), with $r = n - m$.

2. If M is free, it is torsion-free by proposition 12.3.23(4), as R is an integral domain. Conversely, if $m \geq 1$, the class of 1 in the summand $R/(a_1)$ is nonzero, as a_1 is not a unit, and it is killed by $a_1 \neq 0$, so M has a nonzero torsion element. So if M is torsion-free, then $m = 0$ and $M \cong R^r$, and the image of a basis of R^r under such an isomorphism is a basis of M, so M is free.

3. Write $T = R/(a_1) \oplus \cdots \oplus R/(a_m)$. Every element of T is killed by $a_m \neq 0$, since $a_i \big| a_m$ for every i. If a nonzero $s \in R$ kills $(f, t) \in R^r \oplus T$, then $sf = 0$ in R^r forces $f = 0$, as R is an integral domain. So the torsion elements of $R^r \oplus T$ form $0 \oplus T$. An isomorphism maps torsion elements onto torsion elements, since $sz = 0$ exactly when $s\Phi(z) = 0$ for an isomorphism Φ, so $\mathrm{Tor}_R(M) \cong_R T$. If $r \geq 1$, the element $((1, 0, \ldots, 0), 0)$ of $R^r \oplus T$ is not torsion, so M is a torsion module exactly when $r = 0$. Let $r = 0$ and $M \neq 0$, so that $m \geq 1$. Isomorphic modules have the same annihilator, so $\mathrm{Ann}_R(M) = \mathrm{Ann}_R(T)$. An element $c \in (a_m)$ lies in every (a_i), so it kills every summand $R/(a_i)$. Conversely, if c kills T, then it kills the class of 1 in $R/(a_m)$, so $c \in (a_m)$. Hence $\mathrm{Ann}_R(M) = (a_m)$, completing the proof.

In order to prove the uniqueness of this decomposition, we shall show how we can systematically "embed" parts of M into a vector-space and take advantage of the invariant basis number (IBN) of finitely dimensional isomorphic vector spaces, that is if two vector-spaces are isomorphic they are the same dimension.

First, since R is a PID the annihilator of M is a principal ideal. We start by finding the annihilator of M when it is a torsion module:

Lemma 14.1.10: Annihilator of Module Over PID

Let M be a finitely generated torsion module over a PID R with decomposition:

$$R/(a_1) \oplus \cdots R/(a_m)$$

with $a_1 \big| \cdots \big| a_m$. Then $\text{Ann}_R(M) = (a_m)$.

***Proof*:**
$0 \neq r \in \text{Ann}_R(M)$ so that

$$0 = r \cdot (1, ..., 1) = (r, ..., r)$$

which by the given decomposition implies:

$$r \equiv 0 \bmod (a_m) \qquad \Leftrightarrow \qquad r \in (a_m)$$

Hence, $\text{Ann}_R(M) \subseteq (a_m)$. Conversely, if $r \in (a_m)$ and $y \in M$, decomposing y so that

$$y = (y_1, ..., y_m) \qquad y_i \in R/(a_i)$$

since $r \in (a_m) \subseteq (a_i)$, we see that for each a_i:

$$ry_i = 0$$

hence $ry = 0$, and so $r \in \text{Ann}_R(M)$, as we sought to show.

Next, since R is a UFD, $a_m = up_1^{k_1} p_2^{k_2} \cdots p_r^{k_r}$ with u a unit and $p_1, \ldots, p_r$ pairwise non-associate primes. For $i \neq j$, $(p_i^{k_i}) + (p_j^{k_j}) = (d)$ for some d, since R is a PID, and d divides both $p_i^{k_i}$ and $p_j^{k_j}$. As p_i and p_j are non-associate primes, d is a unit, so

$$(p_i^{k_i}) + (p_j^{k_j}) = (1)$$

Hence, by the Chinese Remainder Theorem (theorem 9.6.11): $R/(a_m) \cong R/(p_1^{k_1}) \oplus \cdots \oplus R/(p_r^{k_r})$ Doing this to every a_i gives the following decomposition:

Definition 14.1.11: Elementary Divisor Form Of Modules Over PID's

Let M be a finitely generated module over a PID R. An isomorphism

$$M \cong R^r \oplus R/(p_1^{k_1}) \oplus \cdots \oplus R/(p_s^{k_s})$$

with $p_1, \ldots, p_s$ primes of R (not necessarily distinct) and $k_i \geq 1$, as obtained above, is called a[a] elementary divisor form of M, and the prime powers $p_1^{k_1}, \ldots, p_s^{k_s}$ are called the *elementary divisors* of M.

[a] we cannot yet say "the"

Given M is torsion, since M can be decomposed into modules of the form $R/(p_k^{k_i})$, we may group together these modules to form a decomposition:

$$M \cong N_1 \oplus \cdots \oplus N_s$$

where the elements of N_i are annihilated by a power of a distinct prime p_i.

Theorem 14.1.12: Primary Decomposition Theorem

Let R be a PID and let M be a nonzero torsion R-module (not necessarily finitely generated) whose annihilator is $\mathrm{Ann}_R(M) = (a)$ with $a \neq 0$. Suppose the factorization of a into prime powers in R is

$$a = up_1^{\alpha_1} p_2^{\alpha_2} \cdots p_n^{\alpha_n}$$

with u a unit, $p_1, \ldots, p_n$ pairwise non-associate primes and every $\alpha_i \geq 1$, and let $N_i = \{x \in M | p_i^{\alpha_i} x = 0\}$, $1 \leq i \leq n$. Then N_i is a submodule of M with annihilator $(p_i^{\alpha_i})$ and is the submodule of M of all elements annihilated by some power of p_i. Thus, we get

$$M = N_1 \oplus N_2 \oplus \cdots \oplus N_n$$

If M is finitely generated, then each N_i is the direct sum of finitely many cyclic modules whose annihilators are generated by divisors of $p_i^{\alpha_i}$

***Proof*:**

The ring R is a Unique Factorization Domain (proposition 10.1.33), and its primes are its irreducible elements. If $b, c \in R$ are nonzero and have no common prime factor, then $(b)+(c) = (1)$: since R is a PID, $(b) + (c) = (d)$ for some d, which divides b and c, so $d \neq 0$ has no prime factor and is a unit. The same argument applies to any finite list of nonzero elements with no prime factor common to all of them. Write $q_i = \prod_{j \neq i} p_j^{\alpha_j}$, so that $a = up_i^{\alpha_i} q_i$. A prime dividing a power of p_i is an associate of p_i, and a prime dividing q_i is an associate of some p_j with $j \neq i$, so for every $k \geq 0$ the elements p_i^k and q_i have no common prime factor.

Step 1: N_i is a submodule, made of the elements annihilated by a power of p_i. As R is commutative, $x \mapsto p_i^{\alpha_i} x$ is an R-module homomorphism $M \to M$, and N_i is its kernel, a submodule by theorem 12.2.10(1). If $p_i^k x = 0$ for some $k \geq 0$, write $1 = sp_i^k + tq_i$ with $s, t \in R$. Then

$$p_i^{\alpha_i} x = sp_i^{\alpha_i}(p_i^k x) + tu^{-1}(ax) = 0$$

so $x \in N_i$, and conversely every element of N_i is annihilated by the power $p_i^{\alpha_i}$.

Step 2: $M = N_1 + \cdots + N_n$. No prime divides all of $q_1, \ldots, q_n$: such a prime would be an associate of some p_j, and p_j does not divide q_j. So $1 = s_1 q_1 + \cdots + s_n q_n$ for some $s_i \in R$. For $x \in M$, $x = \sum_i s_i q_i x$, and $p_i^{\alpha_i}(s_i q_i x) = s_i u^{-1}(ax) = 0$, so $s_i q_i x \in N_i$.

Step 3: the sum is direct. Let $y_1 + \cdots + y_n = 0$ with $y_i \in N_i$, and fix j. For $i \neq j$, $p_i^{\alpha_i}$ divides q_j, so $q_j y_i = 0$, and multiplying the relation by q_j gives $q_j y_j = 0$. Writing $1 = sq_j + tp_j^{\alpha_j}$ gives $y_j = sq_j y_j + tp_j^{\alpha_j} y_j = 0$. So every element of $N_1 + \cdots + N_n$ is uniquely a sum of elements of the N_i, and $M = N_1 \oplus \cdots \oplus N_n$ by proposition 12.3.6.

Step 4: the annihilator of N_i. As $p_i^{\alpha_i} \in \mathrm{Ann}_R(N_i)$ and R is a PID, $\mathrm{Ann}_R(N_i) = (d_i)$ with d_i dividing $p_i^{\alpha_i}$, so d_i is an associate of $p_i^{\beta_i}$ for some $0 \leq \beta_i \leq \alpha_i$. The element $c = p_i^{\beta_i} q_i$ annihilates every N_j: for $j \neq i$ because $p_j^{\alpha_j}$ divides q_i, and for $j = i$ because $p_i^{\beta_i}$ annihilates N_i. By Step 2, c annihilates M, so $c \in (a)$, that is, $up_i^{\alpha_i} q_i$ divides $p_i^{\beta_i} q_i$. Cancelling $q_i \neq 0$ in the integral domain R, $p_i^{\alpha_i}$ divides $p_i^{\beta_i}$, and $\beta_i < \alpha_i$ would make the nonunit $p_i^{\alpha_i - \beta_i}$ divide 1. So $\beta_i = \alpha_i$ and $\mathrm{Ann}_R(N_i) = (p_i^{\alpha_i})$.

Step 5: the finitely generated case. By Step 3, $\sum_j y_j \mapsto y_i$ is a well defined surjective R-module homomorphism $M \to N_i$, so N_i is generated by the images of finitely many generators of M. By theorem 14.1.9(1), $N_i \cong R^r \oplus R/(c_1) \oplus \cdots \oplus R/(c_l)$ with $c_1 \big| \cdots \big| c_l$ nonzero nonunits. As N_i is torsion, and nonzero since its annihilator $(p_i^{\alpha_i})$ is not R, $r = 0$ and $\mathrm{Ann}_R(N_i) = (c_l)$ by theorem 14.1.9(3), so $(c_l) = (p_i^{\alpha_i})$ by Step 4, and every c_k divides $p_i^{\alpha_i}$. Each $R/(c_k)$ is cyclic, generated by the class of 1, with annihilator (c_k). The images of these summands under the isomorphism are finitely many cyclic submodules with annihilators (c_k), and N_i is their direct sum, completing the proof.

We give a name to the N_i modules

Definition 14.1.13: p_i-Primary Components

The submodules N_i in the previous theorem are called *p_i-primary components* of M.

With this, we come to the key result:

Lemma 14.1.14: Embedding Module over PID into Vector Space

Let R be a PID, let p be a prime in R, and let F denote the field $R/(p)$. For an R-module M and $j \geq 0$, the set $p^j M = \left\{ p^j x \;\middle|\; x \in M \right\}$ is a submodule of M, and $p^j M/p^{j+1}M$, in particular M/pM, is an F-vector space with $(r + (p)) \cdot \overline{x} = \overline{rx}$. Then, with isomorphisms of F-vector spaces:

1. if $M = R^t$, then $M/pM \cong F^t$
2. if $M = R/(a)$ where a is a nonzero element of R, then
$$M/pM \cong \begin{cases} F & \text{if } p \text{ divides } a \text{ in } R \\ 0 & \text{if } p \text{ does not divide } a \text{ in } R \end{cases}$$
3. if $M = R/(a_1) \oplus R/(a_2) \oplus \cdots \oplus R/(a_k)$ where each a_i is divisible by p, then $M/pM \cong F^k$
4. if $M = R/(p^e)$ with $e \geq 1$, then for $j \geq 1$, $p^{j-1}M/p^j M \cong F$ when $j \leq e$ and $p^{j-1}M/p^j M = 0$ when $j > e$
5. if $M = M_1 \oplus \cdots \oplus M_k$, then $p^{j-1}M/p^j M \cong \bigoplus_l p^{j-1}M_l/p^j M_l$ for every $j \geq 1$, and an isomorphism of R-modules $\varphi\colon M \to M'$ induces isomorphisms $p^{j-1}M/p^j M \to p^{j-1}M'/p^j M'$ for every $j \geq 1$

***Proof*:**
As R is commutative, $p^j M$ is closed under addition and under multiplication by R, since $r(p^j x) = p^j(rx)$. The element p maps $p^j M$ into $p^{j+1}M$, so it acts as zero on $p^j M/p^{j+1}M$, and the action of $r + (p)$ is well defined. An R-linear map between two such quotients is F-linear, so each R-module isomorphism below is an isomorphism of F-vector spaces. Write $\overline{x}$ for classes.

1. The map $R^t \to F^t$, $(r_1, \ldots, r_t) \mapsto (r_1 + (p), \ldots, r_t + (p))$, is R-linear and surjective, with kernel pR^t. By theorem 12.2.10(1), $M/pM \cong F^t$.

2. The image of the ideal (p) under the projection $R \to R/(a)$ is $pM = ((p)+(a))/(a)$, so theorem 12.2.10(3), applied to the submodules $(a) \subseteq (p)+(a)$ of R, gives $M/pM \cong R/((p)+(a))$. As R is a PID, $(p)+(a) = (d)$ for some d dividing p, and p is irreducible (proposition 10.1.7), so d is a unit or an associate of p. If p divides a, then $(a) \subseteq (p)$, so $(d) = (p)$ and $M/pM \cong R/(p) = F$. If p does not divide a, then d, which divides a, is not an associate of p, so d is a unit, $(d) = R$ and $M/pM = 0$.

3. By (5) with $j = 1$ and by (2), $M/pM \cong \bigoplus_l (R/(a_l))/p(R/(a_l)) \cong F^k$.

4. For $i \geq 0$, $p^i M$ is the image of the ideal (p^i) in $R/(p^e)$, that is, $(p^i)+(p^e) = (p^{\min(i,e)})$ modulo (p^e). So theorem 12.2.10(3), applied to the module $(p^{\min(j-1,e)})$ and its submodules $(p^e) \subseteq (p^{\min(j,e)})$, gives $p^{j-1}M/p^jM \cong (p^{\min(j-1,e)})/(p^{\min(j,e)})$. If $j > e$, both exponents are e and this is 0. If $j \leq e$, it is $(p^{j-1})/(p^j)$, and $r \mapsto rp^{j-1} + (p^j)$ is a surjective R-linear map $R \to (p^{j-1})/(p^j)$ with kernel (p): p^j divides rp^{j-1} exactly when p divides r, by cancellation of p^{j-1} in the integral domain R (proposition 9.1.24). By theorem 12.2.10(1), $p^{j-1}M/p^jM \cong R/(p) = F$.

5. Multiplication by p^i acts componentwise, so $p^iM = \bigoplus_l p^iM_l$ for every $i \geq 0$. The map $p^{j-1}M \to \bigoplus_l p^{j-1}M_l/p^jM_l$, $(y_1, \ldots, y_k) \mapsto (\overline{y_1}, \ldots, \overline{y_k})$, is surjective and R-linear with kernel $\bigoplus_l p^jM_l = p^jM$, which gives the first isomorphism by theorem 12.2.10(1). An isomorphism φ satisfies $\varphi(p^iM) = p^i\varphi(M) = p^iM'$, so $y \mapsto \overline{\varphi(y)}$ is a surjective R-linear map $p^{j-1}M \to p^{j-1}M'/p^jM'$, with kernel $p^{j-1}M \cap \varphi^{-1}(p^jM') = p^jM$, and theorem 12.2.10(1) gives the induced isomorphism, completing the proof.

The invariant factors determine the elementary divisors, and conversely:

Lemma 14.1.15: Invariant Factors And Prime Powers

Let R be a PID and let $a_1 \big| a_2 \big| \cdots \big| a_m$ be nonzero nonunits of R. Let $p_1, \ldots, p_s$ be pairwise non-associate primes such that every prime factor of a_m is an associate of some p_j, and write $a_k = u_k p_1^{e_{k1}} \cdots p_s^{e_{ks}}$ with u_k a unit and $e_{kj} \geq 0$ (proposition 10.1.33). Then:

1. $R/(a_k) \cong \bigoplus_{j\,:\,e_{kj} \geq 1} R/(p_j^{e_{kj}})$ as R-modules, for each k

2. $e_{1j} \leq e_{2j} \leq \cdots \leq e_{mj}$ for each j, and m is the largest of the numbers $\#\{k \mid e_{kj} \geq 1\}$, $1 \leq j \leq s$

3. if $b_1 \big| \cdots \big| b_l$ are nonzero nonunits with $b_k = v_k p_1^{f_{k1}} \cdots p_s^{f_{ks}}$, v_k a unit, such that for each j the positive exponents among $e_{1j}, \ldots, e_{mj}$ and among $f_{1j}, \ldots, f_{lj}$ are the same up to order, then $l = m$ and b_k is an associate of a_k for every k

***Proof*:**
The ring R is a Unique Factorization Domain, whose primes are its irreducible elements (proposition 10.1.33, proposition 10.1.7), so the factorizations exist and the exponent of p_j in a product is the sum of its exponents in the factors.

1. Fix k. For $j \neq j'$, the ideals $(p_j^{e_{kj}})$ and $(p_{j'}^{e_{kj'}})$ are comaximal: their sum is (d) for some d dividing both generators, and a prime factor of d would be an associate of p_j and of $p_{j'}$, so d is a unit. The product of these ideals over the j with $e_{kj} \geq 1$ is (a_k), so the Chinese

Remainder Theorem (theorem 9.6.11) gives the isomorphism, induced by $r \mapsto (r + (p_j^{e_{kj}}))_j$ and hence R-linear.

2. Writing $a_{k+1} = ca_k$, the exponent of p_j in a_{k+1} is e_{kj} plus its exponent in c, so $e_{kj} \leq e_{k+1,j}$. Each number $\#\{k \mid e_{kj} \geq 1\}$ is at most m. As a_1 is not a unit, $e_{1j} \geq 1$ for some j, and then $e_{kj} \geq 1$ for every k, so this j gives the number m.

3. By (2), applied to the a_k and to the b_k, both m and l are the largest number of positive exponents of a single p_j, so $l = m$. For each j, the sequences $e_{1j} \leq \cdots \leq e_{mj}$ and $f_{1j} \leq \cdots \leq f_{mj}$ have the same length and the same positive entries, so both consist of zeros followed by these entries in increasing order, and $e_{kj} = f_{kj}$ for all k. Hence $b_k = v_k u_k^{-1} a_k$, completing the proof.

Theorem 14.1.16: Fundamental Theorem of Uniqueness of Factors

Let R be a Principal Ideal Domain. Then two finitely generated R-modules M_1 and M_2 are isomorphic if and only if they have the same rank, that is, the same exponent r of the free part in theorem 14.1.9(1), and the same list of invariant factors up to units, and if and only if they have the same rank and the same elementary divisors (definition 14.1.11) up to associates and order. In particular, the rank, the invariant factors and the elementary divisors of M do not depend on the decomposition used to compute them.

***Proof*:**

If the ranks agree and the invariant factors agree up to units (resp. the elementary divisors agree up to associates and order), then the two direct sums are isomorphic summand by summand, since $R/(c) = R/(c')$ when c and c' are associates (proposition 10.1.4), so $M_1 \cong M_2$. Conversely, let $\varphi\colon M_1 \to M_2$ be an isomorphism, and fix decompositions $M_i \cong R^{r_i} \oplus T_i$, where T_i is the direct sum of the cyclic summands $R/(c)$, $c \neq 0$, of the invariant factor form or of an elementary divisor form.

Step 1: the rank. The torsion elements (definition 12.3.18) of $N = R^r \oplus T$, with T a finite direct sum of modules $R/(c)$ with $c \neq 0$, are the elements of $0 \oplus T$: an element of T is killed by the product of the c's, which is nonzero as R is an integral domain, and if a nonzero s kills (f, t), then $sf = 0$ in R^r forces $f = 0$. So the projection onto R^r induces $N/\mathrm{Tor}_R(N) \cong R^r$ (theorem 12.2.10(1)). An isomorphism maps torsion elements to torsion elements, since $sx = 0$ exactly when $s\varphi(x) = 0$, so φ restricts to an isomorphism $\mathrm{Tor}_R(M_1) \to \mathrm{Tor}_R(M_2)$ and induces $R^{r_1} \cong M_1/\mathrm{Tor}_R(M_1) \cong M_2/\mathrm{Tor}_R(M_2) \cong R^{r_2}$. If R is a field, this is an isomorphism of vector spaces, which maps a basis to a basis, so $r_1 = r_2$ by corollary 13.1.13(2). Otherwise R has a nonzero nonunit, which has a prime factor p, and lemma 14.1.14(5) and (1) give an isomorphism of F-vector spaces $F^{r_1} \cong F^{r_2}$, $F = R/(p)$, so again $r_1 = r_2$.

Step 2: the elementary divisors at a prime p. By Step 1, $T_1 \cong \mathrm{Tor}_R(M_1) \cong \mathrm{Tor}_R(M_2) \cong T_2$. Write T_i through an elementary divisor form, $T_i \cong \bigoplus_l R/(q_{i,l})$ with prime powers $q_{i,l}$, given directly or obtained from the invariant factor form by lemma 14.1.15(1). Fix a prime p and put $T[p^\infty] = \{x \in T \mid p^k x = 0 \text{ for some } k \geq 0\}$, a submodule of T. An isomorphism $T \to T'$ maps $T[p^\infty]$ onto $T'[p^\infty]$. On a summand $R/(q)$ with q a power of an associate of p, say $(q) = (p^e)$, the element p^e acts as 0. On a summand $R/(q)$ with q a power of a prime p' not associate to p, $p^k x = 0$ forces $x = 0$. Indeed $(p^k) + (q) = (d)$ with d dividing p^k and q, and a prime factor of d would be an associate of p and of p', so d is a unit, $1 = sp^k + tq$ for some $s, t \in R$, and

$x = sp^k x + tqx = 0$. As p^k acts componentwise, $T_i[p^\infty]$ corresponds to the sum of the summands $R/(q_{i,l})$ with $(q_{i,l}) = (p^e)$ for some $e \geq 1$. So the modules $P_i = \bigoplus_{l\,:\,(q_{i,l})=(p^{e_{i,l}})} R/(p^{e_{i,l}})$ satisfy $P_1 \cong P_2$.

Step 3: counting. For $P = \bigoplus_{l=1}^{n} R/(p^{e_l})$ with every $e_l \geq 1$ and for $j \geq 1$, lemma 14.1.14(5) and (4) give

$$\dim_F \left(p^{j-1}P/p^j P\right) = \#\{l \mid e_l \geq j\}$$

and by lemma 14.1.14(5) these numbers are the same for isomorphic modules, the dimension of a vector space being well defined (corollary 13.1.13(2)). The number of l with $e_l = j$ is $\#\{l \mid e_l \geq j\} - \#\{l \mid e_l \geq j+1\}$, so P_1 and P_2 have the same exponents $e_{i,l}$ up to order. As p was arbitrary, the elementary divisors of M_1 and M_2 agree up to associates and order.

Step 4: the invariant factors. Let $a_1 \big| \cdots \big| a_m$ and $b_1 \big| \cdots \big| b_l$ be invariant factors of M_1 and of M_2, and choose pairwise non-associate primes $p_1, \ldots, p_s$ such that every prime factor of a_m or of b_l is an associate of some p_j. By lemma 14.1.15(1), the decompositions give elementary divisor forms whose elementary divisors are the $p_j^{e_{kj}}$ and the $p_j^{f_{kj}}$ with positive exponents, in the notation of lemma 14.1.15. By Step 3, for each j the positive exponents of p_j agree up to order, so lemma 14.1.15(3) gives $l = m$ and b_k associate to a_k for every k, completing the proof.

Finally, we can now refer to *the* invariant factor form and *the* elementary divisor form of a finitely generated R-module M for R a PID.

Corollary 14.1.17: Computation Shortcut For Invariant Factors

Let R be a Principal Ideal Domain and let M be a finitely generated R-module, with invariant factors $a_1 \big| \cdots \big| a_m$ (theorem 14.1.9). Then, up to units:

1. The elementary divisors of M are the prime power factors of the invariant factors of M: if $a_k = u_k \prod_j p_j^{e_{kj}}$ with u_k a unit and $p_1, \ldots, p_s$ pairwise non-associate primes, then the elementary divisors are the $p_j^{e_{kj}}$ with $e_{kj} \geq 1$, one for each pair (k, j).
2. The largest invariant factor of M is the product of the largest of the distinct prime powers among the elementary divisors of M, the next largest invariant factor is the product of the largest of the distinct prime powers among the remaining elementary divisors of M, and so on

***Proof*:**
Write $M \cong R^r \oplus R/(a_1) \oplus \cdots \oplus R/(a_m)$ as in theorem 14.1.9(1).

1. Combining the isomorphisms of lemma 14.1.15(1) for $k = 1, \ldots, m$ gives an elementary divisor form of M (definition 14.1.11) whose elementary divisors are the $p_j^{e_{kj}}$ with $e_{kj} \geq 1$. By theorem 14.1.16, the elementary divisors of M do not depend on the decomposition, which proves (1).
2. By lemma 14.1.15(2), $e_{1j} \leq e_{2j} \leq \cdots \leq e_{mj}$ for each j. So the largest power of p_j among the elementary divisors is $p_j^{e_{mj}}$ when $e_{mj} \geq 1$, and the product of these largest powers is $\prod_j p_j^{e_{mj}}$, which is a_m up to the unit u_m. Removing one copy of each $p_j^{e_{mj}}$ leaves the elementary divisors $p_j^{e_{kj}}$ with $k \leq m-1$, which by (1) are those of the module $R/(a_1) \oplus \cdots \oplus R/(a_{m-1})$, with invariant factors $a_1 \big| \cdots \big| a_{m-1}$. Induction on m gives the

remaining invariant factors in decreasing order, and the invariant factors are unique up to units by theorem 14.1.16, completing the proof.

Corollary 14.1.18: The Fundamental Theorem of f.g. Abelian Groups

Let G be a finitely generated abelian group. Then:

$$G \cong \mathbb{Z}^r \times \mathbb{Z}_{n_1} \times \cdots \mathbb{Z}_{n_s}$$

for some integers $r, n_1, ..., n_s$ with the following conditions:

1. $r \geq 0$ and $n_j \geq 2$ for all j, and
2. $n_{i+1} | n_i$ for $1 \leq i \leq s-1$

***Proof*:**
Take $R = \mathbb{Z}$ in theorem 14.1.9. Note that the invariant factors are listed in reverse order in Chapter 5 for ease of computation.

Summary

Any finitely generated module over a PID is fully characterised by its rank $r \in \mathbb{N}$ and a chain of proper nonzero ideals $(a_1), ..., (a_m)$ such that

$$(a_1) \supseteq (a_2) \supseteq \cdots \supseteq (a_m)$$

If M is torsion, (a_m) is the annihilator of M. For a torsion module, the invariant factors also combine into a single ideal, which records their product:

Definition 14.1.19: Characteristic Ideal

Let M be a finitely generated torsion module over a PID, with invariant factors $a_1 | \cdots | a_m$. Then the ideal

$$(a_1 a_2 \cdots a_m)$$

is known as the *characteristic ideal* of M.

The characteristic ideal does not depend on the choice of invariant factors, since these are unique up to units (theorem 14.1.16). It no longer determines M: $\mathbb{Z}/2\mathbb{Z} \oplus \mathbb{Z}/2\mathbb{Z}$ and $\mathbb{Z}/4\mathbb{Z}$ both have characteristic ideal (4). The name "characteristic ideal" is given due to its link to the characteristic polynomial explored in section 13.5, made precise in section 14.3.

14.2 Application: Linear Algebra revisited

With the classification established, it can be put to good use in better understanding linear operators between vector spaces. Recall in section 13.2.2 that it was mentioned that chapter 14 is essentially the theoretical build-up for studying linear operators. We now make good on this promise. In this

section, we define a $k[x]$-module structure on vector-spaces given by a linear operator α, and we shall define a new invariant similar to the characteristic polynomial known as the *minimal polynomial*.

To define this new module, we will have to juggle lot of components. Let k be a field, let x be an indeterminate and let R be the polynomial ring $k[x]$. Let V be an k-module (i.e. a vector space over k). Notice that we are using the same field for V and for $k[x]$. Finally, let α be a linear operator on V, that is a linear operator from V to V where we consider the domain and codomain to be the same object. We shall extend the k-module V (i.e. a vector-space) into an $k[x]$-module using α. To accomplish this, we will take advantage of nice properties of linear operators. First, define

$$\alpha^0 = I \qquad \alpha^n = \alpha \circ \alpha \circ \cdots \circ \alpha \qquad (n \text{ times})$$
$$a_n\alpha^n + a_{n-1}\alpha^{n-1} + \cdots + a_1\alpha + a_0 \in \mathrm{End}_k(V)$$

where I is the identity map from V to V and $\circ$ is function composition (which is well-defined since α maps to V). We will now define the ring action of $k[x]$. Let $p(x) \in k[x]$ be a polynomial. Then define an action of the ring element $p(x)$ on the module element b by:

$$\begin{aligned} p(x) \cdot v &= (a_n\alpha^n + \cdots + a_1\alpha + a_0)(v) \\ &= a_n\alpha^n(v) + \cdots + a_1\alpha(v) + a_0 v \end{aligned}$$

That is, $p(x)$ acts by substituting the linear operator α for x in $p(x)$ and applying the resulting linear operator to v. These rules satisfy the module axioms that make V into an $k[x]$-module. For $p = \sum_i a_i x^i$ and $q = \sum_j b_j x^j$, $(p+q)(\alpha) = p(\alpha) + q(\alpha)$ by definition, and since each α^i is linear and $\alpha^i \circ \alpha^j = \alpha^{i+j}$,

$$p(\alpha) \circ q(\alpha) = \sum_{i,j} a_i b_j \, \alpha^{i+j} = (pq)(\alpha)$$

So $(p+q)\cdot v = p\cdot v + q\cdot v$, $p\cdot(v+w) = p\cdot v + p\cdot w$ as $p(\alpha)$ is linear, $(pq)\cdot v = p\cdot(q\cdot v)$, and $1\cdot v = v$. The same computation applies to an endomorphism of a module over any commutative ring R, giving an $R[x]$-module. Note that k is a subring of $k[x]$, and this new module with constant polynomials acting on it is simply the structure of the vector space. Thus the definition of $k[x]$-module *extends* the action of k to an action of the larger ring $k[x]$.

The key is that $k[x]$ is a PID when k is a field (recall theorem 11.1.1 and its corollaries), and V is a finitely generated $k[x]$-module, generated by any finite k-basis, so theorem 14.1.9 gives:

$$V \cong_{k[x]} (k[x])^n \oplus (k[x]/(f_1)) \oplus \cdots \oplus (k[x]/(f_m))$$

Since V is finite k-dimensional, the action $k[x] \curvearrowright V$ must have rank 0, since if it had rank $n \geq 1$, then the invariant factor $(k[x])^n$ would be nonzero, and it is k-isomorphic to:

$$k[x] \cong k \oplus k \oplus \cdots \oplus k \oplus \cdots$$

which would make V infinite k-dimensional. Thus,

$$V \cong_{k[x]} (k[x]/(f_1)) \oplus \cdots \oplus (k[x]/(f_m))$$

Since the action of $k[x]$ is completely dependent on α, the above proposition may be interpreted as there being a special basis for V that will make the action of α on V rather transparent!

Example 14.2.1: Actions of this Module

1. If $\alpha = 0$, that is, the function maps every element to the identity in V (recall that V is an [abelian] group). Then
$$p(x)v = a_n \cdot 0 + \cdots + a_1 \cdot 0 + a_0 v = a_0 v$$
Thus, with such a linear operator, the $k[x]$-module structure is just like the k-module structure.

2. Another simple example is $\alpha = I$, that is, the identity. Thus
$$p(x)v = a_n v + \cdots + a_0 v = (a_n + \cdots + a_0)v$$
Or simply the sum of coefficient of $p(x)$ act on v

3. Here's a more specific example. Take V to be the affine n-space k^n and let α be the "shift operator"
$$\alpha(x_1, ..., x_n) = (x_2, x_3, ..., x_n, 0)$$
Since V is a vector-space, let e_i be a choice of basis vectors. Then applying this linear operator to only the basis, we get:
$$\alpha^k(e_i) = \begin{cases} e_{i-k} & \text{if } i > k \\ 0 & \text{if } i \leq k \end{cases}$$
So that if $m < n$
$$(a_m x^m + a_{m-1}x^{m-1} + \cdots + a_0)e_n = (0, 0, ..., 0, a_m, a_{m-1}, ..., a_0)$$

4. under what condition is $W \subseteq V$ a $k[x]$-submodule of V with the $k[x]$-action? Since $k[x]$ extends the actions of k, it is necessary that W be a subspace. By the subspace criterion we have that it is necessary that $\alpha(W) \subseteq W$. Not all subspaces satisfy this conditions, if they do they are called α-stable subspaces.

Now, notice that we may take the annihilator of the $k[x]$-module V, that is all elements $p(x) \in k[x]$ such that $p(x) \cdot v = 0$ for all $v \in V$, that is, $p(\alpha) = 0$. Since $k[x]$ is a PID, the annihilator is a principal ideal. It is nonzero: for a field K and $A \in M_n(K)$, the $n^2 + 1$ matrices $I_n, A, \ldots, A^{n^2}$ lie in the vector space $M_n(K)$, which has the basis of the n^2 matrices with a single entry 1 and all other entries 0, so they are linearly dependent (corollary 13.1.13(1)), and a nontrivial relation $\sum_i c_i A^i = 0$ gives a nonzero polynomial $\sum_i c_i x^i$ vanishing at A. The polynomials $f \in K[x]$ with $f(A) = 0$ form an ideal, since $(gf)(A) = g(A)f(A)$ by the computation above with A in place of α, so this ideal is (m) for a nonzero m. Two generators are associates (proposition 10.1.4), so they differ by a unit of $K[x]$, that is, a nonzero constant (proposition 9.2.3(2)), and exactly one generator is monic. This ideal shall be important in contexts where R need not be a field, and hence it is defined more generally:

Definition 14.2.2: Annihilator Ideal and Minimal Polynomial

Let R be an integral domain with field of fractions K, let M be a free R-module with a basis of n elements, and let $\alpha \in \mathrm{End}_R(M)$, with matrix $A \in M_n(R)$ in such a basis. Then the ideal

$$\mathfrak{M}_\alpha = \mathrm{Ann}_{R[x]}(M) = \{f \in R[x] \mid f(\alpha) = 0\}$$

of the $R[x]$-module M on which x acts as α is called the *annihilator ideal* of α. The monic generator m_α of the ideal $\{f \in K[x] \mid f(A) = 0\}$ of $K[x]$ is called the *minimal polynomial* of α. When $R = k$ is a field, $\mathfrak{M}_\alpha = (m_\alpha)$.

The minimal polynomial does not depend on the basis: the matrices of α in two bases are similar (lemma 13.2.16(4)), and $f(PAP^{-1}) = Pf(A)P^{-1}$ for every polynomial f, since $(PAP^{-1})^i = PA^iP^{-1}$. When $R = k$, a polynomial f satisfies $f(\alpha) = 0$ exactly when $f(A) = 0$, since the matrix of $f(\alpha)$ is $f(A)$, by lemma 13.2.16(2) and the linearity of $\alpha \mapsto [\alpha]_\beta$, and an endomorphism is zero exactly when its matrix is.

Recall that $k[x]$ is a PID if and only if k is a field, hence it is natural to work in the setting of $R = k$ is a field to use the above module structure and the classification theorem. The term "minimal polynomial" is because it is the smallest polynomial (with respect to divisibility) that solves the equation $f(\alpha) = 0$. The reader may ask if the characteristic polynomial shares the same property, that is if

$$P_\alpha(\alpha) = 0$$

This is the *Cayley-Hamilton Theorem*, proposition 14.3.10 (the reader may try to use definition 14.1.19 to find a connection between these two polynomials and hence a proof of the above equation). The annihilator ideal shares the same nice invariance properties as the characteristic polynomial:

Lemma 14.2.3: Annihilator Ideal And Similarity

Let α be similar to β. Then $\mathfrak{M}_\alpha = \mathfrak{M}_\beta$

***Proof*:**
Let $\rho \in \mathrm{GL}_R(M)$ such that $\beta = \rho \circ \alpha \circ \rho^{-1}$. Then since when composing conjugates:

$$\beta^k = \rho \circ \alpha^k \circ \rho^{-1}$$

we get that for any $f(x) \in R[x]$,

$$f(\beta) = \rho \circ f(\alpha) \circ \rho^{-1}$$

Hence, $f(\alpha) = 0$ if and only if $f(\beta) = 0$, as we sought to show.

The converse is not necessarily, the case: if $\mathfrak{M}_\alpha = \mathfrak{M}_\beta$, it does not imply that $\alpha \sim \beta$:

Example 14.2.4: Same Annihilators, Not Similar
Over a field k, let A and B be the diagonal matrices with diagonal entries $1, 1, 2$ and $1, 2, 2$, where $2 = 1 + 1 \neq 1$. For $f \in k[x]$, $f(A)$ and $f(B)$ are diagonal with entries $f(1), f(1), f(2)$ and $f(1), f(2), f(2)$, so each vanishes exactly when $f(1) = f(2) = 0$. By proposition 11.3.1, $f(1) = 0$

means $f = (x-1)g$, and then $f(2) = g(2) = 0$ means $x - 2$ divides g. So A and B have the same annihilator ideal, generated by $(x-1)(x-2)$. They are not similar, since their characteristic polynomials $(t-1)^2(t-2)$ and $(t-1)(t-2)^2$ differ (proposition 13.5.2(4)).

On the other hand, annihilators can separate matrices with the same characteristic polynomial, as for the matrices of example 13.5.3: $x - 1$ generates the annihilator of the identity matrix, while it does not annihilate

$$A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$$

since $A - I \neq 0$.

Coming back to the $k[x]$-module on V, we may use it to finally have a converse to establishing two linear operators are similar:

Lemma 14.2.5: Similarity And New Module Structure

Let R be a commutative ring and α, β linear operators on a free R-module M. Then the corresponding $R[x]$-module structures on M are isomorphic if and only if $\alpha \sim \beta$

***Proof*:**
Let M_α, M_β be the corresponding $R[x]$-modules given by α, β respectively. Assume $\alpha \sim \beta$ so that there exists an automorphism ρ of M such that

$$\beta = \rho \circ \alpha \circ \rho^{-1} \Rightarrow \beta \circ \rho = \rho \circ \alpha$$

By induction on i, $\rho \circ \alpha^i = \beta^i \circ \rho$, since $\rho \circ \alpha^{i+1} = \beta^i \circ \rho \circ \alpha = \beta^{i+1} \circ \rho$. As ρ is R-linear, for $f = \sum_i c_i x^i \in R[x]$,

$$\rho(f \cdot v) = \sum_i c_i \rho(\alpha^i(v)) = \sum_i c_i \beta^i(\rho(v)) = f \cdot \rho(v)$$

So $\rho \colon M_\alpha \to M_\beta$ is a bijective $R[x]$-module homomorphism, an isomorphism by lemma 12.2.3, and $M_\alpha \cong_{R[x]} M_\beta$.

Conversely, let $\rho \colon M_\alpha \to M_\beta$ be an isomorphism of $R[x]$-modules. The constants $R \subseteq R[x]$ act on M_α and M_β as on M, so ρ is a bijective R-linear map, an automorphism of M (lemma 12.2.3), and $\rho(\alpha(v)) = \rho(x \cdot v) = x \cdot \rho(v) = \beta(\rho(v))$ for every $v \in M$. So $\beta = \rho \circ \alpha \circ \rho^{-1}$, as we sought to show.

Corollary 14.2.6: Similarity And Isomorphism Classes

Let R be a commutative ring and M a free R-module. Then $\alpha \mapsto M_\alpha$ induces a one-to-one correspondence between the similarity classes of R-linear operators on M and the isomorphism classes of $R[x]$-module structures on M extending its R-module structure.

***Proof*:**
An $R[x]$-module structure on M extending its R-module structure determines $\alpha(v) = x \cdot v$, which is R-linear since $x \cdot (rv) = (xr) \cdot v = (rx) \cdot v = r(x \cdot v)$, and the module axioms give

$(\sum_i c_i x^i) \cdot v = \sum_i c_i \alpha^i(v)$, so the structure is M_α. Distinct operators give distinct structures, since α is recovered as the action of x. By lemma 14.2.5, $\alpha \sim \beta$ if and only if $M_\alpha \cong M_\beta$, so $\alpha \mapsto M_\alpha$ induces the stated bijection, completing the proof.

Exercise 14.2.1

1. Show that the $k[x]$-module on V can be constructed using the universal property of polynomials

14.3 Rational and Jordan Canonical Form

From now on, $R = F$ is a field and V is an n dimensional vector-space over F. We shall use the above results to find a nice representation of a linear operator α on V (also written T), namely a matrix representation for the action of x on the $F[x]$-module V.

By section 14.2, the $F[x]$-module V decomposes as

$$V \cong F[x]/(a_1(x)) \oplus F[x]/(a_2(x)) \oplus \cdots \oplus F[x]/(a_m(x))$$

where $a_1(x), a_2(x), ..., a_m(x)$ are nonzero nonunits of $F[x]$ with the divisibility condition

$$a_1(x) | a_2(x) | \cdots | a_m(x)$$

The units of $F[x]$ are the nonzero constants (proposition 9.2.3(2)), so each a_i has degree at least 1, and replacing a_i by its monic associate does not change the ideal (a_i), so every a_i may be taken monic. Two decompositions of this kind give isomorphic modules, so by theorem 14.1.16 their lists of polynomials agree up to units, and two monic polynomials that differ by a unit are equal. So the monic polynomials $a_1, \ldots, a_m$ are determined by α, and they are called the *invariant factors* of α. The minimal polynomial is the last of them:

Proposition 14.3.1: Finding Minimal Polynomial

Let α be a linear operator on a finite dimensional vector space V over a field F, with invariant factors $a_1 \big| \cdots \big| a_m$. Then the minimal polynomial m_α of α is the largest invariant factor a_m, and all the invariant factors of α divide m_α.

***Proof*:**
By definition 14.2.2, (m_α) is the annihilator of the $F[x]$-module V. Isomorphic modules have the same annihilator, since an isomorphism Φ satisfies $f \cdot \Phi(c) = \Phi(f \cdot c)$, so (m_α) is the annihilator of $F[x]/(a_1) \oplus \cdots \oplus F[x]/(a_m)$, which is (a_m) by lemma 14.1.10. So m_α and a_m generate the same ideal and are associates (proposition 10.1.4), that is, they differ by a nonzero constant (proposition 9.2.3(2)), which is 1 since both are monic. Hence $m_\alpha = a_m$, and every a_i divides a_m, completing the proof.

Be mindful that since everything divides m_α, $a_i(x) \big| m_\alpha(x)$, then $(m_\alpha(x)) \subseteq (a_i(x))$, so it is the *smallest* ideal of all the invariant factors. Be careful then when thinking of the word "minimal", for it is the *biggest* polynomial relative to all the invariant factors. It is smallest when it comes to being the *smallest* polynomial which can annihilate V over $F[x]$ (i.e $m_\alpha(\alpha) = 0$). An invariant factor annihilates V only if it equals a_m.

Each summand $F[x]/(a)$, for a monic $a(x) = x^k + b_{k-1}x^{k-1} + \cdots + b_1 x + b_0$, has the F-basis $1, \overline{x}, \ldots, \overline{x}^{k-1}$, where $\overline{x} = x \bmod (a(x))$, and multiplication by x, which is how α acts, sends each basis element to the next one and $\overline{x}^{k-1}$ to $\overline{x}^k = -b_0 - b_1\overline{x} - \cdots - b_{k-1}\overline{x}^{k-1}$. Its matrix in this basis is therefore

$$\begin{pmatrix} 0 & 0 & \cdots & \cdots & \cdots & -b_0 \\ 1 & 0 & \cdots & \cdots & \cdots & -b_1 \\ 0 & 1 & \cdots & \cdots & \cdots & -b_2 \\ 0 & 0 & \ddots & & & \vdots \\ \vdots & \vdots & & \ddots & & \vdots \\ 0 & 0 & \cdots & \cdots & 1 & -b_{k-1} \end{pmatrix}$$

This matrix has a name:

Definition 14.3.2: Companion Matrix

Let $a(x) = x^k + b_{k-1}x^{k-1} + \cdots + b_1 x + b_0$ be any monic polynomial in $F[x]$. The *companion matrix* of $a(x)$ is the $k \times k$ matrix with 1's down the first subdiagonal, $-b_0, -b_1, .., -b_{k-1}$ down the last column and zeros elsewhere. The companion matrix of $a(x)$ will be denoted $C_{a(x)}$

Lemma 14.3.3: Cyclic Modules And Companion Matrices

Let F be a field, let $a = x^d + b_{d-1}x^{d-1} + \cdots + b_1 x + b_0 \in F[x]$ be monic of degree $d \geq 1$, let $C = F[x]/(a)$, and write $\overline{f}$ for the class of $f \in F[x]$. Then:

1. the classes $\overline{1}, \overline{x}, \ldots, \overline{x}^{d-1}$ form an F-basis of C, and the matrix of multiplication by x on C in this basis is the companion matrix C_a
2. the annihilator of the $F[x]$-module C is (a)
3. $\det(xI_d - C_a) = a$, that is, the characteristic polynomial of C_a is a

***Proof*:**

1. The classes span C: dividing $f \in F[x]$ by a (theorem 11.1.1) gives $f = sa + r$ with $r = 0$ or $\deg r < d$, so $\overline{f} = \overline{r}$ is an F-linear combination of them. They are linearly independent: if $\sum_{l<d} c_l \overline{x}^l = 0$, then a divides $g = \sum_{l<d} c_l x^l$, and a nonzero multiple of a has degree at least d (proposition 9.2.3(1)), so $g = 0$. Multiplication by x sends $\overline{x}^l$ to $\overline{x}^{l+1}$ for $l < d-1$, and $\overline{x}^{d-1}$ to $\overline{x}^d = -b_0\overline{1} - b_1\overline{x} - \cdots - b_{d-1}\overline{x}^{d-1}$, because $\overline{a} = 0$. So its matrix in this basis is the companion matrix C_a (definition 14.3.2).
2. The polynomial a kills every class, and $f \cdot \overline{1} = \overline{f} = 0$ forces $f \in (a)$.
3. Let $N = xI_d - C_a \in M_d(F[x])$. If $d = 1$, then $N = (x + b_0) = a$. Let $d \geq 2$. The nonzero entries of N are $N_{ll} = x$ for $l < d$, $N_{dd} = x + b_{d-1}$, $N_{l+1,l} = -1$ for $l < d$, and $N_{ld} = b_{l-1}$ for $l < d$. For $l = 2, \ldots, d$ add x^{l-1} times row l to row 1. This does not change the determinant (proposition 13.4.9(2), over the commutative ring $F[x]$), and the new first row is $\sum_{l=1}^{d} x^{l-1}$(row l of N). Its entry in column $c < d$ is $x^{c-1}N_{cc} + x^c N_{c+1,c} = x^c - x^c = 0$,

and its entry in column d is

$$\sum_{l=1}^{d-1} x^{l-1} b_{l-1} + x^{d-1}(x + b_{d-1}) = a$$

Moving the first row to the bottom by $d-1$ interchanges of adjacent rows multiplies the determinant by $(-1)^{d-1}$ (proposition 13.4.9(1)). The resulting matrix T consists of rows $2, \ldots, d$ of N followed by $(0, \ldots, 0, a)$, so $T_{lc} = 0$ whenever $l > c$, and its diagonal entries are $-1, \ldots, -1, a$. In the defining sum of $\det(T)$, a permutation $\sigma \neq \mathrm{id}$ has some l with $\sigma(l) < l$, since otherwise $\sigma(d) = d$, then $\sigma(d-1) = d-1$, and so on down to $\sigma = \mathrm{id}$. Its term contains the factor $T_{l\sigma(l)} = 0$. Hence $\det(T) = \prod_l T_{ll} = (-1)^{d-1} a$, and $\det(N) = (-1)^{d-1} \det(T) = a$, completing the proof.

Example 14.3.4: Sanity Check

Let $q(x) = x^4 + x^3 + 5x^2 - 2x + 3 \in \mathbb{Q}[x]$. Its companion matrix is

$$C = \begin{pmatrix} 0 & 0 & 0 & -3 \\ 1 & 0 & 0 & 2 \\ 0 & 1 & 0 & -5 \\ 0 & 0 & 1 & -1 \end{pmatrix}$$

so $Ce_1 = e_2$, $Ce_2 = e_3$, $Ce_3 = e_4$ and $Ce_4 = -3e_1 + 2e_2 - 5e_3 - e_4$. Hence

$$q(C)e_1 = C^4 e_1 + C^3 e_1 + 5C^2 e_1 - 2Ce_1 + 3e_1 = (-3e_1 + 2e_2 - 5e_3 - e_4) + e_4 + 5e_3 - 2e_2 + 3e_1 = 0$$

For $j = 2, 3, 4$ also $q(C)e_j = q(C)C^{j-1}e_1 = C^{j-1}q(C)e_1 = 0$. So $q(C) = 0$, matching lemma 14.3.3(2). This does not yet determine the other invariant factors of an operator, or its minimal polynomial, but it confirms that matrices of this type behave as expected.

The same process applies to each cyclic summand $F[x]/(a_i(x))$, and the bases of the summands together give a basis of V in which α has as matrix the *direct sum* of the companion matrices:

$$\begin{pmatrix} C_{a_1(x)} & & & \\ & C_{a_2(x)} & & \\ & & \ddots & \\ & & & C_{a_m(x)} \end{pmatrix}$$

This matrix is determined by the invariant factors of the $F[x]$-module V, which are unique by theorem 14.1.16. Thus, we can give this matrix a name:

Definition 14.3.5: Rational Cannonical Form

1. A matrix is said to be in *rational canonical form* if it is the direct sum of companion matrices for monic polynomials $a_1(x), a_2(x)..., a_m(x)$ of degree at least one with $a_1(x)|a_2(x)|\cdots|a_m(x)$. The polynomial $a_i(x)$ are called the *invariant factors* of the matrix. Such a matrix is also said to be a *block diagonal* matrix with blocks the companion matrices for $a_i(x)$
2. A *rational canonical form* for a linear transformation T is a matrix representing T which is in rational canonical form

Theorem 14.3.6: Rational Canonical Form

Let V be a finite dimensional vector sapce over the field F and let T be a linear transformation of V, with invariant factors $a_1\big|\cdots\big|a_m$.

1. There is a basis for V with respect to which the matrix T is in rational canonical form, i.e., is a block diagonal matrix whose diagonal blocks are the companion matrices for the invariant factors $a_1(x), a_2(x), ..., a_m(x)$ of T
2. If the matrix of T with respect to some basis is in rational canonical form, with invariant factors $b_1(x)\big|\cdots\big|b_l(x)$, then $l = m$ and $b_i = a_i$ for every i. So T has exactly one rational canonical form, called *the* rational canonical form of T

***Proof*:**

1. Fix an isomorphism of $F[x]$-modules $\Phi\colon F[x]/(a_1)\oplus\cdots\oplus F[x]/(a_m)\to V$. As $F\subseteq F[x]$, Φ is a bijective F-linear map, and $T(\Phi(c)) = x\cdot\Phi(c) = \Phi(x\cdot c)$ for every c, since x acts on V as T. By lemma 14.3.3(1), each summand $F[x]/(a_i)$ has an F-basis in which multiplication by x has matrix C_{a_i}. These bases together form an F-basis of the direct sum, since each element of the direct sum is uniquely the sum of its components, and each component is uniquely a combination of the basis of its summand. Their images under Φ form an F-basis β of V (corollary 13.1.20(5), the two spaces having the same dimension by corollary 13.1.18). Multiplication by x maps each summand into itself, and $T\circ\Phi$ is Φ composed with multiplication by x, so the matrix of T in the basis β is block diagonal with diagonal blocks $C_{a_1},\dots,C_{a_m}$.
2. Let the matrix of T in the basis γ be block diagonal with diagonal blocks $C_{b_1},\dots,C_{b_l}$, let $e_j = \deg b_j \geq 1$, and split γ into consecutive pieces $\gamma_j = (w_{j,0},\dots,w_{j,e_j-1})$ of sizes e_j, with D_j the span of γ_j. Writing $b_j = x^{e_j} + \sum_{s<e_j} c_{j,s}x^s$, the block C_{b_j} says that $T(w_{j,s}) = w_{j,s+1}$ for $s < e_j - 1$ and $T(w_{j,e_j-1}) = -\sum_{s<e_j} c_{j,s}w_{j,s}$. So $w_{j,s} = T^s(w_{j,0})$, the subspace D_j is T-stable, hence an $F[x]$-submodule of V, and $b_j\cdot w_{j,0} = T(w_{j,e_j-1}) + \sum_{s<e_j} c_{j,s}w_{j,s} = 0$. The $F[x]$-module homomorphism $F[x]\to D_j$, $f\mapsto f\cdot w_{j,0}$, is surjective, since its image is a submodule containing every $w_{j,s} = x^s\cdot w_{j,0}$, and its kernel contains (b_j). So $f+(b_j)\mapsto f\cdot w_{j,0}$ is a well defined surjective $F[x]$-module homomorphism $F[x]/(b_j)\to D_j$. Both spaces have dimension e_j over F (lemma 14.3.3(1)), so it is bijective by corollary 13.1.20, an isomorphism of $F[x]$-modules (lemma 12.2.3). As γ is a basis, $(d_1,\dots,d_l)\mapsto d_1+\cdots+d_l$ is a bijective $F[x]$-module homomorphism $D_1\oplus\cdots\oplus D_l\to V$. Hence $V\cong F[x]/(b_1)\oplus\cdots\oplus F[x]/(b_l)$,

and the b_j are nonzero nonunits with $b_1 \mid \cdots \mid b_l$. By theorem 14.1.16, comparing with $V \cong F[x]/(a_1) \oplus \cdots \oplus F[x]/(a_m)$, we get $l = m$ and b_i equal to a_i up to a unit, and $b_i = a_i$ as both are monic, completing the proof.

The rational canonical form of a matrix $A \in M_n(F)$ is that of the linear operator $x \mapsto Ax$ on F^n, and its invariant factors are those of this operator. Similarity is detected by the rational canonical form:

Corollary 14.3.7: RCF And Similarity

Let $\alpha, \beta : V \to V$ be two linear operators on a finite dimensional vector space V over a field F. Then $\alpha \sim \beta$ if and only if they have the same RCF, that is, the same invariant factors. Likewise, two matrices $A, B \in M_n(F)$ are similar if and only if they have the same RCF.

***Proof*:**
By lemma 14.2.5, $\alpha \sim \beta$ if and only if the $F[x]$-modules V_α and V_β, on which x acts as α and as β, are isomorphic. If they are isomorphic, then the decompositions of V_α and V_β into cyclic modules give isomorphic modules, so their invariant factors agree by theorem 14.1.16, both lists being monic. Conversely, if α and β have the same invariant factors $a_1, \ldots, a_m$, then $V_\alpha \cong F[x]/(a_1) \oplus \cdots \oplus F[x]/(a_m) \cong V_\beta$. By theorem 14.3.6, the RCF of an operator is the block diagonal matrix of the companion matrices of its invariant factors, and it determines them, so having the same RCF means having the same invariant factors.

For matrices, let L_A and L_B be the operators $x \mapsto Ax$ and $x \mapsto Bx$ on F^n, whose matrices in the standard basis are A and B. By lemma 13.2.16(3), $A \sim B$ if and only if $L_A \sim L_B$, which by the first part holds if and only if L_A and L_B, that is, A and B, have the same RCF, completing the proof.

The name *rational* reflects that the rational canonical form is computed using only the field operations of F, with no need to find roots of polynomials. In particular it does not change when F is enlarged:

Corollary 14.3.8: RCF Over Extended Fields

Let A and B be two $n \times n$ matrices over a field F and suppose F is a subfield of the field K ($F \subseteq K$, or sometimes written K/F) Then

1. The RCF of A is the same whether it is computed over K or over F. The minimal and characteristic polynomials and the invariant factors of A are the same whether A is considered as a matrix over F or as a matrix over K.
2. The matrices A and B are similar over K if and only if they are similar over F, i.e., there exists an invertible $n \times n$ matrix P with entries from K such that $B = P^{-1}AP$ if and only if there exists an (in general different) invertible $n \times n$ matrix Q with entries from F such that $B = Q^{-1}AQ$

***Proof*:**

1. By theorem 14.3.6(1) and lemma 13.2.16(4), applied to the operator $x \mapsto Ax$ on F^n, there is $P \in \mathrm{GL}_n(F)$ such that $M = P^{-1}AP$ is the rational canonical form of A over F. As $P \in \mathrm{GL}_n(K)$, the matrix M is similar to A over K, so by lemma 13.2.16(4) it is the matrix of the operator $x \mapsto Ax$ on K^n with respect to some basis, and by theorem 14.3.6(2) over K, it is the rational canonical form of A over K. Its diagonal blocks give the invariant factors, so these agree, and so does the minimal polynomial, the largest invariant factor (proposition 14.3.1). The characteristic polynomial $\det(xI_n - A)$ is the same over F and over K, being computed from the entries of $xI_n - A$ in $F[x] \subseteq K[x]$.
2. If $B = Q^{-1}AQ$ with $Q \in \mathrm{GL}_n(F)$, then $Q \in \mathrm{GL}_n(K)$. Conversely, if A and B are similar over K, they have the same rational canonical form over K (corollary 14.3.7), which by (1) is their rational canonical form over F, so they are similar over F by corollary 14.3.7, completing the proof.

The characteristic polynomial is now read off the invariant factors:

Lemma 14.3.9: Characteristic Polynomial For Block Matrix

1. Let R be a commutative ring. If M is the block diagonal matrix
$$M = \begin{pmatrix} A_1 & 0 & \cdots & 0 \\ 0 & A_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & A_k \end{pmatrix}$$
given by the direct sum of square matrices $A_1, A_2, ..., A_k$ over R, then the characteristic polynomial of M is the product of the characteristic polynomials of $A_1, A_2, ..., A_k$
2. Let α be a linear operator on a finite dimensional vector space over a field F, with invariant factors $a_1, \ldots, a_m$. The characteristic polynomial is the product of the invariant factors:
$$P_\alpha(x) = a_1(x)a_2(x)\cdots a_m(x)$$

***Proof*:**

1. The matrix $xI - M$ is block diagonal over the commutative ring $R[x]$, with diagonal blocks $xI - A_1, \ldots, xI - A_k$, so proposition 13.4.8 gives $\det(xI - M) = \prod_i \det(xI - A_i)$.
2. By theorem 14.3.6(1), α has a matrix M that is block diagonal with diagonal blocks $C_{a_1}, \ldots, C_{a_m}$, and $P_\alpha = P_M$ (proposition 13.5.2(4)). By (1) and lemma 14.3.3(3), $P_M = \prod_i \det(xI - C_{a_i}) = a_1 a_2 \cdots a_m$, completing the proof.

Over any commutative ring, a matrix is a root of its characteristic polynomial:

Proposition 14.3.10: Cayley-Hamilton Theorem

Let R be a commutative ring and A an $n \times n$ matrix over R, with characteristic polynomial $P_A(t) = \sum_{i=0}^{n} c_i t^i$. Then

$$P_A(A) = \sum_{i=0}^{n} c_i A^i = 0$$

Consequently $P_\alpha(\alpha) = 0$ for every endomorphism α of a free R-module with a basis of n elements. If $R = F$ is a field, then the minimal polynomial of A divides the characteristic polynomial of A.

***Proof*:**

A matrix with entries in $R[t]$ is uniquely a finite sum $\sum_i X_i t^i$ with $X_i \in M_n(R)$, taking for X_i the matrix of coefficients of t^i in the entries. Choose $N \geq n$ such that every entry of the adjugate $\operatorname{adj}(tI_n - A) \in M_n(R[t])$ (definition 13.4.10) has degree at most N, and write $\operatorname{adj}(tI_n - A) = \sum_{i=0}^{N} B_i t^i$ with $B_i \in M_n(R)$. Put $B_{-1} = B_{N+1} = 0$ and $c_i = 0$ for $n < i \leq N+1$. By proposition 13.4.11(2), over the commutative ring $R[t]$,

$$\operatorname{adj}(tI_n - A)(tI_n - A) = \det(tI_n - A)I_n = P_A(t)I_n$$

The (j,l) entry of the left side is $\sum_m (\sum_i B_i t^i)_{jm}(t\delta_{ml} - a_{ml}) = \sum_i t^{i+1}(B_i)_{jl} - \sum_i t^i (B_i A)_{jl}$, so comparing the coefficient matrices of t^i for $0 \leq i \leq N+1$ gives

$$B_{i-1} - B_i A = c_i I_n$$

Multiplying the ith equation on the right by A^i and adding over i, the left side telescopes:

$$\sum_{i=0}^{N+1} c_i A^i = \sum_{i=0}^{N+1} \left(B_{i-1}A^i - B_i A^{i+1}\right) = \sum_{i=1}^{N+1} B_{i-1}A^i - \sum_{i=0}^{N} B_i A^{i+1} = 0$$

As $c_i = 0$ for $i > n$, this is $P_A(A) = 0$. If A is the matrix of α in a basis β, then the matrix of $P_\alpha(\alpha) = \sum_i c_i \alpha^i$ is $\sum_i c_i A^i = P_A(A) = 0$, by lemma 13.2.16(2) and the linearity of $\alpha \mapsto [\alpha]_\beta$, so $P_\alpha(\alpha) = 0$. If $R = F$ is a field, then P_A lies in the ideal of polynomials vanishing at A, whose monic generator is the minimal polynomial (definition 14.2.2), so the minimal polynomial divides P_A, completing the proof.

Over a field, the rational canonical form gives a second proof of the last statement: the minimal polynomial a_m (proposition 14.3.1) divides the product $a_1 a_2 \cdots a_m$ of the invariant factors, which is the characteristic polynomial (lemma 14.3.9(2)).

In field theory the minimal polynomial of an algebraic element α (definition 19.2.19) is the minimal polynomial of the linear map $\mu_\alpha \colon F(\alpha) \to F(\alpha)$, $v \mapsto \alpha v$: for $f \in F[x]$, $f(\mu_\alpha) = \mu_{f(\alpha)}$, which is zero exactly when $f(\alpha) = 0$.

14.3.1 Converting a square matrix to Rational Canonical Form

Let A be an $n \times n$ matix with entries from a field F

1. First, reduce $xI - A$ to its Smith Normal Form (theorem 13.2.25, which applies since $F[x]$ is a Euclidean domain) by elementary row and column operations, keeping track of the row operations for step (2).

   Let $d_1, d_2, ..., d_m$ be the degrees ofthe monic nonconstant polynomials $a_1(x), ..., a_m(x)$ that are in the respective diagonals

2. Let P' be the $n \times n$ identity matrix. For each row operation of (1), in order, change P' as follows: if rows i and j are interchanged, interchange columns i and j of P'. If $p(x)$ times row j is added to row i, subtract $p(A)$ times column i of P' from column j. If row i is multiplied by a unit u, divide column i of P' by u.

3. Once you have diagonalize $xI - A$ to the Smith Normal Form, the first $n - m$ columns of the matrix P' should be 0, and the remaining m columns of P' are nonzero. For each $i = 1, 2, ...m$, multiply the ith nonzero column of P' successively by $A^0 = I, A^1, A^2, ..., A^{d_i - 1}$, where d_i is the integer in (1) aboce and use the resulting column vectors (in this order) as the next d_i column of an $n \times n$ matrix $P>$ Then $P^{-1}AP$ is in rational canonical form (whose diagonal blocks are the companion matrices for the polynomials $a_1(x), ..., a_m(x)$ in (1))

Example 14.3.11: Finding Rational Canonical Form

1. Let the following 3×3 matrices be over $\mathbb{Q}$:

$$A = \begin{pmatrix} 2 & -2 & 14 \\ 0 & 3 & -7 \\ 0 & 0 & 2 \end{pmatrix} \quad B = \begin{pmatrix} 0 & -4 & 85 \\ 1 & 4 & -30 \\ 0 & 0 & 3 \end{pmatrix} \quad C = \begin{pmatrix} 2 & 2 & 1 \\ 0 & 2 & -1 \\ 0 & 0 & 3 \end{pmatrix}$$

   We can see that each matrix has the same characteirstic polynomial

$$c_A(x) = c_B(x) = c_C(x) = (x-2)^2(x-3)$$

   Since the minimal and characteristic polynomial have the same roots, the only possibilities for the minimal polynomials are $(x-2)(x-3)$ or $(x-2)^2(x-3)$. Since there are only two options, we it will either be one minimal polynomial or the other. Thus, with some computation, we see that $(A - 2I)(A - 3I) = 0$, $(B - 2I)(B - 3I) \neq 0$ (the $1,1$-entry is nonzero) and $(C - 2I)(C - 3I) \neq 0$ (the $1,2$-entry is nonzero). Thus, we get

$$m_A(x) = (x-2)(x-3), \qquad m_B(x) = m_C(x) = (x-2)^2(x-3)$$

   since other invariant factors would need to divide $m_B(c)/m_C(x)$, there must be no other invariatn facors for B and C. Since the invariant factors for A divide the minimal polynomial and have as a product the characteristic polynomial, we see that A has for invariant factors the polynomials, $x - 2, (x-2)(x-3) = x^2 - 5x + 6$. For 2×2 and 3×3 matrices the characteristic and minimal polynomials determine all the invariant factors: the invariant factors other than $m(x)$ divide $m(x)$ and have product $c(x)/m(x)$, of degree at most 2, and if that degree is 2 then $m(x)$ has degree 1 and every invariant factor equals $m(x)$.

   Thus, we can conclude that B and C are similar, and both are not simialr to A. The rational canonical forms are(note $(x-2)^2(x-3) = x^3 - 7x^2 + 16x - 12$)

$$\begin{pmatrix} 2 & 0 & 0 \\ 0 & 0 & -6 \\ 0 & 1 & 5 \end{pmatrix} \quad \begin{pmatrix} 0 & 0 & 12 \\ 1 & 0 & -16 \\ 0 & 1 & 7 \end{pmatrix} \quad \begin{pmatrix} 0 & 0 & 12 \\ 1 & 0 & -16 \\ 0 & 1 & 7 \end{pmatrix}$$

2. In example (1), we managed to conclude the rational canonical form from the minimal polynomials, which we can always do for 3×3 matries and lower. However, that is not the case for matries of higher degree. Before doing such an example, let's practice by going the long way on matrix A from example (1).

   (a) We first start by reducint $xI - A$ in $\mathbb{Q}[x]$

   $$xI - A = \begin{pmatrix} x-2 & 2 & -14 \\ 0 & x-3 & 7 \\ 0 & 0 & x-2 \end{pmatrix}$$

   which we can do by the operations $R_1 + R_2 \mapsto R_1$, $C_1 - C_2 \mapsto C_1$, $-R_1$, $R_2 + (x-3)R_1 \mapsto R_2$, $C_2 + (x-1)C_1 \mapsto C_2$, $C_3 - 7C_1 \mapsto C_3$, $-C_2$, $R_2 - 7R_3 \mapsto R_2$, and finally $R_2 \leftrightarrow R_3$ together with $C_2 \leftrightarrow C_3$, where $R_i + aR_j \mapsto R_i$ adds a times the jth row to the ith row, $-R_i$ multiplies the ith row by -1, $R_i \leftrightarrow R_j$ interchanges two rows, and C denotes the same operations on columns [28, Section 12.2]. After the first six operations the matrix is

   $$\begin{pmatrix} 1 & 0 & 0 \\ 0 & -x^2 + 5x - 6 & 7(x-2) \\ 0 & 0 & x-2 \end{pmatrix}$$

   so, our final result is

   $$\begin{pmatrix} 1 & 0 & 0 \\ 0 & x-2 & 0 \\ 0 & 0 & x^2 - 5x + 6 \end{pmatrix}$$

   This gives us the invariant factors of $x - 2$ and $x^2 - 5x + 6 = (x-2)(x-3)$ for this matrix. Now, let V be a 3 dimensional vector space over $\mathbb{Q}$ with standard basis e_1, e_2, e_3 and let T be the corresponding linear operator (which defines the action of x on V):

   $$\begin{aligned} xe_1 &= T(e_1) = 2e_1 \\ xe_2 &= T(e_2) = -2e_1 + 3e_2 \\ xe_3 &= T(e_3) = 14e_1 - 7e_2 + 2e_3 \end{aligned}$$

   taking the row operations from earlier, $R_1 + R_2 \mapsto R_1$, $-R_1$, $R_2 + (x-3)R_1 \mapsto R_2$, $R_2 - 7R_3 \mapsto R_2$ and $R_2 \leftrightarrow R_3$, and applying them to this basis by the rules of [28, Section 12.2]: interchanging rows i and j interchanges the ith and jth generators, adding $p(x)$ times row j to row i subtracts $p(x)$ times the ith generator from the jth generator, and multiplying row i by a unit u divides the ith generator by u. This gives

   $$\begin{aligned} [e_1, e_2, e_3] &\to [e_1, e_2 - e_1, e_3] \to [-e_1, e_2 - e_1, e_3] \to [-e_1 - (x-3)(e_2 - e_1), e_2 - e_1, e_3] \\ &\to [-e_1 - (x-3)(e_2 - e_1), e_2 - e_1, e_3 + 7(e_2 - e_1)] \end{aligned}$$

   with a final result of

   $$[-e_1 - (x-3)(e_2 - e_1), e_3 + 7(e_2 - e_1), e_2 - e_1]$$

   and applying the formulas for T from earlier, we will get

   $$[0, -7e_1 + 7e_2 + e_3, -e_1 + e_2]$$

which correspond to the elements 1, $x-2$, and x^2-5x+6 we found earlier. Since the first invariant factor has degree 0, we can ignore it. So, we have $f_1 = -7e_1 + 7e_2 + e_3$, and $f_2 = -e_1 + e_2$. These are the $\mathbb{Q}[x]$-module generators for the two cyclic factors of V in its invariant factor decomposition as $\mathbb{Q}[x]$-modules. For the corresponding $\mathbb{Q}$-vector space bases for theset two factros, since $x-2$ has degree 1, and x^2-5x+6 has degree 2, we get f_1, f_2, xf_2 as a basis, where $xf_2 = T(f_2) = -4e_1 + 3e_2$, giving us

$$P = \begin{pmatrix} -7 & -1 & -4 \\ 7 & 1 & 3 \\ 1 & 0 & 0 \end{pmatrix}$$

which we can use to find P^{-1} and get

$$P^{-1}AP = \begin{pmatrix} 2 & 0 & 0 \\ 0 & 0 & -6 \\ 0 & 1 & 5 \end{pmatrix}$$

which is the same result we had in (1).

(b) We can also convert strait into Rational cannonical form. Given the operation for the diagonalization of A, we use them on the identity matrix, by the rules of section 14.3.1:

$$I_3 \xrightarrow{C_2 - C_1 \mapsto C_2} \begin{pmatrix} 1 & -1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \xrightarrow{-C_1} \begin{pmatrix} -1 & -1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \xrightarrow{C_1 - (A-3I)C_2 \mapsto C_1} \begin{pmatrix} 0 & -1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$
$$\xrightarrow{C_3 + 7C_2 \mapsto C_3} \begin{pmatrix} 0 & -1 & -7 \\ 0 & 1 & 7 \\ 0 & 0 & 1 \end{pmatrix} \xrightarrow{C_2 \leftrightarrow C_3} \begin{pmatrix} 0 & -7 & -1 \\ 0 & 7 & 1 \\ 0 & 1 & 0 \end{pmatrix} = P'$$

Here $d_1 = 1$ and $d_2 = 2$, corresponding to the second and third columns of P', and the columns of P are:

$$\begin{pmatrix} -7 \\ 7 \\ 1 \end{pmatrix} \quad \text{and} \quad \begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix} A \begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix} = \begin{pmatrix} -4 \\ 3 \\ 0 \end{pmatrix}$$

which gives the same P as before.

Exercise 14.3.1

1. Show that every square matrix $A \in M_n(k)$ is similar to its transpose A^t. (Hint: by theorem 14.3.6 and corollary 14.3.7 it suffices to treat a companion matrix $C = C_a$ of degree d. Show that $e_d, C^t e_d, \ldots, (C^t)^{d-1} e_d$ is a basis of k^d, and deduce that the rational canonical form of C^t is C_a.)

14.4 Jordan Canonical Form

The RCF was found by taking using the invariant factors. Let us now find a form based on the elementary divisors when the characteristic polynomial factors completely. If the characteristic polynomial does not factor completely, then the field can be enlarged so that every polynomial factors completely (as mentioned in section 13.5 the avid reader may checkout section 19.5.1 for more details on algebraic closure). As we have outlined earlier, the similarity of two linear transformations is independent of the base field, and hence this pursuit does lead to a useful matrix form for deducing similarity.

Given a $\alpha \in \text{End}_k(V)<$ we may obtain the elementary divisor decomposition of the correspondign $k[x]$-module to be:

$$V \cong \bigoplus_{i,j} \frac{k[x]}{(p_i(x)^{r_{ij}})}$$

Then the characteristic polynomial $P_\alpha(x)$ is equal to the product

$$\prod_{i,j} p_i(x)^{r_{ij}}$$

(lemma 14.4.1(1) below). This gives us the following results:

Lemma 14.4.1: Elementary Divisors And Characteristic Polynomial

Let α be a linear operator on a finite dimensional vector space V over a field k, and let

$$V \cong \bigoplus_{i=1}^{s} \bigoplus_{j=1}^{t_i} \frac{k[x]}{(p_i(x)^{r_{ij}})}$$

be an elementary divisor form of the $k[x]$-module V (definition 14.1.11), where $p_1, \ldots, p_s$ are distinct monic irreducible polynomials, $t_i \geq 1$ and $r_{ij} \geq 1$. Then:

1. $P_\alpha(x) = \prod_{i,j} p_i(x)^{r_{ij}}$
2. the minimal polynomial is $m_\alpha(x) = \prod_{i=1}^{s} p_i(x)^{\max_j\{r_{ij}\}}$
3. if the characteristic polynomial factors completely, as

$$P_\alpha(x) = \prod_{i=1}^{s'} (x - \lambda_i)^{m_i}$$

where $\lambda_1, \ldots, \lambda_{s'}$ are the distinct eigenvalues and $m_i \geq 1$ are the algebraic multiplicities, then after renumbering $s' = s$ and

$$p_i(x) = x - \lambda_i \qquad m_i = \sum_j r_{ij}$$

so that the minimal polynomial is

$$m_\alpha(x) = \prod_{i=1}^{s} (x - \lambda_i)^{\max_j\{r_{ij}\}}$$

***Proof*:**

Write $q_{ij} = p_i^{r_{ij}}$ and fix a $k[x]$-module isomorphism $\Phi\colon \bigoplus_{i,j} k[x]/(q_{ij}) \to V$. As $k \subseteq k[x]$, Φ is a bijective k-linear map, and $\alpha(\Phi(c)) = x \cdot \Phi(c) = \Phi(x \cdot c)$ for every c, since x acts on V as α.

1. The bases of lemma 14.3.3(1) of the summands $k[x]/(q_{ij})$ together form a k-basis of the direct sum, and their images under Φ form a k-basis β of V. Multiplication by x maps each summand into itself, and $\alpha \circ \Phi$ is Φ composed with multiplication by x, so by lemma 14.3.3(1) the matrix A of α in the basis β is block diagonal with diagonal blocks $C_{q_{ij}}$. Then $xI - A$ is block diagonal with blocks $xI - C_{q_{ij}}$, and by proposition 13.4.8, applied over $k[x]$, and lemma 14.3.3(3),

$$P_\alpha(x) = \det(xI - A) = \prod_{i,j} \det(xI - C_{q_{ij}}) = \prod_{i,j} p_i(x)^{r_{ij}}$$

The characteristic polynomial does not depend on the basis used to compute it (proposition 13.5.2(4)).

2. By definition 14.2.2, m_α is the monic generator of the annihilator of the $k[x]$-module V. Isomorphic modules have the same annihilator, since an isomorphism Φ of $k[x]$-modules

satisfies $f \cdot \Phi(v) = \Phi(f \cdot v)$. An element of $k[x]$ annihilates a direct sum exactly when it annihilates every summand, since it acts componentwise. So by lemma 14.3.3(2)

$$\mathrm{Ann}_{k[x]}(V) = \bigcap_{i,j}(p_i^{r_{ij}}) = \bigcap_{i=1}^{s}(p_i^{e_i}) \qquad e_i = \max_j\{r_{ij}\}$$

since for fixed i the ideals $(p_i^{r_{ij}})$ are nested and the smallest is $(p_i^{e_i})$. Let $g = \prod_{i=1}^{s} p_i^{e_i}$. Each $p_i^{e_i}$ divides g, so (g) lies in the intersection. Conversely, let $p_i^{e_i}$ divide f for every i. Each p_i is prime in the principal ideal domain $k[x]$ (corollary 11.1.2, proposition 10.1.33), and p_l does not divide p_i for $i \neq l$: otherwise $p_i = up_l$ with u a unit, that is a nonzero constant, and $u = 1$ as both are monic, contradicting $p_i \neq p_l$. If a prime p does not divide a and p^e divides ab, then p^e divides b: for $e \geq 1$, p divides b, say $b = pb'$, and cancelling p in the integral domain $k[x]$ shows p^{e-1} divides ab', so induction on e applies. Now suppose $g_{l-1} = \prod_{i<l} p_i^{e_i}$ divides f, say $f = g_{l-1}h$. As p_l is prime and divides no p_i with $i < l$, it does not divide g_{l-1}, so $p_l^{e_l}$ divides h and g_l divides f. By induction on l, starting from $g_0 = 1$, $g = g_s$ divides f. Hence $\mathrm{Ann}_{k[x]}(V) = (g)$, and since g is monic, $m_\alpha = g$.

3. Each $x - \lambda_l$ is monic of degree 1, hence irreducible and prime in $k[x]$. By (1), $x - \lambda_l$ divides $\prod_{i,j} p_i^{r_{ij}}$, so it divides some p_i, and as in (2) this forces $p_i = x - \lambda_l$. Conversely, each p_i divides $P_\alpha = \prod_l (x - \lambda_l)^{m_l}$, so it divides some $x - \lambda_l$, and then $p_i = x - \lambda_l$, both being monic of degree 1. The p_i are distinct and so are the $x - \lambda_l$, so this matches them bijectively, and after renumbering $s' = s$ and $p_i = x - \lambda_i$. Fix i and put $R_i = \sum_j r_{ij}$. By (1) and by hypothesis, $P_\alpha = (x - \lambda_i)^{R_i} a = (x - \lambda_i)^{m_i} b$, where a and b are products of factors $x - \lambda_l$ with $l \neq i$, so $x - \lambda_i$ divides neither a nor b, as it would then divide some $x - \lambda_l$ with $\lambda_l \neq \lambda_i$. If $R_i < m_i$, cancelling $(x - \lambda_i)^{R_i}$ in the integral domain $k[x]$ gives $a = (x - \lambda_i)^{m_i - R_i} b$, so $x - \lambda_i$ divides a, which is impossible. If $m_i < R_i$, cancelling $(x - \lambda_i)^{m_i}$ gives $(x - \lambda_i)^{R_i - m_i} a = b$, so $x - \lambda_i$ divides b, which is impossible. Hence $m_i = R_i = \sum_j r_{ij}$, and the formula for m_α is (2) with $p_i = x - \lambda_i$, completing the proof.

With this, let's start finding the matrix representation of α. Just like before, let's tart with the cyclic factor, $F[x]/((x - \lambda)^k)$. We will purposefully be choosing our basis so that our representation is particularly simple:

$$(\overline{x} - \lambda)^{k-1}, (\overline{x} - \lambda)^{k-2}, ..., (\overline{x} - \lambda), (\overline{x} - \lambda^)0 = 1$$

this is indeed a basis, since expanding out, simplifying, we get these elements to be a linear combination of $\overline{x}^{k-1}, \overline{x}^{k-2}, ..., \overline{x}, 1$, meaning they do form a basis. Then, we can associate the linear operator T to x, and get

$$\begin{cases} (x - \lambda)^{r-1} \mapsto x(x - \lambda)^{r-1} = \lambda(x - \lambda)^{r-1} + (x - \lambda)^r = \lambda(x - \lambda)^{r-1} \\ (x - \lambda)^{r-2} \mapsto x(x - \lambda)^{r-2} = (x - \lambda)^{r-1} + \lambda(x - \lambda)^{r-1} \\ \vdots \\ 1 \mapsto x = (x - \lambda) + \lambda \end{cases}$$

which in matrix form is be represented as:

$$\begin{pmatrix} \lambda & 1 & & & \\ & \lambda & 1 & & \\ & & \ddots & \ddots & \\ & & & \lambda & 1 \\ & & & & \lambda \end{pmatrix}$$

where the blank spots are all zero. We give a name to this matrix

Definition 14.4.2: Jordan Block

The $k \times k$ matrix with λ along the main diagonal and 1 along the first superdiagonal depicted avoce is called the $k \times k$ *elementary Jordan matrix with eigenvalue* λ, or the *Jordan block of size* k *with eigenvalue* λ

As before, we may combine all of these to get a full representation of α:

Definition 14.4.3: Jordan canonical Form (JCF)

Let α be a linear operator on a finitely dimensional vector space V. Then it's *Jordan Canonical Form* (JCF) is the combination of all the Jordan blocks:

$$\begin{pmatrix} [c|c|c]J_{\lambda_1,(r_{1,j})} & & \\ & \ddots & \\ & & J_{\lambda_s,(r_{s,j})} \end{pmatrix}$$

where $(x - \lambda_i)^{r_{ij}}$ are the elementary divisors of α.

14.4.4 An ambiguity in Uniqueness While the RCF is is completely unique up to isomorphism, since there is no unique way of arranging the eigenvalues, there is no unique way of arranging the Jordan blocks (unlike the invariant blocks which are arranged by the divisibility of the invariant factors)

So far the Jordan canonical form has only been described. What has not yet been said is when an operator *has* one: unlike the rational canonical form, which exists over any field, the Jordan form requires the relevant polynomials to factor into linear pieces. The elementary divisor decomposition of chapter 14 settles the question, and the proof also explains where the superdiagonal 1s come from.

Theorem 14.4.5: Existence of the Jordan Canonical Form

Let k be a field, let V be a finite dimensional k-vector space and let α be a linear operator on V whose minimal polynomial factors into linear factors over k (which holds for every α when k is algebraically closed). Then there is a basis of V in which the matrix of α is in Jordan canonical form, and the multiset of Jordan blocks appearing is uniquely determined by α. Equivalently, a matrix $A \in M_n(k)$ whose minimal polynomial splits over k is similar to a matrix in Jordan canonical form, unique up to the order of its blocks.

***Proof*:**
Regard V as a $k[x]$-module with x acting as α. It is finitely generated, being finite dimensional over k, and it is torsion, since the minimal polynomial $m(x)$ annihilates it. As $k[x]$ is a principal ideal domain, definition 14.1.11 gives a decomposition

$$V \cong \bigoplus_{i=1}^{s} k[x]/\big(p_i(x)^{r_i}\big)$$

with each p_i irreducible, and the multiset of the $p_i^{r_i}$ is uniquely determined by V. There is no free part because V is torsion.

Step 1: the p_i are linear. The polynomial $m(x)$ annihilates V, hence annihilates each summand $k[x]/(p_i^{r_i})$, and therefore $p_i^{r_i}$ divides $m(x)$. By hypothesis $m(x)$ is a product of linear factors, and $k[x]$ is a unique factorization domain, so every irreducible factor of m is linear. Hence each p_i is linear, say $p_i = x - \lambda_i$.

Step 2: each summand is a Jordan block. Fix a summand $W = k[x]/\big((x-\lambda)^r\big)$ and, for $1 \le j \le r$, let $e_j \in W$ be the class of $(x-\lambda)^{r-j}$. These r elements form a k-basis of W, since $1, (x-\lambda), \dots, (x-\lambda)^{r-1}$ is a k-basis of $k[x]/\big((x-\lambda)^r\big)$: the polynomials $(x-\lambda)^d$ for $0 \le d < r$ have distinct degrees $0, 1, \dots, r-1$, so they are a basis of the space of polynomials of degree less than r, which maps isomorphically onto the quotient. Writing $x = \lambda + (x-\lambda)$ and setting $e_0 := 0$, which is consistent because $(x-\lambda)^r = 0$ in W,

$$\alpha(e_j) = x \cdot (x-\lambda)^{r-j} = \lambda(x-\lambda)^{r-j} + (x-\lambda)^{r-j+1} = \lambda e_j + e_{j-1}$$

So the matrix of $\alpha|_W$ in the ordered basis $(e_1, \dots, e_r)$ has λ in every diagonal entry and 1 in every entry of the first superdiagonal, which is the Jordan block $J_{\lambda,r}$ of definition 14.4.2.

Step 3: assembly. Concatenating the bases of the summands gives a basis of V in which the matrix of α is block diagonal with the blocks of Step 2, which is the Jordan canonical form of definition 14.4.3. Uniqueness of the multiset of blocks is the uniqueness of the elementary divisors, since the block $J_{\lambda,r}$ corresponds to the elementary divisor $(x-\lambda)^r$, as we sought to show.

Only the order of the blocks is undetermined, which is the ambiguity recorded in box 14.4.4.

An interesting proeprty the JCF gives us is that it distinguishes algebraic and geometric multiplicity of eigenvalues:

Proposition 14.4.6: Geometric An Algebraic Multiplicity Of Eigenvalues Using JCF

Let α be a linear operator on a finitely dimensional vector space V. Then the geometric multiplicity of an eigenvalue λ of α equals the number of Jordan blocks corresponding to λ in the Jordan canonical form of α, and the algebraic multiplicity equals the number of times the eigenvalue shows up in the Jordan canonical form.

***Proof*:**
The fact that the algebraic multiplicity of the eigenvalue corresponds to the number or times it shows up in the JCF is immediate, hence let's show the number of Jordan blocks equasl the geometric multiplicity. Let:

$$J =: \begin{pmatrix} \lambda & 1 & \cdots & 0 & 0 \\ 0 & \lambda & \cdots & 0 & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \cdots & \lambda & 1 \\ 0 & 0 & \cdots & 0 & \lambda \end{pmatrix}$$

be a Jordan block and $v = \begin{pmatrix} v_1 \\ \vdots \\ v_r \end{pmatrix}$ be an eigenvector corresponding to λ. Then:

$$\lambda \begin{pmatrix} v_1 \\ v_2 \\ \vdots \\ v_r \end{pmatrix} = J \begin{pmatrix} v_1 \\ v_2 \\ \vdots \\ v_r \end{pmatrix} = \begin{pmatrix} \lambda v_1 + v_2 \\ \lambda v_2 + v_3 \\ \vdots \\ \lambda v_r \end{pmatrix}$$

which solving gives us:

$$v_2 = \cdots = v_r = 0$$

Hence, the eigenspace of λ is generated by:

$$v = \begin{pmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix}$$

completing the proof.

Corollary 14.4.7: Diagonalizability And Minimal Polynomial

Assume the characteristic polynomial of $\alpha \in \mathrm{End}_k(V)$ factors completely over k. Then α is diagonalizable iff the minimal polynomial of α has no multiple roots

***Proof*:**
Throughout, write m_α for the minimal polynomial and P_α for the characteristic polynomial. By the Cayley-Hamilton theorem (proposition 14.3.10), $P_\alpha(\alpha) = 0$, so P_α lies in the annihilator ideal (m_α) of definition 14.2.2, that is, $m_\alpha | P_\alpha$. The ring $k[x]$ is a Euclidean domain (theorem 11.1.1), hence a PID (proposition 10.2.7), hence a UFD (proposition 10.1.33). Since P_α factors completely into linear factors, unique factorization shows every irreducible factor of its divisor m_α is linear as well, so m_α also factors completely over k.

($\Rightarrow$) Suppose α is diagonalizable, with a diagonal representation D with diagonal entries $d_1, \ldots, d_n$, and let $\mu_1, \ldots, \mu_s$ be the distinct values among them. Set $f(x) := \prod_{i=1}^{s}(x - \mu_i)$. Fixing the basis giving D, the matrix representation is additive and multiplicative (theorem 13.2.6), so the

matrix representing $f(\alpha)$ is $f(D)$, which is diagonal with entries $f(d_1), \ldots, f(d_n)$, all zero since each d_j is one of the μ_i. Hence $f(\alpha) = 0$, so $f \in (m_\alpha)$, that is, $m_\alpha | f$. If m_α had a multiple root λ, then $(x-\lambda)^2$ would divide f, which is false since the μ_i are distinct. Hence m_α has no multiple roots.

($\Leftarrow$) Suppose m_α has no multiple roots. As shown above, m_α factors into linear factors over k, so theorem 14.4.5 applies: α has a Jordan canonical form. By lemma 14.4.1,

$$m_\alpha(x) = \prod_{i=1}^{s} (x - \lambda_i)^{\max_j\{r_{ij}\}}$$

where the $(x-\lambda_i)^{r_{ij}}$ are the elementary divisors, whose exponents r_{ij} are the sizes of the Jordan blocks. Since m_α has no multiple roots, $\max_j\{r_{ij}\} = 1$ for every i, so every $r_{ij} = 1$ and every Jordan block has size 1. The Jordan canonical form of α is therefore a diagonal matrix, so α admits a diagonal representation and is diagonalizable (definition 13.5.14).

Naturally, the diagonalizability of a matrix also depends on the base field, for example the rotation by $\pi/2$ degrees for $\mathbb{R}^2$ is not diagonalizable, while it is diagonalizable over $\mathbb{C}$

14.4.1 Changing From one Canonical Form to Another

The elementary divisors are the prime power factors of the invariant factors. Conversely, the largest invariant factor is the product of the largest power of each $(x-\lambda)$ among the elementary divisors, the next largest is the product of the largest remaining powers, and so on, with 1 once a root's powers run out. The two canonical forms of a matrix whose characteristic polynomial splits therefore determine each other [28, Section 12.3].

14.4.2 Computation: Convert to JCF

Let A be an $n \times n$ matrix over a field F over which the characteristic polynomial of A splits.

1. Carry out steps (1) and (2) of section 14.3.1, obtaining the invariant factors $a_1(x), \ldots, a_m(x)$ and the nonzero columns $f_1, \ldots, f_m$ of P'.
2. For each i, factor $a_i(x) = (x-\lambda_1)^{e_1} \cdots (x-\lambda_s)^{e_s}$ with $\lambda_1, \ldots, \lambda_s$ distinct, and for each j let $g_{ij} = \prod_{l \neq j} (A - \lambda_l I)^{e_l} f_i$.
3. For each such $g = g_{ij}$, with $\lambda = \lambda_j$ and $e = e_j$, use the column vectors $(A-\lambda I)^{e-1}g, \ldots, (A-\lambda I)g, g$ (in this order) as the next e columns of an $n \times n$ matrix P.

Then $P^{-1}AP$ is in Jordan canonical form, with one Jordan block of size e_j and eigenvalue λ_j for each factor $(x-\lambda_j)^{e_j}$ of each $a_i(x)$, in the order used in step (3) [28, Section 12.3].

14.5 The Jordan-Chevalley Decomposition

A matrix in Jordan canonical form is the sum of two pieces of very different character: its diagonal, which scales, and its superdiagonal of 1s, which shears. These are the two behaviours promised in

chapter 13, and the two pieces even commute, since within each Jordan block the diagonal piece is the scalar matrix λI. What theorem 14.4.5 does not tell us is whether this splitting belongs to the operator or to the basis: perform the same surgery on the matrix of α in some other basis, and the two pieces need not commute and need not agree with the pieces the Jordan basis produces. The goal of this section is to show that the splitting is intrinsic. Every operator whose characteristic polynomial factors completely is the sum of a diagonalizable operator and a nilpotent operator that commute, the pair is unique, and both are polynomial expressions in the operator itself with zero constant term. The polynomial clause is the strongest part of the statement: anything that commutes with α commutes with both pieces, and any subspace that α preserves both pieces preserve. First, names for the two behaviours.

Definition 14.5.1: Semisimple and Nilpotent Operators

Let V be a finite dimensional vector space over a field k and let $\alpha \in \mathrm{End}_k(V)$. Then α is said to be *semisimple* if it is diagonalizable in the sense of definition 13.5.14[a], and α is said to be *nilpotent* if $\alpha^m = 0$ for some $m \geq 1$. The smallest such m is called the *index* of the nilpotent operator α.

[a] over a field where polynomials need not factor completely, the right definition is instead that the minimal polynomial of α is a product of distinct irreducible factors, and corollary 14.4.7 shows the two definitions agree whenever the characteristic polynomial factors completely. Everything in this section happens in that setting, so no distinction arises.

Two lemmas about diagonalizable operators are needed, and both are useful well beyond this section. The first says that diagonalizability survives restriction.

Lemma 14.5.2: Restriction of a Diagonalizable Operator

Let $\alpha \in \mathrm{End}_k(V)$ be diagonalizable with distinct eigenvalues $\mu_1, ..., \mu_r$, and let $W \subseteq V$ be an α-stable subspace (example 14.2.1). Then

$$W = (W \cap E_{\mu_1}) \oplus \cdots \oplus (W \cap E_{\mu_r})$$

and the restriction $\alpha|_W$ is diagonalizable.

***Proof*:**
By theorem 13.5.15, $V = E_{\mu_1} \oplus \cdots \oplus E_{\mu_r}$, hence any $w \in W$ decomposes as $w = w_1 + \cdots + w_r$ with $w_j \in E_{\mu_j}$, and the content of the lemma is that each w_j lies in W. Fix j and consider the polynomial

$$\pi_j(t) = \prod_{l \neq j} \frac{t - \mu_l}{\mu_j - \mu_l} \in k[t]$$

whose denominators are nonzero because the μ_l are distinct. For $v \in E_{\mu_l}$ we have $\alpha v = \mu_l v$, so every polynomial in α acts on E_{μ_l} as its value at μ_l, and π_j takes the value 0 at each μ_l with $l \neq j$ and the value 1 at μ_j. Therefore

$$\pi_j(\alpha)w = \sum_{l=1}^{r} \pi_j(\mu_l) w_l = w_j$$

Since W is α-stable, it is stable under every power of α and under linear combinations of those

powers, so $\pi_j(\alpha)W \subseteq W$ and $w_j \in W$. This proves that W is the sum of the subspaces $W \cap E_{\mu_j}$, and the sum is direct because the E_{μ_j} are independent. Concatenating bases of the $W \cap E_{\mu_j}$ now gives a basis of W consisting of eigenvectors of $\alpha|_W$, which is therefore diagonalizable by theorem 13.5.13, as we sought to show.

The second lemma removes the one ambiguity a diagonalizable operator carries, namely *which* basis diagonalizes it: commuting operators can be made to share one.

Lemma 14.5.3: Simultaneous Diagonalization

Let V be a finite dimensional vector space over a field k and let $\mathcal{F} \subseteq \mathrm{End}_k(V)$ be any family of diagonalizable operators that commute pairwise. Then there is a single basis of V consisting of common eigenvectors of every member of $\mathcal{F}$, so that every member of $\mathcal{F}$ is diagonal in one shared basis. Such a family is called *simultaneously diagonalizable*.

***Proof*:**
We induct on $\dim V$. If every member of $\mathcal{F}$ is a scalar multiple of the identity, then any basis of V diagonalizes all of $\mathcal{F}$ at once, and this in particular settles $\dim V \leq 1$. Otherwise, fix some $\alpha \in \mathcal{F}$ that is not a scalar. By theorem 13.5.15,

$$V = E_{\mu_1} \oplus \cdots \oplus E_{\mu_r}$$

where $\mu_1, ..., \mu_r$ are the distinct eigenvalues of α. Here $r \geq 2$: if α had a single eigenvalue μ, a diagonalizing basis would represent α by the matrix μI, making $\alpha = \mu\mathrm{id}$ a scalar. Consequently each eigenspace satisfies $0 < \dim E_{\mu_j} < \dim V$.

Every $\beta \in \mathcal{F}$ preserves every E_{μ_j}: for $v \in E_{\mu_j}$,

$$\alpha(\beta v) = \beta(\alpha v) = \mu_j(\beta v)$$

so $\beta v \in E_{\mu_j}$. By lemma 14.5.2 each restriction $\beta|_{E_{\mu_j}}$ is diagonalizable, and for fixed j the restrictions commute pairwise because the members of $\mathcal{F}$ do. The inductive hypothesis applied to E_{μ_j} produces a basis of E_{μ_j} of common eigenvectors of all the restrictions. Concatenating these bases over $j = 1, ..., r$ gives a basis of V, and each of its vectors is a common eigenvector of every member of $\mathcal{F}$, completing the proof.

Lemma 14.5.4: Sums and Products of Commuting Semisimple or Nilpotent Operators

Let V be a finite dimensional vector space over a field k and let $a, b \in \mathrm{End}_k(V)$ commute.

1. If a and b are semisimple, then $a + b$, $a - b$ and ab are semisimple
2. If a and b are nilpotent, then $a + b$, $a - b$ and ab are nilpotent

***Proof*:**
(1). By lemma 14.5.3 applied to the family $\{a, b\}$, there is a basis in which both a and b are diagonal, and sums, differences and products of diagonal matrices are diagonal.

(2). Say $a^u = 0$ and $b^v = 0$. Since $ab = ba$, the binomial theorem applies and

$$(a \pm b)^{u+v-1} = \sum_{i=0}^{u+v-1} \binom{u+v-1}{i} (\pm 1)^{u+v-1-i}\, a^i b^{u+v-1-i}$$

In each term either $i \geq u$ or $u + v - 1 - i \geq v$, so every term vanishes and $a \pm b$ is nilpotent. Finally, $(ab)^u = a^u b^u = 0$, using commutativity again, completing the proof.

The remaining input is a decomposition of V into pieces on which α is a scalar plus a nilpotent, and it has already been proven in this chapter under another name. When α is not diagonalizable, the eigenspaces of theorem 13.5.15 are too small to fill out V, and the correct enlargement is to relax "killed by $\alpha - \lambda\mathrm{id}$" to "killed by a power of $\alpha - \lambda\mathrm{id}$".

Definition 14.5.5: Generalized Eigenspace

Let V be a finite dimensional vector space over a field k, let $\alpha \in \mathrm{End}_k(V)$ and let $\lambda \in k$. Then the subspace

$$V_\lambda = \left\{ v \in V \middle| (\alpha - \lambda\mathrm{id})^m v = 0 \text{ for some } m \geq 1 \right\}$$

is called the *generalized eigenspace* of α at λ, and its nonzero elements are called *generalized eigenvectors*.

The eigenspace of definition 13.5.9 is the $m = 1$ part of V_λ, and $V_\lambda \neq 0$ exactly when λ is an eigenvalue: if $(\alpha - \lambda\mathrm{id})^m v = 0$ with $v \neq 0$ and m minimal, then $(\alpha - \lambda\mathrm{id})^{m-1} v$ is an eigenvector for λ. The generalized eigenspaces are exactly the primary components of theorem 14.1.12 for the $k[x]$-module structure that α places on V, so the machinery of this chapter already decomposes V into them.

Corollary 14.5.6: Generalized Eigenspace Decomposition

Let V be a nonzero finite dimensional vector space over a field k and let $\alpha \in \mathrm{End}_k(V)$ have characteristic polynomial

$$P_\alpha(x) = \prod_{i=1}^{r} (x - \lambda_i)^{m_i}$$

factoring completely over k, with $\lambda_1, ..., \lambda_r$ distinct. Then

$$V = V_{\lambda_1} \oplus \cdots \oplus V_{\lambda_r} \qquad \text{where} \qquad V_{\lambda_i} = \ker(\alpha - \lambda_i\mathrm{id})^{m_i}$$

and the restriction of $\alpha - \lambda_i\mathrm{id}$ to V_{λ_i} is nilpotent.

***Proof*:**
Regard V as a $k[x]$-module with x acting as α, as in the proof of theorem 14.4.5. It is a nonzero torsion module, and its annihilator is generated by the minimal polynomial $m_\alpha(x)$

(definition 14.2.2), which by lemma 14.4.1 factors as

$$m_\alpha(x) = \prod_{i=1}^{r} (x - \lambda_i)^{\nu_i}$$

where $\nu_i = \max_j r_{ij}$ and $m_i = \sum_j r_{ij}$ in terms of the elementary divisor exponents, so that $1 \leq \nu_i \leq m_i$ for every i. Theorem 14.1.12 applied to this factorization yields

$$V = N_1 \oplus \cdots \oplus N_r$$

where N_i is the kernel of $(\alpha - \lambda_i \mathrm{id})^{\nu_i}$ and equally the set of all elements killed by *some* power of $\alpha - \lambda_i \mathrm{id}$, which is the definition of V_{λ_i}. The sandwich

$$\ker(\alpha - \lambda_i \mathrm{id})^{\nu_i} \subseteq \ker(\alpha - \lambda_i \mathrm{id})^{m_i} \subseteq V_{\lambda_i} = \ker(\alpha - \lambda_i \mathrm{id})^{\nu_i}$$

obtained from $\nu_i \leq m_i$ forces all three subspaces to coincide, and the middle description is the one in the statement. Finally, $(\alpha - \lambda_i \mathrm{id})^{\nu_i}$ vanishes on V_{λ_i} by construction, so the restriction of $\alpha - \lambda_i \mathrm{id}$ to V_{λ_i} is nilpotent, as we sought to show.

Everything is now assembled.

Theorem 14.5.7: Jordan-Chevalley Decomposition

Let V be a nonzero finite dimensional vector space over a field k and let $\alpha \in \mathrm{End}_k(V)$ be an operator whose characteristic polynomial factors into linear factors over k, which holds for every α when k is algebraically closed, with no restriction on the characteristic of k. Then:

1. there exist $s, n \in \mathrm{End}_k(V)$ with

$$\alpha = s + n \qquad sn = ns$$

   where s is semisimple and n is nilpotent

2. there are polynomials $p(t), q(t) \in k[t]$ with zero constant term such that $s = p(\alpha)$ and $n = q(\alpha)$. In particular, s and n commute with every operator that commutes with α, and every α-stable subspace of V is s-stable and n-stable

3. the pair (s, n) is unique: if $\alpha = s' + n'$ with s' semisimple, n' nilpotent and $s'n' = n's'$, then $s' = s$ and $n' = n$

The expression $\alpha = s + n$ is called the *Jordan-Chevalley decomposition* (or *additive Jordan decomposition*) of α, and s and n are called the *semisimple part* and the *nilpotent part* of α.

***Proof*:**
We first build the polynomial p, then verify that $s = p(\alpha)$ and $n = \alpha - s$ have every property claimed, and prove uniqueness last. Throughout, write

$$P_\alpha(x) = \prod_{i=1}^{r} (x - \lambda_i)^{m_i}$$

with $\lambda_1, ..., \lambda_r$ the distinct eigenvalues of α.

Step 1: constructing p by interpolation. We seek a polynomial $p(t) \in k[t]$ satisfying the congruences

$$p(t) \equiv \lambda_i \bmod (t-\lambda_i)^{m_i} \qquad 1 \leq i \leq r$$

together with $p(0) = 0$, which is the congruence $p(t) \equiv 0 \bmod (t)$. For $i \neq j$ the ideals $\left((t-\lambda_i)^{m_i}\right)$ and $\left((t-\lambda_j)^{m_j}\right)$ of $k[t]$ are comaximal: $t-\lambda_i$ and $t-\lambda_j$ are non-associate primes of $k[t]$, so $\gcd\left((t-\lambda_i)^{m_i}, (t-\lambda_j)^{m_j}\right) = 1$, and a Bézout identity in the Euclidean domain $k[t]$ places 1 in the sum of the two ideals, exactly as in the proof of proposition 11.1.6. The congruence at 0 needs care, and there are two cases according to whether 0 is an eigenvalue.

If no λ_i is 0, then $t = t-0$ is a monic linear polynomial distinct from every $t-\lambda_i$, so by the same argument the ideal (t) is comaximal with each $\left((t-\lambda_i)^{m_i}\right)$. The map of the Chinese Remainder Theorem (theorem 9.6.11) is then surjective onto the product of the $r+1$ quotients, and choosing the preimage of the tuple $(\lambda_1, ..., \lambda_r, 0)$ produces a $p(t)$ satisfying all $r+1$ congruences at once. The last congruence says that t divides $p(t)$, which is exactly the vanishing of the constant term.

If instead $\lambda_{i_0} = 0$ for some necessarily unique i_0, the congruence at 0 cannot be imposed as a separate condition, and does not need to be: the ideals (t) and $\left(t^{m_{i_0}}\right)$ are not comaximal, so theorem 9.6.11 does not apply to the enlarged list of $r+1$ ideals, but the i_0-th congruence already reads

$$p(t) \equiv \lambda_{i_0} = 0 \bmod \left(t^{m_{i_0}}\right)$$

Hence we impose only the original r congruences, whose moduli are pairwise comaximal as shown above, and theorem 9.6.11 again provides a solution $p(t)$. Since $m_{i_0} \geq 1$, divisibility by $t^{m_{i_0}}$ implies divisibility by t, and the constant term of p vanishes in this case too, this time as a consequence of the eigenvalue congruence at $\lambda_{i_0} = 0$ rather than as an extra condition.

In both cases the result is a polynomial $p(t)$ with $p(0) = 0$ such that $(t-\lambda_i)^{m_i}$ divides $p(t) - \lambda_i$ for every i. Set

$$s = p(\alpha) \qquad q(t) = t - p(t) \qquad n = q(\alpha) = \alpha - s$$

and note that $q(0) = 0 - p(0) = 0$ as well.

Step 2: s is semisimple and n is nilpotent. By corollary 14.5.6, $V = V_{\lambda_1} \oplus \cdots \oplus V_{\lambda_r}$ with $V_{\lambda_i} = \ker(\alpha - \lambda_i \mathrm{id})^{m_i}$. Fix i and write $p(t) - \lambda_i = h_i(t)\,(t-\lambda_i)^{m_i}$. Evaluating at α and applying to $v \in V_{\lambda_i}$,

$$(s - \lambda_i \mathrm{id})v = h_i(\alpha)(\alpha - \lambda_i \mathrm{id})^{m_i} v = 0$$

so s acts on V_{λ_i} as the scalar λ_i. A basis of V adapted to the direct sum decomposition therefore consists of eigenvectors of s, and s is semisimple. On V_{λ_i} the operator $n = \alpha - s$ acts as $\alpha - \lambda_i \mathrm{id}$, whose m_i-th power kills V_{λ_i}. Setting $m = \max_i m_i$, the power n^m kills every summand and hence all of V, so n is nilpotent.

This proves clauses (1) and (2): s and n are polynomials in α with zero constant term by Step 1, so each is a linear combination of *positive* powers of α. Such operators commute with α, with each other, and with every operator commuting with α, and they carry any α-stable subspace into itself.

Step 3: uniqueness. Let $\alpha = s' + n'$ with s' semisimple, n' nilpotent and $s'n' = n's'$. Then s' commutes with $s' + n' = \alpha$, and so does n'. An operator commuting with α commutes with every power of α and hence with every polynomial in α, so s' and n' commute with s and with n. Rearranging $s + n = s' + n'$ gives

$$s - s' = n' - n$$

and all four operators commute pairwise. By lemma 14.5.4, the left side is semisimple and the right side is nilpotent, so the operator $\delta = s - s' = n' - n$ is both. Being nilpotent, δ has no eigenvalue besides 0: if $\delta v = \mu v$ with $v \neq 0$ and $\delta^m = 0$, then $0 = \delta^m v = \mu^m v$ forces $\mu = 0$. Being semisimple, δ is represented by a diagonal matrix in some basis, and the diagonal entries of a diagonal matrix are eigenvalues, hence all 0. So $\delta = 0$, that is $s' = s$ and $n' = n$, as we sought to show.

In a Jordan basis the decomposition becomes visible: Step 2 makes s act as λ_i on each generalized eigenspace, which is precisely the diagonal of the Jordan canonical form of theorem 14.4.5, and $n = \alpha - s$ is what remains, the superdiagonal of 1s. Uniqueness then says that the recipe "keep the diagonal of the JCF" computes the semisimple part correctly, while the first example below shows that the same recipe applied in any other basis computes nothing of the sort.

Example 14.5.8: Computing a Jordan-Chevalley Decomposition

1. Take $k = \mathbb{Q}$ and
$$A = \begin{pmatrix} 1 & 1 \\ 0 & 2 \end{pmatrix} \qquad \text{so that} \qquad P_A(t) = (t-1)(t-2)$$
Splitting off the diagonal gives a diagonal piece D and a nilpotent piece N with
$$DN = \begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix}\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \neq \begin{pmatrix} 0 & 2 \\ 0 & 0 \end{pmatrix} = ND$$
so $A = D + N$ is *not* the Jordan-Chevalley decomposition of A. Indeed, the minimal polynomial of A has no multiple roots, so A is already diagonalizable by corollary 14.4.7 and its decomposition is $s = A$ and $n = 0$. The recipe of the proof agrees: 0 is not an eigenvalue, and the congruences
$$p \equiv 1 \bmod (t-1) \qquad p \equiv 2 \bmod (t-2) \qquad p \equiv 0 \bmod (t)$$
are solved by $p(t) = t$, so $s = p(A) = A$.

2. Now take
$$B = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 2 \end{pmatrix} \qquad \text{so that} \qquad P_B(t) = (t-1)^2(t-2)$$
Again 0 is not an eigenvalue, so the proof imposes the congruences
$$p \equiv 1 \bmod (t-1)^2 \qquad p \equiv 2 \bmod (t-2) \qquad p \equiv 0 \bmod (t)$$
The moduli multiply to the degree 4 polynomial $t(t-1)^2(t-2)$, so there is a unique solution of degree at most 3, namely
$$p(t) = t^3 - 3t^2 + 3t$$
All three congruences can be checked directly: $p(t)$ has no constant term, $p(t) - 1 = (t-1)^3$, and $p(2) = 8 - 12 + 6 = 2$. To compute $s = p(B)$, note that on the upper 2×2 Jordan block J the identity $p(t) = (t-1)^3 + 1$ gives $p(J) = (J-I)^3 + I = I$, since $(J-I)^2 = 0$, while on the lower block $p(2) = 2$. Hence
$$s = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 2 \end{pmatrix} \qquad n = B - s = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$
and a direct multiplication confirms $sn = ns$, both products being equal to n.

One clause of the theorem deserves a closing word. Interpolation without the congruence at 0 would already produce commuting semisimple and nilpotent parts, so the zero constant term looks like a free refinement, and Step 1 of the proof shows that it costs nothing. It is nevertheless the clause that transfers the strongest properties of α to s and n. A polynomial in α with constant term $c \neq 0$ acts as $c\mathrm{id}$ plus a combination of positive powers of α, and the summand $c\mathrm{id}$ carries any nonzero subspace onto itself, never into a smaller one. With the constant term gone that obstruction disappears: whenever $W' \subseteq W$ are subspaces with $\alpha(W) \subseteq W'$, every positive power of α also carries W into W', hence so do s and n. The semisimple and nilpotent parts therefore satisfy every containment of the form $\alpha(W) \subseteq W'$ that α satisfies, of which stability is the special case $W' = W$. This stronger inheritance is invisible at the level of a single matrix, and it becomes essential the moment operators are constrained to lie in a subspace of $\mathrm{End}_k(V)$ that does not contain the identity.

15

Tensor Products Of Modules: Linear Algebra III

The results of chapter 13 and chapter 14 describe modules and the linear maps between them, and several of the objects those chapters produced are not linear maps. The determinant of section 13.4 takes n column vectors and is linear in each one separately, not in all of them together. An inner product (section 13.6) takes two vectors and is linear in each. The consequence is that such maps are not morphisms in Vect_k, so no theorem about morphisms applies to them. Multiplication in a ring is the same phenomenon at its most basic. The map $R \times R \to R$ sending (a, b) to ab is additive in each variable when the other is held fixed, and it is not additive in the pair: the product $(a + a')(b + b')$ is $ab + ab' + a'b + a'b'$, not $ab + a'b'$.

This chapter removes the obstruction by changing the domain rather than the map in a natural way. From two modules M and N over a ring R it constructs a single module $M \otimes_R N$, together with a bilinear map $M \times N \to M \otimes_R N$ through which every bilinear map out of $M \times N$ factors, uniquely, as a homomorphism out of $M \otimes_R N$. A question about bilinear maps on $M \times N$ then becomes a question about homomorphisms on $M \otimes_R N$, where the machinery of the previous two chapters applies. Iterating the construction on a single module produces the tensor algebra $\mathcal{T}(M)$ and its two quotients, the symmetric algebra $\mathcal{S}(M)$ and the exterior algebra $\bigwedge(M)$. The determinant reappears in section 16.4 as the map an endomorphism induces on the top exterior power.

15.1 The Tensor Product of Modules

Suppose for a moment that the module $M \otimes_R N$ has been built, along with a map $\iota \colon M \times N \to M \otimes_R N$ whose value at (m, n) is written $m \otimes n$. Asking that ι be additive in each variable forces

$$(m_1 + m_2) \otimes n = m_1 \otimes n + m_2 \otimes n, \qquad m \otimes (n_1 + n_2) = m \otimes n_1 + m \otimes n_2$$

giving relations any construction must impose. The condition governing scalars needs more care. Take R commutative first, with M, N and the target L all R-modules. A map $\varphi\colon M \times N \to L$ that is R-linear in each variable satisfies

$$\varphi(rm, n) = r\varphi(m, n) = \varphi(m, rn)$$

so a scalar may be moved from one argument to the other. Over a non-commutative ring that same conditin is nearly vacuous. Suppose M, N and L are left R-modules and φ is R-linear in each variable. Evaluating $\varphi(rm, sn)$ two ways, once by extracting s from the second argument and once by extracting r from the first, gives

$$(sr)\varphi(m, n) = s\varphi(rm, n) = \varphi(rm, sn) = r\varphi(m, sn) = (rs)\varphi(m, n)$$

so that $(rs - sr)\varphi(m, n) = 0$ for every $m \in M$ and $n \in N$. Any such φ is annihilated by every commutator in R, and over a ring with few central elements this leaves almost nothing.

The repair is to weaken the condition in two places. The target drops from an R-module to an abelian group, so that "pulling out a scalar" is no longer available and cannot be contradicted. And the two modules are taken on opposite sides, M a right R-module and N a left one, so that the surviving requirement $\varphi(mr, n) = \varphi(m, rn)$ moves a scalar across the comma without ever moving it outside. This is the condition the construction can enforce.

Definition 15.1.1: R-Balanced Map

Let R be a ring, let M be a right R-module, let N be a left R-module, and let L be an abelian group written additively. A map $\varphi\colon M \times N \to L$ is *R-balanced*, or *middle linear over R*, if for all $m, m_1, m_2 \in M$, all $n, n_1, n_2 \in N$ and all $r \in R$,

$$\begin{aligned} \varphi(m_1 + m_2, n) &= \varphi(m_1, n) + \varphi(m_2, n) \\ \varphi(m, n_1 + n_2) &= \varphi(m, n_1) + \varphi(m, n_2) \\ \varphi(mr, n) &= \varphi(m, rn) \end{aligned}$$

The third axiom is the only one that mentions R at all, and it is weaker than linearity in either variable, since it never asserts what happens to a scalar that is pulled out of the map, namely as L has no module structure.

Example 15.1.2: R-Balanced Maps

1. Let R be commutative and let M, N, L be R-modules, with $mr := rm$ making M a right R-module. Every map that is R-linear in each variable is R-balanced: $\varphi(mr, n) = \varphi(rm, n) = r\varphi(m, n) = \varphi(m, rn)$. So over a commutative ring the balanced condition is implied by, and strictly weaker than, bilinearity.

2. For any ring R and any right R-module M, the action map $M \times R \to M$ sending (m, r) to mr is R-balanced, where R is viewed as a left module over itself. Additivity in each variable is a module axiom, and the third axiom is associativity of the action: $(ms)r = m(sr)$.

3. Take $R = \mathbb{Z}$. Every $\mathbb{Z}$-module is an abelian group and every abelian group is a $\mathbb{Z}$-module (section 12.1), so a $\mathbb{Z}$-balanced map is exactly a map additive in each variable: the third axiom, $\varphi(mk, n) = \varphi(m, kn)$ for $k \in \mathbb{Z}$, already follows from the first two. Balanced maps over $\mathbb{Z}$ therefore impose no condition beyond biadditivity.

4. The determinant of a 2×2 matrix, read as a map $F^2 \times F^2 \to F$ sending the pair of columns $\big((a,b),(c,d)\big)$ to $ad - bc$, is F-bilinear (section 13.4) and hence F-balanced. It is one of the maps this chapter exists to bring inside linear algebra.

The construction now writes down the freest object carrying such a map. Start with a group in which the pairs (m, n) satisfy nothing whatsoever, then impose the three axioms of definition 15.1.1 by hand. Write $F^{\mathbb{Z}}(A)$ for the free $\mathbb{Z}$-module on a set A (theorem 12.3.11), that is, the free abelian group with basis A.

Applied to the underlying *set* of $M \times N$, this group is enormous. If $R = \mathbb{R}$, $M = \mathbb{R}^2$ and $N = \mathbb{R}^3$, the basis of $F^{\mathbb{Z}}(M \times N)$ is the whole of $\mathbb{R}^2 \times \mathbb{R}^3$, so distinct pairs (m, n) and (m', n') are independent generators even when m and m' differ by a scalar. Nothing that holds in M or in N survives into $F^{\mathbb{Z}}(M \times N)$. Every relation the tensor product satisfies has to be put in by quotienting.

Definition 15.1.3: Tensor Product

Let R be a ring, let M be a right R-module and let N be a left R-module. Let $K \subseteq F^{\mathbb{Z}}(M \times N)$ be the subgroup generated by all elements of the three forms

$$\begin{aligned} &(m_1 + m_2,\ n) - (m_1, n) - (m_2, n) \\ &(m,\ n_1 + n_2) - (m, n_1) - (m, n_2) \\ &(mr,\ n) - (m, rn) \end{aligned}$$

taken over all $m, m_1, m_2 \in M$, all $n, n_1, n_2 \in N$ and all $r \in R$. The *tensor product of M and N over R* is the quotient group

$$M \otimes_R N := F^{\mathbb{Z}}(M \times N)/K$$

The coset of the basis element (m, n) is written $m \otimes n$ and is called a *simple tensor*. The subscript R is dropped when the ring is clear from context.

Killing K is exactly what makes the three defining identities hold in the quotient, so that for all $m, m_i \in M$, $n, n_i \in N$ and $r \in R$,

$$\begin{gathered} (m_1 + m_2) \otimes n = m_1 \otimes n + m_2 \otimes n \qquad m \otimes (n_1 + n_2) = m \otimes n_1 + m \otimes n_2 \\ mr \otimes n = m \otimes rn \end{gathered} \tag{15.1}$$

These three are the only relations available. Anything else true in $M \otimes_R N$ has to be derived from them, and the next lemma is the derivation used most often.

Lemma 15.1.4: Simple Tensors Generate

Let M be a right R-module and N a left R-module. Then

1. $0 \otimes n = 0$ and $m \otimes 0 = 0$ for all $m \in M$, $n \in N$;
2. every element of $M \otimes_R N$ is a finite sum $\sum_{i=1}^{k} m_i \otimes n_i$ of simple tensors;
3. two group homomorphisms $M \otimes_R N \to L$ that agree on every simple tensor are equal.

***Proof*:**

1. By the first identity in (15.1) applied to $m_1 = m_2 = 0$, one has $0 \otimes n = (0+0) \otimes n = 0 \otimes n + 0 \otimes n$, and subtracting $0 \otimes n$ from both sides in the group $M \otimes_R N$ gives $0 \otimes n = 0$. The argument for $m \otimes 0$ is the same using the second identity.

2. The group $F^{\mathbb{Z}}(M \times N)$ is generated by its basis, so every element of it is a finite sum $\sum_i k_i(m_i, n_i)$ with $k_i \in \mathbb{Z}$, and the quotient map is surjective, so every element of $M \otimes_R N$ is a finite sum $\sum_i k_i(m_i \otimes n_i)$. It therefore suffices to see that an integer multiple of a simple tensor is again a simple tensor. For $k \geq 0$ induction on k using the first identity in (15.1) gives $k(m \otimes n) = (km) \otimes n$, the base case $k = 0$ being part (1). For $k < 0$ it suffices to treat $k = -1$, and

$$(-m) \otimes n + m \otimes n = (-m + m) \otimes n = 0 \otimes n = 0$$

gives $(-m) \otimes n = -(m \otimes n)$, so $k(m \otimes n) = (km) \otimes n$ for every $k \in \mathbb{Z}$.

3. Let ψ, ψ' be two such homomorphisms. The set on which two group homomorphisms agree is a subgroup of the domain, and by (2) the simple tensors generate $M \otimes_R N$, so that subgroup is everything, completing the proof.

Part (2) says that every element of the tensor product is a finite sum of simple tensors. It does not say that every element *is* a simple tensor, and that distinction is the most common source of error in computations with tensor products. Box 15.1.6 exhibits an element that is not simple.

The definition of $M \otimes_R N$ was engineered to make ι balanced and to impose nothing further. The following theorem is the statement that it succeeded, and it is the only property of the tensor product used from here on: every later result in this chapter is proved from it rather than from the quotient construction.

Theorem 15.1.5: Universal Property of the Tensor Product

Let R be a ring, let M be a right R-module and let N be a left R-module. The map

$$\iota \colon M \times N \to M \otimes_R N, \qquad \iota(m, n) = m \otimes n$$

is R-balanced, and it is universal among such maps: for every abelian group L and every R-balanced map $\varphi \colon M \times N \to L$ there is a unique group homomorphism $\psi \colon M \otimes_R N \to L$ with $\varphi = \psi \circ \iota$, that is, making

$$\begin{array}{ccc} M \otimes_R N & \xrightarrow{\ \psi\ } & L \\ {\scriptstyle \iota}\uparrow & \nearrow_{\varphi} & \\ M \times N & & \end{array}$$

commute.

***Proof*:**
That ι is R-balanced is precisely the three identities (15.1), which hold because the corresponding elements of $F^{\mathbb{Z}}(M \times N)$ were placed in K.

Existence. Regard $M \times N$ as a set and φ as a map of sets into the abelian group L, which is a $\mathbb{Z}$-module. By the universal property of free modules (theorem 12.3.11, applied with $R = \mathbb{Z}$) there is a group homomorphism $\tilde{\varphi}\colon F^{\mathbb{Z}}(M \times N) \to L$ with $\tilde{\varphi}\big((m,n)\big) = \varphi(m,n)$ on each basis element. Evaluating $\tilde{\varphi}$ on the three families of generators of K gives

$$\begin{aligned}\tilde{\varphi}\big((m_1+m_2,n)-(m_1,n)-(m_2,n)\big) &= \varphi(m_1+m_2,n)-\varphi(m_1,n)-\varphi(m_2,n) = 0\\ \tilde{\varphi}\big((m,n_1+n_2)-(m,n_1)-(m,n_2)\big) &= \varphi(m,n_1+n_2)-\varphi(m,n_1)-\varphi(m,n_2) = 0\\ \tilde{\varphi}\big((mr,n)-(m,rn)\big) &= \varphi(mr,n)-\varphi(m,rn) = 0\end{aligned}$$

each vanishing by the corresponding axiom of definition 15.1.1. A subgroup is generated by any set whose elements it contains together with their sums and inverses, and $\ker\tilde{\varphi}$ is a subgroup containing all three families, so $K \subseteq \ker\tilde{\varphi}$. Since $F^{\mathbb{Z}}(M \times N)$ is abelian, K is normal in it, so theorem 3.1.15 applies and yields a unique map $\psi\colon F^{\mathbb{Z}}(M \times N)/K \to L$ with $\tilde{\varphi} = \psi \circ \pi$, where π is the quotient map. It is a group homomorphism, since for $x, y \in F^{\mathbb{Z}}(M \times N)$,

$$\psi\big(\pi(x)+\pi(y)\big) = \psi\big(\pi(x+y)\big) = \tilde{\varphi}(x+y) = \tilde{\varphi}(x)+\tilde{\varphi}(y) = \psi(\pi(x))+\psi(\pi(y))$$

and π is surjective. Reading the identity $\tilde{\varphi} = \psi \circ \pi$ on the basis element (m,n) gives $\psi(m \otimes n) = \varphi(m,n)$, which is $\varphi = \psi \circ \iota$.

Uniqueness. A homomorphism ψ' with $\varphi = \psi' \circ \iota$ satisfies $\psi'(m \otimes n) = \varphi(m,n) = \psi(m \otimes n)$ for every simple tensor, so $\psi' = \psi$ by lemma 15.1.4, completing the proof.

The theorem is what converts the tensor product from a construction into a tool. To produce a homomorphism out of $M \otimes_R N$ one never returns to cosets of a free group: one writes down a balanced map on $M \times N$, checks three identities, and reads off the homomorphism. To prove two homomorphisms out of $M \otimes_R N$ equal, one checks them on simple tensors. Both moves are used repeatedly below, starting immediately.

15.1.6 Warning: Not Every Tensor Is Simple Let F be a field, let $V = F^2$ with standard basis e_1, e_2, and consider

$$t = e_1 \otimes e_1 + e_2 \otimes e_2 \in V \otimes_F V$$

Then t is not a simple tensor. Suppose $t = v \otimes w$. For linear functionals $f, g \in V^*$ (section 13.3) the map $V \times V \to F$ sending (x,y) to $f(x)g(y)$ is F-bilinear, hence F-balanced, so by theorem 15.1.5 below it induces a homomorphism $\theta_{f,g}\colon V \otimes_F V \to F$ with $\theta_{f,g}(x \otimes y) = f(x)g(y)$. Evaluating on both expressions for t,

$$f(v)g(w) = \theta_{f,g}(t) = f(e_1)g(e_1) + f(e_2)g(e_2)$$

Taking (f,g) to be (e_1^*, e_1^*), then (e_2^*, e_2^*), then (e_1^*, e_2^*) gives in turn $v_1w_1 = 1$, $v_2w_2 = 1$ and $v_1w_2 = 0$, where $v = v_1e_1 + v_2e_2$ and $w = w_1e_1 + w_2e_2$. The first two force v_1 and w_2 to be nonzero, contradicting the third.

A consequence worth recording is that the pair $(M \otimes_R N, \iota)$ is pinned down by the theorem, so the arbitrary choices in definition 15.1.3 leave no trace.

Corollary 15.1.7: Uniqueness of the Tensor Product

Let T be an abelian group and $j\colon M \times N \to T$ an R-balanced map with the property that for every abelian group L and every R-balanced $\varphi\colon M \times N \to L$ there is a unique homomorphism $\psi\colon T \to L$ with $\varphi = \psi \circ j$. Then there is a unique isomorphism $\theta\colon M \otimes_R N \to T$ with $\theta \circ \iota = j$.

***Proof*:**
Applying theorem 15.1.5 to the balanced map j gives a homomorphism $\theta\colon M \otimes_R N \to T$ with $\theta \circ \iota = j$, and applying the hypothesis on T to the balanced map ι gives $\eta\colon T \to M \otimes_R N$ with $\eta \circ j = \iota$. Then $\eta \circ \theta$ is an endomorphism of $M \otimes_R N$ satisfying $(\eta \circ \theta) \circ \iota = \iota$, and so is the identity. By the uniqueness clause of theorem 15.1.5 applied to $\varphi = \iota$, the two agree. Symmetrically $\theta \circ \eta = \mathrm{id}_T$, using the uniqueness clause assumed for T. Hence θ is an isomorphism, and it is the unique one satisfying $\theta \circ \iota = j$ by that same uniqueness clause, as we sought to show.

So far $M \otimes_R N$ is only an abelian group, and definition 15.1.1 was designed to make that unavoidable: nothing in the data of a right module M and a left module N produces an R-action on the tensor product. A module structure has to come from somewhere else, and the smallest extra hypothesis that gives one is a second, compatible action on one of the factors.

Definition 15.1.8: (S, R)-Bimodule

Let R and S be rings. An abelian group M is an (S, R)-*bimodule* if it is a left S-module, a right R-module, and the two actions commute:

$$s(mr) = (sm)r \qquad \text{for all } s \in S,\ r \in R,\ m \in M$$

Example 15.1.9: Bimodules

1. Any ring R is an (R, R)-bimodule over itself, with both actions given by ring multiplication. The compatibility axiom is associativity, $s(mr) = (sm)r$.
2. Let $f\colon R \to S$ be a ring homomorphism. Then S is an (S, R)-bimodule: the left S-action is multiplication in S, and the right R-action is $s \cdot r := sf(r)$. Compatibility reads $s'(sf(r)) = (s's)f(r)$, which is associativity in S. This is the structure that makes extension of scalars along f possible, and it is taken up in section 15.3.
3. As a special case of (2), let $I \subseteq R$ be a two-sided ideal and $f\colon R \to R/I$ the quotient map. Then R/I is an $(R/I, R)$-bimodule.
4. Let R be commutative and M an R-module. Setting $mr := rm$ makes M an (R, R)-bimodule, since the compatibility axiom $s(mr) = (sm)r$ reads $(sr)m = (rs)m$. This is called the *standard* bimodule structure on M, and it is the structure meant whenever a module over a commutative ring is treated as a bimodule without further comment.

Proposition 15.1.10: Module Structure on the Tensor Product

Let M be an (S, R)-bimodule and let N be a left R-module. Then $M \otimes_R N$ is a left S-module under the action determined by

$$s \cdot (m \otimes n) = (sm) \otimes n$$

on simple tensors.

***Proof*:**
Fix $s \in S$ and consider $\lambda_s \colon M \times N \to M \otimes_R N$ defined by $\lambda_s(m, n) = (sm) \otimes n$. It is R-balanced. Additivity in the first variable holds because $s(m_1 + m_2) = sm_1 + sm_2$ and $\otimes$ is additive in its first argument. Additivity in the second is immediate. For the third axiom, the bimodule condition gives $s(mr) = (sm)r$, so

$$\lambda_s(mr, n) = \big(s(mr)\big) \otimes n = \big((sm)r\big) \otimes n = (sm) \otimes rn = \lambda_s(m, rn)$$

the middle equality being the third relation of (15.1). By theorem 15.1.5 there is therefore a unique group endomorphism Λ_s of $M \otimes_R N$ with $\Lambda_s(m \otimes n) = (sm) \otimes n$. Define $s \cdot x := \Lambda_s(x)$ for $x \in M \otimes_R N$. This is additive in x because Λ_s is a homomorphism.

The remaining module axioms are equalities between group homomorphisms $M \otimes_R N \to M \otimes_R N$, so by lemma 15.1.4 each may be checked on simple tensors alone. On $m \otimes n$,

$$\begin{aligned}
\Lambda_{s+s'}(m \otimes n) &= \big((s + s')m\big) \otimes n = (sm + s'm) \otimes n \\
&= (sm) \otimes n + (s'm) \otimes n = (\Lambda_s + \Lambda_{s'})(m \otimes n) \\
\Lambda_{ss'}(m \otimes n) &= \big((ss')m\big) \otimes n = \big(s(s'm)\big) \otimes n \\
&= \Lambda_s\big((s'm) \otimes n\big) = (\Lambda_s \circ \Lambda_{s'})(m \otimes n) \\
\Lambda_1(m \otimes n) &= (1m) \otimes n = m \otimes n
\end{aligned}$$

using in each case only a left S-module axiom for M. Hence $\Lambda_{s+s'} = \Lambda_s + \Lambda_{s'}$, $\Lambda_{ss'} = \Lambda_s \circ \Lambda_{s'}$ and $\Lambda_1 = \mathrm{id}$, which are exactly the axioms making $M \otimes_R N$ a left S-module, completing the proof.

The proof is a template for everything that follows: a map is defined on simple tensors, shown balanced, promoted to a homomorphism by theorem 15.1.5, and its properties are then checked on simple tensors by lemma 15.1.4. No later argument in this chapter returns to the free group.

Specializing to a commutative ring recovers the situation the chapter opened with, in which both factors and the target are modules over the same ring and the universal property is stated for bilinear maps. That case needs the module-theoretic form of bilinearity, which definition 13.3.15 gave only for vector spaces with target the base field.

Definition 15.1.11: R-Bilinear Map

Let R be a commutative ring and let M, N and L be R-modules. A map $\varphi \colon M \times N \to L$ is *R-bilinear* if for each fixed $n \in N$ the map $\varphi(-, n) \colon M \to L$ is an R-module homomorphism, and for each fixed $m \in M$ the map $\varphi(m, -) \colon N \to L$ is an R-module homomorphism.

Theorem 15.1.12: Universal Property Of Tensor Product (Over Commutative Rings)

Let R be a commutative ring and let M, N be R-modules, each carrying its standard bimodule structure. Then $M \otimes_R N$ is an R-module, the map $\iota\colon M \times N \to M \otimes_R N$ is R-bilinear, and for every R-module L and every R-bilinear map $\varphi\colon M \times N \to L$ there is a unique R-module homomorphism $\psi\colon M \otimes_R N \to L$ with $\varphi = \psi \circ \iota$.

***Proof*:**
The R-module structure is proposition 15.1.10 applied with $S = R$, using the standard bimodule structure on M from example 15.1.9(4). Explicitly, $r(m \otimes n) = (rm) \otimes n$. That ι is R-bilinear now follows from (15.1) together with this action: it is additive in each variable, $r(m \otimes n) = (rm) \otimes n$ by definition of the action, and $r(m \otimes n) = (mr) \otimes n = m \otimes rn$ by the third relation.

Let φ be R-bilinear. By example 15.1.2(1) it is R-balanced, so theorem 15.1.5 gives a unique group homomorphism ψ with $\varphi = \psi \circ \iota$. It remains to see that ψ is R-linear. Fix $r \in R$. The two maps $x \mapsto \psi(rx)$ and $x \mapsto r\psi(x)$ are group homomorphisms $M \otimes_R N \to L$, and on a simple tensor

$$\psi\big(r(m \otimes n)\big) = \psi\big((rm) \otimes n\big) = \varphi(rm, n) = r\varphi(m, n) = r\psi(m \otimes n)$$

so they agree on all simple tensors and hence are equal by lemma 15.1.4. Thus ψ is R-linear. Uniqueness is inherited: an R-module homomorphism satisfying $\varphi = \psi \circ \iota$ is in particular a group homomorphism satisfying it, and there is only one of those, as we sought to show.

Two computations show what the construction does to familiar modules, and they point in opposite directions. The first says that tensoring against the ring itself changes nothing; the second says that tensoring two nonzero modules can annihilate both.

Example 15.1.13: Tensoring Against the Ring
Let R be a ring and N a left R-module. Then $R \otimes_R N \cong N$ as left R-modules, where R carries its (R, R)-bimodule structure from example 15.1.9(1).

The action map $\mu\colon R \times N \to N$ with $\mu(r, n) = rn$ is R-balanced: additivity in each variable is a module axiom, and $\mu(rs, n) = (rs)n = r(sn) = \mu(r, sn)$. By theorem 15.1.5 it induces a group homomorphism $\bar{\mu}\colon R \otimes_R N \to N$ with $\bar{\mu}(r \otimes n) = rn$. In the other direction let $\nu\colon N \to R \otimes_R N$ be $\nu(n) = 1 \otimes n$, which is additive. Then $\bar{\mu}(\nu(n)) = \bar{\mu}(1 \otimes n) = n$, so $\bar{\mu} \circ \nu = \mathrm{id}_N$. For the other composite, on a simple tensor

$$\nu\big(\bar{\mu}(r \otimes n)\big) = \nu(rn) = 1 \otimes rn = 1 \cdot r \otimes n = r \otimes n$$

the third equality being the third relation of (15.1) read backwards. So $\nu \circ \bar{\mu}$ agrees with the identity on simple tensors and is the identity by lemma 15.1.4. Finally $\bar{\mu}$ is R-linear, since $\bar{\mu}\big(s(r \otimes n)\big) = \bar{\mu}\big((sr) \otimes n\big) = (sr)n = s(rn) = s\,\bar{\mu}(r \otimes n)$ on simple tensors.

Example 15.1.14: Tensor Products of Cyclic Groups

1. $\mathbb{Z}/2\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z}/3\mathbb{Z} = 0$. Since $3 \equiv 1 \pmod 2$, one has $\bar{a} = 3\bar{a}$ in $\mathbb{Z}/2\mathbb{Z}$ for every a, and therefore

$$\bar{a} \otimes \bar{b} = (3\bar{a}) \otimes \bar{b} = \bar{a} \otimes (3\bar{b}) = 0$$

using $3\bar{b} = \bar{0}$ in $\mathbb{Z}/3\mathbb{Z}$. So every simple tensor vanishes, and by lemma 15.1.4(2) the whole group is zero.

2. $\mathbb{Z}/2\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z}/2\mathbb{Z} \cong \mathbb{Z}/2\mathbb{Z}$. The same manipulation is unavailable here, and in fact $\bar{1} \otimes \bar{1} \neq 0$. To see it, the multiplication map $\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z} \to \mathbb{Z}/2\mathbb{Z}$ sending $(\bar{a}, \bar{b})$ to $\overline{ab}$ is $\mathbb{Z}$-balanced, so it induces Φ with $\Phi(\bar{1} \otimes \bar{1}) = \bar{1} \neq 0$. A homomorphism sends 0 to 0, so $\bar{1} \otimes \bar{1} \neq 0$. Conversely every simple tensor is $\bar{a} \otimes \bar{b}$ with $a, b \in \{0, 1\}$, hence is 0 or $\bar{1} \otimes \bar{1}$, so by lemma 15.1.4(2) the group $\mathbb{Z}/2\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z}/2\mathbb{Z}$ is cyclic, generated by $t = \bar{1} \otimes \bar{1}$. That generator satisfies $2t = (2\bar{1}) \otimes \bar{1} = \bar{0} \otimes \bar{1} = 0$, so it has order dividing 2, and it is nonzero, so its order is exactly 2. Hence $\mathbb{Z}/2\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z}/2\mathbb{Z} \cong \mathbb{Z}/2\mathbb{Z}$, with Φ the isomorphism.

The contrast between the two computations is the first real content of the construction. Both $\mathbb{Z}/2\mathbb{Z}$ and $\mathbb{Z}/3\mathbb{Z}$ are nonzero, and their tensor product is not. So $\otimes_R$ is not a construction that remembers its inputs. The mechanism is visible in the computation: the relation $mr \otimes n = m \otimes rn$ lets a scalar be moved from a factor where it acts invertibly to one where it acts as zero. Section 15.3 turns this from an obstruction into a tool, and the systematic account of which sequences of modules survive tensoring is the subject of section 17.5.

Example 15.1.15: Tensors as Multilinear Maps

Outside algebra, and particularly in physics and in analysis, the word "tensor" usually names a multilinear map rather than an element of a tensor product. In that convention a k-tensor on a vector space V over F *is* a multilinear map $T\colon V^k \to F$, and the tensor product of a k-tensor T and an ℓ-tensor S is defined directly by the formula

$$(T \otimes S)(x_1, \ldots, x_k, y_1, \ldots, y_\ell) = T(x_1, \ldots, x_k) \cdot S(y_1, \ldots, y_\ell)$$

which is again multilinear, now in $k + \ell$ arguments.

The two usages agree when V is finite dimensional. A multilinear map $V^k \to F$ is the same thing as a linear functional on $V^{\otimes k}$, which is the universal property of theorem 15.2.11 below, and for finite-dimensional V the space of such functionals is identified with $(V^*)^{\otimes k}$ by the basis computation of corollary 15.2.6. Under that identification the displayed formula is the image of $T \otimes S$ in the sense of definition 15.1.3. The reason for the analyst's convention is that a multilinear map has a domain and a target, so conditions like continuity, differentiability, or being C^r can be imposed on it. An element of an abstract quotient group cannot carry such conditions until it is realized as a function. This is the form in which tensors appear in *Calculus on Manifolds* [14].

Exercise 15.1.1

1. Let R be a ring and M a right R-module, with R carrying its (R, R)-bimodule structure. Show that $M \otimes_R R \cong M$, and identify the isomorphism on simple tensors.

2. Let $m, n \geq 1$ and let $d = \gcd(m, n)$. Show that

$$\mathbb{Z}/m\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z}/n\mathbb{Z} \cong \mathbb{Z}/d\mathbb{Z}$$

(Suggestion: since $\mathbb{Z}$ is a principal ideal domain, $(d) = (m) + (n)$ by definition 10.1.24, so $d = um + vn$ for some $u, v \in \mathbb{Z}$. Use this to show first that d annihilates the left-hand side.)

3. Let R be a ring, M a right R-module and N a left R-module, and suppose there is an $r \in R$

with $Mr = 0$ and $rN = N$. Show that $M \otimes_R N = 0$. Deduce that $\mathbb{Q} \otimes_{\mathbb{Z}} A = 0$ for every finite abelian group A, and that $\mathbb{Q}/\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Q}/\mathbb{Z} = 0$.

4. Let R be commutative, let M be generated as an R-module by a set X and let N be generated by a set Y. Show that $M \otimes_R N$ is generated as an R-module by the simple tensors $x \otimes y$ with $x \in X$ and $y \in Y$. Give an example showing that these need not be independent.

5. Let M be a right R-module, N a left R-module and L an abelian group. Show that $\varphi \mapsto \psi$ is a bijection from the set of R-balanced maps $M \times N \to L$ onto $\mathrm{Hom}_{\mathbb{Z}}(M \otimes_R N, L)$, and that it commutes with composition by a homomorphism $L \to L'$.

6. Let R be commutative, let A, B, M be R-modules and let $f\colon A \to M$ and $g\colon B \to M$ be R-linear. Show that $(a, b) \mapsto f(a) + g(b)$ is an R-module homomorphism $A \oplus B \to M$, and exhibit f and g for which it is not R-bilinear. Conclude that $A \oplus B$ and $A \otimes_R B$ solve different universal problems.

15.2 Properties of the Tensor Product

With one exception, named where it occurs, no proof below returns to the free abelian group of section 15.1. Everything is derived from theorem 15.1.5 and lemma 15.1.4, by the two moves those results give: to produce a map out of a tensor product, write down a balanced map and invoke the universal property; to prove two maps out of a tensor product equal, check them on simple tensors. What follows records how $\otimes_R$ interacts with each construction the previous chapters gave, in the order in which the proofs depend on one another: homomorphisms, direct sums, free modules, and then the two structural isomorphisms, commutativity and associativity, that make an unbracketed $M_1 \otimes \cdots \otimes M_n$ meaningful. The section closes with the ring structure carried by a tensor product of algebras.

Proposition 15.1.10 produced a left module structure from a second action on the left-hand factor. A second action on the right-hand factor does the mirror-image job, and the two are independent, so both can be imposed at once. The right action is what theorem 15.2.8 needs, since there the middle factor of a triple product is acted on from both sides.

Proposition 15.2.1: Two-Sided Module Structure on a Tensor Product

Let R, S, T be rings, let M be an (S, R)-bimodule and let N be an (R, T)-bimodule. Then $M \otimes_R N$ is an (S, T)-bimodule, with actions determined on simple tensors by

$$s \cdot (m \otimes n) = (sm) \otimes n \qquad \text{and} \qquad (m \otimes n) \cdot t = m \otimes (nt)$$

***Proof*:**

The left S-module structure is proposition 15.1.10, which uses only that M is an (S, R)-bimodule and N a left R-module.

For the right T-action, fix $t \in T$ and consider $\rho_t\colon M \times N \to M \otimes_R N$ given by $\rho_t(m, n) = m \otimes (nt)$. It is R-balanced: additivity in m is additivity of $\otimes$ in its first argument; additivity in n follows because $n \mapsto nt$ is additive and $\otimes$ is additive in its second argument; and for the third axiom

the (R,T)-bimodule condition on N gives $r(nt) = (rn)t$, so

$$\rho_t(mr, n) = (mr) \otimes (nt) = m \otimes r(nt) = m \otimes (rn)t = \rho_t(m, rn)$$

the middle equality being the third relation of (15.1). By theorem 15.1.5 there is a unique group endomorphism P_t of $M \otimes_R N$ with $P_t(m \otimes n) = m \otimes nt$. Set $x \cdot t := P_t(x)$, which is additive in x because P_t is a homomorphism.

The remaining right-module axioms are equalities of group homomorphisms, so by lemma 15.1.4(3) each may be checked on simple tensors:

$$\begin{aligned} P_{t+t'}(m \otimes n) &= m \otimes n(t+t') = m \otimes (nt + nt') = m \otimes nt + m \otimes nt' = (P_t + P_{t'})(m \otimes n) \\ P_{tt'}(m \otimes n) &= m \otimes n(tt') = m \otimes (nt)t' = P_{t'}(m \otimes nt) = (P_{t'} \circ P_t)(m \otimes n) \\ P_1(m \otimes n) &= m \otimes n1 = m \otimes n \end{aligned}$$

The second line gives $x(tt') = (xt)t'$, which is the associativity axiom for a *right* action, the order of composition being reversed exactly as it should be. Finally the two actions commute: $x \mapsto s(xt)$ and $x \mapsto (sx)t$ are group homomorphisms, and on a simple tensor both give $(sm)\otimes(nt)$, so they agree by lemma 15.1.4(3), completing the proof.

Homomorphisms of the two factors induce a homomorphism of the tensor product, and the assignment respects composition. This is what makes $\otimes_R$ usable inside diagrams rather than only on individual modules.

Theorem 15.2.2: The Tensor Product of Two Homomorphisms

Let $\varphi\colon M \to M'$ be a homomorphism of right R-modules and let $\psi\colon N \to N'$ be a homomorphism of left R-modules.

1. There is a unique group homomorphism $\varphi \otimes \psi\colon M \otimes_R N \to M' \otimes_R N'$ with
$$(\varphi \otimes \psi)(m \otimes n) = \varphi(m) \otimes \psi(n)$$
for all $m \in M$ and $n \in N$.
2. If M and M' are (S,R)-bimodules and φ is also a homomorphism of left S-modules, then $\varphi \otimes \psi$ is a homomorphism of left S-modules. In particular, if R is commutative and all four modules carry their standard structures, then $\varphi \otimes \psi$ is R-linear.
3. $\mathrm{id}_M \otimes \mathrm{id}_N = \mathrm{id}_{M \otimes_R N}$, and if $\lambda\colon M' \to M''$ and $\mu\colon N' \to N''$ are further homomorphisms of right and left R-modules respectively, then
$$(\lambda \otimes \mu) \circ (\varphi \otimes \psi) = (\lambda \circ \varphi) \otimes (\mu \circ \psi)$$

***Proof*:**

1. The map $M \times N \to M' \otimes_R N'$ sending (m,n) to $\varphi(m) \otimes \psi(n)$ is R-balanced. Additivity in each variable follows from additivity of φ and ψ together with additivity of $\otimes$ in each argument. For the third axiom, φ is a homomorphism of right R-modules and ψ of left

R-modules, so

$$\varphi(mr)\otimes\psi(n) = \big(\varphi(m)r\big)\otimes\psi(n) = \varphi(m)\otimes r\psi(n) = \varphi(m)\otimes\psi(rn)$$

the middle equality being the third relation of (15.1) in $M'\otimes_R N'$. Theorem 15.1.5 now gives the homomorphism, and its uniqueness.

2. Both $x\mapsto(\varphi\otimes\psi)(sx)$ and $x\mapsto s(\varphi\otimes\psi)(x)$ are group homomorphisms $M\otimes_R N\to M'\otimes_R N'$, so by lemma 15.1.4(3) it suffices to compare them on a simple tensor, where

$$(\varphi\otimes\psi)\big(s(m\otimes n)\big) = (\varphi\otimes\psi)\big((sm)\otimes n\big) = \varphi(sm)\otimes\psi(n) = \big(s\varphi(m)\big)\otimes\psi(n) = s\big(\varphi(m)\otimes\psi(n)\big)$$

using S-linearity of φ in the third step and proposition 15.1.10 in the fourth. When R is commutative and the structures are standard, an R-module homomorphism is S-linear for $S = R$, so the statement applies with $S = R$.

3. All three claims are equalities of group homomorphisms out of $M\otimes_R N$, so lemma 15.1.4(3) reduces each to a check on simple tensors. There $(\mathrm{id}\otimes\mathrm{id})(m\otimes n) = m\otimes n$, and

$$(\lambda\otimes\mu)\big((\varphi\otimes\psi)(m\otimes n)\big) = (\lambda\otimes\mu)\big(\varphi(m)\otimes\psi(n)\big) = \lambda(\varphi(m))\otimes\mu(\psi(n))$$

which is the value of $(\lambda\circ\varphi)\otimes(\mu\circ\psi)$ on $m\otimes n$, completing the proof.

Parts (1) and (3) say precisely that fixing a left R-module N makes $-\otimes_R N$ a functor from right R-modules to abelian groups, and symmetrically for $M\otimes_R -$. The question of how this functor treats exact sequences is left open here and settled in section 17.5. The computation $\mathbb{Z}/2\mathbb{Z}\otimes_{\mathbb{Z}}\mathbb{Z}/3\mathbb{Z} = 0$ of example 15.1.14 is already a warning that it does not treat them well.

The next two results compute a tensor product one summand at a time.

Theorem 15.2.3: Tensor Products of Direct Sums

Let M, M' be right R-modules and let N, N' be left R-modules. There are unique group isomorphisms

$$(M\oplus M')\otimes_R N \cong (M\otimes_R N)\oplus(M'\otimes_R N)$$
$$M\otimes_R (N\oplus N') \cong (M\otimes_R N)\oplus(M\otimes_R N')$$

determined on simple tensors by $(m,m')\otimes n\mapsto(m\otimes n, m'\otimes n)$ and $m\otimes(n,n')\mapsto(m\otimes n, m\otimes n')$ respectively. If M and M' are (S,R)-bimodules, the first is an isomorphism of left S-modules; if R is commutative and all modules carry their standard structures, both are isomorphisms of R-modules.

***Proof*:**
Consider $\alpha\colon (M\oplus M')\times N\to(M\otimes_R N)\oplus(M'\otimes_R N)$ given by $\alpha\big((m,m'),n\big) = (m\otimes n, m'\otimes n)$. It is R-balanced, each axiom holding in both coordinates at once. For the third,

$$\alpha\big((m,m')r,n\big) = \big(mr\otimes n,\ m'r\otimes n\big) = \big(m\otimes rn,\ m'\otimes rn\big) = \alpha\big((m,m'),rn\big)$$

since the action on $M\oplus M'$ is componentwise. By theorem 15.1.5 there is a group homomorphism $\bar\alpha$ on $(M\oplus M')\otimes_R N$ with the stated values.

For the inverse, the maps $\beta_1 \colon M \times N \to (M \oplus M') \otimes_R N$ with $\beta_1(m,n) = (m,0) \otimes n$ and $\beta_2 \colon M' \times N \to (M \oplus M') \otimes_R N$ with $\beta_2(m',n) = (0,m') \otimes n$ are R-balanced, because $m \mapsto (m,0)$ and $m' \mapsto (0,m')$ are homomorphisms of right R-modules. So they induce group homomorphisms $\bar{\beta}_1$ and $\bar{\beta}_2$. Define

$$\beta \colon (M \otimes_R N) \oplus (M' \otimes_R N) \to (M \oplus M') \otimes_R N, \qquad \beta(x,y) = \bar{\beta}_1(x) + \bar{\beta}_2(y)$$

which is a group homomorphism.

On a simple tensor,

$$\beta\big(\bar{\alpha}((m,m') \otimes n)\big) = (m,0) \otimes n + (0,m') \otimes n = \big((m,0) + (0,m')\big) \otimes n = (m,m') \otimes n$$

so $\beta \circ \bar{\alpha} = \mathrm{id}$ by lemma 15.1.4(3). In the other direction, lemma 15.1.4(2) applied in each summand shows that $(M \otimes_R N) \oplus (M' \otimes_R N)$ is generated as a group by the elements $(x,0)$ and $(0,y)$ with x and y simple tensors, so it is enough to evaluate $\bar{\alpha} \circ \beta$ on those. On $(m \otimes n, 0)$,

$$\bar{\alpha}\big(\bar{\beta}_1(m \otimes n)\big) = \bar{\alpha}\big((m,0) \otimes n\big) = (m \otimes n,\, 0 \otimes n) = (m \otimes n,\, 0)$$

by lemma 15.1.4(1), and the computation on $(0, m' \otimes n)$ is the same. Hence $\bar{\alpha} \circ \beta = \mathrm{id}$ and $\bar{\alpha}$ is an isomorphism.

When M, M' are (S,R)-bimodules, so is $M \oplus M'$, and S-linearity of $\bar{\alpha}$ is again an equality of group homomorphisms checked on simple tensors, where both sides give $\big((sm) \otimes n, (sm') \otimes n\big)$. Uniqueness in all cases is lemma 15.1.4(3).

The second isomorphism is the mirror image of the first: interchange the two factors throughout, replacing each appeal to additivity in the first argument of $\otimes$ by additivity in the second, and proposition 15.1.10 by proposition 15.2.1. No new step enters, completing the proof.

Theorem 15.2.4: Arbitrary Direct Sums Commute with Tensor Products

Let $\{M_i\}_{i \in I}$ be a family of right R-modules and let N be a left R-module. Then there is a unique group isomorphism

$$\left(\bigoplus_{i \in I} M_i\right) \otimes_R N \cong \bigoplus_{i \in I} (M_i \otimes_R N)$$

sending $(m_i)_i \otimes n$ to $(m_i \otimes n)_i$. If each M_i is an (S,R)-bimodule it is an isomorphism of left S-modules, and if R is commutative it is an isomorphism of R-modules.

***Proof*:**

Write $M = \bigoplus_{i \in I} M_i$, whose elements are the families $(m_i)_i$ with $m_i = 0$ for all but finitely many i. The assignment $\alpha\big((m_i)_i, n\big) = (m_i \otimes n)_i$ lands in the direct sum rather than the product, since $m_i = 0$ forces $m_i \otimes n = 0$ by lemma 15.1.4(1), so all but finitely many entries vanish. It is R-balanced for the same componentwise reason as in theorem 15.2.3, and so induces $\bar{\alpha}$ on $M \otimes_R N$.

Let $\kappa_j \colon M_j \to M$ be the inclusion of the j-th summand, a homomorphism of right R-modules. Each $\beta_j \colon M_j \times N \to M \otimes_R N$ given by $\beta_j(m,n) = \kappa_j(m) \otimes n$ is therefore R-balanced, inducing

$\bar{\beta}_j$. Define

$$\beta\colon \bigoplus_{i\in I}(M_i \otimes_R N) \to M \otimes_R N, \qquad \beta\big((x_i)_i\big) = \sum_{i\in I} \bar{\beta}_i(x_i)$$

a finite sum because $(x_i)_i$ has finite support, and a group homomorphism because each $\bar{\beta}_i$ is. On a simple tensor, writing $(m_i)_i = \sum_i \kappa_i(m_i)$ as the finite sum it is,

$$\beta\big(\bar{\alpha}((m_i)_i \otimes n)\big) = \sum_i \kappa_i(m_i) \otimes n = \left(\sum_i \kappa_i(m_i)\right) \otimes n = (m_i)_i \otimes n$$

using additivity of $\otimes$ in its first argument, so $\beta \circ \bar{\alpha} = \mathrm{id}$. Conversely $\bigoplus_i (M_i \otimes_R N)$ is generated as a group by the elements $\kappa'_j(x)$ with x a simple tensor of $M_j \otimes_R N$, where κ'_j is the inclusion of the j-th summand, and

$$\bar{\alpha}\big(\beta(\kappa'_j(m \otimes n))\big) = \bar{\alpha}\big(\kappa_j(m) \otimes n\big) = \kappa'_j(m \otimes n)$$

since every entry of $\kappa_j(m)$ other than the j-th is zero and therefore contributes $0 \otimes n = 0$. Hence $\bar{\alpha} \circ \beta = \mathrm{id}$. The bimodule claims and uniqueness are as in theorem 15.2.3, completing the proof.

Theorem 15.2.4 decomposes a direct sum in the *left* argument, and theorem 15.2.3 handles two summands on either side. The remaining case, an arbitrary direct sum in the right argument, is what a free presentation of a module needs, so it is recorded separately.

Corollary 15.2.5: Arbitrary Direct Sums in the Right Argument

Let M be a right R-module and $\{N_i\}_{i\in I}$ a family of left R-modules. Then there is a unique group isomorphism

$$M \otimes_R \left(\bigoplus_{i\in I} N_i\right) \cong \bigoplus_{i\in I}(M \otimes_R N_i)$$

sending $m \otimes (n_i)_i$ to $(m \otimes n_i)_i$. If M is an (S, R)-bimodule it is an isomorphism of left S-modules.

***Proof*:**

Write $N = \bigoplus_{i\in I} N_i$. The assignment $\alpha\big(m, (n_i)_i\big) = (m \otimes n_i)_i$ lands in the direct sum rather than the product, since $n_i = 0$ for all but finitely many i and $m \otimes 0 = 0$ by lemma 15.1.4(1). It is additive in each variable componentwise, and R-balanced because

$$\alpha\big(mr, (n_i)_i\big) = (mr \otimes n_i)_i = (m \otimes rn_i)_i = \alpha\big(m, r \cdot (n_i)_i\big)$$

so it induces $\bar{\alpha}\colon M \otimes_R N \to \bigoplus_i (M \otimes_R N_i)$.

For the inverse, let $\kappa_j\colon N_j \to N$ be the inclusion of the j-th summand, a map of left R-modules. Each $\beta_j\colon M \times N_j \to M \otimes_R N$ given by $\beta_j(m, n) = m \otimes \kappa_j(n)$ is R-balanced, hence induces $\bar{\beta}_j$ on $M \otimes_R N_j$, and these assemble into a map β on the direct sum.

The composites are the identity on simple tensors, which suffices by lemma 15.1.4. Writing $(n_i)_i = \sum_{j\in F} \kappa_j(n_j)$ over the finite set F of nonzero entries,

$$\beta\big(\bar{\alpha}(m \otimes (n_i)_i)\big) = \sum_{j\in F} m \otimes \kappa_j(n_j) = m \otimes \sum_{j\in F} \kappa_j(n_j) = m \otimes (n_i)_i$$

and conversely $\bar{\alpha}\big(\beta(\kappa'_j(m \otimes n))\big) = \bar{\alpha}\big(m \otimes \kappa_j(n)\big) = \kappa'_j(m \otimes n)$, every entry of $\kappa_j(n)$ other than the j-th being zero and so contributing $m \otimes 0 = 0$. The S-linearity claim holds because the left S-action on both sides is $s \cdot (m \otimes n) = (sm) \otimes n$ by proposition 15.2.1, which $\bar{\alpha}$ respects, completing the proof.

Applying theorem 15.2.4 to a free module, where every summand is a copy of the ring, computes the tensor product of two free modules completely. This is the result that makes tensor products calculable in linear algebra: it says that a basis of M and a basis of N determine a basis of $M \otimes_R N$, indexed by pairs.

Corollary 15.2.6: Tensor Products of Free Modules

Let R be a commutative ring, let M be a free R-module with basis $\{m_i\}_{i \in I}$ and let N be a free R-module with basis $\{n_j\}_{j \in J}$. Then $M \otimes_R N$ is a free R-module with basis

$$\{m_i \otimes n_j\}_{(i,j) \in I \times J}$$

In particular $R^s \otimes_R R^t \cong R^{st}$, and if V, W are finite-dimensional vector spaces over a field F then

$$\dim_F(V \otimes_F W) = \dim_F(V) \cdot \dim_F(W)$$

***Proof*:**
Since $\{m_i\}$ is a basis, $M = \bigoplus_{i \in I} Rm_i$ with each $Rm_i \cong R$ as an R-module, the isomorphism carrying m_i to 1 (theorem 12.3.11). Applying theorem 15.2.4, and then example 15.1.13 to each summand after transporting it along $Rm_i \cong R$ by theorem 15.2.2, gives R-module isomorphisms

$$M \otimes_R N \cong \bigoplus_{i \in I} \big(Rm_i \otimes_R N\big) \cong \bigoplus_{i \in I} N$$

under which $m_i \otimes n$ corresponds to the family whose i-th entry is n and whose other entries are 0. Writing $N = \bigoplus_{j \in J} Rn_j \cong \bigoplus_{j \in J} R$ in the same way and substituting,

$$M \otimes_R N \cong \bigoplus_{i \in I} \bigoplus_{j \in J} R \cong \bigoplus_{(i,j) \in I \times J} R$$

and following $m_i \otimes n_j$ through the three isomorphisms shows that it corresponds to the element of $\bigoplus_{I \times J} R$ with 1 in position (i, j) and 0 elsewhere. Those elements form the standard basis of $\bigoplus_{I \times J} R$, and an isomorphism of R-modules carries a basis to a basis, so $\{m_i \otimes n_j\}$ is a basis of $M \otimes_R N$.

The two special cases follow by counting: $I \times J$ has st elements when $|I| = s$ and $|J| = t$, and the dimension of a vector space is the cardinality of any basis, as we sought to show.

This closes the identification promised in example 15.1.15: for finite-dimensional V, a basis of V^* and one of W^* produce a basis of $V^* \otimes_F W^*$, matching the basis of the space of bilinear forms on $V \times W$ under the correspondence of theorem 15.1.12.

Two structural isomorphisms remain. Both are needed before an expression such as $M_1 \otimes M_2 \otimes M_3$ can be written without brackets.

Proposition 15.2.7: Commutativity of the Tensor Product

Let R be a commutative ring and let M, N be R-modules with their standard structures. Then there is a unique R-module isomorphism

$$\tau\colon M \otimes_R N \to N \otimes_R M, \qquad \tau(m \otimes n) = n \otimes m$$

***Proof*:**
The map $M \times N \to N \otimes_R M$ sending (m, n) to $n \otimes m$ is R-bilinear: it is additive in each variable, and for $r \in R$,

$$n \otimes (rm) = (nr) \otimes m = r(n \otimes m)$$

using the third relation of (15.1) and the description of the action in theorem 15.1.12, with the same computation in the other variable. So theorem 15.1.12 gives a unique R-module homomorphism τ with $\tau(m \otimes n) = n \otimes m$. The same construction with the roles of M and N exchanged gives $\tau'\colon N \otimes_R M \to M \otimes_R N$ with $\tau'(n \otimes m) = m \otimes n$, and both composites fix every simple tensor, so both are the identity by lemma 15.1.4(3), as we sought to show.

Theorem 15.2.8: Associativity of the Tensor Product

Let M be a right R-module, let N be an (R, T)-bimodule and let L be a left T-module. Then there is a unique isomorphism of abelian groups

$$(M \otimes_R N) \otimes_T L \cong M \otimes_R (N \otimes_T L)$$

carrying $(m \otimes n) \otimes l$ to $m \otimes (n \otimes l)$. If M is an (S, R)-bimodule it is an isomorphism of left S-modules; if $R = T$ is commutative and all three modules carry their standard structures, it is an isomorphism of R-modules.

***Proof*:**
By proposition 15.2.1, $M \otimes_R N$ is a right T-module with $(m \otimes n)t = m \otimes (nt)$, and $N \otimes_T L$ is a left R-module with $r(n \otimes l) = (rn) \otimes l$. Both tensor products in the statement are therefore defined.

Step 1: a homomorphism for each l. Fix $l \in L$. The map $M \times N \to M \otimes_R (N \otimes_T L)$ sending (m, n) to $m \otimes (n \otimes l)$ is R-balanced. It is additive in m; it is additive in n because $n \mapsto n \otimes l$ is additive and $\otimes$ is additive in its second argument; and

$$(mr) \otimes (n \otimes l) = m \otimes r(n \otimes l) = m \otimes \big((rn) \otimes l\big)$$

which is the third axiom. By theorem 15.1.5 there is a unique group homomorphism $\theta_l\colon M \otimes_R N \to M \otimes_R (N \otimes_T L)$ with $\theta_l(m \otimes n) = m \otimes (n \otimes l)$.

Step 2: assembling over l. The map $(M \otimes_R N) \times L \to M \otimes_R (N \otimes_T L)$ sending (x, l) to $\theta_l(x)$ is T-balanced. It is additive in x because θ_l is a homomorphism. For additivity in l, both $\theta_{l+l'}$ and $\theta_l + \theta_{l'}$ are group homomorphisms out of $M \otimes_R N$, and on a simple tensor both give $m \otimes (n \otimes l) + m \otimes (n \otimes l')$, so they are equal by lemma 15.1.4(3). For the third axiom, $x \mapsto \theta_l(xt)$ and $x \mapsto \theta_{tl}(x)$ are group homomorphisms, and on a simple tensor

$$\theta_l\big((m \otimes n)t\big) = \theta_l(m \otimes nt) = m \otimes \big((nt) \otimes l\big) = m \otimes \big(n \otimes (tl)\big) = \theta_{tl}(m \otimes n)$$

the third equality being the third relation of (15.1) inside $N \otimes_T L$. So theorem 15.1.5 gives a group homomorphism Θ on $(M \otimes_R N) \otimes_T L$ with $\Theta\big((m \otimes n) \otimes l\big) = m \otimes (n \otimes l)$.

Step 3: the inverse. The same two steps run with the outer factors exchanged. Fixing $m \in M$, the map $N \times L \to (M \otimes_R N) \otimes_T L$ sending (n, l) to $(m \otimes n) \otimes l$ is T-balanced, giving $\eta_m \colon N \otimes_T L \to (M \otimes_R N) \otimes_T L$ with $\eta_m(n \otimes l) = (m \otimes n) \otimes l$; and $(m, y) \mapsto \eta_m(y)$ is then R-balanced by the same three checks. This yields a group homomorphism Ξ on $M \otimes_R (N \otimes_T L)$ with $\Xi\big(m \otimes (n \otimes l)\big) = (m \otimes n) \otimes l$.

Step 4: the composites. By lemma 15.1.4(2) every element of $(M \otimes_R N) \otimes_T L$ is a finite sum of terms $x \otimes l$ with $x \in M \otimes_R N$, and each such x is itself a finite sum of simple tensors, so by additivity of $\otimes$ in its first argument every element is a finite sum of terms $(m \otimes n) \otimes l$. Since $\Xi \circ \Theta$ is a group homomorphism fixing each such term, it is the identity. The same argument on the other side gives $\Theta \circ \Xi = \mathrm{id}$.

The S-linearity of Θ, when M is an (S, R)-bimodule, is once more an equality of group homomorphisms verified on the generating terms $(m \otimes n) \otimes l$, where both sides give $(sm) \otimes (n \otimes l)$. And uniqueness holds because a homomorphism is determined by its values on those terms, completing the proof.

15.2.9 Foreshadowing: Monoidal Structure on R-Mod For a commutative ring R, the three isomorphisms

$$(M \otimes_R N) \otimes_R L \cong M \otimes_R (N \otimes_R L), \qquad R \otimes_R M \cong M, \qquad M \otimes_R N \cong N \otimes_R M$$

of theorem 15.2.8, example 15.1.13 and proposition 15.2.7 are not three unrelated facts. They are the associator, the unit isomorphism and the braiding of a structure called a *symmetric monoidal category*, carried by R-Mod with $\otimes_R$ as its product and R as its unit. In that language the R-algebras of definition 12.1.13 are exactly the monoid objects of R-Mod, and the compatibility conditions the three isomorphisms satisfy with one another are the subject of a coherence theorem. The structure is developed in *Category Theory* [15]. Nothing in this chapter depends on it.

Associativity justifies the notation $M_1 \otimes_R \cdots \otimes_R M_n$: define it to mean the left-bracketed product $(\cdots (M_1 \otimes_R M_2) \otimes_R \cdots) \otimes_R M_n$, and write $m_1 \otimes \cdots \otimes m_n$ for the corresponding element. Repeated application of theorem 15.2.8 produces an isomorphism from any other bracketing onto this one matching the corresponding simple tensors, which is what makes the bracket-free notation harmless. With that in place the universal property extends from two factors to n.

Definition 15.2.10: Multilinear Maps

Let R be a commutative ring and let $M_1, \ldots, M_n$ and L be R-modules with their standard structures. A map $\varphi \colon M_1 \times \cdots \times M_n \to L$ is *R-multilinear*, or *n-linear*, if for each index i and each choice of fixed elements $m_k \in M_k$ for $k \neq i$, the map

$$M_i \to L, \qquad m \mapsto \varphi(m_1, \ldots, m_{i-1}, m, m_{i+1}, \ldots, m_n)$$

is an R-module homomorphism. For $n = 2$ and $n = 3$ one says *bilinear* and *trilinear* rather than 2-linear and 3-linear.

This is definition 13.3.15 with the target allowed to be any R-module rather than the base field, and with any number of arguments. For $n = 2$ it is definition 15.1.11.

Theorem 15.2.11: Universal Property for Multilinear Maps

Let R be a commutative ring and let $M_1, \dots, M_n$ and L be R-modules. For every R-multilinear map $\varphi\colon M_1 \times \cdots \times M_n \to L$ there is a unique R-module homomorphism

$$\psi\colon M_1 \otimes_R \cdots \otimes_R M_n \to L, \qquad \psi(m_1 \otimes \cdots \otimes m_n) = \varphi(m_1, \dots, m_n)$$

***Proof*:**

Induction on n. The case $n = 1$ is the identity and $n = 2$ is theorem 15.1.12, so suppose $n \geq 3$ and the statement holds for $n - 1$. Write $P = M_1 \otimes_R \cdots \otimes_R M_{n-1}$, so that by definition of the left-bracketed product the n-fold tensor product is $P \otimes_R M_n$.

First note that P is generated as an R-module by the simple tensors $m_1 \otimes \cdots \otimes m_{n-1}$: this holds for $n - 1 = 1$ trivially, and the inductive step is lemma 15.1.4(2) applied to $P' \otimes_R M_{n-1}$ together with additivity of $\otimes$ in its first argument.

Fix $m_n \in M_n$. The map $(m_1, \dots, m_{n-1}) \mapsto \varphi(m_1, \dots, m_n)$ is $(n-1)$-linear, since fixing all but one of its arguments fixes all but one argument of φ. By the inductive hypothesis there is a unique R-module homomorphism $\psi_{m_n}\colon P \to L$ with $\psi_{m_n}(m_1 \otimes \cdots \otimes m_{n-1}) = \varphi(m_1, \dots, m_n)$.

The map $P \times M_n \to L$ sending (x, m_n) to $\psi_{m_n}(x)$ is R-bilinear. Linearity in x holds because ψ_{m_n} is an R-module homomorphism. For linearity in m_n, the maps $\psi_{m_n + m_n'}$ and $\psi_{m_n} + \psi_{m_n'}$ are R-module homomorphisms out of P agreeing on every $m_1 \otimes \cdots \otimes m_{n-1}$, by linearity of φ in its last argument, and those generate P. So the two are equal. The same argument gives $\psi_{rm_n} = r\psi_{m_n}$.

Theorem 15.1.12 therefore gives a unique R-module homomorphism ψ on $P \otimes_R M_n$ with $\psi(x \otimes m_n) = \psi_{m_n}(x)$, and evaluating at $x = m_1 \otimes \cdots \otimes m_{n-1}$ gives the stated formula. Uniqueness follows because the simple tensors $m_1 \otimes \cdots \otimes m_n$ generate $M_1 \otimes_R \cdots \otimes_R M_n$, by the generation statement above and lemma 15.1.4(2), completing the proof.

One fact about $\otimes_R$ cannot be obtained from the universal property, because it is a statement about how an equation in $M \otimes_R N$ arises rather than about maps out of it. Lemma 15.1.4(2) says every element is a finite sum of simple tensors. The lemma below says that when such a sum vanishes, the vanishing already occurs inside a finitely generated piece of M. Its proof is the one place in this section that returns to the construction in definition 15.1.3.

Lemma 15.2.12: Vanishing Is Witnessed Finitely

Let R be a ring, M a right R-module and N a left R-module, and suppose

$$\sum_{i=1}^{k} m_i \otimes n_i = 0 \qquad \text{in } M \otimes_R N$$

Then there is a *finitely generated* submodule $M_0 \subseteq M$ containing $m_1, \dots, m_k$ such that already

$$\sum_{i=1}^{k} m_i \otimes n_i = 0 \qquad \text{in } M_0 \otimes_R N$$

The same holds with the roles of M and N exchanged.

***Proof*:**

By definition 15.1.3 the group $M \otimes_R N$ is $F^{\mathbb{Z}}(M \times N)/K$, so the hypothesis says that the element $\sum_i (m_i, n_i)$ of $F^{\mathbb{Z}}(M \times N)$ lies in K. Since K is generated by the three displayed forms, there are finitely many generators $g_1, \dots, g_t$ of K and integers $c_1, \dots, c_t$ with

$$\sum_{i=1}^{k} (m_i, n_i) = \sum_{j=1}^{t} c_j \, g_j$$

Each g_j mentions only finitely many elements of M: a generator of the first form mentions m, m' and $m + m'$, one of the second mentions a single m, and one of the third mentions mr and m. Let $S \subseteq M$ be the finite set consisting of $m_1, \dots, m_k$ together with every element of M mentioned by some g_j, and let M_0 be the submodule of M generated by S. It is finitely generated and contains each m_i.

Now form $F^{\mathbb{Z}}(M_0 \times N)$ and the subgroup K_0 generated by the three forms of definition 15.1.3 taken over M_0, N and R, so that $M_0 \otimes_R N = F^{\mathbb{Z}}(M_0 \times N)/K_0$. Every g_j lies in $F^{\mathbb{Z}}(M_0 \times N)$, since all of its M-coordinates belong to $S \subseteq M_0$; and g_j is a generator of K_0, because M_0 is a submodule and so is closed under the addition and the R-action that the three forms use. Hence the displayed identity holds in $F^{\mathbb{Z}}(M_0 \times N)$ and exhibits $\sum_i (m_i, n_i)$ as an element of K_0, which is to say $\sum_i m_i \otimes n_i = 0$ in $M_0 \otimes_R N$.

The statement with M and N exchanged is the same argument applied to the second coordinate.

The lemma does not say that $M_0 \otimes_R N \to M \otimes_R N$ is injective, and in general it is not. What it says is that a *relation* that holds downstairs already holds upstairs, which is what makes it possible to reduce a question about an arbitrary module to the finitely generated case. Theorem 17.5.7 is the use it was recorded for.

A tensor product of quotients of a commutative ring is again a quotient, and the ideal one quotients by is the sum. This is the computation behind most small examples.

Proposition 15.2.13: Tensor Product of Quotient Rings

Let R be a commutative ring and let I, J be ideals of R. Then there is an R-module isomorphism

$$R/I \otimes_R R/J \cong R/(I+J), \qquad \overline{a} \otimes \overline{b} \mapsto \overline{ab}$$

***Proof*:**

The map $R/I \times R/J \to R/(I+J)$ sending $(\overline{a}, \overline{b})$ to $\overline{ab}$ is well defined: if $a - a' \in I$ and $b - b' \in J$ then

$$ab - a'b' = (a-a')b + a'(b-b') \in I + J$$

It is R-bilinear, so theorem 15.1.12 induces an R-module homomorphism Φ with $\Phi(\overline{a} \otimes \overline{b}) = \overline{ab}$.

In the other direction set $\Psi(\overline{r}) = \overline{r} \otimes \overline{1}$. Throughout, a bar in the left-hand factor denotes a class in R/I and a bar in the right-hand factor a class in R/J. To see that Ψ is well defined, let $r - r' \in I + J$ and write $r - r' = a + b$ with $a \in I$ and $b \in J$, so that

$$\overline{r - r'} \otimes \overline{1} = \overline{a} \otimes \overline{1} + \overline{b} \otimes \overline{1}$$

The first term vanishes because $\overline{a} = \overline{0}$ already in R/I. For the second, moving the scalar b across by the third relation of (15.1),

$$\overline{b} \otimes \overline{1} = (b \cdot \overline{1}) \otimes \overline{1} = \overline{1} \otimes (b \cdot \overline{1}) = \overline{1} \otimes \overline{b} = 0$$

since $\overline{b} = \overline{0}$ in R/J. So Ψ is well defined, and it is R-linear.

The composite $\Phi \circ \Psi$ sends $\overline{r}$ to $\overline{r}$. For the other composite,

$$\Psi\big(\Phi(\overline{a} \otimes \overline{b})\big) = \overline{ab} \otimes \overline{1} = (\overline{a} \cdot b) \otimes \overline{1} = \overline{a} \otimes b\overline{1} = \overline{a} \otimes \overline{b}$$

so $\Psi \circ \Phi$ fixes every simple tensor and is the identity by lemma 15.1.4(3), as we sought to show.

Taking $R = \mathbb{Z}$, $I = (m)$ and $J = (n)$ recovers $\mathbb{Z}/m\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z}/n\mathbb{Z} \cong \mathbb{Z}/\gcd(m,n)\mathbb{Z}$, since $(m) + (n) = (\gcd(m,n))$ by definition 10.1.24. That is the second exercise of section 15.1, and both computations of example 15.1.14 are the cases $(m,n) = (2,3)$ and $(m,n) = (2,2)$.

When the two factors are R-algebras rather than bare modules, their multiplications combine into one on the tensor product. This is the construction that will produce the tensor algebra in section 16.3, and it is the source of the coproduct description that closes this section.

Theorem 15.2.14: Tensor Product of R-Algebras

Let R be a commutative ring and let A and B be R-algebras (definition 12.1.13). Then there is a unique R-bilinear multiplication on $A \otimes_R B$ satisfying

$$(a \otimes b)(a' \otimes b') = (aa') \otimes (bb')$$

and it makes $A \otimes_R B$ into an R-algebra with identity $1_A \otimes 1_B$. If A and B are both commutative, so is $A \otimes_R B$.

***Proof*:**
Step 1: the multiplication exists. The map $A \times B \times A \times B \to A \otimes_R B$ sending (a, b, a', b') to $(aa') \otimes (bb')$ is R-multilinear: in each of its four arguments the other three are fixed, and the resulting map is a composite of multiplication in A or in B, which is R-linear in each variable by definition 12.1.13, with $\otimes$, which is R-linear in each argument by theorem 15.1.12. By theorem 15.2.11 it induces an R-module homomorphism

$$\mu' \colon A \otimes_R B \otimes_R A \otimes_R B \to A \otimes_R B, \qquad \mu'(a \otimes b \otimes a' \otimes b') = (aa') \otimes (bb')$$

By theorem 15.2.8 there is an isomorphism $(A \otimes_R B) \otimes_R (A \otimes_R B) \cong A \otimes_R B \otimes_R A \otimes_R B$ matching $(a \otimes b) \otimes (a' \otimes b')$ with $a \otimes b \otimes a' \otimes b'$. Composing it with μ' gives an R-module homomorphism μ on $(A \otimes_R B) \otimes_R (A \otimes_R B)$ with $\mu\big((a \otimes b) \otimes (a' \otimes b')\big) = (aa') \otimes (bb')$. Setting $xy := \mu(x \otimes y)$ defines a map $(A \otimes_R B) \times (A \otimes_R B) \to A \otimes_R B$ which is R-bilinear, being the composite of the R-bilinear ι with the R-linear μ, and which satisfies the displayed formula on simple tensors. It is the unique such: two R-bilinear maps agreeing on pairs of simple tensors induce, by theorem 15.1.12, two homomorphisms on $(A \otimes_R B) \otimes_R (A \otimes_R B)$ agreeing on the simple tensors $(a \otimes b) \otimes (a' \otimes b')$, which generate it, so they coincide.

Step 2: the ring axioms. Distributivity is additivity of the multiplication in each variable, which is part of bilinearity. For associativity, both $(x, y, z) \mapsto (xy)z$ and $(x, y, z) \mapsto x(yz)$ are R-trilinear, being built from bilinear maps, so by theorem 15.2.11 each corresponds to a homomorphism on the triple tensor product, and the two agree if and only if they agree on triples of simple tensors. There

$$\big((a\otimes b)(a'\otimes b')\big)(a''\otimes b'') = \big((aa')a''\big)\otimes\big((bb')b''\big) = \big(a(a'a'')\big)\otimes\big(b(b'b'')\big) = (a\otimes b)\big((a'\otimes b')(a''\otimes b'')\big)$$

by associativity in A and in B. For the identity, $x \mapsto (1_A \otimes 1_B)x$ and the identity map are both additive and agree on simple tensors, so they are equal by lemma 15.1.4(3), and likewise on the other side.

Step 3: the R-algebra structure. The R-module structure is the one $A \otimes_R B$ already carries, and the compatibility $r(xy) = (rx)y = x(ry)$ required by definition 12.1.13 is R-bilinearity of the multiplication. Finally, if A and B are commutative then $(x, y) \mapsto xy$ and $(x, y) \mapsto yx$ are R-bilinear and agree on pairs of simple tensors, since $(aa') \otimes (bb') = (a'a) \otimes (b'b)$, so they are equal by the uniqueness argument of Step 1, completing the proof.

Example 15.2.15: Tensoring Two Copies of $\mathbb{C}$ over $\mathbb{R}$
The base ring matters, and comparing $\mathbb{C} \otimes_{\mathbb{C}} \mathbb{C}$ with $\mathbb{C} \otimes_{\mathbb{R}} \mathbb{C}$ shows how much. The first is $\mathbb{C}$ itself by example 15.1.13. The second has $\mathbb{R}$-dimension 4 by corollary 15.2.6, since $\mathbb{C}$ is free of rank 2 over $\mathbb{R}$, and it is not even a domain:

$$\mathbb{C} \otimes_{\mathbb{R}} \mathbb{C} \cong \mathbb{C} \times \mathbb{C}$$

as $\mathbb{R}$-algebras. To see this, define $\Phi \colon \mathbb{C} \otimes_{\mathbb{R}} \mathbb{C} \to \mathbb{C} \times \mathbb{C}$ on simple tensors by

$$\Phi(z \otimes w) = (zw,\ z\overline{w})$$

This is well defined because $(z, w) \mapsto (zw, z\overline{w})$ is $\mathbb{R}$-bilinear, complex conjugation being $\mathbb{R}$-linear, so theorem 15.1.12 applies. It is a ring homomorphism: on simple tensors

$$\Phi\big((z\otimes w)(z'\otimes w')\big) = \Phi(zz'\otimes ww') = \big(zz'ww',\ zz'\overline{ww'}\big) = (zw\cdot z'w',\ z\overline{w}\cdot z'\overline{w'}) = \Phi(z\otimes w)\Phi(z'\otimes w')$$

and both $(x,y) \mapsto \Phi(xy)$ and $(x,y) \mapsto \Phi(x)\Phi(y)$ are $\mathbb{R}$-bilinear, so agreeing on simple tensors makes them equal. Also $\Phi(1 \otimes 1) = (1,1)$.

For surjectivity, compute Φ on three elements:

$$\Phi(1 \otimes 1) = (1,1), \qquad \Phi(i \otimes 1) = (i,i), \qquad \Phi(i \otimes i) = (i \cdot i,\ i\bar{i}) = (-1,1)$$

The image of Φ is both an $\mathbb{R}$-subspace and a subring. It contains $(1,1) - (-1,1) = (2,0)$ and $(1,1) + (-1,1) = (0,2)$, hence $(1,0)$ and $(0,1)$. And being closed under multiplication it then contains $(1,0)(i,i) = (i,0)$ and $(0,1)(i,i) = (0,i)$. Those four elements form an $\mathbb{R}$-basis of $\mathbb{C} \times \mathbb{C}$, so Φ is surjective. Both sides have $\mathbb{R}$-dimension 4, and a surjective linear map between vector spaces of equal finite dimension is an isomorphism.

So a tensor product of two fields over a common subfield need not be a field, and need not even be an integral domain: $\mathbb{C} \times \mathbb{C}$ has the zero divisors $(1,0)$ and $(0,1)$.

The last result of this section reads the tensor product of algebras backwards, as the answer to a universal problem rather than as a construction. Recall that in R-Mod the direct sum $M \oplus N$ is simultaneously the product and the coproduct of M and N. This is a feature of *finite* families only, since for an infinite index set the product $\prod_i M_i$ and the coproduct $\bigoplus_i M_i$ differ, the latter consisting of the finitely supported families. In commutative R-algebras the two come apart even for two factors. The product is the direct product $A \times B$ with componentwise operations, and the coproduct is the tensor product.

Lemma 15.2.16: Generation of $A \otimes_R B$ as an Algebra

Let R be a commutative ring and A, B be R-algebras. Then $A \otimes_R B$ is generated as an R-algebra by the set $\{a \otimes 1 \mid a \in A\} \cup \{1 \otimes b \mid b \in B\}$.

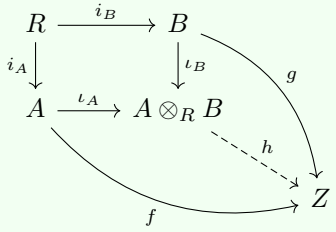
***Proof*:**
By theorem 15.2.14, $(a \otimes 1)(1 \otimes b) = (a \cdot 1) \otimes (1 \cdot b) = a \otimes b$, so every simple tensor is a product of two elements of the stated set. By lemma 15.1.4(2) every element of $A \otimes_R B$ is a finite sum of simple tensors, completing the proof.

Theorem 15.2.17: Universal Property of the Coproduct in R-Alg

Let R be a commutative ring and let A and B be commutative R-algebras, with structure maps $i_A\colon R \to A$ and $i_B\colon R \to B$. Then the maps

$$\iota_A\colon A \to A \otimes_R B, \quad a \mapsto a \otimes 1 \qquad \text{and} \qquad \iota_B\colon B \to A \otimes_R B, \quad b \mapsto 1 \otimes b$$

are R-algebra homomorphisms, and $(A \otimes_R B, \iota_A, \iota_B)$ is the coproduct of A and B in the category of commutative R-algebras: for every commutative R-algebra Z and every pair of R-algebra homomorphisms $f\colon A \to Z$ and $g\colon B \to Z$ there is a unique R-algebra homomorphism $h\colon A \otimes_R B \to Z$ with $h \circ \iota_A = f$ and $h \circ \iota_B = g$:



***Proof*:**

That ι_A is an R-algebra homomorphism is immediate from theorem 15.2.14: it is additive and R-linear because $\otimes$ is R-linear in its first argument, multiplicative because $(a\otimes 1)(a'\otimes 1) = (aa')\otimes 1$, and unital because $\iota_A(1_A) = 1_A \otimes 1_B$. The same holds for ι_B.

Existence. Define $\Phi\colon A \times B \to Z$ by $\Phi(a,b) = f(a)g(b)$. It is R-bilinear: it is additive in each variable because f and g are additive and multiplication in Z distributes, and for $r \in R$,

$$\Phi(ra,b) = f(ra)g(b) = \big(rf(a)\big)g(b) = r\big(f(a)g(b)\big) \qquad \Phi(a,rb) = f(a)\big(rg(b)\big) = r\big(f(a)g(b)\big)$$

using R-linearity of f and of g and the compatibility $r(xy) = (rx)y = x(ry)$ of definition 12.1.13 in Z. By theorem 15.1.12 there is a unique R-module homomorphism $h\colon A \otimes_R B \to Z$ with $h(a \otimes b) = f(a)g(b)$.

The map h is multiplicative. Both $(x,y) \mapsto h(xy)$ and $(x,y) \mapsto h(x)h(y)$ are R-bilinear, and on a pair of simple tensors

$$\begin{aligned} h\big((a \otimes b)(a' \otimes b')\big) &= h(aa' \otimes bb') = f(aa')g(bb') \\ &= f(a)f(a')g(b)g(b') = f(a)g(b)f(a')g(b') \\ &= h(a \otimes b)\, h(a' \otimes b') \end{aligned}$$

the fourth equality using that Z is commutative. Two R-bilinear maps agreeing on pairs of simple tensors are equal, by the argument in Step 1 of theorem 15.2.14, so h is multiplicative. It is unital, since $h(1_A \otimes 1_B) = f(1)g(1) = 1$, and therefore an R-algebra homomorphism. Finally $h(\iota_A(a)) = h(a \otimes 1) = f(a)g(1) = f(a)$ and likewise $h \circ \iota_B = g$, so the diagram commutes.

Uniqueness. Suppose h' is an R-algebra homomorphism with $h' \circ \iota_A = f$ and $h' \circ \iota_B = g$. Then h and h' agree on every $a \otimes 1$ and every $1 \otimes b$. By lemma 15.2.16 those elements generate $A \otimes_R B$ as an R-algebra, and two R-algebra homomorphisms agreeing on a generating set are equal, since the set on which they agree is a subalgebra. Hence $h' = h$, completing the proof.

Every commutative ring is an algebra over $\mathbb{Z}$ in exactly one way, which turns the theorem into a statement about rings with no base ring chosen in advance.

Corollary 15.2.18: Universal Property of the Coproduct in CRing

Let A and B be commutative rings. Then $A \otimes_{\mathbb{Z}} B$, with $\iota_A(a) = a \otimes 1$ and $\iota_B(b) = 1 \otimes b$, is the coproduct of A and B in the category **CRing** of commutative rings.

***Proof*:**
By proposition 9.1.29 there is exactly one ring homomorphism $\mathbb{Z} \to A$ for every ring A, so every commutative ring carries exactly one $\mathbb{Z}$-algebra structure, and its image lies in the centre because A is commutative. Moreover every ring homomorphism $\varphi\colon A \to B$ of commutative rings is automatically a $\mathbb{Z}$-algebra homomorphism: the two ring homomorphisms $\varphi \circ i_A$ and i_B from $\mathbb{Z}$ to B coincide by that same uniqueness. Hence the category of commutative $\mathbb{Z}$-algebras and **CRing** have the same objects and the same morphisms, and theorem 15.2.17 applied with $R = \mathbb{Z}$ gives the claim, completing the proof.

The product $A \times B$ is formed with no reference to a base ring, while the coproduct is formed over one, and in **CRing** the base is forced to be the initial object $\mathbb{Z}$. Over a general base R, the object $A \otimes_R B$ solves the coproduct problem for R-algebras, which is the same as a pushout of $A \leftarrow R \to B$ in commutative rings. Reversing every arrow turns a pushout of rings into a pullback of spaces, and this is the computation that makes $\otimes_R$ the algebraic form of the fibre product of affine schemes in *EYNTKA Algebraic Geometry* [9].

Exercise 15.2.1

1. Show that $\varphi \otimes \mathrm{id}$ need not be injective when φ is. (Take $\varphi\colon \mathbb{Z} \to \mathbb{Z}$ to be multiplication by 2 and tensor with $\mathbb{Z}/2\mathbb{Z}$ over $\mathbb{Z}$.) Conclude that the functor of theorem 15.2.2 does not preserve injections.

2. Let R be commutative, let M be an R-module and let I be an ideal of R. Show that $M \otimes_R R/I \cong M/IM$, and recover proposition 15.2.13 as the case $M = R/J$.

3. Show that every element of $\mathbb{Q} \otimes_{\mathbb{Z}} \mathbb{Q}$ is a simple tensor, and deduce that $\mathbb{Q} \otimes_{\mathbb{Z}} \mathbb{Q} \cong \mathbb{Q}$. (Note that example 15.1.13 does not apply: the left factor is $\mathbb{Q}$, not the base ring $\mathbb{Z}$.)

4. Let R be commutative and A an R-algebra. Show that the isomorphism $A \otimes_R R \cong A$ of example 15.1.13 is an isomorphism of R-algebras.

5. Using the isomorphism of example 15.2.15, find the two nontrivial idempotents of $\mathbb{C} \otimes_{\mathbb{R}} \mathbb{C}$ explicitly as combinations of simple tensors, and verify directly that each squares to itself.

6. Let R be commutative and let M, N be finitely generated R-modules, generated by s and t elements respectively. Show that $M \otimes_R N$ is generated by st elements. Show also that the bound is attained but not always sharp, by computing a case where $M \otimes_R N$ needs fewer.

7. Let F be a field and let V, W be finite-dimensional F-vector spaces. Show that
$$V^* \otimes_F W \to \mathrm{Hom}_F(V, W), \qquad f \otimes w \mapsto \big(v \mapsto f(v)w\big)$$
is an isomorphism of F-vector spaces, and identify where finite-dimensionality is used.

8. Combine theorem 15.2.8 and proposition 15.2.7 into an isomorphism $(A \otimes_R B) \otimes_R (A \otimes_R B) \cong (A \otimes_R A) \otimes_R (B \otimes_R B)$ carrying $(a \otimes b) \otimes (a' \otimes b')$ to $(a \otimes a') \otimes (b \otimes b')$. Use it, together with the maps $A \otimes_R A \to A$ and $B \otimes_R B \to B$ induced by multiplication, to give a second construction of the multiplication of theorem 15.2.14.

15.3 Extension of Scalars

The previous two sections held the ring fixed and varied the modules. This one does the opposite. Given a ring homomorphism $f\colon R \to S$, there are two ways to move a module between the two rings, and they are not symmetric: passing from S to R needs nothing, while passing from R to S requires a construction. That construction is the tensor product, applied to the bimodule of example 15.1.9(2), and everything in this section is a consequence of theorem 15.1.5 rather than a new construction.

The cheap direction first.

Definition 15.3.1: Restriction of Scalars

Let $f\colon R \to S$ be a ring homomorphism and let L be a left S-module. The *restriction of scalars* of L along f is the left R-module L_R with the same underlying abelian group and action

$$r \cdot l := f(r)\, l$$

An S-module homomorphism is in particular an R-module homomorphism $L_R \to L'_R$, so restriction of scalars sends S-modules to R-modules and S-linear maps to R-linear maps.

The module axioms hold because f is a ring homomorphism, exactly as in the construction of an R-algebra from a ring map (section 12.1.2). Nothing is created and nothing is destroyed, only forgotten. The other direction cannot work this way. An R-module N carries no S-action to forget, and there is in general no way to define one on the same underlying set: if $R = \mathbb{Z}$, $S = \mathbb{Q}$ and $N = \mathbb{Z}/2\mathbb{Z}$, then any $\mathbb{Q}$-action on N would make multiplication by 2 invertible on a group killed by 2. A new module has to be built, and the tensor product builds it.

Definition 15.3.2: Extension of Scalars

Let $f\colon R \to S$ be a ring homomorphism and let N be a left R-module. Regard S as an (S, R)-bimodule as in example 15.1.9(2), with left action given by multiplication in S and right action $s \cdot r := sf(r)$. The *extension of scalars* of N along f is the left S-module

$$S \otimes_R N$$

whose S-action, given by proposition 15.1.10, is determined on simple tensors by $s' \cdot (s \otimes n) = (s's) \otimes n$.

No verification is owed here: the module structure was proved once in proposition 15.1.10 and the only input is the bimodule structure on S, which is example 15.1.9(2). The relation that does the work is the third one in (15.1), which now reads

$$sf(r) \otimes n = s \otimes rn$$

so a scalar of R may be moved out of the module and into S, where it becomes invertible whenever $f(r)$ is.

Example 15.3.3: Extension of Scalars of a Free Module
Let $f\colon R \to S$ be a ring homomorphism. Then

$$S \otimes_R R^n \cong S^n$$

as left S-modules, and more generally $S \otimes_R R^{(I)} \cong S^{(I)}$ for a free module on any index set I. Indeed $R^{(I)} = \bigoplus_{i\in I} R$, so theorem 15.2.4 applied on the right-hand factor gives

$$S \otimes_R \bigoplus_{i\in I} R \cong \bigoplus_{i\in I} (S \otimes_R R) \cong \bigoplus_{i\in I} S$$

the second isomorphism being example 15.1.13 in each summand. Following a basis element e_i through, $1 \otimes e_i$ corresponds to the i-th standard generator of $S^{(I)}$. So extension of scalars carries a free R-module of rank n to a free S-module of the same rank: a basis survives, and only the ring it is a basis over changes.

Free modules therefore lose nothing. In general something is lost, and the map that measures the loss is the following.

The assignment $\iota\colon N \to S \otimes_R N$ with $\iota(n) = 1 \otimes n$ is additive, and it is R-linear when $S \otimes_R N$ is regarded as an R-module by restriction:

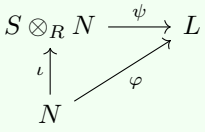
$$\iota(rn) = 1 \otimes rn = 1 \cdot f(r) \otimes n = f(r)\big(1 \otimes n\big) = r \cdot \iota(n)$$

using the third relation and then the definition of the S-action. It need not be injective. For $R = \mathbb{Z}$, $S = \mathbb{Q}$ and N any finite abelian group, $\mathbb{Q} \otimes_{\mathbb{Z}} N = 0$ by the third exercise of section 15.1, so ι is the zero map on a nonzero module.

The next theorem says that although N may not sit inside $S \otimes_R N$, every R-linear map from N into an S-module factors through it, uniquely and S-linearly. It is the defining property of extension of scalars and the reason the construction is the right one.

Theorem 15.3.4: Universal Property of Extension of Scalars

Let $f\colon R \to S$ be a ring homomorphism, let N be a left R-module and let $\iota\colon N \to S \otimes_R N$ be the R-module homomorphism $\iota(n) = 1 \otimes n$. Then for every left S-module L and every R-module homomorphism $\varphi\colon N \to L_R$ there is a unique S-module homomorphism $\psi\colon S \otimes_R N \to L$ with $\varphi = \psi \circ \iota$, that is, making



commute.

***Proof*:**
Existence. The map $S \times N \to L$ sending (s, n) to $s\varphi(n)$ is R-balanced. Additivity in s holds because the S-action on L distributes, and additivity in n because φ is additive. For the third axiom, the right R-action on S is $s \cdot r = sf(r)$ and the R-action on L_R is $r \cdot l = f(r)l$, so

$$(s \cdot r)\varphi(n) = \big(sf(r)\big)\varphi(n) = s\big(f(r)\varphi(n)\big) = s\big(r \cdot \varphi(n)\big) = s\,\varphi(rn)$$

the last equality because φ is R-linear. By theorem 15.1.5 there is a unique group homomorphism $\psi\colon S \otimes_R N \to L$ with $\psi(s \otimes n) = s\varphi(n)$.

It is S-linear. Fix $s' \in S$. The maps $x \mapsto \psi(s'x)$ and $x \mapsto s'\psi(x)$ are group homomorphisms, and on a simple tensor

$$\psi\big(s'(s \otimes n)\big) = \psi\big((s's) \otimes n\big) = (s's)\varphi(n) = s'\big(s\varphi(n)\big) = s'\psi(s \otimes n)$$

so they agree by lemma 15.1.4(3). Finally $\psi(\iota(n)) = \psi(1 \otimes n) = 1 \cdot \varphi(n) = \varphi(n)$, so $\varphi = \psi \circ \iota$.

Uniqueness. As an S-module, $S \otimes_R N$ is generated by the image of ι, since $s \otimes n = s(1 \otimes n) = s\,\iota(n)$ and every element is a finite sum of simple tensors by lemma 15.1.4(2). An S-module homomorphism is determined by its values on a generating set, and any ψ' with $\varphi = \psi' \circ \iota$ agrees with ψ on $\iota(N)$, so $\psi' = \psi$, completing the proof.

Restated without diagrams, the theorem says that R-linear maps out of N into an S-module are the same thing as S-linear maps out of $S \otimes_R N$. That is an adjunction, and it is worth naming as one, because it is the form in which the construction is used downstream.

Corollary 15.3.5: Extension of Scalars Is Left Adjoint to Restriction

Let $f\colon R \to S$ be a ring homomorphism. For every left R-module N and every left S-module L the map

$$\mathrm{Hom}_S\big(S \otimes_R N,\, L\big) \to \mathrm{Hom}_R\big(N,\, L_R\big), \qquad \psi \mapsto \psi \circ \iota$$

is an isomorphism of abelian groups. A pair of functors related this way is called an *adjunction*, and this one is expressed by saying that $S \otimes_R -$, from left R-modules to left S-modules, is *left adjoint* to restriction of scalars. The general theory is [15, Adjoint Functors]. Only the displayed bijection is used below.

***Proof*:**
The map is additive, since $(\psi + \psi') \circ \iota = \psi \circ \iota + \psi' \circ \iota$, and it lands in $\mathrm{Hom}_R(N, L_R)$ because ι is R-linear and ψ is S-linear, hence R-linear after restriction. It is surjective by the existence half of theorem 15.3.4 and injective by the uniqueness half: if $\psi \circ \iota = \psi' \circ \iota$ then both are maps induced by the same φ, so they coincide, as we sought to show.

Two computations show what the construction does. The first is the standard way a real vector space is turned into a complex one.

Example 15.3.6: Complexification
Take $R = \mathbb{R}$, $S = \mathbb{C}$ and f the inclusion. For a real vector space V the *complexification* is the complex vector space

$$V_{\mathbb{C}} := \mathbb{C} \otimes_{\mathbb{R}} V$$

Every element of $V_{\mathbb{C}}$ has the form $1 \otimes v + i \otimes w$ for unique $v, w \in V$. A simple tensor already does, since for $z = x+iy$ with $x, y \in \mathbb{R}$ the third relation of (15.1) gives $z \otimes v = x \otimes v + iy \otimes v = 1 \otimes xv + i \otimes yv$. And a sum of two such collapses to one by additivity of $\otimes$ in its second argument. If V has $\mathbb{R}$-basis $e_1, \ldots, e_n$ then $V_{\mathbb{C}}$ has $\mathbb{C}$-basis $1 \otimes e_1, \ldots, 1 \otimes e_n$ by example 15.3.3, so

$$\dim_{\mathbb{C}}(V_{\mathbb{C}}) = \dim_{\mathbb{R}}(V)$$

A linear map $\varphi \colon V \to W$ extends to $\mathrm{id}_{\mathbb{C}} \otimes \varphi \colon V_{\mathbb{C}} \to W_{\mathbb{C}}$ by theorem 15.2.2, and on the bases above it has the same matrix as φ. The entries are simply now read as complex numbers. This is what permits the standard move of computing eigenvalues of a real matrix over $\mathbb{C}$.

One warning. Complexifying a space that is already complex does not return it. Reading $\mathbb{C}$ itself as a real vector space and complexifying gives $\mathbb{C} \otimes_{\mathbb{R}} \mathbb{C} \cong \mathbb{C} \times \mathbb{C}$ by example 15.2.15, which has $\mathbb{C}$-dimension 2, not 1. The construction depends on the ring the tensor product is taken over, not only on the two spaces.

Example 15.3.7: Induced Representations

Let K be a field, let G be a finite group and let $H \leq G$ be a subgroup. The group rings $K[H]$ and $K[G]$ of definition 9.2.18 are related by the inclusion $K[H] \hookrightarrow K[G]$, so extension of scalars along it carries $K[H]$-modules to $K[G]$-modules. Since a $K[G]$-module is the same data as a representation of G over K, this manufactures a representation of G from one of H.

The dimension is forced. Choosing coset representatives $g_1, \ldots, g_k$ for G/H, where $k = [G : H]$, gives $K[G] = \bigoplus_{i=1}^{k} g_i K[H]$ as a right $K[H]$-module, so $K[G]$ is free of rank k over $K[H]$. If V is a representation of H of dimension d, then theorem 15.2.4 gives

$$K[G] \otimes_{K[H]} V \cong \bigoplus_{i=1}^{k} \left(g_i K[H] \otimes_{K[H]} V \right) \cong V^{\oplus k}$$

as K-vector spaces, so the induced representation has dimension kd. Concretely, $\mathbb{Z}/3\mathbb{Z}$ sits inside S_3 with index 2, so a 10-dimensional representation of $\mathbb{Z}/3\mathbb{Z}$ induces a 20-dimensional representation of S_3.

The construction is the *induced representation*, and its theory, including Frobenius reciprocity, which is corollary 15.3.5 read in this setting, is developed in *Core Representation Theory* [17].

Returning to the map ι, the universal property settles exactly when it is injective, and the answer identifies what part of N can survive into any S-module at all.

Corollary 15.3.8: When the Unit Is Injective

Let $f \colon R \to S$ be a ring homomorphism, let N be a left R-module and let $\iota \colon N \to S \otimes_R N$ be as above. Then every R-module homomorphism from N into (the restriction of) a left S-module vanishes on $\ker \iota$. Consequently N embeds as an R-submodule of some left S-module if and only if ι is injective, in which case $N \cong \iota(N) \subseteq S \otimes_R N$.

***Proof*:**
Let L be a left S-module and $\theta\colon N \to L_R$ an R-module homomorphism. By theorem 15.3.4 there is an S-module homomorphism ψ with $\theta = \psi \circ \iota$, so θ kills $\ker \iota$.

If ι is injective it is itself such an embedding, with image $\iota(N)$. If conversely some $\theta\colon N \to L_R$ is injective, then $\ker \iota \subseteq \ker \theta = 0$ by the previous paragraph, so ι is injective, completing the proof.

For a finite abelian group N and $\mathbb{Z} \hookrightarrow \mathbb{Q}$ this recovers a fact already known: ι is the zero map, so no nonzero element of N survives into any $\mathbb{Q}$-vector space, which is another way of saying that a torsion element cannot live in a space where every nonzero integer acts invertibly.

The adjunction of corollary 15.3.5 is one instance of a general relationship between $\otimes$ and Hom. Stating it here rather than alongside its applications keeps every property of the tensor product in one place. It is used in section 17.5 to compare tensoring with the Hom functors, and in *Homological Algebra* [19] to produce derived functors.

Theorem 15.3.9: Hom-Tensor Adjunction

Let R and S be rings, let A be a right R-module, let B be an (R,S)-bimodule and let C be a right S-module. Give $\mathrm{Hom}_S(B,C)$ the right R-action $(g \cdot r)(b) := g(rb)$. Then there is an isomorphism of abelian groups

$$\mathrm{Hom}_S\big(A \otimes_R B,\, C\big) \cong \mathrm{Hom}_R\big(A,\, \mathrm{Hom}_S(B,C)\big)$$

sending φ to the map $a \mapsto \big(b \mapsto \varphi(a \otimes b)\big)$.

***Proof*:**
First, the objects are what the statement claims. Every right R-module is a $(\mathbb{Z},R)$-bimodule, so proposition 15.2.1, applied with $\mathbb{Z}$ in the role of the left-hand ring and S in the role of the right-hand one, makes $A \otimes_R B$ a right S-module with $(a \otimes b)s = a \otimes (bs)$. The stated action on $\mathrm{Hom}_S(B,C)$ is a right action, since $\big((g \cdot r) \cdot r'\big)(b) = (g \cdot r)(r'b) = g(rr'b) = \big(g \cdot (rr')\big)(b)$.

The forward map. Given $\varphi \in \mathrm{Hom}_S(A \otimes_R B, C)$, define $\Lambda(\varphi)(a)(b) = \varphi(a \otimes b)$. For fixed a the map $\Lambda(\varphi)(a)$ is S-linear: it is additive because $\otimes$ is additive in its second argument, and

$$\Lambda(\varphi)(a)(bs) = \varphi\big(a \otimes bs\big) = \varphi\big((a \otimes b)s\big) = \varphi(a \otimes b)\, s = \Lambda(\varphi)(a)(b)\, s$$

using S-linearity of φ. The map $\Lambda(\varphi)$ is R-linear: it is additive because $\otimes$ is additive in its first argument, and

$$\Lambda(\varphi)(ar)(b) = \varphi(ar \otimes b) = \varphi(a \otimes rb) = \Lambda(\varphi)(a)(rb) = \big(\Lambda(\varphi)(a) \cdot r\big)(b)$$

the second equality being the third relation of (15.1). Finally Λ is additive in φ, directly from the definition.

The backward map. Given $\Psi \in \mathrm{Hom}_R(A, \mathrm{Hom}_S(B,C))$, the map $A \times B \to C$ sending (a,b) to $\Psi(a)(b)$ is R-balanced: it is additive in a because Ψ is additive, additive in b because each $\Psi(a)$ is, and

$$\Psi(ar)(b) = \big(\Psi(a) \cdot r\big)(b) = \Psi(a)(rb)$$

which is the third axiom. By theorem 15.1.5 there is a unique group homomorphism $\Gamma(\Psi)\colon A \otimes_R B \to C$ with $\Gamma(\Psi)(a \otimes b) = \Psi(a)(b)$, and it is S-linear because $x \mapsto \Gamma(\Psi)(xs)$ and $x \mapsto \Gamma(\Psi)(x)s$

are group homomorphisms agreeing on simple tensors, where both give $\Psi(a)(bs)$. Again Γ is additive.

The two are mutually inverse. For $\Psi \in \mathrm{Hom}_R(A, \mathrm{Hom}_S(B,C))$ and all $a \in A$, $b \in B$,

$$\Lambda\big(\Gamma(\Psi)\big)(a)(b) = \Gamma(\Psi)(a \otimes b) = \Psi(a)(b)$$

so $\Lambda(\Gamma(\Psi))(a) = \Psi(a)$ for every a, giving $\Lambda \circ \Gamma = \mathrm{id}$. In the other direction, $\Gamma(\Lambda(\varphi))$ and φ are group homomorphisms out of $A \otimes_R B$ agreeing on every simple tensor, since $\Gamma(\Lambda(\varphi))(a \otimes b) = \Lambda(\varphi)(a)(b) = \varphi(a \otimes b)$, so they are equal by lemma 15.1.4(3). Hence Λ is an isomorphism of abelian groups, completing the proof.

Taking $B = S$, viewed as an (R,S)-bimodule along $f\colon R \to S$, and using $\mathrm{Hom}_S(S,C) \cong C_R$ recovers corollary 15.3.5 in its right-handed form: extension of scalars is the case of the Hom-tensor adjunction in which the bimodule is the ring itself.

Exercise 15.3.1

1. Let $f\colon R \to S$ be a ring homomorphism and let $I \subseteq R$ be an ideal. Show that $S \otimes_R R/I \cong S/f(I)S$ as S-modules, where $f(I)S$ is the ideal of S generated by $f(I)$.
2. Let A be a finitely generated abelian group, so that $A \cong \mathbb{Z}^r \oplus T$ with T finite by chapter 14. Show that $\mathbb{Q} \otimes_{\mathbb{Z}} A \cong \mathbb{Q}^r$ as $\mathbb{Q}$-vector spaces, so that extension of scalars along $\mathbb{Z} \hookrightarrow \mathbb{Q}$ destroys exactly the torsion and $r = \dim_{\mathbb{Q}}(\mathbb{Q} \otimes_{\mathbb{Z}} A)$.
3. Let $R \subseteq S$ be an inclusion of rings and suppose S is free as a right R-module with a basis containing 1. Show that $\iota\colon N \to S \otimes_R N$ is injective for every left R-module N. Where does the argument fail for $\mathbb{Z} \subseteq \mathbb{Q}$?
4. Viewing S as an (R,S)-bimodule along $f\colon R \to S$, show that $\mathrm{Hom}_R(S,L)$ is a left S-module for every left R-module L, and that
$$\mathrm{Hom}_R\big(M_R,\, L\big) \cong \mathrm{Hom}_S\big(M,\, \mathrm{Hom}_R(S,L)\big)$$
for every left S-module M. (So restriction of scalars has a right adjoint as well as a left one. This is *coextension of scalars*.)
5. Let V be a real vector space. Show that $V_{\mathbb{C}} \cong V \oplus V$ as real vector spaces, and describe the complex structure that this decomposition carries in terms of the two summands.
6. Let $f\colon R \to S$ and $g\colon S \to T$ be ring homomorphisms. Show that extension of scalars along $g \circ f$ agrees with extension along f followed by extension along g, that is $T \otimes_S (S \otimes_R N) \cong T \otimes_R N$ naturally in N.
7. Show that theorem 15.3.9 is natural in C: for an S-module homomorphism $h\colon C \to C'$, the two ways of passing from $\mathrm{Hom}_S(A \otimes_R B, C)$ to $\mathrm{Hom}_R(A, \mathrm{Hom}_S(B,C'))$ agree.

15.4 Application: The Kronecker Product

Chapter 13 and chapter 14 computed with matrices: a linear map became an array once bases were chosen, and questions about the map became questions about the array. Nothing in this chapter has

yet been written down that way. The apparatus for doing so is already in place: corollary 15.2.6 says that bases of V and W determine a basis of $V \otimes_R W$, and theorem 15.2.2 says that maps of the factors determine a map of the tensor product. Putting the two together produces a matrix, and it has a name.

Definition 15.4.1: Kronecker Product

Let R be a commutative ring, let $A = (a_{ij}) \in M_{m\times n}(R)$ and let $B \in M_{p\times q}(R)$. The *Kronecker product* $A \otimes B \in M_{mp\times nq}(R)$ is the block matrix

$$A \otimes B := \begin{pmatrix} a_{11}B & a_{12}B & \cdots & a_{1n}B \\ a_{21}B & a_{22}B & \cdots & a_{2n}B \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1}B & a_{m2}B & \cdots & a_{mn}B \end{pmatrix}$$

Equivalently, indexing rows by pairs (i,k) with $1 \le i \le m$, $1 \le k \le p$ and columns by pairs (j,l) with $1 \le j \le n$, $1 \le l \le q$, both ordered lexicographically,

$$(A \otimes B)_{(i,k),(j,l)} = a_{ij}b_{kl}$$

The lexicographic indexing is part of the definition rather than a convenience: a different ordering of the pairs produces a matrix conjugate to this one by a permutation, which is the content of proposition 15.4.3(4) below. Every statement in this section is made with respect to that ordering.

Proposition 15.4.2: The Kronecker Product Represents the Tensor Product of Maps

Let R be a commutative ring and let V, V', W, W' be free R-modules with ordered bases $(e_1,\dots,e_n)$, $(e'_1,\dots,e'_m)$, $(f_1,\dots,f_q)$ and $(f'_1,\dots,f'_p)$ respectively. Let $\varphi\colon V \to V'$ be represented by A and $\psi\colon W \to W'$ by B. Give $V \otimes_R W$ the basis $(e_j \otimes f_l)$ and $V' \otimes_R W'$ the basis $(e'_i \otimes f'_k)$ of corollary 15.2.6, each ordered lexicographically. Then $\varphi \otimes \psi$ is represented by $A \otimes B$.

***Proof*:**
By theorem 15.2.2 and the bilinearity of $\otimes$,

$$(\varphi\otimes\psi)(e_j \otimes f_l) = \varphi(e_j)\otimes\psi(f_l) = \left(\sum_{i=1}^{m} a_{ij}e'_i\right) \otimes \left(\sum_{k=1}^{p} b_{kl}f'_k\right) = \sum_{i=1}^{m}\sum_{k=1}^{p} a_{ij}b_{kl}\,(e'_i \otimes f'_k)$$

By definition 13.2.1 the entry of the representing matrix in row (i,k) and column (j,l) is the coefficient of $e'_i \otimes f'_k$ in the image of $e_j \otimes f_l$, which the display gives as $a_{ij}b_{kl}$. That is the entry prescribed by definition 15.4.1, completing the proof.

Every property of the Kronecker product below is therefore a property of $\otimes$ read in coordinates, and the two viewpoints are used interchangeably from here on.

Proposition 15.4.3: Arithmetic of the Kronecker Product

Let R be a commutative ring.

1. (mixed product rule) $(A\otimes B)(C\otimes D)=(AC)\otimes(BD)$ whenever AC and BD are defined.
2. $I_m\otimes I_p=I_{mp}$.
3. If $A\in M_m(R)$ and $B\in M_p(R)$ are invertible then so is $A\otimes B$, with $(A\otimes B)^{-1}=A^{-1}\otimes B^{-1}$.
4. There is a permutation matrix $P\in M_{mp}(R)$, depending only on m and p, with
$$P(A\otimes B)P^{-1}=B\otimes A$$
for all $A\in M_m(R)$ and $B\in M_p(R)$.

***Proof*:**

1. Both sides are indexed by pairs, and summing over the intermediate index pair (r,s),
$$\begin{aligned}\big((A\otimes B)(C\otimes D)\big)_{(i,k),(j,l)}&=\sum_r\sum_s a_{ir}b_{ks}\,c_{rj}d_{sl}\\&=\left(\sum_r a_{ir}c_{rj}\right)\left(\sum_s b_{ks}d_{sl}\right)=(AC)_{ij}(BD)_{kl}\end{aligned}$$
the middle step using commutativity of R to separate the two sums. The right-hand side is the $(i,k),(j,l)$ entry of $(AC)\otimes(BD)$.
2. $(I_m\otimes I_p)_{(i,k),(j,l)}=\delta_{ij}\delta_{kl}$, which is 1 if $(i,k)=(j,l)$ and 0 otherwise.
3. By (1) and (2), $(A\otimes B)(A^{-1}\otimes B^{-1})=(AA^{-1})\otimes(BB^{-1})=I_m\otimes I_p=I_{mp}$, and the same on the other side.
4. Both $\{(i,k)\}$ and $\{(k,i)\}$ index sets of size mp, so the slot-exchanging bijection $\pi(i,k)=(k,i)$ between them has a permutation matrix P, determined by $Pe_{(i,k)}=e_{(k,i)}$ on standard basis vectors. For any M this gives $\big(PMP^{-1}\big)e_{\pi(y)}=PMe_y=\sum_x M_{xy}e_{\pi(x)}$, that is $\big(PMP^{-1}\big)_{\pi(x),\pi(y)}=M_{x,y}$. Applying this with $M=A\otimes B$,
$$\big(P(A\otimes B)P^{-1}\big)_{(k,i),(l,j)}=(A\otimes B)_{(i,k),(j,l)}=a_{ij}b_{kl}=b_{kl}a_{ij}=(B\otimes A)_{(k,i),(l,j)}$$
again using commutativity, completing the proof.

Part (4) is proposition 15.2.7 in coordinates: the isomorphism $V\otimes W\cong W\otimes V$ becomes, once bases are fixed on both sides, conjugation by the permutation that exchanges the two index slots. The matrix P is the reason the ordering convention in definition 15.4.1 is harmless.

The trace and the determinant of a Kronecker product are determined by those of the factors. Both formulas hold over an arbitrary commutative ring.

Theorem 15.4.4: Trace and Determinant of a Kronecker Product

Let R be a commutative ring, $A \in M_m(R)$ and $B \in M_p(R)$. Then

$$\operatorname{tr}(A \otimes B) = \operatorname{tr}(A)\operatorname{tr}(B) \qquad \text{and} \qquad \det(A \otimes B) = \big(\det A\big)^p \big(\det B\big)^m$$

***Proof*:**

Trace. The diagonal entries of $A \otimes B$ are those indexed by $(i,k) = (j,l)$, that is by $i = j$ and $k = l$, and there the entry is $a_{ii}b_{kk}$. Hence

$$\operatorname{tr}(A \otimes B) = \sum_{i=1}^{m}\sum_{k=1}^{p} a_{ii}b_{kk} = \left(\sum_{i=1}^{m} a_{ii}\right)\left(\sum_{k=1}^{p} b_{kk}\right) = \operatorname{tr}(A)\operatorname{tr}(B)$$

Determinant. By the mixed product rule and proposition 15.4.3(2),

$$A \otimes B = (AI_m) \otimes (I_pB) = (A \otimes I_p)(I_m \otimes B)$$

so $\det(A \otimes B) = \det(A \otimes I_p)\det(I_m \otimes B)$ by proposition 13.4.7. The second factor is immediate: $I_m \otimes B$ is block diagonal with m diagonal blocks, each equal to B, so $\det(I_m \otimes B) = (\det B)^m$ by proposition 13.4.8.

For the first factor, proposition 15.4.3(4) gives a permutation matrix P with $P(A \otimes I_p)P^{-1} = I_p \otimes A$. Determinants are unchanged by conjugation, since proposition 13.4.7 gives $\det(P)\det(P^{-1}) = \det(PP^{-1}) = \det(I) = 1$ and hence $\det(PMP^{-1}) = \det(M)$. Now $I_p \otimes A$ is block diagonal with p diagonal blocks each equal to A, so

$$\det(A \otimes I_p) = \det(I_p \otimes A) = (\det A)^p$$

by proposition 13.4.8 again. Multiplying the two factors gives the claim, as we sought to show.

Note the asymmetry of the exponents: each determinant is raised to the size of the *other* factor. A quick check is $A = B = I$, where both sides are 1, and a sharper one is $m = p = 2$ with $\det A = \det B = 2$, giving $\det(A \otimes B) = 16$ rather than 4.

Rank behaves as simply, though here a field is needed: the argument passes through a complement, and over a general commutative ring a submodule need not be a direct summand.

Proposition 15.4.5: Rank of a Kronecker Product

Let F be a field, $A \in M_{m\times n}(F)$ and $B \in M_{p\times q}(F)$. Then

$$\operatorname{rank}(A \otimes B) = \operatorname{rank}(A) \cdot \operatorname{rank}(B)$$

***Proof*:**

Let $\varphi\colon V \to V'$ and $\psi\colon W \to W'$ be the linear maps represented by A and B, so that $A \otimes B$ represents $\varphi \otimes \psi$ by proposition 15.4.2, and rank is the dimension of the image (definition 13.1.19).

Choose complements $V' = \operatorname{im}\varphi \oplus U$ and $W' = \operatorname{im}\psi \oplus U'$, which exist because F is a field.

Applying theorem 15.2.3 in each variable,

$$V' \otimes_F W' \cong \big(\mathrm{im}\varphi \otimes \mathrm{im}\psi\big) \oplus \big(\mathrm{im}\varphi \otimes U'\big) \oplus \big(U \otimes \mathrm{im}\psi\big) \oplus \big(U \otimes U'\big)$$

and under this decomposition the natural map $\mathrm{im}\varphi \otimes_F \mathrm{im}\psi \to V' \otimes_F W'$ is the inclusion of the first summand, hence injective.

The image of $\varphi \otimes \psi$ is exactly that summand. It is contained in it: every element of $V \otimes_F W$ is a finite sum of simple tensors by lemma 15.1.4(2), and $(\varphi \otimes \psi)(v \otimes w) = \varphi(v) \otimes \psi(w)$ lies in $\mathrm{im}\varphi \otimes \mathrm{im}\psi$. It contains it, because those same elements $\varphi(v) \otimes \psi(w)$ span $\mathrm{im}\varphi \otimes \mathrm{im}\psi$, again by lemma 15.1.4(2) applied there. Therefore

$$\mathrm{rank}(\varphi \otimes \psi) = \dim_F \big(\mathrm{im}\varphi \otimes_F \mathrm{im}\psi\big) = \dim_F(\mathrm{im}\varphi) \cdot \dim_F(\mathrm{im}\psi) = \mathrm{rank}(\varphi)\mathrm{rank}(\psi)$$

the middle equality being corollary 15.2.6, as we sought to show.

Eigenvalues multiply as well. One half of the statement needs nothing at all.

Proposition 15.4.6: Eigenvectors of a Kronecker Product

Let F be a field, $A \in M_m(F)$, $B \in M_p(F)$, and suppose $Av = \lambda v$ and $Bw = \mu w$ with $v \neq 0$ and $w \neq 0$. Then $v \otimes w \neq 0$ and

$$(A \otimes B)(v \otimes w) = \lambda\mu\,(v \otimes w)$$

so $\lambda\mu$ is an eigenvalue of $A \otimes B$.

***Proof*:**
Reading $A \otimes B$ as $\varphi \otimes \psi$ by proposition 15.4.2 and applying theorem 15.2.2,

$$(\varphi \otimes \psi)(v \otimes w) = \varphi(v) \otimes \psi(w) = (\lambda v) \otimes (\mu w) = \lambda\mu\,(v \otimes w)$$

using bilinearity to extract both scalars. For $v \otimes w \neq 0$, extend v to a basis of F^m and w to a basis of F^p. Then $v \otimes w$ is one of the basis elements of $F^m \otimes_F F^p$ furnished by corollary 15.2.6, and a basis element is nonzero, completing the proof.

That accounts for the products of eigenvalues that come from actual eigenvectors. The full spectrum, counted with multiplicity, needs a triangular form, and theorem 14.4.5 gives one exactly when the minimal polynomials split.

Theorem 15.4.7: Spectrum of a Kronecker Product

Let F be a field and let $A \in M_m(F)$ and $B \in M_p(F)$ have minimal polynomials that split into linear factors over F, which holds for all A and B when F is algebraically closed. List the eigenvalues of A as $\lambda_1, \dots, \lambda_m$ and those of B as $\mu_1, \dots, \mu_p$, each repeated according to its algebraic multiplicity. Then the characteristic polynomial of $A \otimes B$ is

$$\prod_{i=1}^{m} \prod_{k=1}^{p} (x - \lambda_i \mu_k)$$

so the eigenvalues of $A \otimes B$, with algebraic multiplicity, are the mp products $\lambda_i \mu_k$.

***Proof*:**

Two general facts are used repeatedly, so both are recorded first.

The determinant of an upper triangular matrix is the product of its diagonal entries. In the defining sum a term $\prod_i t_{i\sigma(i)}$ is nonzero only if $i \leq \sigma(i)$ for every i, and since $\sum_i \big(\sigma(i) - i\big) = 0$ this forces $\sigma(i) = i$ for all i. Only the identity term survives. Consequently an upper triangular T has characteristic polynomial $\prod_i (x - t_{ii})$, since $xI - T$ is again upper triangular.

Similar matrices have the same characteristic polynomial, because $\det\big(xI - PMP^{-1}\big) = \det\big(P(xI - M)P^{-1}\big) = \det(xI - M)$ by proposition 13.4.7, applied over the commutative ring $F[x]$.

Now by theorem 14.4.5 there are invertible P and Q with $A = PJ_AP^{-1}$ and $B = QJ_BQ^{-1}$, where J_A and J_B are in Jordan canonical form and hence upper triangular. Combining the two facts, the characteristic polynomial of A is $\prod_i (x - (J_A)_{ii})$, so the diagonal entries of J_A are exactly the eigenvalues $\lambda_1, \dots, \lambda_m$ of A listed with algebraic multiplicity, and likewise for J_B and the μ_k.

By the mixed product rule and proposition 15.4.3(3),

$$A \otimes B = \big(PJ_AP^{-1}\big) \otimes \big(QJ_BQ^{-1}\big) = (P \otimes Q)\,(J_A \otimes J_B)\,(P \otimes Q)^{-1}$$

so $A \otimes B$ is similar to $J_A \otimes J_B$ and the two have the same characteristic polynomial.

The matrix $J_A \otimes J_B$ is upper triangular for the lexicographic ordering. Its $(i,k),(j,l)$ entry is $(J_A)_{ij}(J_B)_{kl}$, which vanishes unless $i \leq j$ and $k \leq l$. And if (i,k) comes strictly after (j,l) lexicographically then either $i > j$, or $i = j$ and $k > l$, and in both cases one of the two factors is 0. Its diagonal entries are $(J_A)_{ii}(J_B)_{kk} = \lambda_i \mu_k$.

So $J_A \otimes J_B$ is upper triangular with diagonal entries $\lambda_i \mu_k$, and by the first recorded fact its characteristic polynomial is $\prod_{i,k}(x - \lambda_i \mu_k)$, as we sought to show.

Example 15.4.8: Computing with the Kronecker Product

Take $R = \mathbb{Q}$ and

$$A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}, \qquad B = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$$

so that $\det A = -2$, $\mathrm{tr}A = 5$, $\det B = 1$ and $\mathrm{tr}B = 0$. Assembling the blocks $a_{ij}B$,

$$A \otimes B = \begin{pmatrix} 0 & 1 & 0 & 2 \\ -1 & 0 & -2 & 0 \\ 0 & 3 & 0 & 4 \\ -3 & 0 & -4 & 0 \end{pmatrix}$$

Its diagonal entries are all 0, so its trace is $0 = 5 \cdot 0 = \mathrm{tr}(A)\mathrm{tr}(B)$, as theorem 15.4.4 requires. That theorem also predicts

$$\det(A \otimes B) = (\det A)^2(\det B)^2 = (-2)^2 \cdot 1^2 = 4$$

Both A and B are invertible, so both have rank 2, and proposition 15.4.5 predicts $\mathrm{rank}(A \otimes B) = 4$, consistent with the nonzero determinant.

Replacing A by the rank one matrix

$$A' = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$$

gives $\mathrm{rank}(A' \otimes B) = 1 \cdot 2 = 2$, so $A' \otimes B$ is a 4×4 matrix of rank 2. Correspondingly $\det(A' \otimes B) = (\det A')^2(\det B)^2 = 0$. Note that the rank drops to 2 rather than to 3: a Kronecker product cannot have rank 3 at all, since by proposition 15.4.5 its rank is a product of two integers each at most 2.

One identification remains, and it explains a phenomenon recorded without explanation in section 15.1. For finite-dimensional spaces the tensor product of a dual space with another space is the space of linear maps between them.

Theorem 15.4.9: $V^* \otimes W$ and Linear Maps

Let F be a field and let V, W be finite-dimensional F-vector spaces. Then

$$\Theta\colon V^* \otimes_F W \to \mathrm{Hom}_F(V, W), \qquad \Theta(f \otimes w)(v) = f(v)\, w$$

is an isomorphism of F-vector spaces. Under Θ:

1. the simple tensors correspond exactly to the homomorphisms of rank at most one;
2. the least number of simple tensors whose sum is a given $t \in V^* \otimes_F W$ equals $\mathrm{rank}\Theta(t)$.

***Proof*:**

The map $(f, w) \mapsto \big(v \mapsto f(v)w\big)$ is F-bilinear, so theorem 15.1.12 gives the F-linear map Θ.

Let $(e_1, \ldots, e_n)$ be a basis of V with dual basis $(e_1^*, \ldots, e_n^*)$ (section 13.3) and $(w_1, \ldots, w_p)$ a basis of W. By corollary 15.2.6 the np elements $e_i^* \otimes w_k$ form a basis of $V^* \otimes_F W$. Their images are the maps $e_j \mapsto \delta_{ij}w_k$, whose matrices with respect to the two bases are the np elementary matrices having a single 1 in position (k, i). Those form a basis of $\mathrm{Hom}_F(V, W)$. An F-linear map carrying a basis onto a basis is an isomorphism, so Θ is one.

(1). The image of $\Theta(f \otimes w)$ is contained in Fw, so its rank is at most one. Conversely, let θ have rank at most one. If $\theta = 0$ it is $\Theta(0 \otimes 0)$. Otherwise $\mathrm{im}\theta = Fw$ for some $w \neq 0$, and writing $\theta(v) = f(v)w$ defines a map $f\colon V \to F$ which is linear because θ is and w is a basis of its image.

Then $\theta = \Theta(f \otimes w)$.

(2). If $t = \sum_{j=1}^{r} f_j \otimes w_j$ then $\Theta(t) = \sum_j \Theta(f_j \otimes w_j)$ is a sum of r maps of rank at most one. The image of a sum is contained in the sum of the images, and $\dim(X+Y) \le \dim X + \dim Y$, so $\operatorname{rank}\Theta(t) \le r$. Conversely let $\operatorname{rank}\Theta(t) = r$ and choose a basis $u_1, \dots, u_r$ of $\operatorname{im}\Theta(t)$. Every v has $\Theta(t)(v) = \sum_{j=1}^{r} g_j(v) u_j$ for unique scalars $g_j(v)$, and each $g_j \colon V \to F$ so defined is linear because $\Theta(t)$ is and the u_j are independent. Then $\Theta\big(\sum_j g_j \otimes u_j\big) = \Theta(t)$, and Θ is injective, so $t = \sum_{j=1}^{r} g_j \otimes u_j$ is a sum of r simple tensors, completing the proof.

Part (2) settles the question left open in box 15.1.6. Taking $W = V$ of dimension n, the element $\sum_{i=1}^{n} e_i^* \otimes e_i$ corresponds under Θ to the identity map, which has rank n. So it needs exactly n simple tensors and is not itself simple once $n \ge 2$. The witness exhibited there, $e_1 \otimes e_1 + e_2 \otimes e_2$, is this element for $n = 2$ under the identification of V with V^* by the dual basis, and the ad hoc argument given there is replaced by a count of rank.

Exercise 15.4.1

1. Show that $A \otimes B$ and $B \otimes A$ are always similar, and give explicit 2×2 matrices A and B for which they are not equal.
2. Show that $(A \otimes B)^k = A^k \otimes B^k$ for every $k \ge 1$, and deduce that if $A^r = 0$ then $(A \otimes B)^r = 0$ for every B. Conclude that a Kronecker product with a nilpotent factor is nilpotent, with index at most that of the factor.
3. Deduce from the mixed product rule that $A \mapsto A \otimes I_p$ is an injective ring homomorphism $M_m(R) \to M_{mp}(R)$, and that its image commutes elementwise with the image of $B \mapsto I_m \otimes B$.
4. Identify the step in the proof of proposition 15.4.5 that requires a field. Then show it genuinely fails over $\mathbb{Z}$: with $\varphi \colon \mathbb{Z} \to \mathbb{Z}$ multiplication by 2 and ψ the identity on $\mathbb{Z}/2\mathbb{Z}$, show that the natural map $\operatorname{im}\varphi \otimes_{\mathbb{Z}} \mathbb{Z}/2\mathbb{Z} \to \mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z}/2\mathbb{Z}$ is not injective.
5. Let V and W be arbitrary vector spaces over F, not assumed finite dimensional. Show that Θ of theorem 15.4.9 is still injective and that its image is exactly the set of homomorphisms of finite rank. Deduce that Θ is an isomorphism if and only if V or W is finite dimensional.
6. Show that under the isomorphism $V^* \otimes_F V \cong \operatorname{End}_F(V)$ of theorem 15.4.9, the trace of definition 13.4.3 corresponds to the evaluation map determined by $f \otimes v \mapsto f(v)$. (This is the basis-free description of the trace.)

15.5 Summary

Everything in this chapter rests on one universal property: bilinear maps out of $M \times N$ are homomorphisms out of a single module $M \otimes_R N$. The tables below state what follows from it, each entry pointing at the result that establishes it.

Representing objects and adjunctions

Each class of map has a representing object, and in every case it is a tensor product. The representing object is determined up to unique isomorphism by its property (corollary 15.1.7), so the particular

construction used in definition 15.1.3 is not part of the data.

Maps	On	Represented by	Reference
R-balanced, into an abelian group	$M \times N$	$M \otimes_R N$	theorem 15.1.5
R-bilinear, into an R-module	$M \times N$	$M \otimes_R N$	theorem 15.1.12
R-multilinear	$M_1 \times \cdots \times M_n$	$M_1 \otimes_R \cdots \otimes_R M_n$	theorem 15.2.11
R-linear, into an S-module	N	$S \otimes_R N$	theorem 15.3.4
Pairs of R-algebra maps	A, B	$A \otimes_R B$	theorem 15.2.17

The second and third rows require R commutative and the standard structures of example 15.1.9(4). The fifth requires A and B commutative. Two facts are used constantly with these: simple tensors generate, so a homomorphism out of a tensor product is determined by its values on them (lemma 15.1.4), and not every element is itself simple (box 15.1.6).

Two of these are adjunctions.

Left adjoint	Right adjoint	Reference
$S \otimes_R -$	restriction of scalars along $R \to S$	corollary 15.3.5
$- \otimes_R B$	$\mathrm{Hom}_S(B, -)$	theorem 15.3.9

The first is the case $B = S$ of the second.

Module structure

The tensor product of a right module and a left module is only an abelian group. Any further structure comes from a second action on one of the factors.

Hypothesis	Structure on $M \otimes_R N$	Reference
M right R-, N left R-module	abelian group only	definition 15.1.3
M an (S, R)-bimodule	left S-module	proposition 15.1.10
M an (S, R)-, N an (R, T)-bimodule	(S, T)-bimodule	proposition 15.2.1
R commutative, standard structures	R-module	theorem 15.1.12
A, B R-algebras	R-algebra	theorem 15.2.14

What is preserved, and what is not

Statement	Reference
$R \otimes_R N \cong N$	example 15.1.13
$(\bigoplus_i M_i) \otimes_R N \cong \bigoplus_i (M_i \otimes_R N)$	theorem 15.2.4
Free $\otimes$ free is free, on the basis $\{m_i \otimes n_j\}$	corollary 15.2.6
$M \otimes_R N \cong N \otimes_R M$	proposition 15.2.7
$(M \otimes_R N) \otimes_T L \cong M \otimes_R (N \otimes_T L)$	theorem 15.2.8
$(\lambda \otimes \mu) \circ (\varphi \otimes \psi) = (\lambda \circ \varphi) \otimes (\mu \circ \psi)$	theorem 15.2.2
$\mathrm{rank}(A \otimes B) = \mathrm{rank}(A)\mathrm{rank}(B)$ over a field	proposition 15.4.5

The first, fourth and fifth rows are the unit, commutativity and associativity constraints of a symmetric monoidal structure on R-Mod (box 15.2.9).

Fails	**Witness**	**Reference**
Being nonzero	$\mathbb{Z}/2\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z}/3\mathbb{Z} = 0$	example 15.1.14
Injectivity of a map	multiplication by 2 against $\mathbb{Z}/2\mathbb{Z}$	section 17.5
Containing the original module	ι has a kernel	corollary 15.3.8
Being an integral domain	$\mathbb{C} \otimes_{\mathbb{R}} \mathbb{C} \cong \mathbb{C} \times \mathbb{C}$	example 15.2.15

The second row is the one with consequences beyond this chapter: which modules D make $D \otimes_R -$ preserve injections is the definition of flatness, taken up in section 17.5.

In coordinates

Once bases are fixed, corollary 15.2.6 turns each of the constructions above into a matrix identity.

Object	**Coordinate form**	**Reference**
$\varphi \otimes \psi$	the Kronecker product $(a_{ij}B)$	proposition 15.4.2
$\operatorname{tr}(A \otimes B)$	$\operatorname{tr}(A)\operatorname{tr}(B)$	theorem 15.4.4
$\det(A \otimes B)$	$(\det A)^p (\det B)^m$	theorem 15.4.4
Eigenvalues of $A \otimes B$	the products $\lambda_i \mu_k$	theorem 15.4.7
$\operatorname{Hom}_F(V, W)$	$V^* \otimes_F W$	theorem 15.4.9

Computations worth reusing

Identity	**Reference**
$\mathbb{Z}/m\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z}/n\mathbb{Z} \cong \mathbb{Z}/\gcd(m,n)\mathbb{Z}$	proposition 15.2.13
$R/I \otimes_R R/J \cong R/(I+J)$	proposition 15.2.13
$S \otimes_R R^n \cong S^n$	example 15.3.3
$V_{\mathbb{C}} = \mathbb{C} \otimes_{\mathbb{R}} V$, with $\dim_{\mathbb{C}} V_{\mathbb{C}} = \dim_{\mathbb{R}} V$	example 15.3.6
$A \otimes_{\mathbb{Z}} B$ is the coproduct in **CRing**	corollary 15.2.18

16

Multilinear Algebra

Chapter 15 represented the bilinear maps on a product of two modules. Fixing a single module M and allowing more arguments produces further classes of map on M^k: the k-multilinear maps, those left unchanged by permuting the arguments, and those that vanish whenever two arguments coincide. Each class has its own representing object, and the three objects are quotients of one another.

Summing over k turns each family of representing objects into a ring. The tensor algebra $\mathcal{T}(M)$, the symmetric algebra $\mathcal{S}(M)$ and the exterior algebra $\bigwedge(M)$ are graded by the number of arguments, and each carries a universal property of its own, now for homomorphisms into an algebra rather than into a module. The determinant returns at the end as the map an endomorphism induces on the top exterior power, recovering section 13.4 from a universal property rather than from the defining sum.

16.1 Multilinear, Symmetric and Alternating Maps

The k-multilinear maps on M already have a representing object. Definition 15.2.10 defined the class and theorem 15.2.11 produced the module through which every such map factors: an R-linear map out of the k-fold tensor product of M with itself is the same thing as a k-multilinear map on M^k.

Two subclasses of these maps occur constantly, and both are cut out by requiring the map to be insensitive to something. A symmetric map does not see the order of its arguments. An alternating map vanishes on a repetition among them. Each subclass is again represented by a module, and in each case that module is a quotient of the k-fold tensor product by exactly the elements the subclass is required to kill. That is the pattern of this section, applied three times.

Definition 16.1.1: Tensor Powers

Let R be a commutative ring and M an R-module. For $k \geq 1$ the *k-th tensor power* of M is

$$\mathcal{T}^k(M) := \underbrace{M \otimes_R M \otimes_R \cdots \otimes_R M}_{k \text{ factors}}$$

bracketed to the left as in section 15.2, with $\mathcal{T}^0(M) := R$ and $\mathcal{T}^1(M) = M$. Its elements are called *k-tensors*.

In this notation theorem 15.2.11 reads: the map $M^k \to \mathcal{T}^k(M)$ sending $(m_1, \ldots, m_k)$ to $m_1 \otimes \cdots \otimes m_k$ is k-multilinear, and every k-multilinear map on M^k factors through it uniquely by an R-module homomorphism. Nothing further is needed about $\mathcal{T}^k(M)$ here. The two quotients are what this section constructs.

Definition 16.1.2: Symmetric Multilinear Maps

Let R be a commutative ring and let M, N be R-modules. A k-multilinear map $\varphi\colon M^k \to N$ is *symmetric* if

$$\varphi\big(m_{\sigma(1)}, \ldots, m_{\sigma(k)}\big) = \varphi(m_1, \ldots, m_k)$$

for every $\sigma \in S_k$ and all $m_1, \ldots, m_k \in M$.

Definition 16.1.3: Alternating Multilinear Maps

With R, M, N as above, a k-multilinear map $\varphi\colon M^k \to N$ is *alternating* if

$$\varphi(m_1, \ldots, m_k) = 0 \qquad \text{whenever } m_i = m_j \text{ for some } i \neq j$$

The determinant of definition 13.4.5, read as a function of the n columns of a matrix, is the standard example of an alternating n-multilinear map $(R^n)^n \to R$: it is multilinear in the columns and vanishes when two of them agree, which is proposition 13.4.6(3) and (4). A symmetric bilinear form, such as an inner product (section 13.6), is the standard example of a symmetric map.

Alternating maps are often introduced by a sign rule rather than by the vanishing condition. The two agree except in characteristic 2, and the direction that always holds is worth isolating, since it is what makes the exterior power computable.

Lemma 16.1.4: Equivalence Of Alternating Map

Let R be a commutative ring and let $\varphi\colon M^k \to N$ be k-multilinear.

1. If φ is alternating then for every $\sigma \in S_k$,

$$\varphi\big(m_{\sigma(1)}, \ldots, m_{\sigma(k)}\big) = (-1)^{\sigma}\, \varphi(m_1, \ldots, m_k)$$

2. If 2 is a unit in R, the converse holds: a k-multilinear map satisfying the displayed sign rule is alternating.

***Proof*:**
(1). For $\sigma \in S_k$ write $\varphi^\sigma(m_1, \dots, m_k) := \varphi(m_{\sigma(1)}, \dots, m_{\sigma(k)})$. Substituting shows $(\varphi^\sigma)^\tau = \varphi^{\tau\sigma}$, so if the identity $\varphi^\sigma = (-1)^\sigma \varphi$ holds for two permutations it holds for their product:

$$\varphi^{\tau\sigma} = (\varphi^\sigma)^\tau = \big((-1)^\sigma \varphi\big)^\tau = (-1)^\sigma \varphi^\tau = (-1)^\sigma (-1)^\tau \varphi = (-1)^{\tau\sigma} \varphi$$

Since the transpositions generate S_k, it therefore suffices to treat a transposition $\tau = (a\ b)$ with $a < b$.

Fix all arguments except those in positions a and b, and let $f(x, y)$ denote the value of φ with x in position a and y in position b. Then f is bilinear, and $f(x, x) = 0$ for every x because φ is alternating. Expanding,

$$0 = f(x + y,\, x + y) = f(x, x) + f(x, y) + f(y, x) + f(y, y) = f(x, y) + f(y, x)$$

so $f(y, x) = -f(x, y)$, which is exactly the claim for τ.

(2). Suppose the sign rule holds and $m_i = m_j$ with $i \neq j$. Applying the rule to the transposition $\tau = (i\ j)$, the left-hand side is φ evaluated at the same arguments, since interchanging two equal entries changes nothing, while the right-hand side is $-\varphi(m_1, \dots, m_k)$. Hence $2\,\varphi(m_1, \dots, m_k) = 0$, and multiplying by the inverse of 2 gives $\varphi(m_1, \dots, m_k) = 0$, completing the proof.

The hypothesis in (2) cannot be dropped. Over $R = \mathbb{Z}/2\mathbb{Z}$ every sign $(-1)^\sigma$ equals 1, so the sign rule says only that the map is symmetric. And the multiplication map $\varphi\colon \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z} \to \mathbb{Z}/2\mathbb{Z}$ with $\varphi(x, y) = xy$ is symmetric and bilinear but has $\varphi(\overline{1}, \overline{1}) = \overline{1} \neq \overline{0}$, so it is not alternating. In characteristic 2 the vanishing condition of definition 16.1.3 is strictly stronger, and it is the one the exterior power is built from.

Each subclass is now represented, by quotienting $\mathcal{T}^k(M)$ by the elements a map in the subclass is obliged to kill.

Definition 16.1.5: Symmetric Powers

Let $C^k(M) \subseteq \mathcal{T}^k(M)$ be the submodule generated by all elements

$$m_1 \otimes \cdots \otimes m_k - m_{\sigma(1)} \otimes \cdots \otimes m_{\sigma(k)} \qquad \sigma \in S_k,\ m_i \in M$$

The *k-th symmetric power* of M is $\mathcal{S}^k(M) := \mathcal{T}^k(M)/C^k(M)$, and the image of $m_1 \otimes \cdots \otimes m_k$ is written $m_1 m_2 \cdots m_k$. Since $C^0(M) = C^1(M) = 0$, one has $\mathcal{S}^0(M) = R$ and $\mathcal{S}^1(M) = M$.

Theorem 16.1.6: Universal Property of the Symmetric Power

Let R be a commutative ring and M an R-module. The map

$$\varsigma\colon M^k \to \mathcal{S}^k(M), \qquad \varsigma(m_1, \dots, m_k) = m_1 m_2 \cdots m_k$$

is symmetric and k-multilinear, and it is universal among such maps: for every R-module N and every symmetric k-multilinear $\varphi\colon M^k \to N$ there is a unique R-module homomorphism $\overline{\varphi}\colon \mathcal{S}^k(M) \to N$ with $\varphi = \overline{\varphi} \circ \varsigma$.

***Proof*:**
The map ς is the composite of the universal k-multilinear map of theorem 15.2.11 with the quotient map $\pi\colon \mathcal{T}^k(M) \to \mathcal{S}^k(M)$, hence k-multilinear. It is symmetric because the difference $m_1 \otimes \cdots \otimes m_k - m_{\sigma(1)} \otimes \cdots \otimes m_{\sigma(k)}$ lies in $C^k(M)$ by construction and so maps to 0.

Let φ be symmetric and k-multilinear. By theorem 15.2.11 there is a unique R-module homomorphism $\psi\colon \mathcal{T}^k(M) \to N$ with $\psi(m_1 \otimes \cdots \otimes m_k) = \varphi(m_1, \ldots, m_k)$. On a generator of $C^k(M)$,

$$\psi\big(m_1 \otimes \cdots \otimes m_k - m_{\sigma(1)} \otimes \cdots \otimes m_{\sigma(k)}\big) = \varphi(m_1, \ldots, m_k) - \varphi(m_{\sigma(1)}, \ldots, m_{\sigma(k)}) = 0$$

precisely because φ is symmetric. A homomorphism vanishing on a generating set vanishes on the submodule it generates, so $C^k(M) \subseteq \ker\psi$ and ψ factors through the quotient, giving $\overline{\varphi}$ with $\overline{\varphi} \circ \varsigma = \varphi$.

For uniqueness, the products $m_1 \cdots m_k$ generate $\mathcal{S}^k(M)$, being the images of the simple tensors, which generate $\mathcal{T}^k(M)$. Two homomorphisms agreeing on a generating set are equal, as we sought to show.

Definition 16.1.7: Exterior Powers

Let $A^k(M) \subseteq \mathcal{T}^k(M)$ be the submodule generated by all simple tensors

$$m_1 \otimes \cdots \otimes m_k \qquad \text{with } m_i = m_j \text{ for some } i \neq j$$

The *k-th exterior power* of M is $\bigwedge^k(M) := \mathcal{T}^k(M)/A^k(M)$, and the image of $m_1 \otimes \cdots \otimes m_k$ is written $m_1 \wedge \cdots \wedge m_k$, the operation being called the *wedge* or *exterior product*. Since $A^0(M) = A^1(M) = 0$, one has $\bigwedge^0(M) = R$ and $\bigwedge^1(M) = M$.

Theorem 16.1.8: Universal Property of the Exterior Power

Let R be a commutative ring and M an R-module. The map

$$\alpha\colon M^k \to \bigwedge^k(M), \qquad \alpha(m_1, \ldots, m_k) = m_1 \wedge \cdots \wedge m_k$$

is alternating and k-multilinear, and it is universal among such maps: for every R-module N and every alternating k-multilinear $\varphi\colon M^k \to N$ there is a unique R-module homomorphism $\overline{\varphi}\colon \bigwedge^k(M) \to N$ with $\varphi = \overline{\varphi} \circ \alpha$.

***Proof*:**
As before α is k-multilinear, being the composite of the universal k-multilinear map with the quotient map, and it is alternating because a simple tensor with a repeated entry lies in $A^k(M)$ by construction.

Given an alternating k-multilinear φ, let $\psi\colon \mathcal{T}^k(M) \to N$ be the homomorphism given by theorem 15.2.11. Each generator of $A^k(M)$ is a simple tensor $m_1 \otimes \cdots \otimes m_k$ with $m_i = m_j$ for some $i \neq j$, and ψ sends it to $\varphi(m_1, \ldots, m_k)$, which is 0 because φ is alternating. So $A^k(M) \subseteq \ker\psi$ and ψ factors through $\bigwedge^k(M)$. Uniqueness holds because the elements $m_1 \wedge \cdots \wedge m_k$ generate

$\bigwedge^k(M)$, completing the proof.

The sign rule now transfers from maps to elements.

Proposition 16.1.9: Anticommutativity of the Wedge Product

Let M be an R-module. In $\bigwedge^k(M)$,

$$m_1 \wedge \cdots \wedge m_k = 0 \quad \text{whenever } m_i = m_j \text{ for some } i \neq j$$
$$m_{\sigma(1)} \wedge \cdots \wedge m_{\sigma(k)} = (-1)^\sigma \, m_1 \wedge \cdots \wedge m_k$$

for every $\sigma \in S_k$. In particular $m \wedge m' = -\, m' \wedge m$ for all $m, m' \in M$.

***Proof*:**
The first identity is the statement that the generators of $A^k(M)$ die in the quotient. The second is lemma 16.1.4(1) applied to the alternating map α of theorem 16.1.8, whose values are precisely the elements $m_1 \wedge \cdots \wedge m_k$. Taking $k = 2$ and σ the transposition gives the last claim, completing the proof.

Both $\mathcal{S}^k(M)$ and $\bigwedge^k(M)$ have been built as quotients of $\mathcal{T}^k(M)$. They can also be found inside it, as submodules, provided $k!$ is invertible. The comparison rests on an action of the symmetric group.

Proposition 16.1.10: The Symmetric Group Acts on Tensor Powers

Let M be an R-module and $k \geq 1$. For each $\sigma \in S_k$ there is a unique R-module homomorphism $\sigma \cdot (-)$ of $\mathcal{T}^k(M)$ with

$$\sigma \cdot (m_1 \otimes \cdots \otimes m_k) = m_{\sigma^{-1}(1)} \otimes \cdots \otimes m_{\sigma^{-1}(k)}$$

and these assemble into a left action of S_k on $\mathcal{T}^k(M)$ by R-module automorphisms.

***Proof*:**
For fixed σ the assignment $(m_1, \ldots, m_k) \mapsto m_{\sigma^{-1}(1)} \otimes \cdots \otimes m_{\sigma^{-1}(k)}$ is k-multilinear, since each argument appears in exactly one slot of the output, so theorem 15.2.11 gives the homomorphism and its uniqueness.

For the action law, evaluate on a simple tensor. The i-th slot of $\tau \cdot (m_1 \otimes \cdots \otimes m_k)$ holds $m_{\tau^{-1}(i)}$, so the i-th slot of $\sigma \cdot \big(\tau \cdot (m_1 \otimes \cdots \otimes m_k)\big)$ holds $m_{\tau^{-1}(\sigma^{-1}(i))} = m_{(\sigma\tau)^{-1}(i)}$, which is the i-th slot of $(\sigma\tau) \cdot (m_1 \otimes \cdots \otimes m_k)$. Two homomorphisms agreeing on the simple tensors are equal, so $\sigma \cdot (\tau \cdot z) = (\sigma\tau) \cdot z$ for all z. The identity permutation acts as the identity, so each σ acts invertibly, completing the proof.

The inverse in the formula is what makes the assignment a left action rather than a right one. Without it the composition law would come out reversed.

Definition 16.1.11: Symmetric and Alternating Tensors

Let M be an R-module and $k \geq 1$. A tensor $z \in \mathcal{T}^k(M)$ is

1. a *symmetric k-tensor* if $\sigma \cdot z = z$ for every $\sigma \in S_k$;
2. an *alternating k-tensor* if $\sigma \cdot z = (-1)^\sigma z$ for every $\sigma \in S_k$.

These form submodules of $\mathcal{T}^k(M)$, written $\mathcal{T}^k_{\mathrm{sym}}(M)$ and $\mathcal{T}^k_{\mathrm{alt}}(M)$, each being an intersection of kernels of R-module homomorphisms.
The *symmetrization* and *antisymmetrization* operators on $\mathcal{T}^k(M)$ are

$$\mathrm{Sym}(z) := \sum_{\sigma \in S_k} \sigma \cdot z \qquad \text{and} \qquad \mathrm{Alt}(z) := \sum_{\sigma \in S_k} (-1)^\sigma \, \sigma \cdot z$$

Proposition 16.1.12: Symmetric and Alternating Tensors When $k!$ Is Invertible

Let M be an R-module, $k \geq 1$, and suppose $k!$ is a unit in R. Then $\frac{1}{k!}\mathrm{Sym}$ and $\frac{1}{k!}\mathrm{Alt}$ are idempotent endomorphisms of $\mathcal{T}^k(M)$ with images $\mathcal{T}^k_{\mathrm{sym}}(M)$ and $\mathcal{T}^k_{\mathrm{alt}}(M)$, and they induce isomorphisms of R-modules

$$\mathcal{S}^k(M) \xrightarrow{\sim} \mathcal{T}^k_{\mathrm{sym}}(M) \qquad \text{and} \qquad \bigwedge^k(M) \xrightarrow{\sim} \mathcal{T}^k_{\mathrm{alt}}(M)$$

In particular $\mathcal{T}^k(M) = \mathcal{T}^k_{\mathrm{sym}}(M) \oplus \ker \mathrm{Sym}$, and likewise for the alternating part.

***Proof*:**
The symmetric case. Reindexing the sum shows $\tau \cdot \mathrm{Sym}(z) = \mathrm{Sym}(z)$ for every τ, since $\sigma \mapsto \tau\sigma$ permutes S_k. So Sym lands in $\mathcal{T}^k_{\mathrm{sym}}(M)$. If z is already symmetric then every term of the sum equals z, so $\mathrm{Sym}(z) = k!\, z$ and $\frac{1}{k!}\mathrm{Sym}$ restricts to the identity there. Hence $\frac{1}{k!}\mathrm{Sym}$ is idempotent with image exactly $\mathcal{T}^k_{\mathrm{sym}}(M)$, and an idempotent endomorphism splits its domain as image plus kernel.

Writing $\rho = \sigma^{-1}$, the generator $m_1 \otimes \cdots \otimes m_k - m_{\sigma(1)} \otimes \cdots \otimes m_{\sigma(k)}$ of $C^k(M)$ is $z - \rho \cdot z$ for the simple tensor z, and $\mathrm{Sym}(z - \rho \cdot z) = \mathrm{Sym}(z) - \mathrm{Sym}(z) = 0$ by the reindexing above. So $\frac{1}{k!}\mathrm{Sym}$ kills $C^k(M)$ and induces $\Theta\colon \mathcal{S}^k(M) \to \mathcal{T}^k_{\mathrm{sym}}(M)$.

Its inverse is the composite $\mathcal{T}^k_{\mathrm{sym}}(M) \hookrightarrow \mathcal{T}^k(M) \xrightarrow{\pi} \mathcal{S}^k(M)$. Indeed, for a simple tensor z one has $\pi(\sigma \cdot z) = \pi(z)$, since $z - \sigma \cdot z \in C^k(M)$, so

$$\pi\left(\tfrac{1}{k!}\mathrm{Sym}(z)\right) = \tfrac{1}{k!} \sum_{\sigma \in S_k} \pi(\sigma \cdot z) = \pi(z)$$

and both sides are homomorphisms, so this holds for all z. That is one composite. In the other order, $\frac{1}{k!}\mathrm{Sym}$ is the identity on $\mathcal{T}^k_{\mathrm{sym}}(M)$, which is the second. So Θ is an isomorphism.

The alternating case. The same reindexing gives $\tau \cdot \mathrm{Alt}(z) = (-1)^\tau \mathrm{Alt}(z)$, so Alt lands in $\mathcal{T}^k_{\mathrm{alt}}(M)$. And if z is alternating then $(-1)^\sigma \sigma \cdot z = (-1)^\sigma (-1)^\sigma z = z$ for every σ, so $\mathrm{Alt}(z) = k!\, z$. Idempotence and the splitting follow as before.

That Alt kills $A^k(M)$ needs the pairing argument. Let $z = m_1 \otimes \cdots \otimes m_k$ have $m_a = m_b$ with

$a \neq b$ and set $\tau = (a\ b)$. Then $\tau \cdot z = z$, because the two interchanged entries are equal. Pairing each σ with $\sigma\tau$ gives $(\sigma\tau)\cdot z = \sigma\cdot(\tau\cdot z) = \sigma\cdot z$ while $(-1)^{\sigma\tau} = -(-1)^{\sigma}$, so the two terms cancel and $\operatorname{Alt}(z) = 0$. Hence $\frac{1}{k!}\operatorname{Alt}$ induces a map on $\bigwedge^k(M)$.

For the inverse, let $\pi_A \colon \mathcal{T}^k(M) \to \bigwedge^k(M)$ be the quotient map. By proposition 16.1.9, $\pi_A(\sigma\cdot z) = (-1)^{\sigma}\pi_A(z)$ on simple tensors, so

$$\pi_A\left(\tfrac{1}{k!}\operatorname{Alt}(z)\right) = \tfrac{1}{k!}\sum_{\sigma\in S_k}(-1)^{\sigma}(-1)^{\sigma}\pi_A(z) = \pi_A(z)$$

using $(-1)^{\sigma}(-1)^{\sigma} = 1$. And $\frac{1}{k!}\operatorname{Alt}$ is the identity on $\mathcal{T}^k_{\mathrm{alt}}(M)$. So the two maps are mutually inverse, as we sought to show.

Defining an invariant element by averaging over a finite group, which is what $\frac{1}{k!}\operatorname{Sym}$ does, is a technique that recurs whenever the order of the group is invertible in the ring. It is the mechanism behind Maschke's theorem ([17, Maschke's Theorem]), where averaging over a finite group produces an invariant complement to a subrepresentation. When $k!$ is not invertible the comparison genuinely fails, and the exercises give a witness.

Exercise 16.1.1

1. Show that $C^k(M)$ is already generated by the elements $z - \tau\cdot z$ with z a simple tensor and τ a transposition, so that symmetry of a multilinear map need only be checked on transpositions.
2. Let R have characteristic 2. Show that a k-multilinear map satisfies the sign rule of lemma 16.1.4 if and only if it is symmetric, and exhibit a symmetric trilinear map over $\mathbb{Z}/2\mathbb{Z}$ that is not alternating.
3. Show that if M is generated by fewer than k elements then $\bigwedge^k(M) = 0$.
4. Let $\varphi \colon M \to N$ be a surjection of R-modules. Show that the induced maps $\mathcal{S}^k(M) \to \mathcal{S}^k(N)$ and $\bigwedge^k(M) \to \bigwedge^k(N)$ are surjective. (Use theorem 15.2.2 on $\mathcal{T}^k$ first, then check that generators map to generators.)
5. Take $R = \mathbb{Z}$ and $M = \mathbb{Z}/2\mathbb{Z}$. Compute $\mathcal{T}^2(M)$, $\mathcal{S}^2(M)$ and $\bigwedge^2(M)$.
6. Show that $\operatorname{Alt}\circ\operatorname{Sym} = 0$ on $\mathcal{T}^k(M)$ for every $k \geq 2$, over any commutative ring.
7. Continuing the previous exercise with $R = \mathbb{Z}$ and $M = \mathbb{Z}/2\mathbb{Z}$, where $2!$ is not a unit, compute $\mathcal{T}^2_{\mathrm{alt}}(M)$ and compare it with $\bigwedge^2(M)$. Conclude that the invertibility hypothesis in proposition 16.1.12 cannot be dropped: the two modules are not merely joined by a different isomorphism, they are not isomorphic at all.

16.2 Graded Rings and Graded Modules

The tensor powers of definition 16.1.1 come with a bookkeeping device that has so far gone unused. Associativity (theorem 15.2.8) identifies $\mathcal{T}^k(M)\otimes_R\mathcal{T}^l(M)$ with $\mathcal{T}^{k+l}(M)$, so a k-tensor times an l-tensor is a $(k+l)$-tensor: the index adds. A ring carrying an index of this kind is called graded, and the three algebras of section 16.3 are all built by taking a graded ring apart and putting it back

together modulo relations that respect the index. Grading is therefore developed first, in its own right.

Definition 16.2.1: Graded Ring

A ring S is a *graded ring* if it comes with a decomposition

$$S = \bigoplus_{k \geq 0} S_k$$

of its underlying additive group into subgroups such that

$$S_k S_l \subseteq S_{k+l} \qquad \text{for all } k, l \geq 0$$

An element of S_k is *homogeneous of degree* k, and S_k is the *homogeneous component of* S *of degree* k. Every $s \in S$ has a unique expression $s = \sum_k s_k$ with $s_k \in S_k$ and all but finitely many s_k zero. The s_k are the *homogeneous components* of s.

The base ring plays no role in the definition, and the letter S is used for the graded ring throughout to keep it distinct from a ring R that may be acting. Note also that S_k is only required to be an additive subgroup: the multiplicative structure is carried entirely by the containments $S_k S_l \subseteq S_{k+l}$. What those containments force is the following.

Proposition 16.2.2: The Degree-Zero Part

Let $S = \bigoplus_{k \geq 0} S_k$ be a graded ring. Then

1. $1 \in S_0$, and S_0 is a subring of S;
2. each S_k is an S_0-module, and S is an S_0-algebra when S_0 lies in the centre of S;
3. $S_+ := \bigoplus_{k>0} S_k$ is a two-sided ideal of S, and $S/S_+ \cong S_0$ as rings.

***Proof*:**

1. Write $1 = \sum_k e_k$ with $e_k \in S_k$. For $x \in S_l$ the product $x \cdot 1 = \sum_k x e_k$ has $x e_k \in S_{l+k}$, so comparing homogeneous components of degree l on both sides of $x = x \cdot 1$ gives $x = x e_0$. Additivity extends this from homogeneous x to all of S, and the same argument on the other side gives $e_0 x = x$. So e_0 is a two-sided identity, and an identity is unique, so $1 = e_0 \in S_0$. Now $S_0 S_0 \subseteq S_0$ makes S_0 closed under multiplication, it is an additive subgroup by construction, and it contains 1. So it is a subring.

2. $S_0 S_k \subseteq S_k$ and $S_k S_0 \subseteq S_k$, so each S_k is a module over the subring S_0 on either side, and S is their direct sum. The algebra statement is definition 12.1.13, whose hypothesis is exactly that the image of S_0 is central.

3. S_+ is an additive subgroup, and for $x \in S_l$ and $y \in S_k$ with $k > 0$ one has $xy \in S_{k+l}$ with $k + l > 0$. Additivity extends this to all of S, on both sides, so S_+ is a two-sided ideal. The projection $S \to S_0$ sending s to its degree-zero component is additive, sends 1 to 1 by (1), and is multiplicative because the degree-zero component of a product st is $s_0 t_0$, every

other contribution $s_k t_l$ with $k+l=0$ and $k,l \geq 0$ being absent. Its kernel is exactly S_+, completing the proof.

Note that $S_i \subseteq S_j$ is not to be expected for $i < j$: the components are summands, not a filtration. In $R[x]$ graded by degree, $S_1 = Rx$ and $S_2 = Rx^2$ are disjoint apart from 0.

Definition 16.2.3: Graded Modules, Ideals and Homomorphisms

Let $S = \bigoplus_{k\geq 0} S_k$ and $T = \bigoplus_{k\geq 0} T_k$ be graded rings.

1. A left S-module M is *graded* if it comes with a decomposition $M = \bigoplus_{k\geq 0} M_k$ into additive subgroups with $S_k M_l \subseteq M_{k+l}$ for all k, l. Elements of M_k are *homogeneous of degree* k.
2. An ideal $I \subseteq S$ is a *graded ideal*, or a *homogeneous ideal*, if
$$I = \bigoplus_{k\geq 0} (I \cap S_k)$$
Writing $I_k := I \cap S_k$, a graded ideal is in particular a graded S-module.
3. A ring homomorphism $\varphi\colon S \to T$ is *graded* if $\varphi(S_k) \subseteq T_k$ for every k.

Example 16.2.4: Graded Rings

1. Any ring S is graded with $S_0 = S$ and $S_k = 0$ for $k > 0$. This is the *trivial* grading, and it shows the definition imposes nothing by itself.
2. For any ring R, the polynomial ring $R[x]$ is graded by degree:
$$R[x] = \bigoplus_{k\geq 0} Rx^k$$
the degree-k component being the free R-module on the single monomial x^k, and $Rx^k \cdot Rx^l \subseteq Rx^{k+l}$. It is worth being precise about what the components are, since the powers $(x)^k$ of the ideal (x) are *not* a grading: they are nested, $(x) \supseteq (x)^2 \supseteq \cdots$, so their sum is not direct.
3. More generally $R[x_1, \ldots, x_n]$ is graded with k-th component the free R-module on the monomials $x_1^{a_1} \cdots x_n^{a_n}$ with $a_1 + \cdots + a_n = k$. These are the *homogeneous polynomials* of degree k. Here $S_0 = R$, and S_+ is the ideal $(x_1, \ldots, x_n)$ of polynomials with zero constant term.
4. Let R be commutative and $I \subseteq R$ an ideal. The *Rees algebra* of R with respect to I is
$$\mathrm{Rees}_R(I) := \bigoplus_{k\geq 0} I^k = R \oplus I \oplus I^2 \oplus \cdots$$
with multiplication induced by that of R. It is graded because $I^k I^l = I^{k+l}$, so in particular $I^k I^l \subseteq I^{k+l}$. The summands are indexed copies of the ideals I^k, kept formally separate. The sum inside R would not be direct, for the same reason as in (2).[a]

[a] Rees algebras are the algebraic form of a blow-up in algebraic geometry, and are taken up alongside the Artin-Rees lemma in *Commutative Algebra* [16, Artin-Rees Lemma].

Quotienting by a graded ideal is exactly the operation that keeps the grading.

Proposition 16.2.5: Quotient of a Graded Ring by a Graded Ideal

Let $S = \bigoplus_k S_k$ be a graded ring and $I \subseteq S$ a graded ideal, and write $I_k = I \cap S_k$. Then S/I is a graded ring with

$$(S/I)_k \cong S_k/I_k$$

and the quotient map $S \to S/I$ is a graded ring homomorphism.

***Proof*:**
Let $\pi\colon S \to S/I$ be the quotient map and let $(S/I)_k := \pi(S_k)$, an additive subgroup. These generate S/I additively, since π is surjective and the S_k generate S. Their sum is direct: if $\sum_k \pi(s_k) = 0$ with $s_k \in S_k$ and almost all zero, then $\sum_k s_k \in I$, so every $s_k \in I$ by definition 16.2.3(2), whence $\pi(s_k) = 0$ for each k. The containment $(S/I)_k(S/I)_l \subseteq (S/I)_{k+l}$ follows by applying π to $S_kS_l \subseteq S_{k+l}$. So $S/I = \bigoplus_k (S/I)_k$ is a graded ring, and $\pi(S_k) = (S/I)_k$ says exactly that π is graded.

For the displayed isomorphism, the restriction $\pi|_{S_k}\colon S_k \to (S/I)_k$ is a surjective homomorphism of additive groups with kernel $S_k \cap I = I_k$, so $S_k/I_k \cong (S/I)_k$ by theorem 3.1.15, completing the proof.

The same bookkeeping applies to a graded homomorphism, whose kernel and image decompose degree by degree.

Proposition 16.2.6: Kernel and Image of a Graded Homomorphism

Let $\varphi\colon S \to T$ be a graded ring homomorphism and write $\varphi_k := \varphi|_{S_k}\colon S_k \to T_k$. Then

$$\ker\varphi = \bigoplus_{k\geq 0} \ker\varphi_k \qquad \text{and} \qquad \varphi(S) = \bigoplus_{k\geq 0} \varphi(S_k)$$

so $\ker\varphi$ is a graded ideal of S and $\varphi(S)$ is a graded subring of T.

***Proof*:**
If $s \in \ker\varphi$ has homogeneous components s_k, then $\sum_k \varphi(s_k) = 0$ with $\varphi(s_k) \in T_k$, so each $\varphi(s_k) = 0$ by uniqueness of the decomposition in T. Hence $s \in \bigoplus_k \ker\varphi_k$. The reverse inclusion is clear, and the displayed equality shows $\ker\varphi$ is graded.

For the image, $\varphi(S) = \sum_k \varphi(S_k)$ because φ is additive and the S_k span S, and the sum is direct because $\varphi(S_k) \subseteq T_k$ and the T_k are independent. Closure under multiplication is $\varphi(S_k)\varphi(S_l) \subseteq \varphi(S_{k+l})$, which holds because φ is multiplicative, completing the proof.

Those two results give the last of several equivalent ways to recognize a graded ideal.

Proposition 16.2.7: Characterizing Graded Ideals

Let $S = \bigoplus_k S_k$ be a graded ring and $I \subseteq S$ an ideal. The following are equivalent.

1. I is a graded ideal.
2. For every $s \in I$, each homogeneous component of s lies in I.
3. I is generated by homogeneous elements.
4. I is the kernel of a graded ring homomorphism out of S.

***Proof*:**

(1) ⇒ (2). If $I = \bigoplus_k (I \cap S_k)$ then every $s \in I$ is a finite sum $\sum_k t_k$ with $t_k \in I \cap S_k$. That expression is also the decomposition of s into homogeneous components, which is unique, so each component $s_k = t_k$ lies in I.

(2) ⇒ (3). By (2) the set H of homogeneous elements of I is contained in I, and every $s \in I$ is the sum of its homogeneous components, all of which lie in H. So H generates I.

(3) ⇒ (1). Let I be generated by homogeneous elements h_α, say $h_\alpha \in S_{d_\alpha}$. An element $s \in I$ is a finite sum $s = \sum_\alpha a_\alpha h_\alpha$ with $a_\alpha \in S$. Decomposing each $a_\alpha = \sum_j a_{\alpha,j}$ into homogeneous components,

$$s = \sum_\alpha \sum_j a_{\alpha,j} h_\alpha$$

and each $a_{\alpha,j} h_\alpha$ is homogeneous of degree $j + d_\alpha$ and lies in I. Collecting the terms of each fixed degree therefore exhibits the homogeneous components of s as elements of I, so $s \in \bigoplus_k (I \cap S_k)$. The reverse inclusion is automatic, giving (1).

(1) ⇒ (4). By proposition 16.2.5 below, S/I is a graded ring and the quotient map is a graded homomorphism. Its kernel is I.

(4) ⇒ (2). Let $\varphi\colon S \to T$ be graded with kernel I, and let $s \in I$ have homogeneous components s_k. Then $0 = \varphi(s) = \sum_k \varphi(s_k)$ with $\varphi(s_k) \in T_k$, and the decomposition of 0 into homogeneous components of T is unique, so every $\varphi(s_k) = 0$ and every $s_k \in I$, completing the proof.

For prime ideals the grading cuts the work down: only homogeneous elements need to be tested.

Corollary 16.2.8: Prime Graded Ideals

Let S be a graded commutative ring and $I \subseteq S$ a graded ideal with $I \neq S$. Then I is prime if and only if for all *homogeneous* $f, g \in S$ with $fg \in I$, either $f \in I$ or $g \in I$.

***Proof*:**

If I is prime the condition holds, since it is a special case of the definition.

Conversely, suppose the condition holds and let $f, g \in S$ with $fg \in I$ but $f \notin I$ and $g \notin I$. By proposition 16.2.7(2) some homogeneous component of f lies outside I, and only finitely many components are nonzero, so there is a largest index a with $f_a \notin I$. Choose b likewise for g.

Consider the homogeneous component of fg in degree $a+b$, namely $\sum_{i+j=a+b} f_i g_j$. In any term

with $i > a$ one has $f_i \in I$ by maximality of a, so $f_i g_j \in I$. And in any term with $i < a$ one has $j = a + b - i > b$, so $g_j \in I$ and again $f_i g_j \in I$. Every term except $f_a g_b$ therefore lies in I. Since $fg \in I$ and I is graded, the whole degree-$(a+b)$ component lies in I by proposition 16.2.7(2), and subtracting the other terms gives $f_a g_b \in I$. As f_a and g_b are homogeneous, the hypothesis forces $f_a \in I$ or $g_b \in I$, contradicting the choice of a and b. So I is prime, completing the proof.

Finally, the operations on ideals of definition 9.4.1 preserve gradedness.

Proposition 16.2.9: Arithmetic of Homogeneous Ideals

Let S be a graded commutative ring and let $I, J \subseteq S$ be graded ideals. Then each of

$$I + J, \qquad IJ, \qquad I \cap J, \qquad \sqrt{I}$$

is a graded ideal.

***Proof*:**
Throughout, proposition 16.2.7 is used to switch between the definition and the two convenient criteria.

$I + J$ and IJ. By criterion (3) each of I and J has a generating set of homogeneous elements, say H_I and H_J. Then $H_I \cup H_J$ generates $I + J$ and consists of homogeneous elements, and the products $\{hh' | h \in H_I,\ h' \in H_J\}$ generate IJ and are homogeneous, since $S_k S_l \subseteq S_{k+l}$. Criterion (3) applies to both.

$I \cap J$. Let $s \in I \cap J$. Each homogeneous component of s lies in I by criterion (2) applied to I, and in J by the same criterion applied to J, hence in $I \cap J$. Criterion (2) gives the claim.

$\sqrt{I}$. That $\sqrt{I}$ is an ideal is proposition 9.4.3. For criterion (2), let $s \in \sqrt{I}$ and argue by induction on the number of nonzero homogeneous components of s. If that number is 0 there is nothing to prove. Otherwise let d be the largest degree with $s_d \neq 0$, and choose n with $s^n \in I$. The homogeneous component of s^n in the top degree nd is s_d^n, because every other product of n components has total degree strictly less than nd. Since $s^n \in I$ and I is graded, criterion (2) gives $s_d^n \in I$, so $s_d \in \sqrt{I}$. Then $s - s_d \in \sqrt{I}$, as $\sqrt{I}$ is an ideal, and it has one fewer nonzero homogeneous component, so the inductive hypothesis puts all of its components in $\sqrt{I}$. Those together with s_d are the components of s, completing the proof.

Exercise 16.2.1

1. Show that a graded ring S has $S_0 \neq 0$ unless $S = 0$, and give an example of a graded ring in which $S_1 = 0$ but $S_2 \neq 0$.
2. Show that an ideal of $R[x_1, \ldots, x_n]$ is graded if and only if it is generated by homogeneous polynomials, and use this to show that $(1 + x) \subseteq \mathbb{Z}[x]$ is not graded.
3. Show that $(y - x^2) \subseteq k[x, y]$ is not a graded ideal for the standard grading, while $(y,\ y - x^2)$ is. (For the second, identify the ideal.)
4. Let S be a graded ring. Show that S_+ is the unique largest graded ideal I with $I \cap S_0 = 0$, and that S is generated as an S_0-algebra by any set of homogeneous elements whose images generate S_+/S_+^2 as an S_0-module.

5. Show that a graded ideal I is radical if and only if $s_d^n \in I$ implies $s_d \in I$ for every homogeneous s_d and every $n \geq 1$.

6. Show that $I = (x^2 - y^2) \subseteq k[x, y]$ is a graded ideal that is not prime, and exhibit homogeneous $f, g \notin I$ with $fg \in I$. (By corollary 16.2.8 such a homogeneous witness must exist whenever a graded ideal fails to be prime.)

16.3 The Tensor, Symmetric and Exterior Algebras

Section 16.1 produced one module for each degree k. Adding them up produces a ring. The multiplication is concatenation of tensors, which raises the degree by exactly the amount it should, so the result is a graded ring in the sense of definition 16.2.1. And the sum acquires a universal property that the individual degrees do not have, one about homomorphisms into an algebra rather than into a module. The symmetric and exterior algebras are then quotients of it, and the last thing to check is that quotienting the whole and quotienting degree by degree agree.

Definition 16.3.1: Tensor Algebra

Let R be a commutative ring and M an R-module. The *tensor algebra* of M is the R-module

$$\mathcal{T}(M) := \bigoplus_{k \geq 0} \mathcal{T}^k(M) = R \oplus M \oplus \mathcal{T}^2(M) \oplus \cdots$$

equipped with the multiplication determined on homogeneous elements by the composite

$$\mathcal{T}^k(M) \times \mathcal{T}^l(M) \xrightarrow{\iota} \mathcal{T}^k(M) \otimes_R \mathcal{T}^l(M) \xrightarrow{\sim} \mathcal{T}^{k+l}(M)$$

where the second map is the associativity isomorphism of theorem 15.2.8 when $k, l \geq 1$, and is example 15.1.13 when $k = 0$ or $l = 0$, and extended to all of $\mathcal{T}(M)$ by additivity in each variable.

On simple tensors the multiplication is concatenation,

$$(m_1 \otimes \cdots \otimes m_i)(m'_1 \otimes \cdots \otimes m'_j) = m_1 \otimes \cdots \otimes m_i \otimes m'_1 \otimes \cdots \otimes m'_j$$

which is what makes it computable. Note that this is not the multiplication of theorem 15.2.14, which was $(a \otimes b)(a' \otimes b') = aa' \otimes bb'$ and required both factors to be algebras already. Here M is only a module, and the product of two elements of M lands not in M but one degree higher.

Theorem 16.3.2: $\mathcal{T}(M)$ is an R-algebra

Let R be a commutative ring and M an R-module.

1. $\mathcal{T}(M)$ is an associative R-algebra with identity $1 \in R = \mathcal{T}^0(M)$, and it is graded, with $\mathcal{T}^k(M)\mathcal{T}^l(M) \subseteq \mathcal{T}^{k+l}(M)$. It contains M as the homogeneous component of degree 1.
2. (universal property) For every R-algebra A and every R-module homomorphism $\varphi\colon M \to A$ there is a unique R-algebra homomorphism $\Phi\colon \mathcal{T}(M) \to A$ with $\Phi|_M = \varphi$:

$$\begin{array}{ccc} \mathcal{T}(M) & \xrightarrow{\Phi} & A \\ {\scriptstyle\iota}\uparrow & \nearrow_{\varphi} & \\ M & & \end{array}$$

***Proof*:**

1. The multiplication is R-bilinear on each pair of homogeneous components, being a composite of the R-bilinear ι with an R-module isomorphism, and it is extended biadditively, so it distributes over addition and commutes with the R-action. That is the compatibility definition 12.1.13 asks for. The containment $\mathcal{T}^k(M)\mathcal{T}^l(M) \subseteq \mathcal{T}^{k+l}(M)$ holds by construction, which is the grading.

   For associativity, both $(x,y,z) \mapsto (xy)z$ and $(x,y,z) \mapsto x(yz)$ are R-trilinear, so by theorem 15.2.11 each is determined by its values on triples of simple tensors, where both are the concatenation

   $$m_1 \otimes \cdots \otimes m_i \otimes m_1' \otimes \cdots \otimes m_j' \otimes m_1'' \otimes \cdots \otimes m_h''$$

   since concatenation of finite sequences is associative. For the identity, $1 \in R = \mathcal{T}^0(M)$ acts on $\mathcal{T}^k(M)$ through the isomorphism $R \otimes_R \mathcal{T}^k(M) \cong \mathcal{T}^k(M)$ of example 15.1.13, which is the identity map. Finally $\mathcal{T}^1(M) = M$ by definition 16.1.1.

2. Fix A and φ. For each $k \geq 1$ the map

   $$M^k \to A, \qquad (m_1, \ldots, m_k) \mapsto \varphi(m_1)\varphi(m_2)\cdots\varphi(m_k)$$

   is k-multilinear: φ is R-linear and multiplication in A is R-linear in each variable by definition 12.1.13. So theorem 15.2.11 gives a unique R-module homomorphism $\Phi_k\colon \mathcal{T}^k(M) \to A$ with $\Phi_k(m_1 \otimes \cdots \otimes m_k) = \varphi(m_1)\cdots\varphi(m_k)$. Let $\Phi_0\colon R \to A$ be the structure map of the R-algebra A, and let $\Phi := \sum_k \Phi_k$ on the direct sum, which is a well-defined R-module homomorphism because every element has only finitely many nonzero components.

   Φ is multiplicative: both $(x,y) \mapsto \Phi(xy)$ and $(x,y) \mapsto \Phi(x)\Phi(y)$ are R-bilinear, and on a pair of simple tensors both give $\varphi(m_1)\cdots\varphi(m_i)\varphi(m_1')\cdots\varphi(m_j')$, so they agree by theorem 15.1.12. It sends 1 to 1 because Φ_0 does. Restricted to $\mathcal{T}^1(M) = M$ it is φ.

   For uniqueness, $\mathcal{T}(M)$ is generated as an R-algebra by M: the simple tensors $m_1 \otimes \cdots \otimes m_k$ are products of elements of M, and they generate $\mathcal{T}^k(M)$ as an R-module by lemma 15.1.4. An R-algebra homomorphism is determined by its values on an algebra generating set, so any Φ' agreeing with φ on M equals Φ, completing the proof.

Part (2) says that $\mathcal{T}(-)$ converts modules into algebras in the freest possible way: prescribing an algebra homomorphism out of $\mathcal{T}(M)$ needs no more than prescribing a module homomorphism out of M. In categorical language $\mathcal{T}(-)$ is left adjoint to the functor forgetting the multiplication of an R-algebra, which is the same shape of statement as corollary 15.3.5. The general theory is [15, Adjoint Functors].

The two quotients of interest are cut out by degree-two relations. Before taking them, one compatibility has to be settled, since section 16.1 defined $\mathcal{S}^k(M)$ and $\bigwedge^k(M)$ by quotienting $\mathcal{T}^k(M)$ by a *submodule*, while the algebras are obtained by quotienting $\mathcal{T}(M)$ by an *ideal*. The two operations agree, but that is a statement to prove rather than to observe.

Lemma 16.3.3: The Defining Ideals Are Graded, Degree by Degree

Let R be a commutative ring and M an R-module. Let $\mathcal{C}(M) \subseteq \mathcal{T}(M)$ be the two-sided ideal generated by all $m \otimes m' - m' \otimes m$ with $m, m' \in M$, and let $\mathcal{A}(M) \subseteq \mathcal{T}(M)$ be the two-sided ideal generated by all $m \otimes m$ with $m \in M$. Then both are graded ideals, and for every $k \geq 0$

$$\mathcal{C}(M) \cap \mathcal{T}^k(M) = C^k(M), \qquad \mathcal{A}(M) \cap \mathcal{T}^k(M) = A^k(M)$$

where $C^k(M)$ and $A^k(M)$ are the submodules of definition 16.1.5 and definition 16.1.7.

***Proof*:**
Both ideals are generated by homogeneous elements of degree 2, so both are graded by proposition 16.2.7(3), and it remains to identify the degree-k parts. Write $\mathcal{C}(M)_k$ for $\mathcal{C}(M) \cap \mathcal{T}^k(M)$.

The symmetric case, $C^k(M) \subseteq \mathcal{C}(M)_k$. By definition $C^k(M)$ is generated by the differences $z - \sigma \cdot z$ for simple tensors z and $\sigma \in S_k$. Since adjacent transpositions generate S_k, and

$$z - (\tau_1 \cdots \tau_r) \cdot z = \sum_{j=1}^{r} \Big((\tau_{j+1} \cdots \tau_r) \cdot z - (\tau_j \tau_{j+1} \cdots \tau_r) \cdot z \Big)$$

telescopes with each summand of the form $w - \tau \cdot w$ for a simple tensor w and an adjacent transposition τ, it suffices to treat $\tau = (i\ i+1)$. There

$$z - \tau \cdot z = (m_1 \otimes \cdots \otimes m_{i-1}) \big(m_i \otimes m_{i+1} - m_{i+1} \otimes m_i \big) (m_{i+2} \otimes \cdots \otimes m_k)$$

a product in $\mathcal{T}(M)$ of an element of $\mathcal{T}^{i-1}(M)$, a generator of $\mathcal{C}(M)$, and an element of $\mathcal{T}^{k-i-1}(M)$, hence an element of $\mathcal{C}(M)$. And it is homogeneous of degree k.

The symmetric case, $\mathcal{C}(M)_k \subseteq C^k(M)$. An element of $\mathcal{C}(M)$ is a finite sum of products $x\,(m \otimes m' - m' \otimes m)\,y$ with $x, y \in \mathcal{T}(M)$. Expanding x and y into homogeneous components and then into simple tensors, and keeping only the terms of total degree k, every such term is of the form $w - \tau \cdot w$ for a simple tensor $w \in \mathcal{T}^k(M)$ and an adjacent transposition τ, by the display above read backwards. Each lies in $C^k(M)$, and $C^k(M)$ is a submodule, so the sum does too.

The alternating case. The inclusion $\mathcal{A}(M)_k \supseteq A^k(M)$ needs slightly more, since a generator of $A^k(M)$ is a simple tensor whose repeated entries need not be adjacent. Note first that for $u, v \in M$,

$$u \otimes v + v \otimes u = (u+v) \otimes (u+v) - u \otimes u - v \otimes v \in \mathcal{A}(M)$$

so interchanging two adjacent factors of a simple tensor changes it by an element of $\mathcal{A}(M)$, up to sign. Given $z = m_1 \otimes \cdots \otimes m_k$ with $m_a = m_b$ and $a < b$, repeatedly interchanging adjacent factors moves m_b into position $a+1$, so that $z \equiv \pm z'$ modulo $\mathcal{A}(M)$, where z' is a simple tensor with two equal adjacent entries. Then z' factors as a product of an element of $\mathcal{T}^{a-1}(M)$, the generator $m_a \otimes m_a$, and an element of $\mathcal{T}^{k-a-1}(M)$, so $z' \in \mathcal{A}(M)$ and hence $z \in \mathcal{A}(M)$. Being homogeneous of degree k, it lies in $\mathcal{A}(M)_k$.

The reverse inclusion $\mathcal{A}(M)_k \subseteq A^k(M)$ follows as in the symmetric case: the degree-k part of a product $x\,(m \otimes m)\,y$ expands into simple tensors each having two equal adjacent entries, and those are generators of $A^k(M)$, completing the proof.

Corollary 16.3.4: The Quotients Are Graded

With the notation above, $\mathcal{T}(M)/\mathcal{C}(M)$ and $\mathcal{T}(M)/\mathcal{A}(M)$ are graded rings, and their homogeneous components of degree k are $\mathcal{S}^k(M)$ and $\bigwedge^k(M)$ respectively.

***Proof*:**
By lemma 16.3.3 both ideals are graded, so proposition 16.2.5 applies and gives graded quotients with degree-k component $\mathcal{T}^k(M)/(\mathcal{C}(M) \cap \mathcal{T}^k(M))$, which is $\mathcal{T}^k(M)/C^k(M) = \mathcal{S}^k(M)$ by the same lemma and definition 16.1.5. The alternating case is identical, completing the proof.

Definition 16.3.5: Symmetric Algebra

The *symmetric algebra* of an R-module M is the quotient R-algebra

$$\mathcal{S}(M) := \mathcal{T}(M)/\mathcal{C}(M)$$

where $\mathcal{C}(M)$ is the two-sided ideal generated by the elements $m \otimes m' - m' \otimes m$ for $m, m' \in M$. By corollary 16.3.4 it is graded with $\mathcal{S}(M)_k = \mathcal{S}^k(M)$, and the product of the images of $m_1, \ldots, m_k$ is written $m_1 m_2 \cdots m_k$.

Theorem 16.3.6: The Symmetric Algebra Is the Free Commutative Algebra

Let R be a commutative ring and M an R-module. Then $\mathcal{S}(M)$ is a commutative R-algebra, and for every *commutative* R-algebra A and every R-module homomorphism $\varphi\colon M \to A$ there is a unique R-algebra homomorphism $\Phi\colon \mathcal{S}(M) \to A$ restricting to φ on $M = \mathcal{S}^1(M)$.

***Proof*:**
The algebra $\mathcal{S}(M)$ is generated by the image of M, since $\mathcal{T}(M)$ is generated by M and the quotient map is surjective. The images of any two elements of M commute, because $m \otimes m' - m' \otimes m \in \mathcal{C}(M)$. An associative algebra generated by a set of pairwise commuting elements is commutative: the set of elements commuting with everything in the generating set is a subalgebra, so it is the whole algebra, and then the same argument applied a second time gives commutativity outright. So $\mathcal{S}(M)$ is commutative.

Let A be commutative and $\varphi\colon M \to A$ an R-module homomorphism. By theorem 16.3.2 there

is a unique R-algebra homomorphism $\Psi\colon \mathcal{T}(M) \to A$ with $\Psi|_M = \varphi$. On a generator of $\mathcal{C}(M)$,

$$\Psi(m \otimes m' - m' \otimes m) = \varphi(m)\varphi(m') - \varphi(m')\varphi(m) = 0$$

because A is commutative. An algebra homomorphism vanishing on a set of generators of a two-sided ideal vanishes on the ideal, so $\mathcal{C}(M) \subseteq \ker \Psi$ and Ψ factors through $\mathcal{S}(M)$, giving Φ. Uniqueness holds because $\mathcal{S}(M)$ is generated as an R-algebra by the image of M, as we sought to show.

Definition 16.3.7: Exterior Algebra

The *exterior algebra* of an R-module M is the quotient R-algebra

$$\bigwedge(M) := \mathcal{T}(M)/\mathcal{A}(M)$$

where $\mathcal{A}(M)$ is the two-sided ideal generated by the elements $m \otimes m$ for $m \in M$. By corollary 16.3.4 it is graded with $\bigwedge(M)_k = \bigwedge^k(M)$, and the image of $m_1 \otimes \cdots \otimes m_k$ is written $m_1 \wedge \cdots \wedge m_k$.

Proposition 16.3.8: The Exterior Algebra Is the Free Algebra on Squares Zero

Let R be a commutative ring and M an R-module. For every R-algebra A and every R-module homomorphism $\varphi\colon M \to A$ satisfying $\varphi(m)^2 = 0$ for all $m \in M$, there is a unique R-algebra homomorphism $\Phi\colon \bigwedge(M) \to A$ restricting to φ on $M = \bigwedge^1(M)$.

***Proof*:**
As in theorem 16.3.6, theorem 16.3.2 gives $\Psi\colon \mathcal{T}(M) \to A$ with $\Psi|_M = \varphi$, and $\Psi(m \otimes m) = \varphi(m)^2 = 0$ kills every generator of $\mathcal{A}(M)$, hence the ideal. The induced Φ is unique because $\bigwedge(M)$ is generated as an R-algebra by the image of M, completing the proof.

The wedge product is not commutative, and it is not anticommutative either. The sign depends on the degrees. Proposition 16.1.9 gave the rule for elements of degree 1, and the general rule follows from it.

Proposition 16.3.9: Graded Commutativity of the Exterior Algebra

Let $a \in \bigwedge^p(M)$ and $b \in \bigwedge^q(M)$. Then

$$a \wedge b = (-1)^{pq}\, b \wedge a$$

In particular elements of odd degree anticommute and elements of even degree are central among homogeneous elements; the identity $ab = -ba$ fails as soon as one of the two degrees is even.

***Proof*:**
Both $(a,b) \mapsto a \wedge b$ and $(a,b) \mapsto (-1)^{pq} b \wedge a$ are R-bilinear on $\bigwedge^p(M) \times \bigwedge^q(M)$, so by theorem 15.1.12 it suffices to compare them on pairs of products of degree-one elements, which generate each factor. Take $a = m_1 \wedge \cdots \wedge m_p$ and $b = m'_1 \wedge \cdots \wedge m'_q$. Moving m'_1 leftwards past each of $m_1, \ldots, m_p$ in turn costs a factor of -1 each time by proposition 16.1.9, so p sign changes. Doing this for each of the q elements of b in turn gives $(-1)^{pq}$ in total and rearranges $a \wedge b$ into $b \wedge a$. Hence $a \wedge b = (-1)^{pq} b \wedge a$.

For the last sentence, take $p = q = 1$ to get $m \wedge m' = -m' \wedge m$, and $p = 2$, $q = 1$ to get $a \wedge m = m \wedge a$, which is not $-m \wedge a$ unless $2a \wedge m = 0$, completing the proof.

Example 16.3.10: Tensor Algebras

1. Let $R = \mathbb{Z}$ and $M = \mathbb{Q}/\mathbb{Z}$. Then $(\mathbb{Q}/\mathbb{Z}) \otimes_{\mathbb{Z}} (\mathbb{Q}/\mathbb{Z}) = 0$ by the third exercise of section 15.1, so $\mathcal{T}^k(M) = 0$ for every $k \geq 2$ and

$$\mathcal{T}(\mathbb{Q}/\mathbb{Z}) = \mathbb{Z} \oplus \mathbb{Q}/\mathbb{Z}$$

   Addition is componentwise, and the multiplication is

$$(r, \overline{p})(s, \overline{q}) = (rs,\ r\overline{q} + s\overline{p})$$

   since the product of two degree-one elements lands in $\mathcal{T}^2 = 0$. This is the *square-zero extension* of $\mathbb{Z}$ by $\mathbb{Q}/\mathbb{Z}$: the degree-one part is an ideal whose square vanishes.

2. Let $R = \mathbb{Z}$ and $M = \mathbb{Z}/n\mathbb{Z}$. Then $(\mathbb{Z}/n\mathbb{Z}) \otimes_{\mathbb{Z}} (\mathbb{Z}/n\mathbb{Z}) \cong \mathbb{Z}/n\mathbb{Z}$ by proposition 15.2.13, and inductively $\mathcal{T}^k(M) \cong \mathbb{Z}/n\mathbb{Z}$ for every $k \geq 1$, so

$$\mathcal{T}(\mathbb{Z}/n\mathbb{Z}) = \mathbb{Z} \oplus (\mathbb{Z}/n\mathbb{Z}) \oplus (\mathbb{Z}/n\mathbb{Z}) \oplus \cdots \cong \mathbb{Z}[x]/(nx)$$

   the isomorphism sending the degree-k generator to x^k. The relation $nx = 0$ records that the degree-one part is killed by n, and every higher degree inherits it.

3. Take $M = \mathbb{Q}$ as a module over itself. Then $\mathcal{T}^k(\mathbb{Q}) \cong \mathbb{Q}$ for every k by example 15.1.13, so $\mathcal{T}(\mathbb{Q}) \cong \mathbb{Q}[x]$ as graded $\mathbb{Q}$-algebras. Here the tensor algebra happens to be commutative, because M has rank one. For a module of rank at least two it never is, as section 16.4 will show.

Exercise 16.3.1

1. Show that $\mathcal{T}(M)$ is commutative if and only if $\mathcal{S}(M) = \mathcal{T}(M)$, and that this happens if and only if $m \otimes m' = m' \otimes m$ for all $m, m' \in M$.
2. Show that $\mathcal{S}(M)$ satisfies the universal property of theorem 16.3.6 with respect to R-algebra homomorphisms into non-commutative algebras as well, provided the image of M is required to consist of pairwise commuting elements.
3. Let $I \subseteq \mathcal{T}(M)$ be the two-sided ideal generated by $\mathcal{T}^{\geq d}(M) := \bigoplus_{k \geq d} \mathcal{T}^k(M)$ for some $d \geq 1$. Show that $I = \mathcal{T}^{\geq d}(M)$, that it is graded, and identify the quotient $\mathcal{T}(M)/I$ as an R-module.

4. Show that $\bigwedge(M)$ is commutative when $2 = 0$ in R, and give an example over $\mathbb{Z}$ where it is not. (For the example, use the determinant to produce an alternating bilinear map on $\mathbb{Z}^2$ and deduce that $e_1 \wedge e_2$ has infinite order.)

5. Let $\varphi\colon M \to N$ be an R-module homomorphism. Show that it induces graded R-algebra homomorphisms $\mathcal{T}(\varphi)$, $\mathcal{S}(\varphi)$ and $\bigwedge(\varphi)$ between the corresponding algebras, and that each is surjective when φ is.

6. Show that $\mathcal{S}(M \oplus N) \cong \mathcal{S}(M) \otimes_R \mathcal{S}(N)$ as R-algebras, using theorem 16.3.6 together with theorem 15.2.17, which identifies the right-hand side as the coproduct of commutative R-algebras.

16.4 Bases, Ranks and the Determinant

Nothing so far has required M to be free. When it is, all three graded algebras become computable: a basis of M determines a basis of each homogeneous component, and the ranks are the three counts one would guess from thinking of the components as words, unordered words, and sets. The exterior case then delivers the determinant, which reappears not as the defining sum of definition 13.4.5 but as the action of an endomorphism on a rank-one module.

Theorem 16.4.1: Bases of the Tensor Powers

Let R be a commutative ring and let M be a free R-module with basis $\{e_i\}_{i \in I}$. Then $\mathcal{T}^k(M)$ is free with basis

$$\left\{ e_{i_1} \otimes e_{i_2} \otimes \cdots \otimes e_{i_k} \;\middle|\; (i_1, \ldots, i_k) \in I^k \right\}$$

indexed by all k-tuples. If I is finite of size n, then $\mathcal{T}^k(M)$ is free of rank n^k.

***Proof*:**
Induction on k. For $k = 0$ and $k = 1$ the claim is $\mathcal{T}^0(M) = R$ and $\mathcal{T}^1(M) = M$. For the inductive step, $\mathcal{T}^k(M) = \mathcal{T}^{k-1}(M) \otimes_R M$ by definition 16.1.1, and $\mathcal{T}^{k-1}(M)$ is free on the $(k-1)$-tuples by hypothesis, so corollary 15.2.6 makes $\mathcal{T}^k(M)$ free on the products of a basis element of each factor, which are exactly the k-tuples. The rank count is $|I^k| = n^k$, completing the proof.

In particular $\mathcal{T}(M)$ is the free associative R-algebra on the set $\{e_i\}$: its elements are the R-linear combinations of words in the e_i, multiplied by concatenation. For $|I| \geq 2$ it is not commutative, since $e_1 \otimes e_2$ and $e_2 \otimes e_1$ are distinct basis elements, which settles the question raised in example 16.3.10(3).

The symmetric algebra is the polynomial ring, and it is quickest to identify the whole algebra at once and read off the components afterwards.

Theorem 16.4.2: The Symmetric Algebra of a Free Module

Let R be a commutative ring and M free with basis $e_1, \dots, e_n$. Then there is an isomorphism of graded R-algebras

$$\mathcal{S}(M) \;\cong\; R[x_1, \dots, x_n]$$

carrying e_i to x_i. Consequently $\mathcal{S}^k(M)$ is free with basis the monomials

$$e_{i_1} e_{i_2} \cdots e_{i_k}, \qquad 1 \le i_1 \le i_2 \le \cdots \le i_k \le n$$

and has rank $\binom{n+k-1}{k}$.

***Proof*:**
The polynomial ring is a commutative R-algebra, so the R-module homomorphism $M \to R[x_1, \dots, x_n]$ sending e_i to x_i induces, by theorem 16.3.6, an R-algebra homomorphism

$$\Phi \colon \mathcal{S}(M) \longrightarrow R[x_1, \dots, x_n], \qquad \Phi(e_i) = x_i$$

Conversely the universal property of polynomial rings (theorem 9.2.4) gives an R-algebra homomorphism $\Psi \colon R[x_1, \dots, x_n] \to \mathcal{S}(M)$ with $\Psi(x_i) = e_i$, since $\mathcal{S}(M)$ is commutative by theorem 16.3.6.

Each composite is an R-algebra endomorphism fixing a set of algebra generators, hence the identity: $\mathcal{S}(M)$ is generated by the image of M, and $R[x_1, \dots, x_n]$ by the x_i. So Φ is an isomorphism. It is graded because it sends the degree-one generators e_i to the degree-one generators x_i and both algebras are generated in degree one, so it carries a product of k generators to a product of k generators.

Restricting to degree k, $\mathcal{S}^k(M) = \mathcal{S}(M)_k$ by corollary 16.3.4, and the degree-k component of $R[x_1, \dots, x_n]$ is free on the monomials of total degree k (example 16.2.4(3)). Their preimages are the products $e_{i_1} \cdots e_{i_k}$ with indices weakly increasing, one for each monomial. Counting monomials of degree k in n variables is the standard problem of distributing k identical items into n labelled boxes, whose answer is $\binom{n+k-1}{k}$, completing the proof.

The exterior powers need a separate argument, because there is no ambient ring to compare against. Spanning is immediate. Independence is produced by exhibiting, for each proposed basis element, an alternating map that detects it. Those maps are minors, and the fact that a determinant is alternating is proposition 13.4.6. There is no circularity: det is the sum of definition 13.4.5, defined without any reference to exterior algebra, and it is only in theorem 16.4.7 below that the two are identified.

Theorem 16.4.3: Bases of the Exterior Powers

Let R be a commutative ring and M free with basis $e_1, \dots, e_n$. Then $\bigwedge^k(M)$ is free with basis

$$\{\, e_{i_1} \wedge e_{i_2} \wedge \cdots \wedge e_{i_k} \mid 1 \le i_1 < i_2 < \cdots < i_k \le n \,\}$$

so it has rank $\binom{n}{k}$. In particular $\bigwedge^k(M) = 0$ for $k > n$, and $\bigwedge^n(M)$ is free of rank one on $e_1 \wedge \cdots \wedge e_n$. The whole exterior algebra $\bigwedge(M)$ is then free of rank 2^n.

***Proof*:**

Spanning. By theorem 16.4.1 the tensors $e_{i_1} \otimes \cdots \otimes e_{i_k}$ span $\mathcal{T}^k(M)$, so their images span $\bigwedge^k(M)$. If two indices coincide the image is 0 by proposition 16.1.9. Otherwise the same proposition rearranges the factors into strictly increasing order up to a sign. So the displayed set spans.

Independence. Fix a strictly increasing $I = (i_1 < \cdots < i_k)$ and define

$$\varphi_I \colon M^k \to R, \qquad \varphi_I(v_1, \ldots, v_k) = \det\left(\left(v_j\right)_{i_a}\right)_{1 \le a,j \le k}$$

the determinant of the $k \times k$ matrix whose (a, j) entry is the i_a-th coordinate of v_j in the basis $e_1, \ldots, e_n$. Each coordinate is R-linear in v_j, and the determinant is R-linear in each column by proposition 13.4.6(3), so φ_I is k-multilinear. And if $v_j = v_{j'}$ for $j \neq j'$ then the matrix has two equal columns, so φ_I vanishes by proposition 13.4.6(4). Thus φ_I is alternating, and theorem 16.1.8 induces $\overline{\varphi_I} \colon \bigwedge^k(M) \to R$.

Evaluate $\overline{\varphi_I}$ on a proposed basis element $e_J = e_{j_1} \wedge \cdots \wedge e_{j_k}$ with J strictly increasing. The matrix in question has (a, b) entry δ_{i_a, j_b}. If $I = J$ it is the identity matrix and the determinant is 1 by proposition 13.4.6(1). If $I \neq J$ then, both being strictly increasing k-tuples, some i_a lies outside J, and then row a is zero, so the determinant is 0. Hence $\overline{\varphi_I}(e_J) = \delta_{IJ}$.

Now suppose $\sum_J c_J e_J = 0$ is a relation among the proposed basis elements, with $c_J \in R$. Applying $\overline{\varphi_I}$ gives $c_I = 0$ for every I. So the set is independent, and with spanning it is a basis.

Counts. The strictly increasing k-tuples from $\{1, \ldots, n\}$ are the k-element subsets, of which there are $\binom{n}{k}$. For $k > n$ there are none, so $\bigwedge^k(M) = 0$, and for $k = n$ there is exactly one. Summing over k gives $\sum_{k=0}^{n} \binom{n}{k} = 2^n$, completing the proof.

Example 16.4.4: Exterior Algebras of Small Vector Spaces
Let F be a field and V an F-vector space.

1. If $\dim V = 1$ with basis v, then theorem 16.4.3 gives $\bigwedge^0(V) = F$, $\bigwedge^1(V) = V$, and $\bigwedge^i(V) = 0$ for every $i \geq 2$, since there are no strictly increasing i-tuples from a one-element set. So
$$\bigwedge(V) = F \oplus V$$
with every product of two elements of V equal to zero.

2. Now let $\dim V = 2$ with basis v, v'. The only strictly increasing pairs give $\bigwedge^2(V) = F\,(v \wedge v')$, and $\bigwedge^i(V) = 0$ for $i \geq 3$. Computing a general product of two vectors,
$$\begin{aligned}(av + bv') \wedge (cv + dv') &= ac\,(v \wedge v) + ad\,(v \wedge v') + bc\,(v' \wedge v) + bd\,(v' \wedge v') \\ &= (ad - bc)\, v \wedge v'\end{aligned}$$
using $v \wedge v = v' \wedge v' = 0$ and $v' \wedge v = -v \wedge v'$. The coefficient is the determinant of the matrix with those columns, which is theorem 16.4.7 in the case $n = 2$.

3. If $\dim V = 3$ with basis $\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3$, then $\bigwedge^2(V)$ has rank $\binom{3}{2} = 3$, the same as V itself, and
$$v \wedge w = (v_1 w_2 - v_2 w_1)(\mathbf{e}_1 \wedge \mathbf{e}_2) + (v_2 w_3 - v_3 w_2)(\mathbf{e}_2 \wedge \mathbf{e}_3) + (v_1 w_3 - v_3 w_1)(\mathbf{e}_1 \wedge \mathbf{e}_3)$$

Identifying the basis of $\bigwedge^2(V)$ with that of V turns $\wedge$ into the cross product of vector calculus. The coincidence is exactly $\binom{3}{2} = 3$ and happens in no other dimension, which is why the cross product is a three-dimensional phenomenon.

4. Let $R = \mathbb{Z}[x, y]$ and $I = (x, y)$, both regarded as R-modules. The inclusion $\varphi\colon I \hookrightarrow R$ induces $\bigwedge^2(\varphi)\colon \bigwedge^2(I) \to \bigwedge^2(R)$. Now R is free of rank one, so $\bigwedge^2(R) = 0$ by theorem 16.4.3, while $x \wedge y \neq 0$ in $\bigwedge^2(I)$. For the latter, every $f \in I$ has zero constant term, so its coefficients f_x and f_y of x and y are well defined. The map $(f, g) \mapsto f_x g_y - f_y g_x$ into $R/I = \mathbb{Z}$ is R-bilinear, because multiplying f by $h \in R$ scales both coefficients by $h(0, 0)$, and it is alternating. So theorem 16.1.8 induces $\bigwedge^2(I) \to \mathbb{Z}$ sending $x \wedge y$ to 1. Hence $\bigwedge^2(\varphi)$ is not injective even though φ is: the exterior power of an injection need not be an injection, which is the same failure of exactness recorded for $\otimes$ in section 15.5.

Exterior powers turn direct sums into a graded tensor product, in the same way symmetric powers turn them into an ordinary one. The formula below is what makes the exterior algebra computable on a module built from simpler pieces, and it is the tool behind the fact that a stably free module of rank one is free.

Theorem 16.4.5: Exterior Powers of a Direct Sum

Let R be a commutative ring and M, N two R-modules. Then for every $k \geq 0$ there is an isomorphism of R-modules

$$\bigwedge^k(M \oplus N) \;\cong\; \bigoplus_{i+j=k} \bigwedge^i(M) \otimes_R \bigwedge^j(N)$$

natural in M and N, carrying $\alpha \otimes \beta$ to $\alpha \wedge \beta$ when $\bigwedge M$ and $\bigwedge N$ are identified with their images in $\bigwedge(M \oplus N)$ under the maps induced by the inclusions of M and N into $M \oplus N$.

***Proof*:**

Step 1: a twisted product on the tensor product. Put $T = \bigwedge(M) \otimes_R \bigwedge(N)$, graded by total degree, so that T_k is the right-hand side. Define a multiplication on homogeneous elements by

$$(\alpha \otimes \beta) \cdot (\alpha' \otimes \beta') = (-1)^{\deg \beta \, \deg \alpha'} \, (\alpha \wedge \alpha') \otimes (\beta \wedge \beta')$$

and extend bilinearly. This is well defined, both wedge products being bilinear, and it is associative: comparing the two bracketings of a triple product, each produces $(\alpha \wedge \alpha' \wedge \alpha'') \otimes (\beta \wedge \beta' \wedge \beta'')$ with sign exponent $\deg\beta(\deg\alpha' + \deg\alpha'') + \deg\beta' \deg\alpha''$. The element $1 \otimes 1$ is a unit, so T is an R-algebra.

Step 2: a map out of $\bigwedge(M \oplus N)$. Consider the R-module map $\varphi\colon M \oplus N \to T$ sending (m, n) to $m \otimes 1 + 1 \otimes n$, landing in degree one. Then

$$\varphi(m, n)^2 = (m \wedge m) \otimes 1 + m \otimes n + (-1)^{1 \cdot 1} m \otimes n + 1 \otimes (n \wedge n) = 0$$

the two middle terms coming from $(m \otimes 1)(1 \otimes n)$ and $(1 \otimes n)(m \otimes 1)$ with signs $(-1)^0$ and $(-1)^1$, and the outer terms vanishing since $v \wedge v = 0$. So proposition 16.3.8 gives a unique R-algebra map $\Phi\colon \bigwedge(M \oplus N) \to T$ restricting to φ.

Step 3: a map back. The inclusions $M \hookrightarrow M \oplus N$ and $N \hookrightarrow M \oplus N$ induce algebra maps $\iota_M\colon \bigwedge M \to \bigwedge(M \oplus N)$ and ι_N, again by proposition 16.3.8, since the composites $M \to \bigwedge(M \oplus N)$

and $N \to \bigwedge(M \oplus N)$ square to zero. The assignment $(\alpha, \beta) \mapsto \iota_M(\alpha) \wedge \iota_N(\beta)$ is R-bilinear, hence induces $\Psi \colon T \to \bigwedge(M \oplus N)$ on the tensor product, and Ψ respects the twisted multiplication: moving $\iota_N(\beta)$ past $\iota_M(\alpha')$ costs exactly $(-1)^{\deg \beta \deg \alpha'}$ by proposition 16.3.9.

Step 4: the two composites are the identity. Both $\Psi \circ \Phi$ and $\Phi \circ \Psi$ are algebra maps, and on degree-one elements they are the identity: $\Psi(\Phi(m,n)) = \Psi(m \otimes 1) + \Psi(1 \otimes n) = m + n = (m,n)$, and similarly in the other direction. Since $\bigwedge(M \oplus N)$ is generated as an algebra by its degree-one part, the uniqueness clause of proposition 16.3.8 makes $\Psi \circ \Phi$ the identity. And T is generated by the elements $m \otimes 1$ and $1 \otimes n$, which gives the other. Both maps preserve total degree, so restricting to degree k gives the displayed isomorphism, as we sought to show.

Step 5: naturality. Let $u \colon M \to M'$ and $v \colon N \to N'$ be R-module maps, not necessarily isomorphisms. Naturality asserts that the square

$$\begin{array}{ccc} \bigoplus_{i+j=k} \bigwedge^i(M) \otimes_R \bigwedge^j(N) & \longrightarrow & \bigwedge^k(M \oplus N) \\ \downarrow & & \downarrow \\ \bigoplus_{i+j=k} \bigwedge^i(M') \otimes_R \bigwedge^j(N') & \longrightarrow & \bigwedge^k(M' \oplus N') \end{array}$$

commutes, the horizontal maps being $\alpha \otimes \beta \mapsto \alpha \wedge \beta$ and the vertical ones $\bigoplus_{i+j=k} \bigwedge^i(u) \otimes \bigwedge^j(v)$ and $\bigwedge^k(u \oplus v)$. Both composites are R-linear, so it is enough to check them on elements $\alpha \otimes \beta$ with $\alpha = m_1 \wedge \cdots \wedge m_i$ and $\beta = n_1 \wedge \cdots \wedge n_j$ decomposable, since these generate the left-hand module. Going right then down sends $\alpha \otimes \beta$ to $\bigwedge^k(u \oplus v)\big(m_1 \wedge \cdots \wedge m_i \wedge n_1 \wedge \cdots \wedge n_j\big) = u(m_1) \wedge \cdots \wedge u(m_i) \wedge v(n_1) \wedge \cdots \wedge v(n_j)$, and going down then right sends it to $\big(u(m_1) \wedge \cdots \wedge u(m_i)\big) \wedge \big(v(n_1) \wedge \cdots \wedge v(n_j)\big)$, which is the same element. So the square commutes.

Corollary 16.4.6: A Stably Free Module of Rank One Is Free

Let R be a commutative ring and P a finitely generated projective R-module with $P_{\mathfrak{p}} \cong R_{\mathfrak{p}}$ for every prime $\mathfrak{p}$. If $P \oplus R^n \cong R^{n+1}$ for some $n \geq 0$, then $P \cong R$.

***Proof*:**

First, $\bigwedge^i(P) = 0$ for $i \geq 2$. Exterior powers are quotients of tensor powers, so they commute with localization. Hence $\big(\bigwedge^i P\big)_{\mathfrak{p}} \cong \bigwedge^i(P_{\mathfrak{p}}) \cong \bigwedge^i(R_{\mathfrak{p}})$, which vanishes for $i \geq 2$ by theorem 16.4.3. A module vanishing at every prime is zero.

Now apply theorem 16.4.5 with $M = P$ and $N = R^n$ in degree $n+1$. The summand indexed by (i,j) vanishes unless $i \leq 1$ and $j \leq n$, the latter by theorem 16.4.3. Since $i + j = n + 1$, the only survivor is $(i,j) = (1,n)$. Hence

$$R \cong \bigwedge^{n+1}(R^{n+1}) \cong \bigwedge^{n+1}(P \oplus R^n) \cong \bigwedge^1(P) \otimes_R \bigwedge^n(R^n) \cong P$$

using theorem 16.4.3 for the two free top powers, as we sought to show.

The top exterior power of a free module of rank n is free of rank one, so an endomorphism of M acts on it by multiplication by a scalar. That scalar is the determinant, and this is the basis-free description of it.

Theorem 16.4.7: The Determinant as the Top Exterior Power

Let R be a commutative ring, let M be free of rank n, and let $\varphi\colon M \to M$ be an R-module homomorphism with matrix A with respect to some basis. Then

$$\bigwedge^n(\varphi)\colon \bigwedge^n(M) \to \bigwedge^n(M)$$

is multiplication by $\det(A)$. Consequently $\det(A)$ does not depend on the chosen basis, and for endomorphisms φ, ψ of M,

$$\det(\varphi \circ \psi) = \det(\varphi)\det(\psi)$$

***Proof*:**
Let $e_1, \dots, e_n$ be the basis and $A = (a_{ij})$, so $\varphi(e_j) = \sum_i a_{ij} e_i$. By theorem 16.4.3 the module $\bigwedge^n(M)$ is free of rank one on $\omega := e_1 \wedge \cdots \wedge e_n$, so it suffices to compute $\bigwedge^n(\varphi)(\omega) = \varphi(e_1) \wedge \cdots \wedge \varphi(e_n)$.

Expanding each factor and using multilinearity of the wedge,

$$\varphi(e_1) \wedge \cdots \wedge \varphi(e_n) = \sum_f \left(\prod_{j=1}^{n} a_{f(j)\,j} \right) e_{f(1)} \wedge \cdots \wedge e_{f(n)}$$

the sum running over all functions $f\colon \{1, \dots, n\} \to \{1, \dots, n\}$. If f is not injective then two factors coincide and the wedge vanishes by proposition 16.1.9. An injective self-map of a finite set is a permutation, so only $f = \sigma \in S_n$ contributes, and there $e_{\sigma(1)} \wedge \cdots \wedge e_{\sigma(n)} = (-1)^\sigma \omega$ by the same proposition. Therefore

$$\bigwedge^n(\varphi)(\omega) = \left(\sum_{\sigma \in S_n} (-1)^\sigma \prod_{j=1}^{n} a_{\sigma(j)\,j} \right) \omega = \det(A^t)\,\omega = \det(A)\,\omega$$

the last step by proposition 13.4.6(2). Since $\bigwedge^n(\varphi)$ is defined from φ alone, with no basis involved, the scalar it multiplies by is an invariant of φ. So any two bases give matrices with the same determinant.

Finally $\bigwedge^n$ is functorial by theorem 15.2.2, so $\bigwedge^n(\varphi \circ \psi) = \bigwedge^n(\varphi) \circ \bigwedge^n(\psi)$, and composing two multiplications by scalars multiplies the scalars, completing the proof.

The multiplicativity just obtained is proposition 13.4.7 again, now with a reason attached rather than a computation: determinants multiply because $\bigwedge^n$ is a functor into rank-one modules, and endomorphisms of a rank-one free module compose by multiplying their scalars. The Leibniz sum of definition 13.4.5 is what the scalar looks like once a basis is fixed.

Corollary 16.4.8: Invertibility and the Top Exterior Power

Let M be free of rank n over a commutative ring R and φ an endomorphism of M. Then φ is an isomorphism if and only if $\det(\varphi)$ is a unit of R.

***Proof*:**
If φ is an isomorphism with inverse ψ, then $\det(\varphi)\det(\psi) = \det(\mathrm{id}_M) = 1$ by theorem 16.4.7 and proposition 13.4.6(1), so $\det(\varphi)$ is a unit. The converse is proposition 13.4.12(1) applied to the matrix of φ, completing the proof.

Exercise 16.4.1

1. Let M be free of rank n and let $v_1, \dots, v_k \in M$ be linearly dependent. Show that $v_1 \wedge \cdots \wedge v_k = 0$ in $\bigwedge^k(M)$.
2. Let F be a field with $-1 \neq 1$ and V a finite-dimensional F-vector space. Show that $V \otimes_F V \cong \mathcal{S}^2(V) \oplus \bigwedge^2(V)$, and check the dimensions against theorem 16.4.2 and theorem 16.4.3.
3. Verify that $\mathcal{T}^k$, $\mathcal{S}^k$ and $\bigwedge^k$ of a free module of rank n have ranks n^k, $\binom{n+k-1}{k}$ and $\binom{n}{k}$, and show that for $k = 2$ the last two add up to the first. Then show the analogous identity fails for $k = 3$ by computing all three ranks at $n = 2$.
4. Let M be free of rank n and φ an endomorphism. Show that $\bigwedge^k(\varphi)$ acts on the basis of theorem 16.4.3 by the $k \times k$ minors of the matrix of φ, and deduce that φ has rank less than k if and only if $\bigwedge^k(\varphi) = 0$ when R is a field.
5. Let M be free of finite rank n. Show that $\bigwedge(M)$ is finitely generated as an R-module, of rank 2^n, while $\mathcal{S}(M)$ is not finitely generated as an R-module unless $M = 0$.
6. Show that theorem 16.4.3 needs freeness. Take $R = \mathbb{Z}$ and $M = \mathbb{Z}/2\mathbb{Z} \oplus \mathbb{Z}/2\mathbb{Z}$, compute $\mathcal{T}^2(M)$ using theorem 15.2.3, identify the submodule $A^2(M)$, and conclude that $\bigwedge^2(M) \cong \mathbb{Z}/2\mathbb{Z}$. Compare with what $\binom{2}{2} = 1$ would predict for a free module of rank 2.

16.5⋆ The Koszul Complex

⋆ *Optional. This section is an aside; nothing later in the book depends on it.*

The exterior powers of a free module assemble into a complex, and when the ring elements defining it are well enough behaved that complex is exact. The result is the standard source of free resolutions in commutative algebra, and it is the first place where the exterior algebra does work that has nothing to do with volume or orientation.

Nothing later in this book uses it. It is here because the books that do use it are downstream: *Commutative Algebra* [16] and *Homological Algebra* [19] both build on the complex constructed below, and this is the last point in the prerequisite chain at which the exterior powers are available and the statement can be made.

Two definitions are needed, and neither has appeared so far.

Definition 16.5.1: Regular Sequence

Let R be a commutative ring. A sequence $a_1, \dots, a_n \in R$ is a *regular sequence* if

1. a_1 is not a zero divisor in R, and for each $i \geq 2$ the image of a_i is not a zero divisor in $R/(a_1, \dots, a_{i-1})$;
2. $(a_1, \dots, a_n) \neq R$.

An ideal generated by a regular sequence is called a *regular ideal*.

Condition (1) is the one that does the work, and it says the a_i impose genuinely new conditions one after another: each is a non-zero-divisor on the quotient by the previous ones. In $R = k[x_1, \dots, x_n]$ the variables themselves form a regular sequence, since $k[x_1, \dots, x_n]/(x_1, \dots, x_{i-1}) \cong k[x_i, \dots, x_n]$ is a domain in which $x_i \neq 0$. Note that an initial segment $a_1, \dots, a_m$ of a regular sequence is again one, directly from the definition. This is what makes induction on n available below.

The object the complex will produce is a free resolution in the sense of definition 12.3.27: an exact chain complex $\cdots \to F_1 \to F_0 \to M \to 0$ with every F_i free. Every module admits one, since a module is a quotient of a free module and the kernel is again a module. The content of a particular resolution is that it is short, explicit, or both. Over a principal ideal domain chapter 14 produced very short ones, because a submodule of a free module over a PID is free, so the process stops after one step. Over a general ring the process need not stop, and the Koszul complex is an explicit finite resolution available whenever the defining elements form a regular sequence.

The differential is contraction against a linear functional, and it exists for the same reason everything in this chapter exists: it is an alternating map, so theorem 16.1.8 builds it.

Proposition 16.5.2: Contraction Against a Functional

Let R be a commutative ring, F an R-module and $\lambda \in \mathrm{Hom}_R(F, R)$. For each $r \geq 1$ there is a unique R-module homomorphism

$$d_r \colon \bigwedge^{r}(F) \longrightarrow \bigwedge^{r-1}(F)$$

with

$$d_r(v_1 \wedge \cdots \wedge v_r) = \sum_{j=1}^{r} (-1)^{j-1} \lambda(v_j)\; v_1 \wedge \cdots \wedge \widehat{v_j} \wedge \cdots \wedge v_r$$

where $\widehat{v_j}$ means that the factor is omitted. Setting $d_0 = 0$, these satisfy the graded Leibniz rule

$$d(u \wedge w) = d(u) \wedge w + (-1)^p\, u \wedge d(w) \qquad \text{for } u \in \bigwedge^{p}(F)$$

***Proof*:**

Existence. The right-hand side, read as a function of $(v_1, \dots, v_r)$, is r-multilinear: each v_j appears once in each term, linearly, since λ is linear and $\wedge$ is multilinear.

It is alternating. Suppose $v_a = v_b =: v$ with $a < b$. Any term with $j \notin \{a, b\}$ still contains both

v_a and v_b among its factors, so it vanishes by proposition 16.1.9. The two remaining terms are

$$(-1)^{a-1}\lambda(v)\,W_a \qquad \text{and} \qquad (-1)^{b-1}\lambda(v)\,W_b$$

where W_a omits the factor in position a and W_b the one in position b. In W_a the surviving copy of v sits in position $b-1$, and in W_b it sits in position a. Moving it from one to the other is $b-1-a$ interchanges of adjacent factors, so $W_b = (-1)^{b-1-a}W_a$ by proposition 16.1.9. The sum of the two terms is therefore

$$\big((-1)^{a-1} + (-1)^{b-1}(-1)^{b-1-a}\big)\lambda(v)W_a = \big((-1)^{a-1} + (-1)^{a}\big)\lambda(v)W_a = 0$$

So the map is alternating, and theorem 16.1.8 gives d_r and its uniqueness.

Leibniz rule. Both sides are R-bilinear in (u, w), so it suffices to check them on products of degree-one elements, which generate each factor. For $u = v_1 \wedge \cdots \wedge v_p$ and $w = v_{p+1} \wedge \cdots \wedge v_{p+q}$, the defining formula for $d(u \wedge w)$ has $p+q$ terms. The first p of them are exactly the terms of $d(u) \wedge w$. The last q are the terms of $u \wedge d(w)$, except that in $d(u \wedge w)$ the term omitting v_{p+j} carries the sign $(-1)^{p+j-1}$ while in $u \wedge d(w)$ it carries $(-1)^{j-1}$. The two differ by $(-1)^p$, which is the stated factor, completing the proof.

Definition 16.5.3: Koszul Complex

Let R be a commutative ring and $a_1, \dots, a_n \in R$. Let $F = R^n$ with basis $e_1, \dots, e_n$ and let $\lambda\colon F \to R$ be the R-module homomorphism with $\lambda(e_i) = a_i$. The *Koszul complex* $K_\bullet(a_1, \dots, a_n)$ is

$$0 \longrightarrow \bigwedge^{n}(F) \xrightarrow{d_n} \bigwedge^{n-1}(F) \xrightarrow{d_{n-1}} \cdots \xrightarrow{d_2} \bigwedge^{1}(F) \xrightarrow{d_1} \bigwedge^{0}(F) = R$$

with d_r the contraction of proposition 16.5.2. On basis elements,

$$d_r\big(e_{i_1} \wedge \cdots \wedge e_{i_r}\big) = \sum_{j=1}^{r} (-1)^{j-1} a_{i_j}\; e_{i_1} \wedge \cdots \wedge \widehat{e_{i_j}} \wedge \cdots \wedge e_{i_r}$$

Proposition 16.5.4: The Koszul Complex Is a Complex

With the notation above, $d_{r-1} \circ d_r = 0$ for every $r \geq 1$, so $\operatorname{im}(d_r) \subseteq \ker(d_{r-1})$.

***Proof*:**
Both composites are R-linear, so it suffices to evaluate on a basis element $e_I = e_{i_1} \wedge \cdots \wedge e_{i_r}$ with $i_1 < \cdots < i_r$. Applying d_r and then d_{r-1} produces a sum of terms indexed by ordered pairs of distinct positions (j, k), the term being $\pm\, a_{i_j} a_{i_k}$ times the wedge with both factors omitted. Fix $j < k$ and collect the two terms that omit those two positions.

Removing position j first contributes the sign $(-1)^{j-1}$. In the shortened wedge the factor originally at position k now sits at position $k-1$, so removing it contributes $(-1)^{k-2}$, for a total of $(-1)^{j+k-3} = (-1)^{j+k-1}$. Removing position k first contributes $(-1)^{k-1}$, and the factor at position j is unmoved, so removing it contributes $(-1)^{j-1}$, for a total of $(-1)^{j+k-2} = (-1)^{j+k}$.

The two signs are opposite and the coefficients $a_{i_j}a_{i_k}$ agree because R is commutative, so the two terms cancel. Every term cancels in such a pair, completing the proof.

Exactness is where the hypothesis on the a_i enters, and it is the reason regular sequences were defined.

Theorem 16.5.5: Exactness of the Koszul Complex

Let R be a commutative ring and let $a_1, \dots, a_n$ be a regular sequence, with $I = (a_1, \dots, a_n)$. Then $\ker(d_r) = \operatorname{im}(d_{r+1})$ for every $r \geq 1$, and $R/\operatorname{im}(d_1) = R/I$. Consequently

$$0 \longrightarrow \bigwedge^{n}(F) \longrightarrow \cdots \longrightarrow \bigwedge^{1}(F) \longrightarrow R \longrightarrow R/I \longrightarrow 0$$

is a finite free resolution of R/I, of length n.

***Proof*:**
The statement about $R/\operatorname{im}(d_1)$ holds for any a_i: $d_1 = \lambda$ has image $\lambda(F) = (a_1, \dots, a_n) = I$. Freeness of each term is theorem 16.4.3. So only exactness at $r \geq 1$ is at issue, and it is proved by induction on n.

Base case $n = 1$. The complex is $0 \to R \xrightarrow{a_1} R$, and exactness at degree 1 says that multiplication by a_1 is injective, which is the statement that a_1 is not a zero divisor.

Inductive step. Let $n \geq 2$, let $F' \subseteq F$ be free on $e_1, \dots, e_{n-1}$ and let $K'_\bullet$ be the Koszul complex of $a_1, \dots, a_{n-1}$ on F', with differentials d'. An initial segment of a regular sequence is regular, so the inductive hypothesis applies to K'.

By theorem 16.4.3 every basis element of $\bigwedge^r(F)$ either involves e_n or does not, which gives

$$\bigwedge^{r}(F) = \bigwedge^{r}(F') \;\oplus\; \Big(\bigwedge^{r-1}(F') \wedge e_n\Big)$$

so every element is uniquely $x + y \wedge e_n$ with $x \in \bigwedge^r(F')$ and $y \in \bigwedge^{r-1}(F')$. The Leibniz rule of proposition 16.5.2, together with $d(e_n) = a_n$, computes the differential in these coordinates:

$$d_r\big(x + y \wedge e_n\big) = \Big(d'_r(x) + (-1)^{r-1} a_n\, y\Big) + \Big(d'_{r-1}(y)\Big) \wedge e_n \tag{16.1}$$

Suppose now $r \geq 1$ and $d_r(x + y \wedge e_n) = 0$. Comparing the two summands,

$$d'_{r-1}(y) = 0 \qquad \text{and} \qquad d'_r(x) + (-1)^{r-1} a_n y = 0$$

Producing v with $d'_r(v) = y$. If $r \geq 2$ then $r - 1 \geq 1$, so the inductive hypothesis gives exactness of K' at degree $r-1$ and $y = d'_r(v)$ for some $v \in \bigwedge^r(F')$. If $r = 1$ then $y \in \bigwedge^0(F') = R$ and the first equation is vacuous, so a different argument is needed. There the second equation reads $\lambda'(x) + a_n y = 0$ with $\lambda' = d'_1$, so $a_n y \in (a_1, \dots, a_{n-1})$. Because $a_1, \dots, a_n$ is a regular sequence, a_n is not a zero divisor on $R/(a_1, \dots, a_{n-1})$, so $y \in (a_1, \dots, a_{n-1})$. Writing $y = \sum_{i<n} c_i a_i$ and setting $v = \sum_{i<n} c_i e_i \in \bigwedge^1(F')$ gives $d'_1(v) = y$.

Producing u. From $d'_r(x) = -(-1)^{r-1} a_n y = (-1)^r a_n\, d'_r(v)$ one gets

$$d'_r\big(x - (-1)^r a_n v\big) = 0$$

and $r \geq 1$, so the inductive hypothesis gives $x - (-1)^r a_n v = d'_{r+1}(u)$ for some $u \in \bigwedge^{r+1}(F')$.

Conclusion. Apply (16.1) in degree $r+1$ to $u + v \wedge e_n$:

$$d_{r+1}\big(u + v \wedge e_n\big) = \Big(d'_{r+1}(u) + (-1)^r a_n v\Big) + d'_r(v) \wedge e_n = x + y \wedge e_n$$

So every cycle in degree r is a boundary, which is exactness, completing the proof.

Example 16.5.6: The Koszul Complex of the Variables

Take $R = k[x_1, \dots, x_n]$ over a field k and the regular sequence $x_1, \dots, x_n$, so that $I = (x_1, \dots, x_n)$ and $R/I \cong k$. The theorem produces a free resolution of k by free R-modules of ranks $\binom{n}{k}$, of total length n:

$$0 \to R \to R^n \to R^{\binom{n}{2}} \to \cdots \to R^n \to R \to k \to 0$$

For $n = 1$ this is $0 \to k[x] \xrightarrow{x} k[x] \to k \to 0$, which is the observation that x is not a zero divisor. For $n = 2$ it is

$$0 \to R \xrightarrow{\binom{-x_2}{x_1}} R^2 \xrightarrow{(x_1 \;\; x_2)} R \to k \to 0$$

and exactness in the middle says precisely that a pair (f, g) with $fx_1 + gx_2 = 0$ is a multiple of $(-x_2, x_1)$: the only relations among x_1 and x_2 are the obvious ones. Regularity is exactly what makes that true, and it fails for a non-regular sequence, as the exercises show.

The construction is the algebraic origin of several later ones. In *Homological Algebra* [19] the Koszul complex resolves the residue field in the proof of the Hilbert syzygy theorem, in *Commutative Algebra* [16] its homology measures how far a sequence is from being regular, and the same complex reappears in intersection theory, where the exactness above expresses that n hypersurfaces meeting in the expected dimension impose independent conditions.

Exercise 16.5.1

1. Show that x, y is a regular sequence in $k[x, y]$, and that x, x is not. Write out the Koszul complex for x, x and exhibit an element of $\ker(d_1)$ that is not in $\operatorname{im}(d_2)$, so that theorem 16.5.5 genuinely needs its hypothesis.

2. Show that a permutation of a regular sequence need not be regular, using $a_1 = x$, $a_2 = y(1-x)$, $a_3 = z(1-x)$ in $k[x, y, z]$. (Check that a_1, a_2, a_3 is regular but a_2, a_3, a_1 is not.)

3. Show that $d_n \colon \bigwedge^n(F) \to \bigwedge^{n-1}(F)$ is injective for a regular sequence, and identify $\bigwedge^n(F)$ and the image of d_n explicitly when $n = 2$.

4. Show that the Koszul complex of $a_1, \dots, a_n$ is, up to isomorphism, the tensor product over R of the n two-term complexes $0 \to R \xrightarrow{a_i} R \to 0$, in the sense that its degree-r term is the direct sum of the products of r of the left-hand copies with $n - r$ of the right-hand ones. (Only the module structure is asked for, not the differential.)

5. Write out the Koszul complex of x, y, z in $k[x, y, z]$ explicitly, giving the matrices of d_3, d_2 and d_1 with respect to the bases of theorem 16.4.3, and verify $d_2 d_3 = 0$ and $d_1 d_2 = 0$ by direct computation.

16.6 Summary

One module M, four classes of map on its powers, and four objects representing them. The tables below state what was established, each entry pointing at the result that establishes it.

The representing objects

Each class of multilinear map on M^k is represented by a module, and each representing object is a quotient of the tensor power by exactly what the class is required to kill.

Maps on M^k	Represented by	Universal property	Rank
k-multilinear	$\mathcal{T}^k(M)$	theorem 15.2.11	n^k
symmetric k-multilinear	$\mathcal{S}^k(M)$	theorem 16.1.6	$\binom{n+k-1}{k}$
alternating k-multilinear	$\bigwedge^k(M)$	theorem 16.1.8	$\binom{n}{k}$

The ranks are for M free of rank n, and are theorem 16.4.1, theorem 16.4.2 and theorem 16.4.3 respectively. In particular $\bigwedge^k(M) = 0$ once $k > n$, and $\bigwedge^n(M)$ is free of rank one, which is what makes the determinant a scalar.

The graded algebras

Summing the powers over k turns each family into a ring, graded in the sense of definition 16.2.1, and each acquires a universal property about homomorphisms into an algebra.

Algebra	Construction	Universal for maps into	Reference
$\mathcal{T}(M)$	$\bigoplus_k \mathcal{T}^k(M)$	any R-algebra	theorem 16.3.2
$\mathcal{S}(M)$	$\mathcal{T}(M)/\mathcal{C}(M)$	commutative R-algebras	theorem 16.3.6
$\bigwedge(M)$	$\mathcal{T}(M)/\mathcal{A}(M)$	algebras where $\varphi(m)^2 = 0$	proposition 16.3.8

That the two routes to $\mathcal{S}^k(M)$ and $\bigwedge^k(M)$ agree, one quotienting $\mathcal{T}^k(M)$ by a submodule and the other quotienting $\mathcal{T}(M)$ by an ideal, is lemma 16.3.3 and corollary 16.3.4. For M free of rank n the three algebras are the free associative algebra on n generators, the polynomial ring $R[x_1, \dots, x_n]$ (theorem 16.4.2), and a free module of rank 2^n (theorem 16.4.3).

Graded rings

Statement	Reference
$1 \in S_0$; S_0 is a subring; S_+ is an ideal with $S/S_+ \cong S_0$	proposition 16.2.2
Graded $\Leftrightarrow$ components in I $\Leftrightarrow$ homogeneous generators $\Leftrightarrow$ a graded kernel	proposition 16.2.7
S/I is graded, with $(S/I)_k \cong S_k/I_k$	proposition 16.2.5
$\ker\varphi$ and $\mathrm{im}\varphi$ decompose degree by degree	proposition 16.2.6
Primality is testable on homogeneous elements	corollary 16.2.8
$I + J$, IJ, $I \cap J$ and $\sqrt{I}$ are graded	proposition 16.2.9

Signs

The exterior algebra is governed by signs, and three separate statements are involved, which are easy to conflate.

Statement	Reference
Alternating $\Rightarrow$ the sign rule; the converse needs 2 invertible	lemma 16.1.4
$m_{\sigma(1)} \wedge \cdots \wedge m_{\sigma(k)} = (-1)^{\sigma} m_1 \wedge \cdots \wedge m_k$	proposition 16.1.9
$a \wedge b = (-1)^{pq} b \wedge a$ for $a \in \bigwedge^p$, $b \in \bigwedge^q$	proposition 16.3.9
$\mathcal{S}^k(M)$ and $\bigwedge^k(M)$ sit inside $\mathcal{T}^k(M)$ when $k!$ is invertible	proposition 16.1.12

In characteristic 2 the first row is strict: a symmetric bilinear form satisfies the sign rule vacuously without being alternating, so the vanishing condition of definition 16.1.3 is the one the exterior power is built from.

The determinant, and what fails

Statement	Reference
$\bigwedge^n(\varphi)$ is multiplication by $\det(\varphi)$	theorem 16.4.7
det is independent of the basis, and multiplicative	same
φ is an isomorphism $\Leftrightarrow$ $\det(\varphi)$ is a unit	corollary 16.4.8
$\bigwedge^k$ does not preserve injections	example 16.4.4(4)
$\bigwedge^k(M)$ need not be free when M is not	theorem 16.4.3

The second row is the conceptual reading of multiplicativity: determinants multiply because $\bigwedge^n$ is a functor landing in rank-one modules, where composition is multiplication of scalars. The elementary proof from the defining sum is proposition 13.4.7.

The exterior powers of a free module also assemble into a complex, which resolves $R/(a_1, \ldots, a_n)$ whenever the a_i are regular. That construction is section 16.5$\star$, marked as skippable for a reader of this book, and is the form in which *Commutative Algebra* [16] and *Homological Algebra* [19] take up the exterior algebra.

17⋆

Projective, Injective and Flat Modules

⋆ *Optional. This chapter is an aside; nothing later in the book depends on it.*

Given two R-modules A and C, which modules M fit into a short exact sequence

$$0 \to A \to M \to C \to 0\ ?$$

This is the *extension problem*. Section 3.6 asked it for groups and left it open, and theorem 3.7.9 is exactly the reason it matters: the Jordan-Hölder theorem pins down the composition factors of a finite group without saying how they are assembled. For modules there is always at least one answer, $M = A \oplus C$, and for $A = \mathbb{Z}$ and $C = \mathbb{Z}/n\mathbb{Z}$ there are exactly two.

Exactness itself is the tool. A short exact sequence records how M is assembled from A and C, and the three constructions already available ($\mathrm{Hom}_R(D, -)$, $\mathrm{Hom}_R(-, D)$ and $- \otimes_R D$) each turn such a sequence into one that is exact everywhere except at a single spot. Whether the failure at that spot vanishes depends on the module D, and the modules for which it does are the projective, injective and flat modules of the title. What measures the failure when it does not vanish is a module $\mathrm{Ext}^1_R(C, A)$, whose elements turn out to be in bijection with the answers to the question above.

17.1 Exact Sequences, Splitting, and the Extension Problem

Definition 12.2.11 already said what it means for a chain of R-module homomorphisms to be exact: the image of each map is the kernel of the next. Two degenerate cases fix how short an exact sequence can be before it stops carrying information.

Proposition 17.1.1: Injective and Surjective Maps in Exact Sequences

Let A, B and C be R-modules.

1. The sequence $0 \to A \xrightarrow{\psi} B$ is exact at A if and only if ψ is injective.
2. The sequence $B \xrightarrow{\varphi} C \to 0$ is exact at C if and only if φ is surjective.

***Proof*:**

1. The only homomorphism $\alpha\colon 0 \to A$ sends $0 \mapsto 0$, so $\operatorname{im}(\alpha) = 0$. Exactness at A says $\operatorname{im}(\alpha) = \ker(\psi)$, which is therefore the assertion $\ker(\psi) = 0$, and a homomorphism is injective exactly when its kernel is trivial.
2. The only homomorphism $\beta\colon C \to 0$ sends every c to 0, so $\ker(\beta) = C$. Exactness at C says $\operatorname{im}(\varphi) = \ker(\beta) = C$, which is the assertion that φ is surjective.

Consequently $0 \to A \xrightarrow{f} B \to 0$ is exact precisely when f is both injective and surjective, so A and B are isomorphic and nothing has been learned. The first length that carries information is one longer.

Definition 17.1.2: Short Exact Sequence

A *short exact sequence* of R-modules is an exact sequence

$$0 \to A \xrightarrow{\psi} B \xrightarrow{\varphi} C \to 0$$

In this situation B is called an *extension of C by A*.

Unwinding proposition 17.1.1, the displayed sequence is short exact if and only if ψ is injective, φ is surjective and $\operatorname{im}(\psi) = \ker(\varphi)$. The first isomorphism theorem (theorem 12.2.10) then gives $C \cong B/\psi(A)$, while ψ identifies A with the submodule $\psi(A)$. An extension of C by A is therefore a module B containing a copy of A whose quotient by that copy is C, packaged in a form that names the two maps along with the three modules.

17.1.3 The Two Extension Conventions Definition 3.6.3 called G an extension of A by B when $G/A \cong B$, which is the order used in classical group theory and is the reverse of the order fixed above. The convention attached to short exact sequences is the one used for the rest of this chapter: the module being extended is the quotient C, and the module extending it is the submodule A. Both orders are current in the literature, and the safe reading is always the sequence itself.

Example 17.1.4: Short Exact Sequences

1. For any A and C the sequence
$$0 \to A \xrightarrow{\iota} A \oplus C \xrightarrow{\pi} C \to 0, \qquad \iota(a) = (a, 0), \quad \pi(a, c) = c$$
is short exact, since ι is injective, π is surjective, and $\operatorname{im}(\iota) = A \oplus 0 = \ker(\pi)$. Every pair of

modules therefore admits at least one extension.

2. Any homomorphism $\varphi\colon B \to C$ yields
$$0 \to \ker(\varphi) \hookrightarrow B \xrightarrow{\varphi} \operatorname{im}(\varphi) \to 0$$
and when φ is surjective the right-hand term is C itself.

3. Let M be an R-module and S a generating set for M. The homomorphism $\varphi\colon F(S) \to M$ extending the inclusion of S (theorem 12.3.11) is surjective, so with $K = \ker(\varphi)$,
$$0 \to K \hookrightarrow F(S) \xrightarrow{\varphi} M \to 0$$
is short exact. Every module is thus a quotient of a free module, and $F(S)$ is an extension of M by the module K of relations holding among the chosen generators.

Exactness is not confined to modules, nor to length three.

Example 17.1.5: The Centre and the Outer Automorphisms
Section 4.7.1 described the centre $Z(G)$ of a group and its outer automorphism group $\operatorname{Out}(G)$ as opposite ends of one construction, and exactness states this precisely. Let $\sigma\colon G \to \operatorname{Aut}(G)$ send g to conjugation by g, so that $\operatorname{im}(\sigma) = \operatorname{Inn}(G)$ (definition 4.7.3) and $\ker(\sigma) = Z(G)$, and let ρ be the quotient map by $\operatorname{Inn}(G)$, which is normal in $\operatorname{Aut}(G)$ by proposition 4.7.8. Then
$$1 \longrightarrow Z(G) \hookrightarrow G \xrightarrow{\sigma} \operatorname{Aut}(G) \xrightarrow{\rho} \operatorname{Out}(G) \longrightarrow 1$$
is exact: at $Z(G)$ because the inclusion is injective, at G because $\ker(\sigma) = Z(G)$, at $\operatorname{Aut}(G)$ because $\ker(\rho) = \operatorname{Inn}(G) = \operatorname{im}(\sigma)$, and at $\operatorname{Out}(G)$ because ρ is surjective. Cutting the sequence in the middle recovers theorem 4.7.4, the isomorphism $G/Z(G) \cong \operatorname{Inn}(G)$.

Example 17.1.4(1) makes $A \oplus C$ available for every pair. It is not always the only answer.

Example 17.1.6: Two Extensions of $\mathbb{Z}/n\mathbb{Z}$ by $\mathbb{Z}$
Fix $n \geq 2$. Two short exact sequences of $\mathbb{Z}$-modules with the same outer terms are
$$0 \to \mathbb{Z} \xrightarrow{\iota} \mathbb{Z} \oplus \mathbb{Z}/n\mathbb{Z} \xrightarrow{\pi} \mathbb{Z}/n\mathbb{Z} \to 0 \qquad \text{and} \qquad 0 \to \mathbb{Z} \xrightarrow{\cdot n} \mathbb{Z} \xrightarrow{p} \mathbb{Z}/n\mathbb{Z} \to 0$$
where $\iota(z) = (z, \overline{0})$ and $\pi(z, \overline{a}) = \overline{a}$, while $\cdot n$ is multiplication by n and p is reduction modulo n. The first is short exact by example 17.1.4(1). The second is short exact because multiplication by n is injective on $\mathbb{Z}$, reduction is surjective, and $\operatorname{im}(\cdot n) = n\mathbb{Z} = \ker(p)$.

The two middle terms are not isomorphic, since $\mathbb{Z} \oplus \mathbb{Z}/n\mathbb{Z}$ has an element of order n and $\mathbb{Z}$ has no element of finite order other than 0. So $\mathbb{Z}/n\mathbb{Z}$ has at least two genuinely different extensions by $\mathbb{Z}$.

The same phenomenon occurs among finite modules. Both
$$0 \to \mathbb{Z}/2\mathbb{Z} \xrightarrow{\overline{a} \mapsto \overline{2a}} \mathbb{Z}/4\mathbb{Z} \xrightarrow{\bmod 2} \mathbb{Z}/2\mathbb{Z} \to 0$$
$$0 \to \mathbb{Z}/2\mathbb{Z} \xrightarrow{\iota} \mathbb{Z}/2\mathbb{Z} \oplus \mathbb{Z}/2\mathbb{Z} \xrightarrow{\pi} \mathbb{Z}/2\mathbb{Z} \to 0$$
are short exact (for the first, the injection has image $\{\overline{0}, \overline{2}\}$, which is exactly the kernel of reduction modulo 2), and $\mathbb{Z}/4\mathbb{Z} \ncong \mathbb{Z}/2\mathbb{Z} \oplus \mathbb{Z}/2\mathbb{Z}$.

17.1.1 Maps Between Short Exact Sequences

Two extensions of the same pair can differ, and there is more than one reasonable sense in which two of them are the same.

Definition 17.1.7: Homomorphism and Equivalence of Short Exact Sequences

Let

$$0 \to A \xrightarrow{\psi} B \xrightarrow{\varphi} C \to 0 \qquad \text{and} \qquad 0 \to A' \xrightarrow{\psi'} B' \xrightarrow{\varphi'} C' \to 0$$

be short exact sequences of R-modules.

1. A *homomorphism of short exact sequences* from the first to the second is a triple (α, β, γ) of R-module homomorphisms making

$$\begin{array}{ccccccccc} 0 & \longrightarrow & A & \xrightarrow{\psi} & B & \xrightarrow{\varphi} & C & \longrightarrow & 0 \\ & & \downarrow{\scriptstyle \alpha} & & \downarrow{\scriptstyle \beta} & & \downarrow{\scriptstyle \gamma} & & \\ 0 & \longrightarrow & A' & \xrightarrow[\psi']{} & B' & \xrightarrow[\varphi']{} & C' & \longrightarrow & 0 \end{array}$$

   commute, that is $\beta \circ \psi = \psi' \circ \alpha$ and $\gamma \circ \varphi = \varphi' \circ \beta$.

2. The homomorphism is an *isomorphism of short exact sequences* if α, β and γ are all isomorphisms, and the extensions B and B' are then called *isomorphic*.

3. The two sequences are *equivalent* if $A = A'$ and $C = C'$ and there is an isomorphism as in (2) with $\alpha = \mathrm{id}_A$ and $\gamma = \mathrm{id}_C$. The extensions B and B' are then called *equivalent* extensions.

Equivalence is strictly stronger than isomorphism, and the gap between them is what the classification in section 17.6 counts. An isomorphism is free to rebuild the outer terms by any automorphisms it likes. An equivalence holds the outer terms fixed and asks only that the middles be matched compatibly. Two extensions of one fixed pair can accordingly be isomorphic as diagrams while remaining inequivalent as extensions.

Example 17.1.8: Homomorphisms, Isomorphisms and Equivalences

1. Let $n \mid m$ with $n, m > 1$ and put $k = m/n$. Take the second sequence of example 17.1.6 on top and, beneath it, the sequence built from $\ell(\overline{a}) = \overline{na}$ and reduction modulo n:

$$\begin{array}{ccccccccc} 0 & \longrightarrow & \mathbb{Z} & \xrightarrow{\cdot n} & \mathbb{Z} & \xrightarrow{p} & \mathbb{Z}/n\mathbb{Z} & \longrightarrow & 0 \\ & & \downarrow{\scriptstyle \alpha} & & \downarrow{\scriptstyle \beta} & & \downarrow{\scriptstyle \mathrm{id}} & & \\ 0 & \longrightarrow & \mathbb{Z}/k\mathbb{Z} & \xrightarrow[\ell]{} & \mathbb{Z}/m\mathbb{Z} & \xrightarrow[p']{} & \mathbb{Z}/n\mathbb{Z} & \longrightarrow & 0 \end{array}$$

   with α and β the reductions modulo k and modulo m. The bottom row is short exact: ℓ is well defined because $a \equiv a' \pmod{k}$ gives $na \equiv na' \pmod{m}$, injective because $na \equiv 0$

$(\text{mod } m = nk)$ gives $a \equiv 0 \pmod{k}$, and $\text{im}(\ell) = n\mathbb{Z}/m\mathbb{Z} = \ker(p')$. Both squares commute, since $\ell(\alpha(a)) = \overline{na} = \beta(na)$ and $p'(\beta(z)) = \overline{z} = \text{id}(p(z))$.

2. Let p be a prime and, for each $c \in \{1, \dots, p-1\}$, let

$$\mathcal{E}_c\colon \qquad 0 \to \mathbb{Z}/p\mathbb{Z} \xrightarrow{\psi} \mathbb{Z}/p^2\mathbb{Z} \xrightarrow{\varphi_c} \mathbb{Z}/p\mathbb{Z} \to 0, \qquad \psi(\overline{x}) = \overline{px}, \quad \varphi_c(\overline{y}) = \overline{cy}$$

   Each is short exact: ψ is well defined and injective because $px \equiv 0 \pmod{p^2}$ forces $x \equiv 0 \pmod{p}$; φ_c is surjective because c is invertible modulo p; and $\text{im}(\psi) = p\mathbb{Z}/p^2\mathbb{Z} = \ker(\varphi_c)$.

   Any two of them are isomorphic. Taking $\alpha = \text{id}$, $\beta = \text{id}$ and γ multiplication by $c'c^{-1}$ makes both squares of definition 17.1.7 commute, and γ is an automorphism of $\mathbb{Z}/p\mathbb{Z}$ because $c'c^{-1}$ is a unit modulo p.

   No two of them are equivalent. Every endomorphism β of $\mathbb{Z}/p^2\mathbb{Z}$ is multiplication by some t, so $\beta \circ \psi = \psi$ forces $tp \equiv p \pmod{p^2}$, that is $t \equiv 1 \pmod{p}$. Then $\varphi_{c'} \circ \beta$ is multiplication by $c't \equiv c' \pmod{p}$, and requiring it to equal φ_c forces $c' \equiv c \pmod{p}$.

   So $\mathcal{E}_1, \dots, \mathcal{E}_{p-1}$ are $p-1$ pairwise inequivalent extensions of $\mathbb{Z}/p\mathbb{Z}$ by $\mathbb{Z}/p\mathbb{Z}$, all isomorphic to one another, and none of them is $\mathbb{Z}/p\mathbb{Z} \oplus \mathbb{Z}/p\mathbb{Z}$.

3. The two extensions of $\mathbb{Z}/n\mathbb{Z}$ by $\mathbb{Z}$ in example 17.1.6 are not even isomorphic, since an isomorphism of short exact sequences carries B to B' isomorphically and the two middle terms are not isomorphic.

A homomorphism of short exact sequences constrains its own three components: the two outer maps already determine a great deal about the middle one.

Proposition 17.1.9: Short Five Lemma

Let (α, β, γ) be a homomorphism of short exact sequences, in the notation of definition 17.1.7. Then

1. if α and γ are injective, so is β;
2. if α and γ are surjective, so is β;
3. if α and γ are isomorphisms, so is β, and the two sequences are isomorphic.

***Proof*:**
This is the first diagram chase in the book and is written out element by element. A chase is an algorithm with three moves: push an element along an available arrow, use commutativity of a square to identify the two routes around it, and use exactness of a row to move *backwards* along an arrow once the element is known to lie in that arrow's image. The bookkeeping is done by redrawing the relevant square with elements in place of modules, and $\mapsto$ in place of $\to$.

Step 1: β is injective. Let $b \in B$ with $\beta(b) = 0$. The goal is $b = 0$. Chase b around the right-hand

square:

$$\begin{array}{ccc} b & \xmapsto{\varphi} & \varphi(b) \\ {\scriptstyle\beta}\downarrow & & \downarrow{\scriptstyle\gamma} \\ 0 & \xmapsto[\varphi']{} & 0 \end{array}$$

Down then right gives $\varphi'(\beta(b)) = \varphi'(0) = 0$, and right then down gives $\gamma(\varphi(b))$. The square commutes, so $\gamma(\varphi(b)) = 0$. And γ is injective, so $\varphi(b) = 0$, that is $b \in \ker(\varphi)$.

Exactness is what makes the next move available. The top row is exact at B, so $\ker(\varphi) = \operatorname{im}(\psi)$, and there is an $a \in A$ with $\psi(a) = b$. Knowing only that φ kills b, the chase has produced a preimage in A, which is where the hypothesis on α can be used. Now chase a around the left-hand square:

$$\begin{array}{ccc} a & \xmapsto{\psi} & b \\ {\scriptstyle\alpha}\downarrow & & \downarrow{\scriptstyle\beta} \\ \alpha(a) & \xmapsto[\psi']{} & 0 \end{array}$$

Commutativity gives $\psi'(\alpha(a)) = \beta(\psi(a)) = \beta(b) = 0$. The map ψ' is injective by proposition 17.1.1 applied to the bottom row, so $\alpha(a) = 0$, and α is injective by hypothesis, so $a = 0$. Therefore $b = \psi(a) = \psi(0) = 0$.

Step 2: β is surjective. Let $b' \in B'$. The goal is to produce a preimage under β. Both surjectivity hypotheses give a pullback. Since γ is surjective there is a $c \in C$ with $\gamma(c) = \varphi'(b')$, and since φ is surjective by proposition 17.1.1 there is a $b \in B$ with $\varphi(b) = c$. Chasing b gives

$$\begin{array}{ccc} b & \xmapsto{\varphi} & c \\ {\scriptstyle\beta}\downarrow & & \downarrow{\scriptstyle\gamma} \\ \beta(b) & \xmapsto[\varphi']{} & \varphi'(b') \end{array}$$

where the bottom right entry is computed by commutativity:

$$\varphi'(\beta(b)) = \gamma(\varphi(b)) = \gamma(c) = \varphi'(b')$$

Hence $\varphi'(b' - \beta(b)) = 0$, so $b' - \beta(b) \in \ker(\varphi') = \operatorname{im}(\psi')$ by exactness of the bottom row, and $b' - \beta(b) = \psi'(a')$ for some $a' \in A'$. Since α is surjective, $a' = \alpha(a)$ for some $a \in A$, and the left square gives

$$b' - \beta(b) = \psi'(\alpha(a)) = \beta(\psi(a))$$

Therefore $b' = \beta(b) + \beta(\psi(a)) = \beta\big(b + \psi(a)\big)$, so b' lies in the image of β.

Step 3. If α and γ are isomorphisms then Steps 1 and 2 both apply, so β is injective and surjective, hence an isomorphism, and (α, β, γ) is an isomorphism of short exact sequences by definition 17.1.7(2).

Neither chase used the R-action, only the underlying additive groups, and the single subtraction in Step 2 can be written multiplicatively as $b'\beta(b)^{-1}$. The proposition and its proof therefore hold verbatim for short exact sequences of groups, abelian or not.

The next result relates two short exact sequences by splicing their kernels and cokernels into one long exact sequence. Its proof runs the same three moves, and from here on a chase states which move it is making rather than tracking every element.

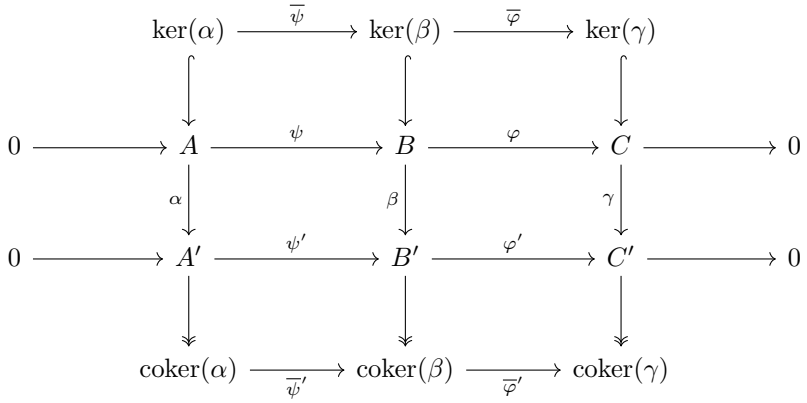
Proposition 17.1.10: Mini Snake Lemma

Let (α, β, γ) be a homomorphism of short exact sequences, in the notation of definition 17.1.7. Then there is an exact sequence of R-modules

$$0 \to \ker(\alpha) \xrightarrow{\overline{\psi}} \ker(\beta) \xrightarrow{\overline{\varphi}} \ker(\gamma) \xrightarrow{d} \operatorname{coker}(\alpha) \xrightarrow{\overline{\psi}'} \operatorname{coker}(\beta) \xrightarrow{\overline{\varphi}'} \operatorname{coker}(\gamma) \to 0$$

in which $\overline{\psi}$ and $\overline{\varphi}$ are the restrictions of ψ and φ, the maps $\overline{\psi}'$ and $\overline{\varphi}'$ are induced by ψ' and φ' on cokernels, and d is the *connecting homomorphism* constructed in the proof.

***Proof*:**
The whole statement is read off the following diagram, in which the two middle rows are the given ones and the outer rows are the sequence being constructed. The connecting homomorphism d is the one arrow not drawn: it runs from the top right corner $\ker(\gamma)$ to the bottom left corner $\operatorname{coker}(\alpha)$, and the route taken to build it (down the right column, left along the bottom row, then up the left column) is the snake the lemma is named for.



Step 1: the four induced maps exist. If $a \in \ker(\alpha)$ then $\beta(\psi(a)) = \psi'(\alpha(a)) = 0$, so ψ carries $\ker(\alpha)$ into $\ker(\beta)$. The same computation with the right square carries $\ker(\beta)$ into $\ker(\gamma)$. Dually $\psi'(\operatorname{im}(\alpha)) \subseteq \operatorname{im}(\beta)$, because $\psi'(\alpha(a)) = \beta(\psi(a))$, so ψ' descends to $\overline{\psi}' : \operatorname{coker}(\alpha) \to \operatorname{coker}(\beta)$. The same computation gives $\overline{\varphi}'$.

Step 2: construction of d. Let $c \in \ker(\gamma)$. Pull back along φ, which is surjective, to get $b \in B$ with $\varphi(b) = c$. Commutativity of the right square gives $\varphi'(\beta(b)) = \gamma(\varphi(b)) = \gamma(c) = 0$, so $\beta(b) \in \ker(\varphi') = \operatorname{im}(\psi')$ by exactness of the bottom row. Since ψ' is injective there is a *unique* $a' \in A'$ with $\psi'(a') = \beta(b)$. Define

$$d(c) := a' + \operatorname{im}(\alpha) \in \operatorname{coker}(\alpha)$$

The three steps just taken, namely pulling back along φ, crossing by β and pulling back along ψ', are the three segments of the route described above.

Step 3: d is a well defined homomorphism. The only choice made was the lift b. If $\varphi(b_1) = c$ as well then $b_1 - b \in \ker(\varphi) = \operatorname{im}(\psi)$, say $b_1 - b = \psi(a)$, so

$$\psi'(a_1') - \psi'(a') = \beta(b_1) - \beta(b) = \beta(\psi(a)) = \psi'(\alpha(a))$$

and injectivity of ψ' gives $a_1' - a' = \alpha(a) \in \operatorname{im}(\alpha)$. The two answers agree in $\operatorname{coker}(\alpha)$. Additivity and R-linearity follow because a lift of $c_1 + rc_2$ may be taken to be $b_1 + rb_2$, and ψ' and β are homomorphisms.

Step 4: exactness, at each of the six interior spots.

At $\ker(\alpha)$. The map $\overline{\psi}$ is a restriction of the injective ψ, so it is injective.

At $\ker(\beta)$. Since $\varphi \circ \psi = 0$ the composite $\overline{\varphi} \circ \overline{\psi}$ is zero, giving $\operatorname{im}(\overline{\psi}) \subseteq \ker(\overline{\varphi})$. Conversely let $b \in \ker(\beta)$ with $\varphi(b) = 0$. Exactness of the top row gives $b = \psi(a)$, and $\psi'(\alpha(a)) = \beta(\psi(a)) = \beta(b) = 0$, so $\alpha(a) = 0$ by injectivity of ψ'. Thus $a \in \ker(\alpha)$ and $b = \overline{\psi}(a)$.

At $\ker(\gamma)$. If $c = \overline{\varphi}(b)$ with $b \in \ker(\beta)$, then b is an admissible lift of c in Step 2 and $\beta(b) = 0 = \psi'(0)$, so $d(c) = 0 + \operatorname{im}(\alpha) = 0$. Conversely suppose $d(c) = 0$, so the element a' of Step 2 is $\alpha(a)$ for some $a \in A$. Then $\beta(b) = \psi'(\alpha(a)) = \beta(\psi(a))$, so $b - \psi(a) \in \ker(\beta)$, while $\varphi(b - \psi(a)) = \varphi(b) = c$. Hence $c = \overline{\varphi}(b - \psi(a))$.

At $\operatorname{coker}(\alpha)$. With c, b and a' as in Step 2,

$$\overline{\psi}'(d(c)) = \psi'(a') + \operatorname{im}(\beta) = \beta(b) + \operatorname{im}(\beta) = 0$$

so $\operatorname{im}(d) \subseteq \ker(\overline{\psi}')$. Conversely let $a' + \operatorname{im}(\alpha)$ lie in $\ker(\overline{\psi}')$, meaning $\psi'(a') = \beta(b)$ for some $b \in B$. Then $\gamma(\varphi(b)) = \varphi'(\beta(b)) = \varphi'(\psi'(a')) = 0$, so $c := \varphi(b)$ lies in $\ker(\gamma)$, and b is a lift of c exhibiting $d(c) = a' + \operatorname{im}(\alpha)$.

At $\operatorname{coker}(\beta)$. Since $\varphi' \circ \psi' = 0$ the composite $\overline{\varphi}' \circ \overline{\psi}'$ is zero. Conversely let $b' + \operatorname{im}(\beta)$ lie in $\ker(\overline{\varphi}')$, so $\varphi'(b') = \gamma(c)$ for some $c \in C$, and write $c = \varphi(b)$ using surjectivity of φ. Then $\varphi'(b' - \beta(b)) = \gamma(c) - \gamma(\varphi(b)) = 0$, so $b' - \beta(b) = \psi'(a')$ for some a', and $b' + \operatorname{im}(\beta) = \overline{\psi}'(a' + \operatorname{im}(\alpha))$.

At $\operatorname{coker}(\gamma)$. The map $\overline{\varphi}'$ is induced by the surjective φ', so it is surjective.

The name records that this is the three-module case. The general snake lemma replaces each of the three modules in a row by a whole chain complex and each kernel and cokernel by homology, and the six-term sequence above becomes a long exact sequence continuing in both directions. That version is [19, Snake Lemma]. The connecting homomorphism is what turns a short exact sequence into a longer one, and section 17.6 uses it to build the long exact sequences of Ext.

17.1.2 Split Sequences

Among all extensions of C by A, the direct sum $A \oplus C$ carries no information beyond its two ends. Example 17.1.6 showed that other extensions exist, so the ones that reduce to the direct sum deserve a name.

Definition 17.1.11: Split Short Exact Sequence

A short exact sequence $0 \to A \xrightarrow{\psi} B \xrightarrow{\varphi} C \to 0$ of R-modules *splits* if $\psi(A)$ has a complement in B, that is if there is a submodule $C' \subseteq B$ with

$$B = \psi(A) \oplus C'$$

The extension B is then called a *split extension* of C by A.

The complement is automatically a copy of C, so the definition need not require it. If $B = \psi(A) \oplus C'$, then φ restricted to C' is injective because $C' \cap \ker(\varphi) = C' \cap \psi(A) = 0$, and surjective because $\varphi(B) = \varphi(\psi(A)) + \varphi(C') = \varphi(C')$. Hence $B \cong \psi(A) \oplus C \cong A \oplus C$.

Definition 17.1.12: Section and Retraction

Let $0 \to A \xrightarrow{\psi} B \xrightarrow{\varphi} C \to 0$ be a short exact sequence of R-modules.

1. A *section* of φ is an R-module homomorphism $\mu\colon C \to B$ with $\varphi \circ \mu = \mathrm{id}_C$. It is also called a *splitting homomorphism* for the sequence.
2. A *retraction* of ψ is an R-module homomorphism $\lambda\colon B \to A$ with $\lambda \circ \psi = \mathrm{id}_A$.

Proposition 17.1.13: Splitting Criteria

For a short exact sequence $0 \to A \xrightarrow{\psi} B \xrightarrow{\varphi} C \to 0$ of R-modules the following are equivalent.

1. The sequence splits.
2. φ has a section $\mu\colon C \to B$.
3. ψ has a retraction $\lambda\colon B \to A$.

***Proof*:**
(1) $\Rightarrow$ (2). Let $B = \psi(A) \oplus C'$. As noted above φ restricts to an isomorphism $\varphi|_{C'} : C' \to C$, so its inverse followed by the inclusion $C' \subseteq B$ is a homomorphism $\mu\colon C \to B$, and $\varphi(\mu(c)) = c$ for every $c \in C$.

(2) $\Rightarrow$ (1). Let μ be a section and put $C' := \mu(C)$, a submodule of B. For any $b \in B$,

$$b = \big(b - \mu(\varphi(b))\big) + \mu(\varphi(b))$$

and $\varphi(b - \mu(\varphi(b))) = \varphi(b) - \varphi(b) = 0$, so the first summand lies in $\ker(\varphi) = \psi(A)$ while the second lies in C'. The sum is direct: if $\mu(c) \in \psi(A) = \ker(\varphi)$ then $0 = \varphi(\mu(c)) = c$, so $\mu(c) = 0$. Hence $B = \psi(A) \oplus C'$.

(1) $\Rightarrow$ (3). Let $B = \psi(A) \oplus C'$ and let $\pi_{\psi(A)}$ be the projection onto the first summand. The map ψ is an isomorphism onto $\psi(A)$ by proposition 17.1.1, so $\lambda := \psi^{-1} \circ \pi_{\psi(A)}$ is defined, and $\lambda(\psi(a)) = \psi^{-1}(\psi(a)) = a$.

(3) $\Rightarrow$ (1). Let λ be a retraction and put $C' := \ker(\lambda)$. For any $b \in B$,

$$b = \psi(\lambda(b)) + \big(b - \psi(\lambda(b))\big)$$

where the first summand lies in $\psi(A)$ and the second lies in C', since $\lambda(b-\psi(\lambda(b))) = \lambda(b)-\lambda(b) = 0$. The sum is direct: if $b = \psi(a)$ and $\lambda(b) = 0$ then $a = \lambda(\psi(a)) = \lambda(b) = 0$, so $b = 0$. Hence $B = \psi(A) \oplus C'$.

Example 17.1.14: Split and Non-split Sequences

1. The sequence $0 \to A \xrightarrow{\iota} A \oplus C \xrightarrow{\pi} C \to 0$ splits, with section $\mu(c) = (0, c)$.
2. Of the two sequences in example 17.1.6, the first splits and the second does not. A section of the second would be a nonzero homomorphism $\mathbb{Z}/n\mathbb{Z} \to \mathbb{Z}$, and the image z of $\overline{1}$ would satisfy $nz = 0$, forcing $z = 0$ and hence the whole map to be zero.
3. The sequence $0 \to \mathbb{Z}/2\mathbb{Z} \to \mathbb{Z}/4\mathbb{Z} \to \mathbb{Z}/2\mathbb{Z} \to 0$ does not split, since a splitting would give $\mathbb{Z}/4\mathbb{Z} \cong \mathbb{Z}/2\mathbb{Z} \oplus \mathbb{Z}/2\mathbb{Z}$, and only the left-hand side has an element of order 4.
4. Over $R = k[x]$ the sequence

   $$0 \to (x) \hookrightarrow k[x] \to k[x]/(x) \to 0$$

   does not split. A section would embed $k[x]/(x)$ as a nonzero submodule of $k[x]$ annihilated by x, and $k[x]$ is an integral domain, so no nonzero element is annihilated by x.

Splitting is a property of the sequence alone, and it is carried along by any isomorphism of sequences. Equivalence is not needed.

Proposition 17.1.15: Splitting Is an Isomorphism Invariant

Let (α, β, γ) be an isomorphism of short exact sequences, in the notation of definition 17.1.7. Then the top sequence splits if and only if the bottom one does.

***Proof*:**
Suppose the top sequence splits and let $\mu\colon C \to B$ be a section of φ, which exists by proposition 17.1.13. Set

$$\mu' := \beta \circ \mu \circ \gamma^{-1}\colon C' \to B'$$

a homomorphism because γ^{-1} is one. Commutativity of the right square reads $\varphi' \circ \beta = \gamma \circ \varphi$, so

$$\varphi' \circ \mu' = \varphi' \circ \beta \circ \mu \circ \gamma^{-1} = \gamma \circ \varphi \circ \mu \circ \gamma^{-1} = \gamma \circ \gamma^{-1} = \mathrm{id}_{C'}$$

Hence μ' is a section of φ' and the bottom sequence splits, again by proposition 17.1.13.

For the converse, note that $(\alpha^{-1}, \beta^{-1}, \gamma^{-1})$ is an isomorphism of short exact sequences from the bottom one to the top one: composing $\beta \circ \psi = \psi' \circ \alpha$ with β^{-1} on the left and α^{-1} on the right gives $\psi \circ \alpha^{-1} = \beta^{-1} \circ \psi'$, and the right square is handled the same way. Applying the first paragraph to it gives the converse.

The vocabulary is now sufficient to state the question this chapter opened with. Given R-modules A and C, write $\mathcal{E}(C, A)$ for the set of equivalence classes of extensions of C by A, in the sense of definition 17.1.7(3). This set is never empty, and all split extensions lie in a single class: if μ is a section of φ, then $B = \psi(A) \oplus \mu(C)$ by the proof of proposition 17.1.13, so $\beta(a, c) := \psi(a) + \mu(c)$ is an isomorphism $A \oplus C \to B$, both ψ and μ being injective. It satisfies $\beta(\iota(a)) = \psi(a)$ and $\varphi(\beta(a, c)) = c = \pi(a, c)$, so $(\mathrm{id}_A, \beta, \mathrm{id}_C)$ is an equivalence with the direct-sum sequence. By example 17.1.8(2) the set can have more than one element.

The *extension problem* for modules asks for $\mathcal{E}(C, A)$. Section 17.6 answers it by constructing a module $\mathrm{Ext}^1_R(C, A)$ together with a bijection $\mathrm{Ext}^1_R(C, A) \leftrightarrow \mathcal{E}(C, A)$ carrying 0 to the class of the split extension.

Exercise 17.1.1

1. For each of the following short exact sequences of $\mathbb{Z}$-modules, decide whether it splits. Exhibit a section when one exists, and prove that none exists otherwise.

   1. $0 \to \mathbb{Z} \xrightarrow{\cdot 3} \mathbb{Z} \to \mathbb{Z}/3\mathbb{Z} \to 0$
   2. $0 \to \mathbb{Z}/2\mathbb{Z} \to \mathbb{Z}/2\mathbb{Z} \oplus \mathbb{Z}/3\mathbb{Z} \to \mathbb{Z}/3\mathbb{Z} \to 0$, with the evident injection and projection
   3. $0 \to \mathbb{Z}/3\mathbb{Z} \to \mathbb{Z}/9\mathbb{Z} \to \mathbb{Z}/3\mathbb{Z} \to 0$, with $\overline{a} \mapsto \overline{3a}$ and reduction modulo 3
   4. $0 \to \mathbb{Z}/2\mathbb{Z} \to \mathbb{Z}/6\mathbb{Z} \to \mathbb{Z}/3\mathbb{Z} \to 0$, with $\overline{a} \mapsto \overline{3a}$ and reduction modulo 3

2. Let p be a prime. Show that every extension of $\mathbb{Z}/p\mathbb{Z}$ by $\mathbb{Z}/p\mathbb{Z}$ has a middle term of order p^2, hence isomorphic to $\mathbb{Z}/p^2\mathbb{Z}$ or to $\mathbb{Z}/p\mathbb{Z} \oplus \mathbb{Z}/p\mathbb{Z}$ by theorem 5.1.6, and that in the second case the sequence splits. In the first case, show the sequence is equivalent to one of the $\mathcal{E}_c$ of example 17.1.8(2). You may use that $\mathbb{Z}/p^2\mathbb{Z}$ has a unique subgroup of order p. Conclude that $\mathcal{E}(\mathbb{Z}/p\mathbb{Z}, \mathbb{Z}/p\mathbb{Z})$ has exactly p elements.

3. Let $m, n \geq 1$. Take both rows of definition 17.1.7 to be $0 \to \mathbb{Z} \xrightarrow{\cdot m} \mathbb{Z} \to \mathbb{Z}/m\mathbb{Z} \to 0$ and take α, β and γ all to be multiplication by n. Verify that this is a homomorphism of short exact sequences, identify all six terms of the sequence of proposition 17.1.10 in terms of $g = \gcd(m, n)$, and compute the connecting homomorphism d explicitly. Then specialise to $m = 4$ and $n = 6$.

4. Let (α, β, γ) be a homomorphism of short exact sequences.

   1. Show that if β is injective then so is α.
   2. Show that if β is surjective then so is γ.
   3. Give an example in which β is an isomorphism while α is not surjective and γ is not injective. *Hint*: one of the two rows may end in the zero module.

5. Show that a short exact sequence $0 \to A \to B \xrightarrow{\varphi} C \to 0$ of R-modules splits whenever C is a free module. *Hint*: choose a basis of C, lift each basis element along φ, and use theorem 12.3.11.

6. Show that if $0 \to A \xrightarrow{\psi} B \xrightarrow{\varphi} C \to 0$ splits then for every R-module D there are isomorphisms of abelian groups
$$\mathrm{Hom}_R(D, B) \cong \mathrm{Hom}_R(D, A) \oplus \mathrm{Hom}_R(D, C)$$
$$\mathrm{Hom}_R(B, D) \cong \mathrm{Hom}_R(A, D) \oplus \mathrm{Hom}_R(C, D)$$

Hint: use the section and the retraction given by proposition 17.1.13.

7. *(The five lemma.)* Consider a commutative diagram of R-modules

$$\begin{array}{ccccccccc} A_1 & \longrightarrow & A_2 & \longrightarrow & A_3 & \longrightarrow & A_4 & \longrightarrow & A_5 \\ \downarrow{\scriptstyle f_1} & & \downarrow{\scriptstyle f_2} & & \downarrow{\scriptstyle f_3} & & \downarrow{\scriptstyle f_4} & & \downarrow{\scriptstyle f_5} \\ B_1 & \longrightarrow & B_2 & \longrightarrow & B_3 & \longrightarrow & B_4 & \longrightarrow & B_5 \end{array}$$

with exact rows. Prove that if f_2 and f_4 are isomorphisms, f_1 is surjective and f_5 is injective, then f_3 is an isomorphism. *Hint*: run the two chases of proposition 17.1.9 one column further in each direction.

17.2 Projective Modules

Example 17.1.14 produced short exact sequences that do not split, so the extension problem has content. The instrument for measuring the failure is already available: fix an R-module D, apply $\mathrm{Hom}_R(D,-)$ to a short exact sequence, and see how much of the sequence survives.

Throughout, ψ_* denotes the map induced on $\mathrm{Hom}_R(D,-)$ by a homomorphism ψ, reserving the primes of definition 17.1.7 for the second row of a diagram.

Proposition 17.2.1: Left Exactness of $\mathrm{Hom}_R(D,-)$

Let D be an R-module.

1. For every R-module homomorphism $\psi\colon L \to M$, the assignment

$$\psi_*\colon \mathrm{Hom}_R(D,L) \to \mathrm{Hom}_R(D,M), \qquad f \mapsto \psi \circ f$$

is a homomorphism of abelian groups.

2. If $0 \to L \xrightarrow{\psi} M \xrightarrow{\varphi} N \to 0$ is a short exact sequence of R-modules, then

$$0 \to \mathrm{Hom}_R(D,L) \xrightarrow{\psi_*} \mathrm{Hom}_R(D,M) \xrightarrow{\varphi_*} \mathrm{Hom}_R(D,N)$$

is exact, for every D. In this situation one says that $\mathrm{Hom}_R(D,-)$ is *left exact*.

Note what part (2) does *not* claim: that φ_* is surjective, so that the sequence may be closed off with a $\to 0$ on the right. That claim is false in general, and the whole section grows out of the failure.

***Proof*:**
(1) For $f, g \in \mathrm{Hom}_R(D,L)$ and $d \in D$,

$$\big(\psi \circ (f+g)\big)(d) = \psi\big(f(d)+g(d)\big) = \psi(f(d)) + \psi(g(d)) = \big(\psi\circ f + \psi \circ g\big)(d)$$

using that ψ is a homomorphism, so $\psi_*(f+g) = \psi_*(f) + \psi_*(g)$.

(2) *Exactness at* $\mathrm{Hom}_R(D,L)$, which by proposition 17.1.1 is the injectivity of ψ_*. Suppose $\psi \circ f = 0$. Then $\psi(f(d)) = 0$ for every $d \in D$, so $f(d) \in \ker(\psi) = 0$ because ψ is injective, and $f = 0$.

Exactness at $\mathrm{Hom}_R(D, M)$, that is $\mathrm{im}(\psi_*) = \ker(\varphi_*)$.

($\subseteq$) For $f \in \mathrm{Hom}_R(D, L)$ we have $\varphi_*(\psi_*(f)) = \varphi \circ \psi \circ f$, and $\varphi \circ \psi = 0$ because $\mathrm{im}(\psi) = \ker(\varphi)$. So $\varphi_*(\psi_*(f)) = 0$.

($\supseteq$) Let $F \in \ker(\varphi_*)$, so $\varphi \circ F = 0$. For each $d \in D$ we have $\varphi(F(d)) = 0$, hence $F(d) \in \ker(\varphi) = \mathrm{im}(\psi)$, so there is an $l \in L$ with $\psi(l) = F(d)$. And l is unique because ψ is injective. Define $F' \colon D \to L$ by letting $F'(d)$ be that unique l, so that $\psi \circ F' = F$ by construction.

It remains to check that F' is a homomorphism, which is the only step here with content. For $d, e \in D$ and $r \in R$,

$$\psi\big(F'(d) + rF'(e)\big) = \psi(F'(d)) + r\psi(F'(e)) = F(d) + rF(e) = F(d + re) = \psi\big(F'(d + re)\big)$$

and ψ is injective, so $F'(d + re) = F'(d) + rF'(e)$. Hence $F' \in \mathrm{Hom}_R(D, L)$ and $F = \psi_*(F') \in \mathrm{im}(\psi_*)$.

Now the failure, computed before anything is named after it. The sequence at hand is the non-split extension of example 17.1.6 with $n = 2$.

Example 17.2.2: $\mathrm{Hom}_R(D, -)$ **Need Not Be Right Exact**
Take $R = \mathbb{Z}$, the short exact sequence

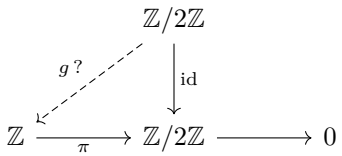
$$0 \to \mathbb{Z} \xrightarrow{\cdot 2} \mathbb{Z} \xrightarrow{\pi} \mathbb{Z}/2\mathbb{Z} \to 0$$

and $D = \mathbb{Z}/2\mathbb{Z}$. Then $\mathrm{Hom}_{\mathbb{Z}}(\mathbb{Z}/2\mathbb{Z}, \mathbb{Z}) = 0$: a homomorphism h sends $\overline{1}$ to some $z \in \mathbb{Z}$ with $2z = h(\overline{2}) = h(\overline{0}) = 0$, so $z = 0$. On the other hand $\mathrm{Hom}_{\mathbb{Z}}(\mathbb{Z}/2\mathbb{Z}, \mathbb{Z}/2\mathbb{Z}) \cong \mathbb{Z}/2\mathbb{Z}$, since such a homomorphism is determined by the image of $\overline{1}$ and both choices are legitimate. So applying $\mathrm{Hom}_{\mathbb{Z}}(D, -)$ gives

$$0 \to 0 \xrightarrow{(\cdot 2)_*} 0 \xrightarrow{\pi_*} \mathbb{Z}/2\mathbb{Z}$$

which is exact, as proposition 17.2.1 promised, and which cannot be extended by $\to 0$ on the right, since π_* is the zero map into a nonzero group.

The obstruction has a name in the language of diagrams. The element of $\mathrm{Hom}_{\mathbb{Z}}(\mathbb{Z}/2\mathbb{Z}, \mathbb{Z}/2\mathbb{Z})$ that is not hit is $\mathrm{id}_{\mathbb{Z}/2\mathbb{Z}}$, and saying it is not hit is saying that the dashed arrow below does not exist:



Indeed the only homomorphism $g \colon \mathbb{Z}/2\mathbb{Z} \to \mathbb{Z}$ is the zero map, and $\pi \circ 0 = 0 \neq \mathrm{id}_{\mathbb{Z}/2\mathbb{Z}}$.

A homomorphism out of D that fails to lift is exactly what stops $\mathrm{Hom}_R(D, -)$ from being exact. The modules for which this never happens are the subject of this section.

Definition 17.2.3: Projective Module

An R-module P is *projective* if the functor $\mathrm{Hom}_R(P, -)$ is exact, that is if for every short exact sequence $0 \to L \to M \to N \to 0$ of R-modules the sequence

$$0 \to \mathrm{Hom}_R(P, L) \to \mathrm{Hom}_R(P, M) \to \mathrm{Hom}_R(P, N) \to 0$$

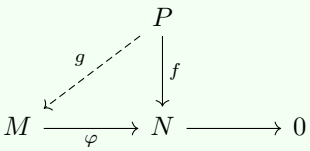
is again short exact.

By proposition 17.2.1 everything in that sequence is automatic except surjectivity on the right, so the definition asks for one thing only: that every homomorphism out of P lift along every surjection. The theorem below makes that precise and adds two further descriptions, one about splitting and one about free modules.

Theorem 17.2.4: Equivalencies for Projective Modules

Let P be an R-module. The following are equivalent.

1. P is projective.
2. *(Lifting.)* For all R-modules M and N, if $\varphi\colon M \to N$ is surjective, then every $f \in \mathrm{Hom}_R(P, N)$ has a lift $g \in \mathrm{Hom}_R(P, M)$ with $\varphi \circ g = f$:



3. *(Splitting.)* Every short exact sequence $0 \to L \to M \to P \to 0$ of R-modules splits.
4. *(Summand.)* P is isomorphic to a direct summand of a free R-module.

***Proof*:**

(1) $\Rightarrow$ (2). Let $\varphi\colon M \to N$ be surjective and put $L = \ker(\varphi)$, so that $0 \to L \to M \xrightarrow{\varphi} N \to 0$ is short exact by example 17.1.4(2). Condition (1) makes $\varphi_*\colon \mathrm{Hom}_R(P, M) \to \mathrm{Hom}_R(P, N)$ surjective, so a given f equals $\varphi_*(g) = \varphi \circ g$ for some g.

(2) $\Rightarrow$ (1). Proposition 17.2.1 gives exactness at the first two spots of the sequence in definition 17.2.3, and (2) applied to the surjection φ says precisely that φ_* is surjective.

(2) $\Rightarrow$ (3). Let $0 \to L \xrightarrow{\psi} M \xrightarrow{\varphi} P \to 0$ be short exact. Then φ is surjective, so applying (2) to $f = \mathrm{id}_P$ produces $\mu \in \mathrm{Hom}_R(P, M)$ with $\varphi \circ \mu = \mathrm{id}_P$. That is a section of φ, so the sequence splits by proposition 17.1.13.

(3) $\Rightarrow$ (4). By example 17.1.4(3), applied with $S = P$ as a generating set, there is a short exact sequence

$$0 \to K \hookrightarrow F \xrightarrow{\pi} P \to 0$$

with F free. By (3) it splits, so $F = K \oplus C'$ for some submodule C', and $\pi|_{C'} : C' \to P$ is an isomorphism by the remark following definition 17.1.11. Hence P is isomorphic to the direct

summand C' of F.

(4) $\Rightarrow$ (2). Suppose $F = P' \oplus K$ with F free and $P' \cong P$. Replacing P by P', we may assume P itself is a summand, and write $\iota\colon P \to F$ for the inclusion and $\rho\colon F \to P$ for the projection, so $\rho \circ \iota = \mathrm{id}_P$.

Let $\varphi\colon M \to N$ be surjective and $f \in \mathrm{Hom}_R(P, N)$. Let $(e_s)_{s\in S}$ be a basis of F. For each s the element $f(\rho(e_s))$ lies in N, so by surjectivity of φ we may choose $m_s \in M$ with $\varphi(m_s) = f(\rho(e_s))$. By theorem 12.3.11 there is a unique homomorphism $G\colon F \to M$ with $G(e_s) = m_s$ for every s, and then $\varphi \circ G$ and $f \circ \rho$ are homomorphisms $F \to N$ agreeing on a basis, hence equal.

Now set $g := G \circ \iota\colon P \to M$. It is a homomorphism, and

$$\varphi \circ g = \varphi \circ G \circ \iota = f \circ \rho \circ \iota = f \circ \mathrm{id}_P = f$$

which is the required lift.

17.2.5 Summand, Not Sum Condition (4) says direct **summand** of a free module, which is weaker than being a free module. A direct summand of M is any submodule admitting a complement, so if $M = A \oplus B \oplus C$ then

$$A,\ B,\ C,\ A \oplus B,\ A \oplus C,\ B \oplus C,\ A \oplus B \oplus C$$

are all direct summands of M, and only the last is M itself. Reading (4) as "direct sum" would collapse the notion of projective onto the notion of free, and the two are genuinely different, as the two examples below show.

Corollary 17.2.6: Free Modules Are Projective

Every free R-module is projective.

***Proof*:**
A free module F is a direct summand of itself, with complement 0, so theorem 17.2.4(4) applies.

The converse fails, and it fails for a reason worth seeing twice. Both witnesses come from splitting a ring into a direct sum of ideals. The summands are then projective by construction and too small to be free.

Example 17.2.7: Projective But Not Free

1. Let $R = \mathbb{Z}/6\mathbb{Z}$. The Chinese remainder theorem gives a ring isomorphism $R \cong \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/3\mathbb{Z}$, which is in particular an isomorphism of R-modules, so the ideal $P = \overline{3}R = \{\overline{0}, \overline{3}\}$ is a direct summand of R, with complement $\overline{4}R = \{\overline{0}, \overline{4}, \overline{2}\}$. Since R is free of rank one over itself, P is projective by theorem 17.2.4(4).

   It is not free. A free R-module of finite rank k has 6^k elements and P has 2, while a free module of infinite rank is infinite. And $2 \neq 6^k$ for every $k \geq 0$.

2. Let F be a field, $n \geq 2$, and $R = M_n(F)$ the ring of $n \times n$ matrices. Let $C = F^n$ be the module of column vectors, a left R-module by matrix multiplication. Writing E_{jj} for the

matrix with a 1 in position (j, j) and zeros elsewhere,

$$R = \bigoplus_{j=1}^{n} RE_{jj}$$

since RE_{jj} consists of the matrices supported in column j and every matrix is the sum of its columns. Each RE_{jj} is isomorphic to C as a left R-module, by reading off column j. So C is a direct summand of the free module R and is projective.

It is not free. Counting dimensions over F, a free R-module of finite rank k has F-dimension kn^2, and one of infinite rank is infinite dimensional. Meanwhile $\dim_F C = n$, which is finite, and $n = kn^2$ has no solution $k \in \mathbb{Z}_{\geq 0}$ once $n \geq 2$.

Condition (4) is structural but awkward to verify, since it requires producing a free module and a complement. The following reformulation is the working criterion, because it asks only for elements of A and homomorphisms out of A, both of which can be exhibited.

Proposition 17.2.8: Dual Basis Lemma

An R-module A is projective if and only if there exist a family $(a_i)_{i\in I}$ of elements of A and a family $(f_i)_{i\in I}$ of homomorphisms $f_i \in \mathrm{Hom}_R(A, R)$ such that, for every $a \in A$,

1. $f_i(a) = 0$ for all but finitely many $i \in I$, and
2. $a = \sum_{i\in I} f_i(a)\, a_i$.

Such a pair of families is called a *dual basis* for A. It may be taken finite exactly when A is finitely generated.

***Proof*:**

($\Rightarrow$) Let A be projective. By example 17.1.4(3) there is a free module F with basis $(e_i)_{i\in I}$ and a surjection $\pi\colon F \to A$, and by theorem 17.2.4(3) the sequence $0 \to \ker(\pi) \to F \xrightarrow{\pi} A \to 0$ splits, so proposition 17.1.13 gives $\sigma\colon A \to F$ with $\pi \circ \sigma = \mathrm{id}_A$.

Put $a_i := \pi(e_i)$, and let $f_i(a)$ be the i-th coordinate of $\sigma(a)$ in the basis (e_i), so that $\sigma(a) = \sum_i f_i(a)e_i$. Each f_i is a homomorphism $A \to R$, being σ followed by a coordinate function, and (1) holds because an element of F has only finitely many nonzero coordinates. Applying π,

$$a = \pi(\sigma(a)) = \pi\Big(\sum_i f_i(a)e_i\Big) = \sum_i f_i(a)\pi(e_i) = \sum_i f_i(a)a_i$$

which is (2).

($\Leftarrow$) Given the two families, let F be free on a basis $(e_i)_{i\in I}$ and define $\pi\colon F \to A$ by $\pi(e_i) = a_i$, using theorem 12.3.11. Define $\sigma\colon A \to F$ by $\sigma(a) = \sum_i f_i(a)e_i$, which is a well defined element of F by (1) and is a homomorphism because each f_i is. Then (2) says exactly that $\pi(\sigma(a)) = a$, so $\pi \circ \sigma = \mathrm{id}_A$. In particular π is surjective, and σ is a section of π, so $0 \to \ker(\pi) \to F \xrightarrow{\pi} A \to 0$ splits by proposition 17.1.13 and A is a direct summand of F by definition 17.1.11. Theorem 17.2.4(4) makes A projective.

For the last sentence: if the families are finite then (2) exhibits $a_1, \ldots, a_n$ as generators of A. Conversely if A is generated by n elements, take $F = R^n$ in the forward direction, so that I is

finite.

Chapter 15 produced an isomorphism $V^* \otimes_F W \cong \mathrm{Hom}_F(V, W)$ for finite dimensional vector spaces (theorem 15.4.9). Over a general commutative ring the same map exists, and the dual basis says exactly when it is an isomorphism.

Corollary 17.2.9: Evaluation and Finitely Generated Projectives

Let R be a commutative ring, A an R-module, and $A^* := \mathrm{Hom}_R(A, R)$. Let

$$\theta\colon A^* \otimes_R A \to \mathrm{End}_R(A), \qquad \theta(f \otimes b)(a) = f(a)b$$

Then A is finitely generated and projective if and only if θ is an isomorphism, and in fact if and only if θ is surjective.

***Proof*:**

The map θ exists because $(f, b) \mapsto \big(a \mapsto f(a)b\big)$ is R-bilinear, so theorem 15.1.12 produces it.

Suppose first that θ is surjective. Then $\mathrm{id}_A = \theta(t)$ for some $t \in A^* \otimes_R A$, and every element of a tensor product is a finite sum of simple tensors, say $t = \sum_{i=1}^{n} f_i \otimes a_i$. Unwinding, $a = \mathrm{id}_A(a) = \sum_{i=1}^{n} f_i(a)a_i$ for every $a \in A$. That is a finite dual basis, so A is finitely generated and projective by proposition 17.2.8.

Conversely let A be finitely generated and projective, with a finite dual basis $(a_i, f_i)_{i=1}^{n}$ from proposition 17.2.8. Two observations do the work. The first is that

$$g = \sum_{i=1}^{n} g(a_i)\, f_i \qquad \text{for every } g \in A^*$$

since evaluating the right-hand side at a gives $\sum_i g(a_i)f_i(a) = g\big(\sum_i f_i(a)a_i\big) = g(a)$, using R-linearity of g.

Surjectivity. Given $\alpha \in \mathrm{End}_R(A)$, put $t = \sum_i f_i \otimes \alpha(a_i)$. Then for every a,

$$\theta(t)(a) = \sum_i f_i(a)\, \alpha(a_i) = \alpha\Big(\sum_i f_i(a)a_i\Big) = \alpha(a)$$

so $\theta(t) = \alpha$.

Injectivity. Let $t = \sum_{j=1}^{m} g_j \otimes b_j$ satisfy $\theta(t) = 0$. Expanding each g_j by the displayed identity and collecting,

$$t = \sum_j \Big(\sum_i g_j(a_i) f_i\Big) \otimes b_j = \sum_i f_i \otimes \Big(\sum_j g_j(a_i) b_j\Big) = \sum_i f_i \otimes \theta(t)(a_i) = 0$$

where the middle step moves the scalars $g_j(a_i) \in R$ across the tensor sign, which is legitimate because the tensor product is over R.

Example 17.2.10: Projective Modules
Throughout, proposition 12.3.36 is what lets Hom_R be computed one summand at a time.

1. If F is a field then every F-module is free (theorem 13.1.8), so every vector space is projective.

2. Any ring R is free of rank one over itself and hence projective. Directly: $\mathrm{Hom}_R(R, M) \cong M$ by $h \mapsto h(1)$, so applying $\mathrm{Hom}_R(R, -)$ to a short exact sequence returns that sequence, which is exact.

3. No nonzero finite abelian group is a projective $\mathbb{Z}$-module. A projective is a direct summand of a free module by theorem 17.2.4(4), hence a submodule of one; a free $\mathbb{Z}$-module has no nonzero element of finite order; and every nonzero element of a nonzero finite group has finite order.

   For $\mathbb{Z}/n\mathbb{Z}$ with $n \geq 2$ the same conclusion is visible in the functor. Applying $\mathrm{Hom}_{\mathbb{Z}}(\mathbb{Z}/n\mathbb{Z}, -)$ to $0 \to \mathbb{Z} \xrightarrow{\cdot n} \mathbb{Z} \xrightarrow{\pi} \mathbb{Z}/n\mathbb{Z} \to 0$ gives

   $$0 \to 0 \to 0 \to \mathbb{Z}/n\mathbb{Z}$$

   since $\mathrm{Hom}_{\mathbb{Z}}(\mathbb{Z}/n\mathbb{Z}, \mathbb{Z}) = 0$ by the argument of example 17.2.2 and $\mathrm{Hom}_{\mathbb{Z}}(\mathbb{Z}/n\mathbb{Z}, \mathbb{Z}/n\mathbb{Z}) \cong \mathbb{Z}/n\mathbb{Z}$. The last map is not surjective.

4. $\mathbb{Q}/\mathbb{Z}$ is not a projective $\mathbb{Z}$-module, being a nonzero torsion module and so not a submodule of a free one. Correspondingly

   $$0 \to \mathbb{Z} \hookrightarrow \mathbb{Q} \to \mathbb{Q}/\mathbb{Z} \to 0$$

   does not split: a section would exhibit $\mathbb{Q}/\mathbb{Z}$ as a submodule of $\mathbb{Q}$, and $\mathbb{Q}$ has no nonzero element of finite order while $\mathbb{Q}/\mathbb{Z}$ is nonzero and torsion.

5. $\mathbb{Q}$ is not a projective $\mathbb{Z}$-module either, and the reason is different: it is torsion-free, so the argument above does not apply. Instead, *every* homomorphism $h\colon \mathbb{Q} \to \mathbb{Z}$ is zero. For if $h(x) \neq 0$ for some x, then for every $n \geq 1$

   $$h(x) = h\Big(n \cdot \frac{x}{n}\Big) = n\, h\Big(\frac{x}{n}\Big)$$

   so n divides $h(x)$ for every $n \geq 1$, forcing $h(x) = 0$.

   Consequently every homomorphism from $\mathbb{Q}$ into a free $\mathbb{Z}$-module $F = \bigoplus_{i \in I} \mathbb{Z}$ is zero, since composing with each coordinate projection gives a homomorphism $\mathbb{Q} \to \mathbb{Z}$. So $\mathbb{Q}$ is not isomorphic to a submodule of a free $\mathbb{Z}$-module, in particular not to a direct summand of one, and theorem 17.2.4(4) fails.

6. Let $0 \to L \xrightarrow{\psi} M \xrightarrow{\varphi} N \to 0$ be short exact with N projective, and write $X^* = \mathrm{Hom}_R(X, R)$. Then

   $$0 \to N^* \xrightarrow{\varphi^*} M^* \xrightarrow{\psi^*} L^* \to 0$$

   is short exact, where $\varphi^*(h) = h \circ \varphi$ and $\psi^*(g) = g \circ \psi$.

   What makes this work is *not* that the modules are projective as arguments of $\mathrm{Hom}_R(-, R)$. Projectivity of L, M and N says nothing directly about a contravariant Hom. It is that N projective forces the sequence to split (theorem 17.2.4(3)), and a split sequence is

preserved by any additive functor. Explicitly, let μ be a section of φ and λ a retraction of ψ (proposition 17.1.13). Then λ^* is a section of ψ^*, since $\psi^*(\lambda^*(g)) = g \circ \lambda \circ \psi = g$, so ψ^* is surjective. And μ^* is a retraction of φ^*, since $\mu^*(\varphi^*(h)) = h \circ \varphi \circ \mu = h$, so φ^* is injective. Exactness in the middle holds because $\psi^*\varphi^*(h) = h \circ \varphi \circ \psi = 0$, and conversely a $g \in M^*$ with $g \circ \psi = 0$ kills $\psi(L) = \ker(\varphi)$ and so factors through $M/\ker(\varphi) \cong N$ by theorem 12.2.10.

7. If P_1 and P_2 are projective then so is $P_1 \oplus P_2$: writing $F_1 = P_1 \oplus K_1$ and $F_2 = P_2 \oplus K_2$ with F_i free, the module $F_1 \oplus F_2$ is free and contains $P_1 \oplus P_2$ as a direct summand with complement $K_1 \oplus K_2$.

Over one class of rings the gap between projective and free closes entirely.

Proposition 17.2.11: Projective Modules Over a PID

Let R be a principal ideal domain. Then every projective R-module is free. Together with corollary 17.2.6, projective and free coincide over a PID.

***Proof*:**
Let P be projective. By theorem 17.2.4(4) there is a free R-module F and a submodule $P' \subseteq F$ with $P \cong P'$, namely a direct summand. By theorem 14.1.6 the submodule P' is free, and hence so is P.

No finiteness hypothesis is needed, which is exactly what theorem 14.1.6 was proved for. The finite rank case (theorem 14.1.5) would give the statement only for finitely generated P. Example 17.2.7 shows that some hypothesis on R is unavoidable, since $\mathbb{Z}/6\mathbb{Z}$ and $M_n(F)$ are not domains. A second class where the two notions agree is the local rings, where every finitely generated projective module is free. That result belongs with the theory of local rings and is proved in *Commutative Algebra* [16, Projective Module over Local Ring].

Two different presentations of the same module by projectives differ by no more than a free summand. This is the result that makes an invariant computed from a projective resolution independent of the resolution chosen, and section 17.6 depends on it.

Proposition 17.2.12: Schanuel's Lemma

Let M be an R-module and let

$$0 \to K_1 \hookrightarrow P_1 \xrightarrow{\pi_1} M \to 0 \qquad \text{and} \qquad 0 \to K_2 \hookrightarrow P_2 \xrightarrow{\pi_2} M \to 0$$

be short exact sequences with P_1 and P_2 projective. Then

$$K_1 \oplus P_2 \cong K_2 \oplus P_1$$

***Proof*:**
Let

$$X = \{(p_1, p_2) \in P_1 \oplus P_2 \mid \pi_1(p_1) = \pi_2(p_2)\}$$

which is a submodule of $P_1 \oplus P_2$, being the kernel of the homomorphism $(p_1, p_2) \mapsto \pi_1(p_1) - \pi_2(p_2)$. Let q_1 and q_2 be the two coordinate projections restricted to X.

q_2 is surjective with kernel K_1. Given $p_2 \in P_2$, surjectivity of π_1 produces $p_1 \in P_1$ with $\pi_1(p_1) = \pi_2(p_2)$, and then $(p_1, p_2) \in X$ maps to p_2. Its kernel consists of the pairs of the form $(p_1, 0)$ that lie in X, and $(p_1, 0) \in X$ says $\pi_1(p_1) = \pi_2(0) = 0$, so the kernel is $K_1 \oplus 0$, which is isomorphic to K_1.

So $0 \to K_1 \to X \xrightarrow{q_2} P_2 \to 0$ is short exact, and P_2 is projective, so it splits by theorem 17.2.4(3), giving $X \cong K_1 \oplus P_2$ by the remark following definition 17.1.11. Exchanging the roles of the two sequences, the same argument applied to q_1 gives $X \cong K_2 \oplus P_1$.

Exercise 17.2.1

1. For each pair, decide whether the module is projective over the indicated ring, and justify the answer.

    1. $\mathbb{Z}/3\mathbb{Z}$ over $\mathbb{Z}/6\mathbb{Z}$
    2. $\mathbb{Z}/2\mathbb{Z}$ over $\mathbb{Z}/4\mathbb{Z}$
    3. $2\mathbb{Z}$ over $\mathbb{Z}$
    4. $\mathbb{Z}[x]/(x)$ over $\mathbb{Z}[x]$
    5. $\mathbb{Z}/2\mathbb{Z} \oplus \mathbb{Z}/3\mathbb{Z}$ over $\mathbb{Z}/6\mathbb{Z}$

2. Let $R = \mathbb{Z}/6\mathbb{Z}$ and $P = \overline{3}R$, the projective module of example 17.2.7(1). Exhibit a dual basis for P consisting of a single pair (a_1, f_1), and verify both conditions of proposition 17.2.8 directly. Then do the same for the complement $\overline{4}R$.

3. Show that the evaluation map θ of corollary 17.2.9 fails to be surjective in each of the following cases, and identify which hypothesis fails.

    1. $R = \mathbb{Z}$ and $A = \mathbb{Q}$
    2. $R = \mathbb{Z}$ and $A = \bigoplus_{k=1}^{\infty} \mathbb{Z}$

4. Take $R = \mathbb{Z}$, $P = \mathbb{Z}$, $M = \mathbb{Z}$, $N = \mathbb{Z}/6\mathbb{Z}$ with φ the reduction map, and $f\colon \mathbb{Z} \to \mathbb{Z}/6\mathbb{Z}$ the homomorphism with $f(1) = \overline{5}$. Determine every lift g of f along φ, and show that the set of lifts is a coset of $\mathrm{Hom}_{\mathbb{Z}}(\mathbb{Z}, 6\mathbb{Z})$ inside $\mathrm{Hom}_{\mathbb{Z}}(\mathbb{Z}, \mathbb{Z})$.

5. Let $(P_i)_{i \in I}$ be a family of R-modules. Show that $\bigoplus_{i \in I} P_i$ is projective if and only if every P_i is projective. *Hint*: for one direction, a direct summand of a direct summand is a direct summand.

6. Let $0 \to L \to M \to N \to 0$ be a short exact sequence of R-modules.

    1. Show that if L and N are projective then so is M.
    2. Give an example with M projective and L not projective. *Hint*: a ring of the form $\mathbb{Z}/p^2\mathbb{Z}$ has a non-projective ideal.

7. Call an R-module M *finitely presented* if there is an exact sequence $R^s \xrightarrow{\alpha} R^t \to M \to 0$ for some $s, t \in \mathbb{N}$. Use proposition 17.2.12 to show that if M is finitely presented then *every* surjection $\varphi\colon R^r \to M$ from a finitely generated free module has finitely generated kernel. Deduce that for a finitely generated M the converse holds too, so that finite presentation may be tested on any one such surjection. *Hint*: the kernel of $R^t \to M$ is $\mathrm{im}(\alpha)$, and a direct summand of a finitely generated module is finitely generated.

17.3 Injective Modules

Section 17.2 fixed a module D and asked what $\mathrm{Hom}_R(D,-)$ does to a short exact sequence. The other slot of Hom_R asks the same question with the arrows reversed: fix D and apply $\mathrm{Hom}_R(-,D)$, which turns a homomorphism $M \to N$ into a homomorphism $\mathrm{Hom}_R(N,D) \to \mathrm{Hom}_R(M,D)$ by composition. A short exact sequence $0 \to L \xrightarrow{\psi} M \xrightarrow{\varphi} N \to 0$ therefore produces

$$0 \longrightarrow \mathrm{Hom}_R(N,D) \xrightarrow{\varphi^*} \mathrm{Hom}_R(M,D) \xrightarrow{\psi^*} \mathrm{Hom}_R(L,D) \longrightarrow 0$$

with the outer terms of the original sequence exchanged. The superscript stars record that the assignment reverses arrows, in contrast to the lower stars of section 17.2.

Proposition 17.3.1: Left Exactness of $\mathrm{Hom}_R(-,D)$

Let D be an R-module.

1. For every R-module homomorphism $\varphi\colon M \to N$, the assignment
$$\varphi^*\colon \mathrm{Hom}_R(N,D) \to \mathrm{Hom}_R(M,D), \qquad f \mapsto f \circ \varphi$$
is a homomorphism of abelian groups.

2. If $0 \to L \xrightarrow{\psi} M \xrightarrow{\varphi} N \to 0$ is a short exact sequence of R-modules, then
$$0 \to \mathrm{Hom}_R(N,D) \xrightarrow{\varphi^*} \mathrm{Hom}_R(M,D) \xrightarrow{\psi^*} \mathrm{Hom}_R(L,D)$$
is exact, for every D. In this situation one says that $\mathrm{Hom}_R(-,D)$ is *left exact*, the word being applied to the sequence $0 \to \mathrm{Hom}_R(N,D) \to \mathrm{Hom}_R(M,D) \to \mathrm{Hom}_R(L,D)$ obtained after the reversal.

As with proposition 17.2.1, part (2) does not claim that the last map is surjective. The two propositions fail at opposite ends of the original sequence: $\mathrm{Hom}_R(D,-)$ can miss homomorphisms into the *right* hand term N, and $\mathrm{Hom}_R(-,D)$ can miss homomorphisms out of the *left* hand term L.

***Proof*:**
(1) For $f, g \in \mathrm{Hom}_R(N,D)$ and $m \in M$,

$$\big((f+g)\circ\varphi\big)(m) = f(\varphi(m)) + g(\varphi(m)) = \big(f\circ\varphi + g\circ\varphi\big)(m)$$

so $\varphi^*(f+g) = \varphi^*(f) + \varphi^*(g)$.

(2) *Exactness at* $\mathrm{Hom}_R(N,D)$, which is the injectivity of φ^*. Suppose $f \circ \varphi = 0$. Every $n \in N$ is $\varphi(m)$ for some m, since φ is surjective, so $f(n) = f(\varphi(m)) = 0$ and $f = 0$.

Exactness at $\mathrm{Hom}_R(M,D)$, that is $\mathrm{im}(\varphi^*) = \ker(\psi^*)$.

($\subseteq$) For $f \in \mathrm{Hom}_R(N,D)$, $\psi^*(\varphi^*(f)) = f \circ \varphi \circ \psi = 0$, since $\varphi \circ \psi = 0$ by $\mathrm{im}(\psi) = \ker(\varphi)$.

($\supseteq$) Let $g \in \mathrm{Hom}_R(M,D)$ with $g\circ\psi = 0$, so g vanishes on $\mathrm{im}(\psi) = \ker(\varphi)$. Let $\pi\colon M \to M/\ker(\varphi)$ be the quotient map. By theorem 12.2.10 the map g factors as $g = \overline{g} \circ \pi$ for a unique $\overline{g}\colon M/\ker(\varphi) \to D$, and φ induces an isomorphism $\overline{\varphi}\colon M/\ker(\varphi) \to N$ with $\overline{\varphi} \circ \pi = \varphi$. Put

$f := \overline{g} \circ \overline{\varphi}^{-1} \in \mathrm{Hom}_R(N, D)$. Then

$$\varphi^*(f) = f \circ \varphi = \overline{g} \circ \overline{\varphi}^{-1} \circ \overline{\varphi} \circ \pi = \overline{g} \circ \pi = g$$

so $g \in \mathrm{im}(\varphi^*)$.

The failure at the other end is visible in the same sequence that example 17.2.2 used, with the roles of the two slots exchanged.

Example 17.3.2: $\mathrm{Hom}_R(-, D)$ Need Not Be Right Exact
Take $R = \mathbb{Z}$, the short exact sequence

$$0 \to \mathbb{Z} \xrightarrow{\cdot 2} \mathbb{Z} \xrightarrow{\pi} \mathbb{Z}/2\mathbb{Z} \to 0$$

and $D = \mathbb{Z}/2\mathbb{Z}$. A homomorphism out of $\mathbb{Z}$ is determined by the image of 1, so $\mathrm{Hom}_{\mathbb{Z}}(\mathbb{Z}, \mathbb{Z}/2\mathbb{Z}) \cong \mathbb{Z}/2\mathbb{Z}$, and likewise $\mathrm{Hom}_{\mathbb{Z}}(\mathbb{Z}/2\mathbb{Z}, \mathbb{Z}/2\mathbb{Z}) \cong \mathbb{Z}/2\mathbb{Z}$. Applying $\mathrm{Hom}_{\mathbb{Z}}(-, D)$ gives

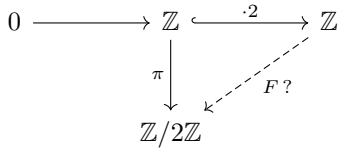
$$0 \to \mathbb{Z}/2\mathbb{Z} \xrightarrow{\pi^*} \mathbb{Z}/2\mathbb{Z} \xrightarrow{(\cdot 2)^*} \mathbb{Z}/2\mathbb{Z}$$

Here $\pi^*(\mathrm{id}_{\mathbb{Z}/2\mathbb{Z}}) = \pi \neq 0$, so π^* is injective and therefore, both groups having two elements, bijective. And $(\cdot 2)^*$ is the zero map: for $g \in \mathrm{Hom}_{\mathbb{Z}}(\mathbb{Z}, \mathbb{Z}/2\mathbb{Z})$ and $x \in \mathbb{Z}$,

$$\big(g \circ (\cdot 2)\big)(x) = g(2x) = 2\, g(x) = 0$$

because every element of $\mathbb{Z}/2\mathbb{Z}$ is killed by 2. The displayed sequence is thus exact, as proposition 17.3.1 promised, and it cannot be closed off with a $\to 0$ on the right, since $(\cdot 2)^*$ has image 0 inside a group of order two.

In the language of diagrams, the homomorphism not in the image of $(\cdot 2)^*$ is π itself, and saying it is not in the image says that π does not *extend* along the injection $\cdot 2$:



Any such F would satisfy $F(2x) = 2F(x) = 0$ while $\pi(x) \neq 0$ for odd x.

The contrast with section 17.2 is worth naming precisely, because the two words are easy to swap. A projective module is one out of which every homomorphism *lifts* along a surjection. An injective module is one into which every homomorphism *extends* along an injection.

Definition 17.3.3: Injective Module

An R-module Q is *injective* if the functor $\mathrm{Hom}_R(-, Q)$ is exact, that is if for every short exact sequence $0 \to L \to M \to N \to 0$ of R-modules the sequence

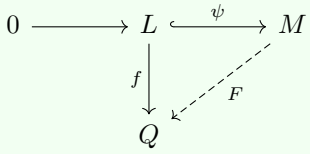
$$0 \to \mathrm{Hom}_R(N, Q) \to \mathrm{Hom}_R(M, Q) \to \mathrm{Hom}_R(L, Q) \to 0$$

is again short exact.

Theorem 17.3.4: Injective Modules Equivalences

Let Q be an R-module. The following are equivalent.

1. Q is injective.
2. *(Extension.)* For all R-modules L and M, if $\psi\colon L \to M$ is injective, then every $f \in \mathrm{Hom}_R(L, Q)$ has an extension $F \in \mathrm{Hom}_R(M, Q)$ with $F \circ \psi = f$:



3. *(Splitting.)* Every short exact sequence $0 \to Q \to M \to N \to 0$ of R-modules splits. Equivalently, whenever Q is a submodule of an R-module M, it is a direct summand of M.

***Proof*:**

(1) $\Rightarrow$ (2). Let $\psi\colon L \to M$ be injective and put $N = M/\psi(L)$, so that $0 \to L \xrightarrow{\psi} M \to N \to 0$ is short exact by proposition 17.1.1 and theorem 12.2.10. Condition (1) makes $\psi^*\colon \mathrm{Hom}_R(M, Q) \to \mathrm{Hom}_R(L, Q)$ surjective, so a given f is $\psi^*(F) = F \circ \psi$ for some F.

(2) $\Rightarrow$ (1). Proposition 17.3.1 gives exactness at the first two spots of the sequence in definition 17.3.3, and (2) applied to the injection ψ says precisely that ψ^* is surjective.

(2) $\Rightarrow$ (3). Let $0 \to Q \xrightarrow{\psi} M \xrightarrow{\varphi} N \to 0$ be short exact, so ψ is injective. Applying (2) to $f = \mathrm{id}_Q$ gives $\lambda \in \mathrm{Hom}_R(M, Q)$ with $\lambda \circ \psi = \mathrm{id}_Q$, a retraction of ψ, so the sequence splits by proposition 17.1.13.

(3) $\Rightarrow$ (2). Let $\psi\colon L \to M$ be injective and $f \in \mathrm{Hom}_R(L, Q)$. The construction that follows glues Q and M along L. Let

$$W = \big\{ \big(f(l),\, -\psi(l)\big) \;\big|\; l \in L \big\} \subseteq Q \oplus M$$

which is a submodule, being the image of the homomorphism $l \mapsto (f(l), -\psi(l))$, and set $X = (Q \oplus M)/W$. Write $\iota\colon Q \to X$ for $q \mapsto \overline{(q, 0)}$ and $j\colon M \to X$ for $m \mapsto \overline{(0, m)}$, both homomorphisms.

The map ι is injective: if $(q, 0) \in W$ then $q = f(l)$ and $\psi(l) = 0$ for some l, and ψ is injective, so $l = 0$ and $q = 0$. Thus $\iota(Q)$ is a submodule of X isomorphic to Q, and by (3) it is a direct

summand, say $X = \iota(Q) \oplus C$. Let $\rho\colon X \to Q$ be the projection onto the first summand followed by ι^{-1}, so that $\rho \circ \iota = \mathrm{id}_Q$.

Put $F := \rho \circ j\colon M \to Q$. For $l \in L$ the element $(f(l), -\psi(l))$ lies in W, so $\overline{(0, \psi(l))} = \overline{(f(l), 0)}$ in X, that is $j(\psi(l)) = \iota(f(l))$. Hence

$$F(\psi(l)) = \rho\big(j(\psi(l))\big) = \rho\big(\iota(f(l))\big) = f(l)$$

so $F \circ \psi = f$.

The name records condition (2): the property concerns extending along injections, exactly as *projective* records lifting along surjections. Condition (3) is the concrete reading: an injective module cannot be enlarged without splitting off, so it is as far from the inside of a bigger module as a submodule can be.

For projective modules theorem 17.2.4 offered a fourth description, as a direct summand of a free module, which reduces every question to free modules. Nothing that simple is available here, and the classification of injective modules over a general ring is genuinely harder. What replaces it is a test that only has to be run on the ideals of R.

Theorem 17.3.5: Baer's Criterion

Let R be a ring and Q an R-module.

1. Q is injective if and only if, for every left ideal I of R, every R-module homomorphism $g\colon I \to Q$ extends to an R-module homomorphism $G\colon R \to Q$.
2. Suppose R is a principal ideal domain. Then Q is injective if and only if $rQ = Q$ for every *nonzero* $r \in R$.

The hypothesis $r \neq 0$ in (2) cannot be dropped: $0 \cdot Q = 0$, so without it the condition would fail for every nonzero module.

***Proof*:**
(1) ($\Rightarrow$) The inclusion $I \hookrightarrow R$ is injective, so theorem 17.3.4(2) extends any $g\colon I \to Q$ to $G\colon R \to Q$.

($\Leftarrow$) Assume the extension property for ideals. It suffices to verify theorem 17.3.4(2), so let $\psi\colon L \to M$ be injective and $f \in \mathrm{Hom}_R(L, Q)$. Replacing L by $\psi(L)$ and f by $f \circ (\psi|^{\psi(L)})^{-1}$, we may assume L is a submodule of M and ψ the inclusion.

Step 1: a poset of partial extensions. Let

$$\mathcal{L} = \{(L', f') \mid L \subseteq L' \subseteq M,\ f' \in \mathrm{Hom}_R(L', Q),\ f'|_L = f\}$$

ordered by $(L', f') \leq (L'', f'')$ when $L' \subseteq L''$ and $f''|_{L'} = f'$. This is a partial order, and $\mathcal{L}$ is nonempty because $(L, f) \in \mathcal{L}$.

Step 2: chains have upper bounds. Let $\{(L_i, f_i)\}_{i \in \mathcal{I}}$ be a chain. The union $\mathfrak{L} = \bigcup_i L_i$ is a submodule of M, since any two of its elements lie in a common L_i by totality of the chain. Define $\mathfrak{f}\colon \mathfrak{L} \to Q$ by $\mathfrak{f}(x) = f_i(x)$ for any i with $x \in L_i$. This is unambiguous because the chain is compatible, and it is a homomorphism because any two elements lie in a common L_i. Then $(\mathfrak{L}, \mathfrak{f}) \in \mathcal{L}$ bounds the chain.

Step 3: a maximal element. By Zorn's lemma (theorem A.2.3) there is a maximal $(M', F) \in \mathcal{L}$.

Step 4: $M' = M$. Suppose not, and choose $m \in M$ with $m \notin M'$. Set

$$I = \{r \in R \mid rm \in M'\}$$

which is a left ideal of R: it contains 0; it is closed under addition because M' is; and if $r \in I$ and $s \in R$ then $(sr)m = s(rm) \in M'$. Define $g\colon I \to Q$ by $g(r) = F(rm)$, which is a homomorphism since F is and since $r \mapsto rm$ is. By hypothesis g extends to $G\colon R \to Q$.

Now let $M'' = M' + Rm$, a submodule of M strictly containing M', and define

$$F'\colon M'' \to Q, \qquad F'(m' + rm) = F(m') + G(r)$$

This is well defined. If $m'_1 + r_1 m = m'_2 + r_2 m$ then $(r_1 - r_2)m = m'_2 - m'_1 \in M'$, so $r_1 - r_2 \in I$ and therefore

$$G(r_1) - G(r_2) = G(r_1 - r_2) = g(r_1 - r_2) = F\big((r_1 - r_2)m\big) = F(m'_2) - F(m'_1)$$

which rearranges to $F(m'_1) + G(r_1) = F(m'_2) + G(r_2)$. That F' is a homomorphism is immediate from the formula, and taking $r = 0$ gives $F'|_{M'} = F$, so in particular $F'|_L = f$ and $(M'', F') \in \mathcal{L}$.

Since $m \in M''$ and $m \notin M'$, the pair (M'', F') is strictly larger than (M', F), contradicting maximality. Hence $M' = M$ and F is the required extension.

(2) ($\Rightarrow$) Let $x \in Q$ and $r \in R$ with $r \neq 0$. Define $g\colon (r) \to Q$ by $g(sr) = sx$. This is well defined because R is an integral domain: if $sr = s'r$ with $r \neq 0$ then $s = s'$. It is a homomorphism, and by part (1) it extends to $G\colon R \to Q$. Then

$$x = g(r) = G(r) = r\,G(1)$$

so $x \in rQ$. As x was arbitrary, $rQ = Q$. Only the domain hypothesis was used here, not that ideals are principal, so this direction holds over any integral domain.

($\Leftarrow$) Assume $rQ = Q$ for every nonzero r, and verify the criterion of part (1). Let I be an ideal of R and $g\colon I \to Q$ a homomorphism. Since R is a principal ideal domain, $I = (r)$ for some r. If $r = 0$ then $I = 0$ and $G = 0$ extends g. If $r \neq 0$, then $g(r) \in Q = rQ$, so there is a $y \in Q$ with $ry = g(r)$. Define $G\colon R \to Q$ by $G(s) = sy$, a homomorphism. For any element sr of I,

$$G(sr) = sry = s\,g(r) = g(sr)$$

so $G|_I = g$. By part (1), Q is injective.

Corollary 17.3.6: Quotients of Injectives Over a PID

Let R be a principal ideal domain, Q an injective R-module and $N \subseteq Q$ a submodule. Then Q/N is injective.

***Proof*:**
For $r \neq 0$ we have $rQ = Q$ by theorem 17.3.5(2), hence

$$r(Q/N) = (rQ + N)/N = (Q + N)/N = Q/N$$

and theorem 17.3.5(2) applies again in the other direction.

The condition appearing in theorem 17.3.5(2) is worth a name of its own, both because it is the only practical test for injectivity over $\mathbb{Z}$ and because it is the source of every concrete injective module in this chapter.

Definition 17.3.7: Divisible Group

Let R be an integral domain. An R-module M is *divisible* if $rM = M$ for every nonzero $r \in R$; equivalently, if for every $x \in M$ and every nonzero $r \in R$ there is a $y \in M$ with $ry = x$.
For $R = \mathbb{Z}$ this says that an abelian group G is divisible when every $x \in G$ is n times some element of G, for every $n \geq 1$. Such a G is called a *divisible group*.

In this language, theorem 17.3.5(2) reads: over a principal ideal domain, injective and divisible are the same condition. Over an integral domain that is not principal only half survives, namely that injective modules are divisible. The reverse implication used principality when it wrote $I = (r)$, and it genuinely fails in general.

Example 17.3.8: Divisible Groups

1. $\mathbb{Q}$ is divisible: given $\frac{a}{b} \in \mathbb{Q}$ and $n \geq 1$, the rational $\frac{a}{nb}$ satisfies $n \cdot \frac{a}{nb} = \frac{a}{b}$. So $\mathbb{Q}$ is an injective $\mathbb{Z}$-module. More generally the field of fractions of any integral domain R is a divisible R-module.

2. Any quotient of a divisible module is divisible, since $r(M/N) = (rM + N)/N = M/N$. In particular $\mathbb{Q}/\mathbb{Z}$ is divisible, hence an injective $\mathbb{Z}$-module. Note that $\mathbb{Q}/\mathbb{Z}$ is a torsion group, so divisibility does not force torsion-freeness. And in a torsion group the element y with $ry = x$ is not unique.

3. Fix a prime p and let

$$\mathbb{Z}_{p^\infty} = \{[x] \in \mathbb{Q}/\mathbb{Z} \mid p^n[x] = 0 \text{ for some } n \geq 0\} = \left\{ \left[\tfrac{a}{p^n}\right] \in \mathbb{Q}/\mathbb{Z} \;\middle|\; a \in \mathbb{Z},\ n \geq 0 \right\}$$

   the p-primary part of $\mathbb{Q}/\mathbb{Z}$, called the *Prüfer p-group*[a]. The two descriptions agree: $p^n\left[\frac{a}{p^n}\right] = [a] = 0$ gives one inclusion, and if $p^n[x] = 0$ with $[x] = \left[\frac{c}{d}\right]$ in lowest terms then $d \mid p^n$, so $[x]$ has the displayed form. It is a subgroup, being the set of elements killed by some power of p.
   It is divisible. Let $[x] = \left[\frac{a}{p^n}\right]$ and let $m \geq 1$, and write $m = p^k u$ with u coprime to p. Since u is invertible modulo p^{n+k}, choose v with $uv \equiv 1 \pmod{p^{n+k}}$, say $uv = 1 + p^{n+k}w$, and set $y = \left[\frac{av}{p^{n+k}}\right] \in \mathbb{Z}_{p^\infty}$. Then

$$my = \left[\frac{p^k uav}{p^{n+k}}\right] = \left[\frac{a(1+p^{n+k}w)}{p^n}\right] = \left[\frac{a}{p^n} + ap^k w\right] = [x]$$

   since $ap^k w \in \mathbb{Z}$. So $\mathbb{Z}_{p^\infty}$ is an injective $\mathbb{Z}$-module.

4. A direct product $\prod_i M_i$ of divisible modules is divisible, since the required y may be chosen coordinatewise, and so is a direct sum. Conversely if $M \oplus N$ is divisible then so are M and N: given $x \in M$ and $r \neq 0$, some (y, z) has $r(y, z) = (x, 0)$, and then $ry = x$.

5. No nonzero finite abelian group is divisible: if $|G| = m$ then $mg = 0$ for every g, so $mG = 0 \neq G$. Combined with theorem 5.1.6, no nonzero finitely generated abelian group is divisible, since a free summand $\mathbb{Z}$ is not divisible either.

[a] It is also $\varinjlim \mathbb{Z}/p^k\mathbb{Z}$ along the inclusions.

Before collecting examples of injective modules, one closure property is needed, and it is the one that fails to be symmetric with the projective case.

Proposition 17.3.9: Summands and Products of Injectives

Let R be a ring.

1. A direct summand of an injective R-module is injective.
2. A direct product $\prod_{i\in I} Q_i$ of R-modules is injective if and only if every Q_i is injective. In particular a finite direct sum of injective modules is injective.

***Proof*:**
(1) Let $Q = E \oplus C$ with Q injective, and let $\iota\colon E \to Q$ and $\rho\colon Q \to E$ be the inclusion and projection, so $\rho \circ \iota = \mathrm{id}_E$. Given an injection $\psi\colon L \to M$ and $f \in \mathrm{Hom}_R(L, E)$, the map $\iota \circ f$ extends to some $H \in \mathrm{Hom}_R(M, Q)$ by theorem 17.3.4(2). Then $F := \rho \circ H$ satisfies $F \circ \psi = \rho \circ \iota \circ f = f$, so theorem 17.3.4(2) holds for E.

(2) ($\Leftarrow$) Let every Q_i be injective, let $\psi\colon L \to M$ be injective and let $f \in \mathrm{Hom}_R(L, \prod_i Q_i)$. Each component $\pi_i \circ f$ extends to $F_i \in \mathrm{Hom}_R(M, Q_i)$, and $F := (F_i)_i$ satisfies $\pi_i \circ F \circ \psi = \pi_i \circ f$ for every i, hence $F \circ \psi = f$.

($\Rightarrow$) Each Q_j is a direct summand of $\prod_i Q_i$, with complement $\prod_{i\neq j} Q_i$, so part (1) applies.

Example 17.3.10: Injective Modules

1. $\mathbb{Z}$ is not an injective $\mathbb{Z}$-module: it is not divisible, since $2y = 1$ has no solution in $\mathbb{Z}$. Equivalently, the sequence $0 \to \mathbb{Z} \xrightarrow{\cdot 2} \mathbb{Z} \to \mathbb{Z}/2\mathbb{Z} \to 0$ does not split (example 17.1.14(2)), so theorem 17.3.4(3) fails. Since $\mathbb{Z}$ is free, free does not imply injective, in contrast with corollary 17.2.6.
2. $\mathbb{Q}$, $\mathbb{Q}/\mathbb{Z}$ and $\mathbb{Z}_{p^\infty}$ are injective $\mathbb{Z}$-modules, by example 17.3.8 and theorem 17.3.5(2). So is $\mathbb{Q} \oplus \mathbb{Q}/\mathbb{Z}$, by proposition 17.3.9(2).
3. No nonzero finitely generated $\mathbb{Z}$-module is injective, by example 17.3.8(5).
4. If $R = F$ is a field then every F-module is injective. Indeed every F-module is free (theorem 13.1.8), and any short exact sequence of F-modules $0 \to Q \to M \to N \to 0$ has N free, hence projective, hence the sequence splits by theorem 17.2.4(3). Theorem 17.3.4(3) follows. Over a field, projective, free and injective all coincide.
5. $\mathbb{Z}/2\mathbb{Z}$ is not an injective $\mathbb{Z}/4\mathbb{Z}$-module. It occurs as the ideal $(\overline{2}) \subseteq \mathbb{Z}/4\mathbb{Z}$, and $\mathbb{Z}/4\mathbb{Z}$ is indecomposable as a module over itself, since its only proper nonzero submodule is $(\overline{2})$ and a complement would have to be a second one. So $(\overline{2})$ is not a direct summand and

theorem 17.3.4(3) fails. Compare exercise set 17.2.1, question 1: the same module is not projective either.

17.3.1 Embedding a Module in an Injective

Example 17.1.4(3) says every module is a quotient of a free, hence of a projective, module. The dual statement is that every module is a submodule of an injective one. It is true, but it is not formal: there is no dual of a free module to hand, and the proof goes through $\mathbb{Z}$ and back.

Lemma 17.3.11: $\mathbb{Z}$-module Submodules of Injective Modules

Every $\mathbb{Z}$-module is a submodule of an injective $\mathbb{Z}$-module.

***Proof*:**
Let M be a $\mathbb{Z}$-module and A a set of generators, so that theorem 12.3.11 gives a surjection $\pi\colon \bigoplus_{a\in A}\mathbb{Z}\to M$ from the free $\mathbb{Z}$-module on A. Writing $K=\ker(\pi)$, theorem 12.2.10 gives $M\cong\bigoplus_{a\in A}\mathbb{Z}/K$.

Now $\bigoplus_{a\in A}\mathbb{Z}$ sits inside $V:=\bigoplus_{a\in A}\mathbb{Q}$, the direct sum of copies of $\mathbb{Q}$ indexed by A, which is divisible by example 17.3.8(1) and (4), hence injective. Since $K\subseteq\bigoplus_{a\in A}\mathbb{Z}\subseteq V$, the quotient V/K is defined, and it is divisible by example 17.3.8(2), hence injective. Finally

$$M\cong\bigoplus_{a\in A}\mathbb{Z}/K\subseteq V/K$$

exhibits M as a submodule of an injective $\mathbb{Z}$-module.

Passing from $\mathbb{Z}$ to an arbitrary ring is done by a change of rings, and the mechanism is the adjunction of theorem 15.3.9. The proof below writes the correspondence out explicitly rather than invoking naturality.

Lemma 17.3.12: Inducing Injective R-modules

Let $f\colon S\to R$ be a homomorphism of commutative rings and let Q be an injective S-module. Regard R as an S-module through f, and give

$$\mathrm{Hom}_S(R,Q)$$

the R-module structure $(r\cdot h)(x):=h(xr)$. Then $\mathrm{Hom}_S(R,Q)$ is an injective R-module.

***Proof*:**
Write $f_*(N)$ for an R-module N regarded as an S-module through f, so that $s\cdot n=f(s)n$.

Step 1: the correspondence. For an R-module N define

$$\alpha\mapsto\widetilde{\alpha},\quad \widetilde{\alpha}(n)=\alpha(n)(1)\qquad\text{and}\qquad \beta\mapsto\widehat{\beta},\quad \widehat{\beta}(n)(r)=\beta(rn)$$

between $\mathrm{Hom}_R\big(N,\mathrm{Hom}_S(R,Q)\big)$ and $\mathrm{Hom}_S\big(f_*(N),Q\big)$. These are the two directions of theorem 15.3.9 applied with N in the role of A, the ring R in the role of the bimodule B and Q in

the role of C, using $N \otimes_R R \cong N$. The verifications are short enough to give directly.

That $\widetilde{\alpha}$ is S-linear: $\alpha(n) \in \mathrm{Hom}_S(R, Q)$ is S-linear on R, whose S-action is $s \cdot x = f(s)x$, so $\alpha(n)(f(s)) = s\,\alpha(n)(1)$. Combining with R-linearity of α,

$$\widetilde{\alpha}(s \cdot n) = \alpha(f(s)n)(1) = \big(f(s) \cdot \alpha(n)\big)(1) = \alpha(n)(f(s)) = s\,\widetilde{\alpha}(n)$$

That $\widehat{\beta}(n)$ is S-linear: $\widehat{\beta}(n)(s \cdot r) = \beta(f(s)rn) = \beta(s \cdot (rn)) = s\,\widehat{\beta}(n)(r)$. That $\widehat{\beta}$ is R-linear: $\widehat{\beta}(r'n)(r) = \beta(rr'n) = \widehat{\beta}(n)(rr') = \big(r' \cdot \widehat{\beta}(n)\big)(r)$.

The two assignments are mutually inverse:

$$\widehat{\widetilde{\alpha}}(n)(r) = \widetilde{\alpha}(rn) = \alpha(rn)(1) = \big(r \cdot \alpha(n)\big)(1) = \alpha(n)(r), \qquad \widetilde{\widehat{\beta}}(n) = \widehat{\beta}(n)(1) = \beta(n)$$

and both are visibly compatible with restriction along a submodule inclusion $L \subseteq N$, in the sense that $\widetilde{\alpha|_L} = \widetilde{\alpha}|_L$.

Step 2: the extension property. Verify theorem 17.3.4(2) for $\mathrm{Hom}_S(R, Q)$. Let $L \subseteq M$ be R-modules and $\alpha \in \mathrm{Hom}_R(L, \mathrm{Hom}_S(R, Q))$. Then $\widetilde{\alpha} \in \mathrm{Hom}_S(f_*(L), Q)$, and $f_*(L) \subseteq f_*(M)$ is still an inclusion of S-modules, since restriction of scalars changes only the action. As Q is an injective S-module, theorem 17.3.4(2) gives $\beta \in \mathrm{Hom}_S(f_*(M), Q)$ with $\beta|_L = \widetilde{\alpha}$.

Put $A := \widehat{\beta} \in \mathrm{Hom}_R(M, \mathrm{Hom}_S(R, Q))$. Then

$$A|_L = \widehat{\beta|_L} = \widehat{\widetilde{\alpha}} = \alpha$$

so A extends α, and $\mathrm{Hom}_S(R, Q)$ is injective.

Corollary 17.3.13: Inducing Injectives Using Divisible Groups

Let R be a commutative ring and D a divisible abelian group. Then $\mathrm{Hom}_{\mathbb{Z}}(R, D)$ is an injective R-module.

***Proof*:**
The group D is an injective $\mathbb{Z}$-module by theorem 17.3.5(2), and there is a unique ring homomorphism $\mathbb{Z} \to R$, so lemma 17.3.12 applies with $S = \mathbb{Z}$.

Lemma 17.3.14: R-module Submodules of Injective Modules

Let R be a commutative ring and M an R-module. Then M is a submodule of an injective R-module.

***Proof*:**
By lemma 17.3.11 there is an injective $\mathbb{Z}$-module D with $M \subseteq D$ as $\mathbb{Z}$-modules. By theorem 17.3.5(2) the group D is divisible. Now compose three injections of R-modules:

$$M \xrightarrow{\sim} \mathrm{Hom}_R(R, M) \hookrightarrow \mathrm{Hom}_{\mathbb{Z}}(R, M) \hookrightarrow \mathrm{Hom}_{\mathbb{Z}}(R, D)$$

The first sends m to the map $x \mapsto xm$, and is an isomorphism of R-modules for the action $(r \cdot h)(x) = h(xr)$, its inverse being $h \mapsto h(1)$. The second is the inclusion of the R-linear maps

among the additive ones, which respects that action. The third is composition with $M \subseteq D$, injective because a map into M is zero exactly when it is zero into D. By corollary 17.3.13 the target $\mathrm{Hom}_{\mathbb{Z}}(R, D)$ is an injective R-module.

Definition 17.3.15: Injective Resolution

Let R be a ring and M an R-module. An *injective resolution* of M is an exact chain complex (definition 12.2.11)

$$0 \longrightarrow M \longrightarrow Q^0 \longrightarrow Q^1 \longrightarrow Q^2 \longrightarrow \cdots$$

in which every Q^i is injective. It has *finite length* k if $Q^i = 0$ for every $i > k$.

Corollary 17.3.16: Existence of Injective Resolutions

Every module over a commutative ring admits an injective resolution.

***Proof*:**
Embed M in an injective Q^0 by lemma 17.3.14, embed the quotient Q^0/M in an injective Q^1, embed $Q^1/(Q^0/M)$ in an injective Q^2, and continue. The composites $Q^i \to Q^i/(\text{previous image}) \hookrightarrow Q^{i+1}$ have image the kernel of the next map by construction, which is exactness.

17.3.2 Injective Hulls

Every module embeds in an injective one, but lemma 17.3.14 produces a wasteful embedding: nothing stops Q from being far larger than M. Asking for the smallest such Q turns out to have a clean answer, and the condition that pins it down is that M should meet every nonzero piece of Q.

Definition 17.3.17: Essential Extension

Let $M \subseteq E$ be R-modules. Then M is *essential* in E, and E is an *essential extension* of M, if

$$N \cap M \neq 0 \qquad \text{for every nonzero submodule } N \subseteq E$$

Equivalently, for every $x \in E$ with $x \neq 0$ there is an $r \in R$ with $rx \in M$ and $rx \neq 0$.

The two formulations agree because Rx is a nonzero submodule whenever $x \neq 0$, and any nonzero N contains such an Rx.

Lemma 17.3.18: Essential Extensions Compose

Let $M \subseteq E \subseteq F$ be R-modules with M essential in E and E essential in F. Then M is essential in F.

***Proof*:**
Let $N \subseteq F$ be a nonzero submodule. Then $N \cap E \neq 0$ because E is essential in F, and $N \cap E$ is a nonzero submodule of E, so $(N \cap E) \cap M \neq 0$ because M is essential in E. That intersection is contained in $N \cap M$.

Theorem 17.3.19: Universal Property of Injective Hulls

Let R be a commutative ring and M an R-module.

1. There is an injective R-module E containing M as an essential submodule. Such an E is called an *injective hull*, or *injective envelope*, of M, and is written $E(M)$.
2. If E and E' are both injective hulls of M, there is an isomorphism $E \to E'$ restricting to the identity on M.
3. If $M \subseteq Q$ with Q injective, there is an injective homomorphism $E(M) \to Q$ restricting to the identity on M, and its image is a direct summand of Q. So $E(M)$ is the smallest injective module containing M.

***Proof*:**

(1) By lemma 17.3.14 choose an injective Q with $M \subseteq Q$.

Step 1: a maximal essential extension inside Q. Let

$$\Sigma = \{E \mid M \subseteq E \subseteq Q \text{ and } M \text{ is essential in } E\}$$

partially ordered by inclusion. It contains M, so it is nonempty. A chain $\{E_i\}$ in Σ has upper bound $\bigcup_i E_i$, which is a submodule of Q containing M, and M is essential in it: a nonzero x in the union lies in some E_i, and $Rx \cap M \neq 0$ there. By Zorn's lemma (theorem A.2.3) there is a maximal $E \in \Sigma$.

Step 2: a maximal complement. Let $\mathcal{C}$ be the set of submodules $C \subseteq Q$ with $C \cap E = 0$, ordered by inclusion. It contains 0, and a chain's union again meets E trivially, so Zorn's lemma gives a maximal $C \in \mathcal{C}$.

Step 3: E is essential in Q/C. First, $E \cap C = 0$, so E maps injectively into Q/C. Let X/C be a nonzero submodule of Q/C, so $C \subsetneq X \subseteq Q$. By maximality of C we have $X \cap E \neq 0$. Choose $0 \neq e \in X \cap E$. Then $e \notin C$, since $E \cap C = 0$, so the class $e + C$ is a nonzero element of X/C lying in the image of E.

Step 4: E is a direct summand of Q. The sum $E + C$ is direct, so there is a projection $p\colon E \oplus C \to E \subseteq Q$. Since $E \oplus C$ is a submodule of Q and Q is injective, theorem 17.3.4(2) extends p to a homomorphism $g\colon Q \to Q$ with $g|_E = \mathrm{id}_E$ and $g(C) = 0$.

Then $\ker(g) \cap E = 0$, because g is the identity on E, and $C \subseteq \ker(g)$. So maximality of C forces $C = \ker(g)$. Hence g induces an isomorphism $Q/C \to \mathrm{im}(g)$, carrying the image of E onto E itself. By Step 3, E is essential in $\mathrm{im}(g)$, and M is essential in E, so M is essential in $\mathrm{im}(g)$ by lemma 17.3.18. Thus $\mathrm{im}(g) \in \Sigma$ and $E \subseteq \mathrm{im}(g)$, and maximality of E gives $E = \mathrm{im}(g)$.

Now $g\colon Q \to Q$ has image E and restricts to the identity on E, so $g \circ g = g$. An idempotent endomorphism splits its domain: every q is $g(q) + (q - g(q))$ with the first term in $\mathrm{im}(g)$ and the second in $\ker(g)$, and an element of $\mathrm{im}(g) \cap \ker(g)$ is $g(y)$ with $0 = g(g(y)) = g(y)$. Therefore $Q = \mathrm{im}(g) \oplus \ker(g) = E \oplus C$.

Step 5. By proposition 17.3.9(1) the summand E of the injective module Q is injective, and M is essential in E by construction.

(2) Let E and E' be injective hulls of M. Since E' is injective and $M \subseteq E$, theorem 17.3.4(2) extends the inclusion $M \hookrightarrow E'$ to a homomorphism $h\colon E \to E'$. Then $\ker(h) \cap M = 0$, because

h is injective on M, and M is essential in E, so $\ker(h) = 0$ and h is injective.

The image $h(E) \cong E$ is injective, so $h(E)$ is a direct summand of E' by theorem 17.3.4(3), say $E' = h(E) \oplus X$. Since $M = h(M) \subseteq h(E)$, the submodule X meets M trivially, and M is essential in E', so $X = 0$. Hence h is an isomorphism, and it is the identity on M by construction.

(3) Let $M \subseteq Q$ with Q injective. Extending the inclusion $M \hookrightarrow Q$ along $M \subseteq E(M)$ gives $h\colon E(M) \to Q$ with $h|_M$ the inclusion. As in (2), $\ker(h) \cap M = 0$ and M essential in $E(M)$ force h to be injective, and $h(E(M)) \cong E(M)$ is injective, hence a direct summand of Q by theorem 17.3.4(3).

Combining (1) and (3): every injective module containing M contains a copy of $E(M)$ over M, and $E(M)$ is injective, so $E(M)$ is a smallest one. Exercise set 17.3.1 below computes $E(\mathbb{Z}) = \mathbb{Q}$ and $E(\mathbb{Z}/p\mathbb{Z}) = \mathbb{Z}_{p^\infty}$.

One consequence of injectivity is worth recording before leaving the section, because it is what makes duality work for abelian groups at all. Section 13.3 defined the dual of a vector space as $V^* = \mathrm{Hom}_k(V, k)$ and found $V \cong V^{**}$ naturally in finite dimensions. Over $\mathbb{Z}$ the same recipe collapses: $\mathrm{Hom}_{\mathbb{Z}}(\mathbb{Z}/n\mathbb{Z}, \mathbb{Z}) = 0$ for every $n \geq 2$, by the argument of example 17.2.2, so dualizing against $\mathbb{Z}$ destroys every torsion group. Replacing $\mathbb{Z}$ by the injective module $\mathbb{Q}/\mathbb{Z}$ repairs it, since a homomorphism $\mathbb{Z}/n\mathbb{Z} \to \mathbb{Q}/\mathbb{Z}$ sends $\overline{1}$ to an element killed by n, and those form the cyclic group $\frac{1}{n}\mathbb{Z}/\mathbb{Z}$, so

$$\mathrm{Hom}_{\mathbb{Z}}(\mathbb{Z}/n\mathbb{Z},\, \mathbb{Q}/\mathbb{Z}) \cong \mathbb{Z}/n\mathbb{Z}$$

The module $\mathbb{Q}/\mathbb{Z}$ is the *dualizing module* for abelian groups. Injectivity is exactly what makes this work: it is what keeps $\mathrm{Hom}_{\mathbb{Z}}(-, \mathbb{Q}/\mathbb{Z})$ exact, so that the dual of a submodule is a quotient of the dual. The isomorphism displayed above is not natural, there being no canonical generator of a cyclic group, but the double dual is. The systematic development, in which these homomorphisms become characters, is [17, Character Theory].

Exercise 17.3.1

1. For each pair, decide whether the module is injective over the indicated ring, and justify the answer.
    1. $\mathbb{Q}$ over $\mathbb{Z}$
    2. $\mathbb{Z}/6\mathbb{Z}$ over $\mathbb{Z}/6\mathbb{Z}$
    3. $\mathbb{Z}/2\mathbb{Z}$ over $\mathbb{Z}/4\mathbb{Z}$
    4. $\mathbb{Q}/\mathbb{Z}$ over $\mathbb{Z}$
    5. $F[x]$ over $F[x]$, for a field F
2. Let $g\colon 6\mathbb{Z} \to \mathbb{Q}/\mathbb{Z}$ be the homomorphism of $\mathbb{Z}$-modules determined by $g(6) = \left[\frac{1}{4}\right]$. Find every extension $G\colon \mathbb{Z} \to \mathbb{Q}/\mathbb{Z}$ of g, and say how many there are. Then explain why theorem 17.3.5 guarantees at least one without any computation.
3. Essential extensions.
    1. Show that $\mathbb{Z}$ is essential in $\mathbb{Q}$ as a $\mathbb{Z}$-module.
    2. Show that $\mathbb{Z}/p\mathbb{Z}$ is essential in $\mathbb{Z}/p^2\mathbb{Z}$.
    3. Show that $\mathbb{Z} \oplus 0$ is *not* essential in $\mathbb{Z} \oplus \mathbb{Z}/2\mathbb{Z}$.

4. Deduce from (a) that $E(\mathbb{Z}) \cong \mathbb{Q}$.

4. Show that every nonzero subgroup of $\mathbb{Z}_{p^\infty}$ contains an element of order p, and deduce that $\mathbb{Z}/p\mathbb{Z}$ is essential in $\mathbb{Z}_{p^\infty}$. Conclude that $E(\mathbb{Z}/p\mathbb{Z}) \cong \mathbb{Z}_{p^\infty}$ as $\mathbb{Z}$-modules. *Hint*: the subgroups of $\mathbb{Z}_{p^\infty}$ are totally ordered by inclusion.

5. Let $0 \to L \to M \to N \to 0$ be a short exact sequence of R-modules.

   1. Show that if L is injective the sequence splits, so $M \cong L \oplus N$.
   2. Show that if L and N are injective then so is M.
   3. Give an example with M injective and N not injective. *Hint*: take $R = \mathbb{Z}/4\mathbb{Z}$, and use example 17.3.10(5). Why does no such example exist over $\mathbb{Z}$?

6. *(The dual of Schanuel's lemma.)* Let

$$0 \to M \to Q_1 \to C_1 \to 0 \qquad \text{and} \qquad 0 \to M \to Q_2 \to C_2 \to 0$$

   be short exact sequences of R-modules with Q_1 and Q_2 injective. Prove that $C_1 \oplus Q_2 \cong C_2 \oplus Q_1$. *Hint*: run the proof of proposition 17.2.12 backwards, replacing the submodule of $P_1 \oplus P_2$ by the quotient of $Q_1 \oplus Q_2$ by the copy of M embedded as $m \mapsto (\iota_1(m), -\iota_2(m))$.

7. Compute injective resolutions.

   1. Show that $0 \to \mathbb{Z} \to \mathbb{Q} \to \mathbb{Q}/\mathbb{Z} \to 0$ is an injective resolution of $\mathbb{Z}$ as a $\mathbb{Z}$-module, of length 1.
   2. Show that multiplication by p on $\mathbb{Z}_{p^\infty}$ is surjective with kernel the subgroup of order p, and deduce that $0 \to \mathbb{Z}/p\mathbb{Z} \to \mathbb{Z}_{p^\infty} \xrightarrow{\cdot p} \mathbb{Z}_{p^\infty} \to 0$ is an injective resolution of $\mathbb{Z}/p\mathbb{Z}$, again of length 1.
   3. Conclude that every $\mathbb{Z}$-module has an injective resolution of length at most 1. *Hint*: corollary 17.3.6.

17.4⋆ Divisible Groups

⋆ *Optional. This section is an aside; nothing later in the book depends on it.*

Example 17.3.8 exhibited $\mathbb{Q}$, $\mathbb{Q}/\mathbb{Z}$ and the Prüfer groups $\mathbb{Z}_{p^\infty}$ as divisible, and theorem 17.3.5(2) made each of them an injective $\mathbb{Z}$-module. This section proves that the list is complete: every divisible group is a direct sum of copies of those, in essentially one way. Since divisible and injective coincide over $\mathbb{Z}$, that is a full classification of the injective $\mathbb{Z}$-modules.

Everything here is about abelian groups, so *group* means *abelian group* throughout, written additively, and $\mathrm{Tor}(G)$ denotes the torsion subgroup of definition 12.3.18.

A subgroup of a divisible group need not be divisible: $\mathbb{Z} \subseteq \mathbb{Q}$ is the standard example. It is therefore not obvious that a group has a largest divisible subgroup, and the first thing to establish is that it does.

Definition 17.4.1: Divisible Subgroup

Let G be an abelian group. Write dG for the subgroup of G generated by all the divisible subgroups of G, that is, the set of finite sums $x_1 + \cdots + x_k$ with each x_i lying in some divisible subgroup. If $dG = 0$, the group G is called *reduced*.

Lemma 17.4.2: dG Is the Largest Divisible Subgroup

Let G be an abelian group. Then dG is divisible, and every divisible subgroup of G is contained in dG.

***Proof*:**
Let $x \in dG$ and $n \geq 1$. Write $x = x_1 + \cdots + x_k$ with $x_i \in D_i$ for divisible subgroups $D_i \subseteq G$. Each D_i is divisible, so there is a $y_i \in D_i$ with $ny_i = x_i$. Then $y := y_1 + \cdots + y_k$ lies in dG and

$$ny = ny_1 + \cdots + ny_k = x_1 + \cdots + x_k = x$$

so $n\,dG = dG$. The second claim holds by construction, since every divisible subgroup is one of the D_i available in the generating set.

Lemma 17.4.3: Splitting Off the Divisible Part

Let G be an abelian group. Then there is a subgroup $H \subseteq G$ with

$$G = dG \oplus H$$

and every such H is reduced, and isomorphic to G/dG.

***Proof*:**
By lemma 17.4.2 the subgroup dG is divisible, hence an injective $\mathbb{Z}$-module by theorem 17.3.5(2), so the short exact sequence

$$0 \to dG \hookrightarrow G \to G/dG \to 0$$

splits by theorem 17.3.4(3). This gives $G = dG \oplus H$ for some subgroup H, with $H \cong G/dG$ by the remark following definition 17.1.11.

Such an H is reduced. The subgroup dH is a divisible subgroup of H, hence a divisible subgroup of G, so $dH \subseteq dG$ by lemma 17.4.2. Also $dH \subseteq H$. Therefore $dH \subseteq dG \cap H = 0$.

So every abelian group is a divisible group plus a reduced one, and the two halves do not interact. Classifying the divisible groups is thus a genuine piece of the classification of abelian groups, and it is the piece that has a clean answer. The route splits G once more, into its torsion part and the rest.

Lemma 17.4.4: The Torsion Subgroup of a Divisible Group

Let G be a divisible abelian group. Then $\mathrm{Tor}(G)$ is divisible, and

$$G \cong \mathrm{Tor}(G) \oplus G/\mathrm{Tor}(G)$$

***Proof*:**
Let $x \in \mathrm{Tor}(G)$ and $n \geq 1$. Since G is divisible there is a $y \in G$ with $ny = x$. Since x is a torsion element there is an $m \geq 1$ with $mx = 0$, and then

$$(mn)y = m(ny) = mx = 0$$

so y is a torsion element and $y \in \mathrm{Tor}(G)$. Hence $n\mathrm{Tor}(G) = \mathrm{Tor}(G)$ for every $n \geq 1$, and $\mathrm{Tor}(G)$ is divisible.

Being divisible, $\mathrm{Tor}(G)$ is an injective $\mathbb{Z}$-module by theorem 17.3.5(2), so the short exact sequence $0 \to \mathrm{Tor}(G) \to G \to G/\mathrm{Tor}(G) \to 0$ splits by theorem 17.3.4(3), and the complement is isomorphic to $G/\mathrm{Tor}(G)$.

The two factors are now treated separately. The quotient turns out to carry a structure it had no right to expect.

Lemma 17.4.5: Structure of $G/\mathrm{Tor}(G)$

Let G be a divisible abelian group. Then $G/\mathrm{Tor}(G)$ carries a unique structure of $\mathbb{Q}$-vector space compatible with its addition, and consequently

$$G/\mathrm{Tor}(G) \cong \bigoplus_{i \in I} \mathbb{Q}$$

for some index set I.

***Proof*:**
Write $A = G/\mathrm{Tor}(G)$.

Step 1: A is torsion-free. This holds for every abelian group G. If $n\overline{x} = \overline{0}$ with $n \geq 1$, then $nx \in \mathrm{Tor}(G)$, so $m(nx) = 0$ for some $m \geq 1$. Then $(mn)x = 0$ with $mn \geq 1$, so $x \in \mathrm{Tor}(G)$ and $\overline{x} = \overline{0}$.

Step 2: A is divisible, being a quotient of the divisible group G (example 17.3.8(2)).

Step 3: division is unambiguous. If $ny = ny'$ with $n \geq 1$ then $n(y - y') = 0$, and A is torsion-free by Step 1, so $y = y'$. Combined with Step 2: for every $x \in A$ and $n \geq 1$ there is exactly one $y \in A$ with $ny = x$.

Step 4: the action. For $\frac{a}{b} \in \mathbb{Q}$ written with $b \geq 1$, and $x \in A$, define $\frac{a}{b} \cdot x$ to be the unique $y \in A$ with $by = ax$, which exists and is unique by Step 3.

This does not depend on the representative. If $\frac{a}{b} = \frac{a'}{b'}$ with $b, b' \geq 1$, so $ab' = a'b$, let y and y' be the elements produced by the two fractions. Then

$$bb'y = b'(by) = b'ax = a'bx = b(b'y') = bb'y'$$

and $bb' \geq 1$, so $y = y'$ by Step 3.

Step 5: the axioms. Write $y = \frac{a}{b}x$, $u = \frac{c}{d}x$ and $y_i = \frac{a}{b}x_i$, so $by = ax$, $du = cx$ and $by_i = ax_i$.

- Additivity in x: $b(y_1 + y_2) = ax_1 + ax_2 = a(x_1 + x_2)$, so $\frac{a}{b}(x_1 + x_2) = y_1 + y_2$.
- Additivity in the scalar: $bd(y + u) = d(by) + b(du) = dax + bcx = (ad + bc)x$, so $\frac{a}{b}x + \frac{c}{d}x = \frac{ad+bc}{bd}x$.
- Associativity: let $v = \frac{a}{b}u$, so $bv = au$. Then $bd\,v = d(bv) = dau = a(du) = acx$, so $\frac{a}{b}\left(\frac{c}{d}x\right) = \frac{ac}{bd}x$.
- Unit: $1 \cdot x$ is the unique y with $y = x$.

So A is a $\mathbb{Q}$-vector space, and the structure is unique because Step 3 leaves no choice about what $\frac{a}{b} \cdot x$ can be.

Step 6. By theorem 13.1.8 the vector space A has a basis, indexed by some set I, so $A \cong \bigoplus_{i \in I} \mathbb{Q}$.

The torsion factor is where the Prüfer groups appear, and the argument has three stages: break the torsion into primary pieces, check each piece is still divisible, and identify a divisible p-group.

Lemma 17.4.6: Structure of $\mathrm{Tor}(G)$

Let G be a divisible abelian group. Then

$$\mathrm{Tor}(G) \cong \bigoplus_{p \text{ prime}} \bigoplus_{i \in I_p} \mathbb{Z}_{p^\infty}$$

for index sets I_p.

***Proof*:**

Write $T = \mathrm{Tor}(G)$, which is divisible by lemma 17.4.4. For a prime p let

$$T_p = \left\{ x \in T \;\middle|\; p^k x = 0 \text{ for some } k \geq 0 \right\}$$

the *p-primary component*, a subgroup of T since $p^k x = p^l y = 0$ gives $p^{k+l}(x - y) = 0$.

Step 1: $T = \bigoplus_p T_p$. This holds for any torsion group. For the sum, let $x \in T$ have order $n = p_1^{n_1} \cdots p_k^{n_k}$ with the p_i distinct primes and $n_i \geq 1$, and set $q_i = n/p_i^{n_i}$. No prime divides every q_i, since p_j does not divide q_j, so $\gcd(q_1, \ldots, q_k) = 1$ and there are integers h_i with

$$h_1 q_1 + h_2 q_2 + \cdots + h_k q_k = 1 \tag{17.1}$$

Put $x_i = h_i q_i x$. Then $p_i^{n_i} x_i = h_i n x = 0$, so $x_i \in T_{p_i}$, and multiplying (17.1) by x gives $x = x_1 + \cdots + x_k$.

For directness, suppose $x \in T_p \cap \sum_{q \neq p} T_q$. Then $p^k x = 0$ for some k, so the order of x is a power of p. On the other hand $x = y_1 + \cdots + y_r$ with $y_j \in T_{q_j}$ for primes $q_j \neq p$, and each y_j has order a power of q_j, so the order of x divides a product of powers of primes different from p. A number that is both a power of p and coprime to p is 1, so $x = 0$.

Step 2: each T_p is divisible. Let $x \in T_p$ and $n \geq 1$. Since T is divisible there is a $y \in T$ with $ny = x$. Decompose $y = \sum_q y_q$ by Step 1. Then $x = ny = \sum_q ny_q$ with $ny_q \in T_q$, and x lies in

T_p, so uniqueness of the decomposition forces $ny_q = 0$ for $q \neq p$ and $ny_p = x$. As $y_p \in T_p$, this shows $nT_p = T_p$.

Step 3: a nonzero divisible p-group contains a copy of $\mathbb{Z}_{p^\infty}$. Let D be a nonzero divisible group in which every element has order a power of p, and pick $g \neq 0$, of order p^k with $k \geq 1$. Then $x_1 := p^{k-1}g$ has order exactly p. Using divisibility choose x_2 with $px_2 = x_1$, then x_3 with $px_3 = x_2$, and so on.

Each x_r has order exactly p^r: repeatedly applying $px_{i+1} = x_i$ gives $p^{r-1}x_r = x_1 \neq 0$ and $p^r x_r = px_1 = 0$. Also $x_r = px_{r+1} \in \langle x_{r+1}\rangle$, so the cyclic groups $\langle x_1\rangle \subseteq \langle x_2\rangle \subseteq \cdots$ form a chain, and $Z := \bigcup_r \langle x_r\rangle$ is a subgroup of D.

Define $\theta\colon Z \to \mathbb{Z}_{p^\infty}$ by $\theta(ax_r) = \left[\frac{a}{p^r}\right]$. It is well defined: if $ax_r = bx_s$ with $r \leq s$, then $x_r = p^{s-r}x_s$, so $(ap^{s-r} - b)x_s = 0$, so p^s divides $ap^{s-r} - b$, and therefore $\left[\frac{a}{p^r}\right] = \left[\frac{ap^{s-r}}{p^s}\right] = \left[\frac{b}{p^s}\right]$. It is a homomorphism by the same computation, surjective because every element of $\mathbb{Z}_{p^\infty}$ has the form $\left[\frac{a}{p^r}\right]$ (example 17.3.8(3)), and injective because $\theta(ax_r) = 0$ says $p^r \mid a$, which says $ax_r = 0$. So $Z \cong \mathbb{Z}_{p^\infty}$.

Step 4: a divisible p-group is a direct sum of copies of $\mathbb{Z}_{p^\infty}$. Let D be such a group and let $\mathcal{T}$ be the set of families of subgroups of D, each member of each family isomorphic to $\mathbb{Z}_{p^\infty}$, whose sum inside D is direct. Order $\mathcal{T}$ by inclusion of families. It contains the empty family. A chain has the union of its families as an upper bound, which is again in $\mathcal{T}$ because directness is a condition on finitely many members at a time, and any finitely many members of the union already lie in one family of the chain. By Zorn's lemma (theorem A.2.3) there is a maximal family $\{Z_j\}_{j\in J}$. Put $H = \bigoplus_{j\in J} Z_j \subseteq D$.

The group H is divisible, being a direct sum of divisible groups (example 17.3.8(4)), hence injective, so $D = H \oplus K$ for some K by theorem 17.3.4(3). The complement K is divisible: given $x \in K$ and $n \geq 1$, divisibility of D gives $d = h + k$ with $nd = x$, and comparing components in $H \oplus K$ gives $nh = 0$ and $nk = x$. Every element of K has order a power of p, so if $K \neq 0$ then Step 3 gives a subgroup $Z \subseteq K$ isomorphic to $\mathbb{Z}_{p^\infty}$. Since $H \cap K = 0$, the family $\{Z_j\}_{j\in J} \cup \{Z\}$ is again in $\mathcal{T}$ and strictly larger, contradicting maximality. So $K = 0$ and $D = H$.

Applying Steps 2 and 4 to each T_p and assembling with Step 1 gives the statement.

Theorem 17.4.7: Classification of Divisible Groups

Let G be a divisible abelian group. Then there are index sets I and I_p, one for each prime p, with

$$G \cong \left(\bigoplus_{p \text{ prime}} \bigoplus_{i\in I_p} \mathbb{Z}_{p^\infty} \right) \oplus \bigoplus_{i\in I} \mathbb{Q}$$

The cardinalities of these index sets are determined by G, namely

$$|I| = \dim_{\mathbb{Q}}\big(G/\mathrm{Tor}(G)\big) \qquad \text{and} \qquad |I_p| = \dim_{\mathbb{F}_p} G[p], \quad \text{where } G[p] = \{g \in G \mid pg = 0\}$$

Since divisible and injective agree over $\mathbb{Z}$ (theorem 17.3.5(2)), this classifies the injective $\mathbb{Z}$-modules.

***Proof*:**
Existence. By lemma 17.4.4, $G \cong \mathrm{Tor}(G) \oplus G/\mathrm{Tor}(G)$. Lemma 17.4.6 identifies the first summand and lemma 17.4.5 the second.

The invariants. Suppose $G \cong \left(\bigoplus_{p \text{ prime}} \bigoplus_{i \in I_p} \mathbb{Z}_{p^\infty}\right) \oplus \bigoplus_{i \in I} \mathbb{Q}$. Each $\mathbb{Z}_{p^\infty}$ is a torsion group and each copy of $\mathbb{Q}$ is torsion-free, so the torsion subgroup of the right-hand side is exactly the first bracket, and

$$G/\mathrm{Tor}(G) \cong \bigoplus_{i \in I} \mathbb{Q}$$

which is a $\mathbb{Q}$-vector space with a basis indexed by I. Since G determines $\mathrm{Tor}(G)$ and hence $G/\mathrm{Tor}(G)$ up to isomorphism, and the cardinality of a basis is an invariant of a vector space (lemma 14.1.1 and the remark following it), $|I|$ is determined by G.

For the primary invariants, note that $G[p]$ is killed by p and so is a vector space over $\mathbb{F}_p$, and that forming $G[p]$ commutes with direct sums, an element of a direct sum being killed by p exactly when each of its components is. Now

$$\mathbb{Q}[p] = 0, \qquad \mathbb{Z}_{q^\infty}[p] = 0 \text{ for } q \neq p, \qquad \mathbb{Z}_{p^\infty}[p] = \left\{ \left[\frac{a}{p}\right] \;\middle|\; a \in \mathbb{Z} \right\} \cong \mathbb{Z}/p\mathbb{Z}$$

The first holds because $\mathbb{Q}$ is torsion-free; the second because every element of $\mathbb{Z}_{q^\infty}$ has order a power of q, and no nonzero such order is divisible by p; the third because $p\left[\frac{a}{p^n}\right] = 0$ forces $p^{n-1} \mid a$, so the element is $\left[\frac{b}{p}\right]$ for some b. Therefore

$$G[p] \cong \bigoplus_{i \in I_p} \mathbb{Z}/p\mathbb{Z}$$

an $\mathbb{F}_p$-vector space with a basis indexed by I_p. As G determines $G[p]$, the cardinality $|I_p|$ is determined by G.

Exercise 17.4.1

1. Compute dG, the largest divisible subgroup, for each of the following.
    1. $G = \mathbb{Z}$
    2. $G = \mathbb{Q}$
    3. $G = \mathbb{Q} \oplus \mathbb{Z}$
    4. $G = \mathbb{Z}/6\mathbb{Z}$
    5. $G = \mathbb{Q}/\mathbb{Z}$
2. Decide which of the following are divisible, and justify each answer.
    1. $\mathbb{R}$ under addition
    2. $\mathbb{R}/\mathbb{Z}$ under addition
    3. $\mathbb{C}^\times$ under multiplication
    4. $\mathbb{Z}[1/p]$ under addition, for a prime p

5. $\mathbb{Q}^\times$ under multiplication

3. For each of the following divisible groups, compute the invariants $|I|$ and $|I_p|$ of theorem 17.4.7.
   1. $\mathbb{Q}/\mathbb{Z}$
   2. $\mathbb{Q} \oplus \mathbb{Q}/\mathbb{Z}$
   3. $\mathbb{Z}_{p^\infty} \oplus \mathbb{Z}_{p^\infty}$
   4. $\mathbb{R}$ under addition. *Hint*: compare cardinalities.

4. Show that $\mathbb{Z}_{p^\infty} \cong \mathbb{Z}[1/p]/\mathbb{Z}$, and that the proper subgroups of $\mathbb{Z}_{p^\infty}$ are exactly the finite cyclic groups $\langle \left[\frac{1}{p^n}\right] \rangle \cong \mathbb{Z}/p^n\mathbb{Z}$ for $n \geq 0$, totally ordered by inclusion. Deduce that $\mathbb{Z}_{p^\infty}$ has no maximal subgroup.

5. Show that every homomorphism $f\colon G \to H$ of abelian groups satisfies $f(dG) \subseteq dH$, and that $G \mapsto dG$ is therefore a functor $\mathbf{Ab} \to \mathbf{Ab}$. *Hint*: the image of a divisible group is divisible.

6. Show that a nonzero divisible group has no maximal subgroup. *Hint*: if M were maximal, G/M would be a simple abelian group, hence $\mathbb{Z}/p\mathbb{Z}$ for some prime p. But G/M is divisible. Deduce that no nonzero divisible group is finitely generated, recovering example 17.3.8(5).

17.5 Flat Modules

The third functor available on a short exact sequence is the tensor product. Fix a right R-module D and apply $D \otimes_R -$, which theorem 15.2.2 turns into a covariant functor from left R-modules to abelian groups, landing in left S-modules when D is an (S, R)-bimodule. Everything below is stated for $D \otimes_R -$. The functor $- \otimes_R D$ on right modules behaves identically, with left and right exchanged throughout.

This functor fails at the opposite end from the two in section 17.2 and section 17.3. Both Hom functors are left exact and can lose surjectivity. Tensoring is *right* exact and can lose injectivity. The reason is the Hom-tensor adjunction of theorem 15.3.9: $D \otimes_R -$ is a left adjoint, and left adjoints preserve the constructions that sit at the right-hand end of a sequence.

Theorem 17.5.1: $D \otimes_R -$ Is Right Exact

Let D be a right R-module and let

$$0 \to L \xrightarrow{\psi} M \xrightarrow{\varphi} N \to 0$$

be a short exact sequence of left R-modules. Then

$$D \otimes_R L \xrightarrow{1\otimes\psi} D \otimes_R M \xrightarrow{1\otimes\varphi} D \otimes_R N \to 0$$

is an exact sequence of abelian groups, and of left S-modules when D is an (S, R)-bimodule. The map $1 \otimes \psi$ need not be injective.

***Proof*:**
Surjectivity of $1 \otimes \varphi$. By lemma 15.1.4(2) every element of $D \otimes_R N$ is a finite sum $\sum_i d_i \otimes n_i$. Each n_i is $\varphi(m_i)$ for some m_i, since φ is surjective, so the element is the image of $\sum_i d_i \otimes m_i$.

$\operatorname{im}(1 \otimes \psi) \subseteq \ker(1 \otimes \varphi)$. By theorem 15.2.2 the composite $(1 \otimes \varphi) \circ (1 \otimes \psi)$ is $1 \otimes (\varphi \circ \psi)$, and $\varphi \circ \psi = 0$ because $\operatorname{im}(\psi) = \ker(\varphi)$.

$\ker(1 \otimes \varphi) \subseteq \operatorname{im}(1 \otimes \psi)$. Write $I = \operatorname{im}(1 \otimes \psi)$. By the previous paragraph $1 \otimes \varphi$ kills I, so it factors as

$$\overline{\varphi} \colon (D \otimes_R M)/I \to D \otimes_R N, \qquad \overline{\varphi}\big(\overline{d \otimes m}\big) = d \otimes \varphi(m)$$

by theorem 12.2.10. It suffices to produce an inverse, for then $\overline{\varphi}$ is injective, which says exactly that $\ker(1 \otimes \varphi) = I$.

Define $\beta \colon D \times N \to (D \otimes_R M)/I$ by $\beta(d, n) = \overline{d \otimes m}$ for any m with $\varphi(m) = n$. This does not depend on the choice: if $\varphi(m) = \varphi(m')$ then $m - m' \in \ker(\varphi) = \operatorname{im}(\psi)$, say $m - m' = \psi(l)$, and then

$$d \otimes m - d \otimes m' = d \otimes \psi(l) = (1 \otimes \psi)(d \otimes l) \in I$$

The map β is R-balanced in the sense of definition 15.1.1: it is additive in d because $\otimes$ is; it is additive in n because a lift of $n + n'$ may be taken to be $m + m'$; and $\beta(dr, n) = \overline{dr \otimes m} = \overline{d \otimes rm} = \beta(d, rn)$, since rm is a lift of rn and by the third relation of (15.1).

By theorem 15.1.5 there is a homomorphism $\overline{\beta} \colon D \otimes_R N \to (D \otimes_R M)/I$ with $\overline{\beta}(d \otimes n) = \overline{d \otimes m}$. Both composites fix every simple tensor, hence are the identity by lemma 15.1.4(3), so $\overline{\varphi}$ is an isomorphism.

Now the failure at the left, computed before anything is named after it, on the same sequence that example 17.2.2 and example 17.3.2 used.

Example 17.5.2: $D \otimes_R -$ Need Not Be Left Exact

1. Take $R = \mathbb{Z}$, the short exact sequence $0 \to \mathbb{Z} \xrightarrow{\cdot 2} \mathbb{Z} \xrightarrow{\pi} \mathbb{Z}/2\mathbb{Z} \to 0$, and $D = \mathbb{Z}/2\mathbb{Z}$. Since $A \otimes_{\mathbb{Z}} \mathbb{Z} \cong A$ for every A, and $\mathbb{Z}/2\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z}/2\mathbb{Z} \cong \mathbb{Z}/2\mathbb{Z}$ by proposition 15.2.13, applying $D \otimes_{\mathbb{Z}} -$ gives

   $$\mathbb{Z}/2\mathbb{Z} \xrightarrow{1 \otimes (\cdot 2)} \mathbb{Z}/2\mathbb{Z} \xrightarrow{1 \otimes \pi} \mathbb{Z}/2\mathbb{Z} \to 0$$

   The first map is zero: for $\overline{a} \in \mathbb{Z}/2\mathbb{Z}$ and $x \in \mathbb{Z}$,

   $$\overline{a} \otimes 2x = \overline{a} \cdot 2 \otimes x = \overline{0} \otimes x = 0$$

   So the sequence is exact where theorem 17.5.1 promises and fails to be exact at the left, the first map being zero on a group of order two.

2. The sharpest version replaces the middle term. The inclusion $\mathbb{Z} \hookrightarrow \mathbb{Q}$ is injective, but $\mathbb{Z}/2\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Q} = 0$, since

   $$\overline{a} \otimes q = \overline{a} \otimes 2 \cdot \tfrac{q}{2} = \overline{a} \cdot 2 \otimes \tfrac{q}{2} = 0$$

   for every q. So $1 \otimes \iota$ maps $\mathbb{Z}/2\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z} \cong \mathbb{Z}/2\mathbb{Z}$ to the zero group, and injectivity fails as badly as it can.

Proposition 17.5.3: Conditions for $A \otimes_R -$ to Be Exact

Let A be a right R-module. The following are equivalent.

1. For every short exact sequence $0 \to L \xrightarrow{\psi} M \xrightarrow{\varphi} N \to 0$ of left R-modules, the sequence
$$0 \to A \otimes_R L \xrightarrow{1\otimes\psi} A \otimes_R M \xrightarrow{1\otimes\varphi} A \otimes_R N \to 0$$
is short exact.
2. For every injective homomorphism $\psi\colon L \to M$ of left R-modules, the map $1 \otimes \psi\colon A \otimes_R L \to A \otimes_R M$ is injective.

***Proof*:**

(1) $\Rightarrow$ (2). Given ψ injective, put $N = M/\psi(L)$, so that $0 \to L \xrightarrow{\psi} M \to N \to 0$ is short exact by proposition 17.1.1 and theorem 12.2.10. Condition (1) makes $1 \otimes \psi$ injective.

(2) $\Rightarrow$ (1). Theorem 17.5.1 gives exactness at $A \otimes_R M$ and at $A \otimes_R N$, and (2) gives injectivity of $1 \otimes \psi$, which is exactness at $A \otimes_R L$ by proposition 17.1.1.

Definition 17.5.4: Flat Module

A right R-module A is *flat* if it satisfies the equivalent conditions of proposition 17.5.3, that is if the functor $A \otimes_R -$ is exact. A left R-module is *flat* when the corresponding statement holds for $- \otimes_R A$.

The side matters only because $\otimes_R$ does. Over a commutative ring the two notions coincide, by proposition 15.2.7, and the qualification can be dropped.

Corollary 17.5.5: Projective Modules Are Flat

Every free R-module is flat, and more generally every projective R-module is flat.

***Proof*:**

First, R itself is flat: the map $R \otimes_R L \to L$ sending $r \otimes l$ to rl is an isomorphism with inverse $l \mapsto 1 \otimes l$, and under it $1 \otimes \psi$ becomes ψ, so $1 \otimes \psi$ is injective whenever ψ is.

Next let $F = \bigoplus_{i\in I} R$ be free. By theorem 15.2.4 there are isomorphisms $F \otimes_R L \cong \bigoplus_{i\in I}(R \otimes_R L) \cong \bigoplus_{i\in I} L$, and similarly for M, carrying $1 \otimes \psi$ to the map $\bigoplus_i \psi$ acting coordinatewise. A direct sum of injective maps is injective, so F is flat.

Finally let P be projective. By theorem 17.2.4(4) there is a free module F and a module K with $F \cong P \oplus K$. By theorem 15.2.3,
$$F \otimes_R L \cong (P \otimes_R L) \oplus (K \otimes_R L)$$
compatibly with M in place of L, and under these identifications $1_F \otimes \psi$ is $(1_P \otimes \psi) \oplus (1_K \otimes \psi)$. A direct sum of maps is injective only if each summand is, so injectivity of $1_F \otimes \psi$ forces injectivity of $1_P \otimes \psi$.

So free implies projective implies flat, and both implications are strict: example 17.2.7 exhibits projective modules that are not free, and example 17.5.8 exhibits a flat $\mathbb{Z}$-module that is not projective. Over a principal ideal domain there is a criterion for flatness as concrete as Baer's is for injectivity.

Proposition 17.5.6: Flat Implies Torsion-Free

Let R be an integral domain and A a flat R-module. Then A is torsion-free.

***Proof*:**
Let $r \in R$ be nonzero. Multiplication by r is an injective R-module homomorphism $R \to R$, since R is a domain, so flatness makes $1 \otimes (\cdot r)\colon A \otimes_R R \to A \otimes_R R$ injective. The map $A \otimes_R R \to A$ sending $a \otimes r$ to ar is an isomorphism, with inverse $a \mapsto a \otimes 1$, and it carries $1 \otimes (\cdot r)$ to multiplication by r on A. Hence $ra = 0$ forces $a = 0$, and A has no nonzero torsion element.

Theorem 17.5.7: Flat Equals Torsion-Free Over a PID

Let R be a principal ideal domain and A an R-module. Then A is flat if and only if A is torsion-free.

***Proof*:**
One direction is proposition 17.5.6, which needs only that R is a domain.

For the converse let A be torsion-free, let $\psi\colon L \to M$ be an injective homomorphism, and let $\xi \in A \otimes_R L$ satisfy $(1 \otimes \psi)(\xi) = 0$. Write $\xi = \sum_{i=1}^{k} a_i \otimes l_i$ using lemma 15.1.4(2), so that

$$\sum_{i=1}^{k} a_i \otimes \psi(l_i) = 0 \qquad \text{in } A \otimes_R M$$

Step 1: pass to a finitely generated piece. By lemma 15.2.12 there is a finitely generated submodule $A_0 \subseteq A$ containing $a_1, \dots, a_k$ with

$$\sum_{i=1}^{k} a_i \otimes \psi(l_i) = 0 \qquad \text{in } A_0 \otimes_R M$$

Step 2: A_0 is flat. A submodule of a torsion-free module is torsion-free, so A_0 is a finitely generated torsion-free module over a principal ideal domain, hence free by theorem 14.1.9(2), hence flat by corollary 17.5.5.

Step 3: conclude. The map $1_{A_0} \otimes \psi\colon A_0 \otimes_R L \to A_0 \otimes_R M$ is therefore injective, and Step 1 says it kills the element $\sum_i a_i \otimes l_i$ of $A_0 \otimes_R L$. So that element is zero in $A_0 \otimes_R L$, and its image under the map $A_0 \otimes_R L \to A \otimes_R L$ induced by the inclusion is ξ. Hence $\xi = 0$, and $1 \otimes \psi$ is injective.

Example 17.5.8: Flat Modules

Throughout, $R = \mathbb{Z}$ unless stated otherwise, so theorem 17.5.7 applies and flat means torsion-free.

1. $\mathbb{Z}$ is flat, being free. For $n \geq 2$ the module $\mathbb{Z}/n\mathbb{Z}$ is not flat, having $\overline{1}$ as a torsion element. Example 17.5.2(1) is that failure for $n = 2$, seen directly.
2. $\mathbb{Q}$ is a flat $\mathbb{Z}$-module, being torsion-free. It is *not* projective, by example 17.2.10(5). So the implication projective $\Rightarrow$ flat is strict, and flatness is genuinely weaker than the other two conditions of this chapter.
3. $\mathbb{Q}/\mathbb{Z}$ is an injective $\mathbb{Z}$-module (example 17.3.10(2)) but is not flat, being a nonzero torsion group. So injective does not imply flat.
4. $\mathbb{Q} \oplus \mathbb{Z}$ is torsion-free, hence flat. It is not projective: a direct summand of a projective module is projective, being a summand of a summand of a free module, and $\mathbb{Q}$ is not projective. It is not injective either, since a direct summand of an injective module is injective by proposition 17.3.9(1) and $\mathbb{Z}$ is not injective (example 17.3.10(1)).
5. A direct sum $\bigoplus_i A_i$ is flat if and only if every A_i is, by the computation in the proof of corollary 17.5.5 applied to theorem 15.2.4.
6. Over a field every module is free (theorem 13.1.8), hence flat. Over a field, free, projective, injective and flat all coincide.
7. Let R be a principal ideal domain and $D \subseteq R \setminus \{0\}$ a multiplicatively closed set containing 1. The ring of fractions $D^{-1}R$ of section 9.5 sits inside the field of fractions of R, which is torsion-free as an R-module, so $D^{-1}R$ is a torsion-free and therefore flat R-module. That localization is flat holds over any commutative ring, and is taken up in *Commutative Algebra* [16, Localization Is Exact, Characterization of Localization].

Corollary 17.5.9: Tensor Products of Projective Modules

Let R be a commutative ring and let P and Q be projective R-modules. Then $P \otimes_R Q$ is projective.

***Proof*:**

Choose P' and Q' with $P \oplus P' = F$ and $Q \oplus Q' = G$ free, using theorem 17.2.4(4). Then $F \otimes_R G$ is free by corollary 15.2.6, and theorem 15.2.3 expands it as

$$F \otimes_R G \cong (P \otimes_R Q) \oplus (P \otimes_R Q') \oplus (P' \otimes_R Q) \oplus (P' \otimes_R Q')$$

So $P \otimes_R Q$ is a direct summand of a free module and theorem 17.2.4(4) applies.

A ring homomorphism $R \to S$ making S flat as an R-module is called a *flat extension*, and the condition is what allows a construction performed over R to be transported to S without losing exactness. Example 17.5.8(7) is the first example. Flat extensions, the local criterion that detects flatness one prime at a time, and the geometric reading of flatness as a family of fibres varying continuously all belong with commutative algebra and are developed in *Commutative Algebra* [16].

Exercise 17.5.1

1. Decide whether each module is flat over the indicated ring, and justify.
   1. $\mathbb{Z}/6\mathbb{Z}$ over $\mathbb{Z}$
   2. $\mathbb{Q}$ over $\mathbb{Z}$
   3. $\mathbb{Z}[1/2]$ over $\mathbb{Z}$
   4. $\mathbb{Z}/2\mathbb{Z}$ over $\mathbb{Z}/6\mathbb{Z}$
   5. $k[x]/(x)$ over $k[x]$, for a field k
2. Let $m, n \geq 1$ and $d = \gcd(m, n)$. Using $\mathbb{Z}/m\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z}/n\mathbb{Z} \cong \mathbb{Z}/d\mathbb{Z}$ (proposition 15.2.13), show directly that $\mathbb{Z}/2\mathbb{Z}$ is not a flat $\mathbb{Z}$-module, by tensoring the injection $\mathbb{Z}/2\mathbb{Z} \to \mathbb{Z}/4\mathbb{Z}$, $\overline{a} \mapsto \overline{2a}$, with $\mathbb{Z}/2\mathbb{Z}$ and computing both sides.
3. Show that a direct sum $\bigoplus_{i \in I} A_i$ of right R-modules is flat if and only if every A_i is flat.
4. Let R be a principal ideal domain.
   1. Show that every submodule of a flat R-module is flat.
   2. Show that a quotient of a flat R-module need not be flat, by exhibiting a surjection from a flat $\mathbb{Z}$-module onto a non-flat one.
5. Let R be a commutative ring and let A and B be flat R-modules. Show that $A \otimes_R B$ is flat. *Hint*: theorem 15.2.8 identifies $(A \otimes_R B) \otimes_R L$ with $A \otimes_R (B \otimes_R L)$.
6. Show that every element of $M \otimes_R N$ lies in the image of $M_0 \otimes_R N_0 \to M \otimes_R N$ for some finitely generated submodules $M_0 \subseteq M$ and $N_0 \subseteq N$. *Hint*: lemma 15.1.4(2). Explain why this is weaker than lemma 15.2.12, and why the weaker statement would not suffice in the proof of theorem 17.5.7.
7. Assemble the three strictness witnesses of this chapter into one picture. Exhibit $\mathbb{Z}$-modules, or modules over a ring of your choice, showing that each of the implications

   $$\text{free} \Rightarrow \text{projective} \Rightarrow \text{flat}$$

   is strict, and that neither implication holds between injective and flat in either direction.

17.6 Ext and Tor

Section 17.1 asked for $\mathcal{E}(C, A)$, the set of equivalence classes of extensions of C by A, and everything since has been about the three functors that a short exact sequence can be pushed through. Each is exact at one end and can fail at the other. Writing $0 \to A \to B \to C \to 0$ for a short exact sequence and D for a fixed module, proposition 17.2.1, proposition 17.3.1 and theorem 17.5.1 say

$$0 \to \mathrm{Hom}_R(D, A) \to \mathrm{Hom}_R(D, B) \to \mathrm{Hom}_R(D, C)$$
$$0 \to \mathrm{Hom}_R(C, D) \to \mathrm{Hom}_R(B, D) \to \mathrm{Hom}_R(A, D)$$
$$A \otimes_R D \to B \otimes_R D \to C \otimes_R D \to 0$$

and in each case the missing end is missing for a reason: something is there, and it is not zero. This section names it. The construction is given in full and the computations below are unconditional, but three of the theorems are stated without proof, because their proofs need machinery that this book does not build. Where that happens it is said explicitly, and the reason is given.

17.6.1 Complexes and Their Homology

Definition 12.2.11 defined an *exact* chain complex. What is needed here is the same object without the exactness, together with a measure of how far from exact it is.

Definition 17.6.1: Chain Complex and Homology

A *chain complex* of R-modules is a family $(M_i)_{i\in\mathbb{Z}}$ together with homomorphisms $d_i\colon M_i \to M_{i-1}$ satisfying $d_i \circ d_{i+1} = 0$ for every i, equivalently $\operatorname{im}(d_{i+1}) \subseteq \ker(d_i)$. Its *$i$-th homology* is the quotient module

$$H_i(M_\bullet) := \ker(d_i)/\operatorname{im}(d_{i+1})$$

A *cochain complex* is the same data with the arrows reversed, $d^i\colon M^i \to M^{i+1}$ and $d^{i+1}\circ d^i = 0$, and its *i-th cohomology* is $H^i(M^\bullet) := \ker(d^i)/\operatorname{im}(d^{i-1})$.

So a complex is exact at M_i exactly when $H_i(M_\bullet) = 0$, and the homology records the failure one degree at a time. This is the whole of the apparatus needed below. The theory of complexes proper is *Homological Algebra* [19].

17.6.2 The Construction

Fix an R-module M and choose a projective resolution of it, which exists by example 17.1.4(3) and corollary 17.2.6: an exact complex

$$\cdots \to P_2 \xrightarrow{d_2} P_1 \xrightarrow{d_1} P_0 \xrightarrow{\varepsilon} M \to 0$$

with every P_i projective. Deleting M leaves the complex $P_\bullet$, which is no longer exact at P_0. Its homology there is M itself. Now apply one of the three functors to $P_\bullet$ and take homology.

Definition 17.6.2: Tor

Let R be a ring, M a right R-module with a chosen projective resolution $P_\bullet$, and N a left R-module. The complex $P_\bullet \otimes_R N$ has differentials $d_i \otimes 1$ and satisfies $(d_i \otimes 1)(d_{i+1} \otimes 1) = (d_i d_{i+1}) \otimes 1 = 0$ by theorem 15.2.2, so it is a chain complex. Define

$$\operatorname{Tor}_i^R(M,N) := H_i\big(P_\bullet \otimes_R N\big) = \ker(d_i \otimes 1)/\operatorname{im}(d_{i+1} \otimes 1)$$

Definition 17.6.3: Ext

With M, $P_\bullet$ as above and N any R-module, the family $\operatorname{Hom}_R(P_i, N)$ with the maps d_{i+1}^* is a cochain complex, since $d_{i+1}^* d_i^* = (d_i d_{i+1})^* = 0$. Define

$$\operatorname{Ext}_R^i(M,N) := H^i\big(\operatorname{Hom}_R(P_\bullet, N)\big) = \ker(d_{i+1}^*)/\operatorname{im}(d_i^*)$$

17.6.4 Two Meanings of Tor The symbol Tor now carries two jobs. Definition 12.3.18 wrote $\mathrm{Tor}_R(M)$, or $\mathrm{Tor}(M)$, for the torsion submodule of M, and definition 17.6.2 writes $\mathrm{Tor}_i^R(M,N)$ for the module just constructed. The subscript decides: an *integer* index means the functor of this section, a *ring* means the torsion submodule, and the functor always takes two arguments. The collision is standard in the literature, and proposition 17.6.7 shows that $\mathrm{Tor}_1^{\mathbb{Z}}(\mathbb{Z}/n\mathbb{Z}, A)$ is the subgroup of A killed by n, so the derived functor really is measuring torsion.

The definitions depend on a choice, and nothing so far says the answer does not. It does not, and that is the first of the three deferred theorems.

Theorem 17.6.5: The Derived Functors Are Well Defined

Let R be a ring and M, N be R-modules.

1. $\mathrm{Tor}_i^R(M,N)$ and $\mathrm{Ext}^i_R(M,N)$ do not depend on the choice of projective resolution of M, up to isomorphism.
2. For Tor, a resolution by flat modules (definition 17.5.4) may be used in place of a projective one, with the same result.
3. $\mathrm{Tor}_i^R(M,N)$ may equally be computed from a projective resolution of the second argument N, and $\mathrm{Ext}^i_R(M,N)$ from an *injective* resolution (definition 17.3.15) of N, in both cases giving the same module.

The proofs are omitted here, and the reason is a genuine gap rather than an economy: all three rest on the comparison theorem, which lifts a map of modules to a map of resolutions uniquely up to chain homotopy, and on the fact that chain homotopic maps induce the same map on homology. Neither statement is available in this book, both being about complexes rather than modules, and they are proved in *Homological Algebra* [19, Unique Projection Resolution, Chain-Homotopy and Cohomology]. Proposition 17.2.12 is the degree-one instance of the comparison theorem: two projective presentations of the same module differ by a free summand, which is why the degree-one answer cannot depend on the presentation.

Every computation below fixes one explicit resolution, so each is a complete calculation on its own terms. Theorem 17.6.5 is what upgrades it from a statement about that resolution to a statement about M.

Proposition 17.6.6: Degree Zero

Let M and N be R-modules. Then

$$\mathrm{Tor}_0^R(M,N) \cong M \otimes_R N \qquad \text{and} \qquad \mathrm{Ext}^0_R(M,N) \cong \mathrm{Hom}_R(M,N)$$

***Proof*:**

Let $P_\bullet \to M$ be a projective resolution. Exactness at P_0 says $P_1 \xrightarrow{d_1} P_0 \xrightarrow{\varepsilon} M \to 0$ is exact.

For Tor, apply $- \otimes_R N$. By theorem 17.5.1 the sequence $P_1 \otimes N \to P_0 \otimes N \to M \otimes N \to 0$ is exact, so $\varepsilon \otimes 1$ identifies $(P_0 \otimes N)/\mathrm{im}(d_1 \otimes 1)$ with $M \otimes_R N$. Since $d_0 = 0$ in the deleted

complex, $\ker(d_0 \otimes 1) = P_0 \otimes N$ and the left-hand side is exactly $\operatorname{Tor}_0^R(M,N)$.

For Ext, apply $\operatorname{Hom}_R(-,N)$. By proposition 17.3.1 the sequence $0 \to \operatorname{Hom}_R(M,N) \xrightarrow{\varepsilon^*} \operatorname{Hom}_R(P_0,N) \xrightarrow{d_1^*} \operatorname{Hom}_R(P_1,N)$ is exact, so ε^* is an isomorphism onto $\ker(d_1^*)$. Since $d_0^* = 0$, that kernel is exactly $\operatorname{Ext}_R^0(M,N)$.

Proposition 17.6.7: Ext and Tor of a Cyclic Group

Let $n \geq 1$ and let A be an abelian group. Write $A[n] = \{a \in A \mid na = 0\}$. Then, as abelian groups,

$$\operatorname{Ext}^i_{\mathbb{Z}}(\mathbb{Z}/n\mathbb{Z}, A) \cong \begin{cases} A[n] & i = 0 \\ A/nA & i = 1 \\ 0 & i \geq 2 \end{cases} \qquad \operatorname{Tor}^{\mathbb{Z}}_i(\mathbb{Z}/n\mathbb{Z}, A) \cong \begin{cases} A/nA & i = 0 \\ A[n] & i = 1 \\ 0 & i \geq 2 \end{cases}$$

***Proof*:**

The sequence $0 \to \mathbb{Z} \xrightarrow{\cdot n} \mathbb{Z} \xrightarrow{\pi} \mathbb{Z}/n\mathbb{Z} \to 0$ of example 17.1.6 is a projective resolution of $\mathbb{Z}/n\mathbb{Z}$, since $\mathbb{Z}$ is free: take $P_0 = P_1 = \mathbb{Z}$, $d_1 = \cdot n$, and $P_i = 0$ for $i \geq 2$. The deleted complex is

$$0 \to \mathbb{Z} \xrightarrow{\cdot n} \mathbb{Z} \to 0$$

concentrated in degrees 1 and 0.

Every homomorphism $\mathbb{Z} \to A$ is determined by the image of 1, so $\operatorname{Hom}_{\mathbb{Z}}(\mathbb{Z}, A) \cong A$ by $h \mapsto h(1)$, and under this identification $d_1^* = (\cdot n)^*$ is multiplication by n on A, since $(h \circ (\cdot n))(1) = h(n) = n\,h(1)$. Likewise $\mathbb{Z} \otimes_{\mathbb{Z}} A \cong A$ by $x \otimes a \mapsto xa$, and $d_1 \otimes 1$ becomes multiplication by n. Both functors therefore turn the deleted complex into

$$0 \to A \xrightarrow{\cdot n} A \to 0$$

with A in degrees 1 and 0 for Tor, and in degrees 0 and 1 for Ext, the arrows of a cochain complex running the other way.

Reading off homology: the kernel of multiplication by n is $A[n]$ and its cokernel is A/nA, and every other module in the complex is zero. For Tor this gives $\operatorname{Tor}_1 = \ker(\cdot n) = A[n]$ and $\operatorname{Tor}_0 = A/nA$. For Ext it gives $\operatorname{Ext}^0 = \ker(\cdot n) = A[n]$ and $\operatorname{Ext}^1 = A/nA$. All higher terms vanish because $P_i = 0$ for $i \geq 2$.

The two degree-zero answers agree with proposition 17.6.6: $\operatorname{Hom}_{\mathbb{Z}}(\mathbb{Z}/n\mathbb{Z}, A)$ is the set of elements of A killed by n, and $\mathbb{Z}/n\mathbb{Z} \otimes_{\mathbb{Z}} A \cong A/nA$ by proposition 15.2.13. Specialising the degree-one answers,

$$\operatorname{Ext}^1_{\mathbb{Z}}(\mathbb{Z}/m\mathbb{Z},\, \mathbb{Z}/n\mathbb{Z}) \cong \mathbb{Z}/\gcd(m,n)\mathbb{Z}, \qquad \operatorname{Ext}^1_{\mathbb{Z}}(\mathbb{Z}/n\mathbb{Z},\, \mathbb{Z}) \cong \mathbb{Z}/n\mathbb{Z}, \qquad \operatorname{Ext}^1_{\mathbb{Z}}(\mathbb{Z}/n\mathbb{Z},\, \mathbb{Q}) = 0$$

the last because $\mathbb{Q} = n\mathbb{Q}$.

Corollary 17.6.8: Vanishing

Let $i \geq 1$.

1. If P is projective then $\operatorname{Ext}^i_R(P, N) = 0$ and $\operatorname{Tor}^R_i(P, N) = 0$ for every N.
2. If F is flat then $\operatorname{Tor}^R_i(M, F) = 0$ for every M.

***Proof*:**
(1) A projective module is its own projective resolution: take $P_0 = P$, $\varepsilon = \mathrm{id}_P$ and $P_i = 0$ for $i \geq 1$, which is exact. The deleted complex has a single nonzero term in degree 0, so every functor applied to it has vanishing homology in degrees $i \geq 1$.

(2) Let $P_\bullet \to M$ be a projective resolution, so the complex $\cdots \to P_1 \to P_0 \to M \to 0$ is exact. Since F is flat, applying $- \otimes_R F$ preserves that exactness, by proposition 17.5.3 applied at each spot. So $P_\bullet \otimes_R F$ is exact except in degree 0, which is to say $\operatorname{Tor}^R_i(M, F) = 0$ for $i \geq 1$.

Part (2) is the sense in which flatness is the condition that makes Tor disappear, just as projectivity kills Ext in the first argument. The dual statement, that $\operatorname{Ext}^i_R(M, Q) = 0$ for Q injective, needs theorem 17.6.5(3): compute Ext from an injective resolution of Q, which may be taken to be Q itself.

17.6.3 What They Repair

Theorem 17.6.9: Long Exact Sequences

Let $0 \to A \to B \to C \to 0$ be a short exact sequence of R-modules and let D be an R-module. Then there are homomorphisms δ_i, δ'_i and ∂_i, called *connecting homomorphisms*, making the following sequences exact.

$$0 \to \operatorname{Hom}_R(D, A) \to \operatorname{Hom}_R(D, B) \to \operatorname{Hom}_R(D, C) \xrightarrow{\delta_0} \operatorname{Ext}^1_R(D, A) \to \cdots$$
$$\cdots \to \operatorname{Ext}^i_R(D, A) \to \operatorname{Ext}^i_R(D, B) \to \operatorname{Ext}^i_R(D, C) \xrightarrow{\delta_i} \operatorname{Ext}^{i+1}_R(D, A) \to \cdots$$

$$0 \to \operatorname{Hom}_R(C, D) \to \operatorname{Hom}_R(B, D) \to \operatorname{Hom}_R(A, D) \xrightarrow{\delta'_0} \operatorname{Ext}^1_R(C, D) \to \cdots$$
$$\cdots \to \operatorname{Ext}^i_R(C, D) \to \operatorname{Ext}^i_R(B, D) \to \operatorname{Ext}^i_R(A, D) \xrightarrow{\delta'_i} \operatorname{Ext}^{i+1}_R(C, D) \to \cdots$$

$$\cdots \to \operatorname{Tor}^R_i(C, D) \xrightarrow{\partial_i} \operatorname{Tor}^R_{i-1}(A, D) \to \operatorname{Tor}^R_{i-1}(B, D) \to \operatorname{Tor}^R_{i-1}(C, D) \to \cdots$$
$$\cdots \to A \otimes_R D \to B \otimes_R D \to C \otimes_R D \to 0$$

These are the sequences the section exists to produce: each short exact sequence, which the three functors truncate, is restored to an exact sequence by the derived functors, running off to infinity in the direction the functor was failing. The proofs are again omitted, and again for a reason that can be named: the connecting homomorphisms are built by the snake lemma applied to a short exact sequence of *complexes* rather than of modules, which is the general form of proposition 17.1.10, and that is developed in *Homological Algebra* [19, Snake Lemma].

Notice how the first sequence recovers a fact already proved. If D is projective then $\operatorname{Ext}^1_R(D, A) = 0$ by

corollary 17.6.8, so the first sequence reads $0 \to \mathrm{Hom}_R(D,A) \to \mathrm{Hom}_R(D,B) \to \mathrm{Hom}_R(D,C) \to 0$, which is definition 17.2.3. In the same way $\mathrm{Tor}_1^R(C,D) = 0$ for D flat turns the third into the short exact sequence of proposition 17.5.3(1). The derived functors do not replace the three module classes. They say what happens when a module is not in them.

17.6.4 The Extension Problem, Answered

Section 17.1 defined $\mathcal{E}(C,A)$ and showed it has at least one element, the class of the split extension, and sometimes more.

Theorem 17.6.10: Ext^1 Classifies Extensions

Let R be a ring and let A and C be R-modules. Then there is a bijection

$$\mathrm{Ext}^1_R(C,A) \longleftrightarrow \mathcal{E}(C,A)$$

between the underlying set of $\mathrm{Ext}^1_R(C,A)$ and the set of equivalence classes of extensions of C by A, under which 0 corresponds to the class of the split extension $A \oplus C$. The addition of $\mathrm{Ext}^1_R(C,A)$ transports to an operation on $\mathcal{E}(C,A)$, the *Baer sum*, making it an abelian group.

This is the theorem the chapter opened with, and it is the last of the three whose proof is omitted. Half of the construction is short, and it uses the following description of Ext^1, which also isolates why the theorem is plausible.

Proposition 17.6.11: Ext^1 as a Cokernel

Let $0 \to K \hookrightarrow P_0 \xrightarrow{\varepsilon} M \to 0$ be exact with P_0 projective, and let N be any R-module. Then

$$\mathrm{Ext}^1_R(M,N) \cong \mathrm{coker}\Big(\mathrm{Hom}_R(P_0,N) \xrightarrow{\text{restrict}} \mathrm{Hom}_R(K,N)\Big)$$

***Proof*:**
Extend the presentation to a projective resolution by choosing a projective P_1 with a surjection $d\colon P_1 \twoheadrightarrow K$ and setting $d_1 = \iota \circ d$, where $\iota\colon K \to P_0$ is the inclusion, and continuing with a resolution of $\ker(d)$.

Since d is surjective, $d^*\colon \mathrm{Hom}_R(K,N) \to \mathrm{Hom}_R(P_1,N)$ is injective by proposition 17.3.1, and its image is $\ker(d_2^*)$: the sequence $P_2 \to P_1 \xrightarrow{d} K \to 0$ is exact, so applying the same proposition identifies $\mathrm{Hom}_R(K,N)$ with the kernel of $\mathrm{Hom}_R(P_1,N) \to \mathrm{Hom}_R(P_2,N)$. Under this identification $\mathrm{im}(d_1^*)$ becomes the image of the restriction map $\mathrm{Hom}_R(P_0,N) \to \mathrm{Hom}_R(K,N)$, because $d_1 = \iota \circ d$ gives $d_1^* = d^* \circ \iota^*$. Hence

$$\mathrm{Ext}^1_R(M,N) = \ker(d_2^*)/\mathrm{im}(d_1^*) \cong \mathrm{Hom}_R(K,N)/\mathrm{im}(\iota^*)$$

which is the stated cokernel.

Now the construction. An extension $0 \to A \to B \xrightarrow{\beta} C \to 0$ together with a projective presentation $0 \to K \to P \xrightarrow{\varepsilon} C \to 0$ produce, by projectivity of P applied to the surjection β, a map $P \to B$ over

C. It carries K into A, and the class of the resulting $K \to A$ in the cokernel of proposition 17.6.11 is the invariant attached to the extension. What is genuinely work is that the assignment is well defined, bijective and additive, and that is deferred to *Homological Algebra* [19], which owes the full correspondence.

The theorem closes the loop with example 17.1.8(2), where the extensions of $\mathbb{Z}/p\mathbb{Z}$ by $\mathbb{Z}/p\mathbb{Z}$ were counted by hand. Proposition 17.6.7 gives

$$\operatorname{Ext}^1_{\mathbb{Z}}(\mathbb{Z}/p\mathbb{Z},\,\mathbb{Z}/p\mathbb{Z}) \cong (\mathbb{Z}/p\mathbb{Z})\big/p(\mathbb{Z}/p\mathbb{Z}) \cong \mathbb{Z}/p\mathbb{Z}$$

which has p elements, matching the $p-1$ pairwise inequivalent nonsplit extensions found there together with the split one. Likewise $\operatorname{Ext}^1_{\mathbb{Z}}(\mathbb{Z}/n\mathbb{Z},\mathbb{Z}) \cong \mathbb{Z}/n\mathbb{Z}$ says there are exactly n equivalence classes of extensions of $\mathbb{Z}/n\mathbb{Z}$ by $\mathbb{Z}$, while example 17.1.6 exhibited only two non-isomorphic middle terms. There is no conflict: equivalence is finer than isomorphism, exactly as definition 17.1.7 warned, and Ext^1 counts the finer relation.

Finally, $\operatorname{Ext}^1_{\mathbb{Z}}(\mathbb{Z}/n\mathbb{Z},\mathbb{Q}) = 0$ says every extension of $\mathbb{Z}/n\mathbb{Z}$ by $\mathbb{Q}$ splits, which is theorem 17.3.4(3) for the injective module $\mathbb{Q}$. And $\operatorname{Tor}^{\mathbb{Z}}_1(\mathbb{Z}/n\mathbb{Z},A) = A[n]$ says the failure of $-\otimes\mathbb{Z}/n\mathbb{Z}$ to preserve an injection is measured by the n-torsion, which is theorem 17.5.7 read one module at a time.

Exercise 17.6.1

1. Using proposition 17.6.7, compute each of the following.
   1. $\operatorname{Ext}^1_{\mathbb{Z}}(\mathbb{Z}/4\mathbb{Z},\mathbb{Z}/6\mathbb{Z})$ and $\operatorname{Tor}^{\mathbb{Z}}_1(\mathbb{Z}/4\mathbb{Z},\mathbb{Z}/6\mathbb{Z})$
   2. $\operatorname{Ext}^1_{\mathbb{Z}}(\mathbb{Z}/n\mathbb{Z},\mathbb{Q}/\mathbb{Z})$ and $\operatorname{Tor}^{\mathbb{Z}}_1(\mathbb{Z}/n\mathbb{Z},\mathbb{Q}/\mathbb{Z})$
   3. $\operatorname{Ext}^1_{\mathbb{Z}}(\mathbb{Z}/n\mathbb{Z},\mathbb{Z}_{p^\infty})$, for a prime p

2. Use theorem 17.6.10 and proposition 17.6.7 to say how many equivalence classes of extensions of $\mathbb{Z}/4\mathbb{Z}$ by $\mathbb{Z}$ there are. Then show that the possible middle terms are $\mathbb{Z}\oplus\mathbb{Z}/4\mathbb{Z}$, $\mathbb{Z}$ and $\mathbb{Z}\oplus\mathbb{Z}/2\mathbb{Z}$, so that strictly fewer isomorphism types occur than there are classes. *Hint*: the extension attached to k has middle term $(\mathbb{Z}\oplus\mathbb{Z})/\langle(4,-k)\rangle$.

3. Show directly from definition 17.6.3 that $\operatorname{Ext}^i_R(M,N) = 0$ for all $i \geq 1$ whenever M admits a projective resolution of length 0, and that $\operatorname{Ext}^i_R(M,N) = 0$ for all $i \geq 2$ whenever M admits one of length 1. Deduce that $\operatorname{Ext}^i_{\mathbb{Z}}(M,N) = 0$ for every $i \geq 2$ and every pair of abelian groups M, N. *Hint*: theorem 14.1.6.

4. Let R be a principal ideal domain. Show that $\operatorname{Tor}^R_1(M,N) = 0$ whenever either argument is torsion-free, and give an example over $R = \mathbb{Z}$ where $\operatorname{Tor}^{\mathbb{Z}}_1(M,N) \neq 0$.

5. Verify the first long exact sequence of theorem 17.6.9 by hand in one case: take $0 \to \mathbb{Z} \xrightarrow{\cdot 2} \mathbb{Z} \to \mathbb{Z}/2\mathbb{Z} \to 0$ and $D = \mathbb{Z}/2\mathbb{Z}$, compute every term using proposition 17.6.7, and check that the sequence you obtain is exact. Identify the connecting map δ_0.

6. Show that $\operatorname{Ext}^1_R(M,N) = 0$ for every N if and only if M is projective. *Hint*: one direction is corollary 17.6.8. For the other, take a projective presentation $0 \to K \to P \to M \to 0$ and use the description of Ext^1 as $\operatorname{coker}\big(\operatorname{Hom}_R(P,N) \to \operatorname{Hom}_R(K,N)\big)$ with $N = K$.

7. Explain why $\operatorname{Tor}_R(M)$ and $\operatorname{Tor}^R_1(M,N)$ are not the same kind of object, and compute both for $M = \mathbb{Z}/6\mathbb{Z}$ over $R = \mathbb{Z}$, taking $N = \mathbb{Z}/4\mathbb{Z}$ in the second.

17.7 Summary

One short exact sequence, three functors that truncate it, three classes of module that stop the truncation, and two families of modules that measure it when it happens. The tables below state what was established, each entry pointing at the result that establishes it.

Short exact sequences

Statement	**Reference**
$0 \to A \to B$ exact $\Leftrightarrow$ injective; $B \to C \to 0$ exact $\Leftrightarrow$ surjective	proposition 17.1.1
Short five lemma: α, γ injective (surjective) $\Rightarrow \beta$ is	proposition 17.1.9
Mini snake lemma: kernels and cokernels splice into one exact sequence	proposition 17.1.10
Splits $\Leftrightarrow$ a section exists $\Leftrightarrow$ a retraction exists	proposition 17.1.13
Splitting is carried along by any isomorphism of sequences	proposition 17.1.15
Equivalence of extensions is strictly finer than isomorphism	example 17.1.8(2)

The last row is what makes the extension problem a counting problem rather than a classification of middle terms, and it is what theorem 17.6.10 counts.

The three functors against exactness

Each functor below is exact at one end of a short exact sequence and can fail at the other. Naming the modules for which it does not fail is what the chapter is for. The first and third are covariant and the second contravariant.

Functor	**Always**	**Exact when**	**Reference**
$\mathrm{Hom}_R(D, -)$	left exact	D projective	proposition 17.2.1, definition 17.2.3
$\mathrm{Hom}_R(-, D)$	left exact	D injective	proposition 17.3.1, definition 17.3.3
$D \otimes_R -$	right exact	D flat	theorem 17.5.1, definition 17.5.4

For the contravariant row, *left exact* refers to the sequence after the outer terms have been exchanged, which is how proposition 17.3.1 states it. The two Hom functors can lose surjectivity. Tensoring can lose injectivity. The failures are at opposite ends because $D \otimes_R -$ is a left adjoint and $\mathrm{Hom}_R(D, -)$ its right adjoint (theorem 15.3.9).

The three classes

Class	Equivalent descriptions	Reference
Projective P	maps out of P lift along surjections; every $0 \to L \to M \to P \to 0$ splits	theorem 17.2.4
	P is a direct summand of a free module; P has a dual basis	proposition 17.2.8
Injective Q	maps into Q extend along injections; every $0 \to Q \to M \to N \to 0$ splits	theorem 17.3.4
	every $g\colon I \to Q$ on a left ideal extends to R	theorem 17.3.5
Flat A	$1 \otimes \psi$ is injective for every injective ψ	proposition 17.5.3

Projective modules have a structural description and injective modules do not, which is why Baer's criterion exists: it replaces a quantifier over all modules by one over the ideals of R.

How the classes relate, and where the implications are strict

Statement	Witness	Reference
free $\Rightarrow$ projective $\Rightarrow$ flat	(always)	corollary 17.2.6, corollary 17.5.5
projective $\not\Rightarrow$ free	$\mathbb{Z}/2\mathbb{Z}$ over $\mathbb{Z}/6\mathbb{Z}$; F^n over $M_n(F)$	example 17.2.7
flat $\not\Rightarrow$ projective	$\mathbb{Q}$ over $\mathbb{Z}$	example 17.5.8(2)
injective $\not\Rightarrow$ flat	$\mathbb{Q}/\mathbb{Z}$ over $\mathbb{Z}$	example 17.5.8(3)
flat $\not\Rightarrow$ injective	$\mathbb{Z}$ over $\mathbb{Z}$	example 17.3.10(1)
free $\not\Rightarrow$ injective	$\mathbb{Z}$ over $\mathbb{Z}$	example 17.3.10(1)
Projective, injective and flat pass to direct summands	(always)	proposition 17.3.9

Over particular rings the classes collapse.

Ring	Collapse	Reference
A field	free = projective = injective = flat	example 17.3.10(4)
A PID	projective = free	proposition 17.2.11
A PID	injective = divisible	theorem 17.3.5(2)
A PID	flat = torsion-free	theorem 17.5.7
An integral domain	injective $\Rightarrow$ divisible, and not conversely	theorem 17.3.5(2)

Resolutions and hulls

Every module is a quotient of a projective one and a submodule of an injective one, so both kinds of resolution exist. The injective side needed real work, having no dual of a free module to hand.

Statement	**Reference**
Every module is a quotient of a free module	example 17.1.4(3)
Every module is a submodule of an injective module	lemma 17.3.14
Injective resolutions exist	corollary 17.3.16
Every module has an injective hull, unique over the module	theorem 17.3.19
Two projective presentations differ by a free summand	proposition 17.2.12
Divisible groups: $\bigoplus_p \bigoplus_{I_p} \mathbb{Z}_{p^\infty} \oplus \bigoplus_I \mathbb{Q}$, invariants determined	theorem 17.4.7

The last row is the classification of injective $\mathbb{Z}$-modules, since divisible and injective agree over $\mathbb{Z}$. It is the one starred division of the chapter, and nothing outside it depends on it.

Ext, Tor, and the answer

Statement	**Reference**
$\operatorname{Tor}_i^R(M,N) = H_i(P_\bullet \otimes_R N)$ and	definition 17.6.2,
$\operatorname{Ext}^i_R(M,N) = H^i(\operatorname{Hom}_R(P_\bullet, N))$	definition 17.6.3
$\operatorname{Tor}_0 = M \otimes_R N$ and $\operatorname{Ext}^0 = \operatorname{Hom}_R(M,N)$	proposition 17.6.6
$\operatorname{Ext}^i_{\mathbb{Z}}(\mathbb{Z}/n\mathbb{Z}, A) = A[n],\ A/nA,\ 0$ and	proposition 17.6.7
$\operatorname{Tor}_i^{\mathbb{Z}}(\mathbb{Z}/n\mathbb{Z}, A) = A/nA,\ A[n],\ 0$	
$\operatorname{Ext}^i(P,N) = 0$ for P projective; $\operatorname{Tor}_i(M,F) = 0$ for F flat	corollary 17.6.8
Independent of the resolution; flat resolutions suffice for Tor	theorem 17.6.5
Long exact sequences in each variable	theorem 17.6.9
$\operatorname{Ext}^1_R(C,A) \leftrightarrow \mathcal{E}(C,A)$, with 0 the split class	theorem 17.6.10

The last row answers the question the chapter opened with, and the row above it is what turns the answer into a computation: $\operatorname{Ext}^1_{\mathbb{Z}}(\mathbb{Z}/p\mathbb{Z}, \mathbb{Z}/p\mathbb{Z}) \cong \mathbb{Z}/p\mathbb{Z}$ has p elements, which is the count example 17.1.8(2) arrived at by hand.

Three of those statements are proved elsewhere. Theorem 17.6.5 and theorem 17.6.9 need the comparison theorem and chain homotopies, and theorem 17.6.10 needs the full construction of the correspondence. The first two are [19, Unique Projection Resolution, Chain-Homotopy and Cohomology], and the third is deferred there with the correspondence. That book takes up the three module classes of this chapter in the setting of an abelian category and takes the derived functors as its subject.

18⋆

Morita Theory

⋆ *Optional. This chapter is an aside; nothing later in the book depends on it.*

The rings R and $M_n(R)$ agree in almost nothing. One may be commutative. The other is not, once $n \geq 2$. Their unit groups are unrelated, and so are their underlying additive groups. Yet every statement about left R-modules translates into a statement about left $M_n(R)$-modules and back again, because the two module categories are equivalent. Chapter 12 met this phenomenon, named it *Morita equivalence*, and deferred it. This chapter settles it.

The result is Morita's theorem: every equivalence R-**Mod** $\to$ S-**Mod** is given by tensoring with a bimodule, and the bimodules that occur are exactly those that are finitely generated, projective, and large enough to reach every module. Most of what the proof needs is already available. Chapter 15 constructed the tensor product of bimodules and proved the Hom-tensor adjunction (theorem 15.3.9). Chapter 17⋆ gave projective modules and the dual basis lemma (proposition 17.2.8). One notion is new, and section 18.1 isolates it before section 18.2 proves the theorem and section 18.3 works out what an equivalence preserves and what it destroys.

Nothing in Part IV uses this chapter, which is why it is marked optional. Its payoff is further out: Morita equivalence is the reduction that makes the representation theory of finite-dimensional algebras tractable, in *Core Representation Theory* [17], and the invariance that makes K_0 computable, in *Algebraic K-Theory* [10].

18.1 Generators and Progenerators

An equivalence of categories carries an object to one with the same categorical description, so the first question is which description singles out R among left R-modules. Three properties do, and each is phrased purely in terms of maps: R is projective (definition 17.2.3), it is finitely generated, and every left R-module is a quotient of a direct sum of copies of R. The first two have names already.

The third does not.

Definition 18.1.1: Generator

Let R be a ring. A left R-module P is a *generator* of R-**Mod** if for every left R-module M there is an index set I and a surjection

$$P^{(I)} := \bigoplus_{i \in I} P \twoheadrightarrow M$$

The module R is a generator: any M is the image of the map $R^{(M)} \to M$ sending the basis element indexed by m to m. So definition 18.1.1 asks of P exactly what makes that argument work for R.

Checking the definition directly means quantifying over every module and every index set, which is unworkable. The next definition gathers the images of all maps $P \to R$ into a single subset of R, and theorem 18.1.4 shows that this one subset already decides the question.

Definition 18.1.2: Trace Ideal

Let P be a left R-module. The *trace ideal* of P is

$$\operatorname{Tr}(P) := \sum_{f \in \operatorname{Hom}_R(P,R)} f(P) \subseteq R$$

the additive subgroup of R generated by the elements $f(p)$ for $f \in \operatorname{Hom}_R(P, R)$ and $p \in P$.

Lemma 18.1.3: The Trace Ideal Is Two-Sided

For every left R-module P, the set $\operatorname{Tr}(P)$ is a two-sided ideal of R.

***Proof*:**
It is an additive subgroup by construction, so only the two multiplications need checking, and since $\operatorname{Tr}(P)$ is generated additively by the elements $f(p)$ it is enough to check them on such an element.

For left multiplication, let $r \in R$. Then $r\,f(p) = f(rp)$ because f is R-linear, and $rp \in P$, so $r\,f(p) \in f(P) \subseteq \operatorname{Tr}(P)$.

For right multiplication, fix $a \in R$ and define $g\colon P \to R$ by $g(x) = f(x)a$. This g is additive, and for $r \in R$ and $x \in P$,

$$g(rx) = f(rx)a = \big(r f(x)\big)a = r\big(f(x)a\big) = r\,g(x)$$

so $g \in \operatorname{Hom}_R(P, R)$. Hence $f(p)a = g(p) \in \operatorname{Tr}(P)$, completing the proof.

Theorem 18.1.4: Characterizations of a Generator

Let P be a left R-module. The following are equivalent.

1. P is a generator of R-**Mod**.
2. $\mathrm{Tr}(P) = R$.
3. R is isomorphic to a direct summand of P^n for some finite $n \geq 1$.
4. The functor $\mathrm{Hom}_R(P, -)\colon R\text{-}\mathbf{Mod} \to \mathbf{Ab}$ is faithful.

***Proof*:**

(1) $\Rightarrow$ (2). Apply definition 18.1.1 to $M = R$, giving a surjection $\pi\colon P^{(I)} \to R$. Write $\iota_i\colon P \to P^{(I)}$ for the inclusion of the i-th summand and set $f_i := \pi \circ \iota_i \in \mathrm{Hom}_R(P, R)$. Since π is surjective there is $x \in P^{(I)}$ with $\pi(x) = 1$, and an element of a direct sum has only finitely many nonzero coordinates, so $x = \sum_{i \in F} \iota_i(p_i)$ for a finite $F \subseteq I$ and elements $p_i \in P$. Then

$$1 = \pi(x) = \sum_{i \in F} f_i(p_i) \in \mathrm{Tr}(P)$$

By lemma 18.1.3 the set $\mathrm{Tr}(P)$ is an ideal, and an ideal containing 1 is the whole ring, so $\mathrm{Tr}(P) = R$.

(2) $\Rightarrow$ (3). By (2) there are $f_1, \ldots, f_n \in \mathrm{Hom}_R(P, R)$ and $p_1, \ldots, p_n \in P$ with $\sum_{i=1}^n f_i(p_i) = 1$. Define

$$\varphi\colon P^n \to R, \quad \varphi(x_1, \ldots, x_n) = \sum_{i=1}^n f_i(x_i), \qquad \sigma\colon R \to P^n, \quad \sigma(r) = (rp_1, \ldots, rp_n)$$

Both are R-linear: φ because each f_i is, and σ because the action on P^n is coordinatewise. For every $r \in R$,

$$\varphi(\sigma(r)) = \sum_{i=1}^n f_i(rp_i) = r \sum_{i=1}^n f_i(p_i) = r$$

so σ is a section of φ. In particular φ is surjective, and proposition 17.1.13 applied to $0 \to \ker(\varphi) \to P^n \xrightarrow{\varphi} R \to 0$ shows that this sequence splits, so $P^n \cong \sigma(R) \oplus \ker(\varphi)$ with $\sigma(R) \cong R$.

(3) $\Rightarrow$ (4). Let $f\colon M \to N$ be a nonzero map of left R-modules. The claim is that $f_* = \mathrm{Hom}_R(P, f)$ is nonzero. Choose $m \in M$ with $f(m) \neq 0$ and let $g\colon R \to M$ be the R-linear map $g(r) = rm$, so $f(g(1)) = f(m) \neq 0$ and hence $f \circ g \neq 0$. By (3) there are R-linear maps $\iota\colon R \to P^n$ and $\rho\colon P^n \to R$ with $\rho \circ \iota = \mathrm{id}_R$. Then

$$(f \circ g \circ \rho) \circ \iota = f \circ g \circ (\rho \circ \iota) = f \circ g \neq 0$$

so $f \circ g \circ \rho\colon P^n \to N$ is nonzero. A map out of a finite direct sum vanishes if and only if its restriction to every summand vanishes, so there is a coordinate inclusion $\kappa_j\colon P \to P^n$ with $f \circ (g \circ \rho \circ \kappa_j) \neq 0$. Setting $h := g \circ \rho \circ \kappa_j \in \mathrm{Hom}_R(P, M)$ gives $f_*(h) \neq 0$.

(4) $\Rightarrow$ (1). Let M be a left R-module and put

$$T := \sum_{h \in \mathrm{Hom}_R(P, M)} h(P) \subseteq M$$

a submodule of M, being a sum of the submodules $h(P)$. Let $q\colon M \to M/T$ be the quotient map. Every $h \in \mathrm{Hom}_R(P, M)$ has $h(P) \subseteq T$, so $q \circ h = 0$. That is, q_* is the zero map on $\mathrm{Hom}_R(P, M)$. Since $\mathrm{Hom}_R(P, -)$ is faithful and q_* agrees with the map induced by the zero homomorphism, it follows that $q = 0$, hence $M/T = 0$ and $T = M$. The map

$$P^{(\mathrm{Hom}_R(P,M))} \to M$$

whose restriction to the summand indexed by h is h therefore has image $T = M$, so it is surjective. As M was arbitrary, P is a generator, completing the proof.

Condition (2) is the one to check in practice, since it trades a quantifier over all modules for a computation inside R. Condition (3) is the form section 18.2 uses: it presents R as a retract of a finite power of P, which is what allows an argument about R to be transported to P. Condition (4) is the categorical reading, that P is a generator exactly when no nonzero map between modules is invisible to P.

Definition 18.1.5: Progenerator

A left R-module P is a *progenerator* if it is finitely generated, projective, and a generator of R-**Mod**.

These are the three properties of R listed at the start of the section, so R is a progenerator. Theorem 18.2.8 will show that the progenerators are precisely the modules that can take its place.

Example 18.1.6: Progenerators and Two Near Misses

1. For any ring R and any $n \geq 1$, the module R^n is a progenerator. It is free, hence projective by corollary 17.2.6, and finitely generated. The coordinate projection $R^n \to R$ is surjective, so $\mathrm{Tr}(R^n) = R$.

2. *Finitely generated and projective, but not a generator.* Let $R = \mathbb{Z}/6\mathbb{Z}$ and $e = \overline{3}$, which satisfies $e^2 = \overline{9} = \overline{3} = e$. Then $R = Re \oplus R(1 - e)$, so $P := Re = \{\overline{0}, \overline{3}\}$ is a direct summand of R, hence projective by theorem 17.2.4(4), and it is generated by one element. Every $f \in \mathrm{Hom}_R(P, R)$ satisfies
$$f(e) = f(e \cdot e) = e\, f(e) \in eR$$
and R is commutative here, so $f(P) = f(Re) = Rf(e) \subseteq Re$, giving $\mathrm{Tr}(P) = Re \neq R$. By theorem 18.1.4 the module P is not a generator. Concretely, every element of a direct sum of copies of P is annihilated by $\overline{2}$, while $\overline{1} \in R$ is not, so no such sum surjects onto R.

3. *A generator that is not projective.* Let $R = \mathbb{Z}$ and $P = \mathbb{Z} \oplus \mathbb{Z}/2\mathbb{Z}$. The projection onto the first coordinate lies in $\mathrm{Hom}_{\mathbb{Z}}(P, \mathbb{Z})$ and has image $\mathbb{Z}$, so $\mathrm{Tr}(P) = \mathbb{Z}$ and P is a generator by theorem 18.1.4. It is not projective: over the principal ideal domain $\mathbb{Z}$ projective and free coincide (proposition 17.2.11), and P has the nonzero torsion element $(0, \overline{1})$, so it is not free.

The two near misses show that neither condition in definition 18.1.5 implies the other, so both are genuine requirements on the bimodule that Morita's theorem produces.

Exercise 18.1.1

1. Let P be a left R-module and $n \geq 1$. Show that $\operatorname{Tr}(P^n) = \operatorname{Tr}(P)$, and deduce that P is a generator if and only if P^n is.
2. Show that every nonzero free left R-module is a generator of R-**Mod**, and that any module with a nonzero free direct summand is a generator.
3. Let $R = \mathbb{Z}/6\mathbb{Z}$. Decide, for each of the R-modules $\mathbb{Z}/2\mathbb{Z}$, $\mathbb{Z}/3\mathbb{Z}$, $\mathbb{Z}/6\mathbb{Z}$ and $\mathbb{Z}/2\mathbb{Z} \oplus \mathbb{Z}/3\mathbb{Z}$, whether it is a progenerator, justifying each answer using theorem 18.1.4.
4. Let P be a projective left R-module. Using the dual basis lemma (proposition 17.2.8), show that $\operatorname{Tr}(P)$ is idempotent, that is $\operatorname{Tr}(P)\operatorname{Tr}(P) = \operatorname{Tr}(P)$.
5. Let P be a generator of R-**Mod**. Show that $\operatorname{Hom}_R(P, M) = 0$ implies $M = 0$. Then show that this consequence does not characterize generators, by exhibiting a ring R and a non-generator P for which it still holds.

18.2 Morita's Theorem

The statement to be proved compares two module categories, so it needs the vocabulary for comparing categories. Functors are available (definition 8.0.4). The maps between functors are not, and the notion of two categories being "the same" rests on them.

Definition 18.2.1: Natural Transformation and Natural Isomorphism

Let $F, G \colon \mathcal{C} \to \mathcal{D}$ be functors. A *natural transformation* $\eta \colon F \Rightarrow G$ assigns to each object C of $\mathcal{C}$ a morphism $\eta_C \colon F(C) \to G(C)$ of $\mathcal{D}$ such that for every morphism $f \colon C \to C'$ of $\mathcal{C}$ the square

$$\begin{array}{ccc} F(C) & \xrightarrow{\eta_C} & G(C) \\ {\scriptstyle F(f)}\downarrow & & \downarrow{\scriptstyle G(f)} \\ F(C') & \xrightarrow[\eta_{C'}]{} & G(C') \end{array}$$

commutes. It is a *natural isomorphism*, written $F \cong G$, if every η_C is an isomorphism.

Definition 18.2.2: Equivalence of Categories

An *equivalence* between categories $\mathcal{C}$ and $\mathcal{D}$ is a pair of functors $F \colon \mathcal{C} \to \mathcal{D}$ and $G \colon \mathcal{D} \to \mathcal{C}$ together with natural isomorphisms $GF \cong \mathrm{id}_{\mathcal{C}}$ and $FG \cong \mathrm{id}_{\mathcal{D}}$. The functor G is a *quasi-inverse* of F, and $\mathcal{C}$ and $\mathcal{D}$ are *equivalent*.

An equivalence is bijective on morphism sets: F induces a bijection $\operatorname{Hom}_{\mathcal{C}}(C, C') \to \operatorname{Hom}_{\mathcal{D}}(FC, FC')$, with inverse obtained by applying G and conjugating by the natural isomorphism $GF \cong \mathrm{id}$. Every property of an object stated purely in terms of morphisms therefore transfers across an equivalence, which is what makes theorem 18.1.4(4) the useful characterization of a generator.

Definition 18.2.3: Morita Equivalence

Rings R and S are *Morita equivalent* if R-**Mod** and S-**Mod** are equivalent categories.

Now fix a left R-module P. Two rings act on it: R on the left, and its own endomorphism ring. Writing endomorphisms on the right makes the second action a right action, and that is the convention the whole section uses.

Proposition 18.2.4: The Bimodules Attached to a Module

Let P be a left R-module and set $S := \mathrm{End}_R(P)^{\mathrm{op}}$ and $Q := \mathrm{Hom}_R(P, R)$.

1. P is an (R, S)-bimodule under $p \cdot s := s(p)$.
2. Q is an (S, R)-bimodule under $(s \cdot q)(p) := q(s(p))$ and $(q \cdot r)(p) := q(p)r$.

***Proof*:**

1. Multiplication in S is $s_1 s_2 = s_2 \circ s_1$, so $p \cdot (s_1 s_2) = (s_2 \circ s_1)(p) = s_2(s_1(p)) = (p \cdot s_1) \cdot s_2$, and $p \cdot \mathrm{id}_P = p$. The action is additive in each variable because endomorphisms are additive. Compatibility with the left action is R-linearity of s:

$$(rp) \cdot s = s(rp) = r\, s(p) = r(p \cdot s)$$

2. First, $s \cdot q$ and $q \cdot r$ lie in Q: the map $p \mapsto q(s(p))$ is a composite of R-linear maps, and $(q \cdot r)(ap) = q(ap)r = a\, q(p)\, r = a\, (q \cdot r)(p)$. The left action is an action because

$$\big((s_1 s_2) \cdot q\big)(p) = q\big((s_1 s_2)(p)\big) = q\big(s_2(s_1(p))\big) = (s_2 \cdot q)(s_1(p)) = \big(s_1 \cdot (s_2 \cdot q)\big)(p)$$

using $(s_1 s_2)(p) = s_2(s_1(p))$ from the opposite multiplication. The right action is one because $(q \cdot (r_1 r_2))(p) = q(p) r_1 r_2 = ((q \cdot r_1) \cdot r_2)(p)$. They commute:

$$\big((s \cdot q) \cdot r\big)(p) = (s \cdot q)(p)\, r = q(s(p))\, r = (q \cdot r)(s(p)) = \big(s \cdot (q \cdot r)\big)(p)$$

completing the proof.

With P an (R, S)-bimodule and Q an (S, R)-bimodule, the two tensor products $P \otimes_S Q$ and $Q \otimes_R P$ are an (R, R)-bimodule and an (S, S)-bimodule respectively (proposition 15.2.1). Each carries an evaluation map to the ring on the corresponding side, and the entire theorem is the assertion that both are isomorphisms.

Proposition 18.2.5: The Morita Maps

With P, S and Q as in proposition 18.2.4, there are well-defined maps

$$\mu\colon P \otimes_S Q \to R, \quad \mu(p \otimes q) = q(p), \qquad \tau\colon Q \otimes_R P \to S, \quad \tau(q \otimes p) = \big[x \mapsto q(x)p\big]$$

the first a homomorphism of (R,R)-bimodules and the second of (S,S)-bimodules, and they satisfy the *associativity identities*

$$\mu(p \otimes q)\, p' = p \cdot \tau(q \otimes p'), \qquad q \cdot \mu(p \otimes q') = \tau(q \otimes p) \cdot q'$$

for all $p, p' \in P$ and $q, q' \in Q$.

***Proof*:**
Well-definedness. The assignment $(p,q) \mapsto q(p)$ is additive in each variable and satisfies $\mu(p{\cdot}s, q) = q(s(p)) = (s \cdot q)(p) = \mu(p, s \cdot q)$, so it is S-balanced and induces μ on $P \otimes_S Q$. For τ, the map $x \mapsto q(x)p$ is R-linear since $q(ax)p = a\, q(x)p$, so it is an element of S. The assignment is additive in each variable and

$$\tau(q \cdot r, p)(x) = (q \cdot r)(x)p = q(x)rp = \tau(q, rp)(x)$$

so it is R-balanced and induces τ.

Bimodule linearity. For μ: $\mu(rp \otimes q) = q(rp) = r\, q(p)$ and $\mu(p \otimes q \cdot r) = (q \cdot r)(p) = q(p)r$. For τ, recall $s_1 s_2 = s_2 \circ s_1$ in S. On the left,

$$\tau(s \cdot q \otimes p)(x) = (s \cdot q)(x)\, p = q(s(x))\, p = \big(\tau(q \otimes p) \circ s\big)(x) = \big(s\, \tau(q \otimes p)\big)(x)$$

and on the right, using R-linearity of s,

$$\tau(q \otimes p \cdot s)(x) = q(x)\, s(p) = s\big(q(x)p\big) = \big(s \circ \tau(q \otimes p)\big)(x) = \big(\tau(q \otimes p)\, s\big)(x)$$

The identities. For the first, $\mu(p \otimes q)p' = q(p)p'$, while $\tau(q \otimes p')$ is the endomorphism $x \mapsto q(x)p'$, so $p \cdot \tau(q \otimes p') = \tau(q \otimes p')(p) = q(p)p'$. For the second, evaluate both sides at $x \in P$:

$$\big(q \cdot \mu(p \otimes q')\big)(x) = q(x)\, q'(p), \qquad \big(\tau(q \otimes p) \cdot q'\big)(x) = q'\big(\tau(q \otimes p)(x)\big) = q'\big(q(x)p\big) = q(x)\, q'(p)$$

as we sought to show.

The two maps measure exactly the two conditions of definition 18.1.5, one each.

Lemma 18.2.6: What the Morita Maps Detect

With notation as above:

1. $\mathrm{im}\mu = \mathrm{Tr}(P)$, so μ is surjective if and only if P is a generator.
2. τ is surjective if and only if P is finitely generated and projective.

***Proof*:**

1. The image of μ is the additive subgroup generated by the elements $q(p)$ for $q \in \mathrm{Hom}_R(P, R)$ and $p \in P$, which is definition 18.1.2 verbatim. The equivalence with being a generator is theorem 18.1.4, (1) $\Leftrightarrow$ (2), since a subgroup of R equals R exactly when the map onto it is surjective.

2. ($\Rightarrow$) If τ is surjective, then $\mathrm{id}_P = \tau(t)$ for some $t \in Q \otimes_R P$, and every element of a tensor product is a finite sum of simple tensors, say $t = \sum_{i=1}^{n} q_i \otimes p_i$. Evaluating at $x \in P$,

$$x = \mathrm{id}_P(x) = \sum_{i=1}^{n} q_i(x)\, p_i$$

which is a finite dual basis, so P is finitely generated and projective by proposition 17.2.8.

($\Leftarrow$) If P is finitely generated and projective, proposition 17.2.8 gives a finite dual basis $(p_i, q_i)_{i=1}^{n}$, and the displayed identity says $\tau\big(\sum_i q_i \otimes p_i\big) = \mathrm{id}_P$. The image of τ is a two-sided ideal of S, being the image of an (S, S)-bimodule map, and an ideal containing the identity is the whole ring, completing the proof.

Surjectivity of the two maps is therefore exactly the progenerator hypothesis. What is not yet clear is injectivity, and the next lemma is the reason the theorem needs no separate argument for it: each map is forced to be injective by its own surjectivity, through the associativity identities that tie it to the other.

Lemma 18.2.7: Surjective Forces Bijective

If μ is surjective then μ is injective. If τ is surjective then τ is injective.

***Proof*:**

Suppose μ is surjective and let $t = \sum_{i=1}^{m} p_i \otimes q_i$ lie in $\ker \mu$, so that $\sum_i q_i(p_i) = 0$. Surjectivity gives elements $a_j \in P$ and $b_j \in Q$ with $1 = \sum_{j=1}^{k} \mu(a_j \otimes b_j)$. Then

$$t = \sum_i \Big(\sum_j \mu(a_j \otimes b_j)\Big) p_i \otimes q_i = \sum_{i,j} \big(\mu(a_j \otimes b_j)\, p_i\big) \otimes q_i = \sum_{i,j} \big(a_j \cdot \tau(b_j \otimes p_i)\big) \otimes q_i$$

using the first associativity identity of proposition 18.2.5. Moving the element of S across the tensor and applying the second identity,

$$t = \sum_{i,j} a_j \otimes \big(\tau(b_j \otimes p_i) \cdot q_i\big) = \sum_{i,j} a_j \otimes \big(b_j \cdot \mu(p_i \otimes q_i)\big) = \sum_j a_j \otimes \Big(b_j \cdot \sum_i \mu(p_i \otimes q_i)\Big)$$

and the inner sum is $\mu(t) = 0$, so $t = 0$. The argument for τ is the same with the roles of the two identities and of (P, R) and (Q, S) interchanged, completing the proof.

Theorem 18.2.8: Morita's Theorem

Let P be a progenerator in R-**Mod**, and set $S = \mathrm{End}_R(P)^{\mathrm{op}}$ and $Q = \mathrm{Hom}_R(P, R)$. Then

$$\mu \colon P \otimes_S Q \xrightarrow{\sim} R, \qquad \tau \colon Q \otimes_R P \xrightarrow{\sim} S$$

are isomorphisms of bimodules, and the functors

$$Q \otimes_R - \colon R\text{-}\mathbf{Mod} \to S\text{-}\mathbf{Mod}, \qquad P \otimes_S - \colon S\text{-}\mathbf{Mod} \to R\text{-}\mathbf{Mod}$$

are mutually quasi-inverse equivalences. In particular R and S are Morita equivalent.

***Proof*:**
Since P is a generator, μ is surjective by lemma 18.2.6(1). Since P is finitely generated and projective, τ is surjective by lemma 18.2.6(2). Both are then injective by lemma 18.2.7, hence bijective, and they are bimodule maps by proposition 18.2.5.

For the functors, let N be a left S-module. Associativity of the tensor product (theorem 15.2.8) and the isomorphism just established give

$$Q \otimes_R \left(P \otimes_S N\right) \cong \left(Q \otimes_R P\right) \otimes_S N \cong S \otimes_S N \cong N$$

and symmetrically, for a left R-module M,

$$P \otimes_S \left(Q \otimes_R M\right) \cong \left(P \otimes_S Q\right) \otimes_R M \cong R \otimes_R M \cong M$$

Each isomorphism in these chains is natural in the module argument, since theorem 15.2.8 is natural and the remaining two steps are the unit isomorphisms of the tensor product. The two chains are therefore natural isomorphisms $(Q \otimes_R -) \circ (P \otimes_S -) \cong \mathrm{id}$ and $(P \otimes_S -) \circ (Q \otimes_R -) \cong \mathrm{id}$, which is definition 18.2.2, completing the proof.

The theorem manufactures an equivalence from every progenerator. The converse says that this is the only source: an equivalence cannot arise any other way, and the progenerator producing it is recovered as the image of the ring.

Theorem 18.2.9: Every Equivalence Is Tensoring with a Progenerator

Let $F \colon R$-**Mod** $\to S$-**Mod** be an equivalence with quasi-inverse G. Put $Q := F(R)$ and $P := G(S)$. Then Q carries a right R-action making it an (S, R)-bimodule, P carries a right S-action making it an (R, S)-bimodule,

$$P \otimes_S Q \cong R \ \text{ as } (R,R)\text{-bimodules}, \qquad Q \otimes_R P \cong S \ \text{ as } (S,S)\text{-bimodules}$$

and $F \cong Q \otimes_R -$. Moreover P is a progenerator in R-**Mod**.

***Proof*:**
Step 1: the right action on Q. For $r \in R$ let $\rho_r \colon R \to R$ be $x \mapsto xr$, which is left R-linear. Since $\rho_{r_1 r_2} = \rho_{r_2} \circ \rho_{r_1}$ and F is a functor, $F(\rho_{r_1 r_2}) = F(\rho_{r_2}) \circ F(\rho_{r_1})$. Setting $q \cdot r := F(\rho_r)(q)$ therefore defines a right R-action on Q, and it commutes with the left S-action because each

$F(\rho_r)$ is a map of S-modules. The same construction gives the right S-action on P.

Step 2: F preserves direct sums and cokernels. An equivalence is bijective on morphism sets, as recorded after definition 18.2.2, and it is essentially surjective, every object of S-**Mod** being isomorphic to $F(G(N))$. A cocone is a direct sum, or a cokernel, exactly when a certain map of morphism sets is bijective for every test object. Bijectivity on morphism sets transports the condition forward and essential surjectivity gives every test object on the far side. So F carries direct sums to direct sums and cokernels to cokernels.

Step 3: $F \cong Q \otimes_R -$. For a left R-module M and $m \in M$ let $\lambda_m \colon R \to M$ be $r \mapsto rm$. The assignment $(q, m) \mapsto F(\lambda_m)(q)$ is additive in each variable and R-balanced, since $\lambda_{rm} = \lambda_m \circ \rho_r$ gives $F(\lambda_{rm})(q) = F(\lambda_m)(q \cdot r)$. It therefore induces $\theta_M \colon Q \otimes_R M \to F(M)$, and θ is natural in M because $f \circ \lambda_m = \lambda_{f(m)}$ for any $f \colon M \to M'$.

At $M = R$ the map θ_R is the isomorphism $Q \otimes_R R \cong Q = F(R)$, since $\lambda_r = \rho_r$. Both $Q \otimes_R -$ and F preserve direct sums, the first by corollary 15.2.5 and the second by Step 2, so $\theta_{R^{(I)}}$ is an isomorphism for every index set I. Now take any M and a free presentation $R^{(J)} \to R^{(I)} \to M \to 0$, which exists because every module is a quotient of a free one and the kernel of that quotient is again such a quotient. Both functors are right exact, again by theorem 17.5.1 and Step 2, so the rows of

$$\begin{array}{ccccccc} Q \otimes_R R^{(J)} & \longrightarrow & Q \otimes_R R^{(I)} & \longrightarrow & Q \otimes_R M & \longrightarrow & 0 \\ \downarrow{\scriptstyle \theta_{R^{(J)}}} & & \downarrow{\scriptstyle \theta_{R^{(I)}}} & & \downarrow{\scriptstyle \theta_M} & & \\ F(R^{(J)}) & \longrightarrow & F(R^{(I)}) & \longrightarrow & F(M) & \longrightarrow & 0 \end{array}$$

are exact and the square commutes by naturality. The two left vertical maps are isomorphisms, and a cokernel is determined up to unique isomorphism by the map it is taken of, so θ_M is an isomorphism.

Step 4: the bimodule isomorphisms. By Step 3 applied to F and to G, the composite $GF \cong P \otimes_S (Q \otimes_R -) \cong (P \otimes_S Q) \otimes_R -$ using theorem 15.2.8. Since $GF \cong \mathrm{id}$, evaluating at R gives $(P \otimes_S Q) \otimes_R R \cong R$, that is $P \otimes_S Q \cong R$. The isomorphism is one of (R, R)-bimodules because the natural isomorphism $GF \cong \mathrm{id}$ commutes with the maps ρ_r, whose images generate the right action. Symmetrically $Q \otimes_R P \cong S$.

Step 5: P is a progenerator. From $P \otimes_S Q \cong R$: the tensor product is generated as an abelian group by simple tensors, so the map $\bigoplus_{q \in Q} P \to P \otimes_S Q$ sending the summand indexed by q to $p \mapsto p \otimes q$ is surjective, exhibiting R as a quotient of a direct sum of copies of P. Hence P is a generator. From $Q \otimes_R P \cong S$: the element of $Q \otimes_R P$ corresponding to 1_S is a *finite* sum $\sum_{i=1}^{n} q_i \otimes p_i$, and unwinding the isomorphism as in lemma 18.2.6(2) yields $x = \sum_i q_i(x) p_i$ for all $x \in P$, a finite dual basis. So P is finitely generated and projective by proposition 17.2.8, completing the proof.

Taken together the two theorems say that Morita equivalence is not an exotic relation to be checked category by category: R and S are Morita equivalent precisely when $S \cong \mathrm{End}_R(P)^{\mathrm{op}}$ for some progenerator P over R. The next example is the case that motivated the whole discussion.

Example 18.2.10: A Ring and Its Matrix Rings

Let R be any ring and $n \geq 1$, and take $P = R^n$, the module of columns. By example 18.1.6(1) it is a progenerator. Its endomorphism ring is computed one summand at a time: an R-linear map $R \to R$ is right multiplication by its value at 1, so $\mathrm{End}_R(R) \cong R^{\mathrm{op}}$, and therefore

$$\mathrm{End}_R(R^n) \cong M_n\big(\mathrm{End}_R(R)\big) \cong M_n(R^{\mathrm{op}}) \cong M_n(R)^{\mathrm{op}}$$

the last isomorphism being transposition, which reverses the order of multiplication. Hence

$$S = \mathrm{End}_R(R^n)^{\mathrm{op}} \cong M_n(R)$$

and theorem 18.2.8 gives an equivalence R-**Mod** $\simeq M_n(R)$-**Mod**. Unwinding it, $Q = \mathrm{Hom}_R(R^n, R) \cong R^n$ as rows, the equivalence R-**Mod** $\to M_n(R)$-**Mod** sends M to the column space M^n with $M_n(R)$ acting by matrix multiplication, and its quasi-inverse sends N to eN where $e = E_{11}$ is the idempotent matrix with a single 1 in the top-left entry.

Nothing about R is recovered by $M_n(R)$ beyond its module category: for $n \geq 2$ the ring $M_n(R)$ is never commutative, so a commutative R is Morita equivalent to noncommutative rings. Section 18.3 makes precise which properties survive.

Exercise 18.2.1

1. Let P be a progenerator over R and $S = \mathrm{End}_R(P)^{\mathrm{op}}$. Show that $\mathrm{Hom}_R(P, -)$ and $Q \otimes_R -$ are naturally isomorphic as functors R-**Mod** $\to S$-**Mod**, where $Q = \mathrm{Hom}_R(P, R)$.
2. Show that Morita equivalence is an equivalence relation on rings.
3. Show that if R and S are Morita equivalent, then so are $M_n(R)$ and $M_n(S)$ for every $n \geq 1$.
4. Show that a commutative ring R and the matrix ring $M_n(R)$ have isomorphic centres for every $n \geq 1$, and explain why example 18.2.10 does not contradict this.
5. Let D be a division ring. Show that every progenerator over D is isomorphic to D^n for some $n \geq 1$, and deduce that the rings Morita equivalent to D are exactly the matrix rings $M_n(D)$.
6. Give an example of two rings that are Morita equivalent but not isomorphic, and an example of two rings that are not Morita equivalent even though both are commutative and have the same cardinality.
7. Let P be a progenerator over R. Show that P is also a progenerator when regarded as a right S-module, where $S = \mathrm{End}_R(P)^{\mathrm{op}}$.

18.3 Morita Invariants and the Basic Algebra

Example 18.2.10 showed that R and $M_n(R)$ have the same module category while differing in nearly every element-level respect. A property of a ring that transfers across every Morita equivalence is called a *Morita invariant*, and the question this section answers is which properties those are. The test is uniform: a property is a Morita invariant when it can be stated in terms of the module category alone, and it fails to be one as soon as a single equivalence moves it.

Commutativity is the basic failure. For $n \geq 2$ the ring $M_n(R)$ is noncommutative whatever R is, so no property implying commutativity survives. What does survive is the part of the multiplication that is visible categorically, and theorem 18.3.2 identifies it exactly: the center.

Proposition 18.3.1: The Center as Endomorphisms of the Identity

Let R be a ring. Sending $c \in Z(R)$ to the family of maps $\eta_M^c \colon M \to M$, $\eta_M^c(m) = cm$, is a ring isomorphism from $Z(R)$ (definition 9.1.8) to the ring of natural transformations $\mathrm{id}_{R\text{-}\mathbf{Mod}} \Rightarrow \mathrm{id}_{R\text{-}\mathbf{Mod}}$, under pointwise addition and composition.

***Proof*:**

The map is well defined. For $c \in Z(R)$ each η_M^c is R-linear, since $\eta_M^c(rm) = c(rm) = (cr)m = (rc)m = r(cm) = r\,\eta_M^c(m)$, which is exactly where centrality is used. It is natural: for $f \colon M \to N$ and $m \in M$,

$$f\big(\eta_M^c(m)\big) = f(cm) = c\,f(m) = \eta_N^c\big(f(m)\big)$$

Injectivity. If $\eta^c = \eta^{c'}$ then in particular $c = \eta_R^c(1) = \eta_R^{c'}(1) = c'$.

Surjectivity. Let η be a natural transformation of the identity and put $c := \eta_R(1)$. For $r \in R$, R-linearity of η_R gives $\eta_R(r) = \eta_R(r \cdot 1) = r\,\eta_R(1) = rc$. Naturality applied to the right multiplication $\rho_r \colon R \to R$, $x \mapsto xr$, which is a map of left R-modules, gives $\eta_R(xr) = \eta_R(x)r$. At $x = 1$ this reads $\eta_R(r) = cr$. Comparing the two expressions, $rc = cr$ for every $r \in R$, so $c \in Z(R)$.

It remains to see that $\eta = \eta^c$ on every module, not only on R. For $m \in M$ let $\lambda_m \colon R \to M$ be $r \mapsto rm$. Naturality gives

$$\eta_M(m) = \eta_M\big(\lambda_m(1)\big) = \lambda_m\big(\eta_R(1)\big) = \lambda_m(c) = cm = \eta_M^c(m)$$

Ring structure. Addition is pointwise on both sides. For multiplication, $\eta_M^c\big(\eta_M^{c'}(m)\big) = c(c'm) = (cc')m = \eta_M^{cc'}(m)$, so composition corresponds to multiplication in $Z(R)$, and $\eta^1 = \mathrm{id}$, completing the proof.

The right-hand side of proposition 18.3.1 mentions only the category R-**Mod**, never the ring R. That is the whole content of the next theorem.

Theorem 18.3.2: The Center Is a Morita Invariant

If R and S are Morita equivalent, then $Z(R) \cong Z(S)$ as rings.

***Proof*:**

Let $F \colon R\text{-}\mathbf{Mod} \to S\text{-}\mathbf{Mod}$ be an equivalence with quasi-inverse G, and let $\alpha \colon GF \cong \mathrm{id}$ and $\beta \colon FG \cong \mathrm{id}$ be the natural isomorphisms of definition 18.2.2. Given a natural transformation η of $\mathrm{id}_{R\text{-}\mathbf{Mod}}$, define

$$\Phi(\eta)_N := \beta_N \circ F\big(\eta_{G(N)}\big) \circ \beta_N^{-1} \colon N \to N \qquad (N \in S\text{-}\mathbf{Mod})$$

Each component is a composite of S-linear maps. It is natural in N: for $g \colon N \to N'$, naturality of β gives $g \circ \beta_N = \beta_{N'} \circ FG(g)$, and naturality of η at $G(g)$ gives $\eta_{G(N')} \circ G(g) = G(g) \circ \eta_{G(N)}$,

so applying F and conjugating,

$$g \circ \Phi(\eta)_N = \beta_{N'} \circ F\big(G(g) \circ \eta_{G(N)}\big) \circ \beta_N^{-1} = \beta_{N'} \circ F\big(\eta_{G(N')} \circ G(g)\big) \circ \beta_N^{-1} = \Phi(\eta)_{N'} \circ g$$

So Φ carries natural transformations of $\mathrm{id}_{R\text{-}\mathbf{Mod}}$ to natural transformations of $\mathrm{id}_{S\text{-}\mathbf{Mod}}$. It preserves addition because F is additive on morphism sets and conjugation is additive, and it preserves composition because the conjugating isomorphisms cancel in the middle:

$$\Phi(\eta)_N \circ \Phi(\eta')_N = \beta_N \circ F\big(\eta_{G(N)}\big) \circ F\big(\eta'_{G(N)}\big) \circ \beta_N^{-1} = \beta_N \circ F\big(\eta_{G(N)} \circ \eta'_{G(N)}\big) \circ \beta_N^{-1}$$

The same construction applied to G gives Ψ in the opposite direction, and $\Psi\Phi = \mathrm{id}$ follows from α being a natural isomorphism by the same cancellation. Hence Φ is a ring isomorphism, and proposition 18.3.1 on each side identifies its source and target with $Z(R)$ and $Z(S)$, completing the proof.

Two consequences follow at once, and together they say that Morita equivalence is a coarser relation than isomorphism on noncommutative rings while collapsing to isomorphism on commutative ones.

Corollary 18.3.3: Commutative Rings Are Morita Rigid

Let R and S be commutative rings. Then R and S are Morita equivalent if and only if $R \cong S$.

***Proof*:**
If $R \cong S$ then the isomorphism induces an equivalence of module categories by restriction of scalars along it. Conversely, if R and S are Morita equivalent then $Z(R) \cong Z(S)$ by theorem 18.3.2, and commutativity gives $Z(R) = R$ and $Z(S) = S$ by definition 9.1.8, so $R \cong S$, as we sought to show.

Example 18.3.4: Morita Equivalence Is Coarser than Isomorphism

Take R any ring and $n \geq 2$. Then R and $M_n(R)$ are Morita equivalent by example 18.2.10 but not isomorphic whenever R is commutative, since $M_n(R)$ is not commutative: the matrix units E_{11} and E_{12} satisfy $E_{11}E_{12} = E_{12}$ while $E_{12}E_{11} = 0$. Consistently with theorem 18.3.2, the center is preserved: $Z(M_n(R))$ consists of the scalar matrices cI with $c \in Z(R)$, so $Z(M_n(R)) \cong Z(R)$.

By corollary 18.3.3 no two non-isomorphic commutative rings are Morita equivalent, so every instance of the phenomenon involves a noncommutative ring.

The remaining business of the section is the construction that produces those equivalences in practice. Example 18.1.6(2) showed that Re is projective and finitely generated for an idempotent e, and computed its trace ideal in a special case. The general computation is what decides when Re is a progenerator.

Definition 18.3.5: Full Idempotent

An idempotent $e \in R$, so $e^2 = e$, is *full* if $ReR = R$, that is if the two-sided ideal generated by e is the whole ring.

Theorem 18.3.6: The Corner Ring of a Full Idempotent

Let $e \in R$ be an idempotent. Then

1. Re is a finitely generated projective left R-module with $\mathrm{Tr}(Re) = ReR$, so Re is a progenerator if and only if e is full;
2. $\mathrm{End}_R(Re)^{\mathrm{op}} \cong eRe$, the *corner ring* of e.

Consequently, if e is full then R and eRe are Morita equivalent.

***Proof*:**

1. Since $e^2 = e$, the element $1 - e$ is also idempotent and $R = Re \oplus R(1-e)$: every r decomposes as $r = re + r(1-e)$, and if x lies in the intersection, written both as $x = re$ and as $x = r'(1-e)$, then multiplying on the right by e gives $x = re \cdot e = xe = r'(1-e)e = 0$, the last step because $(1-e)e = e - e^2 = 0$. So Re is a direct summand of R, hence projective by theorem 17.2.4(4), and it is generated by the single element e.

   For the trace ideal, take $f \in \mathrm{Hom}_R(Re, R)$. Then $f(e) = f(e \cdot e) = e\, f(e)$, so $f(e) \in eR$, and since every element of Re has the form re,

   $$f(Re) = f(Re \cdot e) = R\, f(e) \subseteq ReR$$

   Conversely, for each $a \in R$ the map $f_a \colon Re \to R$ given by $f_a(x) = xa$ is left R-linear with image Rea. Summing over $a \in R$ gives $\mathrm{Tr}(Re) \supseteq \sum_a Rea = ReR$. Hence $\mathrm{Tr}(Re) = ReR$, and theorem 18.1.4 makes Re a generator exactly when $ReR = R$. Combined with the previous paragraph, Re is a progenerator exactly when e is full.

2. A map $\varphi \in \mathrm{End}_R(Re)$ is determined by $\varphi(e)$, since $\varphi(re) = r\varphi(e)$, and $\varphi(e) = \varphi(e \cdot e) = e\, \varphi(e)$ shows $\varphi(e) \in eRe$ (it already lies in Re). Conversely each $x \in eRe$ gives $\varphi_x(re) = rex = rx$, which is left R-linear and lands in Re because $x = ex'e \in Re$. The two assignments are mutually inverse and additive.

   For the multiplication, let $\varphi(e) = ebe$ and $\psi(e) = eae$. Then

   $$(\varphi \circ \psi)(e) = \varphi(eae) = \varphi\big((ea) \cdot e\big) = (ea)\, \varphi(e) = ea\, ebe = (eae)(ebe) = \psi(e)\, \varphi(e)$$

   using $e^2 = e$ in the last-but-one step. So $\varphi \mapsto \varphi(e)$ reverses multiplication, giving $\mathrm{End}_R(Re) \cong (eRe)^{\mathrm{op}}$ and therefore $\mathrm{End}_R(Re)^{\mathrm{op}} \cong eRe$.

For the final claim, if e is full then Re is a progenerator by (1), so theorem 18.2.8 applied to $P = Re$ gives an equivalence between R-**Mod** and $\mathrm{End}_R(Re)^{\mathrm{op}}$-**Mod** $= eRe$-**Mod** by (2), completing the proof.

Theorem 18.3.6 is the form in which Morita theory is used downstream. Passing from R to eRe for a full idempotent shrinks the ring while keeping the module category, and choosing e well makes the smaller ring markedly more tractable. The matrix case is the choice $e = E_{11}$ in $R = M_n(A)$, where $eRe \cong A$ recovers example 18.2.10. In the representation theory of finite-dimensional algebras the same construction selects one idempotent from each isomorphism class of indecomposable projectives

and produces the *basic algebra*, over which every algebra's representation theory can be read. That development, which needs the semisimple structure theory this book does not carry, is in *Core Representation Theory* [17].

Example 18.3.7: The Corner Ring of a Non-Full Idempotent

Fullness is not automatic. Let $R = \mathbb{Z}/6\mathbb{Z}$ and $e = \overline{3}$, as in example 18.1.6(2). Then $ReR = Re = \{\overline{0}, \overline{3}\}$, which is not R, so e is not full. The corner ring is $eRe = \{\overline{0}, \overline{3}\} \cong \mathbb{Z}/2\mathbb{Z}$, and $\mathbb{Z}/6\mathbb{Z}$ is not Morita equivalent to $\mathbb{Z}/2\mathbb{Z}$: both are commutative and not isomorphic, so corollary 18.3.3 rules it out. The construction still produces a ring, but without fullness it produces the wrong one.

One invariant is worth recording separately, because it is the reason *Algebraic K-Theory* [10] cares about this section at all.

Corollary 18.3.8: Finitely Generated Projectives Correspond

Let $F\colon R\text{-}\mathbf{Mod} \to S\text{-}\mathbf{Mod}$ be an equivalence. Then F restricts to a bijection between the isomorphism classes of finitely generated projective R-modules and those of finitely generated projective S-modules, and this bijection preserves direct sums.

***Proof*:**

Projectivity is preserved: M is projective exactly when $\mathrm{Hom}_R(M, -)$ carries surjections to surjections (definition 17.2.3), F is bijective on morphism sets, and F carries surjections to surjections because a map is surjective exactly when its cokernel vanishes and F preserves cokernels by theorem 18.2.9, Step 2.

Being finitely generated is preserved because $F \cong Q \otimes_R -$ with $Q \otimes_R R \cong Q$ (theorem 18.2.9), so F carries R^n to Q^n and a summand of R^n to a summand of Q^n. That Q^n is itself finitely generated over S is theorem 18.2.9 applied with the roles of F and G exchanged, which makes $Q = F(R)$ a progenerator in $S\text{-}\mathbf{Mod}$. The same argument then runs backwards through G. Direct sums are preserved by Step 2 of the same proof, and the quasi-inverse G gives the inverse bijection on isomorphism classes, completing the proof.

Since K_0 of a ring is built from the isomorphism classes of finitely generated projectives under direct sum, corollary 18.3.8 says that K_0 cannot distinguish Morita equivalent rings. That statement, and the group it is about, belong to *Algebraic K-Theory* [10].

Exercise 18.3.1

1. Show that the center of $M_n(R)$ consists of the scalar matrices rI with $r \in Z(R)$, and deduce $Z(M_n(R)) \cong Z(R)$ directly, without using theorem 18.3.2.
2. Decide which of the following are Morita invariants, with proof or counterexample: being a division ring; being commutative; being an integral domain (definition 9.1.21); having exactly two two-sided ideals.
3. Show that if R is commutative then an idempotent $e \in R$ is full if and only if $e = 1$, and give a noncommutative ring with a full idempotent $e \neq 1$.
4. Let $R = M_2(\mathbb{Z})$ and $e = E_{11}$. Verify directly that e is full, identify Re and eRe explicitly, and describe the equivalence $R\text{-}\mathbf{Mod} \simeq \mathbb{Z}\text{-}\mathbf{Mod}$ it produces.

5. Show that if e and f are idempotents of R with $Re \cong Rf$ as R-modules, then $eRe \cong fRf$ as rings.

6. Let R be a ring in which $1 = e_1 + \cdots + e_n$ for orthogonal idempotents e_i, meaning $e_i e_j = 0$ for $i \neq j$. Show that $R \cong \bigoplus_i Re_i$ as left R-modules, and that e_1 is full whenever $Re_1 \cong Re_i$ for every i.

Part IV

Fields

When studying classical algebra, a lot of time and energy is dedicated to solving equations. Famously, for any equation of the form $ax^2 + bx + c = 0$ has solution

$$x = \frac{-b - \sqrt{b^2 - 4ac}}{2a}$$

which was known since at least the time of ancient Egypt (though not in the current algebraic form[1])After a long gap, Niccolò Tartaglia discovers the general equation for the cubic equation in terms by using an "imaginary" number i as a trick to solve the problem of $\sqrt{-n}$ coming up (it will take some more time before i is considered a number and not simply a tool to solve equations). Using this equation, a solution for quartic equation with respect to x was also found. The mission was on in finding a solution to the quintic equation. However, over 200 years, no such equation was found. Abel and Galois have in fact proved that only using $+, -, \times, \div$ and $\sqrt{_}$, it is *impossible* to isolate x!

This is the origin of the study of field theory and Galois theory, and will be the driving motivation in this part of the book, however the study of fields have in modern times moves past this and are now used as a primary tool in the study of algebraic number theory. In [11, Relating Algebra to Number Theory], we will use the language of field theory as the language of algebraic number theory.

[1] It was a geometric proof involving fitting squares of different sizes together

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Field Theory

A reminder that a field F is a unital commutative division ring (definition 9.1.11). Another way of thinking of a field is that it's a set with two binary operations both acting like abelian groups. Yet another way of seeing them is that $F^\times = F - \{0\}$ is an abelian group. In this chapter, we will be mostly thinking about fields as unital commutative division rings, in particular, we are going to develop how the trivial ideal structure of fields (proposition 9.4.10) lets us uniquely define some extensions of a field. We will denote $\mathbb{F}_p := \mathbb{Z}/p\mathbb{Z}$ and usually use the letters k and K to refer to fields.

Furthermore, the complex numbers $\mathbb{C}$ will be referenced for it's nice algebraic properties (ex. $i^2 = 1$ or $z = a + bi$ for any $z \in \mathbb{C}$). If the reader is unfamiliar with these notion, there are many resources online explaining these concepts[1].

19.1 Properties of Fields

First, all fields are rings, and so we can form the category **Fld** with fields as objects and ring homomorphisms as morphisms. We will use the term field homomorphsisms sometimes since the objects of study are fields with the knowledge that they are just ring homomorphisms. In general, the term ring homomorphism will be reserved for when rings will be brought into the discussion as will sometimes happen. Since the objects are fields, it is important to remember that all homomorphisms must be *injective* (see corollary 9.4.13)

[1] A famous book explaining these notions is *Althor's Complex Analysis*, in which the first chapter explores complex numbers

Proposition 19.1.1: Field Homomorphisms

Let $\varphi : k \to K$ be a ring homomorphism of fields. Then φ is injective, and its image $\varphi(k)$ is a subfield of K isomorphic to k. In particular φ is never identically 0.

***Proof*:**
Since $\varphi(1) = 1 \neq 0$ (definition 9.1.5), φ is a nonzero ring homomorphism out of the division ring k, so it is injective by corollary 9.4.13. Its image $\varphi(k)$ is a subring of K (proposition 9.3.3(1)), commutative as K is, with $1 \neq 0$, and every nonzero element of $\varphi(k)$ is $\varphi(a)$ for some $a \neq 0$, with inverse $\varphi(a^{-1}) \in \varphi(k)$, since $\varphi(a)\varphi(a^{-1}) = \varphi(1) = 1$. So $\varphi(k)$ is a subfield of K, and $\varphi\colon k \to \varphi(k)$ is a bijective ring homomorphism, hence an isomorphism by the remark following definition 9.1.5, completing the proof.

Thus, If a homomorphism between fields $\varphi : k \to K$ exists, $\varphi(k) \cong k$. In this way, K can be thought of as extending k:

Definition 19.1.2: Field Extension

Let k and K be fields. If there exists a homomorphism $\varphi : k \to K$ (which as we have shown must be injective), then we say that K *extends* k, or K is a *field extension* of k. We will denote this by $k \subseteq K$, K/k, or

$$\begin{array}{c} F \\ | \\ K \end{array}$$

the field from which we are extending is sometimes called the *base field* or *ground field*.

Note that since fielsd are commutative, field extension $\varphi : k \to F$ makes F a k-algebra.

Example 19.1.3: Field Extensions

1. For any field F, F/F is trivially a field extension, since the identity map is a field homomorphism.
2. Perhaps the most famous field extension is that of $\mathbb{C}/\mathbb{R}$, since $\mathbb{R}$ is directly embedded into $\mathbb{C}$. A perhaps easy mistake to make when first learning about field extensions is that $\mathbb{R}^2/\mathbb{R}$ is not a field extension since $\mathbb{R}^2$ is simply not a field. On the other hand, $\mathbb{C}/\mathbb{Q}$ is also a field extension, since the natural inclusion map $\iota : \mathbb{Q} \to \mathbb{C}$ is a field homomorphism. Similarly, $\mathbb{R}/\mathbb{Q}$ is a field extension.
3. In example 9.1.19, we saw that $\mathbb{Q}(\sqrt{D})$ is a field where D is not a perfect square (i.e. there does not an element $d \in \mathbb{Q}$ s.t. $d^2 \in \mathbb{Q}$). For a concrete examples, let's say $D = 2$. Then $\mathbb{Q}(\sqrt{2})/\mathbb{Q}$ is a field extension. If $D = -1$, then we usually write $\sqrt{-1}$ as i, and we get that $\mathbb{Q}(i)/\mathbb{Q}$ is a field extension. We also have that $\mathbb{C}/\mathbb{Q}(i)$ is a field extension, with the map $\iota : \mathbb{Q}(i) \to \mathbb{C}$ given by mapping $a + bi \mapsto a + bi$. Notice that the map $a + bi \mapsto a - bi$ is also a valid field homomorphism, showing that there need not be a unique map.

   We will soon find a way to define $\mathbb{Q}(\sqrt{D})$ without needing to reference set-builder notation.

4. Let k be a field and $k(x) = \text{Frac}(k[x])$ be the field of fractions of $k[x]$. Then $k(x)/k$ is a field extension.

The notation K/k may be read as "K over k". It does *not* represent the quotient K by k. This can be consider an overload of notation, but since the kernel of a ring homomorphism between fields must *always* be trivial, it is safe to see that K/k would not represent a quotient of the field K by k.

Another thing we can observe is that every field homomorphism $\varphi : F \to K$ can be extended into a ring homomorphism $f^\varphi : F[x] \to K[x]$ through

$$\sum_i a_i x^i \mapsto \sum_i \varphi(a_i) x^i$$

the fact that this is a ring homomorphisms comes straight from the properties of φ. Thus, we can consider this process to be a functor $f\text{—} : \mathbf{Fld} \to \mathbf{Ring}$.

If F/k is any field extension, then we can naturally make F into a *vector-space over* k by making k act on F by multiplication (since $k \subseteq F$, this action is well-defined). Since every field has a prime subfield, every field is at least a vector-space over it's prime-subfield. The dimension of the vector-space is given a name:

Definition 19.1.4: Degree

The *degree* (or *relative degree* or *index*) of a field of extension F/k, denoted $[F : k]$, is the dimension of F as a vector-space over k (i.e. $[F : k] = \dim_k(F)$). The extension is said to be *finite* if $[F : k]$ is finite and is said to be *infinite* otherwise.

Definition 19.1.5: Finite Extension

Let F/k be a field extension. Then F/k is called a *finite extension* if $[F : k]$ is finite, and an *infinite extension* otherwise.

In example 19.1.3, $\mathbb{C}/\mathbb{R}$ is an extension of degree 2, the basis of $\mathbb{C}$ over $\mathbb{R}$ consiss of $\{1, i\}$, and hence a finite extension. The extension $k(x)/k$ is an extension if infinite degree: If it were finite, then any rational function can be written as a sum of a finite number of polynomials up to a constant $c \in k$, but then this collection must have a highest degree term, say n. Then any polynomial of degree $n + 1$ cannot be written as a linear combination of the basis, a contradiction. Hence, it is an infinite extension. Note too that as was demonstrated in theorem 13.1.8, every vector-space has a basis, and so any element $x \in F$ can be represented as a finite linear combination:

$$a_1\beta_1 + a_2\beta_2 + \cdots + a_n\beta_n \qquad a_i \in k,\ \beta_i \in \mathcal{B}$$

We will soon show how to find such a basis.

Since F/k can also be represented as a vector-space over k, we might ask if any field homomorphism/isomorphism can is also a linear transformation. The answer is yes, if we are a bit careful. Let's say that F/k and L/k are two fields extending k, and $\varphi : F \to L$ is a field homomorphism/isomorphism. We will look for a condition under which φ is also a k-vector space homomorphism. Additivity holds because φ is a ring homomorphism. For homogeneity of the scalar, we see that for any $r \in k$ and $m \in F$:

$$\varphi(rm) = \varphi(r)\varphi(m) \overset{!}{=} r\varphi(m)$$

where $\stackrel{!}{=}$ is the condition we need so that φ is a k-vector space homomorphism. In particular, this conditions say's that for any element $r \in k$, $\varphi(r) = r$. This is stronger than saying φ maps k to itself (i.e. $\varphi(k) = k$), rather it is saying that φ is the *identity* on k. Since such field homomorphisms/isomorphsisms are also vector-space homomorphisms, they are given a name:

Definition 19.1.6: Fixed Homomorphisms

Let F/k and L/k be two field extensions of k and $\varphi : F \to L$ be a field homomorphism (resp. isomorphism). Then if φ is the identity on k, φ is said to be a *k-homomorphsm* (resp. *k-isomorphism*). If there exists a k-isomorphisms $A \to B$, then we will say that A is k-isomorphic to B and write $A \cong_k B$.

Note that a fixed homomorhism is in fact a k-algebra homomorphism: if L/k, then there is a natural k-module structure on L given by multiplication.

As was shown in section 13.2, a linear transformation is determined by where it maps it's basis to. Hence, if we have a k-homomorphism, then can deduce where the element map (or resp. construct a k-homomrophism) by finding where the basis maps to (resp. by only choosing where the basis maps to). Even better, this linear transformation respects the multiplicative structure of F (that is $\varphi(ab) = \varphi(a)\varphi(b)$), that if a basis β can be generated multiplicatively and additively from a set S, $\langle S \rangle = \beta$, then a k-homomorphism is only determined by where it sends S. Even better, this makes any k-homomorphism a k-algebra homomrophism, and a field F that extends k (F/k) is a k-algebra.

One more word of warning is in order: Let's say we find two field extensions A/k, B/k such that $A \cong_k B$. Then this implies that $\dim_k A = \dim_k B$, however it does not imply $A = B$. This might seem obvious, however it is an mistake that is easy to make after working with field extensions for awhile (this nuance will be particularly important in theorem 19.5.13).

In most of this and the next chapter, we will focus on finite extensions. Some results for extensions of infinite degree will arise as a natural generalization of the results of finite extensions, however finiteness is used in some key proofs to construct bijections (most importantly, in proposition 20.1.12). Infinite Galois extensions are treated in [11, Fundamental Theorem of Galois Theory, Infinite Extensions].

Characteristic

The following is is perhaps the simplest invariant of a field k (that is, if $\varphi : k \to K$ is a homomorphism, the property we are about to discuss is preserved). Every field has a multiplicative identity $1 \in k$. Since k is a field, it is also a ring, and hence there exists a unique ring homomorphism from $\mathbb{Z}$ to k mapping 1 to 1 (see proposition 9.1.29):

$$\mathbb{Z} \to k,\ n \mapsto \begin{cases} \underbrace{1+1\cdots+1}_{n \text{ times}} & n > 0 \\ -(\underbrace{1+1\cdots+1}_{n \text{ times}}) & n < 0 \\ 0 & n = 0 \end{cases}$$

We will use multiplication notation $n \cdot 1$ as a shorthand [2]. The image $\text{im}(\mathbb{Z})$ of this homomorphism is the smallest subring of k, since every subring contains 1 and hence every $n \cdot 1$.

[2] Note that we can also have defined it as a $\mathbb{Z}$-module on k

Definition 19.1.7: Characteristic

The *characteristic* of a field F, dentoed $ch(F)$, is the order of $\text{im}(\mathbb{Z})$. If $|\text{im}(\mathbb{Z})| < \infty$, we say that F has a *finite characteristic*, and characteristic 0 otherwise. In other words, it is defined as the smallest positive integer n such that $n \cdot 1 = 0$ if such a n exists, and is defined to be 0 otherwise

If n exists, then n needs to be prime. If it were not, then

$$n = n \cdot 1 = (a \cdot 1)(b \cdot 1) = 0$$

meaning either $a \cdot 1$ or $b \cdot 1$ must be zero-divisors, a contradiction since a field has no zero-divisors! Notice that we only need to have an integral domain for the previous proposition, and that the characteristic of the integral on an integral domain will be the same as its field of fractions. For a prime p, $\mathbb{Z}/p\mathbb{Z}$ is a field: if $p \nmid a$, then $\gcd(a, p) = 1$ and Bézout ([24, Extended Euclidean Algorithm (Bézout's Identity)]) gives $ua + vp = 1$, so u is inverse to a modulo p. However, $\mathbb{Z}$ is not a field. Using theorem 9.5.1, we know the smallest field containing $\mathbb{Z}$ would be $\mathbb{Q}$! In both cases, the field this map injects to must be the *smallest* field that contains 1 (which can also be thought of as the field *generated* by 1 in k)

Definition 19.1.8: Prime Subfield

The *prime subfield* of a field k is the subfield k generated by the multiplicative identity 1_F of k. It is (isomorphic to) either $\mathbb{Q}$ (if $\text{ch}(k) = 0$) or k_p (if $\text{ch}(k) = p$).

19.1.9 Note A shorthand of writing the characteristic of a field is the following: if $p \cdot 1 = 0$, then it is sometimes written $p = 0$ to mean a field with characteristic p.

Notice that for a field k of characteristic n ($n \geq 0$), $k/\mathbb{F}_p$ (resp. $k/\mathbb{Q}$) is always a field extension, since we can always map $\mathbb{F}_p$ or $\mathbb{Q}$ to k through the identity map.

Note that for fields of characteristic 0, field homomorphism are automatically a vector-space homomorphism over $\mathbb{Q}$: linearity holds automatically, and for any scalar $c \in \mathbb{Q}$:

$$\varphi(cx) = \varphi(c)\varphi(x) \overset{!}{=} c\varphi(x)$$

where $\overset{!}{=}$ comes from how a field homomorphism must map $\mathbb{Q}$ to itself.

The claim at the beginning of this section should now be checked, namely that if $\text{ch}(k) = n$, $n \geq 0$, and $\varphi : k \to K$ is a homomorphism, then $\text{ch}(K) = n$: φ is injective and $\varphi(m \cdot 1) = m \cdot 1$, so $m \cdot 1 = 0$ in k exactly when $m \cdot 1 = 0$ in K. Since charactersitic is preserved, we can take the subcategory of $\mathbf{Fld}_0$ or $\mathbf{Fld}_p$ where we restric to fields of characteristic p or 0. We will not really do this, however it is inisghtful that doing so will now make these categories have initial objects, namely the respective prime subfield (can you prove this? Can you see why $\mathbf{Fld}$ has no initial objects?). More generally, since homomorphisms can only be injective, we can take $\mathbf{Fld}_k$ to be the category ring homomorphisms as morphisms and all fields that extend k as objects! In most of the following studies, we will fix a field k and study extensions from it, and so field theory can be thought of as studying particular $\mathbf{Fld}_k$ categories

Torsion in Fields

By theorem 11.3.14, every finite subgroup of $F^\times$ is cyclic. Not every element of $F^\times$ has finite order: in $\mathbb{R}$, no positive power 2^n equals 1.

Definition 19.1.10: Root of Unity

Let F be a field and $r \in F$. If there exists an $n \in \mathbb{N}$ such that $r^n = 1$, then we call r an *nth root of unity*.

Example 19.1.11

In $\mathbb{C}$, all elements of the form $e^{\frac{2\pi ik}{n}}$ for $0 \leq k \leq n-1$ are nth roots of unity. In fact, they are all the roots of unity of $\mathbb{C}$. The real numbers $\mathbb{R}$ have only 2 roots of unity of degree 1 and 2 respectively: ± 1.

Notice that not all $e^{\frac{2\pi ik}{n}}$ are generators for the cyclic subgroup they belong to. The generators of these cyclic subgroups are given a name:

Definition 19.1.12: Primitive Root Of Unity

Let F be a field. Then a *primitive nth root of unity* is a generator of the cyclic multiplicative subgroup of all nth roots of unity.

We will return to the study of the roots of unity later. They will be the "building blocks" for [finitely generated] abelian groups in field theory (and Representation theory)[3]

Exercise 19.1.1

1. Let $\zeta_8 = \frac{\sqrt{2}}{2}(1+i) \in \mathbb{C}$. Show that ζ_8 is a primitive 8th root of unity, that $\zeta_8^2 = i$, and that $\zeta_8 + \zeta_8^7 = \sqrt{2}$.
2. (Analysis Problem) Show that the roots of unity are dense in $S^1 = \{z \in \mathbb{C} \mid |z| = 1\}$.

Generating Fields

Let's say $k \subseteq F$ is a subfield of F, and take some subset $A \subseteq F$. We can construct a field using k and A in the same way as we have done all the way back in section 2.3.2: Take all subfield of F containing k and A, and intersect them. Similarly to section 2.3.2, this is equivalent to "closing" the field k with A by taking all possible combinations. There is always at least exists the field K with this which contains this property. Since the intersection of fields is a field (this should be done as an exercise), then we can intersect all fields with this property to get the smallest such subfield.

[3] Remember by the fundamental theory of finitely generated abelian groups that any finitely generated abelian group is the direct product of a free part $\mathbb{Z}^r$ and a torsion part? Notice how the roots of unity can represent any cyclic subgroup?

Definition 19.1.13: Field Generated By Elements Over F

Let F/k be a field extension and S be a set of elements of F. Then the smallest subfield of F containing both k and the elements of S is denoted $k(S)$ and is called *the field generated by S over k*. If S is finite so that $S = \{\alpha_1, \alpha_2, ..., \alpha_n\}$, we usually denote $k(S)$ by $k(\alpha_1, \alpha_2, ..., \alpha_n)$ and call it the field *generated by $\alpha, \alpha_2, ..., \alpha_n$ over k*.

If we have a field extension F/k, and F can be generated by a single element, then we call that a simple extension:

Definition 19.1.14: Simple Extension

Let F/k be a field extension. Then if there exists an element $\alpha \in F$ such that $F = k(\alpha)$, we say that F is a *simple extension*, and call α the *primitive element* of the extension. The field $k(\alpha)$ is sometimes called "k adjoin α".

Be mindful to not confuse the terms "finite generation over k", "finite type k-algebra" [4], or "finite degree over k" (for example, $k(t)/k$ is a simple extension, but certainly not finite as was discussed under the definition of degree).

A similar definition works if we broaden to rings: instead of taking the intersection of all fields containing α and F, we take all rings containing α and F. It is easy to generlize the proof that the intersection of fields is a field to rings. We will usually denote this smallest ring $F[S]$, and if $S = \{\alpha_1, \alpha_2, ..., \alpha_n\}$ is finite, then $F[S] = F[\alpha_1, \alpha_2, ..., \alpha_n]$.

Proposition 19.1.15: Characterizing $F(S)$ and $F[S]$

Let K/F be a field extensions, $S \subseteq K$, and consider both $F(S)$ and $F[S]$. Then:

1. $F[S]$ can be characterized as the set of all finite linear combinations with coefficients in K of finite products of powers of elements of S, that is, polynomials with variables taken from S
2. $F(S)$ can be characterized as the set of all $ab^{-1} \in K$ with $a, b \in F[S]$ with $b \neq 0$. Furthermore, $F(S) \cong \text{Frac}(F[S])$

Note that in the "polynomials" mentioned above, the varaibles S are *not* indeterminates, since they might have relations in K (for example, it might be that for some $s \in S$, $s^2 + 1 = 0$)

***Proof*:**
These proofs are similar to those of section 2.3.2 with appropriate modifications for the added operation of multiplication and inversing.

To close off this section, we introduce a generalization to notion of the smallest field K/k generated by some elements to the smallest field generated with all the elements of another field:

[4] There is a deep connection between those two concepts, which we will explore in *Classical Algebraic Geometry* [23]. As foreshadowing, try seeing if you can think of a finitely generated field extension F/k that is not a finite-type k-algebra (or vice-versa). The answer to this question is given by Zariski's Lemma

Definition 19.1.16: Composite Field

Let K_1 and K_2 be two subfield of K. Then the *composite field* of K_1 and K_2, denoted K_1K_2, is the smallest subfield of K containing both K_1 and K_2. Similarly, the composite of any collection of subfields of K is the smallest subfield containign all the subfields.

Note that like in section 2.3.2, we can describe K_1K_2 as the intersection of all subfields contianing K_1 and K_2 (or any number of subfields).

Example 19.1.17: Composite Fields

The composite field $\mathbb{Q}(\sqrt{2})\mathbb{Q}(\sqrt[3]{2})$ is $\mathbb{Q}(\sqrt[6]{2})$, since thie field contains both these fields, and conversely any field containing both $\sqrt{2}$ and $\sqrt[3]{2}$ contains their quotient, namely $\sqrt[6]{2}$

Exercise 19.1.2

1. Sometimes, we want to consider field extensions without an underlying field. Let E, F be two fields of the same characteristic, say p. Show that $E \otimes_{\mathbb{F}_p} F$ (resp. $E \otimes_{\mathbb{Q}} F$) as a $\mathbb{F}_p$ (resp. $\mathbb{Q}$)-algebra is a field if and only if $[EF : \mathbb{F}_p] = [E : \mathbb{F}_p][F : \mathbb{F}_p]$ (resp. $[EF : \mathbb{Q}] = [E : \mathbb{Q}][F : \mathbb{Q}]$)

19.2 Field Extensions

We now continue our exploration of field extensions. As we saw in proposition 19.1.15, any field K/F such that $K = F(S)$ is the field of fractions of all "polynomials" with variables S and coefficients F. In particular, these polynomials have "solutions" or "roots" characterizing their relations. Similarly to how generators in a group where characterized by a monomial, (for example, D_4 is characterized with the generators a, b satisfying the monomails $ab^{-1}ab = 1$ and $a^2 = b^4 = 1$), generators in fields are characterized by polynomial relationships (for example, $\mathbb{C}/\mathbb{R}$ is characterized by being generated by $\mathbb{R}$ and i where $i^2 = -1$). Just like how any relation of elements within a group can be shifted so all the elements are on the left hand side (ex. writing $a^{-1}bab = 1$ instead of $ab = b^{-1}a$), we can do the same with polynomial relationships (ex. $i^2 + 1 = 0$). In proposition 19.2.24, we will show why it only matters to characterize polynomials, and not rational polynomials (i.e. elements of $p(S)/q(S)$)

As we have seen in chapter 11, every polynomial $p(x) \in F[x]$ can be broken down into irreducible polynomials. We will start finding an easy way of extending any field k which will give us a large class of field and field extensions to work with using irreducible polynomials (recall that $x^2 + 1$ has no solutions in $\mathbb{R}$, but does in $\mathbb{C}$, so is there an extension from $\mathbb{R}$ to $\mathbb{C}$?). We will find that the method used to construct fields will almost create "universal fields" (the same way free groups are sort of the universal groups since any group is the quotient of a free group). This endevaour to create a universal property will be very close in obtaining a universal property for fields, however it will fail in one important way. Due to that failure, it will be important carry around some more information to take into account the lake of universality. In fact, the lack of universal property may be seen less as a failure and more as a peek into the further structure that underly field extensions. Figuring all of this out is the goal of this section

Field Extensions using Polynomial

If we take our field to be $\mathbb{R}$, then $x^2+1=0$ does not have a solution in $\mathbb{R}$. The question then is whether there is a larger field K for which $x^2+1=0$ *has* a solution. More generally, given any irreducible polynomial $p(x) \in k[x]$, does there exists a field K such that K contains a root of $p(x)$[5]?. The answer is yes, and we have all the tools already to answer it. Take a moment and try to find the larger field K which satisfies this.

19.2.1 Taking a moment

Theorem 19.2.2: Kronecker's Theorem

Let k be a field and let $p(x) \in k[x]$ be an irreducible polynomial. Then there exists a field K containing an isomorphic copy of k in which $p(x)$ has a root. Identifying k with this isomorphic copy through i shows that there exists an extension of k in which $p(x)$ has a root. Furthermore, if L/k is a field extension in which $p(x)$ has a root, then there exists a k-homomorphism $j : K \to L$ such that

$$\begin{array}{ccc} K & \xrightarrow{\ j\ } & L \\ {\scriptstyle \iota}\uparrow & \nearrow_{\,i} & \\ k & & \end{array}$$

commutes. Due to this, the field K produced in the following construction is sometimes called the *Kronecker field for* $p(x)$ *over* k.

If in the Kronecker construction we use a *reducible* polynomials, then the resulting object is no longer a field (it contains zero-divisors), but remains to be a k-algebra.

***Proof*:**
Consider

$$K = k[x]/(p(x))$$

Since $k[x]$ is a Principal Ideal Domain (corollary 11.1.2) and since $p(x)$ is irreducible (so prime by proposition 10.1.33), $p(x)$ is maximal (proposition 10.1.32). Then $k[x]/(p(x))$ is a field by proposition 9.4.19. Restricting the canonical projection $\pi : k[x] \to K$ to k gives a homomorphism $\pi|_k : k \to K$, which is injective by proposition 19.1.1, so $\pi(k) \cong k$ is an isomorphic copy of k contained in K. We identify k with its isomorphic image in K and view k as a *subfield* of K. If $\overline{x} = \pi(x)$ denotes the image of x in the quotient K, then

$$\begin{aligned} p(\overline{x}) &= \overline{p(x)} && \text{since } \pi \text{ is a homomorphism} \\ &= p(x) \bmod (p(x)) && \text{in } k[x]/(p(x)) \\ &= 0 \end{aligned}$$

so that K contains a root of the polynomial $p(x)$. Thus, K is an extension of k in which the polynomial $p(x)$ has a root.

[5]We can assume that $p(x)$ is irreducible in $F[x]$, since if any factors of $p(x)$ has a root, then so does $p(x)$

For the second assertion, suppose $p(x)$ has a root α in L. Then the evaluation map $\text{ev}_\alpha : k[x] \to L$ (which we have proved is a ring-homomorphism in theorem 9.2.4), $g(x) \mapsto g(\alpha)$ must map $p(x)$ to 0. Hence, $p(x) \in \ker(\text{ev}_\alpha)$ meaning $(p(x)) \subseteq \ker(\text{ev}_\alpha)$, and by maximality of $(p(x))$ $(p(x)) = \ker(\text{ev}_\alpha)$. Since ideals correspond one-to-one with homomorphisms, we have a homomorphism:

$$j : \frac{k[x]}{(p(x))} \to L$$

and since $\frac{k[x]}{(p(x)} \cong K$, we have:

$$j : K \to L$$

Furthermore, for any $r \in k$, $j(r) = r$. Thus j commutes in the aforementioned diagram, completing the proof.

From now on, when we say *irreducible polynomial*, we will specifically mean *non-linear irreducible polynomials*, since linear irreducible polynomials produce trivial field extensions (i.e. k being extended to sovle for $x - \alpha \in k[x]$ is just k) unless stated otherwise.

The field extension to the Kronecker field K/k is *almost* universal with respect to finding roots to the polynomial $p(x)$. If it were universal, then if there was another field extension F/k such that F has a root of the irreducible polynomial $p(x) \in k[x]$, there would exist a *unique* homomorphism $j : F \to K$ such that the diagarm commutes. However, the homomorphism j need not be unique.

Example 19.2.3: Lack Of Uniqueness

Take $K = \mathbb{Q}[x]/(x^2 - 2)$ and $\mathbb{R}$. Then the polynomial $x^2 - 2$ has a root in $\mathbb{R}$, hence we know there exists a homomorphism:

$$K = \mathbb{Q}[x]/(x^2 - 2) \hookrightarrow \mathbb{R}$$

However, there exists two homomorphisms mapping K to $\mathbb{R}$. Since K extends $\mathbb{Q}$, any homomorphism can be represented by where the roots map: we can map $1 \mapsto 1$ and $\overline{x} \mapsto \sqrt{2}$, or map $1 \mapsto 1$ and $\overline{x} \mapsto -\sqrt{2}$. Both of these are valid homomorphisms, and so adjoints are not universal with respect to roots.

The universality can be recovered if we keep track of which root we want to map to (ex. $\sqrt{2}$), however we would have to make sure that there is only 1 homomorphism given some root (how do we know in more general cases that there cannot be 2 homomorphisms mapping $\overline{x}$ to the same root?). This concern is solved soon in theorem 19.2.27.

Since the aforementioned diagram commutes, notice that for any $r \in k$:

$$r \overset{!}{=} i(r) = j(\iota(r)) = j(r)$$

where ι is the natural inclusion into K, and the image of i is identified with the domain, justifying the $\overset{!}{=}$. Thus, j must *always* fix the base field, i.e. j is a k-homomorphism/isomorphism.

You might be concerned about *which* root of an irreducible polynomial does $\overline{x}$ represent in the previous theorem. For example, $x^2 - 2$ is irreducible over $\mathbb{Q}$, with $\pm\sqrt{2}$. So which root should we pick to represent $\overline{x}$? In fact, we can pick $\overline{x}$ to be represented *any* root of this polynomial: the field extending to contain $\sqrt{2}$ or $-\sqrt{2}$ will be isomorphic. More generally, if $p(x)$ is any irreducible polynomial, then $k[x]/(p(x))$ can be representing *any* choice of the possible roots of $p(x)$[6] and extending k to contain

[6] What it means to "choose" a root here is a bit ambiguous, since we have yet to say there exists a field that makes

that root. We can choose a particular root of $p(x)$, say α (which we define by saying α satisfies some polynomial equation), and isomorphically identify $K \xrightarrow{\sim} k(\alpha)$ (which we will show in more detail in theorem 19.2.8).

Note that to find the root of a polynomial, we required that polynomial to be irreducible. This technique will not obtain us a root of $(x^2-2)(x^3-2) = x^5 - 2x^3 - 2x^2 + 4$, namely because $(x^5 - 2x^3 - 2x^2 + 4)$ is not even a prime ideal (the resulting quotient contains zero-divisors). Section 19.5.2 constructs, for a given polynomial, an extension containing all of its roots, by adjoining roots of irreducible factors one at a time.

Note that we have not shown a way to "find" the root. For example, if we were looking at irreducible polynomial over $\mathbb{R}$ and we pick a root $\alpha \in \mathbb{C}$ that satisfies the polynomial equation, we do not have a computational way of finding it:

Example 19.2.4: Existence, but not construction

Let's say say F is a field extension $F/\mathbb{Q}$. Consider the irreducible polynomial $x^3 - 3x - 1$. over $\mathbb{Q}$ (it has no rationals roots by proposition 11.3.5). Thus, $F = \mathbb{Q}[x]/(x^3 - 3x - 1)$ is a field which has a root! The fact that all the roots of $x^3 - 3x - 1$ are real (and hence being able to identify F with a subfield of $\mathbb{R}$) can be considerd as taken for granted for now. We will have the tools to show that by the end of section 20.1.1. Visually, if we plot $x^3 - 3x - 1$, notice that all the roots are real, and hence $\mathbb{Q}$ can only by roots of $x^3 - 3x - 1$ which are real.

Let's say α is the root, so $\alpha \in \mathbb{R}$. We might hope that we can find which α in $\mathbb{R}$ satisfies the polynomial given that we know α is a root, however there is no *algebraic* way of finding this root! Using Newton-Raphsons approximation method, we can find that it is a real number $x \approx 1.97938...$. Finding these roots is a sub-branch of numerical analysis known as *root-finding algorithms*

For most common roots, we have already developped a symbol to represent them, namely $\sqrt[n]{m}$ is the root of $x^n - m$. More generally, if we extend $\mathbb{Q}[x]/(x^n - m)$ to solve another polynomial of a similar form, say $x^k - l$:

$$\frac{(\mathbb{Q}[x]/(x^n - m))[y]}{(y^k - l)}$$

we can "nest" as many square roots as we want. Since adding and multiplying is closed in a field, any extension of such polynomials produce elements of the form:

$$\sqrt[3]{\frac{3 + \sqrt[9]{40}}{\sqrt[8]{\sqrt{2}\sqrt{10}}}}$$

However, most more complicated polynomials are not given a special symbol, a notable exception being the golden ratio φ which is the root of $x^2 - x - 1$[7]. In section 20.2.4, we will further explore extensions which only add radical elements.

With the extension of a field through an irreducible polynomial constructed, it would be advantageous to be able to easily work with it's elements. Since K is a subfield of k, we can describe K as a k-vector-space!

$p(x)$ factor into linear terms. Such a field is constructed in section 19.5.1

[7]Though, this is not much of an exception since $\varphi = \frac{1+\sqrt{5}}{2}$

Theorem 19.2.5: Basis For Kronecker Field

Let $p(x) \in F[x]$ be an irreducible polynomial of degree n over the field F and let K be the field $F[x]/(p(x))$. Let $\theta = x \bmod (p(x)) \in K$. Then the elements

$$1, \theta, \theta^2, ..., \theta^{n-1}$$

are a basis for K as a vector space over F, so the degree of the extension is n, i.e., $[K : F] = n$. Thus

$$K = \left\{a_0 + a_1\theta + a_2\theta^2 + \cdots + a_{n-1}\theta^{n-1} | a_0, a_1, ..., a_{n-1} \in F\right\}$$

consists of all polynomials of degree $< n$ in θ. In particular, all field extensions of this form have finite degree.

***Proof*:**
Take $K = F[x]/(p(x))$, and consider $a(x) \in F[x]$ to be any polynomial. Then, by the Euclidean algorithm, we get that

$$a(x) = q(x)p(x) + r(x) \qquad q(x), r(x) \in F[x] \text{ with } \deg(r(x)) < n$$

then in K, we get that $[a(x)] = [r(x) + q(x)p(x)] = [r(x)]$, so any polynomial can be represented by it's residue polynomial, which will always be of degree $< n$. Thus, $1, \theta, \theta^2, ..., \theta^{n-1}$ (the image of $1, x, x^2, ..., x^{n-1}$) span the quotient. It remains to show they are linearly independent to conclude they are a basis.

If they were not, then for some $b_i \in F$ (not all 0)

$$b_0 + b_1\theta + \cdots + b_{n-1}\theta^{n-1} = 0$$

that is

$$b_0 + b_1x + \cdots + b_{n-1}x^{n-1} \equiv 0 \bmod (p(x))$$

meaning $p(x) | b_0 + b_1x + \cdots + b_{n-1}x^{n-1}$. However, that's a contradiction, since the degree of $p(x)$ is higher than the polynomial on the right hand side (which is at most $n - 1$)! Thus, these must be linearly independent. Since they also span, they fulfill the condition of a basis.

Since a basis is generated as a k-algebra by θ, any k-homomorphism $\varphi : k(\theta) \to F$ is determined by where it maps θ, making the map particularly easy to determine if we determine where θ maps to (the criterion for which we do in corollary 19.2.29).

As a reminder of arithmetics in $F[x]/(p(x))$:

Corollary 19.2.6: Shortcut On Arithmetics In K

Let K be as in theorem 19.2.2, and let $a(\theta), b(\theta) \in K$ be two polynomials of degree $< N$ in θ. Then addition in K is defined simply bu usual polynomial addition and multiplicatin in K is defined by

$$a(\theta)b(\theta) = r(\theta)$$

where $f(x)$ is the remainder (of degree $< n$) obtaiend after dividing the polynomial $a(x)b(x)$ by $p(x)$ in $F[x]$

***Proof*:**
An exercice in quotient ring of polynomials. Addition multiplication is done as in any quotient ring:

1. addition is usual addition of polynomials, since it does not change the degree
2. For multiplication, one can multiply the terms, then find a representative in the coset with degree $< n$

Note that what is described in corollary 19.2.6 works in any quotient field, including over polynomials which are *not* irreducible. However, the result is not a field, not even an integral domain [8].

Example 19.2.7: Field Extension Through Irreducible Polynomials

1. Let $F = \mathbb{R}$ and $p(x) = x^2 + 1$. Then we know that $x^2 + 1$ does not have a root in $\mathbb{R}$. Thus, consider:
$$K = \mathbb{R}[x]/(x^2 + 1)$$
is an extension field (of degree 2) that contains a root to this polynomial. Let $\theta = \overline{x}$. By theorem 19.2.5, $K/\mathbb{R}$ has degree 2 and any element $t \in K$ can be written as:
$$t = a + b\theta \qquad a, b \in \mathbb{R}$$
If you are comfortable with the arithmetics of $\mathbb{C}$, show that:
$$\mathbb{R}[x]/(x^2 + 1) \cong \mathbb{C}$$
2. Let $F = \mathbb{Q}$ and $p(x) = x^2 + 1$. Then by substitution with Eisntein's Criteron, $x^2 + 1$ is irreducible, hence $K = \mathbb{Q}[x]/(x^2 + 1)$ is a field extension of $\mathbb{Q}$. Since the base field is now $\mathbb{Q}$, any element $t \in K$ is of the form:
$$t = a + b\theta \qquad a, b \in \mathbb{Q}$$
3. Let $F = \mathbb{Q}$ and $p(x) = x^2 - 2$. Then $x^2 - 2$ is irreducible by Eisenstein's, so $F = \mathbb{Q}[x]/(x^2 - 2)$ is a field extension of $\mathbb{Q}$. Letting $K = \mathbb{Q}[x](x^2 + 1)$, show that:
$$F \ncong K$$
that is, extending by different roots of different irreducible polynomial produces distinct fields.

[8] Recall that if $p(x) = a(x)b(x)$, $a(x)$, $b(x)$ not being units, then in $F[x]/(p(x))$, $\overline{a(x)}, \overline{b(x)}$ are both nonzero, but $\overline{a(x)} \cdot \overline{b(x)} = \overline{0}$, meaning $F[x]/(p(x))$ has zero-divisors

4. Let $F = \mathbb{F}_2$ (the finite field with 2 elements) and $p(x) = x^2 + x + 1$. Then $p(x)$ is irreducible since $0^2 + 0 + 1 = 1^1 + 1 + 1 = 1$. Hence, $K = \mathbb{F}_2[x]/(x^2 + x + 1)$ is irreducible of degree 2. Can you make the multiplication table for this field?

Classifying Simple and Transcendental Extensions

Let's now turn our attention to classifying field extensions, given $k \subseteq K$ where we consider k the base field. If $A \subseteq K$, then we already have a notion of field extensions: $k \subseteq k(A)$. We will focus on the special cae where $A = \{\alpha\}$, that is, simple extension with $\alpha \in K$. As an example of this, imagine taking $\mathbb{R} \subseteq \mathbb{C}$, $\alpha = i$, and considering $\mathbb{R}(i)$. Then it is a known fact that $i^2 + 1 = 0$. We might ask if in general, given some $\alpha \in K$, can we find a polynomial that sovles it? In this way, we can characterize the field $k(\alpha)$ without needing to resort to taking the intersection of all subfields of K containing F and α. The answer is in fact, half the time. All simple extensions can be characterized by either being isomorphic to the rational functions over the base field or in terms of a quotient of a polynomial ring, giving us an easy classification of simple extensions:

Theorem 19.2.8: Classifying Simple Extensions

Let K/k be a field, $\alpha \in K$, and $k(\alpha)$. Consider $\text{ev}_\alpha : k[x] \to k(\alpha)$ mapping $f(x) \mapsto f(\alpha)$. Then if ev_α is injective, $k(\alpha)$ is an infinite extension and $k(\alpha) \cong k(x)$, and if ev_α is not injective, then there exists a polynomial $p(x)$ such that $k[x]/(p(x)) \cong k(\alpha)$. Furthermore, $p(x)$ is the polynomial of *minimal degree* that has α as a root.

Note that in this theorem, we are saying that *any* field over K extending k in which $p(x)$ contains a root in K and is irreducible in k contains a subfield isomorphic to the extension of k as we constructed in theorem 19.2.2 and that this field is (up to isomorphism) the smallest extension of k containing such a root.

***Proof*:**
Take the natural homomorphism

$$\text{ev}_\alpha : F[x] \to F(\alpha)(\subseteq K) \qquad a(x) \mapsto a(\alpha)$$

which maps the polynomial $a(x)$ to the polynomial evaluated at α. Note that $\text{ev}_\alpha(F[x])$ is an integral domain, since $F(\alpha)$ is integral domain.

If ev_α is injective, then $F[x]$ can be identified with an subring of $F(\alpha)$ that is an integral domain. By the universal property of field of fraction (theorem 9.5.4), ev_α uniquely extends to a homomorphism

$$\varphi : F(x) \to F(\alpha)$$

Since $\varphi(F(x))$ is a field containing F and α, $\varphi(F(x)) \supseteq F(\alpha)$, and since φ is a function $\varphi(F(x)) \subseteq F(\alpha)$, hence $\varphi(F(x)) = F(\alpha)$. To find the degree $[F(\alpha) : F]$, notice that $1[= \alpha^0], \alpha, \alpha^2, ...$ are all linearly independent over F in $F(\alpha)$ (since $1[x^0], x, x^2, ...$ are all linearly independent all linearly indepndent over F in $F(x)$), hence any basis must at least contain countably many linearly independent elements, and hence the degree must be infinite.

On the other hand, if ev_α is not injective, then $\ker(\text{ev}_\alpha) \neq 0$. The image $\text{ev}_\alpha(F[x])$ is an integral domain, so $\ker(\text{ev}_\alpha)$ is a nonzero prime ideal, hence a maximal ideal since $F[x]$ is a principal ideal domain. The quotient by it is then a field isomorphic to $\text{ev}_\alpha(F[x])$, which is therefore

a subfield of $F(\alpha)$ containing F and α, forcing $\text{ev}_\alpha(F[x]) = F(\alpha)$ and ev_α to be surjective, and since $F[x]$ is a euclidean domain (and hence a PID), any polynomial of minimal degree in $\ker(\text{ev}_\alpha)$ can represent the ideal, that is $(m_{\alpha,k}(x)) = \ker(\text{ev}_\alpha)$ for some minimal degree polynomial $m_{\alpha,k}(x) \in \ker(\text{ev}_\alpha)$. In fact, this minimal polynomial must be unique up to unit: if there where two minimal polynomials where $(m_{\alpha,k}) = (m'_{\alpha,k})$, then they must be equal up to unit, i.e. up to a factor in $k^\times$. Hence, we have that

$$\frac{k[x]}{(m_{\alpha,k}(x))} \cong F(\alpha)$$

By theorem 19.2.5, $[F(\alpha) : F] = \deg m_{\alpha,k}(x)$, and so the extension is finite, completing the proof.

This theorem motivates the following vocabulary:

Definition 19.2.9: Algebraic and Transcendental Elements

Let F be a field and let K be an extension of F. Then:

1. The element $\alpha \in K$ is said to be *algebraic* over F if α is a root of some nonzero polynomial $f(x) \in F[x]$.
2. If α is not algebraic over F (i.e., is not the root of any nonzero polynomial with coefficeints in F) then α is said to be *transcendental over* F.

Dealing with more general transcendental extensions is a straight-forward generlization of the previous thoerem. It naturally generlizes for $k(\alpha_1, ..., \alpha_n) \subseteq K$ where each α_i is transcendental. We first define the following term for the corollary:

Definition 19.2.10: Algebraically Independent

Let R be a k-algebra. Then $\alpha_1, ..., \alpha_n \in R$ are *algebraically independent* over k if the k-algebra homomorphism ev_k (i.e., the evaluation map) which fixes k:

$$\text{ev}_k : k[x_1, ..., x_n] \to R \qquad \text{ev}_k(x_i) = \alpha_i$$

is injective. The set is called *algebraically dependent* otherwise.

The definition is stated for a finite list, and that is the only case a polynomial relation can see: a relation involves finitely many indeterminates however large the set. So the notion extends to arbitrary subsets without any new content.

Definition 19.2.11: Algebraically Independent Set

Let K/k be a field extension. A subset $S \subseteq K$ is *algebraically independent* over k if every finite list of distinct elements of S is algebraically independent over k in the sense of definition 19.2.10. The empty set is algebraically independent.

The analogy with linear algebra now runs further than it first appears. An algebraically independent set behaves like a linearly independent one, a set over which K is algebraic behaves like a spanning

one, and the two notions meet in an object that deserves the name basis.

Definition 19.2.12: Transcendence Basis

Let K/k be a field extension. A subset $B \subseteq K$ is a *transcendence basis* of K over k if B is algebraically independent over k and K is an algebraic extension of $k(B)$.

Existence follows the same route as for vector-space bases: take a maximal independent set. Maximality is exactly what forces the algebraicity.

Theorem 19.2.13: Existence of a Transcendence Basis

Let K/k be a field extension and let $S \subseteq K$ be a subset with K algebraic over $k(S)$. Let $A \subseteq S$ be algebraically independent over k. Then there is a transcendence basis B of K/k with $A \subseteq B \subseteq S$. In particular, taking $S = K$ and $A = \emptyset$, every field extension has a transcendence basis.

***Proof*:**
Order by inclusion the collection of algebraically independent subsets of S containing A. It is nonempty, containing A. A chain in this collection has its union as an upper bound, and that union is algebraically independent because a dependence relation involves finitely many elements and would therefore already lie in some member of the chain. By Zorn's lemma there is a maximal such subset B.

It remains to see that K is algebraic over $k(B)$. Let $s \in S$. If $s \in B$ this is clear, and otherwise $B \cup \{s\}$ is a subset of S containing A that strictly contains B, so by maximality it is algebraically dependent: some finite list of distinct elements of $B \cup \{s\}$ satisfies a nontrivial polynomial relation over k. That list must involve s, since B is independent, so collecting the relation as a polynomial in s with coefficients in $k[B]$ gives a nonzero polynomial over $k(B)$ killing s. Hence s is algebraic over $k(B)$.

So every element of S is algebraic over $k(B)$, whence $k(S)$ is algebraic over $k(B)$, and K is algebraic over $k(S)$ by hypothesis. Algebraicity is transitive in towers, so K is algebraic over $k(B)$ and B is a transcendence basis, as we sought to show.

The substance is that the size of a transcendence basis does not depend on the basis chosen. As in linear algebra, this rests on an exchange argument comparing an independent set against a set over which everything is algebraic.

Lemma 19.2.14: Independent Sets Are No Larger Than Algebraic Generating Sets

Let K/k be a field extension, let $S = \{s_1, \dots, s_n\} \subseteq K$ be finite with K algebraic over $k(S)$, and let $A \subseteq K$ be algebraically independent over k. Then A is finite with $|A| \leq n$.

***Proof*:**
It suffices to show that no $n+1$ distinct elements of A can be algebraically independent. Suppose $a_1, \dots, a_{n+1} \in A$ are distinct. The argument replaces the s_j by the a_i one at a time, and the claim to be maintained after r steps is that, after renumbering the s_j, the field K is algebraic over $k(a_1, \dots, a_r, s_{r+1}, \dots, s_n)$.

Step 1: the base case and the replacement. For $r = 0$ the claim is the hypothesis. Suppose it holds for some $r < n+1$, so K is algebraic over $F_r = k(a_1, \dots, a_r, s_{r+1}, \dots, s_n)$. Then a_{r+1} is algebraic over F_r, so there is a nonzero polynomial relation

$$f\big(a_{r+1};\, a_1, \dots, a_r,\, s_{r+1}, \dots, s_n\big) = 0$$

with f a nonzero polynomial over k in which the first slot occurs. The relation must involve at least one s_j with $j > r$: otherwise it exhibits a nontrivial algebraic relation among $a_1, \dots, a_{r+1}$ over k, contradicting the independence of A. Renumber so that s_{r+1} occurs. Reading the relation as a nonzero polynomial in s_{r+1} over $k(a_1, \dots, a_{r+1}, s_{r+2}, \dots, s_n)$ shows s_{r+1} is algebraic over that field, so F_r is algebraic over $F_{r+1} = k(a_1, \dots, a_{r+1}, s_{r+2}, \dots, s_n)$, and by transitivity K is too. This advances the claim to $r+1$.

Step 2: the contradiction. Running the replacement n times gives that K is algebraic over $k(a_1, \dots, a_n)$. Then a_{n+1} is algebraic over $k(a_1, \dots, a_n)$, which is a nontrivial algebraic relation among distinct elements of A, contradicting independence. Hence A has at most n elements, completing the proof.

Theorem 19.2.15: Transcendence Degree Is Well-Defined

Let K/k be a field extension admitting a finite transcendence basis. Then every transcendence basis of K/k is finite, and any two have the same number of elements.

***Proof*:**
Let B be a finite transcendence basis, of size n, and let B' be any transcendence basis. Since K is algebraic over $k(B)$ and B' is algebraically independent, lemma 19.2.14 gives that B' is finite with $|B'| \le n$. Exchanging the roles of B and B', which is legitimate because B' is now known to be finite and K is algebraic over $k(B')$, gives $n \le |B'|$. Hence $|B'| = n$, as we sought to show.

Definition 19.2.16: Transcendence Degree

Let K/k be a field extension admitting a finite transcendence basis. The *transcendence degree* $\operatorname{trdeg}_k K$ is the common number of elements of its transcendence bases. An extension is algebraic exactly when $\operatorname{trdeg}_k K = 0$, and $\operatorname{trdeg}_k k(x_1, \dots, x_n) = n$ for indeterminates x_i.

Transcendence degree adds along towers, which is what makes it usable as a notion of dimension. The proof is the statement that a transcendence basis of the top over the middle, together with one of the middle over the bottom, is a transcendence basis of the top over the bottom.

Proposition 19.2.17: Transcendence Degree Is Additive in Towers

Let $k \subseteq K \subseteq L$ be fields with L/k admitting a finite transcendence basis. Then

$$\text{trdeg}_k L = \text{trdeg}_k K + \text{trdeg}_K L$$

***Proof*:**
Let B be a transcendence basis of K/k and C a transcendence basis of L/K. Both are finite by lemma 19.2.14, since L is algebraic over $k(B \cup C)$ once the two are combined. The sets B and C are disjoint, an element of C being transcendental over $K \supseteq k(B)$.

Algebraicity. L is algebraic over $K(C)$, and K is algebraic over $k(B)$, so $K(C)$ is algebraic over $k(B)(C) = k(B \cup C)$. By transitivity L is algebraic over $k(B \cup C)$.

Independence. Suppose a nontrivial polynomial relation over k holds among distinct elements of $B \cup C$. Collect it as a polynomial in the elements of C with coefficients in $k[B] \subseteq K$. Those coefficients are not all zero, since the relation is nontrivial and B is algebraically independent over k. So the relation is a nontrivial algebraic relation among elements of C over K, contradicting the independence of C over K.

So $B \cup C$ is a transcendence basis of L/k, and counting gives $\text{trdeg}_k L = |B| + |C| = \text{trdeg}_k K + \text{trdeg}_K L$, completing the proof.

Corollary 19.2.18: Classifying Finitely Generated Transcendental Extensions

Let K/k be a field extensions such that $K = k(\alpha_1, ..., \alpha_n)$ where each element is trancsendental. Then if $k(x_1, ..., x_n)$ is the field of rational functions over the indeterminates x_i[a], then, the set $\alpha_1, ..., \alpha_n$ is algebraically independent, and:

$$k(x_1, ..., x_n) \cong_k k(\alpha_1, ..., \alpha_n)$$

[a] sometimes called *formal variables*

The case for arbitrarily many transcendental elements is the same proof with more book-keeping on notation.

***Proof*:**
This is an immediate generalization of theorem 19.2.8

This shows that in terms of fields, algebraically independent sets do not offer much to our analysis of classification. They act more like a basis with essentially no relation, a resemblance made precise in definition 19.2.12 below. This is not to say that they do not offer any use in Galois theory, where we will correspond group for transcendental extensions, but in terms of field theory their structure is very simple. Thus, we will take more time stuyding algebraically dependent sets, especially since we can use theorem 19.2.2 to study study irreducible polynomials that aquire a root.

Going back to the minimal polynomial that we found in the proof,

Definition 19.2.19: Minimal Polynomial

Let K/k be a field extension and $\alpha \in K$ such that the map ev_α of theorem 19.2.8 is not injective. The *minimal polynomial* $m_{\alpha,k}(x)$ of α over k is the monic polynomial of least degree in $k[x]$ having α as a root.

It exists by theorem 19.2.8, after dividing by the leading coefficient, and it is unique: two monic polynomials of the least degree d with root α differ by a polynomial of degree less than d with root α, which is therefore 0. Every $f \in k[x]$ with $f(\alpha) = 0$ is divisible by $m_{\alpha,k}$: dividing gives $f = qm_{\alpha,k} + r$ with $r = 0$ or $\deg r < \deg m_{\alpha,k}$ (theorem 11.1.1), and $r(\alpha) = 0$ forces $r = 0$ by the minimality of $\deg m_{\alpha,k}$. So $m_{\alpha,k}$ generates $\ker(\mathrm{ev}_\alpha)$.

Proposition 19.2.20: Minimal Polynomial Over A Larger Field

If L/F is an extension of fields and α is algebraic over both F and L, then $m_{\alpha,L}(x)$ divides $m_{\alpha,F}(x)$ in $L[x]$.

***Proof*:**
The polynomial $m_{\alpha,F}$ lies in $F[x] \subseteq L[x]$ and has α as a root. Dividing in $L[x]$ (theorem 11.1.1) gives $m_{\alpha,F} = qm_{\alpha,L} + r$ with $q, r \in L[x]$ and $r = 0$ or $\deg r < \deg m_{\alpha,L}$. Evaluating at α gives $r(\alpha) = 0$, so $r = 0$ by the minimality of $\deg m_{\alpha,L}$ (definition 19.2.19), and $m_{\alpha,L}$ divides $m_{\alpha,F}$ in $L[x]$, completing the proof.

The minimal polynomial will always be irreducible: If it was reducible, then $m_{\alpha,k}(x) = g(x)h(x)$ would produce a polynomial where either $g(\alpha) = 0$ or $h(\alpha) = 0$, contradicting the minimality of $m_{\alpha,k}(x)$. The minimal polynomial can be found from a characteristic polynomial:

Proposition 19.2.21: Finding Minimal Polynomial

Let F/k be a finite field extension of degree n, let $\alpha \in F$, and let $L_\alpha \colon F \to F$ be multiplication by α, $y \mapsto \alpha y$. Then L_α is k-linear, its characteristic polynomial $P_{L_\alpha} \in k[x]$ is monic of degree n with $P_{L_\alpha}(\alpha) = 0$, and the minimal polynomial $m_{\alpha,k}$ divides P_{L_α}.

***Proof*:**
The map L_α is k-linear, since $\alpha(y+y') = \alpha y + \alpha y'$ and $\alpha(cy) = c(\alpha y)$ for $c \in k$, by distributivity and commutativity in F. By proposition 13.5.2(1), computed in a k-basis of F with n elements, P_{L_α} is monic of degree n, and $P_{L_\alpha}(L_\alpha) = 0$ by proposition 14.3.10. For $\beta, \gamma \in F$ and $c \in k$, $L_{\beta\gamma} = L_\beta \circ L_\gamma$, $L_{\beta+\gamma} = L_\beta + L_\gamma$ and $L_c = c\,\mathrm{id}_F$, so $f(L_\alpha) = L_{f(\alpha)}$ for every $f \in k[x]$. Hence $L_{P_{L_\alpha}(\alpha)} = 0$, and $P_{L_\alpha}(\alpha) = L_{P_{L_\alpha}(\alpha)}(1) = 0$. In particular ev_α is not injective, and $m_{\alpha,k}$ divides P_{L_α} by the divisibility property following definition 19.2.19, completing the proof.

Gröbner bases give another way to compute minimal polynomials ([16, Gröbner Bases]).

Example 19.2.22: Minimal Polynomials

1. The minimal polynomial for $\sqrt{2}$ over $\mathbb{Q}$ is $x^2 - 2$ and $\sqrt{2}$ is of degree 2 over $\mathbb{Q}$: $[\mathbb{Q}(\sqrt{2}), \mathbb{Q}] = 2$

2. The minimal polynomial for $\sqrt[3]{2}$ over $\mathbb{Q}$ is x^3-2 and $\sqrt[3]{2}$ is of degree 3 over $\mathbb{Q}$: $[\mathbb{Q}(\sqrt[3]{2}),\mathbb{Q}]=3$
3. similarly, for any $n>1$, the polynomial x^n-2 is irreducible (by Eisenstein's Criterion).
4. The minimal polynomial and the degree of an element α are dependent on the base field! For example, if $F=\mathbb{R}$, then the element $\sqrt[n]{2}$ has degree 1, with minimal polynomial
$$m_{\sqrt[n]{2},\mathbb{R}}(x)=x-\sqrt[n]{2}$$
5. the minimal polynomial for the golden ratio φ is x^2-x-1.
6. In general, most algebraic elements do not have a special symbol representing them, and so we would have to say that some $\alpha\in K$ has minimal polynomial (say) x^5-x-1

As a result of theorem 19.2.8, it also makes sense to ask the context of the existence of an irreducible polynomial what is the degree of an α over F: given $p(x)$ is the polynomial for which α is a root, then it's the degree of $\dim_F F[x]/(p(x))=\dim_F F(\alpha)$. We may write $\deg\alpha$ (or $\deg_F\alpha$) for the degree of α (over F). If $\deg\alpha=1$, then $\alpha\in F$.

Combining the classification of finite simple extensions with corollary 19.2.6, we get to easily represent $F(\alpha)$

Corollary 19.2.23: Representing $F(\alpha)$

Suppose in theorem 19.2.8 that $p(x)$ is of degree n. Then
$$F(\alpha)=\left\{a_0+a_1\alpha+a_2\alpha^2+\cdots+a_{n-1}\alpha^{n-1}|a_0,a_1,...,a_{n-1}\in F\right\}\subseteq K$$

Proposition 19.2.24: Finite Simple Extensions Elements

Let $k(\alpha)/k$ be a finite simple extension. Then:
$$k[\alpha]=k(\alpha)$$

***Proof*:**
The inclusion $k[\alpha]\subseteq k(\alpha)$ holds since $k[\alpha]$ is the smallest subring containing k and α, while $k(\alpha)$ is a field, in particular a ring, containing both.

For the reverse inclusion, consider the evaluation map $\mathrm{ev}_\alpha: k[x]\to k(\alpha)$ of theorem 19.2.8. If ev_α were injective, that theorem would make $k(\alpha)$ an infinite extension of k, contradicting finiteness. Hence ev_α is not injective, and α has a minimal polynomial $p(x)$ over k, say of degree n. By corollary 19.2.23, every element of $k(\alpha)$ has the form
$$a_0+a_1\alpha+a_2\alpha^2+\cdots+a_{n-1}\alpha^{n-1}\qquad a_i\in k$$
which is a polynomial expression in α with coefficients in k, hence lies in $k[\alpha]$ by proposition 19.1.15(1). Therefore $k(\alpha)\subseteq k[\alpha]$, and the two are equal.

This justifies why we only need to care about polynomials, and not rational polynomials.

Example 19.2.25: Simple Extension

1. In $\mathbb{Q}(\sqrt{2})$, $\dim_{\mathbb{Q}} \mathbb{Q}(\sqrt{2}) = 2$, and we can write $\mathbb{Q}(\sqrt{2}) = \left\{a + b\sqrt{2} \mid a, b \in \mathbb{Q}\right\}$. We could also have chosen the other root of $x^2 - 2$, namely $-\sqrt{2}$. However, these two fields are isomrophic: $\mathbb{Q}(\sqrt{2}) \cong \mathbb{Q}(-\sqrt{2})$, showing they are algebraically indistinguishable.
2. In $\mathbb{Q}(\sqrt[3]{2})$, $\dim_{\mathbb{Q}} \mathbb{Q}(\sqrt[3]{2}) = 3$, and we can write $\mathbb{Q}(\sqrt[3]{2}) = \left\{a + b\sqrt[3]{2} + c(\sqrt[3]{2})^2 \mid a, b, c \in \mathbb{Q}\right\}$.
3. Over $\mathbb{R}$, $x^3 - 2$ has no more solutions than the last example. However, over $\mathbb{C}$, there are two more solutions:
$$\zeta_2 = \sqrt[3]{2}\left(\frac{-1 + i\sqrt{3}}{2}\right) \qquad \zeta_3 = \sqrt[3]{2}\left(\frac{-1 - i\sqrt{3}}{2}\right)$$
Thus, $\mathbb{Q}(\zeta_2)$ and $\mathbb{Q}(\zeta_3)$ are field extensions.

Thus, we have fully classified all simple extensions of a field and dealt with transcendental extensions. In following sections, we will show how to classify fields generated by finitely many elements (ex. $k(\alpha, \beta)$) and infinitely many elements. Before continuing into that endeavour, isomorphisms between fields should be given some more attention to be more cear on how to classify fields.

The fields in example 19.2.25 need not be isomorphic (as fields) if the roots solve different polynomials, meaning different irreducible polynomials will produce different extensions:

Example 19.2.26: $\mathbb{Q}(\sqrt{2}) \ncong \mathbb{Q}(\sqrt{5})$

Take $\mathbb{Q}(\sqrt{2})$ and $\mathbb{Q}(\sqrt{5})$. Let's say φ is an isomorphism. Then we know that
$$\varphi(1) = 1$$
since 1 is the only element of order 1 in both fields. Then:
$$\varphi(\sqrt{2}) = a + b\sqrt{5}$$
for some non-zero value in the codomain. Then
$$\varphi(2) = a^2 + 2ab\sqrt{5} + 5b^2$$
but we also know that
$$\varphi(1) = 1 \qquad \varphi(2) = \varphi(1 + 1) = \varphi(1) + \varphi(1) = 1 + 1 = 2$$
so
$$2 = a^2 + 2ab\sqrt{5} + 5b^2 \qquad a, b \in \mathbb{Q}$$
If $a = b = 0$, $2 = 0$, a contradiction. If $a, b \neq 0$, then:
$$\frac{2 - a^2 - 5b^2}{2ab} = \sqrt{5}$$
implying that $\sqrt{5}$ is a rational number, a contradiction. If $a = 0, b \neq 0$, then:
$$\frac{\sqrt{2}}{\sqrt{5}} = b$$
but b is a rational number, a contraidction. If $a \neq 0, b = 0$, then:
$$\sqrt{2} = a$$
but a is a rational number, a contradiction. Thus, these two fields cannot be isomorphic.

However, notice that the extension of to $\mathbb{Q}(\sqrt{2})$ and $\mathbb{Q}(-\sqrt{2})$ were isomorphic, and so was all the extensions of $\mathbb{Q}$ by roots of x^3-2:

$$\mathbb{Q}(\sqrt[3]{2}) \cong \mathbb{Q}\left(\sqrt[3]{2}\left(\frac{-1+i\sqrt{3}}{2}\right)\right) \cong \mathbb{Q}\left(\sqrt[3]{2}\left(\frac{-1-i\sqrt{3}}{2}\right)\right)$$

even though those 3 fields are not equal. In particular, that means that non of these fields contain *all* of the roots of x^3-2. One of the large motifs in the following study of extensions is to find extensions in which for any irreducible (or more generally any polynomial) will contain *all* of the roots.

19.2.1 Isomorphism of Fields Extensions

As we have seen in theorem 19.2.2, If we know that an irreducible polynomial $p(x) \in k[x]$ has a root in a field F. Then there exists a [not necessarily unique] isomorphism φ such that the following diagram commutes:

$$\begin{array}{ccc} k[x]/(p(x)) & \xrightarrow{\varphi} & F \\ {\scriptstyle\iota}\uparrow & \nearrow_{\iota} & \\ k & & \end{array}$$

The fact that φ is not unique gives raise to several interesting results. As we have shown in example 19.2.3, $\mathbb{Q}(\sqrt{2})$ has two homomorphisms into $\mathbb{R}$. In fact, since $-\sqrt{2} \in \mathbb{Q}(\sqrt{2})$, we have that:

$$\begin{array}{ccc} \mathbb{Q}(\sqrt{2}) & \xrightarrow{\varphi} & \mathbb{Q}(-\sqrt{2}) \\ {\scriptstyle\iota}\uparrow & \nearrow_{\iota} & \\ \mathbb{Q} & & \end{array}$$

If we extend the bottom to be the identity map form k to k, then we get a square:

$$\begin{array}{ccc} \mathbb{Q}(\sqrt{2}) & \xrightarrow{\sim} & \mathbb{Q}(-\sqrt{2}) \\ \uparrow & & \uparrow \\ \mathbb{Q} & \xrightarrow{\text{id}} & \mathbb{Q} \end{array}$$

Showing the fields with the two roots of x^2-2 are isomorphic. In fact,$-\sqrt{2} \in \mathbb{Q}(\sqrt{2})$, and so this is in fact a field automorphism (though in general it does not have to be, like $\mathbb{Q}(\sqrt[3]{2})$ and $\mathbb{Q}(\sqrt[3]{2}\zeta_3)$). We have claimed earlier than two fields that have two different roots of the same irreducible polynomial adjoined to them are algebraically indistinguishable. This shows it in the case for x^2-2 over $\mathbb{Q}$, and the following theorem shows the generlizatoin of this result. Note how by keeping track of the root we have in fact recovered a sense of uniqueness:

Theorem 19.2.27: Extending Isomorphisms to Adjoint Fields

Let $k_1 \subseteq F_1 = k_1(\alpha_1)$, $k_2 \subseteq F_2 = k_2(\alpha_2)$ be simple extensions and let $\varphi : k_1 \to k_2$ be an isomorphism of fields. Let $p(x) \in k_1[x]$ be an irreducible polynomial and let $p'(x) \in k_2[x]$ be the irredcibble polynomial obtained by applying the map φ to the coefficients of $p(x)$. Let α_1 be a root of $p(x)$ (in some extension of k_1) and let α_2 be a root of $p'(x)$ (in some extension k_2). Then there is an isomorphism:

$$\sigma : k_1(\alpha_1) \to k_2(\alpha_2) \qquad \alpha_1 \mapsto \alpha_2$$

mapping α_1 to α_2 and extending φ, .i.e., such that σ restricted to F is the isomorphism φ.

In other words, if we have

$$\begin{array}{ccc} k_1(\alpha) & & k_2(\beta) \\ \uparrow & & \uparrow \\ k_1 & \xrightarrow{\sim} & k_2 \end{array}$$

and $\iota(m_{\alpha,k_1})(x)) = m_{\beta,k_2}(x)$ (i.e. it maps the coefficients of the first minimal polynomial to the second one), then we can *uniquely* extend ι to an isomorphism j between the two adjoint fields:

$$\begin{array}{ccc} k_1(\alpha) & \overset{\sim}{\dashrightarrow} & k_2(\beta) \\ \uparrow & & \uparrow \\ k_1 & \xrightarrow{\sim} & k_2 \end{array}$$

such that $j(\alpha) = \beta$. Note that j being an extension of ι means the elements of k_1 are fixed in j.

***Proof*:**
Let $\varphi : k_1 \to k_2$ be an isomorphism of fields. The map φ induces a ring isomorphism:

$$f^\varphi : k_1[x] \to k_2[x]$$

defined by applying φ to the coefficeints of a polynomial in $k_1[x]$. Let $p(x) \in k_1[x]$ be an irreducible polynomial, and let $p'(x) \in k_2[x]$ be the polynomial obtained by applying the map φ to the coefficients of $p(x)$, i.e., the image of $p(x)$ under f^φ. The isomorphism f^φ maps the maximal ideal $(p(x))$ to the ideal $(p'(x))$, so the ideal is also maximal, which shows that $p'(x)$ is also irreducible in $k_2[x]$.

Taking the quotinet by these ideals of the map, we get the following isomorphism of fields

$$k_1[x]/(p(x)) \to k_2[x]/(p'(x))$$

By theorem 19.2.8, the left side is isomorphic to $k_1(\alpha_1)$, and the right side to $k_2(\alpha_2)$, so composing these maps will produce a map for σ! It should also be clear now that the restriction of this isomorphism to k_1 is φ.

This isomorphism will come back big time when we address galois theory. It is sometimes represented in as in the following diagram:

$$\begin{array}{lcc} \sigma: & k_1(\alpha_1) & \xrightarrow{\cong} k_2(\alpha_2) \\ & \big| & \big| \\ \varphi: & k_1 & \xrightarrow{\cong} k_2 \end{array}$$

If we let $\mathrm{id}: k \to k$ be our k-isomorphism, and $m_{\alpha,k}(x)$ be our minimal polynomial with roots α, β, then $\varphi: k(\alpha) \xrightarrow{\sim} k(\beta)$ is also an isomorphism, showing that the roots of a minimal polynomials are algebraically indistinguishable. Furthermore, we can upgrade Kronecker's Theorem so that it gives us a universaly property: instead of L being a field that has a root of $p(x)$, upgrade that to $\alpha \in L$ being a root of $p(x)$. Then there exists a *unique* homomorphism mapping $\overline{x}$ to α (with uniqueness established by the above theorem).

Through this theorem, we get another important insight. If we have a non-trivial finite field extension $k(\alpha)/k$ which has roots of a minimal polynomial $m_{k,\alpha}(x)$, then if any other roots of $m_{k,\alpha}(x)$ are in $k(\alpha)$, there is a k-automorphisms of $k(\alpha)$ (can you see why a k-automorphism is determined by where it sends the roots?). This means we can find elements of the group $\mathrm{Aut}_k(k(\alpha))$ (where the subscript means we fix k), and hence the structure of $k(\alpha)$ [9]. More conretely, $k(\alpha)/k$ represents adding a new element α to the field k which satisfies an algebraic relation $m_{\alpha,k}(\alpha)$ (a polynomial). Finding information about the new field $k(\alpha)$ by looking at the k-automorphisms is essentially studying the properties of the polynomial $m_{\alpha,k}(x)$.

Even better, we will show that this characterizes *all* k-automorphisms, giving us a complete description of these symmetries. These symmetries are at the heart of studying fields in Galois theory. When Galois originally formulated his theory, he looked different ways to permute the roots and noticed that there are often limitation to the permutations of the roots since the roots may be related to each other[10]. For example if α is a root of $x^3 - x - 1$, then the other two roots are:

$$\frac{1}{1-\alpha} \qquad 1 + \frac{1}{\alpha}$$

Similarly, if ζ is a root of $x^n - 1$, then the other roots of $x^n - 1$ are $\zeta^2, \zeta^3, ..., \zeta^{n-1}, 1$. Thus, we take some time to define our terms more precisely, and elaborate further what the automorphisms look like:

Definition 19.2.28: Automorphism and fixed Element

Let K be a field and $\mathrm{Aut}(K)$ be the collection of automorphisms (isomorphisms from k to itself). Then:

1. An auotmorphism $\sigma \in \mathrm{Aut}(K)$ is said to *fix* an element $\alpha \in K$ if $\sigma\alpha = \alpha$. If F is a subset of K, then an automorphism σ is said to *fix* F if it fixes all the elements of F

$$\sigma a = a \qquad \forall a \in F$$

2. If F/k be an extension of fields, then $\mathrm{Aut}(F/k)$ or $\mathrm{Aut}_k(F)$ is the collection of all automorphsism of F which fix k.

[9] Recall that we already used the automorphism group to deduce information about groups, for example $\mathrm{Out}(G) \cong_{\mathbf{Grp}} \mathrm{Aut}(G)$ if and only if G is abelian

[10] Later we shall see that the roots are almost never related to each other, that is most Galois groups are S_n, but it is a good insight to notice that the roots of small polynomials are often related

the previous theorem gives us the following link between $\text{Aut}_k(F)$ and F given that F is finite and simple:

Corollary 19.2.29: Order of Fixed Automorphism Group

Let F/k be a simple finite extension (so $F = k(\alpha)$ for some $\alpha \in F$), and let $m_{k,\alpha}(x) \in k[x]$ be the minimal polynomial of α. Then $|\text{Aut}_k(F)|$ is equal to the number of *distinct* roots of $m_{k,\alpha}(x)$ in F. In particular,

$$|\text{Aut}_k(F)| \leq [k(\alpha) : k] \qquad |\text{Aut}_k(F)| = [k(\alpha) : k]_d$$

where $[k(\alpha) : k]_d$ represents the number of *distinct* roots $k(\alpha)$ contains of the minimal polynomial $m_{k,\alpha})(x)$. Equality holds if and only if $m_{k,\alpha}(x)$ factors over F as a product of *distinct linear* polynomials

This result will be called the *fixed root criterion* when we cover Galois theory.

***Proof*:**
Let $\sigma \in \text{Aut}_k(F)$ be an automorphism fixing k (so for any $t \in k$, $\sigma(t) = t$). Then since every element $t \in F = k(\alpha)$ can be written as a polynomial $t = f(\alpha) \in k[\alpha]$ (recall that $k(\alpha)$ is finite, and so has a basis of the form $\alpha, \alpha^2, ..., \alpha^n$ for some appropriat n), and σ fixes k, all the elements are determined by where α maps to, that is $\sigma(\alpha)$. Then since σ is an automorphism (hence ring homomorphism), given

$$x^n + a_{n-1}x^{n-1} + \cdots + a_1 x + a_0$$

is the minimal polynomial with coefficients from k so that

$$\alpha^n + a_{n-1}\alpha^{n-1} + \cdots + a_1\alpha + a_0 = 0$$

then, applying any automorphism σ to this polynomial will get us

$$\begin{aligned} \sigma(\alpha^n + a_{n-1}\alpha^{n-1} + \cdots + a_1\alpha + a_0) &= \sigma(0) \\ \sigma(\alpha)^n + a_{n-1}\sigma(\alpha)^{n-1} + \cdots + a_1\sigma(\alpha) + \sigma(a_0) &= 0 \qquad \sigma \text{ fixes } a_i \text{ for each } i \end{aligned}$$

since $a_i \in k$ for each i, showing that $\sigma(\alpha)$ is also a root of the minimal polynomial, i.e. σ must map α to some other root of $m_{k,\alpha}$.

As α is the generator, all roots are determined by where ‘ maps to, hence there can be at most $\deg(m_{k,\alpha}(x))$ automorphisms.

Conversely, for any root β of $p(x)$ in $k(\alpha)$, $\beta \in k(\alpha)$, we can take the isomorphism $\text{id} : k \to k$, which by theorem 19.2.27, there exists a unique lift to a root $k(\alpha) \to k(\beta)$ that fixes k. Since $k(\alpha) \cong k(\beta)$, then we can use this isomorphism to define an automorphism from $k(\alpha)$ to $k(\alpha)$, showing us that $\text{Aut}_k(F)$ must have at least as many elements as there are roots of the minimal polynomial in F. If F contains all the roots of $p(x)$ and each root of $p(x)$ is distinct, then $|\text{Aut}_k(F)| = [F : k]$, completing the proof.

Interpreting the above example as a group actions on the set of roots R_F of $m_{\alpha,k}(x)$ in F, $\text{Aut}_k(F) \curvearrowright R_F$, we have that $\text{Aut}_k(F)$ acts *faithfully and transitively* on the set of roots of $p(x)$ in F (the

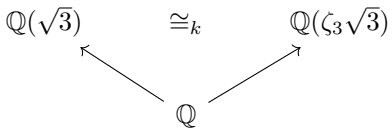
group $\mathrm{Aut}_k(F)$ can be identified with some transitive subgroup of the symmetric group acting on the roots of $p(x)$ in F). This correlation hints at the connection beween fields and groups at the heart of the galois theory, since any collection of automorphisms will always form a group[11]. The main impediment to directly translating this result of more general [finite] field extensions[12] is that the minimal polynomial might not factor into *linear* and *distinct* terms over the simple extension $k(\alpha)/k$[13]. For example, in $\mathbb{Q}(\sqrt[3]{2})$, the polynomial $x^3 - 1$ does not fully factor. The next two sections will encompass finding field extensions in which these conditions are met. In particular, we shall take some time to find fields in which:

1. All polynomials factor (splitting fields)
2. If $f(x) \in k[x]$ is irreducible, then in F/k, either $f(x)$ factors completely or $f(x)$ remains irreducible (normal extensions)
3. Given a polynomial $f(x)$, F/k factors completely (Splitting field)
4. If $f(x)$ is irreducible in k, then in F/k, $f(x)$ factors into linear terms of degree 1 (Separable extensions)

As a final word on relating isomorphisms and extensions, note that if $\dim_k A = \dim_k B$, this does not imply $[B : A] = 1$ or that $A \cong_k B$:

Example 19.2.30: Nuance On Dimension

Let $A = \mathbb{Q}(\sqrt{3})$ and $B = \mathbb{Q}(\zeta_3\sqrt{3})$. Then the two are of the same dimension, both extend $\mathbb{Q}$ and $A \cong_k B$, but $[B : A]$ is not even well defined. To visualize:



Exercise 19.2.1

1. Let $\mathrm{ch}(k) = n$, $n \geq 0$, and $\varphi : k \to K$ be a homomorphism. Then $\mathrm{ch}(K) = n$.
2. Show that the intersection of any collection of subfields of a field is a field. Is the same true for the union? What if the fields where given a partial order through inclusion?
3. Let $N \subseteq M_{2\times 2}(\mathbb{R})$ be a subset where

$$N = \left\{ \begin{pmatrix} a & b \\ -b & a \end{pmatrix} \;\middle|\; a, b \in \mathbb{R} \right\}$$

Then $N \cong \mathbb{C}$.

[11] If you are not convinced, now is good time to check this

[12] Infinite Galois extensions need a topology on the Galois group ([11, Fundamental Theorem of Galois Theory, Infinite Extensions]).

[13] It is possible that a irreducible polynomial factors linearly, but the factors are not distinct, see section 19.6

4. Let E/k and F/k be field extensions inside a common field. Show that if E/k is finite, then so is EF/F. Show that if E/k and F/k are finite, so is EF/k.
5. Suppose that $f(x) \in k[x]$ where $f(x)$ factors into $f(x) = p_1(x)p_2(x)\cdots p_n(x)$ where $p_i(x) \neq p_j(x)$. Show that $k[x]/(f(x))$ is a product of fields (Hint: Chinese Remainder Theorem). If one of the terms had multiplicity greater than 1, would the result still be a product of fields?
6. (If you know Calculus) The $\mathbb{R}$-algebra $\mathbb{R}[x]/(x^2)$ is sometimes the *dual numbers*. Let any element of this degree 2 extension of $\mathbb{R}$ be written in the form $a + b\epsilon$. If $p'(x)$ is the derivtive of $p(x)$, show that $p(a + b\epsilon) = P(a) + bP'(a)\epsilon$, that is, the derivitve "automatically" shows up.
7. Let m, n be square free and $n \neq m$. Show that $\mathbb{Q}(\sqrt{n}) \ncong_{\mathbb{Q}} (\sqrt{m})$.
8. Let ζ_n be a root of unity where n is odd. Show that $\mathbb{Q}(\zeta_n) \cong_{\mathbb{Q}} \mathbb{Q}(\zeta_{2n})$

19.3 Algebraic Extensions

We will now focus some attention on the algebraic elements of an extension. We first give name to elements for which all elements are algebraic or non-algebraic:

Definition 19.3.1: Algebraic and Transcendental Extensions

Let F be a field and let K be an extension of F. Then:

1. The element $\alpha \in K$ is said to be *algebraic* over F if α is a root of some nonzero polynomial $f(x) \in F[x]$.
2. If α is not algebraic over F (i.e., is not the root of any nonzero polynomial with coefficeints in F) then α is said to be *transcendental over* F.
3. The extension K/F is said to be *algebraic* or an *algebraic extension* if every element of K is algebraic over F.
4. The extension K/F is said to be *transcendental* or a *transcendental extension* if every element of K not in F is transcendental over F.

Note that if an element α is algebraic over a field F, then it is algebraic over *any* extension field L of F (if $f(x)$ having α as a root has coefficients in F, then it also has coefficients in L). Note too that the polynomial for which α must be a root need not be irreducible, however it is an immediate consequence that if α is the root of a polynomial $p(x) = (a_1(x))^{n_1} \cdots (a_k(x))^{n_k}$, it is a root for some irreducible polynomial $a_i(x)$.

Example 19.3.2: Algebraic extensions

1. Take $\mathbb{Q}(\sqrt{2})/\mathbb{Q}$. Then every element in $\mathbb{Q}(\sqrt{2})$ is a solution to a polynomial over $\mathbb{Q}$, thus $\mathbb{Q}(\sqrt{2})/\mathbb{Q}$ is algebraic: any element $a + b\sqrt{2}$ is the root of $\left(\frac{x-a}{b}\right)^2 - 2$. If you take $\mathbb{Q}(\sqrt{2})$ as a vector-space over $\mathbb{Q}$, then we see that 1 and $\sqrt{2}$ generate $\mathbb{Q}(\sqrt{2})$. Note that $\sqrt{2}$ is not in $\mathbb{Q}$, but but $\sqrt{2}$ is algebraic over $\mathbb{Q}$).

More generally, is it true that any finite simple extension $F(\alpha)$ of F is an algebraic extension? Is every element of $F(\alpha)$ the root of a polynomial with coefficients in F?

2. $\mathbb{C}$ as an $\mathbb{R}$-vector space is generated by 1 and i. Both of these are root to polynomials over $\mathbb{R}$, in particular $x-1$ and x^2+1. It is a famous result that any $c \in \mathbb{C}$ is a solution of a polynomial over $\mathbb{R}$ (i.e., with coefficients from $\mathbb{R}$). Thus, $\mathbb{C}/\mathbb{R}$ is algebraic.

   The degree of $[\mathbb{C} : \mathbb{R}] = 2$

3. Consider $\mathbb{C}$ again. For any $\alpha \in \mathbb{C}$, is there a polynomial $p(x) \in \mathbb{C}[x]$ for which $p(\alpha) = 0$? In a sense, we are asking if $\mathbb{C}$ is algebraic over itself. [a]

4. For a moment, take for granted that there is no polynomial in $\mathbb{Q}$ such that π is a root (this is Lindemann's theorem, which this book does not prove). Then π is *transcendental* over $\mathbb{Q}$, and so $\mathbb{R}/\mathbb{Q}$ is *not* an algebraic extension.

   Here is an idea to ponder: What is $[\mathbb{R} : \mathbb{Q}]$? The answer will be given in example 19.3.15

[a] The answer is yes! We call a field such that every element of it has a root over itself *algebraically closed*

In these examples, the algebraic extensions all had a finite degree. Is it possible to have an algebraic extension with infinite degree? If K is algebaric over F, then every element of K is a roof a polynomial over F? What about the converse: If K/F is finite, is K algebraic over F? (take a moment to ponder here)

Proposition 19.3.3: degree of Algebraic Extension

The element α is algebraic over F if and only if the simple extension $F(\alpha)/F$ is finite. More precisely, if α is an element of an extension of degree n over F, then α satisfies a polynomial of degree at most n over F and if α satisfies a polynomial of degree n over F, then the degree of $F(\alpha)$ over F is at most n.

***Proof*:**
If α is algebraic over F (so for some $p(x) \in F[x]$, $p(\alpha) = 0$), then the degree of the extension $F(\alpha)/F$ is the degree of the minimal polynomial, making it an extension of finite degree at most n.

Conversely, let α be an element of degree n over F (ex. $[F(\alpha) : F] = n$). Then by theorem 19.2.5, we know that $n+1$ elements will be linearly dependent:

$$c_0 + c_1\alpha + c_2\alpha^2 + \cdots + c_n\alpha^n = 0$$

with $c_0, c_2, ..., c_n \in F$ which are not all 0. Thus, α must be the root of some non-zero polynomial with coefficients from F (of degree $\leq n$), showing α is algebraic over F.

Corollary 19.3.4: Finite Extension Then Algebraic Extension

If the extension K/F is finite, then it is algebaric

***Proof*:**
let K/F be finite and take any $\alpha \in K$. Consider the simple extension $F(\alpha)$. Consider K as an F-vector space, then since the extension is finite, $\dim_F K = n$. Now, $F(\alpha) \subseteq K$ would be a subspace of K ($F(\alpha)$ also being an F-vector sapce), so

$$\dim_F F(\alpha) \leq \dim_F K \qquad \text{or} \qquad [F(\alpha) : F] \leq [K : F]$$

showing the extension is finite, and so by proposition 19.3.3, α is algebraic over F. since α was an artibrary element of K, this means that K is algebaric over F, as we sought to show.

Though not important to us in this chapter, we also have the following characterisation of an algegraic extension more useful in algebraic number theory:

Corollary 19.3.5: Algebraic Extension and Algebras

K/F is algebraic if and only if every sub F-algebra is a field

***Proof*:**
Suppose first that K/F is algebraic, and let $A \subseteq K$ be a sub F-algebra, that is, a subring of K containing F. Let $a \in A$ be nonzero. Since a is algebraic over F, there is a polynomial

$$p(x) = c_n x^n + c_{n-1} x^{n-1} + \cdots + c_1 x + c_0 \in F[x]$$

of minimal degree with $p(a) = 0$. We claim $c_0 \neq 0$: otherwise $p(x) = x\, q(x)$ for some $q \in F[x]$ of smaller degree, and $a\, q(a) = 0$ with $a \neq 0$ in the field K forces $q(a) = 0$, contradicting minimality. Rearranging $p(a) = 0$ gives

$$a \cdot \left(-c_0^{-1}\left(c_n a^{n-1} + c_{n-1} a^{n-2} + \cdots + c_1\right)\right) = 1$$

The element in parentheses is a polynomial expression in a with coefficients in F, hence lies in the ring A. Therefore a is invertible in A, and A is a field.

Conversely, suppose every sub F-algebra of K is a field, and let $\alpha \in K$. The subring $F[\alpha]$ is a sub F-algebra of K, hence a field by hypothesis. If α were transcendental over F, the evaluation map $\mathrm{ev}_\alpha : F[x] \to K$ would be injective, since no nonzero polynomial has α as a root, and its image is $F[\alpha]$. Then $F[x] \cong F[\alpha]$ would be a field. But $F[x]$ is not a field: for any $f \in F[x]$, the degree formula over the integral domain F gives $\deg(x\, f) = 1 + \deg(f) \geq 1$, so $x\, f \neq 1$ and x has no inverse. This contradiction shows α is algebraic over F. Since α was arbitrary, K/F is algebraic.

Note that the converse of corollary 19.3.4 is not true: If K is algebraic over F, this does not mean that K/F has finite degree

Example 19.3.6: Algebraic and infinite degree
We require corollary 19.3.14 which is introduced in a couple of pages, but we can still state the case with the added assumption here to get the idea:

Take $\mathbb{C}/\mathbb{Q}$, and take the set $\overline{\mathbb{Q}}$ to be the set of all algebraic elements over $\mathbb{Q}$. Then by the aforementioned corollary, this is a subfield of $\mathbb{C}$. Since $\sqrt[n]{2}$ is algebraic over $\mathbb{Q}$ (root of the

polynomial $x^n - 2$), we have

$$[\mathbb{Q}(\sqrt[n]{2}) : \mathbb{Q}] = n$$

for every n. Since $\mathbb{Q}(\sqrt[n]{2}) \subseteq \overline{\mathbb{Q}}$, it follows that $\overline{\mathbb{Q}}$ is an algebraic extension of infinite degree.

However, a weaker version of this direction does have a converse, and we will build up to it now. Let's first show that the finite extension of a finite extension remains a finite extension (if you like, the composition of finite extensions is a finite extension)

Theorem 19.3.7: Multiplicity of Finite Extensions (Tower Law)

Let $k \subseteq E \subseteq F$ be fields. Then

$$[F : k] = [F : E][E : k]$$

that is, extension degree multiply each other, where if one side of the equation is infinite, so is the other.

***Proof*:**

The infinite case is clear, so we will deal with the finite case. First, if F is finite dimensional over k, then the subspace E is also finite dimensional. Furthermore, any linear combination of elements of F over the field k naturally is a linear combination over the larger field E. Thus, if $[F : k]$ is finite, so is $[F : E]$ and $[E : k]$.

Conversely, suppose $[F : E]$ and $[E : k]$ are both finite. Let $(e_1, e_2, ..., e_m)$ be a basis for E over k and $(f_1, f_2, ..., f_n)$ be a basis for F over E. I claim that

$$(e_1 f_1, e_1 f_2, ..., e_1 f_n, e_2 f_1, ..., e_m f_n)$$

is a basis for F over k.

Let $g \in F$. Then for appropriate $d_i \in E$,

$$g = \sum_j d_j f_j$$

Next, for each d_j, for appropriate $c_{ij} \in k$, we have

$$d_j = \sum_i c_{ij} e_i$$

Thus, we have:

$$g = \sum_i \sum_j c_{ij} e_i f_j$$

Therefore, the products $e_i f_j$ span F, showing that F must at most be finite dimensional.

To show they form a basis, assume

$$\sum_{i,j} \lambda_{ij} e_i f_j = 0$$

Then:

$$\sum_j \left(\sum_i \lambda_{ij} e_i \right) f_j = 0$$

which, since the the elemnts f_j are linearly independent over E, we have that $\sum_i \lambda_{ij} e_i = 0$. But then, the elements e_i are linearly indepenent over e_i,and so each $\lambda_{ij} = 0$, showing that the original linear combination is also linearly independent, as we sought to show.

Notice how this corollary mirrors Lagranges theorem with respect to the order of groups. This parallel will capture our attention in the next chapter. The following corollary deepends this connection:

Corollary 19.3.8: Degree Extension Divisibility

Suppose L/F is a finite extension and let K be any subfield of L contianing F, $F \subseteq K \subseteq L$. Then $[K : F]$ divides $[L : F]$

***Proof*:**
Since

$$[L : F] = [L : K][K : F]$$

this result holds.

This result gives us a similar power that Lagrange's theorem gives us

Example 19.3.9: Use Of Tower Law

1. We can quickly rule out that $\sqrt{2} \notin \mathbb{Q}(\sqrt[n]{2})$ if n is odd. To see this, if $\sqrt{2} \in \mathbb{Q}(\sqrt[n]{2})$, then $\mathbb{Q}(\sqrt{2}) \subseteq \mathbb{Q}(\sqrt[n]{2})$. Then by what we have just shown:

   $$[\mathbb{Q}(\sqrt[n]{2}) : \mathbb{Q}] = [\mathbb{Q}(\sqrt[n]{2}) : \mathbb{Q}(\sqrt{2})][\mathbb{Q}(\sqrt{2}) : \mathbb{Q}]$$

   However, that would imply that an even number divides an odd number, a contradiction.

2. More generally, we can conclude the following: If K/k is an extension of odd degree and $\alpha \in K$, then α can be written as a polynomial in α^2, that is $\alpha = p(\alpha^2) = \sum_i k_i(\alpha^2)^i$ for $k_i \in k$.

   To see this, we first notice that through simple set theory we must have

   $$k \subseteq k(\alpha^2) \subseteq k(\alpha)$$

   and that $k(\alpha)/k(\alpha^2)$. To find the degree d of this extension, notice that over $k(\alpha^2)$, we get that $x^2 - \alpha^2 \in k(\alpha^2)[x]$, so $d \leq 2$. On the other hand, d must divide $[k(\alpha) : k]$ which is odd, forcing us to have $d = 1$. Thus $k(\alpha^2) = k(\alpha)$, and so $\alpha \in k(\alpha^2)$.

3. Let K/k be an extension with $[K : k]$ prime. Then there is *no* field E with $k \subsetneq E \subsetneq K$, since $[E : k]$ would divide the prime $[K : k]$. This is similar to how all finite groups of prime order are simple.

4. Take $\mathbb{Q} \subseteq \mathbb{R}$ and consider $\mathbb{Q}[x]/(x^3 - 3x - 1) \cong \mathbb{Q}(\alpha)$ as in example 19.2.4. Since $[\mathbb{Q}(\sqrt{2}) : \mathbb{Q}] = 2$, and $[\mathbb{Q}(\alpha) : \mathbb{Q}] = 3$, and $2 \nmid 3$, then $\mathbb{Q}(\sqrt{2}) \not\subseteq \mathbb{Q}(\alpha)$, so $\sqrt{2} \notin \mathbb{Q}(\alpha)$! It is much more difficult to directly show $\sqrt{2}$ cannot be written as a $\mathbb{Q}$-linear combination of $1, \alpha, \alpha^2$.

Continuing to strive towards a partial converse, we define the following generlization of a simple

extension. The key feature about finitely generated fields is that they can be decomposed into simple extensions:

Lemma 19.3.10: Recursive Definition Of Extension Field

$F(\alpha,\beta) = (F(\alpha))(\beta)$, i.e., the field generated over F by α and β is the field generated by β over the field $F(\alpha)$ generated by α.

***Proof*:**
This proof takes complete advantage that by definition, field extensions are the *smallest* field containing F and α (or α and β)

The field $F(\alpha,\beta)$ is the smallest field to contain F,α,β, so $F(\alpha,\beta) \subseteq (F(\alpha))(\beta)$. Conversely, $F((\alpha))(\beta)$ is the smallest field to contain $F(\alpha)$ and β. Since $F(\alpha) \subseteq F(\alpha,\beta)$ and $\beta \in F(\alpha,\beta)$, $(F(\alpha))(\beta) \subseteq F(\alpha,\beta)$, implying

$$F(\alpha,\beta) = (F(\alpha))(\beta)$$

With this, we can break down any finitely generated:

$$K = F(\alpha_1,\alpha_2,...,\alpha_k) = (F(\alpha_1,\alpha_2,...,\alpha_{k-1}))(\alpha_k) = ((\cdots F(\alpha_1))(\alpha_2))\cdots(\alpha_n)$$

Thus, we get the following chain of extensions:

$$F = F_0 \subseteq F_1 \subseteq F_2 \subseteq \cdots \subseteq F_k = K$$

where

$$F_{i+1} = F_i(\alpha_{i+1}) \qquad i = 0,1,...,k-1$$

Importantly, this reduces everything down to lots of simple extensions! Now, suppose that the elements $\alpha_1,\alpha_2,...,\alpha_k$ are algebraic over F of degrees $n_1,n_2,...,n_k$

$$[F(\alpha_1):F] = n_1,\ [F(\alpha_2):F] = n_2 \cdots [F(\alpha_k):F] = n_k$$

By proposition 19.2.20, we can replace F with F_i. Then these can all be characterized in terms of simple extensions, meaning we can use proposition 19.3.3. The relative extension degree $[F_{i+1}:F_i]$ is equal to the degree of the minimal polynomoial of α_{i+1} over F_i, which is at most n_{i+1} (and is equal to n_{i+1} if and only if the minimal polynomial of α_{i+1} over F remains irreducible over F_i). So, by the Tower Law:

$$[K:F] = [F_k:F_{k-1}][F_{k-1}:F_{k-2}]\cdots[F_1:F_0] \leq [F(\alpha_1):F][F(\alpha_2):F]\cdots[F(\alpha_k):F] \tag{19.1}$$

And so $[K:F]$ must be a *finite* extension of degree less than or equal to $n_1n_2\cdots n_k$! We sum up this entire discussion of finite degrees in the following theorem:

Theorem 19.3.11: Degree of Algebraic Extension

The extension K/F is finite if and only if K is generaed by a finite number of algebraic elements over F. More precisely, a field generated over F by a finite number of algebraic elements of degrees $n_1,n_2,...,n_k$ is algegraic of degree $\leq n_1,n_2,...,n_k$

Before, we hade

$$\alpha \text{ algebraic over } F \Leftrightarrow F(\alpha)/F \text{ finite}$$

and when we try to generlize, we loose one side of the implication

$$K \text{ algebraic over } F \Leftarrow K/F \text{ finite}$$

with the counter example 19.3.6 for $\Rightarrow$.

We now restricted the hypothesis to give us the if and only if again:

$$K \text{ generated by finite number of algebraic elements over } F \Leftrightarrow K/F \text{ finite}$$

***Proof*:**
Let the extension K/F be finite, so $a_1, a_2, ..., a_n$ forms a basis for K over F. Then since $F(a_i) \subseteq K$ as a F-vector-space, $[F(a_i) : F] \leq [K : F]$, meaning it has finite degree, and so by proposition 19.3.3, a_i is algebraic. Since a_i was arbitrary, this means that every a_i is algebraic. Thus K is generated by a finite number of algebraic elements over F.

For the converse, let's say K is generated by a finite number of algebraic elements over F. Let

$$\alpha_1, \alpha_2, ..., \alpha_n$$

all be algebraic over F and generate K. Since they generated K, $K = F(\alpha_1, \alpha_2, ..., \alpha_n)$. Then by what we have seen in equation (19.1), we have that K/F is finite.

Using what we have found, it is easier to compute the degree of non-simple extensions:

Example 19.3.12: Computing Degree of non-simple Extension

1. The extension $\mathbb{Q}(\sqrt[6]{2}, \sqrt{2})$ is $\mathbb{Q}(\sqrt[6]{2})$ since $\sqrt{2} \in \mathbb{Q}(\sqrt[6]{2})$. Another way of showing this is that the degree d of $\sqrt{2}$ over $\mathbb{Q}(\sqrt[6]{2})$ is 1. Thus, even though the degree of $\sqrt{2}$ is 2 and the degree of $\sqrt[6]{2}$ is 6, the degre of $\mathbb{Q}(\sqrt[6]{2}, \sqrt{2})$ is 6, not 12.

2. Consider $\mathbb{Q}(\sqrt{2}, \sqrt{3})$. We will proceed to find the degree of this extensions over $\mathbb{Q}$. Since $\sqrt{3}$ is of degree 2 over $\mathbb{Q}$, then the degree of the extension $\mathbb{Q}(\sqrt{2}, \sqrt{3})/\mathbb{Q}(\sqrt{2})$ is at most 2 ($\sqrt{3}$ can at most add another dimension), and *is* 2 if and only if $x^2 - 3$ is irreducible over $\mathbb{Q}(\sqrt{2})$. To see if it is, let's suppose it is reducible. Since $x^2 - 3$ is of degree 2, by proposition 11.3.5, it is irreducible if and only if it has a root, that is, if $\sqrt{3} \in \mathbb{Q}(\sqrt{2})$.

   If $\sqrt{3} \in \mathbb{Q}(\sqrt{2})$, $\sqrt{3} = a + b\sqrt{2}$. From here, it's a lot of manipulation (squaring both sides and checking cases). It will turn out that every possibility will make either $\sqrt{2}$, $\sqrt{3}$ or $\sqrt{6}$ rational, a contradiction.

   Thus, $\deg_{\mathbb{Q}(\sqrt{2})} \mathbb{Q}(\sqrt{2}, \sqrt{3}) = 2$, so

   $$\deg_{\mathbb{Q}} \mathbb{Q}(\sqrt{2}, \sqrt{3}) = 4$$

   and we can find the basis for it by "closing" the set $1, \sqrt{2}, \sqrt{3}$ to produce $1, \sqrt{2}, \sqrt{3}, \sqrt{6}$, giving us

   $$\mathbb{Q}(\sqrt{2}, \sqrt{3}) = \left\{ a + b\sqrt{2} + c\sqrt{3} + d\sqrt{6} | a, b, c, d \in \mathbb{Q} \right\}$$

Corollary 19.3.13: Algebraic Elements are Closed

Suppose α and β are algebraic over F. Then $\alpha \pm \beta$, $\alpha\beta$, α/β (for $\beta \neq 0$, are all algebraic (since α/β is algebraic $1/\alpha$ is also algebraic)

***Proof*:**
All these elements are part of $F(\alpha, \beta)$, which is finite over F by theorem 19.3.11, and so are algebraic by corollary 19.3.4 (in particular, $F(\alpha + \beta)$ is a simple extension, and since $\alpha + \beta \in F(\alpha, \beta)$(and $F \subseteq F(\alpha, \beta)$), we have that $F(\alpha + \beta) \subseteq F(\alpha, \beta)$ by definition, so it is a finite extension, so by corollary 19.3.4 every element will be a root of a polynomial over F).

Corollary 19.3.14: Algebraic Elements Form a Field

Let L/k be an arbitrary extension. Then the collection of elements of k that are algebraic over F form a subfield K of F.

***Proof*:**
By corollary 19.3.13, we get that the collection is closed under $+$ and $\times$. From here, it is easy to verify any extra axiom we want.

Example 19.3.15: Algebraic Closure

1. Consider the extension $\mathbb{C}/\mathbb{Q}$, and let $\overline{\mathbb{Q}}$ represent the subfield of $\mathbb{C}$ containing all elements that are algebraic over $\mathbb{Q}$. Then $\sqrt[n]{2} \in \overline{\mathbb{Q}}$ for all $n \in \mathbb{N}$ (taking the positive roots), so $[\overline{\mathbb{Q}} : \mathbb{Q}] \geq n$ for every $n > 1$. Thus, $\overline{\mathbb{Q}}$ is an *infinite* algebraic extension of $\mathbb{Q}$, called the field of *algebraic numbers*. We have now proved what we have assumed in example 19.3.6.

2. Take $\mathbb{R}$ and the subfield $\overline{\mathbb{Q}}$ consisting of all algebraic elements over $\mathbb{Q}$, and consider $\mathbb{R} \cap \overline{\mathbb{Q}}$. This set is countable: $\mathbb{Q}$ is countable, and $\mathbb{Q}[x]$ consists of the countable union of polynomials of degree n ($S_0 \cup S_1 \cup \cdots$ where S_n contains all polynomials of degree n), so $\mathbb{Q}[x]$ is countable. $\overline{\mathbb{Q}}$ must have a cardinality at least less than $\mathbb{Q}[x]$, and so is also countable.

   Since $\mathbb{R}$ is uncountable, we see that there must be elements in $\mathbb{R}$ which are *not* algebraic (i.e. transcendental) over $\mathbb{Q}$. In fact, the vast majority of elements are transcendental. The subfield $\overline{\mathbb{Q}} \cap \mathbb{R}$ is a *proper* subfield of $\mathbb{R}$.

   Though the majority of elements[a] are transcendental, it is extremely difficult to show that an element *is* transcendental. For example, it was proved that $\pi = 3.1415926...$ was transcendental in 1880, and $e = 2.171828...$ is transcendental in 1870. Proving these are transcendental is very involved. Even combining these elements is tricky to prove: e^π was proven to be transcendental in 1934, however as of writing this book, it is unknown whether π^e is transcendental. In fact, it is also not known whether $\pi + e$ or πe are transcendental.

   We have only a small list of numbers which we *know* to be transcendental, even though the vast majority of them are transcendental [b]. Known transcendental numbers include e^a for a nonzero algebraic number a, a^b where a is algebraic $\neq 0, 1$ and b is algebraic and irrational (ex. $2^{\sqrt{2}}$), and $\sin(a)$ for a nonzero algebraic number a.

[a] In terms of probability, 100% of the elements

[b] If you have taken some statistics/measure theory, then if $\mathbb{A}$ are the algebraic numbers over $\mathbb{Q}$, $\int \chi_{\mathbb{A}} = 0$, i.e., the chances of picking an algebraic number from the set of real number is 0!

We can extend the results of theorem 19.3.11 a bit further to algebraic extensions (finite or infinite)

Theorem 19.3.16: Composing Algebraic extensions

Let $k \subseteq E \subseteq F$ be field extensions. Then F/k is algebraic if and only if F/E and E/k are algebraic.

***Proof*:**
If F/k is algebraic, then *a fortiori* F/E is algebraic (If $\alpha \in F$ are all roots of some polynomial $k[x]$ and $k \subseteq E$, then they are certainly roots in $E[x]$). Since $E \subseteq F$, *a priori* E/k.

For the other way, assume E/k and F/E are algebraic, and consider $\alpha \in F$. Since F/E is algebraic, there exists a $p(x) \in E[x]$ such that

$$p(\alpha) = \alpha^n + e_{n-1}\alpha^{n-1} + \cdots + e_1\alpha + e_0 = 0$$

We thus have that α is algebraic over the subfield $k(e_0, ..., e_{n-1}) \subseteq E$. Hence, by theorem 19.3.11

$$k(e_1, ..., e_{n-1}) \subseteq k(e_1, ..., e_{n-1}, \alpha)$$

is a finite extension. Furthermore, by theorem 19.3.11 again, we have that

$$k \subseteq k(e_1, ..., e_{n-1})$$

is also a finite extension, since each e_i is in E, so is algebraic over k by assumption that E/k is algebaric. Hence, by the Tower Law,

$$k \subseteq k(e_1, ..., e_n, \alpha)$$

is a finite extension as well, and hence α is algebraic over k. Since α was arbitrary, F is algebraic over k, as we sought to show.

To close off this section, we show how our previous discussion applies to composite fields:

Proposition 19.3.17: Degree Of Composite Field

Let K_1 and K_2 be two finite extensions of a field F contained in K. Then

$$[K_1K_2 : F] \leq [K_1 : F][K_2 : F]$$

with equality if and only if an F-basis for one of the fields remains linearly independent over the other field. If $\alpha_1, \alpha_2, ..., \alpha_n$ and $\beta_1, \beta_2, ..., \beta_m$ are bases for K_1 and K_2 over F respectively, then the elements $\alpha_i\beta_j$ for $i = 1, 2, ..., n$ and $j = 1, 2, ..., m$ span K_1K_2 over F

***Proof*:**
Let $n = [K_1 : F]$, $m = [K_2 : F]$, and let $S \subseteq K$ be the F-span of the nm elements $\alpha_i\beta_j$.

The set S is closed under multiplication: $(\alpha_i\beta_j)(\alpha_k\beta_l) = (\alpha_i\alpha_k)(\beta_j\beta_l)$, where $\alpha_i\alpha_k \in K_1$ is an F-combination of the α's and $\beta_j\beta_l \in K_2$ is an F-combination of the β's, so expanding exhibits the product as an F-combination of the $\alpha_i\beta_j$. The general case follows by bilinearity of multiplication. Moreover $K_1 \subseteq S$: given $x \in K_1$, write $x = \sum_i a_i\alpha_i$ with $a_i \in F$ and $1 = \sum_j c_j\beta_j$ with $c_j \in F$ (as $1 \in K_2$), so that $x = x \cdot 1 = \sum_{i,j} a_i c_j\, \alpha_i\beta_j \in S$. Symmetrically $K_2 \subseteq S$, and in particular $1 \in S$. Hence S is a subring of K containing F, and a finite dimensional F-vector space of dimension at most nm.

The ring S is in fact a field. Let $s \in S$ be nonzero and consider the F-linear map $S \to S$, $y \mapsto sy$. It is injective since K is a field, hence by corollary 13.1.20 it is surjective, so $sy = 1$ for some $y \in S$ and s is invertible in S. Since each $\alpha_i\beta_j \in K_1K_2$ we have $S \subseteq K_1K_2$, while S is a subfield of K containing K_1 and K_2, so the minimality in definition 19.1.16 gives $K_1K_2 \subseteq S$. Hence $S = K_1K_2$, the elements $\alpha_i\beta_j$ span K_1K_2 over F, and

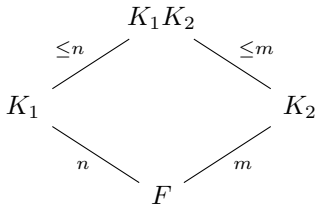
$$[K_1K_2 : F] \leq nm = [K_1 : F][K_2 : F]$$

For the equality criterion, note that every element of $K_1K_2 = S$ can be written $\sum_{i,j} c_{ij}\alpha_i\beta_j = \sum_i \left(\sum_j c_{ij}\beta_j\right)\alpha_i$ with $\sum_j c_{ij}\beta_j \in K_2$, so $\alpha_1, \ldots, \alpha_n$ span K_1K_2 as a K_2-vector space and $[K_1K_2 : K_2] \leq n$. If the α_i are linearly independent over K_2, then $[K_1K_2 : K_2] = n$. If they are dependent over K_2, then, K_2 being a field, some α_i lies in the K_2-span of the others, so $n - 1$ elements span and $[K_1K_2 : K_2] \leq n - 1$. By the Tower Law (theorem 19.3.7),

$$[K_1K_2 : F] = [K_1K_2 : K_2][K_2 : F] = m\,[K_1K_2 : K_2]$$

which equals nm if and only if $[K_1K_2 : K_2] = n$, that is, if and only if the F-basis $\alpha_1, \ldots, \alpha_n$ of K_1 remains linearly independent over K_2. Exchanging the roles of K_1 and K_2 gives the same criterion with the β_j over K_1, proving the statement.

summing up the result in a diagram:



Corollary 19.3.18: Degree of Composite Field and Equality

Suppose that $[K_1 : F] = n$, $[K_2 : F] = m$ in proposition 19.3.17, where n and m are relatively prime $(n, m) = 1$. Then $[K_1K_2 : F] = [K_1 : F][K_2 : F] = nm$

***Proof*:**
In general, the extension is divisible by n and m since K_1 and K_2 are subfields of K_1K_2 (so divisible by its least common multiple). Since $(m, n) = 1$, then $[K_1K_2 : F]$ is divisible by nm. Since by proposition 19.3.17, $[K_1K_2 : F] \leq nm$, we get $[K_1K_2 : F] = nm$

Exercise 19.3.1

1. Let F be a field with $\text{ch}(F) \neq 2$ and let K/F be an extension of degree 2. Show that $K = F(\beta)$ for some $\beta \in K$ with $\beta^2 = D \in F$, where D is not a square in F.
2. Let E/k and F/k be field extensions inside a common field. Show that if E/k is algebraic, then so is EF/F. Show that if E/k and F/k are algebraic, so is EF/k.

19.4 Historical Detour: Straightedge and Compass Construction

For millennia in mathematical history, mathematicians as far back as ancient greeks tried to construct shapes or compute numbers through geometric means by using a straight edge and compass. For example, using a straight edge and compass, there is a way to compute sums, products, quotient (which would have been known as ratios to the ancient greeks), and square-roots. Geometry, mathematicians have figured out how to double the area of the square, take exactly half the angle, and navigate using the stars [14]. These all conformed with the rules of Euclid's geometry, and so straight-edge and compass constructions seemed like a powerful algorithm for geoemtry and arithmetics. However, a few problems were open since at least since the times of the ancient greeks around 400 BC, namely:

1. Given a square, using a straight-edge and compass, can we find a circle with the same area
2. Given a cube, using a straight-edge and compass, can we find another cube that's double it's volume
3. Given an angle, using a straight-edge and compass, can we triscect the angle (i.e. split the angle into 3 equa angles)

These problems do not have much use practically. However the simpliciy in their statement, the ubiquity of using rulers and compasses in many fields of engeneering, navigation, and so on, as well as the belief that the foundation of the world's geometry is inherently euclidean (in particular, that geometric reality followed euclidean rules and were provable using euclid's simple axioms), made these problems the central attention of a huge plethora of mathematicians for over 2 millennia.

With the new tools of field theory, we can characterize the types of numbers a straight-edge and compass can give us:

1. using a straight-dge construction, the 4 basic operations $(+, -, \times, \div)$ can be derivd. Thus, it will form a field.

[14] and an additional instrument to measure the angle between the person and the horizon, and the north star

2. the compass constructions allows the performance of square-roots. Translating this into the cartesian plane, it will be the intersction of a straight-line and a circle, which will sometimes gives polynomials which will have solutions *not* inside the base field, and so we would use an algebraic extension. Such polynomials will always be of degree 2. Hence, if we keep trying to extend these fields, we will always get multiples of 2 as the degree of the extension

From this comes the theorem that feels so mind-blowing that I can state after literal millenia of people trying to solve this problem:

Theorem 19.4.1: Straightedge And Compass Imposibilities

The 3 classical greek straight-edge and compass problems:

1. **Doubling the Cube** Using only straight edge and compasss, construct a cube with precisely twice hte volume of a given cube
2. **Triscecting an Angle** Given a straightedge and compass, triseced any given angle θ
3. **Squaring the circle** Given a straightedge and compass, construct a square whose area is precisely the area of a given circle.

are all answered in the negative

***Proof*:**
Set up coordinates so that the two points given at the start are $(0,0)$ and $(1,0)$. A construction step draws the line through two points already constructed, or the circle centred at a point already constructed whose radius is the distance between two points already constructed. The new points are the points of intersection of these lines and circles. Call $\alpha \in \mathbb{R}$ *constructible* when it is a coordinate of a point reachable by finitely many such steps.

Step 1: one step at a time, the coordinates move up by degree at most 2. Let $F \subseteq \mathbb{R}$ be a subfield containing the coordinates of every point constructed so far. The line through two such points has an equation $ax + by = c$ with $a, b, c \in F$, and the circle centred at such a point through such a point has an equation $x^2 + y^2 + dx + ey + f = 0$ with $d, e, f \in F$, since the square of the radius is a sum of squares of differences of coordinates. There are three ways a new point can appear.

1. Two distinct lines meet, distinctness being what makes the intersection a single point. Solving the two linear equations gives coordinates already in F.
2. A line meets a circle. Solving the linear equation for one coordinate and substituting it into the quadratic leaves a quadratic equation over F for the other coordinate. Adjoining a root of it gives an extension of F of degree 1 or 2, and the first coordinate is an F-linear expression in the second.
3. Two distinct circles meet, distinctness again being what makes the intersection finite. Subtracting one equation from the other cancels $x^2 + y^2$ and leaves a linear equation over F, satisfied by both points of intersection. So the intersection of the two circles is the intersection of one of them with that line, and the previous case applies.

Step 2: a constructible number has degree a power of 2 *over* $\mathbb{Q}$. Running Step 1 along a construction of α produces a tower

$$\mathbb{Q} = F_0 \subseteq F_1 \subseteq \cdots \subseteq F_m \subseteq \mathbb{R}$$

with $[F_{i+1} : F_i] \leq 2$ for every i and $\alpha \in F_m$. By the Tower Law (theorem 19.3.7) applied repeatedly, $[F_m : \mathbb{Q}] = 2^k$ for some $k \geq 0$. Applying the Tower Law once more to $\mathbb{Q} \subseteq \mathbb{Q}(\alpha) \subseteq F_m$ shows $[\mathbb{Q}(\alpha) : \mathbb{Q}]$ divides 2^k, so

$$[\mathbb{Q}(\alpha) : \mathbb{Q}] = 2^j \quad \text{for some } j \geq 0$$

Step 3: (1) doubling the cube. Starting from a cube of side 1, a cube of twice the volume has side $\sqrt[3]{2}$, so the problem asks for $\sqrt[3]{2}$ to be constructible. The polynomial $x^3 - 2$ is irreducible over $\mathbb{Q}$ by Eisenstein's criterion at the prime 2 (corollary 11.3.11), so it is the minimal polynomial of $\sqrt[3]{2}$ and theorem 19.2.8 gives $\mathbb{Q}(\sqrt[3]{2}) \cong \mathbb{Q}[x]/(x^3 - 2)$, whence

$$[\mathbb{Q}(\sqrt[3]{2}) : \mathbb{Q}] = 3$$

which is not a power of 2. By Step 2 the number $\sqrt[3]{2}$ is not constructible.

Step 4: (2) trisecting an angle. An angle θ is constructible from a given one exactly when $\cos\theta$ is a constructible number: the point at distance 1 from the origin at angle θ is $(\cos\theta, \sin\theta)$, and dropping the perpendicular to the horizontal axis produces the point $(\cos\theta, 0)$, while conversely erecting the perpendicular at $(\cos\theta, 0)$ and cutting the unit circle with it recovers the angle. The claim is that no single construction trisects every angle, so it is enough to exhibit one angle it fails on.

Take $\theta = 60$ degrees, itself constructible since $\cos\theta = 1/2$. Write $\beta = \cos 20$ degrees. The triple angle identity $\cos\theta = 4\cos^3(\theta/3) - 3\cos(\theta/3)$ gives

$$4\beta^3 - 3\beta = \frac{1}{2} \qquad \text{that is} \qquad 8\beta^3 - 6\beta - 1 = 0$$

Setting $\alpha = 2\beta$ turns this into $\alpha^3 - 3\alpha - 1 = 0$. The cubic $x^3 - 3x - 1$ is monic with constant term -1, so by the rational roots test (proposition 11.3.6) its only possible rational roots are ± 1, and neither is a root, since $1 - 3 - 1 = -3$ and $-1 + 3 - 1 = 1$. A cubic over a field with no root in that field is irreducible by proposition 11.3.5, so $x^3 - 3x - 1$ is the minimal polynomial of α and $[\mathbb{Q}(\alpha) : \mathbb{Q}] = 3$ by theorem 19.2.8. Since $\alpha = 2\beta$ we have $\mathbb{Q}(\alpha) = \mathbb{Q}(\beta)$, so $[\mathbb{Q}(\beta) : \mathbb{Q}] = 3$ is not a power of 2 and $\beta = \cos 20$ degrees is not constructible. Hence the angle of 60 degrees cannot be trisected.

Step 5: (3) squaring the circle. A square with the area of the circle of radius 1 has side $\sqrt{\pi}$, so the problem asks for $\sqrt{\pi}$ to be constructible. If it were, then $\pi = (\sqrt{\pi})^2$ would lie in the field F_m of Step 2, so $[\mathbb{Q}(\pi) : \mathbb{Q}]$ would divide $[F_m : \mathbb{Q}]$ and in particular be finite.

What remains is the transcendence of π, which this book does not prove. That π is transcendental over $\mathbb{Q}$ is Lindemann's theorem. Its proof is analytic and belongs to a treatment of transcendence. Granting it, π is a root of no nonzero rational polynomial, so $[\mathbb{Q}(\pi) : \mathbb{Q}]$ is infinite by proposition 19.3.3, and in particular is not a power of 2. The circle therefore cannot be squared.

19.5 Closures of Fields

Through theorem 19.2.2, we saw that if we have a irreducible polynomial $f(x) \in k[x]$, we can find a larget field which gives $f(x)$ a root: $f(x) = f'(x)(x-\alpha)$. However, we have only established result for 1 polynomial, and for only 1 root. It would be nice if the set of all roots of every polynomials formed a field, and even better if that field existed for any field k and was unique (up to isomorphism). In such a field, every polynomial would *fully factor* or *split* into linear terms. Through corollary 19.3.14, we found that given K/k the set of algebraic elements $\alpha \in K$ over k forms a field, namely the set of elements from the larger field K which are algebraic over k. In example 19.3.15(1), we where implicitely using the fact that every polynomial over $\mathbb{C}$ reduces to linear terms, allowing us to say that $\overline{\mathbb{Q}}$ to be algebraically closed. It would be much better if the definition would be independent of any larger field, that is, find a field $\overline{k}$ such that the only irreducible polynomials in $\overline{k}[x]$ are of degree 1. Furthermore, if $\overline{k}$ existed for every field and was unique, we can call it *the* algebraic closure of k. This is the first thing proven in this section. Then, we take a closer look at fields for which only some subset $S \subseteq k[x]$ have polynomials that split. These fields will be at the heart of Galois theory, for as we saw in corollary 19.2.29, the group of k-automorphism $\mathrm{Aut}_k(F)$ depends on the number of roots of a polynomial in F. Thus, we take some time constructing such a field.

19.5.1 Algebraic Closure

We start off this section by reminding the reader what an algebraically closed field is:

Definition 19.5.1: Algebraically Closed

Let K be a field. Then K is called algebraically closed if every polynomial in $K[x]$ factors into a product of linear terms.

There are many equivalent ways of defining algebraic closure:

Lemma 19.5.2: Algebraic Closure Equivalences

Let K be a field. then the following are equivalent:

1. K is algebraically closed
2. every maximal ideal in $K[x]$ is of the form $(x-c)$ for some $c \in K$
3. There does not exist nontrivial algebraic extensions of K
4. If L/K is some field extension, and $\alpha \in L$ is algebraic over K, then $\alpha \in K$

***Proof*:**
We prove (1) $\Rightarrow$ (2) $\Rightarrow$ (3) $\Rightarrow$ (4) $\Rightarrow$ (1).

(1) $\Rightarrow$ (2): Let $M \subseteq K[x]$ be a maximal ideal. Since $K[x]$ is a Euclidean domain (theorem 11.1.1) and hence a PID (proposition 10.2.7), $M = (f)$ for some $f \in K[x]$. The polynomial f is nonconstant: a nonzero constant is a unit, giving $M = K[x]$, and $f = 0$ gives $M = (0) \subsetneq (x) \subsetneq K[x]$, in both cases contradicting maximality. By (1), f factors into linear factors, so

$f = (x-c)g$ for some $c \in K$ and $g \in K[x]$. Then $(f) \subseteq (x-c) \subsetneq K[x]$, the last containment being proper since the degree one polynomial $x - c$ is not a unit. Maximality of $M = (f)$ forces $M = (x - c)$.

(2) $\Rightarrow$ (3): Let L/K be an algebraic extension and take any $\alpha \in L$. Since α is algebraic, the evaluation map $\text{ev}_\alpha : K[x] \to K(\alpha)$ is not injective, and theorem 19.2.8 gives $K[x]/(m) \cong K(\alpha)$ where $m = m_{\alpha,K}$ is the minimal polynomial of α (definition 19.2.19). As $K(\alpha)$ is a field, proposition 9.4.19 shows (m) is a maximal ideal, so by (2), $(m) = (x - c)$ for some $c \in K$. Two generators of the same principal ideal in a domain divide one another, so comparing degrees, $m = u(x - c)$ for some unit $u \in K^\times$. Since $m(\alpha) = 0$, we get $u(\alpha - c) = 0$ and hence $\alpha = c \in K$. As α was arbitrary, $L = K$, so there is no nontrivial algebraic extension of K.

(3) $\Rightarrow$ (4): Let L/K be a field extension and $\alpha \in L$ algebraic over K. By theorem 19.2.8 and corollary 19.2.23, $K(\alpha)$ is a finite extension of K, hence algebraic by corollary 19.3.4. By (3), $K(\alpha) = K$, so $\alpha \in K$.

(4) $\Rightarrow$ (1): Let $f \in K[x]$ be nonconstant. We show by induction on $\deg f$ that f factors into linear factors. Since $K[x]$ is a UFD (proposition 10.1.33), f has an irreducible factor p. By Kronecker's theorem (theorem 19.2.2), there is an extension L/K in which p has a root α. Then α is algebraic over K, being a root of p, so $\alpha \in K$ by (4). Dividing f by $x - \alpha$ using theorem 11.1.1 gives $f = (x - \alpha)q + r$ with r constant, and evaluating at α gives $r = f(\alpha) = 0$ since $p(\alpha) = 0$ and $p \mid f$. Hence $f = (x - \alpha)q$ with $\deg q = \deg f - 1$. If q is constant, we are done. Otherwise the induction hypothesis factors q into linear factors, and with it f. Hence K is algebraically closed.

These show that being an algebraically closed field K means that there are no more algebraic extensions L/K, besides trivial ones. Given some field k, there can be many field extensions that are algebraically closed. For example, $\overline{\mathbb{Q}}/\mathbb{Q}$, $\mathbb{C}/\mathbb{Q}$, and $\mathbb{C}(t)/\mathbb{Q}$ are all field extensions of $\mathbb{Q}$ to algebraically closed fields. However, the first extensions seems in some way more simple or "minimal" than the other in that it is defined through only algebraic elements (recall from example 19.3.15(1) that all the elements of $\overline{\mathbb{Q}}$ are algebraic over $\mathbb{Q}$), while $\mathbb{C}$ contains transcendental elements (we have mentioend in example 19.3.15(2) how π and e are not algebraic).

Definition 19.5.3: Algebraic Closure

Let k be any field. Then an *algebraic closure* of k is a field $\overline{k}$ where $\overline{k}/k$ is an *algebaric extension* and $\overline{k}$ is algebraically closed.

Soon, the statement can be re-written with "the algebraic closure" and "the field" instead of "an algebraic closure" and "a field". The field $\overline{k}$ will be the smallest field in which every nonconstant polynomial $p(x) \in k[x]$ will split into linear factors, and since $\overline{k}$ is algebraically closed, the subfield of $\overline{k}$ that contains all the roots of all non-constant polynomials in $k[x]$ is $\overline{k}$ itself. To show the existence of such a field, we first prove that there exists a field extension where every polynomial in $k[x]$ has *at least* one root, and then recursively apply the theorem to get a long chain of field extensions, each containing at least one more root to every polynomial.

Lemma 19.5.4: Adding One Root

Let k be a field. Then there exists a field extension K of k such that every non-constant polynomial of $k[x]$ admits at least 1 root in K.

The following proof is credited to Emil Artin

***Proof*:**
Let $\mathcal{F}$ be a set that is in bijection with all non-constant monic polynomials $F \subseteq k[x]$: $\mathcal{F} = \{t_f\}_{f \in F}$, and let $k[\mathcal{F}]$ be a polynomial ring with indeterminates $t_f \in \mathcal{F}$. Let $I \subseteq k[\mathcal{F}]$ be the set of all polynomials generated by all $f(t_f)$ (that is, the polynomial f that takes in the indeterminate t_f that it is in bijective correspondence).

I claim that I is a proper ideal of $k[\mathcal{F}]$. Indeed, if it were not, then we would have:

$$1 = \sum_i^n a_i f(t_{f_i})$$

with $a_i \in k[\mathcal{F}]$. The identity cannot be possible, for that would lead to imply $1 = 0$. Let's take $k \subseteq F$ where each polynomial $f_1(x), ..., f_n(x)$ has a root $\alpha_1, ..., \alpha_n$ (apply theorem 19.2.2 n times). Then viewing the previous identity to be in $F[\mathcal{F}]$ we get:

$$1 = \sum_i^n a_i f(t_{f_i}) = \sum_i^n a_i f(\alpha_i) = \sum_i^n a_i \cdot 0 = 0$$

a contradiction.

Now, since I is proper, it is contained in a maximal ideal m (proposition 9.4.18). Thus, we have the field extension:

$$k \subseteq K := \frac{k[\mathcal{F}]}{m}$$

By construction every non-constant monic (and so every nonconstant) polynomial $f(x) \in k[x]$ must have a root in K, namely $\overline{t_f}$

Note that we had to use Zorn's lemma to find the maximal ideal m. Very commonly, a statement using Zorn's lemma is equivalent to it (ex. instead of using Zorn's lemma to prove the existence of a maximal ideal, we can use the existence of a maximal ideal to prove Zorn's lemma). We might ask if the existence if the previous result is also equivalent to Zorn's lemma, however this is not the case (it is due to the *compactness theorem of first order logic*, which is known to be weaker than the axiom of choice)

We can now find a field $K_1 \supseteq k$ in which every nonconstant polynomial in $k[x]$ has at least 1 root. We can continue this process by applying lemma 19.5.4 to K_1, which gives a field $K_2 \supseteq K_1$ in which every nonconstant polynomial of $K_1[x]$ has at least 1 root. Continuing this process, we get a chain:

$$k = K_0 \subseteq K_1 \subseteq K_2 \subseteq \cdots$$

in which every nonconstant polynomial of $K_i[x]$ has a root in K_{i+1}. Now, take $L = \bigcup_i K_i$[15]. L can easily be turned into a field through the operation of $a + b$ and $a \cdot b$ where we choose K_i in which $a, b \in K_i$.

[15]This is an example of a *direct limit*

Lemma 19.5.5: Union Of The Root-Adjoining Chain Is Algebraically Closed

Let $k = K_0 \subseteq K_1 \subseteq K_2 \subseteq \cdots$ be a chain of fields in which every nonconstant polynomial of $K_i[x]$ has a root in K_{i+1}, for instance the chain constructed above, and let $L = \bigcup_i K_i$. Then L is algebraically closed.

***Proof*:**
We show by induction on $n \geq 1$ that every polynomial $f \in L[x]$ of degree n is a product of a nonzero constant and n linear factors $x - \alpha$ with $\alpha \in L$. The polynomial f has finitely many coefficients, each lying in some K_j, and the K_j form a chain, so all of them lie in a single K_i, that is, $f \in K_i[x]$. As f is nonconstant, it has a root $\alpha \in K_{i+1} \subseteq L$. Dividing f by $x - \alpha$ in $K_{i+1}[x]$ (theorem 11.1.1) gives $f = (x - \alpha)g + r$ with r constant, and evaluating at α gives $r = f(\alpha) = 0$. So $f = (x - \alpha)g$ with $g \in L[x]$ of degree $n - 1$ (proposition 9.2.3(1)). If $n = 1$, then g is a nonzero constant and $f = g \cdot (x - \alpha)$ has the required form. If $n \geq 2$, the induction hypothesis writes g as a nonzero constant times $n - 1$ linear factors, and with the factor $x - \alpha$ this gives the required form for f. Hence every polynomial in $L[x]$ factors into linear terms, and L is algebraically closed (definition 19.5.1), completing the proof.

Thus, for any field k, there exists an algebreaically closed field L such that L/k. Note that L need not be algebraic closure, since L/k need not be an algebraic extension. However, with an algebraically closed field, we can very simply construct an algebraic closure of k:

Theorem 19.5.6: Existence Of Algebraic Closure

Let k be a field and L/k be a field extension where L is algebraically closed. Then

$$\overline{k} := \{\alpha \in L \mid \alpha \text{ is algebraic over } k \}$$

is an algebraic closure of k

Note how this mimicked $\overline{\mathbb{Q}}$ in example 19.3.15(2) assuming $\mathbb{C}$ is algebraically closed.

***Proof*:**
First, by corollary 19.3.14, $\overline{k}$ is a field, and *a prior* $\overline{k}$ is an algebraic extension of k. To show that k is algeraically closed, let α be algebraic over $\overline{k}$. Then

$$k \subseteq \overline{k} \subseteq \overline{k}(a)$$

is a composition of albegraic extensions, hence $\overline{k}(a)$ is algebraic (theorem 19.3.16), thus α is algebraic over k. But then by definition of $\overline{k}$, $\alpha \in \overline{k}$. Thus, by lemma 19.5.2$\overline{k}$ is algebraically closed, completing the proof

Next, we strive for uniqueness. As we establisehd in theorem 19.2.2, being a field extension is not a universal property, and so the best we an strive for is the existence of a homomorphism, though not necesarrily a unique one. We first establish a lemma:

Lemma 19.5.7: Embedding Into Algebraic Extension

Let L/k be any field extension where L is algebraically closed over k. Let F/k be an algebraic extensions. Then there exists a [not necessarily unique] homomorphism $i : F \to L$ fixing k.

Note that *a-priori*, L need not have any relation to F (ex. we do not know if F is a subset of L, they might only intersect on k).

***Proof*:**

This is a Zorn's lemma argument similar to Baer's Criterion (theorem 17.3.5). Let S be the set of homomorphisms of the form

$$i_K : K \to L$$

The set S is nonempty since $i_k : k \to L \in S$. Define a patial orer on S $i_K \preceq i_{K'}$ if $K \subseteq K' \subseteq L$ and $i_{K'}|_K = i_K$. To apply Zorn's lemma, we need to show that every chain has an upper bound. Let $C \subseteq S$ be a chain, and let K_C be the union of all the domains of elements $i_K \in C$. Then K_C is a field, being the union of a chain of fields (any two elements of K_C lie in a common K), and for every $\alpha \in K_C$ define $i_{K_C}(\alpha)$ to be $i_K(\alpha)$ for appropriate K where $\alpha \in K$. Then i_{K_C} is well-defined: if $\alpha \in K \subseteq K'$ with $i_K, i_{K'} \in C$, then $i_{K'}|_K = i_K$, so the value does not depend on the choice of K, and so we have a homomorphism $i_{K_C} : K_C \to L$ where $i_K \preceq i_{K_C}$ for all $i_K \in C$, showing that i_{K_C} is indeed an upper bound for C. Thus, by Zorn's lemma, Z has an upper bound $i_G : G \to L$ where $k \subseteq G \subseteq L$.

I claim that $G = F$, which will prove the statement. For the sake of contradiction, let's assume this is not the case, so that there exists an element $\alpha \in F \setminus G$. Consider $G(\alpha)/G$. Since $\alpha \in F$ is algebraic over k, α is algebraic over G, and since $\alpha \notin G$, it is the root of an irreducible polynomial $g(x) \in G[x]$. Let $H = i_G(G)$. Then as discussed in the first pages of section 19.2 i_G induces a ring homomorphism

$$f^{i_G} : G[x] \to H[x]$$

Let $h(x) = f^{i_G}(g(x))$. Then $h(x)$ is irreducible over H, and since L is algebraically closed, it must have a root $\beta \in L$.

We are now setup in a situation to apply theorem 19.2.27: we have an isomorphism $i_G : G \to H$ and two simples extensions $G(\alpha), H(\beta)$ with α and β being roots of $g(x)$ and $h(x)$ respectively (with $h(x) = i_G(g(x))$). Thus, i_G lifts to an isomorphism:

$$i_{G(\alpha)} G(\alpha) \to H(\beta) \subseteq L$$

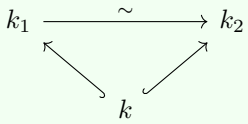
which sends α to β. Then $i_G \preceq i_{G(\alpha)}$, contradicting the maximality of i_G.

Thus, it must be that $G = F$, and so $i_F : F \to L$ is a field homomorphism, as we sought to show.

The fact that the isomorphism is not unique will become important in representing algebraic closures. We first finish off by showing that the algebraic closure is indeed unique up to isomorphism (given a fixed field k)

Theorem 19.5.8: Uniqueness Of Algebraic Closure

Let $k \subseteq \overline{k_1}$ and $k \subseteq \overline{k_2}$ be two algebraic closures of k. Then $\overline{k_1} \cong \overline{k_2}$ where the isomorphism extends the identity on k, or equivalently k_1 is isomorphic to k_2 as extensions. In terms of a commutative diagram, in the following



there always exists an isomorphism from k_1 to k_2

the second condition for the isomorphism is so that the two fields represent the algebraic closure over the common field k. Sometimes, such a isomorphism (resp. homomorphism) is called a k-isomorphism (resp. k-homomorphism)

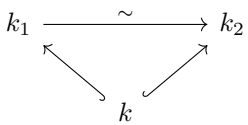
***Proof*:**
Since $\overline{k_1}/k$ is algebraic and $\overline{k_2}$ is algebraically closed over k, by lemma 19.5.7, there exists a homomorphism $i : \overline{k_1} \to \overline{k_2}$. This homomorphism is injective, being a field homomorphism, and surjective since otherwise, $\overline{k_1} \subseteq \overline{k_2}$ would be a non-trivial algebraic extension, contradicting lemma 19.5.2 since $\overline{k_1}$ is algebraically closed. Thus, i must be an isomorphism as we sought to show.

The fact that the isomorphism is not necessarily unique has some important consequences: There is no "cannonical" way of representing the algebraic closure. At first, it might seem harmless that the isomorphism is not unique, as long as the two field are isomorphic. However, this means there is no real cannonical "choice" when it comes to representing algebraic closures.

Take for example $\overline{\mathbb{R}}$, which we know to be $\mathbb{C}$. We can write $\overline{\mathbb{R}}$ as $\mathbb{R}[i]$, which for many mathematicians is "the" algebraic closure of $\mathbb{R}$. However, if we go over to an electrical engineer, we will see that they have also constructed an algebraic closure for $\mathbb{R}$ to represent electrical charge and circuits where they use the letter j (since i is already used for something else), and write it as $\mathbb{R}[j]$. We might think it to be harmless to make the isomorphism $1 \mapsto 1$ and $i \mapsto j$, however the electrical engineers have decided that electrical charge should be represented by a -1. Does this means we should instead have

$$1 \mapsto 1 \qquad i \mapsto -j?$$

Mathematically, there is no good reason to pick one representation over another. In other words, if one person constructs a splitting field L for k and another person constructs a algebraic closure L' for k, then we know that we get:



however, there is no "natural" way to associate an element of L to an element of L'. This often means that we should keep track of the construction of the splitting field we have done, since if we do not we might not be able to communicate to other mathematicians who have done a different

construction (how can you compare your work if you cannot correspond the results you have come up with with the other mathematicians).

The idea there there is no natural choice of isomorphism can be made more rigorous if we ask it in the language of category theory. If we map every field to it's algebraic closure $k \to \overline{k}$, does this define a functor? In order for it to define a functor, we would need that every map $k \to l$ of field defines a map of algebraic closures

$$\begin{array}{ccc} k & \longrightarrow & \overline{k} \\ \downarrow & & \vdots \\ l & \longrightarrow & \overline{l} \end{array}$$

However, it is not clear what map should be chosoen here[16].

19.5.2 Splitting Field and Normal Extensions

Having established that every field has an algebraic closure, meaning we can unambiguously refer to roots of polynomials, we now return to analyzing algebraic extensions. We start off by zoom in on fields that only split some subset of polynomials $S \subseteq k[x]$. In the following sections, we will be working over the case where S is finite. Since S is finite, say $S = \{f_1(x), ..., f_n(x)\}$, then having each $f_i(x)$ split is equivalent to having $f_1(x)f_2(x)\cdots f_n(x)$, thus, we usually have $S = \{f(x)\}$ be one polynomial

Definition 19.5.9: Splitting Field

Let k be a field and $f(x) \in k[x]$ be a polynomial where $\deg(f(x)) = d$. Then a *splitting field* for $f(x)$ over k is a minimal extension F/k such that

$$f(x) = c\prod_{i}^{d}(x - \alpha_i)$$

i.e. $f(x)$ splits completely in $F[x]$, and $f(x)$ does not factor in any proper subfield of F.

Furthermore, by minimality, $F = k(\alpha_1, ..., \alpha_n)$ (i.e. F is generated by the roots of $f(x)$ in F)

The fact that a splitting field exists is immediate from the fact that $\overline{k}$ exists, namely if $f(x)$ is a polynomial, then $F \subseteq \overline{k}$ generated by the roots of $f(x)$ in $\overline{k}$. Furthermore, the splitting field is in fact unique (up to isomorphism), justifying replacing any "a splitting field" in the definition to "the splitting field"

[16] perhaps you can force through a construction using the axiom of choice, but it is certainly not a natural choice

Theorem 19.5.10: Uniqueness Of Splitting Field

Let k be a field and let $f(x) \in k[x]$. Then the splitting field F for $f(x)$ over k is unique up to isomorphism, and $[F : k] \leq (\deg(f))!$ (i.e. the factorial of the degree of f). Furthermore, if $i : k \to k'$ is an isomorphism, $f(x) \in k[x]$ and $g(x) = i(f(x))$, then i extends to an isomorphism from the splitting field of $g(x)$ over k' to the splitting field of $f(x)$ over k:

$$\begin{array}{ccc} F & \overset{\sim}{\dashrightarrow} & F' \\ \uparrow & & \uparrow \\ k & \xrightarrow{\sim} & k' \end{array}$$

***Proof*:**
We will first construct a splitting field explicitly with $[F : k] \leq (\deg(f))!$, and then use that field to show uniqueness. For existence, this is an application of induction on the degree of the polynomial. Degree 1 is trivial, covering the base-case, so let's say the splitting field was constructed for all fields and all polynomials of degree $\deg(f) - 1$. Let $f(x)$ be some polynomial that is not yet fully factored into linear terms. Let $q(x)$ be some irreducible non-linear factor over k. Then

$$k \subseteq F' := \frac{k[x]}{(q(x))}$$

is an extension of degree $\deg q \leq \deg f$ in which $q(x)$ has a root (and so $f(x)$ has a root), say α, and so $q(x) = q'(x)(x - \alpha)$. Then we can define the polynomial $h(x) = f(x)/(x - \alpha) \in F'[x]$ of degree $\deg(f) - 1$. Thus, by the induction hypothesis, there exists a splitting field F for $h(x)$ over F' and

$$[F : F'] \leq (\deg(f) - 1)!$$

Since the factors of $f(x)$ over F are $(x - \alpha)$ and the factors of $h(x)$, it follows that F is a splitting field for $f(x)$, and

$$[F : k] = [F : F'][F' : k] \leq (\deg(f) - 1)!(\deg(f)) = (\deg(f))!$$

To prove that an isomorphism $i : k \to k'$ extends to isomorphisms of splitting fields as stated, let F be the splitting field for $f(x)$, and consider $k \to k' \subseteq \overline{k'}$. Since the extension $k \subseteq F$ is algebraic, by lemma 19.5.7 there exists a k-homomorphism $i : F \to \overline{k'}$. Since F is generated over k by the roots of $f(x)$ in F, we have

$$i(F) = k'(\alpha_1, ..., \alpha_d) \subseteq \overline{k'}$$

where the α_i's are the roots of $g(x) = i(f(x))$ in $\overline{k'}$. Therefore, $L := i(F)$ is independent of the given isomorphism i and the splitting field F. Applying this to i and to the identity $k' \to k'$ completes the proof (since both F and G are isomorphic to L)

Example 19.5.11: Splitting Fields

1. The splitting field of $x^2 - 2$ over $\mathbb{Q}$ is $\mathbb{Q}(\sqrt{2})$, since the two roots are $\pm\sqrt{2}$ and $-\sqrt{2} \in \mathbb{Q}(\sqrt{2})$. The splitting field of $(x - 2)^n$ is also $\mathbb{Q}(\sqrt{2})$ for all $n \in \mathbb{N}_{>0}$, since we only need to add the

root $\sqrt{2}$ (resp. $-\sqrt{2}$) for this polynomial to split.

2. Splitting field for $(x^2-2)(x^2-3)$ over $\mathbb{Q}$ is $\mathbb{Q}(\sqrt{2},\sqrt{3})$. Note that $\mathbb{Q}(\sqrt{2},\sqrt{3})$ is equal to the composite field $\mathbb{Q}(\sqrt{2})\mathbb{Q}(\sqrt{3})$, i.e. $\mathbb{Q}(\sqrt{2})\mathbb{Q}(\sqrt{3}) = \mathbb{Q}(\sqrt{2},\sqrt{3})$.

3. More generally, If $f(x)$ is a polynomial over k such that

$$f(x) = \prod g_i(x)$$

where each $g_i(x)$ is irreduible over k, then the splitting field of $f(x)$ is the composite field of the splitting fields for each $g_i(x)$ since each splitting field is the smallest field containing all the roots of g_i, and the composite field is the smallest field containing all the splitting fields.

4. splitting field of x^3-2 over $\mathbb{Q}$ is not $\mathbb{Q}(\sqrt[3]{2})$, since we are missing two roots, namely:

$$\sqrt[3]{2}\left(\frac{-1+i\sqrt{3}}{2}\right) \qquad \sqrt[3]{2}\left(\frac{-1-i\sqrt{3}}{2}\right)$$

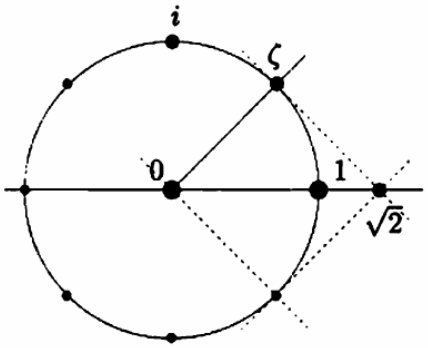
Thus, the splitting field is $\mathbb{Q}(\sqrt[3]{2}, i\sqrt{3}) = \mathbb{Q}(\sqrt[3]{2}, \sqrt{-3})$.

5. The splitting field of x^2+1 over $\mathbb{Q}$ is $\mathbb{Q}(i)$, and the splitting field of the same polynomial x^2+1 over $\mathbb{R}$ is $\mathbb{C}$, showing that the splitting field is dependent on the base field.

6. Let F be the splitting field of x^8-1 over $\mathbb{Q}$. With some knowledge of complex analysis, we see that all the roots are 8th root of unit, that is,

$$\zeta = e^{\frac{2\pi i}{8}}$$

is a generator for the roots (i.e. the primite elements), and so $F = \mathbb{Q}(\zeta)$. Visually, these roots look like so:



In fact, ζ is also the root of x^4+1 (since $\zeta^4 = -1$), and so is the root of the smaller irreducible polynomial x^4+1 over $\mathbb{Q}$. There is no smaller minimal polynomial for which ζ is a root (since smaller powers of ζ is not in $\mathbb{R}$) and so:

$$[\mathbb{Q}(\zeta) : \mathbb{Q}] = 4$$

showing that the degree of the splitting field is much less than $8! = 40'320$. To better undersatnd this field, notice that $\zeta^2 = i$, so $\sqrt{2} = \zeta + \zeta^7$. Thus:

$$\mathbb{Q}(i, \sqrt{2}) \subseteq \mathbb{Q}(\zeta)$$

Conversely, $\zeta = \frac{\sqrt{2}}{2}(1+i) \in \mathbb{Q}(i, \sqrt{2})$, showing that:

$$\mathbb{Q}(i, \sqrt{2}) \supseteq \mathbb{Q}(\zeta)$$

showing that they are in fact equal.

As an exercise, show that $\mathrm{Aut}_{\mathbb{Q}}(\mathbb{Q}(i, \sqrt{2})) = \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$

7. Let $p(x) = x^4 - 1$. this polynomial in fact factors over $\mathbb{Q}$:

$$x^4 - 1 = (x-1)(x+1)(x^2+1)$$

and so the splitting field of $x^4 - 1$ is the splitting field of $x^2 + 1$, which is $\mathbb{Q}(i)$

8. Let $p(x) = x^4 + 2$. This polynomial is irreducible by Eisenstein. Given $\zeta = e^{\frac{2\pi i}{8}}$, then the roots of $p(x)$ are:

$$\sqrt[4]{2}\zeta, \ \sqrt[4]{2}\zeta^3, \ \sqrt[4]{2}\zeta^5, \ \sqrt[4]{2}\zeta^7$$

and so the splitting field is $K = \mathbb{Q}(\sqrt[4]{2}\zeta, \ \sqrt[4]{2}\zeta^3, \ \sqrt[4]{2}\zeta^5, \ \sqrt[4]{2}\zeta^7)$. We can simply the adjoint elements with the following observations:

$$K \subseteq \mathbb{Q}(\zeta, \sqrt[4]{2}) \subseteq \mathbb{Q}(i, \sqrt{2}, \sqrt[4]{2}) = \mathbb{Q}(i, \sqrt[4]{2})$$

so $K \subseteq \mathbb{Q}(i, \sqrt[4]{2})$. For the $\supseteq$ direaction, notice that:

$$\sqrt{2} = \frac{(\sqrt[4]{2}\zeta)^3}{\sqrt[4]{2}\zeta^3} \in K \qquad i = \frac{(\sqrt[4]{2}\zeta)^2}{\sqrt{2}} \in K$$

So $\zeta = \frac{\sqrt{2}}{2}(1+i) \in K$ and $\sqrt[4]{2} = \frac{\sqrt[4]{2}\zeta}{\zeta} \in K$, showing that $K \supseteq \mathbb{Q}(i, \sqrt[4]{2})$. With a little bit of computation, we see that:

$$[\mathbb{Q}(i, \sqrt[4]{2}) : \mathbb{Q}] = 8$$

Overall, small variation in polynomials can create quite large differences in splitting fields (as we have seen with $x^4 + 2$, $x^4 + 1$, $x^4 - 1$). This shows how precise the tools of field theory are in the study of polynomials.

Given a polynomial $f(x)$ over k, we can find a field F in which $f(x)$ factor completely over F. In fact, we have also gained a slightly stronger result for free: if $g(x)$ is irreducible in k has a root in F, then it in fact *also* splits. The idea is that the automorphism group acts transitively on all the roots of an irreducible polynomials (recall the discussion under corollary 19.2.29). If all the roots are inside the field, then the automorphism group will shuffle around all the roots within the field. Hence, if we are given that a single root is inside a field, and that the image of each automorphism in the automorphism group is inside the field, we would have that all the roots are within the field. As the action of the automorphism group $\mathrm{Aut}_k(F) \curvearrowright R_F$ is a evaluation action $\sigma \cdot \alpha = \sigma(\alpha)$ (often called the *galois conjugate* or just *conjugate)*, another way of saying it is that the field is closed under conjugation of the automorphism group. To understand this phenomenon better, we give a name to

fields with this property:

Definition 19.5.12: Normal Extension

Let F/k be an algebraic field extension. Then F/k is called a *normal extension* if for every *irreducible* polynomial $f(x) \in k[x]$, $f(x)$ has a root if and only if $f(x)$ splits as a product of linear factors over F

A normal extension is *not necessarily* the algebraic closure since not necessarily all irreducible polynomials have a root, however the irreducible polynomials over the base field become the polynomials that split completely in the normal extension, constant polynomials, and irreducible polynomials that have no roots in the normal extension (or, they can be thought of as polynomials that have yet to have a root added to the extension).

The name normal is given to this type of extension is due to the fact that most of the nice properties of automorphism groups we have discussed in corollary 19.2.29 apply to normal extensions[17]

Theorem 19.5.13: Finite Normal Extension Equivalences

Let F/k be a field extension. Then the following are equivalent:

1. F is finite and normal
2. F is the splitting field of some polynomial $f(x) \in k[x]$
3. F is finite over k, and every k-homomorphism $\varphi\colon F \to \overline{F}$ satisfies $\varphi(F) = F$

***Proof*:**

For $2 \Rightarrow 3$, let F be the splitting field of some polynomial $f(x) \in k[x]$. It is generated over k by the finitely many roots of $f(x)$, each algebraic, so F is finite over k (theorem 19.3.7). Let $\varphi\colon F \to \overline{F}$ be a k-homomorphism. If α is a root of $f(x)$, then $m_{\alpha,k}(x) \big| f(x)$, and by corollary 19.2.29, $\varphi(\alpha)$ is another root of $m_{\alpha,k}$, and hence of $f(x)$. Since F is a splitting field, it contains all the roots of $f(x)$, and so $\varphi(\alpha) \in F$. Since F is generated by the roots of $f(x)$, $\varphi(F) \subseteq F$. Then φ is an injective k-linear map of the finite-dimensional k-vector space F into itself, hence surjective, so $\varphi(F) = F$.

For $3 \Rightarrow 1$, let $p(x) \in k[x]$ be irreducible with a root $\alpha \in F$, and let $\beta \in \overline{F}$ be a root of $p(x)$. By theorem 19.2.27, there is a k-isomorphism $\sigma\colon k(\alpha) \to k(\beta)$ with $\sigma(\alpha) = \beta$. Regard $\overline{F}$ as an extension of $k(\alpha)$ through σ. As $F/k(\alpha)$ is finite, hence algebraic, lemma 19.5.7 gives a homomorphism $\varphi\colon F \to \overline{F}$ extending σ. It is a k-homomorphism, so $\varphi(F) = F$ and $\beta = \varphi(\alpha) \in F$. Hence $p(x)$ has all its roots in F and splits over F, so F/k is finite and normal.

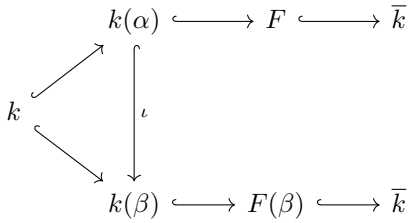
Next, we show $1 \Rightarrow 2$. Assume F is finite and normal. Since it's finite, $F = k(\alpha_1, \alpha_2, ..., \alpha_n)$, where each α_i is algebraic over k. Next, let $m_i(x)$ be the minimal polynomial of α_i. Since F is normal, and each m_i has at least one root (namely α_i), $m_i(\alpha)$ splits. Then $f(x) = m_1(x) \cdots m_n(x)$ is a polynomial that splits over F. But then, F is the smallest field in which $f(x)$ splits (since $F = k(\alpha_1, ..., \alpha_n)$), and so F is the splitting field for $f(x)$.

Finally, we show $2 \Rightarrow 1$: assume F is the splitting field for some polynomial $f(x) \in k[x]$, and

[17] In fact, normal subgroups got their name from normal extensions! The connection will be made in the chapter 20 when discussing Galois Theory in theorem 20.1.23

assume that $p(x) \in k[x]$ is an irreducible polynomial over k such that F contains a root α for $p(x)$. If $p(x)$ has degree less than or equal to 2, we are done, so let's say $\deg p(x) \geq 3$. Since $F \subseteq \overline{k}$, let $\beta \in \overline{k}$ be a different root of $p(x)$. We want to show that $\beta \in F$. This will show that F is normal, and by theorem 19.5.10, F must be finite.

Since $\mathrm{id} : k \to k$ is a field homomorphism, by theorem 19.2.27, there is an isomorphism $i : k(\alpha) \to k(\beta)$ sending $\alpha \mapsto \beta$. Note that $F(\beta) \subseteq \overline{k}$. Visually, we can imagine all of this in the commutative diagram:



Note that we cannot draw the identity map from $\overline{k}$ to $\overline{k}$, since ι maps $\alpha \mapsto \beta$. Now, notice that F is the *splitting field* $f(x)$ over $k(\alpha)$. Indeed, F contains all the roots of $p(x)$ and is generated over k (and hence $k(\alpha)$) by the roots of $p(x)$. Similarly, $F(\beta)$ is the splitting field for $f(x)$ over $k(\beta)$. Then by the uniqueness of the splitting field (theorem 19.5.10), ι extends to an isomorphism $\iota' : F \to F(\beta)$. Now, since ι restricts to the identity on k, ι' is an isomorphism of k-vector spaces, in particular $\dim_k F = \dim_k F(\beta)$ (since splitting fields are always finite extensions). In particular, we get to take advantage of facts about liner algebra, in particular that if the dimensions of two vector-spaces are the same, then any injective linear transformation is in fact an isomrophism.

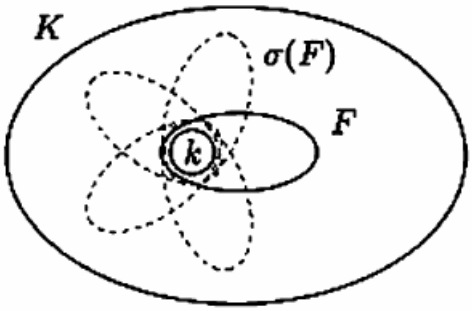
Now, we also have the natural inclusion map $\iota : F \to F(\beta)$ which is both a k-homomorphism and a k-vector spaces. Since F and $F(\beta)$ are k-vector spaces of the same [finite!] dimension, i must *also* be an isomorphism, i.e.

$$[F(\beta) : F] = 1$$

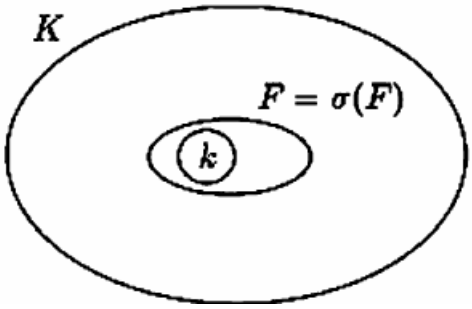
Thus, $\beta \in F$, completing the proof.

An algebraic extension F/k, not necessarily finite, is normal if and only if every k-homomorphism $\varphi\colon F \to \overline{F}$ satisfies $\varphi(F) = F$. The proofs of $3 \Rightarrow 1$ and of $\varphi(F) \subseteq F$ in $2 \Rightarrow 3$ carry over, applied to one irreducible polynomial at a time, and $\varphi(F) \supseteq F$ holds because φ permutes the finitely many roots in F of each such polynomial.

19.5.14 Remark Notice how condition (3) of theorem 19.5.13 allows us to refer to "the" embedding of a Normal extension F/k into $\overline{k}$, since $\varphi(F) = F$ for any F. Recall that $\mathbb{Q}(\sqrt[3]{2})/\mathbb{Q}$ is not normal, and that we have shown that $\mathbb{Q}(\sqrt[3]{2})$ is isomorphic to 3 subfields of $\overline{\mathbb{Q}}$ which do not all overlap:



However, if the extensoin is normal, then there is a notion of "the" embedding since they all overlap:



In this way, having a [algebraic][a] normal extension F/k can be thought of as being more "precise" than a non-normal extension, since any other isomorphic copy of F is just an automorphism of F.

[a] This is to ensure that $F \subseteq \overline{k}$. The algebraic elements of F is also what we care about when we look at k-homomorphisms $F \to \overline{k}$

Example 19.5.15: Consequences
Let's say $p(x)$ is an irreducible polynomial over $\mathbb{Q}$ with a complex root that may be expressed as a polynomial in i and $\sqrt[4]{2}$ over $\mathbb{Q}$. Then *all* roots of $p(x)$ may be expressed as a polynomial in i and $\sqrt[4]{2}$. As we have shown in example 19.5.11, $\mathbb{Q}(\sqrt[4]{2}, i)$ is a splitting field over $\mathbb{Q}$, and hence a normal extension.

Unlike algebraic and finite extensions, normal extensions are more finicky when it comes to composing normal extensions:

Proposition 19.5.16: Tower Law On Normal Extensions

If $L/K/k$ and L/k is normal, then L/K is normal

***Proof*:**
Let $f(x) \in K[x]$ be irreducible with a root $\alpha \in L$. Then $f(x) = m_{\alpha,K}(x)$ is the minimal polynomial over K. Note that

$$m_{\alpha,K}(x) | m_{\alpha,k}(x)$$

And since $m_{\alpha,k}(x)$ splits over L (since L/k is normal), $m_{\alpha,K}$ splits over L, as we sought to show.

Note that this result has little wiggle room:

Example 19.5.17: Low Tolerance for Normal Extension
Consider $L/K/k$ where:

$$\begin{array}{c} \mathbb{Q}(\sqrt[3]{2}, \zeta_3) \\ | \\ \mathbb{Q}(\sqrt[3]{2}) \\ | \\ \mathbb{Q} \end{array}$$

where σ_3 is one of the two other roots of $x^3 - 2$. Then while L/k is normal, K/k is *not* normal here (we have yet to add enough roots). Furthermore, if L/K and K/k is normal, this does not imply that L/k is normal. For example:

$$\begin{array}{c} \mathbb{Q}(\sqrt[4]{2}) \\ | \\ \mathbb{Q}(\sqrt{2}) \\ | \\ \mathbb{Q} \end{array}$$

To see this, notice that each extension is of degree 2, and hence is normal (exercise). However, $\mathbb{Q}(\sqrt[4]{2})/\mathbb{Q}$ is not normal, since 2 of the roots of $x^4 - 2$ are complex, while $\mathbb{Q}(\sqrt[4]{2})$ is contained in the reals, and hence cannot have all the roots.

Notice how this behavior is mimicked by normal subgroups.

Definition 19.5.18: Normal Closure

Let K/k be an algebraic extenion. Then the *normal closure* of K given k is an algebraic extension N/K such that

1. N/k is normal
2. If $N \supseteq N' \supseteq k$, and N'/k is normal, then $N = N'$

Proposition 19.5.19: Existence Of Normal Closure

Let K/k be a finite extension. Then there exists a unique normal closure N up to isomorphism fixing k.

Note the finiteness condition given. It is true for algebraic ones too, but we did not yet prove that

***Proof*:**
Since K is finite, let $K = k(\alpha_1, \alpha_2, ..., \alpha_n)$ where each $\alpha_i \in K$ is algebraic over k. Let

$$f(x) = \prod m_{\alpha_i,k}(x) \in k[x]$$

and let N be the splitting field of $f(x)$ over K. Then $[N : K] < \infty$. But then, N is the splitting field of $f(x)$ over k as $K = k(\alpha_1, \alpha_2, ..., \alpha_n)$. But this implies N/k is normal.

Finally, let's say $k \subseteq N' \subseteq N$ and N'/k is normal. Then by normality, N' has to split each $m_{\alpha_i,k}(x)$, and so splits $f(x)$, and so N/k *is* a normal closure.

Exercise 19.5.1

1. A splitting field of $f \in k[x]$ of degree n has degree at most $n!$ over k (theorem 19.5.10). Show that the splitting field of $x^3 - 2$ over $\mathbb{Q}$ has degree $6 = 3!$.
2. Let E/k and F/k be algebraic extensions inside a common field. Show that if E/k is normal, then so is EF/F. Show that if E/k and F/k are normal, so is EF/k.
3. Let L/K be an algebraic field extension, and $\varphi_K : K \to \overline{F}$ be a homomorphism to an algebraically closed field. Then φ_K extends to a homomorphism $\varphi_L : L \to \overline{F}$ (use Zorn's lemma and a proof similar to lemma 19.5.7 with Zorn being applied to (E, φ_E) where E/K is a subextension of L/K and $\varphi_E : E \to \overline{F}$ extends φ_K)

19.6 Separable Extensions

We have now shown that there exists a field K over F such that a polynomial $f(x)$ splits (or a field $\overline{F}$ such that all polynomials in $F[x]$ split). A pattern has been occuring which you might have not noticed: every time we split an irreducible polynomial completely into linear factors, each factor was *unique*, that is, we had that the multiplicity of every linear term was 1. Does that need to be the case for any polynomial over any field? The answer is no:

Example 19.6.1: Higher Multiplicity

Let p be a prime number, and take the field $\mathbb{F}_p(t)$. Consider the polynomial:

$$p(x) = x^p - t \in \mathbb{F}_p(t)[x]$$

Then $p(x)$ is irreducible: by Eisenstein's criterion, it is irreducible in $\mathbb{F}_p[t][x]$ (since (t) is prime in $\mathbb{F}_p[t]$), and hence it is prime in $\mathbb{F}_p(t)[x]$ by corollary 11.2.2. Let u be a root of $p(x)$ in the extension $L/\mathbb{F}_p(t)$ (L can be the algebraic closure, or the splitting field). Then notice that:

$$x^p - t = (x - u)^p$$

Simply expand the right hand side of the equation. Thus, u has multiplicity p, i.e., the minimal polynomial over $\mathbb{F}_p(t)$ for $u \in L$ vanishes p times at u[a].

[a] $p(x)$ must be the irreducible polynomial, since if the degree was smaller then $p(x)$ would be reducible

Thus, in general, a polynomial $f(x) \in k[x]$ can factorize over a splitting field as:

$$f(x) = (x - \alpha_1)^{n_1}(x - \alpha_2)^{n_2} \cdots (x - \alpha_k)^{n_k}$$

where $\alpha_1, \alpha_2, ..., \alpha_k$ are distinct elements in the splitting fields, and $n_1, n_2, ..., n_k$ is the multiplicity of every root. However, this seems to be a rare occurence given the fields we have worked with. In fact, most field we will work with do not have this property. It will also become important in Galois theory that a polynomial would have to to linear polynomials with multiplicity 1[18]. For those who are curious, see what hapens to the automorphism group $\mathrm{Aut}_k(F)$ of the splitting field F if $(x - \alpha)^n$ is the linear factorization of $f(x)$ over k. Hence, we give a name to polynomials that factor with multiplicity 1:

Definition 19.6.2: Separable Polynomial

A polynomial over k is called *separable* if it factors into linear terms with multiplicity 1 in it's splitting field. Equivalently, it has no multiple roots (i.e., all its' roots are distinct). A polynomial which is not separable is called *inseparable*.

My intuition behind the name is that the roots cannot be completely "separated" from each other if the polynomial is *inseparable*. In Algebraic Geometry, polynomials that have roots with multiplicity higher than 1 (say multiplicity m) shall correspond to a collection of points that shall all be arbitrarily close to each other while not being equal, ostensibly being "inseparable"[19]. Naturally, the choice of splitting field does not affect whether a polynomial is separable since all splitting fields for a polynomial are isomorphic, and isomorphisms preserves roots (theorem 19.2.27). Even better, we can use *any* field in which the chosen polynomial $f(x)$ splits into linear factors (since there will be a homomorphism from the splitting field F to the field K in which $f(x)$ splits). For our purposes, that means that it is just as good to work over the algebraic closure $\overline{k}$.

[18] (the correspondence Galois group is trivial otherwise)

[19] See the plane with two origins for an example of such a geometric object

Definition 19.6.3: Separable Extension

Let F/k be a field extension and $\alpha \in F$. Then:

1. The element α is called *seprable* if α is algebraic and the minimal polynomial corresponding to α is seperable. Otherwise, the element α is called *inseparable*
2. F/K is called a *separable extension* if every element $\alpha \in F$ is separable, and an *inseparable extension* otherwise.

19.6.4 Remark Note that it is not immediately obvious that if α is separable over k that $k(\alpha)/k$ is a separable extension (what if $1+\alpha$ is inseparable?). We will answer this question after we develop more tools to work with separable elements.

Using the fact that separability is independent of the field in which it splits, we can use a clever trick in finding whether a polynomial is separable using a tool usually defined in calculus:

Definition 19.6.5: Derivative

Given a polynomial $f(x) \in k[x]$ where

$$f(x) = a_n x^n + a_{n-1}x^{n-1} + \cdots + a_1 x + a_0$$

then its *derivative* or *formal derivative* is the image of the function $D_x : k[x] \to k[x]$ where:

$$D_x f(x) = n a_n x^{n-1} + (n-1)a_{n-1}x^{n-2} + \cdots + 2a_2 x + a_1$$

or more succinctly

$$D_x\left(\sum_i^n a_i x^i\right) = \sum_i^n i a_i x^{i-1} \tag{19.2}$$

Notice that this derivative acts exactly like the derivative seen in a regular calculus course [20]. In a calculus course, we derive this formula using a limit construction. However, if we just think about the derivative (restricted to polynomials) as a function, then we get a purely algebraic definition of the derivative. In fact, the derivative (and even its generalization with he external operator on differential forms) can be studied purely algebraically; see [9] a deeper exploration of these ideas[21].

Note that the i in equation (19.2) is well-defined since we always have the unique homomorphism $\iota : \mathbb{Z} \to k[x]$ since $\mathbb{Z}$ is initial in **Ring**, and so i is the element associated with $\iota(i)$. The *linearity* and *Leibniz property* of the derivative can both be proven with just this definition:

$$D_x(f(x)+g(x)) = D_x f(x) = D_x g(x)$$
$$D_x(f(x)g(x)) = (D_x f(x))g(x) + f(x)(D_x g(x))$$

We will often abbreviate notation and write $f'(x) := D_x f(x)$ when it is not ambiguous.

[20] A good resources to learn this would be Spivak's Analysis

[21] For example, algebraic differential forms shall be useful to determine the separability of algebraic maps, which is the generalization of the idea that an extension is separable if each element has a nonzero derivative

Proposition 19.6.6: Separable Polynomial Test

A polynomial $f(x)$ has multiple roots α if and only if α is also a root of $D_x f(x)$, i.e., $f(x)$ and $D_x f(x)$ are both divisible by the minimal polynomial for α. In particular, $f(x)$ is separable if and only if it is relatively prime to its derivative

$$(f(x), f'(x)) = 1$$

Noet how $f(x)$ is *not necessarily irreducible*. This means that the derivative test is incredibly powerful, telling us that if $(f(x), f'(x)) = 1$, then any irreducible factor of $f(x)$ but all be distinct and have distinct roots.

***Proof*:**
First, let's say $f(x)$ is inseparable so that it has multiple roots in a splitting field F:

$$f(x) = (x - \alpha)^n g(x)$$

where $n \geq 2$ and $g(x) \in F[x]$. Then:

$$f'(x) = n(x - \alpha)^{n-1} g(x) + (x - \alpha)^n g'(x)$$

showing that $\gcd(f(x), f'(x)) \neq 1$ in $F[x]$. By corollary 11.2.2 to Gauss's Lemma, $\gcd(f(x), f'(x)) \neq 1$ in $k[x]$

Conversely, Assume that $f(x)$ and $f'(x)$ share a root (say $\alpha \in \overline{k}$), so that $\gcd(f(x), f'(x)) \neq 1$ in $k[x]$. Since $f(x)$ has a root, we can write it as:

$$f(x) = (x - \alpha)h(x)$$

for $h(x) \in k[x]$ (or $\overline{k}[x]$?). Then the derivative is:

$$f'(x) = h(x) + (x - \alpha)h'(x)$$

Since $f(x)$ and $f'(x)$ share a root, it must be that $(x - \alpha) \big| h(x)$. But then:

$$(x - \alpha)^2 \big| f(x)$$

showing that $f(x)$ has multiple roots, hence inseparable, as we sought to show.

Using this theorem, we can see that $p(x) = x^p - t$ is inseparable since $p'(x) = px^{p-1} = 0$ (since the characteristic of $\mathbb{F}_p(t)$ is p). Thus, $\gcd(x^p - t, 0) = x^p - t \neq 1$. In fact, this captures a bigger result:

Proposition 19.6.7: Separability and irreducible polynomial

Let k be a field and $f(x) \in k[x]$ be an *inseparable and irreducible* polynomial. Then

$$f'(x) = 0$$

***Proof*:**
Since $f(x)$ is inseparable, by proposition 19.6.6, $f(x)$ and $f'(x)$ share a common irreducible factor $q(x)$, that is $f(x) = q(x)g(x)$ for non-constant $g(x) \in k[x]$. However, f is irreducible, and so $q(x)$ must be an associate of $f(x)$ (and hence be of equal degree). However, that implies that it has degree higher than $f'(x)$ if $f'(x) \neq 0$. Since $q(x) \big| f'(x)$, the only possibility left is that $f'(x) = 0$, as we sought to show.

Corollary 19.6.8: Separability Over Characteristic 0

let k be a field of characteristic 0 and $f(x) \in k[x]$ an irreducible polynomial. Then $f(x)$ is seperable.

***Proof*:**
Since f is irreducible, it must have degree at least 2, and so $f'(x) \neq 0$, showing it is separable by proposition 19.6.7

Thus, every algebraic extension of a characteristic 0 field is a separable extension, and fields in which an irreducible polynomial is inseperable must have characteristic p. Even better, given an inseperable polynomial:

$$f(x) = \sum_{i}^{n} a_i x^i$$

since $f'(x) = 0$:

$$f'(x) = \sum_{i}^{n} i a_i x^{i-1} = 0$$

thus, $ia_i = 0$ for all i, so ia_i is a multiple of p. If i itself is a multiple of p, then certainly $ia_i = 0$, and so it must be that $a_i = 0$ for all indicies that are *not* multiples of p. Therefore, we have reduced inseparable polynomials to polynomials of the form:

$$f(x) = a_0 + a_p x^p + a_{2p} + x^{2p} + \cdots + a_{np} x^{np} = \sum_{k=1}^{n} a_k (x^p)^k$$

Thus, an inseparable polynomial $f(x)$ can be thought of as a polynomial in x^p, $f(x) = g(x^p)$. Using this, we can define the following homomorphism which will be useful to characterize fields which have inseparable polynomials:

Definition 19.6.9: Frobenius Homomorphism

Let k be a field such that $p = \mathrm{ch}(k) > 0$. Then the *Frobenius homomorphism* is a map $k \to k$ defined by

$$x \mapsto x^p$$

It should be verified that this map is indeed a field homomorphism. the Frobenius must be injective, being a homomorphism between fields, however it need not be surjective. Using this homomorphism, we will find all fields in which being separable is identical to being algebraic:

Definition 19.6.10: Perfect Field

Let k be a field. Then k is called a *perfect field* if all irreducible polynomials in $k[x]$ are seperable, that is, every algebraic extension is a separable extension.

Proposition 19.6.11: Frobenius Homomorphism And Separability

Let k be a field. Then k is perfect if and only if either:

1. if $\text{ch}(k) = 0$
2. if $\text{ch}(k) > 0$ and the Frobenius homomorphism is surjective.

***Proof*:**
If $\text{ch}(k) = 0$, then by corollary 19.6.8, any irreducible polynomial is already seperable, so let's say $\text{ch}(k) = p$ for some prime p.

($\Leftarrow$) By our discussion from earlier, we know an inseparable irreducible polynomial must be of the form:

$$f(x) = \sum_{i=1}^{n} a_i (x^p)^i$$

Now, since the Frobenius homomorphism is surjective, for everry a_i, there exists a b_i such that $b_i \mapsto a_i$, that is $(b_i)^p = a_i$. Thus, using the fact that this map is indeed a ring homomorphism, we get:

$$f(x) = \sum_{i=1}^{n} b_i^p (x^p)^i = \sum_{i=1}^{n} (b_i x^i)^p = \left(\sum_{i=1}^{n} b_i x^i \right)^p = g(x)^p$$

where $g(x) = \sum_{i=1}^{n} b_i x^i$. But then $f(x)$ has become reducible! This contradicts our assumption of the irreducibility of $f(x)$, showing us that no such polynomial can exist.

($\Rightarrow$) Take the contra-positive, that $\text{ch}(k) \neq 0$ (say $\text{ch}(k) = p$) and that the Frobenius homomorphism is not surjective. Let's say that φ is the Frobenius homomorphism and that $t \in k \setminus \varphi(k)$. I claim that $x^p - t$ does not split into linear factors. Indeed, emulating what we have done in example 19.6.1, we get $x^p - t$ is irreducible, and that $(x - u)^p = x^p - t$

Here is another one: Suppose that T has characteristic $p > 0$, and that $x \mapsto x^p$ is not an automorphism. Since a ring homomorphism from a field into itself is injective or zero, then $x \mapsto x^p$ is not surjective.

Let $a \in T$ be such that $a \neq b^p$ for every $b \in T$. Let $f(x) = x^p - a$. We prove that f is irreducible and inseparable. If α is a root of $x^p - a$, then $x^p - a = (x - a)^p$, so α is a multiple root of f (with multiplicity p), hence f is inseparable. Now, let $g(x)$ be an irreducible factor of $f(x)$. Note that $\alpha \notin T$, otherwise $a = \alpha^p$, contrary to the assumption. Then $g(x) = (x - \alpha)^k$ for some $1 < k \leq p$. Using the binomial theorem, we have

$$g(x) = (x - \alpha)^k = x^k - k\alpha x^{-1} + \cdots + (-\alpha)^k$$

Therefore, $k\alpha \in T$. Since $\alpha \notin T$, then $k = p$, so $g = f$. Hence, f is irreducible.

Corollary 19.6.12: Finite Fields Are Perfect

let k be a finite field. Then k is perfect

***Proof*:**
Since a finite field must have a $\text{ch}(k) = p > 0$, the Frobenius homomorphism is well-defined. Furthermore, Since k is a field, the homomorphism is injective. But an injective map between two finite sets is surjective, hence the Frobenius homomorphism is surjective, and so by proposition 19.6.11 k is perfect.

Thus, we have shown that over most fields we have worked with so far, they are separable. Inseparable extensions do not behave so well (as we will see, their galois group is [almost] trivial, and hence we get little information about them through galois theory), and so more focus is given in understanding the more common subclass of seperable extensions of a field.

19.6.1 Separability and Embeddings

It should be noted that there *are* separable extensions even if a field is not perfect. For example $x^6 - tx^3 + t \in \mathbb{F}_3(t)[x]$ is irreducible over $\mathbb{F}_3(t)$ and factors into unique linear terms in the splitting field, however the field is not perfect since $x^6 - x^3 + 1$ is irreducible and inseparable. Therefore, we need a better way of finding separable polynomial without relying on nice properties of the ground field. This will be part of the goal of this section.

To characterize separable extensions without relying on k being perfect, we will take advantage that a k-homomorphism is determined by where it maps the roots of minimal polynomials of elements of F. In this way, we get an idea of how many distinct roots a minmial polynomial contains. Recall that we have been able to characterize normal fields through k-homomorphisms: if $\sigma : F \to \overline{k}$ is a k-homomorphism and F/k is normal, then $\sigma(F) = F$, which, as noted in box 19.5.14, shows that for normal extensions we can characterize all k-homomorphisms as in fact being k-automorphisms (in the sense that all isomorphic copies of F are just symmetries of F). In general, separable extensions need not be normal, and hence they might have multiple embeddings into $\overline{k}$. However, understanding all possible k-homomorphisms still gives us a lot of information about our field as was just pointed out. Studying these embeddings will give us some more information about separable extensions, and will allow us to answer the question of whether an extension $k(\alpha)/k$ is separable if α is separable over k, and to see that any algebraic extension can further be broken down into a separable extension followed by a "purely inseparable" extension: F_{sep}/k and F/F_{sep}.

Definition 19.6.13: Separable Degree

Let F/k be a field extension. Then the number of k-homomorphism $F \to \overline{k}$ called the *separable degree* and is denoted

$$[F : k]_s = \text{Hom}_k(F, \overline{k}) = \text{Emb}_k(F)$$

where $\text{Emb}_k(F)$ is shortand for all possible k-embeddings of F into the algebraic closure of F [a]

[a] Note that for any two algebraic closures $\overline{k}_1, \overline{k}_2$, $\text{Hom}(F, \overline{k}_1) \cong \text{Hom}(F, \overline{k}_2)$, making the definition independent of particular algebraic closure, and hence the notation $\text{Emb}_k(F)$ unambiguous

For the curious reader who would want to see the connection between separability and the separable degree and wants a hint before looking at the solution, they may want to think about what happens to the root, since the following theorem is a generalization of theorem 19.2.27.

19.6.14 moment to ponder

Proposition 19.6.15: Separable Degree of Simple Extensions

Let $k \subseteq k(\alpha)$ be a simple algebraic extension. Then $[k(\alpha) : k]_s$ is equal to the number of distincts roots in $\overline{k}$ of the minimal polynomial of α. In particular,

$$[k(\alpha) : k]_s \leq [k(\alpha) : k]$$

with equality if and only if α is separable over k

Notice how this proposition gives us a way to check the separability of the element α without any appeal to k (i.e., without knowing whether k is perfect or not). In fact more is true: $[k(\alpha) : k]_s \mid [k(\alpha) : k]$. The proof of this fact is a bit more laborious than just showing the inequality since it requires the notion of *separable closure*, which is simply the intersection of all fields that are separable, and so we will leave it as an exercise.

***Proof*:**
This proof will be very similar to that of corollary 19.2.29. Consider the collection of k-homomorpihism $\{\iota_r : k(\alpha) \to \overline{k}\}$. By corollary 19.2.29, we can associate each of these homomorphisms with where a root r of the minimal polynomial $m_{\alpha,k}(x)$ maps to, i.e. $\iota_r(\alpha) = r$. We will start by showing that this correspondence is bijective:

$$\{\iota_r : k(\alpha) \to \overline{k}\} \leftrightarrow \{r\}$$

Injectivity comes easily since ι_r fixes k, and so it is completely determined by where α maps to, which necessarily is a root of the minimal polynomial for α. To show surjectivity, consider $\beta \in \overline{k}$ which is some root of the minimal polynomial $m_{\alpha,k}(x)$. Then by lemma 19.5.7, $k(\beta) \subseteq \overline{k}$ exists, and by theorem 19.2.27, there exists a unique isomorphism $k(\alpha) \xrightarrow{\sim} k(\beta)$. Hence, combining with $k(\beta) \subseteq \overline{k}$, we get an extension $k(\alpha) \subseteq \overline{k}$ that maps α to the root β, showing surjectivity.

Thus, we have establisehd a bijection between the roots of the minimal polynomial and the number of embeddings of a simple algebraic extension. Since the degree of α is the degree of the minimal polynomial, which is less than or equal to the number of roots the minimal polynomial contains, we have that:

$$[k(\alpha) : k]_s \leq [k(\alpha) : k]$$

If furthermore $k(\alpha)$ is separable, then the number of roots is the degree of the polynomial, and hence:

$$[k(\alpha) : k]_s = [k(\alpha) : k]$$

completing the proof.

Corollary 19.6.16: Tower Law For Finite Separable Extensions

Furthermore, if $F/E/k$ is a sequence of algebraic extensions, $[F:k]_s$ is finite if and only if $[F:E]_s$ and $[E:k]_s$ is finite, and furthermore:

$$[F:k]_s = [F:E]_s[E:k]_s$$

***Proof*:**
We will show that if $[E:k]_s$ or $[F:E]_s$ are infinite, so must $[F:k]_s$, and then show that if they are both finite, so must $[F:k]_s$ be. Let $\varphi : E \to \overline{k}$ be a k-homomorphism. Then since F/E exists and is algebraic over E (and hence k), by lemma 19.5.7, there exists an exension of φ to $\Phi : F \to \overline{k}$ fixing E (i.e. $\Phi|_E = \varphi$). Furthermore, an extension F into $\overline{E} = \overline{k}$ fixing E extends *a fortiori* the identity on k. Thus, if either $[E:k]_s$ or $[F:E]_s$ is infinite, so must $[F:k]_s$ be.

Conversely if $[F:E]_s$ and $[E:k]_s$ is finite, we will show $[F:k]_s$ is finite and the multiplicity formula holds. This is almost immediate: for any embedding $F \to \overline{k}$ that extends k, we can can first find all extensions from k to E fixing k, of which there are $[E:k]_s$ possible ways of doing so, and then and then extend any chosen embedding from F to $\overline{E} = \overline{k}$ ($F \to \overline{E}$) fixing E in $[F:E]_s$ different ways, giving us a total of $[F:E]_s[E:k]_s$ total possible extensions, completing the proof.

Proposition 19.6.17: Finite Separable Extension Equivalence

Let F/k be a finite extension. Then $[F:k]_s \leq [F:k]$, and the following are equivalent:

1. $F = k(\alpha_1, ..., \alpha_n)$ where each α_i, is separable over k
2. F/k is a separable extension
3. $[F:k]_s = [F:k]$

***Proof*:**
We will first generlize proposition 19.6.15 for finite extensions to show that $[F:k]_s \leq [F:k]$. Since F/k is finite, F is finitely generated over k, so let $F = k(\alpha_1, ..., \alpha_n)$. Using proposition 19.6.15, corollary 19.6.16, and theorem 19.3.7:

$$\begin{aligned}[F:k]_s &= [k(\alpha_1, ..., \alpha_{n-1})(\alpha_n) : k(\alpha_1, ..., \alpha_{n-1})]_s \cdots [k(\alpha_1) : k]_s \\ &\leq [k(\alpha_1, ..., \alpha_{n-1})(\alpha_n) : k(\alpha_1, ..., \alpha_{n-1})] \cdots [k(\alpha_1) : k] \\ &= [F:k]\end{aligned}$$

showing the inequality

($1 \Rightarrow 3$) Since each α_i is separable over k, it is separable over $k(\alpha_1, ..., \alpha_{i-1})$ (the minimal polynomial of α_i over the intermediate field divides the minimal polynomial over the ground field), and so the inequalites in the previous argument are in fact all equalities by corollary 19.6.16

($3 \Rightarrow 2$) Let $[F:k]_s = [F:k]$, take $\alpha \in F$, and consider $k \subseteq k(\alpha) \subseteq F$. Since F is a finite extension, by corollary 19.6.16,

$$[F:k(\alpha)]_s[k(\alpha):k]_s = [F:k]_s = [F:k] = [F:k(\alpha)][k(\alpha):k]$$

Thus, we have two numbers, say a, b such that $a \leq A$, $b \leq B$ and $ab = AB$. This forces $a = A$ and $b = B$, and so we can cancel $[F:k(\alpha)]_s$ and $[F:k(\alpha)]$ on both sides to get:

$$[k(\alpha):k]_s = [k(\alpha):k]$$

which, by proposition 19.6.15 shows that α is separable. Since $\alpha \in F$ was arbitrary, this shows that all elements of F are separable, and hence F/k is a separable extension.

($2 \Rightarrow 1$) Since F/k is separable and F is finite, $F = k(\alpha_1, ..., \alpha_n)$, and each α_i must be separable by definition.

We can finally answer the question posed in remark ref:RemarkSeparableEx: if α is separable over k, then so is $1 + \alpha$, α^2, and any other combination of α, since $k(\alpha)/k$ is separable.

Proposition 19.6.18: Tower Law For Separable Extensions

Let $F/E/k$ be a sequence of algebraic extensions. Then F/k is separable if and only if F/E and E/k is separable.

***Proof*:**
Suppose first that F/k is separable. Then E/k is separable, since every element of E lies in F. And for $\alpha \in F$ the minimal polynomial $m_{\alpha,E}$ divides $m_{\alpha,k}$ in $E[x]$, because $m_{\alpha,k}$ lies in $E[x]$ and has α as a root. A divisor of a polynomial with no repeated roots has none either, so $m_{\alpha,E}$ is separable and F/E is a separable extension.

Conversely, let F/E and E/k both be separable and take $\alpha \in F$. The point is to replace E, which may be enormous, by the finite piece of it that the minimal polynomial of α sees. Write

$$m_{\alpha,E}(x) = x^n + c_{n-1}x^{n-1} + \cdots + c_0 \qquad c_j \in E$$

and put $K = k(c_0, ..., c_{n-1})$. Each c_j lies in E and so is separable over k, whence K/k is a finite separable extension by proposition 19.6.17.

Now $m_{\alpha,E}$ has all its coefficients in K, so α is algebraic over K and $m_{\alpha,K}$ divides $m_{\alpha,E}$ in $K[x]$. The latter is separable because F/E is, so $m_{\alpha,K}$ is separable and $K(\alpha)/K$ is a finite separable extension. Multiplying separable degrees along $K(\alpha)/K/k$ with corollary 19.6.16, and ordinary degrees with theorem 19.3.7,

$$[K(\alpha):k]_s = [K(\alpha):K]_s\,[K:k]_s = [K(\alpha):K]\,[K:k] = [K(\alpha):k]$$

using proposition 19.6.17 on each of the two factors. That same proposition, read backwards, makes $K(\alpha)/k$ separable, so α is separable over k. As $\alpha \in F$ was arbitrary, F/k is separable, as we sought to show.

Note that along with the fact that an extension is normal (implying the minimal polynomial splits), we get a bijection between the roots of the minimal polynomial and the automorphisms fixing the base field. For this reason, this will be one of the definitions (out of many equivalent definitions) of a galois extension.

Exercise 19.6.1

1. Let $K/\mathbb{Q}$ be a finite extension. Show that there exists an injective ring homomorphism $K \hookrightarrow \mathbb{R}^{r_1} \oplus \mathbb{C}^{r_2}$ for appropriate r_1, r_2. This is called the *Archimedean embedding* and the codomain is called the *Minkowski space*. Both are taken up in [11, Archimedean Embedding].

19.6.2 Separability and Simple Extensions

Finally, we show that all finite-degree separable fields (of infinite cardinality) are in fact simple. This means that when we were working with the finite extension F/k, we can always find an α where $F = k(\alpha)$. We will show the case for fields of finite cardinality in the next section.

Proposition 19.6.19: Algebraic And Simple Extensions

Let F/k be an algebraic extension. Then F is simple if and only if the number of distinct intermediate fields $k \subseteq E \subseteq F$ is finite

***Proof*:**

Let F/k be simple, so that $F = k(\alpha)$. Since F is algebraic, let $m_{\alpha,k}(x)$ be the minimal polynomial over k. Embed $k(\alpha)$ into an algebraic closure $\overline{k}$. If E is an intermediary field $k \subseteq E \subseteq F$, then $k(\alpha) = F = E(\alpha)$ is also a simple algebraic extension. Denote $m_{E,\alpha}(x)$ to be the minimal polynomial for $E(\alpha)$ over E. Since $m_{k,\alpha}(x) \in E[x]$ for all E and $m_{k,\alpha}(\alpha) = 0$, we know that each $m_{E,\alpha}(x)$ is a factor of $m_{k,\alpha}(x)$.

I claim that E is determined by $m_{E,\alpha}(x)$. Since $m_{k,\alpha}(x)$ has finitely many factors in $\overline{k}$, this will prove there are only finitely many intermediate fields, completing one direction of the proof.

Proceeding with proving the claim, I will show that E is generated (over k) by the coefficients of $m_{E,\alpha}(x)$. To see this, let E' be the set generated by k and the coefficients of $m_{E,\alpha}(x)$. Then $m_{E,\alpha}(x) \in E'[x]$, and since $m_{E,\alpha}(x)$ is irreducible in E, it is irreducible in E', and in fact minimal. Note too that $E'(\alpha) = k(\alpha) = E(\alpha)$, so $[k(\alpha) : E'] = \deg(m_{E,\alpha}(x)) = [k(\alpha) : E]$

Now, we have $E' \subseteq E \subseteq k(\alpha)$, therefore by the tower law:

$$\deg(m_{E,\alpha}(x)) = [k(\alpha) : E'] = [k(\alpha) : E][E : E'] = \deg(m_{E,\alpha}(x))[E : E']$$

Thus, it must be that $[E : E'] = 1$, hence $E = E'$, showing that E is indeed completely determined by $m_{E,\alpha}(x)$, hence there can only be finiely many intermediate fields.

Conversely, let's assume there are finitely many intermediate field $k \subseteq E \subseteq F$. Then it must be that the extension F/k is finite, or else we could construct an infinite sequene of sub-extensions $k \subseteq k(\alpha_1) \subseteq k(\alpha_1, \alpha_2) \subseteq \cdots F$. Thus, F/k is algebraic. We will deal with the case of F being a finite field in corollary 19.7.15. Thus, for now, assume k is a field of infinite cardinality. We have to show that every $F = k(\alpha_1, ..., \alpha_n)$ is in fact simple.

Arguing induictively, it suffices to show that $k(\alpha, \beta)$ is simple. For every $c \in k$, we have:

$$k \subseteq k(c\alpha + \beta) \subseteq k(\alpha, \beta)$$

Since there are only finitely many intermediate fields, it must be that for some two $c, c' \in k$, $c \neq c$:

$$k(c\alpha + \beta) = k(c'\alpha + \beta)$$

But then:

$$\alpha = \frac{(c'\alpha + \beta) - (c\alpha + \beta)}{c' - c} \in k(c\alpha + \beta)$$
$$\beta = (c\alpha + \beta) - c\alpha \in k(c\alpha + \beta)$$

Thus, $k(\alpha, \beta) \subseteq k(c\alpha + \beta)$. Since $k(c\alpha + \beta) \subseteq k(\alpha, \beta)$, we have an equality, showing that it is indeed simple, completing the proof.

Theorem 19.6.20: Primitive Element Theorem

Let F/k be a finite separable extension. Then F is also a *simple* extension.

***Proof*:**

Arguing inductively, we may take $F = k(\alpha, \beta)$ with α and β separable over k (and hence algebraic). As before, the finite case will be dealt with in corollary 19.7.15.

Let I be the number of k-homomorphisms $\iota : F \to \overline{k}$ (so $|I| = [F : k]_s$). Choose two $\iota, \iota' \in I$ where $\iota \neq \iota$. If x is an indeterminate, then the polynomials

$$\iota(\alpha)x + \iota(\beta) \quad \iota'(\alpha)x + \iota'(\beta)$$

are different (or else $\iota'(\alpha) = \iota(\alpha)$ and $\iota'(\beta) = \iota(\beta)$ so ι, ι' would act in the same way on the whole of $k(\alpha, \beta)$ forcing $\iota = \iota'$)

Therefore, define:

$$f(x) = \prod_{\iota \neq \iota'} ((\iota(\alpha)x + \iota(\beta)) - (\iota'(\alpha)x + \iota'(\beta))) \in \overline{k}[x]$$

The polynomial $f(x)$ cannot be identically 0 since k is infinite. Thus, take $c \in k$ such that $f(c) \neq 0$. So, a distinct map $\iota \in I$ will map $\gamma = c\alpha + \beta$ to distinct elements $\iota(\alpha)c + \iota(\beta)$ (recall $\iota(c) = c$). Since $|I| = [F : k]_s$, and each $\iota(\gamma)$ is a root of the minimal polynomial of γ over k, we have:

$$[F : k]_s \leq [k(\gamma) : k] \leq [F : k]$$

By assumption, the extension is separable, so we have equalities above:

$$[F : k]_s = [k(\gamma) : k] = [F : k]$$

Thus:

$$F = k(\gamma)$$

as we sought to show.

Using this, we can extend corollary 19.2.29 from simple to finite extensions:

Corollary 19.6.21: Order of $\mathrm{Aut}_k(F)$ Of Finite Extension

Let F/k be a finite seprable extension. Then:

$$|\mathrm{Aut}_k(F)| \leq [F : k]$$

with equlaity if and only if F/k is a finite separable normal extension.

***Proof*:**
By theorem 19.6.20, if K is finite and separable, it is simple, and so the inequality immediately holds by corollary 19.2.29. Equality holds if the minimal polynomial splits, which it does over it's splitting field, which is a normal extension by theorem 19.5.13. Conversely, if K/F is normal, then $f(x)$ splits completly in F and has distinct roots since α is separable over k, establishing equalitly once again, completing the proof.

Note that the separability condition is necessary:

Example 19.6.22: Nonseparable Nonsimple Extension
Let $F = \mathbb{F}_p(u, v)$ be the field with two indeterminate variables over $\mathbb{F}_p$, and subfield $k = \mathbb{F}_p(s, t)$ such that $s = u^p$ and $t = v^p$. Then F/k is an algebraic extension. The minimal polynomial of u over k is $x^p - s$, and the minimal polynomial of v over $k(u)$ is $y^p - t$. Thus:

$$[F : k] = p^2$$

Arguing like before, As c ranges in k, we get the intermediate fields:

$$k \subseteq k(cu + v) \subseteq F$$

If any two chocies c, c' get give the same intermediate field, then we can deduce $k(u, v) = k(cu + v)$ as we have argued earlier. This would then give us:

$$[k(cu + v) : k] = p^2$$

However, the minimal polynomial of $cu + v$ has degree p:

$$(cu + v)^p = c^p + st \in k(s, t) = k$$

Thus, since $k = \mathbb{F}_p(s, t)$ is infinite, there are infinitely many intermediate fields, and it follows that F is *not* a simple extension of k by proposition 19.6.19

As a final result, we state the following that shall be handy for [23, Rational Maps and Birational Classification]:

Theorem 19.6.23: Lüroth's Theorem

Let k be a field, let t be transcendental over k, and let E be a field with $k \subsetneq E \subseteq k(t)$. Then there exists $u \in E$ such that

$$E = k(u)$$

***Proof*:**
Since t is transcendental, $k[t]$ is a polynomial ring and every element of $k(t)$ is a quotient of two elements of $k[t]$ (theorem 19.2.8). For a polynomial G in two variables t and X we view $k[t][X] = k[t, X] = k[X][t]$ and write $\deg_t G$ and $\deg_X G$ for its degrees in t and in X.

Step 1: the minimal polynomial of t over E. Choose $u_0 \in E \setminus k$ and write $u_0 = a_0(t)/b_0(t)$ with $a_0, b_0 \in k[t]$ and $b_0 \neq 0$. The polynomial $a_0(X) - u_0 b_0(X) \in E[X]$ has t as a root, and it is nonzero: otherwise its coefficients give $a_{0,l} = u_0 b_{0,l}$ for every l, and an l with $b_{0,l} \neq 0$ gives $u_0 = a_{0,l}/b_{0,l} \in k$. So t is algebraic over E. Since $E \supseteq k$, the field $E(t)$ is $k(t)$. Let $f(X) = X^n + c_{n-1}X^{n-1} + \cdots + c_0 \in E[X]$ be the monic minimal polynomial of t over E. Then $n = [k(t) : E]$, and f divides every polynomial of $E[X]$ having t as a root, since it generates the kernel of evaluation at t (theorem 19.2.8, theorem 19.2.5).

Step 2: a coefficient u of f. Some c_j is not in k, since otherwise f would be a nonzero polynomial over k with the root t. Fix such a j and put $u = c_j \in E$. Write $u = a(t)/b(t)$ with $a, b \in k[t]$, $b \neq 0$ and $\deg b$ as small as possible. Then a and b have no common nonconstant factor, since cancelling one would lower $\deg b$, so in the principal ideal domain $k[t]$ (corollary 11.1.2) the ideal $(a) + (b)$ is (1), that is, $1 = \mu a + \nu b$ for some $\mu, \nu \in k[t]$ (definition 10.1.24). Moreover $a \neq \lambda b$ for every $\lambda \in k$, since otherwise $u = \lambda \in k$. Write $a = \sum_l a_l t^l$ and $b = \sum_l b_l t^l$, and let $m = \max(\deg a, \deg b)$. The polynomial

$$P(X) = a(X) - ub(X) \in k(u)[X]$$

has t as a root, and its coefficient of X^m is $a_m - ub_m \neq 0$: if $b_m \neq 0$, then $a_m - ub_m = 0$ would give $u = a_m/b_m \in k$, and if $b_m = 0$, then $a_m \neq 0$. So P has degree m, and $[k(t) : k(u)] \leq m$ (proposition 19.3.3), as $k(u)(t) = k(t)$. Since $k(u) \subseteq E$, the tower law (theorem 19.3.7) gives

$$[k(t) : k(u)] = [k(t) : E][E : k(u)] = n[E : k(u)]$$

Step 3: clearing denominators. Multiplying f by a common denominator of its coefficients gives elements of the form ef with $e \in k(t)^\times$ lying in $k[t][X]$. Among them choose $F = ef$ whose coefficient of X^n, which is e, has the least degree. Then $e \in k[t]$, and the coefficients $F_0, \ldots, F_n \in k[t]$ of F have no common nonconstant factor π, since $(e/\pi)f$ would have a coefficient of X^n of smaller degree. The coefficient of X^j is $F_j = eu = ea/b$, so $bF_j = ea$ and $e = \mu ea + \nu eb = b(\mu F_j + \nu e)$. Hence b divides e, $\deg F_n = \deg e \geq \deg b$, and $\deg F_j = \deg e - \deg b + \deg a \geq \deg a$, using $a \neq 0$ (proposition 9.2.3(1)). Therefore $\deg_t F \geq m$.

Step 4: F divides $Q = a(X)b(t) - a(t)b(X)$ in $k[t][X]$. We have $Q = b(t)P(X)$. As $P \in k(u)[X] \subseteq E[X]$ has the root t, f divides P in $E[X]$, so $Q = FH$ for some $H \in k(t)[X]$. Write $H = H_0/c$ with $H_0 \in k[t][X]$ and $0 \neq c \in k[t]$ of least possible degree, and suppose that c is

not a nonzero constant. Let π be an irreducible factor of c in $k[t]$. Then (π) is a maximal ideal (proposition 10.1.33, proposition 10.1.32), so $\kappa = k[t]/(\pi)$ is a field (proposition 9.4.19), and reducing coefficients modulo π is a ring homomorphism $k[t][X] \to \kappa[X]$ (theorem 9.2.4). Reducing $cQ = FH_0$ gives $0 = \overline{F}\,\overline{H_0}$ in $\kappa[X]$, an integral domain (proposition 9.2.3(3)). As π does not divide every coefficient of F, $\overline{F} \neq 0$, so $\overline{H_0} = 0$, that is, π divides every coefficient of H_0, and $H = (H_0/\pi)/(c/\pi)$ contradicts the minimality of $\deg c$. So c is a nonzero constant and $H \in k[t][X]$.

Step 5: H is a nonzero constant. The coefficient of X^m in Q is $a_m b(t) - a(t)b_m$, which is nonzero: if $b_m \neq 0$ its vanishing would give $a = (a_m/b_m)b$, which is excluded, and if $b_m = 0$ then $a_m \neq 0$ and it equals $a_m b(t) \neq 0$. So $Q \neq 0$. As a polynomial in t over the integral domain $k[X]$,

$$Q = \sum_l \left(a(X)b_l - a_l b(X)\right) t^l$$

has $\deg_t Q \leq m$, and $\deg_t Q = \deg_t F + \deg_t H$ (proposition 9.2.3(1)). With $\deg_t F \geq m$ this forces $\deg_t H = 0$, that is, $H \in k[X]$. Suppose $\deg H \geq 1$ and let h be an irreducible factor of H in $k[X]$. Then h divides $Q = FH$ in $k[X][t]$, and comparing coefficients of t^l shows that h divides $a(X)b_l - a_l b(X)$ for every l. Let $\kappa' = k[X]/(h)$, a field for the same reason as κ in Step 4, and let θ be the class of X, so that $g(\theta) = 0$ exactly when h divides g. Then $a(\theta)b_l = a_l b(\theta)$ for every l. If $b(\theta) \neq 0$, then $a_l = \lambda b_l$ for every l with $\lambda = a(\theta)/b(\theta)$, and an l with $b_l \neq 0$ shows $\lambda = a_l/b_l \in k$, so $a = \lambda b$, which is excluded. If $b(\theta) = 0$, then $a(\theta)b_l = 0$ for every l, so $a(\theta) = 0$, and h divides both a and b, contradicting $(a) + (b) = (1)$. Hence H is a nonzero constant.

Step 6: conclusion. By Step 5, F is a nonzero constant multiple of Q, so $n = \deg_X F = \deg_X Q$. The coefficient of X^l in Q is $a_l b(t) - a(t)b_l$, which vanishes for $l > m$ and not for $l = m$, so $\deg_X Q = m$. Hence $n = m$, and Step 2 gives $n[E : k(u)] = [k(t) : k(u)] \leq m = n$. So $[E : k(u)] = 1$, that is, $E = k(u)$, as we sought to show.

Exercise 19.6.2

1. Let E/k and F/k be field extensions inside a common field. Show that if E/k is separable, then so is EF/F. Show that if E/k and F/k are separable, so is EF/k.
2. Let F/k is simple so that $F = k(\alpha)$. then if L is the splitting field for a [not necessarily separable] minimal polynomial $m_{\alpha,k}(x)$, $L = \prod_{\sigma \in \mathrm{Emb}_k(F)} \sigma(F)$. If F is normal, then $L = F$, and if F separable, we have an idea of the number of σ exists, namely the degree of $k(\alpha)$ over k)

19.6.3 Important Functional: Resultant and Discriminant

One very useful functional on $R[x]$ that allows us to detect when a polynomial has a root even R need not contain the roots is the following:

Theorem 19.6.24: Resultant

Let R be an integral domain and $f(x), g(x) \in R[x]$ be two polynomials:

$$f(x) = a_n x^n + a_{n-1}x^{n-1} + \cdots + a_1 x + a_0$$
$$g(x) = b_m x^m + b_{m-1}x^{m-1} + \cdots + b_1 x + b_0$$

where $a_n, b_m \neq 0$ so that their degree is the highest order term. Then f and g have a common root if and only if:

$$\text{Res}(f,g) := \det \begin{pmatrix} a_n & 0 & \cdots & 0 & b_m & 0 & \cdots & 0 \\ a_{n-1} & a_n & \cdots & 0 & b_{m-1} & b_m & \cdots & 0 \\ a_{n-2} & a_{n-1} & \cdots & 0 & b_{m-2} & b_{m-1} & \ddots & 0 \\ \vdots & \vdots & \ddots & a_n & \vdots & \vdots & \ddots & b_m \\ a_0 & a_1 & \cdots & \vdots & b_0 & b_1 & \cdots & \vdots \\ 0 & a_0 & \ddots & \vdots & 0 & b_0 & \ddots & \vdots \\ \vdots & \vdots & \ddots & a_{n-1} & \vdots & \vdots & \ddots & b_1 \\ 0 & 0 & \cdots & a_0 & 0 & 0 & \cdots & b_0 \end{pmatrix} = 0$$

is zero.

***Proof*:**

Let $f, g \in R[x]$ be (not necessarily monic) polynomials with degree n and m respectively. If either $f = 0$ or $g = 0$, there is nothing to prove, so assume both are nonzero. Let $\mathcal{P}_n$, $\mathcal{P}_m$ be the free R-modules of polynomial up to degree strictly less than n and strictly less than m respectively. Then $\mathcal{P}_n, \mathcal{P}_m$ have dimension (rank)[a] n, m, so $\mathcal{P}_n \times \mathcal{P}_m$ has dimension $n + m$. Define:

$$\varphi : \mathcal{P}_m \times \mathcal{P}_n \to \mathcal{P}_{n+m} \qquad \varphi(p,q) = fp + gq$$

As R is an integral domain, $\deg(fp + gq) = n + m$, otherwise it may be of a smaller degree and we may get unwanted zero-polynomials. The function φ is linear as:

$$\begin{aligned} \varphi((p_1,q_1) + (p_2,q_2)) &= \varphi(p_1 + q_2, q_1 + q_2) \\ &= f(p_1 + p_2) + g(q_1 + q_2) \\ &= fp_1 + fp_2 + gq_1 + gq_2 \\ &= (fp_1 + gq_1) + (fp_2 + gq_2) \\ &= \varphi(p_1,q_1) + \varphi(p_2,q_2) \end{aligned}$$

Hence, this transformation has a matrix representation. Choosing the (ordered) basis $(x^{n-1}, x^{n-2}, ..., x, 1)$ for $\mathcal{P}_n$ and the (ordered) basis $(y^{m-1}, y^{m-2}, ..., y, 1)$ for $\mathcal{P}_m$, we get the matrix representation:

$$\begin{pmatrix} a_n & 0 & \cdots & 0 & b_m & 0 & \cdots & 0 \\ a_{n-1} & a_n & \cdots & 0 & b_{m-1} & b_m & \cdots & 0 \\ a_{n-2} & a_{n-1} & \cdots & 0 & b_{m-2} & b_{m-1} & \ddots & 0 \\ \vdots & \vdots & \ddots & a_n & \vdots & \vdots & \ddots & b_m \\ a_0 & a_1 & \cdots & \vdots & b_0 & b_1 & \cdots & \vdots \\ 0 & a_0 & \ddots & \vdots & 0 & b_0 & \ddots & \vdots \\ \vdots & \vdots & \ddots & a_{n-1} & \vdots & \vdots & \ddots & b_1 \\ 0 & 0 & \cdots & a_0 & 0 & 0 & \cdots & b_0 \end{pmatrix}$$

Note that $f \notin \mathcal{P}_n$ and $g \notin \mathcal{P}_m$ or else there would always exist $\varphi(-g, f) = -fg + fg = 0$, which would ruin the following argument using the determinant. Let M represent this matrix.

If f, g share a common $\alpha \in \overline{K}$ where $K = \text{Frac}(R)$, then there exists an $h \in \overline{K}[x]$ such that $h(x)f_0(x) = f(x)$ and $h(x)g_0(x) = g(x)$ for appropriate factors $f_0, g_0 \in \overline{K}[x]$. Then $\varphi(-g_0, f_0) = 0$ with $(-g_0, f_0) \neq 0$, so $\det(M) = 0$ by proposition 13.4.13, applied over the integral domain $\overline{K}$.

Conversely, suppose $\det(M) = 0$. By proposition 13.4.13 there exist p_0, q_0, not both zero, such that

$$fp_0 + gq_0 = 0 \qquad \Rightarrow \qquad fp_0 = -gq_0 \tag{19.3}$$

We will use this to show that f, g are not relatively prime. Suppose $(f, g) = 1$ so that there exists s, t such that $ft + sg = 1$. Then $ftp_0 + sgp_0 = p_0$. Substituting $fp_0 = -gq_0$ from equation (19.3) we get:

$$gq_0p_0 + sgp_0 = g(q_0p_0 + sp_0) = p_0$$

As R is an integral domain, $g \big| p_0$ so $p_0 = gh$. Similarly, $q_0 = fr$.

Both of these cause a contradiction. For example, considering $p_0 = gh$, then:

$$m > \deg(p_0) = \deg(gh) = \deg(g) + \deg(h) = m + \epsilon$$

where $\epsilon \in \mathbb{N}$. Thus (f, g) is *not* relatively prime. Then if $K = \text{Frac}(R)$ we have that $\overline{K}[x]$ is a PID and it must be that there exists a unit $h \in \overline{K}[x]$

$$(h) = (f, g)$$

showing f, g have a common root.

[a] Note that as this is over finite dimensions and over commutative rings, the rank is invariant and hence there is no issue when we use the specific rank later in the proof

Definition 19.6.25: Resultant

The determinant in theorem 19.6.24 $\text{Res}(f, g)$ is called the *resultant*. If the base-ring is not clear, we often denote it as $\text{Res}_R(f, g)$ for the desired ring R, and if the indeterminate element is not clear we may denote it as $\text{Res}_x(f, g)$.

The form above is useful as it can be defined over an integral domain R and is better suited for computational purposes as no field extensions are required. The following is often useful when strictly

working over algebraically closed fields:

Proposition 19.6.26: Characterizing Resultant using Algebraically Closed Fields

Let $f, g \in R[x]$ be (not necessarily monic) polynomials over an integral domain R, of degrees n and m with leading coefficients a_n and b_m respectively, and let $\overline{K}$ be the algebraic closure of $K = \mathrm{Frac}(R)$. Then if $\alpha_1, ..., \alpha_n$ are the roots of f and $\beta_1, ..., \beta_m$ the roots of g respectively (listed with multiplicity):

$$\mathrm{Res}(f, g) = a_n^m b_m^n \prod_{i,j} (\alpha_i - \beta_j)$$

***Proof*:**

Over $\overline{K}$ we may write

$$f = a_n \prod_{i=1}^{n} (x - \alpha_i) \qquad g = b_m \prod_{j=1}^{m} (x - \beta_j)$$

The Sylvester matrix of theorem 19.6.24 has entries in $R \subseteq \overline{K}$ and its determinant is the same computed over either ring, so we work over $\overline{K}$ throughout. Note also that $\overline{K}$ is infinite: a finite field $\{c_1, \dots, c_q\}$ is never algebraically closed, since the polynomial $\prod_u (x - c_u) + 1$ has no root in it.

Step 0: a Vandermonde determinant. For scalars $\gamma_1, \dots, \gamma_N$ let $W(\gamma_1, \dots, \gamma_N)$ be the $N \times N$ matrix with entry γ_u^{N-c} in row u and column c, so the columns list decreasing powers $N - 1, \dots, 1, 0$. We claim

$$\det W(\gamma_1, \dots, \gamma_N) = \prod_{u<v} (\gamma_u - \gamma_v)$$

We induct on N, the case $N = 1$ being the empty product 1. For $c = 1, 2, \dots, N-1$ in this order, subtract γ_N times column $c+1$ from column c. By multilinearity and the alternating property (proposition 13.4.6(3), (4)) this does not change the determinant, and when column c is modified, column $c+1$ still holds its original entries. The new entry in row u and column $c \le N-1$ is $\gamma_u^{N-c} - \gamma_N \gamma_u^{N-c-1} = \gamma_u^{N-c-1}(\gamma_u - \gamma_N)$, while column N keeps the entries 1. In particular row N becomes $(0, \dots, 0, 1)$, so in the defining sum (definition 13.4.5) only permutations fixing N contribute, and these are exactly the permutations of $\{1, \dots, N-1\}$ with unchanged sign. Hence the determinant equals that of the top left $(N-1) \times (N-1)$ block, whose row u carries the common factor $(\gamma_u - \gamma_N)$. Factoring it out of each row (proposition 13.4.6(2), (3)) leaves $W(\gamma_1, \dots, \gamma_{N-1})$, and the induction hypothesis gives the claim. In particular, splitting the pairs $u < v$ into pairs of α's, pairs of β's, and mixed pairs,

$$\det W(\alpha_1, \dots, \alpha_n, \beta_1, \dots, \beta_m) = \det W(\alpha_1, \dots, \alpha_n) \det W(\beta_1, \dots, \beta_m) \prod_{i,j} (\alpha_i - \beta_j)$$

Step 1: the identity when $\alpha_1, \dots, \alpha_n, \beta_1, \dots, \beta_m$ are pairwise distinct. Let $\mathcal{P}_d$ denote the polynomials of degree strictly less than d over $\overline{K}$, with the descending monomial bases, and consider the linear maps

$$\Phi : \mathcal{P}_m \times \mathcal{P}_n \to \mathcal{P}_{n+m} \quad \Phi(p, q) = fp + gq \qquad E : \mathcal{P}_{n+m} \to \overline{K}^{n+m} \quad E(h) = (h(\alpha_1), \dots, h(\alpha_n), h(\beta_1), \dots, h(\beta_m))$$

Inspecting the columns of the matrix in theorem 19.6.24, its c-th column for $c \le m$ lists the coefficients of $x^{m-c} f$, and its $(m+c)$-th column for $c \le n$ the coefficients of $x^{n-c} g$, with

respect to the descending powers $x^{n+m-1}, \dots, x, 1$. Hence the Sylvester matrix is the matrix M of Φ, and $\det(M) = \operatorname{Res}(f, g)$. The matrix of E with respect to the descending basis is $W_{\alpha\beta} := W(\alpha_1, \dots, \alpha_n, \beta_1, \dots, \beta_m)$, whose determinant is nonzero by Step 0 since the scalars are pairwise distinct. By theorem 13.2.6 and proposition 13.4.7,

$$\det[E \circ \Phi] = \det(W_{\alpha\beta} M) = \det(W_{\alpha\beta}) \operatorname{Res}(f, g)$$

On the other hand, the matrix of $E \circ \Phi$ can be computed directly: its c-th column for $c \leq m$ is the evaluation vector of $x^{m-c} f$, which vanishes at every α_i and takes the value $\beta_j^{m-c} f(\beta_j)$ at β_j, and its $(m+c)$-th column is the evaluation vector of $x^{n-c} g$, taking the value $\alpha_i^{n-c} g(\alpha_i)$ at α_i and vanishing at every β_j. Hence

$$[E \circ \Phi] = \begin{pmatrix} 0 & G \\ F & 0 \end{pmatrix} \qquad G_{ic} = g(\alpha_i) \alpha_i^{n-c} \qquad F_{jc} = f(\beta_j) \beta_j^{m-c}$$

with G of size $n \times n$ and F of size $m \times m$. Moving the last n columns to the front is a permutation of the columns realized by nm transpositions, which multiplies the determinant by $(-1)^{nm}$ (proposition 13.4.6(2), (5)) and produces the block diagonal matrix with blocks G and F. Hence by proposition 13.4.8 and factoring $g(\alpha_i)$ out of the i-th row of G and $f(\beta_j)$ out of the j-th row of F,

$$\det[E \circ \Phi] = (-1)^{nm} \det(G) \det(F) = (-1)^{nm} \prod_i g(\alpha_i) \prod_j f(\beta_j) \det W(\alpha_1, \dots, \alpha_n) \det W(\beta_1, \dots, \beta_m)$$

Comparing the two computations of $\det[E \circ \Phi]$, cancelling the nonzero factor $\det(W_{\alpha\beta})$ using the factorization from Step 0, we get

$$\operatorname{Res}(f, g) = (-1)^{nm} \frac{\prod_i g(\alpha_i) \prod_j f(\beta_j)}{\prod_{i,j} (\alpha_i - \beta_j)}$$

Substituting $\prod_i g(\alpha_i) = b_m^n \prod_{i,j} (\alpha_i - \beta_j)$ and $\prod_j f(\beta_j) = a_n^m \prod_{i,j} (\beta_j - \alpha_i) = (-1)^{nm} a_n^m \prod_{i,j} (\alpha_i - \beta_j)$ gives

$$\operatorname{Res}(f, g) = a_n^m b_m^n \prod_{i,j} (\alpha_i - \beta_j)$$

Step 2: the general case. Fix a_n and b_m and let $t_1, \dots, t_n, s_1, \dots, s_m$ be indeterminates. Set $f_t = a_n \prod_i (x - t_i)$ and $g_s = b_m \prod_j (x - s_j)$. Each coefficient of f_t and g_s is a polynomial in the t's and s's, every entry of the Sylvester matrix of (f_t, g_s) is such a coefficient or 0, and the determinant is a polynomial in the entries. Hence both

$$D_1 := \operatorname{Res}(f_t, g_s) \qquad D_2 := a_n^m b_m^n \prod_{i,j} (t_i - s_j)$$

lie in the polynomial ring $\overline{K}[t_1, \dots, t_n, s_1, \dots, s_m]$, and by Step 1 they take the same value at every point of $\overline{K}^{n+m}$ with pairwise distinct coordinates. Let E_0 be the product of the differences of all pairs of the $n+m$ variables. Then the polynomial $(D_1 - D_2) E_0$ vanishes at *every* point of $\overline{K}^{n+m}$: where the coordinates are distinct $D_1 - D_2$ vanishes, and elsewhere E_0 does.

A polynomial over $\overline{K}$ in N variables vanishing at every point of $\overline{K}^N$ is the zero polynomial. We induct on N: for $N = 1$ this is corollary 11.3.4 applied against the zero polynomial, using that

$\overline{K}$ is infinite. For $N > 1$, write the polynomial as $\sum_d h_d(x_1, \ldots, x_{N-1})x_N^d$. Fixing any values of the first $N-1$ variables and applying the case $N = 1$ shows each h_d vanishes at every point of $\overline{K}^{N-1}$, so each $h_d = 0$ by induction.

Hence $(D_1 - D_2)E_0 = 0$ in $\overline{K}[t, s]$, which is an integral domain by iterating proposition 9.2.3(3). Since E_0 is a product of nonzero polynomials, $E_0 \neq 0$, and therefore $D_1 = D_2$. Evaluating at $t_i = \alpha_i$ and $s_j = \beta_j$ gives the claimed identity.

One of the more important application is to find whether a polynomial has a multiple root:

Definition 19.6.27: Discriminant

Let $f \in R[x]$ be a polynomial over an integral domain. Then:

$$\Delta(f) = \operatorname{disc}(f) = \frac{(-1)^{n(n-1)/2}}{a_n}\operatorname{Res}(f, f')$$

If f is monic,

$$\Delta(f) = \operatorname{disc}(f) = (-1)^{n(n-1)/2}\operatorname{Res}(f, f')$$

Proposition 19.6.28: Discriminant As A Product

Let $f \in R[x]$ be a polynomial of degree n with leading coefficient a_n over an integral domain, $K = \operatorname{Frac}(R)$, and assume $na_n \neq 0$ in R. Let L/K be the splitting field of f. Then if $\alpha_1, ..., \alpha_n$ are the roots of f (with multiplicity),

$$\Delta(f) = a_n^{2n-2} \prod_{1 \leq i < j \leq n} (\alpha_i - \alpha_j)^2$$

In particular, $\Delta(f) = \prod_{1 \leq i < j \leq n}(\alpha_i - \alpha_j)^2$ when f is monic.

***Proof*:**

Over $L \subseteq \overline{K}$ write $f = a_n \prod_{i=1}^n (x - \alpha_i)$. The hypothesis $na_n \neq 0$ says the derivative $f' = na_n x^{n-1} + \cdots + a_1$ has degree $m := n - 1$ and leading coefficient $b_m = na_n \neq 0$, so proposition 19.6.26 applies to the pair (f, f'): if $\beta_1, \ldots, \beta_{n-1}$ are the roots of f', so that $f' = na_n \prod_j (x - \beta_j)$, then

$$\operatorname{Res}(f, f') = a_n^{n-1}(na_n)^n \prod_{i,j}(\alpha_i - \beta_j) = a_n^{n-1} \prod_{i=1}^{n} \left(na_n \prod_{j=1}^{n-1} (\alpha_i - \beta_j) \right) = a_n^{n-1} \prod_{i=1}^{n} f'(\alpha_i)$$

Writing $f = (x - \alpha_i)h_i$ with $h_i = a_n \prod_{j \neq i}(x - \alpha_j)$, the Leibniz property of the formal derivative gives $f' = h_i + (x - \alpha_i)h_i'$, and evaluating at α_i:

$$f'(\alpha_i) = h_i(\alpha_i) = a_n \prod_{j \neq i}(\alpha_i - \alpha_j)$$

Multiplying over i and pairing the factors $(\alpha_i - \alpha_j)$ and $(\alpha_j - \alpha_i) = -(\alpha_i - \alpha_j)$ for each of the

$n(n-1)/2$ pairs $i<j$:

$$\prod_{i=1}^{n} f'(\alpha_i) = a_n^n \prod_{i\neq j}(\alpha_i-\alpha_j) = a_n^n(-1)^{n(n-1)/2}\prod_{i<j}(\alpha_i-\alpha_j)^2$$

Hence

$$\operatorname{Res}(f,f') = a_n^{2n-1}(-1)^{n(n-1)/2}\prod_{i<j}(\alpha_i-\alpha_j)^2$$

and by definition 19.6.27,

$$\Delta(f) = \frac{(-1)^{n(n-1)/2}}{a_n}\operatorname{Res}(f,f') = (-1)^{n(n-1)}a_n^{2n-2}\prod_{i<j}(\alpha_i-\alpha_j)^2 = a_n^{2n-2}\prod_{1\le i<j\le n}(\alpha_i-\alpha_j)^2$$

since $(-1)^{n(n-1)}=1$, the exponent being even. The monic case is the special case $a_n=1$.

As is immediate after proposition 19.6.6:

Proposition 19.6.29: Environment Title

Let $f\in\mathbb{R}[x]$ be a polynomial. Then if $\Delta(f)=0$, f has a multiple root in the splitting field.

***Proof*:**
immediate.

here)

Separability has a numerical measure that is often more useful than the definition. A finite extension F/k makes F a finite-dimensional k-vector space, so every element of F acts on it by multiplication and has a trace in the sense of definition 13.4.3. That assignment is a k-linear form on F, and the symmetric bilinear form it induces detects separability exactly.

Definition 19.6.30: Trace of a Finite Field Extension

Let F/k be a finite field extension and let $x\in F$. Multiplication by x is a k-linear endomorphism $m_x\colon F\to F$, and the *trace* of x is

$$\operatorname{tr}_{F/k}(x) \;:=\; \operatorname{tr}(m_x)\in k$$

the trace of that endomorphism in the sense of definition 13.4.3. The *trace form* of F/k is the symmetric k-bilinear form

$$\langle x,y\rangle \;:=\; \operatorname{tr}_{F/k}(xy)$$

Proposition 19.6.31: The Trace Form of a Separable Extension Is Nondegenerate

Let F/k be a finite separable field extension. Then the trace form of F/k is nondegenerate: if $x\in F$ satisfies $\operatorname{tr}_{F/k}(xy)=0$ for all $y\in F$, then $x=0$. Equivalently, for any k-basis $e_1,\dots,e_n$ of F there is a dual basis $e_1^*,\dots,e_n^*$ with $\operatorname{tr}_{F/k}(e_ie_j^*)=\delta_{ij}$.

***Proof*:**
Step 1: reduce to a computation in one element. By theorem 19.6.20 the extension is simple, say $F = k(\alpha)$ with α of degree n over k and minimal polynomial m. Then $1, \alpha, \dots, \alpha^{n-1}$ is a k-basis of F, and it suffices to show that the Gram matrix $M = \big(\mathrm{tr}_{F/k}(\alpha^{i+j})\big)_{0 \le i,j \le n-1}$ of the trace form in that basis is invertible: by lemma 13.3.23 a bilinear form on a finite-dimensional space is nondegenerate (definition 13.3.22) exactly when its matrix in a basis is invertible, and the criterion is independent of which basis is used.

Step 2: traces are power sums of the roots. In the basis $1, \alpha, \dots, \alpha^{n-1}$ the matrix of m_α is the companion matrix of m, so the characteristic polynomial of m_α is m itself. Let $\alpha_1, \dots, \alpha_n$ be the roots of m in a splitting field. They are distinct because F/k is separable, so m is a separable polynomial (definition 19.6.3, proposition 19.6.7). The eigenvalues of the companion matrix are exactly the α_r, so those of m_α^j are the α_r^j, and therefore

$$\mathrm{tr}_{F/k}(\alpha^j) \;=\; \mathrm{tr}\big(m_\alpha^j\big) \;=\; \sum_{r=1}^{n} \alpha_r^j$$

for every $j \ge 0$.

Step 3: a Vandermonde determinant. Let $V = (\alpha_r^i)_{0 \le i \le n-1,\ 1 \le r \le n}$ be the Vandermonde matrix of the roots. By Step 2,

$$M_{ij} \;=\; \sum_{r=1}^{n} \alpha_r^i \alpha_r^j \;=\; \big(VV^{\mathsf{T}}\big)_{ij}$$

so $M = VV^{\mathsf{T}}$ and $\det M = (\det V)^2 = \prod_{r<s}(\alpha_r - \alpha_s)^2$. The roots are distinct, so this is nonzero and M is invertible. The dual basis is then obtained by inverting M, completing the proof.

The dual basis is what makes the trace form a finiteness tool: an element whose products with a fixed basis all have trace in a subring is pinned inside the free module spanned by the dual basis. That is the mechanism behind the finiteness of integral closure in *Commutative Algebra* [16, Finiteness of Integral Closure], and behind the finiteness of the ring of integers of a number field in *Algebraic Number Theory* [11].

Both of those applications need one more thing before the trace form can be brought to bear: the extension in question has to be separable over the subfield chosen as a base. A transcendence basis makes K algebraic over $k(B)$, but says nothing about separability, and in characteristic p the two can come apart. Over a perfect field they can always be reconciled.

Definition 19.6.32: Separating Transcendence Basis

Let K/k be a field extension. A transcendence basis B of K/k is *separating* if K is a separable algebraic extension of $k(B)$. The extension K/k is *separably generated* if it admits one.

Theorem 19.6.33: Perfect Base Fields Give Separating Transcendence Bases

Let k be a perfect field and let $K = k(a_1, \dots, a_n)$ be a finitely generated field extension. Then some subset of $\{a_1, \dots, a_n\}$ is a separating transcendence basis of K over k. In particular every finitely generated extension of a perfect field is separably generated.

***Proof*:**
If $\mathrm{char} k = 0$ every algebraic extension is separable (corollary 19.6.8), so any transcendence basis extracted from the generators by theorem 19.2.13 is separating. Assume $\mathrm{char} k = p > 0$ and induct on n, the case $n = 0$ being trivial.

Step 1: dispose of the independent case. If $a_1, \dots, a_n$ are algebraically independent over k they form a transcendence basis and $K = k(a_1, \dots, a_n)$ is separable over itself, so the basis is separating. Otherwise there is a nonzero $f \in k[X_1, \dots, X_n]$ with $f(a_1, \dots, a_n) = 0$, and we may take f irreducible, replacing it by an irreducible factor that still vanishes.

Step 2: perfectness forces a nonzero partial derivative. Suppose every X_i occurred in f only through p-th powers, so that $f = \sum_\alpha c_\alpha X^{p\alpha}$. Since k is perfect, every c_α is a p-th power, say $c_\alpha = d_\alpha^p$, and then

$$f = \sum_\alpha d_\alpha^p X^{p\alpha} = \Big(\sum_\alpha d_\alpha X^\alpha\Big)^p$$

using that raising to the p-th power is a ring homomorphism in characteristic p (proposition 19.6.11). That contradicts irreducibility of f. So some variable occurs in f outside a p-th power, which is to say $\partial f / \partial X_i \neq 0$ for some i. Fix such an i.

Step 3: the i-th generator is separable over the others. Put $F = k(a_1, \dots, \widehat{a_i}, \dots, a_n)$, the subfield generated by the remaining generators. Viewing f as a polynomial in X_i with coefficients in $k[X_1, \dots, \widehat{X_i}, \dots, X_n]$, it is nonconstant in X_i (otherwise $\partial f / \partial X_i = 0$) and irreducible, so up to a unit it is the minimal polynomial of a_i over F. Its derivative in X_i is nonzero and of smaller degree, hence not divisible by the minimal polynomial, so the minimal polynomial has no repeated root by proposition 19.6.6. Thus a_i is separable algebraic over F, and $K = F(a_i)$ is a finite separable extension of F.

Step 4: induct. The field F is generated over k by $n - 1$ elements, so by induction some subset B of $\{a_1, \dots, \widehat{a_i}, \dots, a_n\}$ is a separating transcendence basis of F/k. Then K is algebraic over F, which is algebraic over $k(B)$, so K is algebraic over $k(B)$ and B is a transcendence basis of K/k. Separability passes up the tower $k(B) \subseteq F \subseteq K$ by proposition 19.6.18, both steps being separable, so $K/k(B)$ is separable and B is separating, completing the proof.

The hypothesis is not decorative. Over an imperfect field the theorem fails: if $t \in k$ is not a p-th power, the extension $k(u)$ with $u^p = t$ adjoined to a rational function field produces a finitely generated extension with no separating transcendence basis, and the trace form degenerates accordingly.

Exercise 19.6.3

1. Show that if f, g are polynomials of degree n, m respectively then

$$(x^{m-1}f(x)) \wedge (x^{m-2}f(x)) \wedge \cdots \wedge f(x) \wedge (x^{n-1}g(x) \wedge \cdots \wedge g(x) = \mathrm{Res}(f,g)x^{n+m-1} \wedge x^{n+m-2} \wedge \cdots \wedge 1$$

2. Let $f, g \in R[x]$ have nonzero leading coefficients. Show that $\mathrm{Res}(f,g) = 0$ iff f, g have a common factor of positive degree in $\mathrm{Frac}(R)[x]$.
3. Show that $\mathrm{Res}(f(x+a), g(x+a)) = \mathrm{Res}(f(x), g(x))$ for $a \in R$.
4. Show that $\Delta(fg) = \Delta(f)\Delta(g)\mathrm{Res}(f,g)^2$.
5. Let $f(x) = ax^2 + bx + c$. Show that $\Delta(f) = b^2 - 4ac$.

19.7 Application: Cyclotomic Extensions and Finite Fields

In this section, we use the tools given to us to gain some important examples of field extensions and analyse the corresponding $\mathrm{Aut}_k(F)$, and generalize theorem 19.6.20 to show finite separable extensions of finite fields are in fact simple.

We start by using the tools that were developed to study the important class of polynomials of the form $x^n - 1$. Such extensions will be called *Cyclotomic extensions*. These polynomials represent the "torsion units" and shall show up everywhere when considering solutions to equations as they often add the necessary "symmetry" to find all the solutions to an algebraic equation. They are also important for the continuation of Field Theory to Kummer Theory, which is the study of the case where $\mathrm{Aut}_k(F)$ is an abelian group, and they form an important first step towards Fermat's Last theorem through their application to solutions of Elliptic curves[22].

After having classified Cyclotomic extensions, we shall finish the section with the study of extensions of finite fields. As it turns out, all finite extensions will be splitting fields of polynomials of the form $x(x^{p^d} - 1)$, meaning they will have a very simple structure and automorphism group.

19.7.1 Cyclotomic Polynomials and Fields

We start off with some reminders:

Definition 19.7.1: Roots Of Unity Reminder

We will denote $\zeta_n = e^{\frac{2\pi i}{n}}$, and the cyclic subgroup they generated to be μ_n. Note that:

$$\zeta_n^n = e^{2\pi i} = 1$$

If $\langle \zeta_n^k \rangle = \mu_n$, then this element is called a *primitive root of unity.*

[22] if $\mathrm{ch}(F) \notin 2, 3$, then an elliptic curve is a polynomial of the form $y^2 = x^3 + ax + b$, $a, b \in F$. A more technical definition is given when $\mathrm{ch}(F) \in \{2, 3\}$ that would be presented in a course on elliptic curves

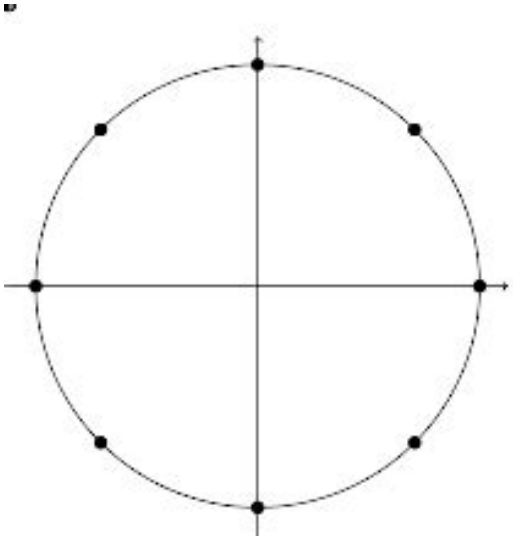
The element ζ_n is always a root of $x^n - 1$, and each ζ_n^k, $1 \leq k \leq n$ where $\gcd(n, k) = 1$ consist of the rest of the roots of $x^n - 1$, for

$$(\zeta_n^k)^n = 1$$

and if $\gcd(n, k) = \ell$, then:

$$(\zeta_n^k)^\ell = 1$$

It is an easy mistake to think that $\zeta_2 = i$, however $\zeta_2 = -1$ and $\zeta_4 = i$. These elements always form a group under multiplication, which we labeled μ_n. This group always lies on the unit circle, for example:



If you know about the euler representation of complex numbers:

$$re^{i\theta} = r(\cos(\theta) + i\sin(\theta))$$

Then $e^{\frac{2\pi i}{n}} = \zeta_n = \cos(2\pi/n) + i\sin(2\pi/n)$. Other identities we get are:

1. $\zeta_3 = \frac{-1+i\sqrt{3}}{2}$
2. $\zeta_5 = \frac{\sqrt{5}-1}{4} + \frac{\sqrt{10+2\sqrt{5}}}{4}i$
3. $\zeta_6 = \frac{1+i\sqrt{3}}{2}$ (to not be confused with ζ_3)
4. $\zeta_8 = \frac{\sqrt{2}}{2} + i\frac{\sqrt{2}}{2}$

These can be deduced using the results from section 19.4. In paticular Gauss proved that primitive nth roots of unity ζ_n can be constructed using the square-roots, additions, subtraction, multiplication, and division of rational numbers, which we saw implies that n is a power of two, or in our case we would require that n is the product of two discint *fermat primes*.

Every root of unity of a root of $x^n - 1$ for some n, however this polynomial is not irreducible, namely $x^n - 1 = (x - 1)p(x)$. The polynomial $p(x)$ will turn out to be irreducible, and hence is given a name:

Definition 19.7.2: Cyclotomic Polynomials

Let:

$$\Phi_n(x) = \prod_{\zeta \text{ primitive } n\text{th root of } 1} (x - \zeta) = \prod_{\substack{1 \leq m \leq n \\ \gcd(m,n)=1}} (x - \zeta_n^m)$$

Then the resulting polynomial is called the *nth cyclotomic polynomial*

As we have seen in the section on number theory ([24, Number Theory]), there are $\varphi(n)$ primitive roots of 1, where φ is Euler's totient function. The term cyclotomic comes from the greek *cyclo* (circle) and *tomos* (divide), owing to how the roots of unity divide the circle into regular sections. Though not obvious, this polynomial has rational coefficients and is irreducible over $\mathbb{Q}$ (which we will show).

Example 19.7.3: Cyclotomic Polynomials

In the case of $n = p$ is a prime, it is simple to see it's irreducible. Every non-identity element of μ_p is a generator, i.e. every pth root of 1 is primitive except 1. In fact, 1 is a root of $x^p - 1$ ($1^p - 1 = 0$), and so we get:

$$\Phi_p(x) = \frac{x^p - 1}{x - 1} = x^{p-1} + x^{p-2} + \cdots + x + 1$$

which is irreducible over $\mathbb{Q}$ using Eisenstein's trick along with shifting (noting that p divides $\binom{n}{k}$). Furthermore, $\mu_p = C_p = \mathbb{Z}/p\mathbb{Z}$.

Aside: for those who know Taylor series and the exponential function, notice how close $\Phi_p(x)$ resembles the pariatal sum for the taylor expansion of e^x:

$$1 + x + \frac{x^2}{2} + \frac{x^3}{3!} + \cdots + \frac{x^p}{p!}$$

the main difference being the coefficient of the partial taylor expansion of e^x being $1/n!$. This reflects on the "rotation" nature of e^x, as well as it's link to roots of unity. However, note that e^x has no roots (even when expended to $\mathbb{C}$, since $e^z \cdot e^{-z} = 1$). Hence, though the partial expansion may have root, even roots that resemble the roots of unity, taylor expansion, the "infinite polynomial representation" need not have roots!

For ease of reference, we record what happens when n is not prime:

Lemma 19.7.4: Cyclotomic Polynomial divisor Representation

For all $n > 0$,

$$x^n - 1 = \prod_{1 \leq d \mid n} \Phi_d(x)$$

***Proof*:**
For all ζ_n^k, $0 \leq k \leq n-1$, $(\zeta_n^k) - 1 = 0$. Hence, $x^n - 1$ splits into products

$$x^n - 1 = \prod_k^n (x - \zeta_k)$$

If $n = dk$ then $1 = \zeta_n^n = (\zeta_n^d)^k$, hence every dth root of unity is an nth root of unity. In particular, every primitive dth root of unity is an nth root of unity. Conversely, if $\zeta_n^d \in \mu_n$, then ζ_n^d forms cyclic subgroup, say μ_d where d is a divisor of n. But then ζ_n^d is a primitive dth root of unity for some divisor d. This establishes:

$$x^n - 1 = \prod_{1 \leq d \Big| n} \left(\prod_{\substack{\zeta \text{ primitive} \\ d\text{th root of 1}}} (x - \zeta) \right) = \prod_{1 \leq d \Big| n} \Phi_d(x)$$

as we sought to show.

Corollary 19.7.5: Integer Coefficients

The cyclotomic polynomials $\Phi_n(x)$ have integer coefficients

***Proof*:**
This is a proof by [strong] induction. The case of $\Phi_1(x) = x - 1$ is immediate, so assume that for $m < n$, Φ_m is a monic polynomial with integer coefficients. Then:

$$f(x) = \prod_{\substack{1 \leq d | n \\ d < n}} \Phi_d(x)$$

is a monic polynomial with degree $< n$. Since $f(x), x^n - 1 \in \mathbb{Z}[x]$ are two monic polynomials with integer ceofficients, we may use the euclidean algorithm to find $q(x), r(x) \in \mathbb{Z}[x]$, $\deg(r(x)) < \deg(f(x))$ or $r(x) = 0$ where

$$x^n - 1 = f(x)q(x) + r(x)$$

Then by lemma 19.7.4 we have

$$x^n - 1 = f(x)\Phi_n(x)$$

Hence, we get

$$f(x)(q(x) - \Phi_n(x)) = r(x)$$

But that implies $\deg(r(x)) > \deg(f(x))$, a contradiction. Thus $r(x) = 0$, giving us $f(x)q(x) = f(x)\Phi_n(x)$, which implies $\Phi_n(x) = q(x) \in \mathbb{Z}[x]$, as we sought to show.

Example 19.7.6: Cyclotomic Polynomial

1. Let $p(x) = x^4 - 1 = \Phi_1(x)\Phi_2(x)\Phi_4(x)$. Then:

$$\Phi_4(x) = \frac{x^4 - 1}{x^2 - 1} = x^2 + 1$$

2. Let $p(x) = x^6 - 1 = \Phi_1(x)\Phi_2(x)\Phi_3(x)\Phi_6(x)$. Then:

$$\Phi_6(x) = \frac{x^6 - 1}{(x^2 - 1)(x^2 + x + 1)} = x^2 - x + 1$$

3. Let $p(x) = x^{12} - 1 = \Phi_1(x)\Phi_2(x)\Phi_3(x)\Phi_4(x)\Phi_6(x)\Phi_{12}(x)$. Then:

$$\Phi_{12}(x) = \frac{x^{12} - 1}{(x^6 - 1)(x^2 + 1)} = x^4 - x^2 + 1$$

It might seem that all the coefficients are 0 or ± 1. However, the first counter-example to this problem is when $n = 105$, and in general the coefficients can be as large as we might want

Proposition 19.7.7: Irreducibility Of Cyclotomic Polynomials

For all $n \in \mathbb{N}_{>0}$, $\Phi_n(x) \in \mathbb{Z}[x]$ is irreducible over $\mathbb{Q}$

Note that since $\Phi_n(x)$ is monic, by Gauss's lemma, irreducibility in $\mathbb{Q}[x]$ is equivalen to ireducibility in $\mathbb{Z}[x]$.

***Proof*:**
Suppose Φ_n were reducible over $\mathbb{Q}$, say $\Phi_n = AB$ with $A, B \in \mathbb{Q}[x]$ nonconstant. By Gauss's Lemma (proposition 11.2.1) we may replace A and B by rational multiples $f, g \in \mathbb{Z}[x]$ with $\Phi_n = fg$. Comparing leading coefficients, the product of the leading coefficients of f and g is 1, so both are ± 1, and changing the sign of both if necessary we may take f and g monic. Shrinking f to one of its monic irreducible factors in $\mathbb{Z}[x]$ and absorbing the rest into g, we may assume in addition that f is irreducible, and it is then irreducible in $\mathbb{Q}[x]$ by corollary 11.2.2.

A root of f and a prime moving it onto g. By definition 19.7.2 the roots of Φ_n are the primitive nth roots of unity, each occurring once, and each is a root of f or of g. Fix a root ζ of f. Every primitive nth root of unity is then a power ζ^m with $\gcd(m, n) = 1$, so choose a root of g and write it as ζ^m with $1 \le m \le n$ and $\gcd(m, n) = 1$. Factor $m = p_1 p_2 \cdots p_r$ into primes, none of which divides n. Along the chain

$$\zeta, \quad \zeta^{p_1}, \quad \zeta^{p_1 p_2}, \quad \ldots, \quad \zeta^{p_1 p_2 \cdots p_r} = \zeta^m$$

every entry is again a primitive nth root of unity, hence a root of f or of g. The first entry is a root of f and the last is a root of g, so for some i the element $\xi = \zeta^{p_1 \cdots p_{i-1}}$ is a root of f while ξ^p is not, where $p = p_i$. Then ξ^p is a root of g, and p does not divide n.

Reduction modulo p. Since f is monic and irreducible in $\mathbb{Q}[x]$ with $f(\xi) = 0$, it is the minimal polynomial of ξ over $\mathbb{Q}$. From $g(\xi^p) = 0$ we get that ξ is a root of $g(x^p)$, so f divides $g(x^p)$ in

$\mathbb{Q}[x]$. Dividing $g(x^p)$ by the monic f inside $\mathbb{Z}[x]$ produces an integer quotient and remainder, and these agree with the quotient and remainder over $\mathbb{Q}$, so the remainder is 0 and

$$g(x^p) = f(x)h(x) \qquad h \in \mathbb{Z}[x]$$

Reduce coefficients modulo p and write $\overline{u}$ for the image of $u \in \mathbb{Z}[x]$ in $\mathbb{F}_p[x]$. The map $u \mapsto u^p$ on $\mathbb{F}_p[x]$ is a ring homomorphism, since $p = 0$ in $\mathbb{F}_p[x]$ and p divides $\binom{p}{i}$ for $0 < i < p$, and it fixes every $c \in \mathbb{F}_p$, since $\mathbb{F}_p^\times$ has order $p-1$ and theorem 3.3.1 gives $c^{p-1} = 1$ for $c \neq 0$. Applying it to $\overline{g}$ therefore gives $\overline{g}(x)^p = \overline{g}(x^p)$, so

$$\overline{g}(x)^p = \overline{f}(x)\overline{h}(x)$$

Both f and g are monic, so $\overline{f}$ and $\overline{g}$ are nonconstant. Let $q \in \mathbb{F}_p[x]$ be an irreducible factor of $\overline{f}$. Then q divides $\overline{g}^p$, and q is prime in the Unique Factorization Domain $\mathbb{F}_p[x]$, so q divides $\overline{g}$. Hence

$$q^2 \big| \overline{f}\ \overline{g} = \overline{\Phi_n}$$

The contradiction. By lemma 19.7.4, Φ_n divides $x^n - 1$ in $\mathbb{Z}[x]$, so q^2 divides $x^n - 1$ in $\mathbb{F}_p[x]$. Write $x^n - 1 = q^2 s$ and differentiate:

$$nx^{n-1} = 2qq's + q^2 s' = q\left(2q's + qs'\right)$$

so q divides nx^{n-1}. Since p does not divide n, the element n is a unit of $\mathbb{F}_p$, so q divides x^{n-1}, and since x is irreducible and $\mathbb{F}_p[x]$ is a Unique Factorization Domain, q is an associate of x. But x does not divide $x^n - 1$, whose constant term is -1. This contradiction shows Φ_n is irreducible over $\mathbb{Q}$.

Definition 19.7.8: Cyclotomic Field

The splitting field $\mathbb{Q}(\zeta_n)$ for the polynomial $x^n - 1$ over $\mathbb{Q}$ is the nth *cyclotomic field*

Proposition 19.7.9: Cyclotomic Field Automorphism

Let $n \geq 1$ and let φ be the Euler totient (definition 1.2.4). Then:

1. $[\mathbb{Q}(\zeta_n) : \mathbb{Q}] = \varphi(n)$
2. for $m \in \mathbb{Z}$, there is an automorphism $\sigma \in \mathrm{Aut}_{\mathbb{Q}}(\mathbb{Q}(\zeta_n))$ with $\sigma(\zeta_n) = \zeta_n^m$ if and only if $\gcd(m, n) = 1$, and it is then unique, denoted σ_m
3. the map
$$j \colon (\mathbb{Z}/n\mathbb{Z})^\times \to \mathrm{Aut}_{\mathbb{Q}}(\mathbb{Q}(\zeta_n)) \qquad [m]_n \mapsto \sigma_m$$
is an isomorphism of groups, where $(\mathbb{Z}/n\mathbb{Z})^\times$ is the group of definition 1.2.5, whose elements are the classes $[m]_n$ of the integers m with $\gcd(m, n) = 1$

In particular,

$$\mathrm{Aut}_{\mathbb{Q}}(\mathbb{Q}(\zeta_n)) \cong (\mathbb{Z}/n\mathbb{Z})^\times$$

***Proof*:**
The number $\zeta_n = e^{2\pi i/n}$ has multiplicative order n, since $e^{2\pi ix} = \cos(2\pi x) + i\sin(2\pi x)$ equals 1 exactly when x is an integer. So $\zeta_n^a = 1$ exactly when $n \big| a$, and $\mu_n = \langle \zeta_n \rangle$ is cyclic of order n (definition 19.7.1). If $m \equiv r \bmod n$, then $\zeta_n^m = \zeta_n^r$, and $\gcd(m,n) = \gcd(r,n)$, since a common divisor of n and one of m, r divides the other.

1. The polynomial Φ_n is monic with integer coefficients (corollary 19.7.5), it is irreducible over $\mathbb{Q}$ (proposition 19.7.7), and $\zeta_n = \zeta_n^1$ is one of its roots (definition 19.7.2). By theorem 19.2.8, $\mathbb{Q}(\zeta_n) \cong \mathbb{Q}[x]/(p)$, where p is a nonconstant polynomial generating the kernel of evaluation at ζ_n. As Φ_n lies in that kernel, p divides Φ_n, and the irreducibility of Φ_n makes the quotient a nonzero constant. So $[\mathbb{Q}(\zeta_n) : \mathbb{Q}] = \deg p = \deg \Phi_n$ (theorem 19.2.5), and $\deg \Phi_n = \varphi(n)$, since Φ_n is the product of the $\varphi(n)$ linear factors $x - \zeta_n^m$ with $1 \le m \le n$ and $\gcd(m,n) = 1$.

2. Uniqueness: every element of $\mathbb{Q}(\zeta_n)$ is a polynomial in ζ_n with rational coefficients (corollary 19.2.23), and a $\mathbb{Q}$-automorphism preserves sums and products and fixes $\mathbb{Q}$, so it is determined by its value at ζ_n.

   Suppose $\gcd(m,n) = 1$, and let $1 \le r \le n$ with $r \equiv m \bmod n$. Then $\zeta_n^m = \zeta_n^r$ with $\gcd(r,n) = 1$ is a root of Φ_n. By theorem 19.2.27, applied to the identity of $\mathbb{Q}$, the irreducible polynomial Φ_n and its roots ζ_n and ζ_n^m, there is an isomorphism $\sigma\colon \mathbb{Q}(\zeta_n) \to \mathbb{Q}(\zeta_n^m)$ fixing $\mathbb{Q}$ with $\sigma(\zeta_n) = \zeta_n^m$. As ζ_n^m generates the cyclic group μ_n of order n (theorem 2.3.11(3), using $\gcd(m,n) = 1$), ζ_n is a power of ζ_n^m, so $\mathbb{Q}(\zeta_n^m) = \mathbb{Q}(\zeta_n)$ and $\sigma \in \mathrm{Aut}_{\mathbb{Q}}(\mathbb{Q}(\zeta_n))$.

   Conversely, let $\sigma \in \mathrm{Aut}_{\mathbb{Q}}(\mathbb{Q}(\zeta_n))$ satisfy $\sigma(\zeta_n) = \zeta_n^m$. Since σ fixes the rational coefficients of Φ_n, $\Phi_n(\zeta_n^m) = \sigma(\Phi_n(\zeta_n)) = 0$. The roots of Φ_n are the ζ_n^r with $1 \le r \le n$ and $\gcd(r,n) = 1$, so $\zeta_n^m = \zeta_n^r$ for such an r. Then $n \big| m - r$, and $\gcd(m,n) = \gcd(r,n) = 1$.

3. If $[m]_n = [m']_n$ with $\gcd(m,n) = 1$, then $\gcd(m',n) = 1$ and $\zeta_n^{m'} = \zeta_n^m$, so $\sigma_{m'} = \sigma_m$ by uniqueness, and j is well defined. For classes $[m]_n$ and $[m']_n$ in $(\mathbb{Z}/n\mathbb{Z})^\times$, the automorphism $\sigma_m \circ \sigma_{m'}$ sends ζ_n to $\sigma_m(\zeta_n)^{m'} = \zeta_n^{mm'}$, so $\gcd(mm',n) = 1$ by (2), and $\sigma_m \circ \sigma_{m'} = \sigma_{mm'}$ by uniqueness. So $j([m]_n[m']_n) = j([m]_n) \circ j([m']_n)$, and j is a homomorphism from the group $(\mathbb{Z}/n\mathbb{Z})^\times$ (definition 1.2.5) to $\mathrm{Aut}_{\mathbb{Q}}(\mathbb{Q}(\zeta_n))$. The map j is injective: if $\sigma_m = \mathrm{id}$, then $\zeta_n^m = \zeta_n$, so $n \big| m - 1$ and $[m]_n = [1]_n$. It is surjective: by (2), every $\sigma \in \mathrm{Aut}_{\mathbb{Q}}(\mathbb{Q}(\zeta_n))$ sends ζ_n to some ζ_n^r with $\gcd(r,n) = 1$, and then $\sigma = \sigma_r = j([r]_n)$ by uniqueness. Hence j is a bijective homomorphism of groups, that is, an isomorphism. In particular $|\mathrm{Aut}_{\mathbb{Q}}(\mathbb{Q}(\zeta_n))| = \varphi(n) = [\mathbb{Q}(\zeta_n) : \mathbb{Q}]$, since $(\mathbb{Z}/n\mathbb{Z})^\times$ has $\varphi(n)$ elements (definition 1.2.5), completing the proof.

19.7.2 Finite Fields

Let F be a finite field of characteristic p. Then as we remarked under definition 19.1.8,

$$F/\mathbb{F}_p$$

where $\mathbb{F}_p = \mathbb{Z}/p\mathbb{Z}$. Since F is finite, let $d = [F : \mathbb{F}_p]$. Since F is a d-dimensional vector-space over $\mathbb{F}_p$, $F \cong_{\mathbf{Vect}} \mathbb{F}_p^d$, hence the order of $|F|$ is a power of p: $|F| = p^d$.

With everything we have learned, we can now classify all finite fields. Interestingly, we get essentially the best possible result we might ask for when classifying objects: for every order p^d, we get a unique field up to isomorphism:

Theorem 19.7.10: Classifying Finite Fields

Let $q = p^d$ be a power of the prime number p. Then the polynomial $x^q - x$ is separable over $\mathbb{F}_p$, and the splitting field of the polynomial $x^q - x$ over $\mathbb{F}_p$ is a field with exactly q elements. Conversely, let F be a field with exactly q elements. Then F is a splitting field for $x^q - x$ over $\mathbb{F}_p$

Hence, all finite extension of finite fields is abelian, and even better cyclic.

***Proof*:**
Let F be the splitting field of $x^q - x$ over $\mathbb{F}_p$. Then necessarily F is finite, and hence it's perfect by corollary 19.6.12, and hence it's a separable extension, meaning F contains q distinct roots of $x^q - x$. We need to show that every element of F is a root (i.e. $|F| = q$). Let R be the set of roots of $x^q - x$ from F. Then I claim that R is a field. Since F is the smallest field containing all the roots, it follows that $R = F$.

We simply need to show it's a subring, and since all nonzero elements are units, it is a field. For any $a, b \in R$, since $a^q - a = 0$, $b^q - b = 0$, then $a^q = a$ and $b^q = b$. Then using the binomial theorem and the characteristic of F, we get:

$$(a - b)^q = a^q + (-1)^q b^q = a - b$$

where $(-1)^q = -1$ if p is odd, and $(-1)^q = 1 = -1$ if $p = 2$. Furthermore, if $b \neq 0$, then:

$$(ab^{-1})^q = a^q (b^q)^{-1} = ab^{-1}$$

showing that R is closed under both operations, hence it is a subfield, hence $R = F$

On the other hand, let F be a field with q elements. Then $|F^\times| = q - 1$, and so by Lagrange Theorem, for any $0 \neq a \in F^\times$, $|a| \big| q - 1$ Thus:

$$a \neq 0 \Rightarrow a^{q-1} = 1 \Rightarrow a^q - a = 0$$

showing that a *must* be a root of $x^q - x$. Furthermore, $0^q - 0 = 0$, showing that $x^q - x$ has q roots in F (i.e. all elements of F). But then, F is the splitting field of $x^q - x$ by the previous logic, as we sought to show.

Corollary 19.7.11: Unique Finite Field For Every Order

For every $q = p^d$, p a prime and $d \in \mathbb{N}_{>0}$, there exists a unique field of order q

***Proof*:**
this follows from theorem 19.7.10 and the uniqueness of the splitting field (theorem 19.5.10)

Definition 19.7.12: Galois Field

Let F be a field of order $q = p^d$. Then F is called the *Galois Field* of order q, and we usually denote it as $\mathbb{F}_q$ or $\mathbb{F}_{p^d}$.

Example 19.7.13: Galois Fields

The polynomial x^4+1 is reducible over *any* field $\mathbb{F}_p$, and hence over any finite field F of characteristic p, which contains $\mathbb{F}_p$ as its prime subfield. When $p = 2$, the binomial coefficients 4, 6 and 4 are even, so $(x+1)^4 = x^4 + 4x^3 + 6x^2 + 4x + 1 = x^4 + 1$ in $\mathbb{F}_2[x]$.

Let p be odd. Every odd integer $2k+1$ satisfies $(2k+1)^2 - 1 = 4k(k+1)$, and $k(k+1)$ is even, so 8 divides $p^2 - 1$. Write $p^2 - 1 = 8s$. In $\mathbb{F}_p[x]$,

$$x^8 - 1 = (x^4 - 1)(x^4 + 1) \qquad x^{p^2-1} - 1 = (x^8)^s - 1 = (x^8 - 1)\sum_{l=0}^{s-1} x^{8l} \qquad x^{p^2} - x = x(x^{p^2-1} - 1)$$

so $x^4 + 1$ divides $x^8 - 1$, which divides $x^{p^2-1} - 1$, which divides $x^{p^2} - x$. Let E be the splitting field of $x^{p^2} - x$ over $\mathbb{F}_p$. It has p^2 elements (theorem 19.7.10), so $[E : \mathbb{F}_p] = 2$, and $x^{p^2} - x$ is a product of linear factors $x - e$ in $E[x]$. A monic irreducible factor q of $x^4 + 1$ in $E[x]$ is prime in the principal ideal domain $E[x]$ (corollary 11.1.2, proposition 10.1.33) and divides the product of these linear factors, so it divides one of them and $q = x - \alpha$ for some $\alpha \in E$. Thus α is a root of $x^4 + 1$ in E, and

$$\mathbb{F}_p \subseteq \mathbb{F}_p(\alpha) \subseteq E$$

so $[\mathbb{F}_p(\alpha) : \mathbb{F}_p]$ divides $[E : \mathbb{F}_p] = 2$ (corollary 19.3.8). The minimal polynomial of α over $\mathbb{F}_p$ therefore has degree 1 or 2 (theorem 19.2.8, theorem 19.2.5), and it divides x^4+1, since it generates the kernel of evaluation at α. Hence $x^4 + 1$ is a product of two nonconstant polynomials in $\mathbb{F}_p[x]$, so it is reducible over $\mathbb{F}_p$, and the same factorization shows that it is reducible over every field containing $\mathbb{F}_p$.

Corollary 19.7.14: Finite Field Extensions

Let p be a prime and let $d \leq e$ be positive integers. Then there is an extension $\mathbb{F}_{p^d} \subseteq \mathbb{F}_{p^e}$ if and only if $d \big| e$. If $d \big| e$, then there is exactly one such extension, in the sense that $\mathbb{F}_{p^e}$ contains a unique copy of $\mathbb{F}_{p^d}$.

Furthermore, all extensions $\mathbb{F}_{p^d} \subseteq \mathbb{F}_{p^e}$ are simple

***Proof*:**

If $\mathbb{F}_{p^d} \subseteq \mathbb{F}_{p^e}$, then $\mathbb{F}_{p^e}$ is a vector space of some dimension r over $\mathbb{F}_{p^d}$, so $p^e = (p^d)^r$ and $d \big| e$.

Conversely, let $e = dr$. Then $p^e - 1 = (p^d - 1)\sum_{j=0}^{r-1} p^{dj}$, so $p^e - 1 = (p^d - 1)s$ for some integer s, and

$$x^{p^e} - x = x(x^{p^e-1} - 1) = x(x^{p^d-1} - 1)\sum_{j=0}^{s-1} x^{(p^d-1)j} = (x^{p^d} - x)\sum_{j=0}^{s-1} x^{(p^d-1)j}$$

Every $a \in \mathbb{F}_{p^e}^\times$ has $a^{p^e-1} = 1$ (theorem 3.3.1), so the p^e elements of $\mathbb{F}_{p^e}$ are p^e distinct roots of $x^{p^e} - x$, which therefore splits over $\mathbb{F}_{p^e}$ with distinct roots. By unique factorization in $\mathbb{F}_{p^e}[x]$, so does its divisor $x^{p^d} - x$, and its set K of roots has p^d elements. Since $a \mapsto a^{p^d}$ is a ring homomorphism (a power of the Frobenius homomorphism $a \mapsto a^p$ of definition 19.6.9, which preserves sums because $p = 0$ in $\mathbb{F}_{p^e}$ and p divides $\binom{p}{i}$ for $0 < i < p$), K is closed under sums and products, and it contains 0, 1 and the inverse of each of its nonzero elements, so K is a subfield with p^d elements, a copy of $\mathbb{F}_{p^d}$ (corollary 19.7.11). A subfield L with p^d elements satisfies $a^{p^d} = a$ for all $a \in L$, again by theorem 3.3.1, so $L \subseteq K$, and $L = K$ as both have p^d elements.

Finally, the multiplicative group of $\mathbb{F}_{p^e}$ is cyclic (theorem 11.3.14), generated by some α. Every nonzero element of $\mathbb{F}_{p^e}$ is a power of α, so $\mathbb{F}_{p^e} = \mathbb{F}_{p^d}(\alpha)$ and the extension is simple.

Corollary 19.7.15: Separable Finite Extensions Of Finite Fields

Let k be a finite field and F/k a finite separable extension. Then F is a simple extension.

***Proof*:**
Most of the proof of this was already done by proposition 19.6.19. If k is a finite field, then so is F, and the extensions are automatically simple by corollary 19.7.14.

Using these results, we can be more gain a lot of information about polynomials over finite fields:

Corollary 19.7.16: Finite Fields And Irreducible Polynomials

Let F be a finite field. Then for all integers $n \geq 1$, there exists irreducible polynomials of degree n in $F[x]$

***Proof*:**
Let $n \geq 1$ and $F = \mathbb{F}_{p^d}$. Then

$$\mathbb{F}_{p^d} \subseteq \mathbb{F}_{p^{dn}}$$

is a finite simple extension, say $\mathbb{F}_{p^{dn}} = \mathbb{F}_{p^d}(\alpha)$. Then sine

$$[\mathbb{F}_{p^{dn}} : \mathbb{F}_{p^d}] = n$$

and α has a minimal polynomial over $\mathbb{F}_{p^d} = F$, we see that we can find an irreducible polynomial of degree n in $F[x]$.

Note how this proof does not necessarily generalize for infinite fields, since we do not have so much control over the degree of the extension (worth pondering over).

Corollary 19.7.17: Factorizing $x^{q^n} - x$

Let $F = \mathbb{F}_q$ be a finite field and $n \geq 1$. Then $x^{q^n} - x$ is the product of all the monic irreducible polynomials in $F[x]$ whose degree divides n.

***Proof*:**
Let p be the characteristic of F, so q is a power of p, and let $N = x^{q^n} - x$. It is separable (theorem 19.7.10, as q^n is a power of p), so it is the product of its distinct monic irreducible factors in $F[x]$. Let $f \in F[x]$ be monic irreducible of degree d, and α a root of f in some extension of F (theorem 19.2.2). Then $F(\alpha) \cong F[x]/(f)$ has q^d elements (theorem 19.2.8), so $\alpha^{q^d} = \alpha$ (theorem 3.3.1 in $F(\alpha)^\times$ when $\alpha \neq 0$, and trivially when $\alpha = 0$). If $d \mid n$, iterating gives $\alpha^{q^n} = \alpha$, so f, the minimal polynomial of α, divides N. Conversely, suppose f divides N, and let L be a splitting field of N over $F(\alpha)$. As in the proof of corollary 19.7.14, the roots of N in L form a subfield K with q^n elements, since $a \mapsto a^{q^n}$ is a ring homomorphism. K contains α and F, whose elements satisfy $a^q = a$. So $F \subseteq F(\alpha) \subseteq K$ with $[K : F] = n$ and $[F(\alpha) : F] = d$, and $d \mid n$ (theorem 19.3.7).

Example 19.7.18: Factorizing Polynomials over Finite Fields
Over $\mathbb{F}_2$, multiplying out gives

$$x^2 - x = x(x+1) \qquad x^4 - x = x(x+1)(x^2+x+1) \qquad x^8 - x = x(x+1)(x^3+x+1)(x^3+x^2+1)$$

The quadratic and the two cubics have no root in $\mathbb{F}_2$, so they are irreducible (proposition 11.3.5). By corollary 19.7.17, $x^4 - x$ is the product of all monic irreducibles of degree 1 or 2, and $x^8 - x$ of all those of degree 1 or 3. So $x^2 + x + 1$ is the only irreducible quadratic over $\mathbb{F}_2$, and $x^3 + x + 1$ and $x^3 + x^2 + 1$ are the only irreducible cubics.

Another result that finiteness brings is that the structure of $\mathrm{Aut}_{\mathbb{F}_p}(\mathbb{F}_{p^d})$ is very simple. We already know that any finite field is perfect, hence all algebraic extensions are separable, and so

$$|\mathrm{Aut}_{\mathbb{F}_p}(\mathbb{F}_{p^d})| = d$$

However, we can say even more about the structure of the group

Proposition 19.7.19: Structure of Aut Of Finite Fields

Let $\mathbb{F}_{p^d}/\mathbb{F}_p$ be a field extension of degree d. Then $\mathrm{Aut}_{\mathbb{F}_p}(\mathbb{F}_{p^d})$ is cyclic, and generated by the Frobenius isomorphism.

***Proof*:**
Let $\varphi\colon \mathbb{F}_{p^d} \to \mathbb{F}_{p^d}$ be the Frobenius homomorphism $x \mapsto x^p$, a ring homomorphism because $p = 0$ in $\mathbb{F}_{p^d}$ and p divides $\binom{p}{i}$ for $0 < i < p$. It is injective, as a field homomorphism, hence surjective, as $\mathbb{F}_{p^d}$ is finite. It fixes $\mathbb{F}_p$: $0^p = 0$, and every $x \in \mathbb{F}_p^\times$ has $x^{p-1} = 1$ (theorem 3.3.1), so $x^p = x$. So $\varphi \in \mathrm{Aut}_{\mathbb{F}_p}(\mathbb{F}_{p^d})$, a group with at most d elements (corollary 19.2.29, as $\mathbb{F}_{p^d} = \mathbb{F}_p(\alpha)$ for a generator α of its multiplicative group, theorem 11.3.14). If $\varphi^e = \mathrm{id}$ with $e \geq 1$, then every element of $\mathbb{F}_{p^d}$ is a root of $x^{p^e} - x$, so $p^d \leq p^e$ (corollary 11.3.2) and $e \geq d$. Hence φ has order d, so it generates $\mathrm{Aut}_{\mathbb{F}_p}(\mathbb{F}_{p^d})$, which is therefore cyclic of order d.

This result is rather interesting: it means that given $(\mathbb{Z}/m\mathbb{Z})^*$ we have a natural relation $p \bmod m$ associated to the Frobenius morphism $x \mapsto x^p$. This relation to prime being modded and the Frobenius homomorphism shall go far in algebraic number theory ([11]) and will fully be realized in class field theory (see [45]).

Every finite extension $\mathbb{F}_{q^r}/\mathbb{F}_q$ of a finite field has a cyclic automorphism group: $\mathrm{Aut}_{\mathbb{F}_q}(\mathbb{F}_{q^r})$ is a subgroup of $\mathrm{Aut}_{\mathbb{F}_p}(\mathbb{F}_{q^r})$, which is cyclic by proposition 19.7.19, and subgroups of cyclic groups are cyclic (theorem 2.3.11). Over $\mathbb{Q}$ this fails, but every finite *abelian* extension of $\mathbb{Q}$ lies in a cyclotomic field $\mathbb{Q}(\zeta_n)$. This is the Kronecker-Weber theorem, proved in [11, Kronecker Weber's Theorem] and also a consequence of class field theory (*Class Field Theory* [45]).

Theorem 19.7.20: Wedderburn's Little Theorem

Every finite division ring is a field.

***Proof*:**
Let D be a finite division ring (definition 9.1.10), and let

$$Z = \{z \in D : zx = xz \text{ for all } x \in D\}$$

be its center. We show that $Z = D$.

Step 1: Z is a field, and $|D| = q^n$, where $q = |Z|$ and $n = \dim_Z D$. The set Z contains 0 and 1 and is closed under sums, negatives and products. If $0 \neq z \in Z$, then for every $x \in D$

$$z^{-1}x = z^{-1}(xz)z^{-1} = z^{-1}(zx)z^{-1} = xz^{-1}$$

so $z^{-1} \in Z$. Hence Z is a commutative division ring, that is, a field, and $q = |Z| \geq 2$. Multiplication on the left makes D a vector space over Z, of finite dimension n because D is finite. A basis $v_1, \ldots, v_n$ (theorem 13.1.8) writes every element of D uniquely as $c_1v_1 + \cdots + c_nv_n$ with $c_i \in Z$, so $|D| = q^n$. The same count gives $|V| = q^{\dim_Z V}$ for every Z-subspace V of D.

Step 2: centralizers. For $x \in D$ put $C(x) = \{y \in D : xy = yx\}$. It contains Z and is closed under sums and products, so it is a Z-subspace of D, and $|C(x)| = q^{d(x)}$ with $d(x) = \dim_Z C(x)$ by Step 1. If $0 \neq y \in C(x)$, then $y^{-1}x = y^{-1}(xy)y^{-1} = y^{-1}(yx)y^{-1} = xy^{-1}$, so $y^{-1} \in C(x)$. Hence, for $x \neq 0$, the centralizer of x in the group $D^\times$ (definition 2.2.7) is $C(x) \setminus \{0\}$, of order $q^{d(x)} - 1$, and by theorem 3.3.1 the number $q^{d(x)} - 1$ divides $|D^\times| = q^n - 1$. This forces $d(x) \big| n$. Write $n = a\,d(x) + r$ with $0 \leq r < d(x)$, where $a \geq 1$ since $d(x) \leq n$. Then

$$q^n - 1 = q^r\big(q^{a\,d(x)} - 1\big) + \big(q^r - 1\big)$$

and $q^{d(x)} - 1$ divides $q^{a\,d(x)} - 1 = \big(q^{d(x)} - 1\big)\big(q^{(a-1)d(x)} + \cdots + q^{d(x)} + 1\big)$, so it divides $q^r - 1$. Since $0 \leq q^r - 1 < q^{d(x)} - 1$, this gives $q^r = 1$, so $r = 0$.

Step 3: the class equation. The center of the group $D^\times$ is $Z \setminus \{0\}$, since an element of $D^\times$ commuting with every element of $D^\times$ also commutes with 0. Let $x_1, \ldots, x_k$ represent the conjugacy classes of $D^\times$ not contained in its center, and put $d_i = d(x_i)$. Each x_i lies outside Z, so $C(x_i) \neq D$ and $d_i < n$, and $d_i \big| n$ by Step 2. Theorem 4.6.3 gives

$$q^n - 1 = (q - 1) + \sum_{i=1}^{k} \frac{q^n - 1}{q^{d_i} - 1}$$

Step 4: a cyclotomic divisibility. Suppose $n > 1$, and write t for the polynomial variable. By lemma 19.7.4,

$$t^n - 1 = \Phi_n(t) \prod_{\substack{1 \leq e \big| n \\ e < n}} \Phi_e(t)$$

and every factor has integer coefficients by corollary 19.7.5. Substituting $t = q$ shows that the integer $\Phi_n(q)$ divides $q^n - 1$. Let d be a divisor of n with $d < n$. By lemma 19.7.4 again, $t^d - 1$ is the product of the $\Phi_e(t)$ over the divisors e of d, and these are divisors of n different from n. Hence

$$t^n - 1 = (t^d - 1)\,\Phi_n(t)\,g(t) \qquad g(t) = \prod_{\substack{e \mid n,\ e \nmid d \\ e < n}} \Phi_e(t) \in \mathbb{Z}[t]$$

and so $(q^n-1)/(q^d-1) = \Phi_n(q)\,g(q)$ is an integer multiple of $\Phi_n(q)$. Applying this to each d_i, the class equation of Step 3 shows that $\Phi_n(q)$ divides $q - 1$. As $q - 1 \geq 1$, this gives $|\Phi_n(q)| \leq q - 1$.

Step 5: the size of $\Phi_n(q)$. By definition 19.7.2, $|\Phi_n(q)|$ is the product of the numbers $|q - \zeta|$ over the primitive nth roots of unity $\zeta \in \mathbb{C}$, of which there is at least one. As $n > 1$, each such ζ satisfies $\zeta \neq 1$ and $|\zeta| = 1$, so $\operatorname{Re}\zeta < 1$ and

$$|q - \zeta|^2 = q^2 - 2q\operatorname{Re}\zeta + 1 > q^2 - 2q + 1 = (q-1)^2$$

Each factor therefore exceeds $q - 1$. With $m \geq 1$ factors, the product exceeds $(q-1)^m$, which is at least $q - 1$ because $q - 1 \geq 1$. This contradicts Step 4, so $n = 1$. Then $D = Z$ is commutative, and D is a field.

Exercise 19.7.1

1. Let $f(x) \in \mathbb{F}_p[x]$ be irreducible of degree n. Show that the field $\mathbb{F}_p[x]/(f(x))$ is a splitting field of $f(x)$ over $\mathbb{F}_p$, and that it is isomorphic to the splitting field of $x^{p^n} - x$ over $\mathbb{F}_p$.
2. Let $k = \mathbb{F}_p$. Show that $f(x^p) = f(x)^p$ for every $f(x) \in k[x]$.

20

Galois Theory

For eons, mathematicians looked for relations between coefficients and roots of a polynomial. The answer is "trivial" for linear polynomials, given:

$$ax + b = 0 \qquad \text{with solution } \alpha, \text{ then } \quad \alpha = -ba^{-1}$$

For a quadratic $x^2 + bx + c$ with roots α and β, then $\alpha + \beta = -b$ and $\alpha\beta = c$[1], and each individual root is given by the quadratic formula:

$$r = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

More generally, a lot of focus was put into finding the roots of polynomials given their coefficients[2], eventually finding the cubic and quartic equation relating the coefficients to roots of cubic and quartic polynomials respectively. However, the effort to generalize to higher degree polynomials brought limited results, in particular even for the next degree (the quintic) it was unclear how its roots can written in terms of it's coefficients and the arithmetic operations of $+, -, \times, \div$ and $\sqrt{-}$. It wasn't until the turn of the 18th century when serious progress has been made on this front. The key insight was to treat the coefficients of a polynomials as determining a dynamical system (that is, the dynamic of how the roots of a polynomial change as we move around the coefficients). In chapter 19, the right context to consider is the algebraic closure of a field; for pedagogical purposes we stick to $\overline{\mathbb{Q}}$ or $\mathbb{C}$ namely since this is the theories starting point and generalizations to $\mathbb{F}_{q^n}$ and other fields/rings is the natural progression.

To demonstrate these dynamics, consider $x^2 + x - 1 = 0$. This equation (by using the quadratic

[1] See section 20.2.3 for much more detail about these relations

[2] This is a little anachronistic, for when the quadratic formula was first discovered in Babylonia and more succinctly by Brahmagupta, the "modern" notion of polynomial and variables where not used, but rather it was shown by relating areas of circles and rectangles

equations) has solutions:

$$x_1 = \frac{-1+\sqrt{5}}{2} x_2 = \frac{-1-\sqrt{5}}{2}$$

If the b and c coefficients are swapped to get $x^2 - x + 1 = 0$, then the solutions are:

$$x_1 = \frac{1+i\sqrt{3}}{2} x_2 = \frac{1-i\sqrt{3}}{2}$$

In other words if (a, b, c) represents the coefficients, then the permutation action $(2\ 3) \in S_3$ does not leave the roots invariant, that is this is *not* invariant to coefficient permutation. Note these dynamics are "discrete", but they are the shadow of a continuous one. Let $\mathcal{P}_n$ denote the space of monic degree-n complex polynomials with distinct roots, that is, $\mathbb{C}^n$ with the discriminant locus removed. Sending a polynomial to its unordered set of roots exhibits the space of *ordered* root tuples as an n-sheeted covering of $\mathcal{P}_n$, so a *loop* of coefficients returns the root set to itself while permuting the individual roots. This is a homomorphism

$$\pi_1(\mathcal{P}_n) \longrightarrow S_n$$

and the permutation above is one of its values: the discrete action is what a continuous loop leaves behind. The group $\pi_1(\mathcal{P}_n)$ is the braid group B_n, and for a generic polynomial the image of this map is all of S_n, the same S_n that will shortly appear as a Galois group. That a fundamental group turns up here is not an accident; see box 20.0.1.

Now, given a general $ax^2 + bx + c = 0$, factoring as $(x - \alpha)(x - \beta)$, then by commutativity:

$$(x - \alpha)(x - \beta) = (x - \beta)(x - \alpha)$$

In other words, applying an S_2 action on the roots leaves the coefficients invariant[3] [4].

The first immediate introduction of dynamics into polynomials is by applying group-actions to the roots. The realization that links roots to coefficients is that, given we can completely split a polynomial:

$$x^n + \cdots + \alpha_1 x + \alpha_0 = (x - a_1)(x - a_2) \cdots (x - a_n)$$

each coefficient of the original polynomial can be represented as a *symmetric function* (see section 20.2.3).

$$\begin{aligned}
\alpha_{n-1} &= -(a_1 + a_2 + \cdots + a_n) \\
\alpha_{n-2} &= a_1 a_2 + a_1 a_3 + \cdots + a_1 a_n + a_2 a_3 + \cdots a_{n-1} a_n \\
&\cdots \\
\alpha_0 &= (-1)^n a_1 a_2 \cdots a_n
\end{aligned}$$

the signs alternating, exactly as in the quadratic case above where $\alpha + \beta = -b$ and $\alpha\beta = c$. These form a natural relationship between the roots and the coefficients. It was noticed that these equations are invariant under permutation action of S_n; if $A = \{A_0, ..., A_{n-1}\}$ represent the above n equations and the action is given by $(r\ s) \cdot a_r = a_s$, then $\sigma \cdot A_i = A_i$ for each A_i and σ, hence the action $S_n \curvearrowright A$ is the trivial action. Thus, there is always a natural group-action.

[3] Concretely, take $x^2 - c$, whose roots are $\pm\sqrt{c}$. As c travels once around the circle $c = re^{i\theta}$, with θ running from 0 to 2π, the roots $\pm\sqrt{r}e^{i\theta/2}$ trade places: a single loop of the coefficient around the discriminant realises the transposition in S_2

[4] As a consequence, there is no continuous function that can take in coefficients and return roots

As any roots of polynomial induce such equations and induce this trivial action, these don't give enough information about polynomials. It shall be shown that roots of a generic polynomial satisfy no further equation, and in these cases there is nothing more that can be done to understand a polynomial via a group-action on its roots. However, there are polynomials whose roots shall have *more* relations the roots satisfy, limiting what group can act on them (i.e. a subgroup of S_n will act trivially, but S_n will not). This chapter shall make rigorous this realization and translate it into the modern language of field theory.

This realization almost managed to show the general quintic is unsolvable:

> "The quintic was almost proven to have no general solutions by radicals by Paolo Ruffini in 1799, whose key insight was to use permutation groups, not just a single permutation. His solution contained a gap, which Cauchy considered minor, though this was not patched until the work of the Norwegian mathematician Niels Henrik Abel, who published a proof in 1824, thus establishing the Abel-Ruffini theorem."[5]

This theory was established with the aim to show there is an unsolvable quintic, but not how to find whether a given quintic is solvable. This is where the work of Évariste Galois comes in. He noticed that the key insight was to work with all possible relationships of the roots of the polynomial, assuming all the roots exist.

Modern uses of Galois Theory

Galois groups are central in both algebraic geometry and number theory, where they serve different purposes in each domain. The galois group of a polynomial:

1. establishes the algebraic relations between the roots and creates a "covering space" that has some similar properties to the covering spaces seen in algebraic topology (see box 20.0.1),
2. establishes the invariance necessary to find a solution over a lower field, in particular if L/K is a field extension and a root of an $L[x]$ polynomial has a root in K, then the root is invariant under the $\mathrm{Gal}_K(L)$-action.

From action on roots to field theory

Take the polynomial:

$$x^4 - 2 = 0$$

which is irreducible over $\mathbb{Q}$ by Eisenstein's criterion (proposition 11.3.10). Adjoining $\sqrt[4]{2}$, $\mathbb{Q}(\sqrt[4]{2})$, the polynomial splits to:

$$(x - \sqrt[4]{2})(x + \sqrt[4]{2})(x^2 + \sqrt{2})$$

Adjoining $\sqrt[4]{2}i$, the polynomial splits completely:

$$(x - \sqrt[4]{2})(x + \sqrt[4]{2})(x - \sqrt[4]{2}i)(x + \sqrt[4]{2}i) \tag{20.1}$$

Already, there is some interesting phenomena at play, for example, another way of getting this splitting is by adjoining $\sqrt[4]{2}$ and then adjoining i (or adjoining i then $\sqrt[4]{2}$). In fact, by just adjoining

[5] taken from Wikipedia for Galois Theory

$\sqrt[4]{2}+i$, we immediately get this splitting. Indeed, certainly $\mathbb{Q}(\sqrt[4]{2}+i) \subseteq \mathbb{Q}(\sqrt[4]{2}, i)$. Conversely, to show $i \in \mathbb{Q}(\sqrt[4]{2}+i)$, letting $a = \sqrt[4]{2}+i$, then:

$$\begin{aligned} a - i &= \sqrt[4]{2} \\ \Leftrightarrow\ (a-i)^4 &= 2 \\ \Leftrightarrow\ (a-i)^4 &= 2 \\ \Leftrightarrow\ a^4 - 4a^3 i - 6a^2 + 4ai + 1 &= 2 \\ \Leftrightarrow\ i(4a - 4a^3) &= 1 - a^4 + 6a^2 \\ \Leftrightarrow\ i &= \frac{1 - a^4 + 6a^2}{4a - 4a^3} \end{aligned}$$

showing that $i \in \mathbb{Q}(\sqrt[4]{2}+i)$, and so $a - i = \sqrt[4]{2} \in \mathbb{Q}(\sqrt[4]{2}+i)$. More generally for any finite separable extension $K/\mathbb{Q}$, such an element always exists and is constructible by the primitive element theorem (theorem 19.6.20).

Let us understand the equations the roots form given by symmetric functions. As $x^4 - 2 = x^4 + 0x^3 + 0x^2 + 0x - 2$, the only meaningful relation obtained by multiplying out the roots in (20.1) is:

$$(\sqrt[4]{2})(-\sqrt[4]{2})(\sqrt[4]{2}i)(-\sqrt[4]{2}i) = -2 \qquad \text{equiv.} \qquad (\sqrt[4]{2})(-\sqrt[4]{2})(\sqrt[4]{2}i)(-\sqrt[4]{2}i) + 2 = 0$$

This equation is invariant under the action of S_4, namely if

$$(a_1, a_2, a_3, a_4) = (\sqrt[4]{2}, \sqrt[4]{2}i, -\sqrt[4]{2}, -\sqrt[4]{2}i)$$

represent the 4 roots, and $A_0 = (\sqrt[4]{2})(-\sqrt[4]{2})(\sqrt[4]{2}i)(-\sqrt[4]{2}i) + 2 = 0$, then:

$$\sigma \cdot A_0 = A_0 \qquad \forall \sigma \in S_4$$

However, there are more equations that can be formed with these roots (using the operations of a field). With the labelling above, the first and third roots are negatives of one another, as are the second and fourth:

$$\begin{aligned} A_1 :\ a_1 + a_3 &= \sqrt[4]{2} + (-\sqrt[4]{2}) = 0 \\ A_2 :\ a_2 + a_4 &= \sqrt[4]{2}i + (-\sqrt[4]{2}i) = 0 \end{aligned}$$

These equations are *not* invariant under the permutation action of S_4:

$$(1\ 2) \cdot A_1 \neq A_1 \qquad \text{(concretely: } (1\ 2) \cdot (a_1 + a_3) = a_2 + a_3 = \sqrt[4]{2}i - \sqrt[4]{2} \neq 0)$$

A subgroup of S_4 must be chosen, namely notice that $D_4 \cong \langle (1\ 2\ 3\ 4), (1\ 3) \rangle \subseteq S_4$[6] keeps them invariant. It does not fix each one separately: the four-cycle carries A_1 to A_2 and back. What it preserves is the *pairing* $\{\{1,3\},\{2,4\}\}$ of the roots into negative pairs, and so the pair of equations as a set:

$$D_4 \cdot A_0 = A_0 \qquad D_4 \cdot \{A_1, A_2\} = \{A_1, A_2\}$$

Indeed D_4 is precisely the stabiliser of that pairing inside S_4, and it has order $8 = [\mathbb{Q}(\sqrt[4]{2}, i) : \mathbb{Q}]$, the first hint that this is the right subgroup. Are there any other equations that we can be considered

[6]The reader may recall exercise set 1.3.1 asks to show S_n is generated by a two cycle and an n-cycle; it was important the 2-cycle was an *adjacent* cycle

which would further restrict the choice of invariant permutation action? To answer this, the above is re-framed in the language of field extensions.

Let us say f is a polynomial over $\mathbb{Q}$ and that α is a root of f. Then the algebraic extension $\mathbb{Q}(\alpha)/\mathbb{Q}$ contains *all* possible polynomial equations that can be generated with elements from $\mathbb{Q}$ and α (recall $\mathbb{Q}(\alpha) = \mathbb{Q}[\alpha]$, see proposition 19.2.24). If any of these polynomials happen to be another root of f, that is $p(\alpha) = \beta$ where $f(\beta) = 0$ and $p(x) \in \mathbb{Q}[x]$, then $\mathbb{Q}(\alpha)/\mathbb{Q}$ contains this relationship between the roots. Let's say one of these relationships is represented as :

$$P(x) = p(x) - \beta \text{ so that } P(\alpha) = p(\alpha) - \beta = 0$$

Then a $\mathbb{Q}$-automorphism (in this case nothing more than a field automorphism) would have to preserve this relationship (it must map roots to roots). Hence, the set of $\mathbb{Q}$-automorphism captures the relationships between α and all possible roots that are $\mathbb{Q}$-polynomials of α. If $K/\mathbb{Q}$ is the splitting field of f, then the $\mathbb{Q}$-automorphisms of K would be restricted to automorphism that must satisfy all possible relationships between roots. Given $K/\mathbb{Q}$ is a splitting field for f, as any $\mathbb{Q}$-automorphisms σ is determined by where the roots map to, it is associated to a permutation in S_n. If there were no relationships (you need to add n roots to split f), then the automorphism group would be isomorphic to S_n. If there is any other type of group, that means there are some relationships between the roots (just like the example of $\mathbb{Q}(\sqrt[4]{2} + i)/\mathbb{Q}$).

The automorphism group of the splitting field above is called the *galois group* and it is the aim of this chapter define the galois group for any field extension L/K, to be able to find some galois groups, to find how frequent certain galois groups are, to show the correspondence between field extension and galois groups, and answer some long-pending mathematical questions such as the solvability of [general] quintic using the newly developed theory.

Another way of thinking about galois group is they are a rough classification tool for field extensions: if two field extensions have different Galois groups they cannot be the same extensions. It is similar in roughness to the fundamental group $\pi_1(X)$ as is often encountered in algebraic topology. To see this roughness in classification, as the splitting field $\mathbb{Q}(\sqrt[4]{2}, i)$ of $x^4 - 2$ is equal to $\mathbb{Q}(\sqrt[4]{2} + i)$, and the minimal polynomial of $\sqrt[4]{2} + i$ is:

$$x^8 + 4x^6 + 2x^4 + 28x^2 + 1$$

then they share the same restrictions on their roots as they have the same galois group, but they certainly also have many differences (for example, these two polynomials have different discriminants, see definition 19.6.27).

20.0.1 Galois Group as Fundamental Group

(Assuming strong familiarity with algebraic geometry and algebraic topology) Under the right framework in algebraic geometry, galois groups can be interpreted as *étale fundamental group* for $\operatorname{Spec} K$,

$$\pi_1^{\text{ét}}(\operatorname{Spec}(K)) \cong \operatorname{Gal}_K(\overline{K})$$

The name *fundamental group* is well-deserved for if X is a connected $\mathbb{C}$-scheme of finite type (i.e. "essentially" a connected variety) and $X(\mathbb{C})$ is its analytification, then:

$$\pi_1^{\text{ét}}(X) \cong \widehat{\pi_1^{\text{top}}(X(\mathbb{C}))}$$

where $\pi_1^{\text{top}}(-)$ is the usual fundamental group and $\widehat{(-)}$ is the profinite completion with respect to its finite-index normal subgroups. See [9] for the details.

20.1 Galois Extension and Galois Group

First, let's the recall the definition of an automorphism and define a new term:

Definition 20.1.1: Automorphism and fixed Element

1. An isomorphism σ of F with itself is called an *automorphism* of F. The collection of automorphisms of F is denoted $\mathrm{Aut}(F)$. If $\alpha \in F$, we write $\sigma\alpha$ for $\sigma(\alpha)$
2. An auotmorphism $\sigma \in \mathrm{Aut}(F)$ is said to *fix* an element $\alpha \in F$ if $\sigma\alpha = \alpha$. If k is a subset of F, then an automorphism σ is said to *fix* F if it fixes all the elements of F

$$\sigma a = a \qquad \forall a \in k$$

3. Let F/k be an extension of fields. Then $\mathrm{Aut}(F)$ is the collection of automorphissm of F, and $\mathrm{Aut}_k(F)$ is collection of automorphsism of F which fix F.

Example 20.1.2: Automorphism and Fixed F

The identity map id is always an automorphism.

Furthermore, the prime field of any field K is always fixed. If P is the prime-field of F, then for any $\sigma \in \mathrm{Aut}(K)$, $\sigma(1) = 1$, since $\sigma(0) = 0$. Thus, $\sigma(a) = a$ for every $a \in P$. This means that $\mathbb{Q}$ and $\mathbb{F}_p$ have only trivial automorphisms

$$\mathrm{Aut}(\mathbb{Q}) = \{\mathrm{id}\}, \mathrm{Aut}(\mathbb{F}_p) = \{\mathrm{id}\}$$

and if P is a prime field of F, then

$$\mathrm{Aut}_P(F) = \mathrm{Aut}(F)$$

Next, we recall a very useful criterion for finding automorphisms of algebraic extensions K/F

Proposition 20.1.3: Fixed Roots Criterion

Let F/k be a field extension and let $\alpha \in F$ be algebraic over k. Then for any $\sigma \in \mathrm{Aut}(F/k)$, $\sigma\alpha$ is a root of the minimal polynomial for α over k, i.e., $\mathrm{Aut}(F/k)$ permutes the roots of irreducible polynomials. Equivalently, any polynomial whith coefficeints in k having α as a root also has $\sigma\alpha$ as a root.
Furthermore, $\mathrm{Aut}_k(k(\alpha))$ acts *transitively* on the set of roots of $m_{k,\alpha}(x)$ (i.e. if β_1, β_2 are roots of $m_{k,\alpha}(x)$, then there exists a $\sigma \in \mathrm{Aut}_k(k(\alpha))$ such that $\sigma\beta_1 = \beta_2$)

***Proof*:**

Let

$$x^n + a_{n-1}x^{n-1} + \cdots + a_1 x + a_0$$

be the minimal polynomial with coefficients from F so that

$$\alpha^n + a_{n-1}\alpha^{n-1} + \cdots + a_1\alpha + a_0 = 0$$

then, applying any automorphism σ to this polynomial will get us

$$\sigma(\alpha^n + a_{n-1}\alpha^{n-1} + \cdots + a_1\alpha + a_0) = \sigma(0)$$
$$\sigma(\alpha)^n + a_{n-1}\sigma(\alpha)^{n-1} + \cdots + a_1\sigma(\alpha) + \sigma(a_0) = 0 \qquad \sigma \text{ fixes } a_i \text{ for each } i$$

since $a_i \in F$ for each i, showing that $\sigma(\alpha)$ is also a root of the minimal polynomial

Transitivty is left as an exercise (recall the appropriate theorem)

Example 20.1.4: Automorphism Groups

1. Take $K = \mathbb{Q}(\sqrt{2})$. Then $\text{Aut}(\mathbb{Q}(\sqrt{2})) = \text{Aut}(\mathbb{Q}(\sqrt{2})/\mathbb{Q})$. If $\sigma \in \text{Aut}(\mathbb{Q}(\sqrt{2}))$, then by the fixed root criterion, it must be that

   $$\sigma(\sqrt{2}) = \pm\sqrt{2}$$

   and since σ must fix everything else in place, this means that we get

   $$\sigma(a + b\sqrt{2}) = a \pm b\sqrt{2}$$

   in other words, we have determined were every other element must map to! Thus, we have the maps

   $$\sqrt{2} \mapsto \sqrt{2} \qquad \sqrt{2} \mapsto \sqrt{-2}$$

   If we let the first map be represetned by 1 and the second by σ, then we get

   $$\text{Aut}(\mathbb{Q}(\sqrt{2})) = \{1, \sigma\} = C_2$$

   that is, $\text{Aut}(\mathbb{Q}(\sqrt{2}))$ is a cyclic group of order 2.

2. Let $K = \mathbb{Q}(\sqrt[3]{2})$. By theorem 19.2.5, every element of $\mathbb{Q}$ can be written as

   $$a + b\sqrt[3]{2} + c(\sqrt[3]{2})^2$$

   If we consider any $\sigma \in \text{Aut}(\mathbb{Q}(\sqrt[3]{2}))$, then we get

   $$a + b\sigma(\sqrt[3]{2}) + c\sigma(\sqrt[3]{2}) \qquad a, b, c \in \mathbb{Q}$$

   and since $\mathbb{Q}(\sqrt[3]{2})$ has non of it's other roots (recall example 19.3.12). Then that means that the only automorphism is the identity, meaning

   $$\text{Aut}(\mathbb{Q}(\sqrt[3]{2})) = \{1\}$$

3. Now let $K = \mathbb{Q}(\sqrt[3]{2}, \sqrt{-3})$, be the splitting field of $x^3 - 2$ in example 19.5.11, which unlike the previous example will insure we have interesting automorphisms since K contains all the roots of $x^3 - 2$. Let's label these roots $\sqrt[3]{2}, \omega, \omega^2$. Then:

   $$[\mathbb{Q}(\sqrt[3]{2}, \sqrt{-3}) : \mathbb{Q}] = [\mathbb{Q}(\sqrt[3]{2}, \sqrt{-3}) : \mathbb{Q}(\sqrt[3]{2})] : [\mathbb{Q}(\sqrt[3]{2}) : \mathbb{Q}] = 2 \cdot 3 = 6$$

   We know that $\mathbb{Q}(\sqrt[3]{2})$ contains one root of $x^3 - 2$ over $\mathbb{Q}$ ($\sqrt[3]{2} \in \mathbb{Q}(\sqrt[3]{2})$), and that $\mathbb{Q}(\sqrt[3]{2}, \omega)$ also contains the other two roots over $\mathbb{Q}$, and in particular over $\mathbb{Q}(\sqrt[3]{2})$ ($\omega, \omega^2 \in \mathbb{Q}(\sqrt[3]{2}, \omega)$.

Next, Since the degree of K is 6, we know it has a basis consisting of 6 elements, namely:

$$1,\ \sqrt[3]{2},\ \sqrt[3]{4},\ \omega,\ \omega\sqrt[3]{2},\ \omega\sqrt[3]{4}$$

Since any $\mathbb{Q}$-automorphism of K is a $\mathbb{Q}$-linear automorphism on K, it suffices to characterise these automorphisms by where by map these basis elements. By the fixed roots criterion, a $\mathbb{Q}$-automorphism σ on K can map $\sqrt[3]{2}$ to:

$$\sqrt[3]{2},\ \omega\sqrt[3]{2},\ \omega^2\sqrt[3]{2}$$

and map ω to:

$$\omega,\ \omega^2$$

Using this information, we can carry through computations to see that $\mathrm{Aut}_{\mathbb{Q}}(K)$ is S_3.

4. More on computing automorphisms groups here: here

Lemma 20.1.5: Field Automorphism Is Functorial

Let $\mathcal{N}_k$ be the category whose objects are the finite normal extensions of k and whose morphisms are the k-homomorphisms. Then $\mathrm{Aut}_k(-)$, sending F to $\mathrm{Aut}_k(F)$ and a morphism $\iota\colon F \to L$ to $\mathrm{Aut}_k(L) \to \mathrm{Aut}_k(F)$, $\varphi \mapsto \iota^{-1} \circ \varphi \circ \iota$, is a contravariant functor $\mathcal{N}_k \to \mathbf{Grp}$.

***Proof*:**
The field homomorphism ι is injective, and $\iota(F)$ is finite and normal over k, being k-isomorphic to F. For $\varphi \in \mathrm{Aut}_k(L)$, φ restricts to a k-homomorphism from $\iota(F)$ into $\overline{L}$, an algebraic closure of $\iota(F)$, so $\varphi(\iota(F)) = \iota(F)$ (theorem 19.5.13) and $\iota^{-1} \circ \varphi \circ \iota \in \mathrm{Aut}_k(F)$. Conjugating by ι preserves composition and identities, so each map is a group homomorphism, and for a second morphism $\kappa\colon L \to M$ and $\psi \in \mathrm{Aut}_k(M)$,

$$(\kappa \circ \iota)^{-1} \circ \psi \circ (\kappa \circ \iota) = \iota^{-1} \circ (\kappa^{-1} \circ \psi \circ \kappa) \circ \iota$$

which is contravariance.

Normality is needed: in $\mathbb{Q}(\sqrt[3]{2}, \omega)$, $\omega = e^{2\pi i/3}$, some $\mathbb{Q}$-automorphism sends $\sqrt[3]{2}$ to $\omega\sqrt[3]{2} \notin \mathbb{Q}(\sqrt[3]{2})$, so it does not restrict to $\mathbb{Q}(\sqrt[3]{2})$. For groups, restriction along an injective homomorphism can fail to exist: for $f\colon \mathbb{Z}/2\mathbb{Z} \to S_3$ with image $\langle (1\,2) \rangle$, conjugation by $(1\,3)$ is an automorphism of S_3 that moves $f(\mathbb{Z}/2\mathbb{Z})$.

Beyond functoriality, $\mathrm{Aut}_k(-)$ preserves more structure, namely the latice of groups and interemetiate fields is [contravariantly] preserved. The following propositions builds up to this result. Since $\mathrm{Aut}_k(F)$ has a relatively easy to find structure, any connections between the structure of $\mathrm{Aut}_k(F)$ (which is a finite group if $[F : k] < \infty$) and F/k would allow us to use the full power of group theory (and better still finite group theory) to analise field extensions. We endevaour to deepen this connection in this chapter. One of the most important connection is that intermediate fields $k \subseteq E \subseteq F$ correspond to subgroups of $\mathrm{Aut}(F/k)$! This is called the *Galois correspondence*, and is what we will be building up for the rest of this section. We start by showing how subgroups of $\mathrm{Aut}_k(F)$ correspond to an intermediate field $k \subseteq E \subseteq F$, and conversely intermediate extension $k \subseteq E \subseteq F$ correspond a subgroup fo $\mathrm{Aut}_k(F)$

Proposition 20.1.6: Fixing F with automorphisms

Let $H \leqslant \mathrm{Aut}_k(F)$ be a subgroup of automorphisms of F (or $H \subseteq \mathrm{Aut}_k(F)$ more generally). Then the collection F^H of elements of F fixed by all the elements of H is a subfield of F extending k, $k \subseteq F^H \subseteq F$. Conversely, Let E be an intermediate field $k \subseteq E \subseteq F$. Then $\mathrm{Aut}_E(F) \leqslant \mathrm{Aut}_k(F)$.

***Proof*:**
Let F^H be the collection of fixed elements by element $h \in H$. Then by definition, every element of k is fixed by H, so F^H is nonempty and $k \subseteq F^H$. Next, for any $a, b \in F^H$

$$h(a \pm b) = h(a) \pm h(b) = a \pm b \qquad h(ab) = h(a)h(b) = ab \qquad h(a^{-1}) = h(a)^{-1} = a^{-1}$$

that is, F^H is closed under the operations of F, and so $F^H \subseteq F$, showing that $k \subseteq F^H \subseteq F$.

For the other direction, let $\mathrm{Aut}_E(F)$ be the set of automorphisms that fix E. This subset is nonempty since $\mathrm{id}_F \in \mathrm{Aut}_E(F)$. If $\sigma \in \mathrm{Aut}_E(F)$, then $\sigma^{-1} \in \mathrm{Aut}_E(F)$: applying σ^{-1} to $\sigma(e) = e$ gives $\sigma^{-1}(e) = e$ for every $e \in E$. Finally, for any $\sigma, \tau \in \mathrm{Aut}_E(F)$, given $e \in E$,

$$\sigma \circ \tau^{-1}(e_1) = \sigma(\tau^{-1}(e)) = \sigma(e) = e$$

showing that $\sigma \circ \tau^{-1} \in \mathrm{Aut}_E(F)$, an so $\mathrm{Aut}_E(F) \leqslant \mathrm{Aut}_k(F)$, completing the proof.

Definition 20.1.7: Fixed Field

Let $H \subseteq \mathrm{Aut}_k(F)$. Then the subfield of F fixed by all the elements of H is called the *fixed field* of H, and is denoted F^H.

Example 20.1.8: Fixed Field

1. The fixed field of $\mathrm{Aut}_{\mathbb{Q}}(\mathbb{Q}(\sqrt{2}))$ is $\mathbb{Q}$. As we have seen in example 20.1.4.
2. The fixed field of $\mathrm{Aut}_{\mathbb{Q}}(\mathbb{Q}(\sqrt[3]{2}))$ is $\mathbb{Q}(\sqrt[3]{2})$, since $\mathrm{Aut}_{\mathbb{Q}}(\sqrt[3]{2}) = \{\mathrm{id}\}$. If we add the other roots of $x^3 - 2$, then the fixed field of $\mathrm{Aut}_{\mathbb{Q}}(\mathbb{Q}(\sqrt[3]{2}, \omega))$ is $\mathbb{Q}$.

Using this vocabulary, we can make the following correspondence betweeen F/k and its intermediate fields with $\mathrm{Aut}_k(F)$ and it's subgroups:

Proposition 20.1.9: Galois Correspondence

Let $k \subseteq F$ be a fields extension. Then there exists a correspondence:

$$\left\{\begin{matrix}\text{Intermediate fields } E \\ k \subseteq E \subseteq F\end{matrix}\right\} \underset{F^H}{\overset{\text{Aut}_E(F)}{\rightleftarrows}} \left\{\begin{matrix}\text{subgroups } H \\ H \subseteq \text{Aut}_k(F)\end{matrix}\right\}$$

sending an intermediate field E to the subgroup $\text{Aut}_E(F)$, and a subgroup H of $\text{Aut}_k(F)$ to the fixed field F^H. Furthermore, we have the following properies:

1. The correspondence is inclusion reversing, that is:
   - if $k \subseteq E_1 \subseteq E_2 \subseteq F$ are two subfield of F extending k , then $\text{Aut}_{E_2}(F) \leqslant \text{Aut}_{E_1}(F)$
   - if $H_1 \leqslant H_2 \leqslant \text{Aut}_k(F)$ are two subgroups of automorphsism with associated fixed fields F^{H_1} and F^{H_2} respectively, then $F^{H_2} \subseteq F^{H_1}$.
2. For all subgroups $H \leqslant \text{Aut}_k(F)$ and intermediate fields $k \subseteq E \subseteq F$
   - $E \subseteq F^{\text{Aut}_E(F)}$
   - $H \subseteq \text{Aut}_{F^H}(F)$
3. If E_1E_2 is the smallest subfield of F containing the two intermediate the E_1 and E_2, and $\langle G_1, G_2\rangle$ denotes the smallest subgroup of $\text{Aut}_k(F)$ containing G_1 and G_2, then:

$$\begin{aligned}\text{Aut}_{E_1E_2}(F) &= \text{Aut}_{E_1}(F) \cap \text{Aut}_{E_2}(F) \\ F^{\langle G_1,G_2\rangle} &= F^{G_1} \cap F^{G_2}\end{aligned}$$

***Proof*:**
The Galois correspondence is an immediate consequence of proposition 20.1.6.

For the inclusion-reversal, consider the extension, $k \subseteq E_1 \subseteq E_2 \subseteq F$. Then $\text{Aut}_{E_2}(F)$ *a fortiori* fixes all the elements of E_1, since $E_1 \subseteq E_2$, and so can only be *smaller* as the number of elements in Aut_{E_2} depend on the number of roots that can be premuted (as shown in the root permutation Criterion). A similar argument can be made for $H_1 \leqslant H_2 \leqslant \text{Aut}_k(F)$.

For the subsets, Take the intermediate field E, $k \subseteq E \subseteq F$. Then $F^{\text{Aut}_E(F)}$ is the set of elements That are fixed by the automorphisms that fix E. Then it is immediate that $E \subseteq F^{\text{Aut}_E(F)}$. Similarly for $H \subseteq \text{Aut}_{F^H}(F)$.

For the minimality statements, If E_1E_2 is the smallest of F containing both E_1 and E_2, then the $\subseteq$ inclusion is immediate (since $\text{Aut}_{E_1}(F) \cap \text{Aut}_{E_2}(F)$ contains only elements that fix E_1 and E_2), while the $\supseteq$ comes from the fact that $\text{Aut}_{E_1}(F)$ and $\text{Aut}_{E_2}(F)$ are in themselves the minimal subgroups of $\text{Aut}_k(F)$ that fix all elements of E_1 and E_2. A similar argument can be made for the second assertion.

It is natural to ask whether the galois correspondence is bijective. Before showing when it is, we will show that it is possible that it is not:

Example 20.1.10: non-bijective Galois Correspondence

Consider $Q(\sqrt[3]{2})/\mathbb{Q}$. Then we know that $[\mathbb{Q}(\sqrt{3})/\mathbb{Q}] = 3$, which is prime, and so the only intermediate fields are $\mathbb{Q}$ and $\mathbb{Q}(\sqrt[3]{2})$. For $\text{Aut}_{\mathbb{Q}}(\mathbb{Q}(\sqrt[3]{2}))$, recall that $\mathbb{Q}(\sqrt[3]{2}) \subseteq \mathbb{R}$, and that $\sqrt[3]{2}$ is the only root of the minimal polynomial within $\mathbb{R}$. Thus, since automorphisms of fields permute roots, $\text{Aut}_{\mathbb{Q}}(\mathbb{Q}(\sqrt[3]{2}))$ can only have one element:

$$\text{Aut}_{\mathbb{Q}}(\mathbb{Q}(\sqrt[3]{2})) = \{e\}$$

Hence, we have:

$$\{\mathbb{Q}, \mathbb{Q}(\sqrt[3]{2})\} \rightleftarrows \{\,\{e\}\,\} = \{\text{Aut}_{\mathbb{Q}}(\mathbb{Q}(\sqrt[3]{2}))\}$$

showing that the function from intermediate fields to subgroups of the automorphism group is not injective with respect to intermediate fields in general[a] (for every intermediate field, we may map it to the same subgroup of $\text{Aut}_k(F)$). In this case, $\mathbb{Q} \subseteq F^{\text{Aut}_{\mathbb{Q}}(\mathbb{Q}(\sqrt[3]{2}))}$ and $\mathbb{Q}(\sqrt[3]{2}) \subseteq F^{\text{Aut}_{\mathbb{Q}}(\mathbb{Q}(\sqrt[3]{2}))}$

[a] Or similarly, the correspondence is not surjective with respect to subgroups of $\text{Aut}_k(F)$

This examples shows that it possible that $E \subsetneq F^{\text{Aut}_E(F)}$. Fortunately, in the finite case, we always have that for any $H \leqslant \text{Aut}_k(F)$, there exists an intermediate field (namely F^H) such that $H = \text{Aut}_{F^H}(F)$ (and hence, the correspondence is at least always surjective with respect to intermediate fields). To prove this result we need the following lemma which shows that the field F^H is essentially as nice as a field as we want. The high-level idea is that automorphisms in $H = \text{Aut}_E(F)$ fix E, and F^H is *defined* to be the field that is fixed by H:

Lemma 20.1.11: Properties Of Fixed Field

Let F/k be a finite extension, and let $H \leqslant \text{Aut}_k(F)$. Then F/F^H is normal and separable. It is furthermore finite and simple.

Note that we are not given that F^H or k is a perfect field, and so separability is a powerful result. Similarly, we do not know beforehand that F is a normal extension of k, yet F is a normal extension of F/F^H. Note too that we can take $H = \{e\}$ so that $F/F^{\{e\}}$ has all the aforementioned properies. As we have seen in the previous example, it is not necessarily the case that $F^{\{e\}} = k$. (in the previous example, we had $F^{\{e\}} = \mathbb{Q}(\sqrt[3]{2})$)

***Proof*:**

First, F/F^H is finite since F/k is finite, so by the tower law:

$$[F : k] = [F : F^H][F^H : k]$$

Next, to show F/F^H is a separable extension, we will show any $\alpha \in F$ is separable over F^H. To do so, we use the following clever trick: Let $\alpha \in F$ be any element of F (possibly an element of F^H). We will construct the minimal polynomial of α, and show that it is separable. Take $\sigma \in H$. Since F is finite over k, α is the root of some minimal polynomial, $m_{\alpha,k}(x)$. By the fixed roots criterion, $h\alpha$ must be a root of the minimal polynomial $m_{\alpha,k}(x)$ of α over k (which might also imply $h\alpha = \alpha$ if no other root are in F). Since $m_{\alpha,k}(x)$ has only finitely many roots, the H-orbit of α (i.e. all possibles distinct images of α under all possible $\sigma \in H$) consists of finitely many elements $\alpha_1, ..., \alpha_n$. Now, the group H acts on the H-orbit by premuting its elements.

Therefore, every element of H keeps the polynomial:

$$q_\alpha(x) = (x-\alpha_1)(x-\alpha_2)\cdots(x-\alpha_n)$$

fixed (where $\alpha_1 = \alpha$). In other words, the coefficients of $q_\alpha(x)$ *must* be in the fixed field F^H: $q_\alpha(x) \in F^H[x]$! Next, we show that $q_\alpha(x) = m_{\alpha,F^H}(x)$, which will show that the minimal polynomial corresponding to α is separable, showing α is separable. To see this, let $g(x)$ be any other polynomial having α as a root. Then:

$$g(\alpha) = b_m\alpha^m + \cdots + b_1\alpha + b_0 = 0$$

and by definition, any $\sigma \in H$ will map α to another root α_i, so:

$$g(\alpha_i) = b_m\alpha_i^m + \cdots + b_1\alpha_i + b_0 = 0$$

meaning $g(x)$ is divisible by $x-\alpha_i$ for every i, and hence $q_\alpha(x)\big|g(x)$. Since the minimal polynomial of α over F^H also has α as a root, we have that $q_\alpha(x)\big|m_{\alpha,f^H}(x)$, and as we have shown before $m_{\alpha,F^H}(x)\big|q_\alpha(x)$. Thus, $q_\alpha(x) = m_{\alpha,F^H}(x)$ up to a constant. Thus, α is separable and since α was arbitrary, F/F^H is a separable extension (Note that it's possible that $\alpha \in F^H$, in which case $q_\alpha(x) = x-\alpha$, which is certainly the minimal polynomial of α over F^H and is separable).

Next, since F/F^H is finite and separable, then by theorem 19.6.20, it simple. Let's say $F = F^H(\beta)$, so $F^H(\beta)/F^H$ is simple. Let $q_\beta(x)$ be define as we have done above (note that $q_\beta(x) = m_{\beta,F^H}(x)$ and that it is not possible that $q_\beta(x) = x-\beta$ unless $F = F^H$)

Finally, since $q_\beta(x) \in F^H[x]$ splits in F, and F is generated over F^H (namely, F is generated by F and β), F is the splitting field of $q_\alpha(x)$ over F^H, and so $F^H \subseteq F$ is a normal extension by theorem 19.5.13, completing the proof (notice again how finiteness was important for this part of the proof)

(checkout other proofs of this that do not require using the fact it's a simple extension, since the primitive roots theorem is sometimes proven using Galois Theory)

This proposition provides us with a couple more results: The polynomial $q_\alpha(x)$ (say of degree n) can be expanded to:

$$q_\alpha(x) = x^n - s_1x^{n-1} + s_2x^{n-1} - \cdots + (-1)^{n-1}s_{n-1}x + (-1)^n s_n$$

where each s_i are functions $s_i(\alpha_1,...,\alpha_n)$ and represent:

$$s_1(\alpha_1,...,\alpha_n) = \alpha_1 + \alpha_2 + \cdots + \alpha_n = \sum_i \alpha_i$$

$$s_2(\alpha_1,...,\alpha_n) = \sum_{i<j} \alpha_i\alpha_j$$

$$s_3(\alpha_1,...,\alpha_n) = \sum_{i<j<k} \alpha_i\alpha_j\alpha_k$$

$$\vdots$$

$$s_n(\alpha_1,...,\alpha_n) = \alpha_1\alpha_2\cdots\alpha_n$$

Since $q_\alpha(x) \in F^H$, then each of these coefficients are in F^H. Furthermore, $\deg(q_\alpha(x)) \leq |H|$

Proposition 20.1.12: $H = \mathrm{Aut}_{F^H}(F)$

Let F/k be a *finite* extension, and let H be a subgroup of $\mathrm{Aut}_k(F)$. Then $|H| = [F : F^H]$ and

$$H = \mathrm{Aut}_{F^H}(F)$$

In particular, the Galois correspondence is surejctive for a finite field (for any subgroup $H \leqslant \mathrm{Aut}_k(F)$, there exists an intermediate field (namely F^H) such that $H = \mathrm{Aut}_{F^H}(F)$)

Intuitively, we want to show that $\mathrm{Aut}_{F^H}(F)$ *only* fixes F^H, and so $H = \mathrm{Aut}_{F^H}(F)$.

***Proof*:**
Let $H \leqslant \mathrm{Aut}_k(F)$. By proposition 20.1.9, we have that $H \leqslant \mathrm{Aut}_{F^H}(F)$, so $|H| \leq |\mathrm{Aut}_{F^H}(F)|$. Since $\mathrm{Aut}_{F^H}(F)$ is finite, verifying that $\geq$ holds will complete the proof.

To see this, recall that we have shown in lemma 20.1.11 that $F = F^H(\alpha)$ for some $\alpha \in F$. Thus, by corollary 19.2.29, $|\mathrm{Aut}_{F^H}(F)|$ equals the number of distinct roots of $m_{\alpha,F^H}(\alpha)$. Though we do not know the minimal polynomial $m_{\alpha,F^H}(x)$, we do know that α is a root of the polynomial $q_\alpha(x)$ defined in lemma 20.1.11. By construction, the number of roots of $q_\alpha(x)$ must be less than or equal $|H|$, giving us:

$$|\mathrm{Aut}_{F^H}(F)| \leq |H|$$

And since $|\mathrm{Aut}_{F^H}(F)| \geq |H|$, we get $|\mathrm{Aut}_{F^H}(F)| = |H|$ and since $H \subseteq \mathrm{Aut}_{F^H}(F)$, then $H = \mathrm{Aut}_{F^H}(F)$, giving the desired equality.

Finally, to show that $[F : F^H] = H$, since F/F^H is finite, separable, and normal, by corollary 19.6.21, $[F : F^H] = |\mathrm{Aut}_{F^H}(F)|$, completing the proof.

A particular consequence of this is that $\mathrm{Aut}_{F^{\mathrm{Aut}_E(F)}}(F) = \mathrm{Aut}_E(F)$, since the intermediate field that maps to $\mathrm{Aut}_{F^{\mathrm{Aut}_E(F)}}(F)$ is $F^{\mathrm{Aut}_E(F)}$, and so the group that it maps to is $\mathrm{Aut}_E(F)$.

Note that F/k being a *finite* extension was necesary for the proof. The Galois correspondence is in fact not necessarily surjective in the infinite case. This correspondence can be recovered given a suitable topology known as the *Krull topology* on $\mathrm{Aut}_k(F)$, which is used to limit the galois correspondence to *closed* subgroups of the automorphism group, which will regain surjectivity ([11, Fundamental Theorem of Galois Theory, Infinite Extensions]).

20.1.1 Galois Correspondence

We finally have reached a point where we can define a Galois extension. the following theorem is sometime also called the *Galois Extension Equivalences*, but it will be called by it's more famous name:

Theorem 20.1.13: Fundamental Theorem of Galois Theory I

Let F/k be a finite field extension. Then the following are equivalent:

1. F is the splitting field of a separable polynomial $f(x) \in k[x]$ over k
2. F/k is separable and normal
3. $|\text{Aut}_k(F)| = [F : k]$
4. $k = F^{\text{Aut}_k(F)}$ is the fixed field of $\text{Aut}_k(F)$
5. F/k is separable, and if K/F is an algebraic extension and $\sigma \in \text{Aut}_k(K)$, then $\sigma(F) = F$
6. The Galois correspondence for F/k is a bijection

***Proof*:**
Many of these have already been shown to be equivalent. We have shown:

- $1 \Leftrightarrow 2$ in theorem 19.5.13
- $2 \Leftrightarrow 3$ in corollary 19.6.21
- $3 \Leftrightarrow 4$ follows from proposition 20.1.12, since $[F : F^{\text{Aut}_k(F)}] = |\text{Aut}_k(F)|$, and since $k \subseteq F^{\text{Aut}_k(F)}$, we have that

$$k = F^{\text{Aut}_k(F)} \text{ if and only if } |\text{Aut}_k(F)| = [F : k]$$

- $2 \Leftrightarrow 5$ follows from the 3rd equivalence of theorem 19.5.13 along with the fact that finite separable extensions are simple (theorem 19.6.20)

Thus, we will show $6 \Rightarrow 4$ and $1 \Rightarrow 6$, which will complete the chain of implications.

Starting with $6 \Rightarrow 4$, assume the galois correspondence is bijective, and consider $E = F^{\text{Aut}_k(F)}$. By proposition 20.1.12, $\text{Aut}_{F^{\text{Aut}_k(F)}}(F) = \text{Aut}_k(F)$, and so E and $F^{\text{Aut}_k(F)}$ map to the same subgroup: $\text{Aut}_k(F) = \text{Aut}_E(F)$. Since the correlation is bijective, it must be that $k = E$, implying 4.

Next, we will show $1 \Rightarrow 6$. Since F/k is a finite extension, by proposition 20.1.12, we already know the Galois correspondence is surjective (i.e., has a right-inverse), so it suffice to show it's injective, or equivalently that it has a left-inverse. To that end, we need to show that every intermediate field E equals the fixed field of the corresponding subgroup $\text{Aut}_E(F)$, that is $E = F^{\text{Aut}_E(F)}$. So, pick some E such that $k \subseteq E \subseteq F$. Since F is the splitting field for some separable polynomial $f(x) \in k[x] \subseteq E[x]$, we see that F is the splitting field of the separable polynomial $f(x)$ over E (i.e., condition (1) holds for F/E too). Since $1 \Leftrightarrow 4$, then $E = F^{\text{Aut}_E}(F)$, thus $1 \Rightarrow 6$ completing the proof.

It can be helpful to have the following visual to keep track of where intermediate fields are with

respect to their galois group:

$$\begin{aligned} \mathrm{Aut}_E(F) &= G \\ E &= F^G \end{aligned}$$

Definition 20.1.14: Galois Extension

Let F/k be a finite extension. Then F is called *Galois* or F/k is called a *Galois extension* if it satisfies any of the conditions in theorem 20.1.13.

Note that we have limited the definition to *finite galois extensions*, due to the requirement to bring in more tools to study the infinite equivalence. Nothing is lost in doing so, since infinite galois extensions will naturally generalize and improve upon the results we will present. From now on, we will take galois extension to mean finite galois extension.

Corollary 20.1.15: Galois Extension Of Char 0

Let F/k be an extension of characteristic 0. Then F is galois over k if and only if F is finite and normal over k

Note that the finite condition was only added since we are focusing on finite Galois extensions.

***Proof*:**
If F is galois then it is *a priori* normal. Conversely, if F is finite and normal, then by corollary 19.6.8, any algebraic extension of F is separable, and hence by theorem 20.1.13, it is Galois.

Definition 20.1.16: Galois Group

Let F/k be a Galois extension. Then the corresponding automrophsim group $\mathrm{Aut}_k(F)$ is called the *galois group* of the extension. We will denote it by $\mathrm{Gal}_k(F)$ or $\mathrm{Gal}(F/k)$

The symbol $\mathrm{Gal}_k(F)$ is simply to emphasize we are working over a Galois extension, and does not reflect any additional structrue or data given to $\mathrm{Aut}_k(F)$.

Definition 20.1.17: Galois Group Of A Polynomial

Let $f(x)$ be a separable polynomial over $k[x]$. Then the *galois group* $\mathrm{Gal}_k(f(x))$ is the galois group of the splitting field of $f(x)$).

Definition 20.1.18: Abelian and Cyclic Extensions

Let F/k be a galois extension. Then F is called a *cyclic* (resp. *abelian*) extension if it's corresponding Galois group is cyclic (resp. abelian)
More generally, if the Galois group G has some property (ex. nilpotent, solvable, free), then the respetive extension is said to have that same property (ex. nilpotent extension, solvable extension, free extension)

Note that any vocabulary already used (like simple extension) will be prioritized over the group theory vocabulary (a simple extension is not related to simple group)

The equivalent conditions for Galois extensions gives us many details on results we have encountered earlier:

20.1.19 Remark

1. For any $H \leqslant \mathrm{Aut}_k(F)$, F/F^H is a Galois extension, and every intermediate field $k \subseteq E \subseteq F$ appears as a fixed field for a unique $H \leqslant \mathrm{Aut}_k(F)$ (If F/k is not Galois, this is not the case, like for $\mathbb{Q}(\sqrt[3]{2})/\mathbb{Q}$). This also means that by the 4th condition of Galois extension equivlences:
$$F^H = F^{\mathrm{Aut}_{F^H}(F)}$$
and since every intermediate field appears as a fixed field, we get that:
$$E = F^{\mathrm{Aut}_E(F)}$$

2. Let F/k be any finite extension. Then $|\mathrm{Aut}_k(F)|$ divides $[F:k]$. Taking any $G = \mathrm{Aut}_k(F)$, then
$$[F:k] = [F:F^G][F^G:k] \overset{!}{=} |\mathrm{Aut}_{F^{\mathrm{Aut}_k(F)}}(F)|[F^G:k] = |\mathrm{Aut}_k(F)|[F^G:k]$$
where $\overset{!}{=}$ comes from theorem 20.1.13(3). If F/k is Galois, then $F^G = k$, meaninig $|\mathrm{Aut}_k(F)|[F^G:k] = |\mathrm{Aut}_k(F)|[k:k] = |\mathrm{Aut}_k(F)|$.

3. Let F/k be a Galois extension. Then $|\mathrm{Aut}_k(F)|$ divides $|\mathrm{Aut}_{F^H}(F)|$. Taking any $H \leqslant \mathrm{Aut}_k(F)$ and $G = \mathrm{Aut}_k(F)$, then
$$|\mathrm{Aut}_k(F)| = [F:k] = [F:F^H][F^H:k] = |\mathrm{Aut}_{F^H}(F)|[F^H:k]$$
Switching up the notation, this means that $|G| = |H|[F^H:k]$, meaning $[F^H:k]$ is the index of H in G: $[F^H:k] = [G:H]$. The fact that it equals to the index means that it corresponds bijectively with the cosets $g\mathrm{Aut}_k(F^H)$ where $g \in \mathrm{Aut}_k(F)$, however it is yet to be shown how these two sets can correspond, only that they are the same size. Theorem 20.1.23 says when this set of cosets forms a group.

4. It may be really difficult to figure out if we have found all intermediate fields of an infinite in general, namely since in general, there are infinitely many possible choices. However, it is relatively easy to find the subgroups of a finite group! Hence, using finite group theory allows us to bring in the powerful tools having that finiteness brings (including Sylow's theorems).

5. Note how the 5th condition is almost the condition for an extension being normal, where just also assumed that the extension is separable. Note that the 5th condition still works if the extension is not separable. This can motivate the idea that we can somhow make separability a superfluous condition given some appropriate generalization, and indeed we can. If we replace Galois Group with an appropriate *algebraic group*, then we can recover the bijective correspondence[a]

6. Given the base field is perfect, for any extension F/k, we can always extend it to its normal closure $N/F/k$, which is normal over k, and so is Galois. Hence, many field theory problems over $\mathbb{Q}$ can be translated "wlog" to Galois theory

[a] algebraic groups are schemes (with with a group structure), for which one of the motivations for their invention was to distinguish roots higher multiplicities at a point

If $E = F^H$, the last observation can be seen visually as:

$$\begin{array}{ccc} F & & \\ | & \} & |H|=[F:E] \\ E & & \\ | & \} & |G:H|=[E:k] \\ k & & \end{array}$$

Example 20.1.20: Galois Groups

1. Let $K = \mathbb{Q}(\sqrt{2})/\mathbb{Q}$. Then since $[K : \mathbb{Q}] = 2.$ and $|\mathrm{Aut}(K/\mathbb{Q})| = 2$ (as we have seen in example 20.1.4), then K is galois with galois group

   $$\mathrm{Gal}(\mathbb{Q}(\sqrt{2})/\mathbb{Q}) = \{1, \sigma\} \cong \mathbb{Z}/2\mathbb{Z}$$

   With the identity automorphism and

   $$a + b\sqrt{2} \mapsto a - b\sqrt{2}$$

2. More generally, let K be any quadratic extension of F $(K = F(\sqrt{D}))$. If $\mathrm{ch}(K) \neq 2$, then K is Galois.

3. The extension $\mathbb{Q}(\sqrt[3]{2})/\mathbb{Q}$ is *not* Galois, since $|\mathrm{Aut}(\sqrt[3]{2})| = 1 < 3 = [\mathbb{Q}(\sqrt[3]{2}) : \mathbb{Q}]$. Notice that $\mathbb{Q} \subseteq \mathbb{Q}(\sqrt[3]{2})$ is not a fixed field, since the only fixed field is $\mathbb{Q}(\sqrt[3]{2})$.

4. Let $K = \mathbb{Q}(\sqrt{2}, \sqrt{3})$ be an extension of $\mathbb{Q}$. Then since K is the splitting field of $(x^2-2)(x^2-3)$, it Galois. To determine its Galois group, we know from example 19.3.12 that $[K : \mathbb{Q}] = 4$, giving us the amount of automorphisms to look for. Since $\sqrt{2} \notin \mathbb{Q}(\sqrt{3})$, by the fixed roots criterion, we see that any automorphism $\sigma \in \mathrm{Aut}_{\mathbb{Q}}(K)$ must map $\sqrt{2}$ to $\pm\sqrt{2}$, and since $\sqrt{3} \notin \mathbb{Q}(\sqrt{2})$, we see that any automorphism $\tau \in \mathrm{Aut}_{\mathbb{Q}}(K)$ must map $\sqrt{3}$ map $\pm\sqrt{3}$.

   Going through defining a σ and a τ by permuting $\sqrt{2}$ and $\sqrt{3}$ we see that:

   $$\mathrm{Gal}(K/\mathbb{Q}) = \{1, \sigma, \tau, \sigma\tau\} \cong V_4 \cong \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$$

   Next, we can use proposition 20.1.9 to find the associated fixed field given the subgroups of

$\mathrm{Gal}(K/\mathbb{Q})$. Using some arithmetic, we see that:

$$\begin{array}{cc} \{1\} & \mathbb{Q}(\sqrt{2},\sqrt{3}) \\ \{1,\sigma\} & \mathbb{Q}(\sqrt{3}) \\ \{1,\sigma\tau\} & \mathbb{Q}(\sqrt{6}) \\ \{1,\tau\} & \mathbb{Q}(\sqrt{2}) \\ \{1,\sigma,\tau,\sigma\tau\} & \mathbb{Q} \end{array}$$

which exhausts all possible intermediate fields of $\mathbb{Q}(\sqrt{2},\sqrt{3})/\mathbb{Q}$

5. Take the splitting field of x^4-2 (say we choose adjoints $\mathbb{Q}(i,\sqrt[4]{2})$). Since $\mathrm{ch}(\mathbb{Q})=0$, the polynomial must be separable, and hence $\mathbb{Q}(i,\sqrt[4]{2})$ must be Galois. Then we know that:

$$[\mathbb{Q}(i,\sqrt[4]{2}):\mathbb{Q}] = [\mathbb{Q}(i,\sqrt[4]{2}):\mathbb{Q}(\sqrt[4]{2})][\mathbb{Q}(\sqrt[4]{2}):\mathbb{Q}] = 4\cdot 2 = 8$$

Telling us the Galois group has 8 elements. Since $i^n\sqrt[4]{2}$ are the 4 roots of x^4-2, we see that

$$\mathrm{Gal}_{\mathbb{Q}}(\mathbb{Q}(i,\sqrt[4]{2})) \leqslant S_4$$

Since $|\mathrm{Gal}_{\mathbb{Q}}(\mathbb{Q}(i,\sqrt[4]{2}))| = 2\cdot 4$, $\mathrm{Gal}_{\mathbb{Q}}(\mathbb{Q}(i,\sqrt[4]{2}))$ has a Sylow 2-subgroup of S_4. By Sylow's 2nd theorem (theorem 4.8.6) this group then *must* be D_4.

This makes finding the automorphisms quite easily. Notice that the automorphism σ:

$$\begin{cases} \sqrt[4]{2} \longrightarrow i\sqrt[4]{2} \\ i \longrightarrow i \end{cases}$$

is cyclic of order 4, and the automorphism τ:

$$\begin{cases} \sqrt[4]{2} \longrightarrow \sqrt[4]{2} \\ i \longrightarrow -i \end{cases}$$

is cyclic of order 2, and that:

$$\sigma\tau = \tau^{-1}\sigma$$

Which gives us that $\mathrm{Gal}_{\mathbb{Q}}(\mathbb{Q}(i,\sqrt[4]{2})) \cong D_4$! We correspond the subgroups with the intermediate fields in example 20.1.22.

6. As we have seen with cyclotomic fields, by proposition 19.7.9,
$$\mathrm{Aut}_{\mathbb{Q}}(\mathbb{Q}(\zeta_n)) \cong (\mathbb{Z}/n\mathbb{Z})^\times$$
giving us a Galois group for every $n \in \mathbb{N}_{>0}$.

7. By theorem 19.7.10, all finite field are Galois, since they are the splitting field over the separable polynomial $x^{p^d} - x$. As we have seen in proposition 19.7.19, the galois group is cyclic of order d:
$$\mathrm{Aut}_{\mathbb{F}_p}(\mathbb{F}_{p^d}) \cong \mathbb{Z}/d\mathbb{Z}$$
and is generated by the frobenius k-automorphism $x \mapsto x^p$

8. The extension $\mathbb{F}_2(x^{1/2})/\mathbb{F}_2(x)$, with $x^{1/2}$ a root of $y^2 - x$, is normal but not separable, and so it is not Galois (theorem 20.1.13(2)). In characteristic 2, $y^2 - x = (y - x^{1/2})^2$, so $\mathbb{F}_2(x^{1/2})$ is a splitting field of $y^2 - x$ over $\mathbb{F}_2(x)$, hence normal (theorem 19.5.13). The polynomial $y^2 - x$ has no root in $\mathbb{F}_2(x)$, since $(g/h)^2 = x$ with $g, h \in \mathbb{F}_2[x]$ would give $2\deg g = 1 + 2\deg h$, so it is irreducible (proposition 11.3.5), the minimal polynomial of $x^{1/2}$, with the repeated root $x^{1/2}$.

The correspondence by theorem 20.1.13 is in fact even stronger: we can correspond the intermediate lattice to the subgroup lattice. The first extension of the theorem is to give the correspondence between the lattices of subgroups of the galois group and of the field extensions. To show two lattices coresond, we need to show that if A_1 and A_2 are two objects, then A_1A_2 (i.e. the smallest object containing both A_1 and A_2) corresponding to B_1B_2, or if the relation is contravariant, A_1A_2 corresponds to $B_1 \cap B_1$. This is what we show with subgroups and intermediate fields:

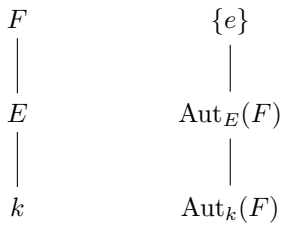
Theorem 20.1.21: Fundamental Theorem Of Galois Theory II

Let F/k be a Galois extension. Then the Galois correspondence is an inclusion-reversing isomorphism between the lattice of intermediate subfield of F/k with the lattice of subgroups of $\mathrm{Aut}_k(F)$.

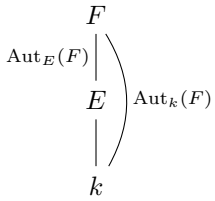
In particular, If E_1, E_2 are intermediate fields and G_1, G_2 are the correspondong subgroups of $\mathrm{Aut}_k(F)$, then $E_1 \cap E_2$ coresponds to $\langle G_1, G_2\rangle$ and E_1E_2 corresponds to $G_1 \cap G_2$

***Proof*:**
From theorem 20.1.13, we get the bijection between the intermediate fields and the subgroups of $\mathrm{Gal}_k(F)$, and from the 3rd property of the Galois Correspondence (proposition 20.1.9), we get that:
$$\mathrm{Aut}_{E_1E_2}(F) = G_1 \cap G_2 \qquad F^{\langle G_1, G_2\rangle} = E_1 \cap E_2$$
completing the proof.

We usually write the lattice of fields with k on the bottom and F on top, while the lattice of groups usualy has $\{e\}$ on top and $\mathrm{Aut}_k(F)$ on bottom to more easily associate the two lattices:

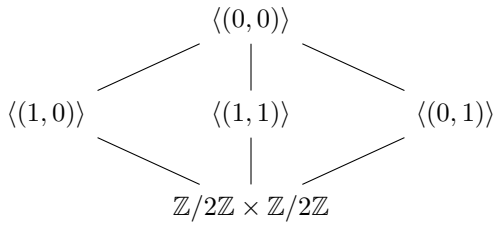


Some authors mark the Galois group though their own edges:

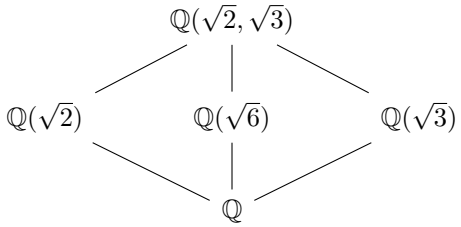


Example 20.1.22: Lattice Correspondence

1. As we have shown in example 20.1.20(3), where we corresponded the subgroups of $\mathrm{Gal}_{\mathbb{Q}}(\mathbb{Q}(\sqrt{2},\sqrt{3}))$ with the intermediate fields $\mathbb{Q} \subseteq E \subseteq \mathbb{Q}(\sqrt{2},\sqrt{3})$, we found that $\mathrm{Gal}_{\mathbb{Q}}(\mathbb{Q}(\sqrt{2},\sqrt{3})) \cong \mathbb{Z}_2 \times \mathbb{Z}_2$ and intermediate fields for each subgroup. We can make this result more precise. If the subgroup lattice is:



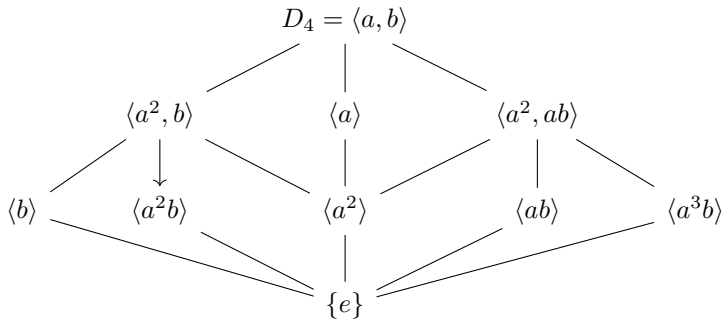
then the corresponding intermediate field lattice is:



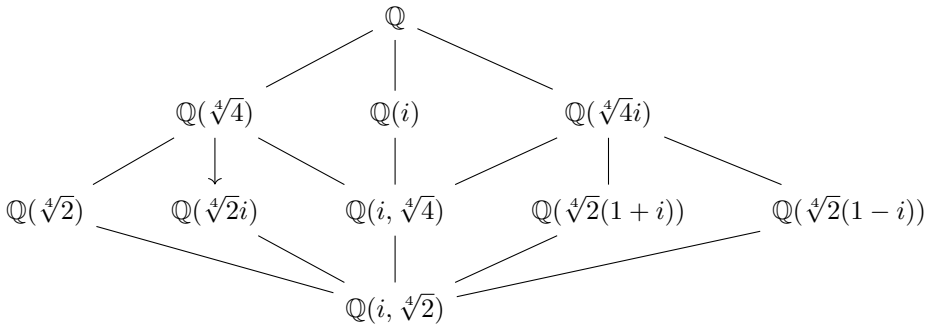
We can futhermore deduce that there are *no* further intermediate extensions $k \subseteq E \subseteq F$.

2. As we have shown in example 20.1.20(4), $x^4 - 2$ has splitting field $\mathbb{Q}(i, \sqrt[4]{2})$ with Galois

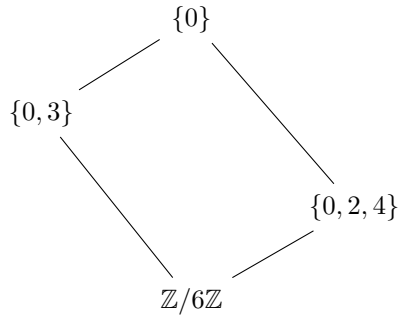
group $\text{Gal}_{\mathbb{Q}}(\mathbb{Q}(i, \sqrt[4]{2})) = D_4$. Then:



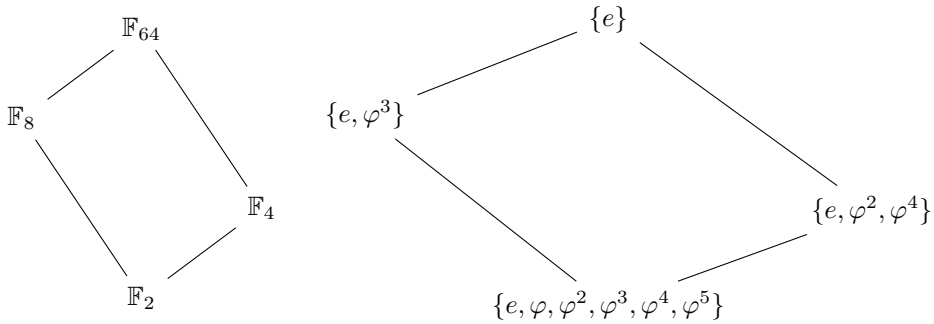
with corresponding intermediate field lattice:



3. Take $\mathbb{F}_{64}/\mathbb{F}_2$, which is a degree 6 extension with Galois group $\text{Aut}_{\mathbb{F}_2}(\mathbb{F}_{64}) \cong \mathbb{Z}/6\mathbb{Z}$. the subgroup lattice is:



which corresponds to:



The next result comes from recalling that if $k \subseteq E \subseteq F$ is a series of field extensions where F/k is normal, this does not imply E/k is normal (as we have discussed in proposition 19.5.16 and example 19.5.17). Intuitively, this can be thought of as E having some but not all the required roots so that F is normal (ex. $\mathbb{Q} \subseteq \mathbb{Q}(\sqrt[3]{2}) \subseteq \mathbb{Q}(\sqrt[3]{2}, \varphi)$). However, it is sometimes the case that the intermediate fields are normal (ex. $Q \subseteq \mathbb{Q}(\sqrt{2}) \subseteq \mathbb{Q}(\sqrt{2}, \sqrt{3})$). We may ask when more generally is an extension E/k also normal, or more generally Galois? The following theorem answers that question:

Theorem 20.1.23: Fundamental Theorem Of Galois Theory III

Let F/k be a galois extension, and let E be an intermediate field. Then E/k is Galois if and only if $\mathrm{Aut}_E(F)$ is *normal* in $\mathrm{Aut}_k(F)$. If this is the case, then

$$\mathrm{Aut}_k(E) = \mathrm{Aut}_k(F)/\mathrm{Aut}_E(F)$$

Hence, *normal* subgroups arise from *normal* extensions. In fact, the term "normal" originate in field theory, and the term normal subgroup was coined after it's field theory counterpart.

The proof essentially comes down to a couple of key observations about the cosets of $\mathrm{Aut}_k(F)/\mathrm{Aut}_E(F)$ and how we can surmise information about them through group actions! We have already observed in box 20.1.19 that E/k has the same cardinality as $\{g\mathrm{Aut}_E(F) \mid g \in \mathrm{Aut}_k(F)\}$. We will make this correspondence more precise.

By theorem 19.5.13, a finite extension E/k is normal if and only if $\iota(E) = E$ for every k-homomorphism $\iota : E \to \overline{k}$.

Let F/K be a Galois extension. Identify $F \subseteq \overline{k}$ within some algebraic closure of $\overline{k}$. Let $k \subseteq E \subseteq F$ be some intermediate field, and consider:

$$I = \left\{ \iota : E \to \overline{k} \;\middle|\; \iota \text{ is a } k\text{-homomorphism} \right\} = \mathrm{Hom}_k(E, \overline{k})$$

We can equivalently see E/k as an arbitrary extension of k where for some $\iota \in I$, $\iota(E) \subseteq F$.

Claim 1: for all $\iota \in I$, $\iota(E) \subseteq F$

Proof. Let ι be a $\iota \in I$ such that $\iota(E) \subseteq F$. Since E is separable (since $\iota(E) \subseteq F$ and F is separable) and finite, is is simple by theorem 19.6.20, Thus $E = k(\alpha)$ for some $\alpha \in \overline{k}$ and α is the root some minimal polynomial over k, $m_{\alpha,k}(x)$. Since $\iota(\alpha)$ determines ι, looking at where it maps is all that

matters. Since $\iota(\alpha)$ is *some* root of the minimum polynomial $m_{\alpha,k}(x)$, and F is normal, F contains *all* roots of the minimal polynomial. But then, for any ι, $\iota(E) \subseteq F$, as we sought to show. □

Notice that

Claim 2: $\mathrm{Aut}_k(F)$ acts on I: for $g \in \mathrm{Aut}_k(F)$ and $\iota \in I$, $g \circ \iota \in I$

Proof. Since $\iota(E) \subseteq F$, and $g : F \to F$, we can intermret ι as $\iota : E \to F$ and compose the functions □

claim 3: $\mathrm{Aut}_k(F)$ is transitive on I (i.e. on $\mathrm{Hom}_k(E, \overline{k})$)

Proof. We need to show that for any $\iota_1, \iota_2 \in I$, $g \circ \iota_1 = \iota_2$. This follows from uniqueness of the splitting field. Since $\iota_1(E)$ and $\iota_2(E)$ each contain k and are each contained in F, and F is a splitting field over k for some polynomial, say $f(x)$ (since F is Galois, it follows from theorem 20.1.13). Hence, *a fortiori*, F must be the splitting field of $f(x)$ over $\iota_1(E)$ and over $\iota_2(E)$. Note that:

$$\iota_2 \circ \iota_1^{-1} : \iota_1(E) \to \iota_2(E)$$

is an k-isomorphism, and hence by the uniqueness of the splitting field, this isomorphism is extended to a k-isomorphism:

$$g : F \to F$$

By construction, $g|_{\iota_1(E)} = \iota_2 \circ \iota_1^{-1}$, or equivalently:

$$g \circ \iota_1 = \iota_2$$

as we sought to show. □

Thus, the galois group $\mathrm{Gal}_k(F)$ acts transitively on $\hom_k(E, \overline{k})$.

Let $\iota_0 \in I$ be the inclusion of E in $\overline{k}$. Its stabilizer in $\mathrm{Aut}_k(F)$ is $\mathrm{Aut}_E(F)$, the subgroup of elements that restrict to the identity on E. Thus, by the Orbit-Stabilizer Theorem, we can deduce the following result:

Let $G = \mathrm{Aut}_k(F)$. Then consdering I as a G-set, I is isomorphic to the set of left-cosets of $\mathrm{Aut}_E(F)$ in $\mathrm{Aut}_k(F)$

20.1.24 key Isomorphism

$$\hom_k(E, \overline{k}) \quad \cong_G \quad \{g\mathrm{Aut}_E(F) \mid g \in \mathrm{Aut}_k(F)\}$$

***Proof*:**
of theorem 20.1.23

Let $G = \mathrm{Aut}_k(F)$ and $I = \left\{\iota : E \to \overline{k} \;\middle|\; \iota \text{ is a } k\text{-homomorphism}\right\}$. Then I is a G-set (Claim 2), transitive by Claim 3. Let $\iota_0 \in I$ be the inclusion $E \subseteq F \subseteq \overline{k}$. The stabilizer of $\iota \in I$ is $\mathrm{Aut}_{\iota(E)}(F)$, since $g \circ \iota = \iota$ exactly when g fixes $\iota(E)$ pointwise. For $g \in G$, $h \circ g \circ \iota = g \circ \iota$ if and only if $(g^{-1}hg) \circ \iota = \iota$, so the stabilizer of $g \circ \iota$ is $g\mathrm{Aut}_{\iota(E)}(F)g^{-1}$. By Claim 3 every $\iota \in I$ equals $g \circ \iota_0$ for some $g \in G$, hence

$$\mathrm{Aut}_{\iota(E)}(F) = g\mathrm{Aut}_E(F)g^{-1}$$

Thus, if $\mathrm{Aut}_E(F)$ is normal, $\mathrm{Aut}_{\iota(E)}(F) = \mathrm{Aut}_E(F)$ for every $\iota \in I$, and the bijectivity of the Galois correspondence for F/k (theorem 20.1.13, item 6) gives $\iota(E) = E$. As E/k is finite and $\overline{k}$ is an algebraic closure of E, E/k is normal by $3 \Rightarrow 1$ of theorem 19.5.13, and it is separable because F/k is, so E/k is Galois (theorem 20.1.13).

Conversely, if E/k is galois, then $\iota(E) = E$ for all ι ($1 \Rightarrow 3$ of theorem 19.5.13, as E/k is finite and normal). Define the homomorphism ρ which takes $g \in \mathrm{Aut}_k(F)$ and restricts it to $E[= \iota(E)]$:

$$\rho : \mathrm{Aut}_k(F) \to \mathrm{Aut}_k(E)$$

This homomorphism is surjective (since the G-action acts transitively on I), and $\ker(\rho)$ is all automorphism that restrict to the identity on E, that is, $\mathrm{Aut}_E(F)$. But then, we know that the kernel of a group homomorphism is normal, and by the 1st isomorphism theorem for groups:

$$\mathrm{Aut}_k(F)/\mathrm{Aut}_E(F) \cong \mathrm{Aut}_k(E)$$

completing the proof.

The correspondence turns the action of $\mathrm{Aut}_k(F)$ on intermediate fields into conjugation of subgroups: if E corresponds to $H = \mathrm{Aut}_E(F)$, then for any $\sigma \in \mathrm{Aut}_k(F)$, $\sigma(E)$ corresponds to $\mathrm{Aut}_{\sigma(E)}(F) = \sigma H \sigma^{-1}$, since $\tau \in \mathrm{Aut}_k(F)$ fixes $\sigma(E)$ pointwise exactly when $\sigma^{-1}\tau\sigma$ fixes E pointwise.

Corollary 20.1.25: Isomorphism Of Categories

Let E/k be a Galois extension. Let $\mathbf{Fld}_k^E$ be the collection of fields F such that $k \subseteq F \subseteq E$ whose morphisms are k-homomorphisms and $\mathcal{O}_G$ where $G = \mathrm{Aut}_k(E)$ be the category whose objects are left G-sets G/H for every $H \leqslant G$ and the morphisms are G-equivariant maps $G/H \to G/K$. Then as categories, $\mathcal{O}_G^{\mathrm{op}} \cong \mathbf{Fld}_k^E$

***Proof*:**
Write $G = \mathrm{Aut}_k(E)$ throughout.

The morphisms of $\mathcal{O}_G$. A G-equivariant map $\psi : G/H \to G/K$ is determined by $\psi(H)$, since $\psi(gH) = g\psi(H)$. Writing $\psi(H) = \gamma K$, the recipe $gH \mapsto g\gamma K$ is well defined exactly when $h\gamma K = \gamma K$ for every $h \in H$, that is exactly when

$$\gamma^{-1} H \gamma \subseteq K$$

and two elements γ, γ' subject to this give the same map if and only if $\gamma K = \gamma' K$. So the morphisms $G/H \to G/K$ are exactly the maps $\psi_\gamma : gH \mapsto g\gamma K$ with $\gamma^{-1}H\gamma \subseteq K$, indexed by the coset γK. Composing, $\psi_\delta(\psi_\gamma(gH)) = \psi_\delta(g\gamma K) = g\gamma\delta L$, so

$$\psi_\delta \circ \psi_\gamma = \psi_{\gamma\delta}$$

and ψ_1 is the identity.

The functor. Send the object G/H to the fixed field E^H, and send the morphism $\psi_\gamma : G/H \to G/K$ to the map

$$\Phi(\psi_\gamma) : E^K \to E^H \qquad x \mapsto \gamma x$$

This lands where claimed: for $x \in E^K$ and $h \in H$ we have $\gamma^{-1}h\gamma \in K$, so $h(\gamma x) = \gamma(\gamma^{-1}h\gamma)x = \gamma x$, and γx is fixed by H. It does not depend on the representative: if $\gamma' = \gamma\kappa$ with $\kappa \in K$ then $\gamma' x = \gamma\kappa x = \gamma x$ for $x \in E^K$. It is a k-homomorphism because γ is. Finally $\Phi(\psi_{\gamma\delta})$ is $x \mapsto \gamma\delta x$, which is $\Phi(\psi_\gamma) \circ \Phi(\psi_\delta)$, and $\Phi(\psi_1) = \mathrm{id}$. So Φ is a functor $\mathcal{O}_G^{\mathrm{op}} \to \mathbf{Fld}_k^E$.

Bijective on objects. Since E/k is Galois, the Galois correspondence is a bijection by theorem 20.1.13(6), with $H \mapsto E^H$ and $F \mapsto \mathrm{Aut}_F(E)$ mutually inverse between the subgroups of G and the intermediate fields of E/k. The objects of $\mathcal{O}_G$ are indexed by the subgroups $H \leqslant G$, so Φ is a bijection on objects.

Bijective on morphisms. For injectivity, suppose ψ_γ and $\psi_{\gamma'}$ are morphisms $G/H \to G/K$ with $\gamma x = \gamma' x$ for all $x \in E^K$. Then $\gamma'^{-1}\gamma$ fixes E^K pointwise, so $\gamma'^{-1}\gamma \in \mathrm{Aut}_{E^K}(E)$, which is K by proposition 20.1.12. Hence $\gamma K = \gamma' K$ and the two morphisms coincide.

For surjectivity, let $\sigma : E^K \to E^H$ be a k-homomorphism. By theorem 20.1.13(1), E is the splitting field over k of a separable $f(x) \in k[x]$. The roots of f generate E over k, hence over any intermediate field, so E is also the splitting field of f over E^K and over $\sigma(E^K)$. The coefficients of f lie in k and σ fixes k, so $\sigma(f) = f$, and theorem 19.5.10 extends the isomorphism $\sigma : E^K \to \sigma(E^K)$ to an isomorphism $\gamma : E \to E$ fixing k, that is to an element $\gamma \in G$. For $x \in E^K$ and $h \in H$ we get $h\gamma x = h\sigma x = \sigma x = \gamma x$, since $\sigma x \in E^H$, so $\gamma^{-1}h\gamma$ fixes E^K pointwise and therefore lies in $\mathrm{Aut}_{E^K}(E) = K$. Thus $\gamma^{-1}H\gamma \subseteq K$, the map $\psi_\gamma : G/H \to G/K$ is a morphism of $\mathcal{O}_G$, and $\Phi(\psi_\gamma) = \sigma$.

A functor that is bijective on objects and bijective on each set of morphisms is an isomorphism of categories, so $\mathcal{O}_G^{\mathrm{op}} \cong \mathbf{Fld}_k^E$.

Example 20.1.26: Application To Lattices

In the lattices of example 20.1.22(2), for $F = \mathbb{Q}(i, \sqrt[4]{2})$, the intermediate fields $\mathbb{Q}(\sqrt[4]{4}) = \mathbb{Q}(\sqrt{2})$, $\mathbb{Q}(i)$, $\mathbb{Q}(\sqrt[4]{4}i) = \mathbb{Q}(i\sqrt{2})$ and $\mathbb{Q}(i, \sqrt[4]{4}) = \mathbb{Q}(i, \sqrt{2})$ are the splitting fields over $\mathbb{Q}$ of the separable polynomials x^2-2, x^2+1, x^2+2 and $(x^2-2)(x^2+1)$, so they are Galois over $\mathbb{Q}$ (theorem 20.1.13(1)). By theorem 20.1.23, the corresponding subgroups $\langle a^2, b\rangle$, $\langle a\rangle$, $\langle a^2, ab\rangle$ and $\langle a^2\rangle$ are normal in D_4. The polynomial $x^4 - 2$ is irreducible over $\mathbb{Q}$ (corollary 11.3.11, with $p = 2$), and the field $\mathbb{Q}(\sqrt[4]{2}) \subseteq \mathbb{R}$ contains its root $\sqrt[4]{2}$ but not its root $i\sqrt[4]{2}$, which is not real. So $\mathbb{Q}(\sqrt[4]{2})/\mathbb{Q}$ is not normal (definition 19.5.12), hence not Galois, and by theorem 20.1.23 the corresponding subgroup $\langle b\rangle$ is not normal in D_4.

20.1.27 Foreshadowing Can give the connection between Galois group/extensions to Algebraic geometry with regular covers and deck transformations

Exercise 20.1.1

1. Let $L = \mathbb{Q}(\zeta_p)$ where p is an odd prime. Show there exists a subfield $K \subseteq L$ such that $K = \mathbb{Q}(\sqrt{p'})$ where
$$p' = \begin{cases} p & p \equiv 1 \bmod 4 \\ -p & p \equiv 2, 3 \bmod 4 \end{cases}$$

2. Let $K/\mathbb{Q}$ be an odd Galois Extension, Show that L must be totally real.

20.1.2 Composite Fields

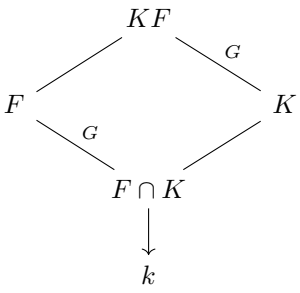
We now take what we proved earlier and do two things with it: we extend the results for composite fields and we classify all cyclic Galois Extensions (given an appropriate restriction on the base field). Earlier exercises (exercises 19.6.2 (1) to 19.5.1 (2) and 19.2.1 (4)) ask to show that:

> If E/k and F/k are field extensions inside a common field (algebraic ones, for normality), and E/k is (finite/algebraic/normal/separable resp.), then EF/F is (finite/algebraic/normal/separable resp). If E/k and F/k are (finite/algebraic/normal/separable resp.) then EF/k is (finite/algebraic/normal/separable resp.)

Since [finite] Galois extensions are finite, normal, and separable, we see immediately that if E/k is Galois, then EF/F is Galois, and if E/k and F/k are Galois, so is EF/k. The natural question to ask is what's the galois group of a composite field and of EF/F:

Proposition 20.1.28: Galois Composite Fields I

Suppose E/k is Galois and F/k is any field extension. Then EF/F is Galois, and $\mathrm{Aut}_F(EF) \cong \mathrm{Aut}_{E\cap F}(E)$. Visually, we see that:



The reader should ponder why does F/k need to be finite for this theorem?

***Proof*:**
Using the Galois extension equivalences, we can quickly prove that EF/k is Galois, namely since E/k is Galois, it is the splitting field of a separable polynomial $p(x) \in k[x] \in F[x]$. The roots of $p(x)$ generate E over k ,and so they generate EF over F, i.e., EF is the splitting field of $p(x)$ over F, thus EF/F is a Galois Extension

To show that $\mathrm{Aut}_F(EF) \cong \mathrm{Aut}_{E\cap F}(E)$, we will first show we can embedd $\mathrm{Aut}_F(EF)$ into $\mathrm{Aut}_k(E)$, and then find that the subgroup of $\mathrm{Aut}_k(E)$ associated to this embedding is $\mathrm{Aut}_{E\cap F}(E)$. For the embedding, consider first any F-automorphism $\sigma : EF \to EF$ (and hence k-automorphism). Then we can restrict σ to E, giving us $\sigma : E \to \sigma(E)$. Since E is Galois, $\sigma(E) = E$. Thus, any element of $\mathrm{Aut}_F(EF)$ get's mapped to an element of $\mathrm{Aut}_k(E)$, that is, we have a natural group homomorphism:

$$\rho : \mathrm{Aut}_F(EF) \to \mathrm{Aut}_k(E)$$

This homomorphism is injective: If the only element that maps to the identity is the identity,

then we are done. Indeed, if $\sigma \in \mathrm{Aut}_k(E)$ is the identity on E, since any element $\mathrm{Aut}_F(EF)$ is the identity on F, the coresponding σ in $\mathrm{Aut}_F(EF)$ must be the identity on EF, showing injectivity.

Thus, $\mathrm{Aut}_F(EF)$ embeds into $\mathrm{Aut}_k(E)$, and so it is associated with some $H \leqslant \mathrm{Aut}_k(E)$. By proposition 20.1.12, $H = \mathrm{Aut}_{E^H}(E)$. It suffices to show that $E^H = E \cap F$. For the $\supseteq$ direction, since every $\sigma \in \mathrm{Aut}_F(EF)$ fixes F, the restriction $\rho(\sigma)$ fixes $E \cap F$, hence $E^H \supseteq E \cap F$. For the $\subseteq$ direction, take $\alpha \in E^H$. Then $\alpha \in E$. By definition, $F(\alpha)$ is in the fixed field of $\mathrm{Aut}_F(EF)$, which is F sinc F is Galois over EF/F (1st Galois Theorem, theorem 20.1.13(4)). Thus, $\alpha \in F$, and so $E^H \subseteq E \cap F$. Therefore, $H = E \cap F$, and:

$$\mathrm{Aut}_F(EF) \cong \mathrm{Aut}_{E\cap F}(E)$$

as we sought to show

Example 20.1.29: Composite Fields Determining Abelian Extensions

1. As we have seen in section 19.7.1, if ζ_n is the nth root of unity, $\mathbb{Q}(\zeta_n)$ is the splitting field of $x^n - 1$ over $\mathbb{Q}$, and hence is Galois since $\mathrm{ch}(\mathbb{Q}) = 0$. By proposition 19.7.9, $\mathrm{Aut}_{\mathbb{Q}}(\mathbb{Q}(\zeta_n)) \cong (\mathbb{Z}/n\mathbb{Z})^\times$.

   Now, let k be any field of characteristic 0. Then the splitting field K/k for $x^n - 1$ is the composite of k and $\mathbb{Q}(\zeta)$, namely $K = k(\zeta)$. Since $\mathbb{Q}(\zeta)/\mathbb{Q}$ is Galois and $k/\mathbb{Q}$ is a field extension, by proposition 20.1.28, we get that:

   $$\mathrm{Aut}_k(k(\zeta)) \cong \mathrm{Aut}_{\mathbb{Q}(\zeta)\cap k}(\mathbb{Q}(\zeta)) \leqslant \mathrm{Aut}_{\mathbb{Q}}(\mathbb{Q}(\zeta)) \cong (\mathbb{Z}/n\mathbb{Z})^\times$$

   showing that $\mathrm{Aut}_k(k(\zeta))$ is some subgroup of $(\mathbb{Z}/n\mathbb{Z})^\times$. in particular, *any* such extension *must* be abelian.

   The fact that every extension of the form $k(\zeta)$ is abelian is quite powerful. In fact, they are relatively easy to classify. The study of Galois extensions that are abelian(which also must contain n distinct roots of unity for some $n \in \mathbb{N}$) is called *Kummer Theory*. We touch a bit of Kummer theory in the following, since we need it for showing no quintic is unsolvable, and will later show that all finite abelian group show up as a field extension of $\mathbb{Q}$ in section 20.4.

2. Generalizing the results from the exercises, we know that if E/k and F/k are Galois, so is EF/k. Can you find a way to link $\mathrm{Gal}_k(EF)$ with $\mathrm{Gal}_k(E)$ and $\mathrm{Gal}_k(F)$? (Hint: Consider $\mathrm{Gal}_k(EF) \to \mathrm{Gal}_k(E) \times \mathrm{Gal}_k(F)$, $\sigma \mapsto (\sigma|_E, \sigma|_F)$)

Corollary 20.1.30: Order of Galois Composite

Let E/k be a galois extension and F/k be any finite extension. Then:

$$[EF : k] = \frac{[E : k][F : k]}{[E \cap F : k]}$$

***Proof*:**
By proposition 20.1.28, EF/F is Galois and $\mathrm{Aut}_F(EF) \cong \mathrm{Aut}_{E\cap F}(E)$. Moreover, $E/(E\cap F)$ is Galois: since E/k is Galois, E is the splitting field over k of some separable polynomial $f(x) \in k[x]$ by theorem 20.1.13(1), and the roots of f generate E over k, hence over the intermediate field $E\cap F$, so E is also the splitting field of the separable polynomial f over $E\cap F$. Applying theorem 20.1.13(3) to the two Galois extensions EF/F and $E/(E\cap F)$:

$$[EF:F] = |\mathrm{Aut}_F(EF)| = |\mathrm{Aut}_{E\cap F}(E)| = [E : E\cap F]$$

By the Tower Law (theorem 19.3.7) applied to $k \subseteq E\cap F \subseteq E$, we get $[E : E\cap F] = [E : k]/[E\cap F : k]$, and applied to $k \subseteq F \subseteq EF$:

$$[EF:k] = [EF:F][F:k] = [E:E\cap F][F:k] = \frac{[E:k][F:k]}{[E\cap F:k]}$$

as we sought to show.

Note that this formula does not hold in general if niether of the extensions is Galois.

Example 20.1.31: Failure of Formula with no Galois Extension
Let $k=\mathbb{Q}$, $E=\mathbb{Q}(\sqrt[3]{2})$, and $F=\mathbb{Q}(\zeta_3\sqrt[3]{2})$), since $EF=\mathbb{Q}(\zeta_3\sqrt[3]{2}), \sqrt{3}$ (with minimal polynomial x^3-2, and hence degree 6 extension) and:

$$6 = [\mathbb{Q}(\zeta_3\sqrt[3]{2})] \neq \frac{[\mathbb{Q}(\sqrt[3]{2}):\mathbb{Q}][\mathbb{Q}(\zeta_3\sqrt[3]{2}):\mathbb{Q}]}{[\mathbb{Q}:\mathbb{Q}]} = 9$$

Using this, we may answer the arguably more intesesting questions of the composite and intersection of fields being galois over the same base field:

Proposition 20.1.32: Galois Composite Fields II

Let $E/k, F/k$ be galois extensions of k. Then:

1. $E\cap F$ is galois over k
2. EF is galois over k. It's galois group is isomorphic to:
$$G = \{(\sigma,\tau) \mid \sigma|_{E\cap F} = \tau|_{E\cap F}\} \leqslant \mathrm{Gal}_k(E)\times\mathrm{Gal}_k(F)$$
3. If $E\cap F = k$, then:
$$\mathrm{Gal}_k(EF) \cong \mathrm{Gal}_k(E)\times\mathrm{Gal}_k(F)$$

***Proof*:**

1. Let $p(x)\in k[x]$ be irreducible and have a root $\alpha\in E\cap F$. Since $\alpha\in E$ and E/k is normal by theorem 20.1.13(2), p splits into linear factors over E, so all roots of p lie in E. The same argument in F puts all roots of p in F, hence all of them lie in $E\cap F$ and p splits over $E\cap F$. By definition 19.5.12, $E\cap F$ is normal over k. It is finite over k, being an intermediate field of the finite extension E/k, and it is separable over k, since each of its

elements lies in E and E/k is separable. By theorem 20.1.13(2), $E \cap F$ is Galois over k.

2. By theorem 20.1.13(1), E is the splitting field over k of a separable $f(x) \in k[x]$ and F is the splitting field over k of a separable $g(x) \in k[x]$. Let $h(x)$ be the product of the distinct monic irreducible factors of fg in $k[x]$. Each such factor divides f or g and so has distinct roots, and distinct monic irreducibles share no root, so h is separable. The roots of h are the roots of f together with those of g, and these generate EF over k, so EF is the splitting field of h over k and EF/k is Galois by theorem 20.1.13(1).

   Since E/k and F/k are Galois, theorem 20.1.13(5) gives $\sigma(E) = E$ and $\sigma(F) = F$ for every $\sigma \in \mathrm{Gal}_k(EF)$, so restriction defines a map

$$\rho : \mathrm{Gal}_k(EF) \to \mathrm{Gal}_k(E) \times \mathrm{Gal}_k(F) \qquad \sigma \mapsto (\sigma|_E , \sigma|_F)$$

   which is a group homomorphism. It is injective: an element of the kernel is the identity on E and on F, hence on the field EF they generate. Its image lies in G, because restricting $\sigma|_E$ and $\sigma|_F$ further to $E \cap F$ gives $\sigma|_{E\cap F}$ both times.

   It remains to count. By part (1), $E \cap F$ is Galois over k, so theorem 20.1.23 applied to $k \subseteq E \cap F \subseteq F$ gives $\mathrm{Gal}_k(F)/\mathrm{Gal}_{E\cap F}(F) \cong \mathrm{Gal}_k(E \cap F)$ through restriction. Hence restriction $\mathrm{Gal}_k(F) \to \mathrm{Gal}_k(E \cap F)$ is surjective with fibres the cosets of $\mathrm{Gal}_{E\cap F}(F)$, so for each $\sigma \in \mathrm{Gal}_k(E)$ the number of $\tau \in \mathrm{Gal}_k(F)$ with $\tau|_{E\cap F} = \sigma|_{E\cap F}$ is $|\mathrm{Gal}_{E\cap F}(F)|$. Therefore

$$|G| = |\mathrm{Gal}_k(E)| \cdot |\mathrm{Gal}_{E\cap F}(F)| = [E : k] \cdot [F : E \cap F] = \frac{[E : k][F : k]}{[E \cap F : k]}$$

   using theorem 20.1.13(3) on the Galois extensions E/k and $F/(E \cap F)$ and the Tower Law (theorem 19.3.7) on $k \subseteq E \cap F \subseteq F$. Here $F/(E \cap F)$ is Galois because F is the splitting field of the separable g over k, hence also over the intermediate field $E \cap F$. On the other side corollary 20.1.30 gives

$$|\mathrm{Gal}_k(EF)| = [EF : k] = \frac{[E : k][F : k]}{[E \cap F : k]}$$

   So ρ is an injection between finite sets of the same size and therefore carries $\mathrm{Gal}_k(EF)$ isomorphically onto G.

3. If $E \cap F = k$ then every element of $\mathrm{Gal}_k(E)$ and of $\mathrm{Gal}_k(F)$ restricts to the identity on $E \cap F$, so the condition cutting out G holds for every pair and G is all of $\mathrm{Gal}_k(E) \times \mathrm{Gal}_k(F)$. Part (2) then reads $\mathrm{Gal}_k(EF) \cong \mathrm{Gal}_k(E) \times \mathrm{Gal}_k(F)$.

Corollary 20.1.33: Galois Closure

Let F/k be a finite separable extension. Then there exists a minimal finite galois extension E/k such that $E/F/k$.

***Proof*:**
Simply take the composite of the splitting fields of the minimal polynomials for the basis of F over k (which is separable since F/k is separable), nad intersect all the galois extensions of k containing F.

Definition 20.1.34: Galois Closure

Let F/K be a finite separable extension. Then the smallest extension $E/F/k$ such that E/k is galois is called the *galois closure* of F over k.

Exercise 20.1.2

1. Show there exists fields E, F such that $[EF : F] \neq [F : E \cap F]$

20.1.3 A bit of Character Theory and Kummer Theory

Let F/K be a galois extension. Then each $\sigma \in \mathrm{Gal}_K(F)$ is also a K-linear isomorphism $F \to F$ of n-dimensional K-vector spaces. Thus, does σ have eigenvalues? Can it be diagonalized? How does each $\sigma \in \mathrm{Gal}_K(F)$ relate to each other? To start:

Lemma 20.1.35: Eigenvalues Are Roots of Unity

Let F/K be a galois extension and $\sigma \in \mathrm{Gal}_K(F)$. Then if σ has an eigenvalue λ, then λ is a root of unity

***Proof*:**
Let σ be as in the statement and assume $\sigma(\alpha) = \lambda\alpha$ where $\lambda \in K$. Then as $\mathrm{Gal}_K(F)$ is a finite group, the exists an n such that $\sigma^n = \mathrm{id}$. Thus:

$$\alpha = \mathrm{id}(\alpha) = \sigma^n(\alpha) = \lambda^n\alpha$$

Then $\lambda^n = 1$, hence λ is a root of unity.

Thus, σ is diagonalizable if K contains enough roots of unity. Such an extension is called a *Kummer extension* (see definition 20.4.2, the definition is delayed due to the need to consider the behaviour over different characteristic which is covered in section 20.4).

Next, we would like to know how each automorphism relates to each other. We shall use a concept from Character theory to frame the discussion:

Definition 20.1.36: Group Character

Let G be a group and L a field. Then a *[linear] character* (denoted χ) is a homomorphism from G to the multiplicative group of L:

$$\chi : G \to L^\times$$

Example 20.1.37: Group Character

For our purposes, any field homomorphism $\sigma : F \to K$ induces a character $\chi_\sigma : F^\times \to K^\times$ (since σ maps the multiplicative group of F into the multiplicative group of K). Note that χ_σ captures essentially all the mapping information of σ, since χ only omits mapping 0 from F to 0 from K, but we can simply deduce that. Thus, if we are working with χ_σ, we are showing that we do not want to remember that the additive structure is preserved.

Definition 20.1.38: Character Linearly Independent

The characters $\chi_1, ..., \chi_n$ of G are *linearly independent* over L if they are linearly independent as functions on G, i.e. for any $a_i \in L$

$$a_1\chi_1 + a_2\chi_2 + \cdots + a_n\chi_n = 0 \quad \text{if and only if} \quad a_i = 0, 1 \leq i \leq n$$

Note that the expression on the left hand side of the above equality is a function on G, and hence it means that the characters are linearly independent if, whenever

$$a_1\chi_1(g) + a_2\chi_2(g) + \cdots + a_n\chi_n(g) = 0 \qquad \text{for all } g \in G$$

then $a_i = 0$ for $1 \leq i \leq n$.

Theorem 20.1.39: Artin's Lemma

Let $\chi_1, ..., \chi_n$ be distinct characters of G with values in L. Then they are linearly independent over L

***Proof*:**

Suppose that the characters are linearly dependent, that is, there exist coefficients $a_1, ..., a_n \in L$, not all zero, with

$$a_1\chi_1 + a_2\chi_2 + \cdots + a_n\chi_n = 0$$

Among all such relations, choose one with the least number $m \geq 1$ of nonzero coefficients. The characters are distinct for any ordering, so we may renumber them so that this relation has $a_i \neq 0$ for $i \leq m$ and $a_i = 0$ for $i > m$. Thus, for any $g \in G$:

$$a_1\chi_1(g) + a_2\chi_2(g) + \cdots + a_m\chi_m(g) = 0 \tag{20.2}$$

Here $m \geq 2$: if $m = 1$, then $a_1\chi_1(g) = 0$ with $\chi_1(g) \in L^\times$, so $a_1 = 0$, which is not the case. Let g_0 be an element of G such that $\chi_1(g_0) \neq \chi_m(g_0)$ (which exists since $\chi_1 \neq \chi_m$). Since equation (20.2) holds for every element of G, we have that:

$$a_1\chi_1(g_0g) + a_2\chi_2(g_0g) + \cdots + a_m\chi_m(g_0g) = 0 \tag{20.3}$$

i.e., as each χ_i is a homomorphism,

$$a_1\chi_1(g_0)\chi_1(g) + a_2\chi_2(g_0)\chi_2(g) + \cdots + a_m\chi_m(g_0)\chi_m(g) = 0 \tag{20.4}$$

Now, multiplying equation (20.2) by $\chi_m(g_0)$ and subtracting equation (20.4), we get:

$$[\chi_m(g_0) - \chi_1(g_0)]a_1\chi_1(g) + \cdots + [\chi_m(g_0) - \chi_{m-1}(g_0)]a_{m-1}\chi_{m-1}(g) = 0$$

which holds for all $g \in G$. This is a relation among $\chi_1, \ldots, \chi_n$ with at most $m - 1$ nonzero coefficients, and its first coefficient $[\chi_m(g_0) - \chi_1(g_0)]a_1$ is nonzero, contradicting the minimality of m. Hence the characters are linearly independent, completing the proof.

Corollary 20.1.40: linear independence of Automorphisms (Dedekind's Lemma)

Let $\sigma_1, \sigma_2, ..., \sigma_n$ be distinct embeddings of K into L. Then they are linearly independent over L as functions on K. In particular, distinct automorphism of a field K are linearly independent over K as functions on K.

***Proof*:**

Each σ_i is an injective ring homomorphism (proposition 19.1.1), so it maps $K^\times$ into $L^\times$, and its restriction $\chi_i \colon K^\times \to L^\times$ is a homomorphism of multiplicative groups, that is, a character of $G = K^\times$ with values in L (definition 20.1.36). The characters $\chi_1, \ldots, \chi_n$ are distinct: if $\chi_i = \chi_j$, then σ_i and σ_j agree on $K^\times$, and also at 0, so $\sigma_i = \sigma_j$ and $i = j$. Now let $a_1, \ldots, a_n \in L$ with $a_1\sigma_1(y) + \cdots + a_n\sigma_n(y) = 0$ for every $y \in K$. Restricting to $y \in K^\times$ gives $a_1\chi_1 + \cdots + a_n\chi_n = 0$, so every $a_i = 0$ by theorem 20.1.39. The case of automorphisms is the case $L = K$, completing the proof.

Independence of automorphisms determines the structure of a Galois extension as a module over its Galois group. A finite Galois extension L/k with group G is a k-vector space of dimension $|G|$ carrying a G-action, so it is a $k[G]$-module of the right size to be free of rank one, and the normal basis theorem says that it is. Equivalently, L has a k-basis permuted simply transitively by G, in contrast with the bases $1, \alpha, \ldots, \alpha^{n-1}$ coming from a primitive element, on which G acts in no controlled way.

Theorem 20.1.41: Normal Basis Theorem

Let L/k be a finite Galois extension with group G, and put $n = [L : k] = |G|$. There exists $x \in L$ whose G-orbit $\{\sigma(x) \mid \sigma \in G\}$ is a k-basis of L. Such a basis is called a *normal basis*, and its existence says exactly that

$$L \cong k[G]$$

as $k[G]$-modules, so that L is free of rank one over the group ring.

***Proof*:**

Write $G = \{\sigma_1, \ldots, \sigma_n\}$ with $\sigma_1 = 1$. The two statements are equivalent: a k-basis of the form $\{\sigma(x) \mid \sigma \in G\}$ is precisely a $k[G]$-basis of L consisting of the single element x.

The criterion used throughout is this. For $x \in L$ the orbit $\{\sigma(x) \mid \sigma \in G\}$ is a k-basis if and only if

$$\det\left(\sigma_i^{-1}\sigma_j(x)\right)_{i,j} \neq 0$$

Indeed, suppose the determinant is non-zero and $\sum_j c_j\sigma_j(x) = 0$ with $c_j \in k$. Applying σ_i^{-1} and using that the c_j lie in k, hence are fixed, gives $\sum_j c_j\sigma_i^{-1}\sigma_j(x) = 0$ for every i. So the matrix annihilates the vector (c_j) and therefore $c = 0$. As there are n elements in the orbit and $\dim_k L = n$, independence is enough. The converse also holds, though it is not needed below: writing $v_j = \sigma_j(x)$ and $\tau_i = \sigma_i^{-1}$, which runs over G, the matrix is $(\tau_i(v_j))$, and if $v = wa$ for a k-basis w and $a \in \mathrm{GL}_n(k)$ then $\det(\tau_i(v_j)) = \det(\tau_i(w_l))\det(a)$, non-zero because $\det(\tau_i(w_l)) \neq 0$ by corollary 20.1.40. It is not the matrix of a k-linear map: for $x \in k^\times$ every entry equals x and the determinant vanishes for $n \geq 2$, so independence of automorphisms is doing the work here too.

Case 1: k infinite. Fix a k-basis $w_1, \dots, w_n$ of L and consider

$$g(X_1, \dots, X_n) = \det\left(\sum_l X_l\, \sigma_i^{-1}\sigma_j(w_l)\right)_{i,j} \in L[X_1, \dots, X_n]$$

This polynomial is not zero. By corollary 20.1.40 the automorphisms in G are linearly independent over L. Hence the $n \times n$ matrix $\big(\sigma(w_l)\big)_{\sigma,l}$ is invertible over L: an L-linear relation among its rows gives $\sum_\sigma a_\sigma \sigma$ vanishing on the k-basis w, hence, that map being k-linear, vanishing on all of L. Choose $(\xi_l) \in L^n$ with $\sum_l \xi_l\, \sigma(w_l) = 1$ for $\sigma = 1$ and 0 for every other $\sigma \in G$. Substituting $X_l = \xi_l$ turns the (i,j) entry into 1 when $\sigma_i^{-1}\sigma_j = 1$, that is when $i = j$, and into 0 otherwise, so $g(\xi) = \det(I_n) = 1$.

Now g has coefficients in L, not in k, so the familiar statement about polynomials over the field one substitutes from does not apply directly. What is needed is the infinite-subset form: *a non-zero polynomial in $L[X_1, \dots, X_n]$ cannot vanish on S^n for an infinite subset $S \subseteq L$.* This follows by induction on n, the case $n = 1$ being that a non-zero one-variable polynomial has finitely many roots (corollary 11.3.2): writing $g = \sum_d g_d(X_1, \dots, X_{n-1})X_n^d$ with some $g_{d_0} \neq 0$, induction gives $s \in S^{n-1}$ with $g_{d_0}(s) \neq 0$, and then $g(s, X_n)$ is a non-zero polynomial in one variable, so it has a non-root in S. Applying this with $S = k$, which is infinite, gives $t_1, \dots, t_n \in k$ with $g(t) \neq 0$. Put $x = \sum_l t_l w_l$. Then $\det(\sigma_i^{-1}\sigma_j(x)) = g(t) \neq 0$ and the criterion applies.

Case 2: k finite. Then L/k is an extension of finite fields, so G is cyclic, generated by the Frobenius σ. Regard L as a module over the polynomial ring $k[T]$ with T acting as σ. Because $\sigma^n = 1$, the minimal polynomial of σ as a k-linear operator on L divides $T^n - 1$, so it has degree at most n. It has degree exactly n: a relation $\sum_{i<n} c_i\sigma^i = 0$ with $c_i \in k$ not all zero would be a non-trivial L-linear dependence among the distinct automorphisms $1, \sigma, \dots, \sigma^{n-1}$, which corollary 20.1.40 forbids.

So the minimal polynomial of σ has degree $n = \dim_k L$. It divides the characteristic polynomial (proposition 14.3.10), which is monic of the same degree (proposition 13.5.2(1)), so the two are equal, and σ then admits a cyclic vector. That last step is theorem 14.3.6 together with proposition 14.3.1, which identifies the minimal polynomial as the largest invariant factor, and lemma 14.3.9(2), which makes the characteristic polynomial the product of the invariant factors: equality of degrees forces a single invariant factor, so $L \cong k[T]/(a_1)$ is cyclic over $k[T]$. Concretely there is $x \in L$ with $L = \operatorname{span}_k\{x, \sigma x, \dots, \sigma^{n-1}x\}$. That spanning set has n elements in an n-dimensional space, so it is a basis, and it is the G-orbit of x, completing the proof.

Proposition 20.1.42: Galois And Cyclic Extensions

Let F/k be an extension of degree m. Assume that k contains a primitive mth root of 1 and $\operatorname{char}(k)$ does not divide m. Then F/k is Galois and cyclic if and only if $F = k(\delta)$ with $\delta^m \in k$.

***Proof*:**
Let $\zeta \in k$ be a primitive m-th root of 1.

($\Leftarrow$) Let's say $F = k(\delta)$ with $\delta^m = c \in k$. Then all m roots of the polynomial $x^m - c$ are in F, in particular,

$$c, \zeta\delta, \zeta^2\delta, ..., \zeta^{m-1}\delta \in F$$

Furthermore, F must be generated by them. Hence F is the splitting field of the separable

polynomial $x^m - c$ in k (separable since $\mathrm{ch}(k)$ does not divide m). Thus, F is Galois

Next, to see that $\mathrm{Aut}_k(F)$ is cyclic, note that for any $\varphi \in \mathrm{Aut}_k(F)$, it is determined by $\varphi(\delta)$, which his necessarily a root of $x^m - c$, and so $\varphi(\delta) = \zeta^i \delta$ for some i determined up to a multiple of m. Then this can be used to define an isomorphism of $\mathrm{Aut}_k(F)$ with $\mathbb{Z}/m\mathbb{Z}$.

($\Rightarrow$) Conversely, assume that F/k is Galois and $\mathrm{Aut}_k(F)$ is cyclic (say, generated by φ). Since the automorphisms φ^i $(1 \leq i \leq m-1)$ are pairwise distinct, by corollary 20.1.40, they are linearly independent. Therefore, there exists an $\alpha \in F$ such that

$$\delta := \alpha + \zeta^{-1}\varphi(\alpha) + \cdots + \zeta^{-(m-2)}\varphi^{m-2}(\alpha) + \zeta^{-(m-1)}\varphi^{m-1}(\alpha) \neq 0$$

Then note that :

$$\varphi(\delta) := \varphi(\alpha) + \zeta^{-1}\varphi^2(\alpha) + \cdots + \zeta^{-(m-2)}\varphi^{m-1}(\alpha) + \zeta^{-(m-1)}\alpha = \zeta\delta$$

since $\zeta \in k$ so $\varphi(\zeta) = \zeta$. Then δ is *not* fixed by any nonidentity element of $\mathrm{Aut}_k(F)$, that is $\mathrm{Aut}_{k(\delta)}(F) = \{e\}$, so $k(\delta) = F$. Furthermore:

$$\varphi(\delta^m) = \varphi(\delta)^m = \zeta^m \delta^m = \delta^m$$

so, δ^m is fixed by φ, and hence by every element of $\mathrm{Aut}_k(F)$. Since F/k is Galois, it must be that $\delta^m \in k$, completing the proof.

20.1.43 Kummer Theory This proposition shall have an alternative proof (theorem 20.4.6) in section 20.4.

Proposition 20.1.44: Kummer Extension and Eigenvalue

Let $F = K(\sqrt[n]{a})$ and let K have enough roots of unity and $\mathrm{char}(K) \nmid n$. Let σ be a generator for $\mathrm{Gal}_K(F)$. Then $\sqrt[n]{a}$ is the eigenvector of σ.

***Proof*:**

Write $\alpha = \sqrt[n]{a}$, so $\alpha^n = a \in K$, and note $\alpha \neq 0$ (otherwise $F = K$ and there is nothing to prove). Since σ fixes K pointwise,

$$\sigma(\alpha)^n = \sigma(\alpha^n) = \sigma(a) = a = \alpha^n$$

so $\zeta := \sigma(\alpha)/\alpha$ satisfies $\zeta^n = 1$, that is, ζ is an n-th root of unity. As K contains the n-th roots of unity, $\zeta \in K$. Hence

$$\sigma(\alpha) = \zeta\alpha \qquad \zeta \in K$$

which says exactly that $\alpha = \sqrt[n]{a}$ is an eigenvector of the K-linear operator $\sigma : F \to F$, with eigenvalue the root of unity ζ, consistent with lemma 20.1.35.

In fact ζ is a primitive m-th root of unity, where $m = |\mathrm{Gal}_K(F)|$ is the order of σ. Iterating gives $\sigma^j(\alpha) = \zeta^j\alpha$ for all j, using that σ fixes $\zeta \in K$. If $\zeta^d = 1$, then σ^d fixes α, and since $F = K(\alpha)$, the automorphism σ^d fixes all of F, so $\sigma^d = \mathrm{id}$ and $m \mid d$. Conversely, $\sigma^m = \mathrm{id}$ gives $\zeta^m = 1$.

Exercise 20.1.3

1. Let $k(t)/k$ be an infinite extension. Determine $\mathrm{Aut}_k(k(t))$
2. Show that for all n, there is an extension F of $\mathbb{Q}$ such that $F/\mathbb{Q}$ is not Galois. Similarly, show that for all n, there exists an extension L such that $\mathrm{Gal}_{\mathbb{Q}}(L) \cong \mathbb{Z}/n\mathbb{Z}$ (Hint: show that for all n, there exists a p such that $p \equiv 1 \bmod n$)

20.2 Application of Galois Theory

20.2.1 Fundamental Theorem of Algebra

Theorem 20.2.1: Fundamental Theorem of Algebra

$\mathbb{C}$ is Algebraically closed

***Proof*:**

Let $f \in \mathbb{C}[x]$ be nonconstant, let $\bar{f}$ have the conjugate coefficients, and let $g = f\bar{f}$. Conjugation is a field automorphism of $\mathbb{C}$, so $\bar{g} = \bar{f}f = g$ and $g \in \mathbb{R}[x]$. Let F be a splitting field of g over $\mathbb{C}$. Since $\mathbb{C} = \mathbb{R}(i)$, F is generated over $\mathbb{R}$ by i and the roots of g, so it is a splitting field of $g(x)(x^2+1)$ over $\mathbb{R}$. For every intermediate field $\mathbb{R} \subseteq E \subseteq F$, F is also a splitting field of $g(x)(x^2+1)$ over E, so F/E is finite and normal (theorem 19.5.13), and it is separable, since the minimal polynomial over E of each element of F is irreducible and $\mathrm{ch}(E) = 0$ (corollary 19.6.8). Hence F/E is Galois (theorem 20.1.13), which gives $|\mathrm{Aut}_E(F)| = [F : E]$. For every subgroup H of $\mathrm{Aut}_{\mathbb{R}}(F)$, $[F : F^H] = |H|$ (proposition 20.1.12).

Let $G = \mathrm{Aut}_{\mathbb{R}}(F)$. Then $|G| = [F : \mathbb{R}] = [F : \mathbb{C}][\mathbb{C} : \mathbb{R}]$ (theorem 19.3.7) is even. Write $|G| = 2^a m$ with m odd, let H be a Sylow 2-subgroup of G (theorem 4.8.6), and let $E = F^H$. Then $[F : E] = |H| = 2^a$, so $[E : \mathbb{R}] = m$. By theorem 19.6.20, $E = \mathbb{R}(\theta)$ for some θ whose minimal polynomial over $\mathbb{R}$ has odd degree m. A real polynomial of odd degree takes both signs, so it has a real root by the intermediate value theorem ([18, The Intermediate Value Theorem]), and being irreducible, it has degree 1. Hence $m = 1$ and G is a 2-group.

The subgroup $P = \mathrm{Aut}_{\mathbb{C}}(F)$ of G therefore has order 2^n for some $n \geq 0$. Every group P' of order 2^r with $r \geq 1$ has a subgroup of index 2. For $r = 1$ the trivial subgroup is one. For $r > 1$, $Z(P')$ is nontrivial (theorem 4.6.5) and contains an element z of order 2 (theorem 4.8.7). The subgroup $\langle z \rangle$ is central, hence normal, and by induction on r, $P'/\langle z \rangle$ has a subgroup of index 2, whose preimage in P' has index 2. Suppose $n \geq 1$, and let Q be a subgroup of index 2 in P. The field $E' = F^Q$ has $[F : E'] = |Q| = 2^{n-1}$, so $[E' : \mathbb{C}] = 2$, and every $\beta \in E' \setminus \mathbb{C}$ has $E' = \mathbb{C}(\beta)$ and a minimal polynomial over $\mathbb{C}$ of degree 2. But every $x^2 + bx + c \in \mathbb{C}[x]$ has the roots $(-b \pm w)/2$ in $\mathbb{C}$, where w is a square root of $b^2 - 4c$. Such a w exists: every $z = s + ti$ with $s, t \in \mathbb{R}$ has the square root $u + iv$, where $u = \sqrt{(|z| + s)/2}$ and $v = \pm\sqrt{(|z| - s)/2}$ with the sign chosen so that $2uv = t$, and real square roots of nonnegative reals, $|z|$ included, exist by the intermediate value theorem again. So no such polynomial is irreducible. Hence $n = 0$, so $[F : \mathbb{C}] = |P| = 1$ and $F = \mathbb{C}$. Then g factors into linear terms in $\mathbb{C}[x]$, and so does its divisor f, as $\mathbb{C}[x]$ is a unique factorization domain.

A topological proof is in [44, Fundamental Theorem Of Algebra].

Exercise 20.2.1

1. Show that every polynomial of $\mathbb{R}[x]$ of degree ≥ 3 is reducible. Show that every irreducible polynomial is linear or is of degree two with negative discriminant, that is, $ax^2 + bx + c$ with $b^2 - 4ac < 0$.

Corollary 20.2.2: No Field Structure On $\mathbb{R}^n$, $n \geq 3$

There does not exist a field structure on $\mathbb{R}^n$ $(n \geq 3)$ such that $\mathbb{R}$ includes into $\mathbb{R}^n$ (i.e. $\mathbb{R}^n$ is a field extension of $\mathbb{R}$)

***Proof*:**
For the sake of contradiction, assume $\mathbb{R} \subseteq \mathbb{R}^n$ is a field extension for $n \geq 3$. Since $\mathbb{R}^n$ is an n-dimensional vector-space, it is a degree n algebraic extension. Hence, by lemma 19.5.7, there is an (injective) homomorphism $\mathbb{R}^n \to \mathbb{C}$ fixing $\mathbb{R}$. However, the dimension of the right hand side is greater than 2, and hence no such homomorphism exists, a contradiction.

Note that we had to be careful in how we state the above corollary. It is not true that there is no field structure on $\mathbb{R}^n$ $(n \geq 3)$. Since $\mathbb{R}^n$ and $\mathbb{R}$ have the same cardinality, we may (by the axiom of choice) find a bijection from $\mathbb{R}^n$ to $\mathbb{R}$. Then we may put a field structure on $\mathbb{R}$ through the bijection. However, the field structure on $\mathbb{R}^n$ would certainly not have $\mathbb{R}$ as a subfield, and it impossible to determine the product of any two elements[7].

Theorem 20.2.3: Frobenius Theorem

Let D be a finite-dimensional associative division algebra over $\mathbb{R}$, with $\mathbb{R}$ contained in the center of D. Then D is isomorphic, as an $\mathbb{R}$-algebra, to exactly one of

1. $\mathbb{R}$, the real numbers
2. $\mathbb{C}$, the complex numbers
3. $\mathbb{H}$, the quaternions

***Proof*:**
Step 1: every element satisfies a real quadratic. For $x \in D$ the subalgebra $\mathbb{R}[x]$ is commutative, finite-dimensional over $\mathbb{R}$, and has no zero divisors, so it is a field: for $0 \neq y \in \mathbb{R}[x]$ multiplication by y is an injective $\mathbb{R}$-linear endomorphism of a finite-dimensional space, hence surjective, so 1 lies in its image and y is invertible. Thus $\mathbb{R}[x]$ is a finite field extension of $\mathbb{R}$, and by theorem 20.2.1 the only such extensions are $\mathbb{R}$ and $\mathbb{C}$. So either $x \in \mathbb{R}$, or $\mathbb{R}[x] \cong \mathbb{C}$ and x satisfies a unique monic real quadratic $t^2 - 2at + b$ with $a^2 < b$.

Step 2: the purely imaginary part is a subspace. Put

$$V = \left\{ x \in D \;\middle|\; x^2 \in \mathbb{R} \text{ and } x^2 \leq 0 \right\}$$

If $x \notin \mathbb{R}$, Step 1 gives $(x - a)^2 = a^2 - b < 0$, so $x - a \in V$ and $D = \mathbb{R} + V$. Also $\mathbb{R} \cap V = 0$, because a real number with $r^2 \leq 0$ is 0. It remains to see that V is closed under addition, and

[7] loosely speaking, if we would be able to explicitly find a bijection from $\mathbb{R}^n$ to $\mathbb{R}$, but that would be a "countable process", but $\mathbb{R}^n$ and $\mathbb{R}$ is uncountable

$\mathbb{R}$-scaling is clear. If $v = \lambda u$ then $(u+v)^2 = (1+\lambda)^2 u^2 \le 0$ is real and $u+v \in V$, so assume u and v are linearly independent. First $1, u, v$ are then linearly independent over $\mathbb{R}$. Take a relation $c_0 + c_1 u + c_2 v = 0$. If $c_2 \neq 0$ then $v = -(c_0 + c_1 u)/c_2$, so $v^2 = (c_0^2 + 2c_0 c_1 u + c_1^2 u^2)/c_2^2$. Since u^2 is real and v^2 is real, $2c_0 c_1 u$ is real and hence $c_0 c_1 = 0$. Then $c_0 = 0$ makes v a multiple of u, and $c_1 = 0$ puts v in $\mathbb{R} \cap V = 0$, each contradicting independence. If instead $c_2 = 0$ and $c_1 \neq 0$ then $u \in \mathbb{R} \cap V = 0$, again a contradiction, so $c_0 = c_1 = c_2 = 0$. Neither $u+v$ nor $u-v$ lies in $\mathbb{R}$: if $u+v = c$ then $v^2 = c^2 - 2cu + u^2 \in \mathbb{R}$ forces $cu \in \mathbb{R}$, so $c = 0$ and $v = -u$, against independence, and the same argument applies to $u - v$. So Step 1 gives $(u+v)^2 = 2\alpha(u+v) - \beta$ and $(u-v)^2 = 2\gamma(u-v) - \delta$ with all four constants real. Adding, and using $(u+v)^2 + (u-v)^2 = 2u^2 + 2v^2 \in \mathbb{R}$,

$$2(\alpha+\gamma)u + 2(\alpha-\gamma)v \in \mathbb{R}$$

so by the independence of $1, u, v$ both $\alpha+\gamma$ and $\alpha-\gamma$ vanish, giving $\alpha = 0$ and $(u+v)^2 = -\beta \in \mathbb{R}$. Since $u+v \notin \mathbb{R}$, Step 1 forces $-\beta < 0$, so $u+v \in V$. Hence V is a subspace and $D = \mathbb{R} \oplus V$.

Step 3: V carries a positive-definite form. For $u, v \in V$ the computation in Step 2 shows $uv + vu = (u+v)^2 - u^2 - v^2 \in \mathbb{R}$, so

$$\langle u, v \rangle = -\tfrac{1}{2}(uv + vu)$$

is a symmetric $\mathbb{R}$-bilinear form on V, and $\langle u, u \rangle = -u^2 > 0$ for $u \neq 0$. It is therefore an inner product, and orthogonality of u and v means exactly $uv = -vu$.

Step 4: the three cases. If $\dim_{\mathbb{R}} V = 0$ then $D = \mathbb{R}$. If $\dim_{\mathbb{R}} V = 1$, choose u with $\langle u, u \rangle = 1$, so $u^2 = -1$ and $D = \mathbb{R} \oplus \mathbb{R}u \cong \mathbb{C}$. Suppose $\dim_{\mathbb{R}} V \ge 2$ and choose orthonormal $i, j \in V$, so that

$$i^2 = j^2 = -1, \qquad ij = -ji$$

Set $k = ij$. Using $ji = -ij$ once,

$$k^2 = (ij)(ij) = i(ji)j = i(-ij)j = -i^2 j^2 = -1$$

so $k \in V$ and $\langle k, k \rangle = 1$. Moreover $ik = i(ij) = i^2 j = -j$ while $ki = (ij)i = i(ji) = i(-ij) = -i^2 j = j$, so $ik + ki = 0$ and $\langle i, k \rangle = 0$. Likewise $jk = j(ij) = (ji)j = -ij^2 = i$ and $kj = (ij)j = ij^2 = -i$, so $\langle j, k \rangle = 0$. Hence i, j, k are orthonormal in V and span a copy of $\mathbb{H}$ inside D. Finally $\dim_{\mathbb{R}} V = 3$: if $w \in V$ were orthogonal to each of i, j, k then $wi = -iw$ and $wj = -jw$, whence

$$wk = w(ij) = (wi)j = -i(wj) = i(jw) = (ij)w = kw$$

so w commutes with k, while orthogonality to k says $wk = -kw$. Then $2wk = 0$, and D has no zero divisors with $k \neq 0$, so $w = 0$. Hence $D = \mathbb{R} \oplus \mathbb{R}i \oplus \mathbb{R}j \oplus \mathbb{R}k \cong \mathbb{H}$. The three algebras have distinct dimensions $1, 2, 4$, so the case is determined by D, completing the proof.

Associativity is doing real work here, and dropping it changes the question rather than merely lengthening the list. The octonions $\mathbb{O}$ are an eight-dimensional real division algebra that is not associative, and Zorn's theorem extends the classification above to $\mathbb{R}$, $\mathbb{C}$, $\mathbb{H}$, $\mathbb{O}$ once "associative" is weakened to *alternative*, meaning that any two elements generate an associative subalgebra. Without some such hypothesis there is no classification, but the *dimension* is still forced: a finite-dimensional real division algebra has dimension 1, 2, 4 or 8. That last statement is not algebra at all. It is

equivalent to asking which spheres are parallelizable, and the known proofs run through topological K-theory and the Adams operations. *Vector Bundles and K-Theory* [48] develops that machinery and proves the theorem it turns on, that a map $S^{4n-1} \to S^{2n}$ of odd Hopf invariant forces $n \in \{1, 2, 4\}$. The passage from a division algebra to such a map uses the Hopf construction, which that book cites rather than builds.

As an exercise, show that $\mathbb{H}$ is not a $\mathbb{C}$-algebra. The quaternions are also the Clifford algebra of the negative definite form $\langle -1, -1\rangle$ on $\mathbb{R}^2$, which is example 22.7.6 in section 22.7.

20.2.2 Construction of Regular n-gon

20.2.3 Symmetric Functions

If $t_1, ..., t_n$ are indeterminates, then the polynomial:

$$P_n(x) := (x - t_1)(x - t_2) \cdots (x - t_n) \in \mathbb{Z}[t_1, ..., t_n][x]$$

is universal in the sense thati every polynomial of degree n with coefficeints in (say) an intergral domain may be obtained from $P_n(x)$ given a suitable choice for $t_1, ..., t_n$, choices form a possibly larger ring. Expanding this polynomial, we get that:

$$P_n(x) = x^n - s_1(t_1, ..., t_n)x^{n-1} + \cdots + (-1)^n s_n(t_1, ..., t_n)$$

where each s_i are the *elementary functions*. If $n = 3$, then:

$$\begin{aligned} s_1(t_1, t_2, t_3) &= t_1 + t_2 + t_3 \\ s_2(t_1, t_2, t_2) &= t_1t_2 + t_1t_3 + t_2t_3 \\ s_3(t_1, t_2, t_3) &= t_1t_2t_3 \end{aligned}$$

The function s_n are called symmetric since they are invariant under permutation (essentially, since $P_n(x)$ is invariant under permutation). The name "elementary" comes because of the following theorem

Theorem 20.2.4: Fundamental Theorem On Symmetric Functions

Let K be a field and let $\varphi \in K(t_1, ..., t_n)$. Then φ is invariant under permutation (i.e. symmetric) if and only if it is a rational function (with coefficient in K) of the elementary symmetric functions $s_1, ..., s_n$.

The functions s_i are viewed as elements of $K[t_1, ..., t_n]$ which we can do unambiguously since $\mathbb{Z}$ is initial in **Ring**.

***Proof*:**
Let $F = K(t_1, ..., t_n)$ and $k = K(s_1, ..., s_n)$ be the subfield generated by the elementary symmetric functions over K. Then F is a splitting field of the separable polynomial $P_n(x)$ over k, i.e. F is Galois and:

$$|\mathrm{Aut}_k(F)| = [F : k] \leq n!$$

Notice that the symmetric group S_n acts (faithfully) on F by permuting $t_1, ..., t_n$ and the action is the identity on $k = K(s_1, ..., s_n)$ since each s_i is symmetric. Thus, S_n can be identified with some subgroup of $\mathrm{Aut}_k(F)$. However,

$$n! = |S_n| \leq |\mathrm{Aut}_k(F)| \leq n!$$

showing that $\mathrm{Aut}_k(F) \cong S_n$

Now, since F/k is Galois, we get $k = F^{S_n}$. This in fact completes the rpoof: if $\varphi \in K(t_1, .., t_n)$, then φ is invariant under the S_n action on $t_1, ..., t_n$ if and only if $\varphi \in k \in K(s_1, ..., s_n)$, as we sought to show.

The theorem above is about rational functions of finitely many variables. Downstream work with formal characteristic-class calculi needs the same statement for formal power series over $\mathbb{Q}$, which does not follow from it, so it is proved here alongside.

The elementary symmetric functions are one basis for the symmetric polynomials, and for counting problems they are the wrong one: the classes that arise geometrically are indexed by partitions, and expressing them in the s_i hides that indexing behind a determinant. The Schur polynomials are the basis adapted to it, and they are recorded here because [20] builds its degeneracy-locus formulas out of them.

Definition 20.2.5: Schur Polynomial

A *partition* $\lambda = (\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_n \geq 0)$ of length at most n is a weakly decreasing tuple of non-negative integers. Write $|\lambda| = \sum_i \lambda_i$. Put $\delta = (n-1, n-2, \ldots, 1, 0)$. The *Schur polynomial* attached to λ in the variables $t_1, \ldots, t_n$ is the bialternant

$$s_\lambda(t_1, \ldots, t_n) \;=\; \frac{\det\left(t_i^{\lambda_j + n - j}\right)_{1 \leq i,j \leq n}}{\det\left(t_i^{n-j}\right)_{1 \leq i,j \leq n}}$$

the denominator being the Vandermonde determinant $\prod_{i<j}(t_i - t_j)$.

Proposition 20.2.6: Schur Polynomials Are a Basis

Each s_λ is a symmetric polynomial, homogeneous of degree $|\lambda|$, and the s_λ for λ of length at most n form a $\mathbb{Z}$-basis of the ring of symmetric polynomials in $t_1, \ldots, t_n$. For the single-row partition $\lambda = (m)$ one has $s_{(m)} = h_m$, the complete homogeneous symmetric polynomial, and for the single-column partition $\lambda = (1^m)$ one has $s_{(1^m)} = s_m$, the m-th elementary symmetric polynomial.

***Proof*:**
The numerator of definition 20.2.5 is alternating: transposing two variables transposes two rows of the matrix and changes the sign of the determinant. The Vandermonde is alternating for the same reason, so the quotient is symmetric. It is a polynomial because an alternating polynomial vanishes on every hyperplane $t_i = t_j$ and is therefore divisible by each $t_i - t_j$, hence by their product, these being pairwise coprime. Degrees subtract to $|\lambda| + \binom{n}{2} - \binom{n}{2} = |\lambda|$.

For the basis claim, order partitions lexicographically. The monomial t^λ occurs in s_λ with

coefficient 1 and every other monomial of s_λ is lexicographically smaller, so the transition matrix from $\{s_\lambda\}$ to the monomial symmetric functions $\{m_\lambda\}$ is unitriangular over $\mathbb{Z}$, hence invertible. And the m_λ are a $\mathbb{Z}$-basis by inspection. The two special cases are the standard evaluations of the bialternant, the second being the observation that $\det(t_i^{\lambda_j+n-j})$ for $\lambda = (1^m)$ factors as the Vandermonde times s_m.

Theorem 20.2.7: Jacobi-Trudi Formulas

For a partition λ with conjugate partition λ',

$$s_\lambda = \det\left(h_{\lambda_i - i+j}\right)_{1\le i,j\le \ell}, \qquad s_\lambda = \det\left(s_{\lambda'_i - i+j}\right)_{1\le i,j\le \ell'}$$

where h_m is the complete homogeneous and s_m the elementary symmetric polynomial, with the conventions $h_0 = s_0 = 1$ and $h_m = s_m = 0$ for $m < 0$, and ℓ, ℓ' are the lengths of λ and λ'.

***Proof Cited*:**
Not reproduced here (QC #3(a)). The clean proof is the Lindström-Gessel-Viennot lattice-path argument: both determinants count families of non-intersecting paths, and the sign-reversing involution on intersecting families cancels every off-diagonal term. That is combinatorics rather than algebra, and belongs with the lattice-path methods of the combinatorics volume. The identity is recorded here because this is where the symmetric functions live.

What the identity is for: it converts a partition-indexed basis element into a determinant in classes one can compute with. In [20] the h_m and s_m become Chern classes of a bundle, and the determinant becomes the Giambelli and Thom-Porteous formulas.

Lemma 20.2.8: Symmetric Power Series in the Chern Roots

Let $\Phi \in \mathbb{Q}[[a_1, \dots, a_r]]$ be invariant under every permutation of $a_1, \dots, a_r$. Then there is a unique $\widetilde{\Phi} \in \mathbb{Q}[[y_1, \dots, y_r]]$ with

$$\Phi(a_1, \dots, a_r) = \widetilde{\Phi}\big(e_1(a), \dots, e_r(a)\big)$$

If Φ is homogeneous of degree d, then $\widetilde{\Phi}$ is a polynomial, isobaric of weight d when y_i is assigned weight i.

***Proof*:**
Existence. Each homogeneous component of Φ is symmetric, and an isobaric polynomial of weight d in the y_i contributes only in degree d after substituting $y_i = e_i$. So it suffices to treat a symmetric homogeneous polynomial P of degree d and assemble the results over d. Order the monomials $a^\nu = a_1^{\nu_1} \cdots a_r^{\nu_r}$ lexicographically, and let a^ν be the largest monomial occurring in a nonzero P, with coefficient λ. Then $\nu_1 \ge \nu_2 \ge \cdots \ge \nu_r$: if $\nu_j < \nu_{j+1}$ for some j, the transposed monomial also occurs in P, by symmetry, and is lexicographically larger. The product

$$e_1^{\nu_1-\nu_2} e_2^{\nu_2-\nu_3} \cdots e_{r-1}^{\nu_{r-1}-\nu_r} e_r^{\nu_r}$$

is symmetric and homogeneous of degree $\sum_j \nu_j = d$, and its lexicographic leading monomial

is a^ν with coefficient 1, since the leading monomial of e_k is $a_1 a_2 \cdots a_k$ and the exponent of a_j in the displayed product is $\sum_{k \geq j}(\nu_k - \nu_{k+1}) = \nu_j$. Subtracting λ times it from P leaves a symmetric homogeneous polynomial of degree d whose leading monomial is strictly smaller. There are finitely many monomials of degree d, so the process terminates and expresses P as an isobaric polynomial of weight d in the e_i.

Uniqueness. It suffices to show that $e_1, \ldots, e_r$ are algebraically independent, since a difference of two representations of Φ is a power series R with $R(e_1, \ldots, e_r) = 0$, whose isobaric components of distinct weights land in distinct degrees in the a_j and so vanish separately. Let $Q = \sum_m \kappa_m y^m$ be a nonzero polynomial. The leading monomial of $e^m = e_1^{m_1} \cdots e_r^{m_r}$ is $a_1^{m_1+\cdots+m_r} a_2^{m_2+\cdots+m_r} \cdots a_r^{m_r}$, and the assignment $m \mapsto (m_1 + \cdots + m_r,\ m_2 + \cdots + m_r,\ \ldots,\ m_r)$ is injective, so distinct exponent vectors m produce distinct leading monomials. Among the finitely many m with $\kappa_m \neq 0$, take the one whose leading monomial is lexicographically largest. Every monomial of every other term is strictly smaller than it, so it survives in $Q(e_1, \ldots, e_r)$ with coefficient $\kappa_m \neq 0$, and $Q(e_1, \ldots, e_r) \neq 0$, as we sought to show.

One important result that was proven in the previous theorem is noted for future reference:

Lemma 20.2.9: S_n Extension

Let K be a field, $t_1, ..., t_n$ indeterminates, and let $s_1, ..., s_n$ be the elementary symmetric functions on $t_1, ..., t_n$. Then the extension:

$$K(s_1, ..., s_n) \subseteq K(t_1, ..., t_n)$$

is Galois, with Galois group S_n

From this, we can prove that every finite group is represented as a galois group:

Theorem 20.2.10: Finite Group Represented by Galois Group

Let G be a finite group. Then there exists a Galois extension F/k such that $\mathrm{Aut}_k(F) \cong G$

***Proof*:**
By theorem 4.5.3, G is isomorphic to a subgroup H of S_n for some n. Let K be any field, $F = K(t_1, \ldots, t_n)$ and $k' = K(s_1, \ldots, s_n)$. By lemma 20.2.9, F/k' is Galois with $\mathrm{Aut}_{k'}(F) = S_n$, acting by permuting the t_i, and F is the splitting field over k' of $\prod_i (x - t_i)$, a polynomial with distinct roots and coefficients $\pm s_j$. Let $k = F^H$. Then F is also the splitting field of that separable polynomial over k, so F/k is Galois (theorem 20.1.13), and $\mathrm{Aut}_k(F) = H \cong G$ (proposition 20.1.12).

It is not known whether every finite group is the Galois group of a Galois extension of $\mathbb{Q}$. This is the *inverse Galois problem* [1, §VII.7]. Every finite abelian group is one, given infinitely many primes in each residue class 1 mod n (exercise 20.2.2 (1)).

Exercise 20.2.2

1. Assume that for every $n \geq 1$ there are infinitely many primes $p \equiv 1 \pmod{n}$ (a special case of Dirichlet's theorem). Show that every finite abelian group G is $\mathrm{Aut}_{\mathbb{Q}}(E)$ for some finite Galois

extension $E/\mathbb{Q}$. (Write $G \cong \mathbb{Z}/n_1\mathbb{Z} \times \cdots \times \mathbb{Z}/n_r\mathbb{Z}$, choose distinct primes $p_i \equiv 1 \pmod{n_i}$, and use $\mathrm{Aut}_{\mathbb{Q}}(\mathbb{Q}(\zeta_N)) \cong (\mathbb{Z}/N\mathbb{Z})^\times$ for $N = p_1 \cdots p_r$ (proposition 19.7.9, corollary 11.3.17) with theorem 20.1.23.)

Corollary 20.2.11

Let D be the discriminant of the field extension. Then:

$$K(s_1, ..., s_n)(\sqrt{D}) \subseteq K(t_1, ..., t_n)$$

is Galois, with Galois group A_n.

20.2.4 Solvability By Radicals

We are finally in a position to prove that an arbitrary degree 5 polynomial cannot be written in terms of it's coefficients using the operations $+, -, \times, \div, \sqrt{}$. Is is famously known, any quadratic polynomial over k ($\mathrm{char}(k) \neq 2$) can be written in terms of:

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

and similar equations exists for degree 3 and 4 polynomials, though they get horrendously messy.

20.2.12 Goal There exists irreducible polynomials of degree $n \geq 5$ that are unsolvable

To avoid complications, we will assume in this section that $\mathrm{char}(k) = 0$.

Definition 20.2.13: Radical Extension

Let F/k be a field extension. Then if $F = k(\alpha_1, \alpha_2, ..., \alpha_n)$ where

$$k \subseteq k(\alpha_1) \subseteq k(\alpha_1, \alpha_2), \subseteq \cdots \subseteq k(\alpha_1, ..., \alpha_n)$$

where for each $i \in \{1, ..., n\}$, there exists an n_i such that $\alpha^{n_i} \in k(\alpha_1, ..., \alpha_{i-1})$, then F/k is called a *radical extension*

Elements of radical extensions are easy to picture: if $k = \mathbb{Q}$, then simply nest as many nth-roots as you please, for example:

$$\sqrt[7]{3 + \sqrt[2]{19}^5 \sqrt[11]{3+2}}$$

it is also clear by definition that a radical extension is the composition of cyclic exensions provided that the appropriate roots of 1 are in k. The next result insures us that any [separable] radical extension is contained in a radical extensions that is *Galois*:

Lemma 20.2.14: Separable And Galois Radical Extensions

Let F/k be a separable radical extension. Then F is conatiend in a Galois radical extension

***Proof*:**
Let F/K be a separable radical extension. Then F is a separable finite extension of K, and so by the primitive root theorem (theorem 19.6.20), there exists an $\alpha \in F$ such that $F = k(\alpha)$. Let L be the splitting field of $k(\alpha)$. Then L is equal to the composite of all the field $\sigma(F)$ for every σ that embeds F into $\overline{k}$:

$$L = \sigma_1(F) \cdots \sigma_n(F)$$

Then the composition of radical extensions is radical, showing that L is radical. Thus, L is the splitting field of a separable polynomial, and so it is Galois, completing the proof.

Lemma 20.2.15: Solvability In Char 0

Let k be a field such that $\text{char}(k) = 0$, and let $f(x) \in k[x]$ be an irreducible polynomial. Then the roots of $f(x)$ can be written in terms of field operations and radicals if and only if the splitting field fo $f(x)$ is contined in a Galois Radical extension

***Proof*:**

($\Leftarrow$) If the splitting field $f(x)$ is contained in a radical extension (not necessarily Galois), then *a fortiori* the roots can be written using field operations and radicals.

($\Rightarrow$) Conversely, assume the roots of the irreducible polynomial $f(x)$ can be written in terms of field operations and radicals . Then any formula for a root t_1 of $f(x)$ can be turned into a formula for t_i by applying an element σ_i of $\text{Aut}_k(\overline{k})$ sending t_1 to t_i (this automorphism exists since $f(x)$ is irreducible). Thus, if the formula exists for one root, a formula exists for all roots. Thus, $k(t_1), ..., k(t_n)$ are each radical extensions and

$$L = k(t_1)k(t_2) \cdots k(t_n)$$

is a radical extension that contains all the roots of $f(x)$. Thus, by minimality of the splitting field, it is contained in a radical extension of k. By lemma 20.2.14, it must also be contained in a galois radical extension, as we sought to show.

Thus, we can characterize when a polynomial is solvable by radicals by when it is contained in a Galois radical extension. Since the extension is Galois, we have a corresponding Galois group to analyze. We will soon see that the appropriate type of group that is needed for $f(x)$ to be solvable is one we have encountered a long time ago:

Definition 20.2.16: Solvable Extension

Let F/K be a Galois extension. Then F/K is a *solvable extension* if $\text{Aut}_K(F)$ is a solvable group

Recall that if G is a solvable group and finite, then G can be decomposed into a subnormal series with cyclic factors:

$$G = G_0 \trianglerighteq G_1 \trianglerighteq G_2 \trianglerighteq \cdots G_n = \{1\}$$

notice how this parrallels the fact that any radical extension is a sequence of cyclic extensions.

Lemma 20.2.17: Relating Radical And Solvable Extensions

Let F/K be a galois extension with $\text{char}(k) = 0$. If F/K has enough roots of 1, then F/K is radical if and only if it is solvable

What is meant by "enough roots of 1" is that K contains primitive mth roots of 1 for a common multiple m of the order of the corresponding cyclic factors.

***Proof*:**
This is a generalization of proposition 20.1.42.

Let F/K be a radical extension, so that:

$$K \subseteq k(\delta_1) \subseteq \cdots \subseteq K(\delta_1, ..., \delta_r) = F$$

where $\delta_i^{m_i} \in K(\delta_1, ..., \delta_{i-1})$. By assumption, K contains primitive m_ith roots of 1 for each i. Next, consider the corresponding subgroups of $G = \text{Aut}_k(F)$:

$$G = G_0 \supseteq G_1 \supseteq \cdots \supseteq G_r = \{e\}$$

where $G_i = \text{Aut}_{K(\delta_1,...,\delta_i)}(F)$. Next, note That F is Galois over each intermediate field (by proposition 19.5.16), and each extension:

$$K(\delta_1, ..., \delta_{i-1}) \subseteq K(\delta_1, ..., \delta_i)$$

satisfies proposition 20.1.42, and hence each G_i is cyclic. By theorem 20.1.23, each G_i is normal (since the automorphism group is abelian) in the previous G_{i-1} with a cyclic quotient, and hence G is solvable.

Conversely, the same argument works in reverse using the converse implication in proposition 20.1.42 to show that an extension with enough roots of 1 is radical.

We were dependent on each extension containing enough roots of unity. In the following, we show that we can drop them:

Proposition 20.2.18: Solvable In Char 0

Let K be a field where $\text{char}(K) = 0$, and let F/K be a Galois extension. Then F/K is solvable if and only if it is containd in a Galois Radical extension

Technically, we only need that $\text{char}(K)$ not divide the relevant degrees of the extensions, however for our purposes it is safe to assume $\text{char}(K) = 0$.

***Proof*:**

($\Rightarrow$) Assume $\text{Aut}_K(F)$ is solvable. Let M be the common multiple of the cyclic quotients and let ζ be a mth root of 1 in an algebraic closure $\overline{K}$ of K. The splitting field $K(\zeta)$ of $x^M - 1$ is Galois over K since $\text{char}(K) = 0$. By proposition 20.1.28, the extension $K(\zeta) \subseteq F(\zeta)$ is also Galois, with Galois group isomorphic to $\text{Aut}_{K(\zeta)\cap F}(F)$. This is a subgroup of the solvable group $\text{Aut}_K(F)$, so $\text{Aut}_{K(\zeta)}(F(\zeta))$ is solvable (proposition 6.1.10).

Since $K(\zeta) \subseteq F(\zeta)$ has enough roots of 1, by lemma 20.2.17, it is radical. Thus, so is $K \subseteq F(\zeta)$ since $K \subseteq K(\zeta)$ is by trivially radical. Thus, F/K is contained in a radical extension.

($\Leftarrow$) Conversely, assume that F/K is Galois and contained in a radical Galois extension L/K. Let M be a common multiple of the order of the cyclic factors in this extensions,and let ζ be a primitive mth root of 1. The composite $L(\zeta)$ is Galois and radical over $K(\zeta)$ (show that if E/K is radical and T/K is any extension, then ET/K is a radical extension), and it has enough roots of 1.

By lemma 20.2.17, $L(\zeta)/K(\zeta)$ is solvable. Then $L(\zeta)/K$ is solvable as well. Indeed, this extension is Galois since it's a composition of galois (exercise 6.13). Applying the fundamental theorem to:

$$K \subseteq K(\zeta) \subseteq L(\zeta)$$

to established that $\mathrm{Aut}_K(K(\zeta))$ is isomorphic to the quotient of $\mathrm{Aut}_K(L(\zeta))$ by its normal subgroup $\mathrm{Aut}_{K(\zeta)}(L(\zeta))$. Both $\mathrm{Aut}_{K\zeta)}(L(\zeta))$ and $\mathrm{Aut}_K(K(\zeta))$ are solvable (being abelian), and it follows that $\mathrm{Aut}_K(L(\zeta))$ is solvable.

Since F is an intermediate field and Galois over T, by the fundamental theorem, $\mathrm{Aut}_K(F)$ is a homomorphic image of $\mathrm{Aut}_K(L(\zeta))$. This shows that $\mathrm{Aut}_K(F)$ is a homomorphic image of a solvable group, hence it is itself solvable, and so we are done.

Corollary 20.2.19: Galois Criterion

Let K be a field such that $\mathrm{char}(K) = 0$ and let $f(x) \in K[x]$ be an irreducible polynomial. Then f(x) is solvable by radicals if and only if its galois group is solvable

***Proof*:**
This is an immediate consequence of lemma 20.2.15 and proposition 20.2.18.

Theorem 20.2.20: Abel's Theorem

Let K be a field of characteristic 0, let $n \geq 5$, let $t_1, \ldots, t_n$ be indeterminates, and let $s_1, \ldots, s_n$ be the elementary symmetric functions on $t_1, \ldots, t_n$. Then the *generic polynomial*

$$f(x) = \prod_{i=1}^{n}(x - t_i) = x^n - s_1 x^{n-1} + s_2 x^{n-2} - \cdots + (-1)^n s_n \in K(s_1, \ldots, s_n)[x]$$

of degree n is not solvable by radicals over $K(s_1, \ldots, s_n)$.

***Proof*:**
Write $E = K(s_1, \ldots, s_n)$. The roots of f are the distinct elements $t_1, \ldots, t_n$, so the splitting field of f over E is $K(t_1, \ldots, t_n)$, and by lemma 20.2.9 its Galois group is S_n, acting by permuting the t_i. This action is transitive, so f is irreducible over E. The roots of a monic factor $g \in E[x]$ of f of positive degree form a nonempty subset of $\{t_1, \ldots, t_n\}$, and every automorphism preserves that subset because it fixes the coefficients of g. Transitivity carries any one root of g to every t_i, so the subset is all of $\{t_1, \ldots, t_n\}$, and $g = f$.

It remains to see that S_n is not solvable. The group A_5 is simple by theorem 4.9.6, and it is not abelian, since the 3-cycles $(1\,2\,3)$ and $(3\,4\,5)$ do not commute: composed in one order they send 3 to 4, and in the other order they send 3 to 1. The commutator subgroup of A_5 is therefore a nontrivial normal subgroup, hence all of A_5. Every term of the derived series of A_5 is then A_5 itself, so the series does not reach $\{1\}$ and A_5 is not solvable. The even permutations of $t_1, \ldots, t_5$ that fix the remaining t_i form a subgroup of S_n isomorphic to A_5, and a subgroup of a solvable group is solvable (proposition 6.1.10), so S_n is not solvable. As $\mathrm{char}(E) = 0$ and f is irreducible over E, the Galois Criterion (corollary 20.2.19) shows that f is not solvable by radicals over E.

Furthermore, we know that S_3 and S_4 *are* solvable, with subnormal series:

$$\{e\} \trianglelefteq A_3 \cong \mathbb{Z}/3\mathbb{Z} \trianglelefteq S_3$$

$$\{e\} \trianglelefteq V_4 \trianglelefteq A_4 \trianglelefteq S_4$$

Hence, over a field of characteristic 0, every polynomial of degree at most 4 is solvable by radicals. The Galois group of an irreducible factor of degree $d \leq 4$ embeds in S_d, hence in S_4, so it is solvable by proposition 6.1.10, and corollary 20.2.19 applies to that factor.

Exercise 20.2.3

1. Let $\alpha \in F$ for an algebraic extension $F/\mathbb{Q}$. In particular, we may think of F as being embedded in $\mathbb{C}$ so that $\alpha = a + bi$. Does there always exist an N such that $\alpha^N \in \mathbb{R}$?

20.3 Galois Groups of Polynomials

In this section, we shall focus on techniques to compute the galois group of polynomial. So far, much of the focus was on showing what properties can be deduced given properties of the galois group, there have not been many technics to find galois groups (for example, can you come up with a non-solvable degree 5 polynomial?). Hence, this section is a compilation of interesting facts and techniques useful for working with galois groups of poylnomials.

Lemma 20.3.1: Transitivity Of Galois Group

Let $f(x) \in k[x]$ be a separable irreucible polynomial of degree n. Then $\mathrm{Gal}_k(f(x))$ acts transitively on the set of roots of $f(x)$ in $\overline{k}$. In particular, $\mathrm{Gal}_k(f(x))$ may be identified with a *transitive* subgroup of the symmetric group S_n.

***Proof*:**

Let $F \subseteq \overline{k}$ be the splitting field of f over k, which contains all the roots of f in $\overline{k}$, so that $\mathrm{Gal}_k(f(x)) = \mathrm{Gal}_k(F)$ by definition 20.1.17. Since f is separable, it has n distinct roots $\alpha_1, \ldots, \alpha_n \in F$.

First, $\mathrm{Gal}_k(F)$ acts on the set of roots: if $\sigma \in \mathrm{Gal}_k(F)$ and $f(\alpha) = 0$, then applying σ to the equation $f(\alpha) = 0$ and using that σ fixes the coefficients of f gives $f(\sigma(\alpha)) = 0$, so σ permutes the roots.

For transitivity, let α and β be two roots of f. The minimal polynomial p of α over k divides f: dividing with theorem 11.1.1 gives $f = qp + r$ with $r = 0$ or $\deg r < \deg p$, and $r(\alpha) = f(\alpha) - q(\alpha)p(\alpha) = 0$ forces $r = 0$ by the minimality of $\deg p$. Since f is irreducible, $f = up$ for a unit u, and the same holds for the minimal polynomial of β. Hence, by theorem 19.2.8 applied to each root, there are isomorphisms $k(\alpha) \cong k[x]/(f) \cong k(\beta)$ fixing k, whose composite $\varphi : k(\alpha) \to k(\beta)$ sends $\alpha \mapsto \overline{x} \mapsto \beta$. Now F is a splitting field of f over $k(\alpha)$ and also over $k(\beta)$, since it is generated over each by the roots of f. By the extension part of theorem 19.5.10 applied to φ (note $\varphi(f) = f$, as φ fixes k), φ extends to an isomorphism $\sigma : F \to F$. Then $\sigma \in \mathrm{Gal}_k(F)$, since it fixes k, and $\sigma(\alpha) = \varphi(\alpha) = \beta$. Hence the action is transitive.

Finally, labelling the n distinct roots gives a homomorphism $\mathrm{Gal}_k(F) \to S_n$ recording the induced permutation. It is injective: if σ fixes every root, then σ fixes k and the generators of $F = k(\alpha_1, \ldots, \alpha_n)$, so $\sigma = \mathrm{id}$. Hence $\mathrm{Gal}_k(f(x))$ is identified with a transitive subgroup of S_n.

By corollary 19.2.29, given α a root of $m_{\alpha,k}(x)$, then the elements of $\mathrm{Gal}_k(m_{\alpha,k}(x))$ are determined by where this root maps. By lemma 20.3.1, the Galois group of a separable irreducible polynomial of degree n is a transitive subgroup of S_n, which limits the possible groups:

1. The only possible group for degree 2 irreducible polynomials is: $\mathbb{Z}/2\mathbb{Z}$
2. The only possible groups for degree 3 irreducible polynomials is:
$$\mathbb{Z}/3\mathbb{Z} \cong A_3, S_3$$
3. The only possible groups for degree 4 irreducible polynomials is:
$$S_4, A_4, D_4, \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}, \mathbb{Z}/4\mathbb{Z}$$

Definition 20.3.2: Galois Conjugate

Let $\alpha \in L$ be an algebraic element of L over k. Then the set of roots of the minimal polynomial $m_{\alpha,k}(x)$ is called the *galois conjugates*, *algebraic conjugates*, or *conjugate elements* of α over k.

The motivation comes from the case of $\alpha \in \mathbb{C}$ which are algebraic over $\mathbb{R}$. In this case, the minimal polynomial is either $x - \alpha$ (in which case it has one root), or it is of degree 2 with galois conjugate $\overline{\alpha}$. We thus see that the galois conjugate is the generalization of the complex conjugate.

For the next result, we recall the following result of the discriminant;

Definition 20.3.3: Discriminant Reminder

Let $f(x) \in k[x]$ be a separable polynomial with roots $\alpha_1, ..., \alpha_n$ in its splitting field. Then the *discriminant* of $f(x)$ is the element $D = \Delta^2$ where
$$\Delta = \prod_{1 \le i < j \le n} (\alpha_i - \alpha_j)$$

Every permutation of the roots fixes D, so D is fixed by the whole Galois group, therefore $D \in k$. Odd permutations move Δ (if $\mathrm{ch}(k) \neq 2$) and even permutations fix it. Tus, Δ is fixed by the Galois group G (i.e. $\Delta \in k$) if and only if $G \subseteq A_n$

Lemma 20.3.4: Discriminant And Galois Group

Let k be a field where $\text{ch}(k) \neq 2$ and let $f(x) \in k[x]$ be a separable polynomial of degree $n \geq 1$ with discriminant D. Then the Galois group of $f(x)$, acting on the n roots, is contained in the alternating group A_n if and only if D is a square in k

***Proof*:**
Let K be the splitting field of f, with the distinct roots $\alpha_1, \ldots, \alpha_n$, and $G = \text{Gal}_k(K)$. An element $\sigma \in G$ fixes the coefficients of f, so $f(\sigma(\alpha_i)) = \sigma(f(\alpha_i)) = 0$, and σ is injective, so $\sigma(\alpha_i) = \alpha_{\pi(i)}$ for a permutation $\pi = \pi_\sigma \in S_n$, through which G acts on the roots. As the roots are distinct, $\Delta \neq 0$. For $i < j$ write $\{\pi(i), \pi(j)\} = \{a, b\}$ with $a < b$. The factor $\alpha_{\pi(i)} - \alpha_{\pi(j)}$ of $\sigma(\Delta) = \prod_{i<j}(\alpha_{\pi(i)} - \alpha_{\pi(j)})$ is $\alpha_a - \alpha_b$ when $\pi(i) < \pi(j)$, and $-(\alpha_a - \alpha_b)$ otherwise, that is, exactly when the factor $(\pi(j) - \pi(i))/(j - i)$ in the definition of $\text{sgn}(\pi)$ is negative. Since $\{i, j\} \mapsto \{\pi(i), \pi(j)\}$ permutes the two-element subsets of $\{1, \ldots, n\}$, $\sigma(\Delta) = \epsilon\Delta$, where ϵ is the sign of the rational number $\text{sgn}(\pi) = \pm 1$ (theorem 1.2.20(1)), so $\sigma(\Delta) = \text{sgn}(\pi_\sigma)\Delta$.

By theorem 20.1.13(1), the extension K/k is Galois, being the splitting field of a separable polynomial, so $\Delta \in k$ if and only if $\sigma(\Delta) = \Delta$ for every $\sigma \in G$ (theorem 20.1.13(4)). As $\Delta \neq 0$ and $2 \neq 0$ in k, $\Delta \neq -\Delta$, so $\sigma(\Delta) = \Delta$ exactly when $\text{sgn}(\pi_\sigma) = 1$. Hence $\Delta \in k$ if and only if G acts through A_n. Finally, $D = \Delta^2$ is a square in k if and only if $\Delta \in k$: if $D = c^2$ with $c \in k$, then $(c - \Delta)(c + \Delta) = 0$ in K, so $c = \pm\Delta$ and $\Delta \in k$, and the converse is clear, completing the proof.

Example 20.3.5: Discriminant

1. Let $f(x) = x^3 + ax^2 + bx + c$ be an irreducible monic cubic over $\mathbb{Q}$. By shifting by $a/3$, we get
$$f(x - a/3) = x^3 + px + q$$
for appropriate p, q. Then it can be computed that:
$$D = -4p^3 - 27q^2$$
The Galois group of an irreducible cubic acts transitively on its three roots (lemma 20.3.1), so it is A_3 or S_3, and by our above lemma it is A_3 if D is a square and S_3 otherwise. For example, the galois group of $x^3 - 2$ is S_3 since $D = -108$ is not a square in $\mathbb{Q}$, while $x^3 - 3x + 1$ has galois group $A_3 \cong \mathbb{Z}/3\mathbb{Z}$ since $D = 81 = 9^2$.
2. Let $f(x) \in \mathbb{Q}[x]$ be irreducible of degree p for prime p with $p - 2$ real roots and 2 non-real complex roots. Then $\text{Gal}_{\mathbb{Q}}(f(x)) = S_p$. Prove this, then show that the galois group of $x^5 - 5x - 1$ over $\mathbb{Q}$ is S_5.

Other useful discriminants are:
$$\begin{aligned} \text{D}(x^3 + ax + b) &= -4a^3 - 27b^2 \\ \text{D}(x^4 + ax + b) &= -27a^4 + 256b^3 \\ \text{D}(x^4 + ax^2 + b) &= 16b(a^2 - 4b)^2 \end{aligned}$$

Exercise 20.3.1

1. Let $f \in \mathbb{Q}[x]$ be an irreducible cubic whose discriminant is a square in $\mathbb{Q}$, and let K be the splitting field of f over $\mathbb{Q}$. Show that the only fields E with $\mathbb{Q} \subseteq E \subseteq K$ are $\mathbb{Q}$ and K.

20.4 Kummer Theory and Artin-Schreier Extensions

As we saw in section 20.2.4, if a field extension F/K is a radical extension, then the Galois group is solvable. The converse is not necessarily true:

Example 20.4.1: Solvable, But Not Radical Extension

Let $f = x^3 - 3x + 1 \in \mathbb{Q}[x]$. Its only possible rational roots are ± 1 (proposition 11.3.6), and $f(1) = -1$ and $f(-1) = 3$, so f is irreducible (proposition 11.3.5). Since $f(-2) = -1$, $f(0) = 1$, $f(1) = -1$ and $f(2) = 3$, it has three real roots, so its splitting field K lies in $\mathbb{R}$. Its Galois group is $A_3 \cong \mathbb{Z}/3\mathbb{Z}$ (example 20.3.5), which is solvable, and $[K : \mathbb{Q}] = 3$ (theorem 20.1.13(3)). Suppose $K = \mathbb{Q}(\alpha_1, \ldots, \alpha_n)$ were a radical extension as in definition 20.2.13. As $[K : \mathbb{Q}] = 3$ is prime, the Tower Law (theorem 19.3.7) shows that exactly one step of the tower is proper, and that step goes from $\mathbb{Q}$ to K, so $K = \mathbb{Q}(\beta)$ with $\beta = \alpha_i$ and $\beta^m \in \mathbb{Q}$ for some i and m. As $K/\mathbb{Q}$ is Galois, hence normal (theorem 20.1.13(2)), the conjugates of β over $\mathbb{Q}$ lie in $K \subseteq \mathbb{R}$, and they are roots of $x^m - \beta^m$, whose only real roots are $\pm\beta$. So β has at most 2 conjugates, while its minimal polynomial has degree 3 and distinct roots (corollary 19.6.8). Hence $K/\mathbb{Q}$ is solvable but not radical.

By doing multiple examples like these, the reader would notice that this seems to be more of a problem of there being a lack of roots of unity in K to insure there are enough solutions to $x^n - a$. In fact, there is no additional issues that arise when the characteristic of K is 0, and the issues in finite characteristic only happen for certain bad primes. We thus explore this converse: that under these mild extra conditions, if the galois group is solvable then the extension is a radical extension. As solvable groups are towers of cyclic groups, it shall suffice to show that if the galois groups in the tower of fields is cyclic, each corresponding field extension is given by a radical $\sqrt[n]{\alpha}$. An interesting consequence of this is that the galois groups of Kummer extensions shall be characterized by certain units of the base field. This shall in fact be true for all abelian extensions (not just Kummer extensions), but the proof of this result is a lot more complicated and was one of the crowning achievements of 20th-century number theory, proved by Artin in 1927. The result is called *Artin reciprocity* and its proof can be found in [45].

Definition 20.4.2: Kummer Extensions and G-Extensions

Let L/K be a field extension. Then L/K is called a *Kummer Extension* if:

1. for some n, K contains μ_n (the roots of $x^n - 1$)
2. L/K is an Abelian extension with exponent n[a]

If in particular we want extensions with Galois groups G, we shall call such extensions G-extensions.

[a] The least common multiple of the orders of the elements

Example 20.4.3: Kummer Extensions

1. Any quadratic extension of $\mathbb{Q}$ is a Kummer extension of degree 2, since $\mathbb{Q}$ have $1, -1$ and has abelian extension. More generally, any multi-quadratic extension is a Kummer extension of degree 2
2. There is no Kummer extensions of degree 3 over $\mathbb{Q}$ since $\mathbb{Q}$ does not have a 3rd root of unity, since it's complex.
3. More generally if K contains μ_n (meaning the characteristic of K does not divide n), then adjoining any nth root of $a \in K$, $L = K(\sqrt[n]{a})$ is a Kummer extension (of degree m dividing n, since the minimal polynomial divides $x^n - a$)

So let's now say that L/K is a Kummer $\mathbb{Z}/n\mathbb{Z}$-extension where the characteristic of K does not divide n. In order to show $L = K(\sqrt[n]{a})$, we must find an appropriate $a \in K^*$. Let us find some necessary conditions. As $\mathrm{Gal}_K(L) = \mathbb{Z}/n\mathbb{Z}$, there is a generator σ such that $\sigma^n = 1$. Then given $x^n = a$, we have the roots:

$$\sqrt[n]{a}, \quad \sqrt[n]{a}\zeta_n, ..., \sqrt[n]{a}\zeta_n^{n-1}$$

note all these roots of unity are distinct and in K by the characteristic condition. Thus, by field theory we know that σ must map $\sqrt[n]{a}$ to another one of these roots, and that this conditions characterizes σ. Without loss of generality we can say $\sigma(\sqrt[n]{a}) = \zeta\sqrt[n]{a}$. Importantly, interpreting L as a K-vector space, $\sqrt[n]{a}$ becomes an *eigenvector* of σ. Note that as $\sigma^n = 1$, the possible eigenvalues must be $1, \zeta_n, ..., \zeta_n^{n-1}$. Thus if σ is diagonalizable, we can find such an a such that $a^n = \alpha \in K$. Exercise 20.4.1 (1) gives a very intuitive computational approach to finding this. We present the following more abstract way of finding it that works more generally so that we can use it later in number theory and cohomology theory in this book:

Theorem 20.4.4: Hilbert's 90th Theorem

Let K be a field of characteristic 0 and L/K be a finite Galois extension such that

$$\mathrm{Gal}_K(L) \cong \mathbb{Z}/n\mathbb{Z}$$

Let σ generate $\mathrm{Gal}_K(L)$, and define the *norm* of $\beta \in L$ to be $\mathrm{Nm}_{L/K}(\beta) = \beta\sigma(\beta)\sigma^2(\beta)\cdots\sigma^{n-1}(\beta)$. Then:

1. For $\beta \in L^\times$ such that $\mathrm{Nm}_{L/K}(\beta) = 1$, there exists an $\alpha \in L^\times$ such that
$$\beta = \frac{\sigma(\alpha)}{\alpha}$$
2. Let $\beta \in L$ such that $\beta + \sigma(\beta) + \cdots + \sigma^{n-1}(\beta) = 0$. Then there exists an $\alpha \in L$ such that
$$\beta = \alpha - \sigma(\alpha)$$

***Proof*:**
The generator σ has order n, so $\sigma^0 = \mathrm{id}, \sigma, \ldots, \sigma^{n-1}$ are n distinct automorphisms of L, and $\sigma^n = \mathrm{id}$.

1. Put $c_0 = 1$ and $c_k = \beta\sigma(\beta)\cdots\sigma^{k-1}(\beta)$ for $1 \leq k \leq n-1$, and for $\theta \in L$ let

$$S(\theta) = c_0\theta + c_1\sigma(\theta) + c_2\sigma^2(\theta) + \cdots + c_{n-1}\sigma^{n-1}(\theta)$$

The distinct automorphisms σ^k are linearly independent over L as functions on L (corollary 20.1.40), and the coefficient $c_0 = 1$ is nonzero, so $S(\theta) \neq 0$ for some $\theta \in L$. Fix such a θ and write $S = S(\theta)$. Applying σ gives

$$\sigma(S) = \sum_{k=0}^{n-1} \sigma(c_k)\sigma^{k+1}(\theta)$$

where $\beta\sigma(c_k) = \beta\sigma(\beta)\cdots\sigma^k(\beta)$ equals c_{k+1} for $k \leq n-2$ and equals $\mathrm{Nm}_{L/K}(\beta) = 1$ for $k = n-1$, while $\sigma^n(\theta) = \theta$. Hence

$$\beta\sigma(S) = c_1\sigma(\theta) + \cdots + c_{n-1}\sigma^{n-1}(\theta) + \theta = S$$

As $S \neq 0$, also $\sigma(S) \neq 0$, and $\alpha = S^{-1} \in L^\times$ satisfies $\frac{\sigma(\alpha)}{\alpha} = \frac{S}{\sigma(S)} = \beta$.

2. An element $\theta \in L$ with $\theta + \sigma(\theta) + \cdots + \sigma^{n-1}(\theta) = 1$ exists: since K has characteristic 0, $n \neq 0$ in K, and $\theta = \frac{1}{n} \in K$ is fixed by every σ^k, so the sum is $n\theta = 1$. Put $b_k = \beta + \sigma(\beta) + \cdots + \sigma^{k-1}(\beta)$ for $1 \leq k \leq n$. Then $b_1 = \beta$, $b_n = 0$ by hypothesis, and $\sigma(b_k) = b_{k+1} - \beta$ for $k \leq n-1$. Let

$$\alpha = \sum_{k=1}^{n-1} b_k\sigma^k(\theta)$$

Then, using $b_n = 0$ and $\sigma^n(\theta) = \theta$,

$$\sigma(\alpha) = \sum_{k=1}^{n-1}(b_{k+1} - \beta)\sigma^{k+1}(\theta) = \sum_{k=2}^{n-1} b_k\sigma^k(\theta) - \beta\left(\theta + \sum_{k=2}^{n-1}\sigma^k(\theta)\right)$$

and subtracting from α, whose first term is $b_1\sigma(\theta) = \beta\sigma(\theta)$,

$$\alpha - \sigma(\alpha) = \beta\left(\theta + \sigma(\theta) + \cdots + \sigma^{n-1}(\theta)\right) = \beta$$

completing the proof.

20.4.5 Foreshadowing In [19, Hilbert's 90th Theorem: Cohomology Upgrade], this theorem is reformulated in purely Homological terms, namely it says that $H^1(G, L^*) = 0$

We now look at some properties of Kummer extensions. Suppose K contains an element ζ of multiplicative order n, so that $\mu_n = \langle\zeta\rangle \subseteq K$, and let $L = K(\alpha)$ where $\alpha^n = \beta \in K^\times$. Then L/K is an abelian Galois extension. As $\alpha^n = \beta \neq 0$, $\alpha \neq 0$, so the n elements $\zeta^i\alpha$, $0 \leq i \leq n-1$, are distinct, and each is a root of $x^n - \beta$. So they are all its roots in L (corollary 11.3.2), $x^n - \beta = \prod_i(x - \zeta^i\alpha)$ by proposition 11.3.1 applied to these distinct roots in turn and a comparison of degrees, and $x^n - \beta$ is separable. They lie in L, and a subfield of L containing K in which $x^n - \beta$ splits contains α, hence equals L. So L is a splitting field of the separable polynomial $x^n - \beta$, and L/K is a finite Galois

extension (theorem 20.1.13(1), definition 20.1.14), finite since α is algebraic over K (theorem 19.2.8). Every $\tau \in \mathrm{Gal}_K(L)$ sends α to a root of $x^n - \beta$, since $\tau(\alpha)^n = \tau(\beta) = \beta$, so $\tau(\alpha)/\alpha = \zeta^i$ for some i, and

$$\varphi : \mathrm{Gal}_K(L) \to \mu_n \qquad \tau \mapsto \frac{\tau(\alpha)}{\alpha}$$

is well defined. It is a homomorphism: $\tau(\alpha)/\alpha \in \mu_n \subseteq K$ is fixed by $\sigma \in \mathrm{Gal}_K(L)$, so

$$\frac{\sigma\tau(\alpha)}{\alpha} = \frac{\sigma\left(\frac{\tau(\alpha)}{\alpha} \cdot \alpha\right)}{\alpha} = \frac{\tau(\alpha)}{\alpha} \cdot \frac{\sigma(\alpha)}{\alpha}$$

It is injective: if $\tau(\alpha) = \alpha$, then τ fixes $K(\alpha) = L$, so $\tau = \mathrm{id}$. So $\mathrm{Gal}_K(L)$ is isomorphic to a subgroup of the cyclic group μ_n, and is abelian. Any nth root $\zeta^j\alpha$ of β gives the same homomorphism, since $\tau(\zeta^j\alpha)/(\zeta^j\alpha) = \tau(\alpha)/\alpha$. Note how important $\mu_n \subseteq K$ is for this homomorphism to work! Without it, $K(\alpha)/K$ need not be Galois: for $n \geq 3$, with $2^{1/n} \in \mathbb{R}$ the positive real root, $\mathbb{Q}(2^{1/n})/\mathbb{Q}$ is not Galois. The polynomial $x^n - 2$ is irreducible over $\mathbb{Q}$ (corollary 11.3.11, with $p = 2$) and has the root $2^{1/n}$ in $\mathbb{Q}(2^{1/n})$, while its root $2^{1/n}e^{2\pi i/n}$ is not real, since $0 < 2\pi/n < \pi$. So $x^n - 2$ does not split into linear factors over the subfield $\mathbb{Q}(2^{1/n})$ of $\mathbb{R}$, since then all its complex roots would be real, and $\mathbb{Q}(2^{1/n})/\mathbb{Q}$ is not normal (definition 19.5.12), hence not Galois (theorem 20.1.13(2)). Using Hilbert's 90th Theorem (theorem 20.4.4), the converse of the statement above is also true:

Theorem 20.4.6: Primitive Element For Cyclic Kummer Extension

Let $\mathrm{char}(K) = 0$, L/K be a finite Galois extension and $\mathrm{Gal}_K(L) \cong \mathbb{Z}/n\mathbb{Z}$. Assume K contains μ_n, the group of nth roots of unity, generated by an element ζ_n of multiplicative order n. Then there exists an $\alpha \in L^\times$ such that

$$L = K(\alpha) \qquad \alpha^n \in K^\times$$

***Proof*:**
Let σ generate $\mathrm{Gal}_K(L)$. As $\zeta_n \in K$ is fixed by σ, its norm (theorem 20.4.4) is $\mathrm{Nm}_{L/K}(\zeta_n) = \zeta_n^n = 1$. Thus, by theorem 20.4.4(1), there exists $\alpha \in L^\times$ such that $\sigma(\alpha) = \zeta_n\alpha$, and then $\sigma^i(\alpha) = \zeta_n^i\alpha$ for every i. The n elements $\zeta_n^i\alpha$ with $0 \leq i \leq n-1$ are distinct, since ζ_n has order n and $\alpha \neq 0$, and each is a root of the minimal polynomial p of α over K, because σ^i fixes the coefficients of p. So $\deg p \geq n$ (corollary 11.3.2), and $[K(\alpha) : K] = \deg p$ (theorem 19.2.8, theorem 19.2.5). As $[L : K] = |\mathrm{Gal}_K(L)| = n$ (theorem 20.1.13(3)) and $K(\alpha)$ is a K-subspace of L, the dimensions force

$$L = K(\alpha)$$

Furthermore,

$$\sigma(\alpha^n) = \sigma(\alpha)^n = \zeta_n^n\alpha^n = \alpha^n$$

Hence, each σ^i fixes α^n, meaning $\alpha^n \in K$ by theorem 20.1.13(4), and $\alpha^n \neq 0$, hence

$$L = K(\alpha) \qquad \alpha^n \in K^\times$$

as we sought to show.

With the existence established, let us now establish to what extent it is unique. Let σ generate $\mathrm{Gal}_K(L)$ and say $L = K(\alpha) = K(\beta)$ with $\alpha^n = a$ and $\beta^n = b$ in $K^\times$. Then α and β are both

eigenvectors of σ: the eigenvalue $\eta = \sigma(\alpha)/\alpha$ satisfies $\eta^n = \sigma(a)/a = 1$, so it lies in μ_n, which contains all nth roots of unity of L since $x^n - 1$ has at most n roots (corollary 11.3.2), and likewise for $\eta' = \sigma(\beta)/\beta$. The eigenvalues may differ, but both have order exactly n: if $\eta^d = 1$, then σ^d fixes α, hence fixes $L = K(\alpha)$, so $n \mid d$. So $\eta' = \eta^r$ for some r with $\gcd(r, n) = 1$ (theorem 2.3.11(3)). Choosing s with $rs \equiv 1 \bmod n$, the power β^s has the eigenvalue $\eta^{rs} = \eta$, the same as α. But then:

$$\sigma\left(\frac{\alpha}{\beta^s}\right) = \frac{\alpha}{\beta^s}$$

and hence $c = \alpha/\beta^s$ is fixed by every σ^i, so it lies in K (theorem 20.1.13(4)). Then $a = \alpha^n = c^n b^s$, so the classes of a and b in $K^\times/(K^\times)^n$ generate the same subgroup. We thus get the following correspondence:

Corollary 20.4.7: Cyclic Kummer Extension Correspondence

Let K be a field of characteristic 0 containing an element ζ of multiplicative order n, so that $\mu_n = \langle\zeta\rangle \subseteq K$, and fix an algebraic closure $\overline{K}$ of K. Write $[c]$ for the class of $c \in K^\times$ in $K^\times/(K^\times)^n$. Then we have the following bijective correspondence

$$\left\{\begin{array}{c} K \subseteq L \subseteq \overline{K},\ L/K \text{ Galois} \\ \mathrm{Gal}_K(L) \cong \mathbb{Z}/n\mathbb{Z} \end{array}\right\} \underset{\psi}{\overset{\varphi}{\rightleftarrows}} \left\{\begin{array}{c} G \subseteq K^\times/(K^\times)^n \\ G \cong \mathbb{Z}/n\mathbb{Z} \end{array}\right\}$$

where $\varphi(L) = \langle[\alpha^n]\rangle$ for any $\alpha \in L^\times$ with $L = K(\alpha)$ and $\alpha^n \in K^\times$, and $\psi(G) = K(\alpha)$ for any $\alpha \in \overline{K}$ with $\alpha^n \in K^\times$ such that $[\alpha^n]$ generates G.

***Proof*:**

Step 1: roots of unity. The n elements $1, \zeta, \ldots, \zeta^{n-1}$ of K are distinct roots of $x^n - 1$, which has at most n roots in $\overline{K}$ (corollary 11.3.2), so they are all of its roots. Likewise, for $\alpha \in \overline{K}^\times$ the n distinct elements $\zeta^i\alpha$ are all the roots of $x^n - \alpha^n$ in $\overline{K}$, and this polynomial has no repeated root.

Step 2: radical extensions. Let $\alpha \in \overline{K}$ with $\beta = \alpha^n \in K^\times$, and let $L = K(\alpha)$. As $\zeta \in K$, the field L contains the roots $\zeta^i\alpha$ of $x^n - \beta$ and is generated by them over K, so L is the splitting field of $x^n - \beta$, which has no repeated root by Step 1. Hence L/K is Galois (theorem 20.1.13, definition 20.1.14). For $\tau \in \mathrm{Gal}_K(L)$, $\tau(\alpha)^n = \tau(\beta) = \beta$, so $\chi(\tau) = \tau(\alpha)/\alpha$ lies in μ_n. For $\tau, \rho \in \mathrm{Gal}_K(L)$, the element $\chi(\rho) \in K$ is fixed by τ, so $(\tau\rho)(\alpha) = \tau(\chi(\rho)\alpha) = \chi(\rho)\chi(\tau)\alpha$, that is, $\chi(\tau\rho) = \chi(\tau)\chi(\rho)$. If $\chi(\tau) = 1$, then τ fixes α and hence $L = K(\alpha)$, so $\tau = \mathrm{id}$. So $\chi\colon \mathrm{Gal}_K(L) \to \mu_n$ is an injective homomorphism, and $\mathrm{Gal}_K(L)$ is cyclic (theorem 2.3.11(1)) of order $m = [L : K]$ (theorem 20.1.13) dividing n (theorem 3.3.1). Let σ generate $\mathrm{Gal}_K(L)$. Then $\chi(\sigma)$ has order m, so $\sigma(\alpha^m) = \chi(\sigma)^m\alpha^m = \alpha^m$. Thus α^m is fixed by $\mathrm{Gal}_K(L) = \langle\sigma\rangle$ and lies in $K^\times$ (theorem 20.1.13(4)), and $\beta^m = (\alpha^m)^n$ gives $[\beta]^m = 1$.

Step 3: ψ is well defined. Let $G \subseteq K^\times/(K^\times)^n$ be cyclic of order n, let $[\beta]$ generate G and let $\alpha \in \overline{K}$ with $\alpha^n = \beta$. The field $K(\alpha)$ depends only on G. Another root of $x^n - \beta$ is $\zeta^i\alpha$ (Step 1), and $K(\zeta^i\alpha) = K(\alpha)$ as $\zeta \in K$. Another representative of $[\beta]$ is $\beta\gamma^n$ with $\gamma \in K^\times$, with the root $\alpha\gamma$, and $K(\alpha\gamma) = K(\alpha)$. Another generator of G is $[\beta]^r$ with $\gcd(r, n) = 1$ (theorem 2.3.11(3)), with the root α^r, and $K(\alpha^r) \subseteq K(\alpha)$. Since $[\beta^r]$ generates G, $[\beta] = [\beta^r]^{r'}$ for some integer r', that is, $\beta = \beta^{rr'}\gamma^n$ for some $\gamma \in K^\times$. Then $\alpha^{rr'}\gamma$ is a root of $x^n - \beta$, so $\alpha^{rr'}\gamma = \zeta^i\alpha$ for some i,

and $\alpha = \zeta^{-i}\gamma(\alpha^r)^{r'} \in K(\alpha^r)$. So $K(\alpha^r) = K(\alpha)$. Moreover, $L = \psi(G) = K(\alpha)$ is Galois with cyclic group of order m dividing n, and $[\beta]^m = 1$ by Step 2. As $[\beta]$ has order n, n divides m, so $m = n$ and $\mathrm{Gal}_K(L) \cong \mathbb{Z}/n\mathbb{Z}$.

Step 4: φ is well defined. Let $L \subseteq \overline{K}$ be Galois over K with $\mathrm{Gal}_K(L)$ cyclic of order n, so $[L : K] = n$. By proposition 20.1.42, $L = K(\delta)$ with $\delta^n \in K$. If $\delta = 0$, then $L = K$ and $n = 1$, and δ may be replaced by 1. So elements $\alpha \in L^\times$ with $L = K(\alpha)$ and $\alpha^n \in K^\times$ exist. Fix one. By Step 2, $\chi\colon \mathrm{Gal}_K(L) \to \mu_n$ is injective between groups of order n, hence bijective. Let $\sigma = \chi^{-1}(\zeta)$, so that $\sigma(\alpha) = \zeta\alpha$ and σ generates $\mathrm{Gal}_K(L)$. We claim that

$$\langle [\alpha^n] \rangle = \left\{ [\gamma^n] \;\middle|\; \gamma \in L^\times,\ \gamma^n \in K^\times \right\}$$

The left side is contained in the right side, since $[\alpha^n]^j = [(\alpha^j)^n]$. Conversely, let $\gamma \in L^\times$ with $c = \gamma^n \in K^\times$. Then $\sigma(\gamma)^n = \sigma(c) = c = \gamma^n$, so $\sigma(\gamma) = \zeta^j\gamma$ for some j (Step 1). The element $e = \gamma\alpha^{-j}$ satisfies $\sigma(e) = e$, so it is fixed by $\mathrm{Gal}_K(L)$ and lies in $K^\times$ (theorem 20.1.13(4)), and $c = e^n(\alpha^n)^j$ gives $[c] = [\alpha^n]^j$. The right side does not involve α, so $\varphi(L)$ does not depend on the choice of α. The class $[\alpha^n]$ has order n: $[\alpha^n]^n = [(\alpha^n)^n] = 1$, and if $[\alpha^n]^j = 1$, then $\alpha^{nj} = e^n$ for some $e \in K^\times$, so $\alpha^j/e \in \mu_n \subseteq K$ and $\alpha^j \in K$, whence $\zeta^j\alpha^j = \sigma(\alpha^j) = \alpha^j$ and $n \mid j$. So $\varphi(L)$ is cyclic of order n.

Step 5: φ and ψ are inverse to each other. Let L be as in Step 4, with $L = K(\alpha)$ and $\alpha^n \in K^\times$. Then $\varphi(L) = \langle [\alpha^n] \rangle$, and by Step 3 $\psi(\varphi(L))$ may be computed with the generator $[\alpha^n]$ and the root α, so $\psi(\varphi(L)) = K(\alpha) = L$. Let G be as in Step 3, with generator $[\beta]$ and $\alpha^n = \beta$. Then $L = \psi(G) = K(\alpha)$ is cyclic of degree n by Step 3, and $\alpha \in L^\times$ satisfies $L = K(\alpha)$ and $\alpha^n \in K^\times$, so by Step 4 $\varphi(\psi(G)) = \langle [\beta] \rangle = G$. Hence φ and ψ are mutually inverse bijections, completing the proof.

Corollary 20.4.8: Pontryagin Dual For Kummer Extensions

Let $\mathbb{Q}(\zeta_n) \subseteq K$, and let $K_n = K(\sqrt[n]{\beta} : \beta \in K^\times) \subseteq \overline{K}$ be the maximal Kummer extension of exponent dividing n (every element of $\mathrm{Gal}_K(K_n)$ has order that divides n). Then there exists an injective natural homomorphism, an isomorphism once the infinite Galois correspondence is available (deferred in the proof below):

$$\mathrm{Gal}_K(K_n) \to \mathrm{Hom}(K^\times/(K^\times)^n, \mu_n)$$

$$\sigma \mapsto \left(\overline{\beta} \mapsto \frac{\sigma(\sqrt[n]{\beta})}{\sqrt[n]{\beta}} \right)$$

***Proof*:**
The displayed map is well-defined. Fix $\beta \in K^\times$ and a root $\alpha \in K_n$ of $x^n - \beta$. Then $(\sigma(\alpha)/\alpha)^n = \sigma(\beta)/\beta = 1$, so $\sigma(\alpha)/\alpha \in \mu_n$. Any other root of $x^n - \beta$ is $\zeta\alpha$ with $\zeta \in \mu_n \subseteq K$, and $\sigma(\zeta\alpha)/(\zeta\alpha) = \sigma(\alpha)/\alpha$ since σ fixes ζ. Likewise, for another representative $\beta\gamma^n$ of $\overline{\beta}$ with $\gamma \in K^\times$, the root $\alpha\gamma$ gives $\sigma(\alpha\gamma)/(\alpha\gamma) = \sigma(\alpha)/\alpha$. The assignment $\overline{\beta} \mapsto \sigma(\sqrt[n]{\beta})/\sqrt[n]{\beta}$ is multiplicative, since a product of n-th roots of β and β' is an n-th root of $\beta\beta'$, so each σ is sent to an element of $\mathrm{Hom}(K^\times/(K^\times)^n, \mu_n)$, and the assignment is a group homomorphism: for $\sigma, \tau \in \mathrm{Gal}_K(K_n)$,

the element $\tau(\alpha)$ is again a root of $x^n - \beta$, so

$$\frac{\sigma\tau(\alpha)}{\alpha} = \frac{\sigma(\tau(\alpha))}{\tau(\alpha)} \cdot \frac{\tau(\alpha)}{\alpha}$$

exhibits the value on $\sigma\tau$ as the product of the values on σ and on τ.

The map is injective: if σ is sent to the trivial homomorphism, then σ fixes every $\sqrt[n]{\beta}$, which is a generating set of K_n over K, so $\sigma = \mathrm{id}$.

Deferred: the surjectivity of this map. The extension K_n/K is infinite whenever $K^\times/(K^\times)^n$ is infinite, and this book develops the Galois correspondence only for finite extensions (see the remark on the Krull topology following proposition 20.1.12). Surjectivity is proven by writing $\mathrm{Gal}_K(K_n)$ as the limit of the groups $\mathrm{Gal}_K(L)$ over the finite Kummer subextensions $L = K(\sqrt[n]{\beta_1}, \ldots, \sqrt[n]{\beta_r})$, identifying each $\mathrm{Gal}_K(L)$ with $\mathrm{Hom}(B/(K^\times)^n, \mu_n)$ for B the subgroup of $K^\times$ generated by $\beta_1, \ldots, \beta_r$ and $(K^\times)^n$ through the finite Galois correspondence, and passing to the limit. That limit description of the full Galois group is exactly the infinite Galois correspondence, which is not available in this book, so we defer this step to the treatment of infinite Galois theory.

We can package the data of all cyclic extensions at once into a single pairing. Its construction needs only the finite computations below, though extracting the full duality from it requires the infinite Galois correspondence, which this book does not develop.

Theorem 20.4.9: Kummer Pairing

Let K contain all roots of unity, let M/K be the extension of K by all possible roots, let n be a positive integer with $\mathrm{char}(K) \nmid n$, and let μ_n be the group of all nth roots of unity. Then there exists a well-defined map:

$$\varphi : \mathrm{Gal}_K(M) \times (K^*/(K^*)^n) \to \mu_n \qquad \varphi(\sigma, \overline{a}) = \frac{\sigma(\sqrt[n]{a})}{\sqrt[n]{a}}$$

which is multiplicative in each variable.

***Proof*:**
Fix $a \in K^*$ and choose a root $\alpha \in M$ of $x^n - a$, which exists since M contains all roots of elements of K. First, the value lands in μ_n:

$$\left(\frac{\sigma(\alpha)}{\alpha}\right)^n = \frac{\sigma(\alpha^n)}{\alpha^n} = \frac{\sigma(a)}{a} = 1$$

since σ fixes $a \in K$. Second, the value does not depend on the choice of root: any other root of $x^n - a$ is $\zeta\alpha$ with $\zeta \in \mu_n \subseteq K$, and since $\sigma(\zeta) = \zeta$,

$$\frac{\sigma(\zeta\alpha)}{\zeta\alpha} = \frac{\zeta\sigma(\alpha)}{\zeta\alpha} = \frac{\sigma(\alpha)}{\alpha}$$

Third, the value does not depend on the representative of $\overline{a} \in K^*/(K^*)^n$: another representative is ab^n with $b \in K^*$, and αb is a root of $x^n - ab^n$, so by the independence shown above we may use it, and $\sigma(\alpha b)/(\alpha b) = \sigma(\alpha)/\alpha$ since $\sigma(b) = b$. Hence φ is well-defined.

For multiplicativity in the first variable, let $\sigma, \tau \in \mathrm{Gal}_K(M)$. The element $\tau(\alpha)$ is again a root of $x^n - a$, so

$$\varphi(\sigma\tau, \overline{a}) = \frac{\sigma(\tau(\alpha))}{\tau(\alpha)} \cdot \frac{\tau(\alpha)}{\alpha} = \varphi(\sigma, \overline{a})\, \varphi(\tau, \overline{a})$$

For multiplicativity in the second variable, if α and β are n-th roots of a and b respectively, then $\alpha\beta$ is an n-th root of ab, and

$$\varphi(\sigma, \overline{ab}) = \frac{\sigma(\alpha\beta)}{\alpha\beta} = \frac{\sigma(\alpha)}{\alpha} \cdot \frac{\sigma(\beta)}{\beta} = \varphi(\sigma, \overline{a})\, \varphi(\sigma, \overline{b})$$

completing the proof.

Definition 20.4.10: Kummer Pairing

Let K contain all roots of unity, let M/K be the extension of K by all possible roots, let n be a positive integer with $\mathrm{char}(K) \nmid n$, and let μ_n be the group of all nth roots of unity. Then the map

$$\varphi : \mathrm{Gal}_K(M) \times (K^*/(K^*)^n) \to \mu_n \qquad \varphi(\sigma, \overline{a}) = \frac{\sigma(\sqrt[n]{a})}{\sqrt[n]{a}}$$

is called the *Kummer Pairing*.

This shows that the galois group and the units are almost dual to each other. A point was made to put μ_n instead of $\mathbb{Z}/n\mathbb{Z}$. These two groups are group isomorphic, however there as a different group-action on μ_n, namely the galois group acts on it non-trivially. The fact there is a non-trivial action is often called the *Tate Twist*.

Artin-Schreier Extensions

Let $\mathrm{char}(K) = p > 0$. In a commutative ring A containing K, the binomial theorem gives

$$(u + v)^p = u^p + v^p \qquad \text{for all } u, v \in A \tag{20.5}$$

since the remaining terms of the expansion are $\binom{p}{l} u^l v^{p-l}$ with $0 < l < p$, and they vanish: the prime p divides $p! = \binom{p}{l}\, l!\, (p-l)!$ and no factor of $l!\,(p-l)!$, so it divides $\binom{p}{l}$, and $p \cdot 1 = 0$ in A. Also $(-1)^p = -1$ in A, for odd p as an odd power and for $p = 2$ as $1 = -1$. So if β is a root of $x^p - a$, $a \in K$, in an extension of K, then $x^p - a = x^p - \beta^p = (x - \beta)^p$, and β is the only root of $x^p - a$. If $\beta \notin K$, every $\tau \in \mathrm{Aut}_K(K(\beta))$ sends β to a root of $x^p - a$, hence to β, so $\mathrm{Aut}_K(K(\beta)) = \{\mathrm{id}\}$ and $K(\beta)/K$ is not Galois (theorem 20.1.13(3)). Furthermore, if L/K is cyclic of degree p with generator σ, and $\gamma \neq 0$ is an eigenvector of the K-linear map σ with eigenvalue $\lambda \in K$, then $\gamma = \sigma^p(\gamma) = \lambda^p \gamma$, so $0 = \lambda^p - 1 = (\lambda - 1)^p$ and $\lambda = 1$. Every eigenvector of σ is then fixed by $\mathrm{Gal}_K(L)$ and lies in K (theorem 20.1.13(4)), losing the method of theorem 20.4.6 to find the element α. Note that there do exist such Galois extensions, for example $\mathbb{F}_{p^p}/\mathbb{F}_p$, whose automorphism group is cyclic of order p (proposition 19.7.19), and hence this is a valid problem.

Definition 20.4.11: Artin-Schreier Extensions

Let K be a field of characteristic $p > 0$. An extension L/K is an *Artin-Schreier extension* if $L = K(\alpha)$ for some $\alpha \notin K$ with $\alpha^p - \alpha \in K$, that is, for a root $\alpha \notin K$ of a polynomial $x^p - x - a$ with $a \in K$ (as compared to $x^n - a$ for Kummer extensions).

The cyclic extensions of degree p are exactly the Artin-Schreier extensions, and the element α with $L = K(\alpha)$ comes from a generalized eigenvector of σ (definition 14.5.5) for the eigenvalue 1:

Theorem 20.4.12: Artin-Schreier Theorem For Cyclic Extensions

Let K be a field of characteristic $p > 0$, and let $\mathbb{F}_p = \{i \cdot 1 \mid 0 \leq i \leq p-1\}$ be its prime field.

1. Let L/K be a finite Galois extension with $\mathrm{Gal}_K(L) \cong \mathbb{Z}/p\mathbb{Z}$, and let σ generate $\mathrm{Gal}_K(L)$. Then there exists $\alpha \in L$ with $\sigma(\alpha) = \alpha + 1$, and every such α satisfies
$$L = K(\alpha) \qquad \alpha^p - \alpha \in K$$
so L/K is an Artin-Schreier extension.

2. Let $a \in K$, let $f = x^p - x - a \in K[x]$, and let α be a root of f in a splitting field E of f over K. Then the roots of f in E are the p distinct elements $\alpha + i$ with $i \in \mathbb{F}_p$. If $\alpha \in K$, then f splits into distinct linear factors over K. If $\alpha \notin K$, then f is irreducible over K, the extension $K(\alpha)/K$ is Galois, and $\mathrm{Gal}_K(K(\alpha))$ is cyclic of order p, generated by an automorphism σ with $\sigma(\alpha) = \alpha + 1$.

***Proof*:**
The p elements $i \cdot 1$ of $\mathbb{F}_p$ are distinct, since p is the least positive integer with $p \cdot 1 = 0$ (definition 19.1.7). Each satisfies $(i{\cdot}1)^p = i{\cdot}1$, by induction on i: $0^p = 0$, and $((i{+}1){\cdot}1)^p = (i{\cdot}1)^p{+}1$ by (20.5).

1. *Step 1: a generalized eigenvector.* Each σ^k fixes K, so it is a K-linear map $L \to L$. The K-linear combinations of the powers of σ form a commutative subring of $\mathrm{End}_K(L)$ containing K, as the maps $c\mathrm{id}$ with $c \in K$, so the binomial theorem and (20.5) apply in it. Let $N = \sigma - \mathrm{id}$. Then
$$N^p = \sigma^p + (-1)^p \mathrm{id} = \sigma^p - \mathrm{id} = 0$$
as $(-1)^p = -1$. The binomial theorem also gives
$$N^{p-1} = \sum_{k=0}^{p-1} \binom{p-1}{k} (-1)^{p-1-k} \sigma^k$$
a combination of the distinct automorphisms $\mathrm{id}, \sigma, \ldots, \sigma^{p-1}$ of L in which σ^{p-1} has coefficient 1, so $N^{p-1} \neq 0$ by corollary 20.1.40. Choose $v \in L$ with $N^{p-1}(v) \neq 0$, and put $w = N^{p-2}(v)$, so that $w = v$ when $p = 2$, and $u = N(w) = N^{p-1}(v)$. Then $u \neq 0$ and $N(u) = N^p(v) = 0$, so w is a generalized eigenvector of σ for the eigenvalue 1 (definition 14.5.5) that is not an eigenvector, and $\sigma(u) = u$. As u is fixed by every power of σ, that is, by $\mathrm{Gal}_K(L)$, it lies in K (theorem 20.1.13(4)).

Step 2: the element α. As $\sigma(w) = w + N(w) = w + u$ and $\sigma(u) = u$, the element w/u of L satisfies $\sigma(w/u) = (w+u)/u = w/u + 1$, so an α as in (1) exists. Let $\alpha \in L$ be any element with $\sigma(\alpha) = \alpha + 1$. By (20.5),

$$\sigma(\alpha^p - \alpha) = (\alpha+1)^p - (\alpha+1) = \alpha^p - \alpha$$

so $\alpha^p - \alpha$ is fixed by $\mathrm{Gal}_K(L) = \langle\sigma\rangle$ and lies in K (theorem 20.1.13(4)). As $\sigma(\alpha) \neq \alpha$, $\alpha \notin K$, so $[K(\alpha) : K] > 1$. By the Tower Law (theorem 19.3.7), $[K(\alpha) : K]$ divides $[L : K]$, which is $|\mathrm{Gal}_K(L)| = p$ by theorem 20.1.13(3). Hence $[K(\alpha) : K] = p$, $[L : K(\alpha)] = 1$, and $L = K(\alpha)$.

2. For $i \in \mathbb{F}_p$, (20.5) and $i^p = i$ give $f(\alpha + i) = f(\alpha) + i^p - i = 0$. The p elements $\alpha + i$ are distinct, and f has at most p roots in E (corollary 11.3.2), so they are all of its roots, and $f = \prod_{i\in\mathbb{F}_p}(x - \alpha - i)$ by proposition 11.3.1 applied to these distinct roots in turn and a comparison of degrees. If $\alpha \in K$, every root $\alpha + i$ lies in K, and f splits into distinct linear factors over K.

   Let $\alpha \notin K$. The field $K(\alpha) \subseteq E$ contains every root $\alpha + i$ of f, and every subfield of E containing K and the roots of f contains α, so $K(\alpha)$ is a splitting field of f over K. As the p roots of f are distinct, f is separable (definition 19.6.2), and $K(\alpha)/K$ is a finite Galois extension (theorem 20.1.13(1), definition 20.1.14), finite since α is algebraic over K (theorem 19.2.8, theorem 19.2.5). Let $G = \mathrm{Gal}_K(K(\alpha))$. Each $\tau \in G$ sends α to a root of f, so $\chi(\tau) = \tau(\alpha) - \alpha$ lies in $\mathbb{F}_p$. For $\tau, \rho \in G$, the element $\chi(\rho) \in \mathbb{F}_p \subseteq K$ is fixed by τ, so $(\tau\rho)(\alpha) = \tau(\alpha + \chi(\rho)) = \alpha + \chi(\tau) + \chi(\rho)$, and $\chi\colon G \to \mathbb{F}_p$ is a homomorphism to the additive group of $\mathbb{F}_p$. It is injective: if $\chi(\tau) = 0$, then τ fixes α, hence fixes $K(\alpha)$, and $\tau = \mathrm{id}$. By theorem 3.3.1, $|G| = |\chi(G)|$ divides $|\mathbb{F}_p| = p$. As $\alpha \notin K$, $|G| = [K(\alpha) : K] > 1$ (theorem 20.1.13(3)), so $|G| = p$ and χ is bijective. So G is cyclic of order p (corollary 3.3.2), and the element $\sigma = \chi^{-1}(1)$, which satisfies $\sigma(\alpha) = \alpha + 1$, generates G, since $\langle\sigma\rangle$ has order dividing p (theorem 3.3.1) and greater than 1.

   Finally, the minimal polynomial m of α over K has degree $[K(\alpha) : K] = p$ (theorem 19.2.8, theorem 19.2.5). The polynomial $f - m$ has the root α and degree less than p, as f and m are both monic of degree p, so $f - m = 0$ by the minimality of $\deg m$ (definition 19.2.19). A factorization $f = gh$ into polynomials of positive degree would give $g(\alpha) = 0$ or $h(\alpha) = 0$, again against the minimality of $\deg m$, so f is irreducible, completing the proof.

In characteristic 0, a Galois extension is solvable exactly when it is contained in a Galois radical extension (proposition 20.2.18). In characteristic p, the radical steps are joined by the steps of theorem 20.4.12: a finite separable extension of K is contained in a Galois extension of K with solvable group if and only if it is contained in the top field of a finite tower $K = E_0 \subseteq E_1 \subseteq \cdots \subseteq E_r$ in which each E_{j+1} is obtained from E_j by adjoining a root of unity, a root of some $x^n - c$ with $c \in E_j$ and $p \nmid n$, or a root of some $x^p - x - c$ with $c \in E_j$ [42, Chapter VI, Theorem 7.2]. This book proves the cyclic steps, proposition 20.1.42 for a degree m prime to p when K contains a primitive mth root of unity and theorem 20.4.12 for degree p, and the characteristic p statement itself is deferred to that reference. There is a similar Artin-Schreier pairing, with values in $\mathbb{F}_p$ and with the additive group $K/\{c^p - c \mid c \in K\}$ in place of $K^\times/(K^\times)^n$ [42, Chapter VI, Theorem 8.3].

Exercise 20.4.1

1. Show that we can find α such that $L = K(\alpha)$ where L is a Kummer $\mathbb{Z}/n\mathbb{Z}$-extension showing all elements can be written as the sum of eigenvalues. (hint: https://www.youtube.com/watch?v=UaeJNQ5x17g&list=PL8yHsr3EFj53Zxu3iRGMYL_89GDMvdkgt&index=15)

21⋆

More Structures on Fields

⋆ *Optional. This chapter is an aside; nothing later in the book depends on it.*

The most common fields the reader will encounter are $\mathbb{Q}, \mathbb{R}$, and $\mathbb{C}$. Each of these fields are natural in some way. Here $\mathbb{Q}$ is initial in $\mathbf{Fld}_0$, $\mathbb{R}$ is the smallest complete ordered field[1], and $\mathbb{C}$ is the smallest complete and algebraically closed field. A key property all three of these fields share is that there is a notion of *size*. For $\mathbb{Q}$ and $\mathbb{R}$, there is even the stronger notion of *ordering* for all the elements. For $\mathbb{R}$ and $\mathbb{C}$, there is a notion of *differentiability*.

Recall that an order can be defined on a set ([24, Order]). An ordered field carries such an order, compatible with addition and multiplication, and an absolute value is a function from a field to the ordered field $\mathbb{R}$ that measures size.

This chapter shall also incorporate some topology from appendix C as it is beneficial for a deeper understanding of the subjects presented in this chapter.

[1] More technically, it is the unique Dedekind-complete ordered field up to isomorphism

21.1 Ordered and Real Fields

Definition 21.1.1: Ordered Field

Let k be a field. Then k is a *totally ordered field* (or *ordered field* for short) if there exists a (total) order $\leq$ on k such that for all $x, y, z \in k$:

1. $x < y$, then $x + z < y + z$
2. If $z > 0$, then if $x < y$, $zx < zy$

If a field k admits a total ordering, then k is said to be a *formally real field*.

Famously, $\mathbb{Q}$ and $\mathbb{R}$ are ordered fields. The reader should prove that $\mathbb{C}$ is *not* an ordered field.

Proposition 21.1.2: Properties Of Ordered Fields

Let k be an ordered field and $x, y, z \in k$. Then

1. $x > 0$ if and only if $-x < 0$ (similarly, $x < 0$ if and only if $-x > 0$)
2. $x > y$ if and only if $x - y > 0$
3. $x < y$ if and only if $-x > -y$
4. if $x > y > 0$, then $y^{-1} > x^{-1} > 0$
5. if $x < y$ and $z < 0$, then $xz > yz$
6. $x^2 \geq 0$, with $x^2 > 0$ for $x \neq 0$ (hence $1 > 0$, since $1 = 1^2 > 0$)

Furthermore, ordered fields have characteristic zero

***Proof*:**
Here $a < b$ means $a \leq b$ and $a \neq b$, and $b > a$ means $a < b$. Since the order is total ([24, Order]), for all a, b exactly one of $a < b$, $a = b$, $b < a$ holds: at least one holds by connectedness, and $a < b$ together with $b < a$ would give $a = b$ by antisymmetry. The relation $<$ is transitive: if $a < b < c$, then $a \leq c$, and $a = c$ would give $a \leq b \leq a$, so $a = b$, which is impossible. The two conditions of definition 21.1.1 are referred to as (O1) and (O2).

1. If $0 < x$, then (O1) with $z = -x$ gives $-x < 0$. If $-x < 0$, then (O1) with $z = x$ gives $0 < x$. Likewise, if $x < 0$, then (O1) with $z = -x$ gives $0 < -x$, and if $0 < -x$, then (O1) with $z = x$ gives $x < 0$.
2. By (O1), $y < x$ gives $0 = y - y < x - y$, and $0 < x - y$ gives $y < x$.
3. By (O1) with $z = -x - y$, $x < y$ gives $-y < -x$, and with $z = x + y$, $-y < -x$ gives $x < y$.
4. First, $a > 0$ implies $a^{-1} > 0$: the element a^{-1} is nonzero, and if $a^{-1} < 0$, then $-a^{-1} > 0$ by (1), and (O2) applied to $0 < a$ with $z = -a^{-1}$ gives $0 < -a^{-1}a = -1$, so $1 < 0$ by (1),

contradicting $1 > 0$ from (6), whose proof below uses only (1). Applying this to $a = x$ and $a = y$ gives $x^{-1} > 0$ and $y^{-1} > 0$, and (O2) applied to $0 < y^{-1}$ with $z = x^{-1}$ gives $x^{-1}y^{-1} > 0$. Now (O2) applied to $y < x$ with $z = x^{-1}y^{-1}$ gives $x^{-1} < y^{-1}$, since k is commutative. Hence $y^{-1} > x^{-1} > 0$.

5. By (1), $-z > 0$, and (O2) applied to $x < y$ with $-z$ gives $-zx < -zy$, using $(-z)x = -(zx)$ (proposition 9.1.9(2)). By (3), $zx > zy$, that is, $xz > yz$.

6. If $x > 0$, then (O2) applied to $0 < x$ with $z = x$ gives $0 = x \cdot 0 < x^2$. If $x < 0$, then $-x > 0$ by (1), so $0 < (-x)^2 = x^2$ (proposition 9.1.9(3)). Since one of $x > 0$, $x = 0$, $x < 0$ holds, $x^2 > 0$ for $x \neq 0$ and $x^2 \geq 0$ for all x. As $1 \neq 0$, $1 = 1^2 > 0$.

For the characteristic, $n \cdot 1 > 0$ for every integer $n \geq 1$, by induction: $1 > 0$ by (6), and if $n \cdot 1 > 0$, then (O1) with $z = 1$ gives $(n + 1) \cdot 1 > 1 > 0$. So $n \cdot 1 \neq 0$ for every $n \geq 1$, and k has characteristic zero (definition 19.1.7), completing the proof.

Note how this proposition immediately shows that $\mathbb{C}$ is not ordered since $-1 = i^2 \not> 0$, and that no finite field is ordered, since ordered fields have characteristic zero while in a finite field two of the elements $m \cdot 1$, $m \geq 1$, coincide, so that some $n \cdot 1$ with $n \geq 1$ vanishes. In fact, this is a characterizing property of non-ordered fields

Proposition 21.1.3: Artin-Schreier Conditions

Let k be a field. Then the following are equivalent:

1. k is formally real, that is, k admits an order making it an ordered field (definition 21.1.1)
2. -1 is *not* a sum of squares of elements of k
3. 0 is *not* a sum of squares of elements of k that are not all zero

***Proof*:**

(1 ⇒ 2) Fix such an order. Every square satisfies $x^2 \geq 0$ (proposition 21.1.2(6)), and a sum of elements $u, v \geq 0$ satisfies $u + v \geq 0$: if $u = 0$ this is $v \geq 0$, and if $u > 0$, then $v < u + v$ by (O1) of definition 21.1.1, and $0 \leq v < u + v$. By induction every sum of squares is ≥ 0, while $-1 < 0$ by proposition 21.1.2(1) and (6). So -1 is not a sum of squares.

(2 ⇒ 3) Suppose $0 = x_1^2 + \cdots + x_m^2$ with the x_i not all zero, and renumber so that $x_m \neq 0$. Then $m \geq 2$, since $x_m^2 \neq 0$ in a field, and dividing $-x_m^2 = x_1^2 + \cdots + x_{m-1}^2$ by x_m^2 gives

$$-1 = \sum_{i=1}^{m-1} \left(\frac{x_i}{x_m} \right)^2$$

contradicting (2).

(3 ⇒ 2) If $-1 = x_1^2 + \cdots + x_m^2$, then $0 = 1^2 + x_1^2 + \cdots + x_m^2$ is a sum of squares of elements that are not all zero, since $1 \neq 0$, contradicting (3).

(2 ⇒ 1) Assume -1 is not a sum of squares, and let T be the set of all sums of squares of elements of k, including $0 = 0^2$. Then $T + T \subseteq T$, $TT \subseteq T$ because

$\left(\sum_i x_i^2\right)\left(\sum_j y_j^2\right) = \sum_{i,j}(x_i y_j)^2$ (proposition 9.1.9(5) and commutativity), and $-1 \notin T$.

Step 1: a maximal cone. Let $\mathcal{P}$ be the set of subsets $P \subseteq k$ with $T \subseteq P$, $P + P \subseteq P$, $PP \subseteq P$ and $-1 \notin P$, ordered by inclusion. It contains T. A chain in $\mathcal{P}$ has an upper bound: T if the chain is empty, and otherwise the union of its members, since two elements of the union lie in a common member, which then contains their sum and product, while -1 lies in no member. By Zorn's Lemma (theorem A.2.3), $\mathcal{P}$ has a maximal element P.

Step 2: $P \cap (-P) = \{0\}$. Here $0 \in T \subseteq P$. If $a \neq 0$ and $a, -a \in P$, then, as $(a^{-1})^2 \in T \subseteq P$,

$$-1 = (-a) \cdot a \cdot (a^{-1})^2 \in P$$

which is impossible.

Step 3: $P \cup (-P) = k$. Let $a \in k$ with $-a \notin P$, and put $P' = \{x + ay \mid x, y \in P\}$. Then $P \subseteq P'$ (take $y = 0$), $P' + P' \subseteq P'$, and $P'P' \subseteq P'$ because

$$(x + ay)(z + at) = (xz + a^2 yt) + a(xt + yz)$$

with $a^2 \in T \subseteq P$. If $-1 = x + ay$ with $x, y \in P$, then $y \neq 0$, since $-1 \notin P$, and

$$-a = (1 + x) \cdot y \cdot (y^{-1})^2$$

lies in P, as $1 = 1^2 \in P$. This contradicts the choice of a, so $-1 \notin P'$ and $P' \in \mathcal{P}$. By maximality $P' = P$, so $a = 0 + a \cdot 1 \in P$. Hence every $a \in k$ lies in P or in $-P$.

Step 4: the order. Define $x \leq y$ when $y - x \in P$. This relation is reflexive as $0 \in P$, antisymmetric by Step 2, transitive since $z - x = (z - y) + (y - x)$, and connected by Step 3, so it is a total order ([24, Order]). If $x < y$, then $(y + z) - (x + z) = y - x$ lies in P and is nonzero, so $x + z < y + z$. If $z > 0$ and $x < y$, then $zy - zx = z(y - x) \in PP \subseteq P$, and it is nonzero as a product of nonzero elements of a field, so $zx < zy$. Hence k is an ordered field (definition 21.1.1), completing the proof.

Since any ordered field is characteristic zero, it must contain a subfield

$$Q = \left\{ \frac{m1}{n1} \;\middle|\; m, n \in \mathbb{Z} \right\}$$

field-isomorphic to $\mathbb{Q}$. In fact, Q is order-field isomorphic, for $n1 > 0$ in Q when $n > 0$ in $\mathbb{Z}$, and $m1/n1 > 0$ in Q when $m/n > 0$ in $\mathbb{Q}$, and so $a1/b1 > c1 > d$ in Q iff $a/b > c/d$ in $\mathbb{Q}$. Thus, we usually identify Q with $\mathbb{Q}$ as an ordered subfield. A property that we may take for granted that an ordered field may have is the following:

Definition 21.1.4: Archimedean Field

Let k be an ordered field. Then k is an *Archimedean field* if every positive element of k is less than a positive integer

Though it may seem like all fields have this property, there do exists field without it, most notably the *hyperreals numbers* which can be thought of as the "order completion" of the reals (think of adding an element ϵ st $0 < \epsilon < x$ for all $x > 0$ and a ω st $0 < x < \omega$ for all $x > 0$). Fortunately, all the ordered fields we shall work with will be Archimedean due to the following classification result:

Theorem 21.1.5: Classification Of Archimedean Fields

Let k be an Archimedean field. Then k is isomorphic to a field:

$$\mathbb{Q} \subseteq k' \subseteq \mathbb{R}$$

***Proof*:**
As noted above, k contains a copy of $\mathbb{Q}$ as an ordered subfield, and we identify the two. The one property of $\mathbb{R}$ used below is its completeness, that every Cauchy sequence of rationals converges in $\mathbb{R}$.

Step 1: $\mathbb{Q}$ is dense in k. Let $x < y$ in k. Since k is Archimedean there are positive integers ℓ, m, n with $-\ell < x$, $y < m$ and $1/(y-x) < n$, so that $0 < 1/n < y - x$. The set of positive integers i with $(i/n) - \ell > x$ is not empty, since $i \geq n(\ell + m)$ gives $(i/n) - \ell \geq m > y > x$, so it has a least element j. If $j \geq 2$ then $((j-1)/n) - \ell \leq x$ by minimality, and if $j = 1$ then $((j-1)/n) - \ell = -\ell < x$. Either way

$$\frac{j}{n} - \ell \leq x + \frac{1}{n} < x + (y - x) = y$$

so $r = (j/n) - \ell$ is a rational number with $x < r < y$.

Step 2: limits inside k. Say that $L = \lim_k x_n$ when $L \in k$ and, for every positive $\epsilon \in \mathbb{Q}$, the inequality $L - \epsilon < x_n < L + \epsilon$ holds for all large n. Such an L is unique: two distinct candidates $L < L'$ would admit a rational ϵ with $0 < 2\epsilon < L' - L$ by Step 1, and then no x_n could be within ϵ of both. The usual arguments give

$$\lim_k (a_n + b_n) = \lim_k a_n + \lim_k b_n \qquad \lim_k (a_n b_n) = \left(\lim_k a_n\right)\left(\lim_k b_n\right)$$

whenever the right hand sides exist, the Archimedean hypothesis being what allows the error bounds to be taken rational. A sequence with a limit in k is Cauchy in the same sense: for positive rational ϵ, $-\epsilon < a_m - a_n < \epsilon$ for all large m, n.

Step 3: the map into $\mathbb{R}$. Every $x \in k$ is a limit of rationals: by Step 1 choose $a_n \in \mathbb{Q}$ with $x - (1/n) < a_n < x + (1/n)$, and then $\lim_k a_n = x$. Such an (a_n) is a Cauchy sequence of rational numbers, so it converges in $\mathbb{R}$. If (b_n) is another sequence of rationals with $\lim_k b_n = x$, then for every positive rational ϵ we get $|a_n - b_n| < \epsilon$ for all large n, so (a_n) and (b_n) have the same

limit in $\mathbb{R}$. Hence

$$\lambda : k \to \mathbb{R} \qquad \lambda(x) = \lim_{n\to\infty} a_n \quad \text{whenever } x = \lim_k a_n \text{ with } a_n \in \mathbb{Q}$$

is well defined, and $\lambda(x) = x$ for $x \in \mathbb{Q}$, taking $a_n = x$ for all n.

Step 4: λ is an order preserving field homomorphism. If $x = \lim_k a_n$ and $y = \lim_k b_n$ with $a_n, b_n \in \mathbb{Q}$, then $x + y = \lim_k(a_n + b_n)$ and $xy = \lim_k(a_n b_n)$ by the limit laws of Step 2, while in $\mathbb{R}$ the limits of $(a_n + b_n)$ and $(a_n b_n)$ are $\lambda(x) + \lambda(y)$ and $\lambda(x)\lambda(y)$. So λ is a ring homomorphism, and it is injective, being a homomorphism of fields that is not identically zero.

Suppose $x > 0$ in k. Since k is Archimedean there is a positive integer m with $x^{-1} < m$, and then $1/m < x$. Choosing the a_n as in Step 3 and discarding the finitely many terms with $a_n \leq 1/m$, we get $\lambda(x) \geq 1/m > 0$. Applied to $y - x$ this shows that $x < y$ in k implies $\lambda(x) < \lambda(y)$, and the converse holds because $x \geq y$ would give $\lambda(x) \geq \lambda(y)$.

Therefore λ is an isomorphism of ordered fields from k onto $k' = \lambda(k)$, and $\mathbb{Q} \subseteq k' \subseteq \mathbb{R}$.

Let us focus on formally real fields. If k is formally real, then so is $k(x)$. For an algebraic extension

Proposition 21.1.6: Algebraic Extension Of Formally Real Field

Let k be a formally real field. Then if $\alpha^2 \in k$ such that $\alpha^2 > 0$, $k(\alpha)$ is a formally real field

***Proof*:**
Let α lie in an extension of k with $a := \alpha^2 \in k$, and fix an order on k witnessing formal reality, with respect to which $a > 0$. If $\alpha \in k$, then $k(\alpha) = k$ and there is nothing to prove, so assume $\alpha \notin k$, in which case α has minimal polynomial $x^2 - a$ and $\{1, \alpha\}$ is a k-basis of $k(\alpha)$.

By proposition 21.1.3, it suffices to show that -1 is not a sum of squares in $k(\alpha)$. Suppose toward a contradiction that

$$-1 = \sum_i (c_i + d_i\alpha)^2 = \sum_i (c_i^2 + d_i^2 a) + 2\left(\sum_i c_i d_i\right)\alpha$$

with $c_i, d_i \in k$, using $\alpha^2 = a$. Comparing the coefficients of 1 in the basis $\{1, \alpha\}$:

$$-1 = \sum_i c_i^2 + a\sum_i d_i^2$$

By proposition 21.1.2(6), every square in the ordered field k is non-negative, so $\sum_i c_i^2 \geq 0$ and $\sum_i d_i^2 \geq 0$, and since $a > 0$, also $a\sum_i d_i^2 \geq 0$. Hence the right hand side is non-negative, while $-1 < 0$ since $1 > 0$ by proposition 21.1.2(6), a contradiction. Therefore -1 is not a sum of squares in $k(\alpha)$, and by proposition 21.1.3, $k(\alpha)$ is formally real.

If an algebraic extension of a formally real field is formally real, then it is called a *real extension.*

Proposition 21.1.7: Odd Extension Of Formally Real Field

Let k be a formally real field. Then every finite extension of odd degree is a formally real field.

***Proof*:**
Fix an order on k witnessing formal reality. The argument is by induction on $n = [E : k]$, run simultaneously over all formally real base fields, so the inductive hypothesis reads: every extension of odd degree less than n of a formally real field is formally real. For $n = 1$ there is nothing to prove, so let $n > 1$ be odd and let $[E : k] = n$.

Suppose first that there is an $\alpha \in E \setminus k$ with $k(\alpha) \neq E$. The Tower Law (theorem 19.3.7) gives $[k(\alpha) : k] \cdot [E : k(\alpha)] = n$, so both factors divide the odd number n and both are smaller than n. By the induction hypothesis $k(\alpha)$ is formally real, and then, applying the hypothesis again over the formally real base $k(\alpha)$, so is E.

So assume $E = k(\alpha)$ for every $\alpha \in E \setminus k$. Fix such an α and let $q \in k[x]$ be its minimal polynomial, of degree n. By corollary 19.2.23 every element of E is $f(\alpha)$ for some $f \in k[x]$ with $\deg f < n$.

Suppose E were not formally real. By proposition 21.1.3 there are $f_i \in k[x]$ with $\deg f_i < n$ and

$$-1 = \sum_i f_i(\alpha)^2$$

If every f_i were constant this would exhibit -1 as a sum of squares in k, against the formal reality of k, so $d = \max_i \deg f_i$ satisfies $1 \leq d \leq n - 1$. The polynomial $1 + \sum_i f_i^2$ vanishes at α, so q divides it and

$$1 + \sum_i f_i^2 = qg$$

for some $g \in k[x]$. Its degree is exactly $2d$: the coefficient of x^{2d} in $\sum_i f_i^2$ is the sum of the squares of the leading coefficients of those f_i of degree d, and that sum is positive by proposition 21.1.2(6), hence nonzero. Therefore

$$\deg g = 2d - n$$

which is odd, since n is odd, and satisfies $\deg g < n$ because $d \leq n - 1$, and $\deg g \geq 1$ because it is a non-negative odd integer.

A product of polynomials of even degree has even degree, so g has an irreducible factor $r \in k[x]$ of odd degree, and $\deg r \leq \deg g < n$. Let β be a root of r in some extension of k. Then r is the minimal polynomial of β, so $[k(\beta) : k] = \deg r$ by theorem 19.2.8, an odd number smaller than n, and the induction hypothesis makes $k(\beta)$ formally real. But r divides g, so $g(\beta) = 0$ and

$$1 + \sum_i f_i(\beta)^2 = q(\beta)g(\beta) = 0$$

exhibits -1 as a sum of squares in $k(\beta)$, contradicting proposition 21.1.3. Hence E is formally real, completing the induction.

Using this, we can find the real closure of a formally real field:

Definition 21.1.8: Closed Real Field

Let k be a formally real field. Then k is said to be *closed* if there is no proper real algebraic extension L/k.

Theorem 21.1.9: Detecting Real Closure

Let k be a formally real field. Then k is real closed if and only if:

1. every positive element of k is a square in k
2. every polynomial of odd degree has a root in k

From these condition, we get that $\overline{k} = k(i)$ where $i^2 = -1$.

***Proof*:**
Fix an order on k witnessing formal reality. Ordered fields have characteristic zero by proposition 21.1.2, so every irreducible polynomial over k or over an extension of k is separable by corollary 19.6.8.

($\Rightarrow$) Suppose k is real closed.

For (1), let $a > 0$ in k and suppose a were not a square. A square root α of a in an algebraic closure then lies outside k, and $k(\alpha)$ is a proper algebraic extension of k which is formally real by proposition 21.1.6. That contradicts definition 21.1.8, so a is a square in k.

For (2), let $f \in k[x]$ have odd degree. Since a product of polynomials of even degree has even degree, f has an irreducible factor p of odd degree. Let β be a root of p, so $[k(\beta) : k] = \deg p$ by theorem 19.2.8. By proposition 21.1.7 the extension $k(\beta)$ is formally real, and it is algebraic over k, so definition 21.1.8 forces $k(\beta) = k$. Hence $\deg p = 1$ and $\beta \in k$ is a root of f.

($\Leftarrow$) Suppose now (1) and (2) hold.

Step A: k has no finite extension of odd degree other than itself. Let $[E : k]$ be odd and take $\alpha \in E$. The minimal polynomial q of α has degree $[k(\alpha) : k]$, which divides $[E : k]$ by the Tower Law (theorem 19.3.7) and is therefore odd. By (2) the polynomial q has a root in k, and being irreducible it must have degree 1, so $\alpha \in k$. Hence $E = k$.

Step B: every element of $C = k(i)$ is a square in C, where $i^2 = -1$. First $i \notin k$: otherwise $-1 = i^2$ would be a square, and $-1 < 0$ contradicts proposition 21.1.2(6). Next, every $a \in k$ is a square in C: for $a \geq 0$ this is (1), and for $a < 0$ we have $-a > 0$, so $-a = y^2$ with $y \in k$ and $a = (iy)^2$. Now take $a + bi$ with $b \neq 0$. We seek $x, y \in k$ with $(x + yi)^2 = a + bi$, that is

$$x^2 - y^2 = a \qquad 2xy = b$$

Note $2 = 1 + 1 > 1 > 0$ in k, by proposition 21.1.2(6) and (O1) of definition 21.1.1 applied to $0 < 1$, so $2 \neq 0$ is invertible. Substituting $y = b/(2x)$ into the first equation turns it into

$$x^4 - ax^2 - \frac{b^2}{4} = 0$$

Read as a quadratic in x^2, its discriminant is $a^2 + b^2 > 0$, hence a square in k by (1), so the

quadratic has roots $s_1, s_2 \in k$ with $s_1 s_2 = -b^2/4 = -(b/2)^2 < 0$ (proposition 21.1.2(6) and (1)). Neither root is 0, and if both were negative, then $s_1 s_2 = (-s_1)(-s_2)$ would be positive by (O2). So one of them is positive, say s_1, and (1) gives $s_1 = x^2$ for some nonzero $x \in k$. Setting $y = b/(2x)$ and using $s_1^2 - a s_1 - b^2/4 = 0$,

$$x^2 - y^2 = s_1 - \frac{b^2}{4s_1} = a \qquad 2xy = b$$

so $a + bi = (x + yi)^2$.

Step C: C has no extension of degree 2. Every quadratic $\alpha x^2 + \beta x + \gamma \in C[x]$ has a root in C, since $\beta^2 - 4\alpha\gamma$ has a square root in C by Step B and 2α is invertible. So no polynomial of degree 2 is irreducible in $C[x]$. If $[E : C] = 2$ and $\alpha \in E \setminus C$, then $[C(\alpha) : C]$ divides 2 and exceeds 1, so the minimal polynomial of α over C would be irreducible of degree 2, which cannot happen.

Step D: $\overline{k} = C$. Let $\alpha \in \overline{k}$ and let E be the splitting field over k of the product of $x^2 + 1$ with the minimal polynomial of α over k. Then $C \subseteq E$, and E/k is finite, normal and separable, hence Galois by theorem 20.1.13(2). Put $G = \mathrm{Gal}_k(E)$, so $|G| = [E : k] = [E : C][C : k] = 2[E : C]$ is even by theorem 20.1.13(3) and the Tower Law.

Let S be a Sylow 2-subgroup of G, which exists by theorem 4.8.6(1), and let $F = E^S$. By proposition 20.1.12, $S = \mathrm{Aut}_F(E)$, so $[E : F] = |S|$ by theorem 20.1.13(3) applied to the Galois extension E/F, and the Tower Law gives $[F : k] = |G|/|S| = [G : S]$, an odd number. Step A forces $F = k$, so $[G : S] = 1$ and G is a 2-group.

Then $H = \mathrm{Gal}_C(E)$ is a subgroup of G, hence a 2-group. Suppose $H \neq 1$. A nontrivial finite group of order 2^m has a subgroup of index 2, by induction on m: this is clear for $m = 1$, and for $m > 1$ the centre is nontrivial by theorem 4.6.5, so corollary 3.3.9 gives a central element z of order 2, the subgroup $\langle z \rangle$ is normal, and a subgroup of index 2 in the smaller 2-group $H/\langle z \rangle$ pulls back to one in H by the Correspondence Theorem (theorem 3.5.7). Taking such a subgroup $H' \leqslant H$ of index 2 and arguing with the Galois correspondence for E/C as above, its fixed field $E^{H'}$ satisfies $[E^{H'} : C] = [H : H'] = 2$, contradicting Step C. Hence $H = 1$ and $E = C$ by theorem 20.1.13(4), so $\alpha \in C$. Therefore $\overline{k} = C = k(i)$.

Step E: k is real closed. By Step D, $[\overline{k} : k] = 2$, so the only proper algebraic extension of k inside $\overline{k}$ is $\overline{k} = k(i)$ itself. It is not formally real, since $-1 = i^2$ is a square there. Any proper algebraic extension L/k embeds in $\overline{k}$ over k by lemma 19.5.7, and formal reality is preserved by isomorphism, so k has no proper real algebraic extension, which is definition 21.1.8.

Due to this, a real closed field can be made into an ordered field in a unique way, namely $x \leq y$ if and only if $y - x = a^2$ for some $a \in k$. It should also be clear that the field of all algebraic real numbers is certainly a real closed field.

Corollary 21.1.10: Irreducible Elements In Real Closed Field

Let k be a real closed field. Then $f \in k[x]$ is irreducible if and only if f has degree 1 or f has degree 2 with no roots

***Proof*:**
($\Leftarrow$) A degree 1 polynomial is irreducible, since in a factorization into two nonunits both factors have degree at least 1, giving total degree at least 2 by proposition 9.2.3(1). A degree 2 polynomial with no root in k is irreducible by proposition 11.3.5.

($\Rightarrow$) Let f be irreducible, so $\deg f \geq 1$. By theorem 21.1.9, $\overline{k} = k(i)$ where $i^2 = -1$. Note $i \notin k$: the field k is formally real, and if $-1 = c^2$ for some $c \in k$, then $-1 \geq 0$ in an order on k by proposition 21.1.2(6), contradicting $1 > 0$. Hence $[\overline{k} : k] = [k(i) : k] = 2$. Since $\overline{k}$ is algebraically closed, f has a root $\alpha \in \overline{k}$, and the Tower Law (theorem 19.3.7) applied to $k \subseteq k(\alpha) \subseteq \overline{k}$ gives $[k(\alpha) : k] \in \{1, 2\}$.

Now f is, up to a unit, the minimal polynomial of α: the minimal polynomial p divides f (dividing f by p leaves a remainder vanishing at α of degree less than $\deg p$, which must be 0), and f is irreducible, so $f = up$ with u a unit. By theorem 19.2.8, $k(\alpha) \cong k[x]/(p)$, so $\deg f = \deg p = [k(\alpha) : k] \leq 2$.

If $\deg f = 2$ and f had a root $c \in k$, then dividing by $x - c$ would factor f into two linear factors, contradicting irreducibility. Hence f has degree 1, or degree 2 with no roots in k.

Finally, we have:

Theorem 21.1.11: Real Closure

Every formally real field has a real closure

***Proof*:**
Let k be formally real and fix an order on it, with $P = \{a \in k \mid a > 0\}$. From definition 21.1.1, P is closed under addition and under multiplication: $x, y > 0$ gives $x + y > y > 0$ and $xy > x \cdot 0 = 0$. Fix an algebraic closure $\overline{k}$, which exists by theorem 19.5.6. By a *real closure* of k we mean an algebraic extension of k that is real closed.

Step 1: adjoining square roots of positive elements keeps the field formally real. Let $a_1, \ldots, a_n \in P$ and put $L_m = k(\sqrt{a_1}, \ldots, \sqrt{a_m}) \subseteq \overline{k}$. We show by induction on m the statement

$$(*_m): \qquad \text{no relation } \sum_j p_j z_j^2 = -1 \text{ holds with } p_j \in P \text{ and } z_j \in L_m$$

For $m = 0$ this is clear, since each $p_j z_j^2 \geq 0$ by proposition 21.1.2(6) and closure of P under multiplication, so the left hand side is nonnegative while $-1 < 0$. Suppose $(*_{m-1})$ and write $\alpha = \sqrt{a_m}$. If $\alpha \in L_{m-1}$ then $L_m = L_{m-1}$ and there is nothing to do, so assume otherwise, in which case $\{1, \alpha\}$ is an L_{m-1}-basis of L_m and $\alpha^2 = a_m$. A relation $\sum_j p_j (c_j + d_j \alpha)^2 = -1$ with $c_j, d_j \in L_{m-1}$ expands to

$$\sum_j p_j \left(c_j^2 + a_m d_j^2\right) + 2\alpha \sum_j p_j c_j d_j = -1$$

and comparing the coefficients of 1 in the basis $\{1, \alpha\}$ leaves

$$\sum_j p_j c_j^2 + \sum_j (p_j a_m) d_j^2 = -1$$

Every p_j and every $p_j a_m$ lies in P, so this contradicts $(*_{m-1})$. This proves $(*_m)$.

Taking all $p_j = 1$ shows -1 is not a sum of squares in L_n. Now let $E \subseteq \overline{k}$ be the subfield generated over k by all square roots of elements of P. A relation expressing -1 as a sum of squares in E involves finitely many elements, hence lies in some L_n, so no such relation exists and E is formally real by proposition 21.1.3.

Step 2: a maximal formally real extension. Let Σ be the set of subfields F with $E \subseteq F \subseteq \overline{k}$ that are formally real, partially ordered by inclusion. It contains E, and the union of a chain in Σ is again in Σ, since a relation exhibiting -1 as a sum of squares in the union involves finitely many elements and so already holds in one member of the chain. By Zorn's lemma Σ has a maximal element R, and R is algebraic over k, being a subfield of $\overline{k}$.

Step 3: R is real closed. Let L/R be a proper algebraic extension. Since $\overline{k}$ is algebraic over k and algebraically closed, it is an algebraic closure of R as well, so lemma 19.5.7 gives a homomorphism $L \to \overline{k}$ fixing R. Its image is a subfield of $\overline{k}$ properly containing R, hence containing E, so maximality of R says the image is not formally real. Formal reality is preserved by isomorphism, so L is not formally real. Therefore R has no proper real algebraic extension and is real closed by definition 21.1.8.

Step 4: the order of R induces that of k. Whatever order R carries, its positive elements are squares by theorem 21.1.9(1) and its nonzero squares are positive by proposition 21.1.2(6), so the positive elements of R are exactly the nonzero squares, so the order on R is determined by the field structure. Every $a \in P$ is a square in $E \subseteq R$, hence positive in R, and if $a < 0$ in k then $-a \in P$ is positive in R, so a is negative in R. Thus the order of R restricts to the given order of k, and R is a real closure of k.

The reader should verify that any two real closures of k are order-field isomorphic.

Finally, we state without proof the following characterization of real-closed fields that show a naturalness to the choice of $\mathbb{R}$

Theorem 21.1.12: Artin-Schreier Theorem

Let $k \neq \overline{k}$ be a field. Then the following are equivalent:

1. k is real closed
2. $[\overline{k} : k]$ is finite
3. there is an upper bound for the degrees of irreducible polynomials in $k[x]$

***Proof*:**
Deferred. The implications (1) ⇒ (2) and (2) ⇒ (3) are immediate. For the first, theorem 21.1.9 gives $\overline{k} = k(i)$ for real closed k, so $[\overline{k} : k] = 2$ is finite. For the second, if $[\overline{k} : k]$ is finite and $p \in k[x]$ is irreducible with a root $\alpha \in \overline{k}$, then $\deg p = [k(\alpha) : k] \leq [\overline{k} : k]$ by theorem 19.2.8 and the Tower Law (theorem 19.3.7). The content is (3) ⇒ (1), and it rests on three inputs, which this book carries only in part:

1. a bound on the degrees of irreducible polynomials in $k[x]$ forces k to be perfect, proved by showing $x^{p^r} - c$ is irreducible for every r when c is not a pth power.

2. if -1 is a square in a field F, then $\overline{F}$ is not a Galois extension of F of prime degree p. This is the substantial step, and it splits by characteristic. For $p \neq \mathrm{ch}(F)$, the pth roots of unity have degree less than p over F, hence lie in F by the Tower Law, and proposition 20.1.42 gives $\overline{F} = F(\alpha)$ with $\alpha^p \in F$. What is missing there is the irreducibility criterion for binomials $x^n - c$ of [42, Chapter VI, Theorem 9.1], applied to $x^{p^2} - \alpha^p$. For $p = \mathrm{ch}(F)$, theorem 20.4.12 gives $\overline{F} = F(\alpha)$ with $\alpha^p - \alpha \in F$, and what is missing is a cyclic extension of F of degree p^2 containing $F(\alpha)$ [42, Chapter VI, Exercise 29], which cannot exist when $[\overline{F} : F] = p$.

3. a Sylow argument on $\mathrm{Gal}(\overline{k}/k)$ of the same shape as the one in Step D of the proof of theorem 21.1.9, which cuts the degree down to a prime and feeds the previous item.

The classifications of cyclic extensions of prime degree that item (2) uses are proposition 20.1.42 and theorem 20.4.12. The irreducibility criterion, the extension of degree p^2 and the Sylow argument are not in this book, and [42, Chapter VI, Corollary 9.3] combines them into a proof that $\overline{k} = k(i)$ whenever $1 < [\overline{k} : k] < \infty$.

A variant of the theorem is worth recording, also without proof. Let G_K denote the absolute Galois group of a field K, and let $\sigma \in G_K$ be an involution, so that $\sigma \neq \mathrm{id}$ and $\sigma \circ \sigma = \mathrm{id}$. Then the fixed field $\overline{K}^{\sigma}$ is real closed. The two statements carry the same content, since the fixed field of an involution sits at degree 2 below $\overline{K}$ and theorem 21.1.12 then applies to it.

21.1.1 The Fields That Carry a Geometry

A vector space over an arbitrary field has no length and no angle, and section 13.6 recorded what producing them takes: an ordering, so that $\langle v, v\rangle$ can be called nonnegative, and a conjugation, so that $\langle v, v\rangle$ can be forced into the ordered field underneath. Both were described there as structure that $\mathbb{R}$ and $\mathbb{C}$ happen to carry. The results of this section say where that structure comes from, and how little freedom there was in it.

Whether a field can support a notion of positivity at all is decided by the field itself. By proposition 21.1.3, k admits a total order exactly when -1 is not a sum of squares, a condition stated in the ring operations alone with no order mentioned.

One step further up, the order and the conjugation both stop being choices. Let k be real closed. Every positive element of k is a square by theorem 21.1.9, and every nonzero square is positive by proposition 21.1.2(6), so the positive elements are exactly the nonzero squares. Hence $x \leq y$ holds if and only if $y - x$ is a square, and the order is recovered from the field operations rather than imposed on them. The same theorem gives $\overline{k} = k(i)$. Since -1 is not a square in a formally real field this extension has degree 2. It is the splitting field of $x^2 + 1$ over k, hence normal, and separable because an ordered field has characteristic zero (proposition 21.1.2). So $k(i)/k$ is Galois, $\mathrm{Gal}(\overline{k}/k)$ has order 2, and its one nontrivial element is $a + bi \mapsto a - bi$. A real closed field therefore comes with a unique order and a unique conjugation, both determined by k.

Corollary 21.1.10 gives the third property. Over a real closed k every irreducible polynomial has degree 1 or 2, so a characteristic polynomial factors into linear and quadratic pieces over k itself and not only over $\overline{k}$.

Theorem 21.1.12 says that no other configuration of this shape exists. If $k \neq \overline{k}$ and $[\overline{k} : k]$ is finite, then k is real closed and the degree is 2. In particular no field has an algebraic closure of degree 3: a field sits at finite distance below its algebraic closure in exactly one way, and that way is the real closed one.

What none of this pins down is $\mathbb{R}$. The field of real algebraic numbers is real closed as well and satisfies every condition above, so the separation between the two is order-completeness rather than an algebraic property. Theorem 21.1.5 gives the algebraic half of that separation by placing every Archimedean field between $\mathbb{Q}$ and $\mathbb{R}$. The geometry built on the order and the conjugation (inner products, orthogonality, adjoints, the spectral theorem) is carried out over $\mathbb{R}$ and $\mathbb{C}$ in *Linear Algebra* [21].

21.2 Absolute Values on General Fields

In an ordered field, $-1 < 0$ and every square is nonnegative, by proposition 21.1.2(1) and (6), so $-1 = i^2$ shows that $\mathbb{C}$ is not an ordered field. The next best thing is some notion of "size". Intuitively, size is a totally ordered concept, and hence the absolute value shall be a function that will capture the intuition of size with codomain $\mathbb{R}$:

Definition 21.2.1: Absolute Value

Let k be a field. Then an *absolute value real valuation* on k is a function $v : k \to \mathbb{R}$ whose image is denoted $|x|_v$ such that

1. (positivity) $|x|_v \geq 0$ for all $x \in k$ and $|x|_v = 0$ if and only if $x = 0$
2. (multiplicative) $|xy|_v = |x|_v|y|_v$
3. (triangle inequality) $|x + y|_v \leq |x|_v + |y|_v$

If the absolute value is unambiguous, $|x|_v$ will be denoted $|x|$. If the image of v is a finitely generated abelian group, then $|-|_v$ is called a *discrete absolute value*

21.2.2 Note The reader should be careful with the definition *real valuation* as there is a different concept called *valuation* that is not a generalization o specialization of the above (see [16, $\mathbb{R}$-Valuation]).

If the reader is familiar with norms, then this can be thought of as the field-equivalent as norms are defined on vector-spaces. In fact, taking a field to be a vector-space over itself, the existence of a absolute value induces a norm. If the reader is familiar with some Lebesgue space theory, they may recall Hölder's inequality stating that:

$$\|fg\|_1 \leq \|f\|_p \|g\|_q \qquad \frac{1}{p} + \frac{1}{q} = 1$$

which contrasts with the strict equality result given above. Recall that equality happens if and only if $\alpha|f|^p = \beta|g|^q$ for $\alpha, \beta \neq 0$, that is they are "power multiples" of each other. This is certainly the case for fields. Furthermore, there is only one absolute value instead of the 3 norms present in Hölder's

inequality. A norm that satisfies the 2nd condition (on an algebra!) is called a *multiplicative norm*, and implies certain geometric invariances in the space.

Example 21.2.3: Absolute Values

1. The usual absolute values on $\mathbb{Q}, \mathbb{R}, \mathbb{C}$ all satisfy the above definition
2. For any field k, there always exists the trivial valuation
$$|x|_t = 1 \qquad x \neq 0$$
This real valuation shall be an example of a *nonarchimedean* real valuation.
3. Let k be any field and take $k(x)$, the simple transcendental extension of k. Then there are two interesting valuations that may be defined. The first is called the infinity valuation:
$$|f/g|_\infty = \begin{cases} 2^{\deg f - \deg g} & f \neq 0 \\ 0 & f = 0 \end{cases}$$
and the 0 valuation:
$$|f/g|_0 = \begin{cases} 2^{\operatorname{ord} f - \operatorname{ord} g} & f \neq 0 \\ 0 & f = 0 \end{cases}$$
where $\operatorname{ord}(f)$ for a polynomial $f(x) = a_0 + a_1 x_1 + \cdots + a_n x^n$ to be the smallest $k \geq 0$ such that $a_k \neq 0$. The choice of 2 was arbitrary in both and can be replaced with any positive constant.
From the 0-valuation, we may define:
$$\left|\frac{f(x)}{g(x)}\right|_a = \left|\frac{f(x-a)}{g(x-a)}\right|_0$$
These valuations when restricted to k are all trivial.

With positivity and the triangle inequality, the absolute value can define a metric (see appendix C) and hence a metric topology on a field:
$$d(x, y) = |x - y|_v$$
For the p-adic case, the idea behind this metric is that $|x - y|$ is small is $x \equiv y \bmod p^n$ for a large n, that is a large power of p divides the difference.

Proposition 21.2.4: Equivalent Topoloies

Let v, w be two absolutes values on a field k. Then the following are equivalent:

1. v, w induce the same topology on k
2. Given $x \in k$, $|x|_v < 1$ if and only if $|x|_w < 1$
3. There exists a real number $n > 0$ such that $|x|_w = |x|_v^n$ for all $x \in k$

***Proof*:**

Throughout we use two consequences of multiplicativity: $|1| = |1 \cdot 1| = |1|^2$ with $|1| \neq 0$ gives $|1| = 1$, and then $|x||x^{-1}| = |1| = 1$ gives $|x^{-1}| = |x|^{-1}$ for $x \neq 0$.

(1) $\Rightarrow$ (2): In the metric topology of an absolute value, $x^j \to 0$ as $j \to \infty$ if and only if $|x^j| = |x|^j \to 0$ in $\mathbb{R}$, which happens if and only if $|x| < 1$. Convergence of a sequence to 0 depends only on the topology, so if the two topologies agree, then $|x|_v < 1$ if and only if $x^j \to 0$ if and only if $|x|_w < 1$.

(2) $\Rightarrow$ (3): First note that (2) also gives: $|x|_v > 1$ if and only if $|x|_w > 1$ (apply (2) to x^{-1}), and hence $|x|_v = 1$ if and only if $|x|_w = 1$. If $|x|_v = 1$ for every $x \neq 0$, then the same holds for w, and $n = 1$ works. Otherwise pick $x_0 \neq 0$ with $|x_0|_v \neq 1$. Replacing x_0 by x_0^{-1} if needed, we may assume $|x_0|_v > 1$, and then $|x_0|_w > 1$. Set

$$n := \frac{\log |x_0|_w}{\log |x_0|_v} > 0$$

Fix $x \neq 0$ and let $\lambda \in \mathbb{R}$ be defined by $|x|_v = |x_0|_v^\lambda$, that is, $\lambda = \log |x|_v / \log |x_0|_v$. For any rational $p/q > \lambda$ with $q > 0$, we have $|x|_v^q < |x_0|_v^p$, so

$$\left|\frac{x^q}{x_0^p}\right|_v < 1 \qquad \overset{(2)}{\Rightarrow} \qquad \left|\frac{x^q}{x_0^p}\right|_w < 1 \qquad \Rightarrow \qquad |x|_w < |x_0|_w^{p/q}$$

and symmetrically, for any rational $p/q < \lambda$ with $q > 0$, we get $|x|_w > |x_0|_w^{p/q}$. Since the rationals are dense in $\mathbb{R}$ and $t \mapsto |x_0|_w^t$ is continuous and increasing, letting p/q approach λ from both sides gives $|x|_w = |x_0|_w^\lambda$. Hence

$$|x|_w = |x_0|_w^\lambda = (|x_0|_v^n)^\lambda = \left(|x_0|_v^\lambda\right)^n = |x|_v^n$$

and for $x = 0$ both sides are 0, proving (3).

(3) $\Rightarrow$ (1): For every $x \in k$ and $\epsilon > 0$, the metric balls satisfy

$$\{y \in k \mid |x - y|_w < \epsilon\} = \{y \in k \mid |x - y|_v^n < \epsilon\} = \left\{y \in k \;\middle|\; |x - y|_v < \epsilon^{1/n}\right\}$$

so every w-ball is a v-ball and, using the exponent $1/n$, every v-ball is a w-ball. The two metrics therefore generate the same topology.

The following can be thought of as the Chinese Remainder Theorem for real valuations

Proposition 21.2.5: Approximation Theorem

Let $|-|_1, ..., |-|_n$ be pairwise inequivalent *nontrivial* valuations for a field K, and let $a_1, \ldots, a_n \in K$. Then for every $\epsilon > 0$ there exists an $x \in K$ such that

$$|x - a_i|_i < \epsilon \qquad \forall 1 \leq i \leq n$$

***Proof*:**
For $n = 1$ the element $x = a_1$ works, so assume $n \geq 2$ throughout.

Two facts about an absolute value are used repeatedly: $|1| = 1$ and $|x^{-1}| = |x|^{-1}$, both proved at the start of proposition 21.2.4, and the reverse triangle inequality $|u + v| \geq \big||u| - |v|\big|$, which follows from the triangle inequality applied to $u = (u + v) - v$.

Step 1: one implication already forces equivalence. Let v, w be nontrivial absolute values on K with $|x|_v < 1 \Rightarrow |x|_w < 1$ for every x. Applying this to x^{-1} gives $|x|_v > 1 \Rightarrow |x|_w > 1$. Suppose $|x|_v = 1$ while $|x|_w \neq 1$. Replacing x by x^{-1} we may assume $|x|_w > 1$. Since v is nontrivial there is a $z \neq 0$ with $|z|_v \neq 1$, and replacing z by z^{-1} we may assume $|z|_v < 1$. Then $|x^m z|_v = |z|_v < 1$ for every $m \geq 0$, so $|x^m z|_w < 1$, that is $|x|_w^m |z|_w < 1$ for every m. This is impossible, since $|x|_w > 1$ makes $|x|_w^m$ exceed $|z|_w^{-1}$ for m large. Hence $|x|_v = 1$ if and only if $|x|_w = 1$, and therefore $|x|_w < 1$ forces $|x|_v < 1$, since $|x|_v \geq 1$ would give $|x|_w \geq 1$. The second condition of proposition 21.2.4 holds, so v and w are equivalent.

Step 2: separating two of them. Let $i \neq j$. Since $|-|_i$ and $|-|_j$ are inequivalent and nontrivial, Step 1 says both implications fail, so there are $a, b \in K$ with

$$|a|_i < 1 \leq |a|_j \qquad |b|_j < 1 \leq |b|_i$$

Then $y = b/a$ satisfies $|y|_i = |b|_i/|a|_i > 1$ and $|y|_j = |b|_j/|a|_j < 1$.

Step 3: separating one from all the others. We claim there is $z \in K$ with $|z|_1 > 1$ and $|z|_j < 1$ for $j = 2, \ldots, n$, and argue by induction on n. The case $n = 2$ is Step 2. So let z satisfy $|z|_1 > 1$ and $|z|_j < 1$ for $2 \leq j \leq n - 1$, and let y be as in Step 2 for the pair $(1, n)$, so $|y|_1 > 1$ and $|y|_n < 1$.

If $|z|_n \leq 1$, take $w = z^m y$. Then $|w|_1 = |z|_1^m |y|_1 > 1$ and $|w|_n = |z|_n^m |y|_n < 1$ for every m, while $|w|_j = |z|_j^m |y|_j < 1$ for $2 \leq j \leq n - 1$ once m is large, because $|z|_j < 1$.

If $|z|_n > 1$, put $t_m = z^m/(1 + z^m)$, which is defined because $|z^m|_1 > 1 = |-1|_1$ forces $z^m \neq -1$. Then

$$|t_m - 1|_1 = \frac{1}{|1 + z^m|_1} \leq \frac{1}{|z|_1^m - 1} \qquad |t_m - 1|_n \leq \frac{1}{|z|_n^m - 1} \qquad |t_m|_j \leq \frac{|z|_j^m}{1 - |z|_j^m}$$

for $2 \leq j \leq n - 1$, using the reverse triangle inequality in each denominator. So $t_m \to 1$ with respect to $|-|_1$ and $|-|_n$, and $t_m \to 0$ with respect to $|-|_2, \ldots, |-|_{n-1}$. Taking $w = t_m y$, the three quantities $|w|_1, |w|_n, |w|_j$ tend to $|y|_1 > 1$, to $|y|_n < 1$ and to 0 respectively, so w has the required properties once m is large.

Step 4: the approximation. Permuting the indices in Step 3, choose for each i an element $z_i \in K$ with $|z_i|_i > 1$ and $|z_i|_j < 1$ for $j \neq i$, and set

$$u_{i,m} = \frac{z_i^m}{1 + z_i^m}$$

which is defined for the reason given in Step 3. The estimates of Step 3 give

$$|u_{i,m} - 1|_i \leq \frac{1}{|z_i|_i^m - 1} \qquad |u_{i,m}|_j \leq \frac{|z_i|_j^m}{1 - |z_i|_j^m} \quad (j \neq i)$$

so $u_{i,m} \to 1$ with respect to $|-|_i$ and $u_{i,m} \to 0$ with respect to every $|-|_j$ with $j \neq i$. Now put

$$x = \sum_{i=1}^{n} a_i u_{i,m}$$

For each fixed i, the triangle inequality gives

$$|x - a_i|_i \leq |a_i|_i \, |u_{i,m} - 1|_i + \sum_{j \neq i} |a_j|_i \, |u_{j,m}|_i$$

and every term on the right tends to 0 as m grows. There are finitely many indices i, so one m makes all n of these bounds smaller than ϵ at once, and that x satisfies $|x - a_i|_i < \epsilon$ for all i.

As mentioned earlier, the trivial valuation is non-Archimedean. The following show that non-Archimedean valuations can be thought of as absolute values induced by *ultra-metrics*

Definition 21.2.6: Archimedean Valuation

Let v be a real valuation on a field k. Then if for all $x \in k$, there exists a $n \in \mathbb{N}$ such that

$$|x|_v < |n|_v$$

the valuation is called an *Archimedean valuation*. On the other hand, if there exists an x such that for all $n \in \mathbb{N}$:

$$|n|_v \leq |x|_v \qquad \forall n \in \mathbb{N}$$

Then v is called a *non-Archimedean valuation*.

Proposition 21.2.7: Characterizing Non-Archimedean Valuation

Let v be a real valuation on a field k. Then the following are equivalent:

1. v is non-Archimedean
2. $|n|_v \leq 1$ for all $n \in \mathbb{N}$
3. $|n|_v \leq 1$ for all $n \in \mathbb{Z}$
4. for all $x, y \in k$,
$$|x + y|_v \leq \max(|x|_v, |y|_v)$$

The final inequality upgrading the triangle inequality is called the *strong triangle inequality* or *ultra-metric inequality*.

Note that if $|x| \neq |y|$ in an ultra-metric , then $|x + y| = \max(|x|, |y|)$!

***Proof*:**

(1) $\Rightarrow$ (2) Let v be a non-Archimedean valuation. For the sake of contradiction, suppose that for some $n \in \mathbb{N}$, $|n|_v > 1$. Then by multiplicity we may have an $|n|_v$ with

an arbitrarily large valuation:

$$|n^k| = |n|_v^k \xrightarrow{k\to\infty} \infty$$

Which would imply there is no $x \in k$ such that $|n|_v \leq |x|_v$ for all $n \in \mathbb{N}$.

(2) $\Rightarrow$ (3) show that $|-1| = |1|$ and use this to conclude.

(3) $\Rightarrow$ (4) Take:

$$\begin{aligned}
|x+y|^n &= \left|\sum_k^n \binom{n}{k} x^k y^{n-k}\right| \\
&\leq \sum_k^n \left|\binom{n}{k}\right| |x|^k |y|^{n-k} \\
&\leq \sum_k^n |x|^k |y|^{n-k} \\
&\leq (n+1)\max(|x|^n, |y|^n)
\end{aligned}$$

Hence

$$|x+y| \leq (n+1)^{1/n} \max(|x|, |y|)$$

As this is true for all n, and $(n+1)^{1/n} \xrightarrow{n\to\infty} 1$, we get the desired result

(4) $\Rightarrow$ (1) by induction we get:

$$|n| = |1+1+\cdots+1| \leq |1| = 1$$

giving the desired result.

Corollary 21.2.8: Values On different characteristic

Let k be a field with characteristic $p \neq 0$. Then there does not exist an Archimedean valuation on k.

If k is a finite field, the only valuation is the trivial valuation.

***Proof*:**

First note that for any valuation v and any root of unity $u \in k$, say $u^d = 1$, multiplicativity gives $|u|_v^d = |u^d|_v = |1|_v$, and $|1|_v = |1 \cdot 1|_v = |1|_v^2$ with $|1|_v \neq 0$ forces $|1|_v = 1$. Since $|u|_v$ is a positive real number with $|u|_v^d = 1$, we get $|u|_v = 1$.

Now let k have characteristic $p \neq 0$ and let v be any valuation on k. The elements $n \cdot 1_k$ for $n \in \mathbb{N}$ all lie in the prime subfield $\mathbb{F}_p \subseteq k$. Every nonzero element u of $\mathbb{F}_p$ satisfies $u^{p-1} = 1$, since the multiplicative group $\mathbb{F}_p^\times$ has order $p-1$ and the order of u divides it by Lagrange's theorem (theorem 3.3.1). Hence $|n|_v = 1$ whenever $n \cdot 1_k \neq 0$, and $|n|_v = 0$ otherwise, so $|n|_v \leq 1 = |1|_v$ for all $n \in \mathbb{N}$. Thus $x = 1$ witnesses that v is non-Archimedean, and no Archimedean valuation exists on k.

For the second claim, let k be a finite field with q elements and v a valuation on k. The group

$k^\times$ has order $q-1$, so every $u \neq 0$ satisfies $u^{q-1} = 1$ by Lagrange's theorem (theorem 3.3.1), and by the first paragraph $|u|_v = 1$. Hence v is the trivial valuation.

In the case of characteristic 0, given $k = \mathbb{Q}$, all the absolute values may be classified. Define p-valuation to be:

$$v_p(m/n) = \begin{cases} p^{-k} & \frac{m}{n} = p^k \frac{m'}{n'} \neq 0,\ p \nmid m', n' \\ 0 & \frac{m}{n} = 0 \end{cases}$$

Theorem 21.2.9: Classification Of Absolute Values On Rationals

Every non-trivial absolute value on $\mathbb{Q}$ is equivalent, in the sense of proposition 21.2.4, to v_p for exactly one prime p, or to the usual absolute value $|-|_\infty$.

This result is due to Ostrowski.

***Proof*:**

Write $|-|$ for the absolute value. Suppose first that it is non-Archimedean, so $|n| \leq 1$ for all $n \in \mathbb{Z}$ (proposition 21.2.7). If $|n| = 1$ for every nonzero $n \in \mathbb{Z}$, then $|m/n| = |m|/|n| = 1$ for all nonzero m/n, and $|-|$ is trivial. So $P = \{n \in \mathbb{Z} \mid |n| < 1\}$ contains a nonzero element. It is an ideal, closed under sums by the strong triangle inequality and under multiples since $|an| \leq |n|$. It is proper since $|1| = 1$, and prime since $|ab| < 1$ with $|a|, |b| \leq 1$ forces $|a| < 1$ or $|b| < 1$. As $\mathbb{Z}$ is a Principal Ideal Domain, $P = (p)$ for a prime p. Let $c = |p| < 1$. If $m/n = p^k m'/n'$ with $p \nmid m', n'$, then $m', n' \notin P$, so $|m'| = |n'| = 1$ and $|m/n| = c^k$. With $s = -\log c/\log p > 0$, this is $v_p(m/n)^s$, so $|-|$ is equivalent to v_p (proposition 21.2.4). The prime is unique, since $|p|_{v_p} < 1$ while $|p|_{v_q} = 1$ for a prime $q \neq p$.

Suppose now that $|-|$ is Archimedean. Let $m, n \geq 2$ be integers and write $m = a_0 + a_1 n + \cdots + a_r n^r$ with $0 \leq a_i < n$ and $r \leq \log m/\log n$. The triangle inequality and $|1| = 1$ give $|a_i| \leq a_i < n$, so

$$|m| \leq (r+1)\, n \max(1, |n|)^r \leq \left(1 + \frac{\log m}{\log n}\right) n \max(1, |n|)^{\log m/\log n}$$

Applying this to m^t and taking t-th roots,

$$|m| \leq \left(1 + t\frac{\log m}{\log n}\right)^{1/t} n^{1/t} \max(1, |n|)^{\log m/\log n}$$

for every $t \geq 1$, and letting $t \to \infty$ gives $|m| \leq \max(1, |n|)^{\log m/\log n}$. Since $|-|$ is Archimedean, some integer $N \geq 2$ has $|N| > 1$ (proposition 21.2.7). If some $n \geq 2$ had $|n| \leq 1$, the inequality with $m = N$ would give $|N| \leq 1$. So $|n| > 1$ for every $n \geq 2$, and the inequality reads $|m|^{1/\log m} \leq |n|^{1/\log n}$. Exchanging m and n gives equality, so $|n| = n^s$ for all $n \geq 2$, with $s = \log|2|/\log 2 > 0$. Since $|-1|^2 = |1| = 1$, $|-1| = 1$, and multiplicativity gives $|x| = |x|_\infty^s$ for every $x \in \mathbb{Q}$. So $|-|$ is equivalent to $|-|_\infty$ (proposition 21.2.4).

In *Commutative Algebra* [16] (valuation rings and DVRs), many of these concepts shall be generalized to more general valuations, namely we have been working with *real* valuations, however there are more generally G-valuations for totally ordered groups G.

In an analysis class, the absolute value is used to define the completion of a ring. A sequence (x_n) in a field k with an absolute value $|-|$ is *Cauchy* if for every real $\epsilon > 0$ there is an N with $|x_n - x_m| < \epsilon$

for all $n, m \geq N$, and k is *complete* if every Cauchy sequence converges in k. The construction of $\mathbb{R}$ from $\mathbb{Q}$ carries over: the Cauchy sequences of k modulo the sequences converging to 0 form a complete field $\widehat{k}$, the *completion* of k. It contains k, as the classes of constant sequences, as a dense subfield, and it carries the absolute value $|(x_n)| = \lim_n |x_n|$ extending that of k (see [49, Chapter II, §4]).

21.2.1 Ostrowski Theorem

Theorem 21.2.10: Ostrowski's Theorem For Complete Fields

Let K be a field complete with respect to an Archimedean absolute value $|-|$. Then there is an isomorphism σ from K onto $\mathbb{R}$ or $\mathbb{C}$ and some $s \in (0, 1]$ with $|a| = |\sigma(a)|^s$ for all $a \in K$, where $\mathbb{R}$ and $\mathbb{C}$ carry their usual absolute values.

The proof reduces to the case where the absolute value restricts to the usual one on $\mathbb{Q}$ (theorem 21.2.9) and is given in [49, Chapter II, Theorem 4.2].

21.3 Derivation on Field

Derivations are the algebraic form of differentiation, and the module of relative differential forms packages every derivation of a ring into a single universal one. The construction is pure commutative algebra and belongs here rather than in any geometry book: it is used by [23] for the canonical class of a curve, by [9] for the sheaf of differentials, and by [8] for invariant differentials on a group variety.

Definition 21.3.1: A-derivation

Let A be a commutative ring, B an A-algebra, and M a B-module. Then an A-derivation of B into M is a map $d : B \to M$ such that

1. $d(b + b') = d(b) + d(b')$
2. $d(bb') = d(b)b' + bd(b')$
3. $d(a) = 0$ for all $a \in A$

Definition 21.3.2: Module of Relative Differential Forms

Let A be a commutative ring, B an A-algebra. Then a module of relative differential form is a B-module $\Omega_{B/A}$ with an A-derivation $d : B \to \Omega_{B/A}$ which satisfies the universal property of relative differential forms: if M is another B-module and $d' : B \to M$ an A-derivation, then there is a unique B-module homomorphism $f : \Omega_{B/A} \to M$ such that the following diagram commutes:

$$\begin{array}{ccc} B & \xrightarrow{d} & \Omega_{B/A} \\ & \searrow^{d'} & \downarrow f \\ & & M \end{array}$$

Certainly $\Omega_{B/A}$ exists via the usual construction: take the free B-module F generated by the primitives $d(b)$ for $b \in B$ and quotient by the submodule generated by the relationships:

$$d\,(b + b') - d(b) - d\,(b') \qquad \text{and} \qquad d\,(bb') - d(b)b' - bd\,(b') \qquad \text{and} \qquad d(a)$$

Then the map $d : B \to \Omega_{B/A}$ is given by $b \mapsto d(b)$. Then it is an exercise in algebra to see that this satisfies the universal property and $\big(\Omega_{B/A}, d\big)$ is unique up to unique isomorphism. The following gives an alternative insightful construction:

Proposition 21.3.3: Natural Differential Structure

Let B be an A-algebra, $\delta : B \otimes_A B \to B$ the diagonal map given by $f(b \otimes b') = bb'$, and let $I = \ker \delta$. Then I/I^2 has the structure of a left B-module, and we can define a map

$$d : B \to I/I^2 \qquad d(b) = 1 \otimes b - b \otimes 1 \bmod I^2$$

In which case $(I/I^2, d)$ is a module of relative differential for B/A.

***Proof*:**

Write $\delta\colon B \otimes_A B \to B$ for the multiplication map and $I = \ker \delta$. Give I/I^2 the B-module structure coming from the left factor, $b \cdot (x \otimes y) = (bx) \otimes y$. The two possible structures agree modulo I^2, since $b \otimes 1 - 1 \otimes b \in I$.

d is an A-derivation. Additivity is clear. For the Leibniz rule, compute in $B \otimes_A B$:

$$d(bb') - b\,d(b') - b'\,d(b) = \big(1 \otimes bb' - bb' \otimes 1\big) - b\big(1 \otimes b' - b' \otimes 1\big) - b'\big(1 \otimes b - b \otimes 1\big)$$

which expands to $-(1 \otimes b - b \otimes 1)(1 \otimes b' - b' \otimes 1)$, a product of two elements of I and so zero in I/I^2. And $d(a) = 1 \otimes a - a \otimes 1 = 0$ for $a \in A$, the tensor being over A.

Universality. Let $d'\colon B \to M$ be an A-derivation. Define $B \otimes_A B \to M$ by $x \otimes y \mapsto x\,d'(y)$, which is A-bilinear and hence well defined. It kills I^2: for $u = \sum x_i \otimes y_i$ and $v = \sum x'_j \otimes y'_j$ in I, meaning $\sum x_i y_i = 0$ and $\sum x'_j y'_j = 0$, expanding uv and applying the Leibniz rule for d' leaves $\big(\sum x_i y_i\big)\big(\sum x'_j d'(y'_j)\big) = 0$. So it descends to $f\colon I/I^2 \to M$, and $f(d(b)) = f(1 \otimes b - b \otimes 1) = d'(b) - b\,d'(1) = d'(b)$ since $d'(1) = 0$. Uniqueness holds because the $d(b)$ generate I/I^2: any $\sum x_i \otimes y_i \in I$ equals $\sum x_i(1 \otimes y_i - y_i \otimes 1)$ modulo nothing, using $\sum x_i y_i = 0$.

So $(I/I^2, d)$ has the universal property of definition 21.3.2, completing the proof.

Example 21.3.4: Relative Differential

1. Take $A = \mathbb{R}$ and $B = \mathbb{R}[x]$. Then $I = \ker \delta$. Then as $\mathbb{R}[x]$ is an integral domain, it can be shown that any element is a linear combination of elements of the form:

   $$f(x) \otimes g(x) - f(x)g(x) \otimes 1 \in I$$

   Then it should be verified any element in I can be written as:

   $$f(x)(1 \otimes g(x) - g(x) \otimes 1)$$

   The reader should then verify that I/I^2 is generated by $f'(x)(x \otimes 1 - 1 \otimes x)$[a]. Let dx represent this element. Notice that this is the image of $d(x)$. Now, let's consider:

   $$d(f(x) = d\left(\sum_i a_i x^i\right) = \sum_i a_i d(x^i)$$

   Hence, let's compute $d(x^i)$. For $i = 2$, we get:

   $$d(x^2) = d(x \cdot x) = xd(x) - xd(x) = 2xdx$$

   Repeating this pattern, we can see that:

   $$d(x^n) = nx^{n-1}dx$$

2. For example, taking $B = \mathbb{R}[x_1, ..., x_n]$, Then $\Omega_{B/A}$ is a free B-module generated by $dx_1, ..., dx_n$.

3. In proposition 21.3.10, we shall see that if E/F is a finitely generated separable extension (of characteristic 0), then

   $$\Omega_{B/A} = \bigoplus_{b \in T} Bdb$$

   where T is a transcendental A-basis for B.

4. What if we take $B = A[\epsilon]/(\epsilon^2)$?

5. Let $A = \mathbb{R}$ and $B = C^\infty(M)$ for some smooth n-manifold M. Then $C^\infty(M)$. Then $\Omega_{C^\infty(M)/\mathbb{R}}$ is a locally free $C^\infty(M)$-module (based off of the patches of M), consisting of the space of sections of the cotangent bundle T^*M.

[a] Hint: what is $(1 \otimes x^2 - x^2 \otimes 1)$. Repeat for higher powers

Proposition 21.3.5: Base Change For Differentials

Let A', B be A-algebras and $B' = B \otimes_A A'$. Then:

$$\Omega_{B'/A'} \cong \Omega_{B/A} \otimes_B B'$$

Furthermore, if S is a multiplicative subset of of S, then:

$$\Omega_{S^{-1}B/A} \cong S^{-1}\Omega_{B/A}$$

***Proof*:**
Both statements are checked against the universal property of definition 21.3.2.

For the base change, a B'-module M receives an A'-derivation from B' exactly when it receives an A-derivation from B that is A'-linear on the second factor, since $B' = B \otimes_A A'$ is generated over A' by the image of B. So $\mathrm{Der}_{A'}(B', M) \cong \mathrm{Der}_A(B, M)$ naturally, and representing both sides gives $\Omega_{B'/A'} \cong \Omega_{B/A} \otimes_B B'$.

For the localization, an A-derivation $d\colon B \to M$ into an $S^{-1}B$-module extends uniquely to $S^{-1}B$ by the quotient rule $d(b/s) = \big(s\,d(b) - b\,d(s)\big)/s^2$, which is forced by the Leibniz rule applied to $b = (b/s) \cdot s$. So $\mathrm{Der}_A(S^{-1}B, M) \cong \mathrm{Der}_A(B, M)$ for such M, and representing gives $\Omega_{S^{-1}B/A} \cong S^{-1}\Omega_{B/A}$, completing the proof.

Proposition 21.3.6: Differentials and Exact Sequence I

Let $A \to B \to C$ be a (not necessarily exact nor a complex) sequence of rings. Then there is an exact sequence of C-modules:

$$\Omega_{B/A} \otimes_B C \to \Omega_{C/A} \to \Omega_{C/B} \to 0$$

***Proof*:**
The maps are the natural ones: $db \otimes c \mapsto c\,db$ and $dc \mapsto dc$, the latter making sense because an A-derivation of C is in particular one killing the image of A.

Exactness is checked by applying $\mathrm{Hom}_C(-, M)$ for an arbitrary C-module M and using definition 21.3.2, which turns the sequence into

$$0 \to \mathrm{Der}_B(C, M) \to \mathrm{Der}_A(C, M) \to \mathrm{Der}_A(B, M)$$

This is exact: a derivation of C over A kills the image of B exactly when it is a derivation over B, which is exactness at the middle term and injectivity on the left. Since $\mathrm{Hom}_C(-, M)$ is left exact and the resulting sequence is exact for every M, the original sequence is exact at $\Omega_{C/A}$ and at $\Omega_{C/B}$, completing the proof.

Proposition 21.3.7: Differentials and Exact Sequence II

Let B be an A-algebra, I be an ideal of B, and $C = B/I$. Then there is a natural exact sequence of C-modules

$$I/I^2 \xrightarrow{\delta} \Omega_{B/A} \otimes_B C \to \Omega_{C/A} \to 0,$$

where for any $b \in I$, if $\bar{b}$ is its image in I/I^2, then $\delta\bar{b} = db \otimes 1$. In particular, I/I^2 has a natural structure of a C-module, and δ is a C-linear map, even though it is defined via the derivation d.

***Proof*:**
First, δ is well defined and C-linear. For $b, b' \in I$ the Leibniz rule gives $d(bb') = b\,db' + b'\,db$, which lies in $I\,\Omega_{B/A}$ and so vanishes in $\Omega_{B/A} \otimes_B C$. Hence δ kills I^2. C-linearity follows because multiplying b by an element of I again lands in I^2.

For exactness, apply $\mathrm{Hom}_C(-, M)$ as in proposition 21.3.6. The sequence becomes

$$0 \to \mathrm{Der}_A(C, M) \to \mathrm{Der}_A(B, M) \to \mathrm{Hom}_C(I/I^2, M)$$

An A-derivation of B into M, where M is a C-module and so killed by I, descends to $C = B/I$ exactly when it vanishes on I. And its restriction to I is precisely the composite with δ. That is exactness at the middle and injectivity on the left, and as before it holds for every M, giving the stated sequence, completing the proof.

Corollary 21.3.8: Finite Generation of Differential Forms

If B is a finitely generated A-algebra or a localization of a finitely generated A-algebra, then $\Omega_{B/A}$ is a finitely generated B-module

***Proof*:**
As B is a finitely generated A-algebra, it is the quotient of a polynomial ring. Hence, it follows from our direct computations from example 21.3.4 and, proposition 21.3.7, and the localization result comes from proposition 21.3.5.

For finite extensions the module of differentials sees exactly one thing, namely separability, and it sees it sharply: a nonzero derivation exists precisely when some element of the extension fails to be separable. The mechanism is already at work on a simple extension $k(\alpha)$, where a k-derivation is pinned down by its value on α and that value is constrained only through the single scalar $f'(\alpha)$.

Lemma 21.3.9: Derivations Detect Separability

Let F/k be a finite field extension. Then the following are equivalent:

1. F/k is a separable extension
2. $\Omega_{F/k} = 0$
3. the only k-derivation of F into F is the zero map

***Proof*:**

(1 ⇒ 2) Let M be any F-module and $D : F \to M$ a k-derivation. Since F/k is finite and separable, theorem 19.6.20 gives $F = k(\beta)$ for some $\beta \in F$, and β is separable over k by definition 19.6.3. Write $g(x) = m_{\beta,k}(x) = \sum_j c_j x^j$. Additivity and the Leibniz rule give $D(\beta^j) = j\beta^{j-1}D(\beta)$, and D kills each $c_j \in k$, so

$$0 = D\big(g(\beta)\big) = \sum_j c_j D(\beta^j) = g'(\beta)D(\beta)$$

Now β is a root of the separable polynomial g, hence not a multiple root, so $g'(\beta) \neq 0$ by proposition 19.6.6. As $g'(\beta)$ is a unit in the field F, this forces $D(\beta) = 0$. Every element of F is a polynomial in β with coefficients in k by proposition 19.2.24, so additivity and the Leibniz rule propagate $D(\beta) = 0$ to $D = 0$. Thus $\mathrm{Der}_k(F, M) = 0$ for every F-module M.

Apply this to $M = \Omega_{F/k}$: the universal derivation d of definition 21.3.2 is itself zero. Both the identity and the zero map $\Omega_{F/k} \to \Omega_{F/k}$ are F-module homomorphisms f with $f \circ d = d$, so the uniqueness clause of definition 21.3.2 forces them to agree, giving $\Omega_{F/k} = 0$.

(2 ⇒ 3) By definition 21.3.2 the k-derivations of F into F are in bijection with the F-module homomorphisms $\Omega_{F/k} \to F$, and the zero module admits only the zero homomorphism.

(3 ⇒ 1) Take the contrapositive: assume F/k is inseparable and produce a nonzero k-derivation of F into F. Since F/k is finite, write $F = k(\alpha_1, ..., \alpha_n)$ and set $F_0 = k$ and $F_i = F_{i-1}(\alpha_i)$, so that $F_n = F$.

First, some α_i is inseparable over F_{i-1}. If instead every α_i were separable over F_{i-1}, then $[F_i : F_{i-1}]_s = [F_i : F_{i-1}]$ for each i by proposition 19.6.15, and multiplying these together using corollary 19.6.16 on the left and theorem 19.3.7 on the right would give $[F : k]_s = [F : k]$, whence F/k is separable by proposition 19.6.17, contrary to assumption. So let i be the *largest* index for which α_i is inseparable over F_{i-1}, and abbreviate

$$E = F_{i-1} \qquad \alpha = \alpha_i \qquad L = F_i = E(\alpha)$$

By maximality of i, each α_j with $j > i$ is separable over F_{j-1}. Running the same product over the tower $L = F_i \subseteq \cdots \subseteq F_n = F$ gives $[F : L]_s = [F : L]$, so F/L is a finite separable extension, again by proposition 19.6.17.

A nonzero E-derivation of L. Let $f(x) = m_{\alpha,E}(x)$. It is irreducible, and it is inseparable because α is inseparable over E (definition 19.6.3), so proposition 19.6.7 gives $f'(x) = 0$. Reading off the leading term of f', this already forces $\mathrm{ch}(k) = p > 0$ and $f \in E[x^p]$, exactly as in the discussion preceding definition 19.6.9. In characteristic 0 there is nothing to prove, since corollary 19.6.8 rules out inseparable irreducibles outright. Regard L as an $E[x]$-module through the evaluation map $\mathrm{ev}_\alpha : E[x] \to L$ of theorem 19.2.8, whose kernel is (f), and consider

$$\partial : E[x] \to L \qquad \partial(h) = h'(\alpha)$$

with h' the formal derivative of definition 19.6.5. It is additive, it kills E, and the Leibniz rule for the formal derivative gives

$$\begin{aligned}\partial(h_1h_2) = (h_1h_2)'(\alpha) &= h_1'(\alpha)h_2(\alpha) + h_1(\alpha)h_2'(\alpha)\\ &= \partial(h_1)h_2(\alpha) + h_1(\alpha)\partial(h_2)\end{aligned}$$

so ∂ is an E-derivation in the sense of definition 21.3.1. It kills the ideal (f): for any $h \in E[x]$,

$$\partial(fh) = (fh)'(\alpha) = f'(\alpha)h(\alpha) + f(\alpha)h'(\alpha) = 0$$

because $f' = 0$ and $f(\alpha) = 0$. Hence ∂ factors through ev_α as an additive map $D : L \to L$, and the displayed Leibniz identity descends, since both sides depend only on the values $h_1(\alpha), h_2(\alpha)$. So D is an E-derivation of L into L with

$$D(\alpha) = \partial(x) = 1$$

and in particular $D \neq 0$.

Extending D across F/L. Since F/L is finite and separable, theorem 19.6.20 gives $F = L(\beta)$ with β separable over L. Put $g(x) = m_{\beta,L}(x)$. As in $(1 \Rightarrow 2)$, $g'(\beta) \neq 0$ by proposition 19.6.6. For $h(x) = \sum_j c_j x^j \in L[x]$ write $h^D(x) = \sum_j D(c_j)x^j$. Expanding a product coefficientwise and applying the Leibniz rule for D to each term gives

$$(h_1h_2)^D = h_1^D h_2 + h_1 h_2^D$$

Set

$$t = -\frac{g^D(\beta)}{g'(\beta)} \in F$$

and regard F as an $L[x]$-module through evaluation at β. Define

$$\Delta : L[x] \to F \qquad \Delta(h) = h^D(\beta) + h'(\beta)\, t$$

Then Δ is additive, and combining the two product rules just recorded,

$$\begin{aligned}\Delta(h_1h_2) &= h_1^D(\beta)h_2(\beta) + h_1(\beta)h_2^D(\beta) + \big(h_1'(\beta)h_2(\beta) + h_1(\beta)h_2'(\beta)\big)t\\ &= \Delta(h_1)h_2(\beta) + h_1(\beta)\Delta(h_2)\end{aligned}$$

On a constant $c \in L$ we get $\Delta(c) = D(c)$, so Δ vanishes on E and a fortiori on k. By the choice of t,

$$\Delta(g) = g^D(\beta) + g'(\beta)t = 0$$

and therefore $\Delta(gh) = \Delta(g)h(\beta) + g(\beta)\Delta(h) = 0$ for every $h \in L[x]$, since $g(\beta) = 0$. So Δ kills the kernel (g) of evaluation at β and descends, by theorem 19.2.8, to an additive map $\widetilde{D} : F \to F$. As before the Leibniz identity descends with it. Thus $\widetilde{D}$ is a k-derivation of F into F restricting to D on L, and

$$\widetilde{D}(\alpha) = D(\alpha) = 1 \neq 0$$

So F carries a nonzero k-derivation into itself, which is the contrapositive of $(3 \Rightarrow 1)$, completing the proof.

Note that the primitive element theorem is used only where it is legitimate, on the separable extension F/L. It says nothing about F/k itself, which by example 19.6.22 need not be simple once separability fails. The whole content of $(3 \Rightarrow 1)$ is that an inseparable step in a tower is a step at which a derivation may be defined freely, and the remaining steps are separable and so carry the derivation along uniquely.

Proposition 21.3.10: Differentials and Field Extensions

Let K/k be a finitely generated field extension. Then:

$$\dim_K \Omega_{K/k} \geq \mathrm{trdeg} K/k$$

where equality holds iff K is separably generated over k.

***Proof Key ideas*:**
This is the separability criterion for field extensions in differential form, and both implications reduce to the exact sequences above: for a separably generated extension a transcendence basis gives a basis of Ω, and conversely a nonzero differential of a purely inseparable generator is killed by the p-th-power relation. A full account is in [35, chapter II.8].

Both *Classical Algebraic Geometry* [23] and *Algebraic Geometry* [9] use it to detect smoothness.

The finite algebraic case is the one this book uses most often, and it is not left to a citation: lemma 21.3.9 proves that a finite extension K/k has $\Omega_{K/k} = 0$ exactly when it is separable. Note that the transcendence-degree statement above genuinely needs the finitely generated hypothesis, whereas the finite case does not.

Proposition 21.3.11: Local Ring and Relative Differential

Let B be a local ring containing a field k isomorphic to $B/\mathfrak{m}$. Then given the map δ from proposition 21.3.7:

$$\delta : \mathfrak{m}/\mathfrak{m}^2 \xrightarrow{\cong} \Omega_{B/k} \otimes_B k$$

***Proof*:**
By proposition 21.3.7 $\mathrm{coker}(\delta) = 0$, so δ is surjective. To show that δ is injective, we shall show the following dual vector space map is surjective:

$$\delta' : \mathrm{Hom}_k(\Omega_{B/k} \otimes k, k) \to \mathrm{Hom}_k(\mathfrak{m}/\mathfrak{m}^2, k)$$

The term on the left is isomorphic to $\mathrm{Hom}_B(\Omega_{B/k}, k)$, which by definition of the differentials, can be identified with the set $\mathrm{Der}_k(B, k)$ of k-derivations of B to k. If $d : B \to k$ is a derivation, then $\delta'(d)$ is obtained by restricting to $\mathfrak{m}$, and noting that $d(\mathfrak{m}^2) = 0$. Now, to show that δ' is surjective, let $h \in \mathrm{Hom}(\mathfrak{m}/\mathfrak{m}^2, k)$. For any $b \in B$, we can write $b = \lambda + c$, $\lambda \in k$, $c \in \mathfrak{m}$, in a unique way. Define $db = h(\bar{c})$, where $\bar{c} \in \mathfrak{m}/\mathfrak{m}^2$ is the image of c. Then one verifies immediately that d is a k-derivation of B to k, and that $\delta'(d) = h$. Thus δ' is surjective, as required.

We finish off this brief summary of the essential results about relative differentials by showing the relation between regular local rings and relative differentials. We require the following lemma:

Lemma 21.3.12: Characterizing Free Modules using Residue and Quotient Field

Let A be a Noetherian local integral domain, with residue field k and quotient field K. Then if M is a finitely generated A-module and $\dim_k M \otimes_A k = \dim_k M \otimes_A K = r$, then M is free of rank r.

***Proof*:**
Since $\dim_k M \otimes k = r$, by Nakayama's lemma M can be generated by r elements. Hence, there is a surjective map $\varphi : A^r \to M$. Let R be its kernel. Then we obtain an exact sequence

$$0 \to R \otimes K \to K^r \to M \otimes K \to 0,$$

As $\dim_K M \otimes K = r$, we have $R \otimes K = 0$. But R is torsion-free, so $R = 0$, and M is isomorphic to A^r, completing the proof.

22⋆

Bilinear, Quadratic and Alternating Forms over a Field

⋆ *Optional. This chapter is an aside; nothing later in the book depends on it.*

A bilinear form is a piece of data attached to a vector space, and section 13.3 said what that data is: a matrix, once a basis is fixed. It did not say when two such data are the same. Two forms should count as the same when some change of basis carries one to the other, and the matrix statement of that relation is $A \mapsto P^t AP$ rather than the $A \mapsto P^{-1}AP$ that governs linear operators. The two relations are genuinely different, and the invariants of the first are not the eigenvalues.

This chapter classifies bilinear forms over a field up to that relation, in the two cases that occur: the symmetric forms, where the answer is subtle and field-dependent, and the alternating ones, where it is not. The answer is controlled by a single group built from the field, the square classes $k^\times/(k^\times)^2$, and this is why the subject sits after Galois theory: for a field of characteristic other than 2 that group is exactly what Kummer theory at $n = 2$ (section 20.4) classifies. Over $\mathbb{C}$ the group is trivial and the classification collapses to the rank; over $\mathbb{R}$ it has two elements and one further invariant appears; over a general field it can be arbitrarily complicated, and the bookkeeping of what survives is the Witt group.

22.1 Quadratic Forms and Symmetric Bilinear Forms

The forms of interest are the symmetric ones, and over a field where 2 is invertible a symmetric bilinear form carries exactly the same information as the single-variable function $x \mapsto \varphi(x, x)$. That function is easier to write down and is what one meets in practice, so the correspondence between the two is recorded first. It fails in characteristic 2, and the failure is not a technicality to be waved past: it is why the whole theory below carries the hypothesis $\operatorname{char} k \neq 2$.

Definition 22.1.1: Quadratic Form over a Field

Let k be a field and V a finite-dimensional k-vector space. A *quadratic form* on V is a function $q\colon V \to k$ such that

1. $q(\lambda v) = \lambda^2 q(v)$ for all $\lambda \in k$ and $v \in V$;
2. the function $b_q\colon V \times V \to k$ defined by

$$b_q(x,y) := q(x+y) - q(x) - q(y)$$

is bilinear.

The form b_q is the *polar form* of q. It is symmetric, since the defining expression is unchanged by exchanging x and y.

Two consequences of the definition are used constantly and are worth extracting at once. Setting $y = x$ in the polar form and applying (1) gives

$$b_q(x,x) = q(2x) - 2q(x) = 4q(x) - 2q(x) = 2q(x)$$

so the polar form recovers q after division by 2, and only then. And if φ is any symmetric bilinear form, the function $q_\varphi(x) := \varphi(x,x)$ satisfies both conditions above, with $b_{q_\varphi} = 2\varphi$ by the same computation. The two constructions are therefore mutually inverse exactly when 2 is invertible.

Proposition 22.1.2: Quadratic Forms and Symmetric Bilinear Forms Correspond

Let k be a field with $\operatorname{char} k \neq 2$ and let V be a finite-dimensional k-vector space. The assignments

$$\varphi \longmapsto q_\varphi, \qquad q \longmapsto \tfrac{1}{2} b_q$$

are mutually inverse bijections between the set of symmetric bilinear forms on V and the set of quadratic forms on V.

***Proof*:**

Step 1: the assignments are well defined. Let φ be symmetric bilinear and put $q_\varphi(x) = \varphi(x,x)$. Then $q_\varphi(\lambda v) = \varphi(\lambda v, \lambda v) = \lambda^2 \varphi(v,v) = \lambda^2 q_\varphi(v)$, which is condition (1). For condition (2), expanding by bilinearity and using symmetry,

$$b_{q_\varphi}(x,y) = \varphi(x+y, x+y) - \varphi(x,x) - \varphi(y,y) = 2\varphi(x,y)$$

which is bilinear. So q_φ is a quadratic form and $b_{q_\varphi} = 2\varphi$. Conversely, if q is a quadratic form then b_q is symmetric bilinear by definition 22.1.1, hence so is $\frac{1}{2} b_q$, using $\operatorname{char} k \neq 2$ to divide.

Step 2: the composites are the identity. Starting from a symmetric φ, Step 1 gives $\frac{1}{2} b_{q_\varphi} = \frac{1}{2} \cdot 2\varphi = \varphi$. Starting from a quadratic form q and writing $\varphi = \frac{1}{2} b_q$, the computation recorded before the proposition gives $b_q(x,x) = 2q(x)$, so

$$q_\varphi(x) = \varphi(x,x) = \tfrac{1}{2} b_q(x,x) = \tfrac{1}{2} \cdot 2q(x) = q(x)$$

Both composites are the identity, so the assignments are mutually inverse bijections, as we sought to show.

Because of the proposition, the words *quadratic form* and *symmetric bilinear form* are used interchangeably below whenever $\operatorname{char} k \neq 2$, and disc, orthogonality and the matrix of a form are attached to either at will. The next example shows what is lost when the hypothesis is dropped, and it is the reason the hypothesis is carried explicitly rather than assumed silently.

Example 22.1.3: Polarization Fails in Characteristic Two

Let $k = \mathbb{F}_2$ and $V = k^2$ with coordinates (x, y).

The map $q \mapsto b_q$ is not injective. Take $q(x, y) = x^2$. Then $q(\lambda v) = \lambda^2 q(v)$ holds, and

$$b_q\big((x, y), (x', y')\big) = (x + x')^2 - x^2 - x'^2 = 2xx' = 0$$

since $2 = 0$ in k. So q and the zero form have the same polar form, yet $q(1, 0) = 1 \neq 0$. More generally $b_q(v, v) = 2q(v) = 0$ for every quadratic form in characteristic 2, so every polar form is alternating and the diagonal information in q is invisible to b_q. Throughout this example *alternating* means the vanishing condition of definition 16.1.3, that $\varphi(v, v) = 0$ for every v. In characteristic 2 that is strictly stronger than the sign rule appearing in definition 13.3.15, since there $-1 = 1$ makes the sign rule say nothing beyond symmetry.

The map $q \mapsto b_q$ is not surjective either. The form $\varphi\big((x, y), (x', y')\big) = xx'$ is symmetric bilinear, but $\varphi\big((1, 0), (1, 0)\big) = 1 \neq 0$, so φ does not vanish on the diagonal and is therefore not the polar form of anything.

Orthogonal bases can fail to exist. Let $\psi\big((x, y), (x', y')\big) = xy' + x'y$, with matrix $\left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$ in the standard basis. It is symmetric, nondegenerate, and nonzero, since $\psi(e_1, e_2) = 1$. But $\psi(v, v) = 2xy = 0$ for every v. Were w_1, w_2 an orthogonal basis, the matrix of ψ in it would be $\operatorname{diag}\big(\psi(w_1, w_1), \psi(w_2, w_2)\big) = \operatorname{diag}(0, 0)$, forcing $\psi = 0$ and contradicting $\psi(e_1, e_2) = 1$. So ψ admits no orthogonal basis at all. Contrast $\varphi(u, v) = xx'$ above, which is already diagonal in the standard basis: failing to be a polar form and failing to be diagonalizable are different failures, and ψ exhibits the second.

All three failures trace to the division by 2 in proposition 22.1.2. In characteristic 2 the quadratic and the bilinear theories separate, and the correct objects are quadratic forms together with their alternating polar forms, and that theory is not developed here.

The classification question can now be posed precisely. Fixing a basis turns a form into a matrix (definition 13.3.18), and changing the basis changes the matrix in a way that must be pinned down before any invariant can be extracted.

Theorem 22.1.4: Change of Basis for the Matrix of a Form

Let V be a finite-dimensional k-vector space with bases $\beta = (\beta_1, \dots, \beta_n)$ and $\beta' = (\beta'_1, \dots, \beta'_n)$, and let $\varphi\colon V \times V \to k$ be a bilinear form. Let P be the $n \times n$ matrix whose j-th column is $[\beta'_j]_\beta$, the coordinate vector of β'_j in the basis β (this is the change of basis matrix of definition 13.2.11). Then P is invertible, $[v]_\beta = P[v]_{\beta'}$ for every $v \in V$, and

$$[\varphi]_{\beta'} = P^t\,[\varphi]_\beta\,P$$

***Proof*:**
The columns of P are the coordinate vectors of a basis, hence linearly independent, so P is invertible. Writing $v = \sum_j c_j \beta'_j$, that is $[v]_{\beta'} = (c_j)_j$, linearity of the coordinate map gives

$$[v]_\beta = \Big[\sum_j c_j \beta'_j\Big]_\beta = \sum_j c_j [\beta'_j]_\beta = P[v]_{\beta'}$$

which is the second claim. For the third, write $A = [\varphi]_\beta$. By lemma 13.3.19, for all $v, w \in V$

$$\varphi(v,w) = [v]_\beta^t A [w]_\beta = \big(P[v]_{\beta'}\big)^t A \big(P[w]_{\beta'}\big) = [v]_{\beta'}^t \big(P^t A P\big)[w]_{\beta'}$$

Taking $v = \beta'_i$ and $w = \beta'_j$ makes the coordinate vectors the standard basis vectors e_i and e_j, so the displayed expression is the (i,j) entry of $P^t A P$. On the other hand that value is $\varphi(\beta'_i, \beta'_j)$, which is the (i,j) entry of $[\varphi]_{\beta'}$ by definition 13.3.18. The two matrices agree entrywise, completing the proof.

Definition 22.1.5: Congruent Matrices

Two matrices $A, B \in M_n(k)$ are *congruent* if $B = P^t A P$ for some invertible $P \in M_n(k)$. Congruence is an equivalence relation: it is reflexive by $P = I$, symmetric because $A = (P^{-1})^t B P^{-1}$, and transitive because $(PQ)^t A (PQ) = Q^t (P^t A P) Q$.

Theorem 22.1.4 says that the matrices of one form in the various bases of V form exactly one congruence class, so a property of matrices is a property of the form precisely when it is congruence-invariant. Similarity, the relation $A \mapsto P^{-1} A P$ appropriate to linear operators, is a different relation, and the invariants of a form are not those of an operator: the determinant, for instance, is not congruence-invariant, but the next definition extracts from it something that is.

Definition 22.1.6: The Square Classes of a Field

Let k be a field. The squares $(k^\times)^2 = \{c^2 \mid c \in k^\times\}$ form a subgroup of the abelian group $k^\times$, and the quotient

$$k^\times / (k^\times)^2$$

is the group of *square classes* of k. Every element has order dividing 2, so it is a vector space over $\mathbb{F}_2$. The class of $a \in k^\times$ is written $a\,(k^\times)^2$, and two elements lie in the same class exactly when their ratio is a square.

The square classes are where the classification lives, and their size measures how far the field is from having all square roots. For $k = \mathbb{C}$, or any algebraically closed field, every element is a square and the group is trivial. For $k = \mathbb{R}$ it has order 2, the classes being the positive and the negative reals. For $k = \mathbb{Q}$ it is infinite. The relevance to forms is immediate from theorem 22.1.4.

Definition 22.1.7: Discriminant of a Nondegenerate Form

Let φ be a nondegenerate symmetric bilinear form (definition 13.3.22) on a finite-dimensional k-vector space V, and let β be any basis. The *discriminant* of φ is the square class

$$\operatorname{disc}(\varphi) := \det\big([\varphi]_\beta\big)\,(k^\times)^2 \in k^\times/(k^\times)^2$$

Proposition 22.1.8: The Discriminant Is Well Defined and Is an Invariant

The class $\operatorname{disc}(\varphi)$ of definition 22.1.7 does not depend on the basis β, and congruent forms have equal discriminants.

***Proof*:**
Nondegeneracy makes $\det([\varphi]_\beta) \neq 0$ by lemma 13.3.23, so the class is taken in $k^\times$ and the definition makes sense. If β' is a second basis then $[\varphi]_{\beta'} = P^t[\varphi]_\beta P$ by theorem 22.1.4, and the multiplicativity of the determinant together with $\det(P^t) = \det(P)$ gives

$$\det\big([\varphi]_{\beta'}\big) = (\det P)^2 \det\big([\varphi]_\beta\big)$$

The two determinants differ by the square $(\det P)^2$ of the nonzero scalar $\det P$, hence lie in the same square class. The same computation, read as a statement about congruent matrices rather than about two bases, gives the second claim, as we sought to show.

22.1.9 Convention: the discriminant is unsigned, and the Witt group will twist it
Definition 22.1.7 takes the determinant with no sign attached. That is the right convention here and the wrong one for the Witt group, and the two sections genuinely differ, so the choice is recorded rather than left implicit.
At a fixed rank n the factor $(-1)^{n(n-1)/2}$ is a constant and carries no information, while multiplicativity over orthogonal sums, $\operatorname{disc}(\varphi \perp \psi) = \operatorname{disc}(\varphi)\operatorname{disc}(\psi)$, holds exactly as stated and is what the classification over a finite field uses.
The Witt group of section 22.6 identifies φ with $\varphi \perp \langle 1, -1\rangle$, and the unsigned discriminant does not survive that identification: adding $\langle 1, -1\rangle$ multiplies the determinant by -1. The *signed* discriminant $(-1)^{n(n-1)/2}\det$ does survive, since the rank rises by two and the sign changes to compensate. The price is that it is multiplicative only up to $(-1)^{mn}$. Each section therefore takes the convention its own problem needs, and section 22.6 states the twist when it introduces it.

The discriminant is the first invariant of the subject that is invisible over $\mathbb{R}$ and $\mathbb{C}$: there the square class group has at most two elements, so the discriminant carries at most one bit, whereas over a general field it is as large as $k^\times/(k^\times)^2$. The notion of sameness it is invariant under deserves its own name.

Definition 22.1.10: Isometry and the Orthogonal Group

Let (V, φ) and (V', φ') be finite-dimensional k-vector spaces equipped with symmetric bilinear forms. An *isometry* $f\colon (V, \varphi) \to (V', \varphi')$ is a linear isomorphism $f\colon V \to V'$ with

$$\varphi'\big(f(x), f(y)\big) = \varphi(x, y) \qquad \text{for all } x, y \in V$$

The two forms are *isometric*, written $\varphi \cong \varphi'$, if such an f exists. The isometries of (V, φ) to itself form a group under composition, the *orthogonal group* $O(\varphi)$ of the form.

Choosing bases turns an isometry into a congruence: if f is an isometry and β is a basis of V, then the matrix of φ' in the basis $f(\beta)$ is $[\varphi]_\beta$, so two forms on spaces of the same dimension are isometric exactly when their matrices are congruent. The classification of forms up to isometry and the classification of symmetric matrices up to congruence are therefore one problem in two languages. The chapter's opening example is a form the reader has already met.

Example 22.1.11: The Trace Form as a Quadratic Form

Let F/k be a finite separable extension with $\operatorname{char} k \neq 2$. The trace form $\langle x, y\rangle = \operatorname{tr}_{F/k}(xy)$ of definition 19.6.30 is a symmetric k-bilinear form on F, viewed as a k-vector space, and it is nondegenerate by proposition 19.6.31. Its associated quadratic form under proposition 22.1.2 is

$$q(x) = \operatorname{tr}_{F/k}(x^2)$$

so F carries a canonical quadratic form built from nothing but its field structure, and definition 22.1.7 attaches to it a square class in $k^\times/(k^\times)^2$.

Take $k = \mathbb{Q}$ and $F = \mathbb{Q}(\sqrt{d})$ with d a squarefree integer, $d \neq 1$. In the basis $\beta = (1, \sqrt{d})$ one computes $\operatorname{tr}(1) = 2$, since multiplication by 1 is the identity of a two-dimensional space. Next $\operatorname{tr}(\sqrt{d}) = 0$, since multiplication by $\sqrt{d}$ sends $1 \mapsto \sqrt{d}$ and $\sqrt{d} \mapsto d$, so its matrix in that basis has both diagonal entries zero. Finally $\operatorname{tr}(d) = 2d$. Hence

$$[\langle\,,\rangle]_\beta = \begin{pmatrix} 2 & 0 \\ 0 & 2d \end{pmatrix}, \qquad \det = 4d$$

and since 4 is a square,

$$\operatorname{disc} = 4d\,(\mathbb{Q}^\times)^2 = d\,(\mathbb{Q}^\times)^2$$

The discriminant of the trace form recovers the square class of d, which is precisely the datum distinguishing the fields $\mathbb{Q}(\sqrt{d})$ from one another. The invariant is doing real work already: over $\mathbb{Q}$ the square class group is infinite, so this single class separates infinitely many non-isometric forms, where over $\mathbb{R}$ or $\mathbb{C}$ it could not have separated more than two.

Exercise 22.1.1

1. Let q be a quadratic form with polar form b_q. Prove the parallelogram identity $q(x+y)+q(x-y) = 2q(x) + 2q(y)$ directly from definition 22.1.1, using no hypothesis on the characteristic. Then explain why the identity carries no information when $\operatorname{char} k = 2$, and say what that has to do with example 22.1.3.

2. Exhibit two matrices in $M_2(\mathbb{Q})$ that are congruent but not similar. Then show that when the

change of basis matrix P of theorem 22.1.4 satisfies $P^t = P^{-1}$ the two relations produce the same matrix, so congruence and similarity can differ only through changes of basis that are not of this kind.

3. Let φ and ψ be nondegenerate symmetric bilinear forms on V and W, and let $\varphi \perp \psi$ denote the form on $V \oplus W$ given by $(\varphi \perp \psi)\big((v,w),(v',w')\big) = \varphi(v,v') + \psi(w,w')$. Show that $\varphi \perp \psi$ is nondegenerate and that $\operatorname{disc}(\varphi \perp \psi) = \operatorname{disc}(\varphi)\operatorname{disc}(\psi)$.

4. Let $F = \mathbb{F}_{q^2}$ and $k = \mathbb{F}_q$ with q odd. Compute the trace form of F/k in a basis of your choosing and determine its discriminant as an element of $k^\times/(k^\times)^2$. Is it the trivial class?

5. Let φ be a symmetric bilinear form on V with radical $R = \operatorname{rad}(\varphi)$ (definition 13.3.22). Show that φ descends to a well-defined form $\bar{\varphi}$ on V/R, that $\bar{\varphi}$ is nondegenerate, and that $\operatorname{disc}(\bar{\varphi})$ is therefore defined even though $\operatorname{disc}(\varphi)$ is not.

22.2 Diagonalization and Orthogonal Bases

A symmetric matrix is congruent to a diagonal one, and the diagonal entries are visible invariants only up to squares. Both halves of that sentence are proved here, and together they reduce the classification problem to a question about the square class group of the field.

Theorem 22.2.1: Every Symmetric Form Has an Orthogonal Basis

Let k be a field with $\operatorname{char} k \neq 2$, let V be a k-vector space of finite dimension n, and let φ be a symmetric bilinear form on V. Then V has a basis $w_1, \dots, w_n$ with $\varphi(w_i, w_j) = 0$ whenever $i \neq j$. Equivalently, every symmetric matrix over k is congruent to a diagonal matrix.

***Proof*:**
Induct on n. For $n = 0$ there is nothing to prove, and for $n = 1$ any basis is orthogonal because the condition $i \neq j$ is vacuous.

Step 1: either φ vanishes identically or some vector is non-isotropic. Suppose $\varphi(v,v) = 0$ for every $v \in V$. Then for all $x, y \in V$,

$$0 = \varphi(x+y, x+y) = \varphi(x,x) + 2\varphi(x,y) + \varphi(y,y) = 2\varphi(x,y)$$

and since $\operatorname{char} k \neq 2$ this forces $\varphi(x,y) = 0$, so φ is the zero form and every basis is orthogonal. This is the only place the characteristic hypothesis is used, and it cannot be dropped: the form ψ of example 22.1.3 is a nonzero symmetric form over $\mathbb{F}_2$ with $\psi(v,v) = 0$ for every v, so it falsifies the implication just proved, and that example shows it admits no orthogonal basis at all.

Otherwise choose $w_1 \in V$ with $a_1 := \varphi(w_1, w_1) \neq 0$.

Step 2: split off w_1. Let $W = \{w_1\}^\perp$, a subspace by lemma 13.3.21. Every $x \in V$ decomposes as

$$x = \frac{\varphi(w_1, x)}{a_1}\, w_1 + \left(x - \frac{\varphi(w_1, x)}{a_1}\, w_1 \right)$$

and the bracketed vector lies in W, since applying $\varphi(w_1, -)$ to it gives $\varphi(w_1, x) - \frac{\varphi(w_1,x)}{a_1} a_1 = 0$. So $V = kw_1 + W$. The sum is direct: if $\lambda w_1 \in W$ then $0 = \varphi(w_1, \lambda w_1) = \lambda a_1$, and $a_1 \neq 0$ forces $\lambda = 0$. Hence $V = kw_1 \oplus W$ and $\dim W = n - 1$.

Step 3: induct. The restriction $\varphi|_{W \times W}$ is a symmetric bilinear form on W, so by the inductive hypothesis W has an orthogonal basis $w_2, \ldots, w_n$. Then $w_1, w_2, \ldots, w_n$ is a basis of V by Step 2, and it is orthogonal: $\varphi(w_i, w_j) = 0$ for $2 \leq i \neq j \leq n$ by the choice of the w_i, and $\varphi(w_1, w_j) = 0$ for $j \geq 2$ because $w_j \in W = \{w_1\}^{\perp}$.

For the matrix formulation, a symmetric $A \in M_n(k)$ is the matrix of a symmetric form on k^n in the standard basis, and an orthogonal basis for that form has diagonal matrix, and by theorem 22.1.4 the two matrices are congruent, completing the proof.

In an orthogonal basis the form is determined by the n scalars $a_i = \varphi(w_i, w_i)$, and the standard notation records exactly that: write

$$\varphi \cong \langle a_1, a_2, \ldots, a_n \rangle$$

for a form admitting an orthogonal basis with those diagonal entries, so that the associated quadratic form is $q(x) = a_1 x_1^2 + \cdots + a_n x_n^2$ in the corresponding coordinates. The notation is an assertion about an isometry class, not about a chosen basis, and the next proposition says how far the entries themselves are from being invariants.

Proposition 22.2.2: Rescaling the Entries of a Diagonalization

Let $\operatorname{char} k \neq 2$. For any $c_1, \ldots, c_n \in k^{\times}$,

$$\langle a_1, \ldots, a_n \rangle \cong \langle a_1 c_1^2, \ldots, a_n c_n^2 \rangle$$

so each diagonal entry is well defined only up to multiplication by a nonzero square. A diagonal entry that is zero stays zero, and the number of zero entries is the dimension of the radical, hence is an invariant.

***Proof*:**

Let $w_1, \ldots, w_n$ be an orthogonal basis realizing $\langle a_1, \ldots, a_n \rangle$ and put $w_i' = c_i w_i$. Each c_i is nonzero, so $w_1', \ldots, w_n'$ is again a basis, and it is orthogonal because $\varphi(w_i', w_j') = c_i c_j \varphi(w_i, w_j) = 0$ for $i \neq j$. Its diagonal entries are $\varphi(w_i', w_i') = c_i^2 a_i$, which gives the displayed isometry. Since $c_i^2 \neq 0$, the entry $c_i^2 a_i$ vanishes exactly when a_i does.

For the last claim, in an orthogonal basis a vector $x = \sum_i x_i w_i$ lies in $\operatorname{rad}(\varphi)$ if and only if $\varphi(w_j, x) = a_j x_j = 0$ for every j, that is if and only if $x_j = 0$ for every j with $a_j \neq 0$. So the radical is spanned by those w_j with $a_j = 0$, and its dimension is the number of zero entries, which is independent of the diagonalization because the radical is, completing the proof.

Corollary 22.2.3: The Discriminant from a Diagonalization

Let $\varphi \cong \langle a_1, \ldots, a_n \rangle$ be nondegenerate, so every $a_i \neq 0$. Then

$$\operatorname{disc}(\varphi) = a_1 a_2 \cdots a_n \, (k^{\times})^2$$

***Proof*:**
In an orthogonal basis the matrix of φ is $\operatorname{diag}(a_1, \dots, a_n)$, whose determinant is $a_1 \cdots a_n$. Apply definition 22.1.7, whose independence of the basis is proposition 22.1.8. Consistently with proposition 22.2.2, replacing each a_i by $a_i c_i^2$ multiplies the product by $(c_1 \cdots c_n)^2$ and so does not change the class, as we sought to show.

Two fields where the square class group is completely known illustrate how much of the classification the discriminant sees. The first is the degenerate case, in the sense that the group is trivial and nothing beyond the rank survives.

Example 22.2.4: Forms over an Algebraically Closed Field
Let k be algebraically closed with $\operatorname{char} k \neq 2$. Every $a \in k^\times$ has a square root, so $(k^\times)^2 = k^\times$ and the square class group of definition 22.1.6 is trivial. Given any symmetric form, diagonalize it by theorem 22.2.1 as $\langle a_1, \dots, a_r, 0, \dots, 0\rangle$ with the $a_i \neq 0$, and apply proposition 22.2.2 with $c_i = a_i^{-1/2}$ to obtain

$$\varphi \cong \langle \underbrace{1, \dots, 1}_{r}, \underbrace{0, \dots, 0}_{n-r} \rangle$$

So a form over an algebraically closed field is determined up to isometry by n and r, and r is the rank of any matrix of φ. The discriminant of a nondegenerate such form is the class of 1 and carries no information, as it must, the group having one element.

Example 22.2.5: Forms over a Finite Field
Let $k = \mathbb{F}_q$ with q odd. The group $\mathbb{F}_q^\times$ is cyclic by theorem 11.3.14, of even order $q - 1$, so the squaring map has kernel $\{\pm 1\}$ of order 2 and image of index 2. Hence

$$\mathbb{F}_q^\times / (\mathbb{F}_q^\times)^2 \cong \mathbb{Z}/2\mathbb{Z}$$

with the two classes the nonzero squares and the non-squares. Unlike the algebraically closed case, the discriminant now carries a genuine bit, and it obstructs isometries that the rank alone would permit.

Take $q = 5$, where the squares in $\mathbb{F}_5^\times = \{1, 2, 3, 4\}$ are $1 = 1^2$ and $4 = 2^2$, so the non-squares are 2 and 3. The forms $\langle 1, 1\rangle$ and $\langle 1, 2\rangle$ on $\mathbb{F}_5^2$ are both nondegenerate of rank 2, but by corollary 22.2.3 their discriminants are the classes of 1 and of 2 respectively, which are distinct in $\mathbb{F}_5^\times / (\mathbb{F}_5^\times)^2$. By proposition 22.1.8 an isometry would force those classes to agree, so

$$\langle 1, 1\rangle \ncong \langle 1, 2\rangle$$

Rank does not classify forms over $\mathbb{F}_5$. Whether rank together with the discriminant does classify them is a genuine question, and it is answered affirmatively in section 22.6 once enough is known about which values a form represents.

Exercise 22.2.1

1. Exhibit two different orthogonal bases for the same form on $\mathbb{Q}^2$ whose diagonal entries are not related by the rescaling of proposition 22.2.2 entry by entry, and reconcile this with that proposition.

2. Let $k = \mathbb{R}$. Determine the square class group, and show that every nondegenerate form is isometric to $\langle 1, \dots, 1, -1, \dots, -1\rangle$. Explain why the discriminant alone does not classify these, and identify the invariant that does. (This book does not prove that the counts are well defined. That is Sylvester's law of inertia, and it needs an order on the scalars.)

3. Over $k = \mathbb{Q}$, show that $\langle 1, 1\rangle$ and $\langle -1, -1\rangle$ have the same rank and the same discriminant and are nevertheless not isometric, by asking which nonzero rationals each represents. Conclude that rank and discriminant together do not classify forms over $\mathbb{Q}$, in contrast with example 22.2.4. Show also that $\langle 1, 1\rangle \cong \langle 2, 2\rangle$ over $\mathbb{Q}$, so a difference of diagonal entries is not by itself evidence of non-isometry.

4. Let $\varphi \cong \langle a_1, \dots, a_n\rangle$ be nondegenerate over k and let $c \in k^\times$ be a value represented by φ, meaning $c = q(v)$ for some $v \in V$. Show that $\varphi \cong \langle c, b_2, \dots, b_n\rangle$ for some b_i, and deduce that the set of represented values is a union of square classes.

22.3 Alternating Forms, the Linear Darboux Theorem, and Lagrangians

Section 22.1 and section 22.2 treated the symmetric forms, where the classification runs through the square classes of k and is not settled by the rank alone. The alternating forms are the second class of bilinear forms for which a classification is available, and there the answer does not depend on the field: a single integer, the codimension of the radical, determines the form up to isometry, and neither the statement nor its proof carries a hypothesis on the characteristic.

The classification is proved in the shape of a basis in which the matrix of the form is one fixed block matrix. That basis then makes visible the subspaces on which an alternating form vanishes identically. When the form is nondegenerate such a subspace has dimension at most half that of the ambient space, and the ones attaining the bound are the Lagrangian subspaces.

Definition 22.3.1: Alternating Bilinear Form

Let k be a field, with no hypothesis on its characteristic, and let V be a finite-dimensional k-vector space. A bilinear form $\varphi\colon V \times V \to k$ (definition 13.3.15) is *alternating* if

$$\varphi(v, v) = 0 \qquad \text{for every } v \in V$$

This is the vanishing condition of definition 16.1.3 read for a 2-multilinear map, and not the sign rule listed in definition 13.3.15. The two conditions differ in characteristic 2, and throughout this section *alternating* means the displayed vanishing condition.

Two alternating forms (V, φ) and (V', φ') are *isometric* if there is a linear isomorphism $f\colon V \to V'$ with $\varphi'\big(f(x), f(y)\big) = \varphi(x, y)$ for all $x, y \in V$. This is the condition of definition 22.1.10 with the hypothesis that the two forms be symmetric dropped, and it is used in that generality below.

One expansion carries the whole section. For $x, y \in V$ the vanishing condition applied to $x + y$ gives

$$0 = \varphi(x + y, x + y) = \varphi(x, x) + \varphi(x, y) + \varphi(y, x) + \varphi(y, y) = \varphi(x, y) + \varphi(y, x)$$

so every alternating form satisfies

$$\varphi(y,x) = -\varphi(x,y) \qquad \text{for all } x, y \in V \tag{22.1}$$

which is lemma 16.1.4(1) for the transposition in S_2, its hypotheses holding here because a field is a commutative ring and both V and k are k-modules. The converse needs 2 to be invertible, and that is lemma 16.1.4(2): over a field with $\operatorname{char} k \neq 2$ a bilinear form satisfying (22.1) is alternating, since setting $y = x$ there gives $2\varphi(x,x) = 0$. Over such a field, therefore, *alternating* and *skew-symmetric* name the same forms, and a form that is both symmetric and alternating is zero: symmetry and (22.1) together give $\varphi(x,y) = -\varphi(x,y)$, hence $2\varphi(x,y) = 0$, hence $\varphi(x,y) = 0$. The classification below and the one in section 22.2 therefore overlap only in the zero form. In characteristic 2 neither statement survives, and $-1 = 1$ makes (22.1) say only that φ is symmetric.

The condition is visible in the matrix of the form. Fix a basis $\beta = (\beta_1, \ldots, \beta_n)$ of V and let $A = (a_{ij}) = [\varphi]_\beta$ be the matrix of definition 13.3.18. If φ is alternating then $a_{ii} = \varphi(\beta_i, \beta_i) = 0$ and, by (22.1), $a_{ji} = -a_{ij}$. Conversely, suppose A has zero diagonal and satisfies $A^t = -A$. For $v = \sum_i x_i \beta_i$, lemma 13.3.19 gives

$$\varphi(v,v) = \sum_{i,j} x_i x_j a_{ij} = \sum_i x_i^2 a_{ii} + \sum_{i<j} x_i x_j \big(a_{ij} + a_{ji}\big) = 0$$

so φ is alternating. Both conditions on A are needed: in characteristic 2 the matrix $\left(\begin{smallmatrix} 1 & 0 \\ 0 & 0 \end{smallmatrix}\right)$ satisfies $A^t = -A$ and has a nonzero diagonal entry.

Example 22.3.2: The Determinant Form, and the Overlap in Characteristic Two

1. Let $V = k^2$ and
$$\varphi(x,y) = x_1 y_2 - x_2 y_1$$
which is the determinant of the matrix with columns x and y. It is bilinear in each column and $\varphi(x,x) = x_1 x_2 - x_2 x_1 = 0$, so it is alternating. Its matrix in the standard basis is $\left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right)$, of determinant 1, so φ is nondegenerate by lemma 13.3.23.

2. Over $k = \mathbb{F}_2$ the form in (1) reads $\varphi(x,y) = x_1 y_2 + x_2 y_1$, which is exactly the form ψ of example 22.1.3. That form is symmetric, nonzero and alternating at once, which is possible only because $\operatorname{char} k = 2$. The example also shows it admits no orthogonal basis, so the conclusion of theorem 22.2.1 genuinely fails for it.

3. Still over $\mathbb{F}_2$, the form $\varphi(x,y) = x_1 y_1$ is symmetric, hence satisfies (22.1) because $-1 = 1$, but $\varphi(e_1, e_1) = 1 \neq 0$. So the sign rule does not imply the vanishing condition when 2 is not invertible, and the hypothesis in lemma 16.1.4(2) cannot be dropped.

The proof of theorem 22.2.1 split off one dimension at a time, and the entry it left behind was pinned down only up to a square (proposition 22.2.2). For an alternating form there is no such entry to leave behind, since $\varphi(v,v) = 0$ for every v, and the smallest piece that carries any information is two-dimensional. On such a piece the one scalar available can be scaled to 1 outright, which is why no square classes enter the answer. The resulting normal form deserves a name before it is proved to exist.

Definition 22.3.3: Symplectic Basis and Symplectic Space

Let k be a field, with no hypothesis on its characteristic, let V be a finite-dimensional k-vector space, and let φ be an alternating bilinear form on V (definition 22.3.1). A *symplectic basis* for φ is a basis

$$e_1, f_1, e_2, f_2, \dots, e_m, f_m$$

of V, necessarily of even cardinality $2m$, such that $\varphi(e_i, f_i) = 1$ for every i, while

$$\varphi(e_i, f_j) = 0 \quad (i \neq j), \qquad \varphi(e_i, e_j) = 0, \qquad \varphi(f_i, f_j) = 0$$

for all i and j. A pair (V, φ) with φ alternating and nondegenerate (definition 13.3.22) is called a *symplectic space*.

Example 22.3.4: The Standard Symplectic Space k^{2m}

Let k be any field and let $e_1, f_1, \dots, e_m, f_m$ be the standard basis of k^{2m}, listed in that order. Define

$$\varphi\Big(\sum_i (x_i e_i + y_i f_i),\ \sum_i (x_i' e_i + y_i' f_i)\Big) = \sum_{i=1}^{m} (x_i y_i' - y_i x_i')$$

Each summand is the form of example 22.3.2(1) in the coordinate pair (x_i, y_i), so φ is bilinear and vanishes when its two arguments agree, hence is alternating. Evaluating on basis vectors gives $\varphi(e_i, f_i) = 1$, while $\varphi(e_i, f_j) = 0$ for $i \neq j$ and $\varphi(e_i, e_j) = \varphi(f_i, f_j) = 0$ for all i and j, because the i-th summand involves only the i-th coordinate pair. So the standard basis is symplectic. The matrix of φ in this basis is block diagonal with m blocks $\left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right)$, so its determinant is 1 and φ is nondegenerate by lemma 13.3.23.

Listing the same basis as $e_1, \dots, e_m, f_1, \dots, f_m$ changes the matrix to $\left(\begin{smallmatrix} 0 & I_m \\ -I_m & 0 \end{smallmatrix}\right)$. The two orderings differ by a permutation matrix P, so by theorem 22.1.4 the matrices differ by $A \mapsto P^t A P$ and the determinant is unchanged, being multiplied by $(\det P)^2 = 1$. This second ordering identifies k^{2m} with $k^m \oplus k^m$ and writes the form as

$$\varphi\big((x, y), (x', y')\big) = x^t y' - y^t x' \tag{22.2}$$

for column vectors $x, y, x', y' \in k^m$, which is the shape used for computations below.

A degenerate example is obtained by padding. On k^{2m+r} with basis $e_1, f_1, \dots, e_m, f_m, z_1, \dots, z_r$, extend φ by declaring $\varphi(z_j, -) = \varphi(-, z_j) = 0$ for every j. The result is alternating, and its radical is $\operatorname{span}(z_1, \dots, z_r)$. To see this, write $v = \sum_i (x_i e_i + y_i f_i) + \sum_j c_j z_j$. Then $\varphi(e_i, v) = y_i$ and $\varphi(f_i, v) = -x_i$ and $\varphi(z_j, v) = 0$, and since φ is linear in its first argument, v lies in the radical (definition 13.3.22) exactly when every x_i and every y_i vanishes.

Every alternating form is one of the forms just written down. This is the classification, and it holds over every field.

Theorem 22.3.5: Linear Darboux Theorem

Let k be a field, with no hypothesis on its characteristic, let V be a k-vector space of finite dimension n, and let φ be an alternating bilinear form on V (definition 22.3.1).

1. There are integers $m, r \geq 0$ with $n = 2m + r$ and a basis
$$e_1, f_1, \ldots, e_m, f_m, z_1, \ldots, z_r$$
of V such that $\varphi(e_i, f_i) = 1$ for every i, while every other value of φ on an ordered pair of basis vectors is 0 except $\varphi(f_i, e_i) = -1$.
2. For any such basis, $\operatorname{rad}(\varphi) = \operatorname{span}(z_1, \ldots, z_r)$. Hence $r = \dim \operatorname{rad}(\varphi)$ and $2m = n - \dim \operatorname{rad}(\varphi)$, so both integers are determined by φ and do not depend on the basis chosen in (1).
3. φ is nondegenerate if and only if $r = 0$, and in that case $n = 2m$ is even and the basis of (1) is a symplectic basis (definition 22.3.3). In particular a symplectic space has even dimension.
4. Let φ' be an alternating form on a k-vector space V' with $\dim V' = \dim V$. Then φ and φ' are isometric if and only if $\dim \operatorname{rad}(\varphi) = \dim \operatorname{rad}(\varphi')$. In particular, for each even n there is exactly one nondegenerate alternating form on an n-dimensional space up to isometry, and none at all when n is odd.

***Proof*:**

1. *Step 1: a pair e, f with $\varphi(e, f) = 1$.* Assume $\varphi \neq 0$ and choose $u, w \in V$ with $c := \varphi(u, w) \neq 0$. Put $e := u$ and $f := c^{-1}w$, so that $\varphi(e, f) = c^{-1}\varphi(u, w) = 1$, and $\varphi(f, e) = -1$ by (22.1). The two are linearly independent: if $\alpha e + \beta f = 0$ then applying $\varphi(e, -)$ gives $\alpha\varphi(e, e) + \beta\varphi(e, f) = \beta = 0$, and applying $\varphi(f, -)$ gives $\alpha\varphi(f, e) = -\alpha = 0$. Write $P = \operatorname{span}(e, f)$, a two-dimensional subspace.

   Step 2: $V = P \oplus P^{\perp}$. The set $P^{\perp}$ is a subspace and equals $\{e, f\}^{\perp}$, both by lemma 13.3.21 applied to the subset $\{e, f\}$, whose span is P. Given $v \in V$, set
$$v' := v - \varphi(e, v)\, f + \varphi(f, v)\, e$$
   Then, using $\varphi(e, e) = \varphi(f, f) = 0$, $\varphi(e, f) = 1$ and $\varphi(f, e) = -1$,
$$\varphi(e, v') = \varphi(e, v) - \varphi(e, v)\cdot 1 + \varphi(f, v)\cdot 0 = 0 \qquad \varphi(f, v') = \varphi(f, v) - \varphi(e, v)\cdot 0 + \varphi(f, v)\cdot(-1) = 0$$
   so $v' \in P^{\perp}$, and $v = \big(\varphi(e, v)f - \varphi(f, v)e\big) + v'$ exhibits v in $P + P^{\perp}$. The sum is direct: if $\alpha e + \beta f \in P^{\perp}$ then $\varphi(e, \alpha e + \beta f) = \beta = 0$ and $\varphi(f, \alpha e + \beta f) = -\alpha = 0$. Hence $V = P \oplus P^{\perp}$ and $\dim P^{\perp} = n - 2$.

   Step 3: induction on n. For $n = 0$ take the empty basis and $m = r = 0$. Let $n \geq 1$ and assume the statement for all spaces of smaller dimension. If $\varphi = 0$, any basis of V serves with $m = 0$ and $r = n$. Otherwise Steps 1 and 2 produce $V = P \oplus P^{\perp}$ with $\dim P^{\perp} = n - 2$.

The restriction of φ to $P^\perp \times P^\perp$ is bilinear and satisfies the vanishing condition, so it is alternating, and the inductive hypothesis gives a basis $e_2, f_2, \dots, e_m, f_m, z_1, \dots, z_r$ of $P^\perp$ with the stated table of values. Set $e_1 := e$ and $f_1 := f$. The concatenated list is a basis of V by Step 2. Its values are as claimed: within $P^\perp$ by the inductive hypothesis, on the pair (e_1, f_1) by Step 1, and for a pair with one entry in $\{e_1, f_1\}$ and the other in $P^\perp$ because $\varphi(e_1, v) = \varphi(f_1, v) = 0$ for $v \in P^\perp$ by the definition of $P^\perp$, whence also $\varphi(v, e_1) = \varphi(v, f_1) = 0$ by (22.1). Counting the list gives $n = 2m + r$.

2. Let $v = \sum_i a_i e_i + \sum_i b_i f_i + \sum_j c_j z_j$. Since φ is linear in its first argument, $v \in \operatorname{rad}(\varphi)$ holds if and only if $\varphi(x, v) = 0$ for x running over the basis. The table of values in (1) gives

$$\varphi(e_i, v) = b_i, \qquad \varphi(f_i, v) = -a_i, \qquad \varphi(z_j, v) = 0$$

for all i and j. So $v \in \operatorname{rad}(\varphi)$ if and only if every a_i and every b_i vanishes, that is if and only if $v \in \operatorname{span}(z_1, \dots, z_r)$. Those z_j are part of a basis, hence independent, so $\dim \operatorname{rad}(\varphi) = r$, and $2m = n - r$ follows from $n = 2m + r$. The radical was defined in definition 13.3.22 without reference to a basis, so r and m are the same for every basis as in (1).

3. By definition 13.3.22, φ is nondegenerate exactly when $\operatorname{rad}(\varphi) = 0$, which by (2) happens exactly when $r = 0$. Then $n = 2m$, and the list of (1) is a basis satisfying the conditions of definition 22.3.3.

4. Suppose first that $\dim \operatorname{rad}(\varphi) = \dim \operatorname{rad}(\varphi')$. By (2) the two forms have bases as in (1) with the same m and the same r, since $\dim V = \dim V'$. Let $f\colon V \to V'$ be the linear map carrying each basis vector of V to the correspondingly named basis vector of V'. It is an isomorphism, being a bijection on bases, and the two bilinear maps $(x, y) \mapsto \varphi'(f(x), f(y))$ and $(x, y) \mapsto \varphi(x, y)$ agree on every ordered pair of basis vectors by construction, hence agree everywhere by bilinearity. So f is an isometry in the sense of definition 22.3.1.

   Conversely let f be an isometry. Its inverse is one as well: putting $x = f^{-1}(x')$ and $y = f^{-1}(y')$ in the defining identity gives $\varphi\big(f^{-1}(x'), f^{-1}(y')\big) = \varphi'(x', y')$. Now let $v \in \operatorname{rad}(\varphi)$ and let $x' \in V'$. Since f is onto, $x' = f(x)$ for some x, and then $\varphi'(x', f(v)) = \varphi(x, v) = 0$. Hence $f\big(\operatorname{rad}(\varphi)\big) \subseteq \operatorname{rad}(\varphi')$, and the same argument for f^{-1} gives $f^{-1}\big(\operatorname{rad}(\varphi')\big) \subseteq \operatorname{rad}(\varphi)$, that is $\operatorname{rad}(\varphi') \subseteq f\big(\operatorname{rad}(\varphi)\big)$. So f restricts to an isomorphism between the two radicals, and their dimensions agree.

   For the last sentence, take φ and φ' nondegenerate of the same dimension n: both radicals are zero, so they are isometric, and the standard form of example 22.3.4 gives one such φ when n is even. When n is odd, (3) forbids a nondegenerate alternating form outright, completing the proof.

The symmetric case behaves differently over the very same field. Example 22.2.5 exhibits two nondegenerate symmetric forms of rank 2 over $\mathbb{F}_5$ that are not isometric, the discriminant separating them, while part (4) makes any two nondegenerate alternating forms on a space of the same dimension isometric, over $\mathbb{F}_5$ or over any other field. No invariant of the discriminant's kind can exist for alternating forms. Indeed, if φ is nondegenerate alternating with symplectic basis β_0, then $[\varphi]_{\beta_0}$ is block diagonal with m blocks $\left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right)$ by definition 22.3.3, so $\det[\varphi]_{\beta_0} = 1$. Any other basis β gives

$[\varphi]_\beta = P^t[\varphi]_{\beta_0}P$ by theorem 22.1.4, whence

$$\det[\varphi]_\beta = (\det P)^2$$

So the determinant of every matrix of φ is a nonzero square, and the square class that definition 22.1.7 extracts in the symmetric case is trivial here for every form and separates nothing.[1]

In odd dimension the theorem is a nonexistence statement: there is no nondegenerate alternating form at all, and every alternating form on a space of odd dimension n has a radical of odd dimension $n - 2m$. Symmetric forms are unconstrained in this respect, since $\langle 1, 1, 1\rangle$ has the identity as its matrix and is nondegenerate in dimension 3 by lemma 13.3.23.

The basis of theorem 22.3.5 exhibits two subspaces of dimension m on which φ vanishes identically, namely $\operatorname{span}(e_1, \dots, e_m)$ and $\operatorname{span}(f_1, \dots, f_m)$. Subspaces with that property have a name, and it is given for an arbitrary bilinear form since the symmetric case needs it too.

Definition 22.3.6: Totally Isotropic Subspace

Let k be a field, with no hypothesis on its characteristic, let V be a finite-dimensional k-vector space, and let φ be a bilinear form on V. A subspace $W \subseteq V$ is *totally isotropic* for φ if

$$\varphi(x, y) = 0 \qquad \text{for all } x, y \in W$$

equivalently if $W \subseteq W^\perp$ in the notation of definition 13.3.20. The shorter term *isotropic subspace* is used below for the same condition.

Example 22.3.7: Totally Isotropic Subspaces in Two Standard Forms

1. Let φ be alternating on V over any field. Every line $W = kv$ is totally isotropic, since $\varphi(\lambda v, \mu v) = \lambda\mu\,\varphi(v, v) = 0$. So an alternating form has isotropic subspaces in abundance, whatever the field.

2. In the standard symplectic space k^{2m} of example 22.3.4, the subspaces $E = \operatorname{span}(e_1, \dots, e_m)$ and $F = \operatorname{span}(f_1, \dots, f_m)$ are totally isotropic, since $\varphi(e_i, e_j) = 0$ and $\varphi(f_i, f_j) = 0$ for all i, j and the condition is bilinear in the two arguments. Each has dimension m, half of $\dim k^{2m}$.

3. Let $k = \mathbb{R}$ and let $\varphi = \langle 1, 1\rangle$ on $\mathbb{R}^2$, that is $\varphi(x, y) = x_1y_1 + x_2y_2$, with associated quadratic form $q(x) = x_1^2 + x_2^2$ under proposition 22.1.2. If W is totally isotropic and $v \in W$, then $q(v) = \varphi(v, v) = 0$, and a sum of two real squares vanishes only when both do, so $v = 0$. The only totally isotropic subspace is 0. So the abundance in (1) is a feature of the alternating case and not of bilinear forms in general.

The subspaces in item (2) have exactly half the dimension of the ambient space, and item (1) produces nothing larger there. Half is in fact the maximum whenever the form is nondegenerate, for symmetric and alternating forms alike, and the bound follows from a dimension count for orthogonal

[1] The name of the theorem is by analogy with Darboux's theorem in symplectic geometry, which states that a closed nondegenerate differential 2-form on a smooth manifold admits, around each point, coordinates in which it is the constant form displayed in example 22.3.4. That theorem is about manifolds and is not proved in this book. The statement here is its pointwise linear-algebra counterpart.

complements. Degeneracy destroys the bound at once: the zero form on any V is alternating, and V itself is then totally isotropic.

Proposition 22.3.8: The Dimension Bound for Isotropic Subspaces

Let k be a field, with no hypothesis on its characteristic, let V be a k-vector space of finite dimension n, and let φ a nondegenerate bilinear form on V (definition 13.3.22).

1. For every subspace $W \subseteq V$,
$$\dim W + \dim W^{\perp} = n$$
2. If W is totally isotropic (definition 22.3.6) then $\dim W \leq \frac{1}{2}n$, with equality if and only if $W = W^{\perp}$.
3. Assume in addition that φ is alternating, so that $n = 2m$ is even by theorem 22.3.5(3). Then every totally isotropic subspace of V is contained in a totally isotropic subspace of dimension m. Such subspaces exist, and the maximal totally isotropic subspaces of V are exactly those of dimension m.

***Proof*:**

1. Let $d = \dim W$ and define $\Psi \colon V \to W^*$ by $\Psi(y)(x) = \varphi(x, y)$ for $x \in W$. Each $\Psi(y)$ is linear on W and Ψ is linear in y, both by bilinearity of φ. By definition 13.3.20, $\Psi(y) = 0$ says exactly that $y \in W^{\perp}$, so $\ker \Psi = W^{\perp}$.

   Ψ is surjective. First, the map $\Theta \colon V \to V^*$ with $\Theta(y) = \varphi(-, y)$ has kernel $\operatorname{rad}(\varphi)$, which is 0 by definition 13.3.22 and the nondegeneracy hypothesis. The dual basis of proposition 13.3.5 is a basis of V^*, so $\dim V^* = \dim V$, while corollary 13.1.18 gives $\dim V = \dim \ker \Theta + \dim \Theta(V) = \dim \Theta(V)$. Hence the image $\Theta(V)$ is a subspace of V^* of the full dimension, so $\Theta(V) = V^*$. Second, every $g \in W^*$ extends to V: choose a basis $w_1, \ldots, w_d$ of W, extend it to a basis $w_1, \ldots, w_n$ of V, and let $\tilde{g} \in V^*$ be the linear map with $\tilde{g}(w_i) = g(w_i)$ for $i \leq d$ and $\tilde{g}(w_i) = 0$ for $i > d$. Then $\tilde{g}$ and g are linear maps on W agreeing on a basis of W, so $\tilde{g}|_W = g$. Combining, given $g \in W^*$ pick y with $\Theta(y) = \tilde{g}$. Then $\Psi(y) = \tilde{g}|_W = g$.

   Now corollary 13.1.18 applied to Ψ gives $n = \dim \ker \Psi + \dim \Psi(V) = \dim W^{\perp} + \dim W^*$, and $\dim W^* = \dim W$ again by proposition 13.3.5, this time for the space W.

2. If W is totally isotropic then $W \subseteq W^{\perp}$ by definition 22.3.6, so $\dim W \leq \dim W^{\perp} = n - \dim W$ by (1), which rearranges to $2 \dim W \leq n$. Equality holds exactly when $\dim W = \dim W^{\perp}$, and a subspace contained in another of the same finite dimension equals it, so equality is equivalent to $W = W^{\perp}$.

3. Let W be totally isotropic with $d = \dim W < m$. By (1), $\dim W^{\perp} = 2m - d > d = \dim W$, and $W \subseteq W^{\perp}$, so there is a vector $v \in W^{\perp}$ with $v \notin W$. Put $W' = W + kv$, of dimension $d + 1$. It is totally isotropic: for $w, w' \in W$ and $\lambda, \mu \in k$,
$$\varphi(w + \lambda v,\ w' + \mu v) = \varphi(w, w') + \mu\, \varphi(w, v) + \lambda\, \varphi(v, w') + \lambda\mu\, \varphi(v, v)$$

and the four terms vanish in turn because W is totally isotropic, because $v \in W^\perp$, because $\varphi(v, w') = -\varphi(w', v) = 0$ by (22.1) and $v \in W^\perp$, and because φ is alternating. Repeating the step $m - d$ times produces a totally isotropic subspace of dimension m containing W.

Existence follows by applying this to $W = 0$, or directly from theorem 22.3.5(3): the span of $e_1, \dots, e_m$ in a symplectic basis is totally isotropic of dimension m. Finally, a totally isotropic subspace of dimension m is maximal among totally isotropic subspaces, since any subspace properly containing it has dimension exceeding the bound in (2). And one of dimension $d < m$ is not maximal, since the construction above properly enlarges it. So maximality is equivalent to having dimension m, as we sought to show.

Part (3) fails for symmetric forms: example 22.3.7(3) gives a nondegenerate symmetric form of rank 2 over $\mathbb{R}$ whose only totally isotropic subspace is 0, so there the maximal one has dimension 0 rather than 1. The alternating case is uniform because $\varphi(v, v) = 0$ holds for every v by definition: enlarging an isotropic W needs a vector of $W^\perp$ outside W on which the form vanishes, and the vanishing is automatic, so the dimension count in (1) gives the vector by itself. The subspaces attaining the bound carry a name.

Definition 22.3.9: Lagrangian Subspace

Let k be a field, with no hypothesis on its characteristic, and let (V, φ) be a symplectic space of dimension $2m$, so φ is alternating and nondegenerate (definition 22.3.3). A *Lagrangian subspace* of (V, φ) is a totally isotropic subspace $L \subseteq V$ with $\dim L = m$.

By proposition 22.3.8(2) this is the same as requiring $L = L^\perp$, and by proposition 22.3.8(3) it is the same as requiring L to be maximal among the totally isotropic subspaces of V. That same part guarantees Lagrangian subspaces exist in every symplectic space.

Example 22.3.10: Lagrangian Subspaces of the Standard Symplectic Space

Work in $k^{2m} = k^m \oplus k^m$ with the form (22.2), $\varphi\big((x, y), (x', y')\big) = x^t y' - y^t x'$.

1. For $S \in M_m(k)$ let $L_S = \{(x, Sx) \mid x \in k^m\}$, the graph of S. It is a subspace of dimension m, the map $x \mapsto (x, Sx)$ being linear and injective. For $x, x' \in k^m$,

$$\varphi\big((x, Sx), (x', Sx')\big) = x^t Sx' - (Sx)^t x' = x^t Sx' - x^t S^t x' = x^t \big(S - S^t\big) x'$$

   Taking x and x' to be the i-th and j-th standard basis vectors of k^m turns the right-hand side into the (i, j) entry of $S - S^t$, so the expression vanishes for all x, x' if and only if $S = S^t$. Hence L_S is Lagrangian exactly when S is symmetric. Taking $S = 0$ recovers $E = \operatorname{span}(e_1, \dots, e_m)$.

2. Conversely, every Lagrangian L with $L \cap F = 0$, where $F = \{(0, y) \mid y \in k^m\}$, is of this form. The projection $\pi(x, y) = x$ restricts to a linear map $L \to k^m$ with kernel $L \cap F = 0$, hence injective, and $\dim L = m$ forces it to be bijective by corollary 13.1.18. Let σ be its inverse and write $\sigma(x) = (x, Sx)$, where S is linear because σ is. Then $L = L_S$, and S is symmetric by (1).

3. For $m = 1$ every line in k^2 is Lagrangian: a line is totally isotropic by example 22.3.7(1) and has dimension $1 = m$. Over $k = \mathbb{F}_q$ the lines of k^2 number $(q^2 - 1)/(q - 1) = q + 1$, so

there are $q+1$ Lagrangians. For $m \geq 2$ having dimension m is not sufficient: inside k^4 the subspace $\operatorname{span}(e_1, f_1)$ has dimension $2 = m$ and is not totally isotropic, since $\varphi(e_1, f_1) = 1$.

Items (1) and (2) together put the Lagrangians meeting F trivially in bijection with the symmetric $m \times m$ matrices, the assignment $S \mapsto L_S$ being injective because L_S determines Sx as the second component of its element with first component x. The symmetric matrices form a k-vector space of dimension $\binom{m+1}{2}$. The two counts agree where they overlap: for $m = 1$ every 1×1 matrix is symmetric, so over $\mathbb{F}_q$ that bijection accounts for q Lagrangians, and the one remaining line, F itself, brings the total to the $q+1$ of item (3).

Exercise 22.3.1

1. Over $k = \mathbb{F}_2$, show that every symmetric bilinear form on k^2 satisfies the sign rule $\varphi(x, y) = -\varphi(y, x)$. Then determine which of the eight symmetric bilinear forms on $\mathbb{F}_2^2$ are alternating in the sense of definition 22.3.1, and say how the count is consistent with lemma 16.1.4(2).

2. Let $\operatorname{char} k \neq 2$ and let φ be a nondegenerate alternating form on a k-vector space of dimension n, with matrix A in some basis. Using $\det(A^t) = \det(A)$ and $A^t = -A$, show directly that n is even. Then explain which step of your argument fails over $\mathbb{F}_2$, and why the conclusion nevertheless still holds there.

3. Let k be any field. Show that a bilinear form φ on k^n is alternating if and only if there are scalars a_{ij} for $1 \leq i < j \leq n$ with

$$\varphi(x, y) = \sum_{i<j} a_{ij}\left(x_i y_j - x_j y_i\right)$$

   and that the a_{ij} are then determined by φ. Deduce that over $k = \mathbb{F}_q$ there are exactly $q^{n(n-1)/2}$ alternating forms on k^n.

4. Let $\operatorname{char} k \neq 2$. Show that every bilinear form φ on V is uniquely $\varphi = s + a$ with s symmetric and a alternating, and identify s and a in terms of φ and $(x, y) \mapsto \varphi(y, x)$. Then show the statement fails over $\mathbb{F}_2$ by exhibiting a bilinear form on $\mathbb{F}_2^2$ that is not such a sum.

5. Let (V, φ) be a symplectic space of dimension $2m$ and let L be a Lagrangian subspace with basis $e_1, \dots, e_m$. Show that there exist $f_1, \dots, f_m \in V$ making $e_1, f_1, \dots, e_m, f_m$ a symplectic basis of (V, φ). (Suggestion: for each i apply proposition 22.3.8(1) to the span of the e_j with $j \neq i$ to produce u_i with $\varphi(e_j, u_i) = 1$ when $i = j$ and 0 otherwise, then replace u_i by $u_i + \sum_j c_{ij} e_j$ for a suitable choice of scalars, noting that this leaves the previous values unchanged.) Deduce that for any two Lagrangian subspaces L and L' of (V, φ) there is an isometry of (V, φ) carrying L onto L'.

6. Let φ be an alternating form on V with radical $R = \operatorname{rad}(\varphi)$. Show that $\bar{\varphi}(x+R, y+R) := \varphi(x, y)$ is a well-defined alternating form on V/R, that it is nondegenerate, and that $\dim V/R$ is even. Compare the resulting statement with theorem 22.3.5(2).

7. Let (V, φ) be a symplectic space of dimension $2m$. Show that there is a chain of subspaces $0 = W_0 \subset W_1 \subset \cdots \subset W_m$ with $\dim W_i = i$ and each W_i totally isotropic, and that W_m is then a Lagrangian subspace. Show also that no such chain can be continued to a totally isotropic W_{m+1}.

22.4 Isotropy, the Hyperbolic Plane, and Witt Decomposition

A diagonalization records the values a form takes on n chosen vectors and says nothing about the vectors the form kills. This section organizes a symmetric form around those vectors instead, the v with $\varphi(v, v) = 0$. Two cases are extreme. A form may have no such vector other than 0. Or it may be spanned by two of them paired against each other, and then it is a single two-dimensional form which is the same over every field. The theorem of this section splits an arbitrary symmetric form into pieces of exactly those two kinds together with its radical, and so isolates the one summand in which the arithmetic of k is still visible.

The number of two-dimensional pieces and the isometry class of the remaining piece are determined by the form, but the argument for that is Witt cancellation and is deferred to section 22.5. What is available here without it is a statement about the subspaces on which the form vanishes identically: the largest dimension such a subspace can have is an invariant by construction, and it counts the two-dimensional pieces in a decomposition built from a subspace attaining it.

Every statement below breaks V into mutually orthogonal subspaces, and the notation for that is fixed first. It has already been used in box 22.1.9.

Definition 22.4.1: Orthogonal Sum of Forms

Let k be a field.

1. Let φ and ψ be bilinear forms on finite-dimensional k-vector spaces V and W. Their *orthogonal sum* is the bilinear form $\varphi \perp \psi$ on $V \oplus W$ given by

$$(\varphi \perp \psi)\big((v, w), (v', w')\big) := \varphi(v, v') + \psi(w, w')$$

   It is symmetric when φ and ψ are.

2. Let φ be a bilinear form on V and let $U_1, U_2 \subseteq V$ be subspaces with $V = U_1 \oplus U_2$ and $\varphi(u_1, u_2) = 0$ for all $u_1 \in U_1$ and $u_2 \in U_2$. Then V is the *internal orthogonal sum* of U_1 and U_2, written $V = U_1 \perp U_2$.

Both constructions iterate to finitely many summands.

The two halves of the definition describe one situation from outside and from inside. If $V = U_1 \perp U_2$ internally, then $(u_1, u_2) \mapsto u_1 + u_2$ is an isometry from $\varphi|_{U_1} \perp \varphi|_{U_2}$ to φ in the sense of definition 22.1.10, because expanding $\varphi(u_1 + u_2, u_1' + u_2')$ by bilinearity leaves only $\varphi(u_1, u_1')$ and $\varphi(u_2, u_2')$, the two cross terms being zero by hypothesis. Concatenating a basis β of U_1 with a basis γ of U_2 gives a basis of V in which every entry $\varphi(\beta_i, \gamma_j)$ vanishes, so by definition 13.3.18 the matrix of φ is block diagonal with blocks $[\varphi|_{U_1}]_\beta$ and $[\varphi|_{U_2}]_\gamma$. The determinant of an orthogonal sum is therefore the product of the determinants of its summands, and by lemma 13.3.23 the sum is nondegenerate exactly when both summands are. The diagonal notation of section 22.2 is an orthogonal sum of lines,

$$\langle a_1, \dots, a_n \rangle = \langle a_1 \rangle \perp \cdots \perp \langle a_n \rangle$$

so theorem 22.2.1 says exactly that every symmetric form is an orthogonal sum of one-dimensional forms.

Definition 22.4.2: Isotropic and Anisotropic Vectors

Let k be a field, let V be a finite-dimensional k-vector space, let φ be a symmetric bilinear form on V, and write $q(v) := \varphi(v, v)$ for the associated quadratic form. A nonzero vector $v \in V$ is *isotropic* if $q(v) = 0$, and *anisotropic* if $q(v) \neq 0$. The form φ is *isotropic* if V contains an isotropic vector, and *anisotropic* otherwise, that is if

$$q(v) = 0 \implies v = 0$$

The zero vector is excluded deliberately: $q(0) = 0$ for every form, so admitting it would make every form isotropic. An anisotropic form is automatically nondegenerate, since a nonzero $v \in \operatorname{rad}(\varphi)$ (definition 13.3.22) satisfies $\varphi(v, x) = 0$ for every x and in particular $q(v) = 0$, which is an isotropic vector. Restricting an anisotropic form to a subspace leaves it anisotropic, since the defining implication is inherited by subsets, and hence leaves it nondegenerate.

A subspace on which φ vanishes identically, $\varphi(w, w') = 0$ for all $w, w' \in W$, is *totally isotropic* (definition 22.3.6). When $\operatorname{char} k \neq 2$ it suffices to check the quadratic form on W: if $q(w) = 0$ for every $w \in W$, then for $w, w' \in W$,

$$0 = q(w + w') = q(w) + 2\varphi(w, w') + q(w') = 2\varphi(w, w')$$

and dividing by 2 gives $\varphi(w, w') = 0$. A totally isotropic subspace is thus one all of whose nonzero vectors are isotropic.

Example 22.4.3: Isotropy of Diagonal Forms

Whether a form is isotropic depends on the field and not only on the diagonal entries.

1. $\langle 1, 1\rangle$ over $\mathbb{R}$ is anisotropic: $x^2 + y^2 = 0$ forces $x^2 = -y^2$, and a sum of two nonnegative reals vanishes only when both do.
2. $\langle 1, 1\rangle$ over $\mathbb{F}_5$ is isotropic. The vector $(2, 1)$ is nonzero and $2^2 + 1^2 = 5 = 0$ in $\mathbb{F}_5$. Over $\mathbb{F}_3$ the squares are 0 and 1, so $x^2 + y^2$ takes only the values $0, 1, 2$ and vanishes only at $x = y = 0$: the same diagonal entries give an anisotropic form.
3. $\langle 1, -1\rangle$ is isotropic over every field, since $q(1, 1) = 1 - 1 = 0$ and $(1, 1) \neq 0$.
4. Let d be a squarefree integer with $d \neq 1$ and let $F = \mathbb{Q}(\sqrt{d})$. By example 22.1.11 the trace form of $F/\mathbb{Q}$ is $\langle 2, 2d\rangle$ in the basis $(1, \sqrt{d})$, with quadratic form $q(x, y) = 2x^2 + 2dy^2$. A zero with $y = 0$ forces $x = 0$, so an isotropic vector has $y \neq 0$, and dividing $x^2 = -dy^2$ by y^2 gives $-d = (x/y)^2$. The trace form is therefore isotropic exactly when $-d$ is a square in $\mathbb{Q}$. A positive squarefree integer is a rational square only when it equals 1, so this happens for $d = -1$ alone: of all the quadratic fields, only $\mathbb{Q}(i)$ has isotropic trace form, and $q(1, 1) = 2 - 2 = 0$ exhibits the vector.

Item (3) is the smallest isotropic form, and it is worth presenting in coordinates that exhibit its two isotropic directions at once. In place of one distinguished isotropic vector it carries two, paired against each other, and that pairing is what keeps the form nondegenerate.

Definition 22.4.4: The Hyperbolic Plane

Let k be a field and let φ be a symmetric bilinear form on a k-vector space U. A *hyperbolic pair* in U is a pair of vectors $e, f \in U$ with

$$\varphi(e,e) = \varphi(f,f) = 0, \qquad \varphi(e,f) = 1$$

The form φ is a *hyperbolic plane* if $\dim U = 2$ and U has a basis which is a hyperbolic pair, so that the matrix of φ in that basis is $\left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$. A hyperbolic plane is written H. A subspace of (V, φ) is *a hyperbolic plane in* V when the restriction of φ to it is one, and the orthogonal sum of m copies,

$$mH := \underbrace{H \perp \cdots \perp H}_{m}$$

is the *hyperbolic space* of rank $2m$, with $0H := 0$.

Writing H for *the* hyperbolic plane is legitimate because any two are isometric: if (e, f) and (e', f') are hyperbolic pairs spanning U and U', then the linear isomorphism carrying $e \mapsto e'$ and $f \mapsto f'$ gives the two forms the same matrix in the corresponding bases, and by lemma 13.3.19 a form is recovered from its matrix in a basis, so the isomorphism is an isometry. That matrix has determinant $-1 \neq 0$, so a hyperbolic plane is nondegenerate by lemma 13.3.23, and its discriminant (definition 22.1.7) is the square class of -1.

Example 22.4.5: The Hyperbolic Plane as a Diagonal Form

Let $\operatorname{char} k \neq 2$ and let H have hyperbolic pair (e, f). Put

$$x := e + \tfrac{1}{2}f, \qquad y := e - \tfrac{1}{2}f$$

Then $x + y = 2e$ and $x - y = f$, and since $2 \neq 0$ these recover e and f, so (x, y) is again a basis of H. Expanding by bilinearity and using $\varphi(e,e) = \varphi(f,f) = 0$ together with $\varphi(e,f) = 1$,

$$q(x) = 2 \cdot \tfrac{1}{2}\varphi(e,f) = 1, \qquad q(y) = -2 \cdot \tfrac{1}{2}\varphi(e,f) = -1$$

$$\varphi(x,y) = -\tfrac{1}{2}\varphi(e,f) + \tfrac{1}{2}\varphi(f,e) = 0$$

so (x, y) is an orthogonal basis and

$$H \cong \langle 1, -1 \rangle$$

The product of the diagonal entries is -1, which agrees through corollary 22.2.3 with the determinant computed above. Over $\mathbb{R}$ this is the indefinite plane $x^2 - y^2$. Over a field containing an i with $i^2 = -1$, proposition 22.2.2 applied with $c_1 = 1$ and $c_2 = i$ gives $\langle 1, -1 \rangle \cong \langle 1, 1 \rangle$. Combined with example 22.2.4, every nondegenerate binary form over an algebraically closed field of characteristic other than 2 is a hyperbolic plane.

The hyperbolic plane was assembled from a pair of isotropic vectors. The next result runs the construction in reverse: a single isotropic vector of a nondegenerate form always has a partner, and the plane the two span splits off as an orthogonal summand. The splitting is what makes the construction iterable, and iterating it is the proof of the decomposition theorem.

Proposition 22.4.6: Every Isotropic Vector Lies in a Hyperbolic Plane

Let k be a field with $\operatorname{char} k \neq 2$, let V be a finite-dimensional k-vector space, and let φ be a nondegenerate symmetric bilinear form on V. Let $v \in V$ be isotropic. Then there is an $f \in V$ such that (v, f) is a hyperbolic pair, and, writing $H_v := \operatorname{span}(v, f)$,

$$V = H_v \perp H_v^{\perp}$$

where H_v is a hyperbolic plane in V and $\varphi|_{H_v^{\perp}}$ is nondegenerate. In particular a nondegenerate symmetric form is anisotropic if and only if V contains no hyperbolic plane.

***Proof*:**

Step 1: producing the partner. An isotropic vector is nonzero, so $v \notin \operatorname{rad}(\varphi) = 0$ and there is a $w_0 \in V$ with $\varphi(v, w_0) \neq 0$. Replacing w_0 by $w := \varphi(v, w_0)^{-1} w_0$ gives $\varphi(v, w) = 1$. Set

$$f := w - \frac{q(w)}{2}\, v$$

which uses $\operatorname{char} k \neq 2$ to divide. Using $q(v) = \varphi(v, v) = 0$,

$$\varphi(v, f) = \varphi(v, w) - \frac{q(w)}{2}\varphi(v, v) = 1$$

and, expanding by bilinearity and symmetry,

$$\varphi(f, f) = \varphi(w, w) - 2 \cdot \frac{q(w)}{2}\varphi(w, v) + \frac{q(w)^2}{4}\varphi(v, v) = q(w) - q(w) = 0$$

so (v, f) is a hyperbolic pair. The two vectors are linearly independent: a relation $f = \lambda v$ would give $1 = \varphi(v, f) = \lambda\varphi(v, v) = 0$, and $v \neq 0$. Hence $\dim H_v = 2$, the matrix of $\varphi|_{H_v}$ in the basis (v, f) is $\left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$, and H_v is a hyperbolic plane in V, nondegenerate by lemma 13.3.23.

Step 2: the splitting. Let $x \in V$ and put

$$x' := x - \varphi(x, f)\, v - \varphi(x, v)\, f$$

Using the values of φ on the pair (v, f),

$$\varphi(x', v) = \varphi(x, v) - \varphi(x, f)\varphi(v, v) - \varphi(x, v)\varphi(f, v) = \varphi(x, v) - \varphi(x, v) = 0$$

$$\varphi(x', f) = \varphi(x, f) - \varphi(x, f)\varphi(v, f) - \varphi(x, v)\varphi(f, f) = \varphi(x, f) - \varphi(x, f) = 0$$

so x' is orthogonal to v and to f, hence to their span by lemma 13.3.21, that is $x' \in H_v^{\perp}$. Since $x - x' \in H_v$, this gives $V = H_v + H_v^{\perp}$. The sum is direct: an element $u \in H_v \cap H_v^{\perp}$ lies in H_v and is orthogonal to every element of H_v, hence lies in $\operatorname{rad}(\varphi|_{H_v})$, which is zero by Step 1. The two summands are orthogonal by the definition of $H_v^{\perp}$ (definition 13.3.20), so $V = H_v \perp H_v^{\perp}$ in the sense of definition 22.4.1.

Step 3: the complement is nondegenerate. Let $x \in \operatorname{rad}\big(\varphi|_{H_v^{\perp}}\big)$, so $\varphi(x, y) = 0$ for every $y \in H_v^{\perp}$. Since $x \in H_v^{\perp}$ and φ is symmetric, $\varphi(x, y) = 0$ for every $y \in H_v$ as well. By Step 2 each element of V is a sum of one vector of each kind, so $\varphi(x, -)$ vanishes on V and $x \in \operatorname{rad}(\varphi) = 0$.

Step 4: the equivalence. If φ is isotropic, Steps 1 to 3 produce a hyperbolic plane in V. Conversely, a hyperbolic plane in V contains a vector e with $\varphi(e,e) = 0$ and $e \neq 0$, the latter because $\varphi(e,f) = 1$ forbids $e = 0$, so φ is isotropic. Hence V contains a hyperbolic plane precisely when φ is isotropic, completing the proof.

The proposition converts one-dimensional information, the existence of a single vector with $q(v) = 0$, into a two-dimensional summand of known isometry type. What is left of φ is the restriction $\varphi|_{H_v^\perp}$, again nondegenerate and of dimension two smaller, so the exchange can be repeated until no isotropic vector remains. In dimension two there is nothing left to repeat, and the statement becomes a criterion.

Example 22.4.7: Binary Forms and Hyperbolic Planes

Let $\operatorname{char} k \neq 2$ and let $\varphi \cong \langle a, b\rangle$ be a nondegenerate form of rank 2, so that $a, b \in k^\times$ by proposition 22.2.2. If φ is isotropic then proposition 22.4.6 puts a hyperbolic plane H_v inside V, and $\dim H_v = 2 = \dim V$ forces $H_v = V$, so $\varphi \cong H$. A nondegenerate binary form is therefore either anisotropic or a hyperbolic plane.

The entries determine which of the two occurs. An isotropic vector for $q(x,y) = ax^2 + by^2$ has both coordinates nonzero, since $y = 0$ forces $ax^2 = 0$ and hence $x = 0$, and symmetrically $x = 0$ forces $y = 0$. Multiplying $ax^2 = -by^2$ by b and dividing by x^2 gives

$$-ab = \left(\frac{by}{x}\right)^2$$

Conversely, if $-ab = c^2$ then $c \neq 0$ because $ab \neq 0$, and the nonzero vector (b, c) satisfies $ab^2 + bc^2 = ab^2 - ab \cdot b = 0$. So $\langle a, b\rangle$ is a hyperbolic plane exactly when $-ab$ is a square, which by corollary 22.2.3 says exactly that $\operatorname{disc}(\varphi)$ is the class of -1. For instance $\langle 1, 1\rangle$ over $\mathbb{F}_5$ has $-ab = -1 = 4 = 2^2$, so it is the hyperbolic plane, whereas over $\mathbb{F}_3$ the element $-1 = 2$ is not a square and the form is anisotropic, matching example 22.4.3.

Iterating proposition 22.4.6 splits a hyperbolic plane off a nondegenerate form as long as one isotropic vector can be found, and the process stops only when none can. A degenerate form has to be handled first: the nonzero vectors of its radical are isotropic and orthogonal to all of V, so no partner exists for them and proposition 22.4.6 does not apply. Splitting the radical off and then iterating gives the decomposition.

Theorem 22.4.8: Witt Decomposition

Let k be a field with $\operatorname{char} k \neq 2$, let V be a finite-dimensional k-vector space and let φ be a symmetric bilinear form on V. Then there are an integer $m \geq 0$, subspaces $H_1, \ldots, H_m \subseteq V$ each of which is a hyperbolic plane in V, and a subspace $V_{\mathrm{an}} \subseteq V$ on which φ is anisotropic, such that

$$V = \operatorname{rad}(\varphi) \perp H_1 \perp \cdots \perp H_m \perp V_{\mathrm{an}}$$

Writing $r = \dim \operatorname{rad}(\varphi)$, this reads

$$\varphi \cong \langle \underbrace{0, \ldots, 0}_{r}\rangle \perp mH \perp \varphi|_{V_{\mathrm{an}}}$$

***Proof*:**
Step 1: splitting off the radical. Write $R := \operatorname{rad}(\varphi)$ and choose a basis of R. By theorem 13.1.8 it extends to a basis of V. Let V' be the span of the added vectors, so that $V = R \oplus V'$. Every element of R is orthogonal to every element of V, in particular to every element of V', so $V = R \perp V'$ in the sense of definition 22.4.1, and $\varphi|_R = 0$.

The restricted form $\varphi|_{V'}$ is nondegenerate. Let $x \in V'$ satisfy $\varphi(x, y) = 0$ for all $y \in V'$. Since also $\varphi(x, y) = 0$ for all $y \in R$, and $V = R + V'$, the map $\varphi(x, -)$ vanishes on V, so $x \in R \cap V' = 0$.

Step 2: splitting hyperbolic planes off a nondegenerate form. It suffices to prove that a nondegenerate symmetric form ψ on a space U decomposes as $U = H_1 \perp \cdots \perp H_m \perp U_{\text{an}}$ with each H_i a hyperbolic plane in U and $\psi|_{U_{\text{an}}}$ anisotropic. Applying this to $\varphi|_{V'}$ and appending Step 1 gives the theorem, because every summand of V' is orthogonal to R.

Induct on $\dim U$. If ψ is anisotropic, take $m = 0$ and $U_{\text{an}} = U$. This covers $\dim U = 0$, where there is nothing to decompose, and $\dim U = 1$: an isotropic vector v would span U, and then $\psi(v, \lambda v) = \lambda\psi(v, v) = 0$ for every λ would put v in $\operatorname{rad}(\psi)$, contradicting nondegeneracy. Otherwise ψ has an isotropic vector, and proposition 22.4.6 gives a hyperbolic plane $H_1 \subseteq U$ with

$$U = H_1 \perp H_1^{\perp}$$

and $\psi|_{H_1^{\perp}}$ nondegenerate. Since $\dim H_1^{\perp} = \dim U - 2$, the inductive hypothesis applies to $H_1^{\perp}$ and gives $H_1^{\perp} = H_2 \perp \cdots \perp H_m \perp U_{\text{an}}$ with each H_i a hyperbolic plane and $\psi|_{U_{\text{an}}}$ anisotropic. Every one of these subspaces lies in $H_1^{\perp}$ and is therefore orthogonal to H_1, so the displayed decomposition of U refines to $U = H_1 \perp H_2 \perp \cdots \perp H_m \perp U_{\text{an}}$.

Step 3: the diagonal reading. Take the chosen basis of R, a hyperbolic pair in each H_i, and an orthogonal basis of V_{an}, which exists by theorem 22.2.1. Their concatenation is a basis of V, and by the block-diagonal description following definition 22.4.1 the matrix of φ in it is the block sum of the zero matrix of size r, m copies of $\left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$, and a diagonal matrix. That is the asserted isometry, completing the proof.

The three summands answer three different questions about φ. The radical is where the form carries no information at all, and its dimension is already known to be an invariant by proposition 22.2.2. The hyperbolic part carries no information beyond its rank, since any two hyperbolic planes over k are isometric, whatever k is. Everything that distinguishes φ from other forms of the same rank is therefore concentrated in $\varphi|_{V_{\text{an}}}$, and how large that summand can be is a property of the field. Over an algebraically closed field it has dimension at most 1: a two-dimensional subspace of an anisotropic space is nondegenerate, hence a hyperbolic plane by example 22.4.5, hence isotropic, which contradicts anisotropy. This recovers example 22.2.4, where the classification collapsed to the rank. Over a finite field the bound is one step weaker, and a counting argument gives it.

Example 22.4.9: Isotropy over a Finite Field
Let $k = \mathbb{F}_q$ with q odd.

Every equation $ax^2 + by^2 = c$ with $a, b \in k^\times$ and $c \in k$ has a solution. As recorded in example 22.2.5, the squaring map on $\mathbb{F}_q^\times$ has kernel $\{\pm 1\}$, which has two elements because q is odd, so there are $(q-1)/2$ nonzero squares and the set $\left\{x^2 \;\middle|\; x \in k\right\}$ has $(q+1)/2$ elements. Multiplication by a is

a bijection of k, so

$$S := \left\{ax^2 \;\middle|\; x \in k\right\}, \qquad T := \left\{c - by^2 \;\middle|\; y \in k\right\}$$

each have $(q+1)/2$ elements. Their sizes add to $q+1 > q = |k|$, so S and T are not disjoint, and a common element gives x, y with $ax^2 = c - by^2$.

Every nondegenerate form of rank at least 3 *over* $\mathbb{F}_q$ *is isotropic.* Diagonalize by theorem 22.2.1 as $\varphi \cong \langle a_1, \ldots, a_n\rangle$ in an orthogonal basis $w_1, \ldots, w_n$. Nondegeneracy makes every a_i nonzero by the last claim of proposition 22.2.2. With $n \geq 3$, solve $a_1x^2 + a_2y^2 = -a_3$ by the previous paragraph and set $v := xw_1 + yw_2 + w_3$, which is nonzero because its third coordinate is 1. Then $q(v) = a_1x^2 + a_2y^2 + a_3 = 0$.

Consequently, in the Witt decomposition of a form over $\mathbb{F}_q$ the anisotropic summand has dimension at most 2. Both values occur: $\langle 1\rangle$ is anisotropic of rank 1, and if $\epsilon \in \mathbb{F}_q^\times$ is a non-square then $\langle 1, -\epsilon\rangle$ has $-ab = \epsilon$ a non-square, so it is anisotropic of rank 2 by example 22.4.7.

Theorem 22.4.8 produces some number m of hyperbolic planes, and nothing in its proof says that a different sequence of choices produces the same number. A quantity that is visibly independent of all choices is available instead, and the rest of the section relates the two.

Definition 22.4.10: The Witt Index

Let k be a field, let V be a finite-dimensional k-vector space and let φ be a symmetric bilinear form on V. The *Witt index* of φ is

$$\nu(\varphi) := \max\left\{\dim W \mid W \subseteq V \text{ a totally isotropic subspace}\right\}$$

The maximum exists: the zero subspace is totally isotropic, so the set of dimensions is nonempty, and it is bounded above by $\dim V$.

The Witt index is an invariant of φ by construction, since it is defined without reference to a basis or a decomposition, and isometries carry totally isotropic subspaces to totally isotropic subspaces of the same dimension. It is zero exactly when φ is anisotropic, since a one-dimensional totally isotropic subspace is the span of an isotropic vector. The next result computes it against theorem 22.4.8.

Proposition 22.4.11: The Witt Index Counts Hyperbolic Planes

Let k be a field with $\operatorname{char} k \neq 2$, let V be a finite-dimensional k-vector space of dimension n and let φ be a nondegenerate symmetric bilinear form on V.

1. Let $W \subseteq V$ be totally isotropic with $\dim W = d$. Then there are hyperbolic planes $H_1, \dots, H_d$ in V, with hyperbolic pairs (e_i, f_i) such that $e_1, \dots, e_d$ is a basis of W, and a subspace U with $\varphi|_U$ nondegenerate, such that

$$V = H_1 \perp \cdots \perp H_d \perp U$$

   In particular $2d \leq n$, which is also proposition 22.3.8(2) applied to the totally isotropic W.

2. In the situation of (1), the form $\varphi|_U$ is anisotropic if and only if W is maximal among the totally isotropic subspaces of V.

3. $\nu(\varphi) \leq n/2$; some decomposition as in theorem 22.4.8 has exactly $\nu(\varphi)$ hyperbolic planes; and every such decomposition has at most $\nu(\varphi)$ of them.

***Proof*:**

1. Induct on d. For $d = 0$ take $U = V$, and there is nothing to prove. Let $d \geq 1$ and choose a basis $w_1, \dots, w_d$ of W. Since W is totally isotropic and $w_1 \neq 0$, the vector w_1 is isotropic, so proposition 22.4.6 gives an f_1 making (w_1, f_1) a hyperbolic pair, with $H_1 = \operatorname{span}(w_1, f_1)$ a hyperbolic plane, $V = H_1 \perp H_1^\perp$ and $\varphi|_{H_1^\perp}$ nondegenerate.

   Consider the linear map $\lambda\colon W \to k$ defined by $\lambda(x) = \varphi(x, f_1)$. It is nonzero, since $\lambda(w_1) = \varphi(w_1, f_1) = 1$, so $\ker\lambda$ has dimension $d-1$. Moreover $\ker\lambda = W \cap H_1^\perp$: every $x \in W$ satisfies $\varphi(x, w_1) = 0$ because W is totally isotropic and $w_1 \in W$, so x lies in $H_1^\perp$ exactly when in addition $\varphi(x, f_1) = 0$, using lemma 13.3.21 to reduce orthogonality to H_1 to orthogonality to the two spanning vectors.

   Set $W_1 := W \cap H_1^\perp$, a totally isotropic subspace of $H_1^\perp$ of dimension $d-1$. The inductive hypothesis, applied to the nondegenerate form $\varphi|_{H_1^\perp}$, gives hyperbolic planes $H_2, \dots, H_d$ in $H_1^\perp$ with hyperbolic pairs (e_i, f_i) whose vectors $e_2, \dots, e_d$ form a basis of W_1, and a subspace U with $\varphi|_U$ nondegenerate and $H_1^\perp = H_2 \perp \cdots \perp H_d \perp U$. All of these lie in $H_1^\perp$ and are therefore orthogonal to H_1, so $V = H_1 \perp H_2 \perp \cdots \perp H_d \perp U$. Finally $w_1 \notin W_1$, because $\lambda(w_1) = 1$ while λ vanishes on W_1, so $w_1, e_2, \dots, e_d$ are d linearly independent vectors of W and hence a basis of it. Setting $e_1 := w_1$ gives the statement. The dimension count is $n = 2d + \dim U \geq 2d$.

2. Write $\mathcal{H} := H_1 \perp \cdots \perp H_d$, so $V = \mathcal{H} \oplus U$ and $W = \operatorname{span}(e_1, \dots, e_d)$ by (1).

   Suppose first that $\varphi|_U$ is anisotropic, and let $W' \supseteq W$ be totally isotropic. Take $x \in W'$ and write $x = \sum_i (a_i e_i + b_i f_i) + u$ with $u \in U$. For each j the vector e_j lies in $W \subseteq W'$, so $\varphi(x, e_j) = 0$. On the other hand the only summand of x not orthogonal to e_j is $b_j f_j$, whence $\varphi(x, e_j) = b_j$. So every b_j vanishes and $x = \sum_i a_i e_i + u$. Expanding $q(x)$ by bilinearity, the terms $\varphi(e_i, e_j)$ vanish for all i, j and the terms $\varphi(e_i, u)$ vanish because $e_i \in \mathcal{H}$ and $u \in U$, leaving $q(x) = q(u)$. As $x \in W'$ is isotropic or zero, $q(u) = 0$, so $u = 0$ by anisotropy and $x \in W$. Hence $W' = W$ and W is maximal.

Suppose instead that $\varphi|_U$ is isotropic, say $u \in U$ is nonzero with $q(u) = 0$. Then $W + ku$ is totally isotropic: expanding $\varphi(w + au,\, w' + a'u)$ leaves $\varphi(w, w') = 0$, the cross terms $\varphi(w, u) = 0$ and $\varphi(u, w') = 0$ since $W \subseteq \mathcal{H}$ and $u \in U$, and $aa'q(u) = 0$. It strictly contains W, because $u \notin \mathcal{H} \supseteq W$, so W is not maximal. The two implications together give the stated equivalence.

3. Applying (1) to a totally isotropic subspace of dimension $\nu(\varphi)$ gives $2\nu(\varphi) \leq n$. That subspace is maximal, being of largest dimension, so (2) makes the corresponding U anisotropic and
$$V = H_1 \perp \cdots \perp H_{\nu(\varphi)} \perp U$$
is a decomposition as in theorem 22.4.8, with radical zero and exactly $\nu(\varphi)$ hyperbolic planes. Conversely, let $V = H'_1 \perp \cdots \perp H'_m \perp V_{\text{an}}$ be any such decomposition, with hyperbolic pair (e'_i, f'_i) in H'_i. The vectors $e'_1, \ldots, e'_m$ lie in distinct summands of a direct sum and are nonzero, hence are linearly independent, and their span is totally isotropic: $\varphi(e'_i, e'_j) = 0$ for $i \neq j$ by orthogonality of the summands and $\varphi(e'_i, e'_i) = 0$ by the definition of a hyperbolic pair. So $m \leq \nu(\varphi)$, completing the proof.

Part (3) does not say that every decomposition has $\nu(\varphi)$ hyperbolic planes. It produces one that does and bounds all the others by that number, and no argument given above closes the gap: comparing two decompositions of the same space requires an isometry between them, and producing one is what Witt cancellation gives. Section 22.5 proves it, and with it every decomposition of a nondegenerate φ in theorem 22.4.8 has $m = \nu(\varphi)$ and an anisotropic summand determined up to isometry. Nondegeneracy is needed there: the Witt index of a degenerate form counts its radical as well, and running the same statement on a complement of the radical gives $m = \nu(\varphi) - \dim \operatorname{rad}(\varphi)$ instead, by exercise 22.4.1 (4). What is established here is that $\nu(\varphi)$ bounds the number of hyperbolic planes in any decomposition, and that the bound is attained when φ is nondegenerate. Over $\mathbb{R}$ that number can be computed directly from a diagonalization.

Example 22.4.12: The Witt Index over the Reals

Let φ be nondegenerate on a real vector space V of dimension n. By theorem 22.2.1 and proposition 22.2.2 applied with $c_i = |a_i|^{-1/2}$, there is a basis $u_1, \ldots, u_p, w_1, \ldots, w_r$ of V, orthogonal for φ, with $q(u_i) = 1$ and $q(w_j) = -1$ and $p + r = n$.

Pairing the vectors two at a time, $\operatorname{span}(u_i, w_i)$ carries the form $\langle 1, -1\rangle$, which is a hyperbolic plane by example 22.4.5. Doing this for $i \leq \min(p, r)$ and collecting the remaining $|p - r|$ basis vectors, which all have the same value of q, gives
$$\varphi \cong \min(p, r)\, H \perp \langle \pm 1, \ldots, \pm 1\rangle$$
with $|p - r|$ entries in the last summand, all equal to $+1$ if $p > r$ and all to -1 if $r > p$. That summand is anisotropic, a sum of $|p - r|$ squares being zero only when each is.

The Witt index is $\min(p, r)$. It is at least that, by the span of the $u_i + w_i$ for $i \leq \min(p, r)$, which are linearly independent and satisfy $q(u_i + w_i) = 1 - 1 = 0$ and $\varphi(u_i + w_i, u_j + w_j) = 0$ for $i \neq j$. For the reverse bound let W be totally isotropic and let $P := \operatorname{span}(u_1, \ldots, u_p)$. A nonzero $x \in P$ has $q(x) = \sum_i x_i^2 > 0$, so x is not isotropic and $W \cap P = 0$, forcing $\dim W \leq n - p = r$. The same argument with $N := \operatorname{span}(w_1, \ldots, w_r)$, where $q(y) = -\sum_j y_j^2 < 0$ for $y \neq 0$, gives $\dim W \leq n - r = p$. Hence $\nu(\varphi) = \min(p, r)$, in agreement with proposition 22.4.11(3).

Since $\nu(\varphi)$ and n depend only on φ, the unordered pair $\{p, r\}$ does too. Which of the two is p is not settled by this argument, and that remaining bit is Sylvester's law of inertia, which needs the

order on $\mathbb{R}$ in a way the present chapter does not use.

Exercise 22.4.1

1. Let $\operatorname{char} k \neq 2$ and let $a \in k^\times$. Show that $\langle 1, -a\rangle$ is isotropic if and only if a is a square in k, and that in that case $\langle 1, -a\rangle \cong H$. Using that $\mathbb{Q}^\times/(\mathbb{Q}^\times)^2$ is infinite, deduce that $\mathbb{Q}$ carries infinitely many pairwise non-isometric anisotropic binary forms.

2. Let $\operatorname{char} k \neq 2$. Show that the hyperbolic plane represents every scalar: for each $c \in k$ there is $x \in H$ with $q(x) = c$. Deduce that a nondegenerate isotropic form on any V represents every element of k.

3. Let $\operatorname{char} k \neq 2$ and let $\varphi \cong \langle a, b\rangle$ be nondegenerate of rank 2. Using example 22.4.7, decide for which pairs (a, b) over $k = \mathbb{Q}$ the form is a hyperbolic plane, and exhibit one anisotropic and one hyperbolic example.

4. Let φ be a symmetric bilinear form on V with radical $R = \operatorname{rad}(\varphi)$, and let $\bar{\varphi}$ be the induced nondegenerate form on V/R from exercise 22.1.1 (5). Show that $\nu(\varphi) = \dim R + \nu(\bar{\varphi})$.

5. Let φ be nondegenerate on V and let $W \subseteq V$ be totally isotropic. Show that W is maximal among totally isotropic subspaces if and only if every isotropic vector of $W^\perp$ already lies in W.

6. Show that $\nu(\varphi \perp \psi) \geq \nu(\varphi) + \nu(\psi)$ for symmetric bilinear forms φ and ψ, and give an example over $\mathbb{R}$ where the inequality is strict.

7. Let q be odd and let φ be a nondegenerate symmetric bilinear form of odd rank n over $\mathbb{F}_q$. Show that $\nu(\varphi) = (n-1)/2$, so the Witt index over a finite field is forced by the rank alone when the rank is odd.

8. Let $V = \mathbb{F}_7^3$ with φ the standard dot product, so $q(x, y, z) = x^2 + y^2 + z^2$. Verify that $v = (3, 2, 1)$ is isotropic, run the construction in the proof of proposition 22.4.6 with $w = (0, 0, 1)$ to produce a partner f, compute a spanning vector of $H_v^\perp$, and write out the resulting decomposition $V = H_v \perp \langle a\rangle$. Is a a square in $\mathbb{F}_7$?

22.5 Witt Cancellation and Witt's Extension Theorem

Theorem 22.4.8 splits a form into an anisotropic part, a sum of hyperbolic planes and the radical, and each step of that construction rests on a choice: which isotropic vector to complete to a hyperbolic plane, and which complement to continue in. Nothing proved so far forces two runs of the construction to return the same number of hyperbolic planes, or isometric anisotropic parts. This section proves that they do. The result that delivers it is a cancellation law for orthogonal sums: a nondegenerate summand common to both sides of an isometry may be deleted. The section then proves Witt's extension theorem, which says that an isometry between two subspaces of a nondegenerate space is always the restriction of an isometry of the whole space. Both rest on one construction, the reflection in an anisotropic vector, which this book has not yet defined.

Throughout, $q(x) = \varphi(x, x)$ is the quadratic form attached to a symmetric bilinear form φ by proposition 22.1.2, and $\varphi \perp \psi$ is the orthogonal sum of definition 22.4.1. One further fact about that operation is used repeatedly below: the bijections $\big(v, (w, x)\big) \mapsto \big((v, w), x\big)$ and $(v, w) \mapsto (w, v)$ leave

the defining expression unchanged, so they are isometries, and an orthogonal sum of several forms may be regrouped and reordered freely without changing its isometry class.

Every theorem below produces an isometry of a space with itself, so those isometries need a name.

Definition 22.5.1: The Orthogonal Group of a Form

Let k be a field, let V be a finite-dimensional k-vector space and let φ be a symmetric bilinear form on V. The *orthogonal group* of φ is

$$O(\varphi) = \{\sigma \colon V \to V \mid \sigma \text{ is an isometry of } (V, \varphi) \text{ with itself}\}$$

where isometry is meant in the sense of definition 22.1.10, and the operation is composition of maps.

That this is a group takes nothing beyond the definition: id_V is an isometry; a composite of linear bijections preserving φ is again a linear bijection preserving φ; and if σ is an isometry then so is σ^{-1}, since $\varphi(\sigma^{-1}x, \sigma^{-1}y) = \varphi(\sigma\sigma^{-1}x, \sigma\sigma^{-1}y) = \varphi(x, y)$ for all $x, y \in V$.

The construction used throughout the section takes an anisotropic vector (definition 22.4.2) and produces the isometry that negates it and fixes everything orthogonal to it. Over $\mathbb{R}$ with the dot product this is the familiar mirror reflection in the hyperplane $\{v\}^{\perp}$, and the formula below is the one from that setting, written so that it makes sense over any field in which $q(v)$ is invertible.

Definition 22.5.2: Reflection in an Anisotropic Vector

Let k be a field with $\operatorname{char} k \neq 2$, let V be a finite-dimensional k-vector space with a symmetric bilinear form φ, and let $v \in V$ be anisotropic, that is $q(v) = \varphi(v, v) \neq 0$. The *reflection in* v is the map $\tau_v \colon V \to V$ given by

$$\tau_v(x) = x - \frac{2\varphi(v, x)}{\varphi(v, v)}\, v$$

The denominator is nonzero precisely because v is anisotropic, which is why the construction is available for anisotropic vectors and for no others.

The properties that make τ_v useful are all direct computations with the defining formula.

Lemma 22.5.3: Reflections Are Involutive Isometries

Let k be a field with $\operatorname{char} k \neq 2$, let V be a finite-dimensional k-vector space with a symmetric bilinear form φ, and let $v \in V$ be anisotropic. Then:

1. $V = kv \oplus \{v\}^{\perp}$;
2. τ_v is linear, $\tau_v(v) = -v$, and $\tau_v(x) = x$ for every $x \in \{v\}^{\perp}$;
3. τ_v preserves φ and satisfies $\tau_v \circ \tau_v = \mathrm{id}_V$. In particular $\tau_v \in O(\varphi)$ and $\tau_v^{-1} = \tau_v$;
4. $\tau_{\lambda v} = \tau_v$ for every $\lambda \in k^{\times}$;
5. $\det \tau_v = -1$. In particular $\tau_v \neq \mathrm{id}_V$, so τ_v has order exactly 2 in $O(\varphi)$.

***Proof*:**
Write $a = q(v) = \varphi(v, v) \neq 0$ throughout. The set $\{v\}^\perp$ is a subspace by lemma 13.3.21.

1. Every $x \in V$ decomposes as

$$x = \frac{\varphi(v,x)}{a}\, v + \left(x - \frac{\varphi(v,x)}{a}\, v\right)$$

and applying $\varphi(v, -)$ to the bracketed vector gives $\varphi(v,x) - \frac{\varphi(v,x)}{a}\, a = 0$, so that vector lies in $\{v\}^\perp$ and $V = kv + \{v\}^\perp$. The sum is direct: if $\lambda v \in \{v\}^\perp$ then $0 = \varphi(v, \lambda v) = \lambda a$, and $a \neq 0$ forces $\lambda = 0$.

2. Linearity holds because $x \mapsto \varphi(v, x)$ is linear and τ_v subtracts a scalar multiple of the fixed vector v. Substituting $x = v$ gives $\tau_v(v) = v - \frac{2a}{a} v = v - 2v = -v$, and if $\varphi(v, x) = 0$ then the subtracted term vanishes, so $\tau_v(x) = x$.

3. For $x, y \in V$, expanding by bilinearity and symmetry,

$$\varphi\big(\tau_v(x), \tau_v(y)\big) = \varphi(x,y) - \frac{2\varphi(v,y)}{a}\varphi(x,v) - \frac{2\varphi(v,x)}{a}\varphi(v,y) + \frac{4\varphi(v,x)\varphi(v,y)}{a^2}\, a$$

The last three terms are $-2c$, $-2c$ and $+4c$ with $c = \varphi(v,x)\varphi(v,y)/a$, so they cancel and $\varphi(\tau_v(x), \tau_v(y)) = \varphi(x, y)$. For the involution, first

$$\varphi\big(v, \tau_v(x)\big) = \varphi(v,x) - \frac{2\varphi(v,x)}{a}\, a = -\varphi(v,x)$$

so

$$\tau_v\big(\tau_v(x)\big) = \tau_v(x) - \frac{2\varphi(v,\tau_v(x))}{a}\, v = x - \frac{2\varphi(v,x)}{a}\, v + \frac{2\varphi(v,x)}{a}\, v = x$$

Hence τ_v is a linear bijection preserving φ, which is an isometry of (V, φ) with itself in the sense of definition 22.1.10, so $\tau_v \in O(\varphi)$ by definition 22.5.1 and $\tau_v^{-1} = \tau_v$.

4. For $\lambda \neq 0$ the vector λv is anisotropic, since $q(\lambda v) = \lambda^2 a \neq 0$, and

$$\tau_{\lambda v}(x) = x - \frac{2\varphi(\lambda v, x)}{\lambda^2 a}\, \lambda v = x - \frac{2\lambda^2 \varphi(v,x)}{\lambda^2 a}\, v = \tau_v(x)$$

5. By (1) choose a basis $v, z_2, \ldots, z_n$ of V with $z_2, \ldots, z_n$ a basis of $\{v\}^\perp$. By (2) the matrix of τ_v in this basis is $\operatorname{diag}(-1, 1, \ldots, 1)$, whose determinant is -1. Since $\operatorname{char} k \neq 2$ this is not 1, so $\tau_v \neq \mathrm{id}_V$, and $\tau_v^2 = \mathrm{id}_V$ from (3) then makes its order exactly 2, completing the proof.

Example 22.5.4: Reflections of $\langle 1, 1\rangle$ over $\mathbb{Q}$
Let $k = \mathbb{Q}$ and let φ be the form on $V = \mathbb{Q}^2$ with matrix the identity in the standard basis e_1, e_2, so $q(x_1, x_2) = x_1^2 + x_2^2$ and $\varphi \cong \langle 1, 1\rangle$.

Take $v = e_1$, with $q(e_1) = 1$. Then $\tau_{e_1}(x) = x - 2\varphi(e_1, x)e_1 = x - 2x_1 e_1$, so $\tau_{e_1}(e_1) = -e_1$ and $\tau_{e_1}(e_2) = e_2$, and the matrix of τ_{e_1} is $\operatorname{diag}(-1, 1)$.

Take instead $v = (1,1)$, with $q(v) = 2 \neq 0$. Here $\tau_v(x) = x - \frac{2\varphi(v,x)}{2} v = x - \varphi(v,x)v$, so

$$\tau_v(e_1) = (1,0) - (1,1) = (0,-1), \qquad \tau_v(e_2) = (0,1) - (1,1) = (-1,0)$$

and the matrix of τ_v is $\left(\begin{smallmatrix} 0 & -1 \\ -1 & 0 \end{smallmatrix}\right)$. Both matrices have determinant -1 and square to the identity, as lemma 22.5.3 requires.

Their product is

$$\begin{pmatrix} 0 & -1 \\ -1 & 0 \end{pmatrix} \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$$

an isometry of determinant 1 whose square is $-\mathrm{id}$ and whose fourth power is id. So $O(\langle 1,1\rangle)$ over $\mathbb{Q}$ contains an element of order 4, which is not a reflection: reflections have order 2 and determinant -1. Products of reflections are therefore strictly more general than reflections, and corollary 22.5.10 will say how much more general.

Reflections are used here for one property: they move a vector to any other vector of the same nonzero value under q. That is how every theorem below produces the isometry it needs, and it takes at most two reflections.

Lemma 22.5.5: Two Reflections Move a Vector to Any Other of the Same Value

Let k be a field with $\operatorname{char} k \neq 2$, let V be a finite-dimensional k-vector space with a symmetric bilinear form φ, and let $u, w \in V$ satisfy $q(u) = q(w) \neq 0$. Then there is a $\sigma \in O(\varphi)$ with $\sigma(u) = w$, and σ can be taken to be a product of at most two reflections.

***Proof*:**

Expanding both terms by bilinearity and using $q(w) = q(u)$,

$$q(u-w) + q(u+w) = \big(q(u) - 2\varphi(u,w) + q(w)\big) + \big(q(u) + 2\varphi(u,w) + q(w)\big) = 4q(u)$$

Since $\operatorname{char} k \neq 2$, the scalar $4 = 2^2$ is nonzero, so $4q(u) \neq 0$ and $q(u-w)$ and $q(u+w)$ do not both vanish.

Case 1: $q(u-w) \neq 0$. Put $v = u - w$, an anisotropic vector. Then

$$q(v) = q(u) - 2\varphi(u,w) + q(w) = 2\big(q(u) - \varphi(u,w)\big) = 2\varphi(v,u)$$

the last equality because $\varphi(v,u) = \varphi(u,u) - \varphi(w,u) = q(u) - \varphi(u,w)$. Hence $2\varphi(v,u)/q(v) = 1$, and

$$\tau_v(u) = u - \frac{2\varphi(v,u)}{q(v)}\, v = u - v = w$$

so $\sigma = \tau_v$ works, and it lies in $O(\varphi)$ by lemma 22.5.3.

Case 2: $q(u-w) = 0$. Then $q(u+w) = 4q(u) \neq 0$ by the displayed identity. The vectors u and $-w$ satisfy $q(-w) = q(w) = q(u) \neq 0$ and $q\big(u - (-w)\big) = q(u+w) \neq 0$, so Case 1 applied to the pair $u, -w$ gives $\tau_{u+w}(u) = -w$. Also $q(w) \neq 0$, so τ_w is defined, and $\tau_w(-w) = -\tau_w(w) = w$ by lemma 22.5.3(2). Therefore

$$(\tau_w \circ \tau_{u+w})(u) = \tau_w(-w) = w$$

and $\sigma = \tau_w \circ \tau_{u+w}$ is a product of two reflections, lying in $O(\varphi)$ because $O(\varphi)$ is closed under composition (definition 22.5.1), as we sought to show.

Both cases occur. In example 22.5.4 the vectors $u = e_1$ and $w = e_2$ have $q(u) = q(w) = 1$ and $q(u - w) = q(1, -1) = 2 \neq 0$, so Case 1 applies and the single reflection $\tau_{(1,-1)}$ carries e_1 to e_2. Indeed $\tau_{(1,-1)}(e_1) = e_1 - \frac{2 \cdot 1}{2}(1, -1) = (0, 1)$. Case 2 is not an artifact of the proof: over $\mathbb{Q}$ with $\varphi \cong \langle 1, 1, -1\rangle$ the vectors $u = e_1$ and $w = (1, 1, 1)$ both have $q = 1$ while $q(u - w) = q(0, -1, -1) = 0$, and there no single reflection suffices: an equality $\tau_v(u) = w$ reads $u - w = \frac{2\varphi(v,u)}{q(v)} v$, and $w \neq u$ makes the coefficient nonzero, so $q(u - w)$ would be a nonzero square multiple of $q(v) \neq 0$ and in particular nonzero, contradicting $q(u - w) = 0$.

Lemma 22.5.5 carries no nondegeneracy hypothesis on φ, which matters immediately: the cancellation theorem is proved for arbitrary φ_1 and φ_2, whose radicals are not controlled, and only the summand being cancelled is required to be nondegenerate. The proof reduces that summand to one dimension, where the lemma does the work.

Theorem 22.5.6: Witt Cancellation

Let k be a field with $\operatorname{char} k \neq 2$. Let ψ be a *nondegenerate* symmetric bilinear form on a finite-dimensional k-vector space, and let φ_1, φ_2 be symmetric bilinear forms on finite-dimensional k-vector spaces. If

$$\psi \perp \varphi_1 \cong \psi \perp \varphi_2$$

then $\varphi_1 \cong \varphi_2$.

***Proof*:**

Step 1: reduction to a one-dimensional ψ. Let ψ live on U with $\dim U = m$. By theorem 22.2.1 the space U has an orthogonal basis $u_1, \dots, u_m$, so $\psi \cong \langle c_1, \dots, c_m\rangle$ with $c_i = \psi(u_i, u_i)$. By the last claim of proposition 22.2.2 the number of indices with $c_i = 0$ is $\dim \operatorname{rad}(\psi)$, which is 0 because ψ is nondegenerate. Hence every $c_i \neq 0$. Regrouping the orthogonal sum,

$$\psi \perp \varphi_j \cong \langle c_1\rangle \perp \big(\langle c_2, \dots, c_m\rangle \perp \varphi_j\big) \qquad (j = 1, 2)$$

So if the theorem holds whenever ψ is one-dimensional and nondegenerate, then cancelling $\langle c_1\rangle$ from the hypothesis gives $\langle c_2, \dots, c_m\rangle \perp \varphi_1 \cong \langle c_2, \dots, c_m\rangle \perp \varphi_2$, and $m - 1$ further cancellations, formally an induction on m with the case $m = 0$ reading $\varphi_1 \cong \varphi_2$, give the theorem. It therefore suffices to treat $\psi = \langle c\rangle$ with $c \in k^\times$.

Step 2: the one-dimensional case. Let φ_j live on W_j and realize $\langle c\rangle \perp \varphi_j$ on $V_j = ke_j \oplus W_j$, where $q(e_j) = c$, $\varphi(e_j, W_j) = 0$, and the form restricted to W_j is φ_j. Write $\varphi^{(j)}$ for the form on V_j. First,

$$W_j = \{e_j\}^\perp \text{ inside } V_j$$

since $W_j \subseteq \{e_j\}^\perp$ by construction, and conversely if $x = \lambda e_j + y$ with $y \in W_j$ satisfies $\varphi^{(j)}(e_j, x) = 0$, then that value is λc, and $c \neq 0$ forces $\lambda = 0$.

Let $f\colon V_1 \to V_2$ be an isometry. Put $u = f(e_1) \in V_2$. Then $q^{(2)}(u) = q^{(1)}(e_1) = c \neq 0$ and $q^{(2)}(e_2) = c$, so lemma 22.5.5 applied inside $(V_2, \varphi^{(2)})$ gives $\sigma \in O(\varphi^{(2)})$ with $\sigma(u) = e_2$. The composite $g = \sigma \circ f\colon V_1 \to V_2$ is again an isometry, and $g(e_1) = e_2$.

Step 3: g carries W_1 onto W_2. For $x \in V_1$,

$$\varphi^{(2)}\big(e_2, g(x)\big) = \varphi^{(2)}\big(g(e_1), g(x)\big) = \varphi^{(1)}(e_1, x)$$

so $x \in \{e_1\}^{\perp}$ if and only if $g(x) \in \{e_2\}^{\perp}$. Since g is bijective, g maps $\{e_1\}^{\perp}$ onto $\{e_2\}^{\perp}$, which by Step 2 says $g(W_1) = W_2$. The restriction $g|_{W_1} \colon W_1 \to W_2$ is a linear isomorphism preserving the forms, that is an isometry $\varphi_1 \cong \varphi_2$, completing the proof.

The theorem is what makes the decomposition of theorem 22.4.8 independent of the choices made in producing it.

Corollary 22.5.7: The Anisotropic Part and the Witt Index Are Invariants

Let k be a field with $\operatorname{char} k \neq 2$ and let φ be a nondegenerate symmetric bilinear form on a finite-dimensional k-vector space. Suppose

$$\varphi \cong \varphi_1 \perp m_1 H \qquad \text{and} \qquad \varphi \cong \varphi_2 \perp m_2 H$$

with φ_1 and φ_2 anisotropic and $m_1, m_2 \geq 0$. Then $m_1 = m_2$ and $\varphi_1 \cong \varphi_2$.

***Proof*:**
By definition 22.4.4 a hyperbolic plane H has a basis e, f with $q(e) = q(f) = 0$ and $\varphi(e, f) = 1$, so its matrix in that basis is $\left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$, of determinant $-1 \neq 0$. By lemma 13.3.23 the form H is nondegenerate. The matrix of an orthogonal sum in the union of two bases is block diagonal with the two matrices as blocks, so its determinant is the product of theirs, and mH is nondegenerate for every $m \geq 0$ by the same lemma. Also e is a nonzero isotropic vector of H.

Assume without loss of generality that $m_1 \leq m_2$. Regrouping the orthogonal sum,

$$\varphi_1 \perp m_1 H \cong \varphi \cong \varphi_2 \perp m_2 H \cong \big(\varphi_2 \perp (m_2 - m_1)H\big) \perp m_1 H$$

Since $m_1 H$ is nondegenerate, theorem 22.5.6 cancels it from the two ends and gives

$$\varphi_1 \cong \varphi_2 \perp (m_2 - m_1)H$$

If $m_2 > m_1$, the right-hand form has the nonzero isotropic vector e of any one of its hyperbolic summands, and the preimage of e under the isometry is nonzero and isotropic for φ_1, since an isometry is injective and preserves q. That contradicts the hypothesis that φ_1 is anisotropic. Hence $m_1 = m_2$, the term $(m_2 - m_1)H$ is the form on the zero space, and $\varphi_1 \cong \varphi_2$, as we sought to show.

So the Witt index of definition 22.4.10 deserves its name. Proposition 22.4.11(3) computes $\nu(\varphi)$ as the number of hyperbolic planes in the decomposition produced by the proof of theorem 22.4.8, and corollary 22.5.7 says that number is the same in every decomposition of that shape. So $m = \nu(\varphi)$ in *any* of them, and the anisotropic part is determined up to isometry by φ alone. Two nondegenerate forms $\varphi \cong \varphi_1 \perp mH$ and $\varphi' \cong \varphi'_1 \perp m'H$, with φ_1 and φ'_1 anisotropic, are therefore isometric if and only if $m = m'$ and $\varphi_1 \cong \varphi'_1$: if $\varphi \cong \varphi'$ then φ has the two decompositions $\varphi \cong \varphi_1 \perp mH$ and $\varphi \cong \varphi'_1 \perp m'H$, so corollary 22.5.7 gives $m = m'$ and $\varphi_1 \cong \varphi'_1$, while conversely $m = m'$ and $\varphi_1 \cong \varphi'_1$ give $\varphi \cong \varphi_1 \perp mH \cong \varphi'_1 \perp m'H \cong \varphi'$. That biconditional is the reduction the classification of forms

over a given field uses.

Cancellation deletes a summand from both sides of an isometry that is already given. The extension theorem is the statement that isometries are plentiful: any isometry defined on a subspace already extends to the whole space, with no compatibility condition beyond preserving the form. The subspace is allowed to be degenerate, and that is the substance of the theorem. For a subspace containing an anisotropic vector, the reduction to a smaller space is the argument already used twice above, whereas a totally isotropic subspace gives that argument nothing to work with. The following lemma is what repairs that case: it manufactures, for an isotropic vector inside a totally isotropic subspace, a partner making it half of a hyperbolic plane, chosen orthogonal to the rest of the subspace.

Lemma 22.5.8: Hyperbolic Partners for Isotropic Vectors

Let k be a field with $\operatorname{char} k \neq 2$, let φ be a nondegenerate symmetric bilinear form on a finite-dimensional k-vector space V, let $U \subseteq V$ be a subspace with $\varphi(x, y) = 0$ for all $x, y \in U$, let $u \in U$ be nonzero, and let $U_1 \subseteq U$ be a subspace with $U = ku \oplus U_1$. Then there exists $u^* \in V$ with

$$q(u^*) = 0, \qquad \varphi(u, u^*) = 1, \qquad \varphi(u^*, x) = 0 \text{ for all } x \in U_1$$

***Proof*:**

Let $u_1, \dots, u_m$ be a basis of U_1, so that $u, u_1, \dots, u_m$ is a basis of U and in particular a linearly independent family in V. Consider the linear map

$$\Phi \colon V \to k^{m+1}, \qquad \Phi(y) = \big(\varphi(u, y), \varphi(u_1, y), \dots, \varphi(u_m, y)\big)$$

Suppose Φ were not surjective. Its image is then a proper subspace of k^{m+1}, hence contained in the kernel of some nonzero linear functional (extend a basis of the image to a basis of k^{m+1} and take the member of the dual basis of proposition 13.3.5 attached to a vector outside the image): there are $\lambda_0, \dots, \lambda_m \in k$, not all zero, with $\lambda_0 \varphi(u, y) + \sum_{i=1}^{m} \lambda_i \varphi(u_i, y) = 0$ for every $y \in V$. Bilinearity rewrites this as $\varphi(z, y) = 0$ for every $y \in V$, where $z = \lambda_0 u + \sum_i \lambda_i u_i$. As φ is symmetric, this says $z \in \operatorname{rad}(\varphi)$ (definition 13.3.22), which is zero by nondegeneracy. And $u, u_1, \dots, u_m$ being linearly independent then forces every $\lambda_j = 0$, contradicting the choice of the λ_j. So Φ is surjective.

Choose $y \in V$ with $\Phi(y) = (1, 0, \dots, 0)$, so $\varphi(u, y) = 1$ and $\varphi(u_i, y) = 0$ for each i. By bilinearity the latter gives $\varphi(x, y) = 0$ for every $x \in U_1$. Since $\operatorname{char} k \neq 2$ the scalar $q(y)/2$ is defined, and we may set

$$u^* := y - \frac{q(y)}{2}\, u$$

Because $u \in U$ and φ vanishes identically on U, we have $q(u) = 0$ and $\varphi(u, x) = 0$ for $x \in U_1$. Hence

$$\varphi(u, u^*) = \varphi(u, y) - \frac{q(y)}{2} q(u) = 1 - 0 = 1$$

$$q(u^*) = q(y) - 2 \cdot \frac{q(y)}{2} \varphi(y, u) + \frac{q(y)^2}{4} q(u) = q(y) - q(y) + 0 = 0$$

using $\varphi(y, u) = \varphi(u, y) = 1$ by symmetry. Finally, for $x \in U_1$ both $\varphi(y, x) = 0$ and $\varphi(u, x) = 0$, so $\varphi(u^*, x) = 0$, completing the proof.

Theorem 22.5.9: Witt's Extension Theorem

Let k be a field with $\operatorname{char} k \neq 2$, let φ be a nondegenerate symmetric bilinear form on a finite-dimensional k-vector space V, let $U \subseteq V$ be a subspace, and let $f\colon U \to V$ be an injective linear map with

$$\varphi\big(f(x), f(y)\big) = \varphi(x, y) \qquad \text{for all } x, y \in U$$

Then there is an $F \in O(\varphi)$ with $F|_U = f$.

***Proof*:**
Induct on $\dim V$. If $\dim V = 0$ then $U = 0$ and $F = \mathrm{id}_V$ serves. So let $\dim V = n \geq 1$ and assume the theorem for every nondegenerate symmetric bilinear form on a space of dimension smaller than n. If $U = 0$, take $F = \mathrm{id}_V$. Assume $U \neq 0$.

Step 1: reduction to the case where φ does not vanish identically on U. Suppose $\varphi(x, y) = 0$ for all $x, y \in U$. Choose $u \in U$ nonzero and a subspace U_1 with $U = ku \oplus U_1$, and let $u_1, \ldots, u_m$ be a basis of U_1. Lemma 22.5.8 gives $u^* \in V$ with $q(u^*) = 0$, $\varphi(u, u^*) = 1$ and $\varphi(u^*, U_1) = 0$.

Put $u' = f(u)$ and $U_1' = f(U_1)$. Since f is injective and preserves φ, the subspace $f(U)$ satisfies $\varphi(x, y) = 0$ for all $x, y \in f(U)$, the vector u' is nonzero, and $f(U) = ku' \oplus U_1'$. A second application of lemma 22.5.8, to the subspace $f(U)$ with the vector u' and the complement U_1', gives $u'^* \in V$ with $q(u'^*) = 0$, $\varphi(u', u'^*) = 1$ and $\varphi(u'^*, U_1') = 0$.

Because $\varphi(u, u^*) = 1$ while $\varphi(u, x) = 0$ for every $x \in U$, the vector u^* is not in U, so $\tilde{U} := U \oplus ku^*$ has $u, u^*, u_1, \ldots, u_m$ as a basis. Define a linear map $\tilde{f}\colon \tilde{U} \to V$ by $\tilde{f}|_U = f$ and $\tilde{f}(u^*) = u'^*$.

The map $\tilde{f}$ preserves φ. By bilinearity it is enough to compare φ on pairs of basis vectors with φ on the pairs of their images, and the six comparisons are:

$$q(u) = 0 = q(u'), \qquad q(u^*) = 0 = q(u'^*), \qquad \varphi(u, u^*) = 1 = \varphi(u', u'^*)$$

$$\varphi(u, u_i) = 0 = \varphi\big(f(u), f(u_i)\big), \qquad \varphi(u^*, u_i) = 0 = \varphi(u'^*, f(u_i)), \qquad \varphi(u_i, u_j) = \varphi\big(f(u_i), f(u_j)\big)$$

the fourth and sixth because f preserves φ, the fifth by the two orthogonality conditions on u^* and u'^*.

The map $\tilde{f}$ is injective. Let $x = \lambda u + \mu u^* + w$ with $w \in U_1$ and $\tilde{f}(x) = 0$, that is $\lambda u' + \mu u'^* + f(w) = 0$. Applying $\varphi(u', -)$ and using $\varphi(u', u') = 0$, $\varphi(u', u'^*) = 1$ and $\varphi(u', f(w)) = \varphi(f(u), f(w)) = \varphi(u, w) = 0$ gives $\mu = 0$. Then $f(\lambda u + w) = 0$, so $\lambda u + w = 0$ by injectivity of f, so $\lambda = 0$ and $w = 0$ because $U = ku \oplus U_1$.

Finally $q(u + u^*) = q(u) + 2\varphi(u, u^*) + q(u^*) = 2 \neq 0$, so φ does not vanish identically on $\tilde{U}$. Any $F \in O(\varphi)$ with $F|_{\tilde{U}} = \tilde{f}$ satisfies $F|_U = f$, so it suffices to prove the theorem for the pair $(\tilde{U}, \tilde{f})$, which is of the kind treated next.

Step 2: the case where φ does not vanish identically on U. Some $u \in U$ has $q(u) \neq 0$: otherwise q vanishes on U and, for $x, y \in U$, $0 = q(x + y) = q(x) + 2\varphi(x, y) + q(y) = 2\varphi(x, y)$, which with $\operatorname{char} k \neq 2$ makes φ vanish identically on U.

Then $q(f(u)) = q(u) \neq 0$, so lemma 22.5.5 gives $\sigma \in O(\varphi)$ with $\sigma(f(u)) = u$. The map $f' := \sigma \circ f$ is again an injective linear map $U \to V$ preserving φ, and $f'(u) = u$.

Set $V_1 = \{u\}^\perp$ and $U' = U \cap V_1$. Four facts about these:

1. $V = ku \oplus V_1$ with $\varphi(u, V_1) = 0$, by lemma 22.5.3(1) applied to (V, φ) and the anisotropic vector u. In particular $\dim V_1 = n - 1$.
2. $U = ku \oplus U'$, by the same part of the same lemma applied to $(U, \varphi|_U)$, whose orthogonal complement of u inside U is $U \cap \{u\}^\perp = U'$.
3. $\varphi|_{V_1}$ is nondegenerate: if $y \in V_1$ satisfies $\varphi(y, V_1) = 0$, then also $\varphi(y, u) = 0$, so $\varphi(y, V) = 0$ by (1) and bilinearity, whence $y \in \operatorname{rad}(\varphi) = 0$.
4. $f'(U') \subseteq V_1$: for $x \in U'$, $\varphi(u, f'(x)) = \varphi(f'(u), f'(x)) = \varphi(u, x) = 0$.

So $f'|_{U'} \colon U' \to V_1$ is an injective linear map from a subspace of V_1 into V_1 preserving $\varphi|_{V_1}$, and $\dim V_1 = n - 1$. The inductive hypothesis gives $G \in O(\varphi|_{V_1})$ with $G|_{U'} = f'|_{U'}$.

Step 3: assembling F. Define $F' \colon V \to V$ by $F'(\lambda u + y) = \lambda u + G(y)$ for $\lambda \in k$ and $y \in V_1$. This is well defined and bijective by (1) and the bijectivity of G. It preserves φ: for $\lambda, \mu \in k$ and $y, z \in V_1$, using $G(y), G(z) \in V_1$ and $\varphi(u, V_1) = 0$ twice,

$$\varphi\big(\lambda u + G(y), \mu u + G(z)\big) = \lambda\mu\, q(u) + \varphi\big(G(y), G(z)\big) = \lambda\mu\, q(u) + \varphi(y, z) = \varphi(\lambda u + y, \mu u + z)$$

So $F' \in O(\varphi)$. It restricts to f' on U: writing $x \in U$ as $x = \lambda u + x'$ with $x' \in U'$ by (2),

$$F'(x) = \lambda u + G(x') = \lambda u + f'(x') = \lambda f'(u) + f'(x') = f'(x)$$

Set $F = \sigma^{-1} \circ F'$, an element of $O(\varphi)$. Then $F|_U = \sigma^{-1} \circ f' = \sigma^{-1} \circ \sigma \circ f = f$, completing the proof.

The hypothesis that φ be nondegenerate on the ambient space is not removable, and it enters the proof twice: in Step 1 through lemma 22.5.8, and in Step 2 as fact (3), where $\operatorname{rad}(\varphi) = 0$ is what makes $\varphi|_{V_1}$ nondegenerate and so permits the inductive hypothesis. What goes wrong without it is direct: an isometry of V carries $\operatorname{rad}(\varphi)$ onto itself, so a form-preserving injection sending a vector outside the radical into the radical extends to no isometry, and exercise 22.5.1 (6) exhibits one. Nondegeneracy of the *subspace* U, on the other hand, is nowhere assumed, and Step 1 is the work of doing without it.

Taking U to be a line already extends lemma 22.5.5 past its restriction to anisotropic vectors. Let φ be nondegenerate, let $c \in k$, and let v, v' be nonzero vectors with $q(v) = q(v') = c$. The map $f \colon kv \to V$ with $f(\lambda v) = \lambda v'$ is linear and injective, and $\varphi(f(\lambda v), f(\mu v)) = \lambda\mu\, q(v') = \lambda\mu\, q(v) = \varphi(\lambda v, \mu v)$, so theorem 22.5.9 extends it to an isometry of V carrying v to v'. So $O(\varphi)$ acts transitively on the nonzero vectors of each fixed value of q, including the value 0, where no reflection was available. The last consequence counts how many reflections it takes to build an arbitrary isometry.

Corollary 22.5.10: Reflections Generate the Orthogonal Group

Let k be a field with $\operatorname{char} k \neq 2$ and let φ be a nondegenerate symmetric bilinear form on a k-vector space V of finite dimension n. Then every $\sigma \in O(\varphi)$ is a product of at most $2n$ reflections τ_v in anisotropic vectors $v \in V$, where the empty product is read as id_V. In particular $O(\varphi)$ is generated by its reflections. The bound $2n$ is not the sharp one. See the discussion after the proof.

***Proof*:**

Induct on n. For $n = 0$ the group $O(\varphi)$ is trivial and id_V is the empty product. Let $n \geq 1$ and assume the statement for nondegenerate forms on spaces of dimension $n - 1$.

First, V contains an anisotropic vector: if $q(x) = 0$ for every $x \in V$, then for all $x, y \in V$

$$0 = q(x + y) = q(x) + 2\varphi(x, y) + q(y) = 2\varphi(x, y)$$

and $\operatorname{char} k \neq 2$ makes φ the zero form, so $\operatorname{rad}(\varphi) = V \neq 0$, contradicting nondegeneracy. Fix such a u, with $q(u) \neq 0$.

Let $\sigma \in O(\varphi)$. Then $q(\sigma(u)) = q(u) \neq 0$, so lemma 22.5.5, applied to the pair $\sigma(u)$ and u, gives $\pi \in O(\varphi)$ with $\pi(\sigma(u)) = u$ and π a product of at most two reflections. Put $\rho = \pi \circ \sigma$, so $\rho(u) = u$.

Set $V_1 = \{u\}^{\perp}$. As in the proof of theorem 22.5.9, $V = ku \oplus V_1$ with $\varphi(u, V_1) = 0$ by lemma 22.5.3(1), $\dim V_1 = n - 1$, and $\varphi|_{V_1}$ is nondegenerate because a vector of V_1 orthogonal to V_1 is also orthogonal to u, hence to V, hence zero. Moreover $\rho(V_1) = V_1$: for $y \in V_1$, $\varphi(u, \rho(y)) = \varphi(\rho(u), \rho(y)) = \varphi(u, y) = 0$, so $\rho(V_1) \subseteq V_1$, and ρ is injective, so the inclusion is an equality by dimension count.

By the inductive hypothesis $\rho|_{V_1}$ is a product $\tau_{v_1}^{V_1} \cdots \tau_{v_r}^{V_1}$ of $r \leq 2(n-1)$ reflections of $(V_1, \varphi|_{V_1})$, with each $v_i \in V_1$ anisotropic. Each v_i is anisotropic in V as well, and the reflection τ_{v_i} of V satisfies $\tau_{v_i}(u) = u$, since $\varphi(v_i, u) = 0$, and restricts on V_1 to $\tau_{v_i}^{V_1}$, the defining formula being the same. Hence $\tau_{v_1} \cdots \tau_{v_r}$ and ρ agree on u and on V_1, so they agree on $V = ku \oplus V_1$, and ρ is a product of at most $2(n-1)$ reflections of V.

Finally $\sigma = \pi^{-1} \circ \rho$, and π^{-1} is a product of at most two reflections, since each reflection is its own inverse by lemma 22.5.3(3) and the inverse of $\tau_v \tau_w$ is $\tau_w \tau_v$. So σ is a product of at most $2 + 2(n-1) = 2n$ reflections, as we sought to show.

The sharp form of this count is the Cartan-Dieudonné theorem, proved in Artin's *Geometric Algebra*: over a field of characteristic other than 2, every isometry of an n-dimensional nondegenerate space is a product of at most n reflections. The bound proved above loses a factor of two at Case 2 of lemma 22.5.5, where two reflections are used in place of one. Recovering the sharp bound requires a separate analysis of the isometries σ for which $q(x - \sigma(x)) = 0$ for every $x \in V$, so that no single reflection returns $\sigma(x)$ to x. Such isometries exist: on $H \perp H$, with the two hyperbolic bases e_1, f_1 and e_2, f_2, the map fixing e_1 and e_2 and sending $f_1 \mapsto f_1 + e_2$ and $f_2 \mapsto f_2 - e_1$ is one. For them the reflection count is governed by $\dim \operatorname{im}(\sigma - \mathrm{id}_V)$ rather than by n. That analysis is not carried out in this book.

Exercise 22.5.1

1. Let φ be a nondegenerate symmetric bilinear form on V with $\dim V = n \geq 1$ and $\operatorname{char} k \neq 2$, and let $\sigma \in O(\varphi)$. Show that if A is the matrix of φ and S the matrix of σ in one common basis, then $S^t A S = A$ (use lemma 13.3.19), and deduce $\det \sigma = \pm 1$ with the help of lemma 13.3.23. Using $\det \tau_v = -1$ from lemma 22.5.3, deduce that $SO(\varphi) := \{\sigma \in O(\varphi) \mid \det \sigma = 1\}$ is a subgroup of index exactly 2, and that the parity of the number of reflections in a factorization of σ given by corollary 22.5.10 does not depend on the factorization.

2. Let $\operatorname{char} k \neq 2$, let φ be a symmetric bilinear form on V, and let $v, w \in V$ be anisotropic. Show that $\tau_v = \tau_w$ if and only if $kv = kw$, so that $v \mapsto \tau_v$ is a bijection from the set of anisotropic lines of V onto a set of involutions in $O(\varphi)$. Then count the reflections in $O(\langle 1, 1\rangle)$ over $\mathbb{F}_5$, by listing the six lines of $\mathbb{F}_5^2$ and deciding which are anisotropic.

3. Show that the hypothesis that ψ be nondegenerate can be removed from theorem 22.5.6: for arbitrary symmetric bilinear forms over a field with $\operatorname{char} k \neq 2$, $\psi \perp \varphi_1 \cong \psi \perp \varphi_2$ implies $\varphi_1 \cong \varphi_2$. Reduce, using theorem 22.2.1, to cancelling $\langle 0\rangle$, and for that case use that an isometry carries the radical of one form onto the radical of the other, together with the description of the radical in an orthogonal basis given by proposition 22.2.2.

4. Over $k = \mathbb{Q}$, show that $\langle 1, 1\rangle$ and $\langle -1, -1\rangle$ are not isometric by asking which nonzero rationals each represents. Deduce from theorem 22.5.6 that $\langle 1, 1, 5\rangle \ncong \langle -1, -1, 5\rangle$, and check that these two forms have the same rank and the same discriminant, so that neither invariant of section 22.1 could have separated them.

5. Let φ be nondegenerate on V with $\operatorname{char} k \neq 2$, and recall the totally isotropic subspaces of definition 22.3.6. Call one *maximal* if no strictly larger subspace of V is totally isotropic. Using theorem 22.5.9, show that any two maximal totally isotropic subspaces of V have the same dimension. Then explain why lemma 22.5.5 gives nothing about isotropic vectors, by exhibiting two nonzero isotropic vectors u, w of $H \perp H$ with $q(u - w) = q(u + w) = 0$.

6. Let $\operatorname{char} k \neq 2$ and let $V = k^3$ carry the form whose matrix in the standard basis is $\operatorname{diag}(1, -1, 0)$. Show that $f\colon k(e_1 + e_2) \to V$ with $f(e_1 + e_2) = e_3$ is an injective linear map preserving the form, and that no isometry of V restricts to it. Conclude that the nondegeneracy hypothesis in theorem 22.5.9 cannot be dropped.

7. Let $\operatorname{char} k \neq 2$ and let $a \in k^\times$. Show that $\langle a, -a\rangle \cong H$, by exhibiting an isotropic vector and applying proposition 22.4.6. Then show that the isometry class of $\langle a, -a\rangle$ does not depend on a, even though the square class of a may vary.

8. Let $\operatorname{char} k \neq 2$ and let V be four-dimensional with basis e_1, f_1, e_2, f_2 and the symmetric bilinear form determined by $\varphi(e_1, f_1) = \varphi(e_2, f_2) = 1$, all other products of two basis vectors being zero, so that $V \cong H \perp H$. Define σ by $\sigma(e_i) = e_i$, $\sigma(f_1) = f_1 + e_2$ and $\sigma(f_2) = f_2 - e_1$. Show that $\sigma \in O(\varphi)$, that $x - \sigma(x)$ lies in the totally isotropic subspace spanned by e_1 and e_2 for every $x \in V$, and hence that $q(x - \sigma(x)) = 0$ for every x. Deduce that no reflection τ_v satisfies $\tau_v(\sigma(x)) = x$ for any x with $\sigma(x) \neq x$, so that Case 2 of lemma 22.5.5 is unavoidable for this σ.

22.6 The Witt Group

Theorem 22.4.8 writes a nondegenerate form as an anisotropic form orthogonal to a number of hyperbolic planes, and corollary 22.5.7 says that the anisotropic part is determined up to isometry. Two forms with isometric anisotropic parts therefore differ only in how many hyperbolic planes are attached to them, and this section discards that number. What remains is an abelian group under $\perp$, and a commutative ring once the tensor product of forms is available.

The construction needs no group completion. The inverse of a class is again the class of a form, because $\varphi \perp (-\varphi)$ is a sum of hyperbolic planes, and that computation is the first item below. The group is then computed for the two families of fields whose square classes are known: the algebraically closed fields, where it has two elements, and the finite fields, where it has four. The finite case answers the question left open at the end of example 22.2.5.

Throughout, k is a field with $\operatorname{char} k \neq 2$ and every form is a symmetric bilinear form on a finite-dimensional k-vector space, identified with its quadratic form through proposition 22.1.2 whenever that is convenient. The orthogonal sum $\varphi \perp \psi$ is definition 22.4.1, and everything used below about it is established there: the block-diagonal Gram matrix, the multiplicativity $\det(\varphi \perp \psi) = \det(\varphi)\det(\psi)$, that the sum is nondegenerate exactly when both summands are, and that $\perp$ descends to isometry classes and is commutative and associative up to isometry. Write $-\varphi$ for the form $(x, y) \mapsto -\varphi(x, y)$ on V. In the diagonal notation of section 22.2,

$$\langle a_1, \dots, a_m \rangle \perp \langle b_1, \dots, b_n \rangle = \langle a_1, \dots, a_m, b_1, \dots, b_n \rangle$$

and $-\langle a_1, \dots, a_n \rangle = \langle -a_1, \dots, -a_n \rangle$. Finally mH denotes the orthogonal sum of m copies of the hyperbolic plane H (definition 22.4.4), with $0H$ the zero form on the zero space.

The identification that drives everything below is that a form and its negative add up to nothing but hyperbolic planes.

Lemma 22.6.1: A Form and Its Negative Sum to Hyperbolic Planes

Let k be a field with $\operatorname{char} k \neq 2$.

1. For every $a \in k^\times$ one has $\langle a, -a \rangle \cong H$. In particular $H \cong \langle 1, -1 \rangle$, so $\det H = -1$.
2. If φ is a nondegenerate form of rank n on V, then $\varphi \perp (-\varphi) \cong nH$.

***Proof*:**

1. By example 22.4.7 a nondegenerate binary form $\langle a, b \rangle$ is a hyperbolic plane exactly when $-ab$ is a square in $k^\times$. Here $b = -a$, so $-ab = a^2$ is a square and $\langle a, -a \rangle \cong H$. Taking $a = 1$ gives $H \cong \langle 1, -1 \rangle$, whose matrix in an orthogonal basis is $\operatorname{diag}(1, -1)$, of determinant -1.
2. Diagonalize $\varphi \cong \langle a_1, \dots, a_n \rangle$ by theorem 22.2.1. Nondegeneracy forces every $a_i \neq 0$: by the last claim of proposition 22.2.2 the number of zero entries is the dimension of the radical, which is 0 here by definition 13.3.22. Then $-\varphi \cong \langle -a_1, \dots, -a_n \rangle$, so

$$\varphi \perp (-\varphi) \cong \langle a_1, \dots, a_n, -a_1, \dots, -a_n \rangle \cong \langle a_1, -a_1 \rangle \perp \cdots \perp \langle a_n, -a_n \rangle$$

the second isometry being the permutation of an orthogonal basis that places each a_i next

to its negative. Applying part (1) to each summand gives $\varphi \perp (-\varphi) \cong nH$, as we sought to show.

Every form of rank n is isometric to a form on k^n, obtained by transporting it along a choice of basis, so the isometry classes of nondegenerate forms over k constitute a set rather than a proper class. That set carries $\perp$ as a commutative associative operation with the zero form as a neutral element, and the lemma says that after hyperbolic planes are discarded every element acquires an inverse.

Definition 22.6.2: Witt Equivalence and the Witt Group

Let k be a field with $\operatorname{char} k \neq 2$. Two nondegenerate symmetric bilinear forms φ and ψ over k, possibly on spaces of different dimensions, are *Witt equivalent*, written $\varphi \sim \psi$, if there are integers $m, n \geq 0$ with

$$\varphi \perp mH \cong \psi \perp nH$$

The *Witt group* $W(k)$ is the set of equivalence classes, the class of φ being written $[\varphi]$, equipped with the operation

$$[\varphi] + [\psi] := [\varphi \perp \psi]$$

That $\sim$ is an equivalence relation and that the operation is well defined are proved in theorem 22.6.3.

Two small computations show what the relation does. Over any k one has $\langle 1 \rangle \perp \langle -1 \rangle \cong H$ by lemma 22.6.1, so $\langle 1, -1 \rangle \sim 0$: the hyperbolic plane is Witt trivial, which is the whole content of the definition. Consequently $\langle 1, 1, -1 \rangle \sim \langle 1 \rangle$, taking $m = 0$ and $n = 1$ against $\psi = \langle 1 \rangle$, since $\langle 1 \rangle \perp H \cong \langle 1, 1, -1 \rangle$. Forms of different ranks can therefore be Witt equivalent, and rank is not an invariant of a Witt class.

Theorem 22.6.3: The Witt Classes Form an Abelian Group

Let k be a field with $\operatorname{char} k \neq 2$. Witt equivalence is an equivalence relation on nondegenerate forms over k, the operation of definition 22.6.2 is well defined on classes, and $W(k)$ is an abelian group whose neutral element is the class $[0] = [H]$ of the zero form and in which $-[\varphi] = [-\varphi]$.

***Proof*:**

Step 1: $\sim$ is an equivalence relation. Reflexivity is the case $m = n = 0$, and symmetry is immediate from the symmetry of $\cong$. For transitivity, suppose $\varphi \perp mH \cong \psi \perp nH$ and $\psi \perp pH \cong \chi \perp qH$. Adding pH to the first isometry and nH to the second, which is legitimate because orthogonal sums of isometric forms are isometric, and rearranging summands by commutativity and associativity of $\perp$, gives

$$\varphi \perp (m+p)H \cong \psi \perp (n+p)H \cong \chi \perp (q+n)H$$

so $\varphi \sim \chi$.

Step 2: the operation is well defined. Let $\varphi \sim \varphi'$ and $\psi \sim \psi'$, say $\varphi \perp mH \cong \varphi' \perp m'H$ and $\psi \perp nH \cong \psi' \perp n'H$. Taking the orthogonal sum of the two isometries and rearranging summands,

$$(\varphi \perp \psi) \perp (m+n)H \cong (\varphi' \perp \psi') \perp (m'+n')H$$

so $\varphi \perp \psi \sim \varphi' \perp \psi'$. Both $\varphi \perp \psi$ and $\varphi' \perp \psi'$ are nondegenerate, since orthogonal sums of nondegenerate forms are, so the sum of two classes is again a class.

Step 3: the group axioms. Commutativity and associativity of $+$ are those of $\perp$, transported to classes by Step 2. The zero form 0 on the zero space is nondegenerate, its radical being the zero space, and $\varphi \perp 0 = \varphi$, so $[0]$ is neutral. Taking $m = 0$ and $n = 1$ in definition 22.6.2 gives $H \perp 0H \cong 0 \perp 1H$, that is $H \sim 0$, so $[H] = [0]$. Finally, if φ has rank n then $\varphi \perp (-\varphi) \cong nH$ by lemma 22.6.1, and $nH \sim 0$ directly from definition 22.6.2, taking the two integers there to be 0 and n, so

$$[\varphi] + [-\varphi] = [\varphi \perp (-\varphi)] = [nH] = [0]$$

which exhibits $[-\varphi]$ as the inverse of $[\varphi]$, completing the proof.

The group is built from forms alone: no formal differences are introduced, and no group completion of the monoid of isometry classes is taken, because the negative of a form is a form and lemma 22.6.1 makes the two cancel. What the construction has cost is the ability to see the rank, since φ and $\varphi \perp H$ have become equal. The next result says that nothing else was lost, and it is where Witt cancellation enters: cancellation plays no part in the group axioms above, and its role is to identify the classes with objects one can list.

Proposition 22.6.4: Each Witt Class Contains Exactly One Anisotropic Form

Let k be a field with $\operatorname{char} k \neq 2$ and let φ, ψ be nondegenerate forms over k, with Witt decompositions $\varphi \cong \varphi_{\text{an}} \perp m_\varphi H$ and $\psi \cong \psi_{\text{an}} \perp m_\psi H$ as in theorem 22.4.8. Then

$$\varphi \sim \psi \quad \text{if and only if} \quad \varphi_{\text{an}} \cong \psi_{\text{an}}$$

Consequently $[\varphi] \mapsto \varphi_{\text{an}}$ is a bijection from $W(k)$ to the set of isometry classes of anisotropic forms over k, and $[\varphi] = 0$ if and only if $\varphi \cong m_\varphi H$.

***Proof*:**
First note that an anisotropic form is nondegenerate: a nonzero vector v of the radical satisfies $\varphi(v, v) = 0$, hence is an isotropic vector (definition 22.4.2), so the radical of an anisotropic form is zero and definition 13.3.22 applies. Both φ_{an} and ψ_{an} are therefore nondegenerate, as is H, which is what the cancellation theorem requires.

Step 1: isometric anisotropic parts give Witt equivalence. If $\varphi_{\text{an}} \cong \psi_{\text{an}}$ then

$$\varphi \perp m_\psi H \cong \varphi_{\text{an}} \perp (m_\varphi + m_\psi)H \cong \psi_{\text{an}} \perp (m_\varphi + m_\psi)H \cong \psi \perp m_\varphi H$$

so $\varphi \sim \psi$ by definition 22.6.2.

Step 2: Witt equivalence gives isometric anisotropic parts. Suppose $\varphi \perp mH \cong \psi \perp nH$. Substituting the two Witt decompositions and writing $a = m_\varphi + m$ and $b = m_\psi + n$,

$$\varphi_{\text{an}} \perp aH \cong \psi_{\text{an}} \perp bH$$

This is exactly the situation corollary 22.5.7 settles: two Witt decompositions of isometric forms have isometric anisotropic parts and equal hyperbolic counts. Applying it gives $a = b$ and $\varphi_{\text{an}} \cong \psi_{\text{an}}$.

Step 3: the two consequences. The two steps say exactly that $[\varphi] \mapsto \varphi_{\mathrm{an}}$ is well defined and injective on $W(k)$. It is surjective because an anisotropic form χ is nondegenerate with $\chi_{\mathrm{an}} = \chi$ and $m_\chi = 0$. For the last claim, $[\varphi] = [0]$ holds if and only if φ_{an} is isometric to the zero form on the zero space, that is if and only if $\varphi \cong m_\varphi H$, completing the proof.

So $W(k)$ is the set of anisotropic forms over k, given an addition that is not orthogonal sum: to add two anisotropic forms, take their orthogonal sum and discard the hyperbolic planes that appear in it. The integer m_φ discarded in the process is the Witt index of definition 22.4.10, the two agreeing by proposition 22.4.11(3), and the pair (anisotropic part, Witt index) is a complete isometry invariant of a nondegenerate form, while the class in $W(k)$ retains only the first half of that pair. The first field to compute is one over which every anisotropic form has rank at most one.

Example 22.6.5: The Witt Group of an Algebraically Closed Field

Let k be algebraically closed with $\operatorname{char} k \neq 2$. The case $k = \mathbb{C}$ is the one to keep in mind. Fix $c \in k$ with $c^2 = -1$, which exists because $x^2 + 1$ has a root.

No anisotropic form has rank 2 or more. Indeed, by example 22.2.4 a nondegenerate form of rank n is isometric to $\langle 1, \dots, 1\rangle$, and if $n \geq 2$ the vector $v = (c, 1, 0, \dots, 0)$ is nonzero and satisfies

$$q(v) = c^2 + 1 = -1 + 1 = 0$$

so the form is isotropic. The anisotropic forms are therefore the zero form and the rank one form $\langle 1\rangle$, the latter being anisotropic because $x^2 = 0$ forces $x = 0$. By proposition 22.6.4,

$$W(k) = \{\, [0],\ [\langle 1\rangle] \,\}$$

has two elements. Its group law is forced: $2[\langle 1\rangle] = [\langle 1, 1\rangle]$, and $\langle 1, 1\rangle \cong \langle 1, -1\rangle$ by proposition 22.2.2, since $-1 \cdot c^2 = 1$ rescales the second entry by the square c^2, so $\langle 1, 1\rangle \cong H$ by lemma 22.6.1 and $2[\langle 1\rangle] = 0$. Hence $W(k) \cong \mathbb{Z}/2\mathbb{Z}$, the isomorphism sending $[\varphi]$ to the rank of φ modulo 2.

Addition is not the only operation available on forms. The tensor product of the underlying spaces carries a form built from the two given ones, and it is compatible with $\perp$ and with Witt equivalence, which upgrades $W(k)$ from a group to a ring. In diagonal coordinates the operation is multiplication of the diagonal entries in all pairs, which is what makes it computable.

Proposition 22.6.6: The Tensor Product of Forms and the Witt Ring

Let k be a field with $\operatorname{char} k \neq 2$, let φ be a symmetric bilinear form on V and ψ one on W.

1. There is exactly one bilinear form $\varphi \otimes \psi$ on $V \otimes_k W$ satisfying
$$(\varphi \otimes \psi)(v \otimes w,\ v' \otimes w') = \varphi(v, v')\,\psi(w, w')$$
for all $v, v' \in V$ and $w, w' \in W$, and it is symmetric.
2. If $\varphi \cong \langle a_1, \ldots, a_m\rangle$ and $\psi \cong \langle b_1, \ldots, b_n\rangle$, then $\varphi \otimes \psi$ is isometric to the diagonal form whose mn entries are the products $a_i b_j$. In particular $\varphi \otimes \psi$ is nondegenerate when φ and ψ are, and isometric inputs produce isometric outputs.
3. The rule $[\varphi] \cdot [\psi] := [\varphi \otimes \psi]$ is well defined and makes $W(k)$ a commutative ring with identity $[\langle 1\rangle]$.

***Proof*:**

1. Fix bases $v_1, \ldots, v_m$ of V and $w_1, \ldots, w_n$ of W. By corollary 15.2.6 the mn vectors $v_i \otimes w_j$ form a basis of $V \otimes_k W$. Let M be the $mn \times mn$ matrix indexed by pairs, with entry $\varphi(v_i, v_{i'})\psi(w_j, w_{j'})$ in position $\big((i,j),(i',j')\big)$, and define $B(x,y) := [x]^t M [y]$ using coordinates in that basis. Taking coordinates is linear and matrix multiplication is bilinear in the two coordinate vectors, so B is a bilinear form.

   To see that B has the stated values, write $v = \sum_i x_i v_i$, $v' = \sum_{i'} x'_{i'} v_{i'}$, $w = \sum_j y_j w_j$ and $w' = \sum_{j'} y'_{j'} w_{j'}$. The map $(v, w) \mapsto v \otimes w$ is bilinear by theorem 15.1.12, so $v \otimes w = \sum_{i,j} x_i y_j\,(v_i \otimes w_j)$, which is the coordinate vector of $v \otimes w$ in the chosen basis. Hence
$$\begin{aligned} B(v \otimes w, v' \otimes w') &= \sum_{i,j,i',j'} x_i y_j x'_{i'} y'_{j'}\,\varphi(v_i, v_{i'})\psi(w_j, w_{j'}) \\ &= \Big(\sum_{i,i'} x_i x'_{i'} \varphi(v_i, v_{i'})\Big)\Big(\sum_{j,j'} y_j y'_{j'} \psi(w_j, w_{j'})\Big) \\ &= \varphi(v, v')\,\psi(w, w') \end{aligned}$$
   the last line by bilinearity of φ and of ψ. For uniqueness, the simple tensors span $V \otimes_k W$ by lemma 15.1.4, and two bilinear forms agreeing on all pairs of vectors from a spanning set agree everywhere, by expanding both arguments. Symmetry follows the same way: the bilinear forms $(x,y) \mapsto B(x,y)$ and $(x,y) \mapsto B(y,x)$ agree on pairs of simple tensors because φ and ψ are symmetric, hence agree.

2. Choose the bases in (1) to be orthogonal bases realizing the two diagonalizations, which exist by theorem 22.2.1, so that $\varphi(v_i, v_{i'}) = 0$ for $i \neq i'$ with $\varphi(v_i, v_i) = a_i$, and likewise for ψ. Then for $(i,j) \neq (i',j')$,
$$(\varphi \otimes \psi)(v_i \otimes w_j,\ v_{i'} \otimes w_{j'}) = \varphi(v_i, v_{i'})\psi(w_j, w_{j'}) = 0$$
   since at least one of the two factors has distinct indices, while the value at $(i,j) = (i',j')$ is $a_i b_j$. So the basis $\{v_i \otimes w_j\}$ is orthogonal for $\varphi \otimes \psi$ with the mn diagonal entries $a_i b_j$,

which is the claim. If all a_i and all b_j are nonzero then so are all a_ib_j, so the matrix of $\varphi\otimes\psi$ in this basis is diagonal with nonzero entries, hence invertible, and lemma 13.3.23 gives nondegeneracy. Finally, if $\varphi' \cong \varphi$ and $\psi' \cong \psi$, then $\varphi' \cong \langle a_1,\dots,a_m\rangle$ and $\psi' \cong \langle b_1,\dots,b_n\rangle$ for the very same entries, since the diagonal notation asserts an isometry class and not a chosen basis. Running the argument just given for those diagonalizations exhibits $\varphi'\otimes\psi'$ and $\varphi\otimes\psi$ as isometric to one and the same diagonal form, so isometric inputs produce isometric outputs. The multiset $\{a_ib_j\}$ is not itself an invariant of the pair of isometry classes, being pinned down only up to multiplying its entries by squares, but the isometry class it names is.

3. Write $\varphi \cong \langle a_1,\dots,a_m\rangle$, $\psi \cong \langle b_1,\dots,b_n\rangle$ and $\chi \cong \langle c_1,\dots,c_p\rangle$, all nondegenerate. By (2) the products a_ib_j and b_ja_i agree, giving $\varphi\otimes\psi \cong \psi\otimes\varphi$; the products $(a_ib_j)c_l$ and $a_i(b_jc_l)$ agree, giving associativity; the entries of $\varphi\otimes(\psi\perp\chi)$ are the a_ib_j together with the a_ic_l, which are the entries of $(\varphi\otimes\psi)\perp(\varphi\otimes\chi)$, giving distributivity over $\perp$; and $\langle 1\rangle\otimes\varphi\cong\varphi$ because the products $1\cdot a_i$ are the a_i.

   It remains to check that the rule respects Witt equivalence. First, $\langle a\rangle\otimes H \cong \langle a\rangle\otimes\langle 1,-1\rangle \cong \langle a,-a\rangle \cong H$, using lemma 22.6.1 at both ends and (2) in the middle, so by distributivity

$$\psi\otimes H \cong (\langle b_1\rangle\otimes H)\perp\cdots\perp(\langle b_n\rangle\otimes H)\cong nH$$

   Now suppose $\varphi\sim\varphi'$, say $\varphi\perp pH\cong\varphi'\perp p'H$. Tensoring with ψ and using that isometric inputs give isometric outputs, then expanding by distributivity and the previous display,

$$(\varphi\otimes\psi)\perp pnH \cong (\varphi'\otimes\psi)\perp p'nH$$

   so $\varphi\otimes\psi\sim\varphi'\otimes\psi$. Commutativity gives the same conclusion in the second variable, so the product of two classes does not depend on the representatives. The ring axioms are the isometries established in the previous paragraph, read on classes, completing the proof.

$W(k)$ is called the *Witt ring* when the multiplication is in play, and the group and the ring have the same underlying set. The multiplication restricts usefully to rank one forms: $\langle a\rangle\otimes\langle b\rangle\cong\langle ab\rangle$ by part (2), so the square classes of k act on $W(k)$ through $[\langle a\rangle]$, and each $[\langle a\rangle]$ is a unit since $\langle a\rangle\otimes\langle a\rangle\cong\langle a^2\rangle\cong\langle 1\rangle$ by proposition 22.2.2.

Two invariants from section 22.1 can now be tested against Witt equivalence. Rank fails, as the computation after definition 22.6.2 shows, but rank modulo 2 survives because a hyperbolic plane has rank 2. The discriminant also fails, and for a reason worth isolating: $\det(\varphi\perp H) = -\det(\varphi)$ by lemma 22.6.1(1), so $\operatorname{disc}(\varphi\perp H)$ and $\operatorname{disc}(\varphi)$ differ by the class of -1, which is nontrivial in most fields. A sign attached to the determinant repairs this, and box 22.1.9 announced it.

Definition 22.6.7: The Signed Discriminant

Let φ be a nondegenerate symmetric bilinear form of rank n over a field k with $\operatorname{char} k \neq 2$. The *signed discriminant* of φ is the square class

$$d_{\pm}(\varphi) := (-1)^{n(n-1)/2}\operatorname{disc}(\varphi) \in k^\times/(k^\times)^2$$

with disc as in definition 22.1.7. For $n=0$ the determinant of the empty matrix is 1, so $d_\pm$ of the zero form is the trivial class.

The definition depends on nothing but the isometry class of φ, since the rank does not and disc does not by proposition 22.1.8. For a diagonal form the sign multiplies the product of the entries: $d_\pm\langle a_1,\dots,a_n\rangle = (-1)^{n(n-1)/2}a_1\cdots a_n$ modulo squares, by corollary 22.2.3. Thus $d_\pm\langle a\rangle = a$, $d_\pm\langle a,b\rangle = -ab$ and $d_\pm\langle a,b,c\rangle = -abc$.

Proposition 22.6.8: Rank Modulo Two and the Signed Discriminant Are Witt Invariants

Let k be a field with $\operatorname{char} k \neq 2$.

1. The assignment $e\big([\varphi]\big) := \dim\varphi + 2\mathbb{Z}$ is a well defined surjective ring homomorphism $e\colon W(k) \to \mathbb{Z}/2\mathbb{Z}$.
2. For nondegenerate φ of rank m and ψ of rank n,
$$d_\pm(\varphi\perp\psi) = (-1)^{mn}\, d_\pm(\varphi)\, d_\pm(\psi)$$
3. Witt equivalent forms have equal signed discriminants, so $d_\pm$ is a well defined map $W(k) \to k^\times/(k^\times)^2$.
4. Restricted to the classes of even rank, $d_\pm$ is a group homomorphism from $\big(\ker e, +\big)$ to $k^\times/(k^\times)^2$.

***Proof*:**

1. If $\varphi \perp mH \cong \psi \perp nH$ then comparing dimensions gives $\dim\varphi + 2m = \dim\psi + 2n$, so $\dim\varphi \equiv \dim\psi \pmod 2$ and e is well defined. It is additive because $\dim(\varphi\perp\psi) = \dim\varphi + \dim\psi$, and multiplicative because $\dim(V\otimes_k W) = \dim V\cdot\dim W$ by corollary 15.2.6. It sends the identity $[\langle 1\rangle]$ to $1+2\mathbb{Z}$, which also gives surjectivity.
2. Both sides are computed from determinants and ranks. The rank of $\varphi\perp\psi$ is $m+n$ and its determinant is $\det(\varphi)\det(\psi)$, by the block form recorded at the start of this section. So the two sides differ only in the power of -1, and
$$\frac{(m+n)(m+n-1)}{2} - \frac{m(m-1)}{2} - \frac{n(n-1)}{2} = \frac{2mn}{2} = mn$$
after expanding the numerator as $m^2+2mn+n^2-m-n-m^2+m-n^2+n$. Raising -1 to that difference gives the stated factor.
3. By lemma 22.6.1(1) the hyperbolic plane has rank 2 and determinant -1, so $d_\pm(H) = (-1)^{2(2-1)/2}\cdot(-1) = (-1)(-1) = 1$ in $k^\times/(k^\times)^2$. Applying (2) with $\psi = H$, where $n=2$ makes $(-1)^{mn} = 1$,
$$d_\pm(\varphi\perp H) = d_\pm(\varphi)\, d_\pm(H) = d_\pm(\varphi)$$
Iterating, $d_\pm(\varphi\perp mH) = d_\pm(\varphi)$ for every $m\geq 0$. Now if $\varphi\perp mH \cong \psi\perp nH$, then the two sides have equal rank and equal discriminant, the latter by proposition 22.1.8, hence equal signed discriminant. Combining the three equalities gives $d_\pm(\varphi) = d_\pm(\psi)$.
4. If m and n are both even then $(-1)^{mn} = 1$, so (2) reads $d_\pm(\varphi\perp\psi) = d_\pm(\varphi)d_\pm(\psi)$. Since addition in $W(k)$ is induced by $\perp$ and the target group is written multiplicatively, this is the homomorphism property on $\ker e$, as we sought to show.

The restriction in (4) is not an artifact of the proof. Over $\mathbb{Q}$ one has $d_\pm\langle 1\rangle = 1$ while $d_\pm\big(\langle 1\rangle \perp \langle 1\rangle\big) = d_\pm\langle 1,1\rangle = -1$, and -1 is not a square in $\mathbb{Q}$, so $d_\pm$ is not additive on all of $W(\mathbb{Q})$. The subgroup where it is additive has a name.

Definition 22.6.9: The Fundamental Ideal

Let k be a field with $\operatorname{char} k \neq 2$. The *fundamental ideal* of $W(k)$ is

$$I(k) := \ker\big(e\colon W(k) \to \mathbb{Z}/2\mathbb{Z}\big)$$

the set of Witt classes of even rank. It is an ideal of the Witt ring, being the kernel of the ring homomorphism of proposition 22.6.8(1), and $W(k)/I(k) \cong \mathbb{Z}/2\mathbb{Z}$.

The classes $[\langle 1, -a\rangle]$ for $a \in k^\times$ lie in $I(k)$ and have signed discriminant $-1\cdot(1\cdot(-a)) = a$, so $d_\pm$ maps $I(k)$ onto the whole square class group. Rank modulo 2 and the signed discriminant are thus two homomorphisms out of $W(k)$ and out of $I(k)$ respectively, and over a finite field they turn out to separate every class from every other.

Proposition 22.6.10: Rank and Discriminant Classify Forms over a Finite Field

Let q be an odd prime power and fix a non-square $\epsilon \in \mathbb{F}_q^\times$.

1. For all $a, b \in \mathbb{F}_q^\times$ and all $c \in \mathbb{F}_q$ there exist $x, y \in \mathbb{F}_q$ with $ax^2 + by^2 = c$.
2. Every nondegenerate form over $\mathbb{F}_q$ of rank at least 3 is isotropic, and every nondegenerate form of rank at least 2 represents every element of $\mathbb{F}_q^\times$.
3. Two nondegenerate forms over $\mathbb{F}_q$ are isometric if and only if they have the same rank and the same discriminant.
4. Up to isometry the anisotropic forms over $\mathbb{F}_q$ are exactly the zero form, $\langle 1\rangle$, $\langle \epsilon\rangle$ and $\langle 1, -\epsilon\rangle$.

***Proof*:**

1. This is proved in example 22.4.9, by the counting argument on $S = \{ax^2 \mid x \in \mathbb{F}_q\}$ and $T = \{c - by^2 \mid y \in \mathbb{F}_q\}$, each of size $(q+1)/2$ in a set of size q.
2. The first claim is example 22.4.9, which deduces it from (1). For the second, let φ be nondegenerate of rank $n \geq 2$ and diagonalize it by theorem 22.2.1 as $\varphi \cong \langle a_1, \dots, a_n\rangle$ in an orthogonal basis $w_1, \dots, w_n$, every a_i being nonzero by the last claim of proposition 22.2.2. Given $c \in \mathbb{F}_q^\times$, part (1) gives $x, y \in \mathbb{F}_q$ with $a_1x^2 + a_2y^2 = c$. Put $v := xw_1 + yw_2$. Then $q(v) = a_1x^2 + a_2y^2 = c$, which is nonzero, so $v \neq 0$ and φ represents c.
3. Isometric forms have the same rank and, by proposition 22.1.8, the same discriminant. For the converse, induct on the common rank n. For $n = 0$ both forms are the zero form on the zero space. For $n = 1$, write $\varphi \cong \langle a\rangle$ and $\psi \cong \langle b\rangle$. Equality of the discriminants means $b = ac^2$ for some $c \in \mathbb{F}_q^\times$ by corollary 22.2.3 and definition 22.1.6, and then $\langle a\rangle \cong \langle b\rangle$ by proposition 22.2.2.

   Let $n \geq 2$. Diagonalizing φ produces a vector v with $c := \varphi(v,v) \neq 0$, and by (2) the form

ψ also represents c, say $\psi(v', v') = c$. A vector of nonzero square splits off: for x in the space of φ,

$$x = \frac{\varphi(v,x)}{c}\, v + \left(x - \frac{\varphi(v,x)}{c}\, v \right)$$

where applying $\varphi(v, -)$ to the bracket gives $\varphi(v,x) - \frac{\varphi(v,x)}{c} c = 0$, so the bracket lies in $\{v\}^\perp$ (definition 13.3.20), a subspace by lemma 13.3.21. And if $\lambda v \in \{v\}^\perp$ then $\lambda c = 0$, so $\lambda = 0$. Hence the space is the direct sum of kv and $\{v\}^\perp$, these are orthogonal, and $\varphi \cong \langle c \rangle \perp \varphi'$ with φ' the restriction to $\{v\}^\perp$. The same argument gives $\psi \cong \langle c \rangle \perp \psi'$. Both φ' and ψ' have rank $n-1$ and are nondegenerate, since $c \det(\varphi') = \det(\varphi) \neq 0$ and likewise for ψ. Their discriminants agree: in the square class group, $\operatorname{disc}(\varphi') = c^{-1} \operatorname{disc}(\varphi) = c^{-1} \operatorname{disc}(\psi) = \operatorname{disc}(\psi')$. By the inductive hypothesis $\varphi' \cong \psi'$, and taking the orthogonal sum with the identity of $\langle c \rangle$ gives $\varphi \cong \psi$.

4. An anisotropic form has rank at most 2 by (2). The zero form is anisotropic vacuously. In rank 1, every $\langle a \rangle$ is anisotropic, and by (3) the isometry class is determined by the discriminant, which is the class of a. The two square classes of $\mathbb{F}_q$ are those of 1 and ϵ by example 22.2.5. In rank 2, the form $\langle a, b \rangle$ is isotropic exactly when $-ab$ is a square: given $ax^2 + by^2 = 0$ with $(x, y) \neq (0, 0)$, the case $y = 0$ would force $ax^2 = 0$ and hence $x = 0$, so $y \neq 0$ and $-ab = a^2(x/y)^2$ is a square. Conversely if $-ab = t^2$ then $(x, y) = (t, a)$ is nonzero and $at^2 + ba^2 = a(t^2 + ab) = 0$. So $\langle a, b \rangle$ is anisotropic exactly when $-ab$ is a non-square, that is when $\operatorname{disc}\langle a, b \rangle = ab$ is the class of $-\epsilon$. By (3) that single condition on the discriminant pins down one isometry class in rank 2, and $\langle 1, -\epsilon \rangle$ is in it, completing the proof.

Part (3) is the affirmative answer promised at the end of example 22.2.5: over $\mathbb{F}_q$ the rank and the discriminant are a complete system of invariants, so the two forms $\langle 1, 1 \rangle$ and $\langle 1, 2 \rangle$ exhibited there over $\mathbb{F}_5$ fail to be isometric for the only reason available. Part (4) turns proposition 22.6.4 into a count.

Example 22.6.11: The Witt Group of a Finite Field

Let q be odd and $\epsilon \in \mathbb{F}_q^\times$ a non-square. By proposition 22.6.4 the elements of $W(\mathbb{F}_q)$ correspond to isometry classes of anisotropic forms, and proposition 22.6.10(4) lists four of them, so

$$W(\mathbb{F}_q) = \{\, [0],\ [\langle 1 \rangle],\ [\langle \epsilon \rangle],\ [\langle 1, -\epsilon \rangle] \,\}$$

has exactly four elements. Which group of order four it is depends on q modulo 4, through whether -1 is a square.

The case $q \equiv 1 \pmod 4$. Then -1 is a square, say $-1 = c^2$: the group $\mathbb{F}_q^\times$ is cyclic by theorem 11.3.14, of order $q - 1$, which is divisible by 4 in this case, so it has an element of order 4, whose square has order 2 and hence equals -1, the unique element of order 2. For any $a \in \mathbb{F}_q^\times$, proposition 22.2.2 gives $\langle a \rangle \cong \langle ac^2 \rangle = \langle -a \rangle$, so every form satisfies $\varphi \cong -\varphi$ after diagonalizing, and

$$2[\varphi] = [\varphi] + [-\varphi] = 0$$

by theorem 22.6.3. A group of order 4 in which every element has order dividing 2 is $\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$.

The case $q \equiv 3 \pmod 4$. Then -1 is a non-square, so $\epsilon = -1$ is an admissible choice, and $\langle 1, 1 \rangle$ is anisotropic by the criterion in the proof of proposition 22.6.10(4), since $-1 \cdot 1 = -1$ is a non-square. Hence $2[\langle 1 \rangle] = [\langle 1, 1 \rangle] \neq 0$ by proposition 22.6.4. On the other hand $\langle 1, 1, 1, 1 \rangle$ and

$2H \cong \langle 1,-1,1,-1\rangle$ both have rank 4 and discriminant the class of 1, so they are isometric by proposition 22.6.10(3), giving $4[\langle 1\rangle] = [2H] = 0$. So $[\langle 1\rangle]$ has order 4 in a group of order 4, and $W(\mathbb{F}_q) \cong \mathbb{Z}/4\mathbb{Z}$.

The invariants separate the four classes. Writing each class through its anisotropic representative and applying proposition 22.6.8, the rank modulo 2 and the signed discriminant take the values

	$[0]$	$[\langle 1\rangle]$	$[\langle \epsilon\rangle]$	$[\langle 1,-\epsilon\rangle]$
e	0	1	1	0
$d_\pm$	1	1	ϵ	ϵ

the last entry because $d_\pm\langle 1,-\epsilon\rangle = -(1\cdot(-\epsilon)) = \epsilon$. The four pairs are distinct, so a Witt class over a finite field is determined by its rank modulo 2 together with its signed discriminant.

A worked instance. Take $q = 5$. The squares in $\mathbb{F}_5^\times$ are 1 and 4, so $\epsilon = 2$ is a non-square and $-1 = 4 = 2^2$ is a square, putting this in the first case: $W(\mathbb{F}_5) \cong \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$, with anisotropic representatives 0, $\langle 1\rangle$, $\langle 2\rangle$ and $\langle 1,-2\rangle = \langle 1,3\rangle$. The two forms of example 22.2.5 land in different classes: $\langle 1,1\rangle \cong \langle 1,-1\rangle \cong H$ is Witt trivial, while $[\langle 1,2\rangle] = [\langle 1\rangle] + [\langle 2\rangle]$ has signed discriminant $-2 = 3$, a non-square, so it is not the class of 0.

Exercise 22.6.1

1. Take $k = \mathbb{F}_7$. List the square classes, determine the four anisotropic forms over $\mathbb{F}_7$ up to isometry, and identify $W(\mathbb{F}_7)$ as an abstract group. Which of the four classes is $3[\langle 1\rangle]$?

2. Let $\operatorname{char} k \neq 2$ and let φ be nondegenerate over k. Show that $[\varphi] = 0$ in $W(k)$ if and only if $\varphi \cong mH$ for some $m \geq 0$, and deduce that a nondegenerate form of odd rank is never Witt trivial. Conclude that $W(k)$ has at least two elements for every such k.

3. Let φ and ψ be nondegenerate forms over k with $[\varphi] = [\psi]$ and $\dim\varphi = \dim\psi$. Show that $\varphi \cong \psi$. Give an example over some field showing that the hypothesis on the dimensions cannot be dropped.

4. Suppose -1 is a square in k, where $\operatorname{char} k \neq 2$. Show that $\varphi \cong -\varphi$ for every form over k, hence that $2[\varphi] = 0$ for every $[\varphi] \in W(k)$, so that $W(k)$ is a vector space over $\mathbb{F}_2$. Check that example 22.6.5 is an instance.

5. Show that $a \mapsto [\langle a\rangle]$ induces a well defined group homomorphism from $k^\times/(k^\times)^2$ to the group of units of the Witt ring, and that it is injective for every field k with $\operatorname{char} k \neq 2$. (For injectivity, identify the anisotropic part of $\langle a\rangle$.)

6. Show that $d_\pm\colon I(k) \to k^\times/(k^\times)^2$ is surjective for every k with $\operatorname{char} k \neq 2$. Then compute $I(\mathbb{F}_q)$ for q odd and show that $d_\pm$ is an isomorphism of groups $I(\mathbb{F}_q) \xrightarrow{\sim} \mathbb{F}_q^\times/(\mathbb{F}_q^\times)^2$.

7. Show that $I(k)$ is generated as an abelian group by the classes $[\langle 1,-a\rangle]$ with a ranging over $k^\times$. (Start by expressing $[\langle a\rangle] - [\langle 1\rangle]$ as one of these generators up to sign.)

8. Show that the form $\langle 1,1,\dots,1\rangle$ of rank n is anisotropic over $\mathbb{Q}$ for every n, and deduce that the classes $n[\langle 1\rangle]$ for $n \geq 0$ are pairwise distinct, so that $W(\mathbb{Q})$ is infinite and $[\langle 1\rangle]$ has infinite order. Contrast this with example 22.6.11.

22.7 The Clifford Algebra

Every construction so far has compared one form with another. This section attaches to a single form an associative algebra, built so that the form becomes the operation of squaring: inside it the equation $v^2 = q(v)$ holds for every $v \in V$. That equation is meaningless in V, where there is no product, so the object carrying it must be an algebra containing V, and section 16.3 is where algebras containing a given vector space are constructed.

The construction below is a quotient of $\mathcal{T}(V)$ by one family of relations, and for $q = 0$ it is the exterior algebra. What is proved here is that the quotient has dimension 2^n rather than collapsing, that maps out of it are prescribed by their restriction to V, and that although the $\mathbb{Z}$-grading of $\mathcal{T}(V)$ does not survive, its reduction modulo 2 does.

Definition 16.3.7 formed $\bigwedge(V)$ by killing the elements $v \otimes v$ of $\mathcal{T}(V)$, which forces $v^2 = 0$ in the quotient. Killing $v \otimes v - q(v)$ instead forces $v^2 = q(v)$. Assume $q \neq 0$ and choose v with $q(v) \neq 0$. The element $v \otimes v - q(v)$ is then a sum of a nonzero term of degree 2 and a nonzero term of degree 0, so it is not homogeneous, and the ideal generated by these elements is not a graded ideal in the sense of definition 16.2.3: were it graded, the two homogeneous components of $v \otimes v - q(v)$ would lie in it separately, so the invertible scalar $q(v)$, and with it 1, would lie in it and the quotient would be 0, contradicting theorem 22.7.5. For $q \neq 0$ the images of the $\mathcal{T}^d(V)$ therefore do not make the quotient a graded ring (definition 16.2.1). Only in the excluded case $q = 0$, where the ideal is the one of definition 16.3.7 and is graded, does the $\mathbb{Z}$-grading survive (example 22.7.2). Proposition 22.7.7 says exactly how much of the grading is left in general.

Definition 22.7.1: Clifford Algebra

Let k be a field with $\operatorname{char} k \neq 2$, let V be a finite-dimensional k-vector space, and let q be a quadratic form on V (definition 22.1.1). Let $I(q) \subseteq \mathcal{T}(V)$ be the two-sided ideal of the tensor algebra (definition 16.3.1) generated by the elements

$$v \otimes v - q(v) \cdot 1 \qquad (v \in V)$$

where $q(v) \cdot 1$ lies in $\mathcal{T}^0(V) = k$. The *Clifford algebra* of q is the quotient k-algebra

$$C(q) := \mathcal{T}(V)/I(q)$$

and $\iota \colon V \to C(q)$ denotes the composite of the inclusion $V = \mathcal{T}^1(V) \subseteq \mathcal{T}(V)$ with the projection $\pi \colon \mathcal{T}(V) \to C(q)$.

Write φ for the symmetric bilinear form attached to q by proposition 22.1.2, so that $\varphi(v, v) = q(v)$. Two identities in $C(q)$ are used in everything below. The first is the defining relation

$$\iota(v)^2 = q(v)$$

which holds because π is an algebra homomorphism, so that $\iota(v)^2 = \pi(v \otimes v) = \pi\big(q(v) \cdot 1\big) = q(v)$, the middle equality holding because the two arguments differ by a generator of $I(q)$. The second identity is obtained by applying the first to $v + w$. Bilinearity of φ gives $q(v + w) = \varphi(v + w, v + w) = q(v) + 2\varphi(v, w) + q(w)$, while expanding the square in $C(q)$ gives

$$\iota(v + w)^2 = \big(\iota(v) + \iota(w)\big)^2 = \iota(v)^2 + \iota(v)\iota(w) + \iota(w)\iota(v) + \iota(w)^2$$

using that ι is linear. Equating the two and cancelling $\iota(v)^2 = q(v)$ and $\iota(w)^2 = q(w)$ leaves

$$\iota(v)\iota(w) + \iota(w)\iota(v) = 2\varphi(v, w) \tag{22.3}$$

Since $\operatorname{char} k \neq 2$, the right-hand side vanishes exactly when v and w are orthogonal (definition 13.3.20), so orthogonality of vectors becomes anticommutation of their images. This is the mechanism that turns a choice of orthogonal basis, which theorem 22.2.1 always gives, into a presentation of $C(q)$ by generators and relations.

Example 22.7.2: The Clifford Algebra of the Zero Form

Let k be a field with $\operatorname{char} k \neq 2$ and take $q = 0$ on a k-vector space V of dimension n. The generators of $I(0)$ are the elements $v \otimes v - 0 = v \otimes v$ for $v \in V$, which are precisely the generators of the ideal $\mathcal{A}(V)$ of definition 16.3.7. The two ideals therefore coincide, and

$$C(0) = \mathcal{T}(V)/\mathcal{A}(V) = \bigwedge(V)$$

not merely up to isomorphism but as the same quotient. Its dimension is

$$\dim_k \bigwedge(V) = \sum_{r=0}^{n} \binom{n}{r} = 2^n$$

since $\bigwedge^r(V)$ is free of rank $\binom{n}{r}$ by theorem 16.4.3 and vanishes for $r > n$. Note that nothing in definition 22.7.1 asked q to be nondegenerate, and for $V \neq 0$ the zero form has radical all of V.

The definition determines the algebra but leaves two questions open: how large it is, and how to produce a homomorphism out of it. The second is answered first, and the answer then gives the first, because the way to bound the dimension of $C(q)$ from below is to map it onto something of known dimension.

Proposition 22.7.3: The Universal Property of the Clifford Algebra

Let k be a field with $\operatorname{char} k \neq 2$, let V be a finite-dimensional k-vector space, and let q be a quadratic form on V.

1. For every k-algebra A and every k-linear map $f\colon V \to A$ satisfying $f(v)^2 = q(v) \cdot 1_A$ for all $v \in V$, there is a unique k-algebra homomorphism $F\colon C(q) \to A$ with $F \circ \iota = f$.
2. Let q' be a quadratic form on V', with associated symmetric bilinear forms φ and φ', and let $g\colon (V, \varphi) \to (V', \varphi')$ be an isometry (definition 22.1.10). Then there is a unique k-algebra isomorphism $C(g)\colon C(q) \to C(q')$ with $C(g)\big(\iota(v)\big) = \iota'\big(g(v)\big)$ for all $v \in V$. In particular isometric forms have isomorphic Clifford algebras.

***Proof*:**

1. *Existence.* Take $R = k$ and $M = V$ in theorem 16.3.2(2): its hypotheses ask for a k-algebra and a k-module homomorphism from V into it, which are A and f. So it gives a unique k-algebra homomorphism $\Psi\colon \mathcal{T}(V) \to A$ with $\Psi|_V = f$. On a generator of $I(q)$,

$$\Psi\big(v \otimes v - q(v) \cdot 1\big) = \Psi(v)\Psi(v) - q(v)\Psi(1) = f(v)^2 - q(v) \cdot 1_A = 0$$

by the hypothesis on f. Every element of $I(q)$ is a finite sum of products $x\gamma y$ with $x, y \in \mathcal{T}(V)$ and γ one of those generators, and $\Psi(x\gamma y) = \Psi(x)\Psi(\gamma)\Psi(y) = 0$, so $I(q) \subseteq \ker \Psi$. Hence

$$F\big(x + I(q)\big) := \Psi(x)$$

is well defined: two representatives of the same class differ by an element of $I(q)$, on which Ψ vanishes. It is k-linear, multiplicative and unital because Ψ is and because $\pi\colon \mathcal{T}(V) \to C(q)$ is a surjective algebra homomorphism with $F \circ \pi = \Psi$. Finally $F \circ \iota = \Psi|_V = f$.

Uniqueness. The algebra $C(q)$ is generated as a k-algebra by $\iota(V)$: the simple tensors $v_1 \otimes \cdots \otimes v_m$ generate $\mathcal{T}^m(V)$ as a k-module by lemma 15.1.4, hence together with 1 they span $\mathcal{T}(V)$, and π is surjective and sends $v_1 \otimes \cdots \otimes v_m$ to $\iota(v_1)\cdots\iota(v_m)$. Now if F_1, F_2 are algebra homomorphisms out of $C(q)$ agreeing on $\iota(V)$, the set on which they agree is closed under sums, products and scalars and contains 1, so it is a subalgebra containing a generating set, hence all of $C(q)$. So F is unique.

2. An isometry satisfies $\varphi'(g(v), g(v)) = \varphi(v, v)$, that is $q'(g(v)) = q(v)$. So the k-linear map $f := \iota' \circ g\colon V \to C(q')$ satisfies

$$f(v)^2 = \iota'\big(g(v)\big)^2 = q'\big(g(v)\big) = q(v)$$

and part (1) gives a unique algebra homomorphism $C(g)$ with $C(g) \circ \iota = \iota' \circ g$. The inverse map g^{-1} is again an isometry, since applying $\varphi'(g(-), g(-)) = \varphi(-,-)$ to the vectors $g^{-1}(x), g^{-1}(y)$ gives $\varphi'(x, y) = \varphi(g^{-1}x, g^{-1}y)$, so the same construction produces $C(g^{-1})\colon C(q') \to C(q)$. The composite $C(g^{-1}) \circ C(g)$ is an algebra homomorphism $C(q) \to C(q)$ carrying $\iota(v)$ to $\iota(g^{-1}g(v)) = \iota(v)$, and the identity of $C(q)$ carries $\iota(v)$ to $\iota(v)$ as well. Both therefore solve the problem of part (1) for $A = C(q)$ and $f = \iota$, whose hypothesis $\iota(v)^2 = q(v)$ holds, so uniqueness forces $C(g^{-1}) \circ C(g) = \mathrm{id}_{C(q)}$. Exchanging the roles of g and g^{-1} gives $C(g) \circ C(g^{-1}) = \mathrm{id}_{C(q')}$, so $C(g)$ is an isomorphism, completing the proof.

Part (1) is the statement that prescribing an algebra homomorphism out of $C(q)$ costs exactly one k-linear map $V \to A$ whose values square to the scalars $q(v)$. The relations impose nothing further. Part (2) is what makes $C(q)$ an invariant of the isometry class rather than of a chosen presentation, so it may be computed from any convenient diagonalization $\langle a_1, \ldots, a_n\rangle$ produced by theorem 22.2.1. The smallest case already shows that the answer depends on the field.

Example 22.7.4: Clifford Algebras of a Line

Let k be a field with $\operatorname{char} k \neq 2$, let $\dim_k V = 1$, say $V = kv$, and let $a := q(v)$, so that $q \cong \langle a\rangle$. Then

$$C(q) \cong k[x]/(x^2 - a)$$

To see this, write $A := k[x]/(x^2 - a)$ and let $\bar{x}$ be the class of x. The map $f\colon V \to A$ sending $\lambda v \mapsto \lambda\bar{x}$ is k-linear, and $f(\lambda v)^2 = \lambda^2\bar{x}^2 = \lambda^2 a = q(\lambda v)$ by definition 22.1.1(1), so proposition 22.7.3(1) gives an algebra homomorphism $F\colon C(q) \to A$ with $F(\iota(v)) = \bar{x}$. In the other direction, evaluation at $\iota(v)$ is an algebra homomorphism $k[x] \to C(q)$, since the powers of $\iota(v)$ commute with one another and k is central in $C(q)$. It kills $x^2 - a$ because $\iota(v)^2 = q(v) = a$, so it factors through a homomorphism $G\colon A \to C(q)$ with $G(\bar{x}) = \iota(v)$. Now $F \circ G$ is an algebra endomorphism of A fixing $\bar{x}$, and the elements it fixes form a subalgebra, which therefore contains the algebra

generated by $\bar{x}$, namely all of A. So $F \circ G = \mathrm{id}_A$. Likewise $G \circ F$ is an algebra endomorphism of $C(q)$ fixing $\iota(V)$ pointwise, hence equals $\mathrm{id}_{C(q)}$ by the uniqueness clause of proposition 22.7.3(1). So F is an isomorphism.

Three cases occur, according to the square class of a (definition 22.1.6).

1. $a = 0$. Then $C(q) \cong k[x]/(x^2)$, the algebra of dual numbers, in agreement with example 22.7.2, which identifies it with $\bigwedge(V)$.
2. $a = c^2$ with $c \in k^\times$. Then $x^2 - a = (x - c)(x + c)$ is a product of two distinct monic irreducibles, distinct because $c \neq -c$ when $\operatorname{char} k \neq 2$, so proposition 11.1.6 gives
$$C(q) \cong k[x]/(x - c) \times k[x]/(x + c) \cong k \times k$$
an algebra with zero divisors: $(\bar{x} - c)(\bar{x} + c) = 0$.
3. $a \in k^\times$ is not a square. Then $C(q)$ is a field, the quadratic extension of k obtained by adjoining a square root of a. Indeed $(u + w\bar{x})(u - w\bar{x}) = u^2 - w^2 a$ for $u, w \in k$, and this is nonzero unless $u = w = 0$: if $w \neq 0$ then $u^2 = w^2 a$ would make $a = (u/w)^2$ a square, and if $w = 0$ then $u \neq 0$ gives $u^2 \neq 0$. So every nonzero element of $C(q)$ has the inverse $(u - w\bar{x})/(u^2 - w^2 a)$.

Over $k = \mathbb{R}$ the three cases are $\mathbb{R}[x]/(x^2)$, $\mathbb{R} \times \mathbb{R}$ and $\mathbb{C}$, the last being $C(\langle -1 \rangle)$.

All three algebras above have dimension $2 = 2^1$, and the same count holds in every dimension. Proving it is the one place where the construction has to be shown not to collapse: nothing said so far rules out $C(q) = 0$, since a quotient by a large ideal can kill everything. The proof below produces an explicit $C(q)$-module of dimension 2^n, and the independence of the monomials of $C(q)$ follows from the independence of that module's basis.

Theorem 22.7.5: The Monomial Basis of $C(q)$

Let k be a field with $\operatorname{char} k \neq 2$, let V be a k-vector space of dimension n, let q be a quadratic form on V, and let $e_1, \dots, e_n$ be an orthogonal basis of V for the associated symmetric bilinear form φ, with $a_i := q(e_i)$. For a subset $S = \{i_1 < i_2 < \cdots < i_r\}$ of $\{1, \dots, n\}$ write
$$e_S := \iota(e_{i_1})\iota(e_{i_2}) \cdots \iota(e_{i_r}) \in C(q), \qquad e_\emptyset := 1$$
Then the 2^n elements e_S form a k-basis of $C(q)$. In particular $\dim_k C(q) = 2^n$ and $\iota \colon V \to C(q)$ is injective.

***Proof*:**
An orthogonal basis exists by theorem 22.2.1, whose hypotheses are the standing ones here. Throughout, ι is suppressed from the notation for products, so e_i denotes $\iota(e_i) \in C(q)$.

Step 1: the relations. By (22.3) applied to $v = w = e_i$ and to $v = e_i$, $w = e_j$ with $i \neq j$,
$$e_i^2 = q(e_i) = a_i, \qquad e_i e_j = -e_j e_i \quad (i \neq j)$$
the second because $\varphi(e_i, e_j) = 0$ for $i \neq j$ by the choice of basis. Also, for $v = \sum_i c_i e_i$, expanding

φ by bilinearity and using orthogonality gives

$$q(v) = \varphi(v,v) = \sum_i c_i^2 a_i \tag{22.4}$$

Step 2: the e_S span $C(q)$. As established in the uniqueness half of the proof of proposition 22.7.3, $C(q)$ is spanned as a k-vector space by 1 together with the products $\iota(v_1)\cdots\iota(v_m)$ with $v_t \in V$. Expanding each v_t in the basis $e_1,\dots,e_n$ and multiplying out, which is legitimate because multiplication in $C(q)$ is k-bilinear, shows that the products $e_{j_1}e_{j_2}\cdots e_{j_m}$ of basis vectors already span.

It remains to reduce such a product to a scalar multiple of some e_S, which is done by induction on m. For $m=0$ the product is $1 = e_\emptyset$. If the indices $j_1,\dots,j_m$ are pairwise distinct, then finitely many interchanges of adjacent factors sort them into increasing order, each interchange multiplying the product by -1 by Step 1, so the product equals $\pm e_S$ with $S = \{j_1,\dots,j_m\}$. If two indices coincide, choose a pair $j_s = j_t = i$ with $s<t$ and $t-s$ minimal. Then $j_u \neq i$ for $s<u<t$, so moving the factor e_{j_t} leftwards past those $t-s-1$ factors changes the product only by a sign and makes the two occurrences of e_i adjacent. Replacing e_ie_i by the scalar a_i produces a scalar multiple of a product of length $m-2$, which the inductive hypothesis handles. Hence the e_S span, and $\dim_k C(q) \le 2^n$.

Step 3: a module of dimension 2^n. Let M be the k-vector space with basis $\{u_S \mid S \subseteq \{1,\dots,n\}\}$, so $\dim_k M = 2^n$. For an index i and a subset S write $S\triangle i$ for the symmetric difference, that is $S\cup\{i\}$ if $i\notin S$ and $S\setminus\{i\}$ if $i\in S$, and set

$$d(i,S) := \#\{l \in S \mid l < i\}, \qquad \sigma(i,S) := (-1)^{d(i,S)}, \qquad c_i(S) := \begin{cases} a_i & i \in S \\ 1 & i \notin S \end{cases}$$

Define $E_i \in \mathrm{End}_k(M)$ on the basis by

$$E_i(u_S) := c_i(S)\,\sigma(i,S)\,u_{S\triangle i}$$

Three facts about these coefficients are what the computation uses. First, $\sigma(i,S\triangle j) = \sigma(i,S)$ whenever $j \ge i$, since $d(i,-)$ counts only elements smaller than i and is therefore unchanged by inserting or deleting j. Second, $\sigma(j,S\triangle i) = -\sigma(j,S)$ whenever $i<j$, since $d(j,-)$ then changes by exactly one. Third, $c_i(S\triangle j) = c_i(S)$ for $j\neq i$, since membership of i is unchanged, while $c_i(S)c_i(S\triangle i) = a_i$, one of the two factors being a_i and the other 1.

Applying E_i twice and using the first and third facts,

$$E_i^2(u_S) = c_i(S)\sigma(i,S)\,c_i(S\triangle i)\sigma(i,S\triangle i)\,u_S = a_i\,\sigma(i,S)^2\,u_S = a_i\,u_S$$

so $E_i^2 = a_i\,\mathrm{id}_M$. For $i<j$, the same facts give

$$E_iE_j(u_S) = c_j(S)\sigma(j,S)\,c_i(S\triangle j)\sigma(i,S\triangle j)\,u_{S\triangle\{i,j\}} = c_i(S)c_j(S)\sigma(i,S)\sigma(j,S)\,u_{S\triangle\{i,j\}}$$

whereas, the second fact now contributing a sign,

$$E_jE_i(u_S) = c_i(S)\sigma(i,S)\,c_j(S\triangle i)\sigma(j,S\triangle i)\,u_{S\triangle\{i,j\}} = -\,c_i(S)c_j(S)\sigma(i,S)\sigma(j,S)\,u_{S\triangle\{i,j\}}$$

so $E_iE_j = -E_jE_i$ for $i\neq j$.

Step 4: the e_S are independent. Let $\rho\colon V \to \mathrm{End}_k(M)$ be the k-linear map with $\rho(e_i) = E_i$. For $v = \sum_i c_i e_i$, expanding the square and grouping the cross terms in pairs,

$$\rho(v)^2 = \sum_i c_i^2 E_i^2 + \sum_{i<j} c_i c_j \big(E_i E_j + E_j E_i\big) = \Big(\sum_i c_i^2 a_i\Big)\mathrm{id}_M = q(v)\,\mathrm{id}_M$$

by Step 3 and (22.4). So proposition 22.7.3(1), applied to the k-algebra $A = \mathrm{End}_k(M)$, yields an algebra homomorphism $\Phi\colon C(q) \to \mathrm{End}_k(M)$ with $\Phi(e_i) = E_i$.

Evaluate $\Phi(e_S)$ at $u_\emptyset$ for $S = \{i_1 < \cdots < i_r\}$, by descending induction on t on the claim that $E_{i_{t+1}} \cdots E_{i_r}(u_\emptyset) = u_T$ with $T = \{i_{t+1}, \ldots, i_r\}$. For $t = r$ the product is empty and the claim reads $u_\emptyset = u_\emptyset$. Given it for t, every element of T exceeds i_t, so $d(i_t, T) = 0$ and $\sigma(i_t, T) = 1$, while $i_t \notin T$ gives $c_{i_t}(T) = 1$. Hence $E_{i_t}(u_T) = u_{T\cup\{i_t\}}$, which is the claim for $t-1$. At $t = 0$ it reads $\Phi(e_S)(u_\emptyset) = u_S$. Now if $\sum_S \lambda_S e_S = 0$ in $C(q)$, applying Φ and evaluating at $u_\emptyset$ gives $\sum_S \lambda_S u_S = 0$ in M, and the u_S are a basis, so every $\lambda_S = 0$.

The e_S therefore span and are independent, so they are a basis and $\dim_k C(q) = 2^n$. The images $\iota(e_1), \ldots, \iota(e_n)$ are the basis elements e_S with $\#S = 1$, hence linearly independent, so the linear map ι carries a basis of V to an independent set and is injective, completing the proof.

Two consequences are recorded before any further computation. First, ι being injective, V is identified with its image from here on and the symbol ι is dropped: a vector $v \in V$ is treated as an element of $C(q)$ with $v^2 = q(v)$. Second, $C(q)$ and $\bigwedge(V)$ have the same dimension for every q, by theorem 22.7.5 and example 22.7.2, and the same monomials index a basis of each. The multiplications differ, since squares of vectors are 0 in one and $q(v)$ in the other. For $n \geq 2$ neither algebra is commutative: $e_1e_2 = -e_2e_1$ in both, and $e_1e_2 \neq 0$ because it is a basis element (theorem 22.7.5 and theorem 16.4.3), so $e_1e_2 = e_2e_1$ would give $2e_1e_2 = 0$ against $\mathrm{char}\, k \neq 2$.

Example 22.7.6: The Quaternions as a Clifford Algebra

Let $V = \mathbb{R}^2$ and $q = \langle -1, -1\rangle$, with orthogonal basis e_1, e_2 and $q(e_1) = q(e_2) = -1$. By theorem 22.7.5 the algebra $C(q)$ has basis $1, e_1, e_2, e_1e_2$. Writing $\omega := e_1e_2$ and using $e_2e_1 = -e_1e_2$ to move factors past one another,

$$e_1^2 = e_2^2 = -1, \qquad \omega^2 = e_1e_2e_1e_2 = -e_1e_1e_2e_2 = -(-1)(-1) = -1$$

$$e_1e_2 = \omega, \qquad e_2\omega = e_2e_1e_2 = -e_1e_2e_2 = e_1, \qquad \omega e_1 = e_1e_2e_1 = -e_1e_1e_2 = e_2$$

These are the relations $i^2 = j^2 = k^2 = -1$, $ij = k$, $jk = i$, $ki = j$ satisfied by the quaternion units, so $C(q)$ and the quaternions $\mathbb{H}$ are the same algebra. Note that $\mathbb{H}$ appears in theorem 20.2.3 as one of the three associative real division algebras, though that theorem gives no basis or multiplication rule, so both are recalled here. To see it as an isomorphism rather than a coincidence of tables, recall that $\mathbb{H}$ is the 4-dimensional $\mathbb{R}$-algebra with basis $1, i, j, k$ and $ij = k = -ji$. Those two relations already force the rest of the table: putting $k = ij$ and using associativity,

$$jk = j(ij) = (ji)j = -ij^2 = i, \qquad ki = (ij)i = i(ji) = -i^2j = j$$

and in the same way $ik = i(ij) = i^2j = -j$ and $kj = (ij)j = ij^2 = -i$, so the products computed in $C(q)$ above are exactly the quaternion products, and they agree with the relations displayed for Q_8 in section 1.2.4. Now let $f\colon V \to \mathbb{H}$ be the $\mathbb{R}$-linear map with $f(e_1) = i$ and $f(e_2) = j$. For $v = c_1e_1 + c_2e_2$,

$$f(v)^2 = c_1^2 i^2 + c_1c_2(ij + ji) + c_2^2 j^2 = -(c_1^2 + c_2^2) = q(v)$$

so proposition 22.7.3(1) gives an algebra homomorphism $F\colon C(q) \to \mathbb{H}$ with $F(e_1) = i$ and $F(e_2) = j$. Its image is a subalgebra containing i, j and hence $ij = k$, so it contains the spanning set $1, i, j, k$ and F is surjective. Both algebras have dimension 4 over $\mathbb{R}$, so

$$C(\langle -1, -1\rangle) \cong \mathbb{H}$$

Compare example 22.7.4, where $C(\langle -1\rangle) \cong \mathbb{C}$: the first two entries of a negative definite real form produce the complex numbers and then the quaternions.

The $\mathbb{Z}$-grading of $\mathcal{T}(V)$ was lost when the inhomogeneous relations were imposed, but the parity of a monomial was unchanged by the reduction in Step 2 of theorem 22.7.5: cancelling a repeated index removes two factors at a time. The following proposition makes that observation into a decomposition of the algebra, obtained not from the monomials but from the universal property, which is what makes it independent of the chosen orthogonal basis.

Proposition 22.7.7: The Parity Grading and the Even Subalgebra

Let k be a field with $\operatorname{char} k \neq 2$, let V be a k-vector space of dimension n, and let q be a quadratic form on V.

1. There is a unique k-algebra automorphism $\alpha\colon C(q) \to C(q)$ with $\alpha(v) = -v$ for all $v \in V$, and $\alpha^2 = \mathrm{id}$.
2. Setting $C^0(q) := \{x \in C(q) \mid \alpha(x) = x\}$ and $C^1(q) := \{x \in C(q) \mid \alpha(x) = -x\}$,
$$C(q) = C^0(q) \oplus C^1(q), \qquad C^\epsilon(q)\, C^\delta(q) \subseteq C^{\epsilon+\delta}(q)$$
the superscripts being read modulo 2. In particular $C^0(q)$ is a subalgebra, the *even part* of $C(q)$, and $V \subseteq C^1(q)$.
3. With the notation of theorem 22.7.5, $C^0(q)$ has basis the e_S with $\#S$ even and $C^1(q)$ has basis the e_S with $\#S$ odd. For $n \geq 1$ both have dimension 2^{n-1}.

***Proof*:**

1. The map $f\colon V \to C(q)$ with $f(v) = -v$ is k-linear and satisfies $f(v)^2 = (-v)^2 = v^2 = q(v)$, so proposition 22.7.3(1) with $A = C(q)$ gives a unique algebra homomorphism α with $\alpha(v) = -v$ on V. Then $\alpha \circ \alpha$ is an algebra homomorphism $C(q) \to C(q)$ restricting on V to $v \mapsto -(-v) = v$, and $\mathrm{id}_{C(q)}$ restricts to that same map. The uniqueness clause of the same proposition, applied to $f = \iota$, forces $\alpha^2 = \mathrm{id}$. Hence α is bijective, so it is an automorphism.
2. For $x \in C(q)$ write
$$x = \tfrac{1}{2}\big(x + \alpha(x)\big) + \tfrac{1}{2}\big(x - \alpha(x)\big)$$
which is legitimate since $\operatorname{char} k \neq 2$. Applying α and using $\alpha^2 = \mathrm{id}$ shows the first summand is fixed by α and the second is negated, so $C(q) = C^0(q) + C^1(q)$. If x lies in both then $x = -x$, so $2x = 0$ and $x = 0$. The sum is direct. For the product rule, let $x \in C^\epsilon(q)$ and $y \in C^\delta(q)$. Then $\alpha(xy) = \alpha(x)\alpha(y) = (-1)^\epsilon(-1)^\delta xy = (-1)^{\epsilon+\delta}xy$, which is the stated containment. Taking $\epsilon = \delta = 0$ shows $C^0(q)$ is closed under multiplication, and $\alpha(1) = 1$ puts 1 in it, so it is a subalgebra. Finally $\alpha(v) = -v$ for $v \in V$ is the statement $V \subseteq C^1(q)$.

3. Since α is an algebra homomorphism, $\alpha(e_S) = \alpha(e_{i_1}) \cdots \alpha(e_{i_r}) = (-1)^r e_S$ for S of size r. Writing an arbitrary $x = \sum_S \lambda_S e_S$ in the basis of theorem 22.7.5,

$$\alpha(x) = \sum_S (-1)^{\#S} \lambda_S e_S$$

and comparing coefficients, $\alpha(x) = x$ holds if and only if $\lambda_S = 0$ for every S of odd size, while $\alpha(x) = -x$ holds if and only if $\lambda_S = 0$ for every S of even size. So the even monomials are a basis of $C^0(q)$ and the odd ones a basis of $C^1(q)$. Their numbers are equal for $n \geq 1$, since

$$\sum_{r \text{ even}} \binom{n}{r} - \sum_{r \text{ odd}} \binom{n}{r} = (1-1)^n = 0$$

by the binomial theorem, and they sum to 2^n, so each is 2^{n-1}, as we sought to show.

The decomposition is a $\mathbb{Z}/2\mathbb{Z}$-grading, and $C^0(q)$ is a subalgebra of half the dimension of $C(q)$. In example 22.7.6 it has basis $1, e_1e_2$ with $(e_1e_2)^2 = -1$, so evaluation at e_1e_2 is a surjective algebra homomorphism $\mathbb{R}[x] \to C^0$ killing $x^2 + 1$, and comparing dimensions gives

$$C^0(\langle -1, -1 \rangle) \cong \mathbb{R}[x]/(x^2+1) \cong \mathbb{C}$$

sitting inside $\mathbb{H}$ as the subalgebra spanned by 1 and one imaginary unit. The general rank-two computation, which recovers $\operatorname{disc}(q)$ of definition 22.1.7 from the even part, is the third exercise below.

Exercise 22.7.1

1. Let $\operatorname{char} k \neq 2$, let q be a quadratic form on V and regard $V \subseteq C(q)$ as in theorem 22.7.5. Show that a vector $v \in V$ is invertible in $C(q)$ if and only if $q(v) \neq 0$, and that its inverse is then $q(v)^{-1}v$. Deduce that if $q(v) = 0$ for some nonzero $v \in V$, then $C(q)$ is not a division algebra.

2. Let $\operatorname{char} k \neq 2$ and let $q = \langle 1, -1 \rangle$ on k^2. Show that $C(q) \cong M_2(k)$. (Send e_1 and e_2 to two explicit matrices, check the hypothesis of proposition 22.7.3(1) for every vector rather than only for e_1 and e_2, and finish with theorem 22.7.5.)

3. Let $\operatorname{char} k \neq 2$ and let $q \cong \langle a, b \rangle$ be nondegenerate of rank 2. Show that $C^0(q) \cong k[x]/(x^2 + ab)$, and that this algebra is a field exactly when $-ab$ is not a square in k. Using corollary 22.2.3, express the condition in terms of $\operatorname{disc}(q)$, and check the answer against example 22.7.6.

4. Let $\operatorname{char} k \neq 2$ and let $q \cong \langle a_1, \ldots, a_n \rangle$ be nondegenerate, so every $a_i \neq 0$, and set $\omega := e_1e_2 \cdots e_n$ for an orthogonal basis as in theorem 22.7.5. Show that $\omega e_i = (-1)^{n-1} e_i \omega$ for every i, and deduce that ω lies in the centre of $C(q)$ if and only if n is odd.

5. Let $\operatorname{char} k \neq 2$ and let $C(g)$ be as in proposition 22.7.3(2). Show that $C(\mathrm{id}_V) = \mathrm{id}_{C(q)}$ and that $C(h \circ g) = C(h) \circ C(g)$ for isometries g and h. Deduce that $g \mapsto C(g)$ is a group homomorphism from the orthogonal group $O(\varphi)$ of definition 22.1.10 to the group of k-algebra automorphisms of $C(q)$, and that every automorphism in its image commutes with the α of proposition 22.7.7 and therefore preserves $C^0(q)$ and $C^1(q)$.

6. Let $\operatorname{char} k \neq 2$ and let $C(q)^{\mathrm{op}}$ denote the k-algebra with the same underlying vector space as $C(q)$ and multiplication $x * y := yx$. Show that there is a unique k-algebra homomorphism

$t\colon C(q) \to C(q)^{\mathrm{op}}$ with $t(v) = v$ for all $v \in V$, that the resulting map on $C(q)$ satisfies $t(xy) = t(y)t(x)$ and $t^2 = \mathrm{id}$, and that $t(e_{i_1}e_{i_2}\cdots e_{i_r}) = (-1)^{r(r-1)/2}e_{i_1}e_{i_2}\cdots e_{i_r}$ for $i_1 < \cdots < i_r$.

7. Let $\operatorname{char} k \neq 2$ and let q be a quadratic form on V. Show that $C^0(q)$ is spanned as a k-vector space by the products $v_1v_2\cdots v_{2m}$ of an even number of vectors of V, and hence that it is generated as a k-algebra by the products vw with $v, w \in V$.